Supplementary Material for "Phases of MUON: When MUON Eclipses SIGNSGD"

Complete statements and proofs

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How to read this supplement. This document contains the complete statements and proofs behind every result of the paper. Its sections, statements and equations carry the prefix S (Section SE, Theorem SE.1, equation (SE.12)); unprefixed numbers (Section 3, Figure 2, equation (5), Theorem B.1) refer to the paper. The appendix of the paper restates each result that the main text uses, explains it, and sketches its proof; each restated result points here for its full proof.

Machine-checked proofs. The statements and proofs of Sections SA–SJ have been formalized in Lean 4 with Mathlib. Every formal target is either proved, or proved assuming explicitly stated results cited from the literature ("certificates"); no axioms beyond Lean's standard three are used. A table mapping each statement to its Lean declaration and status closes this supplement.

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SA Setup: model, momentum, regularized SIGNSVD, normalization

This section fixes the objects that every later section of the appendix uses and proves the exact finite-dimensional identities they start from. It restates the regression model of section 1.1 and the algebra of its risk (appendix SA.2); it defines the algorithm that the appendix analyzes, an exponential moving average of the minibatch gradients with momentum $\beta \in [0, 1)$ followed by the regularized SIGNSVD map at a fixed normalized ridge $\varrho > 0$ (appendix SA.3); it proves the exact one-step recursion of the projected risks under this algorithm (appendix SA.4); it introduces the frozen-state model, in which the buffer is a weighted sum of independent batches, and proves the exact laws of that buffer together with the filter estimates that later sections use (appendix SA.5); and it states the regime assumption and the definition of a deterministic equivalent under which every asymptotic statement of the appendix is made (appendix SA.6). The main statements are propositions SA.1 and SA.4 and corollary SA.1 (exact identities for the algorithm), lemmas SA.2, SA.5 and SA.6 (the frozen buffer and the filter), and Assumption S3 and definition SA.4 (the regime). Nothing in this section is asymptotic except lemma SA.7(iii) and definition SA.4.

SA.1 Conventions

$\|X\|$ is the operator norm, $\|X\|_{\mathrm{F}}$ the Frobenius norm and $\|X\|_*$ the nuclear norm of a matrix X; $\|x\|$ is the Euclidean norm of a vector. Square roots of positive semidefinite matrices are the positive semidefinite roots, and $(\mathrm{Id}+K)^{-1/2}$ for positive definite $\mathrm{Id}+K$ is the positive definite inverse root. For a random scalar ξ and $p \geq 1$, $\|\xi\|_p \overset{\text{def}}{=} (\mathbb{E}\,|\xi|^p)^{1/p}$. For a sub-$\sigma$-field $\mathcal{G}$ we write $\mathbb{E}_{\mathcal{G}} \overset{\text{def}}{=} \mathbb{E}[\,\cdot \mid \mathcal{G}]$ and $\|\xi\|_{2,\mathcal{G}} \overset{\text{def}}{=} (\mathbb{E}_{\mathcal{G}}\,\xi^2)^{1/2}$, and $\mathbb{E}_t \overset{\text{def}}{=} \mathbb{E}[\,\cdot \mid \mathscr{F}_t]$ for the past $\mathscr{F}_t$ of definition SA.1. Expectations of matrices are entrywise. A constant written C may change from line to line; its dependence is stated where it is introduced, and it never depends on the dimension unless the statement says so.

SA.2 Model and risk

Definition SA.1 (Regression model)**.** *Let $N_{\mathrm{out}}, N_{\mathrm{in}}, B$ be positive integers. The features $x_{\mathrm{in}} \in \mathbb{R}^{N_{\mathrm{in}}}$ and $x_{\mathrm{out}} \in \mathbb{R}^{N_{\mathrm{out}}}$ are independent random vectors with finite second moments, centered in every coordinate, with covariances*

$$\Sigma_{\mathrm{in}} \overset{\text{def}}{=} \mathbb{E}[x_{\mathrm{in}} \otimes x_{\mathrm{in}}], \qquad \Sigma_{\mathrm{out}} \overset{\text{def}}{=} \mathbb{E}[x_{\mathrm{out}} \otimes x_{\mathrm{out}}],$$

and with eigendecompositions $\Sigma_{\mathrm{out}} = \sum_{i=1}^{N_{\mathrm{out}}} \mu_i\, u_i \otimes u_i$ and $\Sigma_{\mathrm{in}} = \sum_{j=1}^{N_{\mathrm{in}}} \lambda_j\, v_j \otimes v_j$ in orthonormal eigenbases, all eigenvalues included. For a parameter $W \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ and a fixed teacher $W^\star \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$, the single-sample loss and the population risk are

$$\mathscr{L}(W; (x_{\mathrm{in}}, x_{\mathrm{out}})) \overset{\text{def}}{=} \tfrac{1}{2}\big\langle x_{\mathrm{out}} \otimes x_{\mathrm{in}},\, W - W^\star \big\rangle^2, \qquad \mathscr{R}(W) \overset{\text{def}}{=} \mathbb{E}\left[\mathscr{L}(W; (x_{\mathrm{in}}, x_{\mathrm{out}}))\right]. \tag{SA.1}$$

For $\Delta \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ we write

$$\begin{gathered} \langle V, W \rangle_\Sigma \overset{\text{def}}{=} \mathrm{Tr}\left(\Sigma_{\mathrm{out}} V \Sigma_{\mathrm{in}} W^\top\right), \qquad \|V\|_\Sigma^2 \overset{\text{def}}{=} \langle V, V \rangle_\Sigma, \\ \mathscr{Q}_{ij}(\Delta) \overset{\text{def}}{=} (u_i^\top \Delta v_j)^2, \qquad \mathscr{Q}_i(\Delta) \overset{\text{def}}{=} \|\Delta^\top u_i\|^2. \end{gathered} \tag{SA.2}$$

The past at step t is denoted $\mathscr{F}_t$: it contains the parameter W_t and everything used to compute it, but not the minibatch of step t, which is a family $(x_{\mathrm{in}}^{(a)}, x_{\mathrm{out}}^{(a)})_{a=1}^B$ of independent copies of $(x_{\mathrm{in}}, x_{\mathrm{out}})$ independent of the past. We write

$$\begin{gathered} \Delta_t \overset{\text{def}}{=} W_t - W^\star, \qquad \mathscr{R}(t) \overset{\text{def}}{=} \mathscr{R}(W_t), \qquad d_a^\circ \overset{\text{def}}{=} (x_{\mathrm{out}}^{(a)})^\top \Delta_t x_{\mathrm{in}}^{(a)}, \\ G_{t+1}^\circ \overset{\text{def}}{=} \frac{1}{B} \sum_{a=1}^{B} d_a^\circ\, x_{\mathrm{out}}^{(a)} \otimes x_{\mathrm{in}}^{(a)}, \end{gathered} \tag{SA.3}$$

so that G_{t+1}° is the minibatch gradient eq. (2) at Δ_t; as in the main text we write $\mathscr{Q}_{ij}(t) \overset{\text{def}}{=} \mathscr{Q}_{ij}(\Delta_t)$ and $\mathscr{Q}_i(t) \overset{\text{def}}{=} \mathscr{Q}_i(\Delta_t)$. The data matrices and the residual diagonal of the batch are $X \overset{\text{def}}{=} [x_{\mathrm{in}}^{(1)} \cdots x_{\mathrm{in}}^{(B)}] \in \mathbb{R}^{N_{\mathrm{in}} \times B}$, $Y \overset{\text{def}}{=} [x_{\mathrm{out}}^{(1)} \cdots x_{\mathrm{out}}^{(B)}] \in \mathbb{R}^{N_{\mathrm{out}} \times B}$ and $D^\circ \overset{\text{def}}{=} \mathrm{diag}(d_1^\circ, \ldots, d_B^\circ)$. The superscript $^\circ$ marks physical objects, as opposed to the normalized objects of eq. (SA.10).

In eq. (SA.3), $\mathscr{R}(t)$ is the function $\mathscr{R}$ evaluated at the random matrix W_t; by proposition SA.1(iii) it is the conditional expected loss of a fresh sample given $\mathscr{F}_t$. The main text takes the features Gaussian, and all random-matrix statements of the appendix do so; the identities of this subsection need only second moments.

Assumption S1 (Data-target normalization)**.** *The input and output features are independent and Gaussian, $x_{\rm in} \sim \mathcal{N}(0, \Sigma_{\rm in})$ and $x_{\rm out} \sim \mathcal{N}(0, \Sigma_{\rm out})$, with positive semidefinite covariances $\Sigma_{\rm in} \in \mathbb{R}^{N_{\rm in}\times N_{\rm in}}$ and $\Sigma_{\rm out} \in \mathbb{R}^{N_{\rm out}\times N_{\rm out}}$. They are represented in whitened form as $x_{\rm in} = {\Sigma_{\rm in}}^{1/2} w$ and $x_{\rm out} = {\Sigma_{\rm out}}^{1/2} z$ with $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\rm in}})$ and $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\rm out}})$ independent; the whitened columns of a batch are written $(w_a, z_a)_{a\le B}$.*

Proposition SA.1 (Residual and gradient moments)**.** *In definition SA.1, $\Sigma_{\rm in}$ and $\Sigma_{\rm out}$ are symmetric and positive semidefinite, and the following hold.*

(i) For deterministic $\Delta \in \mathbb{R}^{N_{\rm out}\times N_{\rm in}}$ the residual $d \overset{\rm def}{=} x_{\rm out}^\top \Delta x_{\rm in}$ is centered, and

$$\mathbb{E}\, d^2 = \mathrm{Var}(d) = \mathrm{Tr}\left(\Sigma_{\rm out}\Delta\Sigma_{\rm in}\Delta^\top\right) = \|\Delta\|_\Sigma^2, \qquad \mathscr{R}(W) = \tfrac{1}{2}\|W - W^\star\|_\Sigma^2. \quad \text{(SA.4)}$$

(ii) For deterministic Δ and a batch of B independent copies, the gradient $G^\circ \overset{\rm def}{=} \frac{1}{B}\sum_a d_a^\circ\, x_{\rm out}^{(a)} \otimes x_{\rm in}^{(a)}$ satisfies

$$G^\circ = \frac{1}{B} Y D^\circ X^\top, \qquad \mathbb{E}\, G^\circ = \Sigma_{\rm out}\Delta\Sigma_{\rm in}. \quad \text{(SA.5)}$$

(iii) For the state Δ_t at step t with $\mathbb{E}\,\|\Delta_t\|_{\rm F}^2 < \infty$ and the fresh batch of step t, the residuals and the gradient of eq. (SA.3) satisfy

$$\mathbb{E}_t\; d_a^\circ = 0, \qquad \mathbb{E}_t\,(d_a^\circ)^2 = 2\mathscr{R}(t), \qquad \mathbb{E}_t\; G_{t+1}^\circ = \Sigma_{\rm out}\Delta_t\Sigma_{\rm in}. \quad \text{(SA.6)}$$

Proof. The covariances are symmetric, and $v^\top\Sigma_{\rm in} v = \mathbb{E}[(v^\top x_{\rm in})^2] \ge 0$ for every $v \in \mathbb{R}^{N_{\rm in}}$; the same holds for $\Sigma_{\rm out}$.

Mixed fourth-order moments factorize by independence: for coordinates $i, k \le N_{\rm out}$ and $j, l \le N_{\rm in}$,

$$\mathbb{E}\left[(x_{\rm out})_i (x_{\rm out})_k (x_{\rm in})_j (x_{\rm in})_l\right] = (\Sigma_{\rm out})_{ik}(\Sigma_{\rm in})_{jl}. \quad \text{(SA.7)}$$

The second moment of the residual is the trace. Expanding

$$d^2 = \sum_{i,k\le N_{\rm out}} \sum_{j,l\le N_{\rm in}} \Delta_{ij}\Delta_{kl}\,(x_{\rm out})_i (x_{\rm out})_k (x_{\rm in})_j (x_{\rm in})_l$$

and taking expectations term by term with eq. (SA.7) gives

$$\mathbb{E}\, d^2 = \sum_{i,j,k,l} \Delta_{ij}\Delta_{kl}(\Sigma_{\rm out})_{ik}(\Sigma_{\rm in})_{jl} = \mathrm{Tr}\left(\Sigma_{\rm out}\Delta\Sigma_{\rm in}\Delta^\top\right),$$

the last equality by expanding the trace as $\sum_{i,k,j,l}(\Sigma_{\rm out})_{ik}\Delta_{kl}(\Sigma_{\rm in})_{lj}\Delta_{ij}$ and using the symmetry of $\Sigma_{\rm in}$. The residual is centered because $\mathbb{E}[(x_{\rm out})_i (x_{\rm in})_j] = \mathbb{E}(x_{\rm out})_i\; \mathbb{E}(x_{\rm in})_j = 0$ by independence, and hence $\mathrm{Var}(d) = \mathbb{E}\, d^2$. The formula for $\mathscr{R}$ is eq. (SA.1) with $\Delta = W - W^\star$. This proves (i).

The matrix form of the gradient is finite algebra. Matrix multiplication gives $(Y D^\circ X^\top)_{ij} = \sum_a d_a^\circ (x_{\rm out}^{(a)})_i (x_{\rm in}^{(a)})_j$, which is B times the (i,j) entry of G°. For one sample, the entry $d\,(x_{\rm out})_i (x_{\rm in})_j = \sum_{k,l}\Delta_{kl}(x_{\rm out})_k (x_{\rm out})_i (x_{\rm in})_l (x_{\rm in})_j$ has expectation $\sum_{k,l}(\Sigma_{\rm out})_{ik}\Delta_{kl}(\Sigma_{\rm in})_{lj} = (\Sigma_{\rm out}\Delta\Sigma_{\rm in})_{ij}$ by eq. (SA.7); identical distribution of the samples gives $\mathbb{E}\, G^\circ$. This proves (ii).

Part (iii) follows from (i) and (ii) because the batch of step t is independent of the past: conditionally on $\mathscr{F}_t$ the state Δ_t is fixed and the batch has its unconditional law, so the fixed-state identities of (i) and (ii) at $\Delta = \Delta_t$ are the three conditional expectations. □

Lemma SA.1 (Mode and row decomposition)**.** *For every $\Delta \in \mathbb{R}^{N_{\rm out}\times N_{\rm in}}$,*

$$\mathscr{Q}_i(\Delta) = \sum_{j=1}^{N_{\rm in}} \mathscr{Q}_{ij}(\Delta), \qquad \tfrac{1}{2}\|\Delta\|_\Sigma^2 = \tfrac{1}{2}\sum_{i,j}\mu_i\lambda_j\,\mathscr{Q}_{ij}(\Delta), \quad \text{(SA.8)}$$

and if $\Sigma_{\rm in} = \mathrm{Id}_{N_{\rm in}}$ then $\frac{1}{2}\|\Delta\|_\Sigma^2 = \frac{1}{2}\sum_i \mu_i \mathscr{Q}_i(\Delta)$. Moreover $\|\Delta\|_\Sigma = 0$ if and only if $u_i^\top \Delta v_j = 0$ for every pair (i,j) with $\mu_i\lambda_j > 0$.

Proof. The row identity is Parseval's identity in the basis (v_j): $\Delta^\top u_i = \sum_j (u_i^\top \Delta v_j)\, v_j$, and the squared norm of the right side is $\sum_j (u_i^\top \Delta v_j)^2$.

The risk identity is the eigenexpansion of the trace:

$$\operatorname{Tr}\left(\Sigma_{\text{out}} \Delta \Sigma_{\text{in}} \Delta^\top\right) = \sum_{i,j} \mu_i \lambda_j \operatorname{Tr}\left((u_i \otimes u_i) \Delta (v_j \otimes v_j) \Delta^\top\right) = \sum_{i,j} \mu_i \lambda_j (u_i^\top \Delta v_j)^2.$$

When $\Sigma_{\text{in}} = \mathrm{Id}$ all λ_j equal 1 and the row identity applies. The last claim holds because a finite sum of nonnegative terms vanishes exactly when each term does. □

Whitening and the normalized state. Under Assumption S1 the residual and the gradient are functions of the rotated error $A_t \overset{\text{def}}{=} {\Sigma_{\text{out}}}^{1/2} \Delta_t {\Sigma_{\text{in}}}^{1/2}$:

$$d_a^\circ = z_a^\top A_t w_a, \qquad G_{t+1}^\circ = \frac{1}{B} \sum_{a=1}^{B} d_a^\circ \, {\Sigma_{\text{out}}}^{1/2} (z_a \otimes w_a) {\Sigma_{\text{in}}}^{1/2}, \qquad \mathscr{R}(t) = \tfrac{1}{2} \|A_t\|_{\text{F}}^2, \quad \text{(SA.9)}$$

the last by eq. (SA.4) and $\operatorname{Tr}(\Sigma_{\text{out}} \Delta_t \Sigma_{\text{in}} \Delta_t^\top) = \operatorname{Tr}(A_t A_t^\top)$. When $\mathscr{R}(t) > 0$ we define the normalized state, residuals and gradient

$$\begin{aligned} \widehat{A}_t &\overset{\text{def}}{=} \frac{A_t}{\sqrt{2\mathscr{R}(t)}}, \qquad d_a \overset{\text{def}}{=} \frac{d_a^\circ}{\sqrt{2\mathscr{R}(t)}} = z_a^\top \widehat{A}_t w_a, \\ G_{t+1} &\overset{\text{def}}{=} \frac{G_{t+1}^\circ}{\sqrt{2\mathscr{R}(t)}} = \frac{1}{B} \sum_{a=1}^{B} d_a \, {\Sigma_{\text{out}}}^{1/2} (z_a \otimes w_a) {\Sigma_{\text{in}}}^{1/2}. \end{aligned} \quad \text{(SA.10)}$$

Proposition SA.2 (Normalization)**.** *Assume Assumption S1.*

(i) If $\Delta_t = \Delta$ is deterministic with $\mathscr{R}(t) > 0$, then $\|\widehat{A}_t\|_{\text{F}} = 1$, the normalized residuals are centered with unit variance, and

$$\mathbb{E}\, d_a = 0, \qquad \mathbb{E}\, d_a^2 = 1, \qquad \mathbb{E}\, G_{t+1} = {\Sigma_{\text{out}}}^{1/2} \widehat{A}_t {\Sigma_{\text{in}}}^{1/2} = \frac{\Sigma_{\text{out}} \Delta \Sigma_{\text{in}}}{\sqrt{2\mathscr{R}(t)}}. \quad \text{(SA.11)}$$

For every $c > 0$, replacing Δ by $c\Delta$ with the same batch multiplies $\mathscr{R}(t)$ by c^2 and leaves $\widehat{A}_t$, d_a and G_{t+1} unchanged.

(ii) For the state Δ_t at step t and the fresh batch of step t, on the event $\{\mathscr{R}(t) > 0\}$ the identities eq. (SA.11) hold with $\mathbb{E}_t$ in place of $\mathbb{E}$ and Δ_t in place of Δ.

Proof. Part (i) is proposition SA.1(i),(ii) applied to $\Delta / \|\Delta\|_\Sigma$. Indeed $\|\Delta\|_\Sigma = \sqrt{2\mathscr{R}(t)}$ by eq. (SA.4), so d_a is the residual and G_{t+1} the gradient of the state $\Delta / \|\Delta\|_\Sigma$, whose Σ-norm is 1; the first identity of eq. (SA.11) is the centering, the second is eq. (SA.4), and the third is eq. (SA.5) together with $\Sigma_{\text{out}} \Delta \Sigma_{\text{in}} = {\Sigma_{\text{out}}}^{1/2} A_t {\Sigma_{\text{in}}}^{1/2}$. The identity $\|\widehat{A}_t\|_{\text{F}} = 1$ is eq. (SA.9). Scaling Δ by c scales A_t and every d_a° by c and $\mathscr{R}(t)$ by c^2, so the three quotients are unchanged.

Part (ii) follows from (i) because the batch of step t is independent of the past. □

Every random-matrix statement of the appendix concerns the normalized objects eq. (SA.10). The risk re-enters only through the inverse relations $G_{t+1}^\circ = \sqrt{2\mathscr{R}(t)}\, G_{t+1}$ and $G_{t+1}^\circ G_{t+1}^{\circ\top} = 2\mathscr{R}(t)\, G_{t+1} G_{t+1}^\top$, and through the ridge conversion of proposition SA.3.

SA.3 The algorithm

The regularized spectral map. For $K \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ with singular value decomposition $K = \sum_k s_k\, p_k \otimes r_k$ ($s_k > 0$, orthonormal families (p_k), (r_k)) and $\varrho \geq 0$, the regularized SIGNSVD map is

$$Q_\varrho(K) \overset{\text{def}}{=} \sum_k \frac{s_k}{\sqrt{s_k^2 + \varrho}}\, p_k \otimes r_k, \qquad \text{so that} \qquad Q_\varrho(K) = \left(K K^\top + \varrho\, \mathrm{Id}_{N_{\text{out}}}\right)^{-1/2} K \quad (\varrho > 0). \quad \text{(SA.12)}$$

The sum does not depend on the choice of the decomposition, since $Q_\varrho(K) = K\,\phi_\varrho(K^\top K)$ with $\phi_\varrho(x) = (x+\varrho)^{-1/2}$ on $(0,\infty)$ and $\phi_\varrho(0) \stackrel{\text{def}}{=} 0$ applied through the spectral theorem to $K^\top K$. At $\varrho = 0$ the map is the polar factor $Q_0(K) = \sum_k p_k \otimes r_k$, which is the exact SIGNSVD update eq. (4). In the language of eq. (3), Q_ϱ is the spectral update with $\varphi_\varrho(x) = (x+\varrho)^{-1/2}$. A Newton–Schulz polynomial approximation of $x^{-1/2}$ recovers MUON (section 1.2); no statement of this appendix concerns that approximation.

Lemma SA.2 (Filter bounds and Lipschitz estimate)**.** *Let $K, L \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$.*

(i) For $\varrho \geq 0$: $\|Q_\varrho(K)\| \leq 1$ and $\|Q_\varrho(K)\|_{\text{F}}^2 \leq \operatorname{rank} K \leq \min(N_{\text{out}}, N_{\text{in}})$, with equality $\|Q_0(K)\|_{\text{F}}^2 = \operatorname{rank} K$.

(ii) For $\varrho \geq 0$, if $P \in \mathbb{R}^{N_{\text{out}} \times N_{\text{out}}}$ and $P' \in \mathbb{R}^{N_{\text{in}} \times N_{\text{in}}}$ are orthogonal projections with $K = PKP'$, then $Q_\varrho(K) = PQ_\varrho(K)P'$.

(iii) For $\varrho > 0$,

$$\|Q_\varrho(K) - Q_\varrho(L)\|_{\text{F}} \leq \varrho^{-1/2}\|K - L\|_{\text{F}}. \tag{SA.13}$$

(iv) $Q_\varrho(K) \to Q_0(K)$ as $\varrho \downarrow 0$.

Proof. Part (i) reads off the singular values. By eq. (SA.12) the singular values of $Q_\varrho(K)$ are $s_k/\sqrt{s_k^2+\varrho} \in (0,1]$, one for each positive singular value of K, so the operator norm is at most 1 and the squared Frobenius norm, the sum of their squares, is at most the number of positive singular values, with equality at $\varrho = 0$.

Part (ii) holds because the singular vectors lie in the ranges of the projections. If $K = PKP'$ and $Kr_k = s_k p_k$ with $s_k > 0$, then $p_k = s_k^{-1}PKP'r_k = Pp_k$, and likewise $r_k = s_k^{-1}P'K^\top Pp_k = P'r_k$, so each term $p_k \otimes r_k$ of eq. (SA.12) equals $P(p_k \otimes r_k)P'$.

Part (iii) is a divided-difference bound for a Lipschitz spectral function. Let H, T be real symmetric matrices of the same size with orthonormal eigenbases $Hh_i = x_i h_i$, $Tt_j = y_j t_j$, and let $f : \mathbb{R} \to \mathbb{R}$ be L_f-Lipschitz. Then

$$h_i^\top \big(f(H) - f(T)\big)t_j = \big(f(x_i) - f(y_j)\big)\, h_i^\top t_j, \qquad h_i^\top (H - T)t_j = (x_i - y_j)\, h_i^\top t_j,$$

so $|h_i^\top (f(H) - f(T))t_j| \leq L_f |h_i^\top (H - T)t_j|$ for every pair, including pairs with $x_i = y_j$, where both sides vanish. Summing the squares over i, j, which computes the Frobenius norms in the two orthonormal bases, gives

$$\|f(H) - f(T)\|_{\text{F}} \leq L_f \|H - T\|_{\text{F}}.$$

We apply this to $f(x) \stackrel{\text{def}}{=} x/\sqrt{x^2+\varrho}$, whose derivative is

$$f'(x) = \frac{\varrho}{(x^2+\varrho)^{3/2}} \leq \varrho^{-1/2},$$

so f is $\varrho^{-1/2}$-Lipschitz by the mean value theorem, and to the symmetric dilations

$$\mathcal{D}(K) \stackrel{\text{def}}{=} \begin{pmatrix} 0 & K \\ K^\top & 0 \end{pmatrix}, \qquad f(\mathcal{D}(K)) = \begin{pmatrix} 0 & Q_\varrho(K) \\ Q_\varrho(K)^\top & 0 \end{pmatrix}.$$

The second display holds because, for each positive singular pair, the vectors $(p_k; \pm r_k)/\sqrt{2}$ are eigenvectors of $\mathcal{D}(K)$ with eigenvalues $\pm s_k$, the remaining eigenvalue is 0 with $f(0) = 0$, and f is odd, so that

$$\begin{aligned} f(\mathcal{D}(K)) &= \sum_k f(s_k)\, \frac{(p_k; r_k) \otimes (p_k; r_k) - (p_k; -r_k) \otimes (p_k; -r_k)}{2} \\ &= \sum_k \frac{s_k}{\sqrt{s_k^2+\varrho}} \begin{pmatrix} 0 & p_k \otimes r_k \\ r_k \otimes p_k & 0 \end{pmatrix}. \end{aligned}$$

Since $\|\mathcal{D}(E)\|_{\text{F}}^2 = 2\|E\|_{\text{F}}^2$ for every rectangular E, the divided-difference bound for $\mathcal{D}(K)$ and $\mathcal{D}(L)$ is eq. (SA.13) after cancelling the factor $\sqrt{2}$ on both sides.

Part (iv) holds because each coefficient $s_k/\sqrt{s_k^2+\varrho}$ in eq. (SA.12) increases to 1 as $\varrho \downarrow 0$. □

Lemma SA.3 (Integral representation). *For every $K \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ and $\varrho > 0$,*

$$Q_\varrho(K) = \frac{2}{\pi} \int_0^\infty \left(KK^\top + (\varrho + u^2)\,\mathrm{Id}_{N_{\text{out}}}\right)^{-1} K \, du, \tag{SA.14}$$

and $\int_0^\infty \|(KK^\top + (\varrho+u^2)\,\mathrm{Id})^{-1}K\|_* \, du \le \frac{\pi}{2}\min(N_{\text{out}}, N_{\text{in}})$.

Proof. The identity is a scalar integral applied to each singular value. In the singular value decomposition of K the integrand is

$$\left(KK^\top + (\varrho + u^2)\,\mathrm{Id}_{N_{\text{out}}}\right)^{-1} K = \sum_k \frac{s_k}{s_k^2 + \varrho + u^2}\, p_k \otimes r_k,$$

and for $b > 0$ the substitution $u = \sqrt{b}\, v$ gives

$$\int_0^\infty \frac{du}{b + u^2} = \frac{1}{\sqrt{b}} \int_0^\infty \frac{dv}{1 + v^2} = \frac{\pi}{2\sqrt{b}}.$$

Applying this with $b = s_k^2 + \varrho$ to each term and summing over k gives eq. (SA.14). The nuclear norm of the integrand is the sum of its coefficients, so

$$\int_0^\infty \|(KK^\top + (\varrho+u^2)\,\mathrm{Id})^{-1}K\|_* \, du = \frac{\pi}{2} \sum_{k:\, s_k > 0} \frac{s_k}{\sqrt{s_k^2 + \varrho}} \le \frac{\pi}{2}\min(N_{\text{out}}, N_{\text{in}}). \qquad \square$$

Proposition SA.3 (Scaling, physical ridge and heavy ball). *Let $K \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$, $\varrho \ge 0$ and $a > 0$.*

(i) $Q_\varrho(aK) = Q_{\varrho/a^2}(K)$.

(ii) If $\mathscr{R} > 0$ and $\varrho^\circ \overset{\text{def}}{=} 2\mathscr{R}\,\varrho$, then $Q_{\varrho^\circ}\left(\sqrt{2\mathscr{R}}\,K\right) = Q_\varrho(K)$.

(iii) Let $(G_{t+1})_{t\ge 0}$ be any sequence of matrices, $\beta \in [0,1)$, and let $M_t = \beta M_{t-1} + (1-\beta)G_{t+1}$ and $H_t = \beta H_{t-1} + G_{t+1}$ for $t \ge 0$ with $H_{-1} = M_{-1}/(1-\beta)$. Then $H_t = M_t/(1-\beta)$ for all $t \ge -1$, and $Q_\varrho(H_t) = Q_{(1-\beta)^2\varrho}(M_t)$.

Proof. The singular values of aK are as_k, with the same singular vectors as K, so the singular-value filter of $Q_\varrho(aK)$ is

$$\frac{as_k}{\sqrt{a^2 s_k^2 + \varrho}} = \frac{s_k}{\sqrt{s_k^2 + \varrho/a^2}},$$

which is the filter of $Q_{\varrho/a^2}(K)$; this proves (i). Part (ii) is (i) with $a = \sqrt{2\mathscr{R}}$ and ϱ° in place of ϱ, since $\varrho^\circ/a^2 = \varrho$.

Part (iii) is an induction. If $H_{t-1} = M_{t-1}/(1-\beta)$ then

$$H_t = \beta\frac{M_{t-1}}{1-\beta} + G_{t+1} = \frac{\beta M_{t-1} + (1-\beta)G_{t+1}}{1-\beta} = \frac{M_t}{1-\beta},$$

and the ridge conversion is (i) with $a = (1-\beta)^{-1}$. $\square$

Definition SA.2 (The algorithm). *Fix a momentum $\beta \in [0,1)$, a normalized ridge $\varrho > 0$, a deterministic step sequence $(\eta_t)_{t\ge 0}$ with $\eta_t \ge 0$, an initial parameter W_0 and an initial buffer $M_{-1}^\circ \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$, both given before the first batch is drawn (the default is $M_{-1}^\circ = 0$). Let $t_{\text{stop}} \overset{\text{def}}{=} \inf\{t \ge 0 : \mathscr{R}(t) = 0\} \in \{0, 1, \dots\} \cup \{\infty\}$. The algorithm is, for $t \ge 0$,*

$$\begin{aligned} M_t^\circ &= \beta\, M_{t-1}^\circ + (1-\beta)\, G_{t+1}^\circ, \qquad W_{t+1} = W_t - \eta_t\, \widetilde{G}_{t+1}^\circ, \\ \widetilde{G}_{t+1}^\circ &\overset{\text{def}}{=} \begin{cases} Q_{\varrho_t^\circ}(M_t^\circ), & t < t_{\text{stop}}, \\ 0, & t \ge t_{\text{stop}}, \end{cases} \qquad \varrho_t^\circ \overset{\text{def}}{=} 2\mathscr{R}(t)\,\varrho. \end{aligned} \tag{SA.15}$$

The stopping time matters only for $\beta > 0$. At $\beta = 0$ zero risk is absorbing, because the update at a zero-risk state vanishes by proposition SA.5. For $\beta > 0$ the buffer retains past gradients, so the

update at a zero-risk state need not vanish, and the algorithm is stopped at t_{stop}. The normalized objects eq. (SA.10) are defined at positive risk, so they exist at every time before t_{stop}.

Whether $t < t_{\mathrm{stop}}$ is decided by the past $\mathscr{F}_t$. At $\beta = 0$ the buffer is the current gradient, $M_t^\circ = G_{t+1}^\circ$, and by proposition SA.3(ii) the update is

$$\widetilde{G}_{t+1}^\circ = Q_{2\mathscr{R}(t)\varrho}(G_{t+1}^\circ) = Q_\varrho(G_{t+1}) \qquad (t < t_{\mathrm{stop}}), \tag{SA.16}$$

which is the spectral update eq. (3) with $\varphi(x) = (x + 2\mathscr{R}(t)\varrho)^{-1/2}$, that is, φ_ϱ in the normalized spectral variable $x/(2\mathscr{R}(t))$ up to the factor $(2\mathscr{R}(t))^{-1/2}$ absorbed by $G_{t+1} = G_{t+1}^\circ/\sqrt{2\mathscr{R}(t)}$. Unrolling the buffer recursion gives

$$M_t^\circ = \beta^{t+1} M_{-1}^\circ + \sum_{j=0}^{t} w_j \, G_{t+1-j}^\circ, \qquad w_j \overset{\mathrm{def}}{=} (1-\beta)\beta^j, \tag{SA.17}$$

where G_{t+1-j}° is the gradient computed at the earlier error Δ_{t-j}. The EMA weights satisfy, with the convention $\beta^0 = 1$ at $\beta = 0$ (where $w_0 = 1$ and all other weights vanish),

$$\sum_{j\geq 0} w_j = 1, \qquad s_p \overset{\mathrm{def}}{=} \sum_{j\geq 0} w_j^p = \frac{(1-\beta)^p}{1-\beta^p} \quad (p \geq 2), \qquad s_2 = \frac{1-\beta}{1+\beta}. \tag{SA.18}$$

Remark SA.1 (Fixed normalized versus fixed physical ridge)**.** *The algorithm applies the regularized map at the physical ridge $\varrho_t^\circ = 2\mathscr{R}(t)\varrho$, which decreases with the risk, so that in the normalized variables of eq. (SA.10) the ridge is the constant ϱ; at $\beta = 0$ this is eq. (SA.16). An implementation with a fixed physical ridge $\ell > 0$ applies $Q_\ell(M_t^\circ) = Q_{\ell/(2\mathscr{R}(t))}(M_t^\circ/\sqrt{2\mathscr{R}(t)})$, whose normalized ridge $\ell/(2\mathscr{R}(t))$ varies with the risk. The results of the appendix are stated for a normalized ridge in a fixed compact subset of $(0,\infty)$ (Assumption S3); they cover a fixed physical ridge exactly on the time ranges where $\ell/(2\mathscr{R}(t))$ stays in such a subset, and no statement is made as the normalized ridge tends to 0 or to ∞. The normalized convention is the one under which the frozen-state laws of appendix SA.5 do not depend on the risk, which is what makes the deterministic equivalents of the following sections risk-free.*

Remark SA.2 (Heavy-ball momentum)**.** *The momentum ablation appendix and the* MUON *implementation it describes keep the heavy-ball buffer $H_t = \beta H_{t-1} + G_{t+1}^\circ$ and orthogonalize H_t. By proposition SA.3(iii), with matched initialization, $H_t = M_t^\circ/(1-\beta)$ and $Q_{\varrho'}(H_t) = Q_{(1-\beta)^2\varrho'}(M_t^\circ)$. Every statement of the appendix is for the normalized average M_t°; the heavy-ball version at ridge ϱ' is the same statement at ridge $(1-\beta)^2\varrho'$, and at $\varrho = 0$ the two updates coincide.*

SA.4 The exact risk recursion

Proposition SA.4 (Conditional recursion of the projected risks)**.** *In definition SA.1, fix $t \geq 0$, let Δ_t be the state at step t with $\mathbb{E}\,\|\Delta_t\|_{\mathrm{F}}^2 < \infty$, let S be a random matrix in $\mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ with $\mathbb{E}\,\|\mathsf{S}\|_{\mathrm{F}}^2 < \infty$ (the update, determined by the past and the batch of step t), let $\eta_t \geq 0$ be deterministic, and put $\Delta_{t+1} \overset{\mathrm{def}}{=} \Delta_t - \eta_t\mathsf{S}$. Define*

$$\mathscr{D}_{ij}(t) \overset{\mathrm{def}}{=} (u_i^\top \Delta_t v_j)\; \mathbb{E}_t\left[u_i^\top \mathsf{S} v_j\right], \qquad \mathcal{V}_{ij}(t) \overset{\mathrm{def}}{=} \mathbb{E}_t\left[(u_i^\top \mathsf{S} v_j)^2\right]. \tag{SA.19}$$

Then

$$\mathbb{E}_t\left[\mathscr{Q}_{ij}(\Delta_{t+1})\right] = \mathscr{Q}_{ij}(\Delta_t) - 2\eta_t\,\mathscr{D}_{ij}(t) + \eta_t^2\,\mathcal{V}_{ij}(t), \tag{SA.20}$$

$$\mathbb{E}_t\left[\mathscr{Q}_{i}(\Delta_{t+1})\right] = \mathscr{Q}_{i}(\Delta_t) - 2\eta_t\sum_j \mathscr{D}_{ij}(t) + \eta_t^2\sum_j \mathcal{V}_{ij}(t), \tag{SA.21}$$

$$\mathbb{E}_t\left[\tfrac{1}{2}\|\Delta_{t+1}\|_\Sigma^2\right] = \tfrac{1}{2}\|\Delta_t\|_\Sigma^2 - \eta_t\sum_{i,j}\mu_i\lambda_j\mathscr{D}_{ij}(t) + \frac{\eta_t^2}{2}\sum_{i,j}\mu_i\lambda_j\mathcal{V}_{ij}(t), \tag{SA.22}$$

with the contractions in the last line equal to

$$\sum_{i,j}\mu_i\lambda_j\mathscr{D}_{ij}(t) = \left\langle \Delta_t, \mathbb{E}_t\,\mathsf{S}\right\rangle_\Sigma, \qquad \sum_{i,j}\mu_i\lambda_j\mathcal{V}_{ij}(t) = \mathbb{E}_t\,\|\mathsf{S}\|_\Sigma^2. \tag{SA.23}$$

The quantity $\mathcal{V}_{ij}(t)$ is a conditional second moment, not a centered conditional variance; the distinction matters for momentum and is the subject of the volatility section.

Proof. Write $a_{ij} \overset{\text{def}}{=} u_i^\top \Delta_t v_j$ and $b_{ij} \overset{\text{def}}{=} u_i^\top \mathsf{S} v_j$; the coefficient a_{ij} is determined by the past. The mode identity is the expansion of a square. Pointwise,

$$\mathscr{Q}_{ij}(\Delta_{t+1}) = (a_{ij} - \eta_t b_{ij})^2 = a_{ij}^2 - 2\eta_t a_{ij} b_{ij} + \eta_t^2 b_{ij}^2,$$

and taking $\mathbb{E}_t$ term by term gives eq. (SA.20).

The row and total identities are finite weighted sums of the mode identity. Summing eq. (SA.20) over j with weight 1, and over (i, j) with weight $\mu_i \lambda_j / 2$, and applying lemma SA.1 to Δ_{t+1} and to Δ_t, gives eqs. (SA.21) and (SA.22).

The contraction identities are the eigenexpansion of the pairing. For any $V, W \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$,

$$\langle V, W \rangle_\Sigma = \sum_{i,j} \mu_i \lambda_j \, (u_i^\top V v_j)(u_i^\top W v_j),$$

by the computation in the proof of lemma SA.1 with V and W in the two slots. Taking $(V, W) = (\Delta_t, \mathsf{S})$ and (S, S) and applying $\mathbb{E}_t$ gives eq. (SA.23). □

Corollary SA.1 (Recursion of the algorithm)**.** *In definition SA.2, assume $\mathbb{E}\,\|\Delta_0\|_{\text{F}}^2 < \infty$. Then for every $t \geq 0$ the identities eqs. (SA.20) to (SA.22) hold with $\mathsf{S} = \widetilde{G}_{t+1}^\circ$, with $0 \leq \mathcal{V}_{ij}(t) \leq 1$. In particular*

$$\mathbb{E}_t\left[\mathscr{R}(t+1)\right] - \mathscr{R}(t) = -\eta_t \big\langle \Delta_t, \mathbb{E}_t\, \widetilde{G}_{t+1}^\circ \big\rangle_\Sigma + \frac{\eta_t^2}{2}\, \mathbb{E}_t\, \|\widetilde{G}_{t+1}^\circ\|_\Sigma^2. \tag{SA.24}$$

On $\{t < t_{\text{stop}}\}$ the conditional expectations in eqs. (SA.19) and (SA.24) are obtained by integrating the fresh batch of step t with $(\Delta_t, M_{t-1}^\circ)$ held fixed, because the batch is independent of the past; on $\{t \geq t_{\text{stop}}\}$ both sides of eq. (SA.24) vanish.

Proof. The update satisfies $\|\widetilde{G}_{t+1}^\circ\| \leq 1$ and $\|\widetilde{G}_{t+1}^\circ\|_{\text{F}}^2 \leq \min(N_{\text{out}}, N_{\text{in}})$ by lemma SA.2(i), so $\mathbb{E}\,\|\Delta_t\|_{\text{F}}^2 < \infty$ for every t and proposition SA.4 applies at every t. The identity eq. (SA.24) is eq. (SA.22) rewritten with eq. (SA.23) and eq. (SA.4). The bound $\mathcal{V}_{ij}(t) \leq 1$ follows from $|u_i^\top \widetilde{G}_{t+1}^\circ v_j| \leq \|\widetilde{G}_{t+1}^\circ\| \leq 1$. On $\{t < t_{\text{stop}}\}$ the update is a function of $(\Delta_t, M_{t-1}^\circ)$ and of the batch of step t; on $\{t \geq t_{\text{stop}}\}$ the update vanishes and $\Delta_{t+1} = \Delta_t$. □

Proposition SA.5 (Support event and zero risk)**.** *In definition SA.1 let $P_{\text{out}} \overset{\text{def}}{=} \sum_{\mu_i > 0} u_i \otimes u_i$ and $P_{\text{in}} \overset{\text{def}}{=} \sum_{\lambda_j > 0} v_j \otimes v_j$. There is an event Ω_t of probability one, determined by the batch of step t, on which*

$$x_{\text{out}}^{(a)} = P_{\text{out}} x_{\text{out}}^{(a)}, \qquad x_{\text{in}}^{(a)} = P_{\text{in}} x_{\text{in}}^{(a)} \quad (a \leq B), \qquad G_{t+1}^\circ = P_{\text{out}} G_{t+1}^\circ P_{\text{in}} \tag{SA.25}$$

for every value of Δ_t. On Ω_t, if $\mathscr{R}(t) = 0$ then $d_a^\circ = 0$ for every a and $G_{t+1}^\circ = 0$. If, in definition SA.2, $M_{-1}^\circ = P_{\text{out}} M_{-1}^\circ P_{\text{in}}$, then on $\bigcap_t \Omega_t$ every buffer and every update satisfy $M_t^\circ = P_{\text{out}} M_t^\circ P_{\text{in}}$ and $\widetilde{G}_{t+1}^\circ = P_{\text{out}} \widetilde{G}_{t+1}^\circ P_{\text{in}}$, and $\mathscr{Q}_{ij}(\Delta_t)$ is constant in t for every pair with $\mu_i \lambda_j = 0$.

Proof. A zero eigenvalue is a direction of zero variance. If $\mu_i = 0$ then $\mathbb{E}(u_i^\top x_{\text{out}}^{(a)})^2 = u_i^\top \Sigma_{\text{out}} u_i = 0$, so $u_i^\top x_{\text{out}}^{(a)} = 0$ almost surely, and likewise $v_j^\top x_{\text{in}}^{(a)} = 0$ almost surely when $\lambda_j = 0$. Let Ω_t be the complement of the finite union of these null events over the zero modes and the samples. On Ω_t each $x_{\text{out}}^{(a)} \otimes x_{\text{in}}^{(a)}$ equals $P_{\text{out}}(x_{\text{out}}^{(a)} \otimes x_{\text{in}}^{(a)}) P_{\text{in}}$, which gives the last identity of eq. (SA.25) for every Δ_t.

Zero risk forces zero residuals on Ω_t. By lemma SA.1, $\mathscr{R}(t) = 0$ exactly when $P_{\text{out}} \Delta_t P_{\text{in}} = 0$, since $P_{\text{out}} \Delta_t P_{\text{in}} = \sum_{\mu_i \lambda_j > 0} (u_i^\top \Delta_t v_j)\, u_i \otimes v_j$. On Ω_t each residual is $d_a^\circ = (x_{\text{out}}^{(a)})^\top P_{\text{out}} \Delta_t P_{\text{in}} x_{\text{in}}^{(a)}$, which vanishes, and then $G_{t+1}^\circ = 0$.

The support property propagates through the algorithm by induction. If $M_{t-1}^\circ = P_{\text{out}} M_{t-1}^\circ P_{\text{in}}$ then $M_t^\circ = \beta M_{t-1}^\circ + (1 - \beta) G_{t+1}^\circ$ has the same property on Ω_t, and lemma SA.2(ii) transfers it to $Q_{\varrho_t^\circ}(M_t^\circ)$. For a pair with $\mu_i = 0$ or $\lambda_j = 0$ this gives $u_i^\top \widetilde{G}_{t+1}^\circ v_j = 0$, hence $u_i^\top \Delta_{t+1} v_j = u_i^\top \Delta_t v_j$. □

SA.5 The frozen-state model

In the algorithm, the gradient G°_{t+1-j} of age j in eq. (SA.17) was computed at the earlier error Δ_{t-j}, whose risk differs from $\mathscr{R}(t)$. The frozen-state model replaces every past gradient by a gradient recomputed at the current error, with its own independent batch, and removes the initial buffer; it is the model in which the deterministic equivalents of the following sections are computed. Its relation to the algorithm is the subject of the dynamics section.

Definition SA.3 (Frozen-state model)**.** *Assume Assumption S1. Fix t, condition on $\mathscr{F}_t$, and assume $\mathscr{R}(t) > 0$, so that Δ_t and $\widehat{A}_t$ are deterministic. Let (z_{ja}, w_{ja}), $j \geq 0$, $1 \leq a \leq B$, be independent standard Gaussian vectors in $\mathbb{R}^{N_{\mathrm{out}}} \times \mathbb{R}^{N_{\mathrm{in}}}$, and put*

$$d_{ja} \overset{\mathrm{def}}{=} z_{ja}^{\top} \widehat{A}_t w_{ja}, \qquad G_{t+1-j} \overset{\mathrm{def}}{=} \frac{1}{B} \sum_{a=1}^{B} d_{ja}\, {\Sigma_{\mathrm{out}}}^{1/2} (z_{ja} \otimes w_{ja}) {\Sigma_{\mathrm{in}}}^{1/2},$$
$$M_t \overset{\mathrm{def}}{=} \sum_{j \geq 0} w_j\, G_{t+1-j}. \tag{SA.26}$$

The frozen-state model is the model in which the normalized buffer at time t is M_t and the update applied at step t is $\widetilde{G}_{t+1} \overset{\mathrm{def}}{=} Q_{\varrho}(M_t)$.

By proposition SA.3(ii), $Q_{\varrho}(M_t) = Q_{\varrho^{\circ}_t}(\sqrt{2\mathscr{R}(t)}\, M_t)$, so the frozen-state update is the update eq. (SA.15) with M°_t replaced by $\sqrt{2\mathscr{R}(t)}\, M_t$. The gradients G_{t+1-j}, $j \geq 0$, are independent and identically distributed copies of the normalized gradient G_{t+1} of eq. (SA.10). When lagged buffers $M_{t+\ell}$ are needed, the samples are indexed by time instead of age: independent standard Gaussian pairs (z_{sa}, w_{sa}), $s \in \mathbb{Z}$, $a \leq B$, with G_s built from the samples of time s by eq. (SA.26), $M_{t'} \overset{\mathrm{def}}{=} \sum_{j \geq 0} w_j G_{t'+1-j}$ for every integer t', and the age-j sample of M_t being the sample of time $t+1-j$. The series defining M_t converges by lemma SA.5. The letters G_{t+1} and $\mathbb{E}\, G_{t+1}$ in this subsection refer to the frozen-state model.

Lemma SA.4 (Gaussian covariance of a batch)**.** *Let $A \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ be deterministic with $\|A\|_{\mathrm{F}} = 1$, let $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\mathrm{out}}})$ and $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\mathrm{in}}})$ be independent, and let $X \overset{\mathrm{def}}{=} (z^{\top} A w)\, z \otimes w$. Then $\mathbb{E}\, X = A$, and its covariance tensor $\mathcal{C}_{ij,kl} \overset{\mathrm{def}}{=} \mathrm{Cov}(X_{ij}, X_{kl})$ is*

$$\mathcal{C}_{ij,kl} = \delta_{ik}\delta_{jl} + 2(AA^{\top})_{ik}\delta_{jl} + 2\delta_{ik}(A^{\top}A)_{jl} + A_{ij}A_{kl} + 2A_{il}A_{kj}. \tag{SA.27}$$

Consequently, in the frozen-state model with $A = \widehat{A}_t$, the batches G_{t+1-j} are independent and identically distributed,

$$\mathbb{E}\, G_{t+1} = {\Sigma_{\mathrm{out}}}^{1/2} \widehat{A}_t {\Sigma_{\mathrm{in}}}^{1/2},$$
$$\mathrm{Cov}\big((G_{t+1})_{ij}, (G_{t+1})_{kl}\big) = \frac{1}{B} \sum_{p,q,r,s} ({\Sigma_{\mathrm{out}}}^{1/2})_{ip} ({\Sigma_{\mathrm{in}}}^{1/2})_{jq}\, \mathcal{C}_{pq,rs}\, ({\Sigma_{\mathrm{out}}}^{1/2})_{kr} ({\Sigma_{\mathrm{in}}}^{1/2})_{ls}, \tag{SA.28}$$

and the single-batch Frobenius variance is

$$\mathsf{V}_B \overset{\mathrm{def}}{=} \mathbb{E}\, \|G_{t+1} - \mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2$$
$$= \frac{1}{B} \Big\{ \mathrm{Tr}\, \Sigma_{\mathrm{out}}\, \mathrm{Tr}\, \Sigma_{\mathrm{in}} + 2\, \mathrm{Tr}(\Sigma_{\mathrm{out}} \widehat{A}_t \widehat{A}_t^{\top})\, \mathrm{Tr}\, \Sigma_{\mathrm{in}} + 2\, \mathrm{Tr}\, \Sigma_{\mathrm{out}}\, \mathrm{Tr}(\Sigma_{\mathrm{in}} \widehat{A}_t^{\top} \widehat{A}_t) + 3 \|\, \mathbb{E}\, G_{t+1} \|_{\mathrm{F}}^2 \Big\}. \tag{SA.29}$$

Proof. The mean is computed by independence of z and w and the second moments of standard normals:

$$\mathbb{E}\, X_{ij} = \sum_{p,q} A_{pq}\, \mathbb{E}[z_i z_p]\, \mathbb{E}[w_j w_q] = A_{ij}.$$

The second moment is a sum over nine Wick pairings. Expanding the product,

$$\mathbb{E}[X_{ij} X_{kl}] = \sum_{p,q,r,s} A_{pq} A_{rs}\, \mathbb{E}[z_i z_k z_p z_r]\, \mathbb{E}[w_j w_l w_q w_s],$$

and the fourth moments of independent standard normal coordinates are

$$\mathbb{E}[z_i z_k z_p z_r] = \delta_{ik}\delta_{pr} + \delta_{ip}\delta_{kr} + \delta_{ir}\delta_{kp}, \tag{SA.30}$$

and likewise for w; this identity holds because an index appearing an odd number of times gives zero, two distinct pairs give 1, and four equal indices give $\mathbb{E}\, g^4 = 3$, which is the number of pairings. The product of the two three-term identities has nine terms. The term pairing (i,k) with (j,l) contributes $\delta_{ik}\delta_{jl}\sum_{p,q} A_{pq}^2 = \delta_{ik}\delta_{jl}$. The two terms pairing (i,k) in z with a crossed pairing in w contribute

$$\delta_{ik}\sum_p A_{pj}A_{pl} + \delta_{ik}\sum_p A_{pl}A_{pj} = 2\delta_{ik}(A^\top A)_{jl},$$

and symmetrically the two terms with (j,l) paired in w and a crossed pairing in z contribute $2(AA^\top)_{ik}\delta_{jl}$. The four terms with crossed pairings in both families contribute $A_{ij}A_{kl} + A_{il}A_{kj} + A_{kj}A_{il} + A_{kl}A_{ij} = 2A_{ij}A_{kl} + 2A_{il}A_{kj}$. Subtracting the product of the means $A_{ij}A_{kl}$ gives eq. (SA.27).

The batch statements follow from independence of the samples. The batch G_{t+1} is the average over a of the independent, identically distributed matrices ${\Sigma_{\text{out}}}^{1/2} X_a {\Sigma_{\text{in}}}^{1/2}$, where X_a is built from (z_a, w_a) and $A = \widehat{A}_t$. Its mean is ${\Sigma_{\text{out}}}^{1/2}\widehat{A}_t{\Sigma_{\text{in}}}^{1/2}$, and its covariance is $1/B$ times the covariance of one transformed sample, which is eq. (SA.28) by the bilinearity of the covariance. Different batches use disjoint sets of samples and are therefore independent.

The Frobenius variance is a contraction of the tensor. For one sample,

$$\mathbb{E}\,\|{\Sigma_{\text{out}}}^{1/2}(X-A){\Sigma_{\text{in}}}^{1/2}\|_{\text{F}}^2 = \mathbb{E}\,\text{Tr}\left(\Sigma_{\text{out}}(X-A)\Sigma_{\text{in}}(X-A)^\top\right) = \sum_{i,j,k,l}(\Sigma_{\text{out}})_{ik}(\Sigma_{\text{in}})_{jl}\,\mathcal{C}_{ij,kl},$$

using the symmetry $\mathcal{C}_{ij,kl} = \mathcal{C}_{kl,ij}$ and of the covariances. The five terms of eq. (SA.27) contract to

$$\text{Tr}\,\Sigma_{\text{out}}\,\text{Tr}\,\Sigma_{\text{in}}, \qquad 2\,\text{Tr}(\Sigma_{\text{out}}AA^\top)\,\text{Tr}\,\Sigma_{\text{in}}, \qquad 2\,\text{Tr}\,\Sigma_{\text{out}}\,\text{Tr}(\Sigma_{\text{in}}A^\top A),$$
$$\text{Tr}(\Sigma_{\text{out}}A\Sigma_{\text{in}}A^\top), \qquad 2\,\text{Tr}(\Sigma_{\text{out}}A\Sigma_{\text{in}}A^\top),$$

where the last two use $\sum_{i,j,k,l}(\Sigma_{\text{out}})_{ik}(\Sigma_{\text{in}})_{jl}A_{ij}A_{kl} = \sum_{i,k}(\Sigma_{\text{out}})_{ik}(A\Sigma_{\text{in}}A^\top)_{ik}$ and the same value for $A_{il}A_{kj}$ after exchanging the roles of j and l. Since $\text{Tr}(\Sigma_{\text{out}}A\Sigma_{\text{in}}A^\top) = \|{\Sigma_{\text{out}}}^{1/2}A{\Sigma_{\text{in}}}^{1/2}\|_{\text{F}}^2 = \|\,\mathbb{E}\,G_{t+1}\|_{\text{F}}^2$ and V_B is $1/B$ times the one-sample variance, eq. (SA.29) follows. □

Lemma SA.5 (Stationary EMA law)**.** *In the frozen-state model, index the samples by time as above, and write* $\Sigma_G \overset{\text{def}}{=} \text{Cov}(\text{vec}\,G_{t+1})$ *for the batch covariance eq.* (SA.28)*.*

(i) The series $M_t = \sum_{j\ge 0} w_j G_{t+1-j}$ *and the sample series* $\sum_{j,a} w_j |d_{ja}|\|z_{ja}\|\|w_{ja}\|$ *converge absolutely.*

(ii) For every integer t*,* $M_t = \beta M_{t-1} + (1-\beta)G_{t+1}$*, and the sequence* $(M_t)_{t\in\mathbb{Z}}$ *is strictly stationary.*

(iii) $\mathbb{E}\,M_t = \mathbb{E}\,G_{t+1}$*,* $\text{Cov}(\text{vec}\,M_t) = s_2\,\Sigma_G$*, and* $\mathbb{E}\,\|M_t - \mathbb{E}\,G_{t+1}\|_{\text{F}}^2 = s_2\,\mathsf{V}_B$*.*

(iv) For every lag $\ell \ge 0$*,* $\text{Cov}(\text{vec}\,M_t, \text{vec}\,M_{t+\ell}) = \beta^\ell\, s_2\,\Sigma_G$*.*

(v) For every positive integer T*,*

$$\text{Cov}\left(\sum_{t=1}^{T}\text{vec}(M_t - \mathbb{E}\,G_{t+1})\right) = s_2\Big[T + 2\sum_{\ell=1}^{T-1}(T-\ell)\beta^\ell\Big]\Sigma_G = \Big[T - \frac{2\beta(1-\beta^T)}{1-\beta^2}\Big]\Sigma_G, \tag{SA.31}$$

the two-sided lag sum $\sum_{\ell\in\mathbb{Z}}\text{Cov}(\text{vec}\,M_t, \text{vec}\,M_{t+\ell})$ *converges absolutely and equals* Σ_G*, and* $T^{-1}\,\text{Cov}(\sum_{t\le T}\text{vec}(M_t - \mathbb{E}\,G_{t+1})) \to \Sigma_G$ *as* $T\to\infty$*.*

Proof. Part (i) holds because $\sum_j w_j = 1$ and the batches are identically distributed.

The recursion is a reindexing of the series:

$$\beta M_{t-1} + (1-\beta)G_{t+1} = \sum_{j\geq 0}(1-\beta)\beta^{j+1}G_{t-j} + (1-\beta)G_{t+1} = \sum_{j\geq 1} w_j G_{t+1-j} + w_0 G_{t+1} = M_t.$$

Strict stationarity holds because the batches are independent and identically distributed, so that a common shift of the times is a relabeling of the batches. This proves (ii).

The moments follow from the orthogonality of the centered batches. For $k \leq l$, independence and centering give

$$\mathbb{E}\Big\| \sum_{j=k}^{l} w_j \big(G_{t+1-j} - \mathbb{E}\, G_{t+1}\big)\Big\|_{\mathrm{F}}^2 = \mathsf{V}_B \sum_{j=k}^{l} w_j^2. \tag{SA.32}$$

The mean is $\sum_j w_j\ \mathbb{E}\, G_{t+1} = \mathbb{E}\, G_{t+1}$ by eq. (SA.18); the covariance is $\sum_j w_j^2\, \Sigma_G = s_2 \Sigma_G$ by the same orthogonality; and the Frobenius variance is eq. (SA.32) with $k = 0$, $l \to \infty$. This proves (iii).

The lag covariance follows from the decomposition of a later buffer into the past and a new part. By the reindexing of the series, for $\ell \geq 0$,

$$M_{t+\ell} = \beta^\ell\, M_t + \sum_{r=0}^{\ell-1} w_r\, G_{t+\ell+1-r}, \tag{SA.33}$$

where the finite sum involves only batches of times larger than $t+1$ and is therefore independent of M_t. Centering and bilinearity of the covariance give $\mathrm{Cov}(\mathrm{vec}\, M_t, \mathrm{vec}\, M_{t+\ell}) = \beta^\ell\, \mathrm{Cov}(\mathrm{vec}\, M_t) = \beta^\ell s_2 \Sigma_G$. This proves (iv).

The block covariance is a sum over the T^2 pairs of times. By (iii) and (iv), and the symmetry of the covariance under exchanging the two times,

$$\mathrm{Cov}\Big(\sum_{t=1}^{T} \mathrm{vec}(M_t - \mathbb{E}\, G_{t+1})\Big) = \sum_{t,t'=1}^{T} \beta^{|t-t'|} s_2 \Sigma_G = s_2 \Big[T + 2\sum_{\ell=1}^{T-1}(T-\ell)\beta^\ell\Big]\Sigma_G,$$

which is the first form of eq. (SA.31). The scalar identity

$$\sum_{\ell=1}^{T-1}(T-\ell)\beta^\ell = \sum_{k=1}^{T-1}\sum_{\ell=1}^{k}\beta^\ell = \frac{\beta}{1-\beta}\sum_{k=1}^{T-1}(1-\beta^k) = \frac{T\beta}{1-\beta} - \frac{\beta(1-\beta^T)}{(1-\beta)^2}$$

follows from writing $T-\ell$ as the number of k with $\ell \leq k \leq T-1$ and exchanging the order of summation, and inserting it with $s_2 = (1-\beta)/(1+\beta)$ gives the second form. The two-sided lag sum converges absolutely by the geometric majorant $\beta^{|\ell|}$ and equals

$$s_2\Big(1 + 2\sum_{\ell\geq 1}\beta^\ell\Big)\Sigma_G = \frac{1-\beta}{1+\beta}\cdot\frac{1+\beta}{1-\beta}\,\Sigma_G = \Sigma_G,$$

and the limit of T^{-1} times eq. (SA.31) is the same value. This proves (v). □

Proposition SA.6 (Startup and bias correction)**.** *In the frozen-state model let $K \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ be deterministic and let $(M_t^K)_{t\geq -1}$ be the recursion $M_t^K = \beta M_{t-1}^K + (1-\beta)G_{t+1}$ started from $M_{-1}^K = K$. Then for $t \geq 0$,*

$$M_t^K = \beta^{t+1}K + \sum_{j=0}^{t} w_j G_{t+1-j}, \qquad \mathbb{E}\, M_t^K = \beta^{t+1}K + (1-\beta^{t+1})\,\mathbb{E}\, G_{t+1},$$
$$\mathrm{Cov}(\mathrm{vec}\, M_t^K) = s_2\big(1-\beta^{2(t+1)}\big)\Sigma_G. \tag{SA.34}$$

The bias-corrected zero-start buffer $\widetilde{M}_t \overset{\mathrm{def}}{=} M_t^0/(1-\beta^{t+1})$ has mean $\mathbb{E}\, G_{t+1}$ and covariance $s_2\,\frac{1+\beta^{t+1}}{1-\beta^{t+1}}\Sigma_G$. If M_t^K and the stationary buffer M_t are built from the same batches, then

$$M_t^K - M_t = \beta^{t+1}(K - M_{-1}), \qquad \mathbb{E}\,\|M_t^K - M_t\|_{\mathrm{F}}^2 = \beta^{2(t+1)}\big(\|K - \mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 + s_2 \mathsf{V}_B\big). \tag{SA.35}$$

Proof. The explicit form is an induction on t, the case $t = 0$ being $M_0^K = \beta K + (1-\beta)G_1$. The mean and covariance follow from the finite geometric sums

$$\sum_{j=0}^{t} w_j = 1 - \beta^{t+1}, \qquad \sum_{j=0}^{t} w_j^2 = (1-\beta)^2 \, \frac{1-\beta^{2(t+1)}}{1-\beta^2} = s_2\big(1-\beta^{2(t+1)}\big),$$

and the independence of the batches. Dividing by $1 - \beta^{t+1} > 0$ gives the bias-corrected moments, using $(1-\beta^{2(t+1)})/(1-\beta^{t+1})^2 = (1+\beta^{t+1})/(1-\beta^{t+1})$.

The coupling follows by subtracting the two recursions, whose difference is $M_t^K - M_t = \beta(M_{t-1}^K - M_{t-1})$, and iterating down to $t = -1$. Since $\mathbb{E}\, M_{-1} = \mathbb{E}\, G_{t+1}$ and $\mathbb{E}\,\|M_{-1} - \mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 = s_2 \mathsf{V}_B$ by lemma SA.5(iii), expanding the square gives the second identity of eq. (SA.35). □

Remark SA.3 (The linear buffer and the filtered update)**.** *By lemma SA.5(iii), the raw covariance of the buffer is that of a single batch of size $B/s_2 = B(1+\beta)/(1-\beta)$, and by (v) the two-sided sum of its lag covariances is exactly the covariance of one batch. Both statements concern the linear buffer M_t and say nothing about the nonlinear filtered update $Q_\varrho(M_t)$: the update is not linear in the batches, its lag covariances are not β^ℓ times its covariance, and no effective batch size reproduces its law (the deterministic-equivalent section shows that the weighted fixed-point system depends on the whole weight sequence). The volatility coefficient of the dynamics is the integrated centered lag covariance of the filtered update, which the volatility section identifies.*

Lemma SA.6 (History truncation and sample sensitivity)**.** *In the frozen-state model let $M^{(J)} \overset{\mathrm{def}}{=} \sum_{j=0}^{J} w_j G_{t+1-j}$ for $J \geq 0$, and let $\varrho > 0$.*

(i) $\mathbb{E}\,\big(M_t - M^{(J)}\big) = \beta^{J+1}\, \mathbb{E}\, G_{t+1}$ *and* $\mathbb{E}\,\|M_t - M^{(J)}\|_{\mathrm{F}}^2 = \beta^{2(J+1)}\big(\|\,\mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 + s_2 \mathsf{V}_B\big)$.

(ii) $\mathbb{E}\,\|Q_\varrho(M_t) - Q_\varrho(M^{(J)})\|_{\mathrm{F}}^2 \leq \varrho^{-1}\beta^{2(J+1)}\big(\|\,\mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 + s_2 \mathsf{V}_B\big)$.

(iii) Let $F : \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}} \to \mathbb{R}$ be Lipschitz in Frobenius norm with constant L_F, and let $M_t^{[ja]}$ denote the buffer with the sample (z_{ja}, w_{ja}) replaced by an independent copy. Then

$$\mathbb{E}\,\big|F(M_t) - F(M_t^{[ja]})\big|^2 \leq \frac{2L_F^2\, w_j^2\, \mathsf{V}_B}{B}, \qquad \operatorname{Var} F(M_t) \leq L_F^2\, s_2\, \mathsf{V}_B.$$

(iv) If $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \leq C_\Sigma$ then $\|\,\mathbb{E}\, G_{t+1}\|_{\mathrm{F}} \leq C_\Sigma$ and $\mathsf{V}_B \leq C_\Sigma^2\,(N_{\mathrm{out}}N_{\mathrm{in}} + 2N_{\mathrm{out}} + 2N_{\mathrm{in}} + 3)/B$.

Proof. Part (i) is the tail of the weight sums. The tail $M_t - M^{(J)} = \sum_{j>J} w_j G_{t+1-j}$ has mean $(\sum_{j>J} w_j)\,\mathbb{E}\, G_{t+1} = \beta^{J+1}\,\mathbb{E}\, G_{t+1}$, and by eq. (SA.32) its second moment is

$$\Big(\sum_{j>J} w_j\Big)^2 \|\,\mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 + \mathsf{V}_B \sum_{j>J} w_j^2 = \beta^{2(J+1)}\|\,\mathbb{E}\, G_{t+1}\|_{\mathrm{F}}^2 + \mathsf{V}_B\, s_2\, \beta^{2(J+1)},$$

since $\sum_{j>J} w_j^2 = (1-\beta)^2\beta^{2(J+1)}/(1-\beta^2)$.

Part (ii) is (i) combined with the Lipschitz bound eq. (SA.13).

Part (iii) is the Efron–Stein inequality applied to a truncated history. With $X_{ja} \overset{\mathrm{def}}{=} d_{ja}\, z_{ja} \otimes w_{ja}$ and X'_{ja} the same expression in the independent copy, replacing the sample (j,a) changes M_t by $\frac{w_j}{B}{\Sigma_{\mathrm{out}}}^{1/2}(X_{ja} - X'_{ja}){\Sigma_{\mathrm{in}}}^{1/2}$; by lemma SA.4 the two transformed samples are independent and identically distributed with Frobenius variance $B\mathsf{V}_B$ each, so

$$\mathbb{E}\,\|M_t - M_t^{[ja]}\|_{\mathrm{F}}^2 = \frac{w_j^2}{B^2} \cdot 2B\mathsf{V}_B = \frac{2w_j^2\mathsf{V}_B}{B},$$

and the Lipschitz property of F gives the first bound. For the variance, fix J and regard $F(M^{(J)})$ as a function of the $B(J+1)$ independent samples of ages $0, \dots, J$. The Efron–Stein inequality [13, Theorem 3.1] and the first bound, which holds verbatim for $M^{(J)}$, give

$$\operatorname{Var} F(M^{(J)}) \leq \frac{1}{2}\sum_{j=0}^{J}\sum_{a=1}^{B}\mathbb{E}\,\big|F(M^{(J)}) - F(M^{(J)[ja]})\big|^2 \leq \frac{1}{2}\sum_{j=0}^{J} B\,\frac{2L_F^2 w_j^2 \mathsf{V}_B}{B} \leq L_F^2\, s_2\, \mathsf{V}_B.$$

Letting $J \to \infty$, with $\|F(M_t) - F(M^{(J)})\|_2 \le L_F\big(\mathbb{E}\,\|M_t - M^{(J)}\|_{\mathrm{F}}^2\big)^{1/2} \to 0$ by (i), gives the bound for M_t.

Part (iv) bounds the traces in eq. (SA.29). We have $\|\mathbb{E}\,G_{t+1}\|_{\mathrm{F}} \le \|{\Sigma_{\mathrm{out}}}^{1/2}\|\|\widehat{A}_t\|_{\mathrm{F}}\|{\Sigma_{\mathrm{in}}}^{1/2}\| \le C_\Sigma$, $\operatorname{Tr}\Sigma_{\mathrm{out}} \le C_\Sigma N_{\mathrm{out}}$, $\operatorname{Tr}\Sigma_{\mathrm{in}} \le C_\Sigma N_{\mathrm{in}}$, $\operatorname{Tr}(\Sigma_{\mathrm{out}}\widehat{A}_t\widehat{A}_t^\top) \le C_\Sigma\|\widehat{A}_t\|_{\mathrm{F}}^2 = C_\Sigma$ and $\operatorname{Tr}(\Sigma_{\mathrm{in}}\widehat{A}_t^\top\widehat{A}_t) \le C_\Sigma$; inserting these in eq. (SA.29) gives the bound. □

Lemma SA.7 (Residual law)**.** *Let $A \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ be deterministic with $\|A\|_{\mathrm{F}} = 1$, let $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\mathrm{out}}})$ and $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\mathrm{in}}})$ be independent, and put $d = z^\top A w$. Let $\alpha_1 \ge \alpha_2 \ge \cdots \ge 0$ be the singular values of A and $g \sim \mathcal{N}(0,1)$.*

- *(i) d is symmetric, $\mathbb{E}\,d^2 = 1$, $\mathbb{E}\,d^4 = 3 + 6\sum_k \alpha_k^4 \le 3 + 6\|A\|^2$, and $\mathbb{E}\,|d|^{2p} \le \big((2p-1)!!\big)^2$ for every integer $p \ge 1$.*
- *(ii) $\mathbb{E}\,e^{i\theta d} = \prod_k (1+\theta^2\alpha_k^2)^{-1/2}$ for real θ, and $0 \le \log \mathbb{E}\,e^{i\theta d} + \theta^2/2 \le \theta^4\|A\|^2/4$.*
- *(iii) Along any sequence of such $A = A_B$ with $\|A_B\| \to 0$, $d \Rightarrow g$ in distribution, and $\mathbb{E}\,\psi(d) \to \mathbb{E}\,\psi(g)$ for every continuous ψ with $|\psi(x)| \le 1 + x^2$.*

Proof. The residual reduces to a weighted sum of products of independent normals. Write $A = U\,\mathrm{diag}(\alpha)V^\top$ with U, V orthogonal. Since $U^\top z$ and $V^\top w$ are again independent standard Gaussian vectors, d has the law of

$$d \overset{\text{law}}{=} \sum_k \alpha_k U_k V_k, \qquad U_k, V_k \text{ independent standard normal.}$$

Negating all U_k proves symmetry.

The second and fourth moments follow by expansion. Independence gives $\mathbb{E}\,d^2 = \sum_k \alpha_k^2 = 1$. Each product U_kV_k has fourth moment 9, and expanding the fourth power of the sum leaves only the multiplicities 4 and $2+2$, so

$$\mathbb{E}\,d^4 = 9\sum_k \alpha_k^4 + 6\sum_{k<l}\alpha_k^2\alpha_l^2 = 3\Big(\sum_k \alpha_k^2\Big)^2 + 6\sum_k \alpha_k^4 = 3 + 6\sum_k \alpha_k^4.$$

The bound $\sum_k \alpha_k^4 \le \alpha_1^2 \sum_k \alpha_k^2 = \|A\|^2$ gives the fourth-moment bound.

The $2p$-th moment is bounded by conditioning on w. Given w, the variable d is centered Gaussian with variance $\|Aw\|^2$, so

$$\mathbb{E}\left[|d|^{2p} \mid w\right] = (2p-1)!!\,\|Aw\|^{2p}.$$

In the coordinates above $\|Aw\|^2 = \sum_k \alpha_k^2 V_k^2$ is a convex combination of the V_k^2, so Jensen's inequality for $x \mapsto x^p$ gives

$$\mathbb{E}\,\|Aw\|^{2p} \le \sum_k \alpha_k^2\ \mathbb{E}\,V_k^{2p} = (2p-1)!!,$$

and (i) follows.

The characteristic function is a product. Integrating first in V_k and then in U_k,

$$\mathbb{E}\,e^{i\theta\alpha U_k V_k} = \mathbb{E}\,e^{-\theta^2\alpha^2 U_k^2/2} = (1+\theta^2\alpha^2)^{-1/2},$$

and independence gives the product formula of (ii). For $x \ge 0$,

$$0 \le x - \log(1+x) = \int_0^x \frac{u}{1+u}\,du \le \frac{x^2}{2};$$

summing at $x = \theta^2\alpha_k^2$ and using $\sum_k \alpha_k^2 = 1$ and $\sum_k \alpha_k^4 \le \|A\|^2$ gives the bound of (ii).

Part (iii) is Lévy's continuity theorem: by (ii) the characteristic functions converge pointwise to $e^{-\theta^2/2}$, hence $d \Rightarrow g$ [9, Theorem 26.3], and the convergence of $\mathbb{E}\,\psi(d)$ for $|\psi(x)| \le 1 + x^2$ follows from the bound $\mathbb{E}\,d^4 \le 9$ of (i). □

SA.6 Gram resolvents, the regime, and deterministic equivalents

Gram resolvents of the buffer. In the frozen-state model, for $z \in \mathbb{C} \setminus [0, \infty)$ define

$$R(z) \overset{\text{def}}{=} \left(M_t M_t^\top - z \, \mathrm{Id}_{N_{\text{out}}}\right)^{-1}, \qquad \widetilde{R}(z) \overset{\text{def}}{=} \left(M_t^\top M_t - z \, \mathrm{Id}_{N_{\text{in}}}\right)^{-1}. \tag{SA.36}$$

At $\beta = 0$ one has $M_t = G_{t+1}$ and these are the resolvents of section 2 and fig. 2. By lemma SA.3, the regularized update is an integral of the resolvent along the negative axis,

$$Q_\varrho(M_t) = \frac{2}{\pi} \int_0^\infty R(-\varrho - u^2) \, M_t \, du, \tag{SA.37}$$

which is why every deterministic equivalent of the appendix is stated on the negative axis, $z = -s$ with $s > 0$. The two resolvents are related by the following identities, which we keep in the form of the main text.

Lemma SA.8 (Rectangular resolvent intertwining)**.** *Let $A \in \mathbb{R}^{n_1 \times n_2}$ and $B \in \mathbb{R}^{n_2 \times n_1}$. For $z \in \mathbb{C} \setminus (\sigma(AB) \cup \sigma(BA))$, define*

$$R_{AB}(z) \overset{\text{def}}{=} (AB - z \, \mathrm{Id}_{n_1})^{-1}, \qquad R_{BA}(z) \overset{\text{def}}{=} (BA - z \, \mathrm{Id}_{n_2})^{-1}.$$

Then

$$R_{AB} A = A R_{BA}, \qquad B R_{AB} = R_{BA} B, \tag{SA.38}$$

and consequently

$$A R_{BA} B = \mathrm{Id}_{n_1} + z \, R_{AB}, \qquad B R_{AB} A = \mathrm{Id}_{n_2} + z \, R_{BA}. \tag{SA.39}$$

Proof. The identities

$$(AB - z \, \mathrm{Id}_{n_1}) A = A (BA - z \, \mathrm{Id}_{n_2}), \qquad B (AB - z \, \mathrm{Id}_{n_1}) = (BA - z \, \mathrm{Id}_{n_2}) B$$

are immediate. Multiplying the first on the left by R_{AB} and on the right by R_{BA} gives $R_{AB} A = A R_{BA}$; multiplying the second similarly gives $B R_{AB} = R_{BA} B$. Then

$$A R_{BA} B = (R_{AB} A) B = R_{AB} (AB) = \mathrm{Id}_{n_1} + z R_{AB},$$

and similarly

$$B R_{AB} A = R_{BA} (BA) = \mathrm{Id}_{n_2} + z R_{BA}.$$

□

Applied with $(A, B) = (M_t, M_t^\top)$, whose products have the same nonzero spectrum, the lemma gives for every $z \in \mathbb{C} \setminus [0, \infty)$

$$R(z) M_t = M_t \widetilde{R}(z), \qquad M_t^\top R(z) M_t = \mathrm{Id}_{N_{\text{in}}} + z \widetilde{R}(z), \qquad M_t \widetilde{R}(z) M_t^\top = \mathrm{Id}_{N_{\text{out}}} + z R(z). \tag{SA.40}$$

Spectral measures. The covariances enter the deterministic equivalents only through the two spectral measures

$$\nu_{\text{out}} \overset{\text{def}}{=} \frac{1}{B} \sum_{i=1}^{N_{\text{out}}} \delta_{\mu_i}, \qquad \nu_{\text{in}} \overset{\text{def}}{=} \frac{1}{B} \sum_{k=1}^{N_{\text{in}}} \delta_{\lambda_k}, \tag{SA.41}$$

so that $\frac{1}{B} \operatorname{Tr} \phi(\Sigma_{\text{out}}) = \int \phi \, d\nu_{\text{out}}$ for every function ϕ on $[0, \infty)$, and likewise for Σ_{in}.

The regime. We state the proportional regime of the main text and then the regime under which the random-matrix results of the appendix are proved.

Assumption S2 (Proportional regime)**.** *The dimensions N_{in}, N_{out}, B tend to infinity jointly, with aspect ratios converging to constants:*

$$\frac{N_{\text{in}}}{B} \to \gamma_{\text{in}} \in (0, \infty), \qquad \frac{N_{\text{out}}}{B} \to \gamma_{\text{out}} \in (0, \infty).$$

The covariance operator norms remain uniformly bounded: there exists a constant $C > 0$, independent of N_{in} and N_{out}, such that

$$\|\Sigma_{\text{in}}\|_{\text{op}} \le C, \qquad \|\Sigma_{\text{out}}\|_{\text{op}} \le C.$$

We refer to the joint limit as the proportional regime.

Assumption S3 (Spectrally diffuse proportional regime)**.** *Assumptions S1 and S2 hold, and we fix* $\gamma_\star < \infty$ *and* $C_\Sigma \geq 1$ *with* $\max(N_{\rm in}, N_{\rm out}) \leq \gamma_\star B$, $\|\Sigma_{\rm in}\| \leq C_\Sigma$ *and* $\|\Sigma_{\rm out}\| \leq C_\Sigma$ *along the whole sequence of dimensions; the covariances are deterministic. The momentum* $\beta \in [0, 1)$ *is fixed, and the normalized ridge satisfies* $\varrho \in [\varrho_-, \varrho_+]$ *for fixed* $0 < \varrho_- \leq \varrho_+ < \infty$. *The state is frozen as in definition SA.3, with* $\mathscr{R}(t) > 0$, *and the normalized state is spectrally diffuse:* $\|\widehat{A}_t\| \to 0$.

The two conditions $\|\widehat{A}_t\|_{\rm F} = 1$ and $\|\widehat{A}_t\| \to 0$ say that no single singular direction of the error carries a nonvanishing fraction of the residual variance; by lemma SA.7(iii) they are exactly what makes the normalized residuals asymptotically standard Gaussian. No lower bound on the covariances, no rank condition and no rate for $\|\widehat{A}_t\| \to 0$ is assumed. Under Assumption S3 the spectral measures eq. (SA.41) are positive measures on $[0, C_\Sigma]$ of total mass at most $\gamma_\star$. All limits of the appendix are along a sequence of dimensions satisfying Assumption S3, and all constants are uniform in $\varrho \in [\varrho_-, \varrho_+]$ unless a statement says otherwise; a statement for a fixed ridge is the case $\varrho_- = \varrho_+$.

Deterministic equivalents. We make the underlying convergence mode precise.

Definition SA.4 (Deterministic equivalent)**.** *Let* $M_N(z)$ *be a sequence of random matrices indexed by the dimension parameter* N *and depending on a spectral parameter* $z \in \mathbb{C} \setminus [0, \infty)$. *A deterministic matrix family* $\overline{M}_N(z)$ *of the same size is a* deterministic equivalent *of* $M_N(z)$ *in the proportional regime (Assumption S2) if, for every sequence of deterministic test matrices* T_N *with* $\|T_N\|$ *bounded uniformly in* N,

$$\frac{1}{N}\,\mathrm{Tr}\big[(M_N(z) - \overline{M}_N(z))\,T_N\big] \xrightarrow{\Pr} 0 \qquad \text{as } N \to \infty.$$

Equivalently, $\overline{M}_N(z)$ matches $M_N(z)$ in the normalized trace pairing against any bounded test matrix, up to a vanishing error in probability. All deterministic-equivalent claims of the appendix are statements of this form, with N the appropriate dimension ($N_{\rm in}$, $N_{\rm out}$ or B) determined by context; the deterministic-equivalent section proves them in the stronger form of convergence of the expected matrix in operator norm together with convergence of the trace pairing in mean square, uniformly on compact subsets of the negative axis. Two display-equation symbols make the convergence mode explicit: $A \overset{\rm de}{=} \overline{A}$ (read "$\overline{A}$ is a deterministic equivalent of A") stands for the matrix-valued statement of definition SA.4, and $a \overset{\Pr}{=} b$ (read "a equals b up to $o_{\Pr}(1)$") stands for $a = b + o_{\Pr}(1)$ for scalars normalized so that both sides are $O(1)$. A bare "=" between random and deterministic quantities means exact equality, algebraic or with probability one.

SB Deterministic equivalents for the EMA Gram resolvent

This section proves the first random-matrix input of the appendix. In the frozen-state model of definition SA.3, the exponential moving average M_t of the normalized minibatch gradients is a weighted sum of independent batches, and the resolvent of its Gram matrix $M_t M_t^\top$ on the negative real axis has a deterministic equivalent

$$\overline{R}(z) = \big(\tilde{s}_1(z)\,\Sigma_{\rm out} - z\,\mathrm{Id}_{N_{\rm out}}\big)^{-1}, \qquad z = -s < 0,$$

where $\tilde{s}_1(z)$ is determined by the weighted fixed-point system of definition SB.1. At $\beta = 0$ the system is the one displayed in section 2. The section has three main results. Theorem SB.1 shows that the weighted system has exactly one nonnegative solution and that this solution is continuous in the spectral data and stable under approximate roots. Theorem SB.2 shows that the expected Gram resolvents converge to $\overline{R}$ and $\overline{\widetilde{R}}$ in operator norm, uniformly on compact subsets of the negative axis. Corollary SB.1 gives the normalized-trace form of the closure, with bounded deterministic tests, in expected squared compact supremum.

The section works in the setting of appendix SA: its conventions (appendix SA.1), the frozen-state model of definition SA.3 under Assumption S3, the normalized state $\widehat{A}_t$ of eq. (SA.10), the EMA weights of eq. (SA.18), the Gram resolvents $R(z)$ and $\widetilde{R}(z)$ of eq. (SA.36), and the spectral measures of eq. (SA.41). From that section it also takes the convergence of the buffer series (lemma SA.5(i)), its truncation bound (lemma SA.6(i)) and the residual law (lemma SA.7).

SB.1 The weighted fixed-point system

Scalar functions. Let $w = (w_j)_{j\ge 0}$ be a sequence of nonnegative numbers, not all zero, with $\sum_j w_j < \infty$; write $s_p = \sum_j w_j^p$ for $p \ge 2$. The EMA weights of eq. (SA.18) are the case of interest, and for them we write F_β and L_β for the functions F_w and L_w defined now. For $g \sim \mathcal{N}(0,1)$ and $x, q \ge 0$ put

$$J(x) \stackrel{\text{def}}{=} \mathbb{E}\,\frac{g^2}{1+xg^2}, \qquad F_w(q) \stackrel{\text{def}}{=} \sum_{j\ge 0} w_j^2\, J(w_j^2 q), \qquad L_w(q) \stackrel{\text{def}}{=} \sum_{j\ge 0} w_j\, J(w_j^2 q). \tag{SB.1}$$

Lemma SB.1 (Properties of the scalar functions)**.** *Let w be nonnegative, summable and not all zero.*

(i) J is continuous and strictly decreasing on $[0,\infty)$ with values in $(0,1]$, it is C^∞ on $(0,\infty)$ with one-sided derivatives at 0, and

$$J(0) = 1, \qquad J'(0) = -3.$$

(ii) The series defining F_w and L_w converge uniformly on $[0,\infty)$; F_w and L_w are continuous, C^∞ on $(0,\infty)$ with one-sided derivatives at 0, and

$$F_w(0) = s_2, \qquad F_w'(0) = -3s_4, \qquad L_w(0) = \sum_j w_j, \qquad L_w'(0) = -3s_3.$$

(iii) The function

$$r_w(q) \stackrel{\text{def}}{=} qF_w(q) = \sum_j \mathbb{E}\,\frac{w_j^2 q g^2}{1 + w_j^2 q g^2}$$

is strictly increasing on $[0,\infty)$, with $r_w(0) = 0$ and

$$r_w'(q) = \sum_j w_j^2\ \mathbb{E}\,\frac{g^2}{(1+w_j^2 q g^2)^2} > 0,$$

and $r_w(q) \uparrow \#\{j : w_j > 0\} \in \{1, 2, \dots\} \cup \{\infty\}$ as $q \to \infty$.

Proof. Part (i) is differentiation under the Gaussian integral. The integrand $g^2/(1+xg^2)$ is bounded by g^2 and strictly decreasing in x on $\{g \ne 0\}$, and its x-derivatives satisfy

$$\Big|\frac{\partial^k}{\partial x^k}\frac{g^2}{1+xg^2}\Big| = \frac{k!\, g^{2k+2}}{(1+xg^2)^{k+1}} \le k!\, g^{2k+2} \qquad (x \ge 0),$$

and differentiation under the expectation, dominated by these bounds, gives continuity, strict monotonicity, smoothness on $(0,\infty)$, one-sided derivatives at 0, and

$$J(0) = \mathbb{E}\, g^2 = 1, \qquad J'(0) = -\,\mathbb{E}\, g^4 = -3.$$

Part (ii) is uniform convergence of the series and of their derivatives. The j-th term of F_w is bounded by w_j^2, that of L_w by w_j, and their k-th derivatives in q are bounded, by the chain rule and part (i), by

$$k!\ \mathbb{E}\, g^{2k+2}\, w_j^{2k+2} \qquad \text{and} \qquad k!\ \mathbb{E}\, g^{2k+2}\, w_j^{2k+1}.$$

Since $\sum_j w_j < \infty$ and $w_j \le \sum_k w_k$, all these series converge uniformly on $[0,\infty)$, which gives continuity and smoothness. The values at 0 follow from $J(0) = 1$ and $J'(0) = -3$:

$$F_w(0) = \sum_j w_j^2 = s_2, \qquad F_w'(0) = -3\sum_j w_j^4 = -3s_4,$$

$$L_w(0) = \sum_j w_j, \qquad L_w'(0) = -3\sum_j w_j^3 = -3s_3.$$

Part (iii) is monotone convergence. Termwise,

$$qF_w(q) = \sum_j \mathbb{E}\,\frac{w_j^2 q g^2}{1 + w_j^2 q g^2}, \qquad \frac{\partial}{\partial x}\frac{xg^2}{1+xg^2} = \frac{g^2}{(1+xg^2)^2} > 0 \text{ on } \{g \ne 0\},$$

which gives $r_w(0) = 0$ and the displayed derivative, positive because at least one w_j is positive. As $q \to \infty$ each term with $w_j > 0$ increases to $\Pr(g \ne 0) = 1$ by monotone convergence, and so does the sum. □

Definition SB.1 (Weighted fixed-point system)**.** *Let $(\nu, \tilde{\nu})$ be finite positive Borel measures on $[0, \infty)$ with compact support, let w be as above, and let $z = -s$ with $s > 0$. The weighted fixed-point system at z is the system of equations in the four unknowns $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1) \in [0, \infty)^4$*

$$\begin{aligned} \sigma &= \int \frac{x}{\tilde{s}_1 x - z}\, \nu(dx), & \tilde{\sigma} &= \int \frac{x}{s_1 x - z}\, \tilde{\nu}(dx), \\ q &= -z\, \sigma \tilde{\sigma}, & \tilde{s}_1 &= -z\, \tilde{\sigma}\, F_w(q), \qquad s_1 = -z\, \sigma\, F_w(q). \end{aligned} \tag{SB.2}$$

For the spectral measures of eq. (SA.41) *the first two equations read*

$$\sigma = \frac{1}{B} \operatorname{Tr}\left(\Sigma_{\text{out}} (\tilde{s}_1 \Sigma_{\text{out}} - z\, \mathrm{Id}_{N_{\text{out}}})^{-1}\right), \qquad \tilde{\sigma} = \frac{1}{B} \operatorname{Tr}\left(\Sigma_{\text{in}} (s_1 \Sigma_{\text{in}} - z\, \mathrm{Id}_{N_{\text{in}}})^{-1}\right),$$

and we define the deterministic equivalents

$$\overline{R}(z) \overset{\text{def}}{=} \left(\tilde{s}_1(z)\, \Sigma_{\text{out}} - z\ \mathrm{Id}_{N_{\text{out}}}\right)^{-1}, \qquad \overline{\widetilde{R}}(z) \overset{\text{def}}{=} \left(s_1(z)\, \Sigma_{\text{in}} - z\ \mathrm{Id}_{N_{\text{in}}}\right)^{-1}, \tag{SB.3}$$

where $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1)(z)$ is the solution given by theorem SB.1.

In this appendix $\sigma(z)$, $\tilde{\sigma}(z)$, $\tilde{s}_1(z)$ and $s_1(z)$ always denote the solution of eq. (SB.2); the random traces are written out as $\frac{1}{B} \operatorname{Tr}(\Sigma_{\text{out}} R(z))$. The letters are those of section 2.

Multiplying the last two equations of eq. (SB.2) by σ and $\tilde{\sigma}$ gives the coupling

$$\tilde{s}_1\, \sigma = s_1\, \tilde{\sigma} = q F_w(q) = r_w(q), \tag{SB.4}$$

which at $\beta = 0$ is the third line of the system of section 2 (proposition SB.1).

Under Assumption S3 the pair $(\nu_{\text{out}}, \nu_{\text{in}})$ of eq. (SA.41) belongs to the class

$$\mathcal{K} \overset{\text{def}}{=} \left\{(\nu, \tilde{\nu}) :\ \nu, \tilde{\nu} \text{ finite positive Borel measures on } [0, C_\Sigma],\ \nu([0, C_\Sigma]) \le \gamma_\star,\ \tilde{\nu}([0, C_\Sigma]) \le \gamma_\star\right\}. \tag{SB.5}$$

Theorem SB.1 (The weighted branch)**.** *Let w be nonnegative, summable and not all zero, and let $C_\Sigma, \gamma_\star > 0$.*

(i) (Existence and uniqueness.) For every $(\nu, \tilde{\nu}) \in \mathcal{K}$ and every $s > 0$ the system eq. (SB.2) *at $z = -s$ has exactly one solution in $[0, \infty)^4$. It satisfies*

$$\sigma, \tilde{\sigma} \le \frac{\gamma_\star C_\Sigma}{s}, \qquad \tilde{s}_1, s_1 \le \gamma_\star C_\Sigma\, s_2, \qquad q \le \frac{(\gamma_\star C_\Sigma)^2}{s}. \tag{SB.6}$$

(ii) (Continuity.) Equip $\mathcal{K}$ with the topology of weak convergence against $C([0, C_\Sigma])$ in each coordinate. The solution map $(s, \nu, \tilde{\nu}) \mapsto (\sigma, \tilde{\sigma}, \tilde{s}_1, s_1)$ is continuous on $(0, \infty) \times \mathcal{K}$, and for every compact $I \subset (0, \infty)$ it is uniformly continuous in $s \in I$, uniformly over $(\nu, \tilde{\nu}) \in \mathcal{K}$.

(iii) (Stability under approximate roots.) Let $I \subset (0, \infty)$ be compact with $s_- \overset{\text{def}}{=} \min I$. For every $\epsilon > 0$ there is $\delta > 0$, depending only on $(\epsilon, I, C_\Sigma, \gamma_\star, w)$, with the following property. If $(\nu, \tilde{\nu}) \in \mathcal{K}$, $s \in I$, and $(\sigma', \tilde{\sigma}', a', c') \in [0, \gamma_\star C_\Sigma / s_-]^2 \times [0, \gamma_\star C_\Sigma s_2]^2$ satisfy, with $q' = s \sigma' \tilde{\sigma}'$,

$$\begin{aligned} \left|\sigma' - \int \frac{x}{s + a' x}\, \nu(dx)\right| &\le \delta, & \left|\tilde{\sigma}' - \int \frac{x}{s + c' x}\, \tilde{\nu}(dx)\right| &\le \delta, \\ |a' - s \tilde{\sigma}' F_w(q')| &\le \delta, & |c' - s \sigma' F_w(q')| &\le \delta, \end{aligned}$$

then $|\sigma' - \sigma| + |\tilde{\sigma}' - \tilde{\sigma}| + |a' - \tilde{s}_1| + |c' - s_1| \le \epsilon$, where $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1)$ is the solution of eq. (SB.2) *at $z = -s$ for $(\nu, \tilde{\nu})$.*

(iv) (Smoothness in s.) Fix $(\nu, \tilde{\nu}) \in \mathcal{K}$ with $\int x\, \nu(dx) > 0$ and $\int x\, \tilde{\nu}(dx) > 0$. Then $s \mapsto (\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ is C^∞ on $(0, \infty)$, and $|q'(s)| \le q(s)/s$.

The proof uses a compactness lemma for the class $\mathcal{K}$.

Lemma SB.2 (Compactness of the spectral class)**.** *Every sequence in $\mathcal{K}$ has a subsequence that converges, weakly against $C([0, C_\Sigma])$ in each coordinate, to an element of $\mathcal{K}$. If $\nu_n \to \nu$ weakly in this sense, $s_n \to s > 0$ and $a_n \to a \ge 0$, then $\int \frac{x}{s_n + a_n x}\, \nu_n(dx) \to \int \frac{x}{s + a x}\, \nu(dx)$.*

Proof. By the Riesz representation theorem [56, Theorem 6.19], a finite positive Borel measure ν on $[0, C_\Sigma]$ is the same as a positive bounded linear functional on $C([0, C_\Sigma])$, with norm $\nu([0, C_\Sigma])$. The closed ball of radius $\gamma_\star$ in the dual of $C([0, C_\Sigma])$ is weak-$*$ compact by the Banach–Alaoglu theorem [57, Theorem 3.15] and, since $C([0, C_\Sigma])$ is separable, weak-$*$ metrizable [57, Theorem 3.16], hence weak-$*$ sequentially compact. Positivity of a functional ℓ and the bound $\ell(1) \le \gamma_\star$ are preserved under weak-$*$ limits, so the limit is again in $\mathcal{K}$.

The second claim is continuity of the integrand in its parameters. The function $\phi_{s,a}(x) \stackrel{\text{def}}{=} x/(s+ax)$ is continuous on $[0, C_\Sigma]$, with parameter derivatives

$$\partial_s \phi_{s,a}(x) = -\frac{x}{(s+ax)^2}, \qquad \partial_a \phi_{s,a}(x) = -\frac{x^2}{(s+ax)^2}.$$

For $s_n, s \ge s_\circ > 0$ and $a_n, a \ge 0$ the mean value theorem in (s, a) therefore gives

$$\sup_{x \in [0, C_\Sigma]} |\phi_{s_n,a_n}(x) - \phi_{s,a}(x)| \le \frac{C_\Sigma |s_n - s| + C_\Sigma^2 |a_n - a|}{s_\circ^2}.$$

Splitting the difference of the integrals into a change of integrand and a change of measure,

$$\Big| \int \phi_{s_n,a_n}\, d\nu_n - \int \phi_{s,a}\, d\nu \Big| \le \gamma_\star \, \|\phi_{s_n,a_n} - \phi_{s,a}\|_\infty + \Big| \int \phi_{s,a}\, d\nu_n - \int \phi_{s,a}\, d\nu \Big| \longrightarrow 0,$$

the first term by the displayed bound and the second by weak convergence. □

Proof of theorem SB.1. Throughout, $z = -s$, and we write $r = r_w$ from lemma SB.1(iii). Set $p \stackrel{\text{def}}{=} \nu((0, C_\Sigma])$ and $\tilde{p} \stackrel{\text{def}}{=} \tilde{\nu}((0, C_\Sigma])$.

Degenerate cases. If $p = 0$ the system has the unique solution

$$\sigma = 0, \qquad q = 0, \qquad s_1 = 0, \qquad \tilde{\sigma} = \frac{1}{s} \int x\, \tilde{\nu}(dx), \qquad \tilde{s}_1 = s\tilde{\sigma} F_w(0) = s\tilde{\sigma}\, s_2.$$

Indeed ν is then a multiple of δ_0, so the first equation forces $\sigma = 0$ whatever $\tilde{s}_1$ is, and the remaining equations determine the other unknowns in the displayed order. The case $\tilde{p} = 0$ is symmetric. Assume now $p > 0$ and $\tilde{p} > 0$.

Reduction to one scalar equation. The two trace equations are inverted through the functions

$$H(x) \stackrel{\text{def}}{=} \int \frac{x\lambda}{s + x\lambda}\, \nu(d\lambda), \qquad \widetilde{H}(x) \stackrel{\text{def}}{=} \int \frac{x\lambda}{s + x\lambda}\, \tilde{\nu}(d\lambda) \qquad (x \ge 0). \tag{SB.7}$$

The function H is continuous and strictly increasing on $[0, \infty)$, since $p > 0$ and the integrand is strictly increasing in x for $\lambda > 0$, with

$$H(0) = 0, \qquad H(x) \uparrow p \ \ (x \to \infty)$$

by monotone convergence. Its quotient

$$\frac{H(x)}{x} = \int \frac{\lambda}{s + x\lambda}\, \nu(d\lambda)$$

is strictly decreasing in $x > 0$ and tends to m_1/s as $x \downarrow 0$, where $m_1 \stackrel{\text{def}}{=} \int \lambda\, \nu(d\lambda) > 0$. Let $X : [0, p) \to [0, \infty)$ be the inverse of H. Then X is continuous and strictly increasing, $X(\rho) \to \infty$ as $\rho \uparrow p$, and

$$\frac{X(\rho)}{\rho} = \Big(\frac{H(X(\rho))}{X(\rho)} \Big)^{-1}$$

is strictly increasing on $(0, p)$ with limit s/m_1 at 0^+. The same holds for $\widetilde{H}$, $\widetilde{X}$ and $\tilde{m}_1$.

Every nonnegative solution is determined by q. Let $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1)$ be a solution in $[0, \infty)^4$. Since $p > 0$, the first equation gives $\sigma > 0$; likewise $\tilde{\sigma} > 0$; hence $q > 0$, $F_w(q) > 0$ and $\tilde{s}_1, s_1 > 0$. The coupling eq. (SB.4) reads

$$H(\tilde{s}_1) = \tilde{s}_1 \sigma = r(q), \qquad \widetilde{H}(s_1) = s_1 \tilde{\sigma} = r(q),$$

so $r(q) < \min(p, \tilde{p})$ and

$$\tilde{s}_1 = X(r(q)), \qquad s_1 = \widetilde{X}(r(q)), \qquad \sigma = \frac{r(q)}{\tilde{s}_1}, \qquad \tilde{\sigma} = \frac{r(q)}{s_1}.$$

Substituting into $q = s\sigma\tilde{\sigma}$ gives the scalar equation

$$\Phi(q) \stackrel{\text{def}}{=} \frac{X(r(q))}{r(q)} \cdot \frac{\widetilde{X}(r(q))}{r(q)} - \frac{s}{q} = 0, \qquad q \in Q \stackrel{\text{def}}{=} \{q > 0 : \ r(q) < \min(p, \tilde{p})\}. \tag{SB.8}$$

Conversely, every solution of the scalar equation yields a solution of the system. If $q \in Q$ solves eq. (SB.8), define $\tilde{s}_1$, s_1, σ, $\tilde{\sigma}$ by the last display. The first equation of the system holds because

$$\sigma = \frac{H(\tilde{s}_1)}{\tilde{s}_1} = \int \frac{\lambda}{s + \tilde{s}_1 \lambda} \, \nu(d\lambda),$$

and likewise the second. The crossing equation gives

$$s\sigma\tilde{\sigma} = \frac{s \, r(q)^2}{\tilde{s}_1 s_1} = q,$$

which is the third equation. Finally, the crossing equation and $r = qF_w$ give

$$\tilde{s}_1 s_1 = \frac{s \, r(q)^2}{q} = sqF_w(q)^2,$$

so

$$\tilde{s}_1 = \frac{sqF_w(q)^2}{s_1} = s \, \frac{r(q)}{s_1} \, F_w(q) = s\tilde{\sigma}F_w(q),$$

and symmetrically $s_1 = s\sigma F_w(q)$. Hence nonnegative solutions of eq. (SB.2) are in bijection with solutions of eq. (SB.8).

Exactly one crossing. The scalar equation has exactly one solution because Φ is strictly increasing and changes sign. Since r is continuous and strictly increasing from 0, the set Q is an interval $(0, q_\dagger)$ with $q_\dagger \in (0, \infty]$. On Q the first term of Φ is positive, continuous and nondecreasing, as a product of two positive nondecreasing functions of $r(q)$, while s/q is continuous and strictly decreasing; so Φ is continuous and strictly increasing on Q. At the left endpoint,

$$\lim_{q \downarrow 0} \frac{X(r(q))}{r(q)} \cdot \frac{\widetilde{X}(r(q))}{r(q)} = \frac{s^2}{m_1 \tilde{m}_1} < \infty, \qquad \lim_{q \downarrow 0} \frac{s}{q} = \infty,$$

so $\Phi(q) \to -\infty$ as $q \downarrow 0$. At the right endpoint there are two cases. If $q_\dagger < \infty$ then $r(q) \uparrow \min(p, \tilde{p})$ as $q \uparrow q_\dagger$, so $X(r(q))$ or $\widetilde{X}(r(q))$ tends to infinity while the other factor and s/q stay bounded, and $\Phi(q) \to +\infty$. If $q_\dagger = \infty$ then $s/q \to 0$ while the first term stays at least $s^2/(m_1\tilde{m}_1) > 0$, so $\liminf_{q \to \infty} \Phi(q) > 0$. In both cases the intermediate value theorem and strict monotonicity give exactly one zero. This proves existence and uniqueness.

The a priori bounds eq. (SB.6) follow from the three inequalities

$$\frac{x}{s + ax} \le \frac{x}{s} \le \frac{C_\Sigma}{s} \ \ (x \in [0, C_\Sigma], \ a \ge 0), \qquad F_w \le s_2, \qquad q = s\sigma\tilde{\sigma},$$

the second by lemma SB.1, applied to the two trace equations, to $\tilde{s}_1 = s\tilde{\sigma}F_w(q)$, and to q.

Continuity and stability. We prove (iii) by compactness and uniqueness. Part (ii) is the special case $\delta = 0$ of the same argument with varying $(s, \nu, \tilde{\nu})$. The modulus in $s \in I$ uniform over $\mathcal{K}$ follows from continuity by the same compactness: sequences $s_n, s'_n \in I$ with $|s_n - s'_n| \to 0$ and $(\nu_n, \tilde{\nu}_n) \in \mathcal{K}$ have a subsequence along which $(s_n, \nu_n, \tilde{\nu}_n)$ and $(s'_n, \nu_n, \tilde{\nu}_n)$ converge to the same point of $I \times \mathcal{K}$. Suppose (iii) fails for some $\epsilon > 0$. Then there are sequences $(\nu_n, \tilde{\nu}_n) \in \mathcal{K}$, $s_n \in I$ and tuples $(\sigma'_n, \tilde{\sigma}'_n, a'_n, c'_n)$ in the stated box whose four residuals are at most $\delta_n \to 0$, but with

$$|\sigma'_n - \sigma_n| + |\tilde{\sigma}'_n - \tilde{\sigma}_n| + |a'_n - \tilde{s}_{1,n}| + |c'_n - s_{1,n}| > \epsilon$$

for the exact solutions $(\sigma_n, \tilde{\sigma}_n, \tilde{s}_{1,n}, s_{1,n})$.

All sequences have convergent subsequences. By lemma SB.2 and the compactness of I and of the box, in which the exact solutions also lie by eq. (SB.6), we may pass to a subsequence along which

$$(\nu_n, \tilde{\nu}_n) \to (\nu, \tilde{\nu}) \in \mathcal{K} \text{ weakly}, \qquad s_n \to s \in I,$$
$$(\sigma'_n, \tilde{\sigma}'_n, a'_n, c'_n) \to (\sigma', \tilde{\sigma}', a', c'), \qquad (\sigma_n, \tilde{\sigma}_n, \tilde{s}_{1,n}, s_{1,n}) \to (\sigma, \tilde{\sigma}, a, c).$$

Both limits solve the limit system. By lemma SB.2 and the continuity of F_w, each of the four equations of eq. (SB.2) passes to the limit along the subsequence, with residuals $\delta_n \to 0$ for the approximate tuples and residuals 0 for the exact ones. Hence both limit tuples solve the system at $z = -s$ for $(\nu, \tilde{\nu})$. By uniqueness they coincide, so the displayed distance tends to 0 along the subsequence, which contradicts its being larger than ϵ.

Smoothness in s. Here $p, \tilde{p} > 0$ by hypothesis and the measures are fixed. The dependence on s is removed from the inverse functions by scaling. Write $H(x, s)$ for the function eq. (SB.7) at the spectral parameter s and $X(\rho, s)$ for its inverse. Since

$$H(x, s) = \int \frac{(x/s)\lambda}{1 + (x/s)\lambda}\, \nu(d\lambda) = H(x/s, 1),$$

one has $X(\rho, s) = sX(\rho, 1)$. Put $\mathsf{A}(\rho) \stackrel{\text{def}}{=} X(\rho, 1)$ and $\widetilde{\mathsf{A}}(\rho) \stackrel{\text{def}}{=} \widetilde{X}(\rho, 1)$. The implicit function theorem applied to $H(x, 1) = \rho$, whose x-derivative

$$\partial_x H(x, 1) = \int \frac{\lambda}{(1 + x\lambda)^2}\, \nu(d\lambda)$$

is positive and smooth in x, shows that A is C^∞ on $(0, p)$, and likewise $\widetilde{\mathsf{A}}$.

The crossing equation becomes an implicit equation with a positive q-derivative. Write r for $r(q) = qF_w(q)$. Then eq. (SB.8) reads

$$\frac{s^2 \mathsf{A}(r)\widetilde{\mathsf{A}}(r)}{r^2} = \frac{s}{q},$$

that is, $\mathsf{G}(q, s) = 0$ with

$$\mathsf{G}(q, s) \stackrel{\text{def}}{=} \log s + \log \mathsf{A}(r(q)) + \log \widetilde{\mathsf{A}}(r(q)) - \log q - 2 \log F_w(q).$$

Since $\mathsf{A}(\rho)/\rho$ is nondecreasing, $\mathsf{A}'/\mathsf{A} \geq 1/\rho$, and likewise for $\widetilde{\mathsf{A}}$. With $r' = F_w + qF'_w$ this gives

$$\partial_q \mathsf{G} = \Big(\frac{\mathsf{A}'}{\mathsf{A}} + \frac{\widetilde{\mathsf{A}}'}{\widetilde{\mathsf{A}}}\Big) r' - \frac{1}{q} - \frac{2F'_w}{F_w} \;\geq\; \frac{2r'}{r} - \frac{1}{q} - \frac{2F'_w}{F_w} = \frac{2}{q} + \frac{2F'_w}{F_w} - \frac{1}{q} - \frac{2F'_w}{F_w} = \frac{1}{q} > 0,$$

where the inequality uses $\mathsf{A}'/\mathsf{A} \geq 1/r$ and $r' > 0$, and the first equality uses $r'/r = 1/q + F'_w/F_w$. The implicit function theorem concludes. Since $\partial_s \mathsf{G} = 1/s$ and G is C^∞, the function $q(s)$ is C^∞ with

$$q'(s) = -\frac{\partial_s \mathsf{G}}{\partial_q \mathsf{G}}, \qquad |q'(s)| = \frac{1/s}{\partial_q \mathsf{G}} \leq \frac{q}{s}.$$

The remaining coordinates are the C^∞ compositions

$$\tilde{s}_1 = s\, \mathsf{A}(r(q(s))), \qquad s_1 = s\, \widetilde{\mathsf{A}}(r(q(s))), \qquad \sigma = \frac{r(q(s))}{\tilde{s}_1}, \qquad \tilde{\sigma} = \frac{r(q(s))}{s_1}. \qquad \square$$

The case $\beta = 0$. At $\beta = 0$ the only weight is $w_0 = 1$, so that

$$F_0 = L_0 = J, \qquad \tilde{s}_1 \sigma = s_1 \tilde{\sigma} = qJ(q),$$

the second identity being the coupling eq. (SB.4). The next proposition identifies $qJ(q)$ with the closure function f of section 2.

Proposition SB.1 (No-momentum closure)**.** *For $q > 0$ put $\xi \stackrel{\text{def}}{=} (2q)^{-1/2}$. Then*

$$qJ(q) = \mathbb{E}\, \frac{qg^2}{1 + qg^2} = \mathbb{E}\, \frac{g^2}{g^2 + 2\xi^2} = 1 - \sqrt{\pi}\, \xi\, e^{\xi^2} \operatorname{erfc}(\xi). \tag{SB.9}$$

Proof. The first two equalities are the definition of J and $q = 1/(2\xi^2)$.

The third equality is a Laplace-transform computation. Since

$$\mathbb{E}\,\frac{qg^2}{1+qg^2} = 1 - \mathbb{E}\,\frac{1}{1+qg^2},$$

it suffices to compute the last expectation. It is computed from the Laplace representation and the Gaussian integral

$$\frac{1}{1+qg^2} = \int_0^\infty e^{-u(1+qg^2)}\,du, \qquad \mathbb{E}\,e^{-uqg^2} = (1+2uq)^{-1/2},$$

which give

$$\mathbb{E}\,\frac{1}{1+qg^2} = \int_0^\infty e^{-u}\;\mathbb{E}\,e^{-uqg^2}\,du = \int_0^\infty e^{-u}\Big(1+\frac{u}{\xi^2}\Big)^{-1/2}du.$$

The substitution $u = \rho^2 - \xi^2$ with $\rho \in [\xi, \infty)$ turns the last integral into

$$\int_\xi^\infty e^{-(\rho^2-\xi^2)}\,\frac{\xi}{\rho}\,2\rho\,d\rho = 2\xi e^{\xi^2}\int_\xi^\infty e^{-\rho^2}\,d\rho = \sqrt{\pi}\,\xi\,e^{\xi^2}\,\mathrm{erfc}(\xi). \qquad \square$$

Remark SB.1 (Dictionary to the no-momentum system)**.** *At $\beta = 0$, with $z = -s$, the system eq. (SB.2) for the spectral measures eq. (SA.41) is*

$$\sigma = \frac{1}{B}\,\mathrm{Tr}\,\big(\Sigma_{\mathrm{out}}(\tilde{s}_1\Sigma_{\mathrm{out}} - z\,\mathrm{Id})^{-1}\big), \qquad \tilde{\sigma} = \frac{1}{B}\,\mathrm{Tr}\,\big(\Sigma_{\mathrm{in}}(s_1\Sigma_{\mathrm{in}} - z\,\mathrm{Id})^{-1}\big),$$
$$\tilde{s}_1\sigma = s_1\tilde{\sigma} = qJ(q), \qquad q = -z\sigma\tilde{\sigma}.$$

Put

$$\xi \overset{\mathrm{def}}{=} \frac{1}{\sqrt{-2z\sigma\tilde{\sigma}}} = (2q)^{-1/2}, \qquad f(\xi) \overset{\mathrm{def}}{=} \mathbb{E}\,\frac{V^2}{V^2+2\xi^2} \quad (V \sim \mathcal{N}(0,1)).$$

By proposition SB.1,

$$qJ(q) = f(\xi) = 1 - \sqrt{\pi}\,\xi\,e^{\xi^2}\,\mathrm{erfc}(\xi),$$

which is the third line of the fixed-point system of section 2; the relation $s_1 = \tilde{s}_1\sigma/\tilde{\sigma}$ is its coupling relation. The deterministic equivalents eq. (SB.3) are the ones displayed there. The trace identity

$$\tilde{s}_1\sigma - \frac{N_{\mathrm{out}}}{B} = z\,\frac{1}{B}\,\mathrm{Tr}\,\overline{R}(z)$$

is the sum over the eigenvalues μ of Σ_{out} of

$$\frac{\tilde{s}_1\mu}{\tilde{s}_1\mu - z} - 1 = \frac{z}{\tilde{s}_1\mu - z}.$$

For $\beta > 0$ the single function f is replaced by $qF_\beta(q)$, and the drift section uses L_β in place of J in the source factor.

Proposition SB.2 (Normalization checks)**.** *Let $(\nu, \tilde{\nu}) \in \mathcal{K}$ and $s > 0$.*

(i) (One weight.) Let $w = (\varpi, 0, 0, \dots)$ with $\varpi > 0$, and let $(\sigma_1, \tilde{\sigma}_1, \tilde{s}_{1,1}, s_{1,1})(-s')$ denote the solution for the unit weight $(1, 0, 0, \dots)$ at the spectral parameter $s' = s/\varpi^2$. Then the solution for w at s is

$$\sigma = \varpi^{-2}\sigma_1(-s'), \qquad \tilde{\sigma} = \varpi^{-2}\tilde{\sigma}_1(-s'), \qquad q = \varpi^{-2}q_1(-s'),$$
$$\tilde{s}_1 = \varpi^2\tilde{s}_{1,1}(-s'), \qquad s_1 = \varpi^2 s_{1,1}(-s'),$$

and $\overline{R}_w(-s) = \varpi^{-2}\overline{R}_1(-s')$.

(ii) (K equal weights.) Let $w_j = 1/K$ for $0 \le j < K$ and $w_j = 0$ otherwise. Then $F_w(q) = K^{-1}J(q/K^2)$, and the solution for w at $(\nu, \tilde{\nu})$ is related to the solution $(\sigma', \tilde{\sigma}', \tilde{s}_1', s_1')$ for the unit weight at the measures $(\nu/K, \tilde{\nu}/K)$ by

$$\sigma = K\sigma', \qquad \tilde{\sigma} = K\tilde{\sigma}', \qquad q = K^2q', \qquad \tilde{s}_1 = \tilde{s}_1', \qquad s_1 = s_1'.$$

Proof. In both parts the displayed tuple is checked against the four equations of eq. (SB.2) and then identified with the solution by the uniqueness in theorem SB.1(i).

For (i), the weight function is $F_w(q) = \varpi^2 J(\varpi^2 q)$. The first trace equation holds at s because, for every $x \geq 0$,

$$\frac{x}{s + \varpi^2 \tilde{s}_{1,1} x} = \varpi^{-2} \frac{x}{s' + \tilde{s}_{1,1} x},$$

and the second likewise. The equation for q holds because

$$s\sigma\tilde{\sigma} = \varpi^2 s' \cdot \varpi^{-4} \sigma_1 \tilde{\sigma}_1 = \varpi^{-2} q_1.$$

The equation for $\tilde{s}_1$ holds because

$$s\tilde{\sigma} F_w(q) = \varpi^2 s' \cdot \varpi^{-2} \tilde{\sigma}_1 \cdot \varpi^2 J(\varpi^2 \varpi^{-2} q_1) = \varpi^2 \, s' \tilde{\sigma}_1 J(q_1) = \varpi^2 \tilde{s}_{1,1},$$

and the equation for s_1 symmetrically. The resolvent relation follows from

$$\big(s \, \mathrm{Id} + \varpi^2 \tilde{s}_{1,1} \Sigma_{\mathrm{out}}\big)^{-1} = \varpi^{-2} \big(s' \, \mathrm{Id} + \tilde{s}_{1,1} \Sigma_{\mathrm{out}}\big)^{-1}.$$

For (ii), the weight function is

$$F_w(q) = \sum_{j<K} K^{-2} J(q/K^2) = K^{-1} J(q/K^2).$$

The first trace equation holds because

$$\int \frac{x}{s + \tilde{s}_1' x} \, \nu(dx) = K \int \frac{x}{s + \tilde{s}_1' x} \, \frac{\nu(dx)}{K} = K\sigma',$$

and the second likewise. The equation for q holds because

$$s\sigma\tilde{\sigma} = K^2 s\sigma'\tilde{\sigma}' = K^2 q',$$

and the equation for $\tilde{s}_1$ because

$$s\tilde{\sigma} F_w(q) = s \, K \tilde{\sigma}' \, K^{-1} J(q') = s\tilde{\sigma}' J(q') = \tilde{s}_1',$$

with the equation for s_1 symmetric. □

Remark SB.2 (No scalar effective batch for $\beta > 0$)**.** *By lemma SB.1(ii), the value and the slope of F_w at 0 are s_2 and $-3s_4$. For K equal weights, $s_4/s_2^3 = 1$, so a uniform average over K batches, equivalently a batch of size KB by proposition SB.2(ii), is determined to first order at $q = 0$ by s_2 alone. For the EMA weights with $\beta \in (0, 1)$,*

$$\frac{s_4}{s_2^3} = \frac{(1+\beta)^2}{1+\beta^2} > 1, \qquad \frac{s_3}{s_2^2} = \frac{(1+\beta)^2}{1+\beta+\beta^2} > 1,$$

so no choice of an effective batch size $B_{\mathrm{eff}} = B/s_2$ reproduces the function F_β, nor L_β, beyond the value at $q = 0$. This is why the system is stated with the full weight sequence.

SB.2 Finite identities

The closure theorem is proved by deleting one sample from the buffer. This subsection collects the exact finite-dimensional identities and bounds that the proof uses. Throughout, $s > 0$ and $M \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ is a deterministic or random matrix; we write

$$R_s(M) \stackrel{\mathrm{def}}{=} (MM^\top + s \, \mathrm{Id}_{N_{\mathrm{out}}})^{-1}, \quad \widetilde{R}_s(M) \stackrel{\mathrm{def}}{=} (M^\top M + s \, \mathrm{Id}_{N_{\mathrm{in}}})^{-1}, \quad \mathsf{X}_s(M) \stackrel{\mathrm{def}}{=} R_s(M) M. \tag{SB.10}$$

The dynamics section also needs these objects at complex parameters in the sector

$$\mathbb{S} \stackrel{\mathrm{def}}{=} \{z \in \mathbb{C} : \ \mathrm{Re}\, z > 0, \ |\mathrm{Im}\, z| \leq \mathrm{Re}\, z/2\}, \tag{SB.11}$$

which contains $(0, \infty)$. For $z \in \mathbb{S}$ we define $R_z(M)$, $\widetilde{R}_z(M)$ and $\mathsf{X}_z(M)$ by eq. (SB.10) with z in place of s; the inverses exist by lemma SB.3. Transposes of complex matrices are not conjugated.

Lemma SB.3 (Resolvent identities)**.** *For every M and $s > 0$,*

$$\|R_s(M)\| \le \frac{1}{s}, \qquad \|\widetilde{R}_s(M)\| \le \frac{1}{s}, \qquad \|\mathsf{X}_s(M)\| \le \frac{1}{2\sqrt{s}},$$
$$R_s(M)M = M\widetilde{R}_s(M), \qquad M^\top R_s(M)M = \mathrm{Id}_{N_{\mathrm{in}}} - s\widetilde{R}_s(M),$$
$$R_s(M)MM^\top = \mathrm{Id}_{N_{\mathrm{out}}} - sR_s(M),$$

and for $s, s' > 0$,

$$\|R_s(M) - R_{s'}(M)\| \le \frac{|s - s'|}{ss'}.$$

For $z \in \mathbb{S}$ with $x \stackrel{\mathrm{def}}{=} \mathrm{Re}\, z$, the matrices $MM^\top + z\,\mathrm{Id}_{N_{\mathrm{out}}}$ and $M^\top M + z\,\mathrm{Id}_{N_{\mathrm{in}}}$ are invertible, the three identities hold with z in place of s, and

$$\|R_z(M)\| \le \frac{1}{x}, \qquad \|\widetilde{R}_z(M)\| \le \frac{1}{x}, \qquad \|\mathsf{X}_z(M)\| \le \frac{1}{2\sqrt{x}}, \qquad \frac{|z|}{x} \le \frac{\sqrt{5}}{2}.$$

Proof. The norm bounds follow from a singular value decomposition $M = \sum_k s_k\, p_k \otimes r_k$. In it, the three matrices are

$$R_s(M) = \sum_k \frac{p_k \otimes p_k}{s_k^2 + s} + \frac{\Pi_{\mathrm{out}}}{s}, \qquad \widetilde{R}_s(M) = \sum_k \frac{r_k \otimes r_k}{s_k^2 + s} + \frac{\Pi_{\mathrm{in}}}{s},$$
$$\mathsf{X}_s(M) = \sum_k \frac{s_k}{s_k^2 + s}\, p_k \otimes r_k,$$

where Π_{out} and Π_{in} are the orthogonal projections onto the left and right null spaces of M. The coefficients satisfy

$$\frac{1}{s_k^2 + s} \le \frac{1}{s}, \qquad \frac{s_k}{s_k^2 + s} \le \frac{1}{2\sqrt{s}},$$

the second by the inequality $s_k^2 + s \ge 2s_k\sqrt{s}$ between arithmetic and geometric means.

The intertwining identity follows from the elementary identity

$$(MM^\top + s\,\mathrm{Id}_{N_{\mathrm{out}}})M = M(M^\top M + s\,\mathrm{Id}_{N_{\mathrm{in}}}).$$

Multiplying it by $R_s(M)$ on the left and by $\widetilde{R}_s(M)$ on the right gives $R_s(M)M = M\widetilde{R}_s(M)$.

The second identity follows from the first. Multiplying $R_s(M)M = M\widetilde{R}_s(M)$ on the left by $M^\top$ gives

$$M^\top R_s(M)M = M^\top M\, \widetilde{R}_s(M) = (M^\top M + s\,\mathrm{Id}_{N_{\mathrm{in}}} - s\,\mathrm{Id}_{N_{\mathrm{in}}})\widetilde{R}_s(M) = \mathrm{Id}_{N_{\mathrm{in}}} - s\widetilde{R}_s(M).$$

The third identity is the same computation with $MM^\top$:

$$R_s(M)MM^\top = R_s(M)(MM^\top + s\,\mathrm{Id}_{N_{\mathrm{out}}} - s\,\mathrm{Id}_{N_{\mathrm{out}}}) = \mathrm{Id}_{N_{\mathrm{out}}} - sR_s(M).$$

The Lipschitz bound is the resolvent identity. For $s, s' > 0$,

$$R_s(M) - R_{s'}(M) = R_s(M)\big((MM^\top + s'\,\mathrm{Id}) - (MM^\top + s\,\mathrm{Id})\big)R_{s'}(M) = (s' - s)\, R_s(M)R_{s'}(M),$$

and the norm bound follows from $\|R_s(M)\| \le 1/s$ and $\|R_{s'}(M)\| \le 1/s'$.

The sector statement is the same computation with z in place of s. In the singular value decomposition, $MM^\top + z\,\mathrm{Id}$ and $M^\top M + z\,\mathrm{Id}$ have the real orthonormal eigenvectors p_k, r_k and the null-space vectors, with eigenvalues $s_k^2 + z$ and z, whose real parts are at least $x > 0$; so both matrices are invertible, and $R_z(M)$, $\widetilde{R}_z(M)$, $\mathsf{X}_z(M)$ are the displayed sums with s replaced by z. Since $R_z(M)$ and $\widetilde{R}_z(M)$ are diagonal in real orthonormal bases, their norms are the largest moduli of their eigenvalues, and the singular values of $\mathsf{X}_z(M)$ are $s_k/|s_k^2 + z|$. The bound $|s_k^2 + z| \ge \mathrm{Re}(s_k^2 + z) = s_k^2 + x$ gives

$$\frac{1}{|s_k^2 + z|} \le \frac{1}{x}, \qquad \frac{s_k}{|s_k^2 + z|} \le \frac{s_k}{s_k^2 + x} \le \frac{1}{2\sqrt{x}}.$$

The identities are algebraic and hold verbatim. Finally $|z|^2 = x^2 + (\mathrm{Im}\, z)^2 \le \frac{5}{4}x^2$. □

Lemma SB.4 (Rank-two deletion). *Let $M = M_0 + \mathsf{h}\,\mathsf{u}\otimes\mathsf{v}$ with $M_0 \in \mathbb{R}^{N_{\text{out}}\times N_{\text{in}}}$, $\mathsf{h}\in\mathbb{R}$, $\mathsf{u}\in\mathbb{R}^{N_{\text{out}}}$, $\mathsf{v}\in\mathbb{R}^{N_{\text{in}}}$, and $s>0$. Write $R_0 = R_s(M_0)$, $\widetilde{R}_0 = \widetilde{R}_s(M_0)$, $\mathsf{X}_0 = \mathsf{X}_s(M_0)$, $R = R_s(M)$, and*

$$\mathsf{U} \stackrel{\text{def}}{=} \mathsf{u}^\top R_0 \mathsf{u}, \qquad \mathsf{V} \stackrel{\text{def}}{=} \mathsf{v}^\top \widetilde{R}_0 \mathsf{v}, \qquad \mathsf{C} \stackrel{\text{def}}{=} \mathsf{u}^\top \mathsf{X}_0 \mathsf{v}, \qquad \Lambda \stackrel{\text{def}}{=} (1+\mathsf{h}\mathsf{C})^2 + s\mathsf{h}^2\,\mathsf{U}\mathsf{V}. \tag{SB.12}$$

Then $\Lambda > 0$, and with

$$\alpha_1 \stackrel{\text{def}}{=} \frac{\mathsf{h}(1+\mathsf{h}\mathsf{C})}{\Lambda}, \qquad \alpha_2 \stackrel{\text{def}}{=} \frac{\mathsf{h}^2\,\mathsf{U}}{\Lambda}, \qquad \alpha_3 \stackrel{\text{def}}{=} -\frac{s\mathsf{h}^2\,\mathsf{V}}{\Lambda}, \tag{SB.13}$$

one has

$$R - R_0 = \alpha_3\,(R_0\mathsf{u})\otimes(R_0\mathsf{u}) - \alpha_1\,(R_0\mathsf{u})\otimes(\mathsf{X}_0\mathsf{v}) - \alpha_1\,(\mathsf{X}_0\mathsf{v})\otimes(R_0\mathsf{u}) + \alpha_2\,(\mathsf{X}_0\mathsf{v})\otimes(\mathsf{X}_0\mathsf{v}), \tag{SB.14}$$

$$\mathsf{h}\,RM\mathsf{v} = -\alpha_3\,R_0\mathsf{u} + \alpha_1\,\mathsf{X}_0\mathsf{v}. \tag{SB.15}$$

Moreover, if Λ' denotes the quantity eq. (SB.12) computed from M in place of M_0 and $-\mathsf{h}$ in place of h, then $\Lambda\Lambda' = 1$, and consequently

$$\Lambda^{-1} \le \Big(1 + \frac{3}{4\sqrt{s}}\,|\mathsf{h}|\,\mathsf{N}\Big)^2, \qquad \mathsf{N} \stackrel{\text{def}}{=} \|\mathsf{u}\|^2 + \|\mathsf{v}\|^2. \tag{SB.16}$$

The same holds at $z\in\mathbb{S}$ in place of s, with $x \stackrel{\text{def}}{=} \operatorname{Re} z$, in the following form. Define $R_0, \widetilde{R}_0, \mathsf{X}_0, R$ and $\mathsf{U},\mathsf{V},\mathsf{C},\Lambda,\alpha_1,\alpha_2,\alpha_3$ by the formulas above with z in place of s; they are complex. Then $\Lambda\neq 0$, eqs. (SB.14) and (SB.15) hold, $\Lambda\Lambda' = 1$, and

$$|\Lambda|^{-1} \le \Big(1 + \frac{|\mathsf{h}|\,\mathsf{N}}{\sqrt{x}}\Big)^2. \tag{SB.17}$$

Proof. For $K\in\mathbb{R}^{N_{\text{out}}\times N_{\text{in}}}$ let

$$\mathcal{L}(K) \stackrel{\text{def}}{=} \begin{pmatrix} s\,\mathrm{Id}_{N_{\text{out}}} & K \\ K^\top & -\mathrm{Id}_{N_{\text{in}}} \end{pmatrix}.$$

Block inversion with the Schur complement $s\,\mathrm{Id} + KK^\top$ of the block $-\mathrm{Id}$ gives, for every K,

$$\mathcal{L}(K)^{-1} = \begin{pmatrix} R_s(K) & \mathsf{X}_s(K) \\ \mathsf{X}_s(K)^\top & -\mathrm{Id}_{N_{\text{in}}} + K^\top R_s(K) K \end{pmatrix} = \begin{pmatrix} R_s(K) & \mathsf{X}_s(K) \\ \mathsf{X}_s(K)^\top & -s\widetilde{R}_s(K) \end{pmatrix}, \tag{SB.18}$$

where the second form uses lemma SB.3. The rank-one change of M is a rank-two change of $\mathcal{L}$:

$$\mathcal{L}(M) = \mathcal{L}(M_0) + \mathsf{h}\big(\tilde{\mathsf{u}}\otimes\tilde{\mathsf{v}} + \tilde{\mathsf{v}}\otimes\tilde{\mathsf{u}}\big) = \mathcal{L}(M_0) + PSP^\top,$$

with

$$\tilde{\mathsf{u}} \stackrel{\text{def}}{=} \begin{pmatrix}\mathsf{u}\\ 0\end{pmatrix}, \qquad \tilde{\mathsf{v}} \stackrel{\text{def}}{=} \begin{pmatrix}0\\ \mathsf{v}\end{pmatrix}, \qquad P \stackrel{\text{def}}{=} [\tilde{\mathsf{u}}\ \tilde{\mathsf{v}}], \qquad S \stackrel{\text{def}}{=} \mathsf{h}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}.$$

Assume first $\mathsf{h}\neq 0$. By eq. (SB.18),

$$W \stackrel{\text{def}}{=} P^\top\mathcal{L}(M_0)^{-1}P = \begin{pmatrix}\mathsf{U} & \mathsf{C}\\ \mathsf{C} & -s\mathsf{V}\end{pmatrix}, \qquad S^{-1} + W = \begin{pmatrix}\mathsf{U} & \mathsf{C}+\mathsf{h}^{-1}\\ \mathsf{C}+\mathsf{h}^{-1} & -s\mathsf{V}\end{pmatrix}.$$

The determinant of $S^{-1}+W$ is expressed through Λ:

$$\det(S^{-1}+W) = -s\mathsf{U}\mathsf{V} - \big(\mathsf{C}+\mathsf{h}^{-1}\big)^2 = -\frac{s\mathsf{h}^2\mathsf{U}\mathsf{V} + (1+\mathsf{h}\mathsf{C})^2}{\mathsf{h}^2} = -\frac{\Lambda}{\mathsf{h}^2}.$$

Positivity of Λ follows from the positive definiteness of R_0 and $\widetilde{R}_0$. Indeed $\mathsf{U}\ge 0$ and $\mathsf{V}\ge 0$, with $\mathsf{U}=0$ only if $\mathsf{u}=0$ and $\mathsf{V}=0$ only if $\mathsf{v}=0$. If $\mathsf{u}=0$ or $\mathsf{v}=0$, then C vanishes and $\Lambda = 1$; otherwise $\Lambda \ge s\mathsf{h}^2\mathsf{U}\mathsf{V} > 0$. In particular $S^{-1}+W$ is invertible, and the Sherman–Morrison–Woodbury formula [26, Section 2.1.4] gives

$$\mathcal{L}(M)^{-1} = \mathcal{L}(M_0)^{-1} - \mathcal{L}(M_0)^{-1}P\,\Theta\,P^\top\mathcal{L}(M_0)^{-1},$$

where

$$\Theta \stackrel{\text{def}}{=} (S^{-1}+W)^{-1} = \frac{1}{\Lambda}\begin{pmatrix}s\mathsf{h}^2\mathsf{V} & \mathsf{h}(1+\mathsf{h}\mathsf{C})\\ \mathsf{h}(1+\mathsf{h}\mathsf{C}) & -\mathsf{h}^2\mathsf{U}\end{pmatrix} = \begin{pmatrix}-\alpha_3 & \alpha_1\\ \alpha_1 & -\alpha_2\end{pmatrix}.$$

The two identities are read off from the blocks of the Woodbury identity. By eq. (SB.18), the columns of $\mathcal{L}(M_0)^{-1}P$ are

$$\mathcal{L}(M_0)^{-1}\tilde{\mathsf{u}} = \begin{pmatrix} R_0\mathsf{u} \\ \mathsf{X}_0^\top \mathsf{u} \end{pmatrix}, \qquad \mathcal{L}(M_0)^{-1}\tilde{\mathsf{v}} = \begin{pmatrix} \mathsf{X}_0\mathsf{v} \\ -s\widetilde{R}_0\mathsf{v} \end{pmatrix}.$$

The upper-left block of the Woodbury identity is therefore

$$R - R_0 = -\big[R_0\mathsf{u} \;\; \mathsf{X}_0\mathsf{v}\big] \begin{pmatrix} -\alpha_3 & \alpha_1 \\ \alpha_1 & -\alpha_2 \end{pmatrix} \big[R_0\mathsf{u} \;\; \mathsf{X}_0\mathsf{v}\big]^\top,$$

which, expanded, is eq. (SB.14). The upper-right block is

$$\begin{aligned} RM = \mathsf{X}_0 &+ \alpha_3 \,(R_0\mathsf{u}) \otimes (\mathsf{X}_0^\top \mathsf{u}) + s\alpha_1 \,(R_0\mathsf{u}) \otimes (\widetilde{R}_0\mathsf{v}) \\ &- \alpha_1 \,(\mathsf{X}_0\mathsf{v}) \otimes (\mathsf{X}_0^\top \mathsf{u}) - s\alpha_2 \,(\mathsf{X}_0\mathsf{v}) \otimes (\widetilde{R}_0\mathsf{v}). \end{aligned}$$

Multiplying on the right by v and using the definitions of C and V in eq. (SB.12) gives

$$RM\mathsf{v} = R_0\mathsf{u}\,(\alpha_3\mathsf{C} + s\alpha_1\mathsf{V}) + \mathsf{X}_0\mathsf{v}\,(1 - \alpha_1\mathsf{C} - s\alpha_2\mathsf{V}).$$

The two scalar coefficients simplify by the definitions eq. (SB.13):

$$\alpha_3\mathsf{C} + s\alpha_1\mathsf{V} = \frac{s\mathsf{h}\mathsf{V}}{\Lambda}\big(-\mathsf{h}\mathsf{C} + 1 + \mathsf{h}\mathsf{C}\big) = \frac{s\mathsf{h}\mathsf{V}}{\Lambda} = -\frac{\alpha_3}{\mathsf{h}},$$

$$1 - \alpha_1\mathsf{C} - s\alpha_2\mathsf{V} = \frac{\Lambda - \mathsf{h}\mathsf{C}(1+\mathsf{h}\mathsf{C}) - s\mathsf{h}^2\mathsf{U}\mathsf{V}}{\Lambda} = \frac{(1+\mathsf{h}\mathsf{C})^2 - \mathsf{h}\mathsf{C}(1+\mathsf{h}\mathsf{C})}{\Lambda} = \frac{1+\mathsf{h}\mathsf{C}}{\Lambda} = \frac{\alpha_1}{\mathsf{h}}.$$

Multiplying $RM\mathsf{v}$ by h gives eq. (SB.15). If $\mathsf{h} = 0$, then $M = M_0$, so $R = R_0$ and all three coefficients vanish; both identities hold.

The relation $\Lambda\Lambda' = 1$ is a determinant identity. We factor

$$\mathcal{L}(M) = \mathcal{L}(M_0)\big(\mathrm{Id} + \mathcal{L}(M_0)^{-1}PSP^\top\big).$$

Sylvester's identity $\det(\mathrm{Id} + XY) = \det(\mathrm{Id} + YX)$ [28, Section 0.8.5], applied to the second factor, gives

$$\begin{aligned} \det \mathcal{L}(M) &= \det \mathcal{L}(M_0)\, \det(\mathrm{Id}_2 + SW) = \det \mathcal{L}(M_0)\, \det(S) \det(S^{-1} + W) \\ &= \det \mathcal{L}(M_0)\, (-\mathsf{h}^2)\Big(-\frac{\Lambda}{\mathsf{h}^2}\Big) = \det \mathcal{L}(M_0)\, \Lambda. \end{aligned}$$

Both determinants are nonzero, because the block inverse eq. (SB.18) exists for every K. Exchanging the roles of M and M_0 and replacing h by $-\mathsf{h}$ is the same situation, because $M_0 = M - \mathsf{h}\,\mathsf{u} \otimes \mathsf{v}$; it gives

$$\det \mathcal{L}(M_0) = \det \mathcal{L}(M)\, \Lambda',$$

and the two relations together give $\Lambda\Lambda' = 1$.

The inverse bound follows from the norm bounds of lemma SB.3 applied to M. The primed quantities satisfy

$$0 \le \mathsf{U}' \le \frac{\|\mathsf{u}\|^2}{s}, \qquad 0 \le \mathsf{V}' \le \frac{\|\mathsf{v}\|^2}{s}, \qquad |\mathsf{C}'| \le \frac{\|\mathsf{u}\|\|\mathsf{v}\|}{2\sqrt{s}} \le \frac{\mathsf{N}}{4\sqrt{s}},$$

the last step by $2\|\mathsf{u}\|\|\mathsf{v}\| \le \mathsf{N}$. Using $(1+x)^2 + y^2 \le (1+x+y)^2$ for $x, y \ge 0$ in the last step,

$$\Lambda^{-1} = \Lambda' = (1 - \mathsf{h}\mathsf{C}')^2 + s\mathsf{h}^2\mathsf{U}'\mathsf{V}' \le \Big(1 + \frac{|\mathsf{h}|\mathsf{N}}{4\sqrt{s}}\Big)^2 + \Big(\frac{|\mathsf{h}|\mathsf{N}}{2\sqrt{s}}\Big)^2 \le \Big(1 + \frac{3|\mathsf{h}|\mathsf{N}}{4\sqrt{s}}\Big)^2.$$

At $z \in \mathbb{S}$ the argument is the same with s replaced by z, except for the nonvanishing of Λ and the inverse bound. The block inverse eq. (SB.18) holds with z in place of s, since $z\,\mathrm{Id} + KK^\top$ is invertible by lemma SB.3. Sylvester's identity is used first: directly from the definitions,

$$\mathrm{Id}_2 + SW = \begin{pmatrix} 1 + \mathsf{h}\mathsf{C} & -z\mathsf{h}\mathsf{V} \\ \mathsf{h}\mathsf{U} & 1 + \mathsf{h}\mathsf{C} \end{pmatrix}, \qquad \det(\mathrm{Id}_2 + SW) = \Lambda,$$

so $\det \mathcal{L}(M) = \Lambda \det \mathcal{L}(M_0)$ with both determinants nonzero, and $\Lambda \neq 0$. For $\mathsf{h} \neq 0$ this gives $\det(S^{-1} + W) = -\Lambda/\mathsf{h}^2 \neq 0$, and the Woodbury step, the reading of the blocks and the relation $\Lambda\Lambda' = 1$ are unchanged. For the inverse bound, lemma SB.3 at z applied to M gives $|\mathsf{U}'| \le \|\mathsf{u}\|^2/x$, $|\mathsf{V}'| \le \|\mathsf{v}\|^2/x$ and $|\mathsf{C}'| \le \mathsf{N}/(4\sqrt{x})$, and with $|z| \le \frac{\sqrt{5}}{2}x$ and $\|\mathsf{u}\|^2\|\mathsf{v}\|^2 \le \mathsf{N}^2/4$, writing $t \overset{\text{def}}{=} |\mathsf{h}|\mathsf{N}/\sqrt{x}$,

$$|\Lambda|^{-1} = |\Lambda'| \le |1 - \mathsf{h}\mathsf{C}'|^2 + |z|\,\mathsf{h}^2|\mathsf{U}'||\mathsf{V}'| \le \Big(1 + \frac{t}{4}\Big)^2 + \frac{\sqrt{5}}{8}\,t^2 \le (1+t)^2,$$

the last step by $(1+a)^2 + b^2 \le (1+a+b)^2$ with $a = t/4$, $b = (\sqrt{5}/8)^{1/2}t \le \frac{3}{4}t$. □

Lemma SB.5 (Coefficient bounds). *Let $I \subset (0,\infty)$ be compact with $s_- = \min I$ and put $C_I \overset{\text{def}}{=} \max(1, s_-^{-1})\big(1 + \frac{3}{4\sqrt{s_-}}\big)^3$. In the setting of lemma SB.4 with $\mathsf{h} = \varpi\, d$, $0 \le \varpi \le 1$, $d \in \mathbb{R}$, and $s \in I$,*

$$|\alpha_1| \le C_I\,\varpi\,(1+|d|)^4(1+\mathsf{N})^3, \qquad |\alpha_2| + |\alpha_3| \le C_I\,\varpi^2(1+|d|)^4(1+\mathsf{N})^3, \tag{SB.19}$$

and the same bounds hold for α_1/ϖ and $(\alpha_2,\alpha_3)/\varpi^2$ with the factors ϖ, ϖ^2 removed. Moreover $R - R_0$ has rank at most two and $\|R - R_0\|_* \le 4/s$.

Proof. The three scalars of eq. (SB.12) are bounded by the norm bounds of lemma SB.3 applied to M_0:

$$0 \le \mathsf{U} \le \frac{\mathsf{N}}{s}, \qquad 0 \le s\mathsf{V} \le \mathsf{N}, \qquad |\mathsf{C}| \le \frac{\|\mathsf{u}\|\|\mathsf{v}\|}{2\sqrt{s}} \le \frac{\mathsf{N}}{4\sqrt{s}}.$$

We also use $|\mathsf{h}| \le |d|$, which holds because $\varpi \le 1$, the inverse bound eq. (SB.16), and the elementary inequality

$$1 + c|d|\mathsf{N} \le (1+c)(1+|d|)(1+\mathsf{N}) \qquad (c \ge 0). \tag{SB.20}$$

For α_1 these give

$$|\alpha_1| = \frac{|\mathsf{h}|\,|1+\mathsf{h}\mathsf{C}|}{\Lambda} \le \varpi|d|\Big(1 + \frac{|d|\mathsf{N}}{4\sqrt{s}}\Big)\Big(1 + \frac{3|d|\mathsf{N}}{4\sqrt{s}}\Big)^2 \le \varpi|d|\Big(1 + \frac{3}{4\sqrt{s_-}}\Big)^3(1+|d|)^3(1+\mathsf{N})^3,$$

the first inequality by eq. (SB.16) and $|\mathsf{C}| \le \mathsf{N}/(4\sqrt{s})$, the second by eq. (SB.20) applied to each factor with $s \ge s_-$. For α_2 and α_3 the two numerators are bounded together. Since $s\mathsf{U} \le \|\mathsf{u}\|^2$ and $s\mathsf{V} \le \|\mathsf{v}\|^2$,

$$\mathsf{U} + s\mathsf{V} \le \frac{\|\mathsf{u}\|^2}{s} + \|\mathsf{v}\|^2 \le \max(1, s_-^{-1})\,\mathsf{N}.$$

Since $\alpha_2 \ge 0 \ge \alpha_3$, the first inequality below follows from this bound, eq. (SB.16) and $|\mathsf{h}| \le |d|$, and the second from eq. (SB.20):

$$|\alpha_2| + |\alpha_3| = \frac{\mathsf{h}^2(\mathsf{U} + s\mathsf{V})}{\Lambda} \le \max(1, s_-^{-1})\,\varpi^2 d^2\,\mathsf{N}\Big(1 + \frac{3|d|\mathsf{N}}{4\sqrt{s}}\Big)^2 \le \max(1, s_-^{-1})\Big(1 + \frac{3}{4\sqrt{s_-}}\Big)^2\varpi^2(1+|d|)^4(1+\mathsf{N})^3.$$

Since $|d| \le 1 + |d|$ and $1 + \frac{3}{4\sqrt{s_-}} \ge 1$, both right sides are at most C_I times $\varpi(1+|d|)^4(1+\mathsf{N})^3$, respectively $\varpi^2(1+|d|)^4(1+\mathsf{N})^3$, with the stated C_I. The bounds for the normalized coefficients are the same inequalities with the factor ϖ or ϖ^2 of h or h^2 cancelled before bounding $|\mathsf{h}|$ by $|d|$.

The rank claim follows from eq. (SB.14), whose right side has range in the span of $R_0\mathsf{u}$ and $\mathsf{X}_0\mathsf{v}$. For a matrix of rank at most two the nuclear norm is at most twice the operator norm, so

$$\|R - R_0\|_* \le 2\|R - R_0\| \le 2\big(\|R\| + \|R_0\|\big) \le \frac{4}{s}.$$ □

Lemma SB.6 (Gaussian moment bounds). *Let $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{out}}})$, $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{in}}})$ be independent, let $\|\Sigma_{\text{out}}\|, \|\Sigma_{\text{in}}\| \le C_\Sigma$ and $\max(N_{\text{in}}, N_{\text{out}}) \le \gamma_\star B$, let $\widehat{A}_t \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ with $\|\widehat{A}_t\|_{\text{F}} = 1$, and put $\mathsf{u} = {\Sigma_{\text{out}}}^{1/2} z/\sqrt{B}$, $\mathsf{v} = {\Sigma_{\text{in}}}^{1/2} w/\sqrt{B}$, $\mathsf{N} = \|\mathsf{u}\|^2 + \|\mathsf{v}\|^2$ and $d = z^\top \widehat{A}_t w$. Then for every integer $p \ge 1$*

$$\|\mathsf{N}\|_p \le 2C_\Sigma\gamma_\star\big((2p-1)!!\big)^{1/p}, \qquad \|d\|_{2p} \le \big((2p-1)!!\big)^{1/p},$$
$$\big\|\,|d|\,\|\mathsf{u}\|\|\mathsf{v}\|\big\|_2 \le 15^{2/3}\,C_\Sigma\gamma_\star \le 7\,C_\Sigma\gamma_\star.$$

Also, for any symmetric $A \in \mathbb{R}^{N_{\text{out}} \times N_{\text{out}}}$,

$$\mathbb{E}[z^\top A z] = \operatorname{Tr} A, \qquad \operatorname{Var}(z^\top A z) = 2\|A\|_{\text{F}}^2.$$

Proof. The bound on N reduces to the moments of $\|z\|^2$. The covariance bound gives

$$\|\mathsf{u}\|^2 = \frac{z^\top \Sigma_{\text{out}} z}{B} \le \frac{C_\Sigma \|z\|^2}{B},$$

and Jensen's inequality for the convex function $x \mapsto x^p$, applied to the average of the N_{out} coordinates, gives

$$\mathbb{E}\left(\frac{\|z\|^2}{N_{\text{out}}}\right)^p \le \frac{1}{N_{\text{out}}} \sum_{i=1}^{N_{\text{out}}} \mathbb{E}\, z_i^{2p} = (2p-1)!!.$$

Together,

$$\|\|\mathsf{u}\|^2\|_p \le C_\Sigma \frac{N_{\text{out}}}{B} \left((2p-1)!!\right)^{1/p} \le C_\Sigma \gamma_\star \left((2p-1)!!\right)^{1/p},$$

the same bound holds for $\|\mathsf{v}\|^2$, and the triangle inequality in L^p gives the bound on N.

The bound on d is lemma SA.7(i).

The bound on the product is the generalized Hölder inequality with three exponents equal to 3:

$$\mathbb{E}\left[d^2 \|\mathsf{u}\|^2 \|\mathsf{v}\|^2\right] \le \|d^2\|_3 \, \|\|\mathsf{u}\|^2\|_3 \, \|\|\mathsf{v}\|^2\|_3 \le 15^{2/3} \cdot \left(15^{1/3} C_\Sigma \gamma_\star\right)^2 = 15^{4/3} (C_\Sigma \gamma_\star)^2,$$

where the second inequality uses the bounds on d and on $\|\mathsf{u}\|^2$, $\|\mathsf{v}\|^2$ with $p = 3$:

$$\|d^2\|_3 = \|d\|_6^2 \le (5!!)^{2/3}, \qquad \|\|\mathsf{u}\|^2\|_3, \ \|\|\mathsf{v}\|^2\|_3 \le (5!!)^{1/3} C_\Sigma \gamma_\star.$$

Taking the square root gives the bound $15^{2/3} C_\Sigma \gamma_\star$, which is at most $7 C_\Sigma \gamma_\star$.

The quadratic form is diagonalized by an orthogonal O with $A = O \operatorname{diag}(a) O^\top$:

$$z^\top A z = \sum_i a_i \, (O^\top z)_i^2,$$

a sum of independent terms with means a_i and variances $2a_i^2$, since $O^\top z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{out}}})$ and a standard Gaussian square has variance 2. Summing over i gives

$$\mathbb{E}[z^\top A z] = \sum_i a_i = \operatorname{Tr} A, \qquad \operatorname{Var}(z^\top A z) = 2 \sum_i a_i^2 = 2 \|A\|_{\text{F}}^2. \qquad \square$$

Lemma SB.7 (Bessel bounds)**.** *Let $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{out}}})$ and $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{in}}})$ be independent of each other and of a σ-field $\mathcal{G}$, and let e be a real or complex random variable depending on z, w and $\mathcal{G}$. Then*

$$\begin{gathered} \|\, \mathbb{E}_{\mathcal{G}}[(z \otimes w)\, e]\|_{\text{F}} \le \|e\|_{2,\mathcal{G}}, \qquad \|\, \mathbb{E}_{\mathcal{G}}[(z \otimes z - \mathrm{Id})\, e]\|_{\text{F}} \le \sqrt{2}\, \|e\|_{2,\mathcal{G}}, \\ \|\, \mathbb{E}_{\mathcal{G}}[(w \otimes w - \mathrm{Id})\, e]\|_{\text{F}} \le \sqrt{2}\, \|e\|_{2,\mathcal{G}}. \end{gathered} \tag{SB.21}$$

Proof. Since (z, w) is independent of $\mathcal{G}$, it suffices to prove the inequalities with $\mathbb{E}$ in place of $\mathbb{E}_{\mathcal{G}}$, in the L^2 space of (z, w).

The inequalities are Bessel's inequality for two orthonormal families. By the Gaussian fourth-moment identity eq. (SA.30) and the independence of z and w, the products $z_i w_l$ ($i \le N_{\text{out}}$, $l \le N_{\text{in}}$) form an orthonormal family in L^2, and the variables $(z_i^2 - 1)/\sqrt{2}$ ($i \le N_{\text{out}}$) and $z_i z_k$ ($i < k$) form another orthonormal family.

Bessel's inequality for the first family gives the first bound:

$$\|\, \mathbb{E}[(z \otimes w)\, e]\|_{\text{F}}^2 = \sum_{i,l} |\, \mathbb{E}[z_i w_l e]|^2 \le \mathbb{E}\, |e|^2.$$

Bessel's inequality for the second family gives

$$\sum_{i<k} |\, \mathbb{E}[z_i z_k e]|^2 + \sum_i \frac{|\, \mathbb{E}[(z_i^2 - 1) e]|^2}{2} \le \mathbb{E}\, |e|^2.$$

Each off-diagonal entry of the symmetric matrix $\mathbb{E}[(z \otimes z - \mathrm{Id})\, e]$ occurs twice in its squared Frobenius norm, so

$$\|\, \mathbb{E}[(z \otimes z - \mathrm{Id})\, e]\|_{\text{F}}^2 = 2 \sum_{i<k} |\, \mathbb{E}[z_i z_k e]|^2 + \sum_i |\, \mathbb{E}[(z_i^2 - 1) e]|^2 \le 2\, \mathbb{E}\, |e|^2,$$

which is the second bound. The third bound is the second with w in place of z. $\square$

Lemma SB.8 (Gaussian response)**.** *Let $A \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ be deterministic, $z \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{out}}})$, $w \sim \mathcal{N}(0, \mathrm{Id}_{N_{\text{in}}})$ independent, and $d = z^\top A w$. Let $f \in C^3(\mathbb{R})$ and $\psi \in C^2(\mathbb{R})$ have polynomially bounded derivatives up to the stated orders. Then*

$$\mathbb{E}[(z \otimes w)\, f(d)] = A\ \mathbb{E}\, f'(d) + A\ \mathbb{E}[(w \otimes w)\, f'''(d)]\, A^\top A, \tag{SB.22}$$

$$\|\, \mathbb{E}[(z \otimes w)\, f(d)] - A\ \mathbb{E}\, f'(d)\|_{\mathrm{F}} \le \|f'''\|_\infty \|A\|^2 \|A\|_{\mathrm{F}},$$

$$\mathbb{E}[(z \otimes z - \mathrm{Id})\, \psi(d)] = A\ \mathbb{E}[(w \otimes w)\, \psi''(d)]\, A^\top, \tag{SB.23}$$

$$\|\, \mathbb{E}[(z \otimes z - \mathrm{Id})\, \psi(d)]\|_{\mathrm{F}} \le \|\psi''\|_\infty \|A\| \|A\|_{\mathrm{F}},$$

and both eqs. (SB.22) *and* (SB.23) *hold with (z, w, A) replaced by $(w, z, A^\top)$.*

Proof. The tool is Stein's lemma [62, Lemma 1]: for $g \sim \mathcal{N}(0,1)$ and an absolutely continuous ϕ with $\mathbb{E}\,|\phi'(g)| < \infty$,

$$\mathbb{E}[g\, \phi(g)] = \mathbb{E}\, \phi'(g). \tag{SB.24}$$

We apply it to one coordinate of z or w at a time, conditionally on all other coordinates. The partial derivatives of $d = z^\top A w$ are

$$\partial_{z_i} d = (Aw)_i, \qquad \partial_{w_l} d = (A^\top z)_l.$$

The first identity is obtained by three applications of eq. (SB.24). Applying it to the coordinate z_i gives

$$\mathbb{E}[z_i w_l f(d)] = \mathbb{E}\left[w_l\, \partial_{z_i} f(d)\right] = \mathbb{E}\left[w_l\, (Aw)_i\, f'(d)\right].$$

Applying it to the coordinate w_l, with the product rule, gives

$$\mathbb{E}\left[w_l\, (Aw)_i\, f'(d)\right] = \mathbb{E}\left[\partial_{w_l}\big((Aw)_i f'(d)\big)\right] = A_{il}\ \mathbb{E}\, f'(d) + \mathbb{E}\left[(Aw)_i\, (A^\top z)_l\, f''(d)\right].$$

In matrix form the two steps read

$$\mathbb{E}[(z \otimes w)\, f(d)] = A\ \mathbb{E}\, f'(d) + A\ \mathbb{E}[(w \otimes z)\, f''(d)]\, A. \tag{SB.25}$$

Applying eq. (SB.24) to the coordinate z_l inside the remaining expectation gives

$$\mathbb{E}[w_k z_l f''(d)] = \mathbb{E}\left[w_k\, (Aw)_l\, f'''(d)\right], \qquad \text{that is,} \qquad \mathbb{E}[(w \otimes z)\, f''(d)] = \mathbb{E}[(w \otimes w)\, f'''(d)]\, A^\top.$$

Substituting this into eq. (SB.25) proves eq. (SB.22).

The bound for the first identity has two steps. The matrix $N \overset{\text{def}}{=} \mathbb{E}[(w \otimes w)\, f'''(d)]$ satisfies

$$\|N\| = \sup_{\|x\|=\|y\|=1} \left|\, \mathbb{E}\left[(x^\top w)(y^\top w) f'''(d)\right]\right| \le \|f'''\|_\infty \sup_{\|x\|=\|y\|=1} \mathbb{E}\left[|x^\top w|\, |y^\top w|\right] \le \|f'''\|_\infty,$$

the last step by the Cauchy–Schwarz inequality and $\mathbb{E}(x^\top w)^2 = \|x\|^2$. Then

$$\|A\, N\, A^\top A\|_{\mathrm{F}} \le \|A\|\, \|N\|\, \|A^\top A\|_{\mathrm{F}} \le \|A\|\, \|N\|\, \|A\|\, \|A\|_{\mathrm{F}} \le \|f'''\|_\infty \|A\|^2 \|A\|_{\mathrm{F}},$$

using $\|XY\|_{\mathrm{F}} \le \|X\| \|Y\|_{\mathrm{F}}$ twice.

The second identity is obtained by two applications of eq. (SB.24). Applying it to the coordinate z_i of $z_i \cdot z_k \psi(d)$ gives

$$\mathbb{E}[z_i z_k \psi(d)] = \mathbb{E}\left[\partial_{z_i}\big(z_k \psi(d)\big)\right] = \delta_{ik}\ \mathbb{E}\, \psi(d) + \mathbb{E}\left[z_k\, (Aw)_i\, \psi'(d)\right],$$

that is, in matrix form,

$$\mathbb{E}[(z \otimes z - \mathrm{Id})\, \psi(d)] = A\ \mathbb{E}[(w \otimes z)\, \psi'(d)].$$

The step from $\mathbb{E}[(w \otimes z)\, f''(d)]$ to $\mathbb{E}[(w \otimes w)\, f'''(d)] A^\top$ above, applied with ψ' in place of f'', gives $\mathbb{E}[(w \otimes z)\, \psi'(d)] = \mathbb{E}[(w \otimes w)\, \psi''(d)] A^\top$, which proves eq. (SB.23). Its bound is the same two-step estimate with $\|\psi''\|_\infty$ in place of $\|f'''\|_\infty$ and one factor of $\|A\|$ fewer:

$$\|A\ \mathbb{E}[(w \otimes w)\, \psi''(d)]\, A^\top\|_{\mathrm{F}} \le \|A\|\, \|\psi''\|_\infty\, \|A^\top\|_{\mathrm{F}} = \|\psi''\|_\infty \|A\| \|A\|_{\mathrm{F}}.$$

The last claim follows from the symmetry $d = w^\top A^\top z$: the same computations with the roles of z and w exchanged and A replaced by $A^\top$ give both identities and both bounds for $(w, z, A^\top)$. □

Lemma SB.9 (Nuclear-norm Lipschitz bound)**.** *For $K, K' \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ and $s > 0$, $\|R_s(K) - R_s(K')\|_* \le 2s^{-3/2}\|K - K'\|_*$.*

Proof. The resolvent $R_s(K)$ is a block of the resolvent of a symmetric matrix at a point at distance $\sqrt{s}$ from the real axis. Let

$$\mathcal{D}(K) \stackrel{\mathrm{def}}{=} \begin{pmatrix} 0 & K \\ K^\top & 0 \end{pmatrix}, \qquad \mathcal{Y}(K) \stackrel{\mathrm{def}}{=} \big(\mathcal{D}(K) - i\sqrt{s}\,\mathrm{Id}\,\big)^{-1}.$$

Since $\mathcal{D}(K)$ is real symmetric, its spectrum is real, and $\|\mathcal{Y}(K)\| \le s^{-1/2}$. Block inversion with the Schur complement of the lower-right block $-i\sqrt{s}\,\mathrm{Id}$ shows that the upper-left $N_{\mathrm{out}} \times N_{\mathrm{out}}$ block of $\mathcal{Y}(K)$ is

$$\Big(-i\sqrt{s}\,\mathrm{Id} - K\big(-i\sqrt{s}\big)^{-1}K^\top\Big)^{-1} = \Big(-\frac{i}{\sqrt{s}}\big(KK^\top + s\,\mathrm{Id}\,\big)\Big)^{-1} = i\sqrt{s}\,R_s(K).$$

The difference of the two dilated resolvents is controlled by the resolvent identity:

$$\mathcal{Y}(K) - \mathcal{Y}(K') = \mathcal{Y}(K)\,\mathcal{D}(K' - K)\,\mathcal{Y}(K').$$

The nuclear norm is unitarily invariant, so the ideal property $\|XYZ\|_* \le \|X\|\,\|Y\|_*\,\|Z\|$ [8, Proposition IV.2.4 and (IV.40)] applies; the singular values of $\mathcal{D}(E)$ are those of E, each taken twice, so $\|\mathcal{D}(E)\|_* = 2\|E\|_*$. Hence

$$\|\mathcal{Y}(K) - \mathcal{Y}(K')\|_* \le \|\mathcal{Y}(K)\|\,\|\mathcal{D}(K' - K)\|_*\,\|\mathcal{Y}(K')\| \le \frac{2}{s}\,\|K - K'\|_*.$$

Compression to the upper-left block does not increase the nuclear norm, by the same ideal property with the orthogonal projection P onto the first N_{out} coordinates in place of X and Z. Therefore

$$\sqrt{s}\,\|R_s(K) - R_s(K')\|_* = \|P\big(\mathcal{Y}(K) - \mathcal{Y}(K')\big)P\|_* \le \frac{2}{s}\,\|K - K'\|_*,$$

which is the claim after division by $\sqrt{s}$. □

SB.3 The Gram closure

In the frozen-state model the buffer is a series over samples,

$$M_t = \sum_{j\ge 0}\sum_{a=1}^{B} \mathsf{h}_{ja}\,\mathsf{u}_{ja} \otimes \mathsf{v}_{ja}, \qquad \mathsf{h}_{ja} \stackrel{\mathrm{def}}{=} w_j d_{ja}, \quad \mathsf{u}_{ja} \stackrel{\mathrm{def}}{=} \frac{{\Sigma_{\mathrm{out}}}^{1/2} z_{ja}}{\sqrt{B}}, \quad \mathsf{v}_{ja} \stackrel{\mathrm{def}}{=} \frac{{\Sigma_{\mathrm{in}}}^{1/2} w_{ja}}{\sqrt{B}}, \tag{SB.26}$$

and, with $z = -s$ and the notation eq. (SB.10), the Gram resolvents are

$$R(-s) = R_s(M_t), \qquad \widetilde{R}(-s) = \widetilde{R}_s(M_t).$$

The results of this subsection are stated for a fixed weight sequence $w = (w_j)_{j\ge 0}$ satisfying

$$0 \le w_j \le 1 \quad (j \ge 0), \qquad \sum_{j\ge 0} w_j \le 1, \qquad w_j > 0 \text{ for some } j, \tag{SB.27}$$

where the frozen-state model with weights w is definition SA.3 with w_j in place of the EMA weight of age j in eq. (SA.26). The EMA weights satisfy eq. (SB.27) by eq. (SA.18), and the truncated weights of the drift section do as well. Under eq. (SB.27) one has $s_2 \le 1$ and $s_3 \le 1$, and these two sums are the only way the weights enter the constants below.

Theorem SB.2 (Gram closure)**.** *Under Assumption S3, in the frozen-state model with a weight sequence w satisfying eq. (SB.27), let $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1)(z)$ be the solution of the weighted system eq. (SB.2) for the weights w and the spectral measures eq. (SA.41), and let $\overline{R}, \overline{\widetilde{R}}$ be as in eq. (SB.3). Then for every compact $I \subset (0, \infty)$,*

$$\sup_{s\in I}\Big(\|\,\mathbb{E}\,R(-s) - \overline{R}(-s)\| + \|\,\mathbb{E}\,\widetilde{R}(-s) - \overline{\widetilde{R}}(-s)\|\Big) \longrightarrow 0. \tag{SB.28}$$

Corollary SB.1 (Trace closure with bounded tests)**.** *Under the hypotheses of theorem SB.2, let $I \subset (0,\infty)$ be compact and let $T = T_B \in \mathbb{C}^{N_{\mathrm{out}} \times N_{\mathrm{out}}}$ and $\widetilde{T} = \widetilde{T}_B \in \mathbb{C}^{N_{\mathrm{in}} \times N_{\mathrm{in}}}$ be deterministic with $\|T_B\|, \|\widetilde{T}_B\| \le C_T$ for all B. Then*

$$\mathbb{E} \sup_{s \in I} \Big| \frac{1}{N_{\mathrm{out}}} \operatorname{Tr}\big[T\big(R(-s) - \overline{R}(-s)\big)\big] \Big|^2 \longrightarrow 0, \qquad \mathbb{E} \sup_{s \in I} \Big| \frac{1}{N_{\mathrm{in}}} \operatorname{Tr}\big[\widetilde{T}\big(\widetilde{R}(-s) - \overline{\widetilde{R}}(-s)\big)\big] \Big|^2 \longrightarrow 0. \tag{SB.29}$$

In particular, in mean square and uniformly on compact s-intervals,

$$\frac{1}{B} \operatorname{Tr}\big(\Sigma_{\mathrm{out}} R(-s)\big) \longrightarrow \sigma(-s), \qquad \frac{1}{B} \operatorname{Tr}\big(\Sigma_{\mathrm{in}} \widetilde{R}(-s)\big) \longrightarrow \tilde{\sigma}(-s).$$

The proof of theorem SB.2 occupies the rest of the subsection. We fix a compact $I \subset (0,\infty)$, write $s_- = \min I$, and let C_I denote constants depending only on $(I, C_\Sigma, \gamma_\star)$. Lemmas SB.10 to SB.12 are finite-dimensional: they hold at every dimension in the frozen-state model with a weight sequence in the class eq. (SB.27), $\|\widehat{A}_t\|_{\mathrm{F}} = 1$, $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$ and $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \le \gamma_\star B$, with no condition on $\|\widehat{A}_t\|$, and their constants C_I depend on nothing else. For a sample (j,a) we write $M_0^{(ja)} \overset{\mathrm{def}}{=} M_t - \mathsf{h}_{ja}\, \mathsf{u}_{ja} \otimes \mathsf{v}_{ja}$ for the buffer with that sample deleted, $R_0^{(ja)} = R_s(M_0^{(ja)})$, $\widetilde{R}_0^{(ja)}$, $\mathsf{X}_0^{(ja)}$ accordingly, and $\mathsf{U}_{ja}, \mathsf{V}_{ja}, \mathsf{C}_{ja}, \Lambda_{ja}, \alpha_{1,ja}, \alpha_{2,ja}, \alpha_{3,ja}$ for the quantities of lemma SB.4 with $(M_0, \mathsf{h}, \mathsf{u}, \mathsf{v}) = (M_0^{(ja)}, \mathsf{h}_{ja}, \mathsf{u}_{ja}, \mathsf{v}_{ja})$. The deleted buffer is independent of (z_{ja}, w_{ja}), and $\mathcal{G}_{ja}$ denotes the σ-field generated by all samples other than (j,a). Since samples of the same age are exchangeable, the law of $(M_0^{(ja)}, z_{ja}, w_{ja})$ does not depend on a, and we drop a from the notation when only the law matters. Finally we define the expected resolvents and traces

$$\begin{aligned} \mathcal{P}(s) &\overset{\mathrm{def}}{=} \mathbb{E}\, R(-s), \qquad \mathcal{P}_j(s) \overset{\mathrm{def}}{=} \mathbb{E}\, R_0^{(j1)}, \\ \sigma_B(s) &\overset{\mathrm{def}}{=} \frac{1}{B} \operatorname{Tr}\big(\Sigma_{\mathrm{out}} \mathcal{P}(s)\big), \qquad \tilde{\sigma}_B(s) \overset{\mathrm{def}}{=} \frac{1}{B} \operatorname{Tr}\big(\Sigma_{\mathrm{in}}\, \mathbb{E}\, \widetilde{R}(-s)\big), \end{aligned} \tag{SB.30}$$

and $q_B(s) \overset{\mathrm{def}}{=} s \sigma_B(s) \tilde{\sigma}_B(s)$. Since $\|R(-s)\| \le 1/s$ and $\operatorname{Tr} \Sigma_{\mathrm{out}} \le C_\Sigma N_{\mathrm{out}}$,

$$0 \le \sigma_B, \tilde{\sigma}_B \le \frac{C_\Sigma \gamma_\star}{s}, \qquad 0 \le q_B(s) \le q_I \overset{\mathrm{def}}{=} \frac{(C_\Sigma \gamma_\star)^2}{s_-} \qquad (s \in I). \tag{SB.31}$$

Lemma SB.10 (Variance of normalized traces)**.** *Let $T \in \mathbb{C}^{N_{\mathrm{out}} \times N_{\mathrm{out}}}$ be deterministic with $\|T\| \le C_T$ and put $Y(s) \overset{\mathrm{def}}{=} \frac{1}{B} \operatorname{Tr}\big(T R(-s)\big)$. Then $\operatorname{Var} Y(s) \le C_I C_T^2\, s_2/B$ for $s \in I$. The same bound holds for $\frac{1}{B} \operatorname{Tr}(\widetilde{T} R_s(M_t^\top))$ with $\|\widetilde{T}\| \le C_T$, and for the traces of the resolvents of any buffer with one sample deleted.*

Proof. The variance is controlled by the Efron–Stein inequality for a truncated history, and the truncation is removed by an L^2 limit. For $J \ge 0$ let

$$M^{(J)} \overset{\mathrm{def}}{=} \sum_{j \le J} \sum_{a=1}^{B} \mathsf{h}_{ja}\, \mathsf{u}_{ja} \otimes \mathsf{v}_{ja}, \qquad Y^{(J)} \overset{\mathrm{def}}{=} \frac{1}{B} \operatorname{Tr}\big(T R_s(M^{(J)})\big),$$

so that $Y^{(J)}$ is a function of the $B(J+1)$ independent samples of ages $0, \dots, J$.

The effect of one replaced sample is bounded through the nuclear norm. Replacing the sample (j,a) by an independent copy $(d', \mathsf{u}', \mathsf{v}')$ produces $Y^{(J),(ja)'}$ and changes $M^{(J)}$ by the rank-two matrix $w_j(d\, \mathsf{u} \otimes \mathsf{v} - d'\, \mathsf{u}' \otimes \mathsf{v}')$, whose nuclear norm is at most $w_j(|d| \|\mathsf{u}\| \|\mathsf{v}\| + |d'| \|\mathsf{u}'\| \|\mathsf{v}'\|)$. By trace duality $|\operatorname{Tr}(TX)| \le \|T\| \|X\|_*$ and lemma SB.9,

$$\big|Y^{(J)} - Y^{(J),(ja)'}\big| \le \frac{C_T}{B} \cdot \frac{2}{s^{3/2}}\, w_j\big(|d| \|\mathsf{u}\| \|\mathsf{v}\| + |d'| \|\mathsf{u}'\| \|\mathsf{v}'\|\big),$$

and by lemma SB.6 and the triangle inequality in L^2,

$$\big\|Y^{(J)} - Y^{(J),(ja)'}\big\|_2 \le \frac{C_T}{B} \cdot \frac{2}{s_-^{3/2}}\, w_j \cdot 2 \cdot 7 C_\Sigma \gamma_\star = \frac{C_I C_T\, w_j}{B}.$$

The Efron–Stein inequality [13, Theorem 3.1] then gives

$$\operatorname{Var} Y^{(J)} \le \frac{1}{2}\sum_{j\le J}\sum_{a=1}^{B}\mathbb{E}\left[(Y^{(J)}-Y^{(J),(ja)'})^2\right] \le \frac{1}{2}\sum_{j\le J} B\,\frac{C_I^2 C_T^2 w_j^2}{B^2} \le \frac{C_I^2 C_T^2 s_2}{2B}.$$

The truncation is removed at fixed dimension. By lemma SB.9, trace duality, and $\|E\|_* \le \sqrt{\min(N_{\mathrm{out}},N_{\mathrm{in}})}\|E\|_{\mathrm{F}}$,

$$\left|Y - Y^{(J)}\right| \le \frac{C_T}{B}\cdot\frac{2}{s^{3/2}}\sqrt{\min(N_{\mathrm{out}},N_{\mathrm{in}})}\,\|M_t - M^{(J)}\|_{\mathrm{F}},$$

and $\mathbb{E}\,\|M_t - M^{(J)}\|_{\mathrm{F}}^2 \to 0$ as $J\to\infty$. Indeed, the batches of different ages are independent with the common mean $\mathbb{E}\,G_{t+1}$ and the common Frobenius variance V_B of lemma SA.4, so

$$\mathbb{E}\,\|M_t - M^{(J)}\|_{\mathrm{F}}^2 = \Big(\sum_{j>J} w_j\Big)^2 \|\,\mathbb{E}\,G_{t+1}\|_{\mathrm{F}}^2 + \sum_{j>J} w_j^2\,\mathsf{V}_B,$$

and both tail sums vanish as $J\to\infty$ because $\sum_j w_j \le 1$. (For the EMA weights this is lemma SA.6(i).) Hence $Y^{(J)}\to Y$ in L^2, and

$$\operatorname{Var} Y = \lim_{J\to\infty}\operatorname{Var} Y^{(J)} \le \frac{C_I^2 C_T^2 s_2}{2B}.$$

The input side is the same argument for $M_t^\top = \sum_{j,a}\mathsf{h}_{ja}\,\mathsf{v}_{ja}\otimes\mathsf{u}_{ja}$, and deleting one sample removes one term from the Efron–Stein sum. □

Lemma SB.11 (Concentration of the deleted-sample scalars)**.** *For every age j and $s\in I$,*

$$\|\mathsf{U}_j - \sigma_B(s)\|_2 + \|\mathsf{V}_j - \tilde\sigma_B(s)\|_2 + \|\mathsf{C}_j\|_2 \le \frac{C_I}{\sqrt{B}}. \tag{SB.32}$$

Proof. Write $R_0 = R_0^{(j1)}$, $\mathsf{u} = \mathsf{u}_{j1}$, and so on. Decompose

$$\mathsf{U} - \sigma_B = \Big(\mathsf{u}^\top R_0\mathsf{u} - \frac{\operatorname{Tr}(\Sigma_{\mathrm{out}}R_0)}{B}\Big) + \frac{\operatorname{Tr}\left(\Sigma_{\mathrm{out}}(R_0 - R(-s))\right)}{B} + \Big(\frac{\operatorname{Tr}(\Sigma_{\mathrm{out}}R(-s))}{B} - \mathbb{E}\,\frac{\operatorname{Tr}(\Sigma_{\mathrm{out}}R(-s))}{B}\Big).$$

The first term is a centered Gaussian quadratic form given $\mathcal{G}_{j1}$. Indeed

$$\mathsf{u}^\top R_0\mathsf{u} = z^\top\Big(\frac{{\Sigma_{\mathrm{out}}}^{1/2}R_0{\Sigma_{\mathrm{out}}}^{1/2}}{B}\Big)z.$$

Since R_0 is built from the samples other than $(j,1)$, of which z is independent, lemma SB.6 gives the conditional mean $\operatorname{Tr}(\Sigma_{\mathrm{out}}R_0)/B$ and the conditional variance

$$\frac{2\|{\Sigma_{\mathrm{out}}}^{1/2}R_0{\Sigma_{\mathrm{out}}}^{1/2}\|_{\mathrm{F}}^2}{B^2} \le \frac{2N_{\mathrm{out}}\|{\Sigma_{\mathrm{out}}}^{1/2}R_0{\Sigma_{\mathrm{out}}}^{1/2}\|^2}{B^2} \le \frac{2N_{\mathrm{out}}C_\Sigma^2}{B^2 s^2} \le \frac{2\gamma_\star C_\Sigma^2}{Bs_-^2},$$

using $\|X\|_{\mathrm{F}}^2 \le N_{\mathrm{out}}\|X\|^2$, $\|R_0\|\le 1/s$ and $N_{\mathrm{out}}\le\gamma_\star B$.

The second term is a rank-two correction. By trace duality and lemma SB.5,

$$\frac{\left|\operatorname{Tr}\left(\Sigma_{\mathrm{out}}(R_0 - R(-s))\right)\right|}{B} \le \frac{\|\Sigma_{\mathrm{out}}\|\,\|R_0 - R(-s)\|_*}{B} \le \frac{4C_\Sigma}{Bs_-}.$$

The third term is the fluctuation of a normalized trace. By lemma SB.10 with $T=\Sigma_{\mathrm{out}}$ and $C_T = C_\Sigma$, its L^2 norm is at most $(C_I C_\Sigma^2 s_2/B)^{1/2}$. Adding the three bounds gives $\|\mathsf{U}-\sigma_B\|_2 \le C_I/\sqrt{B}$. The bound for V is the same argument with Σ_{in}, $\widetilde{R}_0$ and the input-side statements of lemmas SB.5 and SB.10.

The bound for C is a conditional second-moment computation. We write

$$\mathsf{C} = \mathsf{u}^\top \mathsf{X}_0 \mathsf{v} = \frac{z^\top \big(\Sigma_{\text{out}}{}^{1/2} \mathsf{X}_0 \Sigma_{\text{in}}{}^{1/2}\big) w}{B}.$$

The matrix X_0 is built from the samples other than $(j,1)$, and $\mathbb{E}[(z^\top A w)^2] = \|A\|_{\text{F}}^2$ for deterministic A. With $\|A\|_{\text{F}}^2 \le \operatorname{rank}(A)\|A\|^2$ and the bound $\|\mathsf{X}_0\| \le (2\sqrt{s})^{-1}$ of lemma SB.3, this gives

$$\mathbb{E}_{\mathcal{G}_{j1}} \mathsf{C}^2 = \frac{\|\Sigma_{\text{out}}{}^{1/2} \mathsf{X}_0 \Sigma_{\text{in}}{}^{1/2}\|_{\text{F}}^2}{B^2} \le \frac{C_\Sigma^2 \min(N_{\text{out}}, N_{\text{in}}) \|\mathsf{X}_0\|^2}{B^2} \le \frac{C_\Sigma^2 \gamma_\star}{4Bs_-}.$$

Taking expectations and square roots gives $\|\mathsf{C}\|_2 \le C_I/\sqrt{B}$. □

Lemma SB.12 (Replacement of the deletion coefficients)**.** *For $x \ge 0$ put $f_x(d) \overset{\text{def}}{=} \frac{d}{1+xd^2}$ and $\psi_x(d) \overset{\text{def}}{=} \frac{d^2}{1+xd^2}$, and for a sample of age j and $s \in I$ define the target coefficients*

$$\check{\alpha}_{1,j} \overset{\text{def}}{=} w_j\, f_{w_j^2 q_B}(d_j), \qquad \check{\alpha}_{2,j} \overset{\text{def}}{=} w_j^2\, \sigma_B\, \psi_{w_j^2 q_B}(d_j), \qquad \check{\alpha}_{3,j} \overset{\text{def}}{=} -s\, w_j^2\, \tilde{\sigma}_B\, \psi_{w_j^2 q_B}(d_j). \tag{SB.33}$$

There is a sequence $e_B \to 0$, depending only on $(I, C_\Sigma, \gamma_\star)$, such that in every frozen-state model with the hypotheses stated before lemma SB.10, whatever its weights, its dimension and its normalized state $\widehat{A}_t$, for every age j and every $s \in I$

$$\|\alpha_{1,j} - \check{\alpha}_{1,j}\|_2 \le w_j\, e_B, \qquad \|\alpha_{2,j} - \check{\alpha}_{2,j}\|_2 \le w_j^2\, e_B, \qquad \|\alpha_{3,j} - \check{\alpha}_{3,j}\|_2 \le w_j^2\, e_B. \tag{SB.34}$$

Proof. Fix j with $w_j > 0$ (for $w_j = 0$ both sides of eq. (SB.34) vanish) and $s \in I$, and abbreviate $\varpi = w_j$, $d = d_j$, $\mathsf{h} = \varpi d$ and

$$\check{D} \overset{\text{def}}{=} 1 + \varpi^2 q_B d^2 \ge 1.$$

We treat α_1 in full; the other two coefficients are handled at the end by the same argument with different numerators.

The deviation of the deleted-sample scalars from their expected values is collected in one random variable,

$$\Xi \overset{\text{def}}{=} s\sigma_B |\tilde{\sigma}_B - \mathsf{V}| + s\mathsf{V}|\sigma_B - \mathsf{U}| + (1+q_B)|\mathsf{C}| + \mathsf{C}^2, \tag{SB.35}$$

whose expectation is small. Indeed $s\sigma_B \le C_\Sigma \gamma_\star$ by eq. (SB.31), and $\|s\mathsf{V}\|_2 \le \||\mathsf{v}|^2\|_2 \le C_I$ by lemmas SB.3 and SB.6; so the Cauchy–Schwarz inequality and lemma SB.11 give

$$\mathbb{E}\,\Xi \le C_\Sigma \gamma_\star \|\tilde{\sigma}_B - \mathsf{V}\|_2 + \|s\mathsf{V}\|_2 \|\sigma_B - \mathsf{U}\|_2 + (1+q_I)\|\mathsf{C}\|_2 + \|\mathsf{C}\|_2^2 \le \frac{C_I}{\sqrt{B}}.$$

The numerators of the differences are linear in Ξ on bounded residuals. Using

$$q_B - s\mathsf{U}\mathsf{V} = s\sigma_B(\tilde{\sigma}_B - \mathsf{V}) + s\mathsf{V}(\sigma_B - \mathsf{U}),$$

direct expansion of $\Lambda = (1+\mathsf{h}\mathsf{C})^2 + s\mathsf{h}^2\mathsf{U}\mathsf{V}$ gives

$$(1+\mathsf{h}\mathsf{C})\check{D} - \Lambda = \varpi^2 d^2\,(q_B - s\mathsf{U}\mathsf{V}) + \varpi^3 d^3 \mathsf{C} q_B - \varpi d\mathsf{C} - \varpi^2 d^2 \mathsf{C}^2, \tag{SB.36}$$

$$\Lambda - \check{D} = 2\varpi d\mathsf{C} + \varpi^2 d^2 \mathsf{C}^2 + \varpi^2 d^2\,(s\mathsf{U}\mathsf{V} - q_B). \tag{SB.37}$$

Fix a level $\mathsf{m} > 0$, which is sent to infinity at the end of the argument. Since $\varpi \le 1$ and $q_B \le q_I$, on the event $\{|d| \le \mathsf{m}\}$ both right sides are bounded in absolute value by $(1+\mathsf{m})^3\,\Xi$.

The normalized difference is small in probability. Let

$$Z \overset{\text{def}}{=} \frac{\alpha_1}{\varpi} - \frac{\check{\alpha}_1}{\varpi} = \frac{d(1+\mathsf{h}\mathsf{C})}{\Lambda} - \frac{d}{\check{D}} = \frac{d\,\big[(1+\mathsf{h}\mathsf{C})\check{D} - \Lambda\big]}{\Lambda \check{D}},$$

and let $E \overset{\text{def}}{=} \{|d| \le \mathsf{m}\} \cap \{\Lambda \ge 1/2\}$. On E, since $\check{D} \ge 1$ and by eq. (SB.36),

$$|Z| \le \frac{|d|\,|(1+\mathsf{h}\mathsf{C})\check{D} - \Lambda|}{\Lambda} \le 2\mathsf{m}(1+\mathsf{m})^3\,\Xi.$$

On the complement, either $|d| > \mathsf{m}$, or $|d| \le \mathsf{m}$ and $\Lambda < 1/2 \le \check{D} - 1/2$, so that $|\Lambda - \check{D}| > 1/2$. Chebyshev's inequality with $\mathbb{E}\, d^2 = 1$, Markov's inequality and eq. (SB.37) give

$$\Pr(E^c) \le \Pr(|d| > \mathsf{m}) + \Pr\left(|d| \le \mathsf{m},\ |\Lambda - \check{D}| > 1/2\right) \le \mathsf{m}^{-2} + 2(1+\mathsf{m})^3\ \mathbb{E}\ \Xi.$$

Combining the two bounds, for every $a > 0$,

$$\Pr(|Z| > a) \le \Pr(E^c) + \Pr\left(E,\ 2\mathsf{m}(1+\mathsf{m})^3\Xi > a\right) \le \mathsf{m}^{-2} + 2(1+\mathsf{m})^3\ \mathbb{E}\ \Xi + \frac{2\mathsf{m}(1+\mathsf{m})^3\ \mathbb{E}\ \Xi}{a}.$$

Convergence in probability is upgraded to L^2 by a uniform fourth moment. By lemma SB.5 with the factor ϖ removed and lemma SB.6, $\|\alpha_1/\varpi\|_4 \le C_I$, and $\|\check{\alpha}_1/\varpi\|_4 \le \|d\|_4 \le C_I$; hence $\|Z\|_4 \le C_I$. Splitting at level a and applying the Cauchy–Schwarz inequality to the second piece,

$$\begin{aligned}\mathbb{E}\, Z^2 &= \mathbb{E}\left[Z^2 \mathbf{1}_{|Z|\le a}\right] + \mathbb{E}\left[Z^2 \mathbf{1}_{|Z|>a}\right] \le a^2 + \|Z\|_4^2\ \Pr(|Z| > a)^{1/2} \\ &\le a^2 + C_I\Big(\mathsf{m}^{-2} + C_I(1+\mathsf{m})^4\big(1 + a^{-1}\big)B^{-1/2}\Big)^{1/2},\end{aligned}$$

where the last step uses $\mathbb{E}\ \Xi \le C_I B^{-1/2}$ and $\mathsf{m} \le 1 + \mathsf{m}$. The right side does not depend on j or on $s \in I$. Taking first $B \to \infty$, then $\mathsf{m} \to \infty$ and then $a \downarrow 0$ shows that

$$\sup_{j,\ s\in I} \frac{\|\alpha_{1,j} - \check{\alpha}_{1,j}\|_2}{w_j} \longrightarrow 0.$$

The other two coefficients follow by the same argument. The normalized differences are

$$Z_2 \overset{\text{def}}{=} \frac{\alpha_2 - \check{\alpha}_2}{\varpi^2} = \frac{d^2\big[\mathsf{U}\check{D} - \sigma_B\Lambda\big]}{\Lambda\check{D}}, \qquad Z_3 \overset{\text{def}}{=} \frac{\alpha_3 - \check{\alpha}_3}{\varpi^2} = -\frac{s\, d^2\big[\mathsf{V}\check{D} - \tilde{\sigma}_B\Lambda\big]}{\Lambda\check{D}},$$

whose numerators satisfy

$$\mathsf{U}\check{D} - \sigma_B\Lambda = (\mathsf{U} - \sigma_B)\check{D} + \sigma_B(\check{D} - \Lambda), \qquad \mathsf{V}\check{D} - \tilde{\sigma}_B\Lambda = (\mathsf{V} - \tilde{\sigma}_B)\check{D} + \tilde{\sigma}_B(\check{D} - \Lambda).$$

On $\{|d| \le \mathsf{m}\}$ these are bounded by $(1 + q_I\mathsf{m}^2)|\mathsf{U} - \sigma_B| + C_I(1+\mathsf{m})^3\Xi$, respectively the same expression with $(\mathsf{V}, \tilde{\sigma}_B)$, and their expectations are at most $C_I(1+\mathsf{m})^3 B^{-1/2}$ by lemma SB.11. The fourth moments of Z_2 and Z_3 are bounded by lemma SB.5 with the factor ϖ^2 removed, together with $\|d^2\|_4 \le C_I$. The event decomposition and the L^2 upgrade are then identical.

The sequence is defined from these bounds, not from the model. Every constant C_I above depends only on $(I, C_\Sigma, \gamma_\star)$, since lemmas SB.5, SB.6 and SB.11 involve neither $\|\widehat{A}_t\|$ nor the weights beyond eq. (SB.27). Let $\epsilon_B(\mathsf{m}, a)$ be the largest of the three right sides obtained for $\mathbb{E}\, Z^2$, $\mathbb{E}\, Z_2^2$ and $\mathbb{E}\, Z_3^2$, and put

$$e_B \overset{\text{def}}{=} \inf_{\mathsf{m}\ge 1,\ 0<a\le 1} \epsilon_B(\mathsf{m}, a)^{1/2}.$$

Then e_B depends only on $(I, C_\Sigma, \gamma_\star)$ and B, it bounds the three normalized errors for every model, every age and every $s \in I$, and $e_B \to 0$ by the order of limits used for α_1. □

Lemma SB.13 (Uniform Gaussian limit of the residual functional)**.** *With ψ_x as in lemma SB.12 and $d = z^\top \widehat{A}_t w$,*

$$r_B \overset{\text{def}}{=} \sup_{0\le x\le q_I} |\,\mathbb{E}\ \psi_x(d) - J(x)| \longrightarrow 0.$$

Proof. Pointwise convergence holds at each x. The function ψ_x is continuous with $0 \le \psi_x(d) \le d^2$, so lemma SA.7(iii) gives

$$\mathbb{E}\ \psi_x(d) \longrightarrow \mathbb{E}\ \psi_x(g) = J(x).$$

The family is uniformly Lipschitz in x. For $x, x' \in [0, q_I]$,

$$|\psi_x(d) - \psi_{x'}(d)| = \frac{d^4\,|x - x'|}{(1 + xd^2)(1 + x'd^2)} \le |x - x'|\, d^4,$$

so, by $\mathbb{E}\, d^4 \le 9$ from lemma SA.7(i) and $\mathbb{E}\, g^4 = 3$,

$$|\,\mathbb{E}\,\psi_x(d) - \mathbb{E}\,\psi_{x'}(d)| \le 9|x - x'|, \qquad |J(x) - J(x')| \le 3|x - x'|.$$

Pointwise convergence of a uniformly Lipschitz family on a compact interval is uniform. Given $\epsilon > 0$, choose a finite $\epsilon/36$-net of $[0, q_I]$ and then B so large that the pointwise error is below $\epsilon/3$ at every point of the net. For every $x \in [0, q_I]$ and its nearest net point x',

$$|\,\mathbb{E}\,\psi_x(d) - J(x)| \le 9|x - x'| + |\,\mathbb{E}\,\psi_{x'}(d) - J(x')| + 3|x - x'| \le \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{12} < \epsilon. \qquad \square$$

Proof of theorem SB.2. We prove the output-side statement. The input side is the same argument applied to $M_t^\top = \sum_{j,a} \mathsf{h}_{ja}\, \mathsf{v}_{ja} \otimes \mathsf{u}_{ja}$, with the roles of $(\Sigma_{\text{out}}, z_{ja}, \widehat{A}_t)$ and $(\Sigma_{\text{in}}, w_{ja}, \widehat{A}_t^\top)$ exchanged; this exchange leaves Assumption S3 and every lemma above invariant, and it produces the same scalars $(\sigma_B, \tilde{\sigma}_B, q_B)$. Fix $s \in I$; all bounds below are uniform in $s \in I$.

Step 1: the deletion identity in expectation. The starting point is the third identity of lemma SB.3,

$$\mathrm{Id} - sR(-s) = R(-s) M_t M_t^\top.$$

By eq. (SB.26), the right side is a sum over samples,

$$R(-s) M_t M_t^\top = \sum_{j \ge 0} \sum_{a=1}^{B} \big(\mathsf{h}_{ja} R(-s) M_t \mathsf{v}_{ja}\big) \otimes \mathsf{u}_{ja},$$

and eq. (SB.15), applied to each sample with the deleted buffer $M_0^{(ja)}$, rewrites each term through the deleted resolvents:

$$R(-s) M_t M_t^\top = \sum_{j \ge 0} \sum_{a=1}^{B} \big(-\alpha_{3,ja} R_0^{(ja)} \mathsf{u}_{ja} + \alpha_{1,ja} \mathsf{X}_0^{(ja)} \mathsf{v}_{ja} \big) \otimes \mathsf{u}_{ja}.$$

Taking expectations and using exchangeability within each age,

$$\mathrm{Id} - s\,\mathcal{P}(s) = B \sum_{j \ge 0} \mathbb{E}\,\big[-\alpha_{3,j}\, R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j) + \alpha_{1,j}\, \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j) \big]. \tag{SB.38}$$

Step 2: the Gram term. Fix j and write

$$\varpi \overset{\text{def}}{=} w_j, \qquad x \overset{\text{def}}{=} \varpi^2 q_B \in [0, q_I], \qquad \psi \overset{\text{def}}{=} \psi_x.$$

Adding and subtracting the target coefficient $\check{\alpha}_{3,j}$ of eq. (SB.33) gives

$$\begin{aligned} B\,\mathbb{E}\,\big[-\alpha_{3,j} R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j) \big] &= B\,\mathbb{E}\,\big[(-\alpha_{3,j} + \check{\alpha}_{3,j}) R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j) \big] \\ &\quad + s\,\varpi^2 \tilde{\sigma}_B \cdot B\,\mathbb{E}\,\big[\psi(d_j)\, R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j) \big]. \end{aligned} \tag{SB.39}$$

The first term of eq. (SB.39) is a replacement error, bounded by the Bessel inequality. Put $e \overset{\text{def}}{=} -\alpha_{3,j} + \check{\alpha}_{3,j}$ and write $\mathcal{G} = \mathcal{G}_{j1}$. We use the two identities

$$B\,\mathsf{u}_j \otimes \mathsf{u}_j = {\Sigma_{\text{out}}}^{1/2} (z_j \otimes z_j) {\Sigma_{\text{out}}}^{1/2}, \qquad \mathbb{E}_{\mathcal{G}}[(z_j \otimes z_j)\, e] = \mathbb{E}_{\mathcal{G}}[(z_j \otimes z_j - \mathrm{Id})\, e] + \mathrm{Id}\; \mathbb{E}_{\mathcal{G}}\, e.$$

With lemma SB.7 for the centered part and $|\,\mathbb{E}_{\mathcal{G}}\, e| \le \|e\|_{2,\mathcal{G}}$ for the second, they give

$$\big\| B\;\mathbb{E}_{\mathcal{G}}[(\mathsf{u}_j \otimes \mathsf{u}_j)\, e] \big\| \le C_\Sigma \big(\sqrt{2} + 1\big) \|e\|_{2,\mathcal{G}}. \tag{SB.40}$$

The matrix $R_0^{(j)}$ is built from the samples other than $(j, 1)$ and satisfies $\|R_0^{(j)}\| \le 1/s$, and $\mathbb{E}\,\|e\|_{2,\mathcal{G}} \le \|e\|_2$ by Jensen's inequality. Hence, by the tower property, eq. (SB.40) and lemma SB.12,

$$\Big\| B\,\mathbb{E}\,\big[e\, R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j) \big] \Big\| = \Big\| \mathbb{E}\,\big[R_0^{(j)}\; B\;\mathbb{E}_{\mathcal{G}}[(\mathsf{u}_j \otimes \mathsf{u}_j)\, e] \big] \Big\| \le \frac{(1 + \sqrt{2}) C_\Sigma}{s_-} \|e\|_2 \le C_I\, \varpi^2 e_B.$$

The second derivative of ψ_x is bounded by 6, by a scaling computation. For $x > 0$,

$$\psi_x(d) = \frac{1}{x}\, \chi(\sqrt{x}\, d), \qquad \chi(y) \overset{\text{def}}{=} \frac{y^2}{1 + y^2},$$

so that

$$\psi_x''(d) = \chi''(\sqrt{x}\, d), \qquad \chi''(y) = \frac{2-6y^2}{(1+y^2)^3}, \qquad |\chi''(y)| \le \frac{2+6y^2}{(1+y^2)^3} \le 6,$$

and at $x = 0$ the second derivative is the constant $\psi_0'' = 2$.

The second term of eq. (SB.39) factorizes by independence and is a Gaussian response. Since $R_0^{(j)}$ is independent of (z_j, w_j), and with $d = z^\top \widehat{A}_t w$,

$$B\ \mathbb{E}\left[\psi(d_j) R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j)\right] = \mathcal{P}_j(s)\, {\Sigma_{\text{out}}}^{1/2}\ \mathbb{E}[(z \otimes z)\, \psi(d)]\, {\Sigma_{\text{out}}}^{1/2},$$
$$\mathbb{E}[(z \otimes z)\, \psi(d)] = \mathrm{Id}\ \mathbb{E}\, \psi(d) + \mathbb{E}[(z \otimes z - \mathrm{Id})\, \psi(d)].$$

The second summand is bounded by eq. (SB.23) with $A = \widehat{A}_t$ and $\|\widehat{A}_t\|_{\mathrm{F}} = 1$:

$$\|\, \mathbb{E}[(z \otimes z - \mathrm{Id})\, \psi(d)]\|_{\mathrm{F}} \le \|\psi_x''\|_\infty \|\widehat{A}_t\| \le 6 \|\widehat{A}_t\|.$$

The first summand equals $\mathrm{Id}\ J(x)$ up to r_B by lemma SB.13. Combining the two summands with $\|\mathcal{P}_j(s)\| \le 1/s_-$, $\|\Sigma_{\text{out}}\| \le C_\Sigma$ and $s\tilde{\sigma}_B \le C_\Sigma \gamma_\star$ from eq. (SB.31),

$$\left\| s\, \varpi^2 \tilde{\sigma}_B \cdot B\ \mathbb{E}\left[\psi(d_j) R_0^{(j)} (\mathsf{u}_j \otimes \mathsf{u}_j)\right] - s\, \varpi^2 \tilde{\sigma}_B\, J(\varpi^2 q_B)\, \mathcal{P}_j(s) \Sigma_{\text{out}} \right\|$$
$$\le s\tilde{\sigma}_B\, \varpi^2\, \frac{C_\Sigma}{s_-} \big(r_B + 6\|\widehat{A}_t\|\big) \le C_I \varpi^2 \big(r_B + \|\widehat{A}_t\|\big).$$

Step 3: the mixed term. The mixed term splits in the same way as the Gram term, with $f \overset{\text{def}}{=} f_x$ and $x = \varpi^2 q_B$:

$$B\ \mathbb{E}\left[\alpha_{1,j} \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right] = B\ \mathbb{E}\left[(\alpha_{1,j} - \check{\alpha}_{1,j}) \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right]$$
$$+ \varpi \cdot B\ \mathbb{E}\left[f(d_j)\, \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right].$$

The replacement error is bounded by the first inequality of lemma SB.7. Put $e \overset{\text{def}}{=} \alpha_{1,j} - \check{\alpha}_{1,j}$, and use

$$B\, \mathsf{v}_j \otimes \mathsf{u}_j = {\Sigma_{\text{in}}}^{1/2} (w_j \otimes z_j) {\Sigma_{\text{out}}}^{1/2}$$

and $\|\mathsf{X}_0^{(j)}\| \le (2\sqrt{s_-})^{-1}$. The tower property, Jensen's inequality and lemma SB.12 give

$$\left\| B\ \mathbb{E}\left[e\, \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right] \right\| \le \frac{1}{2\sqrt{s_-}}\ \mathbb{E} \left\| {\Sigma_{\text{in}}}^{1/2}\ \mathbb{E}_{\mathcal{G}}[(w_j \otimes z_j)\, e]\, {\Sigma_{\text{out}}}^{1/2} \right\|_{\mathrm{F}}$$
$$\le \frac{C_\Sigma}{2\sqrt{s_-}} \|e\|_2 \le C_I\, \varpi\, e_B.$$

The first and third derivatives of $f = f_x$ are bounded, by a scaling computation:

$$\|f'\|_\infty \le 1, \qquad \|f'''\|_\infty \le 36x \le 36 q_I. \tag{SB.41}$$

Indeed, for $x > 0$,

$$f_x(d) = \frac{1}{\sqrt{x}}\, \phi(\sqrt{x}\, d), \qquad \phi(y) \overset{\text{def}}{=} \frac{y}{1+y^2},$$

so that, at $y = \sqrt{x}\, d$,

$$f_x'(d) = \phi'(y) = \frac{1-y^2}{(1+y^2)^2}, \qquad f_x'''(d) = x\, \phi'''(y) = x\, \frac{-6y^4 + 36y^2 - 6}{(1+y^2)^4}.$$

The two numerators are bounded by the elementary inequalities

$$|1 - y^2| \le (1+y^2)^2, \qquad 6y^4 + 36y^2 + 6 \le 36(1+y^2)^2.$$

At $x = 0$ the function f_0 is the identity, whose first derivative is 1 and whose third derivative vanishes.

The main part is a Gaussian response whose leading term carries $\widehat{A}_t^\top$. Independence of $\mathsf{X}_0^{(j)}$ from (z_j, w_j) gives

$$B\ \mathbb{E}\left[f(d_j)\, \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right] = \mathbb{E}\left[\mathsf{X}_0^{(j)}\right] {\Sigma_{\text{in}}}^{1/2}\ \mathbb{E}[(w \otimes z)\, f(d)]\, {\Sigma_{\text{out}}}^{1/2},$$

and eq. (SB.22) applied to $(w, z, \widehat{A}_t^\top)$, with $\|\widehat{A}_t^\top\|_{\mathrm{F}} = 1$ and the derivative bounds eq. (SB.41), gives

$$\|\,\mathbb{E}[(w \otimes z)\, f(d)]\| \le \|\widehat{A}_t^\top\ \mathbb{E}\, f'(d)\| + \|f'''\|_\infty \|\widehat{A}_t\|^2 \le \|\widehat{A}_t\| + 36 q_I \|\widehat{A}_t\|^2.$$

Here the spectral diffuseness $\|\widehat{A}_t\| \to 0$ is used in operator norm: the leading term $\widehat{A}_t^\top\ \mathbb{E}\, f'(d)$ of the mixed response has Frobenius norm $|\,\mathbb{E}\, f'(d)|$ but operator norm at most $\|\widehat{A}_t\|$. Combining the two parts with $\|\,\mathbb{E}[\mathsf{X}_0^{(j)}]\| \le (2\sqrt{s_-})^{-1}$ and $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$,

$$\Big\| B\ \mathbb{E}\left[\alpha_{1,j} \mathsf{X}_0^{(j)} (\mathsf{v}_j \otimes \mathsf{u}_j)\right] \Big\| \le C_I\, \varpi \left(e_B + \|\widehat{A}_t\| + \|\widehat{A}_t\|^2\right).$$

Step 4: summation and the closure identity. Summing the bounds of Steps 2 and 3 over j in eq. (SB.38), with the weight sums of eq. (SA.18),

$$\Big\| \mathrm{Id} - s\,\mathcal{P}(s) - \sum_{j \ge 0} s\, w_j^2 \tilde{\sigma}_B\, J(w_j^2 q_B)\, \mathcal{P}_j(s) \Sigma_{\mathrm{out}} \Big\| \le C_I \left[s_2 (e_B + r_B + \|\widehat{A}_t\|) + e_B + \|\widehat{A}_t\| + \|\widehat{A}_t\|^2 \right].$$

Next we replace $\mathcal{P}_j$ by $\mathcal{P}$, at the cost of one deletion per age:

$$\|\mathcal{P}(s) - \mathcal{P}_j(s)\| \le \frac{C_I\, w_j}{B}. \tag{SB.42}$$

Taking $\mathbb{E}_{\mathcal{G}_{j1}}$ in eq. (SB.14) and then the full expectation writes $\mathcal{P}(s) - \mathcal{P}_j(s)$ as a sum of four terms, each an outer factor $R_0^{(j)}$ or $\mathsf{X}_0^{(j)}$ on either side of a conditional moment of a rank-one matrix. The outer factors satisfy $\|R_0^{(j)}\| \le 1/s_-$ and $\|\mathsf{X}_0^{(j)}\| \le (2\sqrt{s_-})^{-1}$. The conditional moments are bounded by eq. (SB.40) and by its two analogues

$$B\, \|\,\mathbb{E}_{\mathcal{G}}[(\mathsf{v} \otimes \mathsf{v})\, e]\| \le (1 + \sqrt{2}) C_\Sigma \|e\|_{2,\mathcal{G}}, \qquad B\, \|\,\mathbb{E}_{\mathcal{G}}[(\mathsf{u} \otimes \mathsf{v})\, e]\|_{\mathrm{F}} \le C_\Sigma \|e\|_{2,\mathcal{G}},$$

which follow from lemma SB.7 in the same way, applied with $e = \alpha_{3,j}, \alpha_{1,j}, \alpha_{2,j}$. Since (d_j, N_j) are functions of the deleted sample only, lemmas SB.5 and SB.6 bound the conditional L^2 norms of these three coefficients by $C_I w_j^2$, $C_I w_j$ and $C_I w_j^2$. Multiplying the three kinds of bounds gives eq. (SB.42).

The replacement of $\mathcal{P}_j$ by $\mathcal{P}$ costs at most $C_I s_3 / B$. Since $s\tilde{\sigma}_B J \le C_\Sigma \gamma_\star$ and $\|\Sigma_{\mathrm{out}}\| \le C_\Sigma$, eq. (SB.42) gives

$$\Big\| \sum_{j \ge 0} s\, w_j^2 \tilde{\sigma}_B\, J(w_j^2 q_B) \left(\mathcal{P}_j(s) - \mathcal{P}(s)\right) \Sigma_{\mathrm{out}} \Big\| \le \sum_{j \ge 0} C_\Sigma^2 \gamma_\star\, w_j^2\, \frac{C_I w_j}{B} = \frac{C_I s_3}{B}.$$

Writing

$$a_B(s) \overset{\mathrm{def}}{=} s\, \tilde{\sigma}_B(s)\, F_\beta(q_B(s)), \qquad c_B(s) \overset{\mathrm{def}}{=} s\, \sigma_B(s)\, F_\beta(q_B(s)), \tag{SB.43}$$

and recalling $F_\beta(q) = \sum_j w_j^2 J(w_j^2 q)$, we have shown

$$\begin{aligned} E_B(s) &\overset{\mathrm{def}}{=} \mathrm{Id} - s\,\mathcal{P}(s) - a_B(s)\, \mathcal{P}(s) \Sigma_{\mathrm{out}}, \\ \sup_{s \in I} \|E_B(s)\| &\le C_I \left[e_B + r_B + \|\widehat{A}_t\| + \|\widehat{A}_t\|^2 + B^{-1} \right] \longrightarrow 0, \end{aligned} \tag{SB.44}$$

by lemmas SB.12 and SB.13 and Assumption S3.

The closure error controls the distance to the candidate resolvent. Since $\mathcal{P}(s)(s\,\mathrm{Id} + a_B \Sigma_{\mathrm{out}}) = \mathrm{Id} - E_B(s)$ and $a_B \ge 0$,

$$\|\mathcal{P}(s) - (s\,\mathrm{Id} + a_B(s)\Sigma_{\mathrm{out}})^{-1}\| = \|E_B(s)(s\,\mathrm{Id} + a_B \Sigma_{\mathrm{out}})^{-1}\| \le \frac{\|E_B(s)\|}{s_-}. \tag{SB.45}$$

The input-side argument gives in the same way

$$\|\,\mathbb{E}\,\widetilde{R}(-s) - (s\,\mathrm{Id} + c_B(s)\Sigma_{\mathrm{in}})^{-1}\| \le \frac{\|\widetilde{E}_B(s)\|}{s_-}, \qquad \sup_{s \in I} \|\widetilde{E}_B(s)\| \longrightarrow 0.$$

Step 5: identification of the branch. The tuple $(\sigma_B, \tilde{\sigma}_B, a_B, c_B)(s)$ is nonnegative and lies in the box of theorem SB.1(iii) by eq. (SB.31) and $F_\beta \le s_2$. Its third and fourth residuals vanish by eq. (SB.43), and its first residual is

$$\begin{aligned}\Big|\sigma_B - \frac{1}{B}\operatorname{Tr}\left(\Sigma_{\text{out}}(s\operatorname{Id} + a_B\Sigma_{\text{out}})^{-1}\right)\Big| &= \Big|\frac{1}{B}\operatorname{Tr}\left(\Sigma_{\text{out}}(\mathcal{P} - (s\operatorname{Id} + a_B\Sigma_{\text{out}})^{-1})\right)\Big| \\ &\le \frac{\operatorname{Tr}\Sigma_{\text{out}}}{B}\,\frac{\|E_B(s)\|}{s_-} \le \frac{C_\Sigma\gamma_\star}{s_-}\|E_B(s)\|,\end{aligned}$$

with the same bound for the second residual in terms of $\widetilde{E}_B$. Both residuals tend to zero uniformly in $s \in I$. By theorem SB.1(iii), applied with $(\nu, \tilde{\nu}) = (\nu_{\text{out}}, \nu_{\text{in}}) \in \mathcal{K}$,

$$\sup_{s\in I}\Big(|\sigma_B - \sigma| + |\tilde{\sigma}_B - \tilde{\sigma}| + |a_B - \tilde{s}_1| + |c_B - s_1|\Big)(-s) \longrightarrow 0.$$

The conclusion follows from the resolvent identity. For every $s \in I$,

$$\begin{aligned}&\|(s\operatorname{Id} + a_B\Sigma_{\text{out}})^{-1} - (s\operatorname{Id} + \tilde{s}_1\Sigma_{\text{out}})^{-1}\| \\ &\quad = \|(s\operatorname{Id} + a_B\Sigma_{\text{out}})^{-1}\,(\tilde{s}_1 - a_B)\Sigma_{\text{out}}\,(s\operatorname{Id} + \tilde{s}_1\Sigma_{\text{out}})^{-1}\| \le \frac{C_\Sigma|a_B - \tilde{s}_1|}{s_-^2},\end{aligned}$$

and together with eq. (SB.45) and the triangle inequality,

$$\sup_{s\in I}\|\,\mathbb{E}\,R(-s) - \overline{R}(-s)\| \le \sup_{s\in I}\Big(\frac{\|E_B(s)\|}{s_-} + \frac{C_\Sigma|a_B(s) - \tilde{s}_1(-s)|}{s_-^2}\Big) \longrightarrow 0.$$

The input side is identical. □

Proof of corollary SB.1. Put

$$\varepsilon(s) \overset{\text{def}}{=} \frac{1}{N_{\text{out}}}\operatorname{Tr}\big[T\big(R(-s) - \overline{R}(-s)\big)\big].$$

At a fixed $s \in I$ the error splits into a fluctuation and a bias. We use the trace bound

$$\frac{|\operatorname{Tr}(TX)|}{N_{\text{out}}} \le \frac{\|T\|\,\|X\|_*}{N_{\text{out}}} \le \|T\|\,\|X\|,$$

and $N_{\text{out}} \ge B\gamma_{\text{out}}/2$ for all large B. By $(x+y)^2 \le 2x^2 + 2y^2$, lemma SB.10 rescaled from B to N_{out}, and theorem SB.2,

$$\begin{aligned}\mathbb{E}\,|\varepsilon(s)|^2 &\le 2\operatorname{Var}\Big(\frac{\operatorname{Tr}(TR(-s))}{N_{\text{out}}}\Big) + 2\Big|\frac{\operatorname{Tr}[T(\mathcal{P}(s) - \overline{R}(-s))]}{N_{\text{out}}}\Big|^2 \\ &\le \frac{8C_IC_T^2s_2}{\gamma_{\text{out}}^2B} + 2C_T^2\|\mathcal{P}(s) - \overline{R}(-s)\|^2 \longrightarrow 0.\end{aligned}$$

The error is equicontinuous in s, uniformly in the dimension and the realization. By lemma SB.3, for $s, s' \in I$,

$$\|R(-s) - R(-s')\| \le \frac{|s - s'|}{s_-^2},$$

and by the resolvent identity applied to $\overline{R}$,

$$\|\overline{R}(-s) - \overline{R}(-s')\| \le \frac{C_\Sigma|\tilde{s}_1(-s) - \tilde{s}_1(-s')| + |s - s'|}{s_-^2}.$$

By theorem SB.1(ii) the map $s \mapsto \tilde{s}_1(-s)$ has a modulus of continuity ω_I on I that is uniform over $\mathcal{K}$, hence over the dimension. With the trace bound this gives a deterministic modulus $\varkappa$, uniform in the dimension and the realization, such that

$$|\varepsilon(s) - \varepsilon(s')| \le \varkappa(|s - s'|) \qquad (s, s' \in I).$$

The supremum is reduced to a finite net. Let $I_\eta \subset I$ be a finite η-net. Then

$$\sup_{s\in I}|\varepsilon(s)| \le \max_{s\in I_\eta}|\varepsilon(s)| + \varkappa(\eta),$$

and squaring, bounding the maximum by the sum and taking expectations,

$$\mathbb{E}\sup_{s\in I}|\varepsilon(s)|^2 \le 2\sum_{s\in I_\eta}\mathbb{E}\,|\varepsilon(s)|^2 + 2\varkappa(\eta)^2.$$

Taking first the limit in the dimension at fixed η and then $\eta \downarrow 0$ proves the first claim. The input side is identical.

The last claim is the case $T = \Sigma_{\text{out}}\, B/N_{\text{out}}$ and $\widetilde{T} = \Sigma_{\text{in}}\, B/N_{\text{in}}$, admissible since $B/N_{\text{out}} \le 2/\gamma_{\text{out}}$ for large B, for which

$$\frac{1}{N_{\text{out}}}\operatorname{Tr}\big(T\overline{R}(-s)\big) = \frac{1}{B}\operatorname{Tr}\big(\Sigma_{\text{out}}\overline{R}(-s)\big) = \sigma(-s)$$

by definition SB.1. □

SC Drift: the expected regularized update

This section identifies the drift kernel $\mathfrak{d}_{ij}$ of the main text. In the frozen-state model of definition SA.3, the expected regularized update $\mathbb{E}\,Q_\varrho(M_t)$ has a deterministic equivalent in unnormalized Frobenius norm, uniformly in the normalized ridge $\varrho \in [\varrho_-, \varrho_+]$, and its projection on the eigenpair (u_i, v_j) is $\mathfrak{d}(\mu_i,\lambda_j)\,(u_i^\top \Delta_t v_j)/\sqrt{\mathscr{R}(t)}$ with an explicit integral for the kernel function $\mathfrak{d}$. The section has two random-matrix theorems. Theorem SC.1 proves that the expected mixed resolvent $\mathbb{E}[R(-s)M_t]$ converges in Frobenius norm to the candidate $T(s)$ built from the branch of theorem SB.1, uniformly on compact subsets of $(0,\infty)$, and that it satisfies a dimension-independent bound of order $1/s$ for $s \ge 1$. Theorem SC.2 integrates the first statement along the negative axis, with the tail bound controlling large s, and identifies the expected update. Corollary SC.1 reads the result in the eigenbases and defines $\mathfrak{d}$; proposition SC.2 records the one-step risk identity in which the drift enters.

The section works in the setting of appendix SA: the frozen-state model of definition SA.3 under Assumption S3, the normalized state $\widehat{A}_t$ of eq. (SA.10), the EMA weights eq. (SA.18), the Gram resolvents eq. (SA.36), the integral representation lemma SA.3, the residual law lemma SA.7, and the conventions of appendix SA.1. From appendix SB it takes the branch $(\sigma,\tilde\sigma,\tilde s_1, s_1, q)(-s)$ and the deterministic equivalents $\overline{R}(-s) = (\tilde s_1(-s)\Sigma_{\text{out}} + s\,\mathrm{Id})^{-1}$ and $\overline{\widetilde{R}}(-s) = (s_1(-s)\Sigma_{\text{in}} + s\,\mathrm{Id})^{-1}$ of definition SB.1, the source factor $L_\beta(q) = \sum_j w_j J(w_j^2 q)$ of eq. (SB.1), the Gram closure theorem SB.2 with its trace form corollary SB.1, and the finite identities of appendix SB.2: the notation $R_s(M)$, $\widetilde{R}_s(M)$, $\mathsf{X}_s(M) = R_s(M)M$ of eq. (SB.10), the sample decomposition eq. (SB.26) of the buffer, the rank-two deletion lemma SB.4 with its coefficients $\alpha_1,\alpha_2,\alpha_3$ of eq. (SB.13), the coefficient and moment bounds lemmas SB.5 and SB.6, the Bessel bounds lemma SB.7, the Gaussian response lemma SB.8, the replacement of the deletion coefficients lemma SB.12, and the uniform Gaussian limit lemma SB.13. As there, for a sample (j,a) we write $M_0^{(ja)} = M_t - \mathsf{h}_{ja}\,\mathsf{u}_{ja}\otimes\mathsf{v}_{ja}$ for the buffer with that sample deleted, $R_0^{(ja)} = R_s(M_0^{(ja)})$, $\widetilde{R}_0^{(ja)}$ and $\mathsf{X}_0^{(ja)}$ accordingly, $\mathcal{G}_{ja}$ for the σ-field generated by all samples other than (j,a), and $\alpha_{1,ja},\alpha_{2,ja},\alpha_{3,ja}$ for the coefficients of lemma SB.4 with $(M_0,\mathsf{h},\mathsf{u},\mathsf{v}) = (M_0^{(ja)},\mathsf{h}_{ja},\mathsf{u}_{ja},\mathsf{v}_{ja})$; samples of one age are exchangeable, and the index a is dropped when only the law matters. The expected resolvents $\mathcal{P}(s) = \mathbb{E}\,R(-s)$, $\mathcal{P}_j(s) = \mathbb{E}\,R_0^{(j1)}$, the expected traces $\sigma_B(s)$, $\tilde\sigma_B(s)$ and the product $q_B(s) = s\sigma_B\tilde\sigma_B$ are those of eq. (SB.30). Constants: for a compact $I \subset (0,\infty)$ with $s_- = \min I$, C_I denotes a constant depending only on $(I, C_\Sigma, \gamma_\star)$; a constant C without subscript depends only on $(C_\Sigma,\gamma_\star)$; both may change from line to line.

The statements of this section hold for the frozen-state model with any fixed weight sequence $w = (w_j)_{j\ge0}$ in the class

$$0 \le w_j \le 1 \quad (j\ge0), \qquad \sum_{j\ge0} w_j \le 1, \qquad w_j > 0 \text{ for some } j, \tag{SC.1}$$

in place of the EMA weights, with F_β, L_β, s_p, the branch of theorem SB.1 and the candidates below computed for w; every proof uses the weights only through eq. (SC.1). This is the weight class eq. (SB.27) for which theorem SB.2, corollary SB.1 and the lemmas of appendix SB.3 are stated. The EMA weights belong to it by eq. (SA.18) and are the case of interest; statements at $\beta = 0$ concern the EMA weights at $\beta = 0$, that is, $w = (1, 0, 0, \dots)$.

SC.1 The candidates and the main statements

For $s > 0$ the expected mixed resolvent of the frozen-state model and its candidate are

$$\mathcal{H}(s) \stackrel{\text{def}}{=} \mathbb{E}\left[R(-s)M_t\right] = \mathbb{E}\,\mathsf{X}_s(M_t), \qquad T(s) \stackrel{\text{def}}{=} s\,L_\beta\big(q(-s)\big)\,\overline{R}(-s)\,\frac{\Sigma_{\text{out}}\Delta_t\Sigma_{\text{in}}}{\sqrt{2\mathscr{R}(t)}}\,\overline{\widetilde{R}}(-s), \tag{SC.2}$$

where $\Sigma_{\text{out}}\Delta_t\Sigma_{\text{in}}/\sqrt{2\mathscr{R}(t)} = {\Sigma_{\text{out}}}^{1/2}\widehat{A}_t{\Sigma_{\text{in}}}^{1/2} = \mathbb{E}\,G_{t+1}$ by eq. (SA.11). For z in the sector $\mathbb{S}$ of eq. (SB.11) we write likewise $\mathcal{H}(z) \stackrel{\text{def}}{=} \mathbb{E}\,\mathsf{X}_z(M_t)$. The candidate for the expected update is the integral of T along the negative axis,

$$\overline{\widetilde{G}}_\varrho \stackrel{\text{def}}{=} \frac{2}{\pi}\int_0^\infty T(\varrho + u^2)\,du \qquad (\varrho > 0), \tag{SC.3}$$

which theorem SC.2 shows to exist. In the eigenbases of the covariances the candidate is diagonal, which motivates the kernel function

$$\mathfrak{d}(\mu,\lambda) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi}\int_0^\infty \frac{z\,L_\beta(q(-z))\,\mu\lambda}{\big(z + \tilde{s}_1(-z)\mu\big)\big(z + s_1(-z)\lambda\big)}\,du, \qquad z = \varrho + u^2, \quad \mu,\lambda \geq 0. \tag{SC.4}$$

The branch in eq. (SC.4) is the one of the full spectral measures eq. (SA.41), not a branch recomputed at (μ, λ); the function $\mathfrak{d}$ therefore depends on the dimension, on ϱ and on β, and this dependence is suppressed in the notation. The main-text kernel is $\mathfrak{d}_{ij} = \mathfrak{d}(\mu_i, \lambda_j)$ (corollary SC.1).

Theorem SC.1 (Mixed resolvent closure and tail bound)**.** *(i) Under Assumption S3, for every compact $I \subset (0,\infty)$,*

$$\sup_{s\in I}\|\mathcal{H}(s) - T(s)\|_{\text{F}} \longrightarrow 0. \tag{SC.5}$$

(ii) In the frozen-state model under Assumption S1, with $\max(N_{\text{out}}, N_{\text{in}}) \leq \gamma_\star B$, $\|\Sigma_{\text{out}}\|, \|\Sigma_{\text{in}}\| \leq C_\Sigma$, $\|\widehat{A}_t\|_{\text{F}} = 1$ and weights satisfying eq. (SC.1), but without any condition on $\|\widehat{A}_t\|$ and for every finite dimension, there is C depending only on $(C_\Sigma, \gamma_\star)$ such that

$$\|\mathcal{H}(s)\|_{\text{F}} \leq \frac{C}{s} \quad (s \geq 1), \qquad \|T(s)\|_{\text{F}} \leq \frac{C_\Sigma}{s} \quad (s > 0). \tag{SC.6}$$

Moreover, for every $s_\circ > 0$ there is $C_\circ$ depending only on $(s_\circ, C_\Sigma, \gamma_\star)$ such that

$$\|\mathcal{H}(z)\|_{\text{F}} \leq \frac{C_\circ}{\operatorname{Re} z} \qquad (z \in \mathbb{S},\ \operatorname{Re} z \geq s_\circ). \tag{SC.7}$$

Theorem SC.2 (Regularized mean)**.** *Under Assumption S3, the integral eq. (SC.3) converges absolutely in Frobenius norm, with $\|\overline{\widetilde{G}}_\varrho\|_{\text{F}} \leq C_\Sigma/\sqrt{\varrho}$, and*

$$\varepsilon_B \stackrel{\text{def}}{=} \sup_{\varrho\in[\varrho_-,\varrho_+]}\|\,\mathbb{E}\,Q_\varrho(M_t) - \overline{\widetilde{G}}_\varrho\|_{\text{F}} \longrightarrow 0. \tag{SC.8}$$

Consequently, for every deterministic $V \in \mathbb{R}^{N_{\text{out}}\times N_{\text{in}}}$,

$$\sup_{\varrho\in[\varrho_-,\varrho_+]}\left|\left\langle V, \mathbb{E}\,Q_\varrho(M_t) - \overline{\widetilde{G}}_\varrho\right\rangle\right| \leq \|V\|_{\text{F}}\,\varepsilon_B, \qquad \langle V, W\rangle \stackrel{\text{def}}{=} \operatorname{Tr}(VW^\top). \tag{SC.9}$$

Proposition SC.1 (History truncation)**.** *Under Assumption S3, for $J \geq J_w \stackrel{\text{def}}{=} \min\{j : w_j > 0\}$ let $w_j^{(J)} \stackrel{\text{def}}{=} w_j\mathbf{1}_{j\leq J}$, let $M^{(J)} = \sum_{j\leq J} w_j G_{t+1-j}$ be the truncated buffer of lemma SA.6, and let $T^{(J)}$ and $\overline{\widetilde{G}}_\varrho^{(J)}$ be the candidates eqs. (SC.2) and (SC.3) built from the solution of the system eq. (SB.2) for the weights $w^{(J)}$.*

(i) For every fixed J, $\sup_{\varrho\in[\varrho_-,\varrho_+]} \| \mathbb{E}\, Q_\varrho(M^{(J)}) - \overline{\widetilde{G}}_\varrho^{(J)} \|_{\mathrm{F}} \to 0$ as the dimension grows.

(ii) $\lim_{J\to\infty} \sup_B \sup_{\varrho\in[\varrho_-,\varrho_+]} \|\overline{\widetilde{G}}_\varrho - \overline{\widetilde{G}}_\varrho^{(J)} \|_{\mathrm{F}} = 0$, where $\sup_B$ runs over the whole sequence of dimensions.

(iii) Consequently

$$\lim_{J\to\infty} \limsup_{B\to\infty} \sup_{\varrho\in[\varrho_-,\varrho_+]} \| \mathbb{E}\, Q_\varrho(M_t) - \mathbb{E}\, Q_\varrho(M^{(J)}) \|_{\mathrm{F}} = 0. \tag{SC.10}$$

The order of the limits in eq. (SC.10) is part of the statement: the dimension limit is taken first at fixed history length J, and $J \to \infty$ afterwards. No rate in J at fixed dimension is asserted, and the finite-dimensional bound of lemma SA.6(ii), whose right side grows with the dimension, is not used.

Corollary SC.1 (The drift kernel)**.** *Under Assumption S3, let $\widetilde{G}_{t+1} = Q_\varrho(M_t)$ be the frozen-state update and put $(E_\varrho)_{ij} \overset{\text{def}}{=} u_i^\top \big(\mathbb{E}\, \widetilde{G}_{t+1} - \overline{\widetilde{G}}_\varrho \big) v_j$.*

(i) For every pair (i,j) and every $\varrho \in [\varrho_-, \varrho_+]$,

$$\mathbb{E}\left[u_i^\top \widetilde{G}_{t+1} v_j\right] = \frac{\sqrt{2}\,\mathfrak{d}(\mu_i,\lambda_j)}{\sqrt{2\mathscr{R}(t)}}\,(u_i^\top \Delta_t v_j) + (E_\varrho)_{ij} = \frac{\mathfrak{d}(\mu_i,\lambda_j)}{\sqrt{\mathscr{R}(t)}}\,(u_i^\top \Delta_t v_j) + (E_\varrho)_{ij}, \tag{SC.11}$$

and $\sup_{\varrho\in[\varrho_-,\varrho_+]} \big(\sum_{i,j} (E_\varrho)_{ij}^2\big)^{1/2} = \varepsilon_B \to 0$. Consequently the frozen-state drift eq. (SA.19) of the update $\widetilde{G}_{t+1}$ is

$$\mathscr{D}_{ij}(t) = \frac{\mathfrak{d}_{ij}}{\sqrt{\mathscr{R}(t)}}\,\mathscr{Q}_{ij}(t) + (u_i^\top \Delta_t v_j)(E_\varrho)_{ij}, \qquad \mathfrak{d}_{ij} \overset{\text{def}}{=} \mathfrak{d}(\mu_i,\lambda_j), \tag{SC.12}$$

and for every bounded function ϕ on $[0,\infty)^2$,

$$\Big| \sum_{i,j} \phi(\mu_i,\lambda_j)\,(u_i^\top \Delta_t v_j)\,(E_\varrho)_{ij} \Big| \le \|\phi\|_\infty\, \|\Delta_t\|_{\mathrm{F}}\, \varepsilon_B. \tag{SC.13}$$

(ii) In the frozen-state model with $\max(N_{\mathrm{out}}, N_{\mathrm{in}}) \le \gamma_\star B$ and $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$, at every finite dimension and for every $\varrho > 0$, the function $\mathfrak{d}$ is continuous on $[0,\infty)^2$, vanishes when $\mu = 0$ or $\lambda = 0$, and satisfies

$$0 \le \mathfrak{d}(\mu,\lambda) \le \frac{\mu\lambda}{\sqrt{2\varrho}}, \qquad |\partial_\mu \mathfrak{d}(\mu,\lambda)| \le \frac{\lambda}{\sqrt{2\varrho}}, \qquad |\partial_\lambda \mathfrak{d}(\mu,\lambda)| \le \frac{\mu}{\sqrt{2\varrho}}, \tag{SC.14}$$

with one-sided derivatives at 0; in particular, for $\varrho \in [\varrho_-, \varrho_+]$, $\mathfrak{d}$ is Lipschitz on $[0, C_\Sigma]^2$ with constant $C_\Sigma/\sqrt{2\varrho_-}$ in each variable.

In the letters of eq. (6), the first term of eq. (SC.12) is $\mathcal{D}_{ij} = \Omega_{ij}\,\mathfrak{d}_{ij}/\sqrt{\mathcal{R}}$: the drift of the mode (i,j) is the projected risk times the kernel, divided by the square root of the risk. The factor $\sqrt{2}$ in the first form of eq. (SC.11) is the normalization of eq. (SC.4), chosen so that the second form carries no constant.

Proposition SC.2 (One-step risk drift in the frozen state)**.** *Under Assumption S3, let $\eta \ge 0$ be deterministic and put $\Delta^+ \overset{\text{def}}{=} \Delta_t - \eta\,\widetilde{G}_{t+1}$ with $\widetilde{G}_{t+1} = Q_\varrho(M_t)$. Then*

$$\mathbb{E}\left[\mathscr{R}(\Delta^+)\right] - \mathscr{R}(t) = -\eta\,\sqrt{2\mathscr{R}(t)}\,\big\langle \mathbb{E}\, G_{t+1}, \mathbb{E}\, \widetilde{G}_{t+1} \big\rangle + \frac{\eta^2}{2}\,\mathbb{E}\,\|\widetilde{G}_{t+1}\|_\Sigma^2, \tag{SC.15}$$

and the linear term satisfies

$$\Big| \sqrt{2\mathscr{R}(t)}\,\big\langle \mathbb{E}\, G_{t+1}, \mathbb{E}\, \widetilde{G}_{t+1} \big\rangle - \frac{1}{\sqrt{\mathscr{R}(t)}} \sum_{i,j} \mu_i\lambda_j\,\mathfrak{d}_{ij}\,\mathscr{Q}_{ij}(t) \Big| \le \sqrt{2\mathscr{R}(t)}\, C_\Sigma\, \varepsilon_B. \tag{SC.16}$$

Proposition SC.2 is a statement about one step from a frozen state: the history of the buffer is computed at Δ_t, and the risk $\mathscr{R}(t)$ is a fixed number. The replacement error $\eta\sqrt{2\mathscr{R}(t)}C_\Sigma\varepsilon_B$ in the linear term is small relative to the drift scale $\eta\sqrt{\mathscr{R}(t)}$ only; it need not be small in absolute terms along a sequence of physical states whose risk grows. The quadratic term is an exact second moment, bounded by $C_\Sigma^2 \min(N_{\mathrm{out}}, N_{\mathrm{in}})\eta^2/2$ through lemma SA.2(i), and its deterministic equivalent is the subject of the volatility section.

SC.2 Finite identities and bounds

The closure is proved by deleting one sample from the buffer, as the Gram closure was. Two further consequences of lemma SB.4 are needed: the action of the full resolvent on the deleted feature, and the change of the mixed resolvent under the deletion.

The lemmas of this subsection are finite-dimensional. They hold at every dimension in the frozen-state model with weights in eq. (SC.1), $\|\widehat{A}_t\|_{\mathrm{F}} = 1$, $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$ and $\max(N_{\mathrm{out}}, N_{\mathrm{in}}) \le \gamma_\star B$, with no condition on $\|\widehat{A}_t\|$, and their constants depend only on the quantities named in each statement.

Lemma SC.1 (Deletion identities for the mixed resolvent)**.** *In the setting of lemma SB.4, with* $\mathsf{X} \overset{\text{def}}{=} \mathsf{X}_s(M)$,

$$\mathsf{h}\, R\mathsf{u} = \alpha_1\, R_0\mathsf{u} - \alpha_2\, \mathsf{X}_0\mathsf{v}, \tag{SC.17}$$

$$\mathsf{X} - \mathsf{X}_0 = \alpha_3\, R_0(\mathsf{u}\otimes\mathsf{u})\mathsf{X}_0 + s\,\alpha_1\, R_0(\mathsf{u}\otimes\mathsf{v})\widetilde{R}_0 - \alpha_1\, \mathsf{X}_0(\mathsf{v}\otimes\mathsf{u})\mathsf{X}_0 - s\,\alpha_2\, \mathsf{X}_0(\mathsf{v}\otimes\mathsf{v})\widetilde{R}_0. \tag{SC.18}$$

Both identities hold at $z \in \mathbb{S}$ in place of s, with the quantities of lemma SB.4 at z and $\mathsf{X} \overset{\text{def}}{=} \mathsf{X}_z(M)$.

Proof. The first identity is eq. (SB.14) applied to u. Multiplying eq. (SB.14) on the right by u and using the definitions eq. (SB.12) of U and C,

$$R\mathsf{u} = R_0\mathsf{u}\,\big(1 + \alpha_3\mathsf{U} - \alpha_1\mathsf{C}\big) + \mathsf{X}_0\mathsf{v}\,\big(-\alpha_1\mathsf{U} + \alpha_2\mathsf{C}\big).$$

The two scalar coefficients simplify by the definitions eq. (SB.13):

$$1 + \alpha_3\mathsf{U} - \alpha_1\mathsf{C} = \frac{\Lambda - s\mathsf{h}^2\mathsf{U}\mathsf{V} - \mathsf{h}\mathsf{C}(1+\mathsf{h}\mathsf{C})}{\Lambda} = \frac{(1+\mathsf{h}\mathsf{C})^2 - \mathsf{h}\mathsf{C}(1+\mathsf{h}\mathsf{C})}{\Lambda} = \frac{1+\mathsf{h}\mathsf{C}}{\Lambda},$$

$$-\alpha_1\mathsf{U} + \alpha_2\mathsf{C} = \frac{-\mathsf{h}(1+\mathsf{h}\mathsf{C})\mathsf{U} + \mathsf{h}^2\mathsf{U}\mathsf{C}}{\Lambda} = -\frac{\mathsf{h}\mathsf{U}}{\Lambda}.$$

Multiplying by h gives $\mathsf{h}(1+\mathsf{h}\mathsf{C})/\Lambda = \alpha_1$ and $-\mathsf{h}^2\mathsf{U}/\Lambda = -\alpha_2$, which is eq. (SC.17).

The second identity follows from the first and from eq. (SB.14). Since $M = M_0 + \mathsf{h}\,\mathsf{u}\otimes\mathsf{v}$,

$$\mathsf{X} = RM = RM_0 + (\mathsf{h}R\mathsf{u})\otimes\mathsf{v} = \mathsf{X}_0 + (R - R_0)M_0 + (\mathsf{h}R\mathsf{u})\otimes\mathsf{v}.$$

In the middle term we use $\mathsf{u}^\top R_0 M_0 = \mathsf{u}^\top \mathsf{X}_0$ and the identity $\mathsf{X}_0^\top M_0 = M_0^\top R_0 M_0 = \mathrm{Id} - s\widetilde{R}_0$ of lemma SB.3, so that $\mathsf{v}^\top \mathsf{X}_0^\top M_0 = \mathsf{v}^\top - s\,\mathsf{v}^\top\widetilde{R}_0$. Inserting eq. (SB.14),

$$\begin{aligned}(R - R_0)M_0 = {}& \alpha_3\,(R_0\mathsf{u})\otimes(\mathsf{X}_0^\top\mathsf{u}) - \alpha_1\,(R_0\mathsf{u})\otimes\big(\mathsf{v} - s\widetilde{R}_0\mathsf{v}\big)\\ & - \alpha_1\,(\mathsf{X}_0\mathsf{v})\otimes(\mathsf{X}_0^\top\mathsf{u}) + \alpha_2\,(\mathsf{X}_0\mathsf{v})\otimes\big(\mathsf{v} - s\widetilde{R}_0\mathsf{v}\big).\end{aligned}$$

By eq. (SC.17) the last term of X is $(\alpha_1 R_0\mathsf{u} - \alpha_2\mathsf{X}_0\mathsf{v})\otimes\mathsf{v}$, which cancels the two terms of $(R - R_0)M_0$ that carry $\otimes\mathsf{v}$. What remains is eq. (SC.18), since $(R_0\mathsf{u})\otimes(\mathsf{X}_0^\top\mathsf{u}) = R_0(\mathsf{u}\otimes\mathsf{u})\mathsf{X}_0$ and similarly for the other three terms.

At $z \in \mathbb{S}$ the proof is verbatim with s replaced by z: it is algebraic and uses only lemma SB.4 and the identities of lemma SB.3, which hold at z. □

Lemma SC.2 (Conditional moments of the coefficients)**.** *Let $s_\circ > 0$ and $p \ge 1$ be an integer. In the setting of lemma SB.4, let M_0 be a function of the samples other than (j, a), let $(\mathsf{h}, \mathsf{u}, \mathsf{v}) = (\varpi d, \mathsf{u}_{ja}, \mathsf{v}_{ja})$ with $0 \le \varpi \le 1$ and $d = d_{ja}$, and let $\mathcal{G} = \mathcal{G}_{ja}$. Then for every $s \ge s_\circ$,*

$$\|\alpha_1\|_{p,\mathcal{G}} \le C\varpi, \qquad \|\alpha_3\|_{p,\mathcal{G}} \le C\varpi^2, \qquad \|\alpha_2\|_{p,\mathcal{G}} \le \frac{C\varpi^2}{s}, \tag{SC.19}$$

with C depending only on $(p, s_\circ, C_\Sigma, \gamma_\star)$, and $\alpha_2 \ge 0$. The same bounds, with s replaced by $x \overset{\text{def}}{=} \operatorname{Re} z$, hold for the coefficients of lemma SB.4 at every $z \in \mathbb{S}$ with $x \ge s_\circ$, and there $\operatorname{Re}\alpha_2 \ge 0$.

Proof. The pair (d, N) with $\mathsf{N} = \|\mathsf{u}\|^2 + \|\mathsf{v}\|^2$ is a function of the sample (j, a) alone, which is independent of $\mathcal{G}$; the conditional norms are therefore the unconditional norms of the bounds below.

The bounds on α_1 and α_3 are those of lemma SB.5. For $s \geq s_\circ$ apply that lemma with the compact interval $I = [s_\circ, s]$, whose constant $C_I = \max(1, s_\circ^{-1})(1 + \frac{3}{4\sqrt{s_\circ}})^3$ depends only on $s_\circ$:

$$|\alpha_1| \leq C_I \varpi (1 + |d|)^4 (1 + \mathsf{N})^3, \qquad |\alpha_3| \leq C_I \varpi^2 (1 + |d|)^4 (1 + \mathsf{N})^3.$$

The bound on α_2 carries the factor $1/s$ because U does. By lemma SB.3, $0 \leq \mathsf{U} \leq \|\mathsf{u}\|^2/s \leq \mathsf{N}/s$, and by eq. (SB.16) and $|\mathsf{h}| \leq |d|$,

$$\alpha_2 = \frac{\mathsf{h}^2 \mathsf{U}}{\Lambda} \leq \varpi^2 d^2 \, \frac{\mathsf{N}}{s} \Big(1 + \frac{3|d|\mathsf{N}}{4\sqrt{s_\circ}}\Big)^2 \leq \frac{\varpi^2}{s} \Big(1 + \frac{3}{4\sqrt{s_\circ}}\Big)^2 (1 + |d|)^4 (1 + \mathsf{N})^3,$$

the last step by the elementary inequality eq. (SB.20). Positivity of α_2 holds because $\mathsf{U} \geq 0$ and $\Lambda > 0$.

At $z \in \mathbb{S}$ with $x \geq s_\circ$ the coefficients are complex, and the pointwise bounds are obtained from the sector bounds. Put $\kappa \stackrel{\text{def}}{=} s_\circ^{-1/2}$ and $\mathsf{Z} \stackrel{\text{def}}{=} (1 + |d|)^4 (1 + \mathsf{N})^3$. By lemma SB.3 at z, $|\mathsf{U}| \leq \mathsf{N}/x$, $|z\mathsf{V}| \leq \frac{\sqrt{5}}{2}\mathsf{N}$ and $|\mathsf{C}| \leq \mathsf{N}/(4\sqrt{x})$; by eq. (SB.17) and $|\mathsf{h}| \leq |d|$, $|\Lambda|^{-1} \leq (1 + \kappa|d|\mathsf{N})^2$. With eq. (SB.20) for $c = \kappa$ these give

$$|\alpha_1| \leq (1 + \kappa)^3 \varpi \, \mathsf{Z}, \qquad |\alpha_2| \leq \frac{(1 + \kappa)^2 \varpi^2}{x} \, \mathsf{Z}, \qquad |\alpha_3| \leq 2(1 + \kappa)^2 \varpi^2 \, \mathsf{Z}.$$

The sign of α_2 is replaced by that of its real part. Multiplying eq. (SC.17) at z by $\mathsf{u}^\top$ gives $\mathsf{h}\, \mathsf{u}^\top R_z(M) \mathsf{u} = \alpha_1 \mathsf{U} - \alpha_2 \mathsf{C} = \mathsf{h}\mathsf{U}/\Lambda$, so $\alpha_2 = \mathsf{h}^2\, \mathsf{u}^\top R_z(M) \mathsf{u}$. The matrix $R_z(M)$ has real orthonormal eigenvectors and eigenvalues $(s_k^2 + z)^{-1}$ and z^{-1}, all with positive real part, so $\operatorname{Re} \alpha_2 \geq 0$.

The moments of the common factor are bounded by Hölder's inequality and lemma SB.6:

$$\big\|(1 + |d|)^4 (1 + \mathsf{N})^3\big\|_p \leq \|1 + |d|\|_{8p}^4 \, \|1 + \mathsf{N}\|_{6p}^3 \leq \big(1 + \|d\|_{8p}\big)^4 \big(1 + \|\mathsf{N}\|_{6p}\big)^3,$$

and the right side depends only on $(p, C_\Sigma, \gamma_\star)$. □

Lemma SC.3 (Deletion with a bounded multiplier)**.** *Let $s_\circ > 0$. In the setting of lemma SC.2, let h be a scalar random variable depending on the sample (j, a) and on $\mathcal{G} = \mathcal{G}_{ja}$, with $|h| \leq D_h$ for a constant D_h. Then for every $s \geq s_\circ$,*

$$\big\| \mathbb{E}_{\mathcal{G}} \left[h \left(\mathsf{X}_s(M) - \mathsf{X}_s(M_0) \right) \right] \big\|_{\mathrm{F}} \leq \frac{C \, D_h \, \varpi}{s\sqrt{B}}, \tag{SC.20}$$

with C depending only on $(s_\circ, C_\Sigma, \gamma_\star)$. In particular, for a sample of age j and $s \geq s_\circ$,

$$\|\mathcal{H}(s) - \mathcal{H}_j(s)\|_{\mathrm{F}} \leq \frac{C \, w_j}{s\sqrt{B}}, \qquad \mathcal{H}_j(s) \stackrel{\text{def}}{=} \mathbb{E}\, \mathsf{X}_0^{(j1)}. \tag{SC.21}$$

Both bounds, with s replaced by $x \stackrel{\text{def}}{=} \operatorname{Re} z$, hold at every $z \in \mathbb{S}$ with $x \geq s_\circ$, for X_z in place of X_s, for a complex multiplier h with $|h| \leq D_h$, and with $\mathcal{H}(z)$ and $\mathcal{H}_j(z) \stackrel{\text{def}}{=} \mathbb{E}\, \mathsf{X}_z(M_0^{(j1)})$.

Proof. The conditional moments of the rank-one factors are bounded through lemma SB.7. For a scalar e depending on the sample and on $\mathcal{G}$, the identities $B\, \mathsf{u} \otimes \mathsf{u} = {\Sigma_{\text{out}}}^{1/2} (z \otimes z) {\Sigma_{\text{out}}}^{1/2}$, $B\, \mathsf{u} \otimes \mathsf{v} = {\Sigma_{\text{out}}}^{1/2} (z \otimes w) {\Sigma_{\text{in}}}^{1/2}$ and $\mathbb{E}_{\mathcal{G}}[(z \otimes z) e] = \mathbb{E}_{\mathcal{G}}[(z \otimes z - \mathrm{Id}) e] + \mathrm{Id}\, \mathbb{E}_{\mathcal{G}}\, e$, together with $\|\, \mathrm{Id}_{N_{\text{out}}} \|_{\mathrm{F}} = \sqrt{N_{\text{out}}} \leq \sqrt{\gamma_\star B}$ and $|\, \mathbb{E}_{\mathcal{G}}\, e| \leq \|e\|_{2,\mathcal{G}}$, give

$$\big\| \mathbb{E}_{\mathcal{G}}[(\mathsf{u} \otimes \mathsf{u}) e] \big\|_{\mathrm{F}} \leq \frac{C_\Sigma(\sqrt{2} + \sqrt{\gamma_\star B})}{B} \|e\|_{2,\mathcal{G}} \leq \frac{C \|e\|_{2,\mathcal{G}}}{\sqrt{B}},$$

$$\big\| \mathbb{E}_{\mathcal{G}}[(\mathsf{v} \otimes \mathsf{v}) e] \big\|_{\mathrm{F}} \leq \frac{C \|e\|_{2,\mathcal{G}}}{\sqrt{B}}, \qquad \big\| \mathbb{E}_{\mathcal{G}}[(\mathsf{u} \otimes \mathsf{v}) e] \big\|_{\mathrm{F}} = \big\| \mathbb{E}_{\mathcal{G}}[(\mathsf{v} \otimes \mathsf{u}) e] \big\|_{\mathrm{F}} \leq \frac{C_\Sigma \|e\|_{2,\mathcal{G}}}{B}. \tag{SC.22}$$

The four terms of eq. (SC.18) are bounded one by one. The outer factors R_0, $\widetilde{R}_0$ and X_0 are functions of $\mathcal{G}$ with norms at most $1/s$, $1/s$ and $(2\sqrt{s})^{-1}$ by lemma SB.3, and the multiplied coefficients satisfy $\|h\alpha_i\|_{2,\mathcal{G}} \le D_h \|\alpha_i\|_{2,\mathcal{G}}$. With eq. (SC.22) and lemma SC.2 at $p=2$,

$$\begin{aligned}
\big\| \mathbb{E}_{\mathcal{G}}[h\alpha_3\, R_0(\mathsf{u}\otimes\mathsf{u})\mathsf{X}_0]\big\|_{\mathrm{F}} &\le \frac{1}{s}\cdot\frac{1}{2\sqrt{s}}\cdot\frac{C}{\sqrt{B}}\,D_h\,C\varpi^2 \le \frac{CD_h\varpi^2}{s^{3/2}\sqrt{B}},\\
\big\| s\ \mathbb{E}_{\mathcal{G}}[h\alpha_1\, R_0(\mathsf{u}\otimes\mathsf{v})\widetilde{R}_0]\big\|_{\mathrm{F}} &\le s\cdot\frac{1}{s}\cdot\frac{1}{s}\cdot\frac{C_\Sigma}{B}\,D_h\,C\varpi \le \frac{CD_h\varpi}{sB},\\
\big\| \mathbb{E}_{\mathcal{G}}[h\alpha_1\, \mathsf{X}_0(\mathsf{v}\otimes\mathsf{u})\mathsf{X}_0]\big\|_{\mathrm{F}} &\le \frac{1}{4s}\cdot\frac{C_\Sigma}{B}\,D_h\,C\varpi \le \frac{CD_h\varpi}{sB},\\
\big\| s\ \mathbb{E}_{\mathcal{G}}[h\alpha_2\, \mathsf{X}_0(\mathsf{v}\otimes\mathsf{v})\widetilde{R}_0]\big\|_{\mathrm{F}} &\le s\cdot\frac{1}{2\sqrt{s}}\cdot\frac{1}{s}\cdot\frac{C}{\sqrt{B}}\,D_h\,\frac{C\varpi^2}{s} \le \frac{CD_h\varpi^2}{s^{3/2}\sqrt{B}}.
\end{aligned}$$

Since $\varpi^2 \le \varpi$, $s^{-3/2} \le s_\circ^{-1/2}s^{-1}$ and $B^{-1} \le B^{-1/2}$, the sum of the four bounds is at most $CD_h\varpi/(s\sqrt{B})$, which proves eq. (SC.20).

The deletion-mean bound is the case $h=1$. Taking $M = M_t$, $M_0 = M_0^{(j1)}$, $\varpi = w_j$ in eq. (SC.20) and then the full expectation gives eq. (SC.21).

At $z \in \mathbb{S}$ with $x \ge s_\circ$ the proof is the same, with eq. (SC.18) at z, the norms $1/x$, $1/x$ and $(2\sqrt{x})^{-1}$ of R_0, $\widetilde{R}_0$ and X_0 from lemma SB.3, the sector bounds of lemma SC.2, and eq. (SC.22), which holds for complex e since lemma SB.7 does. The factor s of the second and fourth terms becomes z, with $|z| \le \frac{\sqrt{5}}{2}x$, so the four bounds are the displayed ones with s replaced by x, the second and fourth multiplied by $\sqrt{5}/2$. □

The next lemma measures how the conditional mean of the Gram coefficient α_2 reacts to a low-rank change of the minor. For a deterministic $K \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$, a weight $\varpi \in [0,1]$ and $s>0$, or $z \in \mathbb{S}$ in place of s, let

$$\phi_\varpi(K) \overset{\mathrm{def}}{=} \mathbb{E}\ \alpha_2, \tag{SC.23}$$

where α_2 is the coefficient eq. (SB.13) at that parameter, computed from $(M_0, \mathsf{h}, \mathsf{u}, \mathsf{v}) = (K, \varpi d, \mathsf{u}, \mathsf{v})$ with a fresh sample $(d, \mathsf{u}, \mathsf{v}) = (z^\top \widehat{A}_t w, {\Sigma_{\mathrm{out}}}^{1/2} z/\sqrt{B}, {\Sigma_{\mathrm{in}}}^{1/2} w/\sqrt{B})$. Then $\mathbb{E}_{\mathcal{G}_{ja}}\,\alpha_{2,ja} = \phi_{w_j}(M_0^{(ja)})$, because $M_0^{(ja)}$ is a function of $\mathcal{G}_{ja}$ and the sample (j,a) is independent of it.

Lemma SC.4 (Rank sensitivity of the conditional Gram coefficient)**.** *Let $s_\circ > 0$, $s \ge s_\circ$, $\varpi \in [0,1]$, and let $K, K' \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ with $\mathrm{rank}(K-K') \le k$. Then*

$$|\phi_\varpi(K) - \phi_\varpi(K')| \le \frac{C\,\varpi^2\,k}{\sqrt{s}\,B}, \tag{SC.24}$$

with C depending only on $(s_\circ, C_\Sigma, \gamma_\star)$. The same bound, with s replaced by $x \overset{\mathrm{def}}{=} \mathrm{Re}\, z$, holds at every $z \in \mathbb{S}$ with $x \ge s_\circ$.

Proof. Write $\mathsf{U}, \mathsf{V}, \mathsf{C}, \Lambda$ for the scalars eq. (SB.12) computed from K and $\mathsf{U}', \mathsf{V}', \mathsf{C}', \Lambda'$ for those computed from K', with the same sample, and put $\delta\mathsf{U} = \mathsf{U} - \mathsf{U}'$, $\delta\mathsf{V} = \mathsf{V} - \mathsf{V}'$, $\delta\mathsf{C} = \mathsf{C} - \mathsf{C}'$, $\mathsf{h} = \varpi d$.

The resolvent differences have low rank and nuclear norm of order k. The difference $KK^\top - K'K'^\top = (K-K')K^\top + K'(K-K')^\top$ has rank at most $2k$, so

$$R_s(K) - R_s(K') = R_s(K)\big(K'K'^\top - KK^\top\big)R_s(K')$$

has rank at most $2k$ and operator norm at most $2/s$ by lemma SB.3; its nuclear norm is at most $4k/s$. The same holds for $\widetilde{R}_s$. For the mixed resolvent we use the block matrix $\mathcal{L}(K)$ of the proof of lemma SB.4, whose inverse eq. (SB.18) has $\mathsf{X}_s(K)$ as its upper-right block: $\mathcal{L}(K) - \mathcal{L}(K')$ has rank at most $2k$, hence so does $\mathcal{L}(K)^{-1} - \mathcal{L}(K')^{-1} = \mathcal{L}(K)^{-1}(\mathcal{L}(K') - \mathcal{L}(K))\mathcal{L}(K')^{-1}$, and so does its block $\mathsf{X}_s(K) - \mathsf{X}_s(K')$, whose operator norm is at most $2\cdot(2\sqrt{s})^{-1}$; its nuclear norm is at most $2k/\sqrt{s}$.

The differences of the scalars are Gaussian forms of these matrices. With $F \stackrel{\text{def}}{=} {\Sigma_{\text{out}}}^{1/2}(R_s(K) - R_s(K')){\Sigma_{\text{out}}}^{1/2}/B$, a symmetric matrix with $\|F\|_* \le 4C_\Sigma k/(sB)$, we have $\delta\mathsf{U} = z^\top F z$. Diagonalizing $F = O\,\mathrm{diag}(f)O^\top$ and using that $O^\top z$ is standard Gaussian,

$$\|z^\top F z\|_2 \le \sum_i |f_i|\,\|g^2\|_2 = \sqrt{3}\,\|F\|_* \le \frac{4\sqrt{3}\,C_\Sigma k}{sB}.$$

The same bound holds for $\delta\mathsf{V}$. For $\delta\mathsf{C} = z^\top F' w$ with $F' \stackrel{\text{def}}{=} {\Sigma_{\text{out}}}^{1/2}(\mathsf{X}_s(K) - \mathsf{X}_s(K')){\Sigma_{\text{in}}}^{1/2}/B$, a singular value decomposition $F' = \sum_k f'_k\, p_k \otimes r_k$ writes $z^\top F' w = \sum_k f'_k (p_k^\top z)(r_k^\top w)$ as a sum of products of independent standard normals, so

$$\|z^\top F' w\|_2 \le \sum_k f'_k\,\|g\|_2^2 = \|F'\|_* \le \frac{2C_\Sigma k}{\sqrt{s}\,B}.$$

The difference of the coefficients is a rational expression in these differences. From $\alpha_2 = \mathsf{h}^2\mathsf{U}/\Lambda$,

$$\alpha_2 - \alpha'_2 = \mathsf{h}^2\Big[\frac{\delta\mathsf{U}}{\Lambda} - \frac{\mathsf{U}'\,(\Lambda - \Lambda')}{\Lambda\Lambda'}\Big], \qquad \Lambda - \Lambda' = \mathsf{h}\,\delta\mathsf{C}\,\big(2 + \mathsf{h}(\mathsf{C} + \mathsf{C}')\big) + s\mathsf{h}^2\big(\mathsf{V}\,\delta\mathsf{U} + \mathsf{U}'\,\delta\mathsf{V}\big). \tag{SC.25}$$

By lemma SB.3, $0 \le \mathsf{U}, \mathsf{U}' \le \mathsf{N}/s$, $0 \le s\mathsf{V} \le \mathsf{N}$ and $|\mathsf{C}|, |\mathsf{C}'| \le \mathsf{N}/(4\sqrt{s})$; by eq. (SB.16) and $s \ge s_\circ$, $\Lambda^{-1}, \Lambda'^{-1} \le c_\circ^2(1+|d|)^2(1+\mathsf{N})^2$ with $c_\circ = 1 + \frac{3}{4\sqrt{s_\circ}}$. Inserting these bounds in eq. (SC.25), with $|\mathsf{h}| \le |d|$ and $Z \stackrel{\text{def}}{=} (1 + |d|)(1 + \mathsf{N})$,

$$\begin{aligned}|\alpha_2 - \alpha'_2| &\le \varpi^2 d^2\Big[c_\circ^2 Z^2 |\delta\mathsf{U}| + \frac{\mathsf{N}}{s} c_\circ^4 Z^4\Big(|d|\,|\delta\mathsf{C}|\Big(2 + \frac{|d|\mathsf{N}}{2\sqrt{s}}\Big) + d^2\mathsf{N}\big(|\delta\mathsf{U}| + |\delta\mathsf{V}|\big)\Big)\Big] \\ &\le C\varpi^2 Z^8\big(|\delta\mathsf{U}| + |\delta\mathsf{V}| + |\delta\mathsf{C}|\big),\end{aligned}$$

with C depending only on $s_\circ$.

The expectation is bounded by the Cauchy–Schwarz inequality. Since $\phi_\varpi(K) - \phi_\varpi(K') = \mathbb{E}[\alpha_2 - \alpha'_2]$,

$$|\phi_\varpi(K) - \phi_\varpi(K')| \le C\varpi^2\,\|Z^8\|_2\big(\|\delta\mathsf{U}\|_2 + \|\delta\mathsf{V}\|_2 + \|\delta\mathsf{C}\|_2\big) \le C\varpi^2\Big(\frac{8\sqrt{3}C_\Sigma k}{sB} + \frac{2C_\Sigma k}{\sqrt{s}B}\Big),$$

where $\|Z^8\|_2 \le \|1 + |d|\|_{32}^8 \|1 + \mathsf{N}\|_{32}^8 \le C$ by Hölder's inequality and lemma SB.6. Since $s^{-1} \le s_\circ^{-1/2} s^{-1/2}$, this is eq. (SC.24).

At $z \in \mathbb{S}$ with $x \ge s_\circ$ the proof is the same, with s replaced by z in the identities and by x in the bounds, except at three steps. First, by lemma SB.3 at z the resolvent differences have rank at most $2k$ and operator norms at most $2/x$ and $2(2\sqrt{x})^{-1}$, hence nuclear norms at most $4k/x$ and $2k/\sqrt{x}$. Second, F and F' are complex; their real and imaginary parts are real matrices with nuclear norms at most $\|F\|_*$ and $\|F'\|_*$, the part of F being symmetric, and the two Gaussian bounds applied to each part give $\|\delta\mathsf{U}\|_2 \le 2\sqrt{3}\,\|F\|_*$, $\|\delta\mathsf{V}\|_2 \le 2\sqrt{3}\,\|\widetilde{F}\|_*$ for the analogous $\widetilde{F}$, and $\|\delta\mathsf{C}\|_2 \le 2\|F'\|_*$. Third, in eq. (SC.25) the factor s becomes z, the bounds $|z\mathsf{V}|, |z\mathsf{U}'| \le \frac{\sqrt{5}}{2}\mathsf{N}$ and $|\mathsf{U}'| \le \mathsf{N}/x$ replace $s\mathsf{V}, s\mathsf{U}' \le \mathsf{N}$ and $\mathsf{U}' \le \mathsf{N}/s$, and eq. (SB.17) gives $|\Lambda|^{-1}, |\Lambda'|^{-1} \le (1 + s_\circ^{-1/2})^2 Z^2$. The constants change by absolute factors. □

The last finite bound is the covariance between the conditional Gram coefficient of one sample and the deleted mixed resolvent. For a sample of age j and a parameter $s > 0$, or $z \in \mathbb{S}$ in place of s, put

$$\phi_j \stackrel{\text{def}}{=} \mathbb{E}_{\mathcal{G}_{j1}}\,\alpha_{2,j1} = \phi_{w_j}\big(M_0^{(j1)}\big), \qquad b_j \stackrel{\text{def}}{=} \mathbb{E}\,\phi_j = \mathbb{E}\,\alpha_{2,j1}, \qquad \mathcal{C}_j \stackrel{\text{def}}{=} \mathbb{E}\,\big[(\phi_j - b_j)\,\mathsf{X}_0^{(j1)}\big]. \tag{SC.26}$$

By lemma SC.2, $b_j \ge 0$ at real s and $\mathrm{Re}\,b_j \ge 0$ at $z \in \mathbb{S}$.

Lemma SC.5 (Covariance of the Gram coefficient and the mixed resolvent)**.** *Let $s_\circ > 0$. For every age j and every $s \ge s_\circ$,*

$$\|\mathcal{C}_j\|_{\mathrm{F}} \le \frac{C\,w_j^2}{s^{3/2}\sqrt{B}}, \qquad \textit{hence} \qquad \sum_{j \ge 0}\|\mathcal{C}_j\|_{\mathrm{F}} \le \frac{C\,s_2}{s^{3/2}\sqrt{B}}, \tag{SC.27}$$

with C depending only on $(s_\circ, C_\Sigma, \gamma_\star)$. The same bounds, with s replaced by $x \stackrel{\text{def}}{=} \operatorname{Re} z$, hold at every $z \in \mathbb{S}$ with $x \geq s_\circ$.

Proof. The covariance is expanded along a martingale over the remaining samples. Enumerate the samples other than $(j,1)$ as $l = 1, 2, \ldots$, age by age, and let $\mathcal{A}_l$ be the σ-field generated by the first l of them, $\mathcal{A}_0$ the trivial one. Put

$$D_l \stackrel{\text{def}}{=} \mathbb{E}[\phi_j \mid \mathcal{A}_l] - \mathbb{E}[\phi_j \mid \mathcal{A}_{l-1}] \qquad (l \geq 1).$$

Since ϕ_j is bounded by lemma SC.2 and is a function of the enumerated samples, Lévy's upward theorem [74, Theorem 14.2] gives $\mathbb{E}[\phi_j \mid \mathcal{A}_l] \to \phi_j$ almost surely, hence in L^2 by boundedness; that is, $\phi_j - b_j = \sum_{l \geq 1} D_l$ in L^2. At fixed dimension $\|\mathsf{X}_0^{(j1)}\|_{\mathrm{F}} \leq \sqrt{\min(N_{\mathrm{out}}, N_{\mathrm{in}})}/(2\sqrt{s})$, so the Cauchy–Schwarz inequality passes the expansion to the covariance:

$$\mathcal{C}_j = \sum_{l \geq 1} \mathbb{E}\left[D_l \, \mathsf{X}_0^{(j1)}\right]. \tag{SC.28}$$

Each martingale difference is bounded by the rank sensitivity. Let $\phi_j^{[l]}$ denote ϕ_j with the sample l replaced by an independent copy. Conditioning on $\mathcal{A}_l$ after this replacement is the same as conditioning on $\mathcal{A}_{l-1}$, so $D_l = \mathbb{E}[\phi_j - \phi_j^{[l]} \mid \mathcal{A}_l]$. The replacement changes the minor $M_0^{(j1)}$ by a matrix of rank at most two, and lemma SC.4 with $k = 2$ gives, for every value of the samples,

$$|D_l| \leq \sup\left\{|\phi_{w_j}(K) - \phi_{w_j}(K')| : \ \operatorname{rank}(K - K') \leq 2\right\} \leq \frac{2C\, w_j^2}{\sqrt{s}\, B}. \tag{SC.29}$$

Each term of eq. (SC.28) is a deletion term. Let $\mathsf{X}_0^{(j1),[-l]}$ be the mixed resolvent of the buffer with both samples $(j,1)$ and l deleted. Conditionally on all samples other than l, the matrix $\mathsf{X}_0^{(j1),[-l]}$ is fixed, while D_l is a function of the samples up to l with $\mathbb{E}[D_l \mid \mathcal{A}_{l-1}] = 0$; since the sample l is independent of the others, the conditional mean of D_l given all samples other than l is $\mathbb{E}[D_l \mid \mathcal{A}_{l-1}] = 0$. Hence $\mathbb{E}[D_l \mathsf{X}_0^{(j1),[-l]}] = 0$ and

$$\mathbb{E}\left[D_l \, \mathsf{X}_0^{(j1)}\right] = \mathbb{E}\left[D_l \big(\mathsf{X}_0^{(j1)} - \mathsf{X}_0^{(j1),[-l]}\big)\right].$$

If the sample l has age k, this is the situation of lemma SC.3 with $M = M_0^{(j1)}$, the minor $M_0^{(j1),[-l]}$, $\varpi = w_k$ and the multiplier $h = D_l$, bounded by eq. (SC.29); conditioning on all samples other than l and then taking expectations gives

$$\left\| \mathbb{E}\left[D_l \, \mathsf{X}_0^{(j1)}\right] \right\|_{\mathrm{F}} \leq \frac{2Cw_j^2}{\sqrt{s}B} \cdot \frac{Cw_k}{s\sqrt{B}} = \frac{C\, w_j^2\, w_k}{s^{3/2} B \sqrt{B}}.$$

The terms are summed over the enumeration. There are at most B samples of each age k, so

$$\|\mathcal{C}_j\|_{\mathrm{F}} \leq \sum_{k \geq 0} B \, \frac{Cw_j^2 w_k}{s^{3/2} B \sqrt{B}} = \frac{Cw_j^2}{s^{3/2}\sqrt{B}} \sum_{k \geq 0} w_k \leq \frac{Cw_j^2}{s^{3/2}\sqrt{B}},$$

by eq. (SC.1). Summing over j gives the second bound of eq. (SC.27).

At $z \in \mathbb{S}$ with $x \geq s_\circ$ the proof is the same with s replaced by x in the bounds. The variables ϕ_j and D_l are complex: Lévy's theorem is applied to the real and imaginary parts of ϕ_j, eq. (SC.29) follows from lemma SC.4 at z, $\|\mathsf{X}_0^{(j1)}\|_{\mathrm{F}} \leq \sqrt{\min(N_{\mathrm{out}}, N_{\mathrm{in}})}/(2\sqrt{x})$, and the deletion terms are bounded by lemma SC.3 at z with the complex multiplier $h = D_l$. □

SC.3 The response equation and the mixed closure

The expected mixed resolvent is written exactly through one deleted sample per age; the conditional means of the deletion coefficients are then replaced by their Gaussian values. Throughout this subsection $I \subset (0, \infty)$ is compact, $s \in I$, and $q_I = (C_\Sigma \gamma_\star)^2 / s_-$ is the bound eq. (SB.31), which also bounds $q(-s)$ by eq. (SB.6).

Lemma SC.6 (Exact mixed identity)**.** *In the frozen-state model, for every $s > 0$, and for every $z \in \mathbb{S}$ in place of s with $R_0^{(j)} = R_z(M_0^{(j)})$, $\mathsf{X}_0^{(j)} = \mathsf{X}_z(M_0^{(j)})$ and the coefficients at z,*

$$\mathcal{H}(s) = B \sum_{j \geq 0} \mathbb{E}\left[\alpha_{1,j}\, R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j) - \alpha_{2,j}\, \mathsf{X}_0^{(j)}(\mathsf{v}_j \otimes \mathsf{v}_j)\right]. \tag{SC.30}$$

Proof. By the sample decomposition eq. (SB.26),

$$R(-s)M_t = \sum_{j \geq 0} \sum_{a=1}^{B} \left(\mathsf{h}_{ja} R(-s) \mathsf{u}_{ja}\right) \otimes \mathsf{v}_{ja},$$

and eq. (SC.17), applied to each sample with the deleted buffer $M_0^{(ja)}$, rewrites each term as $(\alpha_{1,ja} R_0^{(ja)} \mathsf{u}_{ja} - \alpha_{2,ja} \mathsf{X}_0^{(ja)} \mathsf{v}_{ja}) \otimes \mathsf{v}_{ja}$. Taking expectations and using the exchangeability of the samples of one age gives eq. (SC.30). At $z \in \mathbb{S}$ the same derivation applies to $R_z(M_t)M_t$, with eq. (SC.17) at z. □

Lemma SC.7 (Derivative bounds)**.** *For $x \geq 0$ let $f_x(d) = d/(1 + xd^2)$ and $\psi_x(d) = d^2/(1 + xd^2)$ as in lemma SB.12. Then for all $d \in \mathbb{R}$,*

$$\begin{aligned} \|f_x'\|_\infty \leq 1, \qquad \|f_x'''\|_\infty \leq 36x, \qquad \|\psi_x''\|_\infty \leq 6, \\ |\partial_x f_x(d)| \leq |d|^3, \qquad |\partial_x \psi_x(d)| \leq d^4, \qquad |\partial_x f_x'(d)| \leq 3d^2. \end{aligned} \tag{SC.31}$$

Proof. The first two bounds are eq. (SB.41).

The bound on ψ_x'' is a scaling computation. For $x > 0$,

$$\psi_x(d) = \frac{1}{x}\,\chi(\sqrt{x}\,d), \qquad \chi(y) \stackrel{\text{def}}{=} \frac{y^2}{1+y^2}, \qquad \psi_x''(d) = \chi''(\sqrt{x}\,d), \qquad \chi''(y) = \frac{2 - 6y^2}{(1+y^2)^3},$$

and $|\chi''(y)| \leq (2 + 6y^2)/(1 + y^2)^3 \leq 6$; at $x = 0$ the second derivative is the constant 2.

The three x-derivatives are computed directly:

$$\partial_x f_x(d) = -\frac{d^3}{(1+xd^2)^2}, \qquad \partial_x \psi_x(d) = -\frac{d^4}{(1+xd^2)^2},$$
$$\partial_x f_x'(d) = \partial_x \frac{1 - xd^2}{(1+xd^2)^2} = \frac{-3d^2 + xd^4}{(1+xd^2)^3},$$

and the last numerator is bounded in absolute value by $3d^2(1 + xd^2)$. □

Lemma SC.8 (Uniform Gaussian limit of the derivative functional)**.** *With $d = z^\top \widehat{A}_t w$ as in lemma SB.13,*

$$r_B' \stackrel{\text{def}}{=} \sup_{0 \leq x \leq q_I} |\,\mathbb{E}\, f_x'(d) - J(x)| \longrightarrow 0.$$

Proof. Pointwise convergence holds at each x. The function f_x' is continuous and bounded by 1, so lemma SA.7(iii) gives $\mathbb{E}\, f_x'(d) \to \mathbb{E}\, f_x'(g)$ for $g \sim \mathcal{N}(0,1)$, and Stein's lemma eq. (SB.24) with $\phi = f_x$ identifies the limit:

$$\mathbb{E}\, f_x'(g) = \mathbb{E}\left[g\, f_x(g)\right] = \mathbb{E}\, \frac{g^2}{1 + xg^2} = J(x).$$

The family is uniformly Lipschitz in x. By lemma SC.7 and $\mathbb{E}\, d^2 = 1$,

$$|\,\mathbb{E}\, f_x'(d) - \mathbb{E}\, f_{x'}'(d)| \leq 3|x - x'|, \qquad |J(x) - J(x')| \leq 3|x - x'|,$$

the second because $|J'| \leq \mathbb{E}\, g^4 = 3$ by lemma SB.1(i).

Pointwise convergence of a uniformly Lipschitz family on the compact interval $[0, q_I]$ is uniform, by the finite-net argument in the proof of lemma SB.13 with the constants 3 and 3 in place of 9 and 3. □

The replacement of the deletion coefficients in lemma SB.12 is centered at the expected traces $(\sigma_B, \tilde{\sigma}_B, q_B)$. The candidate $T(s)$ is built from the branch $(\sigma, \tilde{\sigma}, q)(-s)$, and the following lemma recenters the targets there. Put

$$\vartheta_B(I) \stackrel{\text{def}}{=} \sup_{s\in I} \Big(|\sigma_B(s) - \sigma(-s)| + |\tilde{\sigma}_B(s) - \tilde{\sigma}(-s)| \Big). \tag{SC.32}$$

Lemma SC.9 (Candidate-centered targets)**.** *Under Assumption S3, $\vartheta_B(I) \to 0$. For a sample of age j and $s \in I$ define, with $x_j \stackrel{\text{def}}{=} w_j^2 q(-s) \in [0, q_I]$,*

$$\alpha^\star_{1,j} \stackrel{\text{def}}{=} w_j\, f_{x_j}(d_j), \qquad \alpha^\star_{2,j} \stackrel{\text{def}}{=} w_j^2\, \sigma(-s)\, \psi_{x_j}(d_j). \tag{SC.33}$$

Then, with e_B the sequence of lemma SB.12 and $e^\star_B \stackrel{\text{def}}{=} e_B + C_I \vartheta_B(I) \to 0$,

$$\|\alpha_{1,j} - \alpha^\star_{1,j}\|_2 \le w_j\, e^\star_B, \qquad \|\alpha_{2,j} - \alpha^\star_{2,j}\|_2 \le w_j^2\, e^\star_B \qquad (j \ge 0,\ s \in I). \tag{SC.34}$$

Proof. The expected traces converge to the branch. By the last claim of corollary SB.1, $\mathbb{E} \sup_{s\in I} |\frac{1}{B} \operatorname{Tr}(\Sigma_{\text{out}} R(-s)) - \sigma(-s)|^2 \to 0$, and since $\sigma_B(s) = \mathbb{E}\, \frac{1}{B} \operatorname{Tr}(\Sigma_{\text{out}} R(-s))$, Jensen's inequality gives

$$\sup_{s\in I} |\sigma_B(s) - \sigma(-s)| \le \Big(\mathbb{E} \sup_{s\in I} \Big| \frac{1}{B} \operatorname{Tr}(\Sigma_{\text{out}} R(-s)) - \sigma(-s) \Big|^2 \Big)^{1/2} \longrightarrow 0,$$

and likewise for $\tilde{\sigma}_B$. The products are controlled by the bounds eqs. (SB.6) and (SB.31):

$$|q_B - q| = |s\sigma_B(\tilde{\sigma}_B - \tilde{\sigma}) + s\tilde{\sigma}(\sigma_B - \sigma)| \le C_\Sigma \gamma_\star\, \vartheta_B(I).$$

The targets of eq. (SB.33) and eq. (SC.33) differ by the change of center. With $\check{x}_j \stackrel{\text{def}}{=} w_j^2 q_B$, the mean value theorem and lemma SC.7 give

$$|f_{\check{x}_j}(d) - f_{x_j}(d)| \le w_j^2 |q_B - q|\, |d|^3, \qquad |\sigma_B \psi_{\check{x}_j}(d) - \sigma \psi_{x_j}(d)| \le |\sigma_B - \sigma|\, d^2 + \sigma\, w_j^2 |q_B - q|\, d^4.$$

Taking L^2 norms with $\|d\|_6^3 \le 15$, $\|d\|_4^2 \le 3$ and $\|d\|_8^4 \le 105$ from lemma SA.7(i), and $\sigma \le C_\Sigma \gamma_\star / s_-$,

$$\|\check{\alpha}_{1,j} - \alpha^\star_{1,j}\|_2 \le 15\, w_j^3 |q_B - q| \le C_I w_j\, \vartheta_B(I), \qquad \|\check{\alpha}_{2,j} - \alpha^\star_{2,j}\|_2 \le C_I w_j^2\, \vartheta_B(I).$$

The triangle inequality with eq. (SB.34) proves eq. (SC.34). □

Proposition SC.3 (Response equation)**.** *Under Assumption S3, for $s \in I$ let $b_B(s) \stackrel{\text{def}}{=} \sum_{j\ge 0} b_j \ge 0$ with b_j as in eq. (SC.26), and let $\Upsilon_B(s) \stackrel{\text{def}}{=} \mathrm{Id}_{N_{\text{in}}} + b_B(s) \Sigma_{\text{in}}$. Then*

$$\mathcal{H}(s)\, \Upsilon_B(s) = L_\beta\big(q(-s)\big)\, \overline{R}(-s)\ \mathbb{E}\, G_{t+1} + E_B(s), \tag{SC.35}$$

where, with r_B from lemma SB.13, r'_B from lemma SC.8, $e^\star_B$ from lemma SC.9 and $\delta_B(I) \stackrel{\text{def}}{=} \sup_{s\in I} \|\mathcal{P}(s) - \overline{R}(-s)\|$,

$$\sup_{s\in I} \|E_B(s)\|_{\text{F}} \le C_I \Big[e^\star_B + r'_B + \|\widehat{A}_t\| + \|\widehat{A}_t\|^2 + \delta_B(I) + \frac{1}{\sqrt{B}} \Big] \longrightarrow 0, \tag{SC.36}$$

and the coefficient b_B satisfies

$$\sup_{s\in I} |b_B(s) - \sigma(-s) F_\beta(q(-s))| \le s_2 \Big(e^\star_B + \frac{C_\Sigma \gamma_\star}{s_-} r_B \Big) \longrightarrow 0. \tag{SC.37}$$

Proof. Fix $s \in I$; every bound below is uniform in $s \in I$. We start from the exact identity eq. (SC.30) and treat its two terms.

The source term. Fix j, write $\varpi = w_j$, $x = x_j$, $f = f_x$, $\mathcal{G} = \mathcal{G}_{j1}$, and split $\alpha_{1,j} = \alpha^\star_{1,j} + (\alpha_{1,j} - \alpha^\star_{1,j})$. The replacement error is bounded by the Bessel inequality: with $e \stackrel{\text{def}}{=} \alpha_{1,j} - \alpha^\star_{1,j}$, the bound eq. (SC.22), $\|R_0^{(j)}\| \le 1/s_-$, the tower property, Jensen's inequality and lemma SC.9 give

$$\Big\| B\ \mathbb{E}\, \big[e\, R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j) \big] \Big\|_{\text{F}} \le \frac{1}{s_-}\ \mathbb{E} \Big\| B\ \mathbb{E}_{\mathcal{G}}[(\mathsf{u}_j \otimes \mathsf{v}_j) e] \Big\|_{\text{F}} \le \frac{C_\Sigma}{s_-} \|e\|_2 \le C_I \varpi\, e^\star_B.$$

The main part is a Gaussian response. Since $R_0^{(j)}$ is independent of (z_j, w_j) and $B\,\mathsf{u}_j \otimes \mathsf{v}_j = {\Sigma_{\text{out}}}^{1/2}(z_j \otimes w_j){\Sigma_{\text{in}}}^{1/2}$,

$$B\ \mathbb{E}\left[\alpha^\star_{1,j} R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j)\right] = \varpi\, \mathcal{P}_j(s)\, {\Sigma_{\text{out}}}^{1/2}\ \mathbb{E}[(z \otimes w) f(d)]\, {\Sigma_{\text{in}}}^{1/2},$$

and eq. (SB.22) with $A = \widehat{A}_t$, $\|\widehat{A}_t\|_{\text{F}} = 1$ and the bound $\|f'''\|_\infty \le 36 q_I$ of lemma SC.7 give

$$\mathbb{E}[(z \otimes w) f(d)] = \widehat{A}_t\ \mathbb{E}\, f'(d) + N_j, \qquad \|N_j\|_{\text{F}} \le 36 q_I \|\widehat{A}_t\|^2.$$

Since ${\Sigma_{\text{out}}}^{1/2} \widehat{A}_t {\Sigma_{\text{in}}}^{1/2} = \mathbb{E}\, G_{t+1}$, writing $\jmath_j \overset{\text{def}}{=} \mathbb{E}\, f'_{x_j}(d)$, which satisfies $|\jmath_j| \le 1$ and $|\jmath_j - J(x_j)| \le r'_B$,

$$\left\| B\ \mathbb{E}\left[\alpha^\star_{1,j} R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j)\right] - \varpi\, \jmath_j\, \mathcal{P}_j(s)\ \mathbb{E}\, G_{t+1} \right\|_{\text{F}} \le \varpi\, \frac{C_\Sigma}{s_-}\, 36 q_I \|\widehat{A}_t\|^2 \le C_I \varpi \|\widehat{A}_t\|^2.$$

The expected deleted resolvent is replaced by $\mathcal{P}$ and then by $\overline{R}$. By eq. (SB.42) and $\|\mathbb{E}\, G_{t+1}\|_{\text{F}} \le C_\Sigma$ (lemma SA.6(iv)),

$$\|\varpi \jmath_j (\mathcal{P}_j(s) - \mathcal{P}(s))\, \mathbb{E}\, G_{t+1}\|_{\text{F}} \le \frac{C_I \varpi^2}{B}, \qquad \|\varpi \jmath_j (\mathcal{P}(s) - \overline{R}(-s))\, \mathbb{E}\, G_{t+1}\|_{\text{F}} \le C_\Sigma \varpi\, \delta_B(I),$$

and $|\varpi \jmath_j - \varpi J(x_j)| \|\overline{R}(-s)\, \mathbb{E}\, G_{t+1}\|_{\text{F}} \le \varpi\, r'_B\, C_\Sigma / s_-$. Summing over j with eq. (SC.1), and recalling $L_\beta(q) = \sum_j w_j J(w_j^2 q)$,

$$\left\| B \sum_{j \ge 0} \mathbb{E}\left[\alpha_{1,j} R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j)\right] - L_\beta(q(-s))\, \overline{R}(-s)\ \mathbb{E}\, G_{t+1} \right\|_{\text{F}} \le C_I \left[e^\star_B + \|\widehat{A}_t\|^2 + \frac{1}{B} + \delta_B(I) + r'_B \right]. \tag{SC.38}$$

The Gram term. Fix j and write $\psi = \psi_{x_j}$. Conditioning on $\mathcal{G}$ and using $B\,\mathsf{v}_j \otimes \mathsf{v}_j = {\Sigma_{\text{in}}}^{1/2}(w_j \otimes w_j){\Sigma_{\text{in}}}^{1/2}$,

$$B\ \mathbb{E}\left[\alpha_{2,j} \mathsf{X}_0^{(j)}(\mathsf{v}_j \otimes \mathsf{v}_j)\right] = \mathbb{E}\left[\mathsf{X}_0^{(j)}\, {\Sigma_{\text{in}}}^{1/2}\ \mathbb{E}_{\mathcal{G}}[(w_j \otimes w_j)\alpha_{2,j}]\, {\Sigma_{\text{in}}}^{1/2}\right],$$

$$\mathbb{E}_{\mathcal{G}}[(w_j \otimes w_j)\alpha_{2,j}] = \phi_j\ \mathrm{Id} + \Gamma_j,$$

where $\phi_j = \mathbb{E}_{\mathcal{G}}\, \alpha_{2,j}$ is the coefficient eq. (SC.26) and $\Gamma_j \overset{\text{def}}{=} \mathbb{E}_{\mathcal{G}}[(w_j \otimes w_j - \mathrm{Id})\alpha_{2,j}]$. The centered part is small. Splitting $\alpha_{2,j} = \alpha^\star_{2,j} + (\alpha_{2,j} - \alpha^\star_{2,j})$, the Bessel bound lemma SB.7 with Jensen's inequality and lemma SC.9 control the second piece, and for the first, which depends only on the sample $(j, 1)$, eq. (SB.23) for $(w, z, \widehat{A}_t^\top)$ with $\|\psi''\|_\infty \le 6$ gives

$$\mathbb{E}\, \|\Gamma_j\|_{\text{F}} \le \sqrt{2}\, \|\alpha_{2,j} - \alpha^\star_{2,j}\|_2 + \varpi^2 \sigma \left\| \mathbb{E}[(w \otimes w - \mathrm{Id})\psi(d)] \right\|_{\text{F}} \le \varpi^2 \left(\sqrt{2}\, e^\star_B + 6\sigma \|\widehat{A}_t\| \right).$$

With $\|\mathsf{X}_0^{(j)}\| \le (2\sqrt{s_-})^{-1}$ and $\sigma \le C_\Sigma \gamma_\star / s_-$,

$$\left\| B\ \mathbb{E}\left[\alpha_{2,j} \mathsf{X}_0^{(j)}(\mathsf{v}_j \otimes \mathsf{v}_j)\right] - \mathbb{E}\left[\phi_j \mathsf{X}_0^{(j)}\right] \Sigma_{\text{in}} \right\|_{\text{F}} \le \frac{C_\Sigma}{2\sqrt{s_-}}\ \mathbb{E}\, \|\Gamma_j\|_{\text{F}} \le C_I \varpi^2 \left(e^\star_B + \|\widehat{A}_t\| \right).$$

The scalar mean is identified. By lemma SC.9, lemma SB.13 and $\mathbb{E}\, \alpha^\star_{2,j} = \varpi^2 \sigma\ \mathbb{E}\, \psi(d)$,

$$|b_j - \varpi^2 \sigma J(x_j)| \le |\,\mathbb{E}\, \alpha_{2,j} - \mathbb{E}\, \alpha^\star_{2,j}| + \varpi^2 \sigma |\,\mathbb{E}\, \psi(d) - J(x_j)| \le \varpi^2 \left(e^\star_B + \sigma r_B \right), \tag{SC.39}$$

and summing over j with $F_\beta(q) = \sum_j w_j^2 J(w_j^2 q)$ gives eq. (SC.37). The product with the deleted resolvent is split into its mean and its covariance:

$$\mathbb{E}\left[\phi_j \mathsf{X}_0^{(j)}\right] = b_j\, \mathcal{H}_j(s) + \mathcal{C}_j = b_j\, \mathcal{H}(s) + b_j \left(\mathcal{H}_j(s) - \mathcal{H}(s) \right) + \mathcal{C}_j.$$

By lemma SC.2, $b_j \le \|\alpha_{2,j}\|_1 \le C_I \varpi^2$; by eq. (SC.21), $\|\mathcal{H}_j - \mathcal{H}\|_{\text{F}} \le C_I \varpi / \sqrt{B}$; and by lemma SC.5, $\sum_j \|\mathcal{C}_j\|_{\text{F}} \le C_I s_2 / \sqrt{B}$. Summing over j with $\sum_j w_j^3 \le 1$,

$$\left\| B \sum_{j \ge 0} \mathbb{E}\left[\alpha_{2,j} \mathsf{X}_0^{(j)}(\mathsf{v}_j \otimes \mathsf{v}_j)\right] - b_B(s)\, \mathcal{H}(s)\, \Sigma_{\text{in}} \right\|_{\text{F}} \le C_I \left[e^\star_B + \|\widehat{A}_t\| + \frac{1}{\sqrt{B}} \right]. \tag{SC.40}$$

Assembly. Inserting eqs. (SC.38) and (SC.40) into eq. (SC.30) gives $\mathcal{H}(s) = L_\beta(q)\overline{R}(-s)\, \mathbb{E}\, G_{t+1} - b_B(s)\mathcal{H}(s)\Sigma_{\text{in}} + E_B(s)$ with E_B bounded by the sum of the two right sides, which is eq. (SC.36); moving the second term to the left is eq. (SC.35). The convergence to zero holds because $e^\star_B \to 0$ (lemma SC.9), $r_B, r'_B \to 0$ (lemmas SB.13 and SC.8), $\|\widehat{A}_t\| \to 0$ (Assumption S3) and $\delta_B(I) \to 0$ (theorem SB.2). □

Proof of theorem SC.1(i). The response equation is compared with the exact equation of the candidate. The system eq. (SB.2) at $z = -s$ gives $s_1 = s\sigma F_\beta(q)$, so

$$\mathsf{Y}_\star(s) \stackrel{\text{def}}{=} \mathrm{Id}_{N_{\text{in}}} + \sigma(-s) F_\beta(q(-s))\, \Sigma_{\text{in}} = \frac{s\,\mathrm{Id} + s_1(-s)\Sigma_{\text{in}}}{s} = \frac{1}{s}\,\overline{\widetilde{R}}(-s)^{-1},$$

and therefore, by the definition eq. (SC.2) of T,

$$T(s)\,\mathsf{Y}_\star(s) = L_\beta(q(-s))\,\overline{R}(-s)\;\mathbb{E}\,G_{t+1}.$$

Subtracting this from eq. (SC.35),

$$\big(\mathcal{H}(s) - T(s)\big)\mathsf{Y}_B(s) = E_B(s) - T(s)\big(\mathsf{Y}_B(s) - \mathsf{Y}_\star(s)\big).$$

Both matrices Y_B and $\mathsf{Y}_\star$ are the identity plus a positive semidefinite matrix, so their inverses have operator norm at most 1, and $\|\mathsf{Y}_B - \mathsf{Y}_\star\| \le C_\Sigma |b_B - \sigma F_\beta(q)|$. With $\|T(s)\|_{\text{F}} \le C_\Sigma/s_-$ from eq. (SC.6), proved below,

$$\|\mathcal{H}(s) - T(s)\|_{\text{F}} \le \|(\mathcal{H}(s) - T(s))\mathsf{Y}_B(s)\|_{\text{F}}\, \|\mathsf{Y}_B(s)^{-1}\| \le \|E_B(s)\|_{\text{F}} + \frac{C_\Sigma^2}{s_-}|b_B(s) - \sigma F_\beta(q)|,$$

and both terms tend to zero uniformly in $s \in I$ by eqs. (SC.36) and (SC.37). □

SC.4 The dimension-independent tail

Proof of theorem SC.1(ii). The candidate bound uses $L_\beta \le \sum_j w_j \le 1$ (since $J \le 1$), $\|\overline{R}(-s)\|, \|\overline{\widetilde{R}}(-s)\| \le 1/s$ (the matrices $\tilde{s}_1\Sigma_{\text{out}} + s\,\mathrm{Id}$ and $s_1\Sigma_{\text{in}} + s\,\mathrm{Id}$ are at least $s\,\mathrm{Id}$) and $\|\mathbb{E}\,G_{t+1}\|_{\text{F}} \le C_\Sigma$:

$$\|T(s)\|_{\text{F}} \le s \cdot 1 \cdot \frac{1}{s} \cdot C_\Sigma \cdot \frac{1}{s} = \frac{C_\Sigma}{s}.$$

The bound on $\mathcal{H}$ uses the exact identity eq. (SC.30) without any asymptotic replacement, with the finite lemmas at $s_\circ = 1$; no hypothesis on $\|\widehat{A}_t\|$ enters. For the source term, eq. (SC.22), $\|R_0^{(j)}\| \le 1/s$ and lemma SC.2 give

$$\Big\| B\;\mathbb{E}\left[\alpha_{1,j} R_0^{(j)}(\mathsf{u}_j \otimes \mathsf{v}_j)\right]\Big\|_{\text{F}} \le \frac{1}{s}\, C_\Sigma \|\alpha_{1,j}\|_2 \le \frac{C w_j}{s}.$$

For the Gram term we use the decomposition of the proof of proposition SC.3, now without replacing the coefficients: $\mathbb{E}_{\mathcal{G}}[(w_j \otimes w_j)\alpha_{2,j}] = \phi_j\,\mathrm{Id} + \Gamma_j$ with $\mathbb{E}\,\|\Gamma_j\|_{\text{F}} \le \sqrt{2}\|\alpha_{2,j}\|_2 \le C w_j^2/s$ by lemma SB.7 and lemma SC.2, so that

$$\Big\| B\;\mathbb{E}\left[\alpha_{2,j}\mathsf{X}_0^{(j)}(\mathsf{v}_j \otimes \mathsf{v}_j)\right] - \mathbb{E}\left[\phi_j \mathsf{X}_0^{(j)}\right]\Sigma_{\text{in}}\Big\|_{\text{F}} \le \frac{C_\Sigma}{2\sqrt{s}}\;\mathbb{E}\,\|\Gamma_j\|_{\text{F}} \le \frac{C w_j^2}{s^{3/2}},$$

and $\mathbb{E}[\phi_j \mathsf{X}_0^{(j)}] = b_j \mathcal{H}(s) + b_j(\mathcal{H}_j(s) - \mathcal{H}(s)) + \mathcal{C}_j$ with $0 \le b_j \le C w_j^2/s$ (lemma SC.2), $\|\mathcal{H}_j - \mathcal{H}\|_{\text{F}} \le C w_j/(s\sqrt{B})$ (eq. (SC.21)) and $\sum_j \|\mathcal{C}_j\|_{\text{F}} \le C s_2/(s^{3/2}\sqrt{B})$ (lemma SC.5). Summing over j with eq. (SC.1),

$$\|\mathcal{H}(s)\mathsf{Y}_B(s)\|_{\text{F}} \le C\Big[\frac{1}{s} + \frac{1}{s^{3/2}} + \frac{1}{s^2\sqrt{B}} + \frac{1}{s^{3/2}\sqrt{B}}\Big] \le \frac{C}{s} \qquad (s \ge 1,\ B \ge 1),$$

where $\mathsf{Y}_B = \mathrm{Id} + b_B\Sigma_{\text{in}}$ as before. Since $b_B \ge 0$, $\|\mathsf{Y}_B^{-1}\| \le 1$ and $\|\mathcal{H}(s)\|_{\text{F}} \le \|\mathcal{H}(s)\mathsf{Y}_B(s)\|_{\text{F}}\|\mathsf{Y}_B(s)^{-1}\| \le C/s$.

The sector bound eq. (SC.7) is the same argument at $z \in \mathbb{S}$ with $x \stackrel{\text{def}}{=} \mathrm{Re}\,z \ge s_\circ$, using the finite lemmas at $s_\circ$ in their sector forms and replacing s by x in every bound: lemma SC.6 at z, the norms $1/x$ and $(2\sqrt{x})^{-1}$ of $R_0^{(j)}$ and $\mathsf{X}_0^{(j)}$ from lemma SB.3, eq. (SC.22) and lemma SB.7 for complex coefficients, and lemmas SC.2, SC.3 and SC.5 at z. Two steps change. First, b_j is complex, with $\mathrm{Re}\,b_j \ge 0$ and $|b_j| \le \|\alpha_{2,j}\|_1 \le C w_j^2/x$, so the series $b_B = \sum_j b_j$ converges and $\mathrm{Re}\,b_B \ge 0$. Second, Y_B is diagonal in an eigenbasis of Σ_{in} with eigenvalues $1 + b_B\lambda_j$, and $|1 + b_B\lambda_j| \ge \mathrm{Re}(1 + b_B\lambda_j) \ge 1$, so $\|\mathsf{Y}_B^{-1}\| \le 1$. Summing over j,

$$\|\mathcal{H}(z)\|_{\text{F}} \le \|\mathcal{H}(z)\mathsf{Y}_B(z)\|_{\text{F}} \le C\Big[\frac{1}{x} + \frac{1}{x^{3/2}} + \frac{1}{x^2\sqrt{B}} + \frac{1}{x^{3/2}\sqrt{B}}\Big] \le \frac{C_\circ}{x} \qquad (x \ge s_\circ,\ B \ge 1),$$

with $C_\circ$ depending only on $(s_\circ, C_\Sigma, \gamma_\star)$. □

SC.5 The regularized mean and the history truncation

Proof of theorem SC.2. The candidate integral exists. By theorem SB.1(ii) the branch is continuous in s, so $s \mapsto T(s)$ is continuous on $(0,\infty)$, and by eq. (SC.6) and the scalar integral in the proof of lemma SA.3,

$$\frac{2}{\pi}\int_0^\infty \|T(\varrho+u^2)\|_{\mathrm{F}}\,du \le \frac{2C_\Sigma}{\pi}\int_0^\infty \frac{du}{\varrho+u^2} = \frac{C_\Sigma}{\sqrt{\varrho}}.$$

The expected update is the integral of $\mathcal{H}$. By eq. (SA.37),

$$\mathbb{E}\,Q_\varrho(M_t) = \frac{2}{\pi}\int_0^\infty \mathcal{H}(\varrho+u^2)\,du.$$

The difference is split at a cutoff. Fix $\mathsf{R} \ge 1$ and put $I_{\mathsf{R}} \overset{\text{def}}{=} [\varrho_-, \varrho_+ + \mathsf{R}^2]$. For $u \le \mathsf{R}$ and $\varrho \in [\varrho_-, \varrho_+]$ the parameter $s = \varrho + u^2$ lies in I_{R}; for $u \ge \mathsf{R}$ it satisfies $s \ge u^2 \ge 1$, where eq. (SC.6) applies. Hence

$$\begin{aligned}\|\,\mathbb{E}\,Q_\varrho(M_t) - \overline{\widetilde{G}}_\varrho\|_{\mathrm{F}} &\le \frac{2}{\pi}\Big[\mathsf{R}\sup_{s\in I_{\mathsf{R}}}\|\mathcal{H}(s)-T(s)\|_{\mathrm{F}} + (C+C_\Sigma)\int_{\mathsf{R}}^\infty \frac{du}{u^2}\Big]\\ &= \frac{2}{\pi}\Big[\mathsf{R}\sup_{s\in I_{\mathsf{R}}}\|\mathcal{H}(s)-T(s)\|_{\mathrm{F}} + \frac{C+C_\Sigma}{\mathsf{R}}\Big],\end{aligned}$$

uniformly in $\varrho \in [\varrho_-, \varrho_+]$. Taking first the limit superior in the dimension at fixed R, which kills the first term by eq. (SC.5) applied to the compact interval I_{R}, and then $\mathsf{R} \to \infty$, proves eq. (SC.8).

The pairing bound eq. (SC.9) is the Cauchy–Schwarz inequality for the Frobenius pairing. □

Proof of proposition SC.1. Part (i) is theorem SC.2 for the weights $w^{(J)}$, which belong to the class eq. (SC.1) for $J \ge J_w$; the buffer $M^{(J)}$ is the frozen-state buffer with these weights.

Part (ii) is the stability of the branch. Fix a compact $I \subset (0,\infty)$ and let $(\sigma^J, \tilde\sigma^J, a^J, c^J)(-s)$ be the solution of eq. (SB.2) for $w^{(J)}$ at $z = -s$, with $q^J = s\sigma^J\tilde\sigma^J$. It is an approximate solution of the system for the full weights. Its first two residuals vanish, because the trace equations do not involve the weights; its third residual is

$$|a^J - s\tilde\sigma^J F_w(q^J)| = s\tilde\sigma^J\,|F_{w^{(J)}}(q^J) - F_w(q^J)| = s\tilde\sigma^J\sum_{j>J} w_j^2 J(w_j^2 q^J) \le C_\Sigma\gamma_\star\sum_{j>J} w_j^2,$$

by $J \le 1$ and the box eq. (SB.6), and the fourth residual has the same bound. The tuple lies in the box of theorem SB.1(iii) for the full weights, since $s_2^{(J)} \le s_2$. Given $\epsilon > 0$, that theorem provides $\delta > 0$, and J_0 with $C_\Sigma\gamma_\star\sum_{j>J_0} w_j^2 \le \delta$, such that for all $J \ge J_0$, all $s \in I$ and all spectral measures in $\mathcal{K}$, hence uniformly in the dimension,

$$|\sigma^J - \sigma| + |\tilde\sigma^J - \tilde\sigma| + |a^J - \tilde s_1| + |c^J - s_1| \le \epsilon.$$

The candidates are then close. The resolvent identity gives $\|\overline{R}^{(J)}(-s) - \overline{R}(-s)\| \le C_\Sigma\epsilon/s_-^2$ and the same for $\overline{\widetilde{R}}$, and the source factors satisfy

$$|L_{w^{(J)}}(q^J) - L_\beta(q)| \le \sum_{j>J} w_j + |L_\beta(q^J) - L_\beta(q)| \le \sum_{j>J} w_j + 3s_3\,|q^J - q|,$$

by lemma SB.1(ii), with $|q^J - q| \le s\sigma^J|\tilde\sigma^J - \tilde\sigma| + s\tilde\sigma|\sigma^J - \sigma| \le C_\Sigma\gamma_\star\epsilon$. Telescoping the three factors of T with the bounds $\|\overline{R}\|, \|\overline{\widetilde{R}}\| \le 1/s$, $L_\beta \le 1$ and $\|\,\mathbb{E}\,G_{t+1}\|_{\mathrm{F}} \le C_\Sigma$,

$$\sup_{s\in I}\|T^{(J)}(s) - T(s)\|_{\mathrm{F}} \le C_I\Big(\epsilon + \sum_{j>J} w_j\Big) \qquad (J \ge J_0),$$

uniformly in the dimension. Splitting the integral eq. (SC.3) at a cutoff $\mathsf{R} \ge 1$ as in the proof of theorem SC.2, with $\|T^{(J)}(s)\|_{\mathrm{F}}, \|T(s)\|_{\mathrm{F}} \le C_\Sigma/s$,

$$\sup_B \sup_{\varrho\in[\varrho_-,\varrho_+]} \|\overline{\widetilde{G}}_\varrho - \overline{\widetilde{G}}_\varrho^{(J)}\|_{\mathrm{F}} \le \frac{2}{\pi}\Big[\mathsf{R}\,C_{I_{\mathsf{R}}}\Big(\epsilon + \sum_{j>J} w_j\Big) + \frac{2C_\Sigma}{\mathsf{R}}\Big],$$

and choosing first R large, then ϵ small and J large, proves (ii).

Part (iii) is the triangle inequality

$$\|\,\mathbb{E}\,Q_\varrho(M_t)-\mathbb{E}\,Q_\varrho(M^{(J)})\|_{\mathrm{F}} \le \|\,\mathbb{E}\,Q_\varrho(M_t)-\overline{\widetilde{G}}_\varrho\|_{\mathrm{F}}+\|\overline{\widetilde{G}}_\varrho-\overline{\widetilde{G}}_\varrho^{(J)}\|_{\mathrm{F}}+\|\overline{\widetilde{G}}_\varrho^{(J)}-\mathbb{E}\,Q_\varrho(M^{(J)})\|_{\mathrm{F}},$$

in which the first and third terms vanish in the dimension limit at fixed J by theorem SC.2 and part (i), and the second is bounded uniformly in the dimension by a quantity vanishing as $J\to\infty$ by part (ii). □

SC.6 The drift kernel

Proof of corollary SC.1. The candidate is diagonal in the eigenbases. For $s>0$, since $\Sigma_{\mathrm{out}}u_i=\mu_i u_i$ and $\Sigma_{\mathrm{in}}v_j=\lambda_j v_j$,

$$u_i^\top \overline{R}(-s)=\frac{u_i^\top}{s+\tilde{s}_1(-s)\mu_i},\qquad \overline{\widetilde{R}}(-s)v_j=\frac{v_j}{s+s_1(-s)\lambda_j},$$
$$u_i^\top\,\Sigma_{\mathrm{out}}\Delta_t\Sigma_{\mathrm{in}}\,v_j=\mu_i\lambda_j\,(u_i^\top\Delta_t v_j),$$

so that, by eq. (SC.2),

$$u_i^\top T(s)v_j=\frac{s\,L_\beta(q(-s))\,\mu_i\lambda_j}{\big(s+\tilde{s}_1(-s)\mu_i\big)\big(s+s_1(-s)\lambda_j\big)}\cdot\frac{u_i^\top\Delta_t v_j}{\sqrt{2\mathscr{R}(t)}}.$$

Integrating against $\frac{2}{\pi}\,du$ at $s=\varrho+u^2$ and comparing with eq. (SC.4), in which the prefactor is $\sqrt{2}/\pi=(2/\pi)/\sqrt{2}$,

$$u_i^\top\,\overline{\widetilde{G}}_\varrho\,v_j=\frac{\sqrt{2}\,\mathfrak{d}(\mu_i,\lambda_j)}{\sqrt{2\mathscr{R}(t)}}\,(u_i^\top\Delta_t v_j)=\frac{\mathfrak{d}(\mu_i,\lambda_j)}{\sqrt{\mathscr{R}(t)}}\,(u_i^\top\Delta_t v_j).$$

Adding $(E_\varrho)_{ij}$ gives eq. (SC.11), and the Frobenius norm is computed in the orthonormal bases: $\sum_{i,j}(E_\varrho)_{ij}^2=\|\,\mathbb{E}\,\widetilde{G}_{t+1}-\overline{\widetilde{G}}_\varrho\|_{\mathrm{F}}^2$, whose supremum over the ridge is ε_B^2 by theorem SC.2.

The drift is the projection times the projected state. In the frozen-state model the state is deterministic and the update is $\widetilde{G}_{t+1}$, so the drift eq. (SA.19) is $\mathscr{D}_{ij}(t)=(u_i^\top\Delta_t v_j)\;\mathbb{E}[u_i^\top\widetilde{G}_{t+1}v_j]$; multiplying eq. (SC.11) by $u_i^\top\Delta_t v_j$ and using $\mathscr{Q}_{ij}(t)=(u_i^\top\Delta_t v_j)^2$ gives eq. (SC.12). The tested bound eq. (SC.13) is the Cauchy–Schwarz inequality in ℓ^2 over the pairs (i,j), with $\sum_{i,j}(u_i^\top\Delta_t v_j)^2=\|\Delta_t\|_{\mathrm{F}}^2$.

The kernel bounds follow from the integrand. Since $\tilde{s}_1,s_1\ge 0$, $0\le L_\beta\le 1$ and $z\ge\varrho$, the integrand of eq. (SC.4) lies between 0 and $\mu\lambda/z$, and

$$\frac{\sqrt{2}}{\pi}\int_0^\infty\frac{du}{\varrho+u^2}=\frac{\sqrt{2}}{\pi}\cdot\frac{\pi}{2\sqrt{\varrho}}=\frac{1}{\sqrt{2\varrho}},$$

which gives the first bound of eq. (SC.14); the integrand vanishes when $\mu=0$ or $\lambda=0$. Differentiating the integrand in μ,

$$\partial_\mu\frac{z\,L_\beta\,\mu\lambda}{(z+\tilde{s}_1\mu)(z+s_1\lambda)}=\frac{z^2\,L_\beta\,\lambda}{(z+\tilde{s}_1\mu)^2(z+s_1\lambda)}\in\Big[0,\frac{\lambda}{z}\Big],$$

and integrating gives the bound on $\partial_\mu\mathfrak{d}$; the bound on $\partial_\lambda\mathfrak{d}$ is symmetric. Continuity on $[0,\infty)^2$ and the one-sided derivatives at the boundary follow from the same bounds, which are uniform in (μ,λ) on bounded sets. The Lipschitz constant on $[0,C_\Sigma]^2$ is the maximum of the derivative bounds there. □

Corollary SC.2 (Half-anisotropic input)**.** *Under Assumption S3, assume* $\Sigma_{\mathrm{in}}=\mathrm{Id}_{N_{\mathrm{in}}}$, *let* $\varrho\in[\varrho_-,\varrho_+]$, *write* $z=\varrho+u^2$, *and put* $\mathfrak{d}_i\overset{\mathrm{def}}{=}\mathfrak{d}(\mu_i,1)$.

(i) For every pair (i,j) *the kernel is* $\mathfrak{d}_{ij}=\mathfrak{d}_i$, *and*

$$\mathfrak{d}_i=\frac{\sqrt{2}\,B}{\pi\,N_{\mathrm{in}}}\int_0^\infty\frac{z\,L_\beta(q(-z))\,\tilde{\sigma}(-z)\,\mu_i}{z+\tilde{s}_1(-z)\mu_i}\,du.$$

(ii) The row drift of eq. (SA.21) *is, with E_ϱ as in corollary SC.1,*

$$\sum_j \mathscr{D}_{ij}(t) = \frac{\mathfrak{d}_i}{\sqrt{\mathscr{R}(t)}}\,\mathscr{Q}_i(t) + \sum_j (u_i^\top \Delta_t v_j)(E_\varrho)_{ij}.$$

(iii) At $\beta = 0$,

$$\mathfrak{d}_i = \frac{\sqrt{2}\,B}{\pi\,N_{\mathrm{in}}} \int_0^\infty \frac{\tilde{s}_1(-z)\mu_i}{z + \tilde{s}_1(-z)\mu_i}\,du.$$

(iv) At $\beta = 0$, if ρ_i is a probability measure on $[0,\infty)$ with

$$\big(z + \tilde{s}_1(-z)\mu_i\big)^{-1} = \int \frac{\rho_i(dx)}{x+z} \qquad (z>0),$$

then

$$\mathfrak{d}_i = \frac{B}{\sqrt{2}\,N_{\mathrm{in}}} \int_{[0,\infty)} \frac{x}{\sqrt{x+\varrho}}\,\rho_i(dx).$$

Proof. Part (i) follows from the input trace equation. When $\Sigma_{\mathrm{in}} = \mathrm{Id}_{N_{\mathrm{in}}}$ every λ_j equals 1, so $\mathfrak{d}_{ij} = \mathfrak{d}(\mu_i, 1)$ for every j. The input trace equation of eq. (SB.2) becomes algebraic,

$$\tilde{\sigma}(-s) = \frac{N_{\mathrm{in}}/B}{s + s_1(-s)}, \qquad \text{so that} \qquad \frac{1}{s+s_1(-s)} = \frac{B\,\tilde{\sigma}(-s)}{N_{\mathrm{in}}}.$$

Substituting this at $s = z$ into eq. (SC.4) with $\lambda = 1$ gives the formula of (i).

Part (ii) is the sum over j of eq. (SC.12). The identity $\mathscr{Q}_i = \sum_j \mathscr{Q}_{ij}$ of lemma SA.1 collects the first terms, because $\mathfrak{d}_{ij} = \mathfrak{d}_i$ by (i).

Part (iii) is the no-momentum form of the source factor. At $\beta = 0$ the function L_β is J. By remark SB.1, the coupling eq. (SB.4) and $q(-z) = z\sigma(-z)\tilde{\sigma}(-z)$ for $z > 0$,

$$z\,J(q(-z))\,\tilde{\sigma}(-z) = \frac{q(-z)J(q(-z))}{\sigma(-z)} = \tilde{s}_1(-z).$$

Substituting this identity into (i) gives (iii).

Part (iv) integrates the Stieltjes representation in u. By the hypothesis at $z = \varrho + u^2$ and since ρ_i has mass one,

$$\frac{\tilde{s}_1(-z)\mu_i}{z + \tilde{s}_1(-z)\mu_i} = 1 - \int \frac{z\,\rho_i(dx)}{x+z} = \int \frac{x\,\rho_i(dx)}{x+z}.$$

The scalar integral of the proof of lemma SA.3 is

$$\frac{2}{\pi}\int_0^\infty \frac{du}{x+\varrho+u^2} = \frac{1}{\sqrt{x+\varrho}}.$$

Integrating the previous display in u against $\frac{\sqrt{2}\,B}{\pi\,N_{\mathrm{in}}}\,du = \frac{B}{\sqrt{2}\,N_{\mathrm{in}}}\cdot\frac{2}{\pi}\,du$ and inserting the scalar integral gives (iv). □

The kernel $\mathfrak{d}_i$ of the row recursion eq. (14) is the common value $\mathfrak{d}(\mu_i, 1)$ of the $\mathfrak{d}_{ij}$, not their sum over j.

The formal specialization $\varrho \downarrow 0$ of corollary SC.2(iv) is the spectral $\sqrt{x}$-moment

$$\mathfrak{d}_i = \frac{B}{\sqrt{2}\,N_{\mathrm{in}}} \int \sqrt{x}\,\rho_i(dx),$$

in the normalization of the main-text kernel eq. (7), which is that of eq. (SC.4) (remark SC.1).

The hypothesis of corollary SC.2(iv) and the limit $\varrho \downarrow 0$ are not proved here. The hypothesis is the one under which the main text's half-anisotropic section writes its row spectral measure, and the deterministic model of the phase-comparison section adopts the $\varrho = 0$ kernels as assumptions.

When both covariances are the identity, every $\mathfrak{d}_{ij}$ equals $\mathfrak{d}(1,1)$. Appendix SF carries out this specialization, with the constant C_β of eq. (SF.21).

Remark SC.1 (No momentum and the main-text kernel)**.** *At $\beta = 0$ the only weight is $w_0 = 1$, $L_\beta = J$, and the candidate is*

$$T(s) = s\, J(q(-s))\, \overline{R}(-s)\, \frac{\Sigma_{\text{out}} \Delta_t \Sigma_{\text{in}}}{\sqrt{2\mathscr{R}(t)}}\, \overline{\overline{R}}(-s).$$

For the kernel, write for $\mu, \lambda \geq 0$

$$\mathsf{g}_{\mu\lambda}(z) \overset{\text{def}}{=} \frac{-z\, J(q(z))\, \mu\lambda}{\big(\tilde{s}_1(z)\mu - z\big)\big(s_1(z)\lambda - z\big)},$$

so that, at $\beta = 0$,

$$\mathfrak{d}(\mu, \lambda) = \frac{\sqrt{2}}{\pi} \int_0^\infty \mathsf{g}_{\mu\lambda}(-\varrho - u^2)\, du.$$

The main-text formula eq. (7) with the regularized filter $\varphi(y) = (y + 2\mathcal{R}\varrho)^{-1/2}$ reads, after $\sqrt{\mathcal{R}}\,\varphi(2\mathcal{R}x) = (2(x+\varrho))^{-1/2}$ and $\xi^2 f(\xi) = J(q)/2$,

$$\mathfrak{d}_{ij} = \frac{1}{\pi\sqrt{2}} \int_0^\infty \frac{\operatorname{Im}\, \mathsf{g}_{\mu_i\lambda_j}(x + i0^+)}{\sqrt{x+\varrho}}\, dx.$$

Suppose that $\mathsf{g}_{\mu\lambda}$ extends analytically to $\mathbb{C}\setminus[0,\infty)$, that it is the Stieltjes transform $\mathsf{g}_{\mu\lambda}(z) = \int \frac{m(dx)}{x-z}$ of a finite signed measure m on $(0,\infty)$, and that it extends continuously from the upper half-plane to $(0,\infty)$. Then m has the density $\frac{1}{\pi}\operatorname{Im}\, \mathsf{g}_{\mu\lambda}(x + i0^+)$ on $(0,\infty)$, by the Stieltjes inversion formula [1, Theorem 2.4.3] applied to the normalized positive and negative parts of m, and the scalar integral of lemma SA.3 gives

$$\begin{aligned}\frac{\sqrt{2}}{\pi} \int_0^\infty \mathsf{g}_{\mu\lambda}(-\varrho - u^2)\, du &= \frac{\sqrt{2}}{\pi} \int m(dx) \int_0^\infty \frac{du}{x + \varrho + u^2} = \frac{1}{\sqrt{2}} \int \frac{m(dx)}{\sqrt{x+\varrho}} \\ &= \frac{1}{\pi\sqrt{2}} \int_0^\infty \frac{\operatorname{Im}\, \mathsf{g}_{\mu\lambda}(x + i0^+)}{\sqrt{x+\varrho}}\, dx,\end{aligned}$$

so the two expressions of the kernel are the same quantity for every $\varrho > 0$. The analytic continuation of the negative-axis branch and the Stieltjes representation are not proved in this appendix, and the unregularized kernels of the main text, obtained by setting $\varrho = 0$ in eq. (7), are the formal specialization $\varrho \downarrow 0$ of corollary SC.1: no statement of this appendix concerns the limit $\varrho \downarrow 0$, and the phase-comparison section adopts the $\varrho = 0$ kernels as assumptions.

SC.7 One-step drift

Proof of proposition SC.2. The identity eq. (SC.15) is the total-risk line of proposition SA.4 in the frozen state. In the frozen-state model the state Δ_t is deterministic, the update $\mathsf{S} = \widetilde{G}_{t+1}$ has $\mathbb{E}\,\|\mathsf{S}\|_{\text{F}}^2 \leq \min(N_{\text{out}}, N_{\text{in}})$ by lemma SA.2(i), and the conditional expectations of that proposition are plain expectations; its identity eq. (SA.22) with the contractions eq. (SA.23) and $\mathscr{R} = \frac{1}{2}\|\cdot\|_\Sigma^2$ reads

$$\mathbb{E}\left[\mathscr{R}(\Delta^+)\right] - \mathscr{R}(t) = -\eta \left\langle \Delta_t, \mathbb{E}\,\widetilde{G}_{t+1} \right\rangle_\Sigma + \frac{\eta^2}{2}\, \mathbb{E}\,\|\widetilde{G}_{t+1}\|_\Sigma^2.$$

The pairing is rewritten through the mean gradient:

$$\begin{aligned}\left\langle \Delta_t, \mathbb{E}\,\widetilde{G}_{t+1} \right\rangle_\Sigma &= \operatorname{Tr}\left(\Sigma_{\text{out}} \Delta_t \Sigma_{\text{in}}\, (\mathbb{E}\,\widetilde{G}_{t+1})^\top\right) = \left\langle \Sigma_{\text{out}} \Delta_t \Sigma_{\text{in}}, \mathbb{E}\,\widetilde{G}_{t+1} \right\rangle \\ &= \sqrt{2\mathscr{R}(t)}\, \left\langle \mathbb{E}\, G_{t+1}, \mathbb{E}\,\widetilde{G}_{t+1} \right\rangle,\end{aligned}$$

by eq. (SA.11), which gives eq. (SC.15).

The deterministic equivalent of the pairing is read in the eigenbases. By eq. (SC.9) with $V = \mathbb{E}\, G_{t+1}$ and $\|\mathbb{E}\, G_{t+1}\|_{\text{F}} \leq C_\Sigma$,

$$\left| \left\langle \mathbb{E}\, G_{t+1}, \mathbb{E}\,\widetilde{G}_{t+1} \right\rangle - \left\langle \mathbb{E}\, G_{t+1}, \overline{\overline{\widetilde{G}}}_\varrho \right\rangle \right| \leq C_\Sigma \varepsilon_B,$$

and, expanding both matrices in the bases (u_i), (v_j), with $u_i^\top\, \mathbb{E}\, G_{t+1} v_j = \mu_i \lambda_j (u_i^\top \Delta_t v_j)/\sqrt{2\mathscr{R}(t)}$ and eq. (SC.11),

$$\left\langle \mathbb{E}\, G_{t+1}, \overline{\overline{\widetilde{G}}}_\varrho \right\rangle = \sum_{i,j} \frac{\mu_i \lambda_j (u_i^\top \Delta_t v_j)}{\sqrt{2\mathscr{R}(t)}} \cdot \frac{\mathfrak{d}_{ij} (u_i^\top \Delta_t v_j)}{\sqrt{\mathscr{R}(t)}} = \frac{1}{\sqrt{2}\,\mathscr{R}(t)} \sum_{i,j} \mu_i \lambda_j\, \mathfrak{d}_{ij}\, \mathscr{Q}_{ij}(t).$$

Multiplying by $\sqrt{2\mathscr{R}(t)}$ gives eq. (SC.16). □

SD Volatility: same-time and temporal

This section identifies the volatility coefficient of the dynamics. In the exact recursion eq. (SA.20), the quadratic term of step t is the conditional second moment $\mathcal{V}_{ij}(t)$ of the projected update. With momentum, the updates of nearby steps are correlated, because consecutive buffers share most of their batches. Over the B steps of a unit of macroscopic time the quadratic terms therefore accumulate as the integrated centered covariance of the filtered update: the sum, over all lags, of the covariances between the centered updates of two times. The volatility coefficients $\mathfrak{v}_{ij}$ and $\mathfrak{v}_i$ of the appendix are the deterministic equivalents of this integrated covariance on modes and on rows. The paragraph after proposition SD.3 says why the dynamics use them in place of the raw one-step coefficient.

The section has three parts. The first part treats the same-time update. For every pair of covariances the update Gram matrix is an affine function of one Gram resolvent, and theorem SD.1 identifies the expected energy of every row and every column of the update. The second part treats the temporal structure. Theorem SD.2 writes the accumulated covariance over T steps exactly through lag covariances, bounds them geometrically in the lag, and replaces the actual residuals by independent Gaussian weights at every fixed lag. Definition SD.4 gives an explicit moment algorithm for the independent model, theorem SD.3 proves that its unique continuous extension to the filters is the deterministic equivalent of every fixed-lag covariance, and theorem SD.4 sums the lags. The third part evaluates these deterministic objects. Theorem SD.5 gives the lag kernel $\mathfrak{k}_\ell$ in closed form, for every lag, every pair of diagonal tests and every pair of covariances, and corollary SD.3 turns it into an explicit formula for the volatility $\mathfrak{v}_{ij}$ of every mode. Appendix SD.8 assembles the row coefficients $\mathfrak{v}_i$ for $\Sigma_{\text{in}} = \mathrm{Id}$ and the mode coefficients $\mathfrak{v}_{ij}$ in general, and appendix SD.9 recovers from the closed form the same-time totals with two anisotropic covariances, the row coefficients of the half-anisotropic setup and the case without momentum.

SD.1 The volatility in brief

This subsection collects the definitions needed to read the main results of the section, states those results informally, and says where each is proved.

Objects. Freeze the error at Δ_t and let $M_{t'}$, $t' \in \mathbb{Z}$, be the stationary momentum buffer of definition SA.3. At a normalized ridge $\varrho > 0$ the update of step t' and its fluctuation are

$$\mathsf{S}_{t'} = Q_\varrho(M_{t'}), \qquad \mathsf{S}^{\mathrm{c}}_{t'} = \mathsf{S}_{t'} - \mathbb{E}\,\mathsf{S}_{t'},$$

with Q_ϱ the regularized SignSVD map of eq. (SA.12). In the eigenbases (u_i) of Σ_{out} and (v_j) of Σ_{in}, write $(\mathsf{S})_{ij} = u_i^\top \mathsf{S} v_j$ for the entry of mode (i, j).

The volatility of a mode. The actual volatility of mode (i, j) is the long-run variance of its entry,

$$\mathsf{v}_{ij} \overset{\text{def}}{=} \mathbb{E}\left[(\mathsf{S}^{\mathrm{c}}_t)^2_{ij}\right] + 2\sum_{\ell \geq 1} \mathbb{E}\left[(\mathsf{S}^{\mathrm{c}}_t)_{ij}\,(\mathsf{S}^{\mathrm{c}}_{t+\ell})_{ij}\right] = \lim_{T\to\infty} \frac{1}{T}\,\mathbb{E}\Big(\sum_{t'=1}^{T} (\mathsf{S}^{\mathrm{c}}_{t'})_{ij}\Big)^2.$$

It is the noise that T steps inject into mode (i, j), per step, and it is the coefficient of η^2 in the mode recursion. The first term is the same-step variance. The lag terms are present only with momentum: consecutive buffers share most of their batches, so updates at nearby steps are correlated, with correlations decaying like β^ℓ. When $\Sigma_{\text{in}} = \mathrm{Id}$, the main text uses the row sums $\mathsf{v}_i \overset{\text{def}}{=} \sum_j \mathsf{v}_{ij}$, which are the same expression with $(\mathsf{S})_{ij}$ replaced by the row vector $\mathsf{S}^\top u_i$.

The formula. The deterministic volatility is built from the branch $(\sigma, \tilde\sigma, \tilde s_1, s_1, q)(-s)$ of the fixed-point system of appendix SB, evaluated at two spectral parameters $s, u > 0$; primes denote the branch at $-u$. Put

$$\mathsf{U} \overset{\text{def}}{=} \frac{1}{B}\sum_{i=1}^{N_{\text{out}}} \frac{\mu_i^2}{(s + \tilde s_1 \mu_i)(u + \tilde s_1' \mu_i)}, \qquad \mathsf{V} \overset{\text{def}}{=} \frac{1}{B}\sum_{j=1}^{N_{\text{in}}} \frac{\lambda_j^2}{(s + s_1 \lambda_j)(u + s_1' \lambda_j)},$$

and, for a lag $\ell \geq 0$, with the EMA weights $w_j = (1-\beta)\beta^j$ and g a standard Gaussian scalar,

$$b_\ell \overset{\text{def}}{=} \sum_{j\geq 0} \mathbb{E}\,\frac{\beta^\ell w_j^2 g^2}{(1 + q w_j^2 g^2)(1 + \beta^{2\ell} q' w_j^2 g^2)}, \qquad e_\ell \overset{\text{def}}{=} \sum_{j\geq 0} \mathbb{E}\,\frac{\beta^{2\ell} w_j^4 g^4}{(1 + q w_j^2 g^2)(1 + \beta^{2\ell} q' w_j^2 g^2)},$$

$$\Delta_\ell \stackrel{\text{def}}{=} \big(1 - su \,\mathsf{U}\, \tilde\sigma\tilde\sigma' e_\ell\big)\big(1 - su \,\mathsf{V}\, \sigma\sigma' e_\ell\big) - su \,\mathsf{UV}\, b_\ell^2.$$

For eigenvalues μ of Σ_{out} and λ of Σ_{in}, the lag-ℓ kernel of the mode pair and its integral along the negative axis are

$$\mathfrak{h}_\ell(\mu,\lambda;s,u) \stackrel{\text{def}}{=} \frac{su\,\mu\lambda\, b_\ell}{(s+\tilde s_1\mu)(u+\tilde s_1'\mu)(s+s_1\lambda)(u+s_1'\lambda)\,\Delta_\ell}, \qquad \mathfrak{c}_\ell(\mu,\lambda;\varrho) \stackrel{\text{def}}{=} \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \mathfrak{h}_\ell(\mu,\lambda;\varrho{+}x^2,\varrho{+}y^2)\,dx\,dy.$$

The volatility of the mode (i,j) is

$$\mathfrak{v}_{ij}(\varrho) = \frac{1}{B}\Big(\mathfrak{c}_0(\mu_i,\lambda_j;\varrho) + 2\sum_{\ell\ge 1}\mathfrak{c}_\ell(\mu_i,\lambda_j;\varrho)\Big).$$

The first term is the same-step variance and the lag terms are the correlations between nearby steps. Every term is nonnegative.

What is proved.

(i) *Deterministic mode coefficients* (proposition SD.3). There are deterministic numbers $\mathfrak{v}_{ij} = \mathfrak{v}_{ij}(\varrho) \ge 0$, depending on (i,j) only through the eigenvalue pair (μ_i,λ_j) and vanishing when $\mu_i\lambda_j = 0$, that approximate the actual v_{ij} after averaging over modes: for every bounded Lipschitz function ϕ on the spectral rectangle,

$$\frac{1}{B}\sum_{i,j}\phi(\mu_i,\lambda_j)\,\big(\mathsf{v}_{ij} - \mathfrak{v}_{ij}\big) \longrightarrow 0$$

in the proportional regime, uniformly in ϕ with $\|\phi\|_\infty \le 1$, $\operatorname{Lip}\phi \le 1$ and in the ridge on compacts. No statement is made about an individual entry.

(ii) *The formula* (theorem SD.5 and corollary SD.3). The numbers of (i) are given by the formula above, for every pair of positive semidefinite covariances, every $\beta \in [0,1)$ and every $\varrho > 0$. The determinant Δ_ℓ is positive, each lag term $\mathfrak{c}_\ell$ is nonnegative and decays like β^ℓ, and the series converges absolutely. The lag enters only through the two scalars b_ℓ and e_ℓ; no closed form of the lag sum is claimed. The route to the formula is the lag reduction (theorem SD.2), the tree-moment response (theorems SD.3 and SD.4), which identifies the lag kernels with the lag covariances of the actual updates, and the closed-form evaluation of those kernels (theorem SD.5).

(iii) *Same-step energies* (theorem SD.1 and corollary SD.4). For every pair of covariances the expected row and column energies of the update are within a vanishing error of

$$\mathfrak{v}_i^{(0)}(\varrho) = \frac{\tilde s_1(-\varrho)\,\mu_i}{\varrho + \tilde s_1(-\varrho)\,\mu_i}, \qquad \widetilde{\mathfrak{v}}_j^{(0)}(\varrho) = \frac{s_1(-\varrho)\,\lambda_j}{\varrho + s_1(-\varrho)\,\lambda_j},$$

uniformly in i and j, with $\tilde s_1$ and s_1 the scalars of the fixed-point system of appendix SB. In the formula these are the marginals of the same-step term: $B^{-1}\sum_j \mathfrak{c}_0(\mu_i,\lambda_j;\varrho) = \mathfrak{v}_i^{(0)}(\varrho)$ and $B^{-1}\sum_i \mathfrak{c}_0(\mu_i,\lambda_j;\varrho) = \widetilde{\mathfrak{v}}_j^{(0)}(\varrho)$.

(iv) *Specializations* (appendix SD.9). Without momentum, $b_\ell = 0$ for $\ell \ge 1$, so $\mathfrak{v}_{ij} = \mathfrak{c}_0/B$ and the row sums are $\sum_j \mathfrak{v}_{ij} = \mathfrak{v}_i^{(0)}$; for isotropic covariances the total same-step volatility tends, as $\varrho \to 0$, to the rank of the gradient, $\min\{B, N_{\text{in}}, N_{\text{out}}\}$ (proposition SF.2). When $\Sigma_{\text{in}} = \mathrm{Id}$, the row coefficients $\mathfrak{v}_i = \sum_j \mathfrak{v}_{ij}$ used in the main text are given by the two-time row response of proposition SG.3, and $\frac{1}{B}\sum_i |\mathsf{v}_i - \mathfrak{v}_i| \to 0$ (theorem SD.7). With two anisotropic covariances the covariance-weighted same-step total is the two-resolvent kernel of corollary SD.5. In the isotropic case all modes carry the same coefficient, $\mathfrak{v}_{ij} = \mathfrak{v}_{\text{iso}}/(N_{\text{out}}N_{\text{in}})$ (theorem SF.2).

SD.2 Standing setup and finite estimates

The section works in the frozen-state model of definition SA.3, with the samples indexed by time as in the paragraph following that definition, so that $M_{t'} = \sum_{j\ge 0} w_j G_{t'+1-j}$ for every integer t'. All asymptotic statements are made under Assumption S3, with the normalized ridge in $[\varrho_-, \varrho_+]$

and with the eigenbases (u_i) of Σ_{out} and (v_j) of Σ_{in} of definition SA.1; for a matrix V we write $V_{ij} \overset{\text{def}}{=} u_i^\top V v_j$. From appendix SA the section uses the pairing $\langle \cdot, \cdot \rangle_\Sigma$ of eq. (SA.2), the filter bounds and the Lipschitz estimate of lemma SA.2, the integral representation lemma SA.3, the batch variance V_B of lemma SA.4, the stationary law of lemma SA.5, the truncation bounds of lemma SA.6 and the residual law lemma SA.7. From appendix SB it uses the branch $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ and the deterministic equivalents $\overline{R}(-s)$, $\overline{\widetilde{R}}(-s)$ of definition SB.1, the function $r_\beta(q) \overset{\text{def}}{=} qF_\beta(q)$ of lemma SB.1(iii), the notation $R_s(K)$, $\widetilde{R}_s(K)$, $\mathsf{X}_s(K) = R_s(K)K$ of eq. (SB.10), and the results cited where they are used. From appendix SC it uses the regularized mean theorem SC.2.

Constants: for a compact $\mathcal{J} \subset (0, \infty)$, $C_{\mathcal{J}}$ denotes a constant depending only on $(\min \mathcal{J}, \max \mathcal{J}, C_\Sigma, \gamma_\star)$; C_I denotes a constant depending only on $(\varrho_-, C_\Sigma, \gamma_\star)$; C without subscript depends only on $(C_\Sigma, \gamma_\star)$. All may change from line to line. A constant never depends on the dimension, on the lag or on the matrices F, G introduced below, unless the statement says so. A statement at a fixed dimension that involves $\gamma_\star$ or C_Σ assumes $\max(N_{\text{in}}, N_{\text{out}}) \le \gamma_\star B$ and $\|\Sigma_{\text{out}}\|, \|\Sigma_{\text{in}}\| \le C_\Sigma$ at that dimension.

Odd spectral functions. Let $f : \mathbb{R} \to \mathbb{R}$ be odd. For $K \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ with singular value decomposition $K = \sum_k s_k\, p_k \otimes r_k$ we put

$$f^\sharp(K) \overset{\text{def}}{=} \sum_k f(s_k)\, p_k \otimes r_k. \tag{SD.1}$$

The sum does not depend on the decomposition, by the argument following eq. (SA.12). The two filters of the section are

$$\iota_s(x) \overset{\text{def}}{=} \frac{x}{x^2 + s} \quad (s > 0), \qquad \omega_\varrho(x) \overset{\text{def}}{=} \frac{x}{\sqrt{x^2 + \varrho}} \quad (\varrho > 0), \tag{SD.2}$$

for which

$$\iota_s^\sharp(K) = \mathsf{X}_s(K) = \big(KK^\top + s\,\mathrm{Id}\big)^{-1} K, \qquad \omega_\varrho^\sharp(K) = Q_\varrho(K), \tag{SD.3}$$

by eqs. (SA.12) and (SB.10). For odd monomials, $(x^{2a+1})^\sharp(K) = (KK^\top)^a K$.

Lemma SD.1 (Odd spectral functions)**.** *Let f be odd, bounded and Lipschitz on $\mathbb{R}$, with $\|f\|_\infty \overset{\text{def}}{=} \sup_x |f(x)|$ and Lipschitz constant $\mathrm{Lip}(f)$. For all $K, K' \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$,*

$$\begin{gathered} \|f^\sharp(K)\| \le \|f\|_\infty, \qquad \|f^\sharp(K)\|_{\mathrm{F}} \le \sqrt{\min(N_{\text{out}}, N_{\text{in}})}\, \|f\|_\infty, \\ \|f^\sharp(K) - f^\sharp(K')\|_{\mathrm{F}} \le \mathrm{Lip}(f)\, \|K - K'\|_{\mathrm{F}}. \end{gathered} \tag{SD.4}$$

The filters eq. (SD.2) satisfy

$$\|\iota_s\|_\infty = \frac{1}{2\sqrt{s}}, \quad \mathrm{Lip}(\iota_s) \le \frac{1}{s}, \qquad \|\omega_\varrho\|_\infty = 1, \quad \mathrm{Lip}(\omega_\varrho) \le \varrho^{-1/2}, \tag{SD.5}$$

and, for all $s, s' \ge s_\circ > 0$ and $\varrho, \varrho' \ge \varrho_\circ > 0$,

$$\|\iota_s - \iota_{s'}\|_\infty \le \frac{|s - s'|}{s_\circ^{3/2}}, \qquad \|\omega_\varrho - \omega_{\varrho'}\|_\infty \le \frac{|\varrho - \varrho'|}{2\varrho_\circ}. \tag{SD.6}$$

Proof. The first two bounds read off the singular values: the singular values of $f^\sharp(K)$ are the numbers $|f(s_k)| \le \|f\|_\infty$, and there are at most $\min(N_{\text{out}}, N_{\text{in}})$ of them.

The Lipschitz bound is the divided-difference argument in the proof of lemma SA.2(iii). That argument uses only that the function is odd and Lipschitz: for odd f the symmetric dilation satisfies

$$f\big(\mathcal{D}(K)\big) = \begin{pmatrix} 0 & f^\sharp(K) \\ f^\sharp(K)^\top & 0 \end{pmatrix},$$

by the eigenvector computation given there, and the divided-difference bound for $\mathcal{D}(K)$ and $\mathcal{D}(K')$ gives the third bound of eq. (SD.4) after cancelling the factor $\sqrt{2}$.

The constants of the two filters are elementary. The maximum of $x/(x^2 + s)$ is attained at $x = \sqrt{s}$, and

$$\iota_s'(x) = \frac{s - x^2}{(x^2 + s)^2}, \qquad |\iota_s'(x)| \le \frac{1}{x^2 + s} \le \frac{1}{s}.$$

The bounds for ω_ϱ are those of the proof of lemma SA.2(iii). For the parameters,

$$\partial_s \iota_s(x) = -\frac{x}{(x^2+s)^2}, \qquad \partial_\varrho \omega_\varrho(x) = -\frac{x}{2(x^2+\varrho)^{3/2}},$$

whose absolute values are at most $\iota_s(|x|)/(x^2+s) \le s^{-3/2}$ and $1/(2\varrho)$; the mean value theorem in the parameter gives eq. (SD.6). □

Tested contractions. For deterministic $F \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{out}}}$ and $G \in \mathbb{R}^{N_{\mathrm{in}} \times N_{\mathrm{in}}}$, odd bounded functions f_1, f_2, and $K, H \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$, we put

$$\mathcal{K}^{F,G}[f_1, f_2](K, H) \stackrel{\mathrm{def}}{=} \frac{1}{B} \operatorname{Tr}\left[F\, f_1^\sharp(K)\, G\, f_2^\sharp(H)^\top\right], \qquad \mathcal{K}^{F,G}_{s,u}(K, H) \stackrel{\mathrm{def}}{=} \mathcal{K}^{F,G}[\iota_s, \iota_u](K, H). \tag{SD.7}$$

The covariance-weighted case $F = \Sigma_{\mathrm{out}}$, $G = \Sigma_{\mathrm{in}}$ is written without superscript, $\mathcal{K}_{s,u} \stackrel{\mathrm{def}}{=} \mathcal{K}^{\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}}}_{s,u}$; for it, $\mathcal{K}[\omega_\varrho, \omega_{\varrho'}](K, H) = B^{-1}\langle Q_\varrho(K), Q_{\varrho'}(H)\rangle_\Sigma$.

Lemma SD.2 (Finite bounds for tested contractions)**.** *Let $\|F\|, \|G\| \le 1$, let f_1, f_2 be odd, bounded and Lipschitz, and let $K, K', H, H' \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$. Then*

$$|\mathcal{K}^{F,G}[f_1, f_2](K, H)| \le \gamma_\star \|f_1\|_\infty \|f_2\|_\infty, \tag{SD.8}$$

$$|\mathcal{K}^{F,G}[f_1, f_2](K, H) - \mathcal{K}^{F,G}[f_1, f_2](K', H')| \le \sqrt{\frac{\gamma_\star}{B}}\Big(\operatorname{Lip}(f_1)\|f_2\|_\infty \|K - K'\|_{\mathrm{F}} + \|f_1\|_\infty \operatorname{Lip}(f_2) \|H - H'\|_{\mathrm{F}}\Big), \tag{SD.9}$$

and the bilinear dependence on the filters gives

$$|\mathcal{K}^{F,G}[f_1, f_2](K, H) - \mathcal{K}^{F,G}[f_1', f_2'](K, H)| \le \gamma_\star\big(\|f_1 - f_1'\|_\infty \|f_2\|_\infty + \|f_1'\|_\infty \|f_2 - f_2'\|_\infty\big). \tag{SD.10}$$

For the resolvent filters, with $s, u \ge s_\circ > 0$,

$$|\mathcal{K}^{F,G}_{s,u}(K, K) - \mathcal{K}^{F,G}_{s,u}(K', K')| \le \frac{2}{B\, s_\circ^{3/2}} \|K - K'\|_*. \tag{SD.11}$$

Proof. The first bound is trace duality with a rank bound. The matrix $F f_1^\sharp(K) G f_2^\sharp(H)^\top$ has rank at most $\min(N_{\mathrm{out}}, N_{\mathrm{in}}) \le \gamma_\star B$ and operator norm at most $\|f_1\|_\infty \|f_2\|_\infty$ by lemma SD.1, and the absolute trace of a matrix is at most its rank times its operator norm.

The second bound follows by telescoping the two arguments. Changing K to K' changes the trace by at most

$$\frac{1}{B} \|f_1^\sharp(K) - f_1^\sharp(K')\|_{\mathrm{F}}\, \|f_2^\sharp(H)\|_{\mathrm{F}} \le \frac{1}{B} \operatorname{Lip}(f_1) \|K - K'\|_{\mathrm{F}} \cdot \sqrt{\gamma_\star B}\, \|f_2\|_\infty,$$

by the Cauchy–Schwarz inequality for the Frobenius pairing and eq. (SD.4); changing H to H' is symmetric. The third bound is the first bound applied to the two differences.

The nuclear bound is the resolvent identity for the symmetric dilation. The proof of lemma SB.9, read in the upper right block of $(\mathcal{D}(K) - i\sqrt{s}\ \mathrm{Id})^{-1}$, whose upper right block is $\mathsf{X}_s(K)$, gives

$$\|\mathsf{X}_s(K) - \mathsf{X}_s(K')\|_* \le \frac{2}{s} \|K - K'\|_*.$$

With $\|\mathsf{X}_u(K)\| \le (2\sqrt{u})^{-1}$ from lemma SB.3 and trace duality $|\operatorname{Tr}(XY)| \le \|X\|_* \|Y\|$, telescoping the two arguments gives

$$|\mathcal{K}^{F,G}_{s,u}(K, K) - \mathcal{K}^{F,G}_{s,u}(K', K')| \le \frac{1}{B}\Big(\frac{2}{s} \cdot \frac{1}{2\sqrt{u}} + \frac{1}{2\sqrt{s}} \cdot \frac{2}{u}\Big) \|K - K'\|_* \le \frac{2}{B s_\circ^{3/2}} \|K - K'\|_*. \ \square$$

Square truncation of the filter. The regularized update is the integral eq. (SA.14) of $\mathsf{X}_{\varrho+x^2}$ over $x \in (0,\infty)$. We truncate it at $T > 0$:

$$\omega_{\varrho,T}(x) \stackrel{\text{def}}{=} \frac{2}{\pi}\int_0^T \iota_{\varrho+y^2}(x)\,dy, \qquad Q_{\varrho,T}(K) \stackrel{\text{def}}{=} \omega^\sharp_{\varrho,T}(K) = \frac{2}{\pi}\int_0^T \mathsf{X}_{\varrho+y^2}(K)\,dy. \tag{SD.12}$$

Lemma SD.3 (Square truncation)**.** *For $\varrho > 0$, $T > 0$ and $x \in \mathbb{R}$,*

$$0 \le \frac{\omega_{\varrho,T}(x)}{\omega_\varrho(x)} \le 1 \quad (x \neq 0), \qquad |\omega_\varrho(x) - \omega_{\varrho,T}(x)| \le \frac{2|x|}{\pi T}. \tag{SD.13}$$

Consequently, for random $K, H \in \mathbb{R}^{N_{\text{out}}\times N_{\text{in}}}$ with $B^{-1}\,\mathbb{E}\,\|K\|_{\text{F}}^2 \le C_M$ and $B^{-1}\,\mathbb{E}\,\|H\|_{\text{F}}^2 \le C_M$, and for $\|F\|, \|G\| \le 1$,

$$\Big|\,\mathbb{E}\,\mathcal{K}^{F,G}[\omega_\varrho,\omega_{\varrho'}](K,H) - \frac{4}{\pi^2}\int_0^T\!\!\int_0^T \mathbb{E}\,\mathcal{K}^{F,G}_{\varrho+x^2,\varrho'+y^2}(K,H)\,dx\,dy\Big| \le \frac{4\sqrt{\gamma_\star C_M}}{\pi T}. \tag{SD.14}$$

In the frozen-state model, $B^{-1}\,\mathbb{E}\,\|M_t\|_{\text{F}}^2 \le C_M \stackrel{\text{def}}{=} C_\Sigma^2(\gamma_\star+2)^2$.

Proof. The scalar statements are the integral of lemma SA.3 split at T. For $x \neq 0$,

$$\omega_\varrho(x) - \omega_{\varrho,T}(x) = \frac{2}{\pi}\int_T^\infty \frac{x}{x^2+\varrho+y^2}\,dy,$$

which has the sign of x and absolute value at most $\frac{2}{\pi}|x|\int_T^\infty y^{-2}dy = 2|x|/(\pi T)$; the truncated integral has the same sign and a smaller absolute value than the full one, which is $\omega_\varrho(x)$ by the scalar identity in the proof of lemma SA.3.

The matrix bound follows from the scalar one. Applied to the singular values, eq. (SD.13) gives

$$\|Q_{\varrho,T}(K)\|_{\text{F}} \le \|Q_\varrho(K)\|_{\text{F}} \le \sqrt{\gamma_\star B}, \qquad \|Q_\varrho(K) - Q_{\varrho,T}(K)\|_{\text{F}} \le \frac{2}{\pi T}\|K\|_{\text{F}}.$$

Telescoping the two arguments as in the proof of lemma SD.2,

$$\big|\mathcal{K}^{F,G}[\omega_\varrho,\omega_{\varrho'}](K,H) - \mathcal{K}^{F,G}[\omega_{\varrho,T},\omega_{\varrho',T}](K,H)\big| \le \frac{2\sqrt{\gamma_\star}}{\pi T}\Big(\frac{\|K\|_{\text{F}}}{\sqrt{B}} + \frac{\|H\|_{\text{F}}}{\sqrt{B}}\Big),$$

whose expectation is at most $4\sqrt{\gamma_\star C_M}/(\pi T)$ by the Cauchy–Schwarz inequality. The truncated contraction is the double integral of the resolvent contractions, because $Q_{\varrho,T}$ is the integral eq. (SD.12) and the trace is bilinear; this gives eq. (SD.14).

The second-moment bound is lemma SA.5(iii) with the bounds of lemma SA.6(iv):

$$\begin{aligned}\frac{1}{B}\,\mathbb{E}\,\|M_t\|_{\text{F}}^2 = \frac{1}{B}\Big(\|\,\mathbb{E}\,G_{t+1}\|_{\text{F}}^2 + s_2\mathsf{V}_B\Big) &\le \frac{C_\Sigma^2}{B} + \frac{C_\Sigma^2}{B^2}\big(N_{\text{out}}N_{\text{in}} + 2N_{\text{out}} + 2N_{\text{in}} + 3\big)\\ &\le C_\Sigma^2(\gamma_\star+2)^2 = C_M,\end{aligned}$$

using $s_2 \le 1$, $N_{\text{out}}, N_{\text{in}} \le \gamma_\star B$ and $B \ge 1$. □

The filtered updates and their covariances. For $\varrho > 0$ and an integer t' we write

$$\mathsf{S}_{t'}(\varrho) \stackrel{\text{def}}{=} Q_\varrho(M_{t'}), \qquad \mathsf{S}^{\text{c}}_{t'}(\varrho) \stackrel{\text{def}}{=} \mathsf{S}_{t'}(\varrho) - \mathbb{E}\,\mathsf{S}_{t'}(\varrho). \tag{SD.15}$$

By lemma SA.5(ii) the sequence $(\mathsf{S}_{t'}(\varrho))_{t'}$ is stationary, so the quantities below do not depend on t.

Definition SD.1 (Tested lag covariances)**.** *For symmetric deterministic $F \in \mathbb{R}^{N_{\text{out}}\times N_{\text{out}}}$ and $G \in \mathbb{R}^{N_{\text{in}}\times N_{\text{in}}}$, ridges $\varrho, \varrho' > 0$ and a lag $\ell \ge 0$, the tested lag covariance is*

$$C_\ell[F,G](\varrho,\varrho') \stackrel{\text{def}}{=} \frac{1}{B}\,\mathbb{E}\,\text{Tr}\,\big[F\,\mathsf{S}^{\text{c}}_t(\varrho)\,G\,\mathsf{S}^{\text{c}}_{t+\ell}(\varrho')^\top\big], \tag{SD.16}$$

and, when the series converges absolutely, the integrated covariance is

$$\Gamma[F,G](\varrho,\varrho') \stackrel{\text{def}}{=} C_0[F,G](\varrho,\varrho') + \sum_{\ell\ge1}\Big(C_\ell[F,G](\varrho,\varrho') + C_\ell[F,G](\varrho',\varrho)\Big). \tag{SD.17}$$

The covariance-weighted case is written $C_\ell \stackrel{\text{def}}{=} C_\ell[\Sigma_{\text{out}},\Sigma_{\text{in}}]$ and $\Gamma \stackrel{\text{def}}{=} \Gamma[\Sigma_{\text{out}},\Sigma_{\text{in}}]$, so that $C_\ell(\varrho,\varrho') = B^{-1}\,\mathbb{E}\langle\mathsf{S}^{\text{c}}_t(\varrho),\mathsf{S}^{\text{c}}_{t+\ell}(\varrho')\rangle_\Sigma$. When $\Sigma_{\text{in}} = \text{Id}$, the row covariances are

$$C_{i,\ell}(\varrho,\varrho') \stackrel{\text{def}}{=} B\,C_\ell[u_i\otimes u_i,\text{Id}](\varrho,\varrho') = \mathbb{E}\,\big\langle\mathsf{S}^{\text{c}}_t(\varrho)^\top u_i,\ \mathsf{S}^{\text{c}}_{t+\ell}(\varrho')^\top u_i\big\rangle, \tag{SD.18}$$

and Γ_i is their integrated sum, defined as in eq. (SD.17).

The same-time totals are

$$Z(\varrho) \stackrel{\text{def}}{=} \frac{1}{B}\|\mathsf{S}_t(\varrho)\|_\Sigma^2, \qquad Z^{\mathrm{c}}(\varrho) \stackrel{\text{def}}{=} \frac{1}{B}\|\mathsf{S}_t^{\mathrm{c}}(\varrho)\|_\Sigma^2, \tag{SD.19}$$

so that $\mathbb{E}\, Z^{\mathrm{c}}(\varrho) = C_0(\varrho,\varrho)$. In the frozen one-step identity eq. (SC.15) the quadratic term is $\frac{\eta^2}{2}\,\mathbb{E}\,\|Q_\varrho(M_t)\|_\Sigma^2 = \frac{\eta^2 B}{2}\,\mathbb{E}\, Z(\varrho)$.

Lemma SD.4 (Centering)**.** *Under Assumption S3, for all $\varrho,\varrho' \in [\varrho_-,\varrho_+]$, every lag $\ell \geq 0$ and $\|F\|, \|G\| \leq 1$,*

$$\Big|\frac{1}{B}\,\mathbb{E}\,\mathrm{Tr}\left[F\,\mathsf{S}_t(\varrho)\,G\,\mathsf{S}_{t+\ell}(\varrho')^\top\right] - C_\ell[F,G](\varrho,\varrho')\Big| \leq \frac{(C_\Sigma \varrho_-^{-1/2} + \varepsilon_B)^2}{B}, \tag{SD.20}$$

with $\varepsilon_B \to 0$ the sequence of theorem SC.2, and

$$\sup_{\varrho\in[\varrho_-,\varrho_+]} |Z(\varrho) - Z^{\mathrm{c}}(\varrho)| \leq C_\Sigma^2\Big(\frac{4\sqrt{\gamma_\star}}{\sqrt{B}}(C_\Sigma\varrho_-^{-1/2}+\varepsilon_B) + \frac{(C_\Sigma\varrho_-^{-1/2}+\varepsilon_B)^2}{B}\Big). \tag{SD.21}$$

Proof. The mean of the update is bounded by theorem SC.2:

$$\|\,\mathbb{E}\,\mathsf{S}_t(\varrho)\|_{\mathrm{F}} \leq \|\overline{\widetilde{G}}_\varrho\|_{\mathrm{F}} + \varepsilon_B \leq C_\Sigma\varrho_-^{-1/2} + \varepsilon_B.$$

The difference in eq. (SD.20) is $B^{-1}\,\mathrm{Tr}[F\;\mathbb{E}\,\mathsf{S}_t(\varrho)\,G\,(\mathbb{E}\,\mathsf{S}_{t+\ell}(\varrho'))^\top]$, by expanding both factors around their means; its absolute value is at most $B^{-1}\|\,\mathbb{E}\,\mathsf{S}_t(\varrho)\|_{\mathrm{F}}\|\,\mathbb{E}\,\mathsf{S}_{t+\ell}(\varrho')\|_{\mathrm{F}}$.

The sample bound expands the square. Writing $\mathsf{S} = \mathsf{S}_t(\varrho)$ and $\overline{\mathsf{S}} = \mathbb{E}\,\mathsf{S}$,

$$Z(\varrho) - Z^{\mathrm{c}}(\varrho) = \frac{1}{B}\Big(2\langle \mathsf{S}-\overline{\mathsf{S}}, \overline{\mathsf{S}}\rangle_\Sigma + \|\overline{\mathsf{S}}\|_\Sigma^2\Big),$$

and $|\langle V,W\rangle_\Sigma| \leq C_\Sigma^2\|V\|_{\mathrm{F}}\|W\|_{\mathrm{F}}$ with $\|\mathsf{S}-\overline{\mathsf{S}}\|_{\mathrm{F}} \leq 2\sqrt{\gamma_\star B}$ by lemma SA.2(i) gives eq. (SD.21). □

SD.3 Same-time energies of rows and columns

For every pair of covariances, the regularized update Gram matrix is an affine function of one Gram resolvent, and the Gram closure of appendix SB.3 identifies it directly. For $\varrho > 0$ put

$$\mathfrak{v}_i^{(0)}(\varrho) \stackrel{\text{def}}{=} \frac{\tilde{s}_1(-\varrho)\,\mu_i}{\varrho + \tilde{s}_1(-\varrho)\,\mu_i}, \qquad \widetilde{\mathfrak{v}}_j^{(0)}(\varrho) \stackrel{\text{def}}{=} \frac{s_1(-\varrho)\,\lambda_j}{\varrho + s_1(-\varrho)\,\lambda_j}, \qquad \overline{Z}(\varrho) \stackrel{\text{def}}{=} \frac{1}{B}\sum_{i=1}^{N_{\mathrm{out}}} \mu_i\,\mathfrak{v}_i^{(0)}(\varrho) = \frac{1}{B}\,\mathrm{Tr}\,\Sigma_{\mathrm{out}} - \varrho\,\sigma(-\varrho). \tag{SD.22}$$

The second expression of $\overline{Z}$ follows from $\mu_i\mathfrak{v}_i^{(0)} = \mu_i - \varrho\mu_i/(\varrho+\tilde{s}_1\mu_i)$ and the first equation of eq. (SB.2). The letter $\mathfrak{v}^{(0)}$ marks a deterministic equivalent of the raw one-step coefficient.

Theorem SD.1 (Same-time energies of rows and columns)**.** *Assume Assumption S3, let $I = [\varrho_-,\varrho_+]$, and put*

$$e_B \stackrel{\text{def}}{=} \sup_{s\in I}\|\,\mathbb{E}\,R(-s) - \overline{R}(-s)\|, \qquad \tilde{e}_B \stackrel{\text{def}}{=} \sup_{s\in I}\|\,\mathbb{E}\,\widetilde{R}(-s) - \overline{\widetilde{R}}(-s)\|,$$

which tend to zero by theorem SB.2.

(i) The expected update Gram matrices satisfy

$$\begin{aligned}
&\sup_{\varrho\in I}\Big\|\,\mathbb{E}\left[\mathsf{S}_t(\varrho)\mathsf{S}_t(\varrho)^\top\right] - \tilde{s}_1(-\varrho)\Sigma_{\mathrm{out}}\big(\tilde{s}_1(-\varrho)\Sigma_{\mathrm{out}} + \varrho\,\mathrm{Id}\,\big)^{-1}\Big\| \leq \varrho_+\,e_B,\\
&\sup_{\varrho\in I}\Big\|\,\mathbb{E}\left[\mathsf{S}_t(\varrho)^\top\mathsf{S}_t(\varrho)\right] - s_1(-\varrho)\Sigma_{\mathrm{in}}\big(s_1(-\varrho)\Sigma_{\mathrm{in}} + \varrho\,\mathrm{Id}\,\big)^{-1}\Big\| \leq \varrho_+\,\tilde{e}_B.
\end{aligned} \tag{SD.23}$$

(ii) The expected row and column energies satisfy

$$\sup_{\varrho\in I}\max_{1\leq i\leq N_{\mathrm{out}}}\Big|\,\mathbb{E}\,\|\mathsf{S}_t(\varrho)^\top u_i\|^2 - \mathfrak{v}_i^{(0)}(\varrho)\Big| \leq \varrho_+\,e_B, \qquad \sup_{\varrho\in I}\max_{1\leq j\leq N_{\mathrm{in}}}\Big|\,\mathbb{E}\,\|\mathsf{S}_t(\varrho)v_j\|^2 - \widetilde{\mathfrak{v}}_j^{(0)}(\varrho)\Big| \leq \varrho_+\,\tilde{e}_B, \tag{SD.24}$$

$\mathsf{S}_t(\varrho)^\top u_i = 0$ *for every i with $\mu_i = 0$, and* $\mathsf{S}_t(\varrho)v_j = 0$ *for every j with $\lambda_j = 0$.*

(iii) If $\Sigma_{\rm in} = \mathrm{Id}_{N_{\rm in}}$*, the raw and centered totals eq.* (SD.19) *satisfy*

$$\mathbb{E}\sup_{\varrho\in I}|Z(\varrho)-\overline{Z}(\varrho)|^2 \longrightarrow 0, \qquad \mathbb{E}\sup_{\varrho\in I}|Z^{\rm c}(\varrho)-\overline{Z}(\varrho)|^2 \longrightarrow 0, \tag{SD.25}$$

and $\sup_{\varrho\in I}|\,\mathbb{E}\,Z(\varrho)-\overline{Z}(\varrho)| \le \gamma_\star C_\Sigma \varrho_+ e_B$.

(iv) If $\Sigma_{\rm out} = \mathrm{Id}_{N_{\rm out}}$*, (iii) holds with the candidate* $\overline{Z}$ *replaced by* $B^{-1}\operatorname{Tr}\Sigma_{\rm in} - \varrho\,\tilde\sigma(-\varrho)$ *and* e_B *by* $\tilde e_B$.

The row and column bounds eq. (SD.24) control the maximum over the rows and columns of the expected errors. They give no relative accuracy for rows or columns whose coefficient is small, and no statement about the maximum of the random energies. With two anisotropic covariances the covariance-weighted total is not a function of one resolvent; it is identified in corollary SD.5. When $\Sigma_{\rm in} = \mathrm{Id}$, by (iii), the quadratic term of the frozen one-step identity eq. (SC.15) is

$$\frac{\eta^2}{2}\,\mathbb{E}\,\|Q_\varrho(M_t)\|_\Sigma^2 = \frac{\eta^2 B}{2}\Big(\overline{Z}(\varrho)+r_B(\varrho)\Big), \qquad \sup_{\varrho\in I}|r_B(\varrho)| \le \gamma_\star C_\Sigma\varrho_+ e_B. \tag{SD.26}$$

Proof. The proof rests on two exact identities. For $K\in\mathbb{R}^{N_{\rm out}\times N_{\rm in}}$ and $\varrho>0$,

$$Q_\varrho(K)Q_\varrho(K)^\top = KK^\top\big(KK^\top+\varrho\,\mathrm{Id}\,\big)^{-1} = \mathrm{Id}-\varrho\,R_\varrho(K), \qquad Q_\varrho(K)^\top Q_\varrho(K) = \mathrm{Id}-\varrho\,\widetilde{R}_\varrho(K). \tag{SD.27}$$

Indeed, in a singular value decomposition the first left side has the eigenvalue $s_k^2/(s_k^2+\varrho) = 1-\varrho/(s_k^2+\varrho)$ on the left singular vector p_k and vanishes on the left null space of K, where the right side is $1-\varrho/\varrho = 0$. The second identity is the same computation on the right singular vectors.

Part (i) is the expectation of eq. (SD.27). With $K = M_t$ and $R_\varrho(M_t) = R(-\varrho)$,

$$\mathbb{E}\left[\mathsf{S}_t(\varrho)\mathsf{S}_t(\varrho)^\top\right] - \big(\mathrm{Id}-\varrho\overline{R}(-\varrho)\big) = -\varrho\big(\,\mathbb{E}\,R(-\varrho)-\overline{R}(-\varrho)\big),$$

whose norm is at most $\varrho_+ e_B$, and $\mathrm{Id}-\varrho\overline{R}(-\varrho) = \tilde s_1\Sigma_{\rm out}(\tilde s_1\Sigma_{\rm out}+\varrho\,\mathrm{Id})^{-1}$ by eq. (SB.3). The second bound is the same argument with $\widetilde{R}(-\varrho)$, $\overline{\widetilde{R}}(-\varrho) = (\varrho+s_1\Sigma_{\rm in})^{-1}$ and $\tilde e_B$.

Part (ii) tests (i) with unit vectors. Since $\|\mathsf{S}_t^\top u_i\|^2 = u_i^\top\mathsf{S}_t\mathsf{S}_t^\top u_i$ and $u_i^\top(\mathrm{Id}-\varrho\overline{R}(-\varrho))u_i = 1-\varrho/(\varrho+\tilde s_1\mu_i) = \mathfrak{v}_i^{(0)}(\varrho)$, the bound $|u^\top X u|\le\|X\|$ for unit u gives the row bound. The column bound is the same with $\|\mathsf{S}_t v_j\|^2 = v_j^\top\mathsf{S}_t^\top\mathsf{S}_t v_j$ and $v_j^\top(\mathrm{Id}-\varrho\overline{\widetilde{R}}(-\varrho))v_j = \widetilde{\mathfrak{v}}_j^{(0)}(\varrho)$. If $\mu_i = 0$ then ${\Sigma_{\rm out}}^{1/2}u_i = 0$, so $u_i^\top G_{t'} = 0$ for every batch and $u_i^\top M_t = 0$; then lemma SA.2(ii), with the projection $\mathrm{Id}-u_i\otimes u_i$ on the output side, gives $u_i^\top\mathsf{S}_t = 0$. If $\lambda_j = 0$ then ${\Sigma_{\rm in}}^{1/2}v_j = 0$, so $M_t v_j = 0$, and the projection $\mathrm{Id}-v_j\otimes v_j$ on the input side gives $\mathsf{S}_t v_j = 0$.

Part (iii) is the covariance-weighted trace of eq. (SD.27). Since $\Sigma_{\rm in} = \mathrm{Id}$, $\|\mathsf{S}\|_\Sigma^2 = \operatorname{Tr}(\Sigma_{\rm out}\mathsf{S}\mathsf{S}^\top)$, and therefore

$$Z(\varrho)-\overline{Z}(\varrho) = -\varrho\Big(\frac{1}{B}\operatorname{Tr}\big(\Sigma_{\rm out}R(-\varrho)\big)-\sigma(-\varrho)\Big).$$

The last claim of corollary SB.1 gives $\mathbb{E}\sup_{s\in I}|B^{-1}\operatorname{Tr}(\Sigma_{\rm out}R(-s))-\sigma(-s)|^2\to 0$, and the first statement of eq. (SD.25) follows with the factor ϱ_+^2. Taking expectations instead,

$$|\,\mathbb{E}\,Z(\varrho)-\overline{Z}(\varrho)| = \varrho\left|\frac{1}{B}\operatorname{Tr}\big(\Sigma_{\rm out}(\mathbb{E}\,R(-\varrho)-\overline{R}(-\varrho))\big)\right| \le \varrho_+\frac{N_{\rm out}}{B}C_\Sigma e_B \le \gamma_\star C_\Sigma\varrho_+ e_B.$$

The centered statement follows from the raw one and eq. (SD.21), by $(x+y)^2\le 2x^2+2y^2$.

Part (iv) is the same computation on the input side. Since $\Sigma_{\rm out} = \mathrm{Id}$, $\|\mathsf{S}\|_\Sigma^2 = \operatorname{Tr}(\Sigma_{\rm in}\mathsf{S}^\top\mathsf{S})$, so the second identity of eq. (SD.27) gives $Z(\varrho)-B^{-1}\operatorname{Tr}\Sigma_{\rm in}+\varrho\tilde\sigma(-\varrho) = -\varrho\big(B^{-1}\operatorname{Tr}(\Sigma_{\rm in}\widetilde{R}(-\varrho))-\tilde\sigma(-\varrho)\big)$, and the second statement of corollary SB.1 replaces the first. □

Remark SD.1 (The spectral form of the row coefficient)**.** *Let* ρ_i *be the spectral measure of* u_i *for* $M_tM_t^\top$*: with* $M_t = \sum_k s_k\,p_k\otimes r_k$ *and* P_0 *the projection on the left null space of* M_t*,*

$$\rho_i \overset{\rm def}{=} \sum_k (u_i^\top p_k)^2\,\delta_{s_k^2} + \|P_0u_i\|^2\,\delta_0,$$

a random probability measure on $[0,\infty)$ *with* $u_i^\top R(-\varrho)u_i = \int (x+\varrho)^{-1}\rho_i(dx)$. *Testing eq.* (SD.27) *with* u_i *gives, exactly,*

$$\|\mathsf{S}_t(\varrho)^\top u_i\|^2 = \int_{[0,\infty)} \frac{x}{x+\varrho}\,\rho_i(dx) = 1 - \varrho\, u_i^\top R(-\varrho)u_i.$$

This is the spectral formula $\mathfrak{v}_i = \int \varphi(x)^2 x\,\rho_i(dx)$ *used for the half-anisotropic kernels of section 4, for the regularized filter* $\varphi_\varrho(x) = (x+\varrho)^{-1/2}$, *for which* $\varphi_\varrho(x)^2 x = x/(x+\varrho)$; *theorem SD.1(ii) is the statement that its expectation is* $\mathfrak{v}_i^{(0)}(\varrho)$ *up to* $\varrho_+ e_B$. *As* $\varrho \downarrow 0$ *the integrand tends to* $\mathbf{1}_{(0,\infty)}(x)$, *which is the* SIGNSVD *form* $\rho_i((0,\infty))$ *of the main text; no statement of the appendix is made at* $\varrho = 0$. *The coefficient* $\mathfrak{v}_i^{(0)}$ *is the raw same-time coefficient; the integrated coefficient* $\mathfrak{v}_i$ *of the dynamics is identified in theorem SD.7, whose zero-lag term at equal ridges is again* $\mathfrak{v}_i^{(0)}$.

SD.4 The independent model

The temporal analysis compares the actual buffer with a model in which the residuals are replaced by independent Gaussian weights, and compares a fixed finite dimension with its replications. This subsection introduces both and proves the replacement.

Definition SD.2 (Independent product model)**.** *In the frozen-state model with samples indexed by time, let* $(g_{t'a})_{t'\in\mathbb{Z},\,a\le B}$ *be independent standard Gaussian scalars, independent of the features, and put*

$$\begin{aligned} G^{\mathrm{g}}_{t'} &\overset{\mathrm{def}}{=} \frac{1}{B}\sum_{a=1}^{B} g_{t'a}\,{\Sigma_{\mathrm{out}}}^{1/2}(z_{t'a}\otimes w_{t'a}){\Sigma_{\mathrm{in}}}^{1/2}, \\ M^{\mathrm{g}}_{t'} &\overset{\mathrm{def}}{=} \sum_{j\ge 0} w_j G^{\mathrm{g}}_{t'+1-j}, \qquad \mathsf{S}^{\mathrm{g}}_{t'}(\varrho) \overset{\mathrm{def}}{=} Q_\varrho(M^{\mathrm{g}}_{t'}). \end{aligned} \tag{SD.28}$$

The independent model involves no state. Its statements are made under the size conditions of Assumption S3: $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \le \gamma_\star B$, $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$, and $N_{\mathrm{in}}/B \to \gamma_{\mathrm{in}}$, $N_{\mathrm{out}}/B \to \gamma_{\mathrm{out}}$. Since the family $(g_{t'a})$ and its negative have the same law, and odd spectral functions are odd in the matrix, $\mathbb{E}\, f^\sharp(M^{\mathrm{g}}_{t'}) = 0$ for every odd bounded f. The samples of distinct times are independent with centered products, so that

$$\frac{1}{B}\,\mathbb{E}\,\|M^{\mathrm{g}}_{t'}\|_{\mathrm{F}}^2 = \frac{s_2}{B^2}\,\mathrm{Tr}\,\Sigma_{\mathrm{out}}\,\mathrm{Tr}\,\Sigma_{\mathrm{in}} \le C_\Sigma^2\gamma_\star^2. \tag{SD.29}$$

Definition SD.3 (Spectral replication)**.** *Fix a finite dimension* $(N_{\mathrm{out}}, N_{\mathrm{in}}, B, \Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}})$ *and an integer* $N \ge 1$. *The* N*-fold replication is* $(NN_{\mathrm{out}}, NN_{\mathrm{in}}, NB, \Sigma_{\mathrm{out}}\otimes \mathrm{Id}_N, \Sigma_{\mathrm{in}}\otimes \mathrm{Id}_N)$, *with the eigenbases* $(u_i\otimes e_k)$ *and* $(v_j\otimes e_k)$. *A matrix* F *that is diagonal in* (u_i), *respectively* G *diagonal in* (v_j), *is replicated as* $F\otimes \mathrm{Id}_N$, *respectively* $G\otimes \mathrm{Id}_N$. *When an actual model is needed along a replication, the state is* $\widehat{A}_t^{(N)} \overset{\mathrm{def}}{=} r_N^{-1/2}\sum_{k\le r_N} e_k\otimes e'_k$ *with* $r_N = \min(NN_{\mathrm{out}}, NN_{\mathrm{in}})$, *in the replicated eigenbases.*

The spectral measures eq. (SA.41) of a replication are those of the base, since each eigenvalue is repeated N times and normalized by NB. Hence the branch of definition SB.1 and every deterministic quantity of this section built from it is the same at every order N. The replications satisfy the size conditions with the same constants and with $NN_{\mathrm{out}}/(NB)$ and $NN_{\mathrm{in}}/(NB)$ constant; the state $\widehat{A}_t^{(N)}$ has unit Frobenius norm and operator norm $r_N^{-1/2} \to 0$, so the replications with this state satisfy Assumption S3.

SD.4.1 Replacement of the residuals

The actual buffer differs from the independent one through the dependence of each residual $d = z^\top \widehat{A}_t w$ on its own features. The replacement is made sample by sample, for two buffers at once, and with arbitrary bounded tests; this form is used again for the temporal covariances.

Lemma SD.5 (Uniform scalar comparison)**.** *Let* Θ *be a compact metric space and* $\psi : \Theta\times\mathbb{R}\to\mathbb{C}$ *continuous with* $|\psi(\vartheta,x)| \le C_\psi(1+|x|)^4$. *Under Assumption S3, with* $d = z^\top\widehat{A}_t w$ *and* $g\sim\mathcal{N}(0,1)$,

$$\sup_{\vartheta\in\Theta} |\,\mathbb{E}\,\psi(\vartheta,d) - \mathbb{E}\,\psi(\vartheta,g)| \longrightarrow 0.$$

Proof. The tails are uniformly small. By lemma SA.7(i), $\mathbb{E}\, d^8 \le 105^2$, so for $T \ge 1$ and both $X = d$ and $X = g$,

$$\mathbb{E}\left[(1+|X|)^4 \mathbf{1}_{|X|>T}\right] \le \frac{\mathbb{E}(1+|X|)^8}{(1+T)^4} \le \frac{2^7(1+105^2)}{(1+T)^4}.$$

Let χ_T be continuous, equal to 1 on $[-T, T]$, to 0 outside $[-T-1, T+1]$, with values in $[0,1]$. The function $\psi\chi_T$ is uniformly continuous on $\Theta \times [-T-1, T+1]$, so for $\epsilon_\circ > 0$ there is a finite set $\{\vartheta_k\}$ with $\sup_x |\psi\chi_T(\vartheta, x) - \psi\chi_T(\vartheta_k, x)| \le \epsilon_\circ$ for some k, for every ϑ. For each k, $x \mapsto \psi(\vartheta_k, x)\chi_T(x)$ is bounded and continuous, so its expectations converge by lemma SA.7(iii). Hence the limit superior of the supremum is at most $2\epsilon_\circ + 2C_\psi 2^7(1+105^2)(1+T)^{-4}$, for every $\epsilon_\circ > 0$ and $T \ge 1$. □

Lemma SD.6 (Replacement of the residuals)**.** *Under Assumption S3, let $\mathcal{J} \subset (0, \infty)$ be compact. There is a sequence $\varepsilon_B(\mathcal{J}) \to 0$, depending only on $\mathcal{J}$, C_Σ, $\gamma_\star$ and the sequence of states, such that for every lag $\ell \ge 0$*

$$\sup_{s,u \in \mathcal{J}} \sup_{\|F\|, \|G\| \le 1} |\,\mathbb{E}\, \mathcal{K}_{s,u}^{F,G}(M_t, M_{t+\ell}) - \mathbb{E}\, \mathcal{K}_{s,u}^{F,G}(M_t^{\mathrm{g}}, M_{t+\ell}^{\mathrm{g}})| \le 2\varepsilon_B(\mathcal{J}), \tag{SD.30}$$

the supremum running over all deterministic $F \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{out}}}$ and $G \in \mathbb{R}^{N_{\mathrm{in}} \times N_{\mathrm{in}}}$ of norm at most one.

Proof. Paired weights. By eq. (SA.33), $M_{t+\ell} = \beta^\ell M_t + \sum_{r=0}^{\ell-1} w_r G_{t+\ell+1-r}$. Writing $\mathsf{u}_{t'a} \overset{\mathrm{def}}{=} {\Sigma_{\mathrm{out}}}^{1/2} z_{t'a}/\sqrt{B}$ and $\mathsf{v}_{t'a} \overset{\mathrm{def}}{=} {\Sigma_{\mathrm{in}}}^{1/2} w_{t'a}/\sqrt{B}$ as in eq. (SB.26), the pair is

$$(M_t, M_{t+\ell}) = \sum_{t'} \sum_{a=1}^{B} \left(\varpi_{1,t'}, \varpi_{2,t'}\right) d_{t'a}\, \mathsf{u}_{t'a} \otimes \mathsf{v}_{t'a}, \tag{SD.31}$$

with $(\varpi_{1,t'}, \varpi_{2,t'}) = (w_{t+1-t'}, \beta^\ell w_{t+1-t'})$ for $t' \le t+1$, $(0, w_{t+\ell+1-t'})$ for $t+1 < t' \le t+\ell+1$, and $(0,0)$ otherwise. The same holds for the independent model with $g_{t'a}$ in place of $d_{t'a}$. The paired weights have total mass

$$\sum_{t'} \left(\varpi_{1,t'} + \varpi_{2,t'}\right) = (1+\beta^\ell) \sum_{j \ge 0} w_j + \sum_{r < \ell} w_r = 2. \tag{SD.32}$$

One sample: the exact increment. Fix deterministic K, H, a sample (z, w) with u, v as above and $\mathsf{N} \overset{\mathrm{def}}{=} \|\mathsf{u}\|^2 + \|\mathsf{v}\|^2$, weights $\varpi, \varpi' \in [0,1]$, and a scalar e, either $e = d = z^\top \widehat{A}_t w$ or $e = g$ independent of (z, w). For $s > 0$ let $P_s(K) \overset{\mathrm{def}}{=} (\mathcal{D}(K) - i\sqrt{s}\,\mathrm{Id})^{-1}$, where $\mathcal{D}$ is the symmetric dilation. Since $\mathcal{D}(K)^2 = \mathrm{diag}(KK^\top, K^\top K)$,

$$P_s(K) = \begin{pmatrix} i\sqrt{s}\, R_s(K) & \mathsf{X}_s(K) \\ \mathsf{X}_s(K)^\top & i\sqrt{s}\, \widetilde{R}_s(K) \end{pmatrix}, \qquad \|P_s(K)\| \le s^{-1/2},$$

and with $D_F \overset{\mathrm{def}}{=} \mathrm{diag}(F, 0)$, $D_G \overset{\mathrm{def}}{=} \mathrm{diag}(0, G)$, $\mathcal{K}_{s,u}^{F,G}(K, H) = B^{-1}\,\mathrm{Tr}[D_F P_s(K) D_G P_u(H)]$. The update is $\mathcal{D}(K + \varpi e\, \mathsf{u} \otimes \mathsf{v}) = \mathcal{D}(K) + \varpi e\, \mathsf{Z} J_2 \mathsf{Z}^\top$ with $\mathsf{Z} \overset{\mathrm{def}}{=} [(\mathsf{u}; 0)\ (0; \mathsf{v})]$, $\|\mathsf{Z}\|_{\mathrm{F}}^2 = \mathsf{N}$, and $J_2 \overset{\mathrm{def}}{=} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. With $W_s \overset{\mathrm{def}}{=} \mathsf{Z}^\top P_s(K) \mathsf{Z}$, the matrix $\mathrm{Id}_2 + \varpi e J_2 W_s$ has determinant $\Lambda > 0$ of lemma SB.4 with $\mathsf{h} = \varpi e$, because $W_s = \begin{pmatrix} i\sqrt{s}\,\mathsf{U} & \mathsf{C} \\ \mathsf{C} & i\sqrt{s}\,\mathsf{V} \end{pmatrix}$ in the notation eq. (SB.12). The Woodbury formula [26, Section 2.1.4] gives

$$P_s(K + \varpi e\, \mathsf{u} \otimes \mathsf{v}) = P_s - P_s \mathsf{Z}\, \mathcal{C}_s\, \mathsf{Z}^\top P_s, \qquad \mathcal{C}_s \overset{\mathrm{def}}{=} (\mathrm{Id}_2 + \varpi e J_2 W_s)^{-1} \varpi e J_2,$$

and the push-through identity gives $\mathcal{C}_s = \varpi e J_2 - \varpi^2 e^2 J_2\, \mathsf{Z}^\top P_s(K + \varpi e\, \mathsf{u} \otimes \mathsf{v}) \mathsf{Z}\, J_2$, hence

$$\|\mathcal{C}_s\| \le \varpi\left(|e| + e^2 \mathsf{N} s^{-1/2}\right). \tag{SD.33}$$

Let $\mathcal{C}'_u$ be the same matrix for (H, u, ϖ'), and write $P_s = P_s(K)$, $P'_u = P_u(H)$. Expanding the product of the two updated resolvents and using the cyclicity of the trace, the increment multiplied by B is

$$\Phi(e) \overset{\mathrm{def}}{=} -\,\mathrm{tr}\left[\mathcal{C}_s \mathfrak{A}\right] - \mathrm{tr}\left[\mathcal{C}'_u \mathfrak{B}\right] + \mathrm{tr}\left[\mathcal{C}_s \mathfrak{E}\, \mathcal{C}'_u \mathfrak{T}\right], \tag{SD.34}$$

with the 2×2 compressions $\mathfrak{A} = \mathsf{Z}^\top P_s D_G P'_u D_F P_s \mathsf{Z}$, $\mathfrak{B} = \mathsf{Z}^\top P'_u D_F P_s D_G P'_u \mathsf{Z}$, $\mathfrak{E} = \mathsf{Z}^\top P_s D_G P'_u \mathsf{Z}$ and $\mathfrak{T} = \mathsf{Z}^\top P'_u D_F P_s \mathsf{Z}$.

One sample: compression. For a deterministic square matrix $\mathfrak{M}$ of size $N_{\text{out}} + N_{\text{in}}$ with diagonal blocks $\mathfrak{M}_{oo}, \mathfrak{M}_{ii}$, put $\mathfrak{m}(\mathfrak{M}) \overset{\text{def}}{=} \operatorname{diag}(B^{-1} \operatorname{Tr} \Sigma_{\text{out}} \mathfrak{M}_{oo}, B^{-1} \operatorname{Tr} \Sigma_{\text{in}} \mathfrak{M}_{ii})$. The quadratic and bilinear Gaussian moments of lemma SB.6, applied to real and imaginary parts, give

$$\mathbb{E} \, \|\mathsf{Z}^\top \mathfrak{M} \mathsf{Z} - \mathfrak{m}(\mathfrak{M})\|_{\text{F}}^2 \leq \frac{6 \gamma_\star C_\Sigma^2 \|\mathfrak{M}\|^2}{B}. \tag{SD.35}$$

The six matrices P_s, P'_u and those in $\mathfrak{A}, \mathfrak{B}, \mathfrak{E}, \mathfrak{T}$ have norms bounded by $C_{\mathcal{J}}$ for $s, u \in \mathcal{J}$, uniformly in K, H, F, G. The barred single compression is $\mathfrak{m}(P_s) = i \operatorname{diag}(\chi_o, \chi_i)$ with $\chi_o = \sqrt{s} B^{-1} \operatorname{Tr} \Sigma_{\text{out}} R_s(K) \in [0, \gamma_\star C_\Sigma s^{-1/2}]$ and χ_i likewise, so

$$\overline{\mathcal{C}}_s \overset{\text{def}}{=} \big(\operatorname{Id}_2 + \varpi e J_2 \, \mathfrak{m}(P_s) \big)^{-1} \varpi e J_2 = \frac{1}{1 + \varpi^2 e^2 \chi_o \chi_i} \begin{pmatrix} -i\varpi^2 e^2 \chi_i & \varpi e \\ \varpi e & -i\varpi^2 e^2 \chi_o \end{pmatrix},$$

whose denominator is at least one. Let $\overline{\Phi}(e)$ be eq. (SD.34) with every compression replaced by its bar. Telescoping the products, with $\mathcal{C}_s - \overline{\mathcal{C}}_s = -\mathcal{C}_s (W_s - \mathfrak{m}(P_s)) \overline{\mathcal{C}}_s$, eq. (SD.33), $\|\mathsf{Z}^\top \mathfrak{M} \mathsf{Z}\| \leq \mathsf{N} \|\mathfrak{M}\|$, $\|\mathfrak{m}(\mathfrak{M})\| \leq \gamma_\star C_\Sigma \|\mathfrak{M}\|$ and $\varpi \varpi' \leq \varpi$,

$$|\Phi(e) - \overline{\Phi}(e)| \leq C_{\mathcal{J}} (\varpi + \varpi')(1 + |e|)^8 (1 + \mathsf{N})^4 \sum_{\mathfrak{M}} \|\mathsf{Z}^\top \mathfrak{M} \mathsf{Z} - \mathfrak{m}(\mathfrak{M})\|_{\text{F}}.$$

By the Cauchy–Schwarz inequality, eq. (SD.35) and the moment bounds of lemma SA.7(i) and lemma SB.6, $\mathbb{E} \, |\Phi(e) - \overline{\Phi}(e)| \leq C_{\mathcal{J}} (\varpi + \varpi') B^{-1/2}$, for $e = d$ and for $e = g$. No independence between d and the compressions is used.

One sample: the scalar. Put $m \overset{\text{def}}{=} \max(\varpi, \varpi')$. For $m > 0$, $\overline{\Phi}(e)/m$ is a continuous function $\psi(\vartheta, e)$ of e and of the parameter $\vartheta = (m, \varpi/m, \varpi'/m, \chi_o, \chi_i, \chi'_o, \chi'_i,$ entries of $\mathfrak{m}(\mathfrak{A}), \mathfrak{m}(\mathfrak{B}), \mathfrak{m}(\mathfrak{E}), \mathfrak{m}(\mathfrak{T}))$, which ranges in a compact set $\Theta_{\mathcal{J}}$ independent of the dimension, of K, H, F, G and of $s, u \in \mathcal{J}$. The explicit form of $\overline{\mathcal{C}}_s/m$ extends ψ continuously to $m = 0$, and $|\psi| \leq C_{\mathcal{J}} (1 + |e|)^4$. By lemma SD.5, $|\mathbb{E} \, \overline{\Phi}(d) - \mathbb{E} \, \overline{\Phi}(g)| \leq m \, r_B$ with $r_B \to 0$ uniform over $\Theta_{\mathcal{J}}$. Combining with the compression step,

$$\Big| \mathbb{E} \, \mathcal{K}_{s,u}^{F,G}(K + \varpi d \, \mathsf{u} \otimes \mathsf{v}, H + \varpi' d \, \mathsf{u} \otimes \mathsf{v}) - \mathbb{E} \, \mathcal{K}_{s,u}^{F,G}(K + \varpi g \, \mathsf{u} \otimes \mathsf{v}, H + \varpi' g \, \mathsf{u} \otimes \mathsf{v}) \Big| \leq \frac{(\varpi + \varpi') \varepsilon_B(\mathcal{J})}{B}, \tag{SD.36}$$

with $\varepsilon_B(\mathcal{J}) \overset{\text{def}}{=} 2 C_{\mathcal{J}} B^{-1/2} + r_B$, uniformly in K, H, $\|F\|, \|G\| \leq 1$ and $s, u \in \mathcal{J}$.

All samples. Truncate the history at age J, so that finitely many samples enter the pair, and replace the residuals by the independent Gaussians one sample at a time. At each step the other samples are independent of the replaced one and form the minors (K, H), so eq. (SD.36) applies with the paired weights of the sample. Summing over the samples, B per time, with eq. (SD.32), the total error is at most $2\varepsilon_B(\mathcal{J})$, uniformly in J. At fixed dimension the truncated pairs converge to the full pairs in L^2 by lemma SA.6(i) (and the same computation for the independent model), so eq. (SD.9) passes the bound to the full histories. □

Corollary SD.1 (Replacement for the filtered updates)**.** *Under Assumption S3 there is a sequence $\varepsilon'_B \to 0$, independent of the lag, such that for every $\ell \geq 0$*

$$\sup_{\varrho, \varrho' \in I} \sup_{\|F\|, \|G\| \leq 1} |\mathbb{E} \, \mathcal{K}^{F,G}[\omega_\varrho, \omega_{\varrho'}](M_t, M_{t+\ell}) - \mathbb{E} \, \mathcal{K}^{F,G}[\omega_\varrho, \omega_{\varrho'}](M_t^{\text{g}}, M_{t+\ell}^{\text{g}})| \leq \varepsilon'_B. \tag{SD.37}$$

Consequently, for symmetric F, G with $\|F\|, \|G\| \leq 1$, the tested lag covariances of definition SD.1 and their independent counterparts $C_\ell^{\text{g}}[F, G](\varrho, \varrho') \overset{\text{def}}{=} B^{-1} \mathbb{E} \operatorname{Tr}[F \, \mathsf{S}_t^{\text{g}}(\varrho) \, G \, \mathsf{S}_{t+\ell}^{\text{g}}(\varrho')^\top]$ satisfy

$$\sup_{\varrho, \varrho' \in I} |C_\ell[F, G](\varrho, \varrho') - C_\ell^{\text{g}}[F, G](\varrho, \varrho')| \leq \varepsilon'_B + \frac{(C_\Sigma \varrho_-^{-1/2} + \varepsilon_B)^2}{B}. \tag{SD.38}$$

Proof. Fix $T \geq 1$ and put $\mathcal{J}_T \overset{\text{def}}{=} [\varrho_-, \varrho_+ + T^2]$. By lemma SD.3, applied to both models with $C'_M \overset{\text{def}}{=} \max(C_M, C_\Sigma^2 \gamma_\star^2)$ by eq. (SD.29), and lemma SD.6 on the square $[0, T]^2$, the left side of

eq. (SD.37) is at most

$$\frac{4T^2}{\pi^2}\,2\varepsilon_B(\mathcal{J}_T)+\frac{8\sqrt{\gamma_\star C'_M}}{\pi T}.$$

Let ε'_B be the infimum of this expression over $T\geq 1$; its limit superior is at most $8\sqrt{\gamma_\star C'_M}/(\pi T)$ for every T, hence zero. The second statement adds the centering of lemma SD.4; the independent updates have mean zero. □

SD.5 Temporal structure: the lag reduction

Over a block of T consecutive steps, the centered updates accumulate as the sum $\sum_{t'=1}^{T}\mathsf{S}^{\mathrm{c}}_{t'}(\varrho)$. Its tested second moment is an exact triangular sum of the lag covariances of definition SD.1. Put

$$\mathsf{K}_\star \stackrel{\mathrm{def}}{=} C_\Sigma\sqrt{\frac{2\gamma_\star(\gamma_\star^2+4\gamma_\star+3)}{\varrho_-}}. \tag{SD.39}$$

Theorem SD.2 (Lag reduction)**.** *Let F,G be symmetric with $\|F\|,\|G\|\leq 1$, let $I=[\varrho_-,\varrho_+]$ with $\varrho_->0$, and let $\varrho,\varrho'\in I$. Parts (i)–(iii) hold at every finite dimension with $\max(N_{\mathrm{in}},N_{\mathrm{out}})\leq\gamma_\star B$ and $\|\Sigma_{\mathrm{out}}\|,\|\Sigma_{\mathrm{in}}\|\leq C_\Sigma$, and at every normalized state; part (iv) holds under Assumption S3.*

(i) For every integer $T\geq 1$, with $S_T(\varrho)\stackrel{\mathrm{def}}{=}\sum_{t'=1}^{T}\mathsf{S}^{\mathrm{c}}_{t'}(\varrho)$,

$$\begin{aligned}&\frac{1}{BT}\,\mathbb{E}\,\mathrm{Tr}\left[F\,S_T(\varrho)\,G\,S_T(\varrho')^\top\right]\\&\qquad=C_0[F,G](\varrho,\varrho')+\sum_{\ell=1}^{T-1}\Big(1-\frac{\ell}{T}\Big)\Big(C_\ell[F,G](\varrho,\varrho')+C_\ell[F,G](\varrho',\varrho)\Big).\end{aligned} \tag{SD.40}$$

(ii) $|C_0[F,G](\varrho,\varrho')|\leq\gamma_\star$ and $|C_\ell[F,G](\varrho,\varrho')|\leq\mathsf{K}_\star\beta^\ell$ for $\ell\geq 1$; at $\beta=0$, $C_\ell[F,G]=0$ for every $\ell\geq 1$. The same bounds hold for the independent covariances $C^{\mathrm{g}}_\ell[F,G]$ of corollary SD.1, under the size conditions.

(iii) The series eq. (SD.17) converges absolutely, and the left side of eq. (SD.40) differs from $\Gamma[F,G](\varrho,\varrho')$ by at most

$$2\mathsf{K}_\star\Big(\frac{\beta^T}{1-\beta}+\frac{\beta}{T(1-\beta)^2}\Big). \tag{SD.41}$$

(iv) For every fixed $\ell\geq 0$, $\sup_{\varrho,\varrho'\in I}\sup_{F,G}|C_\ell[F,G]-C^{\mathrm{g}}_\ell[F,G]|\to 0$, and $\sup_{\varrho,\varrho'\in I}\sup_{F,G}|\Gamma[F,G]-\Gamma^{\mathrm{g}}[F,G]|\to 0$, where Γ^{g} is the integrated sum of the C^{g}_ℓ and the suprema run over symmetric F,G of norm at most one.

For the covariance-weighted coefficients $C_\ell=C_\ell[\Sigma_{\mathrm{out}},\Sigma_{\mathrm{in}}]$ the bounds of (ii) and (iii) carry the factor C_Σ^2. At $\beta=0$ the integrated covariance is the same-time centered covariance, $\Gamma[F,G]=C_0[F,G]$.

The lag sums in (iv), and again in appendices SD.6 and SD.8, are controlled by one elementary lemma.

Lemma SD.7 (Lag summation)**.** *For each dimension let $(a_\ell(p))_{\ell\geq0}$ and $(b_\ell(p))_{\ell\geq0}$ be real sequences indexed by a parameter p in a set $\mathcal{P}$, with $|a_\ell(p)|+|b_\ell(p)|\leq K\beta^\ell$ for $\ell\geq 1$, uniformly in p and in the dimension, and with $\sup_{p\in\mathcal{P}}|a_\ell(p)-b_\ell(p)|\to 0$ for every fixed $\ell\geq 0$. Then $\sup_{p\in\mathcal{P}}|\sum_{\ell\geq0}(a_\ell(p)-b_\ell(p))|\to 0$.*

Proof. For every $L\geq 0$,

$$\sup_p\Big|\sum_{\ell\geq0}(a_\ell-b_\ell)\Big|\leq\sum_{\ell\leq L}\sup_p|a_\ell-b_\ell|+\frac{K\beta^{L+1}}{1-\beta},$$

and the limit superior of the right side in the dimension is the last term, which tends to zero as $L\to\infty$ (and vanishes for $L\geq 0$ when $\beta=0$). □

Proof of theorem SD.2. Part (i) expands the double sum. For symmetric F, G the trace pairing is symmetric, $\mathrm{Tr}[FXGY^\top] = \mathrm{Tr}[(FXGY^\top)^\top] = \mathrm{Tr}[FYGX^\top]$. By the stationarity of lemma SA.5(ii), the term with times $t' < t''$ equals $B\, C_{t''-t'}[F,G](\varrho,\varrho')$, and by the symmetry the term with $t' > t''$ equals $B\, C_{t'-t''}[F,G](\varrho',\varrho)$. There are $T-\ell$ pairs at lag ℓ, which gives eq. (SD.40).

Part (ii) for $\ell = 0$ is the Cauchy–Schwarz inequality for the Frobenius pairing together with $\mathbb{E}\,\|\mathsf{S}_t^{\mathrm{c}}\|_{\mathrm{F}}^2 \le \mathbb{E}\,\|\mathsf{S}_t\|_{\mathrm{F}}^2 \le \gamma_\star B$, by lemma SA.2(i). For $\ell \ge 1$ we couple with an independent past. Let M_t' be built as M_t from independent batches of the times $t' \le t+1$, independent of everything else, and put

$$M^* \stackrel{\mathrm{def}}{=} \beta^\ell M_t' + \sum_{r=0}^{\ell-1} w_r G_{t+\ell+1-r}.$$

By eq. (SA.33), M^* has the law of $M_{t+\ell}$, it is independent of M_t, and $M_{t+\ell} - M^* = \beta^\ell(M_t - M_t')$. Since $\mathsf{S}_t^{\mathrm{c}}$ is centered and independent of $Q_{\varrho'}(M^*)$,

$$C_\ell[F,G](\varrho,\varrho') = \frac{1}{B}\,\mathbb{E}\,\mathrm{Tr}\Big[F\,\mathsf{S}_t^{\mathrm{c}}(\varrho)\,G\,\big(Q_{\varrho'}(M_{t+\ell}) - Q_{\varrho'}(M^*)\big)^\top\Big].$$

The Cauchy–Schwarz inequality, eq. (SA.13) and $\mathbb{E}\,\|M_t - M_t'\|_{\mathrm{F}}^2 = 2s_2\mathsf{V}_B$ (lemma SA.5(iii)) give

$$|C_\ell[F,G](\varrho,\varrho')| \le \frac{1}{B}\sqrt{\gamma_\star B}\;\varrho_-^{-1/2}\beta^\ell\sqrt{2s_2\mathsf{V}_B} \le \sqrt{\gamma_\star}\;\varrho_-^{-1/2}\beta^\ell\Big(\frac{2\mathsf{V}_B}{B}\Big)^{1/2} \le \mathsf{K}_\star\beta^\ell,$$

using $\mathsf{V}_B/B \le C_\Sigma^2(\gamma_\star^2 + 4\gamma_\star + 3)$ from lemma SA.6(iv) and $B \ge 1$. At $\beta = 0$ the buffers M_t and $M_{t+\ell}$ are built from disjoint batches, hence independent, and the covariance vanishes. For the independent model, the bound at $\ell = 0$ is the same Cauchy–Schwarz inequality, since $\|\mathsf{S}_t^{\mathrm{g}}(\varrho)\|_{\mathrm{F}}^2 \le \min(N_{\mathrm{out}}, N_{\mathrm{in}}) \le \gamma_\star B$ pointwise by lemma SA.2(i). For $\ell \ge 1$ the same coupling applies, with $\mathbb{E}\,\|M_t^{\mathrm{g}} - M_t^{\mathrm{g}\prime}\|_{\mathrm{F}}^2 = 2s_2 B^{-1}\,\mathrm{Tr}\,\Sigma_{\mathrm{out}}\,\mathrm{Tr}\,\Sigma_{\mathrm{in}} \le 2C_\Sigma^2\gamma_\star^2 B$ in place of $2s_2\mathsf{V}_B$.

Part (iii) follows from (ii). The series converges absolutely, and the difference between eq. (SD.40) and eq. (SD.17) consists of the lags $\ell \ge T$ and of the weights ℓ/T for $\ell < T$, so it is at most

$$\sum_{\ell \ge T} 2\mathsf{K}_\star\beta^\ell + \frac{1}{T}\sum_{\ell \ge 1} 2\mathsf{K}_\star\,\ell\beta^\ell = 2\mathsf{K}_\star\Big(\frac{\beta^T}{1-\beta} + \frac{\beta}{T(1-\beta)^2}\Big).$$

Part (iv) for a fixed lag is eq. (SD.38), whose bound does not depend on ℓ. The integrated statement is lemma SD.7 with $p = (\varrho, \varrho', F, G)$, applied to the two orders of the ridges, with the envelopes of (ii). □

SD.6 The tree-moment response

The lag covariances of the independent model are now identified by deterministic quantities. The construction works with finitely many batches and bounded weights, where every moment of the pair of buffers is computed by Gaussian pairings, and then removes both restrictions.

The cutoff model. Fix a lag $\ell \ge 0$, a history cutoff $J \ge 0$ and a scalar cutoff $R > 0$. The batch labels are $\mathcal{A} \stackrel{\mathrm{def}}{=} \{-j : 0 \le j \le J\} \cup \{m : 1 \le m \le \ell\}$: the label $-j$ is the batch of age j at time t, and the label m is the batch of time $t+1+m$. By eq. (SD.31), the paired coefficients are

$$(\varpi_{1,-j}, \varpi_{2,-j}) \stackrel{\mathrm{def}}{=} (w_j, \beta^\ell w_j), \qquad (\varpi_{1,m}, \varpi_{2,m}) \stackrel{\mathrm{def}}{=} (0, w_{\ell-m}), \tag{SD.42}$$

and $\sum_{\alpha\in\mathcal{A}} \varpi_{l,\alpha} \le 1$ for $l = 1, 2$. The truncated Gaussian moments are

$$m_d(R) \stackrel{\mathrm{def}}{=} \mathbb{E}\,[g^d \mathbf{1}_{|g|\le R}] \quad (d \ge 1), \qquad m_0(R) \stackrel{\mathrm{def}}{=} 1, \tag{SD.43}$$

which vanish for odd d. The cutoffs are measured by

$$\tau_R \stackrel{\mathrm{def}}{=} \mathbb{E}\,[g^2\mathbf{1}_{|g|>R}], \qquad \delta_{J,R} \stackrel{\mathrm{def}}{=} \beta^{J+1} + \tau_R^{1/2}, \tag{SD.44}$$

with $\beta^{J+1} = 0$ when $\beta = 0$. For explanation only, the cutoff matrices are

$$\mathsf{K}_l \stackrel{\mathrm{def}}{=} \frac{1}{B}\sum_{\alpha\in\mathcal{A}}\sum_{a=1}^{B} \varpi_{l,\alpha}\, g_{\alpha a}\mathbf{1}_{|g_{\alpha a}|\le R}\,\Sigma_{\mathrm{out}}{}^{1/2}(z_{\alpha a}\otimes w_{\alpha a})\Sigma_{\mathrm{in}}{}^{1/2}, \qquad l = 1, 2, \tag{SD.45}$$

with independent standard Gaussian $g_{\alpha a}$, $z_{\alpha a}$, $w_{\alpha a}$; they are the independent pair $(M_t^{\mathrm{g}}, M_{t+\ell}^{\mathrm{g}})$ with the ages larger than J removed and the weights truncated at R. The definition below contains no expectation.

The tests are diagonal in the eigenbases: $F = \sum_i F_i\, u_i \otimes u_i$ and $G = \sum_j G_j\, v_j \otimes v_j$. For integers $a, b \geq 0$ the word

$$\mathsf{W}_{a,b} \stackrel{\mathrm{def}}{=} F\,(\mathsf{K}_1\mathsf{K}_1^\top)^a \mathsf{K}_1\, G\, \mathsf{K}_2^\top (\mathsf{K}_2\mathsf{K}_2^\top)^b \tag{SD.46}$$

is a cyclic product of $k \stackrel{\mathrm{def}}{=} 2a+2b+2$ factors $X_1, \ldots, X_k$: $X_r = \mathsf{K}_1$ for odd $r \leq 2a+1$, $X_r = \mathsf{K}_1^\top$ for even $r \leq 2a$, $X_r = \mathsf{K}_2^\top$ for even $r \geq 2a+2$, and $X_r = \mathsf{K}_2$ for odd $r \geq 2a+3$. The factor X_r has the label $l_r = 1$ if $r \leq 2a+1$ and $l_r = 2$ otherwise. For odd polynomials $f_1(x) = \sum_a c_a x^{2a+1}$ and $f_2(x) = \sum_b c'_b x^{2b+1}$,

$$\frac{1}{B}\operatorname{Tr}\left[F\, f_1^\sharp(\mathsf{K}_1)\, G\, f_2^\sharp(\mathsf{K}_2)^\top\right] = \sum_{a,b} c_a c'_b\, \frac{1}{B}\operatorname{Tr}\mathsf{W}_{a,b},$$

by eq. (SD.1) and $(x^{2b+1})^\sharp(K)^\top = K^\top (KK^\top)^b$.

Definition SD.4 (Tree moment equations)**.** *Fix (ℓ, J, R), diagonal F, G, and integers $a, b \geq 0$, with $k = 2a+2b+2$ and the factors X_r and labels l_r of eq.* (SD.46)*.*

(1) The polygon. *Take k external vertices $\mathsf{e}_1, \ldots, \mathsf{e}_k$ and k latent vertices $\mathsf{l}_1, \ldots, \mathsf{l}_k$, indices modulo k. The vertex e_r sits between X_r and X_{r+1}; it has the input type if X_r is untransposed and the output type otherwise. The output vertex e_k carries the mark F and the input vertex e_{2a+1} carries the mark G. Each factor X_r contributes a z-edge joining l_r to the output vertex among $\mathsf{e}_{r-1}, \mathsf{e}_r$, and a w-edge joining l_r to the input vertex among them.*

(2) Pairings. *A pairing is a pair (π, π') of a perfect matching π of the k z-edges and a perfect matching π' of the k w-edges. For each matched pair of edges, identify their two latent endpoints and their two external endpoints. The quotient multigraph has one edge for each matched pair, hence k edges, and it is connected. The pairing is* retained *if the quotient is a tree, that is, if it has exactly $k+1$ vertices.*

(3) Values. *For a retained pairing, every vertex class V of the quotient contributes a factor. An output class V containing p_V original external vertices contributes $B^{-1}\sum_i \mu_i^{p_V} F_i^{m_V}$, where $m_V = 1$ if $\mathsf{e}_k \in V$ and $m_V = 0$ otherwise. An input class contributes $B^{-1}\sum_j \lambda_j^{p_V} G_j^{n_V}$, with $n_V = 1$ if $\mathsf{e}_{2a+1} \in V$ and $n_V = 0$ otherwise. A latent class V with set of factors $S_V \subset \{1, \ldots, k\}$ contributes*

$$m_{|S_V|}(R) \sum_{\alpha \in \mathcal{A}} \prod_{r \in S_V} \varpi_{l_r,\alpha}.$$

(4) The response. *$\mathfrak{R}_\ell^{J,R}[F,G](x^{2a+1}, x^{2b+1})$ is the sum, over the retained pairings, of the products of the factors of their classes. It is extended bilinearly to pairs of odd polynomials.*

The definition is a finite computation for each pair (a, b): enumerate the $((k-1)!!)^2$ pairings, form the quotient, keep it if it has $k+1$ vertices, and multiply the factors. Its only inputs are the normalized traces $B^{-1}\sum_i \mu_i^p F_i^m$, $B^{-1}\sum_j \lambda_j^p G_j^n$, the coefficients eq. (SD.42) and the moments eq. (SD.43). Each original external vertex has exponent one because the two factors adjacent to it contribute one covariance square root each. The response is linear in F and in G, since each mark lies in exactly one class. For the shortest word, $a = b = 0$, there is one pairing; it identifies l_1 with l_2, the quotient has the three vertices $\{\mathsf{e}_1\}, \{\mathsf{e}_2\}, \{\mathsf{l}_1, \mathsf{l}_2\}$, and

$$\mathfrak{R}_\ell^{J,R}[F,G](x,x) = \frac{\operatorname{Tr}(\Sigma_{\mathrm{out}}F)}{B}\,\frac{\operatorname{Tr}(\Sigma_{\mathrm{in}}G)}{B}\, m_2(R)\, \beta^\ell \sum_{j \leq J} w_j^2,$$

which tends, as $J, R \to \infty$, to $\beta^\ell s_2 B^{-2}\operatorname{Tr}(\Sigma_{\mathrm{out}}F)\operatorname{Tr}(\Sigma_{\mathrm{in}}G)$, the exact value of $B^{-1}\,\mathbb{E}\operatorname{Tr}[F\, M_t^{\mathrm{g}} G\, M_{t+\ell}^{\mathrm{g}\,\top}]$.

Lemma SD.8 (Moment closure)**.** *Fix (ℓ, J, R) and $a, b \geq 0$. There is C_{W}, depending only on $(a, b, \ell, J, R, C_\Sigma, \gamma_\star)$, such that for all diagonal F, G with $\|F\|, \|G\| \leq 1$,*

$$\left|\frac{1}{B}\,\mathbb{E}\operatorname{Tr}\mathsf{W}_{a,b} - \mathfrak{R}_\ell^{J,R}[F,G](x^{2a+1}, x^{2b+1})\right| \leq \frac{C_{\mathsf{W}}}{B}. \tag{SD.47}$$

The response is unchanged under the replications of definition SD.3.

Proof. Expansion. In the eigenbases the coordinates $z_{\alpha a}(i) \stackrel{\text{def}}{=} u_i^\top z_{\alpha a}$ and $w_{\alpha a}(j) \stackrel{\text{def}}{=} v_j^\top w_{\alpha a}$ are independent standard Gaussian variables indexed by (α, a, i) and (α, a, j), and with $\tilde{g}_{\alpha a} \stackrel{\text{def}}{=} g_{\alpha a} \mathbf{1}_{|g_{\alpha a}| \le R}$,

$$(\mathsf{K}_l)_{ij} = \frac{1}{B} \sum_{\alpha, a} \varpi_{l,\alpha}\, \tilde{g}_{\alpha a}\, \sqrt{\mu_i}\, z_{\alpha a}(i)\, w_{\alpha a}(j) \sqrt{\lambda_j}.$$

Expanding the trace, $B^{-1}\, \mathbb{E}\, \mathrm{Tr}\, \mathsf{W}_{a,b}$ is $B^{-(k+1)}$ times a sum over assignments of an external index (i or j) to each e_r and of a latent index (α_r, a_r) to each l_r. The assignment carries the deterministic weight $\prod_r \varpi_{l_r,\alpha_r} \prod_{\mathsf{e}_r \text{ output}} \mu_{i_r} F_{i_r}^{[r=k]} \prod_{\mathsf{e}_r \text{ input}} \lambda_{j_r} G_{j_r}^{[r=2a+1]}$, since each external vertex receives one square root from each adjacent factor, and the product of three expectations: of the k weights $\tilde{g}_{\alpha_r a_r}$, of the k coordinates of z on the z-edges, and of the k coordinates of w on the w-edges.

Pairings. By the Isserlis formula [29], the expectation of a product of standard Gaussian coordinates is the sum over perfect matchings of the product of the covariances, and the covariance of two coordinates is one if their labels coincide and zero otherwise. Hence the z-expectation is the sum over matchings π of the indicator that each matched pair of z-edges has equal latent and external indices, and likewise for w. Therefore

$$\frac{1}{B}\, \mathbb{E}\, \mathrm{Tr}\, \mathsf{W}_{a,b} = \sum_{(\pi, \pi')} S(\pi, \pi'),$$

where $S(\pi, \pi')$ is $B^{-(k+1)}$ times the sum of the weights over the assignments that are constant on the vertex classes of the quotient of (π, π'), distinct classes being allowed to receive equal indices.

Counting. Let the quotient have v_{e} external and v_{l} latent classes, $v = v_{\mathrm{e}} + v_{\mathrm{l}}$. Each external class ranges over at most $\gamma_\star B$ indices with weight at most C_Σ^k, each latent class over $|\mathcal{A}|B$ indices with $|\prod \varpi| \le 1$, and the $\tilde{g}$-expectation is at most R^k in absolute value. Hence

$$|S(\pi, \pi')| \le B^{-(k+1)} (\gamma_\star B C_\Sigma^k)^{v_{\mathrm{e}}} (|\mathcal{A}|B)^{v_{\mathrm{l}}} R^k \le C_{\mathsf{W}}\, B^{\,v-k-1}.$$

A connected multigraph with k edges has at most $k+1$ vertices: adding its edges one at a time, in an order in which each new edge meets the vertices already reached, each edge adds at most one vertex; and it has exactly $k+1$ vertices if and only if it is a tree. So the pairings that are not retained contribute at most C_{W}/B each.

Trees. Let (π, π') be retained, so $v = k+1$. Split its assignments into those that give distinct latent classes distinct pairs (α, a) and the others. The others have two latent classes with the same sample index, which removes one free factor B from the count, so they contribute at most C_{W}/B. On the first kind, the weights $\tilde{g}$ of distinct classes are independent, and the $|S_V|$ factors of one class carry the same variable, so

$$\mathbb{E} \prod_r \tilde{g}_{\alpha_r a_r} = \prod_{V \text{ latent}} m_{|S_V|}(R).$$

Adding back the assignments of the second kind with this factorized weight costs C_{W}/B again. The resulting sum factorizes over the classes: each external class gives $\sum_i \mu_i^{p_V} F_i^{m_V}$ or $\sum_j \lambda_j^{p_V} G_j^{m_V}$, and each latent class gives $B\, m_{|S_V|}(R) \sum_\alpha \prod_{r \in S_V} \varpi_{l_r,\alpha}$, the factor B from its sample index. With $k+1$ classes the powers of B cancel against $B^{-(k+1)}$, and the result is the value of definition SD.4. There are $((k-1)!!)^2$ pairings, which proves eq. (SD.47).

Replication. Every factor of the value is either a normalized trace $B^{-1} \sum_i \mu_i^p F_i^m$, which is unchanged when the eigenvalues and the entries of F are repeated N times and B is replaced by NB, or does not depend on the dimension. □

Lemma SD.9 (Norm control)**.** *Fix (ℓ, J, R). Let $Z_\alpha \stackrel{\text{def}}{=} [z_{\alpha a}]_{a \le B}$ and $W_\alpha \stackrel{\text{def}}{=} [w_{\alpha a}]_{a \le B}$, let C_0 be the absolute constant of [71, Theorem 4.4.5], and put $T_\star \stackrel{\text{def}}{=} C_0(\sqrt{\gamma_\star} + 2)$, $D_R \stackrel{\text{def}}{=} R C_\Sigma T_\star^2$ and $\mathcal{E} \stackrel{\text{def}}{=} \{\|Z_\alpha\|, \|W_\alpha\| \le T_\star \sqrt{B} \text{ for all } \alpha \in \mathcal{A}\}$. Under the size conditions, $\Pr(\mathcal{E}^c) \le 4|\mathcal{A}| e^{-B}$, on $\mathcal{E}$ both $\|\mathsf{K}_1\|$ and $\|\mathsf{K}_2\|$ are at most D_R, and for every $p \ge 1$, $\mathbb{E}[\|\mathsf{K}_l\|^p \mathbf{1}_{\mathcal{E}^c}] \to 0$.*

Proof. By [71, Theorem 4.4.5] with $t = \sqrt{B}$, an $N_{\text{out}} \times B$ standard Gaussian matrix has norm at most $C_0(\sqrt{N_{\text{out}}} + 2\sqrt{B}) \le T_\star\sqrt{B}$ with probability at least $1 - 2e^{-B}$, and likewise for $N_{\text{in}} \times B$; the union bound gives the probability. On $\mathcal{E}$,

$$\|\mathsf{K}_l\| \le \sum_\alpha \varpi_{l,\alpha} \frac{1}{B} \|{\Sigma_{\text{out}}}^{1/2}\| \|Z_\alpha\| \, R \, \|W_\alpha\| \|{\Sigma_{\text{in}}}^{1/2}\| \le R C_\Sigma T_\star^2,$$

since the coefficient masses are at most one. Finally $\|\mathsf{K}_l\| \le B^{-1} R C_\Sigma \sum_{\alpha,a} \|z_{\alpha a}\| \|w_{\alpha a}\|$, whose moments of every order are bounded by a polynomial in B, and the Cauchy–Schwarz inequality with $\Pr(\mathcal{E}^c) \le 4|\mathcal{A}|e^{-B}$ gives the last claim. □

Lemma SD.10 (Extension to continuous filters)**.** *Fix (ℓ, J, R) and diagonal F, G with $\|F\|, \|G\| \le 1$. Write $\|f\|_D \stackrel{\text{def}}{=} \sup_{|x| \le D} |f(x)|$.*

(i) For odd polynomials f_1, f_2, $|\mathfrak{R}_\ell^{J,R}[F, G](f_1, f_2)| \le \gamma_\star \|f_1\|_{D_R} \|f_2\|_{D_R}$.

(ii) The response extends uniquely to a bilinear functional on pairs of odd continuous functions on $[-D_R, D_R]$ with the same bound. The extension is linear in F and in G and unchanged under replication.

(iii) For odd bounded continuous f_1, f_2 on $\mathbb{R}$, under the size conditions,

$$\sup_{F,G} |\, \mathbb{E}\, \mathcal{K}^{F,G}[f_1, f_2](\mathsf{K}_1, \mathsf{K}_2) - \mathfrak{R}_\ell^{J,R}[F, G](f_1, f_2)| \longrightarrow 0,$$

the supremum over diagonal F, G of norm at most one; the convergence is uniform over $(f_1, f_2) = (\iota_s, \iota_u)$ with s, u in a compact subset of $(0, \infty)$, and over $(f_1, f_2) = (\omega_\varrho, \omega_{\varrho'})$ with $\varrho, \varrho' \in I$.

Proof. For (i), fix the finite dimension and consider its replications. By lemma SD.8 at order N, whose constant does not depend on N, and the replication invariance,

$$\mathfrak{R}_\ell^{J,R}[F, G](f_1, f_2) = \lim_{N \to \infty} \mathbb{E}\, \mathcal{K}^{F^{(N)}, G^{(N)}}[f_1, f_2]\big(\mathsf{K}_1^{(N)}, \mathsf{K}_2^{(N)}\big).$$

On the event $\mathcal{E}$ of lemma SD.9 the singular values of $\mathsf{K}_l^{(N)}$ are at most D_R, so $f_l^\sharp$ only sees f_l on $[-D_R, D_R]$, and eq. (SD.8) bounds the contraction by $\gamma_\star \|f_1\|_{D_R} \|f_2\|_{D_R}$; the same constant D_R serves for every N. Off $\mathcal{E}$ the contraction is bounded by $\gamma_\star$ times a polynomial in $\|\mathsf{K}_1^{(N)}\|, \|\mathsf{K}_2^{(N)}\|$, whose expectation on $\mathcal{E}^c$ tends to zero by lemma SD.9. This proves (i).

For (ii), the Weierstrass theorem [55, Theorem 7.26] gives polynomials P_n converging to an odd continuous f uniformly on $[-D_R, D_R]$, and the odd parts $(P_n(x) - P_n(-x))/2$ converge to f as well. By (i) and bilinearity, the responses of approximating pairs form a Cauchy sequence, and two approximating sequences give the same limit. Linearity in F, G and replication invariance hold for polynomials by definition SD.4 and lemma SD.8, and pass to the limit.

For (iii), let $\epsilon > 0$ and choose odd polynomials p_1, p_2 with $\|f_l - p_l\|_{D_R} \le \epsilon$. On $\mathcal{E}$, eq. (SD.10) (with the sup norms over $[-D_R, D_R]$) bounds $|\mathcal{K}^{F,G}[f_1, f_2] - \mathcal{K}^{F,G}[p_1, p_2]|$ by $\gamma_\star \epsilon(\|f_2\|_\infty + \|p_1\|_{D_R})$. Off $\mathcal{E}$, $|\mathcal{K}[f_1, f_2]| \le \gamma_\star \|f_1\|_\infty \|f_2\|_\infty$ and the polynomial contraction has vanishing expectation by lemma SD.9. The polynomial contraction is within C/B of $\mathfrak{R}_\ell^{J,R}[F, G](p_1, p_2)$ by lemma SD.8, and the latter is within $\gamma_\star \epsilon(\|f_2\|_{D_R} + \|p_1\|_{D_R})$ of $\mathfrak{R}_\ell^{J,R}[F, G](f_1, f_2)$ by (ii). All bounds are uniform over F, G, so the limit superior of the supremum is at most $C\epsilon$. Uniformity in the parameters follows from a finite mesh, since both sides are Lipschitz in the parameters, by eqs. (SD.6) and (SD.10) and (ii). □

Lemma SD.11 (Removal of the cutoffs in the model)**.** *Couple $(\mathsf{K}_1, \mathsf{K}_2)$ with $(M_t^{\text{g}}, M_{t+\ell}^{\text{g}})$ by building both from the same samples. Under the size conditions, with $\delta_{J,R}$ of eq. (SD.44),*

$$\frac{1}{B} \mathbb{E}\, \|M_t^{\text{g}} - \mathsf{K}_1\|_{\text{F}}^2 + \frac{1}{B} \mathbb{E}\, \|M_{t+\ell}^{\text{g}} - \mathsf{K}_2\|_{\text{F}}^2 \le 3 C_\Sigma^2 \gamma_\star^2 \delta_{J,R}^2, \tag{SD.48}$$

and for odd bounded Lipschitz f_1, f_2 and $\|F\|, \|G\| \le 1$,

$$\begin{aligned} &|\, \mathbb{E}\, \mathcal{K}^{F,G}[f_1, f_2](M_t^{\text{g}}, M_{t+\ell}^{\text{g}}) - \mathbb{E}\, \mathcal{K}^{F,G}[f_1, f_2](\mathsf{K}_1, \mathsf{K}_2)| \\ &\qquad \le \sqrt{3}\, C_\Sigma \gamma_\star^{3/2} \big(\operatorname{Lip}(f_1) \|f_2\|_\infty + \|f_1\|_\infty \operatorname{Lip}(f_2)\big)\, \delta_{J,R}. \end{aligned}$$

Proof. The differences are sums of independent centered sample terms. A sample term $B^{-1}{\Sigma_{\text{out}}}^{1/2}(z \otimes w){\Sigma_{\text{in}}}^{1/2}$ has squared Frobenius norm with expectation $B^{-2}\operatorname{Tr}\Sigma_{\text{out}}\operatorname{Tr}\Sigma_{\text{in}} \le C_\Sigma^2\gamma_\star^2$. In $M_t^{\text{g}} - \mathsf{K}_1$ the samples of age $j > J$ carry the weight $w_j g$ and those of age $j \le J$ carry $w_j g \mathbf{1}_{|g|>R}$, so

$$\frac{1}{B}\,\mathbb{E}\,\|M_t^{\text{g}} - \mathsf{K}_1\|_{\text{F}}^2 \le C_\Sigma^2\gamma_\star^2\Big(\sum_{j>J} w_j^2 + \sum_{j\le J} w_j^2\tau_R\Big) \le C_\Sigma^2\gamma_\star^2\delta_{J,R}^2.$$

In $M_{t+\ell}^{\text{g}} - \mathsf{K}_2$ the old samples carry β^ℓ times the same weights, and the fresh samples carry $w_{\ell-m} g\mathbf{1}_{|g|>R}$, whose squared sum is at most τ_R; the two groups are disjoint, which gives at most $2C_\Sigma^2\gamma_\star^2\delta_{J,R}^2$. The contraction bound is eq. (SD.9) and the Cauchy–Schwarz inequality. □

Theorem SD.3 (Deterministic joint moment response)**.** *Fix a finite dimension, a lag $\ell \ge 0$ and diagonal F, G. For odd bounded Lipschitz f_1, f_2, the limit*

$$\mathfrak{R}_\ell[F,G](f_1,f_2) \stackrel{\text{def}}{=} \lim_{J,R\to\infty} \mathfrak{R}_\ell^{J,R}[F,G](f_1,f_2), \tag{SD.49}$$

with J and R tending to infinity independently, exists, and

$$\begin{aligned}&|\mathfrak{R}_\ell[F,G](f_1,f_2) - \mathfrak{R}_\ell^{J,R}[F,G](f_1,f_2)|\\ &\quad\le 2\sqrt{3}\,C_\Sigma\gamma_\star^{3/2}\|F\|\|G\|\big(\operatorname{Lip}(f_1)\|f_2\|_\infty + \|f_1\|_\infty\operatorname{Lip}(f_2)\big)\delta_{J,R}.\end{aligned} \tag{SD.50}$$

The response $\mathfrak{R}_\ell$ is bilinear in (F,G) and in (f_1,f_2), unchanged under replication, and bounded by $\gamma_\star\|F\|\|G\|\|f_1\|_\infty\|f_2\|_\infty$. The lag kernels and the filtered lag coefficients are

$$\mathfrak{k}_\ell[F,G](s,u) \stackrel{\text{def}}{=} \mathfrak{R}_\ell[F,G](\iota_s,\iota_u), \qquad \overline{C}_\ell[F,G](\varrho,\varrho') \stackrel{\text{def}}{=} \mathfrak{R}_\ell[F,G](\omega_\varrho,\omega_{\varrho'}), \tag{SD.51}$$

and $\mathfrak{k}_\ell \stackrel{\text{def}}{=} \mathfrak{k}_\ell[\Sigma_{\text{out}},\Sigma_{\text{in}}]$, $\overline{C}_\ell \stackrel{\text{def}}{=} \overline{C}_\ell[\Sigma_{\text{out}},\Sigma_{\text{in}}]$. Along every sequence of dimensions satisfying the size conditions, for every compact $\mathcal{J} \subset (0,\infty)$,

$$\begin{aligned}\sup_{s,u\in\mathcal{J}}\sup_{F,G}|\,\mathbb{E}\,\mathcal{K}_{s,u}^{F,G}(M_t^{\text{g}},M_{t+\ell}^{\text{g}}) - \mathfrak{k}_\ell[F,G](s,u)| \to 0,\\ \sup_{\varrho,\varrho'\in I}\sup_{F,G}|C_\ell^{\text{g}}[F,G](\varrho,\varrho') - \overline{C}_\ell[F,G](\varrho,\varrho')| \to 0,\end{aligned} \tag{SD.52}$$

the suprema over diagonal F, G of norm at most one. Under Assumption S3 the same holds for the actual model, with $\mathbb{E}\,\mathcal{K}_{s,u}^{F,G}(M_t,M_{t+\ell})$ and $C_\ell[F,G]$ in place of the independent quantities.

Proof. Existence. Fix two pairs of cutoffs (J,R) and (J',R'), and consider the replications of the fixed dimension. By lemma SD.10(iii) along the replications and the replication invariance, each of the two responses is the limit of the expected contraction of its cutoff pair at order N. By lemma SD.11, whose constant does not depend on N, the two cutoff contractions are both within $c\,\delta$ of the contraction of the full independent pair, with $c = \sqrt{3}C_\Sigma\gamma_\star^{3/2}(\operatorname{Lip}(f_1)\|f_2\|_\infty + \|f_1\|_\infty\operatorname{Lip}(f_2))$ for $\|F\|, \|G\| \le 1$. Hence

$$|\mathfrak{R}_\ell^{J,R}[F,G](f_1,f_2) - \mathfrak{R}_\ell^{J',R'}[F,G](f_1,f_2)| \le c\,(\delta_{J,R} + \delta_{J',R'}),$$

the responses form a Cauchy family as $J, R \to \infty$, and the limit exists with eq. (SD.50) (for general norms by bilinearity). Bilinearity, replication invariance and the bound pass to the limit from lemma SD.10(i),(ii).

Closure. Along any sequence of dimensions, the error in eq. (SD.52) is at most the sum of three terms: the full-to-cutoff error $c\,\delta_{J,R}$ of lemma SD.11, the fixed-cutoff error of lemma SD.10(iii), and the cutoff-to-limit error eq. (SD.50). All are uniform over the tests and the parameters, by eq. (SD.5). Taking the limit superior in the dimension at fixed cutoffs, and then $J, R \to \infty$, proves eq. (SD.52); the means of the independent updates vanish, so $C_\ell^{\text{g}}[F,G] = \mathbb{E}\,\mathcal{K}^{F,G}[\omega_\varrho,\omega_{\varrho'}](M_t^{\text{g}},M_{t+\ell}^{\text{g}})$. The actual statements follow from lemma SD.6 and corollary SD.1. □

Remark SD.2 (Other filters)**.** *The proof of eq. (SD.52) uses of the filters only that they are odd, bounded and Lipschitz. For any fixed odd, bounded, Lipschitz f_1, f_2 it therefore gives, with the supremum over diagonal F, G of norm at most one,*

$$\sup_{F,G}|\,\mathbb{E}\,\mathcal{K}^{F,G}[f_1,f_2](M_t^{\text{g}},M_{t+\ell}^{\text{g}}) - \mathfrak{R}_\ell[F,G](f_1,f_2)| \longrightarrow 0,$$

and appendix SG uses this for a fixed filter and for compactly supported resolvent integrals.

Theorem SD.4 (Integrated identification). *Fix a finite dimension, diagonal F, G, and $\varrho, \varrho' \in I$.*

(i) For every $\ell \geq 0$ and $T \geq 1$,

$$\left|\overline{C}_\ell[F,G](\varrho,\varrho') - \frac{4}{\pi^2}\int_0^T\int_0^T \mathfrak{k}_\ell[F,G](\varrho+x^2,\varrho'+y^2)\,dx\,dy\right| \leq \frac{4C_\Sigma\gamma_\star^{3/2}\|F\|\|G\|}{\pi T}, \tag{SD.53}$$

so $\overline{C}_\ell[F,G]$ is the square-truncated double integral of the lag kernel along the negative axis, with an error uniform in the dimension, the lag and the ridges.

(ii) $|\overline{C}_0[F,G]| \leq \gamma_\star\|F\|\|G\|$ and $|\overline{C}_\ell[F,G]| \leq \mathsf{K}_\star\|F\|\|G\|\,\beta^\ell$ for $\ell \geq 1$; at $\beta = 0$, $\overline{C}_\ell[F,G] = 0$ for every $\ell \geq 1$.

(iii) The deterministic integrated coefficient

$$\overline{\Gamma}[F,G](\varrho,\varrho') \stackrel{\text{def}}{=} \overline{C}_0[F,G](\varrho,\varrho') + \sum_{\ell\geq 1}\Big(\overline{C}_\ell[F,G](\varrho,\varrho') + \overline{C}_\ell[F,G](\varrho',\varrho)\Big) \tag{SD.54}$$

converges absolutely, and $\overline{\Gamma}[F,G](\varrho,\varrho) \geq 0$ whenever F and G are positive semidefinite.

(iv) Under Assumption S3, $\sup_{\varrho,\varrho'\in I}\sup_{F,G}|\Gamma[F,G](\varrho,\varrho') - \overline{\Gamma}[F,G](\varrho,\varrho')| \to 0$, the supremum over diagonal F, G of norm at most one, and the finite-time contraction eq. (SD.40) differs from $\overline{\Gamma}[F,G]$ by $o(1)$ plus eq. (SD.41), along every joint sequence of dimensions and block lengths T.

(v) For the covariance-weighted coefficient $\overline{\Gamma} \stackrel{\text{def}}{=} \overline{\Gamma}[\Sigma_{\text{out}},\Sigma_{\text{in}}]$, $\sup_{I^2}|\Gamma - \overline{\Gamma}| \to 0$ under Assumption S3. When $\Sigma_{\text{in}} = \text{Id}$, $\overline{C}_0(\varrho,\varrho) = \overline{Z}(\varrho)$ of eq. (SD.22).

Proof. All deterministic statements are obtained along the replications of the fixed dimension, from the corresponding statements for the independent model at order N, through eq. (SD.52); the marks are replicated with the same norms, and $F^{(N)}, G^{(N)}$ are positive semidefinite when F, G are.

For (i), lemma SD.3 with eq. (SD.29) bounds the difference between $C_\ell^{\mathrm{g},(N)}[F,G]$ and the square integral of $\mathbb{E}\,\mathcal{K}^{F,G}(M_t^{\mathrm{g},(N)}, M_{t+\ell}^{\mathrm{g},(N)})$ by $4\sqrt{\gamma_\star C_\Sigma^2\gamma_\star^2}\|F\|\|G\|/(\pi T)$, uniformly in N. The first statement of eq. (SD.52) on $\mathcal{J}_T = [\varrho_-, \varrho_+ + T^2]$ passes the square integral to that of $\mathfrak{k}_\ell[F,G]$, and the second passes $C_\ell^{\mathrm{g},(N)}$ to $\overline{C}_\ell$.

For (ii), the bounds of theorem SD.2(ii) for the independent model hold at every order N and pass to the limit. At $\beta = 0$ and $\ell \geq 1$, the independent buffers M_t^{g} and $M_{t+\ell}^{\mathrm{g}}$ are independent with centered updates, so $C_\ell^{\mathrm{g}} = 0$ at every order.

For (iii), absolute convergence follows from (ii). For positive semidefinite F, G and every T, the left side of eq. (SD.40) for the independent model is nonnegative, since $\mathrm{Tr}[FSGS^\top] = \|F^{1/2}SG^{1/2}\|_{\mathrm{F}}^2$; by theorem SD.2(iii) the independent integrated coefficient $\Gamma^{\mathrm{g},(N)}[F,G](\varrho,\varrho)$ is nonnegative at every order. Lemma SD.7, with the fixed-lag convergence $C_\ell^{\mathrm{g},(N)} \to \overline{C}_\ell$ as $N \to \infty$ and the envelopes of (ii), gives $\Gamma^{\mathrm{g},(N)} \to \overline{\Gamma}$, which is therefore nonnegative.

For (iv), lemma SD.7 applies to the actual covariances and the deterministic ones, with $p = (\varrho, \varrho', F, G)$, by the actual clause of theorem SD.3 and the envelopes of theorem SD.2(ii) and of (ii). The finite-time statement adds theorem SD.2(iii).

For (v), the first claim is (iv) with $F = \Sigma_{\text{out}}/C_\Sigma$, $G = \Sigma_{\text{in}}/C_\Sigma$. When $\Sigma_{\text{in}} = \text{Id}$, along the replications with the states of definition SD.3, $C_0(\varrho,\varrho) = \mathbb{E}\,Z^{\mathrm{c}}(\varrho)$ tends both to $\overline{Z}(\varrho)$, by theorem SD.1(iii), and to $\overline{C}_0(\varrho,\varrho)$, by theorem SD.3; both are replication invariant. □

The lag kernels $\mathfrak{k}_\ell$ are defined by the infinite system of definition SD.4 and the two limits of lemma SD.10 and theorem SD.3. The next subsection evaluates these limits in closed form.

SD.7 The closed-form lag kernel

The tree response of definition SD.4 can be summed in closed form. This subsection proves that every lag kernel $\mathfrak{k}_\ell[F,G]$ of eq. (SD.51) is an explicit rational expression in the branch of theorem SB.1 at two spectral parameters and in two scalar functions of the momentum. The formula holds for every lag, every pair of diagonal tests and every pair of positive semidefinite covariances, which may be singular and of any shapes. With spectral projections as tests it makes the mode coefficient $\mathfrak{v}_{ij}$ of eq. (SD.80) an explicit series of nonnegative double integrals.

SD.7.1 Statements

Throughout this subsection we fix a finite dimension and constants $\gamma_\star, C_\Sigma$ with

$$\max(N_{\text{out}}, N_{\text{in}}) \le \gamma_\star B, \qquad \|\Sigma_{\text{out}}\|, \|\Sigma_{\text{in}}\| \le C_\Sigma,$$

as in the size conditions. The covariances are positive semidefinite, either may be singular, and no relation between N_{out} and N_{in} is assumed. The tests are diagonal in the eigenbases, $F = \sum_i F_i\, u_i \otimes u_i$ and $G = \sum_j G_j\, v_j \otimes v_j$. The weights are the EMA weights $w_j = (1-\beta)\beta^j$ with $\beta \in [0,1)$, so that $s_2 = \sum_j w_j^2 = (1-\beta)/(1+\beta) \le 1$.

Two-point quantities. For $s, u > 0$ let $(\sigma, \tilde\sigma, \tilde s_1, s_1, q)$ be the branch of theorem SB.1 at $-s$ and $(\sigma', \tilde\sigma', \tilde s_1', s_1', q')$ the branch at $-u$, for the EMA weights and the spectral measures of eq. (SA.41); unprimed quantities refer to s and to the earlier buffer, primed ones to u and to the later buffer. With $\overline{R}(-s) = (s\,\mathrm{Id} + \tilde s_1 \Sigma_{\text{out}})^{-1}$ and $\overline{\widetilde{R}}(-s) = (s\,\mathrm{Id} + s_1 \Sigma_{\text{in}})^{-1}$ of eq. (SB.3), and g a standard Gaussian scalar, put

$$\begin{gathered}
\mathcal{T}^{\text{out}}[F](s,u) \overset{\text{def}}{=} \frac{1}{B}\operatorname{Tr}\left(F\,\Sigma_{\text{out}}\overline{R}(-s)\overline{R}(-u)\right), \quad \mathcal{T}^{\text{in}}[G](s,u) \overset{\text{def}}{=} \frac{1}{B}\operatorname{Tr}\left(G\,\Sigma_{\text{in}}\overline{\widetilde{R}}(-s)\overline{\widetilde{R}}(-u)\right), \\
\mathsf{U} \overset{\text{def}}{=} \mathcal{T}^{\text{out}}[\Sigma_{\text{out}}](s,u), \qquad \mathsf{V} \overset{\text{def}}{=} \mathcal{T}^{\text{in}}[\Sigma_{\text{in}}](s,u), \\
b_\ell \overset{\text{def}}{=} \sum_{j\ge 0} \mathbb{E}\,\frac{\beta^\ell w_j^2 g^2}{(1+qw_j^2g^2)(1+\beta^{2\ell}q'w_j^2g^2)}, \\
e_\ell \overset{\text{def}}{=} \sum_{j\ge 0} \mathbb{E}\,\frac{\beta^{2\ell} w_j^4 g^4}{(1+qw_j^2g^2)(1+\beta^{2\ell}q'w_j^2g^2)}.
\end{gathered} \tag{SD.55}$$

In the notation of eq. (SD.84), $\mathcal{T}^{\text{out}}_p = \mathcal{T}^{\text{out}}[{\Sigma_{\text{out}}}^{p-1}]$ and $\mathcal{T}^{\text{in}}_p = \mathcal{T}^{\text{in}}[{\Sigma_{\text{in}}}^{p-1}]$. Partial fractions give the divided-difference forms

$$b_\ell = \beta^\ell\,\frac{r_\beta(q) - r_\beta(\beta^{2\ell}q')}{q - \beta^{2\ell}q'}, \qquad e_\ell = \beta^{2\ell}\,\frac{F_\beta(q) - F_\beta(\beta^{2\ell}q')}{\beta^{2\ell}q' - q}, \tag{SD.56}$$

with derivatives at coincident arguments (lemma SD.12(d)). The determinant is

$$\Delta_\ell(s,u) \overset{\text{def}}{=} \left(1 - su\,\mathsf{U}\,\tilde\sigma\tilde\sigma'\,e_\ell\right)\left(1 - su\,\mathsf{V}\,\sigma\sigma'\,e_\ell\right) - su\,\mathsf{U}\mathsf{V}\,b_\ell^2. \tag{SD.57}$$

The lag enters only through the two scalars b_ℓ and e_ℓ; the branch, the traces and U, V do not depend on it.

Constants. Put

$$\kappa_\Sigma \overset{\text{def}}{=} \max\left(\frac{\operatorname{Tr}\Sigma_{\text{in}}}{B}\,\|\Sigma_{\text{out}}\|,\ \frac{\operatorname{Tr}\Sigma_{\text{out}}}{B}\,\|\Sigma_{\text{in}}\|\right) \le \gamma_\star C_\Sigma^2, \qquad \theta_\Sigma(t) \overset{\text{def}}{=} \frac{t}{t+\kappa_\Sigma} \quad (t>0).$$

For $0 < \beta < 1$ the function r_β increases to infinity (lemma SB.1(iii)), so $\hat q_\beta \overset{\text{def}}{=} r_\beta^{-1}(\gamma_\star)$ is finite, and we put

$$\epsilon_\beta \overset{\text{def}}{=} \min_{q\in[0,\hat q_\beta]} \min\left\{\frac{w_0^2\ \mathbb{E}\left[g^2(1+qw_0^2g^2)^{-2}\right]}{r_\beta'(q)},\ \frac{w_0^4\ \mathbb{E}\left[g^4(1+qw_0^2g^2)^{-2}\right]}{-F_\beta'(q)}\right\}.$$

Lemma SD.15(c) shows $\epsilon_\beta \in (0,1]$.

Theorem SD.5 (Closed-form lag kernel)**.** *Let $\ell \geq 0$, let F, G be diagonal tests and $s, u > 0$.*

(i) $\Delta_\ell(s,u) > 0$. *If* $\Sigma_{\text{out}} \neq 0$ *and* $\Sigma_{\text{in}} \neq 0$, *then* $\Delta_\ell(s,u) \geq \theta_\Sigma(\min(s,u))^2$; *otherwise* $\Delta_\ell(s,u) = 1$.

(ii) The lag kernel of eq. (SD.51) *is*

$$\mathfrak{k}_\ell[F,G](s,u) = \frac{su\, b_\ell\, \mathcal{T}^{\text{out}}[F](s,u)\, \mathcal{T}^{\text{in}}[G](s,u)}{\Delta_\ell(s,u)}. \tag{SD.58}$$

No ellipticity is assumed. If one covariance vanishes, both sides of eq. (SD.58) vanish. The closed form makes the regularity, the sign and the size of the lag kernels explicit.

Corollary SD.2 (Regularity and bounds)**.** *Let $\ell \geq 0$ and let F, G be diagonal tests.*

(i) $\mathfrak{k}_\ell[F,G]$ *is continuous on* $(0,\infty)^2$. *It is nonnegative if F and G are positive semidefinite. It vanishes identically if* $\beta = 0$ *and* $\ell \geq 1$.

(ii) For all $s, u > 0$,

$$|\mathfrak{k}_\ell[F,G](s,u)| \leq \frac{\gamma_\star \|F\|\|G\|}{4\sqrt{su}}, \qquad |\mathfrak{k}_\ell[F,G](s,u)| \leq \frac{s_2\ \operatorname{Tr}\Sigma_{\text{out}}\ \operatorname{Tr}\Sigma_{\text{in}}\ \|F\|\|G\|\,\beta^\ell}{B^2\, su\, \theta_\Sigma(\min(s,u))^2}.$$

(iii) If $0 < \beta < 1$ *and* $\ell \geq 1$, *then for all* $s, u > 0$,

$$|\mathfrak{k}_\ell[F,G](s,u)| \leq \frac{4 s_2\, \hat{q}_\beta\, \|F\|\|G\|\,\beta^\ell}{\epsilon_\beta^2 \sqrt{su}}.$$

(iv) Consequently $|\mathfrak{k}_\ell[F,G](s,u)| \leq C_\beta^{\text{lag}} \|F\|\|G\|\,\beta^\ell/\sqrt{su}$ *for all* $\ell \geq 0$ *and* $s, u > 0$, *with* $C_\beta^{\text{lag}} \stackrel{\text{def}}{=} \max(\gamma_\star/4,\ 4 s_2 \hat{q}_\beta \epsilon_\beta^{-2})$ *for* $0 < \beta < 1$ *and* $C_0^{\text{lag}} \stackrel{\text{def}}{=} \gamma_\star/4$. *The constant depends only on* $(\beta, \gamma_\star)$.

The closed form contains the two kernels identified earlier: the same-time two-resolvent kernel of eq. (SD.86), and the two-time row response of proposition SG.3. The identities below are identities between explicit expressions.

Proposition SD.1 (Two reductions)**.**

(a) If $\Sigma_{\text{out}} \neq 0$ *and* $\Sigma_{\text{in}} \neq 0$, *then for all* $s, u > 0$ *the right side of eq.* (SD.58) *at* $\ell = 0$, $F = \Sigma_{\text{out}}$, $G = \Sigma_{\text{in}}$ *equals the two-resolvent kernel* $\mathfrak{k}(s,u)$ *of eq.* (SD.86). *Hence* $\mathfrak{k}_0[\Sigma_{\text{out}}, \Sigma_{\text{in}}] = \mathfrak{k}$, *without ellipticity.*

(b) If $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$, *then for every i, every* $\ell \geq 0$ *and all* $s, u > 0$,

$$B\,\mathfrak{k}_\ell[u_i \otimes u_i, \mathrm{Id}_{N_{\text{in}}}](s,u) = \frac{(N_{\text{in}}/B)\, su\, \mu_i\, b_\ell}{(s+\tilde{s}_1\mu_i)(u+\tilde{s}_1'\mu_i)(s+s_1)(u+s_1')\,\Delta_\ell(s,u)}, \qquad \mathsf{V} = \frac{N_{\text{in}}/B}{(s+s_1)(u+s_1')}. \tag{SD.59}$$

For $N_{\text{in}} = N_{\text{out}} = N$ *this is eq.* (SG.16) *with* $\gamma = N/B$; *the identity holds for every pair* $(N_{\text{out}}, N_{\text{in}})$.

The filtered lag coefficients of eq. (SD.51) are double integrals of the lag kernels along the negative axis, and the closed form gives a majorant that makes these integrals absolutely convergent and geometric in the lag.

Theorem SD.6 (Filtered lag coefficients)**.** *Let $\ell \geq 0$, let F, G be diagonal tests, let* $\varrho, \varrho' > 0$ *and* $\varrho_m \stackrel{\text{def}}{=} \min(\varrho, \varrho')$. *The double integral below converges absolutely, and*

$$\overline{C}_\ell[F,G](\varrho,\varrho') = \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \mathfrak{k}_\ell[F,G](\varrho+x^2, \varrho'+y^2)\,dx\,dy, \tag{SD.60}$$

$$|\overline{C}_\ell[F,G](\varrho,\varrho')| \leq \frac{s_2\ \operatorname{Tr}\Sigma_{\text{out}}\ \operatorname{Tr}\Sigma_{\text{in}}\ \|F\|\|G\|\,(1+\kappa_\Sigma/\varrho_m)^2\,\beta^\ell}{B^2\sqrt{\varrho\varrho'}}.$$

Mode kernels. With spectral projections as tests the kernel factors into multiplicities and a nonnegative scalar kernel of the two eigenvalues. For an eigenvalue μ of Σ_{out} and an eigenvalue λ of Σ_{in}, with the projections P_μ, P'_λ and multiplicities r_μ, t_λ of eqs. (SD.78) and (SD.80), put for $s, u > 0$

$$\mathfrak{h}_\ell(\mu, \lambda; s, u) \stackrel{\mathrm{def}}{=} \frac{su\,\mu\lambda\, b_\ell(s,u)}{(s+\tilde{s}_1\mu)(u+\tilde{s}'_1\mu)(s+s_1\lambda)(u+s'_1\lambda)\,\Delta_\ell(s,u)}, \tag{SD.61}$$

and for $\varrho > 0$

$$\mathfrak{c}_\ell(\mu, \lambda; \varrho) \stackrel{\mathrm{def}}{=} \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \mathfrak{h}_\ell(\mu, \lambda; \varrho + x^2, \varrho + y^2)\, dx\, dy. \tag{SD.62}$$

Corollary SD.3 (Mode volatility)**.** *Let $\varrho > 0$, and let μ, λ be eigenvalues of $\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}}$.*

(i) For all $\ell \geq 0$ and $s, u > 0$, $\mathfrak{h}_\ell(\mu, \lambda; s, u) \geq 0$ and

$$\mathfrak{k}_\ell[P_\mu, P'_\lambda](s,u) = \frac{r_\mu t_\lambda}{B^2}\,\mathfrak{h}_\ell(\mu,\lambda;s,u), \qquad \overline{C}_\ell[P_\mu, P'_\lambda](\varrho,\varrho) = \frac{r_\mu t_\lambda}{B^2}\,\mathfrak{c}_\ell(\mu,\lambda;\varrho).$$

(ii) For all $\ell \geq 0$,

$$0 \leq \mathfrak{c}_\ell(\mu, \lambda; \varrho) \leq \frac{\mu\lambda\, s_2\,(1+\kappa_\Sigma/\varrho)^2}{\varrho}\,\beta^\ell.$$

(iii) The mode coefficient eq. (SD.80) is, for every i, j,

$$\begin{aligned} \mathfrak{v}_{ij}(\varrho) &= \frac{1}{B}\Big(\mathfrak{c}_0(\mu_i,\lambda_j;\varrho) + 2\sum_{\ell\geq 1}\mathfrak{c}_\ell(\mu_i,\lambda_j;\varrho)\Big), \\ 0 \leq \mathfrak{v}_{ij}(\varrho) &\leq \frac{\mu_i\lambda_j\, s_2\,(1+\kappa_\Sigma/\varrho)^2}{B\,\varrho}\cdot\frac{1+\beta}{1-\beta}, \end{aligned} \tag{SD.63}$$

the series converging absolutely with nonnegative terms.

In particular every lag term of the mode coefficient is nonnegative, which is stronger than the nonnegativity of the integrated coefficient in theorem SD.4(iii).

The lag sum. The lag enters the closed form only through b_ℓ and e_ℓ. We record this reduction, and the exchange of the lag sum with the double integral. No closed form for the sum over the lags is claimed. Put $c_0 \stackrel{\mathrm{def}}{=} 1$ and $c_\ell \stackrel{\mathrm{def}}{=} 2$ for $\ell \geq 1$, and

$$\phi_q(x) \stackrel{\mathrm{def}}{=} \frac{x}{1+qx^2}, \qquad \psi_q(x) \stackrel{\mathrm{def}}{=} \frac{x^2}{1+qx^2}.$$

Proposition SD.2 (Lag dependence)**.** *Let F, G be diagonal tests and $s, u, \varrho > 0$. With the lag-independent quantities*

$$A_{\mathrm{o}} \stackrel{\mathrm{def}}{=} su\,\mathsf{U}\tilde{\sigma}\tilde{\sigma}', \qquad A_{\mathrm{in}} \stackrel{\mathrm{def}}{=} su\,\mathsf{V}\sigma\sigma', \qquad A_\times \stackrel{\mathrm{def}}{=} su\,\mathsf{U}\mathsf{V},$$

for every $\ell \geq 0$,

$$\mathfrak{k}_\ell[F,G](s,u) = su\,\mathcal{T}^{\mathrm{out}}[F]\,\mathcal{T}^{\mathrm{in}}[G]\,\frac{b_\ell}{1-(A_{\mathrm{o}}+A_{\mathrm{in}})e_\ell + A_{\mathrm{o}}A_{\mathrm{in}}e_\ell^2 - A_\times b_\ell^2},$$

$$b_\ell = \sum_{j\geq 0}\mathbb{E}\left[\phi_q(w_j g)\,\phi_{q'}(w_{j+\ell}g)\right], \qquad e_\ell = \sum_{j\geq 0}\mathbb{E}\left[\psi_q(w_j g)\,\psi_{q'}(w_{j+\ell}g)\right].$$

Moreover $\sum_\ell c_\ell\,\mathfrak{k}_\ell[F,G](\varrho+x^2,\varrho+y^2)$ converges absolutely for all $x, y \geq 0$, and

$$\overline{\Gamma}[F,G](\varrho,\varrho) = \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \sum_{\ell\geq 0} c_\ell\,\mathfrak{k}_\ell[F,G](\varrho+x^2,\varrho+y^2)\,dx\,dy,$$

with $\overline{\Gamma}$ of eq. (SD.54), the double integral converging absolutely.

The proofs occupy the rest of the subsection: scalar lemmas (appendix SD.7.2), the encoding of retained pairings by ordered trees (appendix SD.7.3), the summation of the trees (appendix SD.7.4) and the assembly (appendix SD.7.5).

SD.7.2 Scalar lemmas

The cutoff model of appendix SD.6 and the full EMA model differ only in the random coefficients carried by the batches. One argument covers both if the coefficients are treated abstractly.

Definition SD.5 (Profiles)**.** *A* profile *is a countable family $\xi = (\xi_\alpha)_{\alpha\in\mathcal{A}}$ of real random variables with $\sum_\alpha \mathbb{E}\,\xi_\alpha^2 < \infty$ and $\Pr(\xi_\alpha \neq 0) > 0$ for some α. Its scalar functions are, for $q \geq 0$,*

$$F_\xi(q) \stackrel{\text{def}}{=} \sum_\alpha \mathbb{E}\,\frac{\xi_\alpha^2}{1+q\xi_\alpha^2}, \qquad r_\xi(q) \stackrel{\text{def}}{=} qF_\xi(q).$$

A paired profile *is a countable family $(\xi_{1,\alpha},\xi_{2,\alpha})_{\alpha\in\mathcal{A}}$ such that $\xi_1 = (\xi_{1,\alpha})$ and $\xi_2 = (\xi_{2,\alpha})$ are profiles, $\xi_{1,\alpha}\xi_{2,\alpha} \geq 0$, and $\sum_\alpha \mathbb{E}[\xi_{1,\alpha}^2\xi_{2,\alpha}^2] < \infty$. Its two-point functions are, for $q_1, q_2 \geq 0$,*

$$b(q_1,q_2) \stackrel{\text{def}}{=} \sum_\alpha \mathbb{E}\,\frac{\xi_{1,\alpha}\xi_{2,\alpha}}{(1+q_1\xi_{1,\alpha}^2)(1+q_2\xi_{2,\alpha}^2)}, \qquad e(q_1,q_2) \stackrel{\text{def}}{=} \sum_\alpha \mathbb{E}\,\frac{\xi_{1,\alpha}^2\xi_{2,\alpha}^2}{(1+q_1\xi_{1,\alpha}^2)(1+q_2\xi_{2,\alpha}^2)}.$$

Two paired profiles are used. The *EMA paired profile at lag ℓ* has the labels $j \geq 0$, with $(\xi_{1,j},\xi_{2,j}) = (w_j g_j, \beta^\ell w_j g_j)$, and the labels $m \in \{1,\dots,\ell\}$, with $(\xi_{1,m},\xi_{2,m}) = (0, w_{\ell-m} g_m)$; all g_α are independent standard Gaussians. The first labels are the batches shared by the two buffers and the second the fresh batches seen only by the later buffer. Since $\beta^\ell w_j = w_{j+\ell}$, both marginals have the scalar function F_β of eq. (SB.1), and at $(q_1,q_2) = (q,q')$ the two-point functions are b_ℓ and e_ℓ of eq. (SD.55). The *cutoff paired profile* at (ℓ, J, R) has the labels $\mathcal{A}$ of appendix SD.6 and $\xi_{l,\alpha} \stackrel{\text{def}}{=} \varpi_{l,\alpha}\tilde{g}_\alpha$ for $l = 1, 2$, with the coefficients eq. (SD.42) and one variable $\tilde{g}_\alpha = g_\alpha \mathbf{1}_{|g_\alpha|\leq R}$ per label, shared by the two buffers. Since $\varpi_{l,\alpha} \leq 1$ and $\sum_\alpha \varpi_{l,\alpha} \leq 1$,

$$|\xi_{l,\alpha}| \leq \varpi_{l,\alpha} R, \qquad \sum_\alpha \mathbb{E}\,\xi_{l,\alpha}^2 \leq 1, \qquad F_{\xi_l}(0) \leq 1, \tag{SD.64}$$

and the latent value of a class V in definition SD.4 is $\sum_\alpha \mathbb{E}\prod_{r\in S_V} \xi_{l_r,\alpha}$.

Lemma SD.12 (Profile functions)**.** *Let ξ be a profile.*

(a) F_ξ and r_ξ are continuous on $[0,\infty)$ and real-analytic on $(0,\infty)$. F_ξ is nonincreasing with $0 < F_\xi \leq F_\xi(0) = \sum_\alpha \mathbb{E}\,\xi_\alpha^2$, and $r_\xi(0) = 0$. For $q \geq 0$,

$$r_\xi'(q) = \sum_\alpha \mathbb{E}\,\frac{\xi_\alpha^2}{(1+q\xi_\alpha^2)^2} > 0, \qquad -F_\xi'(q) = \sum_\alpha \mathbb{E}\,\frac{\xi_\alpha^4}{(1+q\xi_\alpha^2)^2} \in [0,\infty],$$

the latter finite for $q > 0$, and

$$r_\xi'(q) + q\big(-F_\xi'(q)\big) = F_\xi(q) \qquad (q > 0). \tag{SD.65}$$

(b) r_ξ increases strictly to $n_\xi \stackrel{\text{def}}{=} \sum_\alpha \Pr(\xi_\alpha \neq 0) \in (0,\infty]$.

(c) For a paired profile, b and e are continuous on $[0,\infty)^2$, real-analytic in each variable on $(0,\infty)$ when the other is fixed, and

$$0 \leq b(q_1,q_2) \leq \min\Big(\sqrt{r_{\xi_1}'(q_1)\,r_{\xi_2}'(q_2)},\ \sum_\alpha \mathbb{E}[\xi_{1,\alpha}\xi_{2,\alpha}]\Big),$$

$$0 \leq e(q_1,q_2) \leq \sqrt{\big(-F_{\xi_1}'(q_1)\big)\big(-F_{\xi_2}'(q_2)\big)}.$$

(d) For the EMA paired profile at lag ℓ, the divided-difference forms eq. (SD.56) hold, and $0 \leq b_\ell \leq s_2\,\beta^\ell$. At $\ell = 0$ and $q = q'$ they read $b_0 = r_\beta'(q)$ and $e_0 = -F_\beta'(q)$.

Proof. For (a), each summand $x^2/(1+qx^2)$ lies in $[0,x^2]$ and is nonincreasing in q. Continuity on $[0,\infty)$ and monotonicity follow by dominated convergence with the majorant $\sum_\alpha \xi_\alpha^2$. For analyticity at $q_0 > 0$, expand, for $|q-q_0| < q_0$,

$$\frac{x^2}{1+qx^2} = \sum_{n\geq 0} \frac{x^2\big(-(q-q_0)x^2\big)^n}{(1+q_0x^2)^{n+1}}, \qquad \Big|\frac{x^2\big(-(q-q_0)x^2\big)^n}{(1+q_0x^2)^{n+1}}\Big| \leq x^2\Big(\frac{|q-q_0|}{q_0}\Big)^n,$$

using $x^2/(1+q_0x^2) \le 1/q_0$. The double family over (α, n) is dominated by $\sum_\alpha \mathbb{E}\,\xi_\alpha^2 \sum_n (|q-q_0|/q_0)^n < \infty$. Hence Fubini's theorem turns F_ξ into a convergent power series in $q - q_0$, and $r_\xi = qF_\xi$ is analytic as well. Termwise differentiation of the series gives the displayed derivatives for $q > 0$, and dominated convergence with the majorant ξ_α^2 gives $r'_\xi(0)$. The derivative r'_ξ is positive because $\Pr(\xi_\alpha \neq 0) > 0$ for some α. The bound $x^4/(1+qx^2)^2 \le x^2/(4q)$ shows that $-F'_\xi(q)$ is finite for $q > 0$. Identity eq. (SD.65) is the sum over α of

$$\frac{x^2}{(1+qx^2)^2} + \frac{qx^4}{(1+qx^2)^2} = \frac{x^2}{1+qx^2}.$$

For (b), $qx^2/(1+qx^2)$ increases to $\mathbf{1}_{x\neq 0}$ as $q \to \infty$. Monotone convergence gives the limit n_ξ, and $r'_\xi > 0$ gives strict monotonicity.

For (c), nonnegativity follows from $\xi_{1,\alpha}\xi_{2,\alpha} \ge 0$. The Cauchy–Schwarz inequality for the sum over α and the expectation, with $X_\alpha = \xi_{1,\alpha}/(1+q_1\xi_{1,\alpha}^2)$ and $Y_\alpha = \xi_{2,\alpha}/(1+q_2\xi_{2,\alpha}^2)$, gives the first bound on b. With $X_\alpha = \xi_{1,\alpha}^2/(1+q_1\xi_{1,\alpha}^2)$ and $Y_\alpha = \xi_{2,\alpha}^2/(1+q_2\xi_{2,\alpha}^2)$ it gives the bound on e. The second bound on b holds because the denominators are at least one. Continuity follows by dominated convergence with the summable majorants $(\xi_1^2+\xi_2^2)/2$ and $\xi_1^2\xi_2^2$. Analyticity in q_1 for fixed q_2 follows as in (a), the factor $1/(1+q_2\xi_2^2) \in (0,1]$ being fixed.

For (d), for $x \ge 0$ and $p \neq p'$ partial fractions give

$$\frac{x}{(1+px)(1+p'x)} = \frac{1}{p-p'}\Big(\frac{px}{1+px} - \frac{p'x}{1+p'x}\Big),$$

$$\frac{x^2}{(1+px)(1+p'x)} = \frac{1}{p-p'}\Big(\frac{x}{1+p'x} - \frac{x}{1+px}\Big).$$

Take $x = w_j^2 g^2$, $p = q$ and $p' = \beta^{2\ell}q'$, multiply by β^ℓ, respectively $\beta^{2\ell}$, take expectations and sum over j; this gives eq. (SD.56). The bound $b_\ell \le \beta^\ell \sum_j w_j^2\,\mathbb{E}\,g^2 = s_2\beta^\ell$ is the second bound of (c). At coincident arguments the formulas are the definitions. □

The next lemma is the branch of theorem SB.1 for an arbitrary profile. It is needed because the cutoff profiles are not of the form covered there, and because the continuation argument below needs real analyticity in the spectral parameter. The proof reduces the system to one monotone scalar equation.

Lemma SD.13 (Branch of a profile)**.** *Let ξ be a profile and $s > 0$. A* branch point *of ξ at s is a quadruple $(\sigma, \tilde\sigma, \tilde s_1, s_1) \in [0,\infty)^4$ with*

$$\begin{aligned} \sigma &= \frac{1}{B}\operatorname{Tr}\Sigma_{\text{out}}(s\,\mathrm{Id} + \tilde s_1\Sigma_{\text{out}})^{-1}, \qquad \tilde\sigma = \frac{1}{B}\operatorname{Tr}\Sigma_{\text{in}}(s\,\mathrm{Id} + s_1\Sigma_{\text{in}})^{-1}, \\ \tilde s_1 &= s\tilde\sigma F_\xi(q), \qquad s_1 = s\sigma F_\xi(q), \end{aligned} \tag{SD.66}$$

where $q \overset{\text{def}}{=} s\sigma\tilde\sigma$.

(i) If $\Sigma_{\text{out}} = 0$, the unique branch point at s is $\sigma = s_1 = 0$, $\tilde\sigma = B^{-1}\operatorname{Tr}\Sigma_{\text{in}}/s$, $\tilde s_1 = B^{-1}\operatorname{Tr}\Sigma_{\text{in}}\,F_\xi(0)$, with $q = 0$. The case $\Sigma_{\text{in}} = 0$ is symmetric.

(ii) If $\Sigma_{\text{out}} \neq 0$ and $\Sigma_{\text{in}} \neq 0$, then eq. (SD.66) has exactly one solution in $[0,\infty)^4$. It lies in $(0,\infty)^4$ and in the box

$$\sigma \le \frac{\operatorname{Tr}\Sigma_{\text{out}}}{Bs}, \qquad \tilde\sigma \le \frac{\operatorname{Tr}\Sigma_{\text{in}}}{Bs}, \qquad \tilde s_1 \le \frac{\operatorname{Tr}\Sigma_{\text{in}}}{B}\,F_\xi(0), \qquad s_1 \le \frac{\operatorname{Tr}\Sigma_{\text{out}}}{B}\,F_\xi(0),$$

and $r_\xi(q) = \tilde s_1\sigma = s_1\tilde\sigma < \min(\operatorname{rank}\Sigma_{\text{out}}, \operatorname{rank}\Sigma_{\text{in}})/B \le \gamma_\star$.

(iii) In case (ii), $s \mapsto (\sigma, \tilde\sigma, \tilde s_1, s_1, q)(s)$ is real-analytic on $(0,\infty)$, and q is strictly decreasing in s.

(iv) (Stability.) Let $\xi^{(n)}$ be profiles with $\sup_{q\ge 0}|F_{\xi^{(n)}}(q) - F_\xi(q)| \to 0$ and $\sup_n F_{\xi^{(n)}}(0) < \infty$. Then for each $s > 0$ the branch points of $\xi^{(n)}$ at s converge to the branch point of ξ at s.

For the EMA weights, eq. (SD.66) *is the system eq.* (SB.2) *at $z = -s$ for the spectral measures of eq.* (SA.41)*, and its branch point is the solution of theorem SB.1.*

Proof. For (i), $\sigma = B^{-1}\operatorname{Tr} 0 = 0$ forces $q = 0$ and $s_1 = 0$. Then $\tilde\sigma = B^{-1}\operatorname{Tr}\Sigma_{\text{in}}/s$ and $\tilde s_1 = s\tilde\sigma F_\xi(0)$.

For (ii), we first introduce the inverse traces. For $X \in \{\Sigma_{\text{out}}, \Sigma_{\text{in}}\}$, nonzero, with eigenvalues (δ_k), put

$$H_X(x) \overset{\text{def}}{=} \frac{1}{B}\sum_k \frac{x\delta_k}{1+x\delta_k} \quad (x \ge 0), \qquad p_X \overset{\text{def}}{=} \frac{\operatorname{rank} X}{B}.$$

Then H_X is real-analytic on $(-1/\|X\|, \infty)$ with $H_X' > 0$, $H_X(0) = 0$ and $H_X(\infty) = p_X$. Let $A_X : [0, p_X) \to [0, \infty)$ be its inverse, which is real-analytic on $(0, p_X)$. The ratio $H_X(x)/x = B^{-1}\operatorname{Tr} X(\operatorname{Id} + xX)^{-1}$ is nonincreasing in x, with limit $B^{-1}\operatorname{Tr} X > 0$ at 0. Hence $y \mapsto y/A_X(y)$ is nonincreasing on $(0, p_X)$, which is the inequality

$$A_X'(y) \ge \frac{A_X(y)}{y}. \tag{SD.67}$$

We write $H_{\text{out}}, A_{\text{out}}, p_{\text{out}}$ for $X = \Sigma_{\text{out}}$ and $H_{\text{in}}, A_{\text{in}}, p_{\text{in}}$ for $X = \Sigma_{\text{in}}$.

Every solution reduces to a root of one scalar equation. Let $(\sigma, \tilde\sigma, \tilde s_1, s_1)$ solve eq. (SD.66). The traces $\sigma, \tilde\sigma$ are positive, being traces of nonzero positive semidefinite matrices against positive definite matrices. Hence $q > 0$, $F_\xi(q) > 0$ and $\tilde s_1, s_1 > 0$. Since

$$\tilde s_1\sigma = s\sigma\tilde\sigma F_\xi(q) = r_\xi(q), \qquad \tilde s_1\sigma = H_{\text{out}}(\tilde s_1/s),$$

we get $r_\xi(q) < p_{\text{out}}$ and $\tilde s_1 = sA_{\text{out}}(r_\xi(q))$; likewise $r_\xi(q) < p_{\text{in}}$ and $s_1 = sA_{\text{in}}(r_\xi(q))$. Substituting into $q = s\sigma\tilde\sigma = s\, r_\xi(q)^2/(\tilde s_1 s_1)$ gives

$$\mathcal{Z}_\xi(q) = s, \qquad \mathcal{Z}_\xi(q) \overset{\text{def}}{=} \frac{r_\xi(q)^2}{q\, A_{\text{out}}(r_\xi(q))\, A_{\text{in}}(r_\xi(q))}, \qquad q \in I_\xi \overset{\text{def}}{=} \{q > 0 : r_\xi(q) < p_{\min}\}, \tag{SD.68}$$

with $p_{\min} \overset{\text{def}}{=} \min(p_{\text{out}}, p_{\text{in}})$.

Conversely, every root gives a solution. Let $q \in I_\xi$ solve eq. (SD.68), and put $\tilde s_1 = sA_{\text{out}}(r_\xi(q))$, $s_1 = sA_{\text{in}}(r_\xi(q))$, $\sigma = r_\xi(q)/\tilde s_1$ and $\tilde\sigma = r_\xi(q)/s_1$. Then

$$\sigma = \frac{H_{\text{out}}(\tilde s_1/s)}{\tilde s_1} = \frac{1}{B}\operatorname{Tr}\Sigma_{\text{out}}(s\operatorname{Id} + \tilde s_1\Sigma_{\text{out}})^{-1},$$

and likewise for $\tilde\sigma$. Next $s\sigma\tilde\sigma = r_\xi^2/(sA_{\text{out}}A_{\text{in}}) = q$ by eq. (SD.68). Finally $s\tilde\sigma F_\xi(q) = r_\xi F_\xi/A_{\text{in}}$, which equals $\tilde s_1 = sA_{\text{out}}$ exactly when $sA_{\text{out}}A_{\text{in}} = r_\xi F_\xi = r_\xi^2/q$, that is, by eq. (SD.68); the same holds for s_1. So the solutions of eq. (SD.66) correspond bijectively to the roots of eq. (SD.68).

The function $\mathcal{Z}_\xi$ is a strictly decreasing analytic bijection of I_ξ onto $(0, \infty)$. It is real-analytic on the interval I_ξ, and by eq. (SD.67) and $r_\xi' > 0$,

$$(\log \mathcal{Z}_\xi)' = \frac{2r_\xi'}{r_\xi} - \frac{1}{q} - r_\xi'\Big(\frac{A_{\text{out}}'}{A_{\text{out}}} + \frac{A_{\text{in}}'}{A_{\text{in}}}\Big) \le -\frac{1}{q} < 0.$$

As $q \downarrow 0$, $r_\xi(q)/A_X(r_\xi(q)) \to B^{-1}\operatorname{Tr} X > 0$ for both X, so $\mathcal{Z}_\xi \to \infty$. At the upper end there are two cases. If $n_\xi \le p_{\min}$, then $I_\xi = (0, \infty)$ and $\mathcal{Z}_\xi(q) \le B^{-2}\operatorname{Tr}\Sigma_{\text{out}}\operatorname{Tr}\Sigma_{\text{in}}/q \to 0$, since $r_\xi/A_X(r_\xi) \le B^{-1}\operatorname{Tr} X$. If $n_\xi > p_{\min}$, then $I_\xi = (0, q_*)$ with $r_\xi(q_*) = p_{\min} = p_X$ for some X; as $q \uparrow q_*$, $A_X(r_\xi(q)) \to \infty$ while the other factors stay bounded, so $\mathcal{Z}_\xi \to 0$. Hence eq. (SD.68) has exactly one root.

The box follows from the equations. The first two bounds are $\Sigma_{\text{out}}(s\operatorname{Id} + \tilde s_1\Sigma_{\text{out}})^{-1} \preceq \Sigma_{\text{out}}/s$ and its input analogue. The last two are $\tilde s_1 = s\tilde\sigma F_\xi(q) \le B^{-1}\operatorname{Tr}\Sigma_{\text{in}}F_\xi(0)$ and its analogue. Finally $r_\xi(q) < p_{\min} \le \max(N_{\text{out}}, N_{\text{in}})/B \le \gamma_\star$.

For (iii), $\mathcal{Z}_\xi' < 0$, so $q(s) = \mathcal{Z}_\xi^{-1}(s)$ is real-analytic with $q'(s) = 1/\mathcal{Z}_\xi'(q(s)) < 0$, by the real-analytic inverse function theorem [36, Section 2.5]. The other coordinates are the analytic expressions of the converse construction.

For (iv), the branch points of $\xi^{(n)}$ lie in the compact box of (ii), whose bounds involve $F_{\xi^{(n)}}(0)$, or are given explicitly by (i). Let x be a limit point along a subsequence, with $q_n \to q$. The trace maps in eq. (SD.66) are continuous on $[0,\infty)^4$, and

$$|F_{\xi^{(n)}}(q_n) - F_\xi(q)| \le \sup_{p\ge 0} |F_{\xi^{(n)}}(p) - F_\xi(p)| + |F_\xi(q_n) - F_\xi(q)| \to 0.$$

So x solves eq. (SD.66) for ξ, and x is its unique branch point. Every subsequence has this limit point, so the whole sequence converges.

For the EMA weights, $F_\xi = F_\beta$ by the definition eq. (SB.1), and the equations are those of eq. (SB.2) at $z = -s$; uniqueness in both results identifies the two solutions. □

The positivity of the determinant is a Perron argument for a nonnegative 2×2 matrix. It is used twice, with and without a gain, so we isolate it.

Lemma SD.14 (A 2×2 Perron bound)**.** *Let L, L_s, L_u be nonnegative 2×2 matrices, let $c \in (0,1]$, and suppose $L_{kl} \le c\sqrt{(L_s)_{kl}(L_u)_{kl}}$ for all k,l. Suppose there are positive vectors v_s, v_u and numbers $\vartheta_s, \vartheta_u \in [0,1]$ with $L_s v_s \le (1-\vartheta_s)v_s$ and $L_u v_u \le (1-\vartheta_u)v_u$ componentwise. Then*

$$\det(\mathrm{Id} - L) \ge \left(1 - c\sqrt{(1-\vartheta_s)(1-\vartheta_u)}\right)^2.$$

Proof. Put $v \stackrel{\text{def}}{=} \left(\sqrt{v_{s,1}v_{u,1}}, \sqrt{v_{s,2}v_{u,2}}\right)$, a positive vector. By the hypothesis and the inequality $\sqrt{xy} + \sqrt{zw} \le \sqrt{(x+z)(y+w)}$,

$$(Lv)_k \le c\sum_l \sqrt{(L_s)_{kl}v_{s,l}\,(L_u)_{kl}v_{u,l}} \le c\sqrt{(L_s v_s)_k\,(L_u v_u)_k} \le c\sqrt{(1-\vartheta_s)(1-\vartheta_u)}\,v_k.$$

In the weighted norm $\|x\|_v \stackrel{\text{def}}{=} \max_k |x_k|/v_k$, the operator norm of the nonnegative matrix L is $\max_k (Lv)_k/v_k \le \eta \stackrel{\text{def}}{=} c\sqrt{(1-\vartheta_s)(1-\vartheta_u)}$. Hence every eigenvalue of L has modulus at most η. The eigenvalues are real, because the discriminant $(L_{11}-L_{22})^2 + 4L_{12}L_{21}$ is nonnegative. Hence $\det(\mathrm{Id}-L) = (1-\lambda_1)(1-\lambda_2) \ge (1-\eta)^2$. □

For a paired profile and $s,u>0$, let $(\sigma,\tilde\sigma,\tilde s_1, s_1)$ be the branch point of ξ_1 at s, with $q = s\sigma\tilde\sigma$, and $(\sigma',\tilde\sigma',\tilde s_1', s_1')$ the branch point of ξ_2 at u, with $q' = u\sigma'\tilde\sigma'$. Form $\mathcal{T}^{\mathrm{out}}[F], \mathcal{T}^{\mathrm{in}}[G], \mathsf{U}, \mathsf{V}$ as in eq. (SD.55) with these branch points, $b \stackrel{\text{def}}{=} b(q,q')$ and $e \stackrel{\text{def}}{=} e(q,q')$, and

$$\begin{aligned} \Delta_\xi &\stackrel{\text{def}}{=} \big(1 - su\,\mathsf{U}\tilde\sigma\tilde\sigma' e\big)\big(1 - su\,\mathsf{V}\sigma\sigma' e\big) - su\,\mathsf{U}\mathsf{V}b^2, \\ \mathfrak{K}_\xi[F,G](s,u) &\stackrel{\text{def}}{=} \frac{su\,b\,\mathcal{T}^{\mathrm{out}}[F]\,\mathcal{T}^{\mathrm{in}}[G]}{\Delta_\xi} \quad (\Delta_\xi \ne 0). \end{aligned} \tag{SD.69}$$

For the EMA paired profile at lag ℓ these are Δ_ℓ and the right side of eq. (SD.58), which we write $\mathfrak{K}_\ell[F,G]$; for the cutoff paired profile at (ℓ, J, R) we write $\Delta^{J,R}$ and $\mathfrak{K}^{J,R}[F,G]$.

Lemma SD.15 (Determinant)**.** *Let (ξ_1,ξ_2) be a paired profile with $F_{\xi_1}(0), F_{\xi_2}(0) \le 1$, and $s,u>0$.*

- *(a) If $\Sigma_{\mathrm{out}} = 0$ or $\Sigma_{\mathrm{in}} = 0$, then $\Delta_\xi = 1$ and $\mathcal{T}^{\mathrm{out}}[F]\,\mathcal{T}^{\mathrm{in}}[G] = 0$ for all diagonal F, G.*
- *(b) If $\Sigma_{\mathrm{out}} \ne 0$ and $\Sigma_{\mathrm{in}} \ne 0$, then $\Delta_\xi \ge \theta_\Sigma(\min(s,u))^2 > 0$.*
- *(c) For the EMA paired profile with $0<\beta<1$ and $\ell \ge 1$, if $\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}} \ne 0$, then $\epsilon_\beta \in (0,1]$ and $\Delta_\ell \ge \left(1 - \sqrt{1-\epsilon_\beta}\right)^2 \ge \epsilon_\beta^2/4$ for all $s,u>0$.*

Proof. For (a), if $\Sigma_{\mathrm{out}} = 0$, then $\mathsf{U} = 0$, $\mathcal{T}^{\mathrm{out}}[F] = 0$, and $\sigma = \sigma' = 0$ by lemma SD.13(i), so $\Delta_\xi = 1$. The case $\Sigma_{\mathrm{in}} = 0$ is symmetric.

For (b), Δ_ξ is the determinant of $\mathrm{Id} - L$ for the nonnegative matrix

$$L \stackrel{\text{def}}{=} \begin{pmatrix} su\,\mathsf{U}\tilde\sigma\tilde\sigma' e & \sqrt{su}\,\mathsf{U}b \\ \sqrt{su}\,\mathsf{V}b & su\,\mathsf{V}\sigma\sigma' e \end{pmatrix}.$$

The comparison matrices are one-parameter versions of L. For one profile $\eta \in \{\xi_1, \xi_2\}$ at one parameter $t \in \{s, u\}$, with its branch point $(\sigma_t, \tilde{\sigma}_t, \tilde{s}_{1,t}, s_{1,t})$ and q_t, put $\overline{R}_t \stackrel{\text{def}}{=} (t\,\mathrm{Id} + \tilde{s}_{1,t}\Sigma_{\text{out}})^{-1}$, $\overline{\widetilde{R}}_t \stackrel{\text{def}}{=} (t\,\mathrm{Id} + s_{1,t}\Sigma_{\text{in}})^{-1}$,

$$\mathsf{U}_t \stackrel{\text{def}}{=} \tfrac{1}{B}\operatorname{Tr}\Sigma_{\text{out}}{}^2\overline{R}_t^2, \quad \mathsf{V}_t \stackrel{\text{def}}{=} \tfrac{1}{B}\operatorname{Tr}\Sigma_{\text{in}}{}^2\overline{\widetilde{R}}_t^2, \quad \Theta_t \stackrel{\text{def}}{=} \tfrac{1}{B}\operatorname{Tr}\Sigma_{\text{out}}\overline{R}_t^2, \quad \widetilde{\Theta}_t \stackrel{\text{def}}{=} \tfrac{1}{B}\operatorname{Tr}\Sigma_{\text{in}}\overline{\widetilde{R}}_t^2,$$

$\mathsf{b}_t \stackrel{\text{def}}{=} r'_\eta(q_t)$, $\mathsf{e}_t \stackrel{\text{def}}{=} -F'_\eta(q_t)$, and

$$L_t \stackrel{\text{def}}{=} \begin{pmatrix} t^2\mathsf{U}_t\tilde{\sigma}_t^2\mathsf{e}_t & t\,\mathsf{U}_t\mathsf{b}_t \\ t\,\mathsf{V}_t\mathsf{b}_t & t^2\mathsf{V}_t\sigma_t^2\mathsf{e}_t \end{pmatrix}.$$

The branch vector is a strict supersolution of L_t:

$$L_t\begin{pmatrix}\sigma_t\\ \tilde{\sigma}_t\end{pmatrix} = \begin{pmatrix}\sigma_t - t\Theta_t\\ \tilde{\sigma}_t - t\widetilde{\Theta}_t\end{pmatrix} \le \big(1 - \theta_\Sigma(t)\big)\begin{pmatrix}\sigma_t\\ \tilde{\sigma}_t\end{pmatrix}. \tag{SD.70}$$

Indeed, by eq. (SD.65), $\mathsf{b}_t + q_t\mathsf{e}_t = F_\eta(q_t)$. So the first component is $t\tilde{\sigma}_t\mathsf{U}_t(\mathsf{b}_t + q_t\mathsf{e}_t) = \tilde{s}_{1,t}\mathsf{U}_t$, and

$$\tilde{s}_{1,t}\mathsf{U}_t = \frac{1}{B}\sum_i \mu_i \frac{\tilde{s}_{1,t}\mu_i}{(t + \tilde{s}_{1,t}\mu_i)^2} = \frac{1}{B}\sum_i \Big(\frac{\mu_i}{t + \tilde{s}_{1,t}\mu_i} - \frac{t\mu_i}{(t + \tilde{s}_{1,t}\mu_i)^2}\Big) = \sigma_t - t\Theta_t.$$

The second component is $s_{1,t}\mathsf{V}_t = \tilde{\sigma}_t - t\widetilde{\Theta}_t$ in the same way. For the inequality, $\tilde{s}_{1,t} \le B^{-1}\operatorname{Tr}\Sigma_{\text{in}}\, F_\eta(0) \le B^{-1}\operatorname{Tr}\Sigma_{\text{in}}$ by lemma SD.13(ii), so

$$t\Theta_t \ge \frac{t}{t + \tilde{s}_{1,t}\|\Sigma_{\text{out}}\|}\,\sigma_t \ge \frac{t}{t + B^{-1}\operatorname{Tr}\Sigma_{\text{in}}\|\Sigma_{\text{out}}\|}\,\sigma_t \ge \theta_\Sigma(t)\,\sigma_t,$$

and likewise $t\widetilde{\Theta}_t \ge \theta_\Sigma(t)\tilde{\sigma}_t$.

The entries of L are at most the geometric means of those of L_s and L_u, where L_s is built from ξ_1 at s and L_u from ξ_2 at u. The Cauchy–Schwarz inequality over i, respectively j, gives $\mathsf{U} \le \sqrt{\mathsf{U}_s\mathsf{U}_u}$ and $\mathsf{V} \le \sqrt{\mathsf{V}_s\mathsf{V}_u}$. Lemma SD.12(c) gives $b \le \sqrt{\mathsf{b}_s\mathsf{b}_u}$ and $e \le \sqrt{\mathsf{e}_s\mathsf{e}_u}$. Hence $L_{kl} \le \sqrt{(L_s)_{kl}(L_u)_{kl}}$. Lemma SD.14 with $c = 1$, $v_t = (\sigma_t, \tilde{\sigma}_t)$ and $\vartheta_t = \theta_\Sigma(t)$ gives

$$\Delta_\xi \ge \Big(1 - \sqrt{(1 - \theta_\Sigma(s))(1 - \theta_\Sigma(u))}\Big)^2 \ge \theta_\Sigma(\min(s,u))^2,$$

since θ_Σ is increasing.

For (c), a lag $\ell \ge 1$ removes the youngest batch from one side of every Cauchy–Schwarz inequality. By lemma SD.13(ii) and the strict monotonicity of r_β, $q, q' \in [0, \hat{q}_\beta)$. Write $b_\ell = \sum_j \mathbb{E}[X_jY_j]$ with

$$X_j = \frac{w_j g}{1 + q w_j^2 g^2}, \qquad Y_j = \frac{w_{j+\ell} g}{1 + q' w_{j+\ell}^2 g^2}.$$

Then $\sum_j \mathbb{E}\,X_j^2 = r'_\beta(q)$ and

$$\sum_j \mathbb{E}\,Y_j^2 = \sum_{k \ge \ell} \mathbb{E}\,\frac{w_k^2 g^2}{(1 + q' w_k^2 g^2)^2} \le r'_\beta(q') - w_0^2\,\mathbb{E}\,\frac{g^2}{(1 + q' w_0^2 g^2)^2} \le (1 - \epsilon_\beta)\, r'_\beta(q'),$$

because the term $k = 0$ is missing. The Cauchy–Schwarz inequality gives $b_\ell \le \sqrt{1 - \epsilon_\beta}\sqrt{\mathsf{b}_s\mathsf{b}_u}$. The same argument with $w_j^2 g^2/(1 + q w_j^2 g^2)$ and $w_{j+\ell}^2 g^2/(1 + q' w_{j+\ell}^2 g^2)$ in place of X_j and Y_j gives $e_\ell \le \sqrt{1 - \epsilon_\beta}\sqrt{\mathsf{e}_s\mathsf{e}_u}$. Every entry of L carries exactly one factor b_ℓ or e_ℓ, so $L_{kl} \le \sqrt{1 - \epsilon_\beta}\sqrt{(L_s)_{kl}(L_u)_{kl}}$. The supersolution eq. (SD.70) holds with $\vartheta_t = 0$. Lemma SD.14 with $c = \sqrt{1 - \epsilon_\beta}$ gives $\Delta_\ell \ge (1 - \sqrt{1 - \epsilon_\beta})^2$, and $1 - \sqrt{1 - \epsilon_\beta} \ge \epsilon_\beta/2$.

It remains to show $\epsilon_\beta \in (0, 1]$. Since $\beta > 0$, infinitely many w_j are positive, so r_β increases to infinity and $\hat{q}_\beta < \infty$. Both ratios in the definition of ϵ_β are continuous and positive on the compact interval $[0, \hat{q}_\beta]$: $w_0 = 1 - \beta > 0$, and r'_β and $-F'_\beta$ are finite, positive and continuous on $[0, \infty)$ by dominated convergence, with the majorants $w_j^2 g^2$ and $w_j^4 g^4$. Each numerator is one summand of its denominator, so $\epsilon_\beta \le 1$. □

SD.7.3 Retained pairings as ordered trees

The retained pairings of definition SD.4 are encoded by typed ordered trees, and the cyclic order of the polygon becomes the contour of the tree. In this subsubsection the word exponents of eq. (SD.46) are written a_1, a_2, so that $k = 2a_1 + 2a_2 + 2$, and we write $\mathsf{e}_0 \stackrel{\text{def}}{=} \mathsf{e}_k$ for the marked output vertex. The polygon positions $0, 1, \ldots, 2k$ are listed along $\mathsf{e}_0, \mathsf{l}_1, \mathsf{e}_1, \ldots, \mathsf{l}_k, \mathsf{e}_k = \mathsf{e}_0$, and the polygon step $t \in \{1, \ldots, 2k\}$ joins the positions $t-1$ and t. The vertex e_r is an output vertex if r is even and an input vertex if r is odd, and the latent vertex l_r carries the buffer label $l_r = 1$ if $r \leq 2a_1 + 1$ and $l_r = 2$ otherwise.

Ordered trees and contours. An *ordered tree* is a rooted finite tree in which the children of every vertex are linearly ordered. Its *contour* $c(0), \ldots, c(2n)$, for n edges, is defined recursively: the contour of a vertex x with children $x_1, \ldots, x_m$ is x, the contour of x_1, x, …, the contour of x_m, x. It crosses every edge once downward and once upward. A vertex with m children is visited $m+1$ times; these visits are its *corners*. An ordered tree is determined by its contour.

Lemma SD.16 (Closed walks on trees)**.** *Let T be a finite tree with n edges and $W = (W_0, \ldots, W_{2n})$ a closed walk in T that traverses every edge exactly twice. Root T at W_0 and order the children of every vertex by the time of their first visit. Then W is the contour of this ordered tree.*

Proof. Each subtree is visited during exactly one time interval. Let $e = \{x, x'\}$ be an edge with x' farther from W_0, and T_e the component of $T - e$ containing x'. The walk starts and ends outside T_e and can enter or leave T_e only through e, and entries and exits alternate. Since e is traversed exactly twice, there is one entry and one exit. So W lies in T_e during one time interval I_e, on which it is a closed walk from x' in T_e traversing every edge of T_e exactly twice.

We argue by induction on n, the case $n = 0$ being trivial. Let $x_1, \ldots, x_m$ be the children of the root $v = W_0$, in the order of first visits, and $e_r = \{v, x_r\}$. The intervals I_{e_r} are disjoint and appear in this order. At every time outside them the walk is at v, since the vertex set is $\{v\} \cup \bigcup_r T_{e_r}$. Consecutive vertices of a walk are adjacent, hence distinct, so the walk is at v exactly once between consecutive intervals, once before the first and once after the last. Thus W is v, the walk on I_{e_1}, v, …, the walk on I_{e_m}, v. By induction each inner walk is the contour of T_{e_r} with its children ordered by first visits, and so W is the contour of T. □

Typed ordered trees. A *typed ordered tree* is an ordered tree whose vertices carry the types o, in or lat, such that every edge joins a latent vertex to an external one (of type o or in), and such that at every latent vertex y with parent x and children $x_1, \ldots, x_m$ the cyclic sequence $(x, x_1, \ldots, x_m)$ alternates between the types o and in; in particular m is odd. An edge to an o vertex is a *z-edge* and an edge to an in vertex a *w-edge*. Given the type of the root, an ordered tree has at most one typing: the children of external vertices are latent, and at a latent vertex the alternation fixes the types of the children. Hence the number of typed ordered trees with n edges and a given root type is at most the Catalan number $\frac{1}{n+1}\binom{2n}{n} \leq 4^n$.

Arcs and the G-vertex. Let $\mathcal{T}_{a_1,a_2}$ be the set of typed ordered trees with $k = 2a_1 + 2a_2 + 2$ edges and root of type o. Mark the contour time $p_* \stackrel{\text{def}}{=} 2(2a_1 + 1)$ and call $c(p_*)$ the *G-vertex*. Contour times in $[0, p_*]$ form *arc 1* and times in $[p_*, 2k]$ form *arc 2*; the time p_* belongs to both. A latent corner at the odd time $2r-1$ carries the buffer label 1 if $2r - 1 < p_*$ and 2 otherwise. Consecutive external visits of the contour alternate in type, because at a latent corner the contour passes between cyclically consecutive neighbours. Hence $c(2r)$ has type o if and only if r is even. In particular the G-vertex has type in, so it is not the root.

Lemma SD.17 (Encoding of retained pairings)**.** *For $a_1, a_2 \geq 0$ there is a bijection between the retained pairings of the word W_{a_1,a_2} in definition SD.4 and $\mathcal{T}_{a_1,a_2}$. Under it the vertex classes are the fibres of the contour c on $\{0, \ldots, 2k-1\}$: the polygon vertex e_r lies in the class of $c(2r)$ and l_r in the class of $c(2r-1)$. Consequently, for the class V of a tree vertex x: $p_V = \deg x$ if x is external; $S_V = \{r : c(2r-1) = x\}$ if x is latent, the labels l_r being the corner labels; $\mathsf{e}_0 \in V$ if and only if x is the root; and $\mathsf{e}_{2a_1+1} \in V$ if and only if x is the G-vertex.*

Proof. From pairings to trees. Let (π, π') be retained, with quotient map ϕ and quotient T. Since T is connected with k edges and $k+1$ vertices, it is a tree. Then $W = (\phi(0), \ldots, \phi(2k))$ is a closed

walk in T: each polygon step is mapped to the quotient edge of its matched pair, and every quotient edge is the image of exactly two steps. By lemma SD.16, W is the contour of T rooted at $\phi(0)$ with children ordered by first visits. The classes are type-homogeneous, since matched z-edges identify output vertices with output vertices and latent vertices with latent vertices, and likewise for w-edges; this types T, with root of type o. At a latent visit W_{2r-1} the walk arrives along step $2r-1$ and leaves along step $2r$, whose external endpoints $\mathsf{e}_{r-1}, \mathsf{e}_r$ have opposite types. So at every latent vertex consecutive corners pass between neighbours of opposite types, which in contour order is the cyclic alternation required. Thus $T \in \mathcal{T}_{a_1,a_2}$.

From trees to pairings. Let $T \in \mathcal{T}_{a_1,a_2}$ have contour c, and pair two polygon steps if and only if they traverse the same tree edge. The external endpoint of step t is at the even one of the positions $t-1, t$. It has type o in T exactly when that position is divisible by 4, which is exactly when the polygon edge is a z-edge. So paired steps are of the same kind, and the pairing is a pair (π, π'). Paired steps traverse their edge in opposite directions, so the identifications made by the pairing only identify positions with equal c-values. Hence c factors through the quotient. The quotient is connected with k edges, so it has at most $k+1$ classes, and it maps onto the $k+1$ vertices of T. Thus it has exactly $k+1$ classes, the pairing is retained, and its classes are the fibres of c.

Bijection and class data. In both directions the matched pairs are the pairs of steps over the same edge, and an ordered tree is determined by its contour, so the two maps are mutually inverse. An external vertex x is visited at exactly $\deg x$ even times in $\{0, \dots, 2k-1\}$: a non-root vertex with m children has $m+1 = \deg x$ visits, and the root with $m = \deg x$ children has $m+1$ visits, one of which is the final time $2k$. The remaining assertions are the definitions of the corner labels and of the G-vertex. □

SD.7.4 Summation of the trees

Assignments and values. Fix (ℓ, J, R), diagonal tests F, G and the cutoff paired profile. For $T \in \mathcal{T}_{a_1,a_2}$, an *assignment* $\mathfrak{a}$ gives an index $i \in \{1, \dots, N_{\text{out}}\}$ to every o vertex, an index $j \in \{1, \dots, N_{\text{in}}\}$ to every in vertex and a label $\alpha \in \mathcal{A}$ to every latent vertex. For an external vertex x write k_x for $\mu_{\mathfrak{a}(x)}$ or $\lambda_{\mathfrak{a}(x)}$, according to its type, and put

$$\operatorname{val}(T, \mathfrak{a}) \stackrel{\text{def}}{=} \prod_{x \text{ external}} \frac{\mathsf{k}_x^{\deg x}}{B} F_{\mathfrak{a}(x)}^{[x=\text{root}]} G_{\mathfrak{a}(x)}^{[x=G\text{-vertex}]} \prod_{y \text{ latent}} \mathbb{E} \prod_{\text{corners c of } y} \xi_{l(\mathsf{c}),\mathfrak{a}(y)},$$

where $l(\mathsf{c})$ is the buffer label of the corner and $x^{[P]}$ is x if P holds and 1 otherwise. By lemma SD.17, the value of the retained pairing corresponding to T in definition SD.4 is $\sum_{\mathfrak{a}} \operatorname{val}(T, \mathfrak{a})$, the product of the class sums.

Corner weights. The resolvent filter is generated by weighting each external corner of arc l by a fixed complex number. With $\zeta_1 \stackrel{\text{def}}{=} (i\sqrt{s})^{-1}$ and $\zeta_2 \stackrel{\text{def}}{=} (i\sqrt{u})^{-1}$ put $Z(T) \stackrel{\text{def}}{=} \zeta_1^{n_1}\zeta_2^{n_2}$, where n_l is the number of even contour times in arc l, the time p_* being counted in both arcs. Then $n_1 = 2a_1 + 2$ and $n_2 = 2a_2 + 2$, so

$$Z(T) = (-1)^{a_1+a_2} s^{-a_1-1} u^{-a_2-1}, \tag{SD.71}$$

which is the coefficient of $x^{2a_1+1} \otimes x^{2a_2+1}$ in the expansion of $\iota_s \otimes \iota_u$.

One-buffer trees. Fix a buffer $l \in \{1, 2\}$, a parameter $t > 0$ and the corner weight $\zeta_l = (i\sqrt{t})^{-1}$; in lemma SD.19 we take $t = s$ for $l = 1$ and $t = u$ for $l = 2$. For an externally rooted typed ordered tree X with n edges and root of type $\mathsf{t} \in \{\text{o}, \text{in}\}$, and an assignment $\mathfrak{a}$, put

$$\operatorname{val}_l(X, \mathfrak{a}) \stackrel{\text{def}}{=} \zeta_l^{n+1}\, \mathsf{k}_{\text{root}}^{\deg_X \text{root}} \prod_{x \text{ external},\ x \neq \text{root}} \frac{\mathsf{k}_x^{\deg_X x}}{B} \prod_{y \text{ latent}} \mathbb{E}\left[\xi_{l,\mathfrak{a}(y)}^{\deg_X y}\right],$$

with degrees taken in X. The exponent $n+1$ is the number of external corners of X when X hangs below a latent vertex: the root then has $\deg_X \text{root} + 1$ corners, every other external vertex x has $\deg_X x$ corners, and $\sum_{x \text{ external}} \deg_X x = n$. The one-buffer sums and their traces are

$$g_{\mathsf{t},l}(i) \stackrel{\text{def}}{=} \sum_X \sum_{\mathfrak{a}:\mathfrak{a}(\text{root})=i} \operatorname{val}_l(X, \mathfrak{a}), \quad m_{\text{o},l} \stackrel{\text{def}}{=} \frac{1}{B}\sum_i \mu_i\, g_{\text{o},l}(i), \qquad m_{\text{in},l} \stackrel{\text{def}}{=} \frac{1}{B}\sum_j \lambda_j\, g_{\text{in},l}(j),$$

the first sum over the trees X with root of type t. Put $\gamma_1 \stackrel{\text{def}}{=} \max(1,\gamma_\star)$, $C_1 \stackrel{\text{def}}{=} \max(1,C_\Sigma)$ and $R_1 \stackrel{\text{def}}{=} \max(1,R)$; let D_R be the radius of lemma SD.9; and put

$$s_R \stackrel{\text{def}}{=} \max\big(64\gamma_1^2 C_1^2 R_1^2,\ 2D_R^2\big), \qquad \delta_t \stackrel{\text{def}}{=} \frac{2\gamma_1 C_1}{\sqrt{t}} \quad (t>0).$$

The one-buffer sums are complex numbers, so they cannot be compared with the real branch by monotonicity. Instead both are fixed points of the same map, which is a contraction on a small polydisc for large parameters.

Lemma SD.18 (One-buffer sums are the branch)**.** *Let $\Sigma_{\text{out}},\Sigma_{\text{in}}\neq 0$, fix $l\in\{1,2\}$ and $t\geq s_R$, use the corner weight $\zeta_l=(i\sqrt{t})^{-1}$, and let $(\sigma_l,\tilde\sigma_l,\tilde s_{1,l},s_{1,l})$ be the branch point of the marginal profile ξ_l at t, with $q_l=t\sigma_l\tilde\sigma_l$. Then the families defining $g_{\mathsf{t},l}$ are absolutely summable,*

$$\sum_X \sum_{\mathfrak{a}:\mathfrak{a}(\text{root})=i} |\operatorname{val}_l(X,\mathfrak{a})| \leq \frac{2}{\sqrt{t}},$$

and

$$g_{\text{o},l}(i)=\frac{-i\sqrt{t}}{t+\tilde s_{1,l}\mu_i}, \qquad g_{\text{in},l}(j)=\frac{-i\sqrt{t}}{t+s_{1,l}\lambda_j}, \qquad m_{\text{o},l}=-i\sqrt{t}\,\sigma_l, \qquad m_{\text{in},l}=-i\sqrt{t}\,\tilde\sigma_l,$$

so that $-m_{\text{o},l}m_{\text{in},l}=q_l$.

Proof. Absolute summability. For X with n edges,

$$\sum_{\mathfrak{a}:\mathfrak{a}(\text{root})=i} |\operatorname{val}_l(X,\mathfrak{a})| \leq t^{-(n+1)/2}\, C_1^{\deg \text{root}} \prod_{x\neq \text{root}} \big(\gamma_\star C_1^{\deg x}\big) \prod_y R_1^{\deg y} \leq t^{-(n+1)/2}(\gamma_1 C_1 R_1)^n.$$

Here $B^{-1}\sum_i \mu_i^p \leq \gamma_\star C_1^p$, and $\sum_\alpha \mathbb{E}\,|\xi_{l,\alpha}|^p \leq R^p \sum_\alpha \varpi_{l,\alpha} \leq R_1^p$ for $p\geq 1$ because $\varpi_{l,\alpha}\leq 1$. There are at most n non-root external vertices, and $\sum_{\text{external}} \deg = \sum_{\text{latent}} \deg = n$. There are at most 4^n trees for each n, and $4\gamma_1 C_1 R_1/\sqrt{t}\leq 1/2$. So the total is at most $t^{-1/2}\sum_n 2^{-n} = 2/\sqrt{t}$.

Recursion. An o-rooted tree X is its root together with an ordered list $(Y_1,\dots,Y_m)$ of latent-rooted subtrees, where $m=\deg_X \text{root}$. Each Y_r is a latent vertex whose children $X_{r,1},\dots,X_{r,2p_r+1}$ have the types $\text{in},\text{o},\dots,\text{in}$, by the alternation with the parent type o, and each $X_{r,c}$ is an arbitrary externally rooted tree of that type. The weights factor accordingly. The corner weights give $\zeta_l^{n(X)+1} = \zeta_l^{m+1}\prod_{r,c}\zeta_l^{n(X_{r,c})+1}$; the root contributes μ_i^m; the latent vertex of Y_r contributes $\sum_\alpha \mathbb{E}\,\xi_{l,\alpha}^{2p_r+2}$; and a grandchild root x' contributes $B^{-1}\mathsf{k}_{x'}\cdot \mathsf{k}_{x'}^{\deg_{X_{r,c}} x'}$. By absolute summability and Fubini's theorem,

$$g_{\text{o},l}(i)=\sum_{m\geq 0}\zeta_l^{m+1}\big(\mu_i\,\eta_{\text{o},l}\big)^m, \qquad \eta_{\text{o},l}\stackrel{\text{def}}{=}\sum_{p\geq 0}\sum_\alpha \mathbb{E}\,\big[\xi_{l,\alpha}^{2p+2}\big]\, m_{\text{in},l}^{p+1} m_{\text{o},l}^p,$$

and symmetrically with $\text{o}\leftrightarrow\text{in}$ and $\mu\leftrightarrow\lambda$. By the first step, $|m_{\mathsf{t},l}|\leq \gamma_\star C_1\cdot 2/\sqrt{t}\leq\delta_t$, so $|\xi_{l,\alpha}^2 m_{\text{o},l}m_{\text{in},l}|\leq R^2\delta_t^2\leq 1/16$. Summing the geometric series,

$$\eta_{\text{o},l}=m_{\text{in},l}\,F_{\xi_l}\big(-m_{\text{o},l}m_{\text{in},l}\big),$$

where F_{ξ_l} extends holomorphically to $|q|<R^{-2}$ by the same series, and $g_{\text{o},l}(i)=\big(i\sqrt{t}-\mu_i m_{\text{in},l}F_{\xi_l}(\cdot)\big)^{-1}$, the ratio being less than one by the bounds below. Hence $(m_{\text{o},l},m_{\text{in},l})$ is a fixed point, in the closed polydisc $\mathbb{B}\stackrel{\text{def}}{=}\{|m_{\text{o}}|,|m_{\text{in}}|\leq\delta_t\}$, of the map

$$\Phi_l(m_{\text{o}},m_{\text{in}})\stackrel{\text{def}}{=}\Big(\frac{1}{B}\sum_i \frac{\mu_i}{i\sqrt{t}-\mu_i m_{\text{in}}F_{\xi_l}(-m_{\text{o}}m_{\text{in}})},\ \frac{1}{B}\sum_j\frac{\lambda_j}{i\sqrt{t}-\lambda_j m_{\text{o}}F_{\xi_l}(-m_{\text{o}}m_{\text{in}})}\Big).$$

Contraction. The map Φ_l sends $\mathbb{B}$ into itself and is a contraction there. On $\mathbb{B}$, $|m_{\text{o}}m_{\text{in}}|\leq\delta_t^2\leq 1/(16R_1^2)$, so $|q\xi^2|\leq 1/16$ inside F_{ξ_l}, and by eq. (SD.64)

$$|F_{\xi_l}|\leq\tfrac{16}{15}, \qquad |F'_{\xi_l}|\leq\big(\tfrac{16}{15}\big)^2R^2, \qquad |\mu_i m_{\text{in}}F_{\xi_l}|\leq\tfrac{32}{15}\,\frac{\gamma_1C_1^2}{\sqrt{t}}\leq\frac{\sqrt{t}}{2},$$

the last because $t \ge 64\gamma_1^2 C_1^2 R_1^2 \ge 64\gamma_1 C_1^2$. Every denominator therefore has modulus at least $\sqrt{t}/2$, and $|\Phi_l| \le 2\gamma_\star C_1/\sqrt{t} \le \delta_t$. The partial derivatives of one summand $\mu/(i\sqrt{t} - \mu m_{\mathrm{in}} F_{\xi_l})$ are

$$\frac{\mu^2\big(F_{\xi_l} - m_{\mathrm{o}} m_{\mathrm{in}} F'_{\xi_l}\big)}{(i\sqrt{t} - \mu m_{\mathrm{in}} F_{\xi_l})^2} \quad \text{in } m_{\mathrm{in}}, \qquad \frac{-\mu^2 m_{\mathrm{in}}^2 F'_{\xi_l}}{(i\sqrt{t} - \mu m_{\mathrm{in}} F_{\xi_l})^2} \quad \text{in } m_{\mathrm{o}}.$$

They are bounded by $C_1^2(\frac{16}{15}+0.08)\cdot 4/t$ and $C_1^2\cdot 0.08\cdot 4/t$, because $\delta_t^2(16/15)^2R^2 \le (16/15)^2/16 < 0.08$. Summing over i with the weight $1/B$ and using the mean value inequality on the convex set $\mathbb{B}$, Φ_l is Lipschitz in the maximum norm with constant at most $5\gamma_1 C_1^2/t \le 1/12$. By the Banach fixed-point theorem, Φ_l has exactly one fixed point in $\mathbb{B}$.

The branch is that fixed point. The point $(-i\sqrt{t}\,\sigma_l, -i\sqrt{t}\,\tilde{\sigma}_l)$ lies in $\mathbb{B}$, since $\sqrt{t}\,\sigma_l \le B^{-1}\operatorname{Tr}\Sigma_{\mathrm{out}}/\sqrt{t} \le \gamma_\star C_1/\sqrt{t}$ by lemma SD.13(ii), and likewise for $\tilde{\sigma}_l$. It is a fixed point. Here $-m_{\mathrm{o}} m_{\mathrm{in}} = t\sigma_l\tilde{\sigma}_l = q_l$, and

$$\frac{\mu_i}{i\sqrt{t} + i\sqrt{t}\,\mu_i\tilde{\sigma}_l F_{\xi_l}(q_l)} = \frac{\mu_i}{i\sqrt{t}\,(t + \tilde{s}_{1,l}\mu_i)/t} = -i\sqrt{t}\,\frac{\mu_i}{t + \tilde{s}_{1,l}\mu_i},$$

whose normalized sum over i is $-i\sqrt{t}\,\sigma_l$ by eq. (SD.66); the second coordinate is the same computation. So the tree sums equal the branch, and the formulas for g follow from $g_{\mathfrak{t},l} = (i\sqrt{t} - \mathsf{k}\,\eta_{\mathfrak{t},l})^{-1}$. □

The path. The two buffers interact only along the path from the root, which carries F, to the G-vertex. Everything hanging off this path lies in one arc and sums to the one-buffer sums of lemma SD.18. The path itself is a product of 2×2 transfer matrices, and its sum is a Neumann series. Figure S1 shows the decomposition.

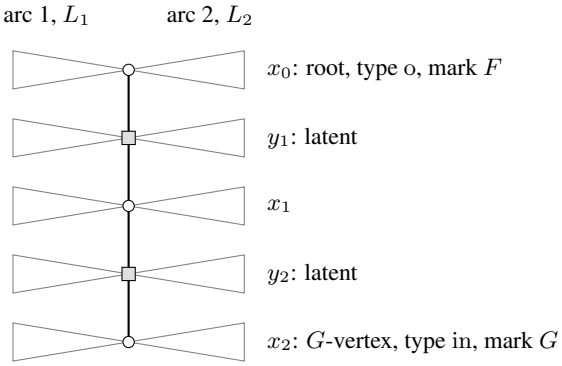


Figure S1: The ancestor path x_0, y_1, x_1, y_2, x_2 of the G-vertex, here with $m = 2$. Each path vertex carries a list L_1 of subtrees visited in arc 1 (left triangles) and a list L_2 visited in arc 2 (right triangles). Subtrees in L_l involve buffer l only. A latent path vertex has $|L_1| + 1$ corners of label 1 and $|L_2| + 1$ of label 2; this is the only place where the two buffers share a sample.

Lemma SD.19 (Path decomposition)**.** *Let $\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}} \ne 0$, and let $s, u \ge s_R$. The family $\{Z(T)\operatorname{val}(T, \mathfrak{a}) : a_1, a_2 \ge 0,\ T \in \mathcal{T}_{a_1,a_2},\ \mathfrak{a}\}$ is absolutely summable, and its sum is*

$$\mathfrak{k}_\ell^{J,R}[F, G](s, u) \stackrel{\mathrm{def}}{=} \mathfrak{R}_\ell^{J,R}[F, G](\iota_s, \iota_u).$$

This sum equals $\mathfrak{K}^{J,R}[F, G](s, u)$ of eq. (SD.69), formed with the branch point of ξ_1 at s and of ξ_2 at u.

Proof. Summability and identification with the kernel. For $T \in \mathcal{T}_{a_1,a_2}$, as in lemma SD.18,

$$\sum_{\mathfrak{a}} |\operatorname{val}(T, \mathfrak{a})| \le \|F\|\|G\|\,\gamma_1(\gamma_1 C_1 R_1)^k.$$

For a latent vertex, $|\sum_\alpha \mathbb{E}\prod \xi| \le R^{\deg}\sum_\alpha \varpi_{l,\alpha} \le R_1^{\deg}$ for either label l that occurs, since all coefficients are at most one. With $|\mathcal{T}_{a_1,a_2}| \le 4^k$ and eq. (SD.71), the total is at most

$$\gamma_1\|F\|\|G\| \sum_{a_1,a_2\ge 0} \Big(\frac{16\gamma_1^2 C_1^2 R_1^2}{s}\Big)^{a_1+1}\Big(\frac{16\gamma_1^2 C_1^2 R_1^2}{u}\Big)^{a_2+1} \le \frac{\gamma_1\|F\|\|G\|}{9}.$$

By lemma SD.17, $\sum_{T\in\mathcal{T}_{a_1,a_2}}\sum_{\mathfrak{a}}\mathrm{val}(T,\mathfrak{a}) = \mathfrak{R}_\ell^{J,R}[F,G](x^{2a_1+1},x^{2a_2+1})$. Since $s > D_R^2$, the series $\iota_s(x)=\sum_a(-1)^a s^{-a-1}x^{2a+1}$ converges uniformly on $[-D_R,D_R]$, and likewise for ι_u. By the bound and the bilinearity of lemma SD.10(i),(ii) and by eq. (SD.71),

$$\begin{aligned}\mathfrak{k}_\ell^{J,R}[F,G](s,u) &= \sum_{a_1,a_2}(-1)^{a_1+a_2}s^{-a_1-1}u^{-a_2-1}\mathfrak{R}_\ell^{J,R}[F,G](x^{2a_1+1},x^{2a_2+1})\\ &= \sum Z(T)\,\mathrm{val}(T,\mathfrak{a}).\end{aligned}$$

Decomposition along the path. For T in $\mathcal{T}\stackrel{\text{def}}{=}\bigcup_{a_1,a_2}\mathcal{T}_{a_1,a_2}$, let $x_0=\mathrm{root},y_1,x_1,\dots,y_m,x_m=G$-vertex be the ancestor path of the G-vertex, with $m\geq 1$, x_r external and y_r latent. For a path vertex $v\neq x_m$ with path child v^+, its list of children is $(L_1(v),v^+,L_2(v))$. For x_m, the time p_* is the corner after its n_G-th child for some $n_G\geq 0$; let $L_1(x_m)$ be its first n_G children and $L_2(x_m)$ the rest. By the recursive definition of the contour, the times before p_* visit exactly the path edges, downward, and the subtrees in the lists L_1; the times after p_* visit the path edges, upward, and the subtrees in the lists L_2. Conversely, the following data assemble to a unique $T\in\mathcal{T}$ with its marked time: the types t_r of the x_r, with $\mathsf{t}_0=\mathrm{o}$ and $\mathsf{t}_m=\mathrm{in}$, and lists $L_1(v),L_2(v)$ of subtrees consistent with the typing. At a latent path vertex y_r the typing requires $(x_{r-1},L_1(y_r),x_r,L_2(y_r))$ to alternate cyclically. So $|L_1(y_r)|$ and $|L_2(y_r)|$ are even if $\mathsf{t}_{r-1}\neq\mathsf{t}_r$ and odd if $\mathsf{t}_{r-1}=\mathsf{t}_r$, and the types inside the lists are forced.

The corners of path vertices split by arc. An external path vertex has $|L_1|+1$ corners in arc 1 and $|L_2|+1$ in arc 2; for the root and the G-vertex the shared corner, at time $0\equiv 2k$ respectively p_*, is counted in both arcs, matching the count in $Z(T)$. A latent path vertex has $|L_1|+1$ corners of label 1 and $|L_2|+1$ of label 2. Subtrees in L_l lie in arc l. Hence $Z(T)\,\mathrm{val}(T,\mathfrak{a})$ is the product of the following local factors, where $\deg_P x\in\{1,2\}$ is the number of path edges at x:

- at an external path vertex x of type t and index i: $B^{-1}\mathsf{k}_i^{\deg_P x}F_i^{[x=x_0]}G_i^{[x=x_m]}\zeta_1^{|L_1|+1}\zeta_2^{|L_2|+1}$, times, for each $Y\in L_l(x)$, the factor k_i and the one-buffer weight of the latent-rooted subtree Y;
- at a latent path vertex with label α: $\mathbb{E}\left[\xi_{1,\alpha}^{|L_1|+1}\xi_{2,\alpha}^{|L_2|+1}\right]$, times, for each $X'\in L_l(y)$ with root x', the factor $B^{-1}\mathsf{k}_{x'}\,\mathrm{val}_l(X',\mathfrak{a})$.

Summation of the local factors. By absolute summability, the sums may be taken factor by factor. At an external path vertex, the recursion of lemma SD.18 sums the subtrees in each list to $\eta_{\mathsf{t},l}$, and the sum over the list lengths gives $\sum_n\zeta_l^{n+1}(\mathsf{k}_i\eta_{\mathsf{t},l})^n = g_{\mathsf{t},l}(i)$. So the external factor sums to

$$\frac{1}{B}\mathsf{k}_i^{\deg_P x}F_i^{[x=x_0]}G_i^{[x=x_m]}\,g_{\mathsf{t},1}(i)\,g_{\mathsf{t},2}(i).$$

At a latent path vertex between the types $(\mathsf{t}_{r-1},\mathsf{t}_r)$, a list of even length $2p$ between opposite types contributes $(m_{\mathrm{o},l}m_{\mathrm{in},l})^p=(-q_l)^p$, and a list of odd length $2p+1$ between two o vertices contributes $m_{\mathrm{in},l}(-q_l)^p$, where $q_l=-m_{\mathrm{o},l}m_{\mathrm{in},l}$ is the branch value of lemma SD.18. Summing the geometric series, which converge because $|q_l\xi^2|\leq 1/16$, the latent factor sums to the entry $\mathsf{Y}_{\mathsf{t}_{r-1}\mathsf{t}_r}$ of

$$\mathsf{Y}\stackrel{\text{def}}{=}\begin{pmatrix}\mathsf{a} & b\\ b & \mathsf{d}\end{pmatrix},\qquad \mathsf{a}\stackrel{\text{def}}{=}m_{\mathrm{in},1}m_{\mathrm{in},2}\,e,\qquad \mathsf{d}\stackrel{\text{def}}{=}m_{\mathrm{o},1}m_{\mathrm{o},2}\,e,$$

indexed by (o,in), with $b=b(q_1,q_2)$ and $e=e(q_1,q_2)$. For example, between opposite types,

$$\sum_\alpha\mathbb{E}\sum_{p_1,p_2\geq 0}\xi_{1,\alpha}^{2p_1+1}\xi_{2,\alpha}^{2p_2+1}(-q_1)^{p_1}(-q_2)^{p_2}=\sum_\alpha\mathbb{E}\,\frac{\xi_{1,\alpha}\xi_{2,\alpha}}{(1+q_1\xi_{1,\alpha}^2)(1+q_2\xi_{2,\alpha}^2)}=b.$$

Put

$$f_{\mathrm{o}}\stackrel{\text{def}}{=}\frac{1}{B}\sum_i F_i\mu_i\,g_{\mathrm{o},1}(i)g_{\mathrm{o},2}(i),\qquad f_{\mathrm{in}}\stackrel{\text{def}}{=}\frac{1}{B}\sum_j G_j\lambda_j\,g_{\mathrm{in},1}(j)g_{\mathrm{in},2}(j),$$

$$p_{\mathrm{o}}\stackrel{\text{def}}{=}\frac{1}{B}\sum_i\mu_i^2\,g_{\mathrm{o},1}(i)g_{\mathrm{o},2}(i),\qquad p_{\mathrm{in}}\stackrel{\text{def}}{=}\frac{1}{B}\sum_j\lambda_j^2\,g_{\mathrm{in},1}(j)g_{\mathrm{in},2}(j),\qquad \mathsf{P}\stackrel{\text{def}}{=}\mathrm{diag}(p_{\mathrm{o}},p_{\mathrm{in}}).$$

Summing over the path length m and the intermediate types,

$$\sum Z(T)\,\mathrm{val}(T,\mathfrak{a}) = f_{\mathrm{o}}\Big[\sum_{m\ge 1}\mathsf{Y}(\mathsf{PY})^{m-1}\Big]_{\mathrm{o,in}} f_{\mathrm{in}} = f_{\mathrm{o}}\big[\mathsf{Y}(\mathrm{Id}-\mathsf{PY})^{-1}\big]_{\mathrm{o,in}} f_{\mathrm{in}}$$
$$= \frac{f_{\mathrm{o}} f_{\mathrm{in}}\, b}{(1-p_{\mathrm{o}}\mathsf{a})(1-p_{\mathrm{in}}\mathsf{d}) - p_{\mathrm{o}}p_{\mathrm{in}}b^2}.$$

The Neumann series converges. Indeed $|g| \le 2/\sqrt{t}$ gives $|p_{\mathsf{t}}| \le 4\gamma_1 C_1^2/\sqrt{su} \le 1/16$, while $|b| \le 1$ and $|\mathsf{a}|, |\mathsf{d}| \le \delta_s\delta_u R^2 \le 1/16$, so $\|\mathsf{PY}\|_\infty < 1/8$. The last equality is 2×2 inversion: the (o,in) entry of $\mathsf{Y}\,\mathrm{adj}(\mathrm{Id}-\mathsf{PY})$ is $\mathsf{a}\,p_{\mathrm{o}}b + b(1-p_{\mathrm{o}}\mathsf{a}) = b$.

Conversion to real variables. By lemma SD.18 at s for $l=1$ and at u for $l=2$,

$$g_{\mathrm{o},1}(i)g_{\mathrm{o},2}(i) = \frac{-\sqrt{su}}{(s+\tilde{s}_1\mu_i)(u+\tilde{s}'_1\mu_i)}, \qquad g_{\mathrm{in},1}(j)g_{\mathrm{in},2}(j) = \frac{-\sqrt{su}}{(s+s_1\lambda_j)(u+s'_1\lambda_j)}.$$

Hence

$$f_{\mathrm{o}} = -\sqrt{su}\,\mathcal{T}^{\mathrm{out}}[F], \quad f_{\mathrm{in}} = -\sqrt{su}\,\mathcal{T}^{\mathrm{in}}[G], \quad p_{\mathrm{o}} = -\sqrt{su}\,\mathsf{U}, \quad p_{\mathrm{in}} = -\sqrt{su}\,\mathsf{V},$$

$\mathsf{a} = -\sqrt{su}\,\tilde{\sigma}\tilde{\sigma}'e$, $\mathsf{d} = -\sqrt{su}\,\sigma\sigma' e$, $q_1 = q$ and $q_2 = q'$. Substituting gives $su\,b\,\mathcal{T}^{\mathrm{out}}[F]\mathcal{T}^{\mathrm{in}}[G]/\Delta^{J,R} = \mathfrak{K}^{J,R}[F,G](s,u)$. □

SD.7.5 Proofs of the statements

Proof of theorem SD.5. The proof computes the cutoff kernel by summing trees for large parameters, continues the identity analytically to all parameters, and removes the cutoffs through the stability of the branch. By eq. (SD.49) and eq. (SD.51), $\mathfrak{k}_\ell[F,G](s,u)$ is the limit of $\mathfrak{k}_\ell^{J,R}[F,G](s,u) = \mathfrak{R}_\ell^{J,R}[F,G](\iota_s,\iota_u)$ as $J,R\to\infty$ independently, and this limit exists by theorem SD.3.

Step 0: degenerate covariances. If $\Sigma_{\mathrm{out}} = 0$, every retained pairing has an output class with value $B^{-1}\sum_i \mu_i^p(\cdot) = 0$, with $p \ge 1$. So $\mathfrak{R}_\ell^{J,R}[F,G]$ vanishes on polynomials, hence on continuous functions, and $\mathfrak{k}_\ell^{J,R} = \mathfrak{k}_\ell = 0$. By lemma SD.15(a) the closed form is 0 and $\Delta_\ell = 1$. The case $\Sigma_{\mathrm{in}} = 0$ is the same with the input class of the marked vertex e_{2a+1}. From now on $\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}} \ne 0$.

Step 1: the cutoff identity for large parameters. By lemma SD.19, $\mathfrak{k}_\ell^{J,R}[F,G] = \mathfrak{K}^{J,R}[F,G]$ on $[s_R,\infty)^2$.

Step 2: the cutoff identity for all parameters. Both sides are real-analytic in each variable on $(0,\infty)$. For the left side, fix u and $s_0 > 0$. For $|s-s_0| < s_0$,

$$\iota_s(x) = \sum_{n\ge 0}(s_0-s)^n \frac{x}{(x^2+s_0)^{n+1}}, \qquad \sup_{x\in\mathbb{R}}\frac{|x|}{(x^2+s_0)^{n+1}} \le \frac{s_0^{-n}}{2\sqrt{s_0}},$$

so the series converges in the supremum norm. By the bound and the bilinearity of lemma SD.10(i),(ii), $s\mapsto \mathfrak{k}_\ell^{J,R}(s,u)$ is a convergent power series around each s_0; the same holds in u. For the right side, the branch of ξ_1 is real-analytic in s with $q_1(s) > 0$, by lemma SD.13(iii). The traces $\mathcal{T}^{\mathrm{out}}[F]$, $\mathcal{T}^{\mathrm{in}}[G]$, U and V are finite sums of reciprocals of positive analytic functions. The functions b and e are analytic in $q_1 > 0$ by lemma SD.12(c). And $\Delta^{J,R} > 0$ by lemma SD.15(b), which applies because $F_{\xi_l}(0) \le 1$ by eq. (SD.64). So $\mathfrak{K}^{J,R}(\cdot,u)$ is real-analytic on $(0,\infty)$, and likewise in u. A real-analytic function of one variable on an interval that vanishes on a subinterval vanishes identically: the set of points where all its derivatives vanish is open by [55, Theorem 8.5], closed, and nonempty. Fix $u \ge s_R$: the two sides agree on $[s_R,\infty)$, hence on $(0,\infty)$. Then fix any $s > 0$ and argue in u. Hence $\mathfrak{k}_\ell^{J,R}[F,G] = \mathfrak{K}^{J,R}[F,G]$ on $(0,\infty)^2$.

Step 3: removal of the cutoffs. The marginal profiles have, since $\beta^\ell w_j = w_{j+\ell}$,

$$F_{\xi_1}(q) = \sum_{j\le J} w_j^2\, \mathbb{E}\,\frac{\tilde{g}^2}{1+qw_j^2\tilde{g}^2}, \qquad F_{\xi_2}(q) = \sum_{k\le J+\ell} w_k^2\, \mathbb{E}\,\frac{\tilde{g}^2}{1+qw_k^2\tilde{g}^2}.$$

Since $0 \le \frac{g^2}{1+xg^2} - \frac{\tilde{g}^2}{1+x\tilde{g}^2} \le g^2\mathbf{1}_{|g|>R}$ for $x \ge 0$,

$$\sup_{q\ge 0}|F_{\xi_l}(q) - F_\beta(q)| \le \sum_{k>J} w_k^2 + s_2\,\tau_R \longrightarrow 0,$$

with τ_R of eq. (SD.44), and $F_{\xi_l}(0) \le 1$. By lemma SD.13(iv) the cutoff branch points converge to the EMA branch points at s and at u, and the traces $\mathcal{T}^{\mathrm{out}}[F]$, $\mathcal{T}^{\mathrm{in}}[G]$, U, V are continuous in the branch coordinates. The fresh labels have $\xi_1 = 0$ and do not contribute to b or e, so

$$b^{J,R}(q_1, q_2) = \sum_{j \le J} \mathbb{E}\, \frac{\beta^\ell w_j^2 \tilde{g}^2}{(1 + q_1 w_j^2 \tilde{g}^2)(1 + q_2 \beta^{2\ell} w_j^2 \tilde{g}^2)},$$

and similarly for $e^{J,R}$. The summands converge pointwise as the cutoffs are removed and the arguments converge, with the summable majorants $\beta^\ell w_j^2 g^2$ and $\beta^{2\ell} w_j^4 g^4$. Dominated convergence over (j, g) gives $b^{J,R} \to b_\ell$ and $e^{J,R} \to e_\ell$. The limit determinant Δ_ℓ is positive by lemma SD.15(b) for the EMA paired profile, whose marginal scalar functions have $F_\beta(0) = s_2 \le 1$. Hence $\mathfrak{K}^{J,R} \to \mathfrak{K}_\ell$ along every sequence with $J, R \to \infty$, which is the double limit. With Step 2 this proves (ii), and (i) is lemma SD.15(a),(b). □

Proof of corollary SD.2. For (i), continuity follows from the closed form. The branch is continuous in the parameter by lemma SD.13(iii), b_ℓ and e_ℓ are jointly continuous in (q, q') by lemma SD.12(c), and $\Delta_\ell > 0$. For positive semidefinite F, G all factors are nonnegative. At $\beta = 0$ and $\ell \ge 1$, $b_\ell = 0$.

For the first bound of (ii), $\|\iota_s\|_\infty = 1/(2\sqrt{s})$ by eq. (SD.5), and the response is bounded by $\gamma_\star \|F\| \|G\| \|f_1\|_\infty \|f_2\|_\infty$ by theorem SD.3. For the second,

$$|\mathcal{T}^{\mathrm{out}}[F]| \le \|F\| \frac{1}{B} \sum_i \frac{\mu_i}{(s + \tilde{s}_1 \mu_i)(u + \tilde{s}_1' \mu_i)} \le \frac{\|F\| \operatorname{Tr} \Sigma_{\mathrm{out}}}{B\, su},$$

likewise $|\mathcal{T}^{\mathrm{in}}[G]| \le \|G\| \operatorname{Tr} \Sigma_{\mathrm{in}}/(B\, su)$, and $b_\ell \le s_2 \beta^\ell$ and $\Delta_\ell \ge \theta_\Sigma(\min(s,u))^2$ by lemma SD.12(d) and theorem SD.5(i).

For (iii), the Cauchy–Schwarz inequality with $\mu_i = \sqrt{\mu_i}\sqrt{\mu_i}$ gives $|\mathcal{T}^{\mathrm{out}}[F]| \le \|F\| \sqrt{\Theta_s \Theta_u'}$, where Θ_s is the quantity of the proof of lemma SD.15 and Θ_u' the same quantity for the primed branch. Termwise $t\Theta_t \le \sigma_t$, so $|\mathcal{T}^{\mathrm{out}}[F]| \le \|F\| \sqrt{\sigma\sigma'/(su)}$, and likewise $|\mathcal{T}^{\mathrm{in}}[G]| \le \|G\| \sqrt{\tilde{\sigma}\tilde{\sigma}'/(su)}$. Since $q = s\sigma\tilde{\sigma}$ and $q' = u\sigma'\tilde{\sigma}'$,

$$su\, |\mathcal{T}^{\mathrm{out}}[F]\, \mathcal{T}^{\mathrm{in}}[G]| \le \|F\| \|G\| \sqrt{\sigma\tilde{\sigma}\sigma'\tilde{\sigma}'} = \|F\| \|G\| \frac{\sqrt{qq'}}{\sqrt{su}} \le \|F\| \|G\| \frac{\hat{q}_\beta}{\sqrt{su}},$$

using $q, q' < \hat{q}_\beta$ from lemma SD.13(ii). With $b_\ell \le s_2 \beta^\ell$ and lemma SD.15(c) this is (iii); if a covariance vanishes, $\mathfrak{k}_\ell = 0$ by Step 0 of the previous proof.

Part (iv) combines the first bound of (ii) for $\ell = 0$, part (iii) for $\ell \ge 1$, and part (i) at $\beta = 0$. □

Proof of proposition SD.1. For (a), take $\ell = 0$, $F = \Sigma_{\mathrm{out}}$ and $G = \Sigma_{\mathrm{in}}$. Then $\mathcal{T}^{\mathrm{out}}[F] = \mathsf{U} = \mathcal{T}_2^{\mathrm{out}}(s,u)$ and $\mathcal{T}^{\mathrm{in}}[G] = \mathsf{V} = \mathcal{T}_2^{\mathrm{in}}(s,u)$ in the notation of eq. (SD.84), and both buffers carry F_β.

On the diagonal $s = u$, lemma SD.12(d) gives $b_0 = r_\beta'(q)$ and $e_0 = -F_\beta'(q)$. So $\Delta_0(s,s) = \Upsilon(s)$ of eq. (SD.85), and the closed form is $s^2 r_\beta'(q) \mathcal{T}_2^{\mathrm{out}}(s,s) \mathcal{T}_2^{\mathrm{in}}(s,s)/\Upsilon(s) = \mathfrak{k}(s,s)$ literally.

Off the diagonal the identity is a rational identity, which we prove without dividing by quantities that might vanish. Let $s \ne u$, write $F_1 = F_\beta(q)$, $F_2 = F_\beta(q')$, $\rho_1 = r_\beta(q)$ and $\rho_2 = r_\beta(q')$, so that $\rho_1 = \tilde{s}_1 \sigma = s_1 \tilde{\sigma}$ and $\rho_2 = \tilde{s}_1' \sigma' = s_1' \tilde{\sigma}'$ by eq. (SB.4), and put $J_{\mathrm{o}} \stackrel{\mathrm{def}}{=} \mathcal{T}_1^{\mathrm{out}}(s,u) > 0$. The resolvent identities

$$\overline{R}(-s) - \overline{R}(-u) = \overline{R}(-s)\big((u-s)\,\mathrm{Id} + (\tilde{s}_1' - \tilde{s}_1)\Sigma_{\mathrm{out}}\big)\overline{R}(-u),$$
$$u\overline{R}(-u) - s\overline{R}(-s) = \overline{R}(-s)\,(u\tilde{s}_1 - s\tilde{s}_1')\Sigma_{\mathrm{out}}\,\overline{R}(-u),$$

traced against Σ_{out}/B, and the analogous identity for Σ_{in}, give

$$\begin{aligned} \sigma - \sigma' &= (u-s)J_{\mathrm{o}} + (\tilde{s}_1' - \tilde{s}_1)\mathsf{U}, \\ u\sigma' - s\sigma = (u\tilde{s}_1 - s\tilde{s}_1')\mathsf{U}, &\qquad u\tilde{\sigma}' - s\tilde{\sigma} = (us_1 - ss_1')\mathsf{V}. \end{aligned} \tag{SD.72}$$

Put

$$\mathsf{A} \stackrel{\mathrm{def}}{=} \frac{u\tilde{s}_1 - s\tilde{s}_1'}{su} = \tilde{\sigma}F_1 - \tilde{\sigma}'F_2, \qquad \mathsf{C} \stackrel{\mathrm{def}}{=} \frac{us_1 - ss_1'}{su} = \sigma F_1 - \sigma' F_2.$$

Lemma SD.12(d) at $\ell = 0$ gives the partial-fraction identities $b_0(q - q') = \rho_1 - \rho_2$ and $e_0(q' - q) = F_1 - F_2$. Expanding with eq. (SD.72), $q = s\sigma\tilde\sigma$ and $q' = u\sigma'\tilde\sigma'$ gives

$$\begin{aligned} su\mathsf{A}\,(u-s)J_{\mathrm{o}} &= (u\tilde s_1 - s\tilde s_1')(\sigma-\sigma') + (\tilde s_1 - \tilde s_1')(u\sigma' - s\sigma) = (u-s)(\rho_1-\rho_2),\\ \mathsf{A}(q'-q)\big(1 - su\,\mathsf{U}\tilde\sigma\tilde\sigma' e_0\big) &= \mathsf{A}(q'-q) - (u\sigma'-s\sigma)\tilde\sigma\tilde\sigma'(F_1-F_2) = (\tilde\sigma'-\tilde\sigma)(\rho_1-\rho_2),\\ \mathsf{C}(q'-q)\big(1 - su\,\mathsf{V}\sigma\sigma' e_0\big) &= (\sigma'-\sigma)(\rho_1-\rho_2). \end{aligned}$$

In the second line the four terms in F_1, F_2 cancel in pairs and leave $s\sigma\tilde\sigma F_1(\tilde\sigma'-\tilde\sigma) + u\sigma'\tilde\sigma' F_2(\tilde\sigma-\tilde\sigma')$; the third line is symmetric. Multiplying the second and third lines and subtracting

$$\mathsf{AC}(q-q')^2 su\,\mathsf{UV} b_0^2 = \frac{(su\mathsf{AU})(su\mathsf{CV})(\rho_1-\rho_2)^2}{su}$$

gives

$$\begin{aligned} \mathsf{AC}(q-q')^2\Delta_0 &= (\rho_1-\rho_2)^2\Big[(\sigma-\sigma')(\tilde\sigma-\tilde\sigma') - \frac{(u\sigma'-s\sigma)(u\tilde\sigma'-s\tilde\sigma)}{su}\Big]\\ &= \frac{(\rho_1-\rho_2)^2(u-s)(q-q')}{su}, \end{aligned}$$

the bracket being $(u-s)(s\sigma\tilde\sigma - u\sigma'\tilde\sigma')/(su)$. By lemma SD.13(iii), $q \neq q'$. Hence $\rho_1 \neq \rho_2$, because r_β is strictly increasing, and

$$\mathsf{AC}(q-q')\Delta_0 = \frac{(\rho_1-\rho_2)^2(u-s)}{su} \neq 0. \tag{SD.73}$$

In particular $su\,\mathsf{AC}(q-q') \neq 0$. Multiply the target identity $su\,b_0(u-s)J_{\mathrm{o}} = (us_1 - ss_1')\Delta_0$ by this number. The left side becomes $su\,\mathsf{C}\,[b_0(q-q')]\,[su\mathsf{A}(u-s)J_{\mathrm{o}}] = su\,\mathsf{C}(\rho_1-\rho_2)^2(u-s)$, by the first line. The right side becomes $(us_1 - ss_1')\,su\,[\mathsf{AC}(q-q')\Delta_0] = su\,\mathsf{C}(\rho_1-\rho_2)^2(u-s)$, by eq. (SD.73) and $us_1 - ss_1' = su\mathsf{C}$. So the target identity holds, and since $\Delta_0 > 0$ and $J_{\mathrm{o}} > 0$,

$$\frac{su\,b_0\,\mathsf{UV}}{\Delta_0} = \mathcal{T}_2^{\mathrm{out}}\,\mathcal{T}_2^{\mathrm{in}}\,\frac{us_1 - ss_1'}{(u-s)\mathcal{T}_1^{\mathrm{out}}} = \mathfrak{k}(s,u),$$

with $s_1 = s_1(-s)$ and $s_1' = s_1(-u)$, which is eq. (SD.86). With theorem SD.5(ii) this gives $\mathfrak{k}_0[\Sigma_{\mathrm{out}}, \Sigma_{\mathrm{in}}] = \mathfrak{k}$.

For (b), with $\Sigma_{\mathrm{in}} = \mathrm{Id}_{N_{\mathrm{in}}}$, $\overline{\overline{R}}(-s) = (s+s_1)^{-1}\,\mathrm{Id}$ and $\overline{\overline{R}}(-u) = (u+s_1')^{-1}\,\mathrm{Id}$, so

$$\mathcal{T}^{\mathrm{in}}[\mathrm{Id}] = \mathsf{V} = \frac{N_{\mathrm{in}}/B}{(s+s_1)(u+s_1')}, \qquad \mathcal{T}^{\mathrm{out}}[u_i \otimes u_i] = \frac{1}{B}\,\frac{\mu_i}{(s+\tilde s_1\mu_i)(u+\tilde s_1'\mu_i)}.$$

Substituting into eq. (SD.58) and multiplying by B gives eq. (SD.59). For $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ the quantities U, V, b_ℓ, e_ℓ and Δ_ℓ are those of eq. (SG.14) with $\gamma = N/B$, symbol by symbol, and the right side is that of eq. (SG.16). □

Proof of theorem SD.6. The closed form gives an integrable majorant, and the square-truncated identity of theorem SD.4(i) passes to the full quadrant. Put

$$M_\ell \overset{\mathrm{def}}{=} s_2\,\frac{\operatorname{Tr}\Sigma_{\mathrm{out}}\,\operatorname{Tr}\Sigma_{\mathrm{in}}}{B^2}\,\|F\|\|G\|\Big(1+\frac{\kappa_\Sigma}{\varrho_m}\Big)^2\beta^\ell.$$

For $s \geq \varrho$ and $u \geq \varrho'$, $\theta_\Sigma(\min(s,u)) \geq \theta_\Sigma(\varrho_m)$ and $\theta_\Sigma(\varrho_m)^{-1} = 1 + \kappa_\Sigma/\varrho_m$, so the second bound of corollary SD.2(ii) gives

$$|\mathfrak{k}_\ell[F,G](s,u)| \leq \frac{M_\ell}{su} \qquad (s \geq \varrho,\ u \geq \varrho'). \tag{SD.74}$$

At $(s,u) = (\varrho + x^2, \varrho' + y^2)$ the right side is integrable on $(0,\infty)^2$, with

$$\int_0^\infty\!\!\int_0^\infty \frac{dx\,dy}{(\varrho+x^2)(\varrho'+y^2)} = \frac{\pi^2}{4\sqrt{\varrho\varrho'}}.$$

So the double integral in eq. (SD.60) converges absolutely, and by dominated convergence it is the limit as $T \to \infty$ of the integrals over $[0,T]^2$. Apply theorem SD.4(i) with the compact interval $I = [\varrho_m, \max(\varrho,\varrho')]$; its error bound tends to zero as $T \to \infty$. This proves eq. (SD.60). Integrating eq. (SD.74) gives $|\overline{C}_\ell[F,G](\varrho,\varrho')| \leq M_\ell/\sqrt{\varrho\varrho'}$, which is the bound. □

Proof of corollary SD.3. For (i),

$$\mathcal{T}^{\mathrm{out}}[P_\mu] = \frac{r_\mu}{B}\,\frac{\mu}{(s+\tilde{s}_1\mu)(u+\tilde{s}'_1\mu)}, \qquad \mathcal{T}^{\mathrm{in}}[P'_\lambda] = \frac{t_\lambda}{B}\,\frac{\lambda}{(s+s_1\lambda)(u+s'_1\lambda)}.$$

Inserting these into eq. (SD.58) gives the factorization of the kernel. Then $\mathfrak{h}_\ell \geq 0$ because $b_\ell \geq 0$ and $\Delta_\ell > 0$. The factorization of $\overline{C}_\ell[P_\mu, P'_\lambda](\varrho,\varrho)$ follows from theorem SD.6 and the definition eq. (SD.62).

For (ii), every denominator factor in eq. (SD.61) other than Δ_ℓ is at least s or u, so

$$\mathfrak{h}_\ell(\mu,\lambda;s,u) \leq \frac{\mu\lambda\, b_\ell}{su\,\Delta_\ell} \leq \frac{\mu\lambda\, s_2\,\beta^\ell}{su\,\theta_\Sigma(\min(s,u))^2},$$

with $\min(s,u) \geq \varrho$ at $(\varrho+x^2, \varrho+y^2)$. Integrating as in the previous proof gives the bound.

For (iii), at equal ridges eq. (SD.54) reads $\overline{\Gamma}[P_\mu, P'_\lambda](\varrho,\varrho) = \sum_\ell c_\ell\, \overline{C}_\ell[P_\mu, P'_\lambda](\varrho,\varrho)$, which converges absolutely by (ii). By eqs. (SD.78) and (SD.80) and part (i),

$$\mathfrak{v}_{ij}(\varrho) = \frac{B\,\overline{\Gamma}[P_{\mu_i}, P'_{\lambda_j}](\varrho,\varrho)}{r_{\mu_i} t_{\lambda_j}} = \frac{B}{r_{\mu_i} t_{\lambda_j}} \cdot \frac{r_{\mu_i} t_{\lambda_j}}{B^2} \sum_{\ell\geq 0} c_\ell\, \mathfrak{c}_\ell(\mu_i,\lambda_j;\varrho),$$

which is eq. (SD.63). The bound follows from (ii) and $\sum_\ell c_\ell \beta^\ell = \frac{1+\beta}{1-\beta}$. □

Proof of proposition SD.2. The branch, $\mathcal{T}^{\mathrm{out}}[F]$, $\mathcal{T}^{\mathrm{in}}[G]$, U and V do not depend on ℓ. Expanding eq. (SD.57) gives

$$\Delta_\ell = 1 - (A_{\mathrm{o}} + A_{\mathrm{in}})e_\ell + A_{\mathrm{o}}A_{\mathrm{in}}e_\ell^2 - A_\times b_\ell^2,$$

and theorem SD.5(ii) then gives the first display. The identity $w_{j+\ell} = \beta^\ell w_j$ turns eq. (SD.55) into the forms with ϕ_q and ψ_q. For the lag sum, eq. (SD.74) with $\varrho = \varrho'$ gives

$$|c_\ell\, \mathfrak{k}_\ell[F,G](\varrho+x^2,\varrho+y^2)| \leq \frac{2M_\ell}{(\varrho+x^2)(\varrho+y^2)}, \qquad \sum_\ell M_\ell < \infty.$$

So the series converges absolutely, its sum is dominated by an integrable function, and Fubini's theorem exchanges $\sum_\ell$ with the double integral in $\overline{\Gamma}[F,G](\varrho,\varrho) = \sum_\ell c_\ell\, \overline{C}_\ell[F,G](\varrho,\varrho)$, each term being given by theorem SD.6. □

SD.8 Mode-resolved coefficients

The tests of theorems SD.3 and SD.4 are arbitrary diagonal matrices of norm at most one. Testing with signs gives statements in a normalized absolute-sum topology over the rows, and testing with spectral projections gives a measure on the spectral rectangle.

Rows, for $\Sigma_{\mathrm{in}} = \mathrm{Id}$. For $\Sigma_{\mathrm{in}} = \mathrm{Id}$ the deterministic row coefficients are

$$\begin{aligned} \overline{C}_{i,\ell}(\varrho,\varrho') &\stackrel{\mathrm{def}}{=} B\,\overline{C}_\ell[u_i \otimes u_i, \mathrm{Id}](\varrho,\varrho'), \\ \mathfrak{v}_i(\varrho,\varrho') &\stackrel{\mathrm{def}}{=} \overline{C}_{i,0}(\varrho,\varrho') + \sum_{\ell\geq 1}\Big(\overline{C}_{i,\ell}(\varrho,\varrho') + \overline{C}_{i,\ell}(\varrho',\varrho)\Big), \end{aligned} \tag{SD.75}$$

and $\mathfrak{v}_i(\varrho) \stackrel{\mathrm{def}}{=} \mathfrak{v}_i(\varrho,\varrho)$. They are the deterministic counterparts of the row covariances $C_{i,\ell}$ and their integrated sums Γ_i of eq. (SD.18).

Theorem SD.7 (Mode-resolved coefficients, $\Sigma_{\mathrm{in}} = \mathrm{Id}$)**.** *Let $\Sigma_{\mathrm{in}} = \mathrm{Id}_{N_{\mathrm{in}}}$.*

(i) Under Assumption S3, for every fixed $\ell \geq 0$,

$$\sup_{\varrho,\varrho'\in I} \frac{1}{B} \sum_{i=1}^{N_{\mathrm{out}}} |C_{i,\ell}(\varrho,\varrho') - \overline{C}_{i,\ell}(\varrho,\varrho')| \longrightarrow 0. \tag{SD.76}$$

(ii) At every finite dimension, $B^{-1}\sum_i |\overline{C}_{i,0}| \leq \gamma_\star$ and $B^{-1}\sum_i |\overline{C}_{i,\ell}| \leq \mathsf{K}_\star \beta^\ell$ for $\ell \geq 1$, and the same bounds hold for $C_{i,\ell}$ under Assumption S3.

(iii) Each series $\mathfrak{v}_i$ converges absolutely, $\mathfrak{v}_i(\varrho) \geq 0$, and under Assumption S3

$$\sup_{\varrho,\varrho' \in I} \frac{1}{B} \sum_{i=1}^{N_{\text{out}}} |\Gamma_i(\varrho,\varrho') - \mathfrak{v}_i(\varrho,\varrho')| \longrightarrow 0. \tag{SD.77}$$

(iv) The coefficients $\overline{C}_{i,\ell}$ and $\mathfrak{v}_i$ depend on i only through μ_i. They vanish when $\mu_i = 0$, and so do $C_{i,\ell}$ and Γ_i. At $\beta = 0$, $\overline{C}_{i,\ell} = 0$ for $\ell \geq 1$. At zero lag and equal ridges, $\overline{C}_{i,0}(\varrho,\varrho) = \mathfrak{v}_i^{(0)}(\varrho)$ of eq. (SD.22).

The theorem is a statement in the normalized absolute-sum topology $B^{-1}\sum_i |\cdot|$, and the same convergence holds with the weights $\mu_i \leq C_\Sigma$. No uniform accuracy for individual rows is claimed: the cutoff error eq. (SD.50) for the rank-one test $u_i \otimes u_i$ is multiplied by B in eq. (SD.75), so the construction controls each row at fixed dimension but not uniformly in the dimension. In the recursion eq. (SA.21) the row coefficients enter through weighted sums over i, and eq. (SD.77) controls exactly such sums with bounded weights.

Proof. For signs $\epsilon = (\epsilon_i) \in \{\pm 1\}^{N_{\text{out}}}$ let $F_\epsilon \stackrel{\text{def}}{=} \sum_i \epsilon_i\, u_i \otimes u_i$, a diagonal symmetric test of norm one. By eq. (SD.18) and the linearity of $\overline{C}_\ell$ in F,

$$\frac{1}{B}\sum_i \epsilon_i \big(C_{i,\ell} - \overline{C}_{i,\ell}\big) = C_\ell[F_\epsilon, \mathrm{Id}] - \overline{C}_\ell[F_\epsilon, \mathrm{Id}],$$

and the left side of eq. (SD.76) is the supremum of the right side over ϵ and the ridges; the actual clause of theorem SD.3 proves (i). The same identity with the bounds of theorem SD.2(ii) and theorem SD.4(ii) proves (ii). For (iii), each $|\overline{C}_{i,\ell}| \leq B\mathsf{K}_\star\beta^\ell$ at fixed dimension, which gives absolute convergence; positivity is theorem SD.4(iii) with $F = u_i \otimes u_i$, $G = \mathrm{Id}$; and eq. (SD.77) is theorem SD.4(iv) for the tests F_ϵ.

For (iv), the tree values of definition SD.4 with $F = u_i \otimes u_i$ contain the mark through the single factor $B^{-1}\mu_i^{p_V}$ with $p_V \geq 1$, which depends on i only through μ_i and vanishes when $\mu_i = 0$; both properties pass to the extension and the limits. The actual row vanishes when $\mu_i = 0$ by theorem SD.1(ii). The case $\beta = 0$ is theorem SD.4(ii). The zero-lag value is eq. (SD.82) of corollary SD.4. □

The joint forcing measure. For an eigenvalue μ of Σ_{out} and an eigenvalue λ of Σ_{in} let $P_\mu \stackrel{\text{def}}{=} \sum_{i:\mu_i=\mu} u_i \otimes u_i$ and $P'_\lambda \stackrel{\text{def}}{=} \sum_{j:\lambda_j=\lambda} v_j \otimes v_j$. On the rectangle $[0, C_\Sigma]^2$ we define the finite signed measures

$$\begin{aligned}
\mathfrak{C}_\ell(\varrho,\varrho') &\stackrel{\text{def}}{=} \sum_{\mu,\lambda} C_\ell[P_\mu, P'_\lambda](\varrho,\varrho')\,\delta_{(\mu,\lambda)}, & \overline{\mathfrak{C}}_\ell(\varrho,\varrho') &\stackrel{\text{def}}{=} \sum_{\mu,\lambda} \overline{C}_\ell[P_\mu, P'_\lambda](\varrho,\varrho')\,\delta_{(\mu,\lambda)},\\
\mathfrak{C}_{\text{int}}(\varrho) &\stackrel{\text{def}}{=} \sum_{\mu,\lambda} \Gamma[P_\mu, P'_\lambda](\varrho,\varrho)\,\delta_{(\mu,\lambda)}, & \zeta_\varrho &\stackrel{\text{def}}{=} \sum_{\mu,\lambda} \overline{\Gamma}[P_\mu, P'_\lambda](\varrho,\varrho)\,\delta_{(\mu,\lambda)}.
\end{aligned} \tag{SD.78}$$

For bounded f, g on $[0, C_\Sigma]$, linearity in the tests gives $\int f \otimes g\, d\mathfrak{C}_\ell = C_\ell[f(\Sigma_{\text{out}}), g(\Sigma_{\text{in}})]$ and the same for the other measures; in entries, $\mathfrak{C}_\ell = B^{-1}\sum_{i,j} \mathbb{E}[(\mathsf{S}_t^{\mathsf{c}})_{ij}(\mathsf{S}_{t+\ell}^{\mathsf{c}})_{ij}]\,\delta_{(\mu_i,\lambda_j)}$. The bounded-Lipschitz norm of a finite signed measure m on the rectangle is $\|m\|_{\text{BL}} \stackrel{\text{def}}{=} \sup\{|\int \phi\, dm| : \|\phi\|_\infty \leq 1,\ |\phi(x) - \phi(y)| \leq |x - y|_1\}$.

Theorem SD.8 (Joint forcing measure)**.**

(i) At every finite dimension, $\|\overline{\mathfrak{C}}_0\|_{\text{TV}} \leq \gamma_\star$ and $\|\overline{\mathfrak{C}}_\ell\|_{\text{TV}} \leq \mathsf{K}_\star\beta^\ell$ for $\ell \geq 1$, and the same holds for $\mathfrak{C}_\ell$ at every normalized state. The atoms of $\overline{\mathfrak{C}}_\ell$ and of ζ_ϱ with $\mu = 0$ or $\lambda = 0$ vanish.

(ii) Under Assumption S3, for every fixed $\ell \geq 0$,

$$\sup_{\varrho,\varrho' \in I} \sup_{\|f\|_\infty, \|g\|_\infty \leq 1} \Big| \int f \otimes g\, d\big(\mathfrak{C}_\ell - \overline{\mathfrak{C}}_\ell\big)\Big| \to 0, \qquad \sup_{\varrho,\varrho' \in I} \|\mathfrak{C}_\ell - \overline{\mathfrak{C}}_\ell\|_{\text{BL}} \to 0.$$

(iii) The forcing measure ζ_ϱ is a nonnegative measure, unchanged under replication, of total mass at most $\gamma_\star + 2\mathsf{K}_\star\beta/(1-\beta)$, and under Assumption S3

$$\sup_{\varrho\in I} \|\mathfrak{C}_{\mathrm{int}}(\varrho) - \zeta_\varrho\|_{\mathrm{BL}} \to 0, \tag{SD.79}$$

together with the product-test form of (ii).

(iv) The finite-time measure

$$\mathfrak{F}_T(\varrho) \stackrel{\mathrm{def}}{=} \frac{1}{BT}\sum_{i,j} \mathbb{E}\Big(\sum_{t'=1}^{T} (\mathsf{S}^{\mathrm{c}}_{t'}(\varrho))_{ij}\Big)^2 \delta_{(\mu_i,\lambda_j)}$$

is nonnegative, and $\|\mathfrak{F}_T(\varrho) - \mathfrak{C}_{\mathrm{int}}(\varrho)\|_{\mathrm{TV}}$ is at most eq. (SD.41).

Proof. For (i), the coupling in the proof of theorem SD.2(ii) works entry by entry: with $\Delta Q \stackrel{\mathrm{def}}{=} Q_{\varrho'}(M_{t+\ell}) - Q_{\varrho'}(M^*)$,

$$\|\mathfrak{C}_\ell\|_{\mathrm{TV}} \le \frac{1}{B}\sum_{i,j} \mathbb{E}\,|(\mathsf{S}^{\mathrm{c}}_t)_{ij}(\Delta Q)_{ij}| \le \frac{1}{B}\big(\mathbb{E}\,\|\mathsf{S}^{\mathrm{c}}_t\|_{\mathrm{F}}^2\big)^{1/2}\big(\mathbb{E}\,\|\Delta Q\|_{\mathrm{F}}^2\big)^{1/2} \le \mathsf{K}_\star\beta^\ell,$$

by the Cauchy–Schwarz inequality for the double sum and the bounds of that proof, and likewise $\|\mathfrak{C}_0\|_{\mathrm{TV}} \le \gamma_\star$. The independent model satisfies the same bounds. At a fixed dimension the measures have finitely many atoms, each the limit along the replications of the corresponding atom of the independent model, by eq. (SD.52); the total variation on a fixed finite set of atoms is continuous, so the bounds pass to $\overline{\mathfrak{C}}_\ell$. If $\mu = 0$, the mark P_0 enters the tree values through $B^{-1}\sum_{i:\mu_i=0}\mu_i^{p_V} = 0$, so the atom vanishes, and likewise for $\lambda = 0$.

For (ii), the product-test statement is the actual clause of theorem SD.3 for the diagonal tests $f(\Sigma_{\mathrm{out}})$ and $g(\Sigma_{\mathrm{in}})$. For the bounded-Lipschitz statement, let $\epsilon > 0$, take a grid $0 = x_0 < \cdots < x_n = C_\Sigma$ of mesh at most ϵ, and the piecewise linear hat functions $a_r \ge 0$ with $a_r(x_s) = \mathbf{1}_{r=s}$, which form a partition of unity. For ϕ in the unit ball of the norm, the interpolant $\phi_\epsilon \stackrel{\mathrm{def}}{=} \sum_{r,s}\phi(x_r,x_s)\,a_r\otimes a_s$ is at each point a convex combination of values of ϕ at points within ℓ^1 distance 2ϵ, so $\|\phi-\phi_\epsilon\|_\infty \le 2\epsilon$, and it is a sum of $(n+1)^2$ product tests with coefficients bounded by one. Hence

$$\Big|\int \phi\, d(\mathfrak{C}_\ell - \overline{\mathfrak{C}}_\ell)\Big| \le (n+1)^2 \sup_{f,g}\Big|\int f\otimes g\, d(\mathfrak{C}_\ell - \overline{\mathfrak{C}}_\ell)\Big| + 2\epsilon\big(\|\mathfrak{C}_\ell\|_{\mathrm{TV}} + \|\overline{\mathfrak{C}}_\ell\|_{\mathrm{TV}}\big),$$

and the limit superior is at most $4\max(\gamma_\star,\mathsf{K}_\star)\epsilon$, for every $\epsilon > 0$.

For (iii), positivity is theorem SD.4(iii) with the positive semidefinite tests P_μ, P'_λ; the mass bound sums the bounds of (i) over the lags at equal ridges; replication invariance is that of the response. For the product-test form of eq. (SD.79), $\int f\otimes g\, d\mathfrak{C}_{\mathrm{int}} = \Gamma[f(\Sigma_{\mathrm{out}}), g(\Sigma_{\mathrm{in}})](\varrho,\varrho)$ and the same holds for ζ_ϱ with $\overline{\Gamma}$, so lemma SD.7 with $p = (\varrho, f, g)$, the product-test statement of (ii) and the envelopes of (i) give it. The bounded-Lipschitz form follows by the interpolation of (ii), with the total variations of $\mathfrak{C}_{\mathrm{int}}$ and ζ_ϱ bounded by the mass bound.

For (iv), the atom of $\mathfrak{F}_T(\varrho)$ at (μ,λ) is the left side of eq. (SD.40) with $F = P_\mu$, $G = P'_\lambda$, $\varrho' = \varrho$, and the estimate in the proof of theorem SD.2(iii), summed over the atoms with the total variation bounds of (i) in place of the envelope, gives the bound. □

Proposition SD.3 (Mode coefficients)**.** *For an eigenvalue μ of Σ_{out} and λ of Σ_{in} let $r_\mu \stackrel{\mathrm{def}}{=} \#\{i : \mu_i = \mu\}$ and $t_\lambda \stackrel{\mathrm{def}}{=} \#\{j : \lambda_j = \lambda\}$, and put*

$$\mathfrak{v}_{ij}(\varrho) \stackrel{\mathrm{def}}{=} \frac{B\,\zeta_\varrho(\{(\mu_i,\lambda_j)\})}{r_{\mu_i}\,t_{\lambda_j}}. \tag{SD.80}$$

Then $\mathfrak{v}_{ij}(\varrho) \ge 0$, it depends on (i,j) only through (μ_i,λ_j), it vanishes when $\mu_i\lambda_j = 0$, and $\zeta_\varrho = B^{-1}\sum_{i,j}\mathfrak{v}_{ij}(\varrho)\,\delta_{(\mu_i,\lambda_j)}$. The actual entry coefficients

$$\mathsf{v}_{ij}(\varrho) \stackrel{\mathrm{def}}{=} \mathbb{E}\,\big((\mathsf{S}^{\mathrm{c}}_t(\varrho))_{ij}\big)^2 + 2\sum_{\ell\ge1}\mathbb{E}\,\big[(\mathsf{S}^{\mathrm{c}}_t(\varrho))_{ij}(\mathsf{S}^{\mathrm{c}}_{t+\ell}(\varrho))_{ij}\big] \tag{SD.81}$$

converge absolutely, are nonnegative, and satisfy $\mathfrak{C}_{\mathrm{int}}(\varrho) = B^{-1}\sum_{i,j}\mathsf{v}_{ij}(\varrho)\,\delta_{(\mu_i,\lambda_j)}$*; consequently, under Assumption S3,*

$$\sup_{\varrho\in I}\Big\|\frac{1}{B}\sum_{i,j}\big(\mathsf{v}_{ij}(\varrho)-\mathfrak{v}_{ij}(\varrho)\big)\delta_{(\mu_i,\lambda_j)}\Big\|_{\mathrm{BL}}\longrightarrow 0.$$

Proof. The disintegration is a sum over the $r_\mu t_\lambda$ pairs (i,j) of a block, and positivity and the null axes are theorem SD.8(i),(iii). Each entry series converges absolutely: the coupling of the proof of theorem SD.2(ii) gives $|\,\mathbb{E}[(\mathsf{S}^{\mathrm{c}}_t)_{ij}(\mathsf{S}^{\mathrm{c}}_{t+\ell})_{ij}]\,|\le(\mathbb{E}\,\|\mathsf{S}^{\mathrm{c}}_t\|_{\mathrm{F}}^2)^{1/2}(\mathbb{E}\,\|\Delta Q\|_{\mathrm{F}}^2)^{1/2}\le B\mathsf{K}_\star\beta^\ell$. Nonnegativity of v_{ij} follows from eq. (SD.40) with $F=u_i\otimes u_i$, $G=v_j\otimes v_j$ and $\varrho'=\varrho$, whose left side is $(BT)^{-1}\,\mathbb{E}(\sum_{t'\le T}(\mathsf{S}^{\mathrm{c}}_{t'})_{ij})^2\ge 0$ and tends to $B^{-1}\mathsf{v}_{ij}$ by theorem SD.2(iii). The identity for $\mathfrak{C}_{\mathrm{int}}$ is the entry form of eq. (SD.78), and the last statement is eq. (SD.79). □

The coefficient $\mathfrak{v}_{ij}$ is the volatility of the mode (i,j) that replaces $\mathcal{V}_{ij}(t)$ in eq. (SA.20) in the dynamics: over a stationary frozen block of T steps of size η, the centered accumulated displacement of the mode has energy $\eta^2\,\mathbb{E}(\sum_{t'\le T}(\mathsf{S}^{\mathrm{c}}_{t'})_{ij})^2$, which differs from $\eta^2 T\,\mathsf{v}_{ij}$ by at most $\eta^2 TB$ times eq. (SD.41), by theorem SD.2(iii) with $F=u_i\otimes u_i$ and $G=v_j\otimes v_j$; and proposition SD.3 identifies v_{ij} with $\mathfrak{v}_{ij}$ in the bounded-Lipschitz distance of the normalized spectral measures. This is the form in which the dynamics section consumes the coefficient; no statement about an individual entry is made.

The dynamics consume the integrated centered coefficient $\mathfrak{v}_{ij}$, and not the raw conditional second moment $\mathcal{V}_{ij}(t)$ of eq. (SA.20), because over many steps the quadratic terms accumulate the lag covariances (theorem SD.2(i)). At $\beta=0$ the lag terms vanish (theorem SD.2(ii)), and the squared mean contributes $O(h^2/B)$ per unit of macroscopic time (lemma SD.4); the raw coefficient is written $\mathfrak{v}^{(0)}$.

Proposition SD.4 (Recovery of the rows)**.** *If* $\Sigma_{\mathrm{in}}=\mathrm{Id}_{N_{\mathrm{in}}}$*, then* $\zeta_\varrho=B^{-1}\sum_i\mathfrak{v}_i(\varrho)\,\delta_{(\mu_i,1)}$*, and* $\mathfrak{v}_{ij}(\varrho)=\mathfrak{v}_i(\varrho)/N_{\mathrm{in}}$*, so that* $\sum_j\mathfrak{v}_{ij}(\varrho)=\mathfrak{v}_i(\varrho)$ *for every* i*.*

Proof. The only eigenvalue of Σ_{in} is 1, with $P_1'=\mathrm{Id}$ and $t_1=N_{\mathrm{in}}$. By the linearity of $\overline{\Gamma}$ in the first test, $\zeta_\varrho(\{(\mu,1)\})=\overline{\Gamma}[P_\mu,\mathrm{Id}](\varrho,\varrho)=\sum_{i:\mu_i=\mu}\overline{\Gamma}[u_i\otimes u_i,\mathrm{Id}](\varrho,\varrho)=r_\mu\,\mathfrak{v}_i(\varrho)/B$ for any i with $\mu_i=\mu$, by theorem SD.7(iv). Inserting this in eq. (SD.80) gives the claims. □

At $\Sigma_{\mathrm{in}}=\mathrm{Id}$ the main text writes $\mathfrak{v}_i=\sum_j\mathfrak{v}_{ij}$ in eq. (14); proposition SD.4 is that relation for the integrated coefficients.

SD.9 Specializations

The closed form of appendix SD.7 contains every volatility coefficient used in the paper. This subsection records the specializations. Each is either a substitution in theorem SD.5 or corollary SD.3, or an identification of the same-step term with the energies of theorem SD.1.

Same-step marginals. Summed over the input modes, the same-step term of the mode formula is the row energy of theorem SD.1; summed over the output modes, it is the column energy.

Corollary SD.4 (Same-step marginals)**.** *Fix a finite dimension and* $\varrho>0$*. For every row* i *and every column* j*,*

$$B\,\overline{C}_0[u_i\otimes u_i,\mathrm{Id}](\varrho,\varrho)=\mathfrak{v}_i^{(0)}(\varrho),\qquad B\,\overline{C}_0[\mathrm{Id},v_j\otimes v_j](\varrho,\varrho)=\widetilde{\mathfrak{v}}_j^{(0)}(\varrho),\tag{SD.82}$$

and, for every eigenvalue μ *of* Σ_{out} *and every eigenvalue* λ *of* Σ_{in}*,*

$$\frac{1}{B}\sum_{j=1}^{N_{\mathrm{in}}}\mathfrak{c}_0(\mu,\lambda_j;\varrho)=\frac{\tilde{s}_1(-\varrho)\,\mu}{\varrho+\tilde{s}_1(-\varrho)\,\mu},\qquad\frac{1}{B}\sum_{i=1}^{N_{\mathrm{out}}}\mathfrak{c}_0(\mu_i,\lambda;\varrho)=\frac{s_1(-\varrho)\,\lambda}{\varrho+s_1(-\varrho)\,\lambda}.\tag{SD.83}$$

Proof. We prove the row statements; the column statements are the same argument with the roles of the two sides exchanged, using the column bound of theorem SD.1(ii).

Dependence on the eigenvalue. The tree values of definition SD.4 with $F = u_i \otimes u_i$ and $G = \mathrm{Id}$ contain the mark through the single factor $B^{-1}\mu_i^{p_V}$ with $p_V \geq 1$. This factor depends on i only through μ_i, and the property passes to the extension and the limits of lemma SD.10 and theorem SD.3. Hence $\overline{C}_{i,0} \stackrel{\mathrm{def}}{=} B\,\overline{C}_0[u_i \otimes u_i, \mathrm{Id}](\varrho, \varrho)$ depends on i only through μ_i.

Replications. Fix the dimension and pass to its replications with the states of definition SD.3. The branch is the same at every order, so $\mathfrak{v}_i^{(0)}(\varrho)$ is the same at every order, and $\overline{C}_{i,0}$ is the same at every order by the replication invariance of theorem SD.3.

The row means are small. By theorem SC.2, $\|(\mathbb{E}\,\mathsf{S}_t)^\top u_i\| \leq \|\overline{\widetilde{G}}_\varrho^{\,\top} u_i\| + \varepsilon_B$. In eq. (SC.2), $u_i^\top \overline{R}(-s) = u_i^\top/(s + \tilde{s}_1\mu_i)$, $\mathbb{E}\,G_{t+1} = {\Sigma_{\mathrm{out}}}^{1/2}\widehat{A}_t{\Sigma_{\mathrm{in}}}^{1/2}$, $L_\beta \leq 1$ and $\|\overline{\widetilde{R}}(-s)\| \leq 1/s$, so that

$$\|T(s)^\top u_i\| \leq s \cdot \frac{1}{s} \cdot \sqrt{C_\Sigma}\,\|\widehat{A}_t\| \cdot \frac{\sqrt{\mu_i}}{s + \tilde{s}_1\mu_i} \leq \frac{C_\Sigma\|\widehat{A}_t\|}{s}.$$

Integrating along eq. (SC.3),

$$\|\overline{\widetilde{G}}_\varrho^{\,\top} u_i\| \leq \frac{2}{\pi}\int_0^\infty \frac{C_\Sigma\|\widehat{A}_t\|}{\varrho + x^2}\,dx = \frac{C_\Sigma\|\widehat{A}_t\|}{\sqrt{\varrho}},$$

which tends to zero along the replications, uniformly in i, because the replicated states have operator norm $r_N^{-1/2}$.

The actual row covariances. The centered row covariance is $C_{i,0} \stackrel{\mathrm{def}}{=} B\,C_0[u_i \otimes u_i, \mathrm{Id}](\varrho, \varrho) = \mathbb{E}\,\|\mathsf{S}_t^\top u_i\|^2 - \|(\mathbb{E}\,\mathsf{S}_t)^\top u_i\|^2$. By the previous step and the row bound of theorem SD.1(ii), $\max_i |C_{i,0} - \mathfrak{v}_i^{(0)}(\varrho)| \to 0$ along the replications.

Closure. For signs $\epsilon = (\epsilon_k)$ indexed by the rows of the N-fold replication, let $F_\epsilon \stackrel{\mathrm{def}}{=} \sum_k \epsilon_k\, u_k \otimes u_k$, a diagonal test of norm one. By linearity in the first test,

$$\frac{1}{NB}\sum_k \epsilon_k\big(C_{k,0} - \overline{C}_{k,0}\big) = C_0[F_\epsilon, \mathrm{Id}] - \overline{C}_0[F_\epsilon, \mathrm{Id}],$$

where k runs over the rows of the replication. The actual clause of theorem SD.3 makes the supremum over ϵ of the right side tend to zero. The N copies of the row i all carry the deterministic coefficient $\overline{C}_{i,0}$ of the base dimension, so the average over these copies of $|C_{k,0} - \overline{C}_{i,0}|$ is at most B times the normalized sum, which tends to zero.

Conclusion. The last two steps give $|\overline{C}_{i,0} - \mathfrak{v}_i^{(0)}(\varrho)| \leq \max_k |C_{k,0} - \mathfrak{v}_i^{(0)}(\varrho)| + N^{-1}\sum_{\text{copies}} |C_{k,0} - \overline{C}_{i,0}| \to 0$, and both sides are independent of N, which proves eq. (SD.82).

The spectral form. Let μ be an eigenvalue of Σ_{out} with multiplicity r_μ. By linearity in the second test and corollary SD.3(i) with theorem SD.6,

$$\overline{C}_0[P_\mu, \mathrm{Id}](\varrho, \varrho) = \sum_\lambda \overline{C}_0[P_\mu, P'_\lambda](\varrho, \varrho) = \frac{r_\mu}{B^2}\sum_{j=1}^{N_{\mathrm{in}}} \mathfrak{c}_0(\mu, \lambda_j; \varrho),$$

the sum over λ running over the distinct eigenvalues of Σ_{in}, each counted with its multiplicity in the sum over j. By linearity in the first test and the first step, the left side is $r_\mu\overline{C}_0[u_i \otimes u_i, \mathrm{Id}](\varrho, \varrho) = r_\mu\mathfrak{v}_i^{(0)}(\varrho)/B$ for any i with $\mu_i = \mu$. Dividing by r_μ/B gives eq. (SD.83). □

The identity eq. (SD.83) is not visible in the closed form: it says that the double integral of the summed kernel collapses to a single resolvent value. Its proof passes through the actual model, where both sides are the expected row energy.

The covariance-weighted same-step total. For $s, u > 0$ and $p \in \{1, 2\}$ put

$$\mathcal{T}_p^{\mathrm{out}}(s,u) \stackrel{\mathrm{def}}{=} \frac{1}{B}\,\mathrm{Tr}\,\big({\Sigma_{\mathrm{out}}}^p\,\overline{R}(-s)\overline{R}(-u)\big), \qquad \mathcal{T}_p^{\mathrm{in}}(s,u) \stackrel{\mathrm{def}}{=} \frac{1}{B}\,\mathrm{Tr}\,\big({\Sigma_{\mathrm{in}}}^p\,\overline{\widetilde{R}}(-s)\overline{\widetilde{R}}(-u)\big), \tag{SD.84}$$

and, with $r'_\beta(q) = \sum_j w_j^2 \; \mathbb{E}\; g^2(1 + w_j^2 q g^2)^{-2}$ the derivative of lemma SB.1(iii) and $q = q(-s)$,

$$\begin{aligned}\Upsilon(s) \stackrel{\text{def}}{=} &\Big(1 + s^2\tilde{\sigma}(-s)^2 F'_\beta(q)\, \mathcal{T}_2^{\text{out}}(s,s)\Big)\Big(1 + s^2\sigma(-s)^2 F'_\beta(q)\, \mathcal{T}_2^{\text{in}}(s,s)\Big) \\ &- s^2 r'_\beta(q)^2\, \mathcal{T}_2^{\text{out}}(s,s)\, \mathcal{T}_2^{\text{in}}(s,s).\end{aligned} \tag{SD.85}$$

The two-resolvent kernel is

$$\mathfrak{k}(s,u) \stackrel{\text{def}}{=} \begin{cases} \mathcal{T}_2^{\text{out}}(s,u)\, \mathcal{T}_2^{\text{in}}(s,u)\, \dfrac{u\, s_1(-s) - s\, s_1(-u)}{(u-s)\, \mathcal{T}_1^{\text{out}}(s,u)}, & s \neq u, \\ \dfrac{s^2\, r'_\beta(q(-s))\, \mathcal{T}_2^{\text{out}}(s,s)\, \mathcal{T}_2^{\text{in}}(s,s)}{\Upsilon(s)}, & s = u, \end{cases} \tag{SD.86}$$

and its total along the negative axis is

$$\overline{Z}_{\mathfrak{k}}(\varrho) \stackrel{\text{def}}{=} \frac{4}{\pi^2} \int_0^\infty \int_0^\infty \mathfrak{k}(\varrho + x^2, \varrho + y^2)\, dx\, dy. \tag{SD.87}$$

These are finite-dimensional deterministic quantities built from the branch of the full spectral measures; they depend on the dimension, on β and on the parameters. The kernel is the same-step closed form for the covariance-weighted tests, written in the variables of a single resolvent pair.

Corollary SD.5 (Same-time volatility, two anisotropic covariances)**.** *Assume the size conditions, and that $\Sigma_{\text{out}} \neq 0$ and $\Sigma_{\text{in}} \neq 0$.*

(i) At every finite dimension, $\Upsilon(s) > 0$ for $s > 0$, and $\mathfrak{k} = \mathfrak{k}_0[\Sigma_{\text{out}}, \Sigma_{\text{in}}]$. In particular $\mathfrak{k}$ is positive, symmetric and continuous on $(0,\infty)^2$, with

$$\mathfrak{k}(s,u) \leq \frac{\gamma_\star C_\Sigma^2}{4\sqrt{su}}. \tag{SD.88}$$

(ii) Under Assumption S3, for every compact $\mathcal{J} \subset (0,\infty)$,

$$\sup_{s,u \in \mathcal{J}} |\, \mathbb{E}\; \mathcal{K}_{s,u}(M_t, M_t) - \mathfrak{k}(s,u)| \longrightarrow 0, \tag{SD.89}$$

and the same holds with M_t replaced by M_t^{g} under the size conditions.

(iii) The integral eq. (SD.87) converges, $\overline{Z}_{\mathfrak{k}}(\varrho) = \overline{C}_0(\varrho,\varrho)$, $0 < \overline{Z}_{\mathfrak{k}}(\varrho) \leq \gamma_\star C_\Sigma^2$, and under Assumption S3

$$\sup_{\varrho \in I} |\, \mathbb{E}\, Z(\varrho) - \overline{Z}_{\mathfrak{k}}(\varrho)| \longrightarrow 0, \qquad \sup_{\varrho \in I} |\, \mathbb{E}\, Z^{\text{c}}(\varrho) - \overline{Z}_{\mathfrak{k}}(\varrho)| \longrightarrow 0. \tag{SD.90}$$

Proof. For (i), proposition SD.1(a) identifies $\mathfrak{k}(s,u)$ with the closed form of theorem SD.5 at $\ell = 0$, $F = \Sigma_{\text{out}}$, $G = \Sigma_{\text{in}}$. At $s = u$ the determinant of that closed form is $\Upsilon(s)$: at coincident arguments the divided differences are $b_0 = r'_\beta(q)$ and $e_0 = -F'_\beta(q)$, and $\mathsf{U} = \mathcal{T}_2^{\text{out}}(s,s)$, $\mathsf{V} = \mathcal{T}_2^{\text{in}}(s,s)$. Hence $\Upsilon(s) = \Delta_0(s,s) > 0$ by theorem SD.5(i). Positivity follows from $b_0 > 0$, $\mathcal{T}^{\text{out}}[\Sigma_{\text{out}}] > 0$, $\mathcal{T}^{\text{in}}[\Sigma_{\text{in}}] > 0$ and $\Delta_0 > 0$. Symmetry holds because at lag zero both buffers are the same matrix, so $\mathfrak{k}_0[F,G](s,u) = \mathfrak{k}_0[F,G](u,s)$ for symmetric tests. Continuity and the bound are corollary SD.2 with $\|\Sigma_{\text{out}}\|, \|\Sigma_{\text{in}}\| \leq C_\Sigma$.

Part (ii) is the first statement of eq. (SD.52) at $\ell = 0$ with the tests $\Sigma_{\text{out}}/C_\Sigma$ and $\Sigma_{\text{in}}/C_\Sigma$, multiplied by C_Σ^2, for the actual and for the independent model.

For (iii), theorem SD.6 at $\ell = 0$ gives the convergence of the integral and $\overline{Z}_{\mathfrak{k}}(\varrho) = \overline{C}_0(\varrho,\varrho)$. The integrand is positive by (i), and $\overline{C}_0(\varrho,\varrho) \leq \gamma_\star C_\Sigma^2$ by theorem SD.4(ii). Since $\mathbb{E}\, Z^{\text{c}}(\varrho) = C_0(\varrho,\varrho)$, the second statement of eq. (SD.90) is theorem SD.3 at $\ell = 0$ with the same tests. The first follows from the second and eq. (SD.21). □

By (iii) and eq. (SD.19), the quadratic term of the frozen one-step identity eq. (SC.15) is $\frac{\eta^2 B}{2}(\overline{Z}_{\mathfrak{k}}(\varrho) + o(1))$, uniformly in $\varrho \in I$. When $\Sigma_{\text{in}} = \text{Id}$, $\overline{Z}_{\mathfrak{k}} = \overline{Z}$ of eq. (SD.22), by theorem SD.4(v).

One isotropic covariance. When $\Sigma_{\rm in} = \mathrm{Id}_{N_{\rm in}}$, the input side has the single eigenvalue 1 with multiplicity $N_{\rm in}$, and $\mathsf{V} = (N_{\rm in}/B)\big((s+s_1)(u+s_1')\big)^{-1}$. Proposition SD.1(b) is the two-time row response eq. (SG.16) of proposition SG.3, and the row coefficients of eq. (SD.75) are

$$\mathfrak{v}_i(\varrho) = N_{\rm in}\,\mathfrak{v}_{ij}(\varrho) = \frac{N_{\rm in}}{B}\Big(\mathfrak{c}_0(\mu_i,1;\varrho) + 2\sum_{\ell\ge 1}\mathfrak{c}_\ell(\mu_i,1;\varrho)\Big), \tag{SD.91}$$

by corollary SD.3 and proposition SD.4. By corollary SD.4, the same-step term is $(N_{\rm in}/B)\,\mathfrak{c}_0(\mu_i,1;\varrho) = \mathfrak{v}_i^{(0)}(\varrho)$, which is theorem SD.7(iv). Appendix SG evaluates these coefficients at zero ridge for power-law spectra.

Two isotropic covariances. When $\Sigma_{\rm out} = \mathrm{Id}_{N_{\rm out}}$ and $\Sigma_{\rm in} = \mathrm{Id}_{N_{\rm in}}$, the only eigenvalue pair is $(1,1)$, and every mode carries the same coefficient. The isotropic coefficient of theorem SF.2 is

$$\mathfrak{v}_{\rm iso}(\varrho) = N_{\rm out}N_{\rm in}\,\mathfrak{v}_{ij}(\varrho) = \frac{N_{\rm out}N_{\rm in}}{B}\Big(\mathfrak{c}_0(1,1;\varrho) + 2\sum_{\ell\ge 1}\mathfrak{c}_\ell(1,1;\varrho)\Big), \tag{SD.92}$$

and by corollary SD.4 its same-step term is $N_{\rm out}\,\tilde{s}_1(-\varrho)/(\varrho+\tilde{s}_1(-\varrho)) = B\psi(\varrho)$, in the notation of appendix SF.5.

No momentum.

Remark SD.3 (No momentum)**.** *At $\beta = 0$ the only weight is $w_0 = 1$, the buffer is the current gradient, $F_0 = L_0 = J$, $r_0'(q) = \mathbb{E}\, g^2(1+qg^2)^{-2}$ and $F_0'(q) = -\,\mathbb{E}\, g^4(1+qg^2)^{-2}$, and the branch is the no-momentum system of remark SB.1. In the closed form, b_ℓ carries the factor β^ℓ, so $b_\ell = 0$ and $\mathfrak{k}_\ell = 0$ for every $\ell \ge 1$; the same-step scalars are the divided differences $b_0 = \big(r_0(q) - r_0(q')\big)/(q-q')$ and $e_0 = -\big(F_0(q) - F_0(q')\big)/(q-q')$. All lag covariances and their deterministic equivalents vanish for $\ell \ge 1$ (theorem SD.2(ii), theorem SD.4(ii)), so $\Gamma = C_0$, $\overline{\Gamma} = \overline{C}_0$, the forcing measure is $\zeta_\varrho = \overline{\mathfrak{C}}_0(\varrho,\varrho)$, and*

$$\mathfrak{v}_{ij}(\varrho) = \frac{1}{B}\,\mathfrak{c}_0(\mu_i,\lambda_j;\varrho), \qquad \sum_{j=1}^{N_{\rm in}}\mathfrak{v}_{ij}(\varrho) = \mathfrak{v}_i^{(0)}(\varrho), \qquad \sum_{i=1}^{N_{\rm out}}\mathfrak{v}_{ij}(\varrho) = \widetilde{\mathfrak{v}}_j^{(0)}(\varrho),$$

by corollaries SD.3 and SD.4, for every pair of covariances. For $\Sigma_{\rm in} = \mathrm{Id}$ this is $\mathfrak{v}_i(\varrho) = \mathfrak{v}_i^{(0)}(\varrho)$ of theorem SD.7(iv), with the spectral form of remark SD.1. The covariance-weighted same-step total is $\overline{C}_0(\varrho,\varrho) = \overline{Z}_{\mathfrak{k}}(\varrho)$ of corollary SD.5, with the kernel eq. (SD.86) formed from F_0' and r_0'.

SE From the algorithm to the deterministic dynamics

This section proves that the algorithm of definition SA.2, run at the constant step $\eta = h/B$, follows the deterministic mode-energy equations of the main text on a positive interval of macroscopic time $\tau = t/B$. The equations are eq. (6) in continuous time, with the drift kernel $\mathfrak{d}_{ij}$ of corollary SC.1 and the volatility coefficient $\mathfrak{v}_{ij}$ of proposition SD.3. The comparison is made in the bounded-Lipschitz distance of spectral energy measures, which place the projected risk $\mathscr{Q}_{ij}$ of the eigenpair (u_i, v_j) at the point (μ_i, λ_j) of the spectral rectangle. Theorem SE.1 is the main result. The half-anisotropic case $\Sigma_{\rm in} = \mathrm{Id}$ is corollary SE.1, and corollary SE.2 passes to dimension-free measure equations when the coefficients and the initial data converge.

The coefficients of the drift and volatility sections are computed in the frozen-state model of definition SA.3: a stationary buffer of independent batches, all computed at one deterministic state whose normalized error has vanishing operator norm. The algorithm differs from it in three ways. Its state moves; its buffer retains the batches of earlier states and an initial buffer; and its ridge follows the risk. The proof compares the algorithm, on blocks of $\lfloor B^{1/8} \rfloor$ consecutive steps, with the frozen-state model at the state of the start of the block. It controls the difference of the expected block increments by a causal expansion in the step, the fluctuations by Gaussian concentration and a martingale argument, and the persistence of a small normalized operator norm by a matrix martingale; it then shows that the risk stays bounded below on a positive interval and passes from fixed tests to the bounded-Lipschitz distance.

SE.1 Standing setup

The algorithm at constant step. Throughout this section the algorithm of definition SA.2 is run with momentum $\beta \in [0,1)$, normalized ridge $\varrho > 0$ and the constant step $\eta_t = \eta = h/B$ with $h > 0$. The features are Gaussian as in Assumption S1, and the whitened samples of the batch of step t are $(z_{t,a}, w_{t,a})_{a \le B}$. For a matrix $E \in \mathbb{R}^{N_{\rm out} \times N_{\rm in}}$ we write the gradient of that batch at E as

$$G^\circ_{t+1}[E] \stackrel{\rm def}{=} \frac{1}{B} \sum_{a=1}^{B} \left(z_{t,a}^\top {\Sigma_{\rm out}}^{1/2} E \, {\Sigma_{\rm in}}^{1/2} w_{t,a} \right) {\Sigma_{\rm out}}^{1/2} (z_{t,a} \otimes w_{t,a}) {\Sigma_{\rm in}}^{1/2}. \tag{SE.1}$$

It is linear in E, its mean is $\Sigma_{\rm out} E \Sigma_{\rm in}$ by eq. (SA.5), and the minibatch gradient of eq. (SA.9) is $G^\circ_{t+1} = G^\circ_{t+1}[\Delta_t]$. For a matrix Δ we write $\mathscr{R}(\Delta) \stackrel{\rm def}{=} \frac{1}{2}\|\Delta\|^2_\Sigma$, so that $\mathscr{R}(t) = \mathscr{R}(\Delta_t)$ by eq. (SA.4), and $\varrho^\circ(\Delta) \stackrel{\rm def}{=} 2\mathscr{R}(\Delta)\varrho$ for the physical ridge. We write S_t for the update applied at step t, the matrix $\widetilde{G}^\circ_{t+1}$ of eq. (SA.15). Before the stopping time $t_{\rm stop}$ of definition SA.2 the algorithm is

$$M^\circ_t = \beta M^\circ_{t-1} + (1-\beta)\, G^\circ_{t+1}[\Delta_t], \qquad \mathsf{S}_t = Q_{\varrho^\circ(\Delta_t)}(M^\circ_t), \qquad \Delta_{t+1} = \Delta_t - \frac{h}{B}\,\mathsf{S}_t. \tag{SE.2}$$

The letter S_t always denotes this actual update; the frozen updates of the volatility section, written there $\mathsf{S}_{t'}(\varrho)$ with their ridge, are written V_r in this section.

Normalized error and diffuseness. For Δ with $\mathscr{R}(\Delta) > 0$ we put

$$\widehat{A}(\Delta) \stackrel{\rm def}{=} \frac{{\Sigma_{\rm out}}^{1/2} \Delta {\Sigma_{\rm in}}^{1/2}}{\sqrt{2\mathscr{R}(\Delta)}}, \qquad \delta(\Delta) \stackrel{\rm def}{=} \|\widehat{A}(\Delta)\|, \tag{SE.3}$$

so that $\|\widehat{A}(\Delta)\|_{\rm F} = 1$ by eq. (SA.9) and $\widehat{A}(\Delta_t) = \widehat{A}_t$. We call $\delta(\Delta)$ the diffuseness of Δ. We write $\mathcal{S}_B$ for the set of all matrices $\widehat{A}(\Delta)$ with $\mathscr{R}(\Delta) > 0$; it is the set of normalized states of the frozen-state model.

Spectral energy measures. Let $\mathbb{K} \stackrel{\rm def}{=} [0, C_\Sigma]^2$. For $\Delta \in \mathbb{R}^{N_{\rm out} \times N_{\rm in}}$ the spectral energy measure of Δ is the positive measure

$$\chi(\Delta) \stackrel{\rm def}{=} \sum_{i,j} \mathscr{Q}_{ij}(\Delta)\, \delta_{(\mu_i, \lambda_j)} \quad \text{on } \mathbb{K}. \tag{SE.4}$$

For bounded functions f, g on $[0, C_\Sigma]$ we write $\mathcal{W}_{f,g} X \stackrel{\rm def}{=} f(\Sigma_{\rm out})\, X\, g(\Sigma_{\rm in})$, a linear map on $\mathbb{R}^{N_{\rm out} \times N_{\rm in}}$ which is selfadjoint for the Frobenius pairing $\langle V, W \rangle = {\rm Tr}(VW^\top)$ and has norm at most $\|f\|_\infty \|g\|_\infty$ for the Frobenius norm. Expanding in the eigenbases,

$$\int f \otimes g \, {\rm d}\chi(\Delta) = \sum_{i,j} f(\mu_i) g(\lambda_j) \mathscr{Q}_{ij}(\Delta) = \langle \Delta, \mathcal{W}_{f,g} \Delta \rangle, \tag{SE.5}$$

where $(f \otimes g)(\mu, \lambda) \stackrel{\rm def}{=} f(\mu) g(\lambda)$. In particular $\chi(\Delta)(\mathbb{K}) = \|\Delta\|^2_{\rm F}$ and $\int \mu\lambda \, {\rm d}\chi(\Delta) = 2\mathscr{R}(\Delta)$ by lemma SA.1. For a finite signed measure m on $\mathbb{K}$ we use the total variation $\|m\|_{\rm TV}$ and the bounded-Lipschitz norm $\|m\|_{\rm BL}$ of the volatility section (the definition following eq. (SD.78)), in which the test functions ϕ satisfy $\|\phi\|_\infty \le 1$ and $|\phi(x) - \phi(y)| \le |x - y|_1$. For a bounded function ϕ on $\mathbb{K}$, $\phi\, m$ denotes the measure with density ϕ with respect to m.

Coefficients. The drift kernel is the function $\mathfrak{d}$ of eq. (SC.4) at the ridge ϱ, with $\mathfrak{d}_{ij} = \mathfrak{d}(\mu_i, \lambda_j)$ (corollary SC.1). The forcing measure is the measure $\zeta \stackrel{\rm def}{=} \zeta_\varrho$ of eq. (SD.78), and the mode coefficients are $\mathfrak{v}_{ij} \stackrel{\rm def}{=} \mathfrak{v}_{ij}(\varrho)$ of eq. (SD.80), so that $\zeta = B^{-1} \sum_{i,j} \mathfrak{v}_{ij}\, \delta_{(\mu_i, \lambda_j)}$ by proposition SD.3. Both are deterministic, depend on the dimension, on β, ϱ and the covariances, and do not depend on the state. By eq. (SC.14) and theorem SD.8(iii), with $\mathsf{K}_\star$ the constant of eq. (SD.39) at $\varrho_- = \varrho$,

$$\begin{gathered} 0 \le \mathfrak{d} \le \mathfrak{d}_{\max} \stackrel{\rm def}{=} \frac{C_\Sigma^2}{\sqrt{2\varrho}} \text{ on } \mathbb{K}, \qquad |\mathfrak{d}(x) - \mathfrak{d}(y)| \le \frac{C_\Sigma}{\sqrt{2\varrho}} |x - y|_1, \\ 0 \le \zeta(\mathbb{K}) \le m_\zeta \stackrel{\rm def}{=} \gamma_\star + \frac{2\mathsf{K}_\star \beta}{1-\beta}. \end{gathered} \tag{SE.6}$$

Conventions. The past $\mathscr{F}_t$ is that of definition SA.1, and $\mathbb{E}_t = \mathbb{E}[\,\cdot \mid \mathscr{F}_t]$. Conditional expectations given $\mathscr{F}_t$ are computed by integrating the batches of the steps $t, t+1, \dots$ with the past held fixed, since these batches are independent of the past. Comparisons with the frozen-state model use auxiliary batches, independent of everything else and drawn anew for each comparison; $\mathbb{E}_t$ integrates them as well. At $\beta = 0$ we use the convention $\beta^0 = 1$, so that $\beta^\ell = 0$ for $\ell \geq 1$. A constant C depends only on $(h, \varrho, \beta, \gamma_\star, C_\Sigma)$ and on the compact sets of states named in the statement where it appears; it never depends on the dimension, on the time or on the block, and it may change from line to line.

SE.2 The deterministic equations

Definition SE.1 (Deterministic mode-energy equations)**.** *For $i \leq N_{\mathrm{out}}$ and $j \leq N_{\mathrm{in}}$, the deterministic mode-energy equations are*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(B^2 \Omega_{ij}(\tau)\big) = -\frac{2h\,\mathfrak{d}_{ij}}{\sqrt{\mathcal{R}(\tau)}}\, B^2 \Omega_{ij}(\tau) + h^2 B\, \mathfrak{v}_{ij}, \qquad \mathcal{R}(\tau) \stackrel{\mathrm{def}}{=} \frac{1}{2}\sum_{i,j} \mu_i \lambda_j\, \Omega_{ij}(\tau), \tag{SE.7}$$

with the initial values $\Omega_{ij}(0) = \mathscr{Q}_{ij}(0) = (u_i^\top \Delta_0 v_j)^2$. Their spectral energy measure is $\chi^{\mathrm{ode}}(\tau) \stackrel{\mathrm{def}}{=} \sum_{i,j} \Omega_{ij}(\tau)\, \delta_{(\mu_i,\lambda_j)}$.

Dividing eq. (SE.7) by B^2 gives

$$\frac{\mathrm{d}\Omega_{ij}}{\mathrm{d}\tau} = -\frac{2h\,\mathfrak{d}_{ij}}{\sqrt{\mathcal{R}}}\, \Omega_{ij} + \frac{h^2}{B}\, \mathfrak{v}_{ij}.$$

With $\eta = h/B$ and B steps per unit of τ, the change over one step is $-2\eta\, \mathfrak{d}_{ij}\Omega_{ij}/\sqrt{\mathcal{R}} + \eta^2 \mathfrak{v}_{ij}$, which is the recursion eq. (6) with the kernels of corollary SC.1 and proposition SD.3. The variables $B^2\Omega_{ij}$ are the ones in which both terms are of order one. The total $\sum_{i,j} \Omega_{ij} = \|\Delta\|_{\mathrm{F}}^2$ is bounded and is spread over at most $\gamma_\star^2 B^2$ pairs, so a typical $B^2\Omega_{ij}$ is of order one; the drift rate $2h\mathfrak{d}_{ij}/\sqrt{\mathcal{R}}$ is bounded by eq. (SE.6) where the risk is bounded below; and since $\zeta = B^{-1}\sum_{i,j} \mathfrak{v}_{ij}\delta_{(\mu_i,\lambda_j)}$ has mass at most m_ζ, the forcing $B\mathfrak{v}_{ij}$ of a typical pair is of order one as well.

The measure form. For a positive measure χ on $\mathbb{K}$ with $\mathcal{R}(\chi) \stackrel{\mathrm{def}}{=} \frac{1}{2}\int \mu\lambda\, \mathrm{d}\chi(\mu,\lambda) > 0$ we define the signed measure

$$\mathsf{F}_B(\chi) \stackrel{\mathrm{def}}{=} -\frac{2h}{\sqrt{\mathcal{R}(\chi)}}\, \mathfrak{d}\, \chi + h^2 \zeta. \tag{SE.8}$$

The atom of $\chi^{\mathrm{ode}}(\tau)$ at a point (μ,λ) is the sum of $\Omega_{ij}(\tau)$ over the pairs with $(\mu_i,\lambda_j) = (\mu,\lambda)$, and the atom of $h^2\zeta$ there is the sum of $h^2\mathfrak{v}_{ij}/B$ over the same pairs. Summing the divided equations over these pairs therefore gives

$$\chi^{\mathrm{ode}}(\tau) = \chi^{\mathrm{ode}}(0) + \int_0^\tau \mathsf{F}_B\big(\chi^{\mathrm{ode}}(\sigma)\big)\, \mathrm{d}\sigma, \tag{SE.9}$$

with $\mathcal{R}(\chi^{\mathrm{ode}}(\tau)) = \mathcal{R}(\tau)$.

Constants and the admissible set. Fix $E_\star < \infty$ and $0 < r_0 \leq r_1 < \infty$. With the constants of eq. (SE.6) put

$$\begin{gathered} C_E \stackrel{\mathrm{def}}{=} \frac{2h\,\mathfrak{d}_{\max}(E_\star+1)}{\sqrt{r_0/2}} + h^2 m_\zeta, \qquad C_{\mathscr{R}} \stackrel{\mathrm{def}}{=} 2h\,\mathfrak{d}_{\max}\sqrt{r_1+1} + \frac{h^2 C_\Sigma^2 m_\zeta}{2}, \\ T_\star \stackrel{\mathrm{def}}{=} \frac{\min(r_0,1)}{8\,(1+C_E+C_{\mathscr{R}})}, \end{gathered} \tag{SE.10}$$

which depend only on $(h, \varrho, \beta, \gamma_\star, C_\Sigma, E_\star, r_0, r_1)$. The admissible measures are

$$\mathfrak{M} \stackrel{\mathrm{def}}{=} \Big\{\chi \text{ positive on } \mathbb{K}:\ \chi(\mathbb{K}) \leq E_\star + 1,\ \frac{r_0}{2} \leq \mathcal{R}(\chi) \leq r_1 + \frac{1}{2}\Big\}. \tag{SE.11}$$

Lemma SE.1 (The field)**.** *There is C_{F}, depending only on $(h, \varrho, C_\Sigma, E_\star, r_0, r_1)$ and on m_ζ, such that for $\chi, \chi' \in \mathfrak{M}$:*

(i) $|\mathsf{F}_B(\chi)(\mathbb{K})| \le \|\mathsf{F}_B(\chi)\|_{\mathrm{TV}} \le C_E$ *and* $\frac{1}{2}\big|\int \mu\lambda \,\mathrm{d}\mathsf{F}_B(\chi)\big| \le C_{\mathscr{R}}$*;*

(ii) $\|\mathsf{F}_B(\chi) - \mathsf{F}_B(\chi')\|_{\mathrm{TV}} \le C_{\mathsf{F}}\|\chi - \chi'\|_{\mathrm{TV}}$*;*

(iii) $\|\mathsf{F}_B(\chi) - \mathsf{F}_B(\chi')\|_{\mathrm{BL}} \le C_{\mathsf{F}}\|\chi - \chi'\|_{\mathrm{BL}}$.

The same statements hold, with C_E, $C_{\mathscr{R}}$, C_{F} depending also on the bounds, on every set defined as $\mathfrak{M}$ with other positive bounds on the mass and the risk.

Proof. The mass bound is eq. (SE.6). On $\mathfrak{M}$,

$$\|\mathsf{F}_B(\chi)\|_{\mathrm{TV}} \le \frac{2h\,\mathfrak{d}_{\max}\,\chi(\mathbb{K})}{\sqrt{\mathcal{R}(\chi)}} + h^2\zeta(\mathbb{K}) \le \frac{2h\,\mathfrak{d}_{\max}(E_\star+1)}{\sqrt{r_0/2}} + h^2 m_\zeta = C_E.$$

The risk bound uses $\mathfrak{d} \le \mathfrak{d}_{\max}$ inside the risk integral. Since $\int \mu\lambda\,\mathfrak{d}\,\mathrm{d}\chi \le \mathfrak{d}_{\max}\int \mu\lambda\,\mathrm{d}\chi = 2\mathfrak{d}_{\max}\mathcal{R}(\chi)$ and $\mu\lambda \le C_\Sigma^2$ on $\mathbb{K}$,

$$\frac{1}{2}\Big|\int \mu\lambda\,\mathrm{d}\mathsf{F}_B(\chi)\Big| \le \frac{h}{\sqrt{\mathcal{R}(\chi)}}\,2\mathfrak{d}_{\max}\mathcal{R}(\chi) + \frac{h^2C_\Sigma^2}{2}\zeta(\mathbb{K}) \le 2h\,\mathfrak{d}_{\max}\sqrt{r_1+1} + \frac{h^2C_\Sigma^2 m_\zeta}{2} = C_{\mathscr{R}}.$$

The Lipschitz bounds follow from one decomposition. The forcing cancels in the difference, and

$$\mathsf{F}_B(\chi) - \mathsf{F}_B(\chi') = -\frac{2h}{\sqrt{\mathcal{R}(\chi)}}\,\mathfrak{d}\,(\chi-\chi') - 2h\Big(\frac{1}{\sqrt{\mathcal{R}(\chi)}} - \frac{1}{\sqrt{\mathcal{R}(\chi')}}\Big)\mathfrak{d}\,\chi'. \tag{SE.12}$$

For $\mathcal{R}, \mathcal{R}' \ge r_0/2$ the mean value theorem gives

$$\Big|\frac{1}{\sqrt{\mathcal{R}}} - \frac{1}{\sqrt{\mathcal{R}'}}\Big| \le \frac{1}{2}\Big(\frac{r_0}{2}\Big)^{-3/2}|\mathcal{R}-\mathcal{R}'|.$$

In total variation, $\|\mathfrak{d}(\chi-\chi')\|_{\mathrm{TV}} \le \mathfrak{d}_{\max}\|\chi-\chi'\|_{\mathrm{TV}}$ and $|\mathcal{R}(\chi)-\mathcal{R}(\chi')| \le \frac{1}{2}C_\Sigma^2\|\chi-\chi'\|_{\mathrm{TV}}$, which with $\|\mathfrak{d}\chi'\|_{\mathrm{TV}} \le \mathfrak{d}_{\max}(E_\star+1)$ proves (ii).

In the bounded-Lipschitz norm we use that multiplication by a bounded Lipschitz function maps the unit ball of test functions into a multiple of it. If $\|\phi\|_\infty \le 1$ and ϕ is 1-Lipschitz for $|\cdot|_1$, then by eq. (SE.6)

$$\|\phi\,\mathfrak{d}\|_\infty \le \mathfrak{d}_{\max}, \qquad |\phi\mathfrak{d}(x) - \phi\mathfrak{d}(y)| \le \Big(\mathfrak{d}_{\max} + \frac{C_\Sigma}{\sqrt{2\varrho}}\Big)|x-y|_1,$$

so $|\int \phi\,\mathfrak{d}\,\mathrm{d}(\chi-\chi')| \le c_1\|\chi-\chi'\|_{\mathrm{BL}}$ with $c_1 \overset{\mathrm{def}}{=} 2\mathfrak{d}_{\max} + C_\Sigma/\sqrt{2\varrho}$. The function $\mu\lambda$ has supremum C_Σ^2 and $|\cdot|_1$-Lipschitz constant C_Σ on $\mathbb{K}$, so $|\mathcal{R}(\chi)-\mathcal{R}(\chi')| \le \frac{1}{2}(2C_\Sigma^2 + C_\Sigma)\|\chi-\chi'\|_{\mathrm{BL}}$. Inserting these two bounds in eq. (SE.12), with $|\int \phi\,\mathfrak{d}\,\mathrm{d}\chi'| \le \mathfrak{d}_{\max}(E_\star+1)$, proves (iii). The last sentence is the same computation with the other bounds. □

Proposition SE.1 (Well-posedness)**.** *At a fixed dimension, let* $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$ *and* $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \le \gamma_\star B$*, and let* Δ_0 *satisfy* $\|\Delta_0\|_{\mathrm{F}}^2 \le E_\star$ *and* $r_0 \le \mathscr{R}(\Delta_0) \le r_1$*. Then:*

(i) the system eq. (SE.7) has a unique solution on $[0, T_\star]$*, and* $\Omega_{ij}(\tau) \ge 0$ *for all* i, j *and* $\tau \le T_\star$*;*

(ii) for $\tau \le T_\star$*,* $|\chi^{\mathrm{ode}}(\tau)(\mathbb{K}) - \|\Delta_0\|_{\mathrm{F}}^2| \le C_E\tau$ *and* $|\mathcal{R}(\tau) - \mathscr{R}(\Delta_0)| \le C_{\mathscr{R}}\tau$*; in particular* $\chi^{\mathrm{ode}}(\tau) \in \mathfrak{M}$*;*

(iii) χ^{ode} *satisfies eq. (SE.9) on* $[0, T_\star]$.

Proof. The vector field is locally Lipschitz where the risk is positive. On the open set $\mathcal{O} \overset{\mathrm{def}}{=} \{(e_{ij}) \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}} : \sum_{i,j}\mu_i\lambda_j e_{ij} > 0\}$, the right side of eq. (SE.7) in the variables $e_{ij} = B^2\Omega_{ij}$ is a product of linear functions of e and of $x \mapsto x^{-1/2}$ evaluated at a positive linear function of e, hence continuously differentiable. The Picard–Lindelöf theorem [65, Theorem 2.2] gives a unique solution on a maximal interval $[0, T_{\max})$ in $\mathcal{O}$.

The solution is nonnegative by variation of constants. Along the solution put $a_{ij}(\tau) \overset{\text{def}}{=} 2h\mathfrak{d}_{ij}/\sqrt{\mathcal{R}(\tau)}$, a continuous function on $[0, T_{\max})$. Then

$$B^2\mathcal{Q}_{ij}(\tau) = e^{-\int_0^\tau a_{ij}}\, B^2\mathcal{Q}_{ij}(0) + h^2 B\,\mathfrak{v}_{ij} \int_0^\tau e^{-\int_\sigma^\tau a_{ij}}\,\mathrm{d}\sigma \geq 0,$$

because $\mathcal{Q}_{ij}(0) \geq 0$ and $\mathfrak{v}_{ij} \geq 0$ by proposition SD.3.

The measure equation eq. (SE.9) holds on $[0, T_{\max})$ by the summation over atoms explained before eq. (SE.10). Testing it with 1 and with $\mu\lambda/2$ gives

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\chi^{\mathrm{ode}}(\tau)(\mathbb{K}) = \mathsf{F}_B(\chi^{\mathrm{ode}}(\tau))(\mathbb{K}), \qquad \frac{\mathrm{d}}{\mathrm{d}\tau}\mathcal{R}(\tau) = \frac{1}{2}\int \mu\lambda\,\mathrm{d}\mathsf{F}_B(\chi^{\mathrm{ode}}(\tau)).$$

The solution cannot leave $\mathfrak{M}$ before $T_\star$. The initial measure satisfies $\chi^{\mathrm{ode}}(0)(\mathbb{K}) = \|\Delta_0\|_{\mathrm{F}}^2 \leq E_\star$ and $\mathcal{R}(0) = \mathscr{R}(\Delta_0) \in [r_0, r_1]$, so it lies in the interior of $\mathfrak{M}$. Let τ_1 be the first time in $[0, T_{\max})$ at which χ^{ode} reaches the boundary of $\mathfrak{M}$, and suppose $\tau_1 \leq T_\star$. On $[0, \tau_1]$ the measure lies in $\mathfrak{M}$, so lemma SE.1(i) bounds both derivatives above, and

$$|\chi^{\mathrm{ode}}(\tau_1)(\mathbb{K}) - \|\Delta_0\|_{\mathrm{F}}^2| \leq C_E T_\star \leq \frac{1}{8}, \qquad |\mathcal{R}(\tau_1) - \mathscr{R}(\Delta_0)| \leq C_{\mathscr{R}} T_\star \leq \frac{\min(r_0, 1)}{8}.$$

Then $\chi^{\mathrm{ode}}(\tau_1)(\mathbb{K}) \leq E_\star + \frac{1}{8}$, $\mathcal{R}(\tau_1) \geq \frac{7}{8}r_0$ and $\mathcal{R}(\tau_1) \leq r_1 + \frac{1}{8}$, so $\chi^{\mathrm{ode}}(\tau_1)$ is in the interior of $\mathfrak{M}$, a contradiction. Hence the solution stays in $\mathfrak{M}$ on $[0, \min(T_{\max}, T_\star))$, with the bounds of (ii).

The solution exists up to $T_\star$. On $[0, \min(T_{\max}, T_\star))$ the vector (e_{ij}) stays in the set $\{e \geq 0 : \sum_{ij} e_{ij} \leq B^2(E_\star + 1),\ \sum \mu_i\lambda_j e_{ij} \geq B^2 r_0\}$, which is a compact subset of $\mathcal{O}$. A solution that stays in a compact subset of the domain on a bounded interval extends beyond it [65, Corollary 2.15], so $T_{\max} > T_\star$. This proves (i), (ii) and (iii). □

SE.3 The main theorem

The actual spectral energy measure at macroscopic time τ is $\chi^{\mathrm{act}}(\tau) \overset{\text{def}}{=} \chi(\Delta_{\lfloor B\tau \rfloor})$.

Theorem SE.1 (Dynamical comparison)**.** *Fix $h > 0$, $\varrho > 0$, $\beta \in [0, 1)$, $\gamma_\star < \infty$, $C_\Sigma \geq 1$, $E_\star < \infty$, $c_M < \infty$ and $0 < r_0 \leq r_1 < \infty$. Along a sequence of dimensions with $B \to \infty$ assume:*

- *(H1) Assumptions S1 and S2 hold with deterministic covariances, $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \leq \gamma_\star B$ and $\|\Sigma_{\mathrm{in}}\|, \|\Sigma_{\mathrm{out}}\| \leq C_\Sigma$;*
- *(H2) the algorithm of definition SA.2 is run with momentum β, normalized ridge ϱ and step $\eta_t = h/B$;*
- *(H3) W_0 is deterministic, $\|\Delta_0\|_{\mathrm{F}}^2 \leq E_\star$, $r_0 \leq \mathscr{R}(0) \leq r_1$, and $\|\widehat{A}_0\| \to 0$;*
- *(H4) the initial buffer M_{-1}° is independent of all batches and $\mathbb{E}\,\|M_{-1}^\circ\|_{\mathrm{F}}^2 \leq c_M B$.*

Let $T_\star$ be given by eq. (SE.10) and let $(\mathcal{Q}_{ij})$ be the solution of eq. (SE.7) of proposition SE.1. Then the following hold.

- *(i) With probability tending to one, $t_{\mathrm{stop}} > BT_\star$, $\inf_{t \leq BT_\star} \mathscr{R}(t) \geq r_0/2$ and $\sup_{t \leq BT_\star} \|\Delta_t\|_{\mathrm{F}}^2 \leq E_\star + 1$.*
- *(ii) $\sup_{\tau \leq T_\star} \|\chi^{\mathrm{act}}(\tau) - \chi^{\mathrm{ode}}(\tau)\|_{\mathrm{BL}} \to 0$ in probability.*
- *(iii) $\sup_{\tau \leq T_\star} \|\mu\lambda\,(\chi^{\mathrm{act}}(\tau) - \chi^{\mathrm{ode}}(\tau))\|_{\mathrm{BL}} \to 0$ and $\sup_{\tau \leq T_\star} |\mathscr{R}(\lfloor B\tau \rfloor) - \mathcal{R}(\tau)| \to 0$ in probability.*
- *(iv) On the event of (i), $\sup_{t \leq BT_\star} \|\widehat{A}_t\| \to 0$ in probability.*

The statement is about aggregates. It compares the actual and deterministic spectral energy measures in a weak topology, and their risks; it does not assert that $B^2\mathscr{Q}_{ij}(t)$ is close to $B^2\mathcal{Q}_{ij}(t/B)$ for every pair. The equations depend on the dimension through the coefficients and the initial data, and no convergence of these is assumed; corollary SE.2 adds it.

Corollary SE.1 (Half-anisotropic input)**.** *Assume the hypotheses of theorem SE.1 with* $\Sigma_{\rm in} = \mathrm{Id}_{N_{\rm in}}$. *Let* $\mathfrak{d}_i \overset{\rm def}{=} \mathfrak{d}(\mu_i, 1)$ *and let* $\mathfrak{v}_i \overset{\rm def}{=} \mathfrak{v}_i(\varrho)$ *be the row coefficient of eq. (SD.75). Then the row equations*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(B\Omega_i(\tau)\big) = -\frac{2h\,\mathfrak{d}_i}{\sqrt{\mathcal{R}(\tau)}}\,B\Omega_i(\tau) + h^2\mathfrak{v}_i, \qquad \mathcal{R}(\tau) = \frac{1}{2}\sum_i \mu_i\Omega_i(\tau), \qquad \Omega_i(0) = \|\Delta_0^\top u_i\|^2, \tag{SE.13}$$

have a unique solution on $[0, T_\star]$, *given by* $\Omega_i = \sum_j \Omega_{ij}$ *with* (Ω_{ij}) *the solution of eq. (SE.7). The row measures* $\chi^{\rm act}_{\rm row}(\tau) \overset{\rm def}{=} \sum_i \mathscr{Q}_i(\lfloor B\tau\rfloor)\,\delta_{\mu_i}$ *and* $\chi^{\rm ode}_{\rm row}(\tau) \overset{\rm def}{=} \sum_i \Omega_i(\tau)\,\delta_{\mu_i}$ *on* $[0, C_\Sigma]$ *satisfy* $\sup_{\tau\le T_\star} \|\chi^{\rm act}_{\rm row}(\tau) - \chi^{\rm ode}_{\rm row}(\tau)\|_{\rm BL} \to 0$ *in probability, and the same holds for the measures weighted by* μ.

Dividing eq. (SE.13) by B gives $\mathrm{d}\Omega_i/\mathrm{d}\tau = -2h\mathfrak{d}_i\Omega_i/\sqrt{\mathcal{R}} + h^2\mathfrak{v}_i/B$, which is the row recursion eq. (14) with B steps per unit of τ. The kernel $\mathfrak{d}_i$ is the common value of the $\mathfrak{d}_{ij}$ (corollary SC.2), and the volatility $\mathfrak{v}_i = \sum_j \mathfrak{v}_{ij}$ is their sum (proposition SD.4).

Corollary SE.2 (Continuum equation)**.** *Assume the hypotheses of theorem SE.1, and write* $\mathfrak{d}_B$, ζ_B, $\chi^{\rm act}_B$, $\chi^{\rm ode}_B$ *for the objects at the dimension indexed by* B.

(i) Every sequence of dimensions has a subsequence along which $\mathfrak{d}_B \to \mathfrak{d}_\infty$ *uniformly on* $\mathbb{K}$, $\zeta_B \to \zeta_\infty$ *weakly and* $\chi^{\rm act}_B(0) \to \chi_0$ *weakly.*

(ii) Along any such subsequence, the measure equation

$$\chi(\tau) = \chi_0 + \int_0^\tau \Big(-\frac{2h\,\mathfrak{d}_\infty\,\chi(\sigma)}{\sqrt{\mathcal{R}(\chi(\sigma))}} + h^2\zeta_\infty\Big)\mathrm{d}\sigma \tag{SE.14}$$

has a unique solution χ *on* $[0, T_\star]$ *among paths with values in* $\mathfrak{M}$ *that are continuous for the bounded-Lipschitz norm, and*

$$\sup_{\tau\le T_\star} \|\chi^{\rm ode}_B(\tau) - \chi(\tau)\|_{\rm BL} \to 0, \qquad \sup_{\tau\le T_\star} \|\chi^{\rm act}_B(\tau) - \chi(\tau)\|_{\rm BL} \to 0 \text{ in probability.}$$

(iii) Along any such subsequence, $\sup_{\tau\le T_\star} |\mathscr{R}(\lfloor B\tau\rfloor) - \mathcal{R}(\chi(\tau))| \to 0$ *in probability.*

Convergence of the forcing measures with the dimension is not proved in the volatility section; it is a hypothesis of (ii), and (i) says that it always holds along a subsequence. If the three sequences converge along the whole sequence of dimensions, the conclusions of (ii) and (iii) hold along it.

Remark SE.1 (Scope)**.** *Theorem SE.1 is a statement about spectral aggregates in a weak topology; it makes no claim about the accuracy of an individual random mode or row. It is proved at a fixed normalized ridge* $\varrho > 0$, *which is the convention of definition SA.2; it covers a fixed physical ridge only on the time ranges described in remark SA.1, and it says nothing about the endpoint* $\varrho = 0$, *the exact* SignSVD *map. It holds on the interval* $[0, T_\star]$, *whose length is fixed by the initial risk, energy and the coefficient bounds, and not for all times. It concerns the regularized map* Q_ϱ *itself; no statement is made about a Newton–Schulz approximation of it.*

Remark SE.2 (No momentum)**.** *At* $\beta = 0$ *the buffer is the current gradient (eq. (SA.16)), the initial buffer plays no role, the drift kernel is that of remark SC.1 with* $L_\beta = J$, *and the forcing measure is the same-time measure* $\zeta = \overline{\mathfrak{C}}_0(\varrho, \varrho)$ *of remark SD.3. Theorem SE.1 then compares the regularized* SignSVD *algorithm with eq. (6) for these kernels. The proof below applies without change; every factor* β^ℓ *with* $\ell \ge 1$ *is zero.*

Organization of the proof. Appendix SE.4 defines the frozen chains attached to a state and collects finite estimates on the regularized map. Appendix SE.5 bounds the expected frozen update and its ridge derivative without any diffuseness. Appendix SE.6 expands the frozen chain to first order in the step and bounds the projected response. Appendix SE.7 transports these estimates to the algorithm: coupling, Taylor transfer, Gaussian concentration, the martingale estimate, the delayed projections, the response of the expected block increment, and the persistence of diffuseness. Appendix SE.8 replaces the frozen quantities by the coefficients of the drift and volatility sections and proves the block residual estimate. Appendix SE.9 shows that the admissible set is not left before $T_\star$, and appendix SE.10 passes to the bounded-Lipschitz norm and proves the three statements above.

SE.4 Frozen chains and finite filter estimates

Definition SE.2 (Frozen chains). *Let $t \geq 0$, let $D, m \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ with $\mathscr{R}(D) > 0$, and let $G_j^*[\cdot]$, $j \geq 0$, be the gradients eq. (SE.1) of auxiliary batches. The stationary chain at (t, D) is*

$$N_{-1} \stackrel{\mathrm{def}}{=} \sum_{j \geq 0} w_j \, G_j^*[D], \qquad N_r \stackrel{\mathrm{def}}{=} \beta N_{r-1} + (1-\beta)\, G_{t+r+1}^\circ[D], \qquad V_r \stackrel{\mathrm{def}}{=} Q_{\varrho^\circ(D)}(N_r),$$

for $r \geq 0$. The same-buffer chain at (t, D, m) is

$$\widehat{N}_{-1} \stackrel{\mathrm{def}}{=} m, \qquad \widehat{N}_r \stackrel{\mathrm{def}}{=} \beta \widehat{N}_{r-1} + (1-\beta)\, G_{t+r+1}^\circ[D], \qquad \widehat{V}_r \stackrel{\mathrm{def}}{=} Q_{\varrho^\circ(D)}(\widehat{N}_r) \qquad (r \geq 0).$$

At a step $t < t_{\mathrm{stop}}$ of the algorithm, the chains at time t are those with $D = \Delta_t$ and $m = M_{t-1}^\circ$. The mean frozen update at D is $\overline{V}(D) \stackrel{\mathrm{def}}{=} \mathbb{E}\, V_0$.

Both chains use the actual batches of the steps $t, t+1, \ldots$; the stationary chain replaces the actual past by an auxiliary past computed at the state D, and the same-buffer chain keeps the actual buffer.

Lemma SE.2 (Laws of the frozen chains). *Let D, m be deterministic with $\mathscr{R}(D) > 0$, and put $C_M \stackrel{\mathrm{def}}{=} C_\Sigma^2(\gamma_\star + 2)^2$.*

(i) The sequence $(N_r/\sqrt{2\mathscr{R}(D)})_{r \geq -1}$ has the law of the buffers $(M_{t'})_{t' \geq t-1}$ of the frozen-state model of definition SA.3 at the normalized state $\widehat{A}(D)$, with the samples indexed by time, and $V_r = Q_\varrho(N_r/\sqrt{2\mathscr{R}(D)})$ is its update. In particular $(V_r)_{r \geq 0}$ is stationary, $\overline{V}(D)$ is the expected frozen update $\mathbb{E}\, Q_\varrho(M_t)$ at $\widehat{A}(D)$, and the lag covariances, the integrated covariance measure $\mathfrak{C}_{\mathrm{int}}(\varrho)$ and the finite-time measures $\mathfrak{F}_T(\varrho)$ of the volatility section at the state $\widehat{A}(D)$ are those of (V_r).

(ii) For $r \geq -1$, $\widehat{N}_r - N_r = \beta^{r+1}(m - N_{-1})$.

(iii) $\mathbb{E}\, \|N_{-1}\|_{\mathrm{F}}^2 \leq 2\mathscr{R}(D)\, C_M B$.

(iv) At a step $t < t_{\mathrm{stop}}$, the statements (i)–(iii) hold conditionally on $\mathscr{F}_t$ with $D = \Delta_t$ and $m = M_{t-1}^\circ$.

Proof. For (i), write $d_a = z_a^\top \widehat{A}(D) w_a$. By eq. (SE.1), $G_{t+1}^\circ[D] = \sqrt{2\mathscr{R}(D)}\, G_{t+1}$, where G_{t+1} is the batch eq. (SA.26) built from the same samples at the normalized state $\widehat{A}(D)$. The auxiliary batches and the batches of the steps $t, t+1, \ldots$ are independent, so $N_r/\sqrt{2\mathscr{R}(D)}$ is the time-indexed buffer of the frozen-state model. By proposition SA.3(ii), $Q_{\varrho^\circ(D)}(N_r) = Q_\varrho(N_r/\sqrt{2\mathscr{R}(D)})$. Stationarity is lemma SA.5(ii), and the covariances of the volatility section are defined from these updates in definition SD.1 and theorem SD.8(iv).

Part (ii) follows by subtracting the two recursions: the difference satisfies $\widehat{N}_r - N_r = \beta(\widehat{N}_{r-1} - N_{r-1})$ and starts from $m - N_{-1}$, as in the proof of proposition SA.6. Part (iii) is (i) together with the bound $B^{-1}\, \mathbb{E}\, \|M_t\|_{\mathrm{F}}^2 \leq C_M$ of lemma SD.3. Part (iv) holds because the batches of the steps $t, t+1, \ldots$ and the auxiliary batches are independent of the past. □

Lemma SE.3 (Moments of one batch). *Let $E \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ be deterministic, put $A_E \stackrel{\mathrm{def}}{=} {\Sigma_{\mathrm{out}}}^{1/2} E {\Sigma_{\mathrm{in}}}^{1/2}$ and $d_a \stackrel{\mathrm{def}}{=} z_a^\top A_E w_a$ for the samples of one batch, and let $G^\circ[E]$ be the gradient eq. (SE.1) of that batch.*

(i) $\mathbb{E}\, \|G^\circ[E]\|_{\mathrm{F}}^2 \leq C_M B \|A_E\|_{\mathrm{F}}^2$.

(ii) $\big(\mathbb{E}\, \|G^\circ[E]\|^2\big)^{1/2} \leq C \log(2B) \|A_E\|_{\mathrm{F}}$ and $\big\| \max_{a \leq B} |d_a| \big\|_6 \leq C \log(2B) \|A_E\|_{\mathrm{F}}$.

(iii) The sample $X_a \stackrel{\mathrm{def}}{=} B^{-1} d_a {\Sigma_{\mathrm{out}}}^{1/2} (z_a \otimes w_a) {\Sigma_{\mathrm{in}}}^{1/2}$ has rank one and $(\mathbb{E}\, \|X_a\|_*^2)^{1/2} \leq 7 C_\Sigma \gamma_\star \|A_E\|_{\mathrm{F}}$.

The same bounds hold conditionally on $\mathscr{F}_t$ for the gradient $G_{t+1}^\circ[E]$ of the batch of step t at a matrix E determined by the past $\mathscr{F}_t$.

Proof. All three bounds are homogeneous of degree one in E, so we may assume $\|A_E\|_{\mathrm{F}} = 1$; then $E \neq 0$, $\widehat{A}(E) = A_E$ and $G^\circ[E]$ is the normalized batch of eq. (SA.10) at the state E, up to the factor $\sqrt{2\mathscr{R}(E)} = 1$.

For (i), lemma SA.4 and lemma SA.6(iv) give

$$\mathbb{E}\,\|G^\circ[E]\|_{\mathrm{F}}^2 = \|\,\mathbb{E}\,G^\circ[E]\|_{\mathrm{F}}^2 + \mathsf{V}_B \le C_\Sigma^2 + \frac{C_\Sigma^2}{B}\big(N_{\mathrm{out}}N_{\mathrm{in}} + 2N_{\mathrm{out}} + 2N_{\mathrm{in}} + 3\big) \le C_\Sigma^2(\gamma_\star + 2)^2 B.$$

For (ii), write $Y \overset{\mathrm{def}}{=} {\Sigma_{\mathrm{out}}}^{1/2}[z_1 \cdots z_B]$ and $X \overset{\mathrm{def}}{=} {\Sigma_{\mathrm{in}}}^{1/2}[w_1 \cdots w_B]$, so that

$$G^\circ[E] = \frac{1}{B}\,Y\,\mathrm{diag}(d_1, \dots, d_B)X^\top, \qquad \|G^\circ[E]\| \le \frac{1}{B}\|Y\|\|X\| \max_a |d_a|.$$

The matrix $[z_1 \cdots z_B]$ has independent standard Gaussian entries, so by [71, Theorem 4.4.5] its operator norm exceeds $C(\sqrt{N_{\mathrm{out}}} + \sqrt{B} + u)$ with probability at most $2e^{-u^2}$; integrating this tail gives $\|\|Y\|\|_6 \le C\sqrt{B}$, and likewise $\|\|X\|\|_6 \le C\sqrt{B}$. For the residuals, lemma SA.7(i) gives $\|d_a\|_{2p} \le ((2p-1)!!)^{1/p} \le 2p$ for every integer $p \ge 1$. Choosing $2p$ to be the least even integer at least $\max(6, \log B)$,

$$\big\|\max_a |d_a|\big\|_6 \le \big\|\max_a |d_a|\big\|_{2p} \le \Big(\sum_{a \le B} \mathbb{E}\,|d_a|^{2p}\Big)^{1/(2p)} \le B^{1/(2p)}\,2p \le C\log(2B).$$

Hölder's inequality with the three exponents 6 gives

$$\mathbb{E}\,\|G^\circ[E]\|^2 \le \frac{1}{B^2}\|\|Y\|\|_6^2\|\|X\|\|_6^2\big\|\max_a |d_a|\big\|_6^2 \le C\log^2(2B).$$

For (iii), $X_a = d_a\,\mathsf{u}_a \otimes \mathsf{v}_a$ with $\mathsf{u}_a = {\Sigma_{\mathrm{out}}}^{1/2}z_a/\sqrt{B}$ and $\mathsf{v}_a = {\Sigma_{\mathrm{in}}}^{1/2}w_a/\sqrt{B}$ as in eq. (SB.26), so $\|X_a\|_* = |d_a|\|\mathsf{u}_a\|\|\mathsf{v}_a\|$, and lemma SB.6 gives the bound. The conditional statement holds because the batch of step t is independent of the past. □

The regularized map is controlled in three norms through the symmetric dilation $\mathcal{D}(K) = \left(\begin{smallmatrix} 0 & K \\ K^\top & 0 \end{smallmatrix}\right)$ of the proof of lemma SA.2. We write $\|\cdot\|_{S_1} = \|\cdot\|_*$ and $\|\cdot\|_{S_\infty} = \|\cdot\|$, and $\mathsf{X}_s(K) = (KK^\top + s\,\mathrm{Id})^{-1}K$ as in eq. (SB.10).

Lemma SE.4 (Schatten sensitivity of the filter)**.** *Let $K, K' \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$, $s > 0$, $\varrho' > 0$ and $p \in \{1, \infty\}$. Then*

- *(i)* $\|\mathsf{X}_s(K) - \mathsf{X}_s(K')\|_{S_p} \le s^{-1}\|K - K'\|_{S_p}$;
- *(ii)* $\|Q_{\varrho'}(K) - Q_{\varrho'}(K')\|_{S_p} \le \varrho'^{-1/2}\|K - K'\|_{S_p}$;
- *(iii)* $\|\partial_{\varrho'}Q_{\varrho'}(K) - \partial_{\varrho'}Q_{\varrho'}(K')\|_* \le \frac{1}{2}\varrho'^{-3/2}\|K - K'\|_*$;
- *(iv) if $\varrho_1, \varrho_2 \ge \varrho_\circ > 0$, then $\|Q_{\varrho_1}(K) - Q_{\varrho_2}(K)\| \le |\varrho_1 - \varrho_2|/(2\varrho_\circ)$ and the same bound multiplied by $\sqrt{\min(N_{\mathrm{out}}, N_{\mathrm{in}})}$ holds in Frobenius norm.*

Proof. The mixed resolvent is a block of a resolvent of the dilation. For $c > 0$, $\mathcal{D}(K)^2 = \mathrm{diag}(KK^\top, K^\top K)$, and by lemma SB.3

$$\mathcal{D}(K)\big(\mathcal{D}(K)^2 + c^2\,\mathrm{Id}\,\big)^{-1} = \begin{pmatrix} 0 & \mathsf{X}_{c^2}(K) \\ \mathsf{X}_{c^2}(K)^\top & 0 \end{pmatrix}.$$

For a real symmetric T the left side is a sum of two resolvents,

$$T(T^2 + c^2\,\mathrm{Id})^{-1} = \frac{1}{2}\Big[(T - ic\,\mathrm{Id})^{-1} + (T + ic\,\mathrm{Id})^{-1}\Big],$$

each of operator norm at most c^{-1}. The resolvent identity $(T - ic)^{-1} - (S - ic)^{-1} = (T - ic)^{-1}(S - T)(S - ic)^{-1}$ and the ideal property $\|XYZ\|_{S_p} \le \|X\|\|Y\|_{S_p}\|Z\|$ [8, Proposition IV.2.4 and (IV.40)] give, for symmetric T, S,

$$\|T(T^2 + c^2)^{-1} - S(S^2 + c^2)^{-1}\|_{S_p} \le c^{-2}\|T - S\|_{S_p}.$$

A matrix $\left(\begin{smallmatrix} 0 & Y \\ Y^\top & 0 \end{smallmatrix}\right)$ has the singular values of Y, each twice, so its S_p norm is $2^{1/p}\|Y\|_{S_p}$, with $2^{1/\infty} = 1$. Applying the last display with $T = \mathcal{D}(K)$, $S = \mathcal{D}(K')$ and $c^2 = s$, the factor $2^{1/p}$ appears on both sides and cancels, which proves (i).

Part (ii) integrates (i). By lemma SA.3, $Q_{\varrho'}(K) = \frac{2}{\pi}\int_0^\infty \mathsf{X}_{\varrho'+u^2}(K)\,\mathrm{d}u$, so

$$\|Q_{\varrho'}(K) - Q_{\varrho'}(K')\|_{S_p} \le \frac{2}{\pi}\int_0^\infty \frac{\mathrm{d}u}{\varrho' + u^2}\,\|K - K'\|_{S_p} = \varrho'^{-1/2}\|K - K'\|_{S_p}.$$

Part (iii) is the same argument for the derivative in the ridge. For a symmetric T, partial fractions give

$$-T(T^2 + c^2)^{-2} = \frac{i}{4c}\Big[(T - ic)^{-2} - (T + ic)^{-2}\Big],$$

and $(T - ic)^{-2} - (S - ic)^{-2} = (T - ic)^{-1}\big[(T - ic)^{-1} - (S - ic)^{-1}\big] + \big[(T - ic)^{-1} - (S - ic)^{-1}\big](S - ic)^{-1}$ has nuclear norm at most $2c^{-3}\|T - S\|_*$. Hence the difference of $-T(T^2 + c^2)^{-2}$ at T and S has nuclear norm at most $c^{-4}\|T - S\|_*$. The derivative of the integrand of lemma SA.3 in the ridge is the upper right block of $-\mathcal{D}(K)(\mathcal{D}(K)^2 + (\varrho' + u^2))^{-2}$; with the block factor cancelled as before,

$$\|\partial_{\varrho'}Q_{\varrho'}(K) - \partial_{\varrho'}Q_{\varrho'}(K')\|_* \le \frac{2}{\pi}\int_0^\infty \frac{\mathrm{d}u}{(\varrho' + u^2)^2}\,\|K - K'\|_* = \frac{1}{2\varrho'^{3/2}}\|K - K'\|_*.$$

Part (iv) is eq. (SD.6) for the filter ω_ϱ of eq. (SD.2) together with the first two bounds of eq. (SD.4). □

Lemma SE.5 (Second-order filter estimates)**.** *Let $0 < \ell_- \le \ell_+ < \infty$. There is C, depending only on (ℓ_-, ℓ_+), such that for $\ell \in [\ell_-, \ell_+]$ and all $N, N', R \in \mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$, with $DQ_\ell(N)[R]$ the derivative of Q_ℓ at N in the direction R:*

- *(i)* $\|DQ_\ell(N)[R]\|_{\mathrm{F}} \le \ell^{-1/2}\|R\|_{\mathrm{F}}$;
- *(ii)* $\|Q_\ell(N + R) - Q_\ell(N) - DQ_\ell(N)[R]\|_{\mathrm{F}} \le C\|R\|\|R\|_{\mathrm{F}}$;
- *(iii)* $\|DQ_\ell(N)[R] - DQ_\ell(N')[R]\|_{\mathrm{F}} \le C\|N - N'\|\|R\|_{\mathrm{F}}$;
- *(iv)* $\|\partial_\ell Q_\ell(N)\| \le C$, $\|\partial_\ell Q_\ell(N)\|_{\mathrm{F}} + \|\partial_\ell^2 Q_\ell(N)\|_{\mathrm{F}} \le C\sqrt{\min(N_{\mathrm{out}}, N_{\mathrm{in}})}$, *and* $\|\partial_\ell Q_\ell(N) - \partial_\ell Q_\ell(N')\|_{\mathrm{F}} \le C\|N - N'\|_{\mathrm{F}}$.

The map $(N, \ell) \mapsto Q_\ell(N)$ is smooth on $\ell > 0$, by the integral representation of lemma SA.3, so the derivatives in the statement exist.

Proof. Part (i) is the Lipschitz estimate eq. (SA.13) read as a bound on directional derivatives.

Parts (ii) and (iii) come from the second derivative of the dilated resolvents. By the proof of lemma SE.4, $Q_\ell(N)$ is the upper right block of

$$f_\ell(\mathcal{D}(N)) = \frac{1}{\pi}\int_0^\infty \Big[(\mathcal{D}(N) - ia\,\mathrm{Id})^{-1} + (\mathcal{D}(N) + ia\,\mathrm{Id})^{-1}\Big]\mathrm{d}u, \qquad a = \sqrt{\ell + u^2}.$$

For $\mathsf{G} = (H \mp ia)^{-1}$ with H symmetric, the second derivative in the directions H_1, H_2 is $\mathsf{G}H_1\mathsf{G}H_2\mathsf{G} + \mathsf{G}H_2\mathsf{G}H_1\mathsf{G}$, whose Frobenius norm is at most $2a^{-3}\|H_1\|\|H_2\|_{\mathrm{F}}$. Since $\|\mathcal{D}(R)\| = \|R\|$, $\|\mathcal{D}(R)\|_{\mathrm{F}} = \sqrt{2}\|R\|_{\mathrm{F}}$ and $\int_0^\infty (\ell + u^2)^{-3/2}\mathrm{d}u = \ell^{-1}$, the second derivative of Q_ℓ satisfies

$$\|D^2 Q_\ell(N)[R_1, R_2]\|_{\mathrm{F}} \le \frac{4\sqrt{2}}{\pi\ell}\|R_1\|\|R_2\|_{\mathrm{F}}.$$

Taylor's formula with integral remainder along $N + \vartheta R$, $\vartheta \in [0, 1]$, gives (ii), and the same formula for $\vartheta \mapsto DQ_\ell(N' + \vartheta(N - N'))[R]$ gives (iii).

Part (iv) concerns singular-value functions. The ridge derivatives of $Q_\ell = \omega_\ell^\sharp$ are $\phi_1^\sharp$ and $\phi_2^\sharp$ in the notation eq. (SD.1), with the odd functions

$$\phi_1(x) = -\frac{x}{2(x^2 + \ell)^{3/2}}, \qquad \phi_2(x) = \frac{3x}{4(x^2 + \ell)^{5/2}},$$

which are bounded by constants depending on ℓ_-, and ϕ_1 has derivative $\phi_1'(x) = (2x^2 - \ell)/(2(x^2 + \ell)^{5/2})$, bounded by $\ell^{-3/2}$. Lemma SD.1 gives the three bounds. □

The last finite tool bounds a covariance by single-coordinate sensitivities.

Lemma SE.6 (Covariance inequality)**.** *Let F, G be square-integrable functions of countably many independent random coordinates $(\xi_l)_{l\ge 1}$, and let $F^{[l]}$, $G^{[l]}$ denote F, G with ξ_l replaced by an independent copy. Then*

$$|\operatorname{Cov}(F,G)| \le \sum_{l\ge 1} \|F - F^{[l]}\|_2 \, \|G - G^{[l]}\|_2 .$$

Proof. Let $\mathcal{A}_l$ be generated by $\xi_1, \dots, \xi_l$ and $D_l F \stackrel{\text{def}}{=} \mathbb{E}[F \mid \mathcal{A}_l] - \mathbb{E}[F \mid \mathcal{A}_{l-1}]$. By Lévy's upward theorem [74, Theorem 14.2], $\mathbb{E}[F \mid \mathcal{A}_l] \to F$ in L^2, so $F - \mathbb{E}\,F = \sum_l D_l F$ in L^2, and likewise for G. Distinct martingale differences are orthogonal, so $\operatorname{Cov}(F,G) = \sum_l \mathbb{E}[D_l F \, D_l G]$. Since the copy of ξ_l is independent, $\mathbb{E}[F^{[l]} \mid \mathcal{A}_l] = \mathbb{E}[F \mid \mathcal{A}_{l-1}]$, hence $D_l F = \mathbb{E}[F - F^{[l]} \mid \mathcal{A}_l]$ and $\|D_l F\|_2 \le \|F - F^{[l]}\|_2$ by Jensen's inequality. The Cauchy–Schwarz inequality applied to each term proves the claim. □

SE.5 Finite mean and susceptibility of the frozen update

The drift section identifies the expected frozen update for diffuse states. Along the algorithm, diffuseness is a conclusion and not a hypothesis, so the comparison needs bounds on the expected update that hold for every normalized state. This subsection proves them. We work in the frozen-state model of definition SA.3 with $\|\widehat{A}_t\|_{\mathrm{F}} = 1$, covariances of norm at most C_Σ and $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \le \gamma_\star B$, and *without any condition on* $\|\widehat{A}_t\|$. We use the notation of the drift section: the samples $(\mathsf{h}_{ja}, \mathsf{u}_{ja}, \mathsf{v}_{ja})$ of eq. (SB.26), the deleted buffers $M_0^{(ja)}$, the coefficients $\alpha_{1,ja}, \alpha_{2,ja}, \alpha_{3,ja}$, the conditioning $\mathcal{G}_{ja}$, the expected mixed resolvent $\mathcal{H}(s) = \mathbb{E}\,\mathsf{X}_s(M_t)$ and its deleted versions $\mathcal{H}_j(s)$, the quantities $\phi_j, b_j, \mathcal{C}_j$ of eq. (SC.26), and the expected resolvents and traces $\mathcal{P}(s), \mathcal{P}_j(s), \sigma_B(s), \tilde{\sigma}_B(s), q_B(s)$ of eq. (SB.30).

For the derivative in the ridge we also need complex spectral parameters. Let

$$\mathbb{S} \stackrel{\text{def}}{=} \{z \in \mathbb{C} : \; x \stackrel{\text{def}}{=} \operatorname{Re} z > 0, \; |\operatorname{Im} z| \le x/2\},$$

and for $z \in \mathbb{S}$ and real K put $R_z(K) \stackrel{\text{def}}{=} (KK^\top + z\,\mathrm{Id})^{-1}$, $\widetilde{R}_z(K) \stackrel{\text{def}}{=} (K^\top K + z\,\mathrm{Id})^{-1}$, $\mathsf{X}_z(K) \stackrel{\text{def}}{=} R_z(K)K$ and $\mathcal{H}(z) \stackrel{\text{def}}{=} \mathbb{E}\,\mathsf{X}_z(M_t)$; at real $z = s$ these are the objects of eq. (SB.10).

Lemma SE.7 (Finite mean and susceptibility)**.** *In the frozen-state model with $\|\widehat{A}_t\|_{\mathrm{F}} = 1$, $\|\Sigma_{\mathrm{out}}\|, \|\Sigma_{\mathrm{in}}\| \le C_\Sigma$ and $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \le \gamma_\star B$, and with no condition on $\|\widehat{A}_t\|$:*

- *(i) for every compact $I \subset (0,\infty)$ there is C_I with $\|\mathcal{H}(z)\|_{\mathrm{F}} \le C_I$ for $z \in \mathbb{S}$ with $\operatorname{Re} z \in I$, and $\|\mathcal{H}(z)\|_{\mathrm{F}} \le C/\operatorname{Re} z$ for $z \in \mathbb{S}$ with $\operatorname{Re} z \ge 1$;*
- *(ii) for real s, $\|\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{H}(s)\|_{\mathrm{F}} \le C_I$ for $s \in I$ and $\|\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{H}(s)\|_{\mathrm{F}} \le C/s^2$ for $s \ge 2$;*
- *(iii) for $0 < \varrho_- \le \varrho' \le \varrho_+$, $\|\mathbb{E}\,Q_{\varrho'}(M_t)\|_{\mathrm{F}} + \|\mathbb{E}\,\partial_{\varrho'} Q_{\varrho'}(M_t)\|_{\mathrm{F}} \le C$, with C depending also on (ϱ_-, ϱ_+);*
- *(iv) for the fixed ridge ϱ there are C and a sequence $r_B \to 0$, depending only on $(\varrho, \gamma_\star, C_\Sigma)$ and on the sequence of dimensions, such that $\|\mathbb{E}\,Q_\varrho(M_t)\| \le C\|\widehat{A}_t\| + r_B$ for every normalized state.*

The constants depend only on $(\gamma_\star, C_\Sigma)$ and on I where indicated, not on $\widehat{A}_t$, on the dimension or on the weights within the class eq. (SB.27). Consequently, for the stationary chain of definition SE.2 at a deterministic D with $r_- \le \mathscr{R}(D) \le r_+$,

$$\begin{aligned} \|\overline{V}(D)\|_{\mathrm{F}} \le C, \qquad \big\|\mathbb{E}\,\partial_\ell Q_\ell(N_0)\big|_{\ell=\varrho^\circ(D)}\big\|_{\mathrm{F}} \le C, \\ \|{\Sigma_{\mathrm{out}}}^{1/2}\overline{V}(D){\Sigma_{\mathrm{in}}}^{1/2}\| \le C\|{\Sigma_{\mathrm{out}}}^{1/2} D {\Sigma_{\mathrm{in}}}^{1/2}\| + Cr_B, \end{aligned} \tag{SE.15}$$

with C depending also on (r_-, r_+).

Part (i) is theorem SC.1(ii) on the sector $\mathbb{S}$, which is what the Cauchy estimate of (ii) requires; part (iv) is the operator-norm statement on which the persistence of diffuseness rests.

Proof. *(i).* By theorem SC.1(ii), in the form eq. (SC.7), for every $s_\circ > 0$ there is $C_\circ$ depending only on $(s_\circ, C_\Sigma, \gamma_\star)$ with $\|\mathcal{H}(z)\|_{\mathrm{F}} \le C_\circ / \operatorname{Re} z$ for $z \in \mathbb{S}$ with $\operatorname{Re} z \ge s_\circ$, in every frozen-state model with the stated hypotheses and weights in the class eq. (SB.27). The case $s_\circ = 1$ is the second bound of (i), and the case $s_\circ = \min I$ gives the first with $C_I = C_\circ / \min I$.

(ii). At fixed dimension $z \mapsto \mathcal{H}(z)$ is holomorphic on $\operatorname{Re} z > 0$, since $z \mapsto \mathsf{X}_z(M_t)$ is. For real $s > 0$ the circle $|\zeta - s| = s/4$ lies in $\mathbb{S}$, because on it $\operatorname{Re}\zeta \ge 3s/4$ and $|\operatorname{Im}\zeta| \le s/4 \le \operatorname{Re}\zeta/3$. Cauchy's estimate for the derivative gives

$$\Big\| \frac{\mathrm{d}}{\mathrm{d}s} \mathcal{H}(s) \Big\|_{\mathrm{F}} \le \frac{4}{s} \max_{|\zeta - s| = s/4} \|\mathcal{H}(\zeta)\|_{\mathrm{F}}.$$

For $s \ge 2$ every point of the circle has real part at least $3s/4 \ge 1$, so (i) bounds the maximum by $4C/(3s)$, which gives C/s^2. For $s \in I$ the circle has real parts in the compact $[\frac{3}{4}\min I, \frac{5}{4}\max I]$, and (i) gives C_I.

(iii). By lemma SA.3 and differentiation of its integrand in the ridge,

$$\mathbb{E}\, Q_{\varrho'}(M_t) = \frac{2}{\pi} \int_0^\infty \mathcal{H}(\varrho' + u^2)\, \mathrm{d}u, \qquad \mathbb{E}\, \partial_{\varrho'} Q_{\varrho'}(M_t) = \frac{2}{\pi} \int_0^\infty \frac{\mathrm{d}\mathcal{H}}{\mathrm{d}s}(\varrho' + u^2)\, \mathrm{d}u.$$

Split both integrals at $u = 2$. On $[0, 2]$ the parameter lies in $I \overset{\text{def}}{=} [\varrho_-, \varrho_+ + 4]$, where (i) and (ii) give C_I; on $[2, \infty)$ it is at least $u^2 \ge 4$, where they give C/u^2 and C/u^4. Both integrals are therefore bounded.

(iv). We work on the real axis. The targets of lemma SB.12 are centered at the expected traces $(\sigma_B, \tilde{\sigma}_B, q_B)$, and its sequence $e_B(I)$ depends only on $(I, C_\Sigma, \gamma_\star)$, so it serves for every normalized state, every dimension and all weights in the class. For a sample of age j, with $\check{x}_j = w_j^2 q_B$ and the targets $\check{\alpha}_{1,j}, \check{\alpha}_{2,j}$ of eq. (SB.33), put

$$\mathsf{e}_B^{(1)}(s) \overset{\text{def}}{=} \sup_j \frac{\|\alpha_{1,j} - \check{\alpha}_{1,j}\|_2}{w_j}, \qquad \mathsf{e}_B^{(2)}(s) \overset{\text{def}}{=} \sup_j \frac{\|\alpha_{2,j} - \check{\alpha}_{2,j}\|_2}{w_j^2}.$$

Then $\sup_{s \in I}(\mathsf{e}_B^{(1)} + \mathsf{e}_B^{(2)}) \le 2 e_B(I)$ on every compact I, and for $s \ge 1$, by lemma SC.2, $|f_x(d)| \le |d|$, $0 \le \psi_x(d) \le d^2$ and $\sigma_B \le C_\Sigma \gamma_\star / s$,

$$\mathsf{e}_B^{(1)}(s) \le C, \qquad \mathsf{e}_B^{(2)}(s) \le \frac{C}{s}. \tag{SE.16}$$

The proof of proposition SC.3, run with these targets and with the expected deleted resolvent $\mathcal{P}_j(s)$ kept in place of $\overline{R}(-s)$, gives an equation for $\mathcal{H}(s)$ in which every error is displayed with its dependence on $\|\widehat{A}_t\|$. In the source term, the replacement of $\alpha_{1,j}$ costs $C_\Sigma w_j \mathsf{e}_B^{(1)}/s$, and eq. (SB.22) with $\|f_x'''\|_\infty \le 36\check{x}_j \le 36 q_B$ (lemma SC.7) costs $36 C_\Sigma w_j q_B \|\widehat{A}_t\|^2/s$. In the Gram term, the centered part costs $C w_j^2 (\mathsf{e}_B^{(2)} + 6\sigma_B \|\widehat{A}_t\|)/\sqrt{s}$, by lemma SB.7 and eq. (SB.23) with $\|\psi''\|_\infty \le 6$. The deletion mean and covariance terms cost $C w_j^3/(s^2\sqrt{B})$ and $C w_j^2/(s^{3/2}\sqrt{B})$, by eq. (SC.21) and lemma SC.5. Summing over j, with $\jmath_j \overset{\text{def}}{=} \mathbb{E}\, f_{\check{x}_j}'(d_j) \in [-1, 1]$ and $\mathsf{Y}_B(s) = \mathrm{Id} + b_B(s)\Sigma_{\mathrm{in}}$ as before,

$$\mathcal{H}(s)\, \mathsf{Y}_B(s) = \sum_j w_j\, \jmath_j\, \mathcal{P}_j(s)\; \mathbb{E}\, G_{t+1} + E_1(s),$$

$$\|E_1(s)\|_{\mathrm{F}} \le C\Big[\frac{\mathsf{e}_B^{(1)}(s)}{s} + \frac{q_B(s)\|\widehat{A}_t\|^2}{s} + \frac{\mathsf{e}_B^{(2)}(s) + \sigma_B(s)\|\widehat{A}_t\|}{\sqrt{s}} + \frac{1}{s\sqrt{B}} \Big] \qquad (s \ge \varrho),$$

with C depending on ϱ. The source has small operator norm: since $\|\mathcal{P}_j(s)\| \le 1/s$, $|\jmath_j| \le 1$, $\sum_j w_j \le 1$ and $\|\mathbb{E}\, G_{t+1}\| = \|{\Sigma_{\mathrm{out}}}^{1/2} \widehat{A}_t {\Sigma_{\mathrm{in}}}^{1/2}\| \le C_\Sigma \|\widehat{A}_t\|$, its operator norm is at most $C_\Sigma \|\widehat{A}_t\|/s$. Since $b_B(s) \ge 0$, $\|\mathsf{Y}_B(s)^{-1}\| \le 1$, and with $q_B, \sigma_B \le C/s$,

$$\|\mathcal{H}(s)\| \le C\Big(\frac{1}{s} + \frac{1}{s^{3/2}} + \frac{1}{s^2}\Big)\|\widehat{A}_t\| + \mathsf{e}_B(s), \qquad \mathsf{e}_B(s) \overset{\text{def}}{=} C\Big[\frac{\mathsf{e}_B^{(1)}(s)}{s} + \frac{\mathsf{e}_B^{(2)}(s)}{\sqrt{s}} + \frac{1}{s\sqrt{B}}\Big], \tag{SE.17}$$

for $s \geq \varrho$. Here $\|\widehat{A}_t\|^2 \leq \|\widehat{A}_t\|$ was used. The error e_B tends to zero uniformly on every compact interval and over all normalized states, and $\mathsf{e}_B(s) \leq C/s$ for $s \geq 1$ by eq. (SE.16).

The operator norm of the expected update is the integral of eq. (SE.17). Splitting at a cutoff $\mathsf{R} \geq 1$,

$$\|\,\mathbb{E}\; Q_\varrho(M_t)\| \leq \frac{2}{\pi}\int_0^\infty \|\mathcal{H}(\varrho+u^2)\|\,\mathrm{d}u \leq C\|\widehat{A}_t\| + \frac{2}{\pi}\Big[\mathsf{R}\sup_{s\in[\varrho,\varrho+\mathsf{R}^2]}\mathsf{e}_B(s) + \frac{C}{\mathsf{R}}\Big].$$

Let r_B be the infimum over $\mathsf{R} \geq 1$ of the bracket. It does not depend on the state, and it tends to zero: for fixed R the first term vanishes as the dimension grows, and the second is small for large R. This proves (iv).

The consequences eq. (SE.15). By lemma SE.2(i), $\overline{V}(D)$ is the expected frozen update at the state $\widehat{A}(D)$, so (iii) bounds its Frobenius norm. By proposition SA.3(i), $Q_\ell(\sqrt{a}K) = Q_{\ell/a}(K)$ for $a > 0$; with $a = 2\mathscr{R}(D)$ and ℓ near $\varrho^\circ(D) = a\varrho$ this gives $\partial_\ell Q_\ell(N_0) = a^{-1}(\partial_{\varrho'}Q_{\varrho'})(N_0/\sqrt{a})$ at $\varrho' = \varrho$, and (iii) bounds it. Finally (iv) and $\|{\Sigma_{\mathrm{out}}}^{1/2}\overline{V}(D){\Sigma_{\mathrm{in}}}^{1/2}\| \leq C_\Sigma\|\overline{V}(D)\|$, $\delta(D) = \|{\Sigma_{\mathrm{out}}}^{1/2}D{\Sigma_{\mathrm{in}}}^{1/2}\|/\sqrt{2\mathscr{R}(D)}$ give the last bound. □

SE.6 Causal response of the frozen chain

Over a block, the algorithm differs from the frozen chain at the start of the block because its state moves by a quantity of order the step. This subsection computes the first-order effect. The expansion is causal: the gradient of a batch acts on updates computed from strictly earlier batches, so the target batch can be deleted from the buffer while the direction in which the buffer is differentiated stays independent of it.

The tangent. Fix $t \geq 0$, a deterministic D with $\mathscr{R}(D) > 0$ and an initial buffer m_0, which will be either the auxiliary buffer N_{-1} or a matrix m. For $\varepsilon \geq 0$ consider the chain

$$\Delta_0^\varepsilon = D, \qquad M_{-1}^\varepsilon = m_0, \qquad M_r^\varepsilon = \beta M_{r-1}^\varepsilon + (1-\beta)\,G_{t+r+1}^\circ[\Delta_r^\varepsilon],$$
$$\mathsf{S}_r^\varepsilon = Q_{\varrho^\circ(\Delta_r^\varepsilon)}(M_r^\varepsilon), \qquad \Delta_{r+1}^\varepsilon = \Delta_r^\varepsilon - \varepsilon\mathsf{S}_r^\varepsilon.$$

For each r it is defined for all ε small enough that the risks $\mathscr{R}(\Delta_k^\varepsilon)$, $k \leq r$, stay positive. At $\varepsilon = 0$ the state is D at every time, and the buffers and updates are those of the frozen chain with initial buffer m_0, which we write $N_r^{m_0}$ and $V_r^{m_0}$: the stationary chain of definition SE.2 for $m_0 = N_{-1}$, and the same-buffer chain for $m_0 = m$. At $\varepsilon = \eta$, $D = \Delta_t$ and $m_0 = M_{t-1}^\circ$ it is the algorithm from step t.

We write $\dot{\Delta}_r, \dot{M}_r, \dot{\mathsf{S}}_r$ for the derivatives at $\varepsilon = 0$, and $\dot{V}_r$, respectively $\dot{\hat{V}}_r$, for $\dot{\mathsf{S}}_r$ when $m_0 = N_{-1}$, respectively $m_0 = m$.

Lemma SE.8 (Causal tangent)**.** *The derivatives at $\varepsilon = 0$ exist, and with $\ell = \varrho^\circ(D)$,*

$$\dot{\Delta}_r = -\sum_{k<r} V_k^{m_0}, \qquad \dot{M}_r = -\sum_{0\leq k<j\leq r} w_{r-j}\,G_{t+j+1}^\circ\big[V_k^{m_0}\big],$$
$$\dot{\mathsf{S}}_r = DQ_\ell(N_r^{m_0})\big[\dot{M}_r\big] - 2\varrho\Big\langle\Sigma_{\mathrm{out}}D\Sigma_{\mathrm{in}}, \sum_{k<r}V_k^{m_0}\Big\rangle\,\partial_\ell Q_\ell(N_r^{m_0}). \tag{SE.18}$$

Proof. The map $(K,\ell) \mapsto Q_\ell(K)$ is smooth on $\ell > 0$, so the chain rule applies at each step. The state is $\Delta_r^\varepsilon = D - \varepsilon\sum_{k<r}\mathsf{S}_k^\varepsilon$, whose derivative at 0 is $-\sum_{k<r}V_k^{m_0}$. Unrolling the buffer, $M_r^\varepsilon = \beta^{r+1}m_0 + \sum_{j\leq r} w_{r-j}G_{t+j+1}^\circ[\Delta_j^\varepsilon]$, and the linearity of G_{t+j+1}° gives $\dot{M}_r = \sum_{j\leq r} w_{r-j}G_{t+j+1}^\circ[\dot{\Delta}_j]$, which is the second formula. The physical ridge is $\varrho^\circ(\Delta) = \varrho\|\Delta\|_\Sigma^2$, whose derivative along Δ_r^ε at 0 is $2\varrho\langle D, \dot{\Delta}_r\rangle_\Sigma = 2\varrho\langle\Sigma_{\mathrm{out}}D\Sigma_{\mathrm{in}}, \dot{\Delta}_r\rangle$. The chain rule for $\mathsf{S}_r^\varepsilon = Q_{\varrho^\circ(\Delta_r^\varepsilon)}(M_r^\varepsilon)$ gives the third formula. □

In $\dot{M}_r$, the gradient of the batch of step $t+j$ acts on updates $V_k^{m_0}$ with $k < j$, which are computed from the initial buffer and the batches of steps before $t+j$. For the stationary chain this direction is independent of the batch of step $t+j$. There is no term in which a batch acts on its own update.

Normalized states and the target sample. Until proposition SE.4 we take D deterministic with $2\mathscr{R}(D) = 1$; then $\widehat{A} \overset{\text{def}}{=} \widehat{A}(D) = {\Sigma_{\text{out}}}^{1/2} D {\Sigma_{\text{in}}}^{1/2}$, $\delta \overset{\text{def}}{=} \delta(D) = \|\widehat{A}\|$, $\varrho^\circ(D) = \varrho$, and the stationary chain has $V_r = Q_\varrho(N_r)$ with N_r the frozen buffer of lemma SE.2(i). In N_r, the sample a of the batch of step $t+j$, $0 \le j \le r$, enters as $w_{r-j} d_{j,a}\, \mathsf{u}_{j,a} \otimes \mathsf{v}_{j,a}$ with $d_{j,a} = z_{t+j,a}^\top \widehat{A} w_{t+j,a}$ and u, v as in eq. (SB.26), and the auxiliary samples enter in the same way. For a direction E we write $C_E \overset{\text{def}}{=} {\Sigma_{\text{out}}}^{1/2} E {\Sigma_{\text{in}}}^{1/2}$ and $e_{j,a} \overset{\text{def}}{=} z_{t+j,a}^\top C_E w_{t+j,a}$, so that $G^\circ_{t+j+1}[E] = \sum_a e_{j,a}\, \mathsf{u}_{j,a} \otimes \mathsf{v}_{j,a}$. A test is a deterministic matrix Π with $\|\Pi\|_{\text{F}} \le c_\Pi$ and $\|\Pi\| \le \pi_\Pi \le c_\Pi$, and we put

$$\epsilon_\Pi \overset{\text{def}}{=} \delta(\pi_\Pi + \delta). \tag{SE.19}$$

For the tests used later, $\Pi = \mathcal{W}_{f,g} D$, one has $\pi_\Pi \le C$ but in general not $\pi_\Pi \to 0$; the factor δ in ϵ_Π is what makes the response small.

Lemma SE.9 (Residual derivatives)**.** *In the setting of lemma SB.4, let $\mathsf{h} = \varpi d$ with $\varpi \in [0,1]$, and regard the coefficients eq. (SB.13) as functions of d with $M_0, \mathsf{u}, \mathsf{v}$ fixed. Then*

$$\partial_d \alpha_1 = \frac{\varpi\big[(1+\mathsf{hC})^2 - s\mathsf{h}^2\mathsf{UV}\big]}{\Lambda^2}, \qquad \partial_d \alpha_2 = \frac{2\varpi \mathsf{hU}(1+\mathsf{hC})}{\Lambda^2}, \qquad \partial_d \alpha_3 = -\frac{2s\varpi \mathsf{hV}(1+\mathsf{hC})}{\Lambda^2}. \tag{SE.20}$$

Let $N = N' + \varpi \sum_{a \le B} d_a\, \mathsf{u}_a \otimes \mathsf{v}_a$, where N' and the vectors $\mathsf{u}_a, \mathsf{v}_a$ do not depend on the scalars d_a, and for each a let the quantities of lemma SB.4 be computed with $(M_0, \mathsf{h}, \mathsf{u}, \mathsf{v}) = (N - \varpi d_a \mathsf{u}_a \otimes \mathsf{v}_a, \varpi d_a, \mathsf{u}_a, \mathsf{v}_a)$, with a superscript (a). Then for real $(e_a)_{a \le B}$,

$$\begin{aligned} &\frac{\mathrm{d}}{\mathrm{d}\vartheta} \mathsf{X}_s\Big(N + \vartheta\varpi \sum_a e_a \mathsf{u}_a \otimes \mathsf{v}_a\Big)\Big|_{\vartheta=0} \\ &\quad = \sum_a e_a \Big[\partial_d \alpha_3^{(a)} R_0^{(a)}(\mathsf{u}_a \otimes \mathsf{u}_a)\mathsf{X}_0^{(a)} + s\, \partial_d \alpha_1^{(a)} R_0^{(a)}(\mathsf{u}_a \otimes \mathsf{v}_a)\widetilde{R}_0^{(a)} \\ &\qquad - \partial_d \alpha_1^{(a)} \mathsf{X}_0^{(a)}(\mathsf{v}_a \otimes \mathsf{u}_a)\mathsf{X}_0^{(a)} - s\, \partial_d \alpha_2^{(a)} \mathsf{X}_0^{(a)}(\mathsf{v}_a \otimes \mathsf{v}_a)\widetilde{R}_0^{(a)}\Big]. \end{aligned} \tag{SE.21}$$

Proof. With $\partial_d \Lambda = 2\varpi \mathsf{C}(1+\mathsf{hC}) + 2s\varpi \mathsf{hUV}$, the quotient rule applied to $\alpha_1 = \mathsf{h}(1+\mathsf{hC})/\Lambda$ gives the numerator

$$\varpi\big[(1+2\mathsf{hC})\Lambda - \mathsf{h}(1+\mathsf{hC})\big(2\mathsf{C}(1+\mathsf{hC}) + 2s\mathsf{hUV}\big)\big] = \varpi\big[(1+\mathsf{hC})^2 - s\mathsf{h}^2\mathsf{UV}\big],$$

after inserting $\Lambda = (1+\mathsf{hC})^2 + s\mathsf{h}^2\mathsf{UV}$ and collecting the terms in $(1+\mathsf{hC})^2$ and in $s\mathsf{h}^2\mathsf{UV}$. For $\alpha_2 = \mathsf{h}^2\mathsf{U}/\Lambda$ the numerator is $\varpi\mathsf{U}[2\mathsf{h}\Lambda - \mathsf{h}^2\partial_d\Lambda/\varpi] = 2\varpi\mathsf{hU}(1+\mathsf{hC})$ by the same insertion, and α_3 is identical with $-s\mathsf{V}$ in place of U.

The directional derivative is the sum of the partial derivatives in the scalars d_a. For each a, the minor $N - \varpi d_a \mathsf{u}_a \otimes \mathsf{v}_a$ does not depend on d_a, and eq. (SC.18) writes $\mathsf{X}_s(N)$ as $\mathsf{X}_0^{(a)}$ plus the four rank-one terms with coefficients $\alpha_i^{(a)}$. Differentiating in d_a replaces each coefficient by its derivative, and the chain rule multiplies by e_a. □

With $e_a = e_{j,a}$, $\varpi = w_{r-j}$ and $N = N_r$, the left side of eq. (SE.21) is $D\mathsf{X}_s(N_r)[w_{r-j} G^\circ_{t+j+1}[E]]$. The derivative coefficients are now replaced by their Gaussian values. For a minor M_0 put

$$\sigma \overset{\text{def}}{=} \frac{1}{B}\operatorname{Tr}\big(\Sigma_{\text{out}} R_s(M_0)\big), \qquad \tilde{\sigma} \overset{\text{def}}{=} \frac{1}{B}\operatorname{Tr}\big(\Sigma_{\text{in}} \widetilde{R}_s(M_0)\big), \qquad x \overset{\text{def}}{=} \varpi^2 s\sigma\tilde{\sigma} \le \frac{(C_\Sigma \gamma_\star)^2}{s}, \tag{SE.22}$$

the traces of the minor itself, and let $f_x(d) = d/(1+xd^2)$ and $\psi_x(d) = d^2/(1+xd^2)$ be as in lemma SB.12.

Lemma SE.10 (Derivative coefficients)**.** *Let $s_\circ > 0$ and $p \ge 1$. In the setting of lemma SE.9, let M_0 be fixed, $(\mathsf{u}, \mathsf{v}) = ({\Sigma_{\text{out}}}^{1/2} z/\sqrt{B}, {\Sigma_{\text{in}}}^{1/2} w/\sqrt{B})$ and $d = z^\top \widehat{A} w$ with (z, w) standard Gaussian and $\|\widehat{A}\|_{\text{F}} = 1$. For $s \ge s_\circ$ and $\varpi \in [0,1]$,*

$$\|\partial_d \alpha_1 - \varpi f_x'(d)\|_p \le \frac{C\varpi}{\sqrt{B}}, \qquad \|\partial_d \alpha_2 - \varpi^2 \sigma \psi_x'(d)\|_p \le \frac{C\varpi^2}{s\sqrt{B}},$$

$$\|\partial_d \alpha_3 + s\varpi^2 \tilde{\sigma} \psi_x'(d)\|_p \le \frac{C\varpi^2}{\sqrt{B}},$$

and $\|\partial_d\alpha_1\|_p \le C\varpi$, $\|\partial_d\alpha_2\|_p \le C\varpi^2/s$, $\|\partial_d\alpha_3\|_p \le C\varpi^2$, *with* C *depending only on* $(p, s_\circ, \gamma_\star, C_\Sigma)$.

Proof. The scalars concentrate at the traces of the minor. Put $\overline{\mathsf{U}} \stackrel{\text{def}}{=} s\mathsf{U}$, $\overline{\mathsf{V}} \stackrel{\text{def}}{=} s\mathsf{V}$ and $\overline{\mathsf{C}} \stackrel{\text{def}}{=} \sqrt{s}\mathsf{C}$. Then $\overline{\mathsf{U}} = z^\top F z$ with $F \stackrel{\text{def}}{=} s{\Sigma_{\text{out}}}^{1/2} R_s(M_0) {\Sigma_{\text{out}}}^{1/2}/B$, whose mean is $\operatorname{Tr} F = s\sigma$ and whose Frobenius norm is at most $C_\Sigma\sqrt{N_{\text{out}}}/B \le C/\sqrt{B}$, since $\|sR_s(M_0)\| \le 1$. The Hanson–Wright inequality [71, Theorem 6.2.1], integrated over the tail, gives $\|\overline{\mathsf{U}} - s\sigma\|_p \le C_p\|F\|_{\mathrm{F}} \le C_p/\sqrt{B}$, and likewise $\|\overline{\mathsf{V}} - s\tilde\sigma\|_p \le C_p/\sqrt{B}$. The bilinear form $\overline{\mathsf{C}} = z^\top F' w$ with $F' \stackrel{\text{def}}{=} \sqrt{s}{\Sigma_{\text{out}}}^{1/2}\mathsf{X}_s(M_0){\Sigma_{\text{in}}}^{1/2}/B$ is, given w, centered Gaussian with variance $\|F'w\|^2$, so $\|\overline{\mathsf{C}}\|_p \le C_p\|F'\|_{\mathrm{F}} \le C_p/\sqrt{B}$, using $\|\sqrt{s}\mathsf{X}_s\| \le 1/2$ from lemma SB.3.

The coefficients are rational in the scaled scalars, with denominators bounded below along an interpolation. Put, for $\vartheta \in [0,1]$,

$$(\mathsf{U}_\vartheta, \mathsf{V}_\vartheta, \mathsf{C}_\vartheta) \stackrel{\text{def}}{=} \big((1-\vartheta)\mathsf{U} + \vartheta\sigma,\ (1-\vartheta)\mathsf{V} + \vartheta\tilde\sigma,\ (1-\vartheta)\mathsf{C}\big), \qquad \Lambda_\vartheta \stackrel{\text{def}}{=} (1+\mathsf{h}\mathsf{C}_\vartheta)^2 + s\mathsf{h}^2\mathsf{U}_\vartheta\mathsf{V}_\vartheta.$$

Since all four of $\mathsf{U}, \mathsf{V}, \sigma, \tilde\sigma$ are nonnegative, $\mathsf{U}_\vartheta\mathsf{V}_\vartheta \ge (1-\vartheta)^2\mathsf{U}\mathsf{V}$, so Λ_ϑ is at least the quantity Λ computed with the residual $(1-\vartheta)d$, and eq. (SB.16) gives $\Lambda_\vartheta^{-1} \le (1 + \frac{3}{4\sqrt{s_\circ}}|d|\mathsf{N})^2$. After division by their leading scales ϖ, ϖ^2/s and ϖ^2, the three expressions eq. (SE.20) are rational functions of $(\overline{\mathsf{U}}, \overline{\mathsf{V}}, \overline{\mathsf{C}})$ and d, for instance $\partial_d\alpha_1/\varpi = [(1 + \varpi d\overline{\mathsf{C}}/\sqrt{s})^2 - \varpi^2 d^2\overline{\mathsf{U}}\,\overline{\mathsf{V}}/s]/\Lambda^2$. Along the interpolation their partial derivatives in the three scaled scalars are bounded by a fixed polynomial in $(1+|d|)(1+\mathsf{N})$, because $|\overline{\mathsf{U}}_\vartheta|, |\overline{\mathsf{V}}_\vartheta| \le \mathsf{N} + C_\Sigma\gamma_\star$, $|\overline{\mathsf{C}}_\vartheta| \le \mathsf{N}/4$ and $s^{-1/2} \le s_\circ^{-1/2}$. At $\vartheta = 1$ the denominator is $\Lambda_1 = 1 + xd^2$, and the three normalized expressions equal $f'_x(d)$, $s\sigma\psi'_x(d)$ and $-s\tilde\sigma\psi'_x(d)$, since $f'_x(d) = (1-xd^2)/(1+xd^2)^2$ and $\psi'_x(d) = 2d/(1+xd^2)^2$. The mean value theorem along ϑ, Hölder's inequality, the concentration bounds of the first paragraph and lemma SB.6 prove the three differences. The same polynomial bounds at $\vartheta = 0$ prove the last three bounds. □

For $x \ge 0$ and $d = z^\top \widehat{A} w$ put

$$b(x) \stackrel{\text{def}}{=} \mathbb{E}\,\psi''_x(d), \qquad N_x \stackrel{\text{def}}{=} \mathbb{E}\left[(w \otimes w)\,\psi''''_x(d)\right].$$

Lemma SE.11 (Gaussian response of the derivative functional)**.** *Let* $\|\widehat{A}\|_{\mathrm{F}} = 1$, $\delta = \|\widehat{A}\|$, *and let* E *be deterministic with* $C_E = {\Sigma_{\text{out}}}^{1/2} E {\Sigma_{\text{in}}}^{1/2}$ *and* $e = z^\top C_E w$. *For* $x, x' \ge 0$, $|b(x)| \le 2$, $|b(x) - b(x')| \le 12|x - x'|$ *and* $\|N_x\| \le 24x$, *and*

$$\mathbb{E}\left[e\,\psi'_x(d)\right] = \langle \widehat{A}, C_E\rangle\, b(x) + \big\langle C_E, \widehat{A}N_x\widehat{A}^\top\widehat{A}\big\rangle, \qquad \big|\big\langle C_E, \widehat{A}N_x\widehat{A}^\top\widehat{A}\big\rangle\big| \le 24x\,\|C_E\|\,\delta. \tag{SE.23}$$

Proof. The scalar bounds are read from

$$\psi''_x(d) = \frac{2(1-3xd^2)}{(1+xd^2)^3}, \qquad \partial_x\psi''_x(d) = -\frac{12d^2(1-xd^2)}{(1+xd^2)^4}, \qquad \psi_x(d) = \frac{1}{x}\Big(1 - \frac{1}{1+xd^2}\Big).$$

The first gives $|\psi''_x| \le 2$; the second gives $|\partial_x\psi''_x(d)| \le 12d^2$, hence the Lipschitz bound with $\mathbb{E}\,d^2 = 1$; and the third gives $\psi''''_x(d) = -x\,\mathsf{g}''''(\sqrt{x}\,d)$ with $\mathsf{g}(y) = (1+y^2)^{-1}$, whose fourth derivative is bounded by its value 24 at $y = 0$. The bound on N_x is the argument for the matrix N in the proof of lemma SB.8.

The identity is eq. (SB.22) with $f = \psi'_x$, paired with C_E, since $\mathbb{E}[e\,\psi'_x(d)] = \langle C_E, \mathbb{E}[(z \otimes w)\psi'_x(d)]\rangle$. The bound on the second term is trace duality and the ideal property: $\|\widehat{A}N_x\widehat{A}^\top\widehat{A}\|_* \le \|\widehat{A}\|\|N_x\|\|\widehat{A}^\top\widehat{A}\|_* = \delta\|N_x\|\|\widehat{A}\|_{\mathrm{F}}^2$. □

In the application $\langle \widehat{A}, C_E\rangle = \langle \Sigma_{\text{out}} D \Sigma_{\text{in}}, E\rangle$ and $\|C_E\| \le C_\Sigma\|E\|$. The remainder is bounded by the operator norm of E; the direction is not replaced by its component orthogonal to D, which need not have bounded operator norm.

Proposition SE.2 (Causal resolvent reduction)**.** *Let* D *be deterministic with* $2\mathscr{R}(D) = 1$, *consider the stationary chain at* (t, D), *let* $0 \le j \le r$, $\varpi \stackrel{\text{def}}{=} w_{r-j}$ *and* $s \ge s_\circ > 0$, *and let* E *be a random matrix such that the batch of step* $t + j$ *is independent of* E *together with all other samples,*

auxiliary batches included, with $\|E\| \le c_E$ *and* $\|E\|_{\mathrm{F}} \le c_E\sqrt{B}$*. For each sample* a *of that batch let* $T_a \stackrel{\mathrm{def}}{=} N_r - \varpi d_{j,a}\mathsf{u}_{j,a}\otimes\mathsf{v}_{j,a}$*, let* $\sigma_a, \tilde\sigma_a, x_a$ *be the quantities eq.* (SE.22) *of* T_a*, and put*

$$F_s(T_a) \stackrel{\mathrm{def}}{=} s\,b(x_a)\Big\{\tilde\sigma_a\big\langle\Pi, R_s(T_a)\Sigma_{\mathrm{out}}\mathsf{X}_s(T_a)\big\rangle + \sigma_a\big\langle\Pi, \mathsf{X}_s(T_a)\Sigma_{\mathrm{in}}\widetilde{R}_s(T_a)\big\rangle\Big\},$$
$$g(E) \stackrel{\mathrm{def}}{=} \langle\Sigma_{\mathrm{out}} D\Sigma_{\mathrm{in}}, E\rangle. \tag{SE.24}$$

Then

$$\Big|\,\mathbb{E}\,\big\langle\Pi, D\mathsf{X}_s(N_r)\big[\varpi G^{\circ}_{t+j+1}[E]\big]\big\rangle + \frac{\varpi^2}{B}\sum_{a\le B}\mathbb{E}\,\big[F_s(T_a)\,g(E)\big]\Big| \le \frac{C\sqrt{B}\,\varpi}{s},$$

with C *depending only on* $(s_\circ, c_\Pi, c_E, \gamma_\star, C_\Sigma)$*.*

Proof. Fix a sample a and condition on all samples other than $(z_{t+j,a}, w_{t+j,a})$ and on E; write $\mathbb{E}_*$ for this conditional expectation, drop the index a, and write $R_0, \widetilde{R}_0, \mathsf{X}_0$ for the resolvents of $T = T_a$. These are fixed under $\mathbb{E}_*$, and the target sample is independent of them and of E. By lemma SE.9 the derivative is the sum over a of the four terms of eq. (SE.21). The norms used are: $\|R_0\|, \|\widetilde{R}_0\| \le 1/s$ and $\|\mathsf{X}_0\| \le (2\sqrt{s})^{-1}$ (lemma SB.3); $\|e\|_{p,*} \le C_p\|C_E\|_{\mathrm{F}} \le C_pC_\Sigma c_E\sqrt{B}$; and the Bessel bounds eq. (SC.22), which give $C\|\cdot\|_{2,*}/\sqrt{B}$ for the quadratic structures $\mathsf{u}\otimes\mathsf{u}$, $\mathsf{v}\otimes\mathsf{v}$ and $C\|\cdot\|_{2,*}/B$ for the mixed ones.

Coefficient errors. Replace $\partial_d\alpha_1, \partial_d\alpha_2, \partial_d\alpha_3$ by $\varpi f'_x(d)$, $\varpi^2\sigma\psi'_x(d)$, $-s\varpi^2\tilde\sigma\psi'_x(d)$. By lemma SE.10 and Hölder's inequality, the errors multiplied by e have conditional L^2 norms at most $C\varpi$, $C\varpi^2/s$ and $C\varpi^2$. With the exterior norms, the four error matrices have Frobenius norms at most

$$\frac{1}{s}\cdot\frac{1}{2\sqrt{s}}\cdot\frac{C\varpi^2}{\sqrt{B}}, \qquad s\cdot\frac{1}{s^2}\cdot\frac{C\varpi}{B}, \qquad \frac{1}{4s}\cdot\frac{C\varpi}{B}, \qquad s\cdot\frac{1}{2\sqrt{s}}\cdot\frac{1}{s}\cdot\frac{C\varpi^2}{s\sqrt{B}},$$

and summing over the B samples gives at most $C(\varpi^2\sqrt{B}s^{-3/2} + \varpi s^{-1}) \le C\sqrt{B}\varpi/s$, using $\varpi^2 \le \varpi$ and $s^{-3/2} \le s_\circ^{-1/2}s^{-1}$.

Mixed and centered target terms. The mixed terms carry $\varpi f'_x(d)e$, with conditional L^2 norm at most $\varpi\|e\|_{2,*} \le C\varpi\sqrt{B}$ since $|f'_x| \le 1$; each has Frobenius norm at most $C\varpi\sqrt{B}/(sB)$, and the B samples give $C\sqrt{B}\varpi/s$. In the quadratic terms we split $\mathsf{u}\otimes\mathsf{u} = B^{-1}{\Sigma_{\mathrm{out}}}^{1/2}(z\otimes z - \mathrm{Id}){\Sigma_{\mathrm{out}}}^{1/2} + B^{-1}\Sigma_{\mathrm{out}}$ and $\mathsf{v}\otimes\mathsf{v} = B^{-1}{\Sigma_{\mathrm{in}}}^{1/2}(w\otimes w - \mathrm{Id}){\Sigma_{\mathrm{in}}}^{1/2} + B^{-1}\Sigma_{\mathrm{in}}$. The centered parts are bounded by lemma SB.7: the scalars $s\varpi^2\tilde\sigma\psi'_x(d)e$ and $s\varpi^2\sigma\psi'_x(d)e$ have conditional L^2 norms at most $C\varpi^2\sqrt{B}$, since $s\sigma, s\tilde\sigma \le C_\Sigma\gamma_\star$ and $|\psi'_x(d)| \le 2|d|$; each centered part has Frobenius norm at most $C\varpi^2\sqrt{B}s^{-3/2}/B$, and the B samples give $C\varpi^2\sqrt{B}s^{-3/2} \le C\sqrt{B}\varpi/s$.

Identity parts. What remains is, for each sample,

$$-\frac{s\varpi^2}{B}\;\mathbb{E}_*\big[e\,\psi'_x(d)\big]\Big\{\tilde\sigma\,R_0\Sigma_{\mathrm{out}}\mathsf{X}_0 + \sigma\,\mathsf{X}_0\Sigma_{\mathrm{in}}\widetilde{R}_0\Big\}.$$

By lemma SE.11, $\mathbb{E}_*[e\psi'_x(d)] = g(E)b(x) + \rho$ with $|\rho| \le 24xC_\Sigma c_E\delta \le C$. Pairing with Π, the term with $g(E)b(x)$ is $-\varpi^2B^{-1}F_s(T)g(E)$. The term with ρ is at most $(s\varpi^2/B)\cdot C\cdot(\tilde\sigma + \sigma)\,c_\Pi\,C_\Sigma\sqrt{\gamma_\star B}\,s^{-3/2}$, because $\|R_0\Sigma_{\mathrm{out}}\mathsf{X}_0\|_{\mathrm{F}}, \|\mathsf{X}_0\Sigma_{\mathrm{in}}\widetilde{R}_0\|_{\mathrm{F}} \le C_\Sigma\sqrt{\gamma_\star B}\,s^{-3/2}/2$; with $s\sigma, s\tilde\sigma \le C$ and the B samples this is at most $C\varpi^2\sqrt{B}s^{-3/2} \le C\sqrt{B}\varpi/s$.

Pairing the error matrices with Π costs the factor c_Π. Taking expectations and summing the contributions proves the claim. □

Lemma SE.12 (The scalar of the minor)**.** *Let* $s \ge s_\circ > 0$*,* $\varpi \in [0,1]$*, and let* $F_s(T)$ *be defined by eq.* (SE.24) *for an arbitrary real matrix* T*, with its own traces. Then*

$$|F_s(T)| \le C\sqrt{B}\,s^{-3/2}, \qquad |F_s(T) - F_s(T')| \le Cs^{-2}\big(\pi_\Pi + B^{-1/2}\big)\|T - T'\|_*,$$

with C *depending only on* $(s_\circ, c_\Pi, \gamma_\star, C_\Sigma)$*.*

Proof. The size bound uses $|b| \le 2$, $s\sigma, s\tilde\sigma \le C_\Sigma\gamma_\star$ and $|\langle \Pi, R\Sigma_{\text{out}}\mathsf{X}\rangle| \le c_\Pi \|R\Sigma_{\text{out}}\mathsf{X}\|_{\mathrm{F}} \le c_\Pi C_\Sigma\sqrt{\gamma_\star B}\, s^{-3/2}/2$, and the same for the second pairing.

The sensitivity is obtained by changing one factor at a time; put $E \stackrel{\text{def}}{=} T - T'$. By lemma SB.9, $\|R_s(T) - R_s(T')\|_* \le 2s^{-3/2}\|E\|_*$, the same for $\widetilde{R}_s$, and by lemma SE.4(i), $\|\mathsf{X}_s(T) - \mathsf{X}_s(T')\|_* \le s^{-1}\|E\|_*$. With trace duality $|\operatorname{Tr}(XY)| \le \|X\|\|Y\|_*$,

$$\big|\langle \Pi, R\Sigma_{\text{out}}\mathsf{X}\rangle - \langle \Pi, R'\Sigma_{\text{out}}\mathsf{X}'\rangle\big| \le \pi_\Pi C_\Sigma\Big(\frac{2}{s^{3/2}}\cdot\frac{1}{2\sqrt{s}} + \frac{1}{s}\cdot\frac{1}{s}\Big)\|E\|_* = \frac{2\pi_\Pi C_\Sigma}{s^2}\|E\|_*,$$

and the same for the second pairing; with the bounded prefactors this contributes $C\pi_\Pi s^{-2}\|E\|_*$. The traces satisfy $|\sigma - \sigma'| \le C_\Sigma B^{-1}\|R_s(T) - R_s(T')\|_* \le 2C_\Sigma B^{-1}s^{-3/2}\|E\|_*$, and the same for $\tilde\sigma$; multiplied by s and by the pairing of size $C\sqrt{B}s^{-3/2}$ this contributes $CB^{-1/2}s^{-2}\|E\|_*$. Finally $|x - x'| \le s(\sigma|\tilde\sigma - \tilde\sigma'| + \tilde\sigma'|\sigma - \sigma'|) \le CB^{-1}s^{-3/2}\|E\|_*$, so $|b(x) - b(x')| \le 12|x - x'|$ contributes $s\cdot CB^{-1}s^{-3/2}\cdot Cs^{-1}\cdot C\sqrt{B}s^{-3/2}\|E\|_* = CB^{-1/2}s^{-3}\|E\|_*$, which is at most $Cs_\circ^{-1}B^{-1/2}s^{-2}\|E\|_*$. □

Proposition SE.3 (Scalar contraction)**.** *In the setting of proposition SE.2, let $0 \le k < j$ and $g_k \stackrel{\text{def}}{=} g(V_k) = \langle \Sigma_{\text{out}} D\Sigma_{\text{in}}, V_k\rangle$. Then for every sample a of the batch of step $t + j$,*

$$\big|\,\mathbb{E}\,\big[F_s(T_a)\, g_k\big]\big| \le C\Big\{\sqrt{B}\, s^{-3/2} + \big(1 + B\epsilon_\Pi\big)s^{-2}\beta^{r-k}\Big\},$$

with C depending only on $(s_\circ, c_\Pi, \varrho, \gamma_\star, C_\Sigma)$. No independence between V_k and the traces of the minor is used.

Proof. The mean product is bounded by the finite mean. $|\,\mathbb{E}\, g_k| \le \|\Sigma_{\text{out}} D\Sigma_{\text{in}}\|_{\mathrm{F}}\|\overline{V}(D)\|_{\mathrm{F}} \le C$ by eq. (SE.15), and $|\,\mathbb{E}\, F_s(T_a)| \le C\sqrt{B}s^{-3/2}$ by lemma SE.12.

The covariance is bounded by lemma SE.6, with the individual samples of all batches, the auxiliary ones included, as coordinates. A sample of the batch of relative time v enters N_k with weight w_{k-v} if $v \le k$ and does not enter it if $v > k$; it enters T_a with weight w_{r-v}, except the target sample, which is absent from T_a. Replacing it by an independent copy changes N_k and T_a by rank-two matrices with nuclear L^2 norms at most $14C_\Sigma\gamma_\star w_{k-v}$ and $14C_\Sigma\gamma_\star w_{r-v}$, by lemma SE.3(iii). By trace duality, lemma SE.4(ii) and $\|\Sigma_{\text{out}} D\Sigma_{\text{in}}\| \le C_\Sigma\delta$,

$$\|g_k - g_k^{[l]}\|_2 \le C\delta\, w_{k-v}, \qquad \|F_s(T_a) - F_s(T_a)^{[l]}\|_2 \le Cs^{-2}\big(\pi_\Pi + B^{-1/2}\big)w_{r-v},$$

the second by lemma SE.12. Only the samples with $v \le k$ change both, and for them $w_{r-v} = \beta^{r-k}w_{k-v}$. With B samples per batch and $\sum_v w_{k-v}^2 \le s_2 \le 1$,

$$|\operatorname{Cov}(F_s(T_a), g_k)| \le Cs^{-2}\big(B\delta\pi_\Pi + \sqrt{B}\delta\big)\beta^{r-k} \le Cs^{-2}\big(1 + B\epsilon_\Pi\big)\beta^{r-k},$$

using $\sqrt{B}\delta \le \frac{1}{2}(1 + B\delta^2)$ and $\delta\pi_\Pi + \delta^2 = \epsilon_\Pi$. Adding the product of the means proves the claim. □

Lemma SE.13 (Ridge susceptibility covariance)**.** *Let D be deterministic with $2\mathscr{R}(D) = 1$, consider the stationary chain at (t, D) and let $k, r \ge 0$. Then*

$$\big|\,\mathbb{E}\,\big[g_k\,\big\langle \Pi, \partial_\ell Q_\ell(N_r)\big|_{\ell=\varrho}\big\rangle\big]\big| \le C\big(1 + B\delta\pi_\Pi\,\beta^{|r-k|}\big).$$

Proof. The means are bounded by eq. (SE.15): $|\,\mathbb{E}\, g_k| \le C$ and $|\,\mathbb{E}\langle \Pi, \partial_\ell Q_\ell(N_r)\rangle| \le c_\Pi\|\,\mathbb{E}\,\partial_\ell Q_\ell(N_r)\|_{\mathrm{F}} \le C$. For the covariance, a sample of relative time v changes $\langle \Pi, \partial_\ell Q_\ell(N_r)\rangle$ by at most $\pi_\Pi\frac{1}{2}\varrho^{-3/2}\cdot 14C_\Sigma\gamma_\star w_{r-v}$ in L^2, by lemma SE.4(iii) and trace duality, and changes g_k by at most $C\delta w_{k-v}$ as in the previous proof. The shared samples are those with $v \le \min(k, r)$, and lemma SE.6 gives $CB\delta\pi_\Pi\sum_{v\le\min(k,r)} w_{k-v}w_{r-v} \le CB\delta\pi_\Pi\beta^{|r-k|}$. □

Proposition SE.4 (Stationary causal tangent)**.** *Let D be deterministic with $r_- \le \mathscr{R}(D) \le r_+$, let Π be a test, and let $\dot V_r$ be the tangent of lemma SE.8 for the stationary chain at (t, D), with its initial buffer held fixed. Then for $r \ge 0$,*

$$\eta\,|\,\mathbb{E}\langle \Pi, \dot V_r\rangle| \le C\Big\{\frac{r}{\sqrt{B}} + \frac{1}{B} + \epsilon_\Pi\Big\}, \qquad \epsilon_\Pi = \delta(D)\big(\pi_\Pi + \delta(D)\big),$$

with C depending only on (c_Π, r_-, r_+) and the fixed parameters.

Proof. The statement reduces to $2\mathscr{R}(D) = 1$. Put $a \stackrel{\text{def}}{=} \sqrt{2\mathscr{R}(D)}$. Since $G^\circ[\Delta]/a = G^\circ[\Delta/a]$, $\varrho^\circ(\Delta) = a^2\varrho^\circ(\Delta/a)$ and $Q_{a^2\ell}(aK) = Q_\ell(K)$, the chain of (D, m_0) at step ε has the updates of the chain of $(D/a, m_0/a)$ at step ε/a, and $\eta\dot V_r(D) = (\eta/a)\dot V_r(D/a)$. The step h/a lies in a compact subset of $(0,\infty)$ determined by (r_-, r_+), and $\delta(D/a) = \delta(D)$. We therefore assume $2\mathscr{R}(D) = 1$.

By lemma SE.8 and the integral representation $DQ_\varrho(N)[R] = \frac{2}{\pi}\int_0^\infty D\mathsf{X}_{\varrho+u^2}(N)[R]\,\mathrm{d}u$,

$$\begin{aligned}\mathbb{E}\langle \Pi, \dot V_r\rangle = &-\frac{2}{\pi}\int_0^\infty \sum_{0\le k<j\le r} \mathbb{E}\left\langle \Pi, D\mathsf{X}_{\varrho+u^2}(N_r)\big[w_{r-j}G^\circ_{t+j+1}[V_k]\big]\right\rangle \mathrm{d}u\\ &- 2\varrho\sum_{k<r}\mathbb{E}\left[g_k\langle \Pi, \partial_\ell Q_\ell(N_r)\rangle\right].\end{aligned}$$

The matrix part is bounded by the reduction and the scalar contraction. For $k < j$ the direction V_k is determined by the auxiliary batches and the batches up to step $t + k$, so the batch of step $t + j$ is independent of V_k together with all other samples, with $\|V_k\| \le 1$ and $\|V_k\|_{\mathrm F} \le \sqrt{\gamma_\star B}$. By propositions SE.2 and SE.3 with $s_\circ = \varrho$,

$$\left|\,\mathbb{E}\left\langle \Pi, D\mathsf{X}_s(N_r)[\varpi G^\circ_{t+j+1}[V_k]]\right\rangle\right| \le \frac{C\sqrt{B}\varpi}{s} + C\varpi^2\Big\{\sqrt{B}s^{-3/2} + (1 + B\epsilon_\Pi)s^{-2}\beta^{r-k}\Big\},$$

with $\varpi = w_{r-j}$. The integrals of s^{-1}, $s^{-3/2}$ and s^{-2} against $\mathrm{d}u$ at $s = \varrho + u^2$ are finite. The causal sums are

$$\sum_{j\le r} w_{r-j}\, j \le r, \qquad \sum_{j\le r} w_{r-j}^2 \sum_{k<j}\beta^{r-k} \le \frac{1}{1-\beta}\sum_{j\le r} w_{r-j}^2\beta^{r-j+1} \le 1,$$

the second being zero at $\beta = 0$. The matrix part is therefore at most $C(\sqrt{B}\,r + 1 + B\epsilon_\Pi)$.

The ridge part is bounded by lemma SE.13: its absolute value is at most $C\sum_{k<r}(1 + B\delta\pi_\Pi\beta^{r-k}) \le C(r + B\epsilon_\Pi)$.

Adding the two parts and multiplying by $\eta = h/B$ proves the claim. □

SE.7 Transport, concentration and diffuseness along the algorithm

From here on the hypotheses (H1)–(H4) of theorem SE.1 are in force.

Blocks and admissible states. For $0 < r_- \le r_+$ and $E < \infty$ let $\mathcal{K}(r_-, r_+, E) \stackrel{\text{def}}{=} \{\Delta :\ r_- \le \mathscr{R}(\Delta) \le r_+,\ \|\Delta\|_{\mathrm F}^2 \le E\}$, and for $\mathcal{K} = \mathcal{K}(r_-, r_+, E)$ let $\mathcal{K}' \stackrel{\text{def}}{=} \mathcal{K}(r_-/2, 2r_+, 2E + 2)$. The block length and the prefix length are

$$L \stackrel{\text{def}}{=} \lfloor B^{1/8}\rfloor, \qquad \mathsf{q} \stackrel{\text{def}}{=} \lceil C_0\log(2B)\rceil, \qquad C_0 \stackrel{\text{def}}{=} \max\Big(1, \frac{2}{\log(1/\beta)}\Big) \tag{SE.25}$$

($C_0 = 1$ at $\beta = 0$), so that $\beta^{\mathsf q} \le (2B)^{-2}$, hence $\sqrt{B}\beta^{\mathsf q} \le 1$ and $\sqrt{B}L\beta^{\mathsf q} \to 0$, while $\mathsf q \le L$ for large B. The block starts are $t_b \stackrel{\text{def}}{=} bL$, $b \ge 0$. The admissible set for block starts and its enlargements are

$$\mathcal{A} \stackrel{\text{def}}{=} \mathcal{K}\Big(\frac{r_0}{2}, r_1 + \frac{1}{2}, E_\star + 1\Big), \qquad \mathcal{A}^+ \stackrel{\text{def}}{=} \mathcal{A}', \qquad \mathcal{A}^{++} \stackrel{\text{def}}{=} (\mathcal{A}^+)',$$

and $\tau^+ \stackrel{\text{def}}{=} \inf\{t \ge 0 :\ \Delta_t \notin \mathcal{A}^+\}$. Constants called C in this subsection may depend on $(E_\star, r_0, r_1, c_M)$.

Lemma SE.14 (Movement). *(i) For $t \le t' < t_{\mathrm{stop}}$, $\|\Delta_{t'} - \Delta_t\| \le h(t'-t)/B$, $\|\Delta_{t'} - \Delta_t\|_{\mathrm F} \le h\sqrt{\gamma_\star}\,(t'-t)/\sqrt{B}$ and $|\sqrt{2\mathscr{R}(\Delta_{t'})} - \sqrt{2\mathscr{R}(\Delta_t)}| \le C_\Sigma h\sqrt{\gamma_\star}\,(t'-t)/\sqrt{B}$.*

(ii) For every $\mathcal{K} = \mathcal{K}(r_-, r_+, E)$ there is B_0 such that for $B \ge B_0$: if $\Delta_t \in \mathcal{K}$, then $t + 4L < t_{\mathrm{stop}}$ and $\Delta_{t'} \in \mathcal{K}'$ for all $t' \ge 0$ with $|t' - t| \le 4L$.

Proof. For (i), $\Delta_{t'} - \Delta_t = -\eta \sum_{t \le u < t'} \mathsf{S}_u$, and $\|\mathsf{S}_u\| \le 1$, $\|\mathsf{S}_u\|_{\mathrm{F}} \le \sqrt{\min(N_{\mathrm{out}}, N_{\mathrm{in}})} \le \sqrt{\gamma_\star B}$ by lemma SA.2(i). The risk bound is the reverse triangle inequality for $\sqrt{2\mathscr{R}(\Delta)} = \|{\Sigma_{\mathrm{out}}}^{1/2} \Delta {\Sigma_{\mathrm{in}}}^{1/2}\|_{\mathrm{F}}$, with $\|{\Sigma_{\mathrm{out}}}^{1/2} E {\Sigma_{\mathrm{in}}}^{1/2}\|_{\mathrm{F}} \le C_\Sigma \|E\|_{\mathrm{F}}$.

For (ii), choose B_0 with $4C_\Sigma h\sqrt{\gamma_\star} L/\sqrt{B} \le \min\big(1, (1 - 2^{-1/2})\sqrt{2r_-}\big)$ for $B \ge B_0$. Since $\mathscr{R}(\Delta_t) \ge r_- > 0$, $t < t_{\mathrm{stop}}$, so the bounds of (i) hold between t' and t for $t' < t$. Forward, induction on t' shows that $\mathscr{R}(\Delta_{t'}) \ge r_-/2 > 0$ for $t \le t' \le t + 4L$, so $t' < t_{\mathrm{stop}}$. In both directions $\|\Delta_{t'}\|_{\mathrm{F}} \le \sqrt{E} + 1$ and $\sqrt{2\mathscr{R}(\Delta_{t'})}$ differs from $\sqrt{2\mathscr{R}(\Delta_t)}$ by at most $(1 - 2^{-1/2})\sqrt{2r_-}$, which gives $\Delta_{t'} \in \mathcal{K}'$. □

We fix B_0 such that lemma SE.14(ii) holds for $\mathcal{K} = \mathcal{A}$ and for $\mathcal{K} = \mathcal{A}^+$, and such that $\mathsf{q} \le L$; all statements below are for $B \ge B_0$. In particular, if all block starts $t_{b'}$ with $b' \le b$ lie in $\mathcal{A}$, then every $t \le t_b + L$ is within L steps of one of them, so $\Delta_t \in \mathcal{A}^+$ and $\tau^+ > t_b + L$.

Lemma SE.15 (Stopped buffer moments)**.** *Let $\mathcal{K} = \mathcal{K}(r_-, r_+, E)$, and let τ be a time such that whether $t < \tau$ is decided by the past $\mathscr{F}_t$, and such that $\Delta_t \in \mathcal{K}$ for $t < \tau$. Then for every $t \ge 0$,*

$$\mathbb{E}\left[\mathbf{1}_{t \le \tau} \|M^\circ_{t-1}\|_{\mathrm{F}}^2\right] \le 2c_M B + 4C_M r_+ B, \qquad \mathbb{E}\left[\mathbf{1}_{t \le \tau} \|M^\circ_{t-1}\|^2\right] \le 2\beta^{2t} c_M B + C \log^2(2B).$$

Proof. The stopped-input buffer $\widehat{M}_t \overset{\mathrm{def}}{=} \beta \widehat{M}_{t-1} + (1-\beta)\mathbf{1}_{t<\tau} G^\circ_{t+1}[\Delta_t]$, $\widehat{M}_{-1} \overset{\mathrm{def}}{=} M^\circ_{-1}$, coincides with M°_t for $t < \tau$, so $\mathbf{1}_{t \le \tau} M^\circ_{t-1} = \mathbf{1}_{t \le \tau} \widehat{M}_{t-1}$. Since $\mathbf{1}_{t<\tau}$ and Δ_t are determined by the past, lemma SE.3(i) applied conditionally gives $\mathbb{E}\,\|\mathbf{1}_{t<\tau} G^\circ_{t+1}[\Delta_t]\|_{\mathrm{F}}^2 \le C_M B \cdot 2r_+$. Minkowski's inequality for the unrolled recursion gives

$$\big(\mathbb{E}\,\|\widehat{M}_{t-1}\|_{\mathrm{F}}^2\big)^{1/2} \le \beta^t \sqrt{c_M B} + \sum_j w_j \sqrt{2C_M r_+ B},$$

and squaring with $(x+y)^2 \le 2x^2 + 2y^2$ proves the first bound. The second is the same computation with lemma SE.3(ii) and $\mathbb{E}\,\|M^\circ_{-1}\|^2 \le \mathbb{E}\,\|M^\circ_{-1}\|_{\mathrm{F}}^2$. □

Lemma SE.16 (Coupling)**.** *Let $\mathcal{K} = \mathcal{K}(r_-, r_+, E)$ with B_0 as in lemma SE.14. At a step t, on the event $\{\Delta_t \in \mathcal{K}\}$, let the chains at time t be those of definition SE.2 with $D = \Delta_t$ and $m = M^\circ_{t-1}$. Then for $0 \le r \le 4L$:*

(i) $\mathbb{E}_t\,\|M^\circ_{t+r} - \widehat{N}_r\|_{\mathrm{F}}^2 + \mathbb{E}_t\,\|\mathsf{S}_{t+r} - \widehat{V}_r\|_{\mathrm{F}}^2 \le Cr^2$ *and* $\mathbb{E}_t\,\|M^\circ_{t+r} - \widehat{N}_r\|^2 \le Cr^2 \log^2(2B)/B$;

(ii) $\mathbb{E}_t\,\|\widehat{V}_r - V_r\|_{\mathrm{F}}^2 \le C\beta^{2(r+1)}(\|m\|_{\mathrm{F}}^2 + B)$ *and* $\mathbb{E}_t\,\|\widehat{V}_r - V_r\|^2 \le C\beta^{2(r+1)}(\|m\|^2 + \log^2(2B))$;

(iii) $\mathbb{E}_t\,\big\|\sum_{r<L}(\mathsf{S}_{t+r} - V_r)\big\|_{\mathrm{F}}^2 \le C(L^4 + B + \|m\|_{\mathrm{F}}^2)$;

(iv) $\mathbb{E}_t\,\big\|\sum_{r<L}(\mathsf{S}_{t+r} - V_r)\big\| \le C(\|m\| + \log(2B)) + CL^2 \log(2B)/\sqrt{B}$.

The constant C depends only on $\mathcal{K}$ and the fixed parameters.

Proof. By lemma SE.14(ii) the states Δ_{t+r}, $r \le 4L$, lie in $\mathcal{K}'$, so the physical ridges are at least ϱr_-.

For (i), subtracting the recursions of the algorithm and of the same-buffer chain, which start from the same buffer m, gives

$$M^\circ_{t+r} - \widehat{N}_r = \sum_{j \le r} w_{r-j}\, G^\circ_{t+j+1}\big[\Delta_{t+j} - \Delta_t\big].$$

The matrix $\Delta_{t+j} - \Delta_t$ is determined by the past and the batches before step $t + j$, and its weighted Frobenius norm is at most $C_\Sigma h\sqrt{\gamma_\star} j/\sqrt{B}$ by lemma SE.14(i). Lemma SE.3(i),(ii) applied conditionally, and Minkowski's inequality with $\sum_j w_j \le 1$, give the two bounds on $M^\circ_{t+r} - \widehat{N}_r$. For the updates,

$$\mathsf{S}_{t+r} - \widehat{V}_r = \big[Q_{\varrho^\circ(\Delta_{t+r})}(M^\circ_{t+r}) - Q_{\varrho^\circ(\Delta_t)}(M^\circ_{t+r})\big] + \big[Q_{\varrho^\circ(\Delta_t)}(M^\circ_{t+r}) - Q_{\varrho^\circ(\Delta_t)}(\widehat{N}_r)\big].$$

The ridges differ by $2\varrho|\mathscr{R}(\Delta_{t+r}) - \mathscr{R}(\Delta_t)| \le Cr/\sqrt{B}$, so by lemma SE.4(iv) the first bracket has Frobenius norm at most $\sqrt{\gamma_\star B} \cdot Cr/\sqrt{B} = Cr$. The second has Frobenius norm at most $(\varrho r_-)^{-1/2}\|M^\circ_{t+r} - \widehat{N}_r\|_{\mathrm{F}}$ by eq. (SA.13).

For (ii), lemma SE.2(ii) gives $\widehat{V}_r - V_r = Q_{\varrho^\circ(\Delta_t)}(\widehat{N}_r) - Q_{\varrho^\circ(\Delta_t)}(N_r)$ with $\widehat{N}_r - N_r = \beta^{r+1}(m - N_{-1})$. By lemma SE.4(ii), lemma SE.2(iii) and, for the operator norm, $(\mathbb{E}\,\|N_{-1}\|^2)^{1/2} \le \sum_j w_j\, C\log(2B)\sqrt{2r_+}$ from lemma SE.3(ii), both bounds follow.

Parts (iii) and (iv) sum (i) and (ii) over $r < L$ with Minkowski's inequality. In Frobenius norm the sum of Cr is at most CL^2 and the sum of $C\beta^{r+1}(\|m\|_{\mathrm{F}} + \sqrt{B})$ is at most $C(\|m\|_{\mathrm{F}} + \sqrt{B})$. In operator norm the ridge bracket of (i) has norm at most $Cr/\sqrt{B}$ by lemma SE.4(iv), the matrix bracket has expected norm at most $Cr\log(2B)/\sqrt{B}$ by lemma SE.4(ii) and (i), and the startup difference has expected norm at most $C\beta^{r+1}(\|m\| + \log(2B))$. □

Lemma SE.17 (Stationary block sums and the startup cross term)**.** *Let D, m be deterministic with $r_- \le \mathscr{R}(D) \le r_+$, consider the chains at (t, D, m), and put $S \overset{\text{def}}{=} \sum_{r<L} V_r$, $A \overset{\text{def}}{=} \sum_{r<L}(\widehat{V}_r - V_r)$ and $a \overset{\text{def}}{=} 1 + \|m\|_{\mathrm{F}}/\sqrt{B}$. Then*

(i) $\mathbb{E}\,\|S - L\overline{V}(D)\|_{\mathrm{F}}^2 \le CBL$;

(ii) for every linear map $\mathcal{W}$ on $\mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ of Frobenius norm at most one, $|\,\mathbb{E}\langle S, \mathcal{W}A\rangle| \le Ca\big(B + L\sqrt{B}\big)$.

Proof. For (i), by lemma SE.2(i) the lag covariances of (V_r) are those of the volatility section at the state $\widehat{A}(D)$. With the tests $F = \mathrm{Id}$, $G = \mathrm{Id}$ in definition SD.1 and the envelopes of theorem SD.2(ii),

$$\mathbb{E}\,\|S - L\overline{V}(D)\|_{\mathrm{F}}^2 = B\sum_{r,r'<L} C_{|r-r'|}[\mathrm{Id},\mathrm{Id}](\varrho,\varrho) \le BL\Big(\gamma_\star + \frac{2\mathsf{K}_\star\beta}{1-\beta}\Big).$$

For (ii), write $A_r \overset{\text{def}}{=} \widehat{V}_r - V_r$. By the proof of lemma SE.16(ii), $\|A_r\|_{L^2(\mathrm{F})} \le Ca\sqrt{B}\beta^{r+1}$, where $\|X\|_{L^2(\mathrm{F})} \overset{\text{def}}{=} (\mathbb{E}\,\|X\|_{\mathrm{F}}^2)^{1/2}$. Expand $\mathbb{E}\langle S, \mathcal{W}A\rangle = \sum_{u,r<L}\mathbb{E}\langle V_u, \mathcal{W}A_r\rangle$ and center V_u at $\overline{V} \overset{\text{def}}{=} \overline{V}(D)$. The mean part is at most $\sum_{u,r}\|\overline{V}\|_{\mathrm{F}}\|A_r\|_{L^2(\mathrm{F})} \le CLa\sqrt{B}$, by eq. (SE.15). For $u \le r$, the Cauchy–Schwarz inequality and $\|V_u - \overline{V}\|_{\mathrm{F}} \le 2\sqrt{\gamma_\star B}$ give $|\,\mathbb{E}\langle V_u - \overline{V}, \mathcal{W}A_r\rangle| \le CaB\beta^{r+1}$, and $\sum_r (r+1)\beta^{r+1} \le C$. For $u > r$, let V_u' be the update obtained by replacing the auxiliary batches and the batches of the steps up to $t + r$ by independent copies. It has the law of V_u, it is independent of A_r, which depends only on m, the auxiliary batches and the batches up to step $t + r$, and by the lag decomposition eq. (SA.33) its buffer differs from N_u by β^{u-r} times the difference of two buffers with second moments at most CB. Hence $\mathbb{E}\langle V_u - \overline{V}, \mathcal{W}A_r\rangle = \mathbb{E}\langle V_u - V_u', \mathcal{W}A_r\rangle$ is at most $C\sqrt{B}\beta^{u-r}\cdot Ca\sqrt{B}\beta^{r+1}$, and summing over $u > r$ and then over r gives CaB. □

Lemma SE.18 (Taylor transfer)**.** *In the setting of lemma SE.16, let $\dot{\widehat{V}}_r$ be the tangent of lemma SE.8 for the same-buffer chain at $(t, \Delta_t, M_{t-1}^\circ)$. On $\{\Delta_t \in \mathcal{K}\}$, for $0 \le r \le 4L$,*

$$\mathbb{E}_t\,\big\|\mathsf{S}_{t+r} - \widehat{V}_r - \eta\,\dot{\widehat{V}}_r\big\|_{\mathrm{F}} \le \frac{Cr^2\log(2B)}{\sqrt{B}}.$$

Proof. Write $D = \Delta_t$, $\ell = \varrho^\circ(D)$, $E_r \overset{\text{def}}{=} \Delta_{t+r} - D$, $R_r \overset{\text{def}}{=} M_{t+r}^\circ - \widehat{N}_r$ and $s_r \overset{\text{def}}{=} \varrho^\circ(\Delta_{t+r}) - \ell$. By lemma SE.14 and lemma SE.16(i),

$$\|E_r\|_{\mathrm{F}} \le \frac{Cr}{\sqrt{B}}, \qquad |s_r| \le \frac{Cr}{\sqrt{B}}, \qquad \mathbb{E}_t\,\|R_r\|_{\mathrm{F}}^2 \le Cr^2, \qquad \mathbb{E}_t\,\|R_r\|^2 \le \frac{Cr^2\log^2(2B)}{B}.$$

The update is expanded to first order. Since $\mathsf{S}_{t+r} = Q_{\ell+s_r}(\widehat{N}_r + R_r)$,

$$\begin{aligned}\mathsf{S}_{t+r} - \widehat{V}_r = {}& \big[Q_\ell(\widehat{N}_r + R_r) - Q_\ell(\widehat{N}_r)\big] + s_r\,\partial_\ell Q_\ell(\widehat{N}_r)\\ &+ s_r\big[\partial_\ell Q_\ell(\widehat{N}_r + R_r) - \partial_\ell Q_\ell(\widehat{N}_r)\big]\\ &+ \big[Q_{\ell+s_r}(\widehat{N}_r + R_r) - Q_\ell(\widehat{N}_r + R_r) - s_r\partial_\ell Q_\ell(\widehat{N}_r + R_r)\big].\end{aligned}$$

By lemma SE.5(ii) the first bracket is $DQ_\ell(\widehat{N}_r)[R_r]$ up to $C\|R_r\|\|R_r\|_{\mathrm{F}}$; by (iv) the third term is at most $C|s_r|\|R_r\|_{\mathrm{F}}$ and the last bracket at most $C\sqrt{B}\,s_r^2$. The Cauchy–Schwarz inequality bounds the conditional expectations of these three errors by $Cr^2\log(2B)/\sqrt{B}$.

The arguments are replaced by their tangents. Put $E_r^{(1)} \stackrel{\mathrm{def}}{=} \eta\dot{\Delta}_r = -\eta\sum_{k<r}\widehat{V}_k$, while $E_r = -\eta\sum_{k<r}\mathsf{S}_{t+k}$. By (i) of lemma SE.16, $(\mathbb{E}_t\,\|E_r - E_r^{(1)}\|_{\mathrm{F}}^2)^{1/2} \le \eta\sum_{k<r}Ck \le Cr^2/B$. The tangent of the buffer is $\eta\dot{\widehat{N}}_r = \sum_{j\le r} w_{r-j}G^\circ_{t+j+1}[E_j^{(1)}]$, so $R_r - \eta\dot{\widehat{N}}_r = \sum_{j\le r} w_{r-j}G^\circ_{t+j+1}[E_j - E_j^{(1)}]$, and lemma SE.3(i) applied conditionally gives $(\mathbb{E}_t\,\|R_r - \eta\dot{\widehat{N}}_r\|_{\mathrm{F}}^2)^{1/2} \le C\sqrt{B}\cdot Cr^2/B$. The ridge is quadratic in the state, $\varrho^\circ(D+E) - \varrho^\circ(D) = \varrho(2\langle D, E\rangle_\Sigma + \|E\|_\Sigma^2)$, so

$$s_r - \eta\dot{\ell}_r = 2\varrho\langle D, E_r - E_r^{(1)}\rangle_\Sigma + \varrho\|E_r\|_\Sigma^2, \qquad \mathbb{E}_t\,|s_r - \eta\dot{\ell}_r| \le \frac{Cr^2}{B},$$

where $\dot{\ell}_r = 2\varrho\langle D, \dot{\Delta}_r\rangle_\Sigma$ is the ridge tangent of lemma SE.8. By lemma SE.5(i),(iv), replacing R_r by $\eta\dot{\widehat{N}}_r$ in $DQ_\ell(\widehat{N}_r)[\cdot]$ costs $\ell^{-1/2}Cr^2/\sqrt{B}$, and replacing s_r by $\eta\dot{\ell}_r$ in front of $\partial_\ell Q_\ell(\widehat{N}_r)$ costs $C\sqrt{B}\cdot Cr^2/B$. By lemma SE.8, $\eta\dot{\widehat{V}}_r = DQ_\ell(\widehat{N}_r)[\eta\dot{\widehat{N}}_r] + \eta\dot{\ell}_r\,\partial_\ell Q_\ell(\widehat{N}_r)$, which completes the proof. □

Lemma SE.19 (Projection variance)**.** *Let D be deterministic with $r_- \le \mathscr{R}(D) \le r_+$, consider the stationary chain at (t, D), and let $T \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ be deterministic. Then*

$$\mathrm{Var}\langle T, V_r\rangle \le C\log^2(2B)\|T\|_{\mathrm{F}}^2, \qquad \mathrm{Var}\left\langle T, \sum_{r<L} V_r\right\rangle \le CL^2\log^2(2B)\|T\|_{\mathrm{F}}^2.$$

No bound on $\|T\|$ and no diffuseness of D is assumed.

Proof. First keep only the batches of the relative times $r-J,\dots,r$, auxiliary batches included. The resulting update is a smooth function f of finitely many independent standard Gaussian coordinates, namely the whitened samples, and the Gaussian Poincaré inequality [13, Theorem 3.20] gives $\mathrm{Var}\,f \le \mathbb{E}\,\|\nabla f\|^2$.

The gradient is computed by the chain rule. Write $A_D \stackrel{\mathrm{def}}{=} {\Sigma_{\mathrm{out}}}^{1/2} D {\Sigma_{\mathrm{in}}}^{1/2}$, so $\|A_D\|_{\mathrm{F}}^2 = 2\mathscr{R}(D) \le 2r_+$, and let $\mathsf{C} \stackrel{\mathrm{def}}{=} DQ_\ell(N)^*[T]$ be the adjoint of the derivative of the filter applied to T, at $\ell = \varrho^\circ(D)$; by lemma SE.5(i), $\|\mathsf{C}\|_{\mathrm{F}} \le \ell^{-1/2}\|T\|_{\mathrm{F}}$ pointwise. A sample (z, w) of a batch of weight ϖ enters the buffer as $\varpi B^{-1}(z^\top A_D w){\Sigma_{\mathrm{out}}}^{1/2}(z\otimes w){\Sigma_{\mathrm{in}}}^{1/2}$, so with $\mathsf{q}_a \stackrel{\mathrm{def}}{=} z_a^\top {\Sigma_{\mathrm{out}}}^{1/2}\mathsf{C}\,{\Sigma_{\mathrm{in}}}^{1/2} w_a$ and $d_a = z_a^\top A_D w_a$,

$$\nabla_{z_a} f = \frac{\varpi}{B}\big[(A_D w_a)\,\mathsf{q}_a + d_a\,{\Sigma_{\mathrm{out}}}^{1/2}\mathsf{C}\,{\Sigma_{\mathrm{in}}}^{1/2} w_a\big],$$
$$\nabla_{w_a} f = \frac{\varpi}{B}\big[(A_D^\top z_a)\,\mathsf{q}_a + d_a\,{\Sigma_{\mathrm{in}}}^{1/2}\mathsf{C}^\top{\Sigma_{\mathrm{out}}}^{1/2} z_a\big].$$

With $Y \stackrel{\mathrm{def}}{=} {\Sigma_{\mathrm{out}}}^{1/2}[z_1\cdots z_B]$ and $X \stackrel{\mathrm{def}}{=} {\Sigma_{\mathrm{in}}}^{1/2}[w_1\cdots w_B]$, pointwise matrix inequalities give

$$\sum_a \mathsf{q}_a^2 \le \max_a \|{\Sigma_{\mathrm{in}}}^{1/2} w_a\|^2\,\|\mathsf{C}\|_{\mathrm{F}}^2\|Y\|^2,$$
$$\sum_a d_a^2\|{\Sigma_{\mathrm{out}}}^{1/2}\mathsf{C}\,{\Sigma_{\mathrm{in}}}^{1/2} w_a\|^2 \le C_\Sigma \max_a d_a^2\,\|\mathsf{C}\|_{\mathrm{F}}^2\|X\|^2,$$

and the transposed bounds. Hence the squared gradient over one batch is at most

$$\frac{C\varpi^2}{B^2}\|\mathsf{C}\|_{\mathrm{F}}^2\Big[\max_a\|A_D w_a\|^2\max_a\|w_a\|^2\|Y\|^2 + \max_a\|A_D^\top z_a\|^2\max_a\|z_a\|^2\|X\|^2 + \max_a d_a^2\big(\|X\|^2+\|Y\|^2\big)\Big].$$

In every fixed L^p norm, $\max_a\|A_D w_a\|^2$ and $\max_a\|A_D^\top z_a\|^2$ are at most $C_p\log(2B)$, $\max_a\|w_a\|^2$, $\max_a\|z_a\|^2$, $\|X\|^2$ and $\|Y\|^2$ are at most C_pB, and $\max_a d_a^2$ is at most $C_p\log^2(2B)$. These follow

from $\|\|A_D w\|^2\|_q \le \sum_k s_k(A_D)^2 \|g^2\|_q \le Cq\|A_D\|_{\mathrm{F}}^2$ (a diagonalization and Minkowski's inequality), from $\|\|w\|^2\|_q \le C(N_{\mathrm{in}}+q)$, from the maximum over B samples taken with $q \asymp \log(2B)$ as in the proof of lemma SE.3(ii), and from the operator norm moments there. Hölder's inequality, with no independence between C and the samples, bounds the expected squared gradient over the batch by $C\varpi^2 \log^2(2B)\|T\|_{\mathrm{F}}^2$. Summing over the batches with $\sum_j w_j^2 \le 1$ gives the first bound for the truncated update. As $J \to \infty$ the truncated update converges to V_r in L^2, by lemma SA.6(ii), so the bound passes to V_r.

The second bound is the Cauchy–Schwarz inequality $\operatorname{Var} \sum_{r<L} Y_r \le L \sum_{r<L} \operatorname{Var} Y_r$. □

The block martingale. For bounded f, g on $[0, C_\Sigma]$ write $\mathcal{W} = \mathcal{W}_{f,g}$ and $X_{f,g}(\Delta) \stackrel{\mathrm{def}}{=} \int f \otimes g \, \mathrm{d}\chi(\Delta) = \langle \Delta, \mathcal{W}\Delta \rangle$. Since $\mathcal{W}$ is selfadjoint, for every Δ, S,

$$X_{f,g}(\Delta - \eta\mathsf{S}) = X_{f,g}(\Delta) - 2\eta\langle \mathcal{W}\Delta, \mathsf{S}\rangle + \eta^2 \langle \mathsf{S}, \mathcal{W}\mathsf{S}\rangle. \tag{SE.26}$$

Lemma SE.20 (Block martingale)**.** *Let $b_\star$ be a block index such that whether $b < b_\star$ is decided by the past $\mathscr{F}_{t_b}$, and such that $\Delta_{t_b} \in \mathcal{A}$ for $b < b_\star$; put $I_b \stackrel{\mathrm{def}}{=} \mathbf{1}_{b<b_\star}$. Let $\|f\|_\infty, \|g\|_\infty \le 1$ and*

$$\xi_b \stackrel{\mathrm{def}}{=} X_{f,g}(\Delta_{t_{b+1}}) - X_{f,g}(\Delta_{t_b}) - \mathbb{E}_{t_b}\big[X_{f,g}(\Delta_{t_{b+1}}) - X_{f,g}(\Delta_{t_b})\big].$$

Then for every $T > 0$,

$$\mathbb{E} \max_{j \le TB/L} \Big| \sum_{b<j} I_b \xi_b \Big|^2 \le C(T+1)\Big\{ \frac{L\log^2(2B)}{B} + \frac{L^3}{B} + \frac{1}{L} \Big\}.$$

Proof. Fix a block b with $I_b = 1$ and write $t = t_b$, $D = \Delta_t$, $m = M_{t-1}^\circ$. There is no stop in the block, by lemma SE.14(ii), so $\Delta_{t+L} = D - \eta\Sigma_{\mathsf{S}}$ with $\Sigma_{\mathsf{S}} \stackrel{\mathrm{def}}{=} \sum_{r<L} \mathsf{S}_{t+r}$, and eq. (SE.26) for the block gives

$$X_{f,g}(\Delta_{t+L}) - X_{f,g}(D) = -2\eta\langle \mathcal{W}D, S\rangle - 2\eta\langle \mathcal{W}D, \mathcal{E}\rangle + \eta^2\langle \Sigma_{\mathsf{S}}, \mathcal{W}\Sigma_{\mathsf{S}}\rangle,$$

where $S = \sum_{r<L} V_r$ is the stationary sum of the chains at time t and $\mathcal{E} \stackrel{\mathrm{def}}{=} \Sigma_{\mathsf{S}} - S$. The left side does not depend on the auxiliary batches, and conditionally on $\mathscr{F}_t$ its centered version is the sum of the centered versions of the three terms. Hence $\mathbb{E}_t\, \xi_b^2 \le 3\big(4\eta^2 \operatorname{Var}_t\langle \mathcal{W}D, S\rangle + 4\eta^2\, \mathbb{E}_t\langle \mathcal{W}D, \mathcal{E}\rangle^2 + \eta^4\, \mathbb{E}_t\langle \Sigma_{\mathsf{S}}, \mathcal{W}\Sigma_{\mathsf{S}}\rangle^2\big)$.

The three terms are bounded as follows. By lemma SE.19, conditionally on $\mathscr{F}_t$ with $T = \mathcal{W}D$ and $\|\mathcal{W}D\|_{\mathrm{F}} \le \|D\|_{\mathrm{F}} \le C$, the first is at most $C\eta^2 L^2 \log^2(2B)$. By lemma SE.16(iii), the second is at most $C\eta^2(L^4 + B + \|m\|_{\mathrm{F}}^2)$, and $\mathbb{E}[I_b\|m\|_{\mathrm{F}}^2] \le CB$ by lemma SE.15 with $\tau = \tau^+$, since $I_b \le \mathbf{1}_{t_b \le \tau^+}$. The third is at most $\eta^4(L^2\gamma_\star B)^2 = CL^4/B^2$, because $\|\Sigma_{\mathsf{S}}\|_{\mathrm{F}} \le L\sqrt{\gamma_\star B}$. Altogether

$$\mathbb{E}\big[I_b \xi_b^2\big] \le C\Big\{ \frac{L^2\log^2(2B)}{B^2} + \frac{L^4}{B^2} + \frac{1}{B} \Big\}.$$

The variables $I_b\xi_b$ are martingale differences for the pasts at the block starts, so $\mathbb{E}(\sum_{b<j} I_b\xi_b)^2 = \sum_{b<j} \mathbb{E}[I_b\xi_b^2]$, and Doob's L^2 maximal inequality [74, Theorem 14.11] bounds the expected squared maximum over $j \le TB/L$ by four times the sum over the at most $TB/L + 1$ blocks. □

Delayed projections. The linear term of a block increment is the projection of the updates on the state. Its first q updates are not close to stationary, because they still depend on the buffer inherited from the previous block. They are controlled by freezing q steps earlier.

Lemma SE.21 (Delayed projections and prefixes)**.** *Let $\mathcal{W}$ be a linear map on $\mathbb{R}^{N_{\mathrm{out}} \times N_{\mathrm{in}}}$ with $\|\mathcal{W}\|_{\mathrm{F}\to\mathrm{F}} \le c_W$.*

(i) For every $u \ge \mathsf{q}$, $\mathbb{E}\big[\mathbf{1}_{u-\mathsf{q}<\tau^+} |\langle \mathcal{W}\Delta_u, \mathsf{S}_u\rangle|\big] \le C\mathsf{q}$.

(ii) For every $t \ge \mathsf{q}$, $\mathbb{E}\big[\mathbf{1}_{t<\tau^+} \big|\sum_{r<\mathsf{q}} \langle \mathcal{W}\Delta_t, \mathsf{S}_{t+r}\rangle\big|\big] \le C\mathsf{q}^2$.

(iii) For every $t \ge \mathsf{q}$, with the chains at time t and $0 \le r \le \mathsf{q}$, $\mathbb{E}\big[\mathbf{1}_{t<\tau^+} |\langle \Sigma_{\mathrm{out}}\Delta_t\Sigma_{\mathrm{in}}, \widehat{V}_r - V_r\rangle|\big] \le \min\big(C\mathsf{q},\ C\sqrt{B}\beta^{r+1}\big)$, and for every $r \ge 0$ the second bound holds.

Consequently, with $\Omega_\beta(\epsilon) \stackrel{\text{def}}{=} \sum_{r\ge 0}\min(\epsilon,\beta^{r+1})$ *and* $\epsilon_B \stackrel{\text{def}}{=} \mathsf{q}/\sqrt{B}$,

$$\frac{1}{\sqrt{B}}\sum_{r<L}\mathbb{E}\left[\mathbf{1}_{t<\tau^+}|\langle\Sigma_{\text{out}}\Delta_t\Sigma_{\text{in}},\widehat{V}_r - V_r\rangle|\right] \le C\big(\Omega_\beta(\epsilon_B)+\beta^{\mathsf{q}}\big), \tag{SE.27}$$

and $\Omega_\beta(\epsilon)\le C\epsilon(1+\log(1/\epsilon))$ *for* $0<\epsilon\le 1$, *with* $\Omega_0 = 0$.

Proof. For (i), put $a \stackrel{\text{def}}{=} u-\mathsf{q}$ and $D\stackrel{\text{def}}{=}\Delta_a$, and consider the chains at time a, with stationary updates $V_r^{(a)}$ and same-buffer updates $\widehat{V}_r^{(a)}$ built from $m = M_{a-1}^\circ$. On $\{a<\tau^+\}$, $D\in\mathcal{A}^+$, and by lemma SE.14(ii) the states up to step u lie in $\mathcal{A}^{++}$; the risk stays bounded below on this window even if it crosses τ^+. Decompose

$$\langle\mathcal{W}\Delta_u,\mathsf{S}_u\rangle = \langle\mathcal{W}D,V_{\mathsf{q}}^{(a)}\rangle + \langle\mathcal{W}D,\mathsf{S}_u - V_{\mathsf{q}}^{(a)}\rangle + \langle\mathcal{W}(\Delta_u - D),\mathsf{S}_u\rangle.$$

By eq. (SE.15) and lemma SE.19 with $T=\mathcal{W}D$, $\mathbb{E}_a|\langle\mathcal{W}D,V_{\mathsf{q}}^{(a)}\rangle| \le C + C\log(2B)$. By lemma SE.16(i),(ii) with $r=\mathsf{q}$, $\mathbb{E}_a|\langle\mathcal{W}D,\mathsf{S}_u - V_{\mathsf{q}}^{(a)}\rangle| \le C\mathsf{q} + C\beta^{\mathsf{q}+1}(\|m\|_{\text{F}}+\sqrt{B})$. By lemma SE.14(i), $|\langle\mathcal{W}(\Delta_u-D),\mathsf{S}_u\rangle| \le c_W\|\Delta_u - D\|_{\text{F}}\|\mathsf{S}_u\|_{\text{F}} \le C\mathsf{q}$. The indicator $\mathbf{1}_{a<\tau^+}$ is decided by the past at a; multiplying by it, taking expectations and using $\mathbb{E}[\mathbf{1}_{a<\tau^+}\|m\|_{\text{F}}]\le C\sqrt{B}$ from lemma SE.15 gives at most $C\log(2B)+C\mathsf{q}+C\sqrt{B}\beta^{\mathsf{q}}\le C\mathsf{q}$.

For (ii), write $\langle\mathcal{W}\Delta_t,\mathsf{S}_{t+r}\rangle = \langle\mathcal{W}\Delta_{t+r},\mathsf{S}_{t+r}\rangle + \langle\mathcal{W}(\Delta_t-\Delta_{t+r}),\mathsf{S}_{t+r}\rangle$. Since $t+r-\mathsf{q}\le t$, $\mathbf{1}_{t<\tau^+}\le\mathbf{1}_{t+r-\mathsf{q}<\tau^+}$, and (i) with $u=t+r$ bounds the first term by $C\mathsf{q}$; the second is at most Cr by lemma SE.14(i). Summing over $r<\mathsf{q}$ proves (ii).

For (iii), let $D=\Delta_t$ and write $\Sigma_D\stackrel{\text{def}}{=}\Sigma_{\text{out}}D\Sigma_{\text{in}}$. The first bound follows from

$$|\langle\Sigma_D,\widehat{V}_r\rangle| \le |\langle\Sigma_{\text{out}}\Delta_{t+r}\Sigma_{\text{in}},\mathsf{S}_{t+r}\rangle| + |\langle\Sigma_{\text{out}}(D-\Delta_{t+r})\Sigma_{\text{in}},\mathsf{S}_{t+r}\rangle| + |\langle\Sigma_D,\widehat{V}_r-\mathsf{S}_{t+r}\rangle|,$$

whose three terms have expectations, with the indicator, at most $C\mathsf{q}$ by (i) for $\mathcal{W}X=\Sigma_{\text{out}}X\Sigma_{\text{in}}$, at most Cr by lemma SE.14(i), and at most Cr by lemma SE.16(i); together with $\mathbb{E}_t|\langle\Sigma_D,V_r\rangle|\le C\log(2B)$, from eq. (SE.15) and lemma SE.19. The second bound is lemma SE.16(ii) with lemma SE.15.

For eq. (SE.27), use the minimum of the two bounds of (iii) for $r\le\mathsf{q}$, divided by $\sqrt{B}$, and the second bound for $r>\mathsf{q}$, whose sum is at most $C\beta^{\mathsf{q}}$. For the bound on Ω_β, let r_ϵ be the least r with $\beta^{r+1}\le\epsilon$; then $r_\epsilon\le\log(1/\epsilon)/\log(1/\beta)$, and $\Omega_\beta(\epsilon)\le r_\epsilon\epsilon+\sum_{r\ge r_\epsilon}\beta^{r+1}\le r_\epsilon\epsilon+\epsilon/(1-\beta)$. □

The response of the expected block increment. The causal tangent of the same-buffer chain is transported to the stationary chain. The difference of the two chains is the startup difference $A_k=\widehat{V}_k-V_k$, whose projection on $\Sigma_{\text{out}}D\Sigma_{\text{in}}$ is controlled by the delayed projections.

Proposition SE.5 (Tangent transport)**.** *Let* $t\ge\mathsf{q}$, *let the chains at time* t *be those of definition SE.2 with* $D=\Delta_t$ *and* $m=M_{t-1}^\circ$, *and let* Π_t *be a matrix determined by the past* $\mathscr{F}_t$ *with* $\|\Pi_t\|_{\text{F}},\|\Pi_t\|\le c_\Pi$. *Then for* $0\le r<L$,

$$\eta\,\mathbb{E}\left[\mathbf{1}_{t<\tau^+}\big|\,\mathbb{E}_t\langle\Pi_t,\dot{\widehat{V}}_r-\dot{V}_r\rangle\big|\right] \le C\Big\{\frac{r}{\sqrt{B}}+\Omega_\beta(\epsilon_B)+\beta^{\mathsf{q}}+\sqrt{B}\,r\beta^{r+1}\Big\}.$$

Proof. Conditionally on $\mathscr{F}_t$, the matrices D, m and Π_t are fixed, and by the scaling in the proof of proposition SE.4 we may take $2\mathscr{R}(D)=1$. By lemma SE.8, with $A_k\stackrel{\text{def}}{=}\widehat{V}_k-V_k$, $\ell=\varrho$, $\dot\ell_r^\wedge\stackrel{\text{def}}{=}-2\varrho\sum_{k<r}\langle\Sigma_D,\widehat{V}_k\rangle$ and $\dot\ell_r\stackrel{\text{def}}{=}-2\varrho\sum_{k<r}\langle\Sigma_D,V_k\rangle$,

$$\dot{\widehat{V}}_r-\dot{V}_r = \big[DQ_\ell(\widehat{N}_r)-DQ_\ell(N_r)\big]\big[\dot{\widehat{N}}_r\big] - DQ_\ell(N_r)\Big[\sum_{0\le k<j\le r}w_{r-j}G^\circ_{t+j+1}[A_k]\Big]$$
$$+\dot\ell_r^\wedge\big[\partial_\ell Q_\ell(\widehat{N}_r)-\partial_\ell Q_\ell(N_r)\big] + \big(\dot\ell_r^\wedge-\dot\ell_r\big)\,\partial_\ell Q_\ell(N_r).$$

The change of the derivative operators. By lemma SE.5(iii) and lemma SE.2(ii), the first term has Frobenius norm at most $C\beta^{r+1}\|m-N_{-1}\|_{\text{F}}\|\dot{\widehat{N}}_r\|_{\text{F}}$. Since $\|\widehat{V}_k\|_{\text{F}}\le\sqrt{\gamma_\star B}$, lemma SE.3(i) gives

$(\mathbb{E}_t \|G^\circ_{t+j+1}[\widehat{V}_k]\|_{\mathrm{F}}^2)^{1/2} \le CB$, so $(\mathbb{E}_t \|\dot{\widehat{N}}_r\|_{\mathrm{F}}^2)^{1/2} \le CBr$. By the Cauchy–Schwarz inequality, lemma SE.2(iii) and lemma SE.15, the first term contributes at most $\eta \cdot C\beta^{r+1}\sqrt{B} \cdot CBr = C\sqrt{B}r\beta^{r+1}$. The third term is at most $C\sqrt{B}r \cdot C\beta^{r+1}\|m - N_{-1}\|_{\mathrm{F}}$ by lemma SE.5(iv) and $|\dot{\ell}_r^\wedge| \le Cr\sqrt{B}$, and contributes at most $\eta \cdot CBr\beta^{r+1} \le Cr\beta^{r+1}$.

The change of the argument. For $k < j$ the direction A_k depends only on m, the auxiliary batches and the batches up to step $t+k$, so the batch of step $t+j$ is independent of A_k together with all other samples; moreover $\|A_k\| \le 2$ and $\|A_k\|_{\mathrm{F}} \le 2\sqrt{\gamma_\star B}$. By the integral representation of DQ_ℓ, proposition SE.2 with $E = A_k$ and the size bound of lemma SE.12,

$$\Big|\, \mathbb{E}_t \left\langle \Pi_t, DQ_\ell(N_r)\big[w_{r-j} G^\circ_{t+j+1}[A_k]\big] \right\rangle \Big| \le C\sqrt{B}\, w_{r-j} + C\sqrt{B}\, w_{r-j}^2\ \mathbb{E}_t\, |\langle \Sigma_D, A_k\rangle|,$$

after integration over $s = \varrho + u^2$. Summing over $0 \le k < j \le r$ and multiplying by η, the second term contributes at most $Cr/\sqrt{B} + CB^{-1/2}\sum_{k<r} \mathbb{E}_t\, |\langle \Sigma_D, A_k\rangle|$. The fourth term is at most $C\sqrt{B} \cdot 2\varrho \sum_{k<r} |\langle \Sigma_D, A_k\rangle|$ by lemma SE.5(iv), and contributes at most $CB^{-1/2}\sum_{k<r} |\langle \Sigma_D, A_k\rangle|$. Multiplying by $\mathbf{1}_{t<\tau^+}$, taking expectations and applying eq. (SE.27) proves the claim. □

Proposition SE.6 (Post-prefix response)**.** *Let $\delta_B \downarrow 0$ be a deterministic sequence. Let $t \ge \mathsf{q}$, let Π_t be determined by $\mathscr{F}_t$ with $\|\Pi_t\|_{\mathrm{F}}, \|\Pi_t\| \le c_\Pi$, and let $\mathfrak{E}_t \overset{\mathrm{def}}{=} \{t < \tau^+,\ \delta(\Delta_t) \le \delta_B\}$. Then*

$$\mathbb{E}\left[\mathbf{1}_{\mathfrak{E}_t}\Big|\frac{1}{L}\sum_{r=\mathsf{q}}^{L-1} \big\langle \Pi_t, \mathbb{E}_t\, \mathsf{S}_{t+r} - \overline{V}(\Delta_t)\big\rangle\Big|\right] \le \rho_B,$$

$$\rho_B \overset{\mathrm{def}}{=} C\Big\{\frac{L^2\log(2B)}{\sqrt{B}} + \frac{L}{\sqrt{B}} + \frac{1}{B} + \delta_B(c_\Pi + \delta_B) + \Omega_\beta(\epsilon_B) + \beta^{\mathsf{q}} + \sqrt{B}\, L\beta^{\mathsf{q}}\Big\},$$

and $\rho_B \to 0$.

Proof. For $\mathsf{q} \le r < L$, since $\mathbb{E}_t\, V_r = \overline{V}(\Delta_t)$ by lemma SE.2,

$$\mathbb{E}_t\, \mathsf{S}_{t+r} - \overline{V}(\Delta_t) = \mathbb{E}_t\left[\mathsf{S}_{t+r} - \widehat{V}_r - \eta\dot{\widehat{V}}_r\right] + \mathbb{E}_t\left[\widehat{V}_r - V_r\right] + \eta\ \mathbb{E}_t\left[\dot{\widehat{V}}_r - \dot{V}_r\right] + \eta\ \mathbb{E}_t\, \dot{V}_r.$$

Paired with Π_t, the four terms are bounded by lemma SE.18 (by $Cr^2\log(2B)/\sqrt{B}$), by lemma SE.16(ii) with lemma SE.15 (by $C\sqrt{B}\beta^{r+1}$ in expectation), by proposition SE.5, and by proposition SE.4 applied conditionally on $\mathscr{F}_t$ (by $C(r/\sqrt{B} + 1/B + \epsilon_{\Pi_t})$, with $\epsilon_{\Pi_t} \le \delta_B(c_\Pi + \delta_B)$ on $\mathfrak{E}_t$). Averaging over $\mathsf{q} \le r < L$ and using $\beta^{r+1} \le \beta^{\mathsf{q}}$ gives ρ_B. Each term of ρ_B tends to zero by the choice eq. (SE.25) and lemma SE.21. □

Lemma SE.22 (Quadratic block response)**.** *In the setting of lemma SE.16 with $\mathcal{K} = \mathcal{A}$, let $S = \sum_{r<L} V_r$, $\mathcal{E} = \sum_{r<L} \mathsf{S}_{t+r} - S$ and $a \overset{\mathrm{def}}{=} 1 + \|M^\circ_{t-1}\|_{\mathrm{F}}/\sqrt{B}$, and let $\mathcal{W}$ be selfadjoint with $\|\mathcal{W}\|_{\mathrm{F}\to\mathrm{F}} \le 1$. On $\{\Delta_t \in \mathcal{A}\}$,*

$$\frac{1}{LB}\Big|\, \mathbb{E}_t\left[2\langle S, \mathcal{W}\mathcal{E}\rangle + \langle \mathcal{E}, \mathcal{W}\mathcal{E}\rangle\right]\Big| \le C\Big\{\frac{a+a^2}{L} + \frac{a}{\sqrt{B}} + \frac{L^2}{B} + \frac{L^{3/2}}{\sqrt{B}} + \frac{aL}{\sqrt{B}} + \frac{L^3}{B}\Big\}.$$

Proof. Write $\mathcal{E} = A + J$ with $A = \sum_{r<L}(\widehat{V}_r - V_r)$ and $J = \sum_{r<L}(\mathsf{S}_{t+r} - \widehat{V}_r)$. Since $\mathcal{W}$ is selfadjoint,

$$2\langle S, \mathcal{W}\mathcal{E}\rangle + \langle \mathcal{E}, \mathcal{W}\mathcal{E}\rangle = 2\langle S, \mathcal{W}A\rangle + \langle A, \mathcal{W}A\rangle + 2\langle S + A, \mathcal{W}J\rangle + \langle J, \mathcal{W}J\rangle.$$

By lemma SE.17(ii) applied conditionally, the first term is at most $Ca(B + L\sqrt{B})$. By lemma SE.16(ii), the second is at most Ca^2B. By lemma SE.17(i), eq. (SE.15) and lemma SE.16, $(\mathbb{E}_t\, \|S + A\|_{\mathrm{F}}^2)^{1/2} \le CL + C\sqrt{BL} + Ca\sqrt{B}$ and $(\mathbb{E}_t\, \|J\|_{\mathrm{F}}^2)^{1/2} \le CL^2$, so the last two terms are at most $C(L + \sqrt{BL} + a\sqrt{B})L^2 + CL^4$. Dividing by LB gives the claim. □

Persistence of diffuseness. The operator norm of the weighted state is controlled by a matrix martingale inequality and the operator-norm mean bound.

Lemma SE.23 (Matrix martingale)**.** *Let $Z_0, Z_1, \dots$ be random selfadjoint $d \times d$ matrices such that Z_b is determined by the past at block $b+1$, has conditional mean zero given the past at block b, and satisfies $\|Z_b\| \le c$. Then for $x > 0$ and every integer $N \ge 1$,*

$$\Pr\Big(\max_{j \le N} \Big\| \sum_{b<j} Z_b \Big\| \ge x\Big) \le 2dN \exp\Big(-\frac{x^2}{2Nc^2}\Big).$$

Proof. A centered increment has a subgaussian matrix moment generating function. For $u > 0$ and $|y| \le c$, convexity gives $e^{uy} \le \cosh(uc) + (y/c)\sinh(uc)$, so by the spectral theorem $e^{uZ_b} \preceq \cosh(uc)\,\mathrm{Id} + \sinh(uc) Z_b/c$; conditioning on the past at block b and using $\cosh(uc) \le e^{u^2c^2/2}$ gives $\mathbb{E}_b\, e^{uZ_b} \preceq e^{u^2c^2/2}\,\mathrm{Id}$.

The trace exponential of the sum is bounded by induction. Put $S_j \stackrel{\text{def}}{=} \sum_{b<j} Z_b$. The Golden–Thompson inequality [8, Theorem IX.3.7 and (IX.19)] gives $\operatorname{Tr} e^{uS_{j+1}} \le \operatorname{Tr}\big(e^{uS_j} e^{uZ_j}\big)$. Conditioning on the past at block j, and using $\operatorname{Tr}(PQ) \le \operatorname{Tr}(PQ')$ for $P \succeq 0$ and $Q \preceq Q'$,

$$\mathbb{E}\operatorname{Tr} e^{uS_{j+1}} \le e^{u^2c^2/2}\,\mathbb{E}\operatorname{Tr} e^{uS_j} \le d\, e^{(j+1)u^2c^2/2}.$$

Markov's inequality applied to $\operatorname{Tr} e^{uS_j} \ge e^{u\lambda_{\max}(S_j)}$ with $u = x/(jc^2)$ gives $\Pr(\lambda_{\max}(S_j) \ge x) \le d\, e^{-x^2/(2jc^2)} \le d\, e^{-x^2/(2Nc^2)}$ for $j \le N$. The same bound for $-S_j$ and a union bound over $j \le N$ prove the claim. ☐

Lemma SE.24 (Persistence of diffuseness)**.** *Let $b_c \stackrel{\text{def}}{=} \inf\{b \ge 0 : \Delta_{t_b} \notin \mathcal{A}\}$. For every fixed $T > 0$,*

$$\max\big\{\delta(\Delta_t) :\ t \le t_{b_c} \wedge TB\big\} \longrightarrow 0 \quad \textit{in probability}.$$

No diffuseness beyond the initial condition (H3) is used.

Proof. All states up to t_{b_c} lie in $\mathcal{A}^+$, so their risk is at least $r_0/4$ and δ is defined there. Put $Y_b \stackrel{\text{def}}{=} {\Sigma_{\text{out}}}^{1/2} \Delta_{t_{b\wedge b_c}} {\Sigma_{\text{in}}}^{1/2}$ and $J \stackrel{\text{def}}{=} \lceil TB/L \rceil$. For $b < b_c$ write $t = t_b$, $D = \Delta_t$, $m = M^\circ_{t-1}$ and $\mathcal{E} = \sum_{r<L} \mathsf{S}_{t+r} - \sum_{r<L} V_r$ as before. Then $Y_{b+1} - Y_b = -\eta {\Sigma_{\text{out}}}^{1/2}\big(\sum_{r<L} \mathsf{S}_{t+r}\big){\Sigma_{\text{in}}}^{1/2}$ and

$$\mathbb{E}_t\big[Y_{b+1} - Y_b\big] = -\eta {\Sigma_{\text{out}}}^{1/2}\big(L\overline{V}(D) + \mathbb{E}_t\,\mathcal{E}\big){\Sigma_{\text{in}}}^{1/2}.$$

The drift. By eq. (SE.15) and lemma SE.16(iv),

$$\|\,\mathbb{E}_t[Y_{b+1} - Y_b]\| \le C\eta L\big(\|Y_b\| + r_B\big) + C\eta\big(\|m\| + \log(2B) + L^2\log(2B)/\sqrt{B}\big).$$

By lemma SE.15 with $\tau = \tau^+$, $\mathbb{E}[\mathbf{1}_{b<b_c}\|M^\circ_{t_b-1}\|] \le \sqrt{2c_M B}\,\beta^{t_b} + C\log(2B)$. Hence the sum $\mathsf{E}_B \stackrel{\text{def}}{=} \sum_{b<J} \mathbf{1}_{b<b_c} C\eta(\|M^\circ_{t_b-1}\| + \log(2B) + L^2\log(2B)/\sqrt{B})$ satisfies

$$\begin{aligned}\mathbb{E}\,\mathsf{E}_B &\le \frac{Ch}{B}\Big(\frac{\sqrt{B}}{1-\beta^L} + (J+1)\Big(\log(2B) + \frac{L^2\log(2B)}{\sqrt{B}}\Big)\Big)\\ &\le C\Big(\frac{1}{\sqrt{B}} + \frac{(T+1)\log(2B)}{L} + \frac{(T+1)L\log(2B)}{\sqrt{B}}\Big) \to 0.\end{aligned}$$

The fluctuation. Put $Z_b \stackrel{\text{def}}{=} \mathbf{1}_{b<b_c}(Y_{b+1} - Y_b - \mathbb{E}_{t_b}[Y_{b+1} - Y_b])$. Since $\|\mathsf{S}\| \le 1$, $\|Y_{b+1} - Y_b\| \le C_\Sigma \eta L$ and the same holds for its conditional mean, so $\|Z_b\| \le 2C_\Sigma hL/B$. The symmetric dilations $\mathcal{D}(Z_b)$ have dimension $N_{\text{out}} + N_{\text{in}} \le 2\gamma_\star B$ and the same norms, and lemma SE.23 gives, for every $x > 0$,

$$\Pr\Big(\max_{j \le J} \Big\| \sum_{b<j} Z_b \Big\| \ge x\Big) \le 4\gamma_\star BJ \exp\Big(-\frac{x^2 B}{8C_\Sigma^2 h^2 (T+1)L}\Big) \to 0.$$

Gronwall. For $j \le J$, adding the drift and the fluctuation and using $\eta LJ \le h(T+1)$,

$$\|Y_j\| \le \mathfrak{a}_B + \frac{ChL}{B}\sum_{b<j}\|Y_b\|, \qquad \mathfrak{a}_B \stackrel{\text{def}}{=} \|Y_0\| + Ch(T+1)r_B + \mathsf{E}_B + \max_{j \le J}\Big\|\sum_{b<j} Z_b\Big\|.$$

Induction on j gives $\|Y_j\| \le \mathfrak{a}_B(1+ChL/B)^j \le \mathfrak{a}_B e^{Ch(T+1)}$. Here $\|Y_0\| = \sqrt{2\mathscr{R}(0)}\|\widehat{A}_0\| \to 0$ by (H3), $r_B \to 0$ by lemma SE.7, and the last two terms tend to zero in probability, so $\max_{j\le J}\|Y_j\| \to 0$ in probability. Within a block the weighted state moves by at most $C_\Sigma hL/B$ in operator norm, and $\delta(\Delta_t) \le \|{\Sigma_{\mathrm{out}}}^{1/2}\Delta_t{\Sigma_{\mathrm{in}}}^{1/2}\|/\sqrt{r_0/2}$ for $t \le t_{b_c}$, which proves the claim. □

SE.8 Block freezing

The drift and volatility sections state their deterministic equivalents along sequences of diffuse frozen states. Along the algorithm the states are random and change from block to block, so the equivalents are needed uniformly over all states of small diffuseness.

Lemma SE.25 (Uniformity over diffuse states)**.** *Assume (H1). For each dimension let $e_B \ge 0$ be a function on the set $\mathcal{S}_B$ of normalized states such that $e_B(\widehat{A}_B) \to 0$ along every subsequence of dimensions and every choice of $\widehat{A}_B \in \mathcal{S}_B$ along it with $\|\widehat{A}_B\| \to 0$. Then for every sequence $\delta_B \to 0$,*

$$\bar{e}_B \overset{\text{def}}{=} \sup\left\{e_B(\widehat{A}) :\ \widehat{A} \in \mathcal{S}_B,\ \|\widehat{A}\| \le \delta_B\right\} \longrightarrow 0,$$

with the convention that the supremum of the empty set is zero.

Proof. Suppose not. Then there are $\epsilon > 0$ and a subsequence of dimensions at which $\bar{e}_B > \epsilon$, and at each of them a state $\widehat{A}_B \in \mathcal{S}_B$ with $\|\widehat{A}_B\| \le \delta_B$ and $e_B(\widehat{A}_B) > \epsilon/2$. Along this subsequence $\|\widehat{A}_B\| \to 0$ while $e_B(\widehat{A}_B)$ does not tend to zero, a contradiction. □

We apply this lemma to two errors of the frozen-state model at the state $\widehat{A}$ and the fixed ridge ϱ. The first is the mean error $e_B^{(1)}(\widehat{A}) \overset{\text{def}}{=} \|\,\mathbb{E}\,Q_\varrho(M_t) - \overline{\widetilde{G}}_\varrho\|_{\mathrm{F}}$, the quantity ε_B of theorem SC.2 at $\varrho_- = \varrho_+ = \varrho$. The second is the forcing error $e_B^{(2)}(\widehat{A}) \overset{\text{def}}{=} \sup_{\|f\|_\infty,\|g\|_\infty\le 1}\left|\int f\otimes g\,\mathrm{d}\big(\mathfrak{C}_{\mathrm{int}}(\varrho)-\zeta\big)\right|$ of theorem SD.8(iii). Along any subsequence of dimensions and any frozen states along it with $\|\widehat{A}_B\| \to 0$, Assumption S3 holds under (H1), and these theorems give $e_B^{(i)}(\widehat{A}_B) \to 0$. For a sequence $\delta_B \to 0$ we write $\bar{e}_B^{(i)}$ for the suprema of lemma SE.25.

Lemma SE.26 (Block residual)**.** *Let b_c be as in lemma SE.24 and $\tau_c \overset{\text{def}}{=} t_{b_c}/B$. For every $T > 0$ and all bounded f, g on $[0, C_\Sigma]$,*

$$\sup_{\tau\le T}\left|\int f\otimes g\,\mathrm{d}\chi^{\mathrm{act}}(\tau\wedge\tau_c) - \int f\otimes g\,\mathrm{d}\chi^{\mathrm{act}}(0) - \int_0^{\tau\wedge\tau_c}\int f\otimes g\,\mathrm{d}\mathsf{F}_B\big(\chi^{\mathrm{act}}(\sigma)\big)\,\mathrm{d}\sigma\right| \longrightarrow 0$$

in probability. The process and the integral are stopped at the same time.

For $\sigma < \tau_c$ the state $\Delta_{\lfloor B\sigma\rfloor}$ lies in $\mathcal{A}^+$, so $\mathsf{F}_B(\chi^{\mathrm{act}}(\sigma))$ is defined. By eq. (SE.8) the integrand is $-2h\mathscr{R}(\lfloor B\sigma\rfloor)^{-1/2}\int f\otimes g\,\mathfrak{d}\,\mathrm{d}\chi^{\mathrm{act}}(\sigma) + h^2\int f\otimes g\,\mathrm{d}\zeta$.

Proof. We may assume $\|f\|_\infty, \|g\|_\infty \le 1$. Write $X = X_{f,g}$, $\mathcal{W} = \mathcal{W}_{f,g}$ and $J \overset{\text{def}}{=} \lceil TB/L\rceil$.

The block identity. Let $t = t_b$ be a block start with $\Delta_t \in \mathcal{A}$, write $D = \Delta_t$, and let the chains at time t be as in definition SE.2, with $S = \sum_{r<L} V_r$ and $\mathcal{E} = \sum_{r<L}\mathsf{S}_{t+r} - S$. The block form of eq. (SE.26) and the decomposition $\mathbb{E}_t\langle S, \mathcal{W}S\rangle = L^2\langle\overline{V}, \mathcal{W}\overline{V}\rangle + \mathbb{E}_t\langle S - L\overline{V}, \mathcal{W}(S - L\overline{V})\rangle$, with $\overline{V} = \overline{V}(D)$, give the exact identity

$$\begin{aligned}\mathbb{E}_t\left[X_{f,g}(\Delta_{t+L}) - X_{f,g}(D)\right] = &-2\eta\Big\langle\mathcal{W}D, \sum_{r<L}\mathbb{E}_t\,\mathsf{S}_{t+r}\Big\rangle + \eta^2L^2\langle\overline{V}, \mathcal{W}\overline{V}\rangle\\ &+\eta^2LB\int f\otimes g\,\mathrm{d}\mathfrak{F}_L(D) + \eta^2\,\mathbb{E}_t\left[2\langle S, \mathcal{W}\mathcal{E}\rangle + \langle\mathcal{E}, \mathcal{W}\mathcal{E}\rangle\right],\end{aligned} \tag{SE.28}$$

where $\mathfrak{F}_L(D)$ is the finite-time measure of theorem SD.8(iv) at the state $\widehat{A}(D)$ and $T = L$. Indeed, in the eigenbases, $\mathbb{E}_t\langle S - L\overline{V}, \mathcal{W}(S - L\overline{V})\rangle = \sum_{i,j} f(\mu_i)g(\lambda_j)\,\mathbb{E}_t\left((S - L\overline{V})_{ij}\right)^2$, which by lemma SE.2(i) is LB times the integral of $f\otimes g$ against $\mathfrak{F}_L(D)$.

The diffuse cutoff. By lemma SE.24 with $T+1$ in place of T, the maximum X_B of $\delta(\Delta_t)$ over $t \le t_{b_c} \wedge (T+1)B$ tends to zero in probability. Choose a deterministic $\delta_B \downarrow 0$ with $\Pr(\mathsf{X}_B > \delta_B) \to 0$, which is possible because $\Pr(\mathsf{X}_B > \epsilon) \to 0$ for every $\epsilon > 0$. Let $b_d \stackrel{\text{def}}{=} \inf\{b : \delta(\Delta_{t_b}) > \delta_B\}$, $b_\star \stackrel{\text{def}}{=} b_c \wedge b_d$ and $I_b \stackrel{\text{def}}{=} \mathbf{1}_{b<b_\star}$. With probability tending to one $b_\star \wedge (J+1) = b_c \wedge (J+1)$, so it suffices to prove the claim with $\tau_\star \stackrel{\text{def}}{=} t_{b_\star}/B$ in place of τ_c. Since $\Delta_0 \in \mathcal{A}$, and $\delta(\Delta_0) \le \delta_B$ for large B because $\delta(\Delta_0) \le \mathsf{X}_B$ is deterministic, we have $b_\star \ge 1$.

The drift of a block. For $1 \le b < J$ with $I_b = 1$, put $t = t_b$, $D = \Delta_t$ and

$$\Lambda_b \stackrel{\text{def}}{=} \mathbb{E}_t\big[X(\Delta_{t+L}) - X(D)\big] - \frac{L}{B}\int f \otimes g \,\mathrm{d}\mathsf{F}_B\big(\chi(D)\big).$$

Since $(L/B)\mathsf{F}_B(\chi(D)) = -2\eta L \mathscr{R}(D)^{-1/2}\mathfrak{d}\chi(D) + \eta^2 LB\zeta$, eq. (SE.28) writes Λ_b as the sum of six terms:

$$\begin{aligned}\Lambda_b = \; & -2\eta\sum_{r<\mathsf{q}}\big\langle \mathcal{W}D, \mathbb{E}_t\,\mathsf{S}_{t+r} - \overline{V}\big\rangle - 2\eta\sum_{r=\mathsf{q}}^{L-1}\big\langle \mathcal{W}D, \mathbb{E}_t\,\mathsf{S}_{t+r} - \overline{V}\big\rangle \\ & + \eta^2\,\mathbb{E}_t\big[2\langle S, \mathcal{W}\mathcal{E}\rangle + \langle \mathcal{E}, \mathcal{W}\mathcal{E}\rangle\big] + \eta^2 L^2\langle \overline{V}, \mathcal{W}\overline{V}\rangle \\ & - 2\eta L\Big(\langle \mathcal{W}D, \overline{V}\rangle - \frac{1}{\sqrt{\mathscr{R}(D)}}\int f\otimes g\,\mathfrak{d}\,\mathrm{d}\chi(D)\Big) + \eta^2 LB\int f\otimes g\,\mathrm{d}\big(\mathfrak{F}_L(D) - \zeta\big).\end{aligned}$$

We bound $\mathbb{E}[I_b|\cdot|]$ of each term, using $t \ge L \ge \mathsf{q}$, $I_b \le \mathbf{1}_{\mathfrak{E}_t}$ for the event of proposition SE.6, and $\|\mathcal{W}D\|_{\mathrm{F}}, \|\mathcal{W}D\| \le \|D\|_{\mathrm{F}} \le C$.

(a) The prefix term is at most $2\eta\,\mathbb{E}[\mathbf{1}_{t<\tau^+}|\sum_{r<\mathsf{q}}\langle \mathcal{W}D, \mathsf{S}_{t+r}\rangle|] + 2\eta\mathsf{q}\|D\|_{\mathrm{F}}\|\overline{V}\|_{\mathrm{F}} \le C\mathsf{q}^2/B$, by lemma SE.21(ii) and eq. (SE.15).

(b) The post-prefix term is at most $2\eta L\rho_B$, by proposition SE.6 with $\Pi_t = \mathcal{W}\Delta_t$.

(c) The quadratic response is at most $\eta^2 LB\,C\,\mathbb{E}[I_b\,\mathsf{r}_B(a_b)]$, by lemma SE.22, where $\mathsf{r}_B(a)$ is its right side and $a_b = 1 + \|M^\circ_{t-1}\|_{\mathrm{F}}/\sqrt{B}$. By lemma SE.15, $\mathbb{E}[I_b(a_b + a_b^2)] \le C$, so $\mathbb{E}[I_b\mathsf{r}_B(a_b)] \le C\bar{\mathsf{r}}_B$ with $\bar{\mathsf{r}}_B \stackrel{\text{def}}{=} L^{-1} + B^{-1/2} + L^2/B + L^{3/2}/\sqrt{B} + L/\sqrt{B} + L^3/B \to 0$.

(d) The coherent mean term is at most $\eta^2 L^2 C = (L/B)\cdot Ch^2 L/B$.

(e) The drift replacement is at most $2\eta L\|D\|_{\mathrm{F}}\,\bar{e}_B^{(1)}$. Indeed, by lemma SE.2(i) and corollary SC.1 at the state $\widehat{A}(D)$, $u_i^\top \overline{V} v_j = \mathfrak{d}_{ij}(u_i^\top D v_j)/\sqrt{\mathscr{R}(D)} + (E_\varrho)_{ij}$, so $\langle \mathcal{W}D, \overline{V}\rangle - \mathscr{R}(D)^{-1/2}\int f\otimes g\,\mathfrak{d}\,\mathrm{d}\chi(D) = \sum_{i,j} f(\mu_i)g(\lambda_j)(u_i^\top D v_j)(E_\varrho)_{ij}$, which eq. (SC.13) bounds by $\|D\|_{\mathrm{F}} e_B^{(1)}(\widehat{A}(D))$, and $\delta(D) \le \delta_B$.

(f) The forcing replacement is at most $\eta^2 LB\big(\|\mathfrak{F}_L(D) - \mathfrak{C}_{\mathrm{int}}(\varrho)\|_{\mathrm{TV}} + \bar{e}_B^{(2)}\big)$, and the first norm is at most $2\mathsf{K}_\star(\beta^L/(1-\beta) + \beta/(L(1-\beta)^2))$ by theorem SD.8(iv).

With $\eta L = hL/B$ and $\eta^2 LB = h^2 L/B$, every bound is L/B times a quantity that tends to zero uniformly in b, except (a), which is (L/B) times $C\mathsf{q}^2/L$. Summing over the at most TB/L blocks,

$$\mathbb{E}\sum_{1\le b<J} I_b|\Lambda_b| \le C(T+1)\Big[\frac{\mathsf{q}^2}{L} + \rho_B + \bar{\mathsf{r}}_B + \frac{L}{B} + \bar{e}_B^{(1)} + \bar{e}_B^{(2)} + \beta^L + \frac{1}{L}\Big] \longrightarrow 0.$$

Decomposition of the path. For $1 \le j \le J$,

$$X(\Delta_{t_{j\wedge b_\star}}) - X(\Delta_{t_1}) = \sum_{1\le b<j} I_b\xi_b + \sum_{1\le b<j} I_b\Big[\frac{L}{B}\int f\otimes g\,\mathrm{d}\mathsf{F}_B\big(\chi(\Delta_{t_b})\big) + \Lambda_b\Big],$$

with ξ_b as in lemma SE.20, whose bound tends to zero. The Riemann sum is compared with the integral. For $t_b \le t' < t_{b+1}$ with $b < b_\star$, the Cauchy–Schwarz inequality over the pairs (i,j) and lemma SE.14(i) give

$$\|\chi(\Delta_{t'}) - \chi(\Delta_{t_b})\|_{\mathrm{TV}} = \sum_{i,j}|\mathscr{Q}_{ij}(\Delta_{t'}) - \mathscr{Q}_{ij}(\Delta_{t_b})| \le \|\Delta_{t'} - \Delta_{t_b}\|_{\mathrm{F}}\big(\|\Delta_{t'}\|_{\mathrm{F}} + \|\Delta_{t_b}\|_{\mathrm{F}}\big) \le \frac{CL}{\sqrt{B}}.$$

By lemma SE.1(ii) on the set of measures with the bounds of $\mathcal{A}^+$, the integral of $\int f\otimes g\,\mathrm{d}\mathsf{F}_B(\chi^{\mathrm{act}}(\sigma))$ over $[t_b/B, t_{b+1}/B)$ differs from the Riemann term by at most $(L/B)CL/\sqrt{B}$, and the sum over the blocks by at most $C(T+1)L/\sqrt{B}$.

The first block and intermediate times. No stop occurs in the first block, and by eq. (SE.26), $|X(\Delta_{t_1}) - X(\Delta_0)| \le (2\|\Delta_0\|_{\mathrm{F}} + \|\Delta_L - \Delta_0\|_{\mathrm{F}})\|\Delta_L - \Delta_0\|_{\mathrm{F}} \le C(L/\sqrt{B} + L^2/B)$, while the integral over $[0, L/B]$ is at most CL/B by lemma SE.1(i). For a time τ between two block starts, the same bounds control the change of both sides over at most L steps. Collecting the bounds proves the claim. □

SE.9 A positive interval

Lemma SE.27 (No exit before $T_\star$)**.** *Let b_c be as in lemma SE.24. Then* $\Pr(t_{b_c} \le BT_\star + L) \to 0$. *Moreover, with probability tending to one, for all $t \le BT_\star$,*

$$|\|\Delta_t\|_{\mathrm{F}}^2 - \|\Delta_0\|_{\mathrm{F}}^2| \le \frac{1}{4}, \qquad |\mathscr{R}(t) - \mathscr{R}(0)| \le \frac{\min(r_0, 1)}{4};$$

in particular $t_{\mathrm{stop}} > BT_\star$ and $\chi^{\mathrm{act}}(\tau) \in \mathfrak{M}$ for $\tau \le T_\star$.

Proof. Apply lemma SE.26 with $T = T_\star + 1$ to the two tests $f = g = 1$ and $f(\mu) = \mu/2$, $g(\lambda) = \lambda$, whose integrals against $\chi(\Delta)$ are $\|\Delta\|_{\mathrm{F}}^2$ and $\mathscr{R}(\Delta)$. The integrands are bounded on the stopped interval. For $\sigma < \tau_c$ the measure $\chi^{\mathrm{act}}(\sigma)$ is within total variation $CL/\sqrt{B}$ of the measure $\chi(\Delta_{t_b})$ of an admissible block start, which lies in $\mathfrak{M}$ because $\mathcal{A}$ has exactly the bounds of $\mathfrak{M}$. By lemma SE.1(i), and (ii) on the enlarged set,

$$\left|\mathsf{F}_B(\chi^{\mathrm{act}}(\sigma))(\mathbb{K})\right| \le C_E + \frac{CL}{\sqrt{B}}, \qquad \frac{1}{2}\left|\int \mu\lambda\,\mathrm{d}\mathsf{F}_B(\chi^{\mathrm{act}}(\sigma))\right| \le C_{\mathscr{R}} + \frac{CL}{\sqrt{B}}.$$

Hence there are random variables $o_B \to 0$ in probability such that for all $\tau \le T_\star + 1$,

$$|\|\Delta_{\lfloor B(\tau\wedge\tau_c)\rfloor}\|_{\mathrm{F}}^2 - \|\Delta_0\|_{\mathrm{F}}^2| \le C_E(\tau\wedge\tau_c) + o_B, \qquad |\mathscr{R}(\lfloor B(\tau\wedge\tau_c)\rfloor) - \mathscr{R}(0)| \le C_{\mathscr{R}}(\tau\wedge\tau_c) + o_B.$$

Suppose that $t_{b_c} \le BT_\star + L$, so that $\tau_c \le T_\star + L/B$. At $\tau = \tau_c$ the state $\Delta_{t_{b_c}}$ is not in $\mathcal{A}$: its energy exceeds $E_\star + 1$, or its risk is below $r_0/2$ or above $r_1 + \frac{1}{2}$. Since $\|\Delta_0\|_{\mathrm{F}}^2 \le E_\star$ and $\mathscr{R}(0) \in [r_0, r_1]$, the energy or the risk then deviates from its initial value by at least 1, $r_0/2$ or $\frac{1}{2}$. But by eq. (SE.10) the deviations are at most $C_E(T_\star + L/B) + o_B \le \frac{1}{8} + o(1) + o_B$ and $C_{\mathscr{R}}(T_\star + L/B) + o_B \le \min(r_0, 1)/8 + o(1) + o_B$. Hence $\Pr(t_{b_c} \le BT_\star + L) \le \Pr(o_B > \min(r_0, 1)/16) \to 0$ for large B.

On the event $\{t_{b_c} > BT_\star + L\} \cap \{o_B \le \min(r_0, 1)/16\}$, whose probability tends to one, the same bounds hold at every $t \le BT_\star$ with the right sides at most $\frac{1}{4}$ and $\min(r_0, 1)/4$. Then $\mathscr{R}(t) \ge \frac{3}{4}r_0 > 0$, so $t < t_{\mathrm{stop}}$, and $\chi^{\mathrm{act}}(\tau) \in \mathfrak{M}$. □

SE.10 The bounded-Lipschitz passage and the proofs

Proof of theorem SE.1. Part (i) is lemma SE.27, since $\|\Delta_0\|_{\mathrm{F}}^2 + \frac{1}{4} \le E_\star + 1$ and $\mathscr{R}(0) - \min(r_0, 1)/4 \ge r_0/2$. Part (iv) is lemma SE.24 with $T = T_\star + 1$ on the event $\{t_{b_c} > BT_\star + L\}$.

For (ii), consider the residual signed measure

$$\mathcal{R}_B(\tau) \stackrel{\mathrm{def}}{=} \chi^{\mathrm{act}}(\tau\wedge\tau_c) - \chi^{\mathrm{act}}(0) - \int_0^{\tau\wedge\tau_c} \mathsf{F}_B\big(\chi^{\mathrm{act}}(\sigma)\big)\,\mathrm{d}\sigma \qquad (\tau \le T_\star).$$

Its total variation is bounded by a deterministic constant: the two measures have masses at most $2E_\star + 4$, and the field has total variation at most C on the stopped interval, by the argument in the proof of lemma SE.27. The residual tends to zero against products of hat functions. Let $\epsilon > 0$, let $0 = x_0 < \cdots < x_n = C_\Sigma$ be a grid of mesh at most ϵ, and let $a_0, \ldots, a_n$ be the piecewise linear hat functions with $a_r(x_s) = \mathbf{1}_{r=s}$, which form a partition of unity. By lemma SE.26 applied to the finitely many tests $a_r \otimes a_s$,

$$\sup_{\tau \le T_\star} \sum_{r,s \le n} \left|\int a_r \otimes a_s\,\mathrm{d}\mathcal{R}_B(\tau)\right| \longrightarrow 0 \quad \text{in probability.}$$

Every test ϕ of the unit ball of the bounded-Lipschitz norm is close to its interpolant $\phi_\epsilon \stackrel{\text{def}}{=} \sum_{r,s} \phi(x_r, x_s)\, a_r \otimes a_s$: at each point ϕ_ϵ is a convex combination of values of ϕ at points within ℓ^1 distance 2ϵ, so $\|\phi - \phi_\epsilon\|_\infty \le 2\epsilon$. Since $|\phi(x_r, x_s)| \le 1$,

$$\|\mathcal{R}_B(\tau)\|_{\mathrm{BL}} \le \sum_{r,s} \Big| \int a_r \otimes a_s \,\mathrm{d}\mathcal{R}_B(\tau) \Big| + 2\epsilon \|\mathcal{R}_B(\tau)\|_{\mathrm{TV}},$$

and since ϵ is arbitrary, $\sup_{\tau \le T_\star} \|\mathcal{R}_B(\tau)\|_{\mathrm{BL}} \to 0$ in probability.

Gronwall's inequality closes the argument. On the event of lemma SE.27, $\tau_c > T_\star$ and $\chi^{\mathrm{act}}(\tau) \in \mathfrak{M}$ for $\tau \le T_\star$, and by proposition SE.1 so is $\chi^{\mathrm{ode}}(\tau)$. Subtracting eq. (SE.9) from the definition of $\mathcal{R}_B$, with $\chi^{\mathrm{ode}}(0) = \chi^{\mathrm{act}}(0)$, and using lemma SE.1(iii),

$$\|\chi^{\mathrm{act}}(\tau) - \chi^{\mathrm{ode}}(\tau)\|_{\mathrm{BL}} \le \sup_{\sigma \le T_\star} \|\mathcal{R}_B(\sigma)\|_{\mathrm{BL}} + C_{\mathsf{F}} \int_0^\tau \|\chi^{\mathrm{act}}(\sigma) - \chi^{\mathrm{ode}}(\sigma)\|_{\mathrm{BL}} \,\mathrm{d}\sigma,$$

and Gronwall's inequality [65, Lemma 2.7] gives $\sup_{\tau \le T_\star} \|\chi^{\mathrm{act}}(\tau) - \chi^{\mathrm{ode}}(\tau)\|_{\mathrm{BL}} \le e^{C_{\mathsf{F}} T_\star} \sup_\sigma \|\mathcal{R}_B(\sigma)\|_{\mathrm{BL}}$. This proves (ii).

For (iii), multiplication by $\mu\lambda$ maps the unit ball of the bounded-Lipschitz norm into $C_\Sigma(C_\Sigma+1)$ times itself, because $\|\mu\lambda\,\phi\|_\infty \le C_\Sigma^2$ and $\mu\lambda\phi$ is $(C_\Sigma^2 + C_\Sigma)$-Lipschitz for $|\cdot|_1$; and $\mathscr{R}(\lfloor B\tau \rfloor) - \mathcal{R}(\tau) = \frac{1}{2}\int \mu\lambda \,\mathrm{d}(\chi^{\mathrm{act}}(\tau) - \chi^{\mathrm{ode}}(\tau))$. □

Proof of corollary SE.1. When $\Sigma_{\mathrm{in}} = \mathrm{Id}$, every λ_j equals 1, so $\mathfrak{d}_{ij} = \mathfrak{d}_i$, and by proposition SD.4 $\mathfrak{v}_{ij} = \mathfrak{v}_i / N_{\mathrm{in}}$ with $\sum_j \mathfrak{v}_{ij} = \mathfrak{v}_i$. Summing eq. (SE.7) over j and dividing by B, the functions $\mathcal{Q}_i \stackrel{\text{def}}{=} \sum_j \mathcal{Q}_{ij}$ satisfy eq. (SE.13), with $\mathcal{R} = \frac{1}{2}\sum_{i,j} \mu_i \mathcal{Q}_{ij} = \frac{1}{2}\sum_i \mu_i \mathcal{Q}_i$ and $\mathcal{Q}_i(0) = \|\Delta_0^\top u_i\|^2$ by lemma SA.1. The row system is locally Lipschitz where its risk is positive, so its solution is unique by the Picard–Lindelöf theorem, as in the proof of proposition SE.1.

The measures $\chi^{\mathrm{act}}(\tau)$ and $\chi^{\mathrm{ode}}(\tau)$ are supported on the segment $\{\lambda = 1\}$ of $\mathbb{K}$, and their images under $(\mu, \lambda) \mapsto \mu$ are $\chi^{\mathrm{act}}_{\mathrm{row}}(\tau)$ and $\chi^{\mathrm{ode}}_{\mathrm{row}}(\tau)$, by $\mathscr{Q}_i = \sum_j \mathscr{Q}_{ij}$. For a signed measure m on $\mathbb{K}$ supported on that segment, with image $\tilde{m}$, $\|m\|_{\mathrm{BL}} = \|\tilde{m}\|_{\mathrm{BL}}$: a test ϕ on $\mathbb{K}$ restricts to the test $\phi(\cdot, 1)$ on $[0, C_\Sigma]$, and a test ψ on $[0, C_\Sigma]$ extends to the test $\phi(\mu, \lambda) \stackrel{\text{def}}{=} \psi(\mu)$ on $\mathbb{K}$, which is 1-Lipschitz for $|\cdot|_1$. Theorem SE.1(ii),(iii) therefore give the claims, since $\mu\lambda = \mu$ on the segment. □

Proof of corollary SE.2. (i), compactness. By eq. (SE.6) the functions $\mathfrak{d}_B$ are uniformly bounded and uniformly Lipschitz on $\mathbb{K}$, so the Arzelà–Ascoli theorem [55, Theorem 7.25] gives a uniformly convergent subsequence. The measures ζ_B are positive with masses at most m_ζ, and $\chi^{\mathrm{act}}_B(0)$ are positive with masses at most $E_\star$. Along a further subsequence the masses converge; if a limit mass is zero the measures tend to zero weakly, and otherwise the normalized measures are probability measures on the compact $\mathbb{K}$, hence tight, and Prokhorov's theorem [10, Theorem 5.1] gives a weakly convergent subsequence.

(ii), the limit data. Fix a subsequence as in (i). The limit $\mathfrak{d}_\infty$ satisfies the bounds eq. (SE.6), which pass to uniform limits; ζ_∞ is positive with mass at most m_ζ; and χ_0 is positive, with $\chi_0(\mathbb{K}) \le E_\star$ and $\mathcal{R}(\chi_0) = \lim \mathscr{R}(\Delta_0) \in [r_0, r_1]$, since 1 and $\mu\lambda$ are continuous on $\mathbb{K}$. The field $\mathsf{F}_\infty(\chi) \stackrel{\text{def}}{=} -2h\mathcal{R}(\chi)^{-1/2}\mathfrak{d}_\infty\chi + h^2\zeta_\infty$ therefore satisfies lemma SE.1 with the same constants, whose proof uses only these bounds.

Weak convergence of positive measures on $\mathbb{K}$ with bounded masses is convergence in the bounded-Lipschitz norm. Indeed, if $c_B \to c$ are the masses and $c > 0$, the normalized measures converge weakly, and Dudley's theorem [22, Theorem 11.3.3] gives convergence of their bounded-Lipschitz distance, which dominates ours up to a factor two; the difference of masses contributes $|c_B - c|$. If $c = 0$ the norm is at most the mass. Hence $\|\zeta_B - \zeta_\infty\|_{\mathrm{BL}} \to 0$ and $\|\chi^{\mathrm{act}}_B(0) - \chi_0\|_{\mathrm{BL}} \to 0$.

(ii), the Cauchy bound. For two dimensions B, B' of the subsequence write $\chi_B = \chi^{\mathrm{ode}}_B$, which lies in $\mathfrak{M}$ on $[0, T_\star]$ by proposition SE.1. Adding and subtracting $\mathsf{F}_B(\chi_{B'})$,

$$\|\mathsf{F}_B(\chi_{B'}) - \mathsf{F}_{B'}(\chi_{B'})\|_{\mathrm{BL}} \le \frac{2h(E_\star + 1)}{\sqrt{r_0/2}} \|\mathfrak{d}_B - \mathfrak{d}_{B'}\|_\infty + h^2 \|\zeta_B - \zeta_{B'}\|_{\mathrm{BL}},$$

and lemma SE.1(iii) with Gronwall's inequality give

$$\sup_{\tau\le T_\star}\|\chi_B(\tau)-\chi_{B'}(\tau)\|_{\mathrm{BL}}\le e^{C_{\mathsf{F}}T_\star}\Big(\|\chi_B(0)-\chi_{B'}(0)\|_{\mathrm{BL}}+CT_\star\|\mathfrak{d}_B-\mathfrak{d}_{B'}\|_\infty + h^2T_\star\|\zeta_B-\zeta_{B'}\|_{\mathrm{BL}}\Big),$$

which tends to zero as $B, B' \to \infty$ along the subsequence.

(ii), the limit path. For each τ, the measures $\chi_B(\tau)$ lie in $\mathfrak{M}$, so every subsequence has a weakly convergent subsequence, by Prokhorov's theorem as in (i), and its limit is a limit in the bounded-Lipschitz norm. By the Cauchy bound all these limits coincide; call the common limit $\chi(\tau)$. The convergence is uniform in $\tau \le T_\star$, by the Cauchy bound, and $\chi(\tau) \in \mathfrak{M}$, because mass and risk are integrals of continuous functions. Each χ_B is Lipschitz in total variation with constant C_E, by lemma SE.1(i), so χ is continuous for the bounded-Lipschitz norm.

(ii), the equation. For a test ϕ of the unit ball and $\sigma \le T_\star$,

$$\Big|\int\phi\,\mathrm{d}\big(\mathsf{F}_B(\chi_B(\sigma))-\mathsf{F}_\infty(\chi(\sigma))\big)\Big|\le C_{\mathsf{F}}\|\chi_B(\sigma)-\chi(\sigma)\|_{\mathrm{BL}} + \frac{2h(E_\star+1)}{\sqrt{r_0/2}}\|\mathfrak{d}_B-\mathfrak{d}_\infty\|_\infty+h^2\|\zeta_B-\zeta_\infty\|_{\mathrm{BL}},$$

which tends to zero uniformly in σ. Passing to the limit in eq. (SE.9) tested with ϕ gives eq. (SE.14) tested with ϕ. Bounded Lipschitz functions are dense in the continuous functions on $\mathbb{K}$ for the uniform norm, so eq. (SE.14) holds against every continuous test.

(ii), uniqueness and the actual measures. If χ' is another solution with values in $\mathfrak{M}$, continuous for the bounded-Lipschitz norm, then $\|\chi(\tau)-\chi'(\tau)\|_{\mathrm{BL}} \le C_{\mathsf{F}}\int_0^\tau\|\chi(\sigma)-\chi'(\sigma)\|_{\mathrm{BL}}\mathrm{d}\sigma$, and Gronwall's inequality gives $\chi=\chi'$. The statement for χ_B^{act} follows from theorem SE.1(ii) and the triangle inequality.

(iii). As in the proof of theorem SE.1(iii), $|\mathscr{R}(\lfloor B\tau\rfloor)-\mathcal{R}(\chi(\tau))| \le \frac{1}{2}(C_\Sigma^2+C_\Sigma)\|\chi_B^{\mathrm{act}}(\tau)-\chi(\tau)\|_{\mathrm{BL}}$. □

SF The isotropic case

This section specializes the theorems of appendices SA to SE to isotropic data, $\Sigma_{\mathrm{in}} = \mathrm{Id}_{N_{\mathrm{in}}}$ and $\Sigma_{\mathrm{out}} = \mathrm{Id}_{N_{\mathrm{out}}}$, which is the setting of section 3. Both spectral measures are then point masses, the two trace equations of the fixed-point system become algebraic, the drift kernel $\mathfrak{d}_{ij}$ and the volatility coefficient $\mathfrak{v}_{ij}$ each take one value, and the deterministic mode-energy equations close on the risk alone. The closed equation is the main text's scalar recursion eq. (8).

SF.1 Main results in brief

The main text uses the following results of this section. Write $\gamma = N/B$ in the square case $N_{\mathrm{in}} = N_{\mathrm{out}} = N$, and $B_{\mathrm{eff}} = B(1+\beta)/(1-\beta)$.

(1) *The risk follows a scalar equation* (proposition SF.4 and corollary SF.1). With isotropic data every mode has the same drift and volatility, so the mode-energy equations close on the risk:

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\tau} = -2h\,\mathfrak{d}_{\mathrm{iso}}\sqrt{\mathcal{R}}+\frac{h^2}{2B}\,\mathfrak{v}_{\mathrm{iso}},$$

and the risk of the actual algorithm tracks its solution on a positive time interval. This is the main text's recursion eq. (8).

(2) *The drift constant* (eq. (SF.21) and proposition SF.5). At zero ridge the drift is a single number $C_\beta(\gamma)$, given by an explicit integral over the fixed-point root.

(3) *Its two batch limits* (theorems SF.4, SF.7 and SF.11 and proposition SF.12). At large batch, $C_\beta \approx \frac{8}{3\pi}\sqrt{B_{\mathrm{eff}}/(2N)}$: momentum acts as the effective batch B_{eff}. At small batch with momentum, $C_\beta \to 1/(2\sqrt{\pi})$, a constant, whereas without momentum $C_0 \approx B/(\sqrt{\pi}N) \to 0$; the limit $\beta \downarrow 0$ is therefore discontinuous at zero ridge.

(4) *The volatility constant* (definition SF.1, lemma SF.10, and theorems SF.8 and SF.11). Without momentum the zero-ridge volatility is $\min\{B, N\}$, the rank of the gradient. With momentum it is NV_β at large batch, with an explicit factor $1 < V_\beta < (1+\beta)/(1-\beta)$, and $N^2/(2B)$ at small batch.

(5) *Momentum as a rescaled step* (remark SF.1). At large batch, momentum at step h is no momentum at step $h\sqrt{B_{\text{eff}}/B}$, with the volatility reduced by the factor $c_\beta = s_2 V_\beta < 1$ ($0.826\ldots$ at $\beta = \frac{1}{2}$, tending to $0.7776\ldots$ as $\beta \uparrow 1$).

(6) *A small ridge at small batch* (theorems SF.13 and SF.14). With momentum and $B_{\text{eff}} < N$, a ridge ϱ limits the number of past batches the update whitens to about $\log(1/\varrho)/(2\log(1/\beta))$. The drift and the volatility then follow explicit crossover laws between the fixed-ridge and the zero-ridge regimes. This is the mechanism by which a few Newton–Schulz steps can beat exact whitening (Appendix SL).

The remaining statements support these. Appendices SF.3 to SF.6 specialize the general theorems of appendices SA to SE; nothing there is reproved, and each proof displays the specialization. The later subsections carry their own proofs: the participation representation of the drift (appendix SF.8), the exact constant for $N \geq B$ and at aspect ratio one (proposition SF.9 and theorem SF.6), the spectral form at $\beta = 0$ (proposition SF.7), the lag profile of the volatility (theorems SF.9, SF.10 and SF.12), and the order of the limits in ϱ, β and γ (proposition SF.11).

SF.2 Standing setup

Throughout the section the features are Gaussian as in Assumption S1, with $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$ and $\Sigma_{\text{out}} = \mathrm{Id}_{N_{\text{out}}}$. Every eigenvalue μ_i and λ_j equals 1, and any orthonormal bases (u_i) of $\mathbb{R}^{N_{\text{out}}}$ and (v_j) of $\mathbb{R}^{N_{\text{in}}}$ are eigenbases. At each dimension we write

$$\gamma_{\text{in}} \overset{\text{def}}{=} \frac{N_{\text{in}}}{B}, \qquad \gamma_{\text{out}} \overset{\text{def}}{=} \frac{N_{\text{out}}}{B}$$

for the two aspect ratios; along a sequence of dimensions they converge, by Assumption S2, and the limits are the aspect ratios of that assumption. The spectral measures eq. (SA.41) are the point masses

$$\nu_{\text{out}} = \gamma_{\text{out}}\,\delta_1, \qquad \nu_{\text{in}} = \gamma_{\text{in}}\,\delta_1. \tag{SF.1}$$

The pairing of eq. (SA.2) is the Frobenius pairing. With identity covariances,

$$\langle V, W\rangle_\Sigma = \mathrm{Tr}\left(VW^\top\right), \qquad \mathscr{R}(t) = \tfrac{1}{2}\|\Delta_t\|_{\text{F}}^2 = \tfrac{1}{2}\sum_{i,j}\mathscr{Q}_{ij}(t), \qquad \widehat{A}_t = \frac{\Delta_t}{\|\Delta_t\|_{\text{F}}}, \tag{SF.2}$$

by eqs. (SA.4), (SA.9) and (SA.10). In particular the diffuseness $\|\widehat{A}_t\|$ of Assumption S3 is the ratio $\|\Delta_t\|/\|\Delta_t\|_{\text{F}}$ of the operator and Frobenius norms of the error. The covariance bound of that assumption holds with $C_\Sigma = 1$.

The momentum $\beta \in [0, 1)$, the normalized ridge $\varrho > 0$ and the EMA weights are those of definition SA.2. For $s > 0$, $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ denotes the solution of the weighted system eq. (SB.2) for the EMA weights and the measures eq. (SF.1), given by theorem SB.1, and

$$\psi(s) \overset{\text{def}}{=} q(-s)\,F_\beta\big(q(-s)\big) \qquad (s > 0) \tag{SF.3}$$

is the common value of the coupling eq. (SB.4). At $\beta = 0$ we write, as in definition SI.2,

$$\Phi(v) \overset{\text{def}}{=} vJ(v) = \mathbb{E}\,\frac{vg^2}{1+vg^2} \qquad (v \geq 0,\ g \sim \mathcal{N}(0,1)), \tag{SF.4}$$

so that $\psi(s) = \Phi(q(-s))$ at $\beta = 0$.

SF.3 The isotropic closure

Proposition SF.1 (Isotropic closure)**.** *Let $\beta \in [0, 1)$, $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$ and $s > 0$, and let $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q) = (\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ be the solution of eq. (SB.2) for the measures eq. (SF.1).*

(i) The branch satisfies

$$\sigma = \frac{\gamma_{\text{out}}}{s+\tilde{s}_1}, \qquad \tilde{\sigma} = \frac{\gamma_{\text{in}}}{s+s_1}, \qquad \tilde{s}_1\sigma = s_1\tilde{\sigma} = \psi(s), \tag{SF.5}$$

and the two anchors and their consequences

$$\psi(s) = \gamma_{\text{out}} - s\sigma = \gamma_{\text{in}} - s\tilde{\sigma}, \qquad \tilde{\sigma} - \sigma = \frac{\gamma_{\text{in}} - \gamma_{\text{out}}}{s}, \qquad (s+\tilde{s}_1)(s+s_1) = \frac{s\,\gamma_{\text{in}}\gamma_{\text{out}}}{q}. \tag{SF.6}$$

(ii) The number $q = q(-s)$ is the unique $q > 0$ with $qF_\beta(q) < \min(\gamma_{\text{in}}, \gamma_{\text{out}})$ and

$$s\,q = \big(\gamma_{\text{out}} - qF_\beta(q)\big)\big(\gamma_{\text{in}} - qF_\beta(q)\big). \tag{SF.7}$$

(iii) At $\beta = 0$, with $\xi \overset{\text{def}}{=} (2q)^{-1/2} = (2s\sigma\tilde{\sigma})^{-1/2}$,

$$\psi(s) = \Phi(q) = f(\xi) = 1 - \sqrt{\pi}\,\xi\, e^{\xi^2}\,\operatorname{erfc}(\xi). \tag{SF.8}$$

Proof. Part (i) is the system eq. (SB.2) at point masses. For $\nu = \gamma_{\text{out}}\delta_1$ the first trace equation is

$$\sigma = \int \frac{x}{\tilde{s}_1 x + s}\,\gamma_{\text{out}}\delta_1(dx) = \frac{\gamma_{\text{out}}}{\tilde{s}_1 + s},$$

and the second is the same computation for $\gamma_{\text{in}}\delta_1$. The third identity of eq. (SF.5) is the coupling eq. (SB.4). Multiplying the first identity by $s + \tilde{s}_1$ gives

$$s\sigma + \tilde{s}_1\sigma = \gamma_{\text{out}},$$

which is the first anchor, and the second anchor is the same computation on the input side. Subtracting the anchors gives the second identity of eq. (SF.6). The product of the first two identities of eq. (SF.5) and $q = s\sigma\tilde{\sigma}$ give

$$(s+\tilde{s}_1)(s+s_1) = \frac{\gamma_{\text{in}}\gamma_{\text{out}}}{\sigma\tilde{\sigma}} = \frac{s\,\gamma_{\text{in}}\gamma_{\text{out}}}{q}.$$

Part (ii) eliminates σ and $\tilde{\sigma}$. By the anchors,

$$\sigma = \frac{\gamma_{\text{out}} - \psi(s)}{s}, \qquad \tilde{\sigma} = \frac{\gamma_{\text{in}} - \psi(s)}{s},$$

and inserting both into $q = s\sigma\tilde{\sigma}$ gives eq. (SF.7). Since $\tilde{s}_1 \geq 0$, the first identity of eq. (SF.5) gives $\sigma > 0$, so $\psi(s) < \gamma_{\text{out}}$; likewise $\psi(s) < \gamma_{\text{in}}$, and $q > 0$.

Conversely, let $q' > 0$ satisfy $q'F_\beta(q') < \min(\gamma_{\text{in}}, \gamma_{\text{out}})$ and eq. (SF.7), and put $\psi' \overset{\text{def}}{=} q'F_\beta(q')$ and

$$\sigma' \overset{\text{def}}{=} \frac{\gamma_{\text{out}} - \psi'}{s}, \qquad \tilde{\sigma}' \overset{\text{def}}{=} \frac{\gamma_{\text{in}} - \psi'}{s}, \qquad a' \overset{\text{def}}{=} \frac{\psi'}{\sigma'}, \qquad c' \overset{\text{def}}{=} \frac{\psi'}{\tilde{\sigma}'}.$$

These four numbers are nonnegative. The tuple solves the system eq. (SB.2) at $z = -s$. Indeed the first trace equation holds because

$$s + a' = s + \frac{s\,\psi'}{\gamma_{\text{out}} - \psi'} = \frac{s\,\gamma_{\text{out}}}{\gamma_{\text{out}} - \psi'} = \frac{\gamma_{\text{out}}}{\sigma'},$$

the second holds by the same computation, the equation $q' = s\sigma'\tilde{\sigma}'$ is eq. (SF.7), and

$$s\,\tilde{\sigma}'\,F_\beta(q') = \frac{s\,\tilde{\sigma}'\,\psi'}{q'} = \frac{\psi'}{\sigma'} = a',$$

with the equation for c' symmetric. The uniqueness in theorem SB.1(i) gives $q' = q$.

Part (iii) is proposition SB.1: at $\beta = 0$ the only weight is $w_0 = 1$, so $F_\beta = J$ and $\psi(s) = qJ(q) = \Phi(q)$. □

The no-momentum form of eq. (SF.5) at $z = -s$ is the system

$$\begin{aligned} \sigma(z) = \frac{\gamma_{\text{out}}}{\tilde{s}_1(z) - z}, \qquad \tilde{\sigma}(z) = \frac{\gamma_{\text{in}}}{s_1(z) - z}, \\ \tilde{s}_1(z)\sigma(z) = s_1(z)\tilde{\sigma}(z) = f(\xi(z)), \qquad \xi(z) = \frac{1}{\sqrt{-2z\sigma(z)\tilde{\sigma}(z)}}, \end{aligned} \tag{SF.9}$$

which is the specialization of the system of fig. 2. The companion denominator of $\tilde{\sigma}$ is $s_1 - z$.

SF.4 The drift

Theorem SF.1 (Isotropic drift)**.** *Work under Assumption S3 with* $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$ *and* $\Sigma_{\text{out}} = \mathrm{Id}_{N_{\text{out}}}$*, let* $\widetilde{G}_{t+1} = Q_\varrho(M_t)$ *be the frozen-state update, and let* $\varepsilon_B \to 0$ *be the sequence of theorem SC.2.*

(i) Every kernel $\mathfrak{d}_{ij}$ *of corollary SC.1 equals* $\mathfrak{d}_{\text{iso}} \overset{\text{def}}{=} \mathfrak{d}(1,1)$*, and, with* $z = \varrho + u^2$*,*

$$\mathfrak{d}_{\text{iso}} = \frac{\sqrt{2}}{\pi\,\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty q(-z)\, L_\beta\big(q(-z)\big)\, du, \qquad 0 \le \mathfrak{d}_{\text{iso}} \le \frac{1}{\sqrt{2\varrho}}. \tag{SF.10}$$

(ii) At $\beta = 0$*, with* $z = \varrho + u^2$*,*

$$\mathfrak{d}_{\text{iso}} = \frac{\sqrt{2}}{\pi\,\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty \psi(z)\, du = \frac{\sqrt{2}}{\pi\,\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty \big(\gamma_{\text{out}} - z\,\sigma(-z)\big)\, du. \tag{SF.11}$$

(iii) The candidate eq. (SC.3) is a multiple of the error, and the expected update is

$$\overline{\widetilde{G}}_\varrho = \frac{\mathfrak{d}_{\text{iso}}}{\sqrt{\mathscr{R}(t)}}\,\Delta_t, \qquad \mathbb{E}\,\widetilde{G}_{t+1} = \frac{\mathfrak{d}_{\text{iso}}}{\sqrt{\mathscr{R}(t)}}\,\Delta_t + E_\varrho, \qquad \sup_{\varrho\in[\varrho_-,\varrho_+]} \|E_\varrho\|_{\text{F}} = \varepsilon_B. \tag{SF.12}$$

(iv) The total drift is

$$\sum_{i,j} \mathscr{D}_{ij}(t) = \big\langle \Delta_t, \mathbb{E}\,\widetilde{G}_{t+1} \big\rangle_{\text{F}} = 2\,\mathfrak{d}_{\text{iso}}\sqrt{\mathscr{R}(t)} + \langle \Delta_t, E_\varrho\rangle_{\text{F}},$$
$$|\langle \Delta_t, E_\varrho\rangle_{\text{F}}| \le \sqrt{2\mathscr{R}(t)}\,\varepsilon_B. \tag{SF.13}$$

Proof. The theorem is corollary SC.1 and theorem SC.2 at $\mu_i = \lambda_j = 1$. By corollary SC.1(i), $\mathfrak{d}_{ij} = \mathfrak{d}(\mu_i, \lambda_j) = \mathfrak{d}(1,1)$ for every pair. At $\mu = \lambda = 1$ the denominator of the kernel eq. (SC.4) is the product computed in eq. (SF.6), so that

$$\frac{z\,L_\beta(q)}{(z+\tilde{s}_1)(z+s_1)} = \frac{z\,L_\beta(q)\,q}{z\,\gamma_{\text{in}}\gamma_{\text{out}}} = \frac{q\,L_\beta(q)}{\gamma_{\text{in}}\gamma_{\text{out}}},$$

with the branch evaluated at $-z$. Inserting this in eq. (SC.4) gives the integral of eq. (SF.10). The bound is eq. (SC.14) at $\mu = \lambda = 1$.

Part (ii) is part (i) with $L_\beta = J$. Then $qJ(q) = \psi(z)$, and $\psi(z) = \gamma_{\text{out}} - z\sigma(-z)$ by eq. (SF.6).

Part (iii) evaluates the candidate. With identity covariances, $\overline{R}(-s) = (s+\tilde{s}_1)^{-1}\,\mathrm{Id}_{N_{\text{out}}}$ and $\overline{\widetilde{R}}(-s) = (s+s_1)^{-1}\,\mathrm{Id}_{N_{\text{in}}}$, so the candidate eq. (SC.2) is

$$T(s) = \frac{s\,L_\beta(q)}{(s+\tilde{s}_1)(s+s_1)}\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}} = \frac{q\,L_\beta(q)}{\gamma_{\text{in}}\gamma_{\text{out}}}\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}}.$$

Integrating against $\frac{2}{\pi}\,du$ at $s = \varrho + u^2$ and comparing with eq. (SF.10) gives

$$\overline{\widetilde{G}}_\varrho = \sqrt{2}\,\mathfrak{d}_{\text{iso}}\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}} = \frac{\mathfrak{d}_{\text{iso}}}{\sqrt{\mathscr{R}(t)}}\,\Delta_t.$$

The second and third statements of eq. (SF.12) are the definition $E_\varrho \overset{\text{def}}{=} \mathbb{E}\,\widetilde{G}_{t+1} - \overline{\widetilde{G}}_\varrho$ and eq. (SC.8).

Part (iv) pairs part (iii) with the error. The first identity of eq. (SF.13) is eq. (SA.23) with $\mu_i\lambda_j = 1$. By eq. (SF.12) and eq. (SF.2),

$$\big\langle \Delta_t, \mathbb{E}\,\widetilde{G}_{t+1} \big\rangle_{\text{F}} = \frac{\mathfrak{d}_{\text{iso}}}{\sqrt{\mathscr{R}(t)}}\,\|\Delta_t\|_{\text{F}}^2 + \langle \Delta_t, E_\varrho\rangle_{\text{F}} = 2\,\mathfrak{d}_{\text{iso}}\sqrt{\mathscr{R}(t)} + \langle \Delta_t, E_\varrho\rangle_{\text{F}},$$

and the Cauchy–Schwarz inequality with $\|\Delta_t\|_{\text{F}} = \sqrt{2\mathscr{R}(t)}$ bounds the last term. □

Part (iii) says that in the isotropic case the deterministic equivalent of the expected regularized update is a scalar multiple of the error itself, not only diagonal in some eigenbasis. The projected drift is therefore $\mathscr{D}_{ij}(t) = \mathfrak{d}_{\text{iso}}\mathscr{Q}_{ij}(t)/\sqrt{\mathscr{R}(t)}$ up to the error of eq. (SC.12), for every pair of unit vectors (u_i, v_j).

SF.5 The volatility

Theorem SF.2 (Isotropic volatility)**.** *Work under Assumption S3 with* $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$ *and* $\Sigma_{\text{out}} = \mathrm{Id}_{N_{\text{out}}}$*, let* $I = [\varrho_-, \varrho_+]$*, let* $\mathsf{S}_t(\varrho) = Q_\varrho(M_t)$ *be the frozen update of eq.* (SD.15)*, and let* $e_B \to 0$ *be the sequence of theorem SD.1.*

(i) (Same time.) For every row i *and every column* j*,*

$$\sup_{\varrho\in I}\Big|\,\mathbb{E}\,\|\mathsf{S}_t(\varrho)^\top u_i\|^2 - \frac{\psi(\varrho)}{\gamma_{\text{out}}}\Big| \le \varrho_+ e_B, \qquad \sup_{\varrho\in I}\Big|\,\mathbb{E}\,\|\mathsf{S}_t(\varrho)v_j\|^2 - \frac{\psi(\varrho)}{\gamma_{\text{in}}}\Big| \le \varrho_+ e'_B, \tag{SF.14}$$

where $e'_B \overset{\text{def}}{=} \sup_{s\in I}\|\,\mathbb{E}\,\widetilde{R}(-s) - \overline{\widetilde{R}}(-s)\| \to 0$*, and*

$$\mathbb{E}\sup_{\varrho\in I}\Big|\frac{1}{B}\|\mathsf{S}_t(\varrho)\|_{\text{F}}^2 - \psi(\varrho)\Big|^2 \longrightarrow 0. \tag{SF.15}$$

(ii) (Integrated.) There is a number $\mathfrak{v}_{\text{iso}}(\varrho) \ge 0$*, depending on the dimension,* β *and* ϱ*, such that the mode coefficients of proposition SD.3 and the row coefficients of theorem SD.7 are*

$$\mathfrak{v}_{ij}(\varrho) = \frac{\mathfrak{v}_{\text{iso}}(\varrho)}{N_{\text{out}}N_{\text{in}}}, \qquad \mathfrak{v}_i(\varrho) = \frac{\mathfrak{v}_{\text{iso}}(\varrho)}{N_{\text{out}}}, \qquad \zeta_\varrho = \frac{\mathfrak{v}_{\text{iso}}(\varrho)}{B}\,\delta_{(1,1)}, \tag{SF.16}$$

for all i, j*, and* $\mathfrak{v}_{\text{iso}}(\varrho) = \sum_{i,j}\mathfrak{v}_{ij}(\varrho) = B\,\overline{\Gamma}(\varrho,\varrho)$ *with* $\overline{\Gamma} = \overline{\Gamma}[\mathrm{Id}_{N_{\text{out}}}, \mathrm{Id}_{N_{\text{in}}}]$ *of eq.* (SD.54)*.*

(iii) (Lag expansion.) With the deterministic lag covariances $\overline{C}_\ell = \overline{C}_\ell[\mathrm{Id}_{N_{\text{out}}}, \mathrm{Id}_{N_{\text{in}}}]$ *of theorem SD.4,*

$$\mathfrak{v}_{\text{iso}}(\varrho) = B\,\psi(\varrho) + 2B\sum_{\ell\ge 1}\overline{C}_\ell(\varrho,\varrho), \qquad |\overline{C}_\ell(\varrho,\varrho)| \le \mathsf{K}_\star\beta^\ell, \tag{SF.17}$$

with $\mathsf{K}_\star$ *of eq.* (SD.39) *at* $C_\Sigma = 1$*. At* $\beta = 0$*,*

$$\mathfrak{v}_{\text{iso}}(\varrho) = B\,\psi(\varrho) = B\,\Phi\big(q(-\varrho)\big) = B\big(\gamma_{\text{out}} - \varrho\,\sigma(-\varrho)\big). \tag{SF.18}$$

(iv) (Actual coefficient.) The actual integrated covariance $\Gamma = \Gamma[\mathrm{Id}_{N_{\text{out}}}, \mathrm{Id}_{N_{\text{in}}}]$ *of eq.* (SD.17) *satisfies* $\sup_{\varrho\in I}|\Gamma(\varrho,\varrho) - \mathfrak{v}_{\text{iso}}(\varrho)/B| \to 0$*.*

Proof. Part (i) is theorem SD.1 at $\mu_i = 1$ and $\lambda_j = 1$. By eq. (SF.5), the same-time coefficient eq. (SD.22) is the same for every row,

$$\mathfrak{v}_i^{(0)}(\varrho) = \frac{\tilde{s}_1(-\varrho)}{\varrho + \tilde{s}_1(-\varrho)} = \frac{\tilde{s}_1\sigma}{\gamma_{\text{out}}} = \frac{\psi(\varrho)}{\gamma_{\text{out}}},$$

and eq. (SD.24) is the first bound of eq. (SF.14). The column bound of eq. (SD.24) (theorem SD.1(ii)), with the column coefficient

$$\frac{s_1(-\varrho)}{\varrho + s_1(-\varrho)} = \frac{s_1\tilde{\sigma}}{\gamma_{\text{in}}} = \frac{\psi(\varrho)}{\gamma_{\text{in}}},$$

gives the second bound. The candidate total of eq. (SD.22) is

$$\overline{Z}(\varrho) = \frac{1}{B}\,\mathrm{Tr}\,\mathrm{Id}_{N_{\text{out}}} - \varrho\,\sigma(-\varrho) = \gamma_{\text{out}} - \varrho\,\sigma(-\varrho) = \psi(\varrho),$$

by eq. (SF.6), and $\|\cdot\|_\Sigma = \|\cdot\|_{\text{F}}$, so eq. (SD.25) is eq. (SF.15).

Part (ii) is propositions SD.3 and SD.4 with one eigenvalue on each side. The only atom of the measures of eq. (SD.78) is $(1,1)$, with multiplicities $r_1 = N_{\text{out}}$ and $t_1 = N_{\text{in}}$. Put $\mathfrak{v}_{\text{iso}}(\varrho) \overset{\text{def}}{=} B\,\zeta_\varrho(\{(1,1)\})$. Then eq. (SD.80) gives the first identity of eq. (SF.16), and proposition SD.4 gives

$$\mathfrak{v}_i(\varrho) = N_{\text{in}}\,\mathfrak{v}_{ij}(\varrho) = \frac{\mathfrak{v}_{\text{iso}}(\varrho)}{N_{\text{out}}}.$$

The atom of ζ_ϱ at $(1,1)$ is, by eq. (SD.78) with $P_1 = \mathrm{Id}_{N_{\mathrm{out}}}$ and $P_1' = \mathrm{Id}_{N_{\mathrm{in}}}$, equal to $\overline{\Gamma}(\varrho,\varrho)$, and summing the first identity of eq. (SF.16) over the $N_{\mathrm{out}}N_{\mathrm{in}}$ pairs gives $\sum_{i,j}\mathfrak{v}_{ij} = \mathfrak{v}_{\mathrm{iso}}$. Nonnegativity is theorem SD.8(iii).

Part (iii) expands $\overline{\Gamma}$. By eq. (SD.54) at equal ridges,

$$\overline{\Gamma}(\varrho,\varrho) = \overline{C}_0(\varrho,\varrho) + 2\sum_{\ell\geq 1}\overline{C}_\ell(\varrho,\varrho),$$

and $\overline{C}_0(\varrho,\varrho) = \overline{Z}(\varrho) = \psi(\varrho)$ by theorem SD.4(v) and part (i). The bound on the lag terms is theorem SD.4(ii) with $\|F\| = \|G\| = 1$. At $\beta = 0$ the lag terms vanish by the same statement, and the three forms of eq. (SF.18) are eqs. (SF.4) and (SF.6).

Part (iv) is theorem SD.4(v). □

The main text writes $\mathfrak{v}_{ij} = \mathfrak{v}/N^2$ in the square case $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ (section 3). By eq. (SF.16) this is the convention $\mathfrak{v} = \mathfrak{v}_{\mathrm{iso}}$ in general, with N^2 replaced by $N_{\mathrm{out}}N_{\mathrm{in}}$. The coefficient $\mathfrak{v}_{ij}$ of the volatility section is a per-step coefficient: it replaces the conditional second moment $\mathcal{V}_{ij}(t)$ of eq. (SA.20), in which the step enters as η^2. Summed over the pairs, the quadratic term of the total recursion eq. (SA.22) is therefore $\frac{\eta^2}{2}\sum_{i,j}\mathfrak{v}_{ij} = \frac{\eta^2}{2}\mathfrak{v}_{\mathrm{iso}}$, which is the volatility term $\frac{\eta^2}{2}\mathfrak{v}$ of eq. (8). For $\beta > 0$ the integrated coefficient $\mathfrak{v}_{\mathrm{iso}}$ differs from the same-time value $B\psi(\varrho)$ by the lag terms of eq. (SF.17). Each lag term is given in closed form by corollary SD.3 at the single eigenvalue pair $(1,1)$,

$$\overline{C}_\ell(\varrho,\varrho) = \frac{N_{\mathrm{out}}N_{\mathrm{in}}}{B^2}\,\mathfrak{c}_\ell(1,1;\varrho),$$

which is eq. (SD.92); no closed form of their sum is claimed, and proposition SF.10 evaluates the lag coefficients at zero ridge in participation coordinates.

Proposition SF.2 (Zero-ridge isotropic volatility without momentum)**.** *Let $\beta = 0$ and $m_\gamma \overset{\mathrm{def}}{=} \min\{1,\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}\}$. Then $\psi(\varrho)\to m_\gamma$ as $\varrho\downarrow 0$, and*

$$\lim_{\varrho\downarrow 0}\mathfrak{v}_{\mathrm{iso}}(\varrho) = B\,m_\gamma = \min\{B,N_{\mathrm{in}},N_{\mathrm{out}}\}.$$

Proof. The function ψ lies below m_γ. Indeed $\psi < \min(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}})$ by proposition SF.1(ii), and $\psi < 1$ by lemma SF.1(ii).

The function ψ cannot stay a fixed distance below m_γ. Suppose that $\psi(\varrho_k) \leq m_\gamma - \delta$ for some $\delta > 0$ along a sequence $\varrho_k \downarrow 0$. Then eq. (SF.7) gives

$$q(-\varrho_k) = \frac{(\gamma_{\mathrm{out}} - \psi(\varrho_k))(\gamma_{\mathrm{in}} - \psi(\varrho_k))}{\varrho_k} \geq \frac{\delta^2}{\varrho_k} \longrightarrow \infty,$$

and $\psi(\varrho_k) = \Phi(q(-\varrho_k)) \to 1 \geq m_\gamma$ by lemma SF.1(iv), a contradiction. Hence $\psi(\varrho)\to m_\gamma$.

The limit of the volatility follows from eq. (SF.18), which gives $\mathfrak{v}_{\mathrm{iso}}(\varrho) = B\psi(\varrho)$ at $\beta = 0$. Since $\gamma_{\mathrm{in}} = N_{\mathrm{in}}/B$ and $\gamma_{\mathrm{out}} = N_{\mathrm{out}}/B$, the product $B\,m_\gamma$ is $\min\{B,N_{\mathrm{in}},N_{\mathrm{out}}\}$. □

The main text's value $\min\{B,N\}$ is, in general, the exact identity $\|Q_0(G_{t+1})\|_{\mathrm{F}}^2 = \operatorname{rank} G_{t+1} \leq \min\{B,N_{\mathrm{out}},N_{\mathrm{in}}\}$ of lemma SA.2(i) for a gradient of full rank. No statement of this appendix is made at $\varrho = 0$.

Definition SF.1 (Zero-ridge isotropic volatility)**.** *Let $\beta \in [0,1)$ and let $\mathfrak{v}_{\mathrm{iso}}(\varrho)$ be the number of theorem SF.2(ii) at momentum β. The zero-ridge isotropic volatility at momentum β is*

$$\mathfrak{v}_{\mathrm{iso},\beta} \overset{\mathrm{def}}{=} \lim_{\varrho\downarrow 0}\mathfrak{v}_{\mathrm{iso}}(\varrho),$$

the limit being taken at this fixed β, whenever it exists.

Like $\mathfrak{v}_{\mathrm{iso}}(\varrho)$, the number $\mathfrak{v}_{\mathrm{iso},\beta}$ depends on the dimension. The limit is taken at a fixed momentum β. At $\beta = 0$ it exists, and

$$\mathfrak{v}_{\mathrm{iso},0} = \min\{B,N_{\mathrm{in}},N_{\mathrm{out}}\},$$

by proposition SF.2. For $\beta > 0$ the next proposition gives the limit of the same-time part and, in the square case, the existence of the limit.

Proposition SF.3 (Zero-ridge limits with momentum). *Let $\beta \in (0,1)$ and $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$, put $\gamma_\wedge \overset{\text{def}}{=} \min(\gamma_{\text{in}}, \gamma_{\text{out}})$ and $\gamma_\vee \overset{\text{def}}{=} \max(\gamma_{\text{in}}, \gamma_{\text{out}})$, and let K_β be the unique $q > 0$ with $qF_\beta(q) = \gamma_\wedge$.*

(i) As $\varrho \downarrow 0$, $\psi(\varrho) \to \gamma_\wedge$, and for every $\varrho > 0$

$$\gamma_\wedge - \psi(\varrho) \le \frac{\varrho\, K_\beta}{\gamma_\vee - \gamma_\wedge} \quad (\gamma_{\text{in}} \neq \gamma_{\text{out}}), \qquad \gamma_\wedge - \psi(\varrho) \le \sqrt{\varrho\, K_\beta} \quad (\gamma_{\text{in}} = \gamma_{\text{out}}).$$

In particular the same-time row and column coefficients $\psi(\varrho)/\gamma_{\text{out}}$ and $\psi(\varrho)/\gamma_{\text{in}}$ of eq. (SF.14) tend to $\gamma_\wedge/\gamma_{\text{out}}$ and $\gamma_\wedge/\gamma_{\text{in}}$, and the same-time total $B\psi(\varrho)$ of eq. (SF.15) tends to $\min\{N_{\text{in}}, N_{\text{out}}\}$.

(ii) Let $N_{\text{in}} = N_{\text{out}} = N$ and $\gamma = N/B$. For $\ell \ge 0$ let $\overline{C}_{1,\ell}(\varrho)$ be the row lag coefficient of eq. (SG.2) for the spectrum $\mu_1 = \cdots = \mu_N = 1$, which is the same for every row. Then $\overline{C}_{1,0}(\varrho) = \psi(\varrho)/\gamma$; for every $\ell \ge 0$ the limit $\overline{C}_{1,\ell}(0) \overset{\text{def}}{=} \lim_{\varrho\downarrow 0} \overline{C}_{1,\ell}(\varrho)$ exists and is nonnegative, with $\overline{C}_{1,0}(0) = 1$; and the limit of definition SF.1 exists and equals

$$\mathfrak{v}_{\text{iso},\beta} = N\Big(1 + 2\sum_{\ell\ge 1} \overline{C}_{1,\ell}(0)\Big) \ge N.$$

(iii) If moreover $N > B$, then $\mathfrak{v}_{\text{iso},\beta} \ge N > B = \mathfrak{v}_{\text{iso},0}$.

Proof. For part (i), K_β is well defined because every EMA weight is positive, so that $q \mapsto qF_\beta(q)$ is an increasing bijection of $[0,\infty)$ by lemma SB.1(iii). By proposition SF.1(ii), $q(-\varrho)F_\beta(q(-\varrho)) = \psi(\varrho) < \gamma_\wedge$, so $q(-\varrho) < K_\beta$, and eq. (SF.7) gives

$$\big(\gamma_{\text{out}} - \psi(\varrho)\big)\big(\gamma_{\text{in}} - \psi(\varrho)\big) = \varrho\, q(-\varrho) \le \varrho\, K_\beta.$$

Both factors are positive. For $\gamma_{\text{in}} \neq \gamma_{\text{out}}$ the larger one is at least $\gamma_\vee - \gamma_\wedge$, and for $\gamma_{\text{in}} = \gamma_{\text{out}}$ the two coincide; this gives the two bounds, and $\psi(\varrho) \to \gamma_\wedge$. The coefficients are those of theorem SF.2(i), and $B\gamma_\wedge = \min\{N_{\text{in}}, N_{\text{out}}\}$.

Part (ii) is proposition SG.4 at $\mu_1 = \cdots = \mu_N = 1$, which is the case $\Sigma_{\text{out}} = \Sigma_{\text{in}} = \mathrm{Id}_N$ of the setup of the power-law section; the kernel eq. (SG.16) depends on the row only through μ_i, so every row has the same coefficients. The zero-lag coefficient is the raw coefficient $\mathfrak{v}_1^{(0)}(\varrho)$ (theorem SD.7(iv)), which is $\psi(\varrho)/\gamma$ by the proof of theorem SF.2(i). By theorem SF.2(ii) the row coefficient is $\mathfrak{v}_1(\varrho) = \mathfrak{v}_{\text{iso}}(\varrho)/N$, and by proposition SG.4(ii) it converges to $1 + 2\sum_{\ell\ge 1} \overline{C}_{1,\ell}(0) \ge 1$.

Part (iii) combines (ii) with proposition SF.2, which gives $\mathfrak{v}_{\text{iso},0} = \min\{B, N\} = B$. □

The limits $\varrho \downarrow 0$ and $\beta \downarrow 0$ therefore do not commute for the volatility when $N > B$, by (iii), and they do not commute for the same-time total when $B < \min\{N_{\text{in}}, N_{\text{out}}\}$: its zero-ridge limit is $\min\{N_{\text{in}}, N_{\text{out}}\}$ for every $\beta > 0$ and $\min\{B, N_{\text{in}}, N_{\text{out}}\}$ at $\beta = 0$. At small aspect ratio $\mathfrak{v}_{\text{iso},\beta} = NV_\beta(1 + O(\gamma))$ with the explicit factor V_β of theorem SF.8; at large aspect ratio $\mathfrak{v}_{\text{iso},\beta} = (N^2/(2B))(1 + O(B/N))$ at every fixed β, with explicit constants (theorem SF.11(ii)).

SF.6 The scalar equation and the dynamics

Proposition SF.4 (Scalar reduction). *Let $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$ and $\Sigma_{\text{out}} = \mathrm{Id}_{N_{\text{out}}}$, let $\beta \in [0,1)$, $\varrho > 0$ and $h > 0$, and write $\mathfrak{d}_{\text{iso}}$ and $\mathfrak{v}_{\text{iso}} = \mathfrak{v}_{\text{iso}}(\varrho)$ for the numbers of theorems SF.1 and SF.2.*

(i) The deterministic mode-energy equations eq. (SE.7) close on the risk: if (Ω_{ij}) solves them on an interval on which $\mathcal{R} > 0$, then $\mathcal{R} = \frac{1}{2}\sum_{i,j} \Omega_{ij}$ solves

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}\tau} = -2h\,\mathfrak{d}_{\text{iso}}\sqrt{\mathcal{R}} + \frac{h^2}{2B}\,\mathfrak{v}_{\text{iso}}, \qquad \mathcal{R}(0) = \mathscr{R}(0), \tag{SF.19}$$

and eq. (SF.19) has at most one solution with $\mathcal{R} > 0$ on any interval $[0,T]$.

(ii) Under Assumption S3 and for $\varrho \in [\varrho_-, \varrho_+]$, one step of size $\eta \geq 0$ from a frozen state, $\Delta^+ = \Delta_t - \eta\, Q_\varrho(M_t)$, satisfies

$$\begin{aligned} \mathbb{E}\, \mathscr{R}(\Delta^+) - \mathscr{R}(t) &= -2\eta\, \mathfrak{d}_{\mathrm{iso}} \sqrt{\mathscr{R}(t)} + \frac{\eta^2 B}{2}\, \psi(\varrho) + r_t, \\ |r_t| &\leq \eta \sqrt{2\mathscr{R}(t)}\, \varepsilon_B + \frac{\eta^2 B}{2}\, \gamma_\star \varrho_+ e_B, \end{aligned} \tag{SF.20}$$

with ε_B of theorem SC.2 and e_B of theorem SD.1.

Proof. Part (i) sums eq. (SE.7) over the pairs. By theorem SF.1(i), $\mathfrak{d}_{ij} = \mathfrak{d}_{\mathrm{iso}}$ for every pair, and by theorem SF.2(ii), $\sum_{i,j} \mathfrak{v}_{ij} = \mathfrak{v}_{\mathrm{iso}}$. The risk of eq. (SE.7) is $\mathcal{R} = \frac{1}{2}\sum_{i,j} \Omega_{ij}$, because $\mu_i \lambda_j = 1$. Summing eq. (SE.7) therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(2B^2\mathcal{R}\big) = -\frac{2h\, \mathfrak{d}_{\mathrm{iso}}}{\sqrt{\mathcal{R}}}\, 2B^2\mathcal{R} + h^2 B\, \mathfrak{v}_{\mathrm{iso}},$$

and dividing by $2B^2$ gives eq. (SF.19). The initial value is $\frac{1}{2}\sum_{i,j} \mathscr{Q}_{ij}(0) = \mathscr{R}(0)$ by eq. (SF.2). The right side of eq. (SF.19) is a continuously differentiable function of $\mathcal{R}$ on $(0, \infty)$, so the Picard–Lindelöf theorem [65, Theorem 2.2] gives uniqueness.

Part (ii) is proposition SC.2 with the quadratic term eq. (SD.26), both at $C_\Sigma = 1$. By eqs. (SC.15) and (SC.16), the linear term differs from

$$\frac{\eta}{\sqrt{\mathscr{R}(t)}} \sum_{i,j} \mathfrak{d}_{ij}\, \mathscr{Q}_{ij}(t) = \frac{\eta\, \mathfrak{d}_{\mathrm{iso}}}{\sqrt{\mathscr{R}(t)}}\, 2\mathscr{R}(t) = 2\eta\, \mathfrak{d}_{\mathrm{iso}} \sqrt{\mathscr{R}(t)}$$

by at most $\eta\sqrt{2\mathscr{R}(t)}\, \varepsilon_B$. By eq. (SD.26) and $\overline{Z}(\varrho) = \psi(\varrho)$ (theorem SF.2(i)), the quadratic term differs from $\frac{\eta^2 B}{2}\psi(\varrho)$ by at most $\frac{\eta^2 B}{2}\gamma_\star \varrho_+ e_B$. □

With $\eta = h/B$ and $t = B\tau$, the chain rule turns eq. (SF.19) into

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}t} = -2\eta\, \mathfrak{d}_{\mathrm{iso}} \sqrt{\mathcal{R}} + \frac{\eta^2}{2}\, \mathfrak{v}_{\mathrm{iso}},$$

which is the equation eq. (SI.11) of the common-drift model of proposition SI.2 with $\mathfrak{d} = \mathfrak{d}_{\mathrm{iso}}$ and $\mathfrak{v}_{\mathrm{tot}} = \mathfrak{v}_{\mathrm{iso}}$, and the continuous-time form of the main text's recursion eq. (8). In the deterministic row model of definition SI.1 the isotropic case is the case of N_{out} rows with $\mu_i = 1$, $\mathfrak{d}_i = \mathfrak{d}_{\mathrm{iso}}$ and $\mathfrak{v}_i = \mathfrak{v}_{\mathrm{iso}}/N_{\mathrm{out}}$. In the frozen one-step identity eq. (SF.20) the quadratic term is the same-time value $B\psi(\varrho)$, which equals $\mathfrak{v}_{\mathrm{iso}}$ at $\beta = 0$; over many steps the integrated coefficient $\mathfrak{v}_{\mathrm{iso}}$ replaces it (theorem SD.2(i)), and the next corollary is the statement that does this for the algorithm.

Corollary SF.1 (Isotropic dynamics)**.** *Let $\Sigma_{\mathrm{in}} = \mathrm{Id}_{N_{\mathrm{in}}}$ and $\Sigma_{\mathrm{out}} = \mathrm{Id}_{N_{\mathrm{out}}}$, fix $h > 0$, $\varrho > 0$, $\beta \in [0, 1)$, $\gamma_\star < \infty$, $E_\star < \infty$, $c_M < \infty$ and $0 < r_0 \leq r_1 < \infty$, and along a sequence of dimensions with $B \to \infty$ assume:*

- *(a) Assumption S2 holds and $\max(N_{\mathrm{in}}, N_{\mathrm{out}}) \leq \gamma_\star B$;*
- *(b) the algorithm of definition SA.2 is run with momentum β, normalized ridge ϱ and step $\eta_t = h/B$;*
- *(c) W_0 is deterministic, $\|\Delta_0\|_{\mathrm{F}}^2 \leq E_\star$, $r_0 \leq \mathscr{R}(0) \leq r_1$ and $\|\Delta_0\|/\|\Delta_0\|_{\mathrm{F}} \to 0$;*
- *(d) the initial buffer M_{-1}° is independent of all batches and $\mathbb{E}\,\|M_{-1}^\circ\|_{\mathrm{F}}^2 \leq c_M B$.*

Let $T_\star$ be given by eq. (SE.10) at $C_\Sigma = 1$. Then eq. (SF.19) has a unique solution $\mathcal{R}$ on $[0, T_\star]$, and

- *(i) with probability tending to one, $t_{\mathrm{stop}} > BT_\star$ and $\inf_{t \leq BT_\star} \mathscr{R}(t) \geq r_0/2$;*
- *(ii) $\sup_{\tau \leq T_\star} |\mathscr{R}(\lfloor B\tau \rfloor) - \mathcal{R}(\tau)| \to 0$ in probability;*
- *(iii) on the event of (i), $\sup_{t \leq BT_\star} \|\Delta_t\|/\|\Delta_t\|_{\mathrm{F}} \to 0$ in probability.*

Proof. Hypotheses (a)–(d) are (H1)–(H4) of theorem SE.1 at identity covariances. In (H1) the features are Gaussian by the standing setup, and the covariances are deterministic with $\|\Sigma_{\text{in}}\| = \|\Sigma_{\text{out}}\| = 1$, so $C_\Sigma = 1$. In (H3) the diffuseness is $\|\widehat{A}_0\| = \|\Delta_0\|/\|\Delta_0\|_{\text{F}}$ by eq. (SF.2). By proposition SE.1 the equations eq. (SE.7) have a unique solution on $[0, T_\star]$ with $\mathcal{R} \geq r_0/2$, and by proposition SF.4(i) its risk is the unique solution of eq. (SF.19). Conclusions (i), (ii) and (iii) are theorem SE.1(i), the risk statement of (iii), and (iv). □

Of the four hypotheses of theorem SE.1, the isotropic case makes the covariance conditions of (H1) vacuous. The proportional regime, the bound $\gamma_\star$, the algorithm, the bounds on the initial energy and risk, the initial diffuseness and, for $\beta > 0$, the moment of the initial buffer remain; at $\beta = 0$ the buffer hypothesis (H4) is vacuous (remark SE.2). The spectral energy measure of eq. (SE.4) is the single atom $\|\Delta\|_{\text{F}}^2\, \delta_{(1,1)} = 2\mathscr{R}(\Delta)\, \delta_{(1,1)}$, whose bounded-Lipschitz norm is its mass, so the measure statement theorem SE.1(ii) is again (ii) of the corollary.

SF.7 The drift constant at zero ridge

Let $\beta \in [0, 1)$, $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$, and for $t > 0$ let $q(-t)$ be the branch of proposition SF.1 at momentum β. The zero-ridge drift constant at momentum β is

$$C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}}) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi\, \gamma_{\text{in}} \gamma_{\text{out}}} \int_0^\infty q(-u^2)\, L_\beta\big(q(-u^2)\big)\, du, \qquad C_\beta(\gamma) \stackrel{\text{def}}{=} C_\beta(\gamma, \gamma). \tag{SF.21}$$

It is the integral eq. (SF.10) at $\varrho = 0$. At $\beta = 0$, $qJ(q) = \psi$ by eq. (SF.8), so C_0 is the integral eq. (SF.11) at $\varrho = 0$. The square case $C_\beta(N/B)$ is the constant of eq. (9). The branch, and hence the integrand, depends on $(\gamma_{\text{in}}, \gamma_{\text{out}})$ and β, and this dependence is suppressed.

Lemma SF.1 (The Gaussian scalar)**.** *For $v > 0$:*

(i) $\Phi(v) = f\big((2v)^{-1/2}\big)$*;*

(ii) Φ *is strictly increasing, with* $\Phi'(v) = \mathbb{E}\, g^2(1 + vg^2)^{-2} \in (0, 1]$*, and* $0 < \Phi(v) < 1$*;*

(iii) $0 \leq v - \Phi(v) \leq 3v^2$*;*

(iv) $\Phi(v)/v \to 1$ *as* $v \downarrow 0$ *and* $\Phi(v) \to 1$ *as* $v \to \infty$*.*

In particular, for real $\xi > 0$,

$$\Big| f(\xi) - \frac{1}{2\xi^2} \Big| \leq \frac{3}{4\xi^4}. \tag{SF.22}$$

Proof. Part (i) is proposition SB.1. Parts (ii) and (iv) are lemma SB.1 for the single weight $w = (1, 0, 0, \dots)$, for which $r_w = \Phi$ and $\#\{j : w_j > 0\} = 1$: part (iii) of that lemma gives the monotonicity, the derivative and the limit 1 at infinity, and $\Phi(v)/v = J(v) \to J(0) = 1$ by its part (i). The derivative is at most $\mathbb{E}\, g^2 = 1$, and $\Phi < 1$ because the integrand of eq. (SF.4) is below 1. Part (iii) is the identity

$$v - \Phi(v) = \mathbb{E}\left[vg^2 - \frac{vg^2}{1 + vg^2}\right] = \mathbb{E}\, \frac{v^2 g^4}{1 + vg^2} \in \big[0,\, v^2\, \mathbb{E}\, g^4\big] = \big[0,\, 3v^2\big].$$

The bound eq. (SF.22) is (iii) at $v = 1/(2\xi^2)$, by (i). □

Proposition SF.5 (The zero-ridge constant)**.** *Let $\beta \in [0, 1)$ and $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$. Put $K_0 \stackrel{\text{def}}{=} 1$, and for $\beta > 0$ let K_β be the unique $q > 0$ with $qF_\beta(q) = \min(\gamma_{\text{in}}, \gamma_{\text{out}})$.*

(i) For every $t > 0$,

$$0 < q(-t)\, L_\beta\big(q(-t)\big) \leq \min\left(K_\beta,\, \frac{\gamma_{\text{in}}\gamma_{\text{out}}}{t}\right),$$

so the integral eq. (SF.21) converges and $0 < C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}}) < \infty$. Moreover $C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}}) = C_\beta(\gamma_{\text{out}}, \gamma_{\text{in}})$, and

$$C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{1}{\sqrt{2}\, \pi\, \gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty t^{-1/2}\, q(-t)\, L_\beta\big(q(-t)\big)\, \mathrm{d}t. \tag{SF.23}$$

(ii) Let $\mathfrak{d}_{\mathrm{iso}}(\varrho)$ be the kernel eq. (SF.10) at momentum β, for the measures eq. (SF.1). Then, at this fixed β, $\mathfrak{d}_{\mathrm{iso}}(\varrho) \to C_\beta(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$ as $\varrho \downarrow 0$.

Proof. For part (i), K_β is well defined: for $\beta > 0$ every EMA weight is positive, so by lemma SB.1(iii) the map $q \mapsto qF_\beta(q)$ is a continuous strictly increasing bijection of $[0, \infty)$. The integrand is positive because $q(-t) > 0$ by proposition SF.1(ii) and $L_\beta > 0$. By lemma SB.1(i), $J \leq 1$, so

$$L_\beta(q) = \sum_{j \geq 0} w_j \, J(w_j^2 q) \leq \sum_{j \geq 0} w_j = 1, \qquad q \, L_\beta(q) \leq q.$$

The first identity of eq. (SF.5) gives $\sigma(-t) \leq \gamma_{\mathrm{out}}/t$ and $\tilde{\sigma}(-t) \leq \gamma_{\mathrm{in}}/t$, so that

$$q(-t) = t \, \sigma(-t) \, \tilde{\sigma}(-t) \leq \frac{\gamma_{\mathrm{in}} \gamma_{\mathrm{out}}}{t}.$$

At $\beta = 0$, $qJ(q) = \Phi(q) < 1 = K_0$ by lemma SF.1(ii). For $\beta > 0$, proposition SF.1(ii) gives $q(-t)F_\beta(q(-t)) < \min(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}) = K_\beta F_\beta(K_\beta)$, so $q(-t) < K_\beta$ by monotonicity, and $qL_\beta(q) \leq q < K_\beta$. The integrand of eq. (SF.21) is therefore at most $\min(K_\beta, \gamma_{\mathrm{in}}\gamma_{\mathrm{out}}/u^2)$. The equation eq. (SF.7) is symmetric in $(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$ for every β, so $q(-t)$ is unchanged when the two ratios are exchanged, which gives the symmetry. The substitution $t = u^2$ gives eq. (SF.23).

Part (ii) is a limit under the integral. By theorem SF.1(i),

$$\mathfrak{d}_{\mathrm{iso}}(\varrho) = \frac{\sqrt{2}}{\pi \, \gamma_{\mathrm{in}} \gamma_{\mathrm{out}}} \int_0^\infty q(-\varrho - u^2) \, L_\beta\big(q(-\varrho - u^2)\big) \, du.$$

The function $t \mapsto q(-t)$ is continuous on $(0, \infty)$ by theorem SB.1(iv), and L_β is continuous by lemma SB.1(ii), so the integrand converges to that of eq. (SF.21) for every $u > 0$. By part (i) it is at most $\min(K_\beta, \gamma_{\mathrm{in}}\gamma_{\mathrm{out}}/u^2)$. Dominated convergence gives the limit. □

The statements of appendices SC to SE hold at a fixed positive ridge, so $C_\beta(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$ is the drift of the zero-ridge algorithm only as a formal limit (corollary SC.1 and remark SE.1): part (ii) is a statement about the deterministic kernel, and no statement of the appendix concerns the algorithm at $\varrho = 0$.

The bounds, the symmetry and the form eq. (SF.23) hold at every $\beta \in [0, 1)$. The limit of part (ii) is taken at a fixed β. The limits $\varrho \downarrow 0$ and $\beta \downarrow 0$ commute at a positive ridge (proposition SF.11(i)) but not at zero ridge: in the square case $\lim_{\beta \downarrow 0} C_\beta(\gamma)$ differs from $C_0(\gamma)$ for every $\gamma > 1$ (proposition SF.12). The asymptotics of C_β at every momentum are in appendix SF.11. The spectral form, the measure pair, the limiting laws of appendix SF.9 and the value at aspect ratio one are proved at $\beta = 0$. In the rest of this subsection, $\beta = 0$, and we write $C(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}) \stackrel{\mathrm{def}}{=} C_0(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$ and $C(\gamma) \stackrel{\mathrm{def}}{=} C_0(\gamma)$; the same convention holds in appendices SF.9 and SF.10. For $t > 0$ the branch at $\beta = 0$ satisfies, by eqs. (SF.6) and (SF.8),

$$q(-t)J\big(q(-t)\big) = \psi(t) = \gamma_{\mathrm{out}} - t\sigma(-t) = \gamma_{\mathrm{in}} - t\tilde{\sigma}(-t) = \Phi\big(t\sigma(-t)\tilde{\sigma}(-t)\big) = f\big(\xi(-t)\big), \tag{SF.24}$$

so that the integrand of eqs. (SF.21) and (SF.23) at $\beta = 0$ is ψ.

The spectral form. The constant is a half-moment of a spectral measure when the branch is a Stieltjes transform. For a finite nonnegative measure ν on $[0, \infty)$ we write its transform on the negative axis as

$$S_\nu(-t) \stackrel{\mathrm{def}}{=} \int \frac{\nu(\mathrm{d}x)}{x + t} \qquad (t > 0).$$

Definition SF.2 (Isotropic measure pair)**.** *Let $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 0$. An isotropic measure pair at $(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$ is a pair of finite nonnegative measures $\mathfrak{m}_{\mathrm{out}}$ and $\mathfrak{m}_{\mathrm{in}}$ on $[0, \infty)$, of masses γ_{out} and γ_{in}, whose transforms $\sigma(-t) \stackrel{\mathrm{def}}{=} S_{\mathfrak{m}_{\mathrm{out}}}(-t)$ and $\tilde{\sigma}(-t) \stackrel{\mathrm{def}}{=} S_{\mathfrak{m}_{\mathrm{in}}}(-t)$ satisfy, for every $t > 0$,*

$$\gamma_{\mathrm{out}} - t\,\sigma(-t) = \gamma_{\mathrm{in}} - t\,\tilde{\sigma}(-t) = \Phi\big(t\,\sigma(-t)\,\tilde{\sigma}(-t)\big). \tag{SF.25}$$

A pair says that the no-momentum isotropic closure is the transform of two measures on the half-line. For every $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 0$ exactly one pair exists, by proposition SF.6 below. At $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = \gamma$ a pair

is the measure of the scalar measure hypothesis of definition SI.2 in the phase-comparison section: a measure μ_γ of that definition gives the pair $\mathfrak{m}_{\mathrm{out}} = \mathfrak{m}_{\mathrm{in}} = \mu_\gamma$, and conversely the two measures of a pair coincide when $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}}$, by lemma SF.3(iii). That hypothesis therefore holds at every $\gamma > 0$. The next two lemmas are the facts about such pairs that the existence proof and the asymptotic statements use.

Lemma SF.2 (Negative-axis transforms). *(i) Let ν_j $(j \geq 1)$ and ν be probability measures on $[0,\infty)$ with $S_{\nu_j}(-t) \to S_\nu(-t)$ for every $t > 0$. Then $\nu_j \to \nu$ weakly. If moreover $\sup_j \int x\,\nu_j(\mathrm{d}x) < \infty$ and $\int x\,\nu(\mathrm{d}x) < \infty$, then*

$$\int \sqrt{x}\,\nu_j(\mathrm{d}x) \longrightarrow \int \sqrt{x}\,\nu(\mathrm{d}x).$$

(ii) Two finite nonnegative measures on $[0,\infty)$ with equal masses and equal transforms at every $-t$, $t > 0$, are equal.

Proof. Part (i) first extends the convergence off the negative axis. For $k \geq 1$ let $D_k \stackrel{\mathrm{def}}{=} \{z \in \mathbb{C} : \operatorname{dist}(z,[0,\infty)) > 1/k\}$, a connected open set. For a probability measure ν on $[0,\infty)$ the function $S_\nu(z) \stackrel{\mathrm{def}}{=} \int (x-z)^{-1}\nu(\mathrm{d}x)$ is analytic on $\mathbb{C} \setminus [0,\infty)$, because for $|z - z_0| < \operatorname{dist}(z_0,[0,\infty))$ the expansion

$$\frac{1}{x-z} = \sum_{n\geq 0} \frac{(z-z_0)^n}{(x-z_0)^{n+1}}$$

converges uniformly in $x \geq 0$; and $|S_\nu(z)| \leq k$ on D_k. The points $-t$ with $t > 1/k$ lie in D_k and accumulate at $-2/k \in D_k$. Vitali's convergence theorem [66, §5.21], whose hypotheses are the uniform bound on D_k and the convergence on this set, gives that S_{ν_j} converges at every point of D_k to an analytic function on D_k. That function equals S_ν on the negative axis in D_k, so it equals S_ν on D_k by the identity theorem [66, §2.6]. Since $\bigcup_k D_k = \mathbb{C} \setminus [0,\infty)$,

$$S_{\nu_j}(z) \longrightarrow S_\nu(z) \qquad (z \in \mathbb{C} \setminus \mathbb{R}).$$

Theorem 2.4.4(c) of [1], applied to the deterministic measures ν_j with the limit S_ν, the Stieltjes transform of the probability measure ν, gives $\nu_j \to \nu$ weakly.

The half-moments converge by truncation. For $L > 0$ and $x > L$ one has $\sqrt{x} \leq x/\sqrt{L}$, so

$$0 \leq \int \sqrt{x}\,\nu_j(\mathrm{d}x) - \int \min(\sqrt{x},\sqrt{L})\,\nu_j(\mathrm{d}x) \leq \int_{(L,\infty)} \sqrt{x}\,\nu_j(\mathrm{d}x) \leq \frac{1}{\sqrt{L}} \sup_k \int x\,\nu_k(\mathrm{d}x),$$

and the same bound holds for ν. The truncated integrand is bounded and continuous, so its integrals converge by weak convergence, and letting $L \to \infty$ proves the claim.

Part (ii) is part (i) for a constant sequence. If the common mass is zero there is nothing to prove. Otherwise, after dividing both measures by their mass, the constant sequence $\nu_j \stackrel{\mathrm{def}}{=} \nu'$ has the transforms of ν, so $\nu' \to \nu$ weakly by part (i), that is, $\nu' = \nu$. □

Lemma SF.3 (Moments and atoms). *Let $(\mathfrak{m}_{\mathrm{out}}, \mathfrak{m}_{\mathrm{in}})$ be an isotropic measure pair at $(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}})$.*

(i) The transforms are the branch: $\sigma(-t)$ and $\tilde{\sigma}(-t)$ in eq. (SF.25) are those of proposition SF.1 at $\beta = 0$, and $\psi(t) = \int \frac{\lambda}{\lambda+t}\,\mathfrak{m}_{\mathrm{out}}(\mathrm{d}\lambda)$.

(ii) The first moments are

$$\int \lambda\,\mathfrak{m}_{\mathrm{out}}(\mathrm{d}\lambda) = \int \lambda\,\mathfrak{m}_{\mathrm{in}}(\mathrm{d}\lambda) = \gamma_{\mathrm{in}}\gamma_{\mathrm{out}}, \qquad \text{and} \qquad \int \sqrt{\lambda}\,\mathfrak{m}_{\mathrm{out}}(\mathrm{d}\lambda) \leq \gamma_{\mathrm{out}}\sqrt{\gamma_{\mathrm{in}}}.$$

(iii) $\mathfrak{m}_{\mathrm{in}} - \mathfrak{m}_{\mathrm{out}} = (\gamma_{\mathrm{in}} - \gamma_{\mathrm{out}})\,\delta_0$.

(iv) If $\gamma_{\mathrm{in}} > 1$ and $\gamma_{\mathrm{out}} > 1$, then $\mathfrak{m}_{\mathrm{out}} = (\gamma_{\mathrm{out}} - 1)\delta_0 + \eta$ and $\mathfrak{m}_{\mathrm{in}} = (\gamma_{\mathrm{in}} - 1)\delta_0 + \eta$ for a probability measure η on $(0,\infty)$.

(v) For every finite nonnegative measure ν on $[0,\infty)$,

$$\int \sqrt{x}\,\nu(\mathrm{d}x) = \frac{1}{\pi}\int_0^\infty t^{-1/2}\int \frac{x}{x+t}\,\nu(\mathrm{d}x)\,\mathrm{d}t. \tag{SF.26}$$

Proof. For part (i), fix $t>0$ and put $q' \overset{\text{def}}{=} t\sigma(-t)\tilde\sigma(-t)$. The measures have positive masses, so $\sigma(-t),\tilde\sigma(-t)>0$, and eq. (SF.25) gives $\Phi(q')=\gamma_{\text{out}}-t\sigma(-t)<\gamma_{\text{out}}$ and likewise $\Phi(q')<\gamma_{\text{in}}$. Inserting $\sigma(-t)=(\gamma_{\text{out}}-\Phi(q'))/t$ and $\tilde\sigma(-t)=(\gamma_{\text{in}}-\Phi(q'))/t$ in the definition of q' gives

$$t\,q' = \big(\gamma_{\text{out}}-\Phi(q')\big)\big(\gamma_{\text{in}}-\Phi(q')\big),$$

which is eq. (SF.7) at $\beta=0$. By the uniqueness in proposition SF.1(ii), $q'=q(-t)$, and then the anchors eq. (SF.6) identify $\sigma(-t)$ and $\tilde\sigma(-t)$. The formula for ψ follows from $\gamma_{\text{out}}-t\sigma(-t)=\int\big(1-\frac{t}{\lambda+t}\big)\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)$.

Part (ii) reads the first moment off the large-t behaviour of ψ. By dominated convergence, $t\sigma(-t)\to\gamma_{\text{out}}$ and $t\tilde\sigma(-t)\to\gamma_{\text{in}}$ as $t\to\infty$. Hence $q(-t)\to 0$ and $t\,q(-t)\to\gamma_{\text{in}}\gamma_{\text{out}}$, and by lemma SF.1(iii),

$$|t\,\psi(t)-t\,q(-t)| \le 3t\,q(-t)^2 \longrightarrow 0,$$

so $t\psi(t)\to\gamma_{\text{in}}\gamma_{\text{out}}$. On the other hand,

$$t\,\psi(t) = \int \frac{t\lambda}{\lambda+t}\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)$$

increases to $\int\lambda\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)$ by monotone convergence. The same argument with $\psi(t)=\gamma_{\text{in}}-t\tilde\sigma(-t)$ gives the first moment of $\mathfrak{m}_{\text{in}}$. The Cauchy–Schwarz inequality gives

$$\int\sqrt{\lambda}\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda) \le \Big(\gamma_{\text{out}}\int\lambda\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)\Big)^{1/2} = \gamma_{\text{out}}\sqrt{\gamma_{\text{in}}}.$$

Part (iii) compares two measures of equal mass. By eq. (SF.6), $\tilde\sigma(-t)-\sigma(-t)=(\gamma_{\text{in}}-\gamma_{\text{out}})/t$, so the measures $\mathfrak{m}_{\text{in}}+\gamma_{\text{out}}\delta_0$ and $\mathfrak{m}_{\text{out}}+\gamma_{\text{in}}\delta_0$ have the same transform at every $-t$:

$$\tilde\sigma(-t)+\frac{\gamma_{\text{out}}}{t} = \sigma(-t)+\frac{\gamma_{\text{in}}}{t}.$$

Both have mass $\gamma_{\text{in}}+\gamma_{\text{out}}$, so they are equal by lemma SF.2(ii).

Part (iv) identifies the mass away from zero. By lemma SF.1(ii), $\psi<1$, so by eq. (SF.7)

$$q(-t) = \frac{(\gamma_{\text{out}}-\psi(t))(\gamma_{\text{in}}-\psi(t))}{t} \ge \frac{(\gamma_{\text{out}}-1)(\gamma_{\text{in}}-1)}{t} \longrightarrow \infty \qquad (t\downarrow 0),$$

and $\psi(t)=\Phi(q(-t))\to 1$ by lemma SF.1(iv). By part (i) and dominated convergence, $\psi(t)\to\mathfrak{m}_{\text{out}}((0,\infty))$ as $t\downarrow 0$. Hence $\mathfrak{m}_{\text{out}}((0,\infty))=1$, the atom of $\mathfrak{m}_{\text{out}}$ at 0 is $\gamma_{\text{out}}-1$, and η is the restriction of $\mathfrak{m}_{\text{out}}$ to $(0,\infty)$. Part (iii) gives the decomposition of $\mathfrak{m}_{\text{in}}$.

Part (v) is a scalar integral. For $x>0$ the substitution $t=xv^2$ gives

$$\int_0^\infty t^{-1/2}\,\frac{x}{x+t}\,\mathrm{d}t = 2\sqrt{x}\int_0^\infty\frac{\mathrm{d}v}{1+v^2} = \pi\sqrt{x},$$

and for $x=0$ both sides vanish. Integrating against $\nu(\mathrm{d}x)$ gives eq. (SF.26). □

Proposition SF.6 (Existence of the isotropic measure pair)**.** *For every $\gamma_{\text{in}}>0$ and $\gamma_{\text{out}}>0$ there is exactly one isotropic measure pair $(\mathfrak{m}_{\text{out}},\mathfrak{m}_{\text{in}})$ at $(\gamma_{\text{in}},\gamma_{\text{out}})$.*

Proof. Uniqueness is lemma SF.3(i) with lemma SF.2(ii). By the former, the transforms of any pair at $(\gamma_{\text{in}},\gamma_{\text{out}})$ are the branch $\sigma(-t)$ and $\tilde\sigma(-t)$ of proposition SF.1 at $\beta=0$. The masses are fixed, so two pairs have equal masses and equal transforms.

Existence: the pair is a limit of expected Gram spectral measures. Fix dimensions with $B\to\infty$, $N_{\text{out}}/B\to\gamma_{\text{out}}$ and $N_{\text{in}}/B\to\gamma_{\text{in}}$, identity covariances, $\beta=0$, and the frozen states $\widehat{A}_t = k^{-1/2}\sum_{l\le k}u_l\otimes v_l$ with $k=\min(N_{\text{out}},N_{\text{in}})$. Then $\|\widehat{A}_t\|_{\text{F}}=1$ and $\|\widehat{A}_t\|=k^{-1/2}\to 0$, so

Assumption S3 holds, and $M_t = G_{t+1}$. Let ζ_B be the expected spectral measure of $G_{t+1}G_{t+1}^\top$ normalized by B,

$$\int \phi \,\mathrm{d}\zeta_B \overset{\text{def}}{=} \frac{1}{B}\,\mathbb{E}\,\mathrm{Tr}\,\phi\big(G_{t+1}G_{t+1}^\top\big), \qquad \zeta_B([0,\infty)) = \frac{N_{\text{out}}}{B}, \qquad S_{\zeta_B}(-t) = \frac{1}{B}\,\mathbb{E}\,\mathrm{Tr}\,R(-t).$$

By corollary SB.1 with $\Sigma_{\text{out}} = \mathrm{Id}_{N_{\text{out}}}$, and the continuity of the branch in the ratios (theorem SB.1(ii)),

$$S_{\zeta_B}(-t) \longrightarrow \sigma(-t) \qquad (t > 0),$$

where σ is the branch at the limiting ratios. The first moments are bounded: by lemma SA.4 with identity covariances and $\|\,\mathbb{E}\,G_{t+1}\|_{\mathrm{F}} = \|\widehat{A}_t\|_{\mathrm{F}} = 1$,

$$\int x\,\zeta_B(\mathrm{d}x) = \frac{1}{B}\,\mathbb{E}\,\|G_{t+1}\|_{\mathrm{F}}^2 = \frac{1}{B} + \frac{N_{\text{out}}N_{\text{in}} + 2N_{\text{out}} + 2N_{\text{in}} + 3}{B^2} \longrightarrow \gamma_{\text{in}}\gamma_{\text{out}}.$$

The normalized measures $\zeta_B/\zeta_B([0,\infty))$ are therefore tight, since $\zeta([L,\infty)) \le L^{-1}\int x\,\mathrm{d}\zeta$, and by Prokhorov's theorem [10, Theorem 5.1] a subsequence converges weakly. The limit times γ_{out} is a measure $\mathfrak{m}_{\text{out}}$ of mass γ_{out} whose transform is $\sigma(-t)$, because $x \mapsto (x+t)^{-1}$ is bounded and continuous on $[0,\infty)$. The same argument for $G_{t+1}^\top G_{t+1}$, along a further subsequence, gives $\mathfrak{m}_{\text{in}}$ of mass γ_{in} with transform $\tilde{\sigma}(-t)$. By eq. (SF.24), the two transforms satisfy eq. (SF.25). □

The pair is built from Gram matrices of the frozen-state model and is not identified with the drift of the zero-ridge algorithm, so $C(\gamma_{\text{in}}, \gamma_{\text{out}})$ remains the formal limit $\varrho \downarrow 0$ of the ridge coefficient (proposition SF.5(ii)).

Proposition SF.7 (Spectral form of the constant)**.** *Let $(\mathfrak{m}_{\text{out}}, \mathfrak{m}_{\text{in}})$ be the isotropic measure pair at $(\gamma_{\text{in}}, \gamma_{\text{out}})$ of proposition SF.6. Then*

$$C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{1}{\sqrt{2}\,\gamma_{\text{in}}\gamma_{\text{out}}} \int \sqrt{\lambda}\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda) = \frac{1}{\sqrt{2}\,\gamma_{\text{in}}\gamma_{\text{out}}} \int \sqrt{\lambda}\,\mathfrak{m}_{\text{in}}(\mathrm{d}\lambda). \tag{SF.27}$$

For $b > 0$ let $\widehat{\mathfrak{m}}_b$ be the image of $\mathfrak{m}_{\text{out}}$ under $\lambda \mapsto \lambda/b$, a measure of mass γ_{out} with transform $S_{\widehat{\mathfrak{m}}_b}(-t) = b\,\sigma(-bt)$, and with density $b\rho(bx)$ when $\mathfrak{m}_{\text{out}}$ has density ρ. Then

$$C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{\sqrt{b}}{\sqrt{2}\,\gamma_{\text{in}}\gamma_{\text{out}}} \int \sqrt{x}\,\widehat{\mathfrak{m}}_b(\mathrm{d}x). \tag{SF.28}$$

In particular, at $b = \gamma_{\text{in}}\gamma_{\text{out}}$,

$$C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{1}{\sqrt{2\gamma_{\text{in}}\gamma_{\text{out}}}} \int \sqrt{x}\,\widehat{\mathfrak{m}}_{\gamma_{\text{in}}\gamma_{\text{out}}}(\mathrm{d}x).$$

Proof. By eq. (SF.23), lemma SF.3(i) and the half-moment formula eq. (SF.26),

$$C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{1}{\sqrt{2}\,\pi\,\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty t^{-1/2} \int \frac{\lambda}{\lambda + t}\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)\,\mathrm{d}t = \frac{1}{\sqrt{2}\,\gamma_{\text{in}}\gamma_{\text{out}}} \int \sqrt{\lambda}\,\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda).$$

The second form of eq. (SF.27) holds because the two measures differ by an atom at 0 (lemma SF.3(iii)), where $\sqrt{\lambda}$ vanishes. For the rescaled measure,

$$S_{\widehat{\mathfrak{m}}_b}(-t) = \int \frac{\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)}{\lambda/b + t} = b \int \frac{\mathfrak{m}_{\text{out}}(\mathrm{d}\lambda)}{\lambda + bt} = b\,\sigma(-bt),$$

the density is the change of variables $x = \lambda/b$, and $\sqrt{\lambda} = \sqrt{b}\,\sqrt{\lambda/b}$ gives eq. (SF.28). □

SF.8 The participation representation

In this subsection the momentum $\beta \in [0,1)$ is arbitrary. The drift integral eq. (SF.10) is an integral of one scalar root. This subsection reparametrizes the root by its participation level and integrates by parts once, which isolates the main term of the drift.

We write $r_\beta(q) \overset{\text{def}}{=} qF_\beta(q)$ for the function of lemma SB.1(iii) at the EMA weights, and $s_k = \sum_{j\ge 0} w_j^k$ for $k \ge 2$, so that $s_k = (1-\beta)^k/(1-\beta^k)$ for $\beta > 0$ and $s_k = 1$ at $\beta = 0$. Put

$$\gamma_\wedge \overset{\text{def}}{=} \min(\gamma_{\text{in}}, \gamma_{\text{out}}), \qquad \gamma_\vee \overset{\text{def}}{=} \max(\gamma_{\text{in}}, \gamma_{\text{out}}),$$

$$\mathsf{W}(m) \stackrel{\text{def}}{=} \sqrt{(\gamma_{\text{in}} - m)(\gamma_{\text{out}} - m)} \qquad (0 \le m \le \gamma_\wedge),$$

and, for $q \ge 0$ and $s > 0$,

$$\Lambda_\beta(q) \stackrel{\text{def}}{=} \int_0^q x^{-1/2} L_\beta(x)\,\mathrm{d}x, \qquad \mathsf{G}_\beta(s) \stackrel{\text{def}}{=} q(-s)\, L_\beta\big(q(-s)\big),$$

where $q(-s)$ is the root of proposition SF.1(ii) at momentum β. By theorem SF.1(i), for $\varrho > 0$,

$$\mathfrak{d}_{\text{iso}}(\varrho) = \frac{\sqrt{2}}{\pi\,\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^\infty \mathsf{G}_\beta(\varrho + u^2)\,\mathrm{d}u, \tag{SF.29}$$

and we put $\mathfrak{d}_{\text{iso}}(0) \stackrel{\text{def}}{=} C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}})$, the same integral at $\varrho = 0$ (eq. (SF.21)), which is its limit by proposition SF.5(ii). When the momentum and the aspect ratios have to be shown we write $\mathfrak{d}_{\text{iso}}(\varrho; \beta, \gamma_{\text{in}}, \gamma_{\text{out}})$. The square case is $\gamma_{\text{in}} = \gamma_{\text{out}} = \gamma$.

Lemma SF.4 (Scalar facts)**.** *Let $\beta \in [0, 1)$.*

(i) For $q \ge 0$,

$$r_\beta(q) = \sum_{j:\, w_j > 0} \Phi(w_j^2 q), \qquad r'_\beta(q) = \sum_{j:\, w_j > 0} w_j^2\, \Phi'(w_j^2 q) > 0.$$

For $\beta > 0$, r_β is an increasing bijection of $[0, \infty)$ onto itself; at $\beta = 0$, $r_0 = \Phi$ is an increasing bijection of $[0, \infty)$ onto $[0, 1)$.

(ii) For $x > 0$, $\Phi(x) \le \min(1, x)$, the derivative Φ' is decreasing, and

$$1 - \Phi(x) \le \sqrt{\frac{\pi}{2}}\, x^{-1/2}, \qquad x\,\Phi'(x) \le \min\Big(x, \sqrt{\frac{\pi}{2}}\, x^{-1/2}\Big).$$

(iii) For $x > 0$, $J(x) \le \min(1, 1/x)$ and $-3 \le J'(x) \le 0$.

(iv) For $q, z \ge 0$,

$$\Lambda_\beta(q) = \sum_{j:\, w_j > 0} \Lambda_0(w_j^2 q), \qquad \Lambda_0(z) \le 2\sqrt{z},$$

for $z > 0$,

$$0 \le \sqrt{2\pi} - \Lambda_0(z) \le 2z^{-1/2},$$

and $\Lambda_0(z) \uparrow \sqrt{2\pi}$ as $z \to \infty$.

(v) $F_\beta \le s_2$ and $L_\beta \le 1$, and $q \mapsto qL_\beta(q)$ is strictly increasing.

Proof. Part (i) is termwise. Each term of $r_\beta(q) = \sum_j w_j^2 q\, J(w_j^2 q)$ is $\Phi(w_j^2 q)$ by eq. (SF.4). The derivative and its positivity are lemma SB.1(iii), with $\mathbb{E}\, g^2(1 + vg^2)^{-2} = \Phi'(v)$ from lemma SF.1(ii). By the same lemma, r_β increases to the number of positive weights, which is infinite for $\beta > 0$ and equal to 1 at $\beta = 0$.

Part (ii) uses the Gaussian density. The bound $\Phi \le 1$ is lemma SF.1(ii), and $\Phi(x) = xJ(x) \le x$ because $J \le 1$. The derivative $\Phi'(x) = \mathbb{E}\, g^2(1 + xg^2)^{-2}$ is decreasing in x. Since the standard Gaussian density is at most $(2\pi)^{-1/2}$,

$$1 - \Phi(x) = \mathbb{E}\,\frac{1}{1 + xg^2} \le \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{\mathrm{d}y}{1 + xy^2} = \sqrt{\frac{\pi}{2}}\, x^{-1/2}.$$

Finally $x\Phi'(x) = \mathbb{E}[xg^2(1+xg^2)^{-2}]$ is at most $\mathbb{E}[xg^2] = x$, and at most $\mathbb{E}[(1+xg^2)^{-1}] = 1 - \Phi(x)$ because $xg^2/(1 + xg^2) \le 1$.

Part (iii) reads off the integrand: $g^2/(1+xg^2)$ is at most g^2 and at most $1/x$, and $J'(x) = -\,\mathbb{E}\, g^4(1 + xg^2)^{-2} \in [-\,\mathbb{E}\, g^4, 0] = [-3, 0]$.

Part (iv) is a substitution. In the j-th term of L_β the substitution $y = w_j^2 x$ gives

$$\int_0^q x^{-1/2} w_j J(w_j^2 x)\,\mathrm{d}x = \int_0^{w_j^2 q} y^{-1/2} J(y)\,\mathrm{d}y = \Lambda_0(w_j^2 q).$$

The bound $\Lambda_0(z) \le \int_0^z y^{-1/2}\mathrm{d}y = 2\sqrt{z}$ uses $J \le 1$. The full integral $\int_0^\infty y^{-1/2}J(y)\,\mathrm{d}y = \sqrt{2\pi}$ is computed in the proof of lemma SG.2, and by (iii)

$$\sqrt{2\pi} - \Lambda_0(z) = \int_z^\infty y^{-1/2}J(y)\,\mathrm{d}y \le \int_z^\infty y^{-3/2}\,\mathrm{d}y = 2z^{-1/2}.$$

Part (v) uses $J \le 1$ again: $F_\beta \le \sum_j w_j^2 = s_2$ and $L_\beta \le \sum_j w_j = 1$. Finally $qL_\beta(q) = \sum_{j:\,w_j>0} \Phi(w_j^2 q)/w_j$, a sum of strictly increasing functions. □

A larger link function produces a smaller root. This comparison is the tool for every two-sided bound below.

Lemma SF.5 (Root comparison)**.** *Let r be continuous and strictly increasing on $[0,\infty)$ with $r(0) = 0$, let $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}, s > 0$, and put $D_r \stackrel{\mathrm{def}}{=} \{q \ge 0 :\ r(q) < \gamma_\wedge\}$.*

(i) The function

$$G_r(q) \stackrel{\mathrm{def}}{=} s\,q - \big(\gamma_{\mathrm{out}} - r(q)\big)\big(\gamma_{\mathrm{in}} - r(q)\big)$$

is strictly increasing on the interval D_r and has exactly one zero $q_r(s)$ there. The map $s \mapsto q_r(s)$ is continuous and strictly decreasing.

(ii) Let r_1, r_2 be two such functions, with zeros $q_1 = q_{r_1}(s)$ and $q_2 = q_{r_2}(s)$ at the same s. If $r_1(q_i) \le r_2(q_i)$ for $i = 1$ or for $i = 2$, then $q_2 \le q_1$.

For $r = r_\beta$ the zero is the root of proposition SF.1(ii): $q_{r_\beta}(s) = q(-s)$.

Proof. For part (i), the map $x \mapsto (\gamma_{\mathrm{out}} - x)(\gamma_{\mathrm{in}} - x)$ is positive and decreasing on $[0, \gamma_\wedge)$, so G_r is strictly increasing on D_r, and $G_r(0) = -\gamma_{\mathrm{in}}\gamma_{\mathrm{out}} < 0$. If $\sup D_r < \infty$, then $r(q) \to \gamma_\wedge$ as $q \uparrow \sup D_r$ by continuity, and G_r tends to $s \sup D_r > 0$. Otherwise $G_r(q) \ge sq - \gamma_{\mathrm{in}}\gamma_{\mathrm{out}} \to \infty$. In both cases G_r has exactly one zero in D_r. For $s < s'$ the function at s' exceeds the function at s by $(s' - s)q > 0$ at $q = q_r(s) > 0$, so the zero at s' is smaller. For continuity, fix s and $0 < \delta < q_r(s)$. At s the function is negative at $q_r(s) - \delta$, and it is positive at $q_r(s) + \delta$ unless that point lies outside D_r. For a fixed q the value $G_r(q)$ is continuous in s, so the same signs hold at every s' near s, and the zero at s' lies within δ of $q_r(s)$.

For part (ii) with $i = 2$, $r_1(q_2) \le r_2(q_2) < \gamma_\wedge$, so $q_2 \in D_{r_1}$, and since $(\gamma_{\mathrm{out}} - x)(\gamma_{\mathrm{in}} - x)$ decreases in x,

$$G_{r_1}(q_2) \le G_{r_2}(q_2) = 0.$$

Hence $q_2 \le q_1$ by part (i). With $i = 1$, either $r_2(q_1) \ge \gamma_\wedge$, and then $q_1 \ge \sup D_{r_2} > q_2$, or $q_1 \in D_{r_2}$ and $G_{r_2}(q_1) \ge G_{r_1}(q_1) = 0$, which gives $q_1 \ge q_2$.

The last sentence is the uniqueness of proposition SF.1(ii). □

The next lemma makes the participation scale explicit. For $\beta > 0$ the level $r_\beta(q)$ counts the ages j with $w_j^2 q \ge 1$, up to explicit errors.

Lemma SF.6 (Participation count)**.** *Let $\beta \in (0,1)$ and put*

$$\mathsf{n}_\beta(q) \stackrel{\mathrm{def}}{=} \frac{\log\big((1-\beta)^2 q\big)}{2\log(1/\beta)}, \qquad \mathsf{c}_\beta \stackrel{\mathrm{def}}{=} \frac{\sqrt{\pi/2}}{1-\beta}, \qquad \mathsf{c}'_\beta \stackrel{\mathrm{def}}{=} 1 + \frac{1}{1-\beta^2},$$

$$\mathsf{e}_\beta^- \stackrel{\mathrm{def}}{=} \beta^2\,\Phi'(1), \qquad \mathsf{e}_\beta^+ \stackrel{\mathrm{def}}{=} \frac{\sqrt{\pi/2}}{1-\beta} + \frac{1}{1-\beta^2}.$$

(i) If $(1-\beta)^2 q \ge 1$, then $\mathsf{n}_\beta(q) - \mathsf{c}_\beta < r_\beta(q) \le \mathsf{n}_\beta(q) + \mathsf{c}'_\beta$. Moreover $r_\beta\big((1-\beta)^{-2}\big) \le \mathsf{c}'_\beta$.

(ii) For every $q > 0$, $0 < q\,r'_\beta(q) \le \mathsf{e}_\beta^+$. If $(1-\beta)^2 q \ge 1$, then $q\,r'_\beta(q) \ge \mathsf{e}_\beta^-$.

The value $\Phi'(1) = \mathbb{E}[g^2/(1+g^2)^2] = 0.15568\ldots$ enters only e_β^-.

Proof. Put $x_j \stackrel{\text{def}}{=} w_j^2 q = (1-\beta)^2 q\,\beta^{2j}$. Then $r_\beta(q) = \sum_j \Phi(x_j)$ and $q\,r'_\beta(q) = \sum_j x_j \Phi'(x_j)$ by lemma SF.4(i).

For part (i), let $(1-\beta)^2 q \geq 1$ and $n \stackrel{\text{def}}{=} \lfloor \mathsf{n}_\beta(q) \rfloor$. Since $(1-\beta)^2 q = \beta^{-2\mathsf{n}_\beta(q)}$, we have $x_j = \beta^{2(j-\mathsf{n}_\beta(q))}$, so $x_j \geq 1$ exactly for $j \leq n$, and

$$x_j^{-1/2} = \beta^{\mathsf{n}_\beta(q)-j} \leq \beta^{n-j} \qquad (j \leq n).$$

The lower bound uses $\Phi(x) \geq 1 - \sqrt{\pi/2}\,x^{-1/2}$ (lemma SF.4(ii)) on the terms $j \leq n$:

$$r_\beta(q) \geq \sum_{j \leq n} \Big(1 - \sqrt{\frac{\pi}{2}}\,x_j^{-1/2}\Big) \geq n + 1 - \sqrt{\frac{\pi}{2}} \sum_{k \geq 0} \beta^k = n + 1 - \mathsf{c}_\beta > \mathsf{n}_\beta(q) - \mathsf{c}_\beta.$$

The upper bound uses $\Phi \leq 1$ on the terms $j \leq n$ and $\Phi(x) \leq x$ on the others. Since $x_{n+1} < 1$ and $x_j = \beta^{2(j-n-1)} x_{n+1}$,

$$r_\beta(q) \leq n + 1 + \sum_{j > n} x_j < n + 1 + \frac{1}{1-\beta^2} \leq \mathsf{n}_\beta(q) + \mathsf{c}'_\beta.$$

At $q = (1-\beta)^{-2}$ one has $\mathsf{n}_\beta(q) = 0$, which gives the second claim.

For part (ii), split the sum at $x_j = 1$ and use $x\Phi'(x) \leq \min(x, \sqrt{\pi/2}\,x^{-1/2})$. The numbers $x_j^{-1/2}$ over the terms with $x_j \geq 1$, taken from the largest index down, are at most $1, \beta, \beta^2, \ldots$, and the numbers x_j over the terms with $x_j < 1$, taken from the smallest index up, are below $1, \beta^2, \beta^4, \ldots$. The two geometric sums give e^+_β, and positivity is lemma SF.4(i). If $x_0 \geq 1$, then $x_{n+1} \in [\beta^2, 1)$, and since Φ' is decreasing,

$$q\,r'_\beta(q) \geq x_{n+1}\Phi'(x_{n+1}) \geq \beta^2 \Phi'(1) = \mathsf{e}^-_\beta.$$

□

Two constants control the error of the participation representation. They both rest on one geometric sum.

Lemma SF.7 (Two constants)**.** *Let* $\beta \in (0,1)$.

(i) If $y_0 > 0$ *and* $y_j = y_0 \beta^j$, *then* $\sum_{j \geq 0} \min(y_j, 1/y_j) < 2/(1-\beta)$.

(ii) $\mathsf{g}_\beta \stackrel{\text{def}}{=} \sup_{q>0} \sqrt{q}\,L_\beta(q) < \dfrac{2}{1-\beta}$.

(iii) $\mathsf{g}'_\beta \stackrel{\text{def}}{=} \sup_{q>0} \Big|\Lambda_\beta(q) - \sqrt{2\pi}\,r_\beta(q)\Big| < \dfrac{2\pi}{1-\beta}$.

At $\beta = 0$, $\sqrt{q}\,J(q) \leq 1$ *and* $|\Lambda_0(q) - \sqrt{2\pi}\,\Phi(q)| \leq \pi$ *for every* $q > 0$.

Proof. Part (i) splits the sum at the last index with $y_j \geq 1$, and the bound is uniform in y_0. If there is no such index, every $y_j < 1$ and the sum is $y_0/(1-\beta) < 1/(1-\beta)$. Otherwise let j_0 be that index and $y \stackrel{\text{def}}{=} y_{j_0}$. Then $1 \leq y < 1/\beta$, because $y_{j_0+1} = \beta y < 1$, and

$$\sum_{j \leq j_0} \frac{1}{y_j} = \frac{1}{y} \sum_{k=0}^{j_0} \beta^k \leq \frac{1}{(1-\beta)\,y}, \qquad \sum_{j > j_0} y_j = \frac{\beta y}{1-\beta}.$$

The function $y \mapsto 1/y + \beta y$ is convex and equals $1+\beta$ at both ends of $[1, 1/\beta]$. Hence

$$\sum_{j \geq 0} \min\Big(y_j, \frac{1}{y_j}\Big) \leq \frac{1+\beta}{1-\beta} < \frac{2}{1-\beta}$$

for every $y_0 > 0$, and the suprema in (ii) and (iii) inherit the strict inequality.

Part (ii) is part (i) with $y_j = w_j\sqrt{q}$. By lemma SF.4(iii),

$$\sqrt{q}\,L_\beta(q) = \sum_j y_j\,J(y_j^2) \le \sum_j \min\Big(y_j, \frac{1}{y_j}\Big).$$

Part (iii) rests on a pointwise bound. Put $\varphi \stackrel{\text{def}}{=} \Lambda_0 - \sqrt{2\pi}\,\Phi$. By lemma SF.4(i),(iv),

$$\Lambda_\beta(q) - \sqrt{2\pi}\,r_\beta(q) = \sum_j \varphi(w_j^2 q).$$

We claim that $|\varphi(z)| \le \pi\min(\sqrt{z}, z^{-1/2})$. For $z \le 1$ the two terms $\Lambda_0(z) \le 2\sqrt{z}$ and $\sqrt{2\pi}\,\Phi(z) \le \sqrt{2\pi}\,z \le \sqrt{2\pi}\,\sqrt{z}$ are nonnegative, so $|\varphi(z)| \le \sqrt{2\pi}\,\sqrt{z} < \pi\sqrt{z}$. For $z \ge 1$,

$$\varphi(z) = \sqrt{2\pi}\,\big(1 - \Phi(z)\big) - \big(\sqrt{2\pi} - \Lambda_0(z)\big)$$

is a difference of two nonnegative terms, bounded by $\sqrt{2\pi}\sqrt{\pi/2}\,z^{-1/2} = \pi z^{-1/2}$ and by $2z^{-1/2}$ (lemma SF.4(ii),(iv)), so $|\varphi(z)| \le \pi z^{-1/2}$. Part (i) with $y_j = w_j\sqrt{q}$ gives the bound.

At $\beta = 0$ the same pointwise bounds give $\sqrt{q}\,J(q) \le \min(\sqrt{q}, 1/\sqrt{q}) \le 1$ and $|\varphi(q)| \le \pi$. □

The participation level of the root at s is $r_\beta(q(-s)) = \psi(s)$. The next lemma changes variables from s to this level.

Lemma SF.8 (Participation identity)**.** *Let $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$, and assume either $\beta \in (0,1)$, or $\beta = 0$ and $\gamma_\wedge < 1$. For $0 \le m \le \gamma_\wedge$ put $Q_m \stackrel{\text{def}}{=} r_\beta^{-1}(m)$,*

$$\mathsf{i}_\beta(m) \stackrel{\text{def}}{=} \sqrt{Q_m}\,L_\beta(Q_m) + \tfrac{1}{2}\Lambda_\beta(Q_m),$$

$$\widetilde{\mathcal{I}}(H) \stackrel{\text{def}}{=} \int_0^H \big(-\mathsf{W}'(m)\big)\,\mathsf{i}_\beta(m)\,\mathrm{d}m + \tfrac{1}{2}\mathsf{W}(H)\,\Lambda_\beta(Q_H),$$

and put $m_\varrho \stackrel{\text{def}}{=} \psi(\varrho)$ for $\varrho > 0$ and $m_0 \stackrel{\text{def}}{=} \gamma_\wedge$. Then for every $\varrho \ge 0$,

$$\widetilde{\mathcal{I}}(m_\varrho) \le \int_0^\infty \mathsf{G}_\beta(\varrho + u^2)\,\mathrm{d}u \le \widetilde{\mathcal{I}}(m_\varrho) + \mathsf{W}(m_\varrho)\sqrt{Q_{m_\varrho}}\,L_\beta(Q_{m_\varrho}), \tag{SF.30}$$

and the last term vanishes at $\varrho = 0$, so that equality holds there. If $\beta = 0$ and $\gamma_\wedge \ge 1$, the same holds with $0 \le m < 1$, $Q_m = \Phi^{-1}(m)$, $m_0 \stackrel{\text{def}}{=} 1$ and $\Lambda_0(Q_1) \stackrel{\text{def}}{=} \sqrt{2\pi}$; the boundary term of $\widetilde{\mathcal{I}}(1)$ is then $\frac{1}{2}\mathsf{W}(1)\sqrt{2\pi}$.

Proof. The proof changes variables to the participation level and integrates by parts once. Let $m_\star$ be $\gamma_\wedge$ in the first two cases and 1 in the last. In the first two cases $Q_{\gamma_\wedge} < \infty$, by lemma SF.4(i) for $\beta > 0$ and because $\Phi^{-1}(\gamma_\wedge) < \infty$ for $\gamma_\wedge < 1$. The map $m \mapsto Q_m$ is continuously differentiable on $(0, m_\star)$ with $\mathrm{d}Q_m/\mathrm{d}m = 1/r'_\beta(Q_m)$, by lemma SF.4(i). Put

$$s(m) \stackrel{\text{def}}{=} \frac{\mathsf{W}(m)^2}{Q_m} \qquad (0 < m < m_\star).$$

It is a continuously differentiable, strictly decreasing bijection of $(0, m_\star)$ onto $(0, \infty)$: W decreases and Q_m increases, $s(m) \to \infty$ as $m \downarrow 0$, and $s(m) \to 0$ as $m \uparrow m_\star$, because $\mathsf{W}(\gamma_\wedge) = 0$ in the first two cases and $Q_m \to \infty$ in the last. Since

$$s(m)\,Q_m = \mathsf{W}(m)^2 = \big(\gamma_{\text{out}} - r_\beta(Q_m)\big)\big(\gamma_{\text{in}} - r_\beta(Q_m)\big),$$

lemma SF.5(i) gives $q(-s(m)) = Q_m$ and $\psi(s(m)) = m$. In particular $s(m_\varrho) = \varrho$ for $\varrho > 0$.

The function G_β is nonincreasing, because $q(-s)$ decreases in s (lemma SF.5(i)) and $qL_\beta(q)$ increases in q (lemma SF.4(v)). The ridge is removed first. The substitution $s = \varrho + u^2$ gives

$$\int_0^\infty \mathsf{G}_\beta(\varrho + u^2)\,\mathrm{d}u = \int_\varrho^\infty \mathsf{G}_\beta(s)\,\frac{\mathrm{d}s}{2\sqrt{s - \varrho}}.$$

The kernel $\frac{1}{2\sqrt{s-\varrho}} - \frac{1}{2\sqrt{s}}$ is nonnegative on (ϱ, ∞) with integral $\sqrt{\varrho}$, and $\mathsf{G}_\beta \le \mathsf{G}_\beta(\varrho)$ there. Hence the integral lies between

$$\mathcal{I}_\star(\varrho) \stackrel{\text{def}}{=} \int_\varrho^\infty \mathsf{G}_\beta(s)\,\frac{\mathrm{d}s}{2\sqrt{s}} \qquad \text{and} \qquad \mathcal{I}_\star(\varrho) + \sqrt{\varrho}\,\mathsf{G}_\beta(\varrho),$$

and, at $s(m_\varrho) = \varrho$,

$$\sqrt{\varrho}\,\mathsf{G}_\beta(\varrho) = \frac{\mathsf{W}(m_\varrho)}{\sqrt{Q_{m_\varrho}}}\,Q_{m_\varrho}L_\beta(Q_{m_\varrho}) = \mathsf{W}(m_\varrho)\sqrt{Q_{m_\varrho}}\,L_\beta(Q_{m_\varrho}).$$

The change of variables $s = s(m)$, $m \in (0, m_\varrho)$, transforms $\mathcal{I}_\star(\varrho)$. Differentiating $s = \mathsf{W}^2/Q_m$,

$$\begin{aligned}\mathsf{G}_\beta(s(m))\,\frac{-s'(m)}{2\sqrt{s(m)}} &= Q_m L_\beta(Q_m)\Big(\frac{-\mathsf{W}'(m)}{\sqrt{Q_m}} + \frac{\mathsf{W}(m)}{2Q_m^{3/2}\,r'_\beta(Q_m)}\Big)\\ &= \sqrt{Q_m}\,L_\beta(Q_m)\big(-\mathsf{W}'(m)\big) + \frac{\mathsf{W}(m)}{2}\,\frac{\mathrm{d}}{\mathrm{d}m}\Lambda_\beta(Q_m),\end{aligned}$$

since $\frac{\mathrm{d}}{\mathrm{d}m}\Lambda_\beta(Q_m) = Q_m^{-1/2}L_\beta(Q_m)/r'_\beta(Q_m)$. An integration by parts on $[0, H]$, with $\Lambda_\beta(Q_0) = 0$, turns the last term into

$$\int_0^H \frac{\mathsf{W}(m)}{2}\,\frac{\mathrm{d}}{\mathrm{d}m}\Lambda_\beta(Q_m)\,\mathrm{d}m = \tfrac{1}{2}\mathsf{W}(H)\Lambda_\beta(Q_H) + \int_0^H \big(-\mathsf{W}'(m)\big)\,\tfrac{1}{2}\Lambda_\beta(Q_m)\,\mathrm{d}m,$$

so that $\mathcal{I}_\star(\varrho) = \widetilde{\mathcal{I}}(m_\varrho)$ for $\varrho > 0$. At $\varrho = 0$ the boundary term $\sqrt{\varrho}\,\mathsf{G}_\beta(\varrho)$ is absent, and we let $H \uparrow m_\star$ by monotone convergence. In the first two cases $\mathsf{W}(m_\star) = 0$ and $\Lambda_\beta(Q_{m_\star}) < \infty$, so the boundary term of $\widetilde{\mathcal{I}}$ vanishes. In the last case $\Lambda_0(Q_H) \to \sqrt{2\pi}$ by lemma SF.4(iv), which gives the boundary term $\frac{1}{2}\mathsf{W}(1)\sqrt{2\pi}$. □

The main term of $\widetilde{\mathcal{I}}$ is $\frac{\sqrt{2\pi}}{2}\int_0^H \mathsf{W}$, because $\Lambda_\beta(Q_m)$ is $\sqrt{2\pi}\,m$ up to the bounded error of lemma SF.7(iii). This gives the drift as a fixed function of the participation level.

Theorem SF.3 (Master drift estimate)**.** *Let $\beta \in (0,1)$, $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$ and $\varrho \ge 0$, with the convention $\psi(0) \stackrel{\text{def}}{=} \gamma_\wedge$.*

(i) The drift satisfies

$$\Big|\mathfrak{d}_{\text{iso}}(\varrho) - \frac{1}{\sqrt{\pi}\,\gamma_{\text{in}}\gamma_{\text{out}}}\int_0^{\psi(\varrho)} \mathsf{W}(m)\,\mathrm{d}m\Big| \le \frac{\sqrt{2}\,(\mathsf{g}_\beta + \mathsf{g}'_\beta/2)}{\pi\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}} < \frac{2.32}{(1-\beta)\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}}. \tag{SF.31}$$

(ii) In the square case, with $\Xi_0(x) \stackrel{\text{def}}{=} \pi^{-1/2}\big(x - \frac{1}{2}x^2\big)$,

$$\Big|\mathfrak{d}_{\text{iso}}(\varrho) - \Xi_0\Big(\frac{\psi(\varrho)}{\gamma}\Big)\Big| < \frac{2.32}{(1-\beta)\gamma}. \tag{SF.32}$$

Proof. For part (i) write $H \stackrel{\text{def}}{=} \psi(\varrho) = m_\varrho$ and $\Lambda_\beta(Q_m) = \sqrt{2\pi}\,m + k(m)$, so that $|k| \le \mathsf{g}'_\beta$ by lemma SF.7(iii) and $r_\beta(Q_m) = m$. An integration by parts gives

$$\int_0^H m\big(-\mathsf{W}'(m)\big)\,\mathrm{d}m + H\,\mathsf{W}(H) = \int_0^H \mathsf{W}(m)\,\mathrm{d}m.$$

By lemma SF.8, therefore,

$$\int_0^\infty \mathsf{G}_\beta(\varrho + u^2)\,\mathrm{d}u = \frac{\sqrt{2\pi}}{2}\int_0^H \mathsf{W}(m)\,\mathrm{d}m + R,$$

$$R = \int_0^H \big(-\mathsf{W}'(m)\big)\Big[\sqrt{Q_m}\,L_\beta(Q_m) + \frac{k(m)}{2}\Big]\mathrm{d}m + \frac{1}{2}\mathsf{W}(H)\,k(H) + E_\varrho,$$

where $0 \le E_\varrho \le \mathsf{g}_\beta\,\mathsf{W}(H)$ by eq. (SF.30), whose last term is at most $\mathsf{g}_\beta\,\mathsf{W}(H)$.

Since $-\mathsf{W}' \ge 0$ and $0 \le \sqrt{Q_m}L_\beta(Q_m) \le \mathsf{g}_\beta$,

$$|R| \le \Big(\mathsf{g}_\beta + \frac{\mathsf{g}'_\beta}{2}\Big)\big(\mathsf{W}(0) - \mathsf{W}(H)\big) + \Big(\mathsf{g}_\beta + \frac{\mathsf{g}'_\beta}{2}\Big)\mathsf{W}(H) = \Big(\mathsf{g}_\beta + \frac{\mathsf{g}'_\beta}{2}\Big)\sqrt{\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}.$$

Multiplying by $\sqrt{2}/(\pi\gamma_{\mathrm{in}}\gamma_{\mathrm{out}})$ (eq. (SF.29)) and using $\sqrt{2}\cdot\sqrt{2\pi}/(2\pi) = \pi^{-1/2}$ gives the first inequality of eq. (SF.31). By lemma SF.7(ii),(iii), $\mathsf{g}_\beta + \mathsf{g}'_\beta/2 < (2+\pi)/(1-\beta)$, and $\sqrt{2}(2+\pi)/\pi = 2.3146\ldots$.

Part (ii) is part (i) at $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = \gamma$, where

$$\frac{1}{\gamma^2}\int_0^{\psi}(\gamma - m)\,\mathrm{d}m = \frac{\psi}{\gamma} - \frac{1}{2}\Big(\frac{\psi}{\gamma}\Big)^2.$$

□

SF.9 Asymptotics in the aspect ratios

In this subsection and the next, $\beta = 0$ and $C = C_0$, as in appendix SF.7. The small-aspect limit is a Marchenko–Pastur-type law, which we write down explicitly with its atom at zero.

Lemma SF.9 (The limiting law)**.** *For $r > 0$ put*

$$a_r \stackrel{\text{def}}{=} \frac{(1-\sqrt{r})^2}{r}, \qquad b_r \stackrel{\text{def}}{=} \frac{(1+\sqrt{r})^2}{r}, \qquad \omega_r \stackrel{\text{def}}{=} \Big(1 - \frac{1}{r}\Big)_+.$$

The measure

$$\mathsf{M}_r(\mathrm{d}x) \stackrel{\text{def}}{=} \omega_r\,\delta_0(\mathrm{d}x) + \frac{\sqrt{(b_r - x)(x - a_r)}}{2\pi x}\,\mathbf{1}_{(a_r,b_r)}(x)\,\mathrm{d}x \tag{SF.33}$$

is a probability measure with first moment $1/r$. For $t > 0$ its transform $m_r(t) \stackrel{\text{def}}{=} S_{\mathsf{M}_r}(-t)$ is the unique positive root m of

$$t\,m^2 + \Big(t + \frac{1}{r} - 1\Big)m - 1 = 0, \tag{SF.34}$$

namely, with $A_r \stackrel{\text{def}}{=} 1 + 1/r$ and $D_r \stackrel{\text{def}}{=} 2/\sqrt{r}$,

$$m_r(t) = \frac{1 - 1/r - t + \sqrt{(A_r + t)^2 - D_r^2}}{2t}. \tag{SF.35}$$

Proof. One integral carries the whole computation. Write $a = a_r$, $b = b_r$, $A = A_r = (a+b)/2$ and $D = D_r = (b-a)/2$, and for $t \ge 0$ put

$$I(t) \stackrel{\text{def}}{=} \int_a^b \frac{\sqrt{(b-x)(x-a)}}{x+t}\,\mathrm{d}x.$$

The substitution $x = A + D\cos\vartheta$, $\vartheta \in [0,\pi]$, gives $\sqrt{(b-x)(x-a)} = D\sin\vartheta$. With $c \stackrel{\text{def}}{=} A + t \ge D$ and the identity

$$\frac{D^2\sin^2\vartheta}{c + D\cos\vartheta} = c - D\cos\vartheta - \frac{c^2 - D^2}{c + D\cos\vartheta},$$

together with $\int_0^\pi (c + D\cos\vartheta)^{-1}\mathrm{d}\vartheta = \pi/\sqrt{c^2 - D^2}$ for $c > D$, obtained by the substitution $v = \tan(\vartheta/2)$, this gives

$$I(t) = \pi\Big(A + t - \sqrt{(A+t)^2 - D^2}\Big), \tag{SF.36}$$

for $t > 0$, and for $t = 0$ by monotone convergence.

The mass and the first moment follow. Since $A^2 - D^2 = (1 - 1/r)^2$, the continuous part of M_r has mass

$$\frac{I(0)}{2\pi} = \frac{1 + 1/r - |1 - 1/r|}{2} = \min\Big(1, \frac{1}{r}\Big),$$

and adding ω_r gives 1. The first moment is

$$\frac{1}{2\pi}\int_a^b \sqrt{(b-x)(x-a)}\,\mathrm{d}x = \frac{D^2}{2\pi}\int_0^\pi \sin^2\vartheta\,\mathrm{d}\vartheta = \frac{D^2}{4} = \frac{1}{r}.$$

The transform is a difference of two values of I. By the partial fractions $\frac{1}{x(x+t)} = \frac{1}{t}\big(\frac{1}{x} - \frac{1}{x+t}\big)$,

$$m_r(t) = \frac{\omega_r}{t} + \frac{I(0) - I(t)}{2\pi t},$$

and inserting eq. (SF.36) with $2\omega_r - |1 - 1/r| = 1 - 1/r$ gives eq. (SF.35). Since

$$\Big(t + \frac{1}{r} - 1\Big)^2 + 4t = (A+t)^2 - D^2,$$

eq. (SF.35) is the root of eq. (SF.34) with the positive square root. The product of the two roots is $-1/t < 0$, so it is the only positive root. □

Theorem SF.4 (Small aspect ratios)**.** *Let $\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)} \to 0$ with $r_j \overset{\mathrm{def}}{=} \gamma_{\mathrm{out}}^{(j)}/\gamma_{\mathrm{in}}^{(j)} \to r \in (0,\infty)$, and let $(\mathfrak{m}_{\mathrm{out}}^{(j)}, \mathfrak{m}_{\mathrm{in}}^{(j)})$ be the isotropic measure pair at $(\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)})$ of proposition SF.6. Let P_j be the image of the probability measure $\mathfrak{m}_{\mathrm{out}}^{(j)}/\gamma_{\mathrm{out}}^{(j)}$ under $\lambda \mapsto \lambda/\gamma_{\mathrm{out}}^{(j)}$, and write $\mathsf{m}_j(t) \overset{\mathrm{def}}{=} S_{\mathsf{P}_j}(-t)$.*

(i) For every $t > 0$,

$$\Big|t\,\mathsf{m}_j(t)^2 + \Big(t + \frac{1}{r_j} - 1\Big)\mathsf{m}_j(t) - 1\Big| \le \frac{3\gamma_{\mathrm{out}}^{(j)}}{r_j^2 t^2}. \tag{SF.37}$$

(ii) $\mathsf{P}_j \to \mathsf{M}_r$ weakly, and $\int\sqrt{x}\,\mathsf{P}_j(\mathrm{d}x) \to J_r \overset{\mathrm{def}}{=} \int\sqrt{x}\,\mathsf{M}_r(\mathrm{d}x) > 0$.

(iii) The constant satisfies

$$C\big(\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)}\big) = \frac{r_j}{\sqrt{2\gamma_{\mathrm{out}}^{(j)}}}\int\sqrt{x}\,\mathsf{P}_j(\mathrm{d}x) \sim \frac{r}{\sqrt{2\gamma_{\mathrm{out}}^{(j)}}}\,J_r,$$
$$J_r = \int_{a_r}^{b_r}\frac{\sqrt{(b_r - x)(x - a_r)}}{2\pi\sqrt{x}}\,\mathrm{d}x. \tag{SF.38}$$

In particular $J_1 = 8/(3\pi)$, and $C(\gamma) \sim 8/(3\pi\sqrt{2\gamma})$ as $\gamma \downarrow 0$.

Proof. We drop the index j from $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}, \mathfrak{m}_{\mathrm{out}}, \sigma, \tilde\sigma$ and keep it on r_j, P_j and m_j. Fix $t > 0$. The transform of P_j is the branch at $-\gamma_{\mathrm{out}}t$:

$$\mathsf{m}_j(t) = \frac{1}{\gamma_{\mathrm{out}}}\int\frac{\mathfrak{m}_{\mathrm{out}}(\mathrm{d}\lambda)}{\lambda/\gamma_{\mathrm{out}} + t} = \sigma(-\gamma_{\mathrm{out}}t),$$

and by eq. (SF.6) the companion is

$$\tilde\sigma(-\gamma_{\mathrm{out}}t) = \sigma(-\gamma_{\mathrm{out}}t) + \frac{\gamma_{\mathrm{in}} - \gamma_{\mathrm{out}}}{\gamma_{\mathrm{out}}t} = \mathsf{m}_j(t) + \frac{1/r_j - 1}{t}.$$

Both transforms are bounded by the mass over $\gamma_{\mathrm{out}}t$: since P_j is a probability measure, $\mathsf{m}_j(t) \le 1/t$, and $\tilde\sigma(-\gamma_{\mathrm{out}}t) \le \gamma_{\mathrm{in}}/(\gamma_{\mathrm{out}}t) = 1/(r_j t)$. The argument of Φ in the closure is therefore small:

$$v_j \overset{\mathrm{def}}{=} \gamma_{\mathrm{out}}t\,\sigma(-\gamma_{\mathrm{out}}t)\,\tilde\sigma(-\gamma_{\mathrm{out}}t) = \gamma_{\mathrm{out}}\,\mathsf{m}_j\Big(t\,\mathsf{m}_j + \frac{1}{r_j} - 1\Big) \le \frac{\gamma_{\mathrm{out}}}{r_j t}.$$

For part (i), the closure eq. (SF.25) at $\gamma_{\mathrm{out}}t$ reads

$$\gamma_{\mathrm{out}}\big(1 - t\,\mathsf{m}_j\big) = \Phi(v_j) = v_j - e_j, \qquad 0 \le e_j \le 3v_j^2,$$

by lemma SF.1(iii). Dividing by γ_{out} and inserting v_j,

$$t\,\mathsf{m}_j^2 + \Big(t + \frac{1}{r_j} - 1\Big)\mathsf{m}_j - 1 = \frac{e_j}{\gamma_{\mathrm{out}}},$$

and $e_j/\gamma_{\text{out}} \le 3v_j^2/\gamma_{\text{out}} \le 3\gamma_{\text{out}}/(r_j^2 t^2)$.

For part (ii), the residual eq. (SF.37) tends to zero and $\mathsf{m}_j(t) \in [0, 1/t]$. Every limit point m of $(\mathsf{m}_j(t))_j$ is therefore a nonnegative root of eq. (SF.34), and $m \neq 0$ because the constant term is -1. By lemma SF.9, $m = m_r(t)$, so $\mathsf{m}_j(t) \to S_{\mathsf{M}_r}(-t)$ for every $t > 0$. The first moments are bounded: by lemma SF.3(ii),

$$\int x \, \mathsf{P}_j(\mathrm{d}x) = \frac{1}{\gamma_{\text{out}}^2} \int \lambda \, \mathfrak{m}_{\text{out}}(\mathrm{d}\lambda) = \frac{\gamma_{\text{in}}\gamma_{\text{out}}}{\gamma_{\text{out}}^2} = \frac{1}{r_j},$$

and M_r has first moment $1/r$. Lemma SF.2(i) gives the weak convergence and the convergence of the half-moments. The limit J_r is positive because M_r gives positive mass $\min(1, 1/r)$ to $(a_r, b_r) \subset (0, \infty)$.

For part (iii), eq. (SF.27) and the change of variables $\lambda = \gamma_{\text{out}} x$ give

$$C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{1}{\sqrt{2}\,\gamma_{\text{in}}\gamma_{\text{out}}} \gamma_{\text{out}}^{3/2} \int \sqrt{x} \, \mathsf{P}_j(\mathrm{d}x) = \frac{r_j}{\sqrt{2\gamma_{\text{out}}}} \int \sqrt{x} \, \mathsf{P}_j(\mathrm{d}x),$$

and part (ii) gives the asymptotic equivalence. The atom of M_r at 0 does not contribute to J_r. For $r = 1$, $a_1 = 0$ and $b_1 = 4$, so

$$J_1 = \frac{1}{2\pi} \int_0^4 \sqrt{4 - x} \, \mathrm{d}x = \frac{1}{2\pi} \cdot \frac{16}{3} = \frac{8}{3\pi}.$$

□

Proposition SF.8 (Reciprocity). *(i) Let $(\mathfrak{m}_{\text{out}}, \mathfrak{m}_{\text{in}})$ be the isotropic measure pair at $(\gamma_{\text{in}}, \gamma_{\text{out}})$, let $b = \gamma_{\text{in}}\gamma_{\text{out}}$, and let P be the image of $\mathfrak{m}_{\text{out}}/\gamma_{\text{out}}$ under $\lambda \mapsto \lambda/\gamma_{\text{out}}$. Then P is the image of $\widehat{\mathfrak{m}}_b/\gamma_{\text{out}}$ under $x \mapsto \gamma_{\text{in}} x$. If $\widehat{\mathfrak{m}}_b$ has a density $\widehat{\rho}$, then P has the density $\widehat{\rho}(x/\gamma_{\text{in}})/(\gamma_{\text{in}}\gamma_{\text{out}})$. Its transform and the constant are*

$$S_{\mathsf{P}}(-t) = \sigma(-\gamma_{\text{out}} t) = \frac{1}{\gamma_{\text{in}}\gamma_{\text{out}}} S_{\widehat{\mathfrak{m}}_b}\Big(-\frac{t}{\gamma_{\text{in}}}\Big), \qquad C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \frac{\sqrt{\gamma_{\text{out}}}}{\sqrt{2}\,\gamma_{\text{in}}} \int \sqrt{x} \, \mathsf{P}(\mathrm{d}x).$$

(ii) For every $r > 0$, $J_{1/r} = r^{3/2} J_r$, and the right side of eq. (SF.38) is unchanged when γ_{in} and γ_{out} are exchanged (with r replaced by $1/r$).

(iii) For every $r > 0$ the endpoint pairs $\{(1 \pm \sqrt{r})^2/r\}$ and $\{(1 \pm 1/\sqrt{r})^2\}$ are equal.

Proof. For part (i), the image of λ under the two maps is $\lambda/(\gamma_{\text{in}}\gamma_{\text{out}})$ followed by $\gamma_{\text{in}}\lambda/(\gamma_{\text{in}}\gamma_{\text{out}}) = \lambda/\gamma_{\text{out}}$. The density is the change of variables with the Jacobian $1/\gamma_{\text{in}}$ and the mass normalization $1/\gamma_{\text{out}}$. The transform is the first computation of the proof of theorem SF.4 together with $S_{\widehat{\mathfrak{m}}_b}(-t/\gamma_{\text{in}}) = b\,\sigma(-\gamma_{\text{out}} t)$ from proposition SF.7, and the constant is the computation of theorem SF.4(iii) with $r_j = \gamma_{\text{out}}/\gamma_{\text{in}}$, since $r_j/\sqrt{\gamma_{\text{out}}} = \sqrt{\gamma_{\text{out}}}/\gamma_{\text{in}}$.

For part (ii), the endpoints at $1/r$ are $a_{1/r} = r\,a_r$ and $b_{1/r} = r\,b_r$. Writing ρ_r for the density of the continuous part of M_r, direct substitution gives

$$\rho_{1/r}(ry) = \frac{\sqrt{r^2(b_r - y)(y - a_r)}}{2\pi r y} = \rho_r(y),$$

so that

$$J_{1/r} = \int \sqrt{x}\,\rho_{1/r}(x)\,\mathrm{d}x = \int \sqrt{ry}\,\rho_r(y)\,r\,\mathrm{d}y = r^{3/2} J_r.$$

Exchanging the ratios replaces (r, γ_{out}) by $(1/r, \gamma_{\text{in}})$ in eq. (SF.38), and with $\gamma_{\text{out}} = r\gamma_{\text{in}}$,

$$\frac{1/r}{\sqrt{2\gamma_{\text{in}}}} J_{1/r} = \frac{\sqrt{r}\,J_r}{\sqrt{2\gamma_{\text{in}}}} = \frac{r\,J_r}{\sqrt{2\gamma_{\text{out}}}}.$$

Part (iii) is the identity $(1 \pm \sqrt{r})^2/r = (1/\sqrt{r} \pm 1)^2$, whose right side is $(1 \pm 1/\sqrt{r})^2$ for the same choice of sign. □

Theorem SF.5 (Large aspect ratios). *Let $\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)} \to \infty$, without a condition on their ratio, and let $(\mathfrak{m}_{\mathrm{out}}^{(j)}, \mathfrak{m}_{\mathrm{in}}^{(j)})$ be the isotropic measure pair at $(\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)})$ of proposition SF.6. For j with $\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)} > 1$ write $\mathfrak{m}_{\mathrm{out}}^{(j)} = (\gamma_{\mathrm{out}}^{(j)} - 1)\delta_0 + \eta_j$ as in lemma SF.3(iv), let E_j be the image of the probability measure η_j under $\lambda \mapsto \lambda/(\gamma_{\mathrm{in}}^{(j)}\gamma_{\mathrm{out}}^{(j)})$, and put $\epsilon_j \overset{\mathrm{def}}{=} 1/\gamma_{\mathrm{in}}^{(j)} + 1/\gamma_{\mathrm{out}}^{(j)}$. Let X be the law of g^2 for $g \sim \mathcal{N}(0,1)$, which has density, first moment and half-moment*

$$\frac{e^{-x/2}}{\sqrt{2\pi x}}\,\mathbf{1}_{x>0}, \qquad \int x\,\mathsf{X}(\mathrm{d}x) = 1, \qquad \int \sqrt{x}\,\mathsf{X}(\mathrm{d}x) = \sqrt{\frac{2}{\pi}}. \tag{SF.39}$$

Then $\mathsf{E}_j \to \mathsf{X}$ weakly,

$$0 \le \sqrt{\frac{2}{\pi}} - \int \sqrt{x}\,\mathsf{E}_j(\mathrm{d}x) \le \frac{4}{\pi}\sqrt{\epsilon_j}, \tag{SF.40}$$

and

$$C\big(\gamma_{\mathrm{in}}^{(j)}, \gamma_{\mathrm{out}}^{(j)}\big) = \frac{1}{\sqrt{2\gamma_{\mathrm{in}}^{(j)}\gamma_{\mathrm{out}}^{(j)}}} \int \sqrt{x}\,\mathsf{E}_j(\mathrm{d}x) \sim \frac{1}{\sqrt{\pi\gamma_{\mathrm{in}}^{(j)}\gamma_{\mathrm{out}}^{(j)}}}. \tag{SF.41}$$

In particular $C(\gamma) \sim 1/(\sqrt{\pi}\,\gamma)$ as $\gamma \to \infty$.

Proof. We drop the index j from the aspect ratios and the branch, write $b = \gamma_{\mathrm{in}}\gamma_{\mathrm{out}}$, and fix j with $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 1$. For $t > 0$ put

$$\mathsf{s}_j(t) \overset{\mathrm{def}}{=} S_{\mathsf{E}_j}(-t), \qquad \mathsf{H}_j(t) \overset{\mathrm{def}}{=} 1 - t\,\mathsf{s}_j(t) = \int \frac{x}{x+t}\,\mathsf{E}_j(\mathrm{d}x) \in [0,1].$$

The closure at bt is a closure for E_j. By the decomposition of $\mathfrak{m}_{\mathrm{out}}$,

$$bt\,\sigma(-bt) = (\gamma_{\mathrm{out}} - 1) + t\,\mathsf{s}_j(t) = \gamma_{\mathrm{out}} - \mathsf{H}_j(t),$$

and by lemma SF.3(iv) the same computation on the input side gives $bt\,\tilde{\sigma}(-bt) = \gamma_{\mathrm{in}} - \mathsf{H}_j(t)$. The closure eq. (SF.25) at bt therefore reads

$$\mathsf{H}_j(t) = \Phi\big(v_j(t)\big), \qquad v_j(t) \overset{\mathrm{def}}{=} bt\,\sigma(-bt)\,\tilde{\sigma}(-bt) = \frac{1}{t}\Big(1 - \frac{\mathsf{H}_j(t)}{\gamma_{\mathrm{out}}}\Big)\Big(1 - \frac{\mathsf{H}_j(t)}{\gamma_{\mathrm{in}}}\Big).$$

The argument $v_j(t)$ is close to $1/t$. Since $0 \le \mathsf{H}_j \le 1$ and $(1-x)(1-y) \ge 1 - x - y$ for $x, y \ge 0$,

$$0 \le \frac{1}{t} - v_j(t) \le \frac{\epsilon_j}{t}.$$

The Gaussian law is the fixed point at $v = 1/t$. Indeed

$$\Phi(1/t) = \mathbb{E}\,\frac{g^2}{g^2 + t} = 1 - t\,S_{\mathsf{X}}(-t) = \int \frac{x}{x+t}\,\mathsf{X}(\mathrm{d}x).$$

Since Φ is increasing with slope at most 1 and takes values in $[0,1]$ (lemma SF.1(ii)),

$$0 \le \Phi(1/t) - \mathsf{H}_j(t) \le \min\Big(1, \frac{\epsilon_j}{t}\Big), \qquad |\mathsf{s}_j(t) - S_{\mathsf{X}}(-t)| \le \frac{\epsilon_j}{t^2}. \tag{SF.42}$$

The law X has the stated density by the change of variables $x = u^2$ in the Gaussian density, first moment $\mathbb{E}\,g^2 = 1$, and half-moment $\mathbb{E}\,|g| = \sqrt{2/\pi}$. The first moment of E_j is

$$\int x\,\mathsf{E}_j(\mathrm{d}x) = \frac{1}{b}\int \lambda\,\eta_j(\mathrm{d}\lambda) = \frac{1}{b}\int \lambda\,\mathfrak{m}_{\mathrm{out}}(\mathrm{d}\lambda) = 1,$$

by lemma SF.3(ii). Since $\epsilon_j \to 0$, eq. (SF.42) and lemma SF.2(i) give $\mathsf{E}_j \to \mathsf{X}$ weakly.

The quantitative bound integrates eq. (SF.42). By the half-moment formula eq. (SF.26) for X and for E_j,

$$\sqrt{\frac{2}{\pi}} - \int \sqrt{x}\,\mathsf{E}_j(\mathrm{d}x) = \frac{1}{\pi}\int_0^\infty t^{-1/2}\big(\Phi(1/t) - \mathsf{H}_j(t)\big)\,\mathrm{d}t,$$

whose integrand lies between 0 and $t^{-1/2}\min(1,\epsilon_j/t)$. Splitting at $t=\epsilon_j$,

$$\int_0^{\epsilon_j} t^{-1/2}\,\mathrm{d}t + \epsilon_j \int_{\epsilon_j}^{\infty} t^{-3/2}\,\mathrm{d}t = 2\sqrt{\epsilon_j} + 2\sqrt{\epsilon_j},$$

which proves eq. (SF.40).

Finally, eq. (SF.27), the vanishing of $\sqrt{\lambda}$ on the atom, and the change of variables $\lambda = bx$ give

$$C(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}) = \frac{1}{\sqrt{2}\,b}\int \sqrt{\lambda}\,\eta_j(\mathrm{d}\lambda) = \frac{\sqrt{b}}{\sqrt{2}\,b}\int \sqrt{x}\,\mathsf{E}_j(\mathrm{d}x),$$

which is the identity of eq. (SF.41); the equivalence follows from eq. (SF.40), since $\sqrt{2/\pi}/\sqrt{2b} = 1/\sqrt{\pi b}$. □

For aspect ratios at least one the participation representation of appendix SF.8 gives the constant exactly.

Proposition SF.9 (The constant for $N \geq B$)**.** *For every* $\gamma \geq 1$,

$$C(\gamma) = \frac{\gamma-1}{\sqrt{\pi}\,\gamma^2} + \frac{C(1)}{\gamma^2}.$$

Proof. Apply lemma SF.8 at $\beta = 0$, $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = \gamma \geq 1$ and $\varrho = 0$. In the square case $-\mathsf{W}' = 1$ and $\mathsf{W}(1) = \gamma - 1$, and at $\beta = 0$ the integrand $\mathsf{G}_0(u^2)$ is $\psi(u^2)$ by eq. (SF.24), so

$$\int_0^{\infty} \psi(u^2)\,\mathrm{d}u = \mathcal{I}_1 + \frac{(\gamma-1)\sqrt{2\pi}}{2}, \qquad \mathcal{I}_1 \stackrel{\mathrm{def}}{=} \int_0^1 \mathsf{i}_0(m)\,\mathrm{d}m.$$

The levels $Q_m = \Phi^{-1}(m)$, and hence $\mathcal{I}_1$, do not depend on γ. At $\gamma = 1$ the second term vanishes, and eq. (SF.21) gives $\mathcal{I}_1 = \pi C(1)/\sqrt{2}$. Multiplying by $\sqrt{2}/(\pi\gamma^2)$ and using $\sqrt{2}\cdot\sqrt{2\pi}/(2\pi) = \pi^{-1/2}$ gives the formula. □

With the value of $C(1)$ in theorem SF.6, proposition SF.9 is explicit. It is consistent with theorem SF.5, since $\gamma C(\gamma) \to \pi^{-1/2}$. On $\gamma \geq 1$ it supersedes the interpolation $[\pi\gamma(9\pi/32+\gamma)]^{-1/2}$ of eq. (9), which remains an approximation on $\gamma < 1$ only.

SF.10 The value at aspect ratio one

Theorem SF.6 (The constant at aspect ratio one)**.** *The constant eq.* (SF.21) *at* $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = 1$ *is*

$$C(1) = \frac{1 + \log(1+\sqrt{2})/\sqrt{2}}{2\sqrt{\pi}}, \tag{SF.43}$$

and

$$C(1) > \frac{3}{4\sqrt{\pi}} > \left[\pi\left(1+\frac{9\pi}{32}\right)\right]^{-1/2}. \tag{SF.44}$$

The constant $C(1)$ of definition SI.2, whose hypothesis holds at $\gamma = 1$ by proposition SF.6, is the constant eq. (SF.43), by proposition SF.7 and the remark following definition SF.2. Numerically $C(1) \approx 0.4579$, while the interpolation of eq. (9) at $N/B = 1$ is $[\pi(1+9\pi/32)]^{-1/2} \approx 0.4111$.

Proof. The proof changes variables from t to the closure variable ξ and then evaluates a Gaussian double integral.

At $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = 1$ the two transforms coincide, $\sigma = \tilde{\sigma}$, by eq. (SF.6), and the closure eq. (SF.24) reads

$$\psi(t) = 1 - t\,\sigma(-t) = \Phi\big(t\,\sigma(-t)^2\big).$$

Put $\xi(t) \stackrel{\mathrm{def}}{=} (2t\sigma(-t)^2)^{-1/2}$ and, for $\xi > 0$,

$$\bar{f}(\xi) \stackrel{\mathrm{def}}{=} 1 - f(\xi) = \mathbb{E}\,\frac{2\xi^2}{V^2 + 2\xi^2} \qquad (V \sim \mathcal{N}(0,1)).$$

By lemma SF.1(i), $\psi(t) = f(\xi(t))$, so that $t\sigma(-t) = \bar f(\xi(t))$. Dividing the square of this identity by $t\sigma(-t)^2 = 1/(2\xi(t)^2)$ gives

$$\sqrt{t} = \sqrt{2}\,\xi(t)\,\bar f(\xi(t)). \tag{SF.45}$$

The map $\xi \mapsto \sqrt{2}\,\xi\bar f(\xi)$ is a smooth increasing bijection of $(0,\infty)$. Indeed $\bar f(\xi) = 1 - \Phi(1/(2\xi^2))$ is smooth and strictly increasing, by lemma SF.1(ii) and lemma SB.1(ii), and it tends to 0 as $\xi \downarrow 0$ and to 1 as $\xi \to \infty$, by lemma SF.1(iv). By eq. (SF.45), $t \mapsto \xi(t)$ is its inverse composed with $\sqrt{t}$.

The constant becomes an integral in ξ. By eq. (SF.23) at $\gamma_{\rm in} = \gamma_{\rm out} = 1$ and the substitution $t = 2\xi^2\bar f(\xi)^2$, for which $t^{-1/2}\,\mathrm{d}t = 2\sqrt{2}\,(\xi\bar f)'\,\mathrm{d}\xi$,

$$C(1) = \frac{1}{\sqrt{2}\,\pi}\int_0^\infty t^{-1/2}\psi(t)\,\mathrm{d}t = \frac{2}{\pi}\int_0^\infty \big(1-\bar f(\xi)\big)\big(\xi\bar f(\xi)\big)'\,\mathrm{d}\xi.$$

Two integrations by parts simplify the integrand. Since

$$(1-\bar f)(\xi\bar f)' = \big[(1-\bar f)\,\xi\bar f\big]' + \xi\bar f\bar f', \qquad \xi\bar f\bar f' = \tfrac12\big[\xi(\bar f^2-1)\big]' + \tfrac12\big(1-\bar f^2\big),$$

and the boundary terms vanish,

$$C(1) = \frac{1}{\pi}\int_0^\infty \big(1-\bar f(\xi)^2\big)\,\mathrm{d}\xi. \tag{SF.46}$$

The boundary terms are at most ξ and ξ in absolute value near 0, and at most $1/(2\xi)$ and $1/\xi$ near infinity, because $1-\bar f(\xi) = f(\xi) = \Phi(1/(2\xi^2)) \le 1/(2\xi^2)$ by lemma SF.1(iii).

The square of $\bar f$ is a Gaussian double expectation. Let U, V be independent standard Gaussian variables, and put $\mathsf{a} \stackrel{\rm def}{=} |U|/\sqrt{2}$ and $\mathsf{b} \stackrel{\rm def}{=} |V|/\sqrt{2}$. Then $\bar f(\xi) = \mathbb{E}[\xi^2/(\xi^2+\mathsf{a}^2)]$, and independence gives

$$1-\bar f(\xi)^2 = \mathbb{E}\Big[1 - \frac{\xi^4}{(\xi^2+\mathsf{a}^2)(\xi^2+\mathsf{b}^2)}\Big] = \mathbb{E}\Big[\frac{(\mathsf{a}^2+\mathsf{b}^2)\xi^2+\mathsf{a}^2\mathsf{b}^2}{(\xi^2+\mathsf{a}^2)(\xi^2+\mathsf{b}^2)}\Big].$$

For $\mathsf{a} \ne \mathsf{b}$, which holds with probability one, partial fractions give

$$\frac{(\mathsf{a}^2+\mathsf{b}^2)\xi^2+\mathsf{a}^2\mathsf{b}^2}{(\xi^2+\mathsf{a}^2)(\xi^2+\mathsf{b}^2)} = \frac{1}{\mathsf{a}^2-\mathsf{b}^2}\Big(\frac{\mathsf{a}^4}{\xi^2+\mathsf{a}^2} - \frac{\mathsf{b}^4}{\xi^2+\mathsf{b}^2}\Big),$$

and with $\int_0^\infty (\xi^2+c^2)^{-1}\mathrm{d}\xi = \pi/(2c)$,

$$\int_0^\infty \frac{(\mathsf{a}^2+\mathsf{b}^2)\xi^2+\mathsf{a}^2\mathsf{b}^2}{(\xi^2+\mathsf{a}^2)(\xi^2+\mathsf{b}^2)}\,\mathrm{d}\xi = \frac{\pi}{2}\,\frac{\mathsf{a}^3-\mathsf{b}^3}{\mathsf{a}^2-\mathsf{b}^2} = \frac{\pi}{2}\,\frac{\mathsf{a}^2+\mathsf{a}\mathsf{b}+\mathsf{b}^2}{\mathsf{a}+\mathsf{b}}.$$

Inserting this in eq. (SF.46) gives

$$C(1) = \frac12\,\mathbb{E}\,\frac{\mathsf{a}^2+\mathsf{a}\mathsf{b}+\mathsf{b}^2}{\mathsf{a}+\mathsf{b}} = \frac{1}{2\sqrt{2}}\,\mathbb{E}\,\frac{U^2+|UV|+V^2}{|U|+|V|}.$$

The expectation is computed in polar coordinates. Write $(|U|,|V|) = R(\cos\vartheta,\sin\vartheta)$. By the Gaussian density in polar coordinates, R has density $re^{-r^2/2}$ on $(0,\infty)$, ϑ is uniform on $[0,\pi/2]$, and the two are independent. An integration by parts gives

$$\mathbb{E}\,R = \int_0^\infty r^2 e^{-r^2/2}\,\mathrm{d}r = \int_0^\infty e^{-r^2/2}\,\mathrm{d}r = \sqrt{\frac{\pi}{2}},$$

and the ratio is $R(1+\cos\vartheta\sin\vartheta)/(\cos\vartheta+\sin\vartheta)$, so

$$C(1) = \frac{1}{2\sqrt{2}}\sqrt{\frac{\pi}{2}}\cdot\frac{2}{\pi}\int_0^{\pi/2}\frac{1+\cos\vartheta\sin\vartheta}{\cos\vartheta+\sin\vartheta}\,\mathrm{d}\vartheta = \frac{1}{2\sqrt{\pi}}\int_0^{\pi/2}\frac{1+\cos\vartheta\sin\vartheta}{\cos\vartheta+\sin\vartheta}\,\mathrm{d}\vartheta.$$

Since $1+\cos\vartheta\sin\vartheta = \frac12\big((\cos\vartheta+\sin\vartheta)^2+1\big)$ and $\int_0^{\pi/2}(\cos\vartheta+\sin\vartheta)\,\mathrm{d}\vartheta = 2$, the last integral is

$$1+\frac12\int_0^{\pi/2}\frac{\mathrm{d}\vartheta}{\cos\vartheta+\sin\vartheta} = 1+\frac{1}{2\sqrt{2}}\int_{\pi/4}^{3\pi/4}\frac{\mathrm{d}\vartheta'}{\sin\vartheta'} = 1+\frac{1}{2\sqrt{2}}\log\frac{\tan(3\pi/8)}{\tan(\pi/8)} = 1+\frac{\log(1+\sqrt{2})}{\sqrt{2}},$$

using $\cos\vartheta + \sin\vartheta = \sqrt{2}\sin(\vartheta + \pi/4)$ with $\vartheta' = \vartheta + \pi/4$, the antiderivative $\log\tan(\vartheta'/2)$ of $1/\sin\vartheta'$, and $\tan(3\pi/8) = 1 + \sqrt{2} = 1/\tan(\pi/8)$. This proves eq. (SF.43).

For eq. (SF.44), the derivative of $\log(t + \sqrt{1+t^2})$ is $(1+t^2)^{-1/2}$, so

$$\log(1+\sqrt{2}) = \int_0^1 \frac{\mathrm{d}t}{\sqrt{1+t^2}} > \frac{1}{\sqrt{2}},$$

because the integrand exceeds $1/\sqrt{2}$ on $[0,1)$. Hence $C(1) > (1+\frac{1}{2})/(2\sqrt{\pi}) = 3/(4\sqrt{\pi})$. The second inequality is equivalent to $16/9 < 1 + 9\pi/32$, that is, to $\pi > 224/81$, which holds because $224/81 < 3 < \pi$. □

At $\beta = 0$ the main text's recursion eq. (8) is eq. (SF.19) in step time (the display after proposition SF.4), with $\mathfrak{d} = \mathfrak{d}_{\mathrm{iso}}$, $\mathfrak{v} = \mathfrak{v}_{\mathrm{iso}}$ and $\mathfrak{v}_{ij} = \mathfrak{v}/(N_{\mathrm{out}}N_{\mathrm{in}})$. The table of section 3 gives the formal values at $\varrho = 0$: the drift $C(N/B)$, which is the limit of proposition SF.5(ii), and the volatility $\min\{B, N\}$ of proposition SF.2. At momentum $\beta > 0$ the formal values are $C_\beta(N/B)$ and $\mathfrak{v}_{\mathrm{iso},\beta}$ (proposition SF.3), with the asymptotics of appendix SF.11.

The interpolation $[\pi\gamma(9\pi/32 + \gamma)]^{-1/2}$ of eq. (9) has the endpoint behaviour of $C(\gamma)$. As $\gamma \downarrow 0$ it is asymptotic to

$$\frac{\sqrt{32}}{3\pi\sqrt{\gamma}} = \frac{8}{3\pi\sqrt{2\gamma}},$$

the limit of theorem SF.4, and as $\gamma \to \infty$ it is asymptotic to $1/(\sqrt{\pi}\gamma)$, the limit of theorem SF.5. At $\gamma = 1$ it is strictly below $C(1)$, by eq. (SF.44). On $\gamma \geq 1$ it is superseded by the exact formula of proposition SF.9.

The SIGNSGD side of the isotropic comparison is proposition SJ.4, and theorem SI.6 compares the two times to reach a target risk, with its value at $\gamma = 1$ in corollary SI.1, which uses theorem SF.6.

SF.11 Asymptotics with momentum

This subsection gives the drift and volatility coefficients at a fixed momentum in the limits of small and large aspect ratios, and the order of the limits in ϱ, β and the aspect ratio. Every statement concerns the deterministic coefficient functions $\mathfrak{d}_{\mathrm{iso}}(\varrho)$, $\psi(\varrho)$ and $\overline{C}_{1,\ell}(\varrho)$ and their limits at $\varrho = 0$ (proposition SF.5(ii), proposition SF.3); none concerns the algorithm at zero ridge. The notation is that of appendix SF.8. At large aspect ratio three regimes are distinguished by the participation count

$$\mathfrak{p}_\beta(\varrho) \stackrel{\mathrm{def}}{=} \frac{\log(1/\varrho)}{2\log(1/\beta)} \qquad (\beta \in (0,1),\ 0 < \varrho < 1), \tag{SF.47}$$

which is, up to one, the number of ages $j \geq 0$ with $\beta^{2j} \geq \varrho$: zero ridge, where $\mathfrak{p}_\beta = \infty$ (theorem SF.11); a fixed ridge with $\mathfrak{p}_\beta(\varrho)$ small against $\gamma_\wedge$ (theorem SF.13); and the crossover $\mathfrak{p}_\beta(\varrho) \asymp \gamma$ (theorem SF.14).

Small aspect ratios. At small aspect ratios the root stays where r_β is nearly linear with slope s_2. The closure then becomes the linear closure with the aspect ratios multiplied by s_2. For $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}, s > 0$ let $Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(s)$ be the root $Y < \gamma_\wedge$ of $sY = (\gamma_{\mathrm{out}} - Y)(\gamma_{\mathrm{in}} - Y)$, which is lemma SF.5 with $r(q) = q$, explicitly

$$Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(s) = \frac{2\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}{\gamma_{\mathrm{in}} + \gamma_{\mathrm{out}} + s + \sqrt{(\gamma_{\mathrm{in}} + \gamma_{\mathrm{out}} + s)^2 - 4\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}},$$

and for $\varrho \geq 0$ put

$$\mathfrak{d}_{\mathrm{lin}}(\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}; \varrho) \stackrel{\mathrm{def}}{=} \frac{\sqrt{2}}{\pi\,\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}} \int_0^\infty Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(\varrho + v^2)\,\mathrm{d}v. \tag{SF.48}$$

Theorem SF.7 (Small aspect ratios with momentum)**.** *Let $\beta \in [0,1)$ and $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 0$, and assume $\varepsilon_\beta \stackrel{\mathrm{def}}{=} 6\gamma_\wedge(1-\beta^2) \leq \frac{1}{2}$.*

(i) For every $\varrho \geq 0$ there is $\vartheta \in \left[1 - \varepsilon_\beta, (1-\varepsilon_\beta)^{-1/2}\right]$ with

$$\mathfrak{d}_{\mathrm{iso}}(\varrho; \beta, \gamma_{\mathrm{in}}, \gamma_{\mathrm{out}}) = \vartheta\ \mathfrak{d}_{\mathrm{lin}}(s_2\gamma_{\mathrm{in}}, s_2\gamma_{\mathrm{out}}; \varrho).$$

(ii) With $r \stackrel{\text{def}}{=} \gamma_{\text{out}}/\gamma_{\text{in}}$ *and* J_r *of theorem SF.4,*

$$\mathfrak{d}_{\text{lin}}(\gamma_{\text{in}}, \gamma_{\text{out}}; 0) = \frac{r\, J_r}{\sqrt{2\gamma_{\text{out}}}}, \qquad \mathfrak{d}_{\text{lin}}(\gamma, \gamma; 0) = \frac{8}{3\pi\sqrt{2\gamma}}.$$

In particular, in the square case $C_\beta(\gamma) = 8\vartheta/(3\pi\sqrt{2s_2\gamma})$ *with* ϑ *of (i) at* $\varrho = 0$.

(iii) If $6\gamma_\wedge \le \frac{1}{2}$, *then for every* $\beta \in [0, 1)$

$$\big(1 - 6\gamma_\wedge\big)^{3/2} \le \frac{C_\beta(\gamma_{\text{in}}, \gamma_{\text{out}})}{s_2^{-1/2}\, C_0(\gamma_{\text{in}}, \gamma_{\text{out}})} \le \big(1 - 6\gamma_\wedge\big)^{-3/2}.$$

(iv) For every $\varrho \ge 0$,

$$(1 - \varepsilon_\beta)\big(1 - 6s_2\gamma_\wedge\big)^{1/2} \le \frac{\mathfrak{d}_{\text{iso}}(\varrho; \beta, \gamma_{\text{in}}, \gamma_{\text{out}})}{\mathfrak{d}_{\text{iso}}(\varrho; 0, s_2\gamma_{\text{in}}, s_2\gamma_{\text{out}})} \le (1 - \varepsilon_\beta)^{-1/2}\big(1 - 6s_2\gamma_\wedge\big)^{-1}.$$

Part (iv) says that, at the same normalized ridge, momentum acts on the drift as the batch replacement $B \mapsto B/s_2 = B(1+\beta)/(1-\beta)$, since $s_2\gamma_{\text{in}} = N_{\text{in}}/(B/s_2)$ and $s_2\gamma_{\text{out}} = N_{\text{out}}/(B/s_2)$; this effective batch is a conclusion, with the stated relative error. The small parameter is $\gamma_\wedge(1-\beta^2)$, so the theorem also covers a fixed aspect ratio with $\beta \uparrow 1$ (proposition SF.13). At $\beta = 0$, parts (i) and (ii) at $\varrho = 0$ give $C(\gamma_{\text{in}}, \gamma_{\text{out}}) = \vartheta\, r J_r/\sqrt{2\gamma_{\text{out}}}$ with $\vartheta \in [1 - 6\gamma_\wedge, (1 - 6\gamma_\wedge)^{-1/2}]$, a rate for the limit of theorem SF.4.

Proof. The proof sandwiches the root between two linear roots and then integrates.

Constants. For $\beta > 0$ a direct computation from $s_k = (1-\beta)^k/(1-\beta^k)$ gives

$$\frac{s_4}{s_2^2} = \frac{1-\beta^2}{1+\beta^2}, \qquad \frac{s_3}{s_2} = \frac{1-\beta^2}{1+\beta+\beta^2},$$

so that $\varepsilon_1 \stackrel{\text{def}}{=} 6\gamma_\wedge s_4/s_2^2$ and $\varepsilon_2 \stackrel{\text{def}}{=} 6\gamma_\wedge s_3/s_2$ are both at most ε_β. At $\beta = 0$ all power sums equal 1 and $\varepsilon_1 = \varepsilon_2 = \varepsilon_0$.

Expansions. By lemma SG.1(i) for $\beta > 0$, and by lemma SF.1(iii) and $J(x) \ge 1 - 3x$ at $\beta = 0$,

$$s_2 q\Big(1 - 3\frac{s_4}{s_2} q\Big) \le r_\beta(q) \le s_2 q, \qquad 1 - 3s_3 q \le L_\beta(q) \le 1 \qquad (q \ge 0), \tag{SF.49}$$

where the upper bounds are lemma SF.4(v). The bound $J(x) \ge 1 - 3x$ is $(1+t)^{-1} \ge 1 - t$ under the expectation.

A priori bound. Put $\bar q \stackrel{\text{def}}{=} 2\gamma_\wedge/s_2$. By eq. (SF.49),

$$r_\beta(\bar q) \ge 2\gamma_\wedge\big(1 - \varepsilon_1\big) \ge \gamma_\wedge.$$

Since $r_\beta(q(-s)) < \gamma_\wedge$ and r_β is increasing, $q(-s) < \bar q$ for every $s > 0$.

Sandwich. For $c > 0$ the root of lemma SF.5 with $r(q) = cq$ is $q_c(s) = c^{-1} Y_{\gamma_{\text{in}}, \gamma_{\text{out}}}(s/c)$, because $Y = cq$ turns the equation into $(s/c)Y = (\gamma_{\text{out}} - Y)(\gamma_{\text{in}} - Y)$. Lemma SF.5(ii) with $r_1 = r_\beta$, $r_2(q) = s_2 q$ and $i = 1$ applies by the upper bound in eq. (SF.49), and gives $q_{s_2}(s) \le q(-s)$. Next put $c \stackrel{\text{def}}{=} s_2(1 - \varepsilon_1)$. By the a priori bound,

$$c\, q(-s) \le s_2\, q(-s)\Big(1 - 3\frac{s_4}{s_2}\bar q\Big) \le r_\beta\big(q(-s)\big),$$

so lemma SF.5(ii) with $r_1(q) = cq$, $r_2 = r_\beta$ and $i = 2$ gives $q(-s) \le q_c(s)$.

The factor L_β. By eq. (SF.49) and $q(-s) < \bar q$,

$$1 - \varepsilon_2 \le 1 - 3s_3\bar q \le L_\beta\big(q(-s)\big) \le 1.$$

Integration. The function $Y_{\gamma_{\text{in}}, \gamma_{\text{out}}}$ is decreasing and $c \le s_2$, so the substitution $u = \sqrt{c}\, v$ gives

$$\int_0^\infty q_c(\varrho + u^2)\, \mathrm{d}u = c^{-1/2} \int_0^\infty Y_{\gamma_{\text{in}}, \gamma_{\text{out}}}\Big(\frac{\varrho}{c} + v^2\Big)\mathrm{d}v \le c^{-1/2} \int_0^\infty Y_{\gamma_{\text{in}}, \gamma_{\text{out}}}\Big(\frac{\varrho}{s_2} + v^2\Big)\mathrm{d}v,$$

and in the same way

$$\int_0^\infty q_{s_2}(\varrho+u^2)\,\mathrm{d}u = s_2^{-1/2}\int_0^\infty Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}\Big(\frac{\varrho}{s_2}+v^2\Big)\mathrm{d}v.$$

The scaling $Y_{t\gamma_{\mathrm{in}},t\gamma_{\mathrm{out}}}(ts) = t\,Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(s)$ and the substitution $v=\sqrt{s_2}\,w$ give

$$\mathfrak{d}_{\mathrm{lin}}(s_2\gamma_{\mathrm{in}},s_2\gamma_{\mathrm{out}};\varrho) = s_2^{-1/2}\,\frac{\sqrt{2}}{\pi\,\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}\int_0^\infty Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}\Big(\frac{\varrho}{s_2}+w^2\Big)\mathrm{d}w. \tag{SF.50}$$

Inserting the sandwich and the bounds on L_β in eq. (SF.29) (at $\varrho=0$, in eq. (SF.21)) proves (i) with $\vartheta\in[1-\varepsilon_2,(1-\varepsilon_1)^{-1/2}]$.

Closed form. Put $y(x)\overset{\mathrm{def}}{=}Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(\gamma_{\mathrm{out}}x)/\gamma_{\mathrm{out}}$, so that

$$x\,y=(1-y)\Big(\frac{1}{r}-y\Big),\qquad y<\min\Big(1,\frac{1}{r}\Big).$$

Let m_r be the transform of the law M_r of lemma SF.9 and $\tilde y(x)\overset{\mathrm{def}}{=}1-x\,m_r(x)=\int\frac{\lambda}{\lambda+x}\,\mathsf{M}_r(\mathrm{d}\lambda)$. Substituting $m=(1-\tilde y)/x$ in eq. (SF.34) and expanding gives $x\tilde y=(1-\tilde y)(1/r-\tilde y)$. Moreover $\tilde y<\mathsf{M}_r((0,\infty))=1-\omega_r=\min(1,1/r)$. By the uniqueness in lemma SF.5(i), $\tilde y=y$. The half-moment formula eq. (SF.26) for M_r then gives

$$\int_0^\infty y(w^2)\,\mathrm{d}w=\frac12\int_0^\infty t^{-1/2}\,y(t)\,\mathrm{d}t=\frac{\pi}{2}\,J_r,\qquad \int_0^\infty Y_{\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}}(v^2)\,\mathrm{d}v=\gamma_{\mathrm{out}}^{3/2}\int_0^\infty y(w^2)\,\mathrm{d}w,$$

the second by $v=\sqrt{\gamma_{\mathrm{out}}}\,w$. Hence $\mathfrak{d}_{\mathrm{lin}}(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}};0)=\frac{\sqrt2}{\pi\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}\gamma_{\mathrm{out}}^{3/2}\frac{\pi}{2}J_r=rJ_r/\sqrt{2\gamma_{\mathrm{out}}}$. In the square case $r=1$, and $y(v^2)=e^{-2\vartheta'}$ at $v=2\sinh\vartheta'$, since $4\sinh^2\vartheta'\,e^{-2\vartheta'}=(1-e^{-2\vartheta'})^2$. Therefore

$$\int_0^\infty y(v^2)\,\mathrm{d}v=\int_0^\infty e^{-2\vartheta'}\,2\cosh\vartheta'\,\mathrm{d}\vartheta'=\int_0^\infty\big(e^{-\vartheta'}+e^{-3\vartheta'}\big)\mathrm{d}\vartheta'=\frac43,$$

and $\mathfrak{d}_{\mathrm{lin}}(\gamma,\gamma;0)=\frac{\sqrt2}{\pi\gamma^2}\gamma^{3/2}\frac43=\frac{8}{3\pi\sqrt{2\gamma}}$. The last claim of (ii) is (i) at $\varrho=0$ with eq. (SF.50).

Ratios. For (iii), part (i) at $\varrho=0$ and eq. (SF.50) give $C_\beta=\vartheta_\beta\,s_2^{-1/2}\mathfrak{d}_{\mathrm{lin}}(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}};0)$ and $C_0=\vartheta_0\,\mathfrak{d}_{\mathrm{lin}}(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}};0)$, with $\vartheta_\beta,\vartheta_0\in[1-6\gamma_\wedge,(1-6\gamma_\wedge)^{-1/2}]$ because $\varepsilon_\beta\le6\gamma_\wedge$. Dividing gives (iii). For (iv), part (i) at momentum 0 and aspect ratios $(s_2\gamma_{\mathrm{in}},s_2\gamma_{\mathrm{out}})$ gives $\mathfrak{d}_{\mathrm{iso}}(\varrho;0,s_2\gamma_{\mathrm{in}},s_2\gamma_{\mathrm{out}})=\vartheta'\,\mathfrak{d}_{\mathrm{lin}}(s_2\gamma_{\mathrm{in}},s_2\gamma_{\mathrm{out}};\varrho)$ with $\vartheta'\in[1-6s_2\gamma_\wedge,(1-6s_2\gamma_\wedge)^{-1/2}]$; its hypothesis holds because $s_2\le1-\beta^2$, so that $6s_2\gamma_\wedge\le\varepsilon_\beta\le\frac12$. Dividing by part (i) at momentum β gives (iv). □

The integrated volatility at small aspect ratio is amplified by an explicit constant, which lies strictly between 1 and $1/s_2$. For $n\ge0$ and $0\le z\le1$ put

$$\begin{gathered}k_n\overset{\mathrm{def}}{=}\frac{1}{2n+1}+\frac{1}{2n+3},\qquad \Omega(z)\overset{\mathrm{def}}{=}\frac{4}{\pi^2}\sum_{n\ge0}k_n^2\,z^{2n+1},\\ V_\beta\overset{\mathrm{def}}{=}1+2\sum_{\ell\ge1}\Omega(\beta^\ell)\qquad(\beta\in(0,1)).\end{gathered}\tag{SF.51}$$

Lemma SF.10 (The volatility factor)**.** $\Omega(1)=1$, *and for* $\beta\in(0,1)$

$$V_\beta=\frac{4}{\pi^2}\sum_{n\ge0}k_n^2\,\frac{1+\beta^{2n+1}}{1-\beta^{2n+1}},\qquad 1<V_\beta<\frac{1}{s_2}=\frac{1+\beta}{1-\beta},$$

$$\lim_{\beta\uparrow1}s_2V_\beta=\frac14+\frac{1}{\pi^2}+\frac{7\zeta(3)}{2\pi^2}=0.77759\ldots.$$

Proof. The value at one is a sum over odd integers. With $x=2n+1$, $k_n^2=x^{-2}+2/(x(x+2))+(x+2)^{-2}$, and

$$\sum_{n\ge0}k_n^2=\frac{\pi^2}{8}+2\cdot\frac12+\Big(\frac{\pi^2}{8}-1\Big)=\frac{\pi^2}{4},$$

by $\sum_{x\ \mathrm{odd}} x^{-2} = \pi^2/8$ and the telescoping sum $\sum_{x\ \mathrm{odd}} \frac{2}{x(x+2)} = \sum_{x\ \mathrm{odd}} \left(\frac{1}{x} - \frac{1}{x+2}\right) = 1$. Hence $\Omega(1) = 1$.

The series form sums the geometric series in ℓ: $1+2\sum_{\ell\geq 1}\beta^{\ell x} = (1+\beta^x)/(1-\beta^x)$, and the weights $\frac{4}{\pi^2}k_n^2$ sum to 1. The map $z \mapsto (1+z)/(1-z)$ is increasing, $1 < (1+\beta^x)/(1-\beta^x) \leq (1+\beta)/(1-\beta)$, with strict inequality for $n \geq 1$, and $k_1 > 0$. This gives $1 < V_\beta < 1/s_2$.

For the limit, put $x = 2n+1$. Then

$$s_2\,\frac{1+\beta^x}{1-\beta^x} = \frac{1-\beta}{1+\beta}\cdot\frac{1+\beta^x}{1-\beta^x} \leq \frac{2(1-\beta)}{1-\beta^x} \leq 2, \qquad s_2\,\frac{1+\beta^x}{1-\beta^x} \longrightarrow \frac{1}{x} \quad (\beta\uparrow 1),$$

so dominated convergence over n gives $\lim s_2 V_\beta = \frac{4}{\pi^2}\sum_n k_n^2/x$. Partial fractions give

$$\frac{k_n^2}{x} = \frac{1}{x^3} + \frac{1}{x^2} - \frac{1}{2x} + \frac{1}{2(x+2)} + \frac{1}{4x} - \frac{1}{4(x+2)} - \frac{1}{2(x+2)^2},$$

and summing over odd x, with $\sum_{x\ \mathrm{odd}} x^{-3} = \frac{7}{8}\zeta(3)$ and the two telescoping sums $-\frac{1}{2}$ and $\frac{1}{4}$,

$$\sum_{n\geq 0}\frac{k_n^2}{2n+1} = \frac{7}{8}\zeta(3) + \frac{\pi^2}{8} - \frac{1}{2} + \frac{1}{4} - \frac{1}{2}\Big(\frac{\pi^2}{8} - 1\Big) = \frac{7}{8}\zeta(3) + \frac{\pi^2}{16} + \frac{1}{4}.$$

Multiplying by $4/\pi^2$ gives the limit. □

For example $V_{1/2} = 2.4775\ldots$, while $1/s_2 = 3$ at $\beta = \frac{1}{2}$.

The volatility theorems use the lag kernel of the power-law section at the spectrum $\mu_1 = \cdots = \mu_N = 1$. In the square isotropic case it collapses to an expression in the two roots.

Lemma SF.11 (Isotropic lag response)**.** *Let $\beta \in (0,1)$, $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B$. For $s, s' > 0$ put $q \overset{\mathrm{def}}{=} q(-s)$ and $q' \overset{\mathrm{def}}{=} q(-s')$, let b_ℓ and e_ℓ be the numbers of eq.* (SG.14) *for these roots, and put*

$$X_\ell \overset{\mathrm{def}}{=} \frac{q\,q'\,e_\ell}{\gamma}, \qquad Z_\ell \overset{\mathrm{def}}{=} \frac{\sqrt{qq'}\,b_\ell}{\gamma}, \qquad \Delta_\ell \overset{\mathrm{def}}{=} (1-X_\ell)^2 - Z_\ell^2 \qquad (\ell \geq 0).$$

(i) $0 \leq b_\ell \leq \beta^\ell s_2$, $0 \leq e_\ell \leq 3\beta^{2\ell}s_4$, $0 \leq X_\ell < 1$ *and* $0 < \Delta_\ell \leq 1$.

(ii) The row lag kernel eq. (SG.1) *at $\mu_1 = \cdots = \mu_N = 1$ is the same for every row and equals*

$$\mathfrak{k}_{1,\ell}(s,s') = \frac{b_\ell\,q\,q'}{\gamma^3\,\Delta_\ell} \;\geq\; \frac{b_\ell\,q\,q'}{\gamma^3},$$

and for every $\varrho \geq 0$ the row lag coefficient is

$$\overline{C}_{1,\ell}(\varrho) = \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \mathfrak{k}_{1,\ell}(\varrho+x^2, \varrho+y^2)\,\mathrm{d}x\,\mathrm{d}y, \tag{SF.52}$$

where $\overline{C}_{1,\ell}(0)$ is the limit of proposition SF.3(ii).

Proof. Part (ii) is proposition SG.3 at $\mu_i = 1$. In the square case the two anchors of eq. (SF.6) give $\sigma = \tilde{\sigma}$ and $\tilde{s}_1 = s_1$, and $q = s\sigma^2$, so that $s + \tilde{s}_1 = s + s_1 = \gamma/\sigma$. The scalars of eq. (SG.14) become $\mathsf{U} = \mathsf{V} = \sigma\sigma'/\gamma$, and

$$ss'\,\mathsf{U}\tilde{\sigma}\tilde{\sigma}' = ss'\,\mathsf{V}\sigma\sigma' = \frac{ss'\sigma^2\sigma'^2}{\gamma} = \frac{qq'}{\gamma}, \qquad ss'\,\mathsf{U}\mathsf{V} = \frac{qq'}{\gamma^2}.$$

Hence the determinant of eq. (SG.14) is Δ_ℓ, and eq. (SG.16) reads

$$\mathfrak{k}_{1,\ell}(s,s') = \frac{\gamma\,ss'\,b_\ell}{(\gamma/\sigma)^2(\gamma/\sigma')^2\,\Delta_\ell} = \frac{b_\ell\,qq'}{\gamma^3\Delta_\ell}.$$

The integral eq. (SF.52) is eq. (SG.2) for $\varrho > 0$, where the kernel is nonnegative so that the limit $T \to \infty$ is the integral over $(0,\infty)^2$, and eq. (SG.22) at $\varrho = 0$. The lower bound on the kernel is $\Delta_\ell \leq 1$ from part (i).

For part (i), every factor $(1+\cdots)^{-1}$ in the series eq. (SG.14) lies in $(0,1]$, so

$$0 \le b_\ell \le \beta^\ell \sum_j w_j^2\ \mathbb{E}\, g^2 = \beta^\ell s_2, \qquad 0 \le e_\ell \le \beta^{2\ell} \sum_j w_j^4\ \mathbb{E}\, g^4 = 3\beta^{2\ell} s_4.$$

For X_ℓ put $T(q) \overset{\text{def}}{=} -F'_\beta(q) = \sum_j w_j^4\ \mathbb{E}[g^4(1+w_j^2 q g^2)^{-2}]$. The Cauchy–Schwarz inequality over j and g gives

$$e_\ell \le \sqrt{T(q)}\,\sqrt{\beta^{4\ell} T(\beta^{2\ell} q')},$$

and since $\beta^\ell w_j = w_{j+\ell}$,

$$\beta^{4\ell} T(\beta^{2\ell} q') = \sum_{j\ge 0} w_{j+\ell}^4\ \mathbb{E}\,\frac{g^4}{(1+w_{j+\ell}^2 q' g^2)^2} \le T(q').$$

Hence $X_\ell \le \sqrt{X(q)X(q')}$ with $X(q) \overset{\text{def}}{=} q^2 T(q)/\gamma$. Since $r'_\beta = F_\beta - qT > 0$ (lemma SF.4(i)),

$$X(q) = \frac{r_\beta(q) - q\, r'_\beta(q)}{\gamma} < \frac{r_\beta(q)}{\gamma} = \frac{\psi(s)}{\gamma} < 1,$$

by proposition SF.1(ii). Thus $0 \le X_\ell < 1$, and $\Delta_\ell \le (1-X_\ell)^2 \le 1$. Positivity of Δ_ℓ is proposition SG.3. □

Theorem SF.8 (Small-aspect integrated volatility)**.** *Let* $\beta \in (0,1)$, $N_{\text{in}} = N_{\text{out}} = N$ *and* $\gamma = N/B \le (1+\beta^2)/96$. *Then for every* $\ell \ge 1$

$$\left|\overline{C}_{1,\ell}(0) - \Omega(\beta^\ell)\right| \le \frac{96\gamma}{1+\beta^2}\,\Omega(\beta^\ell),$$

and the zero-ridge volatility of definition SF.1 satisfies

$$\left|\frac{\mathfrak{v}_{\text{iso},\beta}}{N} - V_\beta\right| \le \frac{96\gamma}{1+\beta^2}\,(V_\beta - 1), \qquad \left|\frac{\mathfrak{v}_{\text{iso},\beta}}{N\,V_\beta} - 1\right| \le 96\gamma.$$

The amplification of the integrated volatility at small aspect ratio is therefore V_β, strictly smaller than the factor $1/s_2$ of the drift's effective batch (lemma SF.10); the lag profile is $\Omega(\beta^\ell)$, not β^ℓ. The bound is uniform in β for $\gamma \le 1/96$.

Proof. The proof scales the zero-ridge double integral to the linear root and compares the integrands pointwise.

Scaling. Write $q_s \overset{\text{def}}{=} q(-s)$ and $\bar\gamma \overset{\text{def}}{=} s_2\gamma$. For $\xi > 0$ put $y_\gamma(\xi) \overset{\text{def}}{=} s_2\, q_{\bar\gamma\xi^2}/\gamma$, and let y be the root of $xy = (1-y)^2$, $y < 1$, the case $r = 1$ of the closed form in the proof of theorem SF.7. The sandwich of that proof with $\gamma_\wedge = \gamma$, which uses only $\varepsilon_1 \le \frac12$, together with $q_c(s) = c^{-1} Y_{\gamma,\gamma}(s/c)$ and the monotonicity of $Y_{\gamma,\gamma}$, gives

$$y(\xi^2) \le y_\gamma(\xi) \le (1-\varepsilon_1)^{-1} y(\xi^2), \qquad q_s \le \bar q = \frac{2\gamma}{s_2}, \qquad \varepsilon_1 = 6\gamma\frac{s_4}{s_2^2}.$$

Fix $\ell \ge 1$, put $r \overset{\text{def}}{=} \beta^\ell$, and in eq. (SF.52) at $\varrho = 0$ substitute $x = \sqrt{\bar\gamma}\,\xi$ and $y = \sqrt{\bar\gamma}\,\xi'$. With $\mathsf{Y} \overset{\text{def}}{=} y_\gamma(\xi) y_\gamma(\xi')$, $\tilde b \overset{\text{def}}{=} b_\ell/(r s_2)$ and $\eta \overset{\text{def}}{=} X_\ell$, one has $q\,q' = \gamma^2 \mathsf{Y}/s_2^2$ and $Z_\ell^2 = r^2 \tilde b^2 \mathsf{Y}$, so that

$$\overline{C}_{1,\ell}(0) = \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \frac{r\,\tilde b\,\mathsf{Y}}{(1-\eta)^2 - r^2\tilde b^2 \mathsf{Y}}\,\mathrm{d}\xi\,\mathrm{d}\xi'.$$

Pointwise bounds. Put $\delta \overset{\text{def}}{=} 2\varepsilon_1 = 12\gamma s_4/s_2^2 = 12\gamma(1-\beta^2)/(1+\beta^2)$. The inequality $\frac{1}{(1+a)(1+c)} \ge 1-a-c$ for $a, c \ge 0$, inserted in the series eq. (SG.14), gives

$$b_\ell \ge r\big(s_2 - 3s_4(q + r^2 q')\big) \ge r s_2(1-\delta),$$

so $\tilde b \in [1-\delta, 1]$ by lemma SF.11(i). By the same lemma and $q, q' \le \bar q$, $0 \le \eta \le 3r^2 s_4 \bar q^2/\gamma \le \delta$. With $\mathsf{Y}_0 \stackrel{\text{def}}{=} y(\xi^2)y(\xi'^2) \le 1$, the sandwich gives $\mathsf{Y}_0 \le \mathsf{Y} \le (1-\delta)^{-2}\mathsf{Y}_0$. The hypothesis on γ is equivalent to $4\delta \le (1-\beta^2)/2$. The integrand increases in $\tilde b$, in Y and in η, its denominator being $\Delta_\ell > 0$. Put

$$\mathfrak{f} \stackrel{\text{def}}{=} \frac{r\,\mathsf{Y}_0}{1 - r^2\mathsf{Y}_0}, \qquad 1 - r^2\mathsf{Y}_0 \ge 1 - \beta^2.$$

For the upper bound, evaluate at $\tilde b = 1$, $\mathsf{Y} = (1-\delta)^{-2}\mathsf{Y}_0$ and $\eta = \delta$, and use $(1-\delta)^4 \ge 1 - 4\delta$ and $(1-x)^{-1} \le 1 + 2x$ for $x \le \frac12$:

$$\text{integrand} \le \frac{r\mathsf{Y}_0}{(1-\delta)^4 - r^2\mathsf{Y}_0} \le \frac{r\mathsf{Y}_0}{1 - r^2\mathsf{Y}_0 - 4\delta} \le \mathfrak{f}\Big(1 + \frac{8\delta}{1-\beta^2}\Big).$$

For the lower bound, evaluate at $\tilde b = 1-\delta$, $\mathsf{Y} = \mathsf{Y}_0$ and $\eta = 0$, and use $1 - r^2(1-\delta)^2\mathsf{Y}_0 \le 1 - r^2\mathsf{Y}_0 + 2\delta$:

$$\text{integrand} \ge \frac{r(1-\delta)\mathsf{Y}_0}{1 - r^2(1-\delta)^2\mathsf{Y}_0} \ge \mathfrak{f}\,(1-\delta)\Big(1 - \frac{2\delta}{1-\beta^2}\Big) \ge \mathfrak{f}\Big(1 - \frac{3\delta}{1-\beta^2}\Big).$$

Limit kernel. Expanding $\mathfrak{f} = \sum_{n\ge0} r^{2n+1}\mathsf{Y}_0^{n+1}$ and integrating termwise, with $\xi = 2\sinh\vartheta'$ as in the proof of theorem SF.7,

$$\frac{4}{\pi^2}\iint \mathfrak{f}\,\mathrm{d}\xi\,\mathrm{d}\xi' = \frac{4}{\pi^2}\sum_{n\ge0} r^{2n+1}\Big(\int_0^\infty y(\xi^2)^{n+1}\mathrm{d}\xi\Big)^2,$$

$$\int_0^\infty y(\xi^2)^{n+1}\mathrm{d}\xi = \int_0^\infty e^{-2(n+1)\vartheta'}\,2\cosh\vartheta'\,\mathrm{d}\vartheta' = k_n.$$

The integral of $\mathfrak{f}$ is therefore $\Omega(\beta^\ell)$, and $8\delta/(1-\beta^2) = 96\gamma/(1+\beta^2)$ gives the lag bound.

Summation. By proposition SF.3(ii), $\mathfrak{v}_{\mathrm{iso},\beta}/N = 1 + 2\sum_{\ell\ge1}\overline{C}_{1,\ell}(0)$, and $V_\beta = 1 + 2\sum_{\ell\ge1}\Omega(\beta^\ell)$. Summing the lag bounds gives the first volatility bound, and dividing by $V_\beta \ge 1$ gives the second. □

Large aspect ratios at zero ridge. At zero ridge with $\beta > 0$ every row participates, $\psi(0) = \gamma_\wedge$ (proposition SF.3(i)), so the master estimate makes the drift of order one, and the integrated volatility grows linearly in the aspect ratio. An explicit lower bound for the lag coefficients at every ridge, in which the participation count of lemma SF.6 is effective, comes first; it is used at a fixed ridge (theorem SF.13). With the constants of that lemma put

$$\mathsf{c}''_\beta \stackrel{\text{def}}{=} \frac{1}{\log(1/\beta)} + \mathsf{c}_\beta + \mathsf{c}'_\beta, \qquad \mathsf{k}_\beta \stackrel{\text{def}}{=} \frac{\mathsf{e}^-_\beta\,(e^2-1)\,e^{-3}}{\pi^2\,\mathsf{e}^+_\beta}. \tag{SF.53}$$

Lemma SF.12 (Lag lower bound)**.** *Let $\beta \in (0,1)$, $N_{\mathrm{in}} = N_{\mathrm{out}} = N$, $\gamma = N/B$, $\varepsilon \ge 0$ and $\ell \ge 1$. Put $m_\varepsilon \stackrel{\text{def}}{=} \psi(\varepsilon)$ for $\varepsilon > 0$ and $m_0 \stackrel{\text{def}}{=} \gamma$. Then*

$$\overline{C}_{1,\ell}(\varepsilon) \ge \frac{\mathsf{k}_\beta}{\gamma^3}\int_{\mathsf{c}'_\beta}^{m_\varepsilon - \ell - \mathsf{c}''_\beta} \big(\gamma - \eta - \mathsf{c}''_\beta\big)\big(\gamma - \eta - \ell - \mathsf{c}''_\beta\big)\,\mathrm{d}\eta,$$

where the right side is zero if the range is empty. For $\varepsilon > 0$ the right side is at least $\frac{\mathsf{k}_\beta}{\gamma^3}(\gamma - m_\varepsilon)^2\big(m_\varepsilon - \ell - \mathsf{c}''_\beta - \mathsf{c}'_\beta\big)_+$.

Proof. The proof bounds the determinant by one, changes variables to the two roots, and integrates over a thin region where the participation count is effective. Write $r = r_\beta$ and $Q \stackrel{\text{def}}{=} r^{-1}(m_\varepsilon)$, so that $Q = q(-\varepsilon)$ for $\varepsilon > 0$.

Change of variables. By lemma SF.11,

$$\overline{C}_{1,\ell}(\varepsilon) \ge \frac{4}{\pi^2\gamma^3}\int_0^\infty\!\!\int_0^\infty b_\ell\, q\, q'\,\mathrm{d}x\,\mathrm{d}y, \qquad q = q(-\varepsilon - x^2), \quad q' = q(-\varepsilon - y^2).$$

In the square case the closure eq. (SF.7) inverts as $s(q) = (\gamma - r(q))^2/q$, so each axis is parametrized by its root $q \in (0, Q)$, with $x = \sqrt{s(q) - \varepsilon}$. Since $\sqrt{s(q)} = (\gamma - r(q))\, q^{-1/2}$ and $r' > 0$,

$$\Big|\frac{\mathrm{d}x}{\mathrm{d}q}\Big| \geq \Big|\frac{\mathrm{d}\sqrt{s(q)}}{\mathrm{d}q}\Big| \geq \frac{\gamma - r(q)}{2\, q^{3/2}}.$$

By eq. (SG.15), $b_\ell = \beta^\ell\big(r(q) - r(\beta^{2\ell} q')\big)/\big(q - \beta^{2\ell} q'\big)$. The substitution $p = \beta^{2\ell} q'$ in the second variable therefore gives

$$\overline{C}_{1,\ell}(\varepsilon) \geq \frac{1}{\pi^2 \gamma^3} \iint D(q,p)\, \big(\gamma - r(q)\big)\big(\gamma - r(p\,\beta^{-2\ell})\big)\, \mathrm{d}q\, \mathrm{d}p, \qquad D(q,p) \stackrel{\mathrm{def}}{=} \frac{r(q) - r(p)}{(q-p)\sqrt{qp}} \geq 0,$$

over the region $r(q) < m_\varepsilon$, $r(p\,\beta^{-2\ell}) < m_\varepsilon$.

Restriction. Keep only $p \geq (1-\beta)^{-2}$ and $q = p\, e^\delta$ with $\delta \in [0, 2]$. By lemma SF.6(ii), $r'(x) \geq \mathsf{e}_\beta^-/x$ for $x \geq p$, so $r(q) - r(p) \geq \mathsf{e}_\beta^-\, \delta$, and with $(e^\delta - 1)/\delta \leq e^\delta$,

$$D(q,p) \geq \frac{\mathsf{e}_\beta^-\, \delta}{p^2 (e^\delta - 1)\, e^{\delta/2}} \geq \frac{\mathsf{e}_\beta^-\, e^{-3}}{p^2}.$$

Since $\mathsf{n}_\beta(q) = \mathsf{n}_\beta(p) + \delta/(2\log(1/\beta))$ and $\mathsf{n}_\beta(p\beta^{-2\ell}) = \mathsf{n}_\beta(p) + \ell$, lemma SF.6(i) gives

$$r(q) < r(p) + \mathsf{c}''_\beta, \qquad r(p\,\beta^{-2\ell}) < r(p) + \ell + \mathsf{c}''_\beta.$$

Put $\eta \stackrel{\mathrm{def}}{=} r(p)$ and keep only $\eta \leq m_\varepsilon - \ell - \mathsf{c}''_\beta$. Then both constraints of the region hold, and the two brackets are at least $\gamma - \eta - \mathsf{c}''_\beta$ and $\gamma - \eta - \ell - \mathsf{c}''_\beta$, which are nonnegative. Integrating over $q = pe^\delta$, $\mathrm{d}q = pe^\delta \mathrm{d}\delta$,

$$\int_p^{pe^2} D(q,p)\, \mathrm{d}q \geq \frac{\mathsf{e}_\beta^-\, e^{-3}(e^2 - 1)}{p}.$$

Finally, by lemma SF.6(ii), $\mathrm{d}\eta = r'(p)\, \mathrm{d}p \leq \mathsf{e}_\beta^+\, \mathrm{d}p/p$, so $\mathrm{d}p/p \geq \mathrm{d}\eta/\mathsf{e}_\beta^+$. The condition $p \geq (1-\beta)^{-2}$ holds when $\eta \geq \mathsf{c}'_\beta$, because $r((1-\beta)^{-2}) \leq \mathsf{c}'_\beta$ (lemma SF.6(i)). Collecting the factors gives the bound.

The case $\varepsilon > 0$. The integrand decreases in η, and at the upper endpoint it is $(\gamma - m_\varepsilon + \ell)(\gamma - m_\varepsilon) \geq (\gamma - m_\varepsilon)^2$. □

The lag coefficients in participation coordinates. The two-sided laws for the integrated volatility at large aspect ratio rest on an exact rewriting of the lag coefficients of lemma SF.11 in the participation scale. From here to the end of the proof of theorem SF.12, $\beta \in (0,1)$, $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B$. We write $r = r_\beta$ and $Q_m = r^{-1}(m)$ as in lemma SF.8, and $\psi(0) \stackrel{\mathrm{def}}{=} \gamma$ as in theorem SF.3, so that $Q_{\psi(\varrho)} = q(-\varrho)$ for $\varrho > 0$ and $Q_{\psi(0)} = Q_\gamma$. For $y > 0$ put

$$\mathsf{e}_\beta(y) \stackrel{\mathrm{def}}{=} y\, r'(y), \qquad \mathsf{A}(y) \stackrel{\mathrm{def}}{=} \gamma - r(y), \qquad \mathsf{j}(y) \stackrel{\mathrm{def}}{=} \mathsf{A}(y) + 2\, \mathsf{e}_\beta(y),$$

$$\mathsf{n}_\beta^-(y) \stackrel{\mathrm{def}}{=} \max\big(0, -\mathsf{n}_\beta(y)\big),$$

with n_β of lemma SF.6. The number $\mathsf{e}_\beta(y)$ is the slope of r in the variable $\log y$, and $\mathsf{A}(y)$ is the residual participation at the root y. For $y \neq z$ put

$$\mathcal{Z}(y,z) \stackrel{\mathrm{def}}{=} \sqrt{yz}\, \frac{r(y) - r(z)}{y - z}, \qquad \mathcal{X}(y,z) \stackrel{\mathrm{def}}{=} \frac{z\, r(y) - y\, r(z)}{z - y}, \qquad D(y,z) \stackrel{\mathrm{def}}{=} \frac{r(y) - r(z)}{(y-z)\sqrt{yz}},$$

$$\widehat{\Delta}(y,z) \stackrel{\mathrm{def}}{=} \Big(1 - \frac{\mathcal{X}(y,z)}{\gamma}\Big)^2 - \Big(\frac{\mathcal{Z}(y,z)}{\gamma}\Big)^2, \qquad \mathsf{A}_\vee(y,z) \stackrel{\mathrm{def}}{=} \max\big(\mathsf{A}(y), \mathsf{A}(z)\big),$$

extended continuously to $y = z$, and for an integer $\ell \geq 1$ write $z_\ell \stackrel{\mathrm{def}}{=} \beta^{-2\ell} z$. For $\varrho \geq 0$ and $0 < y < Q_{\psi(\varrho)}$ put

$$\mathsf{u}_\varrho(y) \stackrel{\mathrm{def}}{=} \Big(1 - \frac{\varrho\, y}{\mathsf{A}(y)^2}\Big)^{-1/2},$$

so that $\mathsf{u}_0 \equiv 1$. The constants are

$$\bar{\mathsf{e}}_\beta \stackrel{\text{def}}{=} \max\Big(1,\ \frac{1}{4} + \frac{1}{2\log(1/\beta)}\Big), \qquad \mathsf{c}_\beta^+ \stackrel{\text{def}}{=} \mathsf{c}_\beta + \mathsf{c}'_\beta, \qquad \gamma_\beta^\circ \stackrel{\text{def}}{=} \mathsf{c}'_\beta + 1,$$

with c_β and c'_β of lemma SF.6, and $\Phi(1) = 0.34432\ldots$.

Letting $z \to y$ in the definitions gives the diagonal values

$$\begin{aligned} \mathcal{Z}(y,y) &= \mathsf{e}_\beta(y), \qquad \mathcal{X}(y,y) = r(y) - \mathsf{e}_\beta(y), \\ \gamma^2 \widehat{\Delta}(y,y) &= \big(\mathsf{A}(y) + \mathsf{e}_\beta(y)\big)^2 - \mathsf{e}_\beta(y)^2 = \mathsf{A}(y)\,\mathsf{j}(y). \end{aligned} \tag{SF.54}$$

At $y = z$ the weight $\mathsf{j}(y)\mathsf{j}(z_\ell)/(\gamma^2\widehat{\Delta}(y,z))$ of proposition SF.10 below is therefore $\mathsf{j}(z_\ell)/\mathsf{A}(y)$. The lag profile of theorem SF.10 comes from this diagonal value.

Lemma SF.13 (Participation slope)**.** *For every $y > 0$,*

$$\mathsf{e}_\beta(y) \le \frac{1}{4} + \frac{1}{2\log(1/\beta)} \le \bar{\mathsf{e}}_\beta.$$

Proof. The slope is a lattice sum of the logistic density $\mathsf{l}(v) \stackrel{\text{def}}{=} e^v/(1+e^v)^2$, which is unimodal, with maximum $\frac{1}{4}$ and integral 1 over $\mathbb{R}$. Put $x_j \stackrel{\text{def}}{=} w_j^2 y$. By lemma SF.4(i) and the identity $t/(1+t)^2 = \mathsf{l}(\log t)$ for $t > 0$,

$$\mathsf{e}_\beta(y) = \sum_{j\ge 0} x_j \Phi'(x_j) = \sum_{j\ge 0} \mathbb{E}\,\frac{x_j g^2}{(1+x_j g^2)^2} = \mathbb{E}\sum_{j\ge 0} \mathsf{l}\Big(\log\big((1-\beta)^2 y g^2\big) - 2j\log(1/\beta)\Big).$$

For fixed $g \neq 0$ the arguments of l lie on a lattice of spacing $2\log(1/\beta)$.

A lattice sum of a unimodal function is at most its maximum plus its integral divided by the spacing. Let $F \ge 0$ be unimodal with mode v_*, and let the points p form a lattice of spacing h. Each point $p < v_*$ other than the largest such point satisfies

$$F(p) \le \frac{1}{h}\int_p^{p+h} F,$$

because F is nondecreasing on $[p, p+h]$. Each point $p > v_*$ other than the smallest such point satisfies $F(p) \le h^{-1}\int_{p-h}^{p} F$ for the same reason. These cells are disjoint. The two remaining points $p_- \le v_* \le p_+ = p_- + h$ satisfy $F \ge \min(F(p_-), F(p_+))$ on $[p_-, p_+]$, so that

$$F(p_-) + F(p_+) \le \max F + \frac{1}{h}\int_{p_-}^{p_+} F.$$

Summing gives $\sum_p F(p) \le \max F + h^{-1}\int_{\mathbb{R}} F$. With $F = \mathsf{l}$ and $h = 2\log(1/\beta)$ the inner sum is at most $\frac{1}{4} + 1/(2\log(1/\beta))$ for every $g \neq 0$, and taking the expectation proves the lemma. □

Lemma SF.14 (Shifts of the participation level)**.** *Let $Q, y, z > 0$ and let $k \ge 0$ be an integer.*

(i) $r(\beta^{-2k}Q) - r(Q) = \sum_{i=1}^{k} \Phi\big((1-\beta)^2\beta^{-2i}Q\big) \in [0,k]$.

(ii) $r(\beta^{-2k}Q) \ge \mathsf{n}_\beta(Q) + k - \mathsf{c}_\beta$.

(iii) $r(Q) \le \mathsf{n}_\beta(Q) + \mathsf{c}'_\beta + \mathsf{n}_\beta^-(Q)$.

(iv) $|r(y) - r(z)| \le \bar{\mathsf{e}}_\beta\,|\log(y/z)|$.

(v) If $\gamma \ge \gamma_\beta^\circ$ and $r(\beta^{-2}z) < \gamma$, then $\mathsf{A}(z) > \Phi(1)$.

Proof. For (i), reindexing the sum of lemma SF.4(i) gives

$$r(\beta^{-2k}Q) = \sum_{j\ge 0} \Phi\big((1-\beta)^2\beta^{2(j-k)}Q\big),$$

and the k terms with $j < k$ are the extra terms. Each lies in $[0,1]$.

For (ii), suppose first that $(1-\beta)^2\beta^{-2k}Q \ge 1$. Then lemma SF.6(i) at $\beta^{-2k}Q$ gives

$$r(\beta^{-2k}Q) > \mathsf{n}_\beta(\beta^{-2k}Q) - \mathsf{c}_\beta = \mathsf{n}_\beta(Q) + k - \mathsf{c}_\beta.$$

Otherwise $\mathsf{n}_\beta(Q) + k = \mathsf{n}_\beta(\beta^{-2k}Q) < 0 \le r(\beta^{-2k}Q)$.

For (iii), if $(1-\beta)^2 Q \ge 1$ the bound is the upper bound of lemma SF.6(i), with $\mathsf{n}^-_\beta(Q) = 0$. Otherwise $\mathsf{n}_\beta(Q) + \mathsf{n}^-_\beta(Q) = 0$, and $r(Q) \le r((1-\beta)^{-2}) \le \mathsf{c}'_\beta$ by the same lemma.

Part (iv) holds because $\mathrm{d}r/\mathrm{d}\log y = \mathsf{e}_\beta(y) \le \bar{\mathsf{e}}_\beta$ by lemma SF.13.

For (v), part (i) with $k = 1$ gives

$$\mathsf{A}(z) = \gamma - r(z) > r(\beta^{-2}z) - r(z) = \Phi\big((1-\beta)^2\beta^{-2}z\big).$$

If $(1-\beta)^2\beta^{-2}z \ge 1$, the right side is at least $\Phi(1)$, because Φ is increasing. Otherwise $z < (1-\beta)^{-2}$, so that $r(z) \le \mathsf{c}'_\beta$ by lemma SF.6(i), and $\mathsf{A}(z) \ge \gamma - \mathsf{c}'_\beta \ge 1 > \Phi(1)$. □

Lemma SF.15 (Kernel bounds)**.** *Let $y \ne z$, put $\delta \stackrel{\text{def}}{=} |\log(y/z)|$ and $r_\wedge \stackrel{\text{def}}{=} \min(r(y), r(z))$, and write $d_j \stackrel{\text{def}}{=} w_j g$ with $g \sim \mathcal{N}(0,1)$.*

(i) The two kernels are the series

$$\mathcal{Z}(y,z) = \sqrt{yz}\sum_{j\ge 0} \mathbb{E}\,\frac{d_j^2}{(1+yd_j^2)(1+zd_j^2)}, \qquad \mathcal{X}(y,z) = yz\sum_{j\ge 0} \mathbb{E}\,\frac{d_j^4}{(1+yd_j^2)(1+zd_j^2)}.$$

(ii) $0 \le \mathcal{Z}(y,z) \le \min\big(\bar{\mathsf{e}}_\beta\,\delta/(2\sinh(\delta/2)),\ s_2\sqrt{yz}\big)$, *and* $\delta/(2\sinh(\delta/2)) \le \min\big(1, (1+\delta)e^{-\delta/2}\big)$.

(iii) $\max(0, r_\wedge - \bar{\mathsf{e}}_\beta) \le \mathcal{X}(y,z) \le r_\wedge$.

(iv) $\mathcal{X}(y,z) + \mathcal{Z}(y,z) \le \sqrt{r(y)\,r(z)}$.

(v) If $\mathsf{A}(y) > 0$ and $\mathsf{A}(z) > 0$, then, with $\mathsf{A}_\vee = \mathsf{A}_\vee(y,z)$,

$$\gamma - \mathcal{X} \ge \mathsf{A}_\vee, \qquad \gamma - \mathcal{X} - \mathcal{Z} \ge \max\Big(\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta,\ \frac{\mathsf{A}_\vee}{2}\Big) > 0,$$

$$\mathsf{A}_\vee \max\Big(\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta,\ \frac{\mathsf{A}_\vee}{2}\Big) \le \gamma^2\widehat{\Delta}(y,z) \le \big(\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta\big)^2.$$

By continuity, (ii)–(v) also hold at $y = z$.

Proof. Part (i) is partial fractions. By lemma SF.4(i) and $F_\beta(y) = r(y)/y$,

$$r(y) = \sum_j \mathbb{E}\,\frac{yd_j^2}{1+yd_j^2}, \qquad F_\beta(y) = \sum_j \mathbb{E}\,\frac{d_j^2}{1+yd_j^2}.$$

The difference quotient of the first series is the series of $\mathcal{Z}/\sqrt{yz}$. The second form follows from $z\,r(y) - y\,r(z) = yz\big(F_\beta(y) - F_\beta(z)\big)$ and

$$\frac{d_j^2}{1+yd_j^2} - \frac{d_j^2}{1+zd_j^2} = \frac{(z-y)\,d_j^4}{(1+yd_j^2)(1+zd_j^2)}.$$

For (ii), nonnegativity is read from (i). The first bound uses $|r(y) - r(z)| \le \bar{\mathsf{e}}_\beta\delta$ (lemma SF.14(iv)) and $\sqrt{yz}/|y-z| = 1/(2\sinh(\delta/2))$. The second bounds every factor $(1+\cdots)^{-1}$ in (i) by one,

which leaves $\sqrt{yz}\sum_j w_j^2\,\mathbb{E}\,g^2 = s_2\sqrt{yz}$. The bound $\delta/(2\sinh(\delta/2)) \le 1$ is $\sinh x \ge x$. Since $2\sinh(\delta/2) = e^{\delta/2}(1-e^{-\delta})$, the bound $\delta/(2\sinh(\delta/2)) \le (1+\delta)e^{-\delta/2}$ is equivalent to $(1+\delta)e^{-\delta} \le 1$, that is to $e^{\delta} \ge 1+\delta$.

For (iii), take $z < y$, so that $y = ze^{\delta}$; the case $y < z$ is symmetric. Then

$$\mathcal{X}(y,z) = r(z) - \frac{r(y)-r(z)}{e^{\delta}-1}, \qquad 0 \le \frac{r(y)-r(z)}{e^{\delta}-1} \le \frac{\bar{\mathsf{e}}_\beta\,\delta}{e^{\delta}-1} \le \bar{\mathsf{e}}_\beta,$$

and $r(z) = r_\wedge$. Nonnegativity of $\mathcal{X}$ is read from (i).

For (iv), put $v(y) \stackrel{\text{def}}{=} \big(yd^2, \sqrt{y}\,d\big)/(1+yd^2) \in \mathbb{R}^2$ for a real number d. Then (i) reads

$$\mathcal{X}(y,z) + \mathcal{Z}(y,z) = \sum_j \mathbb{E}\,\big\langle v(y), v(z)\big\rangle\Big|_{d=d_j}, \qquad |v(y)|^2 = \frac{yd^2}{1+yd^2},$$

and the second identity summed over j and averaged over g is $r(y)$. The Cauchy–Schwarz inequality in j and g gives (iv).

For (v), part (iii) gives $\gamma - \mathcal{X} \ge \gamma - r_\wedge = \mathsf{A}_\vee$. With $\mathcal{Z} \le \bar{\mathsf{e}}_\beta$ from (ii), this gives $\gamma - \mathcal{X} - \mathcal{Z} \ge \mathsf{A}_\vee - \bar{\mathsf{e}}_\beta$. By (iv) and the arithmetic-geometric mean inequality,

$$\gamma - \mathcal{X} - \mathcal{Z} \ge \gamma - \tfrac{1}{2}\big(r(y)+r(z)\big) = \tfrac{1}{2}\big(\mathsf{A}(y)+\mathsf{A}(z)\big) \ge \frac{\mathsf{A}_\vee}{2}.$$

The determinant factors as

$$\gamma^2\widehat{\Delta} = (\gamma - \mathcal{X} - \mathcal{Z})(\gamma - \mathcal{X} + \mathcal{Z}),$$

and $\gamma - \mathcal{X} + \mathcal{Z} \ge \mathsf{A}_\vee$, which gives the lower bound. For the upper bound, (iii) gives $0 \le \gamma - \mathcal{X} \le \gamma - r_\wedge + \bar{\mathsf{e}}_\beta = \mathsf{A}_\vee + \bar{\mathsf{e}}_\beta$, so that $\gamma^2\widehat{\Delta} \le (\gamma - \mathcal{X})^2 \le (\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta)^2$. □

In the representation below, the two roots of the lag kernel are the levels y and z_ℓ, and the determinant depends only on the pair (y,z).

Proposition SF.10 (Lag coefficients in participation coordinates)**.** *Let $\varrho \ge 0$ and let $\ell \ge 1$ be an integer. Then $\widehat{\Delta}(y,z) > 0$ whenever $y < Q_{\psi(\varrho)}$ and $z_\ell < Q_{\psi(\varrho)}$, and*

$$\overline{C}_{1,\ell}(\varrho) = \frac{1}{\pi^2\gamma^3}\int_0^\infty\!\!\int_0^\infty D(y,z)\,\mathbf{1}\{y < Q_{\psi(\varrho)},\ z_\ell < Q_{\psi(\varrho)}\}\,\frac{\mathsf{j}(y)\,\mathsf{u}_\varrho(y)\,\mathsf{j}(z_\ell)\,\mathsf{u}_\varrho(z_\ell)}{\widehat{\Delta}(y,z)}\,\mathrm{d}y\,\mathrm{d}z. \tag{SF.55}$$

The lag ℓ enters eq. (SF.55) only through the weight $\mathsf{j}(z_\ell)\mathsf{u}_\varrho(z_\ell)$ and the constraint $z_\ell < Q_{\psi(\varrho)}$.

Proof. The proof writes the two scalars of the isotropic lag kernel as divided differences of r at the pair (y,z), and then changes variables from the spectral parameters to the roots.

The kernel at a pair of roots. By lemma SF.11(ii), the lag kernel at (s,s') is $b_\ell\,qq'/(\gamma^3\Delta_\ell)$ with $q = q(-s)$ and $q' = q(-s')$. Put $y \stackrel{\text{def}}{=} q$ and $z \stackrel{\text{def}}{=} \beta^{2\ell}q'$. The divided-difference forms eq. (SG.15) and $F_\beta(x) = r(x)/x$ give

$$\sqrt{qq'}\,b_\ell = \sqrt{yz}\,\frac{r(y)-r(z)}{y-z} = \mathcal{Z}(y,z), \qquad qq'\,e_\ell = yz\,\frac{F_\beta(y)-F_\beta(z)}{z-y} = \mathcal{X}(y,z).$$

The determinant of lemma SF.11 is therefore $\Delta_\ell = \widehat{\Delta}(y,z)$, which is positive by lemma SF.11(i).

Change of variables. In the square case the closure eq. (SF.7) inverts as $s = \mathsf{s}(q) \stackrel{\text{def}}{=} \mathsf{A}(q)^2/q$. The map s is continuously differentiable and strictly decreasing from $(0, Q_\gamma)$ onto $(0,\infty)$, with

$$\mathsf{s}'(y) = \frac{-2y\,r'(y)\,\mathsf{A}(y) - \mathsf{A}(y)^2}{y^2} = -\frac{\mathsf{A}(y)\,\mathsf{j}(y)}{y^2},$$

and $q(-\mathsf{s}(y)) = y$ by the uniqueness of proposition SF.1(ii). For $\varrho > 0$ one has $\mathsf{s}(Q_{\psi(\varrho)}) = \varrho$, by the closure at $s = \varrho$. Hence $x = \sqrt{\mathsf{s}(y) - \varrho}$ is a decreasing bijection of $(0, Q_{\psi(\varrho)})$ onto $(0,\infty)$ for every $\varrho \ge 0$, with

$$\Big|\frac{\mathrm{d}x}{\mathrm{d}y}\Big| = \frac{\mathsf{A}(y)\,\mathsf{j}(y)}{2y^2\sqrt{\mathsf{A}(y)^2/y - \varrho}} = \frac{\mathsf{j}(y)\,\mathsf{u}_\varrho(y)}{2y^{3/2}}.$$

Apply this substitution in both variables of eq. (SF.52), with $y = q(-\varrho - x^2)$ and $y' = q(-\varrho - x'^2)$. Since the kernel is $b_\ell\, yy'/(\gamma^3\widehat{\Delta})$,

$$\begin{aligned}\overline{C}_{1,\ell}(\varrho) &= \frac{4}{\pi^2}\iint_{y,y'<Q_{\psi(\varrho)}} \frac{b_\ell\, yy'}{\gamma^3\,\widehat{\Delta}}\cdot\frac{\mathsf{j}(y)\mathsf{u}_\varrho(y)\,\mathsf{j}(y')\mathsf{u}_\varrho(y')}{4\,(yy')^{3/2}}\,\mathrm{d}y\,\mathrm{d}y' \\ &= \frac{1}{\pi^2\gamma^3}\iint_{y,y'<Q_{\psi(\varrho)}} \frac{b_\ell\,\mathsf{j}(y)\mathsf{u}_\varrho(y)\,\mathsf{j}(y')\mathsf{u}_\varrho(y')}{\widehat{\Delta}\,\sqrt{yy'}}\,\mathrm{d}y\,\mathrm{d}y'.\end{aligned}$$

The lag shift. Substitute $y' = \beta^{-2\ell}z = z_\ell$. Then

$$\frac{b_\ell\,\mathrm{d}y'}{\sqrt{yy'}} = \beta^\ell\,\frac{r(y)-r(z)}{y-z}\cdot\frac{\beta^{-2\ell}\,\mathrm{d}z}{\beta^{-\ell}\sqrt{yz}} = D(y,z)\,\mathrm{d}z,$$

which is eq. (SF.55). □

The next lemma averages the divided difference D over the intermediate participation level. Under this averaging the level enters only through the base point of a fixed random map, and r' is absorbed by the change of variables $\mathrm{d}\eta = r'(s)\,\mathrm{d}s$.

Lemma SF.16 (Layer averaging)**.** *Let $\nu_\star$ be the measure on $\{+,-\}\times(0,\infty)^2$ whose density in (a,c) on each of the two copies is*

$$\frac{1}{\pi^2}\cdot\frac{1}{2\sinh\big((a+c)/2\big)},$$

write $\mathbb{E}_\star$ for the integral against it, and put $\tau \stackrel{\text{def}}{=} a+c$,

$$\Pi_+(s;a,c) \stackrel{\text{def}}{=} (se^a, se^{-c}), \qquad \Pi_-(s;a,c) \stackrel{\text{def}}{=} (se^{-c}, se^a).$$

(i) $\nu_\star$ is a probability measure, and for every nonnegative function Ψ on $(0,\infty)^2$,

$$\frac{1}{\pi^2}\int_0^\infty\!\!\int_0^\infty D(y,z)\,\Psi(y,z)\,\mathrm{d}y\,\mathrm{d}z = \int_0^\infty \mathbb{E}_\star\left[\Psi\big(\Pi(Q_\eta;\cdot)\big)\right]\mathrm{d}\eta. \tag{SF.56}$$

(ii) The moments $\mathsf{M}_k \stackrel{\text{def}}{=} \mathbb{E}_\star\,\tau^k$ and $\mathsf{M}_e \stackrel{\text{def}}{=} \mathbb{E}_\star\,e^{\tau/4}$ are

$$\mathsf{M}_1 = \frac{28\,\zeta(3)}{\pi^2} = 3.41023\ldots, \quad \mathsf{M}_2 = 2\pi^2,$$

$$\mathsf{M}_3 = \frac{1488\,\zeta(5)}{\pi^2} = 156.333\ldots, \quad \mathsf{M}_e = 2 + \frac{16\,\mathrm{G}}{\pi^2} = 3.48491\ldots,$$

where $\mathrm{G} = 0.91596\ldots$ is Catalan's constant.

(iii) If $(y,z) = \Pi(s;a,c)$, then $|\log(y/s)| \le \tau$, $|\log(z/s)| \le \tau$, $|\mathsf{A}(y) - \mathsf{A}(s)| \le \bar{\mathsf{e}}_\beta\tau$, $|\mathsf{A}(z) - \mathsf{A}(s)| \le \bar{\mathsf{e}}_\beta\tau$ and $z \ge se^{-\tau}$.

Proof. For (i), write the divided difference as an integral of r' over the intermediate level,

$$D(y,z) = \frac{1}{|y-z|\sqrt{yz}}\int_{\min(y,z)}^{\max(y,z)} r'(s)\,\mathrm{d}s,$$

and exchange the order of integration. For fixed s, the region $z < s < y$ is the image of $(0,\infty)^2$ under $\Pi_+(s;\cdot)$, with $\mathrm{d}y\,\mathrm{d}z = yz\,\mathrm{d}a\,\mathrm{d}c$ and

$$\frac{yz}{|y-z|\sqrt{yz}} = \frac{e^{(a-c)/2}}{e^a - e^{-c}} = \frac{1}{2\sinh(\tau/2)}.$$

The region $y < s < z$ is the image under $\Pi_-(s;\cdot)$, with the same factor. Finally $\eta = r(s)$ is a continuously differentiable increasing bijection of $(0,\infty)$ onto itself (lemma SF.4(i)), with $\mathrm{d}\eta = r'(s)\,\mathrm{d}s$ and $s = Q_\eta$. This gives eq. (SF.56), once $\nu_\star$ is shown to have mass one, which is the case $k = 0$ below.

For (ii), the density depends on (a,c) only through τ, and the set $\{a+c\in \mathrm{d}\tau\}$ has length $\tau\,\mathrm{d}\tau$. Hence, for $k\ge 0$,

$$\mathbb{E}_\star\,\tau^k=\frac{2}{\pi^2}\int_0^\infty \frac{\tau^{k+1}}{2\sinh(\tau/2)}\,\mathrm{d}\tau.$$

Expanding $1/(2\sinh(\tau/2))=\sum_{n\ge0}e^{-(n+1/2)\tau}$ and integrating term by term gives, for $p>0$,

$$\int_0^\infty \frac{\tau^p}{2\sinh(\tau/2)}\,\mathrm{d}\tau=\Gamma(p+1)\sum_{n\ge0}\big(n+\tfrac12\big)^{-p-1}=2^{p+1}\,\Gamma(p+1)\big(1-2^{-p-1}\big)\,\zeta(p+1).$$

At $p=1,2,3,4$ this is $\pi^2/2$, $14\zeta(3)$, π^4 and $744\zeta(5)$. Multiplying by $2/\pi^2$ gives $\mathbb{E}_\star\,1=1$ and the stated $\mathsf{M}_1,\mathsf{M}_2,\mathsf{M}_3$. For M_e the same expansion gives

$$\mathsf{M}_e=\frac{2}{\pi^2}\sum_{n\ge0}\int_0^\infty \tau\,e^{-(n+1/4)\tau}\,\mathrm{d}\tau=\frac{2}{\pi^2}\sum_{n\ge0}\big(n+\tfrac14\big)^{-2}=\frac{32}{\pi^2}\sum_{n\ge0}\frac{1}{(4n+1)^2},$$

and $\sum_{n\ge0}(4n+1)^{-2}=\pi^2/16+\mathrm{G}/2$, the sum of $\sum_{n\ge0}(2n+1)^{-2}=\pi^2/8$ and $\mathrm{G}=\sum_{n\ge0}\big((4n+1)^{-2}-(4n+3)^{-2}\big)$, halved.

For (iii), (y,z) is (se^{a},se^{-c}) or (se^{-c},se^{a}) with $a,c\ge0$, which gives the bounds on the logarithms and on z. Lemma SF.14(iv) turns them into the bounds on A. □

The low levels, where $(1-\beta)^2z<1$, are controlled by the moments of n_β^- under the averaging.

Lemma SF.17 (Low levels)**.** *For $\eta>0$ let $(y,z)=\Pi(Q_\eta;a,c)$ as in lemma SF.16, and put $\sigma\overset{\text{def}}{=}\log\big((1-\beta)^2Q_\eta\big)$. Then*

$$\mathsf{n}_\beta^-(z)\le\frac{(\tau-\sigma)_+}{2\log(1/\beta)},$$

and $\mathsf{L}_k\overset{\text{def}}{=}\int_0^\infty\mathbb{E}_\star\big[\mathsf{n}_\beta^-(z)^k\big]\,\mathrm{d}\eta$ satisfies $\mathsf{L}_k\le\bar{\mathsf{L}}_k$ for $k=1,2$, where

$$\bar{\mathsf{L}}_1\overset{\text{def}}{=}\frac{1}{2\log(1/\beta)}\Big(\frac{\mathsf{M}_1+1}{1-\beta^2}+\frac{\bar{\mathsf{e}}_\beta\mathsf{M}_2}{2}\Big),\qquad \bar{\mathsf{L}}_2\overset{\text{def}}{=}\frac{1}{4\log^2(1/\beta)}\Big(\frac{\mathsf{M}_2+2\mathsf{M}_1+2}{1-\beta^2}+\frac{\bar{\mathsf{e}}_\beta\mathsf{M}_3}{3}\Big).$$

Proof. The function n_β^- is nonincreasing and $z\ge Q_\eta e^{-\tau}$ (lemma SF.16(iii)), which gives the first claim.

For the moments, change variables from η to σ. Since $\eta=r(Q_\eta)$,

$$\mathrm{d}\eta=r'(Q_\eta)\,\mathrm{d}Q_\eta=\mathsf{e}_\beta(Q_\eta)\,\mathrm{d}\sigma.$$

Two bounds on the slope are used. For $\sigma<0$, $\mathsf{e}_\beta(s)\le s_2\,s=e^\sigma/(1-\beta^2)$, because $r'\le s_2$ (lemma SF.4(i),(ii)) and $s_2/(1-\beta)^2=1/(1-\beta^2)$. For $\sigma\ge0$, $\mathsf{e}_\beta\le\bar{\mathsf{e}}_\beta$ (lemma SF.13). Hence, for $\tau\ge0$ and $k=1,2$,

$$\int_{\mathbb{R}}(\tau-\sigma)_+^k\,\mathsf{e}_\beta\,\mathrm{d}\sigma\le\frac{1}{1-\beta^2}\int_0^\infty(\tau+v)^ke^{-v}\,\mathrm{d}v+\bar{\mathsf{e}}_\beta\,\frac{\tau^{k+1}}{k+1},$$

with $\int_0^\infty(\tau+v)e^{-v}\mathrm{d}v=\tau+1$ and $\int_0^\infty(\tau+v)^2e^{-v}\mathrm{d}v=\tau^2+2\tau+2$. Dividing by $(2\log(1/\beta))^k$ and integrating against $\nu_\star$ with lemma SF.16(ii) gives the bounds. □

The lag sum and the lag profile at zero ridge. Put

$$\mathsf{K}_0\overset{\text{def}}{=}11\bar{\mathsf{e}}_\beta+\frac{48\,\bar{\mathsf{e}}_\beta^3}{\Phi(1)^2},\qquad \mathsf{K}_1\overset{\text{def}}{=}8+\frac{32\,\bar{\mathsf{e}}_\beta^2}{\Phi(1)^2},\qquad \mathsf{K}_2\overset{\text{def}}{=}\frac{2}{\Phi(1)}+\frac{4\,\bar{\mathsf{e}}_\beta}{\Phi(1)^2},$$

$$\mathsf{K}_\beta^-\overset{\text{def}}{=}\frac52\,\bar{\mathsf{e}}_\beta(\mathsf{M}_1+1),\qquad \mathsf{K}_\beta^+\overset{\text{def}}{=}\frac{\bar{\mathsf{e}}_\beta\mathsf{M}_1}{2}+\mathsf{K}_0+\mathsf{K}_1\mathsf{c}_\beta^++2\mathsf{K}_2(\mathsf{c}_\beta^+)^2+\mathsf{K}_1\bar{\mathsf{L}}_1+2\mathsf{K}_2\bar{\mathsf{L}}_2.\quad\text{(SF.57)}$$

Theorem SF.9 (Lag sum at zero ridge)**.** *Let $\beta\in(0,1)$, $N_{\mathrm{in}}=N_{\mathrm{out}}=N$ and $\gamma=N/B\ge\gamma_\beta^\circ$. Then*

$$-\mathsf{K}_\beta^-\le\sum_{\ell\ge1}\overline{C}_{1,\ell}(0)-\frac{\gamma}{4}\le\mathsf{K}_\beta^+.$$

Proof. The proof sums over the lag first, then averages over the participation level, and compares the integrand with $\mathsf{A}(s)/2$ pointwise. The errors are linear or quadratic in τ and n_β^-, which have finite moments.

Step 1: the summed representation. By proposition SF.10 at $\varrho = 0$, summed over ℓ, and lemma SF.16(i),

$$\sum_{\ell \geq 1} \overline{C}_{1,\ell}(0) = \frac{1}{\gamma} \int_0^\gamma \mathbb{E}_\star \left[\Psi\big(\Pi(Q_\eta; \cdot)\big) \right] \mathrm{d}\eta, \tag{SF.58}$$

where

$$\Psi(y,z) \stackrel{\mathrm{def}}{=} \mathbf{1}\{y < Q_\gamma\} \frac{\mathsf{j}(y)\,\mathsf{T}(z)}{\gamma^2\, \widehat{\Delta}(y,z)}, \qquad \mathsf{T}(z) \stackrel{\mathrm{def}}{=} \sum_{\ell \geq 1} \mathbf{1}\{z_\ell < Q_\gamma\}\, \mathsf{j}(z_\ell).$$

The range of η is $(0,\gamma)$, because $\Psi(y,z) \neq 0$ forces $y < Q_\gamma$ and $z < z_1 < Q_\gamma$, and the base point $s = Q_\eta$ lies between y and z. Write $x \stackrel{\mathrm{def}}{=} \mathsf{A}(s) = \gamma - \eta$. Then

$$\frac{1}{\gamma} \int_0^\gamma \frac{x}{2}\, \mathrm{d}\eta = \frac{\gamma}{4}.$$

Step 2: bounds on T. Put $L_z \stackrel{\mathrm{def}}{=} \gamma - \mathsf{n}_\beta(z) + \mathsf{c}_\beta$ and $E_z \stackrel{\mathrm{def}}{=} \mathsf{c}_\beta^+ + \mathsf{n}_\beta^-(z)$. A term of $\mathsf{T}(z)$ with $z_\ell < Q_\gamma$ has $r(z_\ell) < \gamma$, and lemma SF.14(ii) gives $r(z_\ell) \geq \mathsf{n}_\beta(z) + \ell - \mathsf{c}_\beta$. Hence $\ell < L_z$ and, by lemma SF.13,

$$\mathsf{j}(z_\ell) \leq \gamma - r(z_\ell) + 2\bar{\mathsf{e}}_\beta \leq L_z - \ell + 2\bar{\mathsf{e}}_\beta.$$

By lemma SF.14(iii), $L_z \leq \mathsf{A}(z) + E_z \leq \mathsf{A}_\vee + E_z$. If $\mathsf{T}(z) > 0$, then $L_z > 1$, and since $\sum_{1 \leq \ell < L}(L - \ell) \leq L^2/2$ and $t \mapsto t^2/2 + 2\bar{\mathsf{e}}_\beta t$ increases on $[0,\infty)$,

$$\mathsf{T}(z) \leq \frac{L_z^2}{2} + 2\bar{\mathsf{e}}_\beta L_z \leq \frac{(\mathsf{A}_\vee + E_z)^2}{2} + 2\bar{\mathsf{e}}_\beta(\mathsf{A}_\vee + E_z); \tag{SF.59}$$

the bound also holds when $\mathsf{T}(z) = 0$. Conversely, every $1 \leq \ell < \mathsf{A}(z)$ has $r(z_\ell) \leq r(z) + \ell < \gamma$ by lemma SF.14(i), and $\mathsf{j}(z_\ell) \geq \mathsf{A}(z_\ell) \geq \mathsf{A}(z) - \ell$. Comparing the decreasing summand with an integral,

$$\mathsf{T}(z) \geq \sum_{1 \leq \ell < \mathsf{A}(z)} \big(\mathsf{A}(z) - \ell\big) \geq \frac{(\mathsf{A}(z) - 1)_+^2}{2}. \tag{SF.60}$$

Step 3: upper bound. On the set $\{\Psi > 0\}$ one has $\mathsf{A}(y) > 0$, and $\mathsf{T}(z) > 0$ forces $r(\beta^{-2}z) < \gamma$, so that $\mathsf{A}(z) > \Phi(1)$ by lemma SF.14(v). Thus $\mathsf{A}_\vee \geq \Phi(1)$ there. With $\mathsf{j}(y) \leq \mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta$, eq. (SF.59) and the lower bound of lemma SF.15(v),

$$\Psi \leq \frac{(\mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta)\big((\mathsf{A}_\vee + E_z)^2/2 + 2\bar{\mathsf{e}}_\beta(\mathsf{A}_\vee + E_z)\big)}{\mathsf{A}_\vee \max(\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta, \mathsf{A}_\vee/2)}.$$

Suppose first that $\mathsf{A}_\vee \geq 2\bar{\mathsf{e}}_\beta$. Then

$$\frac{\mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta}{\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta} = 1 + \frac{3\bar{\mathsf{e}}_\beta}{\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta} \leq 1 + \frac{6\bar{\mathsf{e}}_\beta}{\mathsf{A}_\vee} \leq 4.$$

The second factor divided by $\mathsf{A}_\vee$ is $\mathsf{A}_\vee/2 + E_z + E_z^2/(2\mathsf{A}_\vee) + 2\bar{\mathsf{e}}_\beta + 2\bar{\mathsf{e}}_\beta E_z/\mathsf{A}_\vee$. Applying the factor $1 + 6\bar{\mathsf{e}}_\beta/\mathsf{A}_\vee$ to its first term and the factor 4 to the rest, and using $2\bar{\mathsf{e}}_\beta/\mathsf{A}_\vee \leq 1$ and $\mathsf{A}_\vee \geq \Phi(1)$,

$$\Psi \leq \frac{\mathsf{A}_\vee}{2} + 3\bar{\mathsf{e}}_\beta + 4\Big(E_z + \frac{E_z^2}{2\mathsf{A}_\vee} + 2\bar{\mathsf{e}}_\beta + \frac{2\bar{\mathsf{e}}_\beta E_z}{\mathsf{A}_\vee}\Big) \leq \frac{\mathsf{A}_\vee}{2} + 11\bar{\mathsf{e}}_\beta + 8E_z + \frac{2E_z^2}{\Phi(1)}.$$

Suppose instead that $\Phi(1) \leq \mathsf{A}_\vee < 2\bar{\mathsf{e}}_\beta$. Then $\mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta < 4\bar{\mathsf{e}}_\beta$, the second factor of the numerator is at most $6\bar{\mathsf{e}}_\beta^2 + 4\bar{\mathsf{e}}_\beta E_z + E_z^2/2$, and the denominator is at least $\Phi(1)^2/2$, so that

$$\Psi \leq \frac{8\bar{\mathsf{e}}_\beta}{\Phi(1)^2}\Big(6\bar{\mathsf{e}}_\beta^2 + 4\bar{\mathsf{e}}_\beta E_z + \frac{E_z^2}{2}\Big).$$

In both cases, and also where $\Psi = 0$,

$$\Psi \leq \frac{\mathsf{A}_\vee}{2} + \mathsf{K}_0 + \mathsf{K}_1 E_z + \mathsf{K}_2 E_z^2.$$

Now $\mathsf{A}_\vee \le x + \bar{\mathsf{e}}_\beta \tau$ by lemma SF.16(iii), and $E_z^2 \le 2(\mathsf{c}_\beta^+)^2 + 2\,\mathsf{n}_\beta^-(z)^2$. Therefore

$$\Psi \le \frac{x}{2} + \frac{\bar{\mathsf{e}}_\beta \tau}{2} + \mathsf{K}_0 + \mathsf{K}_1 \mathsf{c}_\beta^+ + 2\mathsf{K}_2 (\mathsf{c}_\beta^+)^2 + \mathsf{K}_1\, \mathsf{n}_\beta^-(z) + 2\mathsf{K}_2\, \mathsf{n}_\beta^-(z)^2.$$

Integrate against $\gamma^{-1} \mathrm{d}\eta\, \nu_\star$ over $\eta \in (0, \gamma)$. The first term gives $\gamma/4$, the second $\bar{\mathsf{e}}_\beta \mathsf{M}_1/2$, the constants give themselves, and the last two terms give at most $(\mathsf{K}_1 \bar{\mathsf{L}}_1 + 2\mathsf{K}_2 \bar{\mathsf{L}}_2)/\gamma$ by lemma SF.17, which is at most $\mathsf{K}_1 \bar{\mathsf{L}}_1 + 2\mathsf{K}_2 \bar{\mathsf{L}}_2$ because $\gamma \ge 1$.

Step 4: lower bound. By eq. (SF.60), $\mathsf{j}(y) \ge \mathsf{A}(y)$ and the upper bound of lemma SF.15(v),

$$\Psi \ge \mathbf{1}\{y < Q_\gamma\}\, \frac{\mathsf{A}(y)\, (\mathsf{A}(z) - 1)_+^2}{2(\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta)^2}.$$

Put $\upsilon \stackrel{\mathrm{def}}{=} \bar{\mathsf{e}}_\beta(\tau + 1)$. Since $\bar{\mathsf{e}}_\beta \ge 1$, lemma SF.16(iii) gives

$$\mathsf{A}(y) \ge x - \upsilon, \qquad \mathsf{A}(z) - 1 \ge x - \upsilon, \qquad \mathsf{A}_\vee + \bar{\mathsf{e}}_\beta \le x + \upsilon.$$

If $x > \upsilon$, then $\mathsf{A}(y) > 0$, that is $y < Q_\gamma$, and with $t \stackrel{\mathrm{def}}{=} \upsilon/x < 1$,

$$\Psi \ge \frac{(x-\upsilon)^3}{2(x+\upsilon)^2} = \frac{x}{2}\,(1-t)\Big(\frac{1-t}{1+t}\Big)^2 \ge \frac{x}{2}\,(1-t)(1-2t)^2.$$

For $t < \frac{1}{5}$ this is at least $\frac{x}{2}(1-3t)(1-2t) \ge \frac{x}{2}(1-5t) = \frac{x-5\upsilon}{2}$. For $t \ge \frac{1}{5}$, and for $x \le \upsilon$, the right side $(x-5\upsilon)/2$ is negative. Hence $\Psi \ge (x-5\upsilon)/2$ everywhere, and integration as in Step 3, with $\mathbb{E}_\star\, \upsilon = \bar{\mathsf{e}}_\beta(\mathsf{M}_1 + 1)$, gives

$$\sum_{\ell \ge 1} \overline{C}_{1,\ell}(0) \ge \frac{\gamma}{4} - \frac{5}{2}\, \bar{\mathsf{e}}_\beta(\mathsf{M}_1 + 1).$$

□

The lag sum is $\gamma/4$ because the individual lag coefficients follow a fixed profile in the variable ℓ/γ, whose integral over $[0,1]$ is $\frac{1}{4}$. Put

$$\mathcal{G}(x) \stackrel{\mathrm{def}}{=} \begin{cases} 1 - x + x \log x, & 0 \le x \le 1, \\ 0, & x \ge 1, \end{cases}$$

with $\mathcal{G}(0) = 1$, and

$$\mathsf{K}_3 \stackrel{\mathrm{def}}{=} 14\bar{\mathsf{e}}_\beta + \frac{32\,\bar{\mathsf{e}}_\beta^2}{\Phi(1)}, \qquad \mathsf{K}_4 \stackrel{\mathrm{def}}{=} 4 + \frac{8\,\bar{\mathsf{e}}_\beta}{\Phi(1)}, \qquad \mathsf{K}_5 \stackrel{\mathrm{def}}{=} 4 + \frac{4 + 8\bar{\mathsf{e}}_\beta}{\Phi(1)} + \frac{8\bar{\mathsf{e}}_\beta + 32\bar{\mathsf{e}}_\beta^2}{\Phi(1)^2},$$

$$\mathsf{K}_\beta^{\mathrm{pr}} \stackrel{\mathrm{def}}{=} \big(\bar{\mathsf{e}}_\beta \mathsf{M}_1 + 1\big) + \frac{2(\mathsf{K}_3 + \mathsf{K}_4 \mathsf{c}_\beta^+)(\Phi(1) + \bar{\mathsf{e}}_\beta \mathsf{M}_1)}{\Phi(1)} \Big(1 + \log \frac{1}{2\Phi(1)}\Big) + \frac{\mathsf{K}_4 \bar{\mathsf{L}}_1}{\Phi(1)},$$

$$\delta_\beta \stackrel{\mathrm{def}}{=} \min\Big(\frac{1}{8\bar{\mathsf{e}}_\beta}, \frac{\log(1/\beta)}{8}\Big), \qquad \mathsf{K}_\beta^{\mathrm{dec}} \stackrel{\mathrm{def}}{=} \mathsf{K}_5 \Big(1 + \mathsf{c}_\beta^+ + \frac{1}{\delta_\beta}\Big) \mathsf{M}_e \Big(1 + \frac{8}{7(1-\beta^2)} + 8\bar{\mathsf{e}}_\beta\Big) e^{\delta_\beta \mathsf{c}_\beta^+}.$$

Theorem SF.10 (Lag profile at zero ridge)**.** *Let $\beta \in (0,1)$, $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B \ge \gamma_\beta^\circ$. For every integer $\ell \ge 1$,*

$$-\frac{4\bar{\mathsf{e}}_\beta(\mathsf{M}_1 + 1)}{\gamma}\Big(1 + \log^+ \frac{\gamma}{4\bar{\mathsf{e}}_\beta}\Big) \le \overline{C}_{1,\ell}(0) - \mathcal{G}\Big(\frac{\ell}{\gamma}\Big) \le \frac{\mathsf{K}_\beta^{\mathrm{pr}}(1 + \log\gamma)}{\gamma},$$

$$\overline{C}_{1,\ell}(0) \le \mathsf{K}_\beta^{\mathrm{dec}}\, e^{-\delta_\beta(\ell - \gamma)}.$$

If $\ell_\gamma/\gamma \to x \in [0,\infty)$ along integer pairs (N,B), then $\overline{C}_{1,\ell_\gamma}(0) \to \mathcal{G}(x)$. At the corner the profile vanishes quadratically,

$$\mathcal{G}(x) = \tfrac{1}{2}(1-x)^2 + \tfrac{1}{6}(1-x)^3 + O\big((1-x)^4\big) \qquad (x \uparrow 1),$$

and beyond it the lag coefficients decay exponentially in $\ell - \gamma$.

Proof. The proof is the unsummed version of the proof of theorem SF.9: the same pointwise bounds, with $\mathsf{T}(z)$ replaced by a single weight, compare the integrand with $(1-\ell/x)_+$, whose integral over the level is exactly $\mathcal{G}(\ell/\gamma)$.

Step 1: representation. By proposition SF.10 at $\varrho = 0$ and lemma SF.16(i),

$$\overline{C}_{1,\ell}(0) = \frac{1}{\gamma}\int_0^\gamma \mathbb{E}_\star[\Psi_\ell]\,\mathrm{d}\eta, \qquad \Psi_\ell \stackrel{\text{def}}{=} \mathbf{1}\{y < Q_\gamma,\ z_\ell < Q_\gamma\}\,\frac{\mathsf{j}(y)\,\mathsf{j}(z_\ell)}{\gamma^2\,\widehat{\Delta}(y,z)},$$

with $(y,z) = \Pi(Q_\eta;\cdot)$. With $x = \gamma - \eta$, direct integration gives

$$\frac{1}{\gamma}\int_0^\gamma \Big(1-\frac{\ell}{x}\Big)_+ \mathrm{d}\eta = \frac{1}{\gamma}\int_\ell^\gamma \Big(1-\frac{\ell}{x}\Big)\mathrm{d}x = \mathcal{G}\Big(\frac{\ell}{\gamma}\Big).$$

One more integral recurs below. For $Y > 0$,

$$\frac{1}{\gamma}\int_0^\gamma \min\Big(1,\frac{Y}{x}\Big)\mathrm{d}\eta \le \frac{Y}{\gamma}\Big(1+\log^+\frac{\gamma}{Y}\Big), \tag{SF.61}$$

and the right side is nondecreasing in Y.

Step 2: upper bound. As in Steps 2 and 3 of the proof of theorem SF.9, $\mathsf{A}_\vee \ge \Phi(1)$ on $\{\Psi_\ell > 0\}$, and

$$\mathsf{j}(z_\ell)\,\mathbf{1}\{z_\ell < Q_\gamma\} \le (\mathsf{A}_\vee + E_z - \ell)_+ + 2\bar{\mathsf{e}}_\beta,$$

so that

$$\Psi_\ell \le \frac{(\mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta)\big((\mathsf{A}_\vee + E_z - \ell)_+ + 2\bar{\mathsf{e}}_\beta\big)}{\mathsf{A}_\vee \max(\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta, \mathsf{A}_\vee/2)}.$$

If $\mathsf{A}_\vee \ge 2\bar{\mathsf{e}}_\beta$, this is at most

$$\Big(1+\frac{6\bar{\mathsf{e}}_\beta}{\mathsf{A}_\vee}\Big)\Big[\Big(1-\frac{\ell}{\mathsf{A}_\vee}\Big)_+ + \frac{E_z + 2\bar{\mathsf{e}}_\beta}{\mathsf{A}_\vee}\Big] \le \Big(1-\frac{\ell}{\mathsf{A}_\vee}\Big)_+ + \frac{14\bar{\mathsf{e}}_\beta + 4E_z}{\mathsf{A}_\vee}.$$

If $\mathsf{A}_\vee < 2\bar{\mathsf{e}}_\beta$, it is at most $8\bar{\mathsf{e}}_\beta(4\bar{\mathsf{e}}_\beta + E_z)/(\Phi(1)\mathsf{A}_\vee)$. Hence

$$\Psi_\ell \le \Big(1-\frac{\ell}{\mathsf{A}_\vee}\Big)_+ + \frac{\mathsf{K}_3 + \mathsf{K}_4 E_z}{\mathsf{A}_\vee}.$$

The first term is compared with its value at x. The function $A \mapsto (1-\ell/A)_+$ is nondecreasing, and $(1-\ell/(x+\delta))_+ - (1-\ell/x)_+ \le \min(1, \delta/x)$ for $\delta \ge 0$; with $\mathsf{A}_\vee \le x + \bar{\mathsf{e}}_\beta\tau$ this gives

$$\Big(1-\frac{\ell}{\mathsf{A}_\vee}\Big)_+ \le \Big(1-\frac{\ell}{x}\Big)_+ + \min\Big(1,\frac{1+\bar{\mathsf{e}}_\beta\tau}{x}\Big).$$

The second term is controlled by

$$\frac{1}{\mathsf{A}_\vee} \le \frac{1}{\Phi(1)}\min\Big(1,\frac{2(\Phi(1)+\bar{\mathsf{e}}_\beta\tau)}{x}\Big):$$

if $\bar{\mathsf{e}}_\beta\tau \le x/2$, then $\mathsf{A}_\vee \ge \mathsf{A}(y) \ge x/2$, and otherwise the minimum is one. Hence

$$\Psi_\ell \le \Big(1-\frac{\ell}{x}\Big)_+ + \min\Big(1,\frac{1+\bar{\mathsf{e}}_\beta\tau}{x}\Big) + \frac{\mathsf{K}_3 + \mathsf{K}_4\mathsf{c}_\beta^+}{\Phi(1)}\min\Big(1,\frac{2(\Phi(1)+\bar{\mathsf{e}}_\beta\tau)}{x}\Big) + \frac{\mathsf{K}_4\,\mathsf{n}_\beta^-(z)}{\Phi(1)}.$$

Integrate with eq. (SF.61). For $Y = 1 + \bar{\mathsf{e}}_\beta\tau \ge 1$ the logarithm is at most $\log\gamma$. For $Y = 2(\Phi(1)+\bar{\mathsf{e}}_\beta\tau) \ge 2\Phi(1)$ it is at most $\log\gamma + \log(1/(2\Phi(1)))$, and $1 + a + b \le (1+a)(1+b)$ for $a, b \ge 0$. Then take $\mathbb{E}_\star$, and bound the last term by lemma SF.17.

Step 3: lower bound. If $\mathsf{A}(z) > \ell$, then $r(z_\ell) \le r(z) + \ell < \gamma$ and $\mathsf{j}(z_\ell) \ge \mathsf{A}(z) - \ell$. With lemma SF.15(v),

$$\Psi_\ell \ge \mathbf{1}\{y < Q_\gamma\}\,\frac{\mathsf{A}(y)\,(\mathsf{A}(z)-\ell)_+}{(\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta)^2}.$$

With $\upsilon = \bar{\mathsf{e}}_\beta(\tau+1)$ as before, $\mathsf{A}(y) \ge x - \upsilon$, $\mathsf{A}(z) - \ell \ge x - \ell - \upsilon$ and $\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta \le x + \upsilon$. If $x - \ell > 4\upsilon$, put $t \stackrel{\text{def}}{=} \upsilon/x < \frac{1}{4}$. Then $y < Q_\gamma$, and

$$\Psi_\ell \ge \frac{1-t}{(1+t)^2}\Big(1-\frac{\ell}{x}-t\Big) \ge (1-2t)(1-t)\Big(1-\frac{\ell}{x}-t\Big) \ge (1-3t)\Big(1-\frac{\ell}{x}-t\Big) \ge 1-\frac{\ell}{x}-4t.$$

If $x-\ell \le 4\upsilon$, then $(1-\ell/x)_+ \le \min(1, 4\upsilon/x)$. Hence

$$\Psi_\ell \ge \Big(1-\frac{\ell}{x}\Big)_+ - \min\Big(1, \frac{4\upsilon}{x}\Big)$$

in all cases. Integrate with eq. (SF.61), using $4\upsilon \ge 4\bar{\mathsf{e}}_\beta$ in the logarithm and $\mathbb{E}_\star[4\upsilon] = 4\bar{\mathsf{e}}_\beta(\mathsf{M}_1+1)$.

Step 4: decay. Keeping the indicator $\mathbf{1}\{\ell < L_z\}$ in Step 2, and using $L_z - \ell + 2\bar{\mathsf{e}}_\beta \le \mathsf{A}_\vee + E_z + 2\bar{\mathsf{e}}_\beta$ in both cases of $\mathsf{A}_\vee$, gives

$$\Psi_\ell \le \mathsf{K}_5\,(1+E_z)\,\mathbf{1}\{\ell < L_z\}.$$

Also $L_z \le \mathsf{A}_\vee + E_z \le \gamma + \bar{\mathsf{e}}_\beta\tau + \mathsf{c}_\beta^+ + \mathsf{n}_\beta^-(z)$. With $\delta = \delta_\beta$, the bounds $\mathbf{1}\{\ell < L_z\} \le e^{\delta(L_z-\ell)}$ and $(1+\mathsf{c}_\beta^+ + n)e^{\delta n} \le (1+\mathsf{c}_\beta^+ + \delta^{-1})e^{2\delta n}$ for $n \ge 0$ give

$$\Psi_\ell \le \mathsf{K}_5\Big(1+\mathsf{c}_\beta^+ + \frac{1}{\delta}\Big)e^{\delta \mathsf{c}_\beta^+}\, e^{-\delta(\ell-\gamma)}\, e^{\tau/8}\, e^{(\tau-\sigma)_+/8},$$

because $\delta\bar{\mathsf{e}}_\beta \le \frac{1}{8}$ and $2\delta\,\mathsf{n}_\beta^-(z) \le (\tau-\sigma)_+/8$, with $\sigma = \log((1-\beta)^2 Q_\eta)$ (lemma SF.17). Finally, with $\mathrm{d}\eta = \mathsf{e}_\beta\,\mathrm{d}\sigma$ and the two bounds on e_β of the proof of lemma SF.17,

$$\begin{aligned}\int_0^\gamma \mathbb{E}_\star\left[e^{\tau/8}e^{(\tau-\sigma)_+/8}\right]\mathrm{d}\eta &\le \gamma\,\mathsf{M}_e + \mathbb{E}_\star\Big[e^{\tau/8}\int_{\mathbb{R}}\big(e^{(\tau-\sigma)_+/8}-1\big)\mathsf{e}_\beta\,\mathrm{d}\sigma\Big]\\ &\le \mathsf{M}_e\Big(\gamma + \frac{8}{7(1-\beta^2)} + 8\bar{\mathsf{e}}_\beta\Big),\end{aligned}$$

where the part $\sigma < 0$ of the inner integral is at most $8e^{\tau/8}/(7(1-\beta^2))$ and the part $0 \le \sigma < \tau$ is at most $8\bar{\mathsf{e}}_\beta e^{\tau/8}$. Dividing by $\gamma \ge 1$ gives the decay bound. □

The zero-ridge laws at large aspect ratio.

Theorem SF.11 (Zero ridge, large aspect ratios)**.** *Let $\beta \in (0,1)$ and $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 0$, put $\iota \overset{\mathrm{def}}{=} \gamma_\wedge/\gamma_\vee \in (0,1]$ and*

$$\mathsf{F}(\iota) \overset{\mathrm{def}}{=} \frac{1}{\iota}\int_0^\iota \sqrt{(1-x)(\iota-x)}\,\mathrm{d}x.$$

(i) The zero-ridge constant satisfies

$$\Big|C_\beta(\gamma_{\mathrm{in}},\gamma_{\mathrm{out}}) - \frac{\mathsf{F}(\iota)}{\sqrt{\pi}}\Big| \le \frac{2.32}{(1-\beta)\sqrt{\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}},$$

where $\mathsf{F}(1) = \frac{1}{2}$ and, for $\iota < 1$,

$$\mathsf{F}(\iota) = \frac{1}{\iota}\Big[\frac{(1+\iota)\sqrt{\iota}}{4} + \frac{(1-\iota)^2}{8}\log\frac{1-\sqrt{\iota}}{1+\sqrt{\iota}}\Big].$$

In particular, in the square case,

$$\Big|C_\beta(\gamma) - \frac{1}{2\sqrt{\pi}}\Big| \le \frac{2.32}{(1-\beta)\gamma} \qquad (\gamma > 0).$$

(ii) Let $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B \ge \gamma_\beta^\circ$. The zero-ridge volatility of definition SF.1 satisfies, with $\mathsf{K}_\beta^\pm$ of eq. (SF.57),

$$\frac{1-2\mathsf{K}_\beta^-}{\gamma} \le \frac{\mathfrak{v}_{\mathrm{iso},\beta}}{N\gamma} - \frac{1}{2} \le \frac{1+2\mathsf{K}_\beta^+}{\gamma}.$$

In particular $\mathfrak{v}_{\mathrm{iso},\beta} = \frac{N^2}{2B}\big(1+O_\beta(B/N)\big)$, and $\mathfrak{v}_{\mathrm{iso},\beta}/(N\gamma) \to \frac{1}{2}$ as $\gamma \to \infty$ at every fixed $\beta \in (0,1)$.

The square-case limit $1/(2\sqrt{\pi})$ involves no logarithm and does not depend on β. The zero-ridge drift therefore does not decay with the aspect ratio, in contrast with $C_0(\gamma) \sim 1/(\sqrt{\pi}\gamma)$ (theorem SF.5); only the rate depends on β. At zero ridge the same-time total is $\min\{N_{\mathrm{in}}, N_{\mathrm{out}}\}$ (proposition SF.3(i)). By part (ii) the integrated total is $N^2/(2B)$ at leading order, half of N times the aspect ratio, for every $\beta \in (0,1)$; the leading constant does not depend on β, and the constants $\mathsf{K}_\beta^\pm$ of the error do. Without momentum the integrated total is $\min\{B, N\}$ (proposition SF.2).

Proof. For (i), apply theorem SF.3 at $\varrho = 0$, where $\psi(0) = \gamma_\wedge$, and substitute $m = \gamma_\vee x$:

$$\frac{1}{\sqrt{\pi}\,\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}\int_0^{\gamma_\wedge}\sqrt{(\gamma_\vee - m)(\gamma_\wedge - m)}\,\mathrm{d}m = \frac{\gamma_\vee^2}{\sqrt{\pi}\,\gamma_\vee^2\iota}\int_0^{\iota}\sqrt{(1-x)(\iota-x)}\,\mathrm{d}x = \frac{\mathsf{F}(\iota)}{\sqrt{\pi}}.$$

The closed form follows by differentiating the antiderivative

$$\frac{2x-1-\iota}{4}\sqrt{(1-x)(\iota-x)} - \frac{(1-\iota)^2}{8}\log\left|2x-1-\iota+2\sqrt{(1-x)(\iota-x)}\right| \qquad (0 \le x < \iota < 1)$$

and evaluating it at $x = \iota$ and $x = 0$, using $(1-\sqrt{\iota})^2/(1-\iota) = (1-\sqrt{\iota})/(1+\sqrt{\iota})$. At $\iota = 1$ the integral is $\int_0^1 (1-x)\,\mathrm{d}x = \frac{1}{2}$.

For (ii), proposition SF.3(ii) gives $\mathfrak{v}_{\mathrm{iso},\beta}/N = 1 + 2\sum_{\ell\ge1}\overline{C}_{1,\ell}(0)$, that is

$$\frac{\mathfrak{v}_{\mathrm{iso},\beta}}{N\gamma} - \frac{1}{2} = \frac{1}{\gamma}\Big(1 + 2\Big(\sum_{\ell\ge1}\overline{C}_{1,\ell}(0) - \frac{\gamma}{4}\Big)\Big),$$

and theorem SF.9 bounds the inner bracket. □

Comparison with SIGNSGD. The volatility coefficients of the two algorithms are compared in one convention. For each algorithm, the coefficient is the factor of $\eta^2/2$ in its one-step risk identity at identity covariances, where the norm $\|\cdot\|_\Sigma$ is the Frobenius norm (eq. (SF.2)). For SIGNSGD this factor is $\mathbb{E}\,\|\operatorname{sign}(G^\circ)\|_{\mathrm{F}}^2 = N_{\mathrm{in}}N_{\mathrm{out}}$ (proposition SJ.2). For the algorithm of this section it is $\mathfrak{v}_{\mathrm{iso}}(\varrho)$, which is B times the integrated covariance of the centred update (theorem SF.2(ii),(iv)), and its zero-ridge value is $\mathfrak{v}_{\mathrm{iso},\beta}$. In the square case theorem SF.11(ii) gives, for every fixed $\beta \in (0,1)$,

$$\frac{\mathfrak{v}_{\mathrm{iso},\beta}}{N_{\mathrm{in}}N_{\mathrm{out}}} = \frac{1}{2B}\Big(1 + O_\beta\Big(\frac{B}{N}\Big)\Big). \tag{SF.62}$$

The two coefficients differ in definition in two ways. SIGNSGD keeps no buffer, so its coefficient has no lag terms. Its value $N_{\mathrm{in}}N_{\mathrm{out}}$ is an uncentred second moment, which exceeds the centred one by $\|\,\mathbb{E}\operatorname{sign}(G^\circ)\|_{\mathrm{F}}^2$; this difference is of the size of the drift term, and it is not estimated in this appendix.

In the rectangular case $N_{\mathrm{in}} \ne N_{\mathrm{out}}$ the large-aspect law of the integrated volatility with momentum is open: proposition SF.10 rests on the square-case lag response of lemma SF.11, and no rectangular analogue of that response is available.

The integrated volatility at positive ridge. At a positive ridge the representation eq. (SF.55) carries the factor $\mathsf{u}_\varrho \ge 1$, and the participation is cut at $\psi(\varrho)$ instead of γ. For $\varrho > 0$ and $\ell \ge 1$ write

$$\overline{C}_{1,\ell}(\varrho) = \overline{C}_\ell^{\,\circ}(\varrho) + \overline{C}_\ell^{\,\sharp}(\varrho),$$

where $\overline{C}_\ell^{\,\circ}(\varrho)$ is the right side of eq. (SF.55) with both factors u_ϱ replaced by 1, and $\overline{C}_\ell^{\,\sharp}(\varrho) \ge 0$ is the same integral with the factor $\mathsf{u}_\varrho(y)\mathsf{u}_\varrho(z_\ell) - 1 \ge 0$. The excess $\overline{C}_\ell^{\,\sharp}$ costs $O((1+\psi)/\gamma)$ in total. Put

$$\mathsf{K}_6 \stackrel{\mathrm{def}}{=} \max\Big(8,\ \frac{32\,\bar{\mathsf{e}}_\beta^2}{\Phi(1)^2}\Big), \qquad \mathsf{K}_{\mathrm{J}} \stackrel{\mathrm{def}}{=} \frac{3.2 + 2\log 2}{1-\beta^2} + \big(9.6 + 4e^{1/2}\log 2\big)\bar{\mathsf{e}}_\beta,$$

$$\mathsf{K}_{\mathrm{ex}}^{\flat} \stackrel{\mathrm{def}}{=} \frac{\mathsf{K}_6}{\pi^2}\Big[\frac{\bar{\mathsf{e}}_\beta}{2\log(1/\beta)}\Big(\frac{\pi^4}{6} + 56\,\zeta(3)\log 2\Big) + 4\log 2\Big(\bar{\mathsf{e}}_\beta(\pi^2 + 2\log 2)(\mathsf{c}_\beta + 1) + \mathsf{K}_{\mathrm{J}}\Big(3.64 + \frac{3}{\log(1/\beta)}\Big)\Big)\Big],$$

$$\mathsf{K}_{\mathrm{ex}}^{\sharp} \stackrel{\mathrm{def}}{=} \frac{\mathsf{K}_6}{\pi^2}\,4\log 2\;\bar{\mathsf{e}}_\beta\big(\pi^2 + 2\log 2\big).$$

Lemma SF.18 (Ridge excess)**.** *Let $\gamma \ge \gamma_\beta^\circ$ and $\varrho > 0$. Then*

$$\sum_{\ell\ge1}\overline{C}_\ell^{\,\sharp}(\varrho) \le \frac{\mathsf{K}_{\mathrm{ex}}^{\flat} + \mathsf{K}_{\mathrm{ex}}^{\sharp}\,\psi(\varrho)}{\gamma}.$$

Proof. The ridge factor exceeds one only within a bounded distance of the top level $\log Q_{\psi(\varrho)}$ in logarithmic scale, and the lag lattice meets that window once per period of length $2\log(1/\beta)$; there are about $\psi(\varrho)$ periods. Write $\psi = \psi(\varrho)$, $b \stackrel{\text{def}}{=} \log(1/\beta)$, $U \stackrel{\text{def}}{=} \log Q_\psi$, $u \stackrel{\text{def}}{=} \log y$, $v \stackrel{\text{def}}{=} \log z$ and $v_\ell \stackrel{\text{def}}{=} v + 2b\ell = \log z_\ell$. Then $D(y,z)\,\mathrm{d}y\,\mathrm{d}z = \mathcal{Z}(y,z)\,\mathrm{d}u\,\mathrm{d}v$.

The ridge factor. The closure at $s = \varrho$ gives $\varrho\, Q_\psi = (\gamma - \psi)^2$, and $\mathsf{A}(y) \geq \gamma - \psi$ for $y < Q_\psi$. Hence $\varrho y/\mathsf{A}(y)^2 \leq y/Q_\psi = e^{-(U-u)}$, and

$$\mathsf{u}_\varrho(e^u) \leq 1 + \varpi(U-u), \qquad \varpi(\zeta) \stackrel{\text{def}}{=} \left(1 - e^{-\zeta}\right)^{-1/2} - 1 \leq \frac{1}{e^\zeta - 1} \qquad (\zeta > 0).$$

The substitution $\sqrt{1-e^{-\zeta}} = t$ gives $\varpi(\zeta)\,\mathrm{d}\zeta = 2\,\mathrm{d}t/(1+t)$, so that

$$\int_0^\infty \varpi = 2\log 2, \qquad \int_0^\infty \zeta\,\varpi(\zeta)\,\mathrm{d}\zeta \leq \int_0^\infty \frac{\zeta}{e^\zeta - 1}\,\mathrm{d}\zeta = \frac{\pi^2}{6}.$$

Moreover ϖ is decreasing, and $\varpi(1) < 0.6$.

The other factors. On the support of $\overline{C}_\ell^{\,\sharp}$ one has $z_1 \leq z_\ell < Q_\psi < Q_\gamma$, so $\mathsf{A}_\vee \geq \Phi(1)$ by lemma SF.14(v). Also $\mathsf{j}(y) \leq \mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta$ and $\mathsf{j}(z_\ell) \leq \mathsf{A}(z_\ell) + 2\bar{\mathsf{e}}_\beta \leq \mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta$. The two cases $\mathsf{A}_\vee \geq 2\bar{\mathsf{e}}_\beta$ and $\mathsf{A}_\vee < 2\bar{\mathsf{e}}_\beta$ of lemma SF.15(v) give

$$\frac{\mathsf{j}(y)\,\mathsf{j}(z_\ell)}{\gamma^2\,\widehat{\Delta}(y,z)} \leq \mathsf{K}_6.$$

The same bound at $z_\ell = e^{v_\ell}$ gives

$$\mathsf{u}_\varrho(y)\,\mathsf{u}_\varrho(z_\ell) - 1 \leq \varpi(U-u) + \big(1 + \varpi(U-u)\big)\,\varpi(U - v_\ell),$$

and summing over ℓ gives

$$\sum_{\ell \geq 1} \overline{C}_\ell^{\,\sharp}(\varrho) \leq \frac{\mathsf{K}_6}{\pi^2\gamma} \iint_{u<U} \mathcal{Z}\left[\varpi(U-u)\,\mathsf{n}(v) + \big(1 + \varpi(U-u)\big)\,\mathsf{W}_\varrho(v)\right]\mathrm{d}u\,\mathrm{d}v,$$

$$\mathsf{n}(v) \stackrel{\text{def}}{=} \#\{\ell \geq 1 :\ v_\ell < U\} \leq \frac{(U-v)_+}{2b}, \qquad \mathsf{W}_\varrho(v) \stackrel{\text{def}}{=} \sum_{\ell \geq 1,\ v_\ell < U} \varpi(U - v_\ell).$$

The gaps $U - v_\ell$ form an arithmetic progression of step $2b$ starting at some $\zeta_0(v) \in (0, 2b]$, and ϖ is decreasing, so that

$$\mathsf{W}_\varrho(v) \leq \left[\varpi\big(\zeta_0(v)\big) + \frac{\log 2}{b}\right]\mathbf{1}\{v < U - 2b\}.$$

The bracket is a $2b$-periodic function of v whose integral over one period is at most $\int_0^\infty \varpi + 2\log 2 = 4\log 2$.

The term with n. By lemma SF.15(ii), $\mathcal{Z} \leq \bar{\mathsf{e}}_\beta\,\mathsf{h}(|u-v|)$ with $\mathsf{h}(w) \stackrel{\text{def}}{=} w/(2\sinh(w/2)) \leq 1$, and the computation in the proof of lemma SF.16(ii) gives $\int_{\mathbb{R}} \mathsf{h} = \pi^2$ and $\int_{\mathbb{R}} |w|\,\mathsf{h}(w)\,\mathrm{d}w = 28\,\zeta(3)$. With $(U-v)_+ \leq (U-u) + |u-v|$ for $u < U$, this term is at most

$$\frac{\bar{\mathsf{e}}_\beta}{2b}\Big(\pi^2 \int_0^\infty \zeta\varpi(\zeta)\,\mathrm{d}\zeta + 28\,\zeta(3)\int_0^\infty \varpi\Big) \leq \frac{\bar{\mathsf{e}}_\beta}{2b}\Big(\frac{\pi^4}{6} + 56\,\zeta(3)\log 2\Big).$$

The term with W_ϱ*, for* $v \geq V_0 \stackrel{\text{def}}{=} -2\log(1-\beta)$. Put $I(v) \stackrel{\text{def}}{=} \int_{u<U} \mathcal{Z}\,(1 + \varpi(U-u))\,\mathrm{d}u$. The same bound on $\mathcal{Z}$ gives $I(v) \leq \bar{\mathsf{e}}_\beta(\pi^2 + 2\log 2)$. If $(1-\beta)^2 Q_\psi \geq 1$, the lower bound of lemma SF.6(i) at Q_ψ gives

$$U - V_0 = \log\big((1-\beta)^2 Q_\psi\big) = 2b\,\mathsf{n}_\beta(Q_\psi) < 2b\,(\psi + \mathsf{c}_\beta),$$

so that $[V_0, U - 2b]$ is covered by at most $\psi + \mathsf{c}_\beta + 1$ periods; otherwise this interval is empty. This part is therefore at most $4\log 2\;\bar{\mathsf{e}}_\beta(\pi^2 + 2\log 2)(\psi + \mathsf{c}_\beta + 1)$.

The term with W_ϱ*, for* $v < V_0$. Put $w_1 \stackrel{\text{def}}{=} V_0 - v > 0$; we show $I(v) \leq \mathsf{K}_{\mathrm{J}}(1 + w_1)e^{-w_1/2}$. On $u < V_0$, lemma SF.15(ii) and $s_2 e^{V_0} = 1/(1-\beta^2)$ give

$$\mathcal{Z} \leq s_2 e^{(u+v)/2} = \frac{e^{-(V_0-u)/2}\,e^{-w_1/2}}{1-\beta^2}.$$

Splitting $1+\varpi$ into 1, ϖ on $U-u\ge 1$ (where $\varpi<0.6$) and ϖ on $U-u<1$ (where $e^{-(V_0-u)/2}\le 1$ and $\int\varpi\le 2\log 2$), the part $u<V_0$ of $I(v)$ is at most $(2+1.2+2\log 2)\,e^{-w_1/2}/(1-\beta^2)$. On $u\ge V_0$ put $w\stackrel{\text{def}}{=}u-v\ge w_1$; then $\mathcal{Z}\le \bar{\mathsf{e}}_\beta(1+w)e^{-w/2}$, and

$$\int_{w_1}^{\infty}(1+w)e^{-w/2}\,\mathrm{d}w=(6+2w_1)\,e^{-w_1/2}\le 6(1+w_1)\,e^{-w_1/2}.$$

The same splitting gives at most $(6+3.6)\bar{\mathsf{e}}_\beta(1+w_1)e^{-w_1/2}$ from 1 and from ϖ on $U-u\ge 1$, and at most $2\log 2\;\bar{\mathsf{e}}_\beta\sup_{w\ge w_1}(1+w)e^{-w/2}$ from ϖ on $U-u<1$. The function $(1+w)e^{-w/2}$ is at most $2e^{-1/2}$, is at least 1 on $[0,1]$ and decreases on $[1,\infty)$, so this supremum is at most $2e^{1/2}(1+w_1)e^{-w_1/2}$. Adding the parts gives the bound on $I(v)$.

The function $f(w)\stackrel{\text{def}}{=}(1+w)e^{-w/2}$ is unimodal on $(0,\infty)$, with $\sup f=2e^{-1/2}<1.214$ and $\int_0^\infty f=6$. Cut $(-\infty,V_0)$ into consecutive intervals of length $2b$ ending at V_0. On each, W_ϱ has integral at most $4\log 2$, and $I\le \mathsf{K}_{\mathrm{J}}\sup f$ over the corresponding interval of w_1. For a unimodal $f\ge 0$ the sum of its suprema over consecutive intervals of length h is at most $3\sup f+h^{-1}\int f$: an interval that neither contains the mode nor is adjacent to the interval containing it has its supremum at the end nearest the mode, which is at most h^{-1} times the integral of f over the next interval towards the mode, and the three remaining intervals contribute at most $\sup f$ each. With $h=2b$ this part is at most $4\log 2\;\mathsf{K}_{\mathrm{J}}\,(3.64+3/b)$.

Adding the three parts and multiplying by $\mathsf{K}_6/(\pi^2\gamma)$ gives the lemma. □

Define, with $\bar{\mathsf{L}}_2$ of lemma SF.17,

$$\mathsf{K}_7\stackrel{\text{def}}{=}3\bar{\mathsf{e}}_\beta+\frac{3\bar{\mathsf{e}}_\beta^2}{\Phi(1)}+\big(\bar{\mathsf{e}}_\beta+\mathsf{c}_\beta^+ +1\big)\Big(1+\frac{3\bar{\mathsf{e}}_\beta}{\Phi(1)}\Big)+\frac{3\big(\bar{\mathsf{e}}_\beta^2+(\mathsf{c}_\beta^+)^2+1\big)}{2\Phi(1)},$$

$$\mathsf{K}_8\stackrel{\text{def}}{=}6\bar{\mathsf{e}}_\beta+\frac{6\bar{\mathsf{e}}_\beta^2}{\Phi(1)}+4\mathsf{K}_7+\frac{8\bar{\mathsf{e}}_\beta}{\Phi(1)}\big(3\bar{\mathsf{e}}_\beta+\mathsf{K}_7\big),$$

$$\mathsf{K}_\beta^\circ\stackrel{\text{def}}{=}6\bar{\mathsf{e}}_\beta(\mathsf{M}_1+1)+\frac{3}{2},\qquad \mathsf{K}_\beta^\flat\stackrel{\text{def}}{=}\mathsf{K}_8\bar{\mathsf{L}}_2+\mathsf{K}_{\mathrm{ex}}^\flat,\qquad \mathsf{K}_\beta^\sharp\stackrel{\text{def}}{=}\mathsf{K}_8(1+\mathsf{M}_2)+\mathsf{K}_{\mathrm{ex}}^\sharp,$$

and the crossover function

$$\mathcal{P}(x)\stackrel{\text{def}}{=}\frac{1}{2}-\frac{1}{2}(1-x)^2+(1-x)^2\log(1-x)\qquad(0\le x<1),\qquad \mathcal{P}(1)\stackrel{\text{def}}{=}\frac{1}{2}.\tag{SF.63}$$

Theorem SF.12 (Integrated volatility at positive ridge)**.** *Let $\beta\in(0,1)$, $N_{\mathrm{in}}=N_{\mathrm{out}}=N$, $\gamma=N/B\ge\gamma_\beta^\circ$ and $\varrho>0$, and put $\psi=\psi(\varrho)\in(0,\gamma)$. Then*

$$-\frac{\psi}{\gamma}\,\mathsf{K}_\beta^\circ\le\sum_{\ell\ge 1}\overline{C}_{1,\ell}(\varrho)-\frac{\gamma}{2}\,\mathcal{P}\Big(\frac{\psi}{\gamma}\Big)\le\frac{\mathsf{K}_\beta^\flat+\mathsf{K}_\beta^\sharp\psi}{\gamma},$$

and the integrated volatility of theorem SF.2(ii) satisfies

$$-\frac{2\psi\,\mathsf{K}_\beta^\circ}{\gamma}\le\frac{\mathfrak{v}_{\mathrm{iso}}(\varrho)}{N}-\frac{\psi}{\gamma}-\gamma\,\mathcal{P}\Big(\frac{\psi}{\gamma}\Big)\le\frac{2(\mathsf{K}_\beta^\flat+\mathsf{K}_\beta^\sharp\psi)}{\gamma}.$$

The function $\mathcal{P}$ is continuous on $[0,1]$, with $\mathcal{P}(x)=x^2-x^3/3+O(x^4)$ as $x\downarrow 0$ and $\mathcal{P}(1)=\frac{1}{2}$. It interpolates between the fixed-ridge regime, where $\gamma\,\mathcal{P}(\psi/\gamma)$ is close to ψ^2/γ (corollary SF.2(iii)), and the zero-ridge rate $\mathfrak{v}_{\mathrm{iso},\beta}/(N\gamma)\to\frac{1}{2}$ of theorem SF.11(ii) (theorem SF.14).

Proof. The main part $\overline{C}_\ell^{\,\circ}$ is treated as at zero ridge, with the participation cut at ψ, and the excess is lemma SF.18.

Step 0: the target integral. For $0\le X\le a$ put $\mathsf{f}_a(X)\stackrel{\text{def}}{=}X-X^2/(2a)$; then $0\le\partial_X\mathsf{f}_a\le 1$ and $0\le\partial_a\mathsf{f}_a\le\frac{1}{2}$. For $0\le\eta\le\psi$ put $X\stackrel{\text{def}}{=}\psi-\eta$, $x\stackrel{\text{def}}{=}\gamma-\eta$ and $\mathsf{H}(\eta)\stackrel{\text{def}}{=}\mathsf{f}_x(X)$. With $p\stackrel{\text{def}}{=}\psi/\gamma$ and

$c \stackrel{\text{def}}{=} 1-p$, the substitution $\eta = \gamma y'$ turns $\gamma^{-1}\int_0^{\psi} \mathsf{H}$ into $\gamma\int_0^{p}\big[(p-y')-(p-y')^2/(2(1-y'))\big]\mathrm{d}y'$, and the substitution $1-y'=t'$ gives

$$\int_0^p \frac{(p-y')^2}{2(1-y')}\,\mathrm{d}y' = \frac{1}{2}\int_c^1 \Big(t' - 2c + \frac{c^2}{t'}\Big)\mathrm{d}t' = \frac{1-c^2}{4} - c(1-c) - \frac{c^2}{2}\log c.$$

Hence

$$\frac{1}{\gamma}\int_0^{\psi} \mathsf{H}(\eta)\,\mathrm{d}\eta = \gamma\Big[\frac{p^2}{2} - \frac{1-c^2}{4} + c(1-c) + \frac{c^2}{2}\log c\Big] = \frac{\gamma}{2}\,\mathcal{P}(p).$$

Step 1: the summed representation of the main part. As in Step 1 of the proof of theorem SF.9,

$$\sum_{\ell\ge 1} \overline{C}_\ell^{\circ}(\varrho) = \frac{1}{\gamma}\int_0^{\psi} \mathbb{E}_\star\,\Psi_\varrho\,\mathrm{d}\eta,$$

where

$$\Psi_\varrho \stackrel{\text{def}}{=} \mathbf{1}\{y < Q_\psi\}\,\frac{\mathsf{j}(y)\,\mathsf{T}_\varrho(z)}{\gamma^2\widehat{\Delta}(y,z)}, \qquad \mathsf{T}_\varrho(z) \stackrel{\text{def}}{=} \sum_{\ell\ge 1}\mathbf{1}\{z_\ell < Q_\psi\}\,\mathsf{j}(z_\ell).$$

The range is $\eta < \psi$ because s lies between y and z, both below Q_ψ. On $\{\Psi_\varrho > 0\}$, $\mathsf{A}_\vee \ge \Phi(1)$ by lemma SF.14(v), since $z_1 < Q_\psi < Q_\gamma$.

Step 2: the main part, upper bound. The terms of $\mathsf{T}_\varrho(z)$ have $\ell < L_z^\varrho \stackrel{\text{def}}{=} \psi - \mathsf{n}_\beta(z) + \mathsf{c}_\beta = L_z - (\gamma-\psi)$, by lemma SF.14(ii) and $r(z_\ell) < \psi$, and each is at most $L_z - \ell + 2\bar{\mathsf{e}}_\beta$. For an integer $n < L_z^\varrho \le L_z$,

$$\sum_{\ell=1}^{n}(L_z - \ell) \le nL_z - \frac{n^2}{2} \le L_z^\varrho L_z - \frac{(L_z^\varrho)^2}{2},$$

so that

$$\mathsf{T}_\varrho(z) \le \mathsf{b}(L_z^\varrho) + 2\bar{\mathsf{e}}_\beta L_z^\varrho, \qquad \mathsf{b}(\zeta) \stackrel{\text{def}}{=} \zeta\Big(\frac{\zeta}{2} + \gamma - \psi\Big),$$

where b is increasing on $[0,\infty)$. By lemma SF.14(iii),(iv) and lemma SF.16(iii), $L_z^\varrho \le X + Y$ with $Y \stackrel{\text{def}}{=} \bar{\mathsf{e}}_\beta\tau + \mathsf{c}_\beta^+ + \mathsf{n}_\beta^-(z)$. Since $\gamma - \psi = x - X$,

$$\mathsf{b}(X+Y) = x\,\mathsf{H} + Yx + \frac{Y^2}{2}.$$

From $\mathsf{A}_\vee \ge \mathsf{A}(y) \ge x - \bar{\mathsf{e}}_\beta\tau$ and $\mathsf{A}_\vee \ge \Phi(1)$,

$$\frac{x}{\mathsf{A}_\vee} \le 1 + \frac{\bar{\mathsf{e}}_\beta\tau}{\Phi(1)}, \qquad \frac{x\,\mathsf{H}}{\mathsf{A}_\vee} = \mathsf{H} + \mathsf{H}\,\frac{x-\mathsf{A}_\vee}{\mathsf{A}_\vee} \le \mathsf{H} + \bar{\mathsf{e}}_\beta\tau\Big(1 + \frac{\bar{\mathsf{e}}_\beta\tau}{\Phi(1)}\Big),$$

using $\mathsf{H} \le X \le x$. Hence

$$\frac{\mathsf{T}_\varrho(z)}{\mathsf{A}_\vee} \le \mathsf{H} + \mathsf{E}_1, \qquad \mathsf{E}_1 \stackrel{\text{def}}{=} \bar{\mathsf{e}}_\beta\tau\Big(1+\frac{\bar{\mathsf{e}}_\beta\tau}{\Phi(1)}\Big) + \Big(1+\frac{\bar{\mathsf{e}}_\beta\tau}{\Phi(1)}\Big)Y + \frac{Y^2}{2\Phi(1)} + 2\bar{\mathsf{e}}_\beta\Big(1+\frac{\bar{\mathsf{e}}_\beta\tau}{\Phi(1)}\Big) + \frac{2\bar{\mathsf{e}}_\beta Y}{\Phi(1)}.$$

By lemma SF.15(v) and $\mathsf{j}(y) \le \mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta$,

$$\Psi_\varrho \le \frac{\mathsf{A}_\vee + 2\bar{\mathsf{e}}_\beta}{\max(\mathsf{A}_\vee - \bar{\mathsf{e}}_\beta, \mathsf{A}_\vee/2)}\cdot\frac{\mathsf{T}_\varrho(z)}{\mathsf{A}_\vee}.$$

If $\mathsf{A}_\vee \ge 2\bar{\mathsf{e}}_\beta$, this is at most $(1+6\bar{\mathsf{e}}_\beta/\mathsf{A}_\vee)(\mathsf{H}+\mathsf{E}_1) \le \mathsf{H} + 6\bar{\mathsf{e}}_\beta(1+\bar{\mathsf{e}}_\beta\tau/\Phi(1)) + 4\mathsf{E}_1$. If $\mathsf{A}_\vee < 2\bar{\mathsf{e}}_\beta$, it is at most $(8\bar{\mathsf{e}}_\beta/\Phi(1))(\mathsf{H}+\mathsf{E}_1)$; since $8\bar{\mathsf{e}}_\beta/\Phi(1) \ge 1$ and $\mathsf{H} \le x \le \mathsf{A}_\vee + \bar{\mathsf{e}}_\beta\tau < 2\bar{\mathsf{e}}_\beta + \bar{\mathsf{e}}_\beta\tau$, it is at most $\mathsf{H} + (8\bar{\mathsf{e}}_\beta/\Phi(1))(2\bar{\mathsf{e}}_\beta + \bar{\mathsf{e}}_\beta\tau + \mathsf{E}_1)$. Put $\mathsf{R} \stackrel{\text{def}}{=} 1 + \tau^2 + \mathsf{n}_\beta^-(z)^2$, which dominates 1, τ, $\mathsf{n}_\beta^-(z)$, their squares and their product. Then

$$Y \le \big(\bar{\mathsf{e}}_\beta + \mathsf{c}_\beta^+ + 1\big)\mathsf{R}, \qquad Y^2 \le 3\big(\bar{\mathsf{e}}_\beta^2 + (\mathsf{c}_\beta^+)^2 + 1\big)\mathsf{R}, \qquad \tau Y \le \big(\bar{\mathsf{e}}_\beta + \mathsf{c}_\beta^+ + 1\big)\mathsf{R},$$

hence $\mathsf{E}_1 \le \mathsf{K}_7\mathsf{R}$, and in both cases

$$\Psi_\varrho \le \mathsf{H} + \mathsf{K}_8\,\mathsf{R}.$$

Integrating over $\eta \in (0,\psi)$, with $\int_0^{\psi} \mathbb{E}_\star\, \mathsf{R}\, \mathrm{d}\eta \le \psi(1+\mathsf{M}_2)+\bar{\mathsf{L}}_2$ by lemma SF.16(ii) and lemma SF.17, gives

$$\sum_{\ell\ge 1} \overline{C}^{\circ}_{\ell}(\varrho) \le \frac{\gamma}{2}\,\mathcal{P}\Big(\frac{\psi}{\gamma}\Big) + \frac{\mathsf{K}_8\big(\psi(1+\mathsf{M}_2)+\bar{\mathsf{L}}_2\big)}{\gamma}.$$

Step 3: the main part, lower bound. Put $X_z \overset{\text{def}}{=} \psi - r(z) \le \mathsf{A}(z)$. Every $1 \le \ell < X_z$ has $r(z_\ell) \le r(z) + \ell < \psi$ by lemma SF.14(i), and $\mathsf{j}(z_\ell) \ge \mathsf{A}(z) - \ell$. With $n \overset{\text{def}}{=} \lceil X_z \rceil - 1 \ge X_z - 1$ and $\mathsf{A} = \mathsf{A}(z)$,

$$\mathsf{T}_{\varrho}(z) \ge \sum_{\ell=1}^{n} (\mathsf{A}-\ell) = \int_0^n (\mathsf{A}-t)\,\mathrm{d}t - \frac{n}{2} \ge \mathsf{A}\,\mathsf{f}_{\mathsf{A}}(X_z) - \frac{3}{2}\,\mathsf{A},$$

where the last step uses $n \ge X_z - 1$, $\partial_X \mathsf{f}_{\mathsf{A}} \le 1$ and $n \le \mathsf{A}$; the final bound also holds when $0 < X_z < 1$, where $\mathsf{A}\,\mathsf{f}_{\mathsf{A}}(X_z) < \mathsf{A}$. Using $\mathsf{A}(y)\mathsf{A}(z) \le \mathsf{A}_\vee^2$ and lemma SF.15(v),

$$\Psi_{\varrho} \ge \mathbf{1}\{y < Q_\psi\}\Big[\frac{\mathsf{A}(y)\,\mathsf{A}(z)}{(\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta)^2}\,\mathsf{f}_{\mathsf{A}(z)}(X_z) - \frac{3}{2}\Big].$$

Put $\upsilon \overset{\text{def}}{=} \bar{\mathsf{e}}_\beta(\tau+1)$. If $X \le \upsilon$, then $\mathsf{H} \le X \le \upsilon$ and $\mathsf{H} - 6\upsilon - \frac{3}{2} < 0 \le \Psi_\varrho$. Otherwise $X_z \ge X - \upsilon > 0$ and $\psi - r(y) \ge X - \upsilon > 0$, so that $y < Q_\psi$, and $\mathsf{A}(z) \ge x - \upsilon$. Since $\mathsf{f}_a(X)$ is nondecreasing in a and in X, moving first X to $X-\upsilon$ at $a = x$ and then a to $x-\upsilon$ at $X-\upsilon$ gives

$$\mathsf{f}_{\mathsf{A}(z)}(X_z) \ge \mathsf{f}_{x-\upsilon}(X-\upsilon) \ge \mathsf{H} - \upsilon - \frac{\upsilon}{2} \ge \mathsf{H} - 2\upsilon.$$

The ratio $\mathsf{A}(y)\mathsf{A}(z)/(\mathsf{A}_\vee + \bar{\mathsf{e}}_\beta)^2$ lies between $((x-\upsilon)/(x+\upsilon))^2 \ge 1 - 4\upsilon/x$ and 1. If $\mathsf{H} \ge 2\upsilon$ and $4\upsilon \le x$, the product is at least $(1-4\upsilon/x)(\mathsf{H}-2\upsilon) \ge \mathsf{H} - 6\upsilon$, because $\mathsf{H} \le x$. If $\mathsf{H} < 2\upsilon$ or $4\upsilon > x \ge \mathsf{H}$, then $\mathsf{H} - 6\upsilon < 0$. In all cases

$$\Psi_{\varrho} \ge \mathsf{H} - 6\upsilon - \frac{3}{2},$$

and integration over $(0,\psi)$ gives $\sum_\ell \overline{C}^{\circ}_{\ell}(\varrho) \ge \frac{\gamma}{2}\mathcal{P}(\psi/\gamma) - \frac{\psi}{\gamma}\big(6\bar{\mathsf{e}}_\beta(\mathsf{M}_1+1)+\frac{3}{2}\big)$. Finally $\overline{C}_{1,\ell}(\varrho) \ge \overline{C}^{\circ}_{\ell}(\varrho)$.

Step 4: the excess and the total. Lemma SF.18 bounds $\sum_\ell \overline{C}^{\sharp}_{\ell}(\varrho)$, and together with Step 2 it gives the upper bound. For the volatility, eq. (SG.2) at $\mu_1 = \cdots = \mu_N = 1$, theorem SF.2(ii) and proposition SF.3(ii) give

$$\frac{\mathfrak{v}_{\mathrm{iso}}(\varrho)}{N} = \overline{C}_{1,0}(\varrho) + 2\sum_{\ell\ge 1}\overline{C}_{1,\ell}(\varrho), \qquad \overline{C}_{1,0}(\varrho) = \frac{\psi}{\gamma},$$

and the second display of the theorem follows from the first. □

Large aspect ratios at a fixed ridge. At a fixed positive ridge only about

$$\widehat{\mathfrak{p}}_\beta(\varrho) \overset{\text{def}}{=} \frac{\log\big((1-\beta)^2\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}/\varrho\big)}{2\log(1/\beta)} = \mathfrak{p}_\beta(\varrho) + \frac{\log\big((1-\beta)\sqrt{\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}\big)}{\log(1/\beta)} \tag{SF.64}$$

participation levels are available, and the drift is this count divided by $\sqrt{\pi\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}$. In the notation of lemma SF.6, $\widehat{\mathfrak{p}}_\beta(\varrho) = \mathsf{n}_\beta(\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}/\varrho)$.

Theorem SF.13 (Fixed ridge, large aspect ratios)**.** *Let $\beta \in (0,1)$, $\varrho > 0$ and $\gamma_{\mathrm{in}}, \gamma_{\mathrm{out}} > 0$, write $\widehat{\mathfrak{p}} = \widehat{\mathfrak{p}}_\beta(\varrho)$, and assume*

$$\widehat{\mathfrak{p}} \ge \frac{2\mathsf{c}'_\beta}{\log(1/\beta)\,\gamma_\wedge}, \qquad \widehat{\mathfrak{p}} + \mathsf{c}'_\beta \le \frac{\gamma_\wedge}{2}.$$

Put $e_\beta \overset{\text{def}}{=} \max\big(\mathsf{c}'_\beta,\ \mathsf{c}_\beta + 2(\widehat{\mathfrak{p}} + \mathsf{c}'_\beta)/(\log(1/\beta)\gamma_\wedge)\big)$. Then

$$\big|\psi(\varrho) - \widehat{\mathfrak{p}}\big| \le e_\beta, \qquad \Big|\sqrt{\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}}\,\mathfrak{d}_{\mathrm{iso}}(\varrho) - \frac{\widehat{\mathfrak{p}}}{\sqrt{\pi}}\Big| \le \frac{1}{\sqrt{\pi}}\Big(e_\beta + \frac{(\widehat{\mathfrak{p}}+\mathsf{c}'_\beta)^2}{\gamma_\wedge}\Big) + \frac{2.32}{1-\beta}.$$

If moreover $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B$, then

$$\mathfrak{v}_{\mathrm{iso}}(\varrho) \ge B\,\psi(\varrho) + \mathsf{k}_\beta B\Big(1 - \frac{\psi(\varrho)}{\gamma}\Big)^2\big(\psi(\varrho) - \mathsf{c}''_\beta - \mathsf{c}'_\beta - 1\big)_+^2.$$

Proof. Let $H \overset{\text{def}}{=} \psi(\varrho)$, so that $Q_H = q(-\varrho)$ and the closure reads $\varrho\, Q_H = \mathsf{W}(H)^2$. For $0 \le m \le \gamma_\wedge$,

$$\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}\Big(1 - \frac{m}{\gamma_\wedge}\Big) \le \mathsf{W}(m) \le \sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}.$$

Suppose first $(1-\beta)^2 Q_H \ge 1$. By lemma SF.6(i), $H \in (\mathsf{n}_\beta(Q_H) - \mathsf{c}_\beta,\ \mathsf{n}_\beta(Q_H) + \mathsf{c}'_\beta]$, where

$$\mathsf{n}_\beta(Q_H) = \frac{\log\big((1-\beta)^2 \mathsf{W}(H)^2/\varrho\big)}{2\log(1/\beta)} = \widehat{\mathfrak{p}} + \frac{\log\big(\mathsf{W}(H)/\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}\big)}{\log(1/\beta)} \le \widehat{\mathfrak{p}}.$$

The upper bound gives $H \le \widehat{\mathfrak{p}} + \mathsf{c}'_\beta \le \gamma_\wedge/2$. Then $\log(1 - H/\gamma_\wedge) \ge -2H/\gamma_\wedge$, so

$$H > \widehat{\mathfrak{p}} - \mathsf{c}_\beta - \frac{2(\widehat{\mathfrak{p}} + \mathsf{c}'_\beta)}{\log(1/\beta)\,\gamma_\wedge}.$$

Suppose instead $(1-\beta)^2 Q_H < 1$. Then $H \le r_\beta((1-\beta)^{-2}) \le \mathsf{c}'_\beta$, and

$$\varrho = \frac{\mathsf{W}(H)^2}{Q_H} > (1-\beta)^2\, \mathsf{W}(H)^2 \ge (1-\beta)^2 \gamma_{\text{in}}\gamma_{\text{out}}\Big(1 - \frac{\mathsf{c}'_\beta}{\gamma_\wedge}\Big)^2,$$

so $\widehat{\mathfrak{p}} < -\log(1 - \mathsf{c}'_\beta/\gamma_\wedge)/\log(1/\beta) \le 2\mathsf{c}'_\beta/(\log(1/\beta)\gamma_\wedge)$, which the hypothesis excludes. This proves the bound on $\psi(\varrho)$.

For the drift, the bounds on W give

$$\frac{\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}}{\gamma_{\text{in}}\gamma_{\text{out}}} \int_0^H \mathsf{W}(m)\,\mathrm{d}m \in \Big[H - \frac{H^2}{\gamma_\wedge},\ H\Big],$$

whose distance to $\widehat{\mathfrak{p}}$ is at most $e_\beta + (\widehat{\mathfrak{p}} + \mathsf{c}'_\beta)^2/\gamma_\wedge$. Theorem SF.3, multiplied by $\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}$, gives the drift bound.

For the volatility, proposition SF.3(ii) gives $\mathfrak{v}_{\text{iso}}(\varrho) = N\big(\overline{C}_{1,0}(\varrho) + 2\sum_{\ell\ge1}\overline{C}_{1,\ell}(\varrho)\big)$ with $N\overline{C}_{1,0}(\varrho) = NH/\gamma = BH$. By lemma SF.12 with $\varepsilon = \varrho$ and $\sum_{\ell\ge1}(X-\ell)_+ \ge \int_1^\infty (X-x)_+\mathrm{d}x = (X-1)_+^2/2$, where $X \overset{\text{def}}{=} H - \mathsf{c}''_\beta - \mathsf{c}'_\beta$,

$$2N\sum_{\ell\ge1}\overline{C}_{1,\ell}(\varrho) \ge \frac{N\mathsf{k}_\beta}{\gamma^3}(\gamma - H)^2(X-1)_+^2 = \mathsf{k}_\beta B\Big(1 - \frac{H}{\gamma}\Big)^2(X-1)_+^2.$$

□

Corollary SF.2 (Fixed-ridge laws)**.** *Let $\beta \in (0,1)$.*

(i) Along $\gamma_\wedge \to \infty$ and $\varrho > 0$ with $\widehat{\mathfrak{p}}_\beta(\varrho) \to \infty$ and $\widehat{\mathfrak{p}}_\beta(\varrho)/\gamma_\wedge \to 0$,

$$\mathfrak{d}_{\text{iso}}(\varrho) \sim \frac{\log\big((1-\beta)^2\gamma_{\text{in}}\gamma_{\text{out}}/\varrho\big)}{2\sqrt{\pi}\,\log(1/\beta)\,\sqrt{\gamma_{\text{in}}\gamma_{\text{out}}}},$$

and $\psi(\varrho) - \widehat{\mathfrak{p}}_\beta(\varrho)$ stays bounded.

(ii) In the square case, for fixed $\varrho > 0$, the quantity

$$\gamma\,\mathfrak{d}_{\text{iso}}(\varrho) - \frac{1}{\sqrt{\pi}}\Big[\mathfrak{p}_\beta(\varrho) + \frac{\log((1-\beta)\gamma)}{\log(1/\beta)}\Big]$$

stays bounded as $\gamma \to \infty$. In particular $\gamma\,\mathfrak{d}_{\text{iso}}(\varrho) \to \infty$.

(iii) In the square case, along $\gamma \to \infty$ and $\varrho > 0$ with $\psi(\varrho) \to \infty$ and $\psi(\varrho)/\gamma \to 0$, and in particular at a fixed $\varrho > 0$,

$$\mathfrak{v}_{\text{iso}}(\varrho) = B\,\psi(\varrho)^2\Big(1 + O_\beta\Big(\frac{1}{\psi(\varrho)} + \frac{\psi(\varrho)}{\gamma}\Big)\Big).$$

Proof. In parts (i) and (ii) the hypotheses of theorem SF.13 hold eventually, because $\widehat{\mathfrak{p}}_\beta(\varrho) \to \infty$ and $\widehat{\mathfrak{p}}_\beta(\varrho) = o(\gamma_\wedge)$; in part (ii) $\widehat{\mathfrak{p}}_\beta(\varrho)$ is the bracket, which is of order $\log\gamma$. Then e_β stays bounded and $(\widehat{\mathfrak{p}} + \mathsf{c}'_\beta)^2/\gamma_\wedge = o(\widehat{\mathfrak{p}})$, and the theorem gives both statements.

For part (iii), at a fixed ridge the bound $|\psi(\varrho) - \widehat{\mathfrak{p}}_\beta(\varrho)| \le e_\beta$ of theorem SF.13, with $\widehat{\mathfrak{p}}_\beta(\varrho)$ of order $\log\gamma$, gives $\psi(\varrho) \to \infty$ and $\psi(\varrho)/\gamma \to 0$. The function $(\mathcal{P}(x) - x^2)/x^3$ is continuous on $(0,1]$ and tends to $-\frac{1}{3}$ as $x \downarrow 0$, by eq. (SF.63) and $\log(1-x) = -x - x^2/2 - x^3/3 + O(x^4)$. Hence $|\mathcal{P}(x) - x^2| \le Cx^3$ on $[0,1]$ for an absolute constant C, and

$$N\gamma\,\mathcal{P}\Big(\frac{\psi}{\gamma}\Big) = B\gamma^2\mathcal{P}\Big(\frac{\psi}{\gamma}\Big) = B\psi^2\Big(1 + O\Big(\frac{\psi}{\gamma}\Big)\Big) \qquad (\psi = \psi(\varrho)).$$

For $\gamma \ge \gamma^\circ_\beta$ the second display of theorem SF.12, multiplied by N and with $N/\gamma = B$, gives $\mathfrak{v}_{\mathrm{iso}}(\varrho) = B\psi + B\gamma^2\mathcal{P}(\psi/\gamma) + O_\beta\big(B(1+\psi)\big)$, which is the claim. □

At a fixed ridge the drift therefore decays like $\log\gamma/\gamma$, not like $1/\gamma$, with leading term $\log\gamma/(\sqrt{\pi}\log(1/\beta)\gamma)$. The same-time resolved rank is $B\psi(\varrho) = B\,\widehat{\mathfrak{p}}_\beta(\varrho) + O(B)$. It differs from $B\,\mathfrak{p}_\beta(\varrho)$ by $B\log((1-\beta)\sqrt{\gamma_{\mathrm{in}}\gamma_{\mathrm{out}}})/\log(1/\beta)$, which is not negligible at a fixed ridge. The integrated total is $B\psi(\varrho)^2$ at leading order, with an error of order $B(1+\psi(\varrho))$ (corollary SF.2(iii), theorem SF.12); the explicit lower bound of theorem SF.13 holds at every γ.

The crossover. When $\mathfrak{p}_\beta(\varrho)$ is of the order of γ, the resolved fraction ψ/γ converges, and eq. (SF.32) turns this into a limit for the drift itself.

Theorem SF.14 (Crossover)**.** *Let $\beta \in (0,1)$ be fixed, let $N_{\mathrm{in}} = N_{\mathrm{out}} = N$ and $\gamma = N/B \to \infty$, and let $\varrho_\gamma \in (0,1)$ satisfy $\mathfrak{p}_\beta(\varrho_\gamma)/\gamma \to c \in (0,\infty)$. Then $\psi(\varrho_\gamma)/\gamma \to c \wedge 1$, the same-time total satisfies $B\psi(\varrho_\gamma)/N \to c \wedge 1$,*

$$\mathfrak{d}_{\mathrm{iso}}(\varrho_\gamma) \longrightarrow \Xi(c) \stackrel{\mathrm{def}}{=} \frac{(c\wedge 1)\big(1 - \frac{1}{2}(c\wedge 1)\big)}{\sqrt{\pi}},$$

and, with $\mathcal{P}$ of eq. (SF.63),

$$\frac{\mathfrak{v}_{\mathrm{iso}}(\varrho_\gamma)}{N\gamma} \longrightarrow \mathcal{P}(c\wedge 1).$$

The function Ξ interpolates between the two other regimes. As $c \to 0$, $\Xi(c) \sim c/\sqrt{\pi}$, the fixed-ridge law of corollary SF.2 with $\psi \approx c\gamma$; for $c \ge 1$, $\Xi(c) = 1/(2\sqrt{\pi})$, the zero-ridge limit of theorem SF.11. It reaches this value at $c = 1$, that is at $\varrho \approx \beta^{2\gamma}$, and it is continuously differentiable there. In particular $\gamma\,\mathfrak{d}_{\mathrm{iso}}(\varrho_\gamma) \to \infty$. The order-one object in the window is $\mathfrak{d}_{\mathrm{iso}}$ itself. The volatility limit $\mathcal{P}(c\wedge 1)$ interpolates in the same way: $\mathcal{P}(c) \sim c^2$ as $c \to 0$, which is the fixed-ridge law $B\psi^2$ of corollary SF.2(iii) with $\psi \approx c\gamma$, and $\mathcal{P}(c\wedge 1) = \frac{1}{2}$ for $c \ge 1$, which is the zero-ridge law of theorem SF.11(ii).

Proof. Write $H \stackrel{\mathrm{def}}{=} \psi(\varrho_\gamma)$ and $\mathfrak{p} \stackrel{\mathrm{def}}{=} \mathfrak{p}_\beta(\varrho_\gamma) \to \infty$. The case $(1-\beta)^2 Q_H < 1$ is excluded for large γ: as in the proof of theorem SF.13, it forces $H \le \mathsf{c}'_\beta$ and $\varrho_\gamma > (1-\beta)^2(\gamma - \mathsf{c}'_\beta)^2$, hence $\mathfrak{p} < -\log((1-\beta)(\gamma - \mathsf{c}'_\beta))/\log(1/\beta) < 0$ eventually. Since $Q_H = (\gamma - H)^2/\varrho_\gamma$, lemma SF.6(i) gives

$$\mathfrak{p} + \frac{\log\big((1-\beta)(\gamma - H)\big)}{\log(1/\beta)} - \mathsf{c}_\beta < H \le \mathfrak{p} + \frac{\log\big((1-\beta)\gamma\big)}{\log(1/\beta)} + \mathsf{c}'_\beta.$$

If $c < 1$, the upper bound gives $H \le c\gamma + o(\gamma)$, so eventually $H \le (1-\delta)\gamma$ with $\delta \stackrel{\mathrm{def}}{=} (1-c)/2$, and then the lower bound gives $H > \mathfrak{p} + \log((1-\beta)\delta\gamma)/\log(1/\beta) - \mathsf{c}_\beta = c\gamma + o(\gamma)$. If $c \ge 1$ and $H \le (1-\delta)\gamma$ along a subsequence for some $\delta \in (0,1)$, the lower bound gives $H > (1-\delta/2)\gamma$ eventually along it, a contradiction; since $H < \gamma$, $H/\gamma \to 1$. The drift limit follows from eq. (SF.32) and the continuity of Ξ_0, and the same-time total is $B\psi/N = \psi/\gamma$.

For the volatility, theorem SF.12 applies for large γ, since $\gamma \ge \gamma^\circ_\beta$ eventually. Dividing its second display by γ and using $H < \gamma$,

$$\Big|\frac{\mathfrak{v}_{\mathrm{iso}}(\varrho_\gamma)}{N\gamma} - \mathcal{P}\Big(\frac{H}{\gamma}\Big)\Big| \le \frac{H}{\gamma^2} + \frac{2\big(\mathsf{K}^\flat_\beta + (\mathsf{K}^\sharp_\beta + \mathsf{K}^\circ_\beta)H\big)}{\gamma^2} \longrightarrow 0,$$

and $\mathcal{P}(H/\gamma) \to \mathcal{P}(c\wedge 1)$ because $\mathcal{P}$ is continuous on $[0,1]$. □

The order of the limits. The limits $\beta \downarrow 0$, $\varrho \downarrow 0$ and $\gamma \to \infty$ do not commute. At a fixed positive ridge the limit $\beta \downarrow 0$ is continuous but does not commute with $\gamma \to \infty$; at zero ridge it is discontinuous as soon as $\gamma > 1$.

Proposition SF.11 (Order of limits at positive ridge)**.**

(i) For fixed $\gamma_{\text{in}}, \gamma_{\text{out}} > 0$ *and* $\varrho > 0$, $\lim_{\beta\downarrow 0} \mathfrak{d}_{\text{iso}}(\varrho; \beta, \gamma_{\text{in}}, \gamma_{\text{out}}) = \mathfrak{d}_{\text{iso}}(\varrho; 0, \gamma_{\text{in}}, \gamma_{\text{out}})$.

(ii) For $\gamma \geq 2$ *and* $\varrho \geq 0$,

$$\left|\gamma\, \mathfrak{d}_{\text{iso}}(\varrho; 0, \gamma, \gamma) - \frac{1}{\sqrt{\pi}}\right| \leq \frac{\sqrt{2}\left(1 + \sqrt{2\pi}/2 + \sqrt{\varrho}\right)}{\pi\gamma}.$$

(iii) For every $\varrho > 0$, $\lim_{\gamma\to\infty} \lim_{\beta\downarrow 0} \gamma\, \mathfrak{d}_{\text{iso}}(\varrho; \beta, \gamma, \gamma) = \pi^{-1/2}$, *whereas for every fixed* $\beta \in (0, 1)$, $\gamma\, \mathfrak{d}_{\text{iso}}(\varrho; \beta, \gamma, \gamma) \to \infty$ *as* $\gamma \to \infty$.

Part (iii) has no version uniform in β. The error bound of eq. (SF.32), multiplied by γ, stays bounded as $\beta \downarrow 0$, so the non-uniformity lies in the main term. By theorem SF.13, at fixed $\beta > 0$ the product $\gamma\, \mathfrak{d}_{\text{iso}}$ grows like $\widehat{\mathfrak{p}}_\beta(\varrho)/\sqrt{\pi}$, and $\widehat{\mathfrak{p}}_\beta(\varrho)/\log\gamma \to 1/\log(1/\beta)$, a rate that vanishes as $\beta \downarrow 0$. Part (iii) is stated for $\varrho > 0$; at $\varrho = 0$ the inner limit is given by proposition SF.12 below and is not C_0.

Proof. For (i), write $q_\beta(s)$ for the root at momentum β. The link functions converge pointwise: $w_0 = 1 - \beta \to 1$ and Φ is continuous, and

$$\sum_{j\geq 1} \Phi(w_j^2 q) \leq q \sum_{j\geq 1} w_j^2 = q\big(s_2 - (1-\beta)^2\big) \longrightarrow 0,$$

so $r_\beta(q) \to \Phi(q)$. By lemma SF.4(iii), $|J'| \leq 3$, $J \leq 1$ and $\sum_{j\geq 1} w_j = \beta$ give

$$|L_\beta(q) - J(q)| \leq |(1-\beta)J(w_0^2 q) - J(q)| + \beta \leq 3q\big(1 - w_0^2\big) + 2\beta.$$

The roots converge. Fix $s > 0$ and $\delta \in (0, q_0(s))$ with $\Phi(q_0(s) + \delta) < \gamma_\wedge$. The function G of lemma SF.5(i) at $r = \Phi$ is negative at $q_0(s) - \delta$ and positive at $q_0(s) + \delta$; by the pointwise convergence the same holds at $r = r_\beta$ for small β, both points lying in D_{r_β}, so $|q_\beta(s) - q_0(s)| < \delta$. With the bound on $L_\beta - J$ and $|J'| \leq 3$, the integrand of eq. (SF.29) converges pointwise. It is at most $q_\beta(s) \leq \gamma_{\text{in}}\gamma_{\text{out}}/s \leq \min(\gamma_{\text{in}}\gamma_{\text{out}}/\varrho, \gamma_{\text{in}}\gamma_{\text{out}}/u^2)$ at $s = \varrho + u^2$, by lemma SF.4(v) and the proof of proposition SF.5(i), and dominated convergence gives (i).

For (ii), apply lemma SF.8 with $\beta = 0$ and $\gamma_{\text{in}} = \gamma_{\text{out}} = \gamma \geq 1$, where $-\mathsf{W}' = 1$ and $\mathsf{W}(1) = \gamma - 1$, and write $h_\varrho \overset{\text{def}}{=} \psi(\varrho) < 1$. For the lower bound, drop the integral term of $\widetilde{\mathcal{I}}$:

$$\int_0^\infty \mathsf{G}_0(\varrho + u^2)\, \mathrm{d}u \geq \tfrac{1}{2}(\gamma - h_\varrho)\Lambda_0(Q_{h_\varrho}) \geq \tfrac{1}{2}(\gamma - 1)\Lambda_0(Q_{h_\varrho}),$$

and use $\sqrt{2\pi} - \Lambda_0(Q) \leq 2Q^{-1/2}$ (lemma SF.4(iv)) with $Q_{h_\varrho} = (\gamma - h_\varrho)^2/\varrho \geq (\gamma-1)^2/\varrho$; at $\varrho = 0$ the boundary term is $\frac{1}{2}(\gamma-1)\sqrt{2\pi}$ directly. This gives

$$\int_0^\infty \mathsf{G}_0(\varrho + u^2)\, \mathrm{d}u \geq \frac{(\gamma-1)\sqrt{2\pi}}{2} - \sqrt{\varrho}.$$

For the upper bound, the integrand is nonincreasing in ϱ, and at $\varrho = 0$ lemma SF.7 at $\beta = 0$ ($\sqrt{q}J(q) \leq 1$) and $\Lambda_0 \leq \sqrt{2\pi}$ give

$$\int_0^\infty \mathsf{G}_0(\varrho + u^2)\, \mathrm{d}u \leq \int_0^1 \Big(1 + \frac{\sqrt{2\pi}}{2}\Big)\mathrm{d}m + \frac{(\gamma-1)\sqrt{2\pi}}{2}.$$

Multiplying by $\sqrt{2}/(\pi\gamma)$, with $\frac{\sqrt{2}}{\pi\gamma} \cdot \frac{(\gamma-1)\sqrt{2\pi}}{2} = \frac{1}{\sqrt{\pi}} - \frac{1}{\sqrt{\pi}\gamma}$ and $\pi^{-1/2} \leq \sqrt{2}(1 + \sqrt{2\pi}/2)/\pi$, gives (ii).

For (iii), (i) at each γ and then (ii) give the first limit, and corollary SF.2(ii) gives the second. □

Proposition SF.12 (Zero ridge as the momentum vanishes)**.** *In the square case write $\gamma = n + y$ with an integer $n \geq 0$ and $y \in [0,1)$, and put $y^2 C_0(y) \stackrel{\text{def}}{=} 0$ at $y = 0$. Then*

$$\lim_{\beta \downarrow 0} C_\beta(\gamma) = C^\sharp(\gamma) \stackrel{\text{def}}{=} \frac{n\, C_0(1) + y^2 C_0(y)}{\gamma^2} + \frac{n(n-1)/2 + ny}{\sqrt{\pi}\,\gamma^2}.$$

In particular $C^\sharp(\gamma) = C_0(\gamma)$ for $\gamma \leq 1$, while for $\gamma \geq 1$

$$\gamma^2\big(C^\sharp(\gamma) - C_0(\gamma)\big) = (n-1)\, C_0(1) + y^2 C_0(y) + \frac{(n-1)(n-2)/2 + (n-1)y}{\sqrt{\pi}},$$

which is strictly positive for every $\gamma > 1$. Finally $C^\sharp(\gamma) \to 1/(2\sqrt{\pi})$ as $\gamma \to \infty$.

The zero-ridge drift constant is therefore discontinuous at $\beta = 0$ for every $\gamma > 1$, and the limits $\beta \downarrow 0$ and $\varrho \downarrow 0$ do not commute there; for example $C^\sharp(2) = 0.37000\ldots$ while $C_0(2) = 0.25552\ldots$. For $\beta > 0$ every row eventually participates at zero ridge ($\psi \to \gamma$, proposition SF.3(i)), whereas at $\beta = 0$ at most one unit of participation is available ($\Phi < 1$). As $\beta \downarrow 0$ the extra participation is supplied by the ages $j \geq 1$ at ever larger roots.

Proof. The proof is dominated convergence in the participation representation. By lemma SF.8 at $\varrho = 0$, in the square case where $-\mathsf{W}' = 1$ and $\mathsf{W}(\gamma) = 0$,

$$C_\beta(\gamma) = \frac{\sqrt{2}}{\pi\gamma^2} \int_0^\gamma \mathsf{i}_\beta(m)\, \mathrm{d}m \qquad (\beta \in (0,1)).$$

Domination. For $\beta \leq \frac{1}{2}$, lemma SF.7 gives $\mathsf{g}_\beta < 4$ and $\Lambda_\beta(Q_m) \leq \sqrt{2\pi}\, m + \mathsf{g}'_\beta < \sqrt{2\pi}\,\gamma + 4\pi$, so $0 \leq \mathsf{i}_\beta \leq 4 + \frac{1}{2}(\sqrt{2\pi}\gamma + 4\pi)$ on $(0,\gamma)$.

Pointwise limit. Fix a non-integer $m \in (k, k+1)$ with $m < \gamma$, and put $x_j \stackrel{\text{def}}{=} w_j^2 Q_m$, so that $\sum_j \Phi(x_j) = m$ and $x_{j+1} = \beta^2 x_j$. Take a subsequence $\beta \downarrow 0$ along which $x_k \to L \in [0,\infty]$. If $L = \infty$, then $x_j \geq x_k \to \infty$ for all $j \leq k$, so $\liminf \sum_j \Phi(x_j) \geq k + 1 > m$, which is impossible. If $L = 0$, then

$$\sum_{j \geq k} \Phi(x_j) \leq \sum_{j \geq k} x_j \leq \frac{x_k}{1 - \beta^2} \longrightarrow 0,$$

so $\limsup \sum_j \Phi(x_j) \leq k < m$, which is also impossible. Hence $L \in (0,\infty)$. Then $x_j = \beta^{-2(k-j)} x_k \to \infty$ for $j < k$, and $\sum_{j>k} x_j \leq x_k \beta^2/(1-\beta^2) \to 0$, so $\Phi(L) = m - k$. Since Φ is strictly increasing, $L = \Phi^{-1}(m-k)$, independently of the subsequence. By lemma SF.4(iv),

$$\mathsf{i}_\beta(m) = \sum_j \Big[\sqrt{x_j}\, J(x_j) + \tfrac{1}{2}\Lambda_0(x_j)\Big].$$

For $j < k$, $\sqrt{x_j} J(x_j) \leq x_j^{-1/2} \to 0$ and $0 \leq \sqrt{2\pi} - \Lambda_0(x_j) \leq 2x_j^{-1/2} \to 0$. For $j > k$, $\sqrt{x}\, J(x) + \frac{1}{2}\Lambda_0(x) \leq 2\sqrt{x}$, so

$$\sum_{j>k} \Big[\sqrt{x_j}\, J(x_j) + \tfrac{1}{2}\Lambda_0(x_j)\Big] \leq 2 \sum_{j>k} \sqrt{x_j} \leq \frac{2\sqrt{x_k}\,\beta}{1-\beta} \longrightarrow 0.$$

The term $j = k$ converges to $\mathsf{i}_0(m-k) = \sqrt{L}\, J(L) + \frac{1}{2}\Lambda_0(L)$, the integrand at $\beta = 0$ and level $m - k$. Hence $\mathsf{i}_\beta(m) \to \frac{k}{2}\sqrt{2\pi} + \mathsf{i}_0(m-k)$ for almost every $m \in (0,\gamma)$.

Evaluation. By dominated convergence,

$$\lim_{\beta \downarrow 0} C_\beta(\gamma) = \frac{\sqrt{2}}{\pi\gamma^2}\Big[\sum_{k<n}\Big(\frac{k}{2}\sqrt{2\pi} + \int_0^1 \mathsf{i}_0\Big) + \frac{n}{2}\sqrt{2\pi}\, y + \int_0^y \mathsf{i}_0\Big].$$

For $0 < y < 1$, lemma SF.8 at $\beta = 0$ and $\gamma_{\text{in}} = \gamma_{\text{out}} = y$ gives $\int_0^y \mathsf{i}_0 = \pi y^2 C_0(y)/\sqrt{2}$, and the proof of proposition SF.9 gives $\int_0^1 \mathsf{i}_0 = \pi C_0(1)/\sqrt{2}$. Substituting these and $\sqrt{2}\cdot\sqrt{2\pi}/(2\pi) = \pi^{-1/2}$

gives the formula for $C^\sharp$. For $\gamma \le 1$ one has $n = 0$ and $y = \gamma$, or $n = 1$ and $y = 0$, and $C^\sharp = C_0$. For $\gamma \ge 1$, subtracting $\gamma^2 C_0(\gamma) = C_0(1) + (n + y - 1)/\sqrt{\pi}$ (proposition SF.9) and using $n(n-1)/2 - n + 1 = (n-1)(n-2)/2$ gives the displayed difference. Every term is nonnegative; the first is positive if $n \ge 2$, and the second is positive if $n = 1$ and $y > 0$. Finally $C^\sharp(\gamma) = n^2/(2\sqrt{\pi}\gamma^2) + O(n/\gamma^2) \to 1/(2\sqrt{\pi})$. □

Proposition SF.13 (Aspect ratio one with momentum)**.** *Let $\beta \in (0,1)$. Then*

$$C_\beta(1) = \frac{\sqrt{2}}{\pi} \int_0^1 \Big[\sqrt{Q_m}\, L_\beta(Q_m) + \tfrac{1}{2}\Lambda_\beta(Q_m) \Big] \mathrm{d}m,$$

$C_\beta(1) \to C_0(1)$ as $\beta \downarrow 0$, and if $1 - \beta^2 \le \frac{1}{12}$, then

$$C_\beta(1) = \frac{8\vartheta}{3\pi\sqrt{2s_2}} \qquad \text{for some} \quad \vartheta \in \Big[1 - 6(1-\beta^2),\ \big(1 - 6(1-\beta^2)\big)^{-1/2} \Big].$$

Proof. The representation is lemma SF.8 with $\gamma_{\mathrm{in}} = \gamma_{\mathrm{out}} = 1$, $-\mathsf{W}' = 1$ and $\mathsf{W}(1) = 0$. The limit is proposition SF.12 with $n = 1$ and $y = 0$, where $C^\sharp(1) = C_0(1)$. The last claim is theorem SF.7(i),(ii) with $\gamma_\wedge = 1$, where $\varepsilon_\beta = 6(1-\beta^2) \le \frac{1}{2}$. □

The $\beta \downarrow 0$ limit at aspect ratio one is continuous because $\gamma = 1$ is not larger than one; at $\beta \uparrow 1$ the constant grows like $8/(3\pi\sqrt{2s_2})$, the effective-batch law of theorem SF.7.

SF.12 Remarks

Remark SF.1 (Momentum as a change of batch and of step)**.** *In the oversampled square case $N < B$, with $\gamma = N/B$ small, the small-aspect theorems describe momentum at the level of the scalar equation eq.* (SF.19)*, read with its formal zero-ridge coefficients. For every step h, with $h' \overset{\mathrm{def}}{=} h/\sqrt{s_2} = h\sqrt{(1+\beta)/(1-\beta)}$,*

$$-2h\, C_\beta \sqrt{\mathcal{R}} + \frac{h^2}{2B}\, \mathfrak{v}_{\mathrm{iso},\beta} = -2h' \big(\sqrt{s_2}\, C_\beta\big)\sqrt{\mathcal{R}} + \frac{h'^2}{2B} \big(s_2\, \mathfrak{v}_{\mathrm{iso},\beta}\big).$$

On the drift, $\sqrt{s_2}\, C_\beta(\gamma) = C_0(\gamma)(1 + O(\gamma))$ by theorem SF.7(iii): momentum acts as the effective batch $B/s_2 = B(1+\beta)/(1-\beta)$. On the volatility the factor is V_β, not $1/s_2$: $\mathfrak{v}_{\mathrm{iso},0} = N$ and $\mathfrak{v}_{\mathrm{iso},\beta} = N V_\beta (1 + O(\gamma))$ (theorem SF.8), so that $s_2\, \mathfrak{v}_{\mathrm{iso},\beta} = s_2 V_\beta\, \mathfrak{v}_{\mathrm{iso},0}(1 + O(\gamma))$ with $s_2 V_\beta \in (s_2, 1)$ (lemma SF.10). Momentum at step h is therefore no momentum at step h' with the volatility multiplied by $s_2 V_\beta < 1$. The step-rescaling reading is only approximate in the oversampled isotropic case: momentum carries less noise than the rescaled step, by the factor $s_2 V_\beta$, which is $0.826\ldots$ at $\beta = \frac{1}{2}$ and tends to $0.7776\ldots$ as $\beta \uparrow 1$. The main text writes $B_{\mathrm{eff}} \overset{\mathrm{def}}{=} B/s_2$ for the effective batch and $c_\beta \overset{\mathrm{def}}{=} s_2 V_\beta$ for this factor, so that $\mathfrak{v}_{\mathrm{iso},\beta} = c_\beta N B_{\mathrm{eff}}/B \,\big(1 + O(\gamma)\big)$. The factor $1/s_2$ for the volatility of corollary SG.1(ii) belongs to a different regime, the resolved rows of a power-law spectrum with $\gamma \to \infty$; that regime and the oversampled isotropic one are disjoint, and neither statement specializes to the other.

SG Power-law coefficient profiles with momentum

This section computes the half-anisotropic drift and volatility coefficients of appendices SC and SD for the power-law output covariance $\mu_i = c_N i^{-\alpha}$, in the thin-batch regime in which N, B and the aspect ratio $\gamma = N/B$ all tend to infinity, at a fixed momentum $\beta \in (0,1)$. For each finite dimension the coefficients are evaluated at zero normalized ridge, as limits of the positive-ridge coefficients of those sections. The main result, theorem SG.1, gives explicit approximations of the drift coefficient $\mathfrak{d}_i$ and of the integrated volatility coefficient $\mathfrak{v}_i$, with relative errors tending to zero uniformly over all rows $n_N \le i \le N$ for every sequence $n_N \to \infty$. Both coefficients have a crossover between a resolved window, where

$$\sqrt{2}\, \mathfrak{d}_i \sim \sqrt{\frac{\mu_i}{\gamma s_2}}, \qquad \mathfrak{v}_i \sim \frac{1}{s_2},$$

and an unresolved window, where both are governed by explicit bulk profiles; theorems SG.2 and SG.3 and corollary SG.1 locate the crossovers and give the endpoint laws, and proposition SG.6 compares the crossover indices with the resolution threshold $i_\#$ of eq. (15).

Two conventions of notation are used. The letter κ denotes only the resolution scalar of the main text's half-anisotropic table, defined in proposition SG.6. The target exponent ς of the main text does not appear in this section, because every statement here concerns the coefficients and none concerns the initial data.

Every statement of the section is a statement about deterministic coefficient functions at a finite dimension and about their asymptotics. None is a statement about the random algorithm at zero ridge or in the thin-batch regime; remark SG.3 explains how the deterministic model of the phase-comparison section uses the profiles.

SG.1 Standing setup

Dimensions and covariances. Throughout the section $N_{\mathrm{out}} = N_{\mathrm{in}} = N$, the input covariance is $\Sigma_{\mathrm{in}} = \mathrm{Id}_N$, and the output covariance is $\Sigma_{\mathrm{out}} = \sum_{i=1}^{N} \mu_i \, u_i \otimes u_i$ with $\mu_1 \geq \cdots \geq \mu_N > 0$. The aspect ratio is $\gamma \stackrel{\mathrm{def}}{=} N/B$. In appendix SG.2 the spectrum is an arbitrary positive one; from appendix SG.3 on it is the power law $\mu_i = c_N i^{-\alpha}$. The momentum $\beta \in (0,1)$ is fixed, the weights are $w_j = (1-\beta)\beta^j$, and

$$s_p = \sum_{j \geq 0} w_j^p = \frac{(1-\beta)^p}{1-\beta^p} \quad (p \geq 2), \qquad s_2 = \frac{1-\beta}{1+\beta}, \qquad c_0 \stackrel{\mathrm{def}}{=} \mathbb{E}\,|g| = \sqrt{\frac{2}{\pi}},$$

with $g \sim \mathcal{N}(0,1)$. The no-momentum case $\beta = 0$ is treated in remark SG.1.

The coefficients at a finite dimension. At a fixed dimension, the objects of appendices SB to SD are deterministic functions of the spectrum, of B, of β and of the normalized ridge $\varrho > 0$. They are the branch $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ of theorem SB.1 for the measures $\nu = B^{-1} \sum_i \delta_{\mu_i}$ and $\tilde{\nu} = \gamma\, \delta_1$ of eq. (SA.41); the drift coefficient

$$\mathfrak{d}_i(\varrho) \stackrel{\mathrm{def}}{=} \mathfrak{d}(\mu_i, 1)$$

of eq. (SC.4), which is the row coefficient of corollary SC.2; the lag kernels $\mathfrak{k}_\ell$ of theorem SD.3; the row lag coefficients $\overline{C}_{i,\ell}(\varrho) \stackrel{\mathrm{def}}{=} \overline{C}_{i,\ell}(\varrho, \varrho)$ and the integrated row coefficient $\mathfrak{v}_i(\varrho) \stackrel{\mathrm{def}}{=} \mathfrak{v}_i(\varrho, \varrho)$ of eq. (SD.75); and the raw coefficient $\mathfrak{v}_i^{(0)}(\varrho)$ of eq. (SD.22), which equals $\overline{C}_{i,0}(\varrho)$ by theorem SD.7(iv). The constructions of appendix SD.6 are made at a fixed finite dimension and use the size conditions only through constants $\gamma_\star \geq \gamma$ and $C_\Sigma \geq \mu_1$ of that dimension, along its replications (definition SD.3). The row lag kernel is

$$\mathfrak{k}_{i,\ell}(s,s') \stackrel{\mathrm{def}}{=} B\, \mathfrak{k}_\ell[u_i \otimes u_i, \mathrm{Id}_N](s,s') \qquad (s,s' > 0). \tag{SG.1}$$

By theorem SD.4(i) with $F = u_i \otimes u_i$ and $G = \mathrm{Id}_N$, multiplied by B, at the fixed dimension and with $T \to \infty$,

$$\overline{C}_{i,\ell}(\varrho) = \lim_{T \to \infty} \frac{4}{\pi^2} \int_0^T \int_0^T \mathfrak{k}_{i,\ell}(\varrho + x^2, \varrho + y^2)\, \mathrm{d}x\, \mathrm{d}y, \qquad \mathfrak{v}_i(\varrho) = \overline{C}_{i,0}(\varrho) + 2 \sum_{\ell \geq 1} \overline{C}_{i,\ell}(\varrho). \tag{SG.2}$$

Scalar functions. The functions J, F_β and L_β are those of eq. (SB.1) for the EMA weights, and $r_\beta(q) \stackrel{\mathrm{def}}{=} q F_\beta(q)$ is the function of lemma SB.1(iii). Since infinitely many weights are positive, r_β increases strictly from 0 to ∞, and we put

$$Q_m \stackrel{\mathrm{def}}{=} r_\beta^{-1}(m), \qquad E(m) \stackrel{\mathrm{def}}{=} Q_m\, r_\beta'(Q_m) \qquad (m > 0). \tag{SG.3}$$

Asymptotic conventions. Asymptotic statements are made along sequences of dimensions indexed by N, with $B = B_N$, $\gamma = \gamma_N$ and c_N depending on N; a row sequence is a sequence of integers $i = i_N \in \{1, \ldots, N\}$. We write $a \sim b$ when $a/b \to 1$, and $a \asymp b$ when a/b is bounded above and below by positive constants. Unless a statement says otherwise, constants c, C depend only on (α, β) and may change from line to line.

SG.2 The coefficients at finite spectrum and zero ridge

In this subsection the spectrum $\mu_1 \geq \cdots \geq \mu_N > 0$ and the dimension are fixed. The half-anisotropic branch is parametrized by one real coordinate, the participation $m \in (0, \gamma)$, in which the drift becomes a single integral and the two-time volatility kernel becomes explicit. Both coefficients then extend to zero normalized ridge.

Definition SG.1 (Participation coordinates)**.** *The participation function and its inverse are*

$$m_N(\Lambda) \stackrel{\text{def}}{=} \frac{1}{B} \sum_{i=1}^{N} \frac{\Lambda \mu_i}{1 + \Lambda \mu_i} \quad (\Lambda \geq 0), \qquad \Lambda_N \stackrel{\text{def}}{=} m_N^{-1} : (0, \gamma) \to (0, \infty). \tag{SG.4}$$

For $0 < m < \gamma$ put

$$\begin{gathered} f_i(m) \stackrel{\text{def}}{=} \frac{\Lambda_N(m)\mu_i}{1 + \Lambda_N(m)\mu_i}, \qquad s_N(m) \stackrel{\text{def}}{=} \frac{(\gamma - m)\, m}{Q_m\, \Lambda_N(m)}, \\ \vartheta_N(m) \stackrel{\text{def}}{=} \frac{1}{E(m)} + \frac{\Lambda_N'(m)}{\Lambda_N(m)} + \frac{1}{\gamma - m} - \frac{1}{m}. \end{gathered} \tag{SG.5}$$

The function m_N is strictly increasing from 0 to γ, because each of its N terms is, so Λ_N is well defined, increasing and smooth.

Proposition SG.1 (The branch in participation coordinates)**.**

(i) The function s_N is a smooth strictly decreasing bijection of $(0, \gamma)$ onto $(0, \infty)$, and $-\frac{\mathrm{d}}{\mathrm{d}m} \log s_N(m) = \vartheta_N(m) > 0$.

(ii) For $s > 0$ let $m = m(s) \in (0, \gamma)$ be the solution of $s_N(m) = s$. Then

$$\begin{gathered} q(-s) = Q_m, \qquad \tilde{s}_1(-s) = s\, \Lambda_N(m), \qquad s_1(-s) = \frac{s\, m}{\gamma - m}, \\ \sigma(-s) = \frac{m}{s\, \Lambda_N(m)}, \qquad \tilde{\sigma}(-s) = \frac{\gamma - m}{s}, \qquad \frac{\tilde{s}_1(-s)\mu_i}{s + \tilde{s}_1(-s)\mu_i} = f_i(m). \end{gathered} \tag{SG.6}$$

(iii) For every $\varrho > 0$,

$$\mathfrak{d}_i(\varrho) = \frac{\sqrt{2}}{\pi\gamma} \int_0^\infty \frac{L_\beta(Q_{m(\varrho+u^2)})}{F_\beta(Q_{m(\varrho+u^2)})}\, f_i\big(m(\varrho + u^2)\big)\, \mathrm{d}u, \qquad \mathfrak{v}_i^{(0)}(\varrho) = f_i\big(m(\varrho)\big). \tag{SG.7}$$

Proof. Part (i) rests on the monotonicity of $\Lambda_N(m)/m$. Since $m_N(\Lambda)/\Lambda$ is decreasing in Λ, its inverse relation makes $\Lambda_N(m)/m$ nondecreasing, so that

$$\frac{\Lambda_N'(m)}{\Lambda_N(m)} \geq \frac{1}{m}.$$

The logarithmic derivative of s_N is, by $\frac{\mathrm{d}}{\mathrm{d}m} \log Q_m = 1/E(m)$,

$$\frac{\mathrm{d}}{\mathrm{d}m} \log s_N(m) = -\frac{1}{\gamma - m} + \frac{1}{m} - \frac{1}{E(m)} - \frac{\Lambda_N'(m)}{\Lambda_N(m)} = -\vartheta_N(m),$$

and the preceding inequality gives $\vartheta_N(m) \geq 1/(\gamma - m) + 1/E(m) > 0$. At the left endpoint, $\Lambda_N(m) \sim Bm/\sum_i \mu_i$ and $Q_m \sim m/s_2$ as $m \downarrow 0$, by lemma SB.1(ii), so that

$$s_N(m) \sim \frac{\gamma\, s_2}{m} \cdot \frac{1}{B} \sum_i \mu_i \longrightarrow \infty.$$

At the right endpoint, $\Lambda_N(m) \to \infty$ and $Q_m \to Q_\gamma < \infty$, so $s_N(m) \to 0$.

Part (ii) uses the coupling eq. (SB.4). Let $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)$ be the branch at $-s$ and put $m \stackrel{\text{def}}{=} \tilde{s}_1\sigma = s_1\tilde{\sigma} = r_\beta(q)$, so that $q = Q_m$. The second trace equation of eq. (SB.2) for $\tilde{\nu} = \gamma\delta_1$ reads $\tilde{\sigma} = \gamma/(s + s_1)$, hence

$$s\tilde{\sigma} = \gamma - s_1\tilde{\sigma} = \gamma - m, \qquad \tilde{s}_1 = s\tilde{\sigma}\, F_\beta(q) = (\gamma - m)\, F_\beta(Q_m).$$

Put $\Lambda \stackrel{\text{def}}{=} \tilde{s}_1/s$. The first trace equation gives

$$m = \tilde{s}_1\sigma = \frac{1}{B}\sum_i \frac{\tilde{s}_1\mu_i}{s+\tilde{s}_1\mu_i} = m_N(\Lambda),$$

so $\Lambda = \Lambda_N(m)$, and, with $F_\beta(Q_m) = m/Q_m$,

$$s = \frac{\tilde{s}_1}{\Lambda} = \frac{(\gamma-m)F_\beta(Q_m)}{\Lambda_N(m)} = s_N(m).$$

By (i) this determines $m = m(s)$. The remaining identities of eq. (SG.6) are $s_1 = m/\tilde{\sigma}$, $\sigma = m/\tilde{s}_1$ and $\tilde{s}_1\mu_i/(s+\tilde{s}_1\mu_i) = \Lambda\mu_i/(1+\Lambda\mu_i)$.

Part (iii) is a computation in these coordinates. By (ii),

$$s + s_1 = \frac{s\gamma}{\gamma - m}, \qquad \frac{\mu_i}{s+\tilde{s}_1\mu_i} = \frac{f_i(m)}{\tilde{s}_1} = \frac{f_i(m)}{(\gamma-m)F_\beta(Q_m)},$$

so the integrand of eq. (SC.4) at $(\mu,\lambda) = (\mu_i, 1)$ is

$$\frac{s\,L_\beta(q)\,\mu_i}{(s+\tilde{s}_1\mu_i)(s+s_1)} = \frac{L_\beta(Q_m)}{\gamma\,F_\beta(Q_m)}\,f_i(m), \qquad m = m(s),$$

which gives the first identity with $s = \varrho + u^2$. The second is the last identity of eq. (SG.6) at $s = \varrho$, compared with eq. (SD.22). □

The weight functions. The analysis of the participation integrals uses the behaviour of the EMA weight functions at small and large arguments. For $Q > 0$ let

$$n(Q) \stackrel{\text{def}}{=} \#\{j \geq 0 :\ w_j^2 Q \geq 1\}$$

be the number of ages whose weight is resolved at the scale Q; it is 0 when $(1-\beta)^2 Q < 1$, and otherwise $n(Q) = \lfloor \log((1-\beta)^2 Q)/(2|\log\beta|)\rfloor + 1$.

Lemma SG.1 (Weight functions)**.**

(i) For $q \geq 0$,

$$F_\beta(q) = s_2 - 3s_4 q + R_F(q), \quad 0 \leq R_F \leq 15 s_6 q^2,$$
$$L_\beta(q) = 1 - 3s_3 q + R_L(q), \quad 0 \leq R_L \leq 15 s_5 q^2.$$

Consequently $r_\beta(q) = s_2 q + O(q^2)$, and $Q_m \sim m/s_2$ and $E(m) \sim m$ as $m \downarrow 0$.

(ii) For $q > 0$, $|qF'_\beta(q)| \leq F_\beta(q)$ and $|qL'_\beta(q)| \leq L_\beta(q)$. For $m > 0$, $E(m) \leq m$.

(iii) There are $0 < c_\beta \leq C_\beta < \infty$ such that, for $Q \geq 1$,

$$|r_\beta(Q) - n(Q)| \leq C_\beta, \qquad c_\beta \leq \sqrt{Q}\,L_\beta(Q) \leq C_\beta, \qquad c_\beta \leq Q\,r'_\beta(Q) \leq C_\beta.$$

Consequently $E(m) \asymp \min(m,1)$ for all $m > 0$, and $|\log Q_m - 2|\log\beta|\,m| \leq C_\beta$ for $m \geq 1$.

Proof. Part (i) is a second-order expansion. With $t = xg^2$ and $(1+t)^{-1} = 1 - t + t^2/(1+t)$, the Gaussian moments $\mathbb{E}\,g^2 = 1$, $\mathbb{E}\,g^4 = 3$, $\mathbb{E}\,g^6 = 15$ give

$$J(x) = 1 - 3x + \rho(x), \qquad 0 \leq \rho(x) = \mathbb{E}\,\frac{x^2 g^6}{1+xg^2} \leq 15x^2.$$

Inserting this in $F_\beta(q) = \sum_j w_j^2 J(w_j^2 q)$ and $L_\beta(q) = \sum_j w_j J(w_j^2 q)$, with $\sum_j w_j = 1$, gives the two expansions. The consequences follow from $r_\beta(q) = qF_\beta(q)$ and $r'_\beta(q) \to s_2$ as $q \downarrow 0$.

Part (ii) is a termwise bound. Since $J'(x) = -\,\mathbb{E}\,g^4(1+xg^2)^{-2}$ and $xg^4(1+xg^2)^{-2} \leq g^2(1+xg^2)^{-1}$,

$$|xJ'(x)| \leq J(x) \qquad (x > 0).$$

Each term of $qF'_\beta(q)$ and $qL'_\beta(q)$ is $w_j^2 \cdot (w_j^2 q)J'(w_j^2 q)$, respectively $w_j \cdot (w_j^2 q)J'(w_j^2 q)$, which proves the two bounds. Since $F'_\beta \le 0$,

$$q\, r'_\beta(q) = r_\beta(q) + q^2 F'_\beta(q) \le r_\beta(q),$$

and at $q = Q_m$ this is $E(m) \le m$.

Part (iii) compares each sum with the count of resolved ages. Put $\mathsf{f}(x) \stackrel{\text{def}}{=} xJ(x) = \mathbb{E}\, xg^2(1+xg^2)^{-1}$, so that $r_\beta(Q) = \sum_j \mathsf{f}(w_j^2 Q)$. Then $\mathsf{f}(x) \le \min(x, 1)$, and the bound $(2\pi)^{-1/2}$ on the Gaussian density gives

$$1 - \mathsf{f}(x) = \mathbb{E}\,\frac{1}{1+xg^2} \le \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{\mathrm{d}y}{1+xy^2} = \sqrt{\frac{\pi}{2}}\, x^{-1/2}.$$

The ages $j < n(Q)$ have $w_j\sqrt{Q} \ge 1$, and the numbers $(w_j\sqrt{Q})^{-1}$ for $j = n(Q)-1, n(Q)-2, \dots$ are at most $1, \beta, \beta^2, \dots$; the ages $j \ge n(Q)$ have $w_j^2 Q < 1$, and the numbers $w_j^2 Q$ for $j = n(Q), n(Q)+1, \dots$ are at most $1, \beta^2, \dots$. Hence

$$|r_\beta(Q) - n(Q)| \le \sqrt{\frac{\pi}{2}} \sum_{j<n(Q)} \frac{1}{w_j\sqrt{Q}} + \sum_{j \ge n(Q)} w_j^2 Q \le \sqrt{\frac{\pi}{2}}\,\frac{1}{1-\beta} + \frac{1}{1-\beta^2}.$$

The same splitting bounds the other two sums from above. Indeed $\sqrt{Q}L_\beta(Q) = \sum_j \phi(w_j^2 Q)$ with $\phi(x) = \sqrt{x}\, J(x) \le \min(\sqrt{x}, x^{-1/2})$, using $J \le 1$ and $J(x) \le 1/x$; and $Qr'_\beta(Q) = \sum_j \mathsf{k}(w_j^2 Q)$ with $\mathsf{k}(x) = \mathbb{E}\, xg^2(1+xg^2)^{-2} \le \min(\mathsf{f}(x), 1-\mathsf{f}(x)) \le \min(x, \sqrt{\pi/2}\, x^{-1/2})$. For the lower bounds, note that for $Q \ge 1$ some age has $w_j^2 Q \in [(1-\beta)^2, \beta^{-2}]$: the age 0 if $(1-\beta)^2 Q < 1$, and the age $n(Q)-1$ otherwise. The continuous positive functions ϕ and k are bounded below on this compact interval, which gives c_β.

The consequences follow. For $m \ge 1$ one has $Q_m \ge 1$, because $r_\beta(1) \le s_2 < 1$, so $E(m) = Q_m r'_\beta(Q_m) \asymp 1$. For $0 < m \le 1$ the ratio $E(m)/m = Q_m r'_\beta(Q_m)/r_\beta(Q_m)$ is continuous and positive, with limit 1 at $m = 0$ by (i), so $E(m) \asymp m$. Finally $r_\beta(Q_m) = m$ and the first bound of (iii) give $|n(Q_m) - m| \le C_\beta$, and the explicit form of n gives the bound on $\log Q_m$. □

The large-argument behaviour of $\sqrt{q}\, L_\beta(q)$ oscillates with the geometric spacing of the weights and has no limit. Its average in the participation coordinate has one, and this is the form in which it enters the drift.

Lemma SG.2 (Averaging in the participation coordinate)**.** *Put $\mathcal{A}(m) \stackrel{\text{def}}{=} \sqrt{Q_m}\, L_\beta(Q_m)/E(m)$ for $m > 0$. For every $m_0 > 0$, $\mathcal{A} \asymp 1$ on $[m_0, \infty)$ with constants depending on (β, m_0), and*

$$\sup_{t \ge m_0} \Big| \int_{m_0}^{t} \big(\mathcal{A}(m) - \sqrt{2\pi}\big)\, \mathrm{d}m \Big| \le C_{\beta, m_0}. \tag{SG.8}$$

Consequently, for every $[a, c] \subset [m_0, \infty)$ and every continuously differentiable W on $[a, c]$,

$$\Big| \int_a^c \big(\mathcal{A}(m) - \sqrt{2\pi}\big) W(m)\, \mathrm{d}m \Big| \le C_{\beta, m_0} \Big(|W(a)| + |W(c)| + \int_a^c |W'(m)|\, \mathrm{d}m \Big). \tag{SG.9}$$

Proof. The two-sided bound on $\mathcal{A}$ is lemma SG.1(iii) for $m \ge 1$ and continuity and positivity on $[m_0, 1]$.

The average is computed in the variable q. Since $\mathrm{d}m = r'_\beta(q)\, \mathrm{d}q$ at $m = r_\beta(q)$ and $E(m) = qr'_\beta(q)$,

$$\int_{m_0}^{t} \mathcal{A}(m)\, \mathrm{d}m = \int_{q_0}^{Q_t} \sqrt{q}\, L_\beta(q)\, \frac{\mathrm{d}q}{q} = \sum_{j \ge 0} \int_{w_j^2 q_0}^{w_j^2 Q_t} x^{-1/2} J(x)\, \mathrm{d}x, \qquad q_0 \stackrel{\text{def}}{=} Q_{m_0},$$

by the substitution $x = w_j^2 q$ in the j-th term of L_β. The full integral of each term is

$$\int_0^\infty x^{-1/2} J(x)\, \mathrm{d}x = \mathbb{E}\Big[|g| \int_0^\infty \frac{y^{-1/2}}{1+y}\, \mathrm{d}y \Big] = \pi\, \mathbb{E}\,|g| = \sqrt{2\pi},$$

by the substitution $y = xg^2$ and the Beta integral $\int_0^\infty y^{-1/2}(1+y)^{-1}\mathrm{d}y = \pi$ [21, (5.12.3)]. The incomplete terms are controlled by $x^{-1/2}J(x) \le \min(x^{-1/2}, x^{-3/2})$: for $j < n(Q_t)$ the missing parts are at most $2w_j\sqrt{q_0}$ and $2(w_j\sqrt{Q_t})^{-1}$, and for $j \ge n(Q_t)$ the whole term is at most $2w_j\sqrt{Q_t}$. Summing these geometric bounds as in the proof of lemma SG.1(iii),

$$\Big|\int_{m_0}^{t} \mathcal{A}(m)\,\mathrm{d}m - \sqrt{2\pi}\,n(Q_t)\Big| \le 2\sqrt{q_0} + \frac{4}{1-\beta}.$$

For $t \ge 1$, lemma SG.1(iii) replaces $n(Q_t)$ by $r_\beta(Q_t) = t$ up to C_β, and for $m_0 \le t \le 1$ both integrals are bounded. This proves eq. (SG.8).

The weighted bound is an integration by parts. With $\Phi_0(v) \stackrel{\text{def}}{=} \int_{m_0}^{v}(\mathcal{A} - \sqrt{2\pi})\,\mathrm{d}m$,

$$\int_a^c \big(\mathcal{A} - \sqrt{2\pi}\big)W\,\mathrm{d}m = \Phi_0(c)W(c) - \Phi_0(a)W(a) - \int_a^c \Phi_0 W'\,\mathrm{d}m$$

[55, Theorem 6.22], and $|\Phi_0| \le C_{\beta,m_0}$ by eq. (SG.8). □

The drift at zero ridge. By proposition SG.1(i), the participation $m(s)$ tends to γ as $s \downarrow 0$. Put

$$\begin{gathered} F_\star \stackrel{\text{def}}{=} F_\beta(Q_\gamma), \qquad z_1 \stackrel{\text{def}}{=} s_N\big(m_N(1)\big), \qquad D_i \stackrel{\text{def}}{=} \frac{1}{\mu_i}\sqrt{\frac{1+\mu_1}{\gamma F_\star}}, \\ \mathcal{G}_i(s) \stackrel{\text{def}}{=} \frac{L_\beta(Q_{m(s)})}{F_\beta(Q_{m(s)})}\,f_i\big(m(s)\big). \end{gathered} \tag{SG.10}$$

Proposition SG.2 (Drift at zero ridge)**.**

(i) As $s \downarrow 0$, $Q_{m(s)} \to Q_\gamma < \infty$ and $s\,\Lambda_N(m(s))^2 \to F_\star\,B^{-1}\sum_i \mu_i^{-1}$. For $0 < s \le z_1$,

$$1 - \mathfrak{v}_i^{(0)}(s) = \frac{1}{1+\Lambda_N(m(s))\mu_i} \le D_i\sqrt{s}. \tag{SG.11}$$

(ii) The integral

$$\mathfrak{d}_i(0) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi\gamma}\int_0^\infty \mathcal{G}_i(u^2)\,\mathrm{d}u \tag{SG.12}$$

is finite, and for every $\varrho > 0$

$$\max_{1\le i\le N} |\mathfrak{d}_i(\varrho) - \mathfrak{d}_i(0)| \le \frac{C_\beta}{\gamma}\sqrt{\varrho}\,\big(1+\sqrt{Q_\gamma}\big), \tag{SG.13}$$

with C_β depending only on β.

Proof. Part (i) reads the endpoint off the participation coordinate. As $s \downarrow 0$, $m(s) \to \gamma$ and $\Lambda \stackrel{\text{def}}{=} \Lambda_N(m(s)) \to \infty$, and

$$\gamma - m_N(\Lambda) = \frac{1}{B}\sum_i \frac{1}{1+\Lambda\mu_i}, \qquad \Lambda\big(\gamma - m_N(\Lambda)\big) \longrightarrow \frac{1}{B}\sum_i \frac{1}{\mu_i}.$$

Since $s\Lambda = (\gamma - m)F_\beta(Q_m)$ by eq. (SG.5), the product $s\Lambda^2$ has the stated limit. For the deficit, F_β is decreasing, so $F_\beta(Q_m) \ge F_\star$, and $\gamma - m_N(\Lambda) \ge \gamma/(1+\Lambda\mu_1)$. If $s \le z_1$ then $\Lambda \ge 1$, and

$$s = \frac{(\gamma-m)F_\beta(Q_m)}{\Lambda} \ge \frac{\gamma F_\star}{\Lambda(1+\Lambda\mu_1)} \ge \frac{\gamma F_\star}{(1+\mu_1)\Lambda^2}.$$

Hence $1/(1+\Lambda\mu_i) \le 1/(\Lambda\mu_i) \le D_i\sqrt{s}$, and the identity in eq. (SG.11) is eq. (SG.7).

Part (ii) bounds the variation of $\mathcal{G}_i$. As s decreases from ∞ to 0, $q = Q_{m(s)}$ increases from 0 to Q_γ, and $\mathcal{G}_i = \mathsf{R}(q)\,f_i(r_\beta(q))$ with $\mathsf{R} \stackrel{\text{def}}{=} L_\beta/F_\beta$. The factor $f_i(r_\beta(q))$ increases from 0 to 1, so

$$\int_0^\infty \Big|\frac{\mathrm{d}\mathcal{G}_i}{\mathrm{d}s}\Big|\,\mathrm{d}s \le \sup_{[0,Q_\gamma]}\mathsf{R} + \int_0^{Q_\gamma} |\mathsf{R}'(q)|\,\mathrm{d}q.$$

On $[0,1]$, R and R' are bounded by constants depending on β, because $F_\beta \geq F_\beta(1) > 0$, $L_\beta \leq 1$, $|F'_\beta| \leq 3s_4$ and $|L'_\beta| \leq 3s_3$ (lemma SB.1). On $[1,\infty)$, lemma SG.1(ii),(iii) and $r_\beta(q) \geq r_\beta(1)$ give

$$\mathsf{R}(q) = \frac{\sqrt{q}\cdot\sqrt{q}L_\beta(q)}{r_\beta(q)} \leq C_\beta\sqrt{q}, \qquad |\mathsf{R}'(q)| \leq \frac{|L'_\beta|}{F_\beta} + \frac{L_\beta|F'_\beta|}{F_\beta^2} \leq \frac{2L_\beta(q)}{qF_\beta(q)} = \frac{2\sqrt{q}L_\beta(q)}{\sqrt{q}\,r_\beta(q)} \leq \frac{C_\beta}{\sqrt{q}}.$$

Integrating, the variation of $\mathcal{G}_i$ is at most $C_\beta(1+\sqrt{Q_\gamma})$.

The ridge shift is a pairing of this variation with a bounded function. The integrand of eq. (SG.12) is bounded, by $\sup\mathsf{R}$, and for large s it is at most $\mathsf{R}\,\Lambda\mu_i \leq C\gamma s_2\mu_i/s$, since $\Lambda\mu_i = \tilde{s}_1\mu_i/s$ and $\tilde{s}_1 = (\gamma-m)F_\beta \leq \gamma s_2$; so the integral is finite. With $\delta_\varrho(t) \overset{\text{def}}{=} \sqrt{(t-\varrho)_+} - \sqrt{t}$, the substitutions $t = \varrho + u^2$ and $t = u^2$ give

$$\frac{\pi\gamma}{\sqrt{2}}\big(\mathfrak{d}_i(\varrho) - \mathfrak{d}_i(0)\big) = \int_0^\infty \mathcal{G}_i(t)\,\delta'_\varrho(t)\,\mathrm{d}t = -\int_0^\infty \delta_\varrho(t)\,\mathcal{G}'_i(t)\,\mathrm{d}t,$$

by integration by parts on $[0,\varrho]$ and $[\varrho,T]$ [55, Theorem 6.22] and $T\to\infty$, the boundary terms vanishing because $\delta_\varrho(0)=0$, δ_ϱ is continuous and $|\delta_\varrho(T)\mathcal{G}_i(T)| \leq \sqrt{\varrho}\,C\gamma s_2\mu_i/T$. Since $|\delta_\varrho| \leq \sqrt{\varrho}$, the variation bound gives eq. (SG.13). □

The two-time row response. The volatility needs the lag kernel of eq. (SG.1). When $\Sigma_{\text{in}} = \mathrm{Id}$ this kernel has a closed form in terms of the branch at the two spectral parameters. For $s, s' > 0$ write $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)$ and $(\sigma', \tilde{\sigma}', \tilde{s}'_1, s'_1, q')$ for the branch at $-s$ and at $-s'$, and for an integer $\ell \geq 0$ put

$$\mathsf{U} \overset{\text{def}}{=} \frac{1}{B}\sum_{j=1}^N \frac{\mu_j^2}{(s+\tilde{s}_1\mu_j)(s'+\tilde{s}'_1\mu_j)}, \qquad \mathsf{V} \overset{\text{def}}{=} \frac{\gamma}{(s+s_1)(s'+s'_1)},$$

$$b_\ell \overset{\text{def}}{=} \sum_{j\geq 0}\mathbb{E}\,\frac{\beta^\ell w_j^2 g^2}{(1+qw_j^2g^2)(1+\beta^{2\ell}q'w_j^2g^2)}, \qquad e_\ell \overset{\text{def}}{=} \sum_{j\geq 0}\mathbb{E}\,\frac{\beta^{2\ell}w_j^4g^4}{(1+qw_j^2g^2)(1+\beta^{2\ell}q'w_j^2g^2)},$$

$$\Delta_\ell \overset{\text{def}}{=} \big(1 - ss'\,\mathsf{U}\tilde{\sigma}\tilde{\sigma}'e_\ell\big)\big(1 - ss'\,\mathsf{V}\sigma\sigma'e_\ell\big) - ss'\,\mathsf{U}\mathsf{V}\,b_\ell^2. \tag{SG.14}$$

Partial fractions in q give the divided-difference forms

$$b_\ell = \beta^\ell\,\frac{r_\beta(q) - r_\beta(\beta^{2\ell}q')}{q - \beta^{2\ell}q'}, \qquad e_\ell = \beta^{2\ell}\,\frac{F_\beta(q) - F_\beta(\beta^{2\ell}q')}{\beta^{2\ell}q' - q}, \tag{SG.15}$$

with derivatives at coincident arguments.

Proposition SG.3 (Two-time row response)**.** *For every finite positive spectrum, every $\ell \geq 0$ and all $s, s' > 0$, one has $\Delta_\ell > 0$ and*

$$\mathfrak{k}_{i,\ell}(s,s') = \frac{\gamma\,ss'\,\mu_i\,b_\ell}{(s+\tilde{s}_1\mu_i)(s'+\tilde{s}'_1\mu_i)(s+s_1)(s'+s'_1)\,\Delta_\ell} \geq 0. \tag{SG.16}$$

The proposition is the case $\Sigma_{\text{in}} = \mathrm{Id}_N$ of the closed-form lag kernel of theorem SD.5, recorded in proposition SD.1(b), and positivity of Δ_ℓ is theorem SD.5(i). With an isotropic input side the input traces of the closed form are scalar, which gives V and the factor $\gamma/((s+s_1)(s'+s'_1))$, and the rank-one output test $u_i \otimes u_i$ contributes $\mu_i/(B(s+\tilde{s}_1\mu_i)(s'+\tilde{s}'_1\mu_i))$.

The volatility at zero ridge. The lag coefficients are now compared at two ridges. With the constants of eq. (SG.10) put

$$A_N \overset{\text{def}}{=} \Big(2\gamma s_2 \sum_{j=1}^N \mu_j\Big)^{1/2}. \tag{SG.17}$$

Lemma SG.3 (Two ridges and the lag envelope)**.** *For $0 < \eta \leq \varrho$ and every $\ell \geq 0$,*

$$|\overline{C}_{i,\ell}(\eta) - \overline{C}_{i,\ell}(\varrho)| \leq 2\sqrt{1-\mathfrak{v}_i^{(0)}(\varrho)}, \qquad |\overline{C}_{i,\ell}(\varrho)| \leq 1, \tag{SG.18}$$

and for every $\ell \geq 1$,

$$|\overline{C}_{i,\ell}(\varrho)| \leq \frac{A_N\,\beta^\ell}{\sqrt{\varrho}}. \tag{SG.19}$$

Proof. All three bounds are read from the independent model along the replications. At the n-fold replication of definition SD.3, let M_t^{g} be the buffer of the independent model of definition SD.2, and for the n copies $u_i \otimes e_k$ of the row u_i put

$$a_k(\varrho) \overset{\text{def}}{=} Q_\varrho(M_t^{\mathrm{g}})^\top (u_i \otimes e_k), \qquad b_k(\varrho) \overset{\text{def}}{=} Q_\varrho(M_{t+\ell}^{\mathrm{g}})^\top (u_i \otimes e_k),$$

$$\mathsf{a}^{(n)}(\varrho, \varrho') \overset{\text{def}}{=} \frac{1}{n} \sum_{k=1}^{n} \mathbb{E}\langle a_k(\varrho), b_k(\varrho')\rangle.$$

The independent updates have mean zero, so $\mathsf{a}^{(n)}(\varrho, \varrho')$ is B times the independent covariance $C_\ell^{\mathrm{g}}[F, \mathrm{Id}]$ at order n, for the replicated test $F = (u_i \otimes u_i) \otimes \mathrm{Id}_n$ of norm one. By eq. (SD.52) and the replication invariance of theorem SD.3,

$$\mathsf{a}^{(n)}(\varrho, \varrho) \longrightarrow \overline{C}_{i,\ell}(\varrho) \qquad (n \to \infty),$$

and at $\ell = 0$ the limit is $\overline{C}_{i,0}(\varrho) = \mathfrak{v}_i^{(0)}(\varrho)$.

Each row of a regularized update has norm at most one, because $Q_\varrho(K)Q_\varrho(K)^\top \preceq \mathrm{Id}$; this gives the second bound of eq. (SG.18). For the first, $0 \le \omega_\varrho \le \omega_\eta \le 1$ on $[0, \infty)$ for the filters of eq. (SD.2), so

$$(\omega_\eta - \omega_\varrho)^2 \le (1 - \omega_\varrho)^2 \le 1 - \omega_\varrho^2,$$

and functional calculus gives $(Q_\eta - Q_\varrho)(Q_\eta - Q_\varrho)^\top(K) \preceq \mathrm{Id} - Q_\varrho(K)Q_\varrho(K)^\top$. Hence $\mathbb{E}\,\|a_k(\eta) - a_k(\varrho)\|^2 \le 1 - \mathbb{E}\,\|a_k(\varrho)\|^2$, and the same holds for b_k, which has the same law. Writing $\langle a(\eta), b(\eta)\rangle - \langle a(\varrho), b(\varrho)\rangle = \langle a(\eta) - a(\varrho), b(\eta)\rangle + \langle a(\varrho), b(\eta) - b(\varrho)\rangle$ and applying the Cauchy–Schwarz inequality to the average over k,

$$|\mathsf{a}^{(n)}(\eta, \eta) - \mathsf{a}^{(n)}(\varrho, \varrho)| \le 2\Big(1 - \frac{1}{n}\sum_k \mathbb{E}\,\|a_k(\varrho)\|^2\Big)^{1/2}.$$

The average in the bracket is $\mathsf{a}^{(n)}(\varrho, \varrho)$ at $\ell = 0$, which tends to $\mathfrak{v}_i^{(0)}(\varrho)$, and the first bound follows as $n \to \infty$.

The envelope is the coupling of the proof of theorem SD.2(ii). Let M^* be built from an independent past, so that M^* is independent of M_t^{g}, has the law of $M_{t+\ell}^{\mathrm{g}}$, and $M_{t+\ell}^{\mathrm{g}} - M^* = \beta^\ell(M_t^{\mathrm{g}} - M_t')$ with M_t' an independent copy of M_t^{g}. The term with M^* has zero mean, so the Cauchy–Schwarz inequality over the copies and eq. (SA.13) give

$$|\mathsf{a}^{(n)}(\varrho, \varrho)| \le \frac{1}{n}\Big(n\ \mathbb{E}\,\|Q_\varrho(M_{t+\ell}^{\mathrm{g}}) - Q_\varrho(M^*)\|_{\mathrm{F}}^2\Big)^{1/2} \le \frac{\beta^\ell}{\sqrt{n\varrho}}\Big(\mathbb{E}\,\|M_t^{\mathrm{g}} - M_t'\|_{\mathrm{F}}^2\Big)^{1/2}.$$

At order n the last expectation is $2s_2(nB)^{-1}\,\mathrm{Tr}(\Sigma_{\mathrm{out}} \otimes \mathrm{Id}_n)\,nN = 2n\gamma s_2 \sum_j \mu_j = nA_N^2$, which gives eq. (SG.19) in the limit. □

In participation coordinates, with f_i, Λ_N and ϑ_N of definition SG.1 and with $q_k \overset{\text{def}}{=} Q_{m_k}$, put for $m, m_1, m_2 \in (0, \gamma)$

$$W_i(m) \overset{\text{def}}{=} \frac{\sqrt{\mu_i \Lambda_N(m)}}{\sqrt{m}\,(1 + \mu_i \Lambda_N(m))}\sqrt{1 - \frac{m}{\gamma}}\,\vartheta_N(m), \qquad \mathsf{T}_N(m_1, m_2) \overset{\text{def}}{=} \frac{1}{B}\sum_{j=1}^{N} f_j(m_1) f_j(m_2),$$

$$\mathsf{D}_\ell \overset{\text{def}}{=} \sum_{a \ge 0} \mathbb{E}\,\frac{\beta^\ell \sqrt{q_1 q_2}\, w_a^2 g^2}{(1 + q_1 w_a^2 g^2)(1 + \beta^{2\ell} q_2 w_a^2 g^2)}, \qquad \mathsf{M}_\ell \overset{\text{def}}{=} \sum_{a \ge 0} \mathbb{E}\,\frac{\beta^{2\ell} q_1 q_2\, w_a^4 g^4}{(1 + q_1 w_a^2 g^2)(1 + \beta^{2\ell} q_2 w_a^2 g^2)},$$

$$\Delta_\ell^N \overset{\text{def}}{=} \Big(1 - \frac{\mathsf{T}_N \mathsf{M}_\ell}{m_1 m_2}\Big)\Big(1 - \frac{\mathsf{M}_\ell}{\gamma}\Big) - \frac{\mathsf{T}_N \mathsf{D}_\ell^2}{\gamma\, m_1 m_2}, \tag{SG.20}$$

all evaluated at (m_1, m_2); for a negative lag $-\ell$ the three functions $\mathsf{D}, \mathsf{M}, \Delta^N$ are those of the lag ℓ at (m_2, m_1).

Proposition SG.4 (Volatility at zero ridge)**.** *Let* $0 < \bar{z} \le z_1$ *and* $E_i \overset{\text{def}}{=} A_N \bar{z}^{-1/2} + 2\sqrt{D_i}\,\bar{z}^{1/4}$.

(i) For every $\ell \geq 0$ the limit $\overline{C}_{i,\ell}(0) \stackrel{\text{def}}{=} \lim_{\varrho \downarrow 0} \overline{C}_{i,\ell}(\varrho)$ exists, $\overline{C}_{i,0}(0) = 1$, and $|\overline{C}_{i,\ell}(\varrho)| \leq E_i\, \beta^{\ell/3}$ for $0 \leq \varrho \leq \bar{z}$ and $\ell \geq 1$.

(ii) The series $\mathfrak{v}_i(0) \stackrel{\text{def}}{=} 1 + 2\sum_{\ell \geq 1} \overline{C}_{i,\ell}(0)$ converges absolutely, $\mathfrak{v}_i(\varrho) \to \mathfrak{v}_i(0)$ as $\varrho \downarrow 0$, and for every integer $H \geq 0$ and $0 < \varrho \leq \bar{z}$,

$$|\mathfrak{v}_i(\varrho) - \mathfrak{v}_i(0)| \leq 2(1+2H)\sqrt{D_i}\,\varrho^{1/4} + \frac{4E_i\,\beta^{(H+1)/3}}{1-\beta^{1/3}}. \tag{SG.21}$$

(iii) For every $\ell \geq 0$,

$$\overline{C}_{i,\ell}(0) = \frac{4}{\pi^2}\int_0^\infty\!\!\int_0^\infty \mathfrak{k}_{i,\ell}(x^2,y^2)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{\pi^2}\int_0^\gamma\!\!\int_0^\gamma W_i(m_1)W_i(m_2)\,\frac{\mathsf{D}_\ell}{\Delta_\ell^N}\,\mathrm{d}m_1\,\mathrm{d}m_2, \tag{SG.22}$$

all terms are nonnegative, and $\mathfrak{v}_i(0) = \sum_{\ell\in\mathbb{Z}} \overline{C}_{i,\ell}(0) \geq 1$ with $\overline{C}_{i,-\ell}(0) \stackrel{\text{def}}{=} \overline{C}_{i,\ell}(0)$, which is the integral eq. (SG.22) with the kernels of the lag $-\ell$.

Proof. Part (i) combines the two bounds of lemma SG.3 with the deficit eq. (SG.11). For $0 < \eta \leq \varrho \leq z_1$,

$$|\overline{C}_{i,\ell}(\eta) - \overline{C}_{i,\ell}(\varrho)| \leq 2\sqrt{D_i}\,\varrho^{1/4},$$

so the family is Cauchy as $\varrho \downarrow 0$, uniformly in ℓ, and the limit satisfies $|\overline{C}_{i,\ell}(0) - \overline{C}_{i,\ell}(\varrho)| \leq 2\sqrt{D_i}\,\varrho^{1/4}$. At $\ell = 0$ the limit is $\lim_{\varrho\downarrow 0}\mathfrak{v}_i^{(0)}(\varrho) = 1$, by eq. (SG.11). For the envelope fix $\ell \geq 1$ and put $\varrho_\ell \stackrel{\text{def}}{=} \bar{z}\,\beta^{4\ell/3}$. If $\varrho \leq \varrho_\ell$, comparison with ϱ_ℓ and eq. (SG.19) give

$$|\overline{C}_{i,\ell}(\varrho)| \leq \frac{A_N\beta^\ell}{\sqrt{\varrho_\ell}} + 2\sqrt{D_i}\,\varrho_\ell^{1/4} = \big(A_N\bar{z}^{-1/2} + 2\sqrt{D_i}\,\bar{z}^{1/4}\big)\beta^{\ell/3} = E_i\beta^{\ell/3}.$$

If $\varrho_\ell < \varrho \leq \bar{z}$, eq. (SG.19) alone gives $A_N\beta^\ell\varrho^{-1/2} \leq A_N\beta^\ell\varrho_\ell^{-1/2} \leq E_i\beta^{\ell/3}$.

Part (ii) splits the lag sum at H. The first $1+2H$ terms of $\mathfrak{v}_i(\varrho) - \mathfrak{v}_i(0)$ are bounded by the comparison of part (i), and the remaining ones by the envelope:

$$|\mathfrak{v}_i(\varrho) - \mathfrak{v}_i(0)| \leq (1+2H)\,2\sqrt{D_i}\,\varrho^{1/4} + 4E_i\sum_{\ell>H}\beta^{\ell/3},$$

which is eq. (SG.21). Letting $\varrho \downarrow 0$ and then $H \to \infty$ gives the convergence.

Part (iii) compares the regularized filter with a truncated resolvent filter at zero ridge. For $0 < d < T$ let $f_{d,T}(x) \stackrel{\text{def}}{=} \frac{2}{\pi}\int_d^T \iota_{y^2}(x)\,\mathrm{d}y$, an odd function with $0 \leq f_{d,T} \leq 1$ on $[0,\infty)$ and Lipschitz constant at most $2/(\pi d)$. For $x > 0$,

$$0 \leq 1 - f_{d,T}(x) = \frac{2}{\pi}\arctan\frac{d}{x} + \frac{2}{\pi}\arctan\frac{x}{T} \leq \frac{2}{\pi}\arctan\frac{d}{x} + \frac{2x}{\pi T},$$
$$\Big(\frac{2}{\pi}\arctan\frac{d}{x}\Big)^2 \leq \frac{2d^2}{x^2+d^2} = 2\big(1-\omega_{d^2}(x)^2\big),$$

the last inequality by the bound 1 when $x \leq d$ and $\arctan(d/x) \leq d/x$ when $x > d$. Both ω_η and $f_{d,T}$ vanish at $x = 0$, and on $x > 0$, $|\omega_\eta - f_{d,T}| \leq (1-\omega_\eta) + (1-f_{d,T})$. Squaring and using $(1-\omega_\eta)^2 \leq 1 - \omega_\eta^2$, functional calculus gives

$$\big(Q_\eta(K) - f_{d,T}^\sharp(K)\big)\big(Q_\eta(K) - f_{d,T}^\sharp(K)\big)^\top$$
$$\preceq 2\big(\mathrm{Id} - Q_\eta(K)Q_\eta(K)^\top\big) + 8\big(\mathrm{Id} - Q_{d^2}(K)Q_{d^2}(K)^\top\big) + \frac{16}{\pi^2T^2}KK^\top.$$

The argument of lemma SG.3 applies to the pair of filters $(\omega_\eta, f_{d,T})$: the closure eq. (SD.52) holds for the fixed odd bounded Lipschitz filter $f_{d,T}$ by remark SD.2, and the independent model has the exact row moment $\mathbb{E}\,(M_t^{\mathrm{g}}M_t^{\mathrm{g}\,\top})_{ii} = \gamma s_2\mu_i$. In the limit,

$$\Big|\overline{C}_{i,\ell}(\eta) - B\,\mathfrak{R}_\ell[u_i\otimes u_i, \mathrm{Id}](f_{d,T}, f_{d,T})\Big| \leq 2\Big(2\big(1-\mathfrak{v}_i^{(0)}(\eta)\big) + 8\big(1-\mathfrak{v}_i^{(0)}(d^2)\big) + \frac{16\gamma s_2\mu_i}{\pi^2T^2}\Big)^{1/2},$$

where $\mathfrak{R}_\ell$ is the response of eq. (SD.49). By the bilinearity and the sup-norm bound of the response (theorem SD.3), and the uniform continuity of $y \mapsto \iota_{y^2}$ in the supremum norm on $[d, T]$, the response of $f_{d,T}$ is the integral of the responses of the resolvent filters,

$$B\,\mathfrak{R}_\ell[u_i \otimes u_i, \mathrm{Id}](f_{d,T}, f_{d,T}) = \frac{4}{\pi^2}\int_d^T\!\!\int_d^T \mathfrak{k}_{i,\ell}(x^2, y^2)\,\mathrm{d}x\,\mathrm{d}y.$$

Let $\eta \downarrow 0$, then $d \downarrow 0$ and $T \to \infty$. The right side of the error bound tends to zero by eq. (SG.11), the left side tends to $\overline{C}_{i,\ell}(0)$ by (i), and the double integrals of the nonnegative kernel (proposition SG.3) increase to the integral over $(0,\infty)^2$. This proves the first equality of eq. (SG.22).

The second equality is the change of variables $x^2 = s_N(m_1)$, $y^2 = s_N(m_2)$. By proposition SG.1(i), $\mathrm{d}x = \frac{1}{2}\sqrt{s_N(m_1)}\,\vartheta_N(m_1)\,\mathrm{d}m_1$ in absolute value, so

$$\frac{4}{\pi^2}\iint \mathfrak{k}_{i,\ell}(x^2, y^2)\,\mathrm{d}x\,\mathrm{d}y = \frac{1}{\pi^2}\int_0^\gamma\!\!\int_0^\gamma \mathfrak{k}_{i,\ell}(s_1', s_2')\sqrt{s_1' s_2'}\,\vartheta_N(m_1)\vartheta_N(m_2)\,\mathrm{d}m_1\,\mathrm{d}m_2, \qquad s_k' \overset{\mathrm{def}}{=} s_N(m_k).$$

In the coordinates of eq. (SG.6) at $s = s_N(m)$, writing $\Lambda = \Lambda_N(m)$ and using $\tilde{s}_1 = s\Lambda = (\gamma - m)m/Q_m$,

$$\frac{s}{s + s_1} = \frac{\gamma - m}{\gamma}, \qquad \frac{\sqrt{\mu_i}\,\sqrt{s}}{s + \tilde{s}_1\mu_i} = \frac{\sqrt{\mu_i}}{\sqrt{s}\,(1 + \Lambda\mu_i)} = \sqrt{\frac{Q_m}{\gamma - m}}\cdot\frac{\sqrt{\mu_i\Lambda}}{\sqrt{m}\,(1 + \Lambda\mu_i)}.$$

Inserting these in eq. (SG.16), multiplied by $\sqrt{ss'}$, gives

$$\mathfrak{k}_{i,\ell}(s_1', s_2')\sqrt{s_1' s_2'}\,\vartheta_N(m_1)\vartheta_N(m_2) = W_i(m_1)W_i(m_2)\,\frac{\sqrt{q_1 q_2}\,b_\ell}{\Delta_\ell}.$$

The scalars of Δ_ℓ transform in the same way. Since $\mu_j/(s + \tilde{s}_1\mu_j) = f_j(m)/\tilde{s}_1$, $s\tilde{\sigma} = \gamma - m$ and $\sigma = m/\tilde{s}_1 = Q_m/(\gamma - m)$,

$$ss'\,\mathsf{U}\tilde{\sigma}\tilde{\sigma}' = \frac{\mathsf{T}_N\,q_1 q_2}{m_1 m_2}, \qquad ss'\,\mathsf{V}\sigma\sigma' = \frac{q_1 q_2}{\gamma}, \qquad ss'\,\mathsf{UV} = \frac{\mathsf{T}_N\,q_1 q_2}{\gamma\,m_1 m_2}.$$

With $\mathsf{D}_\ell = \sqrt{q_1 q_2}\,b_\ell$ and $\mathsf{M}_\ell = q_1 q_2 e_\ell$ this gives $\Delta_\ell = \Delta_\ell^N$, which proves the second equality.

For the lag sum, the negative lags are defined by the transposed kernels, whose integrals over the symmetric square equal those of the positive lags. Absolute convergence and the bound $\mathfrak{v}_i(0) \geq \overline{C}_{i,0}(0) = 1$ follow from (i) and the nonnegativity of every term. □

Definition SG.2 (Coefficients at zero normalized ridge)**.** *At a finite dimension with positive spectrum, the zero-ridge drift coefficient $\mathfrak{d}_i(0)$ is the integral eq. (SG.12), and the zero-ridge lag and integrated volatility coefficients $\overline{C}_{i,\ell}(0)$ and $\mathfrak{v}_i(0)$ are those of proposition SG.4. By propositions SG.2 and SG.4 they are the limits as $\varrho \downarrow 0$ of the coefficients $\mathfrak{d}_i(\varrho)$, $\overline{C}_{i,\ell}(\varrho)$ and $\mathfrak{v}_i(\varrho)$ of appendices SC and SD.*

The zero-ridge coefficients are limits of deterministic coefficient functions taken at each fixed finite spectrum, before any dimension limit; this is a statement about the coefficients, not a stochastic theorem about the algorithm at zero ridge. At a positive ridge the coefficients are the deterministic equivalents of corollary SC.1 and theorem SD.7 under the proportional regime; no deterministic equivalent is asserted at zero ridge or for a growing aspect ratio.

Lemma SG.4 (Spectral amplitude)**.** *Write $\mathfrak{d}_i(\varrho;\mu)$, $\overline{C}_{i,\ell}(\varrho;\mu)$ and $\mathfrak{v}_i(\varrho;\mu)$ for the coefficients of the spectrum $\mu = (\mu_j)$. For every $c > 0$ and $\varrho \geq 0$,*

$$\mathfrak{d}_i(\varrho; c\mu) = \sqrt{c}\,\mathfrak{d}_i(\varrho/c;\mu), \qquad \overline{C}_{i,\ell}(\varrho; c\mu) = \overline{C}_{i,\ell}(\varrho/c;\mu), \qquad \mathfrak{v}_i(\varrho; c\mu) = \mathfrak{v}_i(\varrho/c;\mu).$$

Proof. For the drift, $m_N(\Lambda; c\mu) = m_N(c\Lambda;\mu)$, so $\Lambda_N(m; c\mu) = \Lambda_N(m;\mu)/c$, the functions f_i are unchanged, and $s_N(m; c\mu) = c\,s_N(m;\mu)$. In eq. (SG.7) for $c\mu$ the participation at $\varrho + u^2$ is therefore the participation for μ at $\varrho/c + (u/\sqrt{c})^2$, and the substitution $u = \sqrt{c}\,u'$ gives the first identity; the case $\varrho = 0$ is the same computation in eq. (SG.12). For the volatility, the buffer of the independent model with covariance $c\Sigma_{\mathrm{out}}$ has the law of $\sqrt{c}\,M_t^{\mathrm{g}}$, and $Q_\varrho(\sqrt{c}K) = Q_{\varrho/c}(K)$ because $\sqrt{c}x/\sqrt{cx^2 + \varrho} = x/\sqrt{x^2 + \varrho/c}$. The independent row covariances of the two spectra therefore coincide at every order of replication, and so do their limits $\overline{C}_{i,\ell}$ (theorem SD.3). The integrated and zero-ridge statements follow by summation and by the limits of proposition SG.4. □

SG.3 The power-law regime: statements

Assumption S4 (Thin-batch power law)**.** *The exponent $\alpha > 0$ and the momentum $\beta \in (0,1)$ are fixed. Along a sequence of positive integers N and $B = B_N$,*

$$N \to \infty, \qquad B \to \infty, \qquad \gamma = \frac{N}{B} \to \infty,$$

and the spectrum is $\mu_i = c_N i^{-\alpha}$ for $1 \le i \le N$, with amplitudes $c_N > 0$, with $\Sigma_{\text{in}} = \mathrm{Id}_N$ and $N_{\text{out}} = N_{\text{in}} = N$.

Constants. Under Assumption S4 put

$$\begin{aligned} a_N &\stackrel{\text{def}}{=} \sqrt{c_N}\, N^{-\alpha/2}, \qquad K_\alpha \stackrel{\text{def}}{=} \frac{\pi}{\alpha \sin(\pi/\alpha)}, \qquad i_{\text{c}} \stackrel{\text{def}}{=} \frac{B}{K_\alpha}, \\ b_N &\stackrel{\text{def}}{=} \sqrt{c_N}\,\big(\gamma\, i_{\text{c}}^{\alpha}\big)^{-1/2}, \qquad d_\alpha \stackrel{\text{def}}{=} \frac{c_0\,\pi}{\alpha \cos(\pi/(2\alpha))}, \end{aligned} \tag{SG.23}$$

where K_α, i_{c}, b_N and d_α are used only for $\alpha > 1$.

Definition SG.3 (Profiles)**.** *Let $S_\alpha(V) \stackrel{\text{def}}{=} \int_0^1 V(V + t^\alpha)^{-1}\mathrm{d}t$ for $V > 0$, a strictly increasing bijection onto $(0,1)$, and $V_\alpha \stackrel{\text{def}}{=} S_\alpha^{-1}$. For $0 < x \le 1$ and $0 < p < 1$ put*

$$f_x(p) \stackrel{\text{def}}{=} \frac{V_\alpha(p)}{V_\alpha(p) + x^\alpha}, \qquad U_\alpha(p) \stackrel{\text{def}}{=} \frac{V_\alpha(p)}{p(1-p)}.$$

The bulk profiles are

$$\begin{aligned} \Psi_\alpha(x) &\stackrel{\text{def}}{=} c_0 \int_0^1 \sqrt{\frac{1-p}{p\, V_\alpha(p)}}\; f_x(p)\,\mathrm{d}p, \\ C_{\alpha,x}(\tau) &\stackrel{\text{def}}{=} \int_0^{1-\tau} \frac{f_x(p+\tau) - f_x(p)}{\tau} \sqrt{\frac{U_\alpha(p)}{U_\alpha(p+\tau)}}\,\mathrm{d}p \quad (0 < \tau < 1), \\ \Gamma_\alpha(x) &\stackrel{\text{def}}{=} 2\int_0^1 C_{\alpha,x}(\tau)\,\mathrm{d}\tau. \end{aligned} \tag{SG.24}$$

For $\alpha > 1$, the crossover profiles are, for $y, X > 0$,

$$\begin{aligned} \Phi_{\alpha,\beta}(y) \stackrel{\text{def}}{=} \frac{1}{\pi}\int_0^\infty & \sqrt{q}\, L_\beta(q)\, r_\beta(q)^{-\frac{\alpha+1}{2}} \\ & \times \frac{y\, r_\beta(q)^\alpha}{1 + y\, r_\beta(q)^\alpha}\Big(1 + (\alpha-1)\frac{q\, r_\beta'(q)}{r_\beta(q)}\Big)\frac{\mathrm{d}q}{q}, \end{aligned} \tag{SG.25}$$

$$\begin{aligned} G_{\alpha,\beta}(X) \stackrel{\text{def}}{=} \frac{1}{\pi^2}\sum_{\ell\in\mathbb{Z}} \int_0^\infty\int_0^\infty & \frac{\sqrt{\zeta_X(u)\big(1-\zeta_X(u)\big)\zeta_X(v)\big(1-\zeta_X(v)\big)}}{\sqrt{uv}} \\ & \times \frac{\mathsf{D}_\ell(u,v)\,\vartheta_\infty(u)\vartheta_\infty(v)}{\Delta_\ell^\infty(u,v)}\,\mathrm{d}u\,\mathrm{d}v, \end{aligned} \tag{SG.26}$$

where D_ℓ and M_ℓ are the functions of eq. (SG.20), which depend on $(m_1, m_2) = (u, v)$ only through (Q_u, Q_v), and

$$\begin{aligned} \zeta_X(u) &\stackrel{\text{def}}{=} \frac{u^\alpha}{X^\alpha + u^\alpha}, \qquad \vartheta_\infty(u) \stackrel{\text{def}}{=} \frac{1}{E(u)} + \frac{\alpha-1}{u}, \\ t_\alpha(u,v) &\stackrel{\text{def}}{=} \frac{v^\alpha u - u^\alpha v}{v^\alpha - u^\alpha}, \qquad \Delta_\ell^\infty \stackrel{\text{def}}{=} 1 - \frac{t_\alpha(u,v)\,\mathsf{M}_\ell(u,v)}{uv}, \end{aligned}$$

with $t_\alpha(u,u) \stackrel{\text{def}}{=} (\alpha-1)u/\alpha$, and with the convention of eq. (SG.20) for negative lags.

The profiles $\Psi_\alpha, \Gamma_\alpha$ do not depend on β; the crossover profiles $\Phi_{\alpha,\beta}, G_{\alpha,\beta}$ keep the full EMA weight functions.

Proposition SG.5 (The profiles)**.**

(i) *Ψ_α and Γ_α are finite, positive and continuous on $(0,1]$, Ψ_α is bounded below by a positive constant there, and as $x \downarrow 0$*

$$\Psi_\alpha(x) \sim \begin{cases} \dfrac{c_0}{\sqrt{1-\alpha}} \log(x^{-\alpha}), & \alpha < 1, \\ \dfrac{2}{3} c_0 \big[\log(1/x)\big]^{3/2}, & \alpha = 1, \\ d_\alpha \sqrt{K_\alpha}\, x^{-(\alpha-1)/2}, & \alpha > 1, \end{cases}$$
$$\Gamma_\alpha(x) \sim \begin{cases} \dfrac{x^\alpha \big[\log(x^{-\alpha})\big]^2}{1-\alpha}, & \alpha < 1, \\ \dfrac{4}{9} x \big[\log(1/x)\big]^3, & \alpha = 1, \\ \dfrac{4K_\alpha^2}{\alpha - 1}\, x, & \alpha > 1. \end{cases} \tag{SG.27}$$

(ii) *For $\alpha > 1$, $\Phi_{\alpha,\beta}$ and $G_{\alpha,\beta}$ are finite, positive and continuous on $(0,\infty)$, and*

$$\Phi_{\alpha,\beta}(y) \sim \sqrt{y/s_2}\ (y \to \infty), \quad \Phi_{\alpha,\beta}(y) \sim d_\alpha\, y^{\frac{\alpha-1}{2\alpha}}\ (y \downarrow 0),$$
$$G_{\alpha,\beta}(X) \to \frac{1}{s_2}\ (X \downarrow 0), \quad G_{\alpha,\beta}(X) \sim \frac{4K_\alpha}{\alpha - 1} X\ (X \to \infty). \tag{SG.28}$$

Approximants. The approximating coefficients are, for $0 < \alpha \le 1$,

$$\sqrt{2}\,\widehat{\mathfrak{d}}_i \stackrel{\text{def}}{=} \sqrt{\frac{\mu_i}{\gamma s_2}} + a_N \Psi_\alpha\Big(\frac{i}{N}\Big), \qquad \widehat{\mathfrak{v}}_i \stackrel{\text{def}}{=} \frac{1}{s_2} + \frac{2c_0}{\sqrt{s_2}} \sqrt{\gamma \Gamma_\alpha\Big(\frac{i}{N}\Big)} + \gamma \Gamma_\alpha\Big(\frac{i}{N}\Big), \tag{SG.29}$$

and, for $\alpha > 1$,

$$\sqrt{2}\,\widehat{\mathfrak{d}}_i \stackrel{\text{def}}{=} b_N\, \Phi_{\alpha,\beta}\Big(\Big(\frac{i_{\mathrm{c}}}{i}\Big)^\alpha\Big)\, \frac{\Psi_\alpha(i/N)}{d_\alpha \sqrt{K_\alpha}\,(i/N)^{-(\alpha-1)/2}}, \qquad \widehat{\mathfrak{v}}_i \stackrel{\text{def}}{=} G_{\alpha,\beta}\Big(\frac{i}{i_{\mathrm{c}}}\Big)\, \frac{\Gamma_\alpha(i/N)}{4K_\alpha^2 (i/N)/(\alpha-1)}. \tag{SG.30}$$

The second factors in eq. (SG.30) tend to one when $i/N \to 0$, by eq. (SG.27).

Theorem SG.1 (The coefficient pair)**.** *Under Assumption S4, for every sequence of integers $1 \le n_N \le N$ with $n_N \to \infty$,*

$$\sup_{n_N \le i \le N} \max\Big\{\Big|\frac{\mathfrak{d}_i(0)}{\widehat{\mathfrak{d}}_i} - 1\Big|,\ \Big|\frac{\mathfrak{v}_i(0)}{\widehat{\mathfrak{v}}_i} - 1\Big|\Big\} \longrightarrow 0. \tag{SG.31}$$

The covariance of the row is $\mu_i = c_N i^{-\alpha}$ exactly, so theorem SG.1 describes the whole triple $(\mu_i, \mathfrak{d}_i(0), \mathfrak{v}_i(0))$ that enters the row equations eq. (SE.13). It does not require the windows of corollary SG.1 to be nonempty. The rows $i < n_N$ are excluded because the approximation is asymptotic in i.

Theorem SG.2 (Drift crossover)**.** *Assume Assumption S4 and let $i = i_N \to \infty$ be a row sequence.*

(i) *Let $0 < \alpha \le 1$ and put*

$$d \stackrel{\text{def}}{=} \frac{c_0}{\sqrt{1-\alpha}}, \quad i_{\mathrm{c},\mathfrak{d}} \stackrel{\text{def}}{=} \Big[\frac{BN^{\alpha-1}}{s_2 d^2 \log^2\gamma}\Big]^{1/\alpha}, \quad a_{\mathfrak{d}} \stackrel{\text{def}}{=} d\, a_N \log\gamma \quad (\alpha < 1);$$
$$d \stackrel{\text{def}}{=} \frac{2}{3} c_0, \quad i_{\mathrm{c},\mathfrak{d}} \stackrel{\text{def}}{=} \frac{B}{s_2 d^2 \log^3\gamma}, \quad a_{\mathfrak{d}} \stackrel{\text{def}}{=} d\, a_N (\log\gamma)^{3/2} \quad (\alpha = 1). \tag{SG.32}$$

If $i/i_{\mathrm{c},\mathfrak{d}} \to x \in (0,\infty)$, then $\sqrt{2}\,\mathfrak{d}_i(0)/a_{\mathfrak{d}} \to 1 + x^{-\alpha/2}$. If $i/i_{\mathrm{c},\mathfrak{d}} \to 0$, then $\sqrt{2}\,\mathfrak{d}_i(0) \sim \sqrt{\mu_i/(\gamma s_2)}$. If $i/i_{\mathrm{c},\mathfrak{d}} \to \infty$, then $\sqrt{2}\,\mathfrak{d}_i(0) \sim a_N \Psi_\alpha(i/N)$, and if moreover $i/N \to 0$,

$$\sqrt{2}\,\mathfrak{d}_i(0) \sim d\, a_N \log\big((N/i)^\alpha\big)\ (\alpha < 1), \qquad \sqrt{2}\,\mathfrak{d}_i(0) \sim d\, a_N \big[\log(N/i)\big]^{3/2}\ (\alpha = 1).$$

(ii) Let $\alpha > 1$. If $i/i_{\mathrm{c}} \to x \in (0,\infty)$, then $\sqrt{2}\,\mathfrak{d}_i(0)/b_N \to \Phi_{\alpha,\beta}(x^{-\alpha})$. If $i/i_{\mathrm{c}} \to 0$, then $\sqrt{2}\,\mathfrak{d}_i(0) \sim \sqrt{\mu_i/(\gamma s_2)}$. If $i/i_{\mathrm{c}} \to \infty$, then $\sqrt{2}\,\mathfrak{d}_i(0) \sim a_N \Psi_\alpha(i/N)$, and if moreover $i/N \to 0$,

$$\sqrt{2}\,\mathfrak{d}_i(0) \sim d_\alpha \sqrt{c_N K_\alpha}\, N^{-1/2}\, i^{-(\alpha-1)/2}.$$

Theorem SG.3 (Volatility crossover and bulk profile)**.** *Assume Assumption S4 and let $i = i_N$ be a row sequence.*

(i) (Bulk.) If $i/N \to x \in (0,1]$ and $\ell = \ell_N$ are integers with $\ell/\gamma \to \tau \in (0,1)$, then

$$\overline{C}_{i,\ell}(0) \longrightarrow C_{\alpha,x}(\tau), \qquad \frac{\mathfrak{v}_i(0)}{\gamma} \longrightarrow \Gamma_\alpha(x),$$

locally uniformly in $x \in (0,1]$ and $\tau \in (0,1)$.

(ii) Let $0 < \alpha \le 1$, let $i \to \infty$, and put

$$i_{\mathrm{c},\mathfrak{v}} \overset{\mathrm{def}}{=} c_0^{2/\alpha}\, i_{\mathrm{c},\mathfrak{d}} = \begin{cases} N\big[(1-\alpha)/(s_2\gamma\log^2\gamma)\big]^{1/\alpha}, & \alpha < 1, \\ 9B/(4s_2\log^3\gamma), & \alpha = 1. \end{cases} \tag{SG.33}$$

If $i/i_{\mathrm{c},\mathfrak{v}} \to y \in (0,\infty)$, then $s_2\,\mathfrak{v}_i(0) \to 1 + 2c_0 y^{\alpha/2} + y^\alpha$, uniformly for y in compact subsets of $(0,\infty)$. If $i/i_{\mathrm{c},\mathfrak{v}} \to 0$, then $\mathfrak{v}_i(0) \to 1/s_2$. If $i/i_{\mathrm{c},\mathfrak{v}} \to \infty$, then $\mathfrak{v}_i(0) \sim \gamma\Gamma_\alpha(i/N)$, and if moreover $i/N \to 0$,

$$\mathfrak{v}_i(0) \sim \frac{\gamma(i/N)^\alpha\big[\log((N/i)^\alpha)\big]^2}{1-\alpha} \quad (\alpha<1), \qquad \mathfrak{v}_i(0) \sim \frac{4}{9}\,\frac{i}{B}\,\big[\log(N/i)\big]^3 \quad (\alpha = 1).$$

(iii) Let $\alpha > 1$ and $i \to \infty$. If $i/i_{\mathrm{c}} \to X_0 \in (0,\infty)$, then $\mathfrak{v}_i(0) \to G_{\alpha,\beta}(X_0)$. If $i/i_{\mathrm{c}} \to 0$, then $\mathfrak{v}_i(0) \to 1/s_2$. If $i/i_{\mathrm{c}} \to \infty$, then $\mathfrak{v}_i(0) \sim \gamma\Gamma_\alpha(i/N)$, and if moreover $i/N \to 0$,

$$\mathfrak{v}_i(0) \sim \frac{4K_\alpha^2}{\alpha-1}\,\frac{i}{B}.$$

In (ii) the two crossover indices of the drift and of the volatility differ by the fixed factor $c_0^{2/\alpha} = (2/\pi)^{1/\alpha}$: each equates the leading head and tail terms of its own approximant. In common coordinates the two crossovers read as follows.

Corollary SG.1 (Crossover windows and joint endpoints)**.** *Assume Assumption S4, let $i = i_N \to \infty$ be a row sequence, and let $i_\star \overset{\mathrm{def}}{=} i_{\mathrm{c},\mathfrak{d}}$ for $\alpha \le 1$ and $i_\star \overset{\mathrm{def}}{=} i_{\mathrm{c}}$ for $\alpha > 1$.*

(i) If $\alpha \le 1$ and $i/i_\star \to x \in (0,\infty)$, then

$$\frac{\sqrt{2}\,\mathfrak{d}_i(0)}{a_{\mathfrak{d}}} \to 1 + x^{-\alpha/2}, \qquad s_2\,\mathfrak{v}_i(0) \to 1 + 2x^{\alpha/2} + c_0^{-2}x^\alpha, \qquad \frac{\mu_i}{c_N i_\star^{-\alpha}} \to x^{-\alpha}.$$

(ii) The joint endpoint equivalents are

	$i/i_\star \to 0$	$i/i_\star \to \infty$
$\sqrt{2}\,\mathfrak{d}_i(0)$	$\sqrt{\mu_i/(\gamma s_2)}$	$a_N\Psi_\alpha(i/N)$
$\mathfrak{v}_i(0)$	$1/s_2$	$\gamma\Gamma_\alpha(i/N)$
μ_i	$c_N i^{-\alpha}$ *(exact)*	$c_N i^{-\alpha}$ *(exact)*

(SG.34)

(iii) One has $i_\star/N \to 0$. There are row sequences $i \to \infty$ in both windows $i/i_\star \to 0$ and $i/i_\star \to \infty$ if and only if

$$\frac{BN^{\alpha-1}}{\log^2\gamma} \to \infty \ \ (\alpha<1), \qquad \frac{B}{\log^3\gamma} \to \infty \ \ (\alpha=1), \qquad B \to \infty \ \ (\alpha>1), \tag{SG.35}$$

and then $1 \ll i_\star \ll N$.

(iv) For $\alpha > 1$, equating the two mesoscopic endpoint equivalents gives the indices $i_{\mathrm{c}}/(s_2 d_\alpha^2)$ for the drift and $B(\alpha-1)/(4s_2K_\alpha^2)$ for the volatility, whose ratio is $(\alpha-1)d_\alpha^2/(4K_\alpha)$; all indices of the corollary are independent of c_N.

The main text's half-anisotropic table states its resolution index through the effective batch $B(1+\beta)/(1-\beta)$, and at $\beta = 0$ through the resolution scalar of the no-momentum participation equation. The relation of that scalar to the crossover indices of the momentum coefficients is the following.

Proposition SG.6 (Reference scales)**.** *Assume Assumption S4. For $\gamma > 1$ let $\kappa \overset{\text{def}}{=} \Lambda_N(1)$ be the unique solution of*

$$\sum_{i=1}^{N} \frac{\kappa\mu_i}{1+\kappa\mu_i} = B,$$

and put $i_\# \overset{\text{def}}{=} (c_N\kappa)^{1/\alpha}$. Then

$$c_N\kappa \sim \begin{cases} (1-\alpha)BN^{\alpha-1}, & \alpha < 1, \\ B/\log\gamma, & \alpha = 1, \\ (B/K_\alpha)^\alpha, & \alpha > 1, \end{cases} \qquad \frac{i_{\mathrm{c},\mathfrak{d}}}{i_\#} \sim \begin{cases} \big(\pi/(2s_2\log^2\gamma)\big)^{1/\alpha}, & \alpha < 1, \\ 9\pi/(8s_2\log^2\gamma), & \alpha = 1, \end{cases} \qquad \frac{i_{\mathrm{c}}}{i_\#} \to 1 \ \ (\alpha > 1), \tag{SG.36}$$

and $i_{\mathrm{c},\mathfrak{v}}/i_\# \sim (s_2\log^2\gamma)^{-1/\alpha}$ for $\alpha < 1$ and $\sim 9/(4s_2\log^2\gamma)$ for $\alpha = 1$.

For $c_N = 1$ the index $i_\# = \kappa^{1/\alpha}$ is the threshold of eq. (15). For $\alpha > 1$ the momentum crossover sits at that threshold. For $\alpha \le 1$ it sits below it by the factor $(\log\gamma)^{-2/\alpha}$, up to constants; remark SG.2 explains why no fixed effective batch removes this factor.

Corollary SG.2 (Nonnegative weighted sums)**.** *Assume Assumption S4. Let $I_N \subset \{n_N, \dots, N\}$ with $n_N \to \infty$, and let $a_{i,N} \ge 0$, $i \in I_N$, be deterministic weights with $\sum_{i\in I_N} a_{i,N} > 0$ for each N. Then*

$$\frac{\sum_{i\in I_N} a_{i,N}\,\mathfrak{d}_i(0)}{\sum_{i\in I_N} a_{i,N}\,\widehat{\mathfrak{d}}_i} \longrightarrow 1, \qquad \frac{\sum_{i\in I_N} a_{i,N}\,\mathfrak{v}_i(0)}{\sum_{i\in I_N} a_{i,N}\,\widehat{\mathfrak{v}}_i} \longrightarrow 1.$$

The zero-ridge results transfer to small positive ridges. With the constants of eqs. (SG.10) and (SG.17) at each N, put

$$\bar{z}_N \overset{\text{def}}{=} \min(z_1, 1), \qquad D_\star \overset{\text{def}}{=} \max_i D_i = D_N, \qquad E_\star \overset{\text{def}}{=} A_N \bar{z}_N^{-1/2} + 2\sqrt{D_\star}\,\bar{z}_N^{1/4}. \tag{SG.37}$$

Proposition SG.7 (Negligible ridge)**.** *Assume Assumption S4. Let $0 < \varrho_N \le \bar{z}_N$ and let $H_N \ge 0$ be integers with*

$$\frac{\varrho_N\, Q_\gamma}{c_N N^{-\alpha}\gamma^2} \to 0, \qquad (1+H_N)\sqrt{D_\star}\,\varrho_N^{1/4} \to 0, \qquad E_\star\,\beta^{H_N/3} \to 0. \tag{SG.38}$$

Then theorems SG.1 to SG.3, corollaries SG.1 and SG.2 hold with $\mathfrak{d}_i(\varrho_N)$, $\overline{C}_{i,\ell}(\varrho_N)$ and $\mathfrak{v}_i(\varrho_N)$ in place of the zero-ridge coefficients. Such sequences (ϱ_N, H_N) exist.

The window eq. (SG.38) is sufficient and not sharp: the quantities Q_γ and $D_\star$ grow exponentially in γ by lemma SG.1(iii). It is a comparison of deterministic coefficients and gives no statement about the algorithm at these ridges.

Remark SG.1 (No momentum)**.** *At $\beta = 0$ the weights are $w_0 = 1$, so $F_\beta = L_\beta = J$ and $r_0(q) = qJ(q)$ increases from 0 to 1, not to ∞; the participation is confined to $m < 1$, and at zero ridge $m(s) \to 1$ when $\gamma > 1$. The computation of proposition SG.1(ii), which uses only the equations of the branch, then gives $\Lambda_N(m(s)) \to \kappa$, the root of proposition SG.6, instead of infinity, and by eq. (SG.7)*

$$\lim_{\varrho\downarrow 0} \mathfrak{v}_i^{(0)}(\varrho) = \frac{\kappa\mu_i}{1+\kappa\mu_i} = \frac{1}{1+(i/i_\#)^\alpha}, \qquad \sum_i \lim_{\varrho\downarrow 0} \mathfrak{v}_i^{(0)}(\varrho) = B.$$

All lag coefficients with $\ell \ge 1$ vanish (theorem SD.7(iv)), so $\mathfrak{v}_i = \mathfrak{v}_i^{(0)}$, and the volatility crossover at $\beta = 0$ is this exact profile, with critical index $i_\#$ and the equivalents eq. (SG.36). The drift at $\beta = 0$ is the single integral eq. (SG.7) with $L_\beta/F_\beta = 1$; its two endpoint laws, on either side of $i_\#$, are proposition SG.8 below. These no-momentum profiles are obtained by setting $\beta = 0$ before the ridge and dimension limits. For every fixed $\beta > 0$, r_β is unbounded, $\mathfrak{v}_i^{(0)}(0) = 1$ for every row (proposition SG.2), and the profiles of theorem SG.1 hold; neither family is a limit of the other under an exchange of $\beta \downarrow 0$ with the ridge and dimension limits.

The endpoint laws of the no-momentum coefficients are the following. At $\beta = 0$ the weight functions are $F_0 = L_0 = J$, and the participation coordinate of definition SG.1 is used with $r_0(q) = qJ(q)$, which increases from 0 to 1. Put

$$A_\alpha \stackrel{\text{def}}{=} c_0 = \sqrt{\frac{2}{\pi}} \quad (0 < \alpha \le 1), \qquad A_\alpha \stackrel{\text{def}}{=} \frac{2\alpha}{\pi(\alpha+1)} \int_0^\infty q^{-3/2} \big(qJ(q)\big)^{\frac{\alpha+1}{2}} \,\mathrm{d}q \quad (\alpha > 1), \tag{SG.39}$$

which is finite and positive, since $qJ(q) \le \min(q, 1)$ and $\alpha > 1$.

Proposition SG.8 (No-momentum endpoints)**.** *Let $\beta = 0$, let $\alpha > 0$ be fixed, and let $N, B \to \infty$ with $\gamma = N/B \to \infty$, $\Sigma_{\text{in}} = \mathrm{Id}_N$, $N_{\text{out}} = N_{\text{in}} = N$ and $\mu_i = i^{-\alpha}$. For $\gamma > 1$ let κ and $i_\# = \kappa^{1/\alpha}$ be as in proposition SG.6. At each such dimension the limits $\mathfrak{d}_i(0) \stackrel{\text{def}}{=} \lim_{\varrho \downarrow 0} \mathfrak{d}_i(\varrho)$ and $\mathfrak{v}_i(0) \stackrel{\text{def}}{=} \lim_{\varrho \downarrow 0} \mathfrak{v}_i(\varrho)$ exist, and:*

(i) for every i,

$$\mathfrak{v}_i(0) = \frac{\kappa\mu_i}{1+\kappa\mu_i} = \frac{1}{1+(i/i_\#)^\alpha}, \qquad \sum_{i=1}^N \mathfrak{v}_i(0) = B;$$

(ii) for every row sequence $1 \le i = i_N \le N$,

$$\frac{i}{i_\#} \to 0 \implies \mathfrak{d}_i(0) \sim \sqrt{\frac{\mu_i}{2\gamma}}, \qquad \frac{i}{i_\#} \to \infty \implies \mathfrak{d}_i(0) \sim A_\alpha \, \mu_i \sqrt{\frac{\kappa}{2\gamma}}.$$

Proof. The participation coordinate applies at $\beta = 0$. The computation of proposition SG.1(ii),(iii) uses only the equations of the branch, so with $m(s) \in (0,1)$ defined by $s_N(m(s)) = s$,

$$\mathfrak{v}_i^{(0)}(\varrho) = f_i\big(m(\varrho)\big), \qquad \mathfrak{d}_i(\varrho) = \frac{\sqrt{2}}{\pi\gamma} \int_0^\infty f_i\big(m(\varrho + u^2)\big) \,\mathrm{d}u.$$

The proof of proposition SG.1(i) shows that s_N is strictly decreasing on $(0,1)$; it tends to ∞ as $m \downarrow 0$, and to 0 as $m \uparrow 1$, because then $Q_m \to \infty$ while $\Lambda_N(m) \to \Lambda_N(1) = \kappa$ and $\gamma - m \to \gamma - 1 > 0$. Hence $m(s) \to 1$ as $s \downarrow 0$, and $\Lambda_N(m(s)) \le \kappa$ for all s.

Part (i) is the zero-lag coefficient. At $\beta = 0$ all lag coefficients with $\ell \ge 1$ vanish (theorem SD.7(iv)), so $\mathfrak{v}_i(\varrho) = \mathfrak{v}_i^{(0)}(\varrho) = f_i(m(\varrho)) \to \kappa\mu_i/(1+\kappa\mu_i)$, and $\kappa\mu_i = (i_\#/i)^\alpha$. The sum is $Bm_N(\kappa) = B$.

The drift has a zero-ridge limit. The function $t \mapsto f_i(m(t))$ is nonincreasing with values in $[0,1]$, and $f_i \le \Lambda_N \mu_i = \tilde{s}_1 \mu_i / t \le \gamma\mu_i/t$ because $\tilde{s}_1 = (\gamma - m)J(Q_m) \le \gamma$. The integration by parts in the proof of proposition SG.2(ii), with $L_\beta/F_\beta = 1$ and total variation at most one, gives

$$|\mathfrak{d}_i(\varrho) - \mathfrak{d}_i(0)| \le \frac{\sqrt{2}}{\pi\gamma}\sqrt{\varrho}, \qquad \mathfrak{d}_i(0) = \frac{\sqrt{2}}{\pi\gamma} \int_0^\infty f_i\big(m(u^2)\big) \,\mathrm{d}u.$$

The participation scale at small participation is a power of m. Put $\nu \stackrel{\text{def}}{=} 1$ for $\alpha \le 1$ and $\nu \stackrel{\text{def}}{=} \alpha$ for $\alpha > 1$. For fixed $m \in (0,1)$,

$$\frac{\Lambda_N(m)}{\kappa} \longrightarrow m^\nu, \qquad m\frac{\Lambda_N'(m)}{\Lambda_N(m)} \longrightarrow \nu. \tag{SG.40}$$

Indeed, by eq. (SG.42), $\Lambda_N(m)/\kappa = V_N(m/\gamma)/V_N(1/\gamma)$, and lemma SG.6(i) gives $S_\alpha(V_N(p)) = p(1 + O(1/(Bm)))$ at $p = m/\gamma$; the small-V laws of lemma SG.5(i),(iii) then give the first limit, and $VS_N'(V) = \mathsf{T}(V) + O(N^{-1})$ with $\mathsf{T}/S_\alpha \to 1/\nu$ at small V gives the second, as in the proof of lemma SG.6(iv). Moreover $\Lambda_N(m) \le m\kappa$ for $m \le 1$, because $\Lambda_N(m)/m$ is nondecreasing.

The resolved endpoint is the head computation of lemma SG.11. Let $i/i_\# \to 0$, so $\kappa\mu_i \to \infty$, and fix $\delta \in (0,1)$. By eq. (SG.40), $\Lambda_N(\delta)\mu_i \ge c_\delta \kappa \mu_i \to \infty$, so $s_N(\delta)/(\gamma\mu_i) \le C_\delta/(\Lambda_N(\delta)\mu_i) \to 0$. On $m \le \delta$, lemma SG.1(i) at $\beta = 0$ gives $\tilde{s}_1/\gamma = (1 - m/\gamma)J(Q_m) = 1 + O(\delta) + o(1)$, so that part of the integral is

$$\int_{\sqrt{s_N(\delta)}}^\infty \frac{\tilde{s}_1 \mu_i}{u^2 + \tilde{s}_1 \mu_i} \,\mathrm{d}u = \frac{\pi}{2}\sqrt{\gamma\mu_i}\,\big(1 + O(\delta) + o(1)\big),$$

and the part $u < \sqrt{s_N(\delta)}$ is at most $\sqrt{s_N(\delta)} = o(\sqrt{\gamma\mu_i})$. Multiplying by $\sqrt{2}/(\pi\gamma)$ and letting $\delta \downarrow 0$ gives $\mathfrak{d}_i(0) \sim \sqrt{\mu_i/(2\gamma)}$.

The unresolved endpoint is linear in μ_i. Let $i/i_\# \to \infty$, so $\kappa\mu_i \to 0$. Since $\Lambda_N(m(t)) \le \kappa$, $f_i = \Lambda_N \mu_i \,(1 + O(\kappa\mu_i))$ uniformly in t, and the substitution $u^2 = s_N(m)$ of lemma SG.8 gives

$$\frac{\mathfrak{d}_i(0)}{\mu_i\sqrt{\kappa/(2\gamma)}} = \big(1+o(1)\big)\,\frac{1}{\pi}\int_0^1 \sqrt{\frac{\Lambda_N(m)}{\kappa}}\sqrt{1-\frac{m}{\gamma}}\,\sqrt{\frac{m}{Q_m}}\;\vartheta_N(m)\,\mathrm{d}m.$$

The integrand converges for each $m \in (0,1)$, by eq. (SG.40), to $m^{(\nu+1)/2}Q_m^{-1/2}\big(1/E(m) + (\nu-1)/m\big)$. It is bounded by $m\,Q_m^{-1/2}\big(1/E(m) + C/m\big)$, using $\Lambda_N(m) \le m\kappa$ and lemma SG.7; this bound is $O(m^{-1/2})$ near 0, where $Q_m \asymp m \asymp E(m)$, and bounded near 1, where $E(m)\sqrt{Q_m}$ stays bounded below because $Q\,r_0'(Q) = \mathbb{E}\; Qg^2(1+Qg^2)^{-2} \ge c\,Q^{-1/2}$ for $Q \ge 1$ (the Gaussian density is at least c on $|g| \le 1$). The integrals therefore converge to the integral of the limit, which in the variable $q = Q_m$, with $m = r_0(q)$, $\mathrm{d}m = r_0'(q)\,\mathrm{d}q$ and $E = qr_0'(q)$, is

$$\frac{1}{\pi}\int_0^\infty \Big(r_0(q)^{\frac{\nu+1}{2}}\,q^{-3/2} + (\nu-1)\,r_0(q)^{\frac{\nu-1}{2}}\,r_0'(q)\,q^{-1/2}\Big)\mathrm{d}q.$$

For $\nu = 1$ this is $\pi^{-1}\int_0^\infty q^{-1/2}J(q)\,\mathrm{d}q = \sqrt{2\pi}/\pi = c_0$, by the computation in the proof of lemma SG.2. For $\nu = \alpha > 1$, the second term is $\frac{2(\alpha-1)}{\alpha+1}\int q^{-1/2}\,\mathrm{d}\big(r_0^{(\alpha+1)/2}\big) = \frac{\alpha-1}{\alpha+1}\int_0^\infty q^{-3/2}r_0^{(\alpha+1)/2}\,\mathrm{d}q$, by integration by parts [55, Theorem 6.22] on $[\epsilon, T]$, whose boundary terms vanish as $\epsilon \downarrow 0$, $T \to \infty$ because $r_0(q) \le \min(q,1)$; the total is $\frac{1}{\pi}\big(1 + \frac{\alpha-1}{\alpha+1}\big)\int_0^\infty q^{-3/2}r_0^{(\alpha+1)/2}\,\mathrm{d}q = A_\alpha$. □

In the variables of the main text's table, $\kappa\mu_i \to \infty$ and $\kappa\mu_i \to 0$ are the two sides of $i_\#$. The kernel $\mathfrak{d}_i$ carries the normalization of eq. (SC.4), so both endpoints carry the factor $1/\sqrt{2}$ relative to the half-moment $\gamma^{-1}\int\sqrt{x}\,\rho_i(\mathrm{d}x)$ of corollary SC.2. For $\alpha > 1$ the constant is A_α and not c_0.

Remark SG.2 (Why the reference scale is not the no-momentum threshold)**.** *For $\alpha \le 1$, along rows with $i/i_\# \to x \in (0,\infty)$, the momentum drift satisfies $\sqrt{\gamma\kappa}\,\sqrt{2}\,\mathfrak{d}_i(0) \to \infty$. Indeed $i/i_{\mathrm{c},\mathfrak{d}} \to \infty$ and $i/N \to 0$ by proposition SG.6, so theorem SG.2(i) gives $\sqrt{2}\,\mathfrak{d}_i(0) \sim d\,a_N \log\gamma$ for $\alpha < 1$ and $\sim d\,a_N(\log\gamma)^{3/2}$ for $\alpha = 1$, while the no-momentum normalization is $(\gamma\kappa)^{-1/2} \sim a_N/\sqrt{1-\alpha}$, respectively $a_N(\log\gamma)^{1/2}$, by eq. (SG.36). The extra logarithm comes from the growth $\log Q_m \sim 2|\log\beta|\,m$ of the momentum participation (lemma SG.1(iii)). A fixed effective batch size multiplies $i_\#$ and $(\gamma\kappa)^{-1/2}$ by fixed factors and cannot remove a logarithm, which is why the crossover indices of theorem SG.2 are not effective-batch versions of $i_\#$. For $\alpha > 1$ the logarithm is absent and the reference scale $i_{\mathrm{c}} = B/K_\alpha$ is asymptotic to $i_\#$.*

Remark SG.3 (Scope for the dynamics)**.** *The comparison of theorem SE.1 and of its row form corollary SE.1 holds at a fixed positive normalized ridge and in proportional dimensions, with the restrictions of remark SE.1. The profiles of this section concern zero (or negligible, proposition SG.7) ridge and a diverging aspect ratio, which are outside those hypotheses. The deterministic model of the phase-comparison section inserts the coefficient pair of theorem SG.1, with the exact covariance $\mu_i = c_N i^{-\alpha}$, into the row equations eq. (SE.13) as adopted coefficients; that insertion is a modeling assumption and not a consequence of appendix SE. For nonnegative aggregates such as the noise constant of that model, corollary SG.2 transfers the relative accuracy of theorem SG.1 to weighted sums over rows $i \ge n_N$; it makes no statement about the rows $i < n_N$ or about trajectories.*

SG.4 Power-law participation

From here on Assumption S4 holds. The participation function of a power law is a Riemann sum of the continuum function S_α of definition SG.3. Put

$$S_N(V) \overset{\text{def}}{=} \frac{1}{N}\sum_{j=1}^{N}\frac{V}{V+(j/N)^\alpha}, \qquad V_N \overset{\text{def}}{=} S_N^{-1}, \qquad \mathsf{T}(V) \overset{\text{def}}{=} V S_\alpha'(V) = \int_0^1 \frac{Vt^\alpha}{(V+t^\alpha)^2}\,\mathrm{d}t. \tag{SG.41}$$

With $V = c_N \Lambda N^{-\alpha}$ one has $\Lambda\mu_j = V(j/N)^{-\alpha}$, hence $m_N(\Lambda) = \gamma S_N(V)$ and

$$\Lambda_N(m) = \frac{N^\alpha}{c_N}\, V_N\Big(\frac{m}{\gamma}\Big) \qquad (0 < m < \gamma). \tag{SG.42}$$

Lemma SG.5 (Continuum participation)**.**

(i) As $V \downarrow 0$, $S_\alpha(V) \sim V/(1-\alpha)$ for $\alpha < 1$, $S_1(V) = V\log(1+1/V) \sim V\log(1/V)$, and $S_\alpha(V) \sim K_\alpha V^{1/\alpha}$ for $\alpha > 1$. As $V \to \infty$, $1 - S_\alpha(V) \sim ((\alpha+1)V)^{-1}$.

(ii) $\mathsf{T}(V) \asymp S_\alpha(V)\big(1 - S_\alpha(V)\big)$ for $V > 0$.

(iii) As $p \downarrow 0$, $V_\alpha(p) \sim (1-\alpha)p$ for $\alpha < 1$, $V_1(p) \sim p/\log(1/p)$, and $V_\alpha(p) \sim (p/K_\alpha)^\alpha$ for $\alpha > 1$; and $V_\alpha(p) \asymp (1-p)^{-1}$ for $p \geq 1/2$.

Proof. Part (i) is a scaling of the integral. For $\alpha < 1$, $V^{-1}S_\alpha(V) = \int_0^1 (V + t^\alpha)^{-1}\mathrm{d}t$ increases to $\int_0^1 t^{-\alpha}\mathrm{d}t = (1-\alpha)^{-1}$ as $V \downarrow 0$. For $\alpha = 1$ the integral is explicit. For $\alpha > 1$ the substitution $t = V^{1/\alpha}u$ gives

$$S_\alpha(V) = V^{1/\alpha}\int_0^{V^{-1/\alpha}} \frac{\mathrm{d}u}{1+u^\alpha}, \qquad \int_0^\infty \frac{\mathrm{d}u}{1+u^\alpha} = \frac{1}{\alpha}\int_0^\infty \frac{t^{1/\alpha - 1}}{1+t}\,\mathrm{d}t = \frac{\pi}{\alpha\sin(\pi/\alpha)} = K_\alpha,$$

by the Beta integral and the reflection formula [21, (5.12.3), (5.5.3)]. As $V \to \infty$, $1 - S_\alpha(V) = \int_0^1 t^\alpha (V + t^\alpha)^{-1}\mathrm{d}t$ and $V(1 - S_\alpha(V)) \to \int_0^1 t^\alpha \mathrm{d}t$.

Part (ii) compares the ratio at the two ends. The same substitutions give $\mathsf{T}/S_\alpha \to 1$ as $V \downarrow 0$ for $\alpha \leq 1$, $\mathsf{T}/S_\alpha \to 1/\alpha$ for $\alpha > 1$, and $\mathsf{T}/(1 - S_\alpha) \to 1$ as $V \to \infty$. Both T and $S_\alpha(1 - S_\alpha)$ are continuous and positive on $(0, \infty)$, so their ratio is bounded above and below.

Part (iii) inverts (i). □

Lemma SG.6 (Finite participation)**.**

(i) $0 \leq S_\alpha(V) - S_N(V) \leq 1/N$ and $|VS'_N(V) - \mathsf{T}(V)| \leq 1/(2N)$ for $V > 0$.

(ii) There is C such that, for $M \geq 1$, $B \geq 2$ and $M/\gamma \leq p \leq 1 - M/\gamma$,

$$\Big|\log\frac{V_N(p)}{V_\alpha(p)}\Big| \leq \frac{C}{BM}.$$

(iii) $V_N(p) \asymp (1-p)^{-1}$ for $p \geq 1/2$, uniformly in N.

(iv) If $\alpha > 1$, $Bm \to \infty$ and $m/\gamma \to 0$, then $c_N\Lambda_N(m) \sim (Bm/K_\alpha)^\alpha$ and $m\Lambda'_N(m)/\Lambda_N(m) \to \alpha$, locally uniformly in m on compact subsets of $(0,\infty)$ when m is held fixed.

(v) If $\alpha > 1$, then $c_N\Lambda_N(m) \asymp (Bm)^\alpha$ for $1/B \leq m \leq \gamma/2$ and $c_N\Lambda_N(m) \asymp Bm$ for $0 < m \leq 1/B$.

Proof. Part (i) is a Riemann-sum comparison. The function $t \mapsto V(V + t^\alpha)^{-1}$ decreases from 1 on $[0,1]$, so $S_N(V)$ is a lower Riemann sum of $S_\alpha(V)$ that differs from the upper one by at most $1/N$. The function $t \mapsto Vt^\alpha(V + t^\alpha)^{-2}$ vanishes at 0, increases to its maximum $1/4$ and then decreases, so its total variation is at most $1/2$, and its Riemann sum differs from its integral by at most $1/(2N)$.

Part (ii) follows from (i) and lemma SG.5(ii). By (i), $S_\alpha(V_N(p)) \in [p, p + 1/N]$, so $V_N(p) = V_\alpha(p')$ with $p \leq p' \leq p + 1/N$, and

$$\log\frac{V_\alpha(p')}{V_\alpha(p)} = \int_p^{p'} \frac{\mathrm{d}p''}{\mathsf{T}(V_\alpha(p''))} \leq \frac{1}{N}\sup_{p''\in[p,p']} \frac{C}{p''(1-p'')}.$$

On the stated range, $p'' \geq M/\gamma$ and $1 - p'' \geq M/\gamma - 1/N \geq M/(2\gamma)$, so $p''(1-p'') \geq M/(4\gamma)$, and the right side is at most $4C\gamma/(NM) = 4C/(BM)$.

Part (iii) bounds the deficit. Since $1 - S_N(V) = N^{-1}\sum_j (j/N)^\alpha (V + (j/N)^\alpha)^{-1}$ and $N^{-1}\sum_j (j/N)^\alpha$ is bounded above and below, $1 - S_N(V) \asymp \min(1, 1/V)$ for V bounded below, which inverts to the claim.

Part (iv) inverts (i) at small participation. With $p = m/\gamma$, $S_\alpha(V_N(p)) = p(1 + O(1/(Np))) = p(1 + O(1/(Bm)))$, so $V_N(p) \sim (p/K_\alpha)^\alpha$ by lemma SG.5(iii), and eq. (SG.42) gives $c_N \Lambda_N(m) \sim (Np/K_\alpha)^\alpha = (Bm/K_\alpha)^\alpha$. For the derivative, $\Lambda'_N/\Lambda_N = \gamma^{-1}/(V S'_N(V))$ at $V = V_N(p)$, and by (i) and lemma SG.5(ii) at small V, $V S'_N(V) = \mathsf{T}(V) + O(N^{-1}) = \alpha^{-1} p\,(1 + o(1))$, so $m\Lambda'_N(m)/\Lambda_N(m) \to \alpha$.

Part (v) compares the sum $Bm_N(\Lambda) = \sum_j n^\alpha (n^\alpha + j^\alpha)^{-1}$, $n = (c_N\Lambda)^{1/\alpha}$, with integrals. If $n \le 1$, comparison with $\sum_j j^{-\alpha} < \infty$ gives $Bm_N \asymp n^\alpha$. If $1 \le n \le N$, the terms $j \le n$ give at least $\lfloor n \rfloor/2$, and splitting the sum at n and comparing the terms $j > n$ with $\int_n^\infty n^\alpha t^{-\alpha} \mathrm{d}t = n/(\alpha - 1)$ gives at most $C_\alpha n$. The case $n > N$ does not occur when $m \le \gamma/2$, because then every term would exceed $1/2$. Since $Bm_N = Bm$, these give $c_N\Lambda \asymp Bm$ for $Bm \le 1$ and $c_N\Lambda = n^\alpha \asymp (Bm)^\alpha$ for $Bm \ge 1$, the bounded transition region included. □

Lemma SG.7 (Elasticity)**.** *There is $c_\alpha > 0$, independent of N, B, c_N and Λ, such that*

$$c_\alpha\, m_N(\Lambda)\Big(1 - \frac{m_N(\Lambda)}{\gamma}\Big) \le \Lambda m'_N(\Lambda) \le m_N(\Lambda), \qquad \frac{1}{m} \le \frac{\Lambda'_N(m)}{\Lambda_N(m)} \le \frac{1}{c_\alpha}\Big(\frac{1}{m} + \frac{1}{\gamma - m}\Big).$$

Consequently $\vartheta_N(m) \le C\big(1 + m^{-1} + (\gamma - m)^{-1}\big)$, and $E(m)\vartheta_N(m) \ge 1$.

Proof. Write $n \stackrel{\text{def}}{=} (c_N\Lambda)^{1/\alpha}$ and $u_j \stackrel{\text{def}}{=} (n/j)^\alpha = \Lambda\mu_j$, so that

$$Bm_N(\Lambda) = \sum_j \frac{u_j}{1 + u_j}, \qquad B\Lambda m'_N(\Lambda) = \sum_j \frac{u_j}{(1 + u_j)^2}.$$

The upper bound holds term by term. For the lower bound put

$$S \stackrel{\text{def}}{=} \sum_j \frac{u_j}{1 + u_j} = Bm_N, \qquad T \stackrel{\text{def}}{=} \sum_j \frac{1}{1 + u_j} = N - S, \qquad F \stackrel{\text{def}}{=} \sum_j \frac{u_j}{(1 + u_j)^2} = B\Lambda m'_N,$$

so that $ST/N = Bm_N(1 - m_N/\gamma)$ and the lower bound reads $c_\alpha ST/N \le F$. Let $K \stackrel{\text{def}}{=} \#\{j : u_j > 1\}$. Since u_j decreases in j, these are the indices $j \le K$. A term with $u_j \le 1$ has $u_j/(1 + u_j) \le 2u_j/(1 + u_j)^2$, and a term with $u_j \ge 1$ has $1/(1 + u_j) \le 2u_j/(1 + u_j)^2$. Bounding the remaining terms by one gives

$$S \le 2F + K, \qquad T \le 2F + N - K.$$

Since $S, T \le N$, it follows that $ST/N \le \min(S, T) \le 2F + \min(K, N - K)$.

A window count bounds $\min(K, N - K)$ by F. Put $\delta \stackrel{\text{def}}{=} 2^\alpha/(1 + 2^\alpha)^2$. For $2^{-\alpha} \le u \le 2^\alpha$,

$$\frac{u}{(1 + u)^2} - \delta = \frac{(2^\alpha - u)(2^\alpha u - 1)}{(1 + u)^2(1 + 2^\alpha)^2} \ge 0,$$

so $F \ge \delta\,\#\{j : 2^{-\alpha} \le u_j \le 2^\alpha\}$. If $2K \le N$, then $u_{K+1} \le 1$ gives $n \le K + 1$. The at least $K/2$ indices $(K + 1)/2 \le j \le K$ then have $1 < u_j \le ((K + 1)/j)^\alpha \le 2^\alpha$, so $K \le 2F/\delta$. If $2K > N$, then $u_K > 1$ gives $n > K$. The $N - K$ indices $K < j \le N$ then have $2^{-\alpha} < (K/j)^\alpha < u_j \le 1$, because $j \le N < 2K$, so $N - K \le F/\delta$. In both cases $\min(K, N - K) \le 2F/\delta$, and therefore

$$\frac{ST}{N} \le \Big(2 + \frac{2}{\delta}\Big)F.$$

This is the lower bound with $c_\alpha \stackrel{\text{def}}{=} \delta/(2\delta + 2)$.

Inverting gives the second display, whose lower bound is the monotonicity of $\Lambda_N(m)/m$ shown in the proof of proposition SG.1; with $1/(m(1 - m/\gamma)) = 1/m + 1/(\gamma - m)$. The bound on ϑ_N follows with $E(m) \asymp \min(m, 1)$ (lemma SG.1(iii)), and $E\vartheta_N \ge 1$ because $\Lambda'_N/\Lambda_N \ge 1/m$. □

Proof of proposition SG.6. At $m = 1$, eq. (SG.42) gives $V \overset{\text{def}}{=} c_N \kappa N^{-\alpha} = V_N(1/\gamma)$, and lemma SG.6(i) gives $S_\alpha(V) \in [\gamma^{-1}, \gamma^{-1} + N^{-1}] = \gamma^{-1}[1, 1 + B^{-1}]$. Hence $S_\alpha(V) \sim 1/\gamma \to 0$, and lemma SG.5(iii) gives

$$c_N \kappa = N^\alpha V \sim N^\alpha \cdot \begin{cases} (1-\alpha)\gamma^{-1}, \\ (\gamma \log \gamma)^{-1}, \\ (K_\alpha \gamma)^{-\alpha}, \end{cases}$$

which is the first equivalence of eq. (SG.36). The ratios follow from eqs. (SG.32) and (SG.33) with $d^2 = c_0^2/(1-\alpha)$ for $\alpha < 1$ and $d^2 = \frac{4}{9} c_0^2 = 8/(9\pi)$ for $\alpha = 1$:

$$\frac{i_{\mathrm{c},\mathfrak{d}}^\alpha}{c_N \kappa} \sim \frac{B N^{\alpha-1}}{s_2 d^2 \log^2 \gamma \cdot (1-\alpha) B N^{\alpha-1}} = \frac{\pi}{2 s_2 \log^2 \gamma} \quad (\alpha < 1),$$

$$\frac{i_{\mathrm{c},\mathfrak{d}}}{i_\#} \sim \frac{B/(s_2 d^2 \log^3 \gamma)}{B/\log \gamma} = \frac{9\pi}{8 s_2 \log^2 \gamma} \quad (\alpha = 1),$$

$i_{\mathrm{c}} = B/K_\alpha \sim i_\#$ for $\alpha > 1$, and $i_{\mathrm{c},\mathfrak{v}} = c_0^{2/\alpha} i_{\mathrm{c},\mathfrak{d}}$ gives the last statement. □

SG.5 Proof of the drift statements

The drift at zero ridge is first written in the participation coordinate. Put

$$\mathcal{B}_i(m) \overset{\text{def}}{=} \sqrt{\frac{\gamma - m}{m \, \Lambda_N(m)}} \, f_i(m) \qquad (0 < m < \gamma). \tag{SG.43}$$

Lemma SG.8 (Drift in the participation coordinate)**.** *For every finite positive spectrum,*

$$\sqrt{2}\, \mathfrak{d}_i(0) = \frac{1}{\pi\gamma} \int_0^\gamma \mathcal{A}(m)\, \mathcal{B}_i(m)\, E(m) \vartheta_N(m) \, \mathrm{d}m. \tag{SG.44}$$

Under Assumption S4, with $x = i/N$ and $m = \gamma p$,

$$\mathcal{B}_i(\gamma p) = a_N \sqrt{\frac{1-p}{p\, V_N(p)}} \, \frac{V_N(p)}{V_N(p) + x^\alpha}. \tag{SG.45}$$

Proof. In eq. (SG.12) substitute $u^2 = s_N(m)$. By proposition SG.1(i), $\mathrm{d}u = \frac{1}{2}\sqrt{s_N(m)}\, \vartheta_N(m)\, \mathrm{d}m$ in absolute value, and with $F_\beta(Q_m) = m/Q_m$,

$$\frac{L_\beta(Q_m)}{F_\beta(Q_m)} \sqrt{s_N(m)} = \frac{Q_m L_\beta(Q_m)}{m} \sqrt{\frac{(\gamma - m)m}{Q_m \Lambda_N(m)}} = \sqrt{Q_m}\, L_\beta(Q_m) \sqrt{\frac{\gamma - m}{m \Lambda_N(m)}},$$

which gives eq. (SG.44), since $\sqrt{Q_m} L_\beta(Q_m) = \mathcal{A}(m) E(m)$. For the second identity, eq. (SG.42) gives $\Lambda_N \mu_i = V_N(p) x^{-\alpha}$, so

$$\frac{f_i}{\sqrt{\Lambda_N}} = \frac{\sqrt{\Lambda_N}\, \mu_i}{1 + \Lambda_N \mu_i} = \frac{\sqrt{c_N} N^{\alpha/2} \sqrt{V_N}\, i^{-\alpha}}{1 + V_N x^{-\alpha}} = a_N \frac{\sqrt{V_N}}{V_N + x^\alpha},$$

and $\sqrt{(\gamma - m)/m} = \sqrt{(1-p)/p}$. □

By eq. (SG.45) and $\sqrt{2\pi}/\pi = c_0$, $(\pi\gamma)^{-1}\sqrt{2\pi} \int_{\eta\gamma}^\gamma \mathcal{B}_i \, \mathrm{d}m$ is the profile Ψ_α with V_N in place of V_α and the integral restricted to $p \geq \eta$. For $0 < \eta < 1$ we write

$$\Psi_\alpha^{[\eta]}(x) \overset{\text{def}}{=} c_0 \int_\eta^1 \sqrt{\frac{1-p}{p V_\alpha(p)}} \, f_x(p) \, \mathrm{d}p,$$

and, for $M > 0$, $\mathfrak{d}_i^{>M}$ and $\mathfrak{d}_i^{\leq M}$ for the parts of eq. (SG.44) over $m > M$ and $m \leq M$, divided by $\sqrt{2}$.

Lemma SG.9 (The profile Ψ_α)**.** *Ψ_α is finite, positive and continuous on $(0,1]$, bounded below by a positive constant, and satisfies the first equivalences of eq.* (SG.27)*.*

Proof. The bounds of lemma SG.5(iii) control the integrand $\psi_x(p) \overset{\text{def}}{=} \sqrt{(1-p)/(pV_\alpha)}\, V_\alpha/(V_\alpha + x^\alpha)$. Near $p=1$, $V_\alpha \asymp (1-p)^{-1}$ gives $\psi_x \le C(1-p)$. Near $p=0$, $\psi_x \le x^{-\alpha}\sqrt{V_\alpha/p}$, which is bounded for $\alpha \le 1$ up to a logarithm and is $O(p^{(\alpha-1)/2})$ for $\alpha > 1$. These bounds are locally uniform in x, which gives finiteness and continuity. On $p \in [1/4, 1/2]$, $\psi_x \ge c$ uniformly in $x \le 1$, which gives the lower bound.

The small-x equivalents come from small p. For $\alpha < 1$, $V_\alpha(p) \sim (1-\alpha)p$, so on a fixed small interval $(0, p_0)$

$$\psi_x(p) = \frac{1+o(1)}{\sqrt{1-\alpha}} \cdot \frac{1}{p + x^\alpha/(1-\alpha)}, \qquad \int_0^{p_0} \frac{\mathrm{d}p}{p + x^\alpha/(1-\alpha)} = \log\left(x^{-\alpha}\right)(1+o(1)),$$

the error $o(1)$ being uniform as $p_0 \downarrow 0$; the part $p > p_0$ is bounded. For $\alpha = 1$, $V_1(p) \sim p/\log(1/p)$ gives $\psi_x(p) = (1+o(1))\sqrt{\log(1/p)}\,/\,(p + x\log(1/p))$. Put $p_x \overset{\text{def}}{=} x\log(1/x)$ and $\omega \overset{\text{def}}{=} \log(1/x)$. On $p \ge \omega p_x$, $x\log(1/p)/p \le 1/\omega \to 0$, so the integrand is $(1+o(1))\sqrt{\log(1/p)}/p$, with integral $\frac{2}{3}[\log(1/x)]^{3/2}(1+o(1))$ because $\log\omega = o(\log(1/x))$. On $[p_x, \omega p_x]$ the integrand is at most $\sqrt{\log(1/p)}/p$, with integral at most $\sqrt{\log(1/p_x)}\,\log\omega = o([\log(1/x)]^{3/2})$. Below p_x, the substitution $p = p_x u$ bounds the contribution by $C\sqrt{\log(1/x)}\int_0^1 u^{-1/2}(1+|\log u|)^{1/2}\mathrm{d}u$. For $\alpha > 1$, substitute $p = K_\alpha x u$ on a fixed small interval. Since $V_\alpha(K_\alpha x u) \sim x^\alpha u^\alpha$,

$$\psi_x(K_\alpha x u)\, K_\alpha x\, \mathrm{d}u = (1+o(1))\sqrt{K_\alpha}\, x^{-(\alpha-1)/2}\, \frac{u^{(\alpha-1)/2}}{1+u^\alpha}\, \mathrm{d}u,$$

$$\int_0^\infty \frac{u^{(\alpha-1)/2}}{1+u^\alpha}\, \mathrm{d}u = \frac{\pi}{\alpha\cos(\pi/(2\alpha))},$$

by [21, (5.12.3), (5.5.3)], and the tails are controlled by the bounds $Cu^{(\alpha-1)/2}$ near 0 and $Cu^{-(\alpha+1)/2}$ near ∞, the latter integrable because $\alpha > 1$. Multiplying by c_0 gives $d_\alpha\sqrt{K_\alpha}x^{-(\alpha-1)/2}$. □

Lemma SG.10 (Averaged tail)**.** *There is C such that, for every fixed $M \ge M_0(\alpha,\beta)$,*

$$\limsup_{N\to\infty} \sup_{1\le i\le N} \left| \frac{\sqrt{2}\,\mathfrak{d}_i^{>M}}{a_N \Psi_\alpha^{[M/\gamma]}(i/N)} - 1 \right| \le \frac{C}{M}.$$

Proof. The tail integrand is $\mathcal{A}\mathcal{B}_i \cdot E\vartheta_N$, and the factor $E\vartheta_N$ is close to one inside $[M, \gamma - M]$. Indeed

$$E(m)\vartheta_N(m) - 1 = E(m)\Big(\frac{\Lambda_N'(m)}{\Lambda_N(m)} - \frac{1}{m} + \frac{1}{\gamma - m}\Big) \ge 0,$$

and lemmas SG.1 and SG.7(iii) bound it by $C(m^{-1} + (\gamma-m)^{-1}) \le 2C/M$ there. The function $\mathcal{B}_i$ varies slowly there:

$$\frac{\mathcal{B}_i'(m)}{\mathcal{B}_i(m)} = -\frac{1}{2(\gamma-m)} - \frac{1}{2m} + \Big(\frac{1}{2} - f_i(m)\Big)\frac{\Lambda_N'(m)}{\Lambda_N(m)}, \qquad \Big|\frac{\mathcal{B}_i'}{\mathcal{B}_i}\Big| \le C\Big(\frac{1}{m} + \frac{1}{\gamma-m}\Big),$$

uniformly in i.

The averaging error is controlled by lemma SG.2 with $W = \mathcal{B}_i$ on $[M, \gamma - M]$ and $m_0 = 1$. It is at most

$$C\Big(\mathcal{B}_i(M) + \mathcal{B}_i(\gamma - M) + \int_M^{\gamma - M} \mathcal{B}_i(m)\Big(\frac{1}{m} + \frac{1}{\gamma - m}\Big)\mathrm{d}m\Big).$$

The logarithmic-derivative bound gives $\mathcal{B}_i(M) \le (C/M)\int_M^{2M} \mathcal{B}_i$, and the integral term is at most $(2C/M)\int_M^\gamma \mathcal{B}_i$. Near the top, eq. (SG.45) and lemma SG.6(iii) give

$$\mathcal{B}_i(m) \le C\, a_N\, \frac{\gamma - m}{\gamma} \qquad (\gamma - M < m < \gamma),$$

uniformly in i, and $E\vartheta_N \le C(1+(\gamma-m)^{-1})$ there, so the boundary value $\mathcal{B}_i(\gamma-M)$ and the whole part $m>\gamma-M$ of the integral contribute $O(a_N(M+M^2)/\gamma^2)$ after division by $\pi\gamma$. On the other hand, the part $\gamma/4<m<\gamma/2$ gives

$$\frac{\sqrt{2\pi}}{\pi\gamma}\int_{\gamma/4}^{\gamma/2}\mathcal{B}_i\,\mathrm{d}m \ge c\,a_N,$$

because $V_N(p)/(V_N(p)+x^\alpha)\ge V_N(1/4)/(V_N(1/4)+1)$ there. Hence replacing $E\vartheta_N$ by one and $\mathcal{A}$ by $\sqrt{2\pi}$ changes the tail by a relative error at most $C/M+o(1)$, and the tail becomes a_N times the profile with V_N in place of V_α and the cutoff $p\ge M/\gamma$.

The replacement of V_N by V_α is uniform in the row. The integrand of the profile depends on V through $\sqrt{V}/(V+x^\alpha)$, whose logarithmic derivative in $\log V$ lies in $[-\frac12,\frac12]$, so lemma SG.6(ii) changes it by a factor $1+O(1/(BM))$ on $[M/\gamma,1-M/\gamma]$; the interval $p>1-M/\gamma$ is negligible for both functions by the bound near the top. □

Lemma SG.11 (Head). *For every fixed $M>0$ and all large N, $\sqrt{2}\,\mathfrak{d}_i^{\le M}\le C_M\sqrt{\mu_i/\gamma}$ for every i. If a row sequence satisfies $m_N(1/\mu_i)\to0$, then $\sqrt{2}\,\mathfrak{d}_i^{\le M}\sim\sqrt{\mu_i/(\gamma s_2)}$.*

Proof. In the variable u of eq. (SG.12), the head is the part where the participation is at most M, and there $L_\beta/F_\beta\le C_M$ and, by eq. (SG.6),

$$c_M\gamma\le\tilde s_1=(\gamma-m)F_\beta(Q_m)\le s_2\gamma\qquad(\gamma>2M).$$

Hence the integrand is at most $C_M\gamma\mu_i/(u^2+c_M\gamma\mu_i)$, whose integral is $C'_M\sqrt{\gamma\mu_i}$, and the factor $\sqrt{2}\cdot\sqrt{2}/(\pi\gamma)$ gives the bound.

Assume now $m_N(1/\mu_i)\to0$ and fix $\delta\in(0,M)$. Then $\Lambda_N(\delta)\mu_i\to\infty$: otherwise, along a subsequence $\Lambda_N(\delta)\mu_i\le L_0$ with $L_0\ge1$, and since $m_N(\Lambda/L_0)\ge m_N(\Lambda)/L_0$ by the concavity of m_N,

$$m_N(1/\mu_i)\ge m_N\big(\Lambda_N(\delta)/L_0\big)\ge\delta/L_0.$$

Consequently $s_N(\delta)/(\gamma\mu_i)\le C_\delta/(\Lambda_N(\delta)\mu_i)\to0$. On the range $m\le\delta$, which is $u^2\ge s_N(\delta)$, lemma SG.1(i) gives

$$\frac{\tilde s_1}{\gamma s_2}=1+O(\delta)+o(1),\qquad\frac{L_\beta}{F_\beta}=\frac{1+O(\delta)}{s_2},$$

so that part of the integral is

$$\frac{1+O(\delta)}{s_2}\int_{\sqrt{s_N(\delta)}}^{\infty}\frac{\gamma s_2\mu_i(1+O(\delta))}{u^2+\gamma s_2\mu_i(1+O(\delta))}\,\mathrm{d}u=\frac{\pi}{2}\sqrt{\frac{\gamma\mu_i}{s_2}}\big(1+O(\delta)+o(1)\big),$$

the missing interval $u<\sqrt{s_N(\delta)}$ having relative size $O(\sqrt{s_N(\delta)/(\gamma\mu_i)})=o(1)$. The range $\delta<m\le M$ has $u\le\sqrt{s_N(\delta)}$ and a bounded integrand, so it contributes $o(\sqrt{\gamma\mu_i})$. Multiplying by $2/(\pi\gamma)$ and letting $\delta\downarrow0$ proves the claim. □

Lemma SG.12 (Cutoff estimates). *Let $0<\alpha\le1$, $c_N=1$, and let $i\to\infty$ with $x=i/N\to0$. Put $\nu\overset{\text{def}}{=}1$ for $\alpha<1$ and $\nu\overset{\text{def}}{=}3/2$ for $\alpha=1$, and*

$$T\overset{\text{def}}{=}\log(x^{-\alpha})\ (\alpha<1),\quad T\overset{\text{def}}{=}\log(1/x)\ (\alpha=1),\qquad W\overset{\text{def}}{=}\frac{\sqrt{\mu_i/(\gamma s_2)}}{a_N}=s_2^{-1/2}e^{(T-\log\gamma)/2}.$$

Then $\Psi_\alpha(x)\sim dT^\nu$ with d of eq. (SG.32), and for fixed M:

(i) if $W/(\log\gamma)^\nu\to z\in(0,\infty)$, then $\Psi_\alpha^{[M/\gamma]}(x)\sim d(\log\gamma)^\nu\sim\Psi_\alpha(x)$;

(ii) if $W/(\log\gamma)^\nu\to\infty$, then $\Psi_\alpha(x)+\Psi_\alpha^{[M/\gamma]}(x)=o(W)$;

(iii) if $T\le\log\gamma+C_1\log\log\gamma$ for a constant C_1, then $\Psi_\alpha^{[M/\gamma]}(x)\sim\Psi_\alpha(x)$; this applies when $W/(\log\gamma)^\nu\to0$, and then $W=o(\Psi_\alpha(x))$.

Proof. The equivalence $\Psi_\alpha(x) \sim dT^\nu$ is lemma SG.9. By lemma SG.5(iii), on a fixed small interval $0 < p < p_0$ the integrand of Ψ_α, without the factor c_0, is comparable, uniformly in $0 < x < p_0$, to

$$\frac{1}{p + x^\alpha} \quad (\alpha < 1), \qquad \frac{\sqrt{\mathcal{L}(p)}}{p + x\mathcal{L}(p)} \quad (\alpha = 1), \qquad \mathcal{L}(p) \stackrel{\text{def}}{=} 1 + \log(1/p),$$

and above p_0 all integrals are bounded uniformly in x.

For (i), $W \asymp (\log\gamma)^\nu$ gives $T = \log\gamma + 2\nu\log\log\gamma + O(1)$. Then x^α, and for $\alpha = 1$ also $x\mathcal{L}(p)$ for $p \geq M/\gamma$, is $o(M/\gamma)$, so on $p \geq M/\gamma$ the integrand of Ψ_α is that of Ψ_α at $x = 0$ to leading order, and its integral from M/γ is $d(\log\gamma)^\nu(1 + o(1))$, as in the proof of lemma SG.9. Since $T \sim \log\gamma$, this is also $\Psi_\alpha(x)(1 + o(1))$.

For (ii), $T = \log\gamma + 2\log W + \log s_2$, so $T^\nu \leq C((\log\gamma)^\nu + (\log W)^\nu) = o(W)$, and $\Psi_\alpha^{[M/\gamma]} \leq \Psi_\alpha \leq C(1 + T^\nu)$.

For (iii), let $R_{\text{rm}} \stackrel{\text{def}}{=} \Psi_\alpha(x) - \Psi_\alpha^{[M/\gamma]}(x)$ be the removed part, with $\eta \stackrel{\text{def}}{=} M/\gamma$. For $\alpha < 1$ the majorant gives

$$R_{\text{rm}} \leq C\log\left(1 + \eta x^{-\alpha}\right) = C\log\left(1 + Me^{T - \log\gamma}\right).$$

For $\alpha = 1$ put $s_x \stackrel{\text{def}}{=} x(1 + T)$, which is comparable to the solution of $p = x\mathcal{L}(p)$. If $\eta \leq s_x$, the bound $\sqrt{\mathcal{L}}/(x\mathcal{L}) \leq 1/(x\sqrt{\mathcal{L}(\eta)})$ gives $R_{\text{rm}} \leq C\eta/(x\sqrt{\mathcal{L}(\eta)}) \leq C(1 + T)/\sqrt{\mathcal{L}(\eta)}$. If $\eta > s_x$, the part below s_x is at most $C\sqrt{1 + T}$, and the rest is at most $C[\mathcal{L}(s_x)^{3/2} - \mathcal{L}(\eta)^{3/2}]$. Two cases remain. If $T < \frac{1}{2}\log\gamma$, then $\Psi_\alpha(x) \geq c$ and the bounds give $R_{\text{rm}} \leq CM\gamma^{-1/2}$. If $T \geq \frac{1}{2}\log\gamma$, the hypothesis gives $T \asymp \log\gamma$ and $T - \log\gamma \leq C_1\log\log\gamma$, and the bounds give $R_{\text{rm}} = O_M(1 + \log\log\gamma)$ for $\alpha < 1$ and $R_{\text{rm}} = O_M(\sqrt{\log\gamma}\,(1 + \log\log\gamma))$ for $\alpha = 1$, using $\mathcal{L}(s_x) - \mathcal{L}(\eta) \leq T - \log\gamma + \log M$. Both are $o(T^\nu)$. Finally, if $W/(\log\gamma)^\nu \to 0$, then $e^{(T - \log\gamma)/2} = o((\log\gamma)^\nu)$, which is the hypothesis with $C_1 = 2\nu$, and $W = o((\log\gamma)^\nu)$ is $o(\Psi_\alpha(x))$ when $T \geq \frac{1}{2}\log\gamma$, while $W \leq Ce^{-\log\gamma/4} \to 0$ and $\Psi_\alpha(x) \geq c$ when $T < \frac{1}{2}\log\gamma$. □

Proof of the drift part of theorem SG.1 and of theorem SG.2(i). By lemma SG.4, $\mathfrak{d}_i(0)$ and $\widehat{\mathfrak{d}}_i$ both scale by $\sqrt{c_N}$ when the amplitude is changed, and the indices of theorem SG.2 do not depend on c_N; so we take $c_N = 1$. Let $\alpha \leq 1$. If the supremum in eq. (SG.31) for the drift did not tend to zero, there would be a row sequence $i \geq n_N$, hence $i \to \infty$, along which the ratio stays away from one; so it suffices to prove $\mathfrak{d}_i(0) \sim \widehat{\mathfrak{d}}_i$ along every row sequence $i \to \infty$, and we may pass to subsequences.

Suppose $x = i/N \to x_0 \in (0, 1]$. The head is $O_M(\sqrt{\mu_i/\gamma}) = o(a_N)$ by lemma SG.11, the tail is $a_N\Psi_\alpha^{[M/\gamma]}(x)(1 + O(1/M) + o(1))$ by lemma SG.10, and $\Psi_\alpha^{[M/\gamma]}(x) \to \Psi_\alpha(x_0) > 0$. Letting $M \to \infty$ gives $\sqrt{2}\,\mathfrak{d}_i(0) \sim a_N\Psi_\alpha(x) \sim \sqrt{2}\,\widehat{\mathfrak{d}}_i$.

Suppose $x \to 0$, and pass to a subsequence along which $W/(\log\gamma)^\nu \to z \in [0, \infty]$, in the notation of lemma SG.12; note $\sqrt{2}\,\widehat{\mathfrak{d}}_i = a_N(W + \Psi_\alpha(x))$. If $z = 0$, the head is $O_M(a_N W) = o(a_N\Psi_\alpha(x))$, and the tail is $a_N\Psi_\alpha(x)(1 + O(1/M) + o(1))$ by lemma SG.10 and lemma SG.12(iii). If $z > 0$, then $m_N(1/\mu_i) \to 0$. Indeed $m_N(1/\mu_i) = \gamma S_N(x^\alpha) \leq \gamma S_\alpha(x^\alpha)$, which is $\asymp \gamma x^\alpha = 1/(s_2 W^2)$ for $\alpha < 1$ and $\sim \gamma x\log(1/x) = T/(s_2 W^2)$ for $\alpha = 1$, and both tend to zero. Hence the head is $a_N W(1 + o(1))$ by lemma SG.11, and the tail is $a_N\Psi_\alpha(x)(1 + O(1/M) + o(1))$ if $z < \infty$ and $o(a_N W)$ if $z = \infty$, by lemma SG.10 and lemma SG.12(i),(ii). In all cases $\sqrt{2}\,\mathfrak{d}_i(0) = a_N(W + \Psi_\alpha(x))(1 + O(1/M) + o(1))$, and $M \to \infty$ concludes.

For theorem SG.2(i), the definitions eq. (SG.32) give, at $i/i_{\text{c},\mathfrak{d}} \to x_0 \in (0, \infty)$,

$$W = d(\log\gamma)^\nu x_0^{-\alpha/2}(1 + o(1)), \qquad T = \log\gamma + O(\log\log\gamma),$$

so $\Psi_\alpha(x) \sim d(\log\gamma)^\nu$ and $\sqrt{2}\,\mathfrak{d}_i(0)/a_{\mathfrak{d}} \to 1 + x_0^{-\alpha/2}$. If the ratio tends to zero, $W/(\log\gamma)^\nu \to \infty$ and lemma SG.12(ii) makes the head term dominant. If it tends to infinity, $W/(\log\gamma)^\nu \to 0$ along the mesoscopic rows, lemma SG.12(iii) makes the tail term dominant, and for bulk rows the first case of this proof applies; lemma SG.9 gives the explicit forms. □

For $\alpha > 1$ the crossover is at the reference index $i_{\text{c}} = B/K_\alpha$, and its profile keeps the momentum.

Lemma SG.13 (The supercritical drift profile). *Let $\alpha > 1$ and $p_\alpha \stackrel{\text{def}}{=} (\alpha-1)/(2\alpha)$. The function $\Phi_{\alpha,\beta}$ of eq. (SG.25) is finite, positive and continuous on $(0,\infty)$, and*

$$\Phi_{\alpha,\beta}(y) \sim \sqrt{y/s_2} \quad (y \to \infty), \qquad \Phi_{\alpha,\beta}(y) \sim d_\alpha y^{p_\alpha} \quad (y \downarrow 0).$$

Under Assumption S4, for every row sequence $i \to \infty$ with $i/N \to 0$,

$$\frac{\sqrt{2}\,\mathfrak{d}_i(0)}{b_N} \sim \Phi_{\alpha,\beta}\big((i_{\mathrm{c}}/i)^\alpha\big).$$

Proof. The profile has an exact single-integral form. Put $t_\infty(m) \stackrel{\text{def}}{=} m^{1-\alpha}/Q_m$ for $m > 0$. Since $\frac{\mathrm{d}}{\mathrm{d}m}\log t_\infty = (1-\alpha)/m - 1/E(m) \le -\alpha/m$ by $E(m) \le m$, the map t_∞ decreases from ∞ to 0, with $-\frac{\mathrm{d}}{\mathrm{d}m}\log t_\infty = \vartheta_\infty(m)$. Writing $m(t)$ for its inverse, the computation of lemma SG.8 with $\Lambda = m^\alpha$ and $\gamma - m$ replaced by γ gives

$$\begin{aligned}\Phi_{\alpha,\beta}(y) &= \frac{2}{\pi}\int_0^\infty \frac{L_\beta(Q_{m(u^2)})}{F_\beta(Q_{m(u^2)})}\,\frac{y\,m(u^2)^\alpha}{1+y\,m(u^2)^\alpha}\,\mathrm{d}u \\ &= \frac{1}{\pi}\int_0^\infty \mathcal{A}(m)E(m)\vartheta_\infty(m)\,m^{-\frac{\alpha+1}{2}}\,\frac{ym^\alpha}{1+ym^\alpha}\,\mathrm{d}m.\end{aligned} \tag{SG.46}$$

In the second form the integrand is $O(y\,m^{\alpha/2-1})$ near $m = 0$, by $\mathcal{A}E = \sqrt{Q_m}L_\beta \asymp \sqrt{m}$ and $E\vartheta_\infty \le \alpha$, and $O(m^{-(\alpha+1)/2})$ at infinity, which is integrable because $\alpha > 1$; these bounds are locally uniform in y and give finiteness and continuity. Positivity is clear.

The large-y endpoint is the head computation of lemma SG.11 in the first form. On $m \le \delta$, $t_\infty(m) = s_2 m^{-\alpha}(1+O(\delta))$ and $L_\beta/F_\beta = s_2^{-1}(1+O(\delta))$, so the integrand is $s_2^{-1}ys_2/(u^2+ys_2)$ up to factors $1+O(\delta)$, with integral $\frac{\pi}{2}\sqrt{y/s_2}(1+O(\delta))$; the range $m > \delta$ is a bounded interval of u with a bounded integrand, which is $o(\sqrt{y})$.

The small-y endpoint is an averaging in the second form. The term $(\alpha-1)E/m$ of $E\vartheta_\infty = 1+(\alpha-1)E/m$ contributes $O(y)$ from $m \le 1$ and, since $E \le C$, at most

$$Cy^{\frac{\alpha+1}{2\alpha}}\int_{y^{1/\alpha}}^\infty w^{-\frac{\alpha+3}{2}}\,\frac{w^\alpha}{1+w^\alpha}\,\mathrm{d}w$$

from $m \ge 1$ after $m = y^{-1/\alpha}w$, which is $o(y^{p_\alpha})$ because $\alpha > 1$. In the remaining term substitute $m = y^{-1/\alpha}w$:

$$\frac{1}{\pi}\int \mathcal{A}(m)\,m^{-\frac{\alpha+1}{2}}\,\frac{ym^\alpha}{1+ym^\alpha}\,\mathrm{d}m = \frac{y^{p_\alpha}}{\pi}\int_0^\infty \mathcal{A}(y^{-1/\alpha}w)\,\frac{w^{(\alpha-1)/2}}{1+w^\alpha}\,\mathrm{d}w.$$

On $w \in [\delta,\delta^{-1}]$, eq. (SG.9) with $W(m) = m^{-(\alpha+1)/2}ym^\alpha/(1+ym^\alpha)$, whose boundary values and variation on the corresponding m-interval are $O_\delta(y^{p_\alpha+1/\alpha})$, replaces $\mathcal{A}$ by $\sqrt{2\pi}$ with an error $o(y^{p_\alpha})$. The ranges $w < \delta$ and $w > \delta^{-1}$ contribute $O(\delta^{(\alpha+1)/2})$ and $O(\delta^{(\alpha-1)/2})$ times y^{p_α}, by the bounds on $\mathcal{A}$. With $\sqrt{2\pi}/\pi = c_0$ and the integral of lemma SG.9, the limit is $d_\alpha y^{p_\alpha}$.

For the joint statement put $y \stackrel{\text{def}}{=} (i_{\mathrm{c}}/i)^\alpha$ and normalize eq. (SG.44) by b_N. Compact ranges of m converge to the second form of eq. (SG.46). Indeed, by lemma SG.6(iv), $\Lambda_N(m)\mu_i \to ym^\alpha$, $m\Lambda_N'/\Lambda_N \to \alpha$, so that $E\vartheta_N \to E\vartheta_\infty$, and

$$\frac{\mathcal{B}_i(m)}{\pi\gamma b_N} = \frac{\sqrt{\gamma-m}\,\mu_i\,\Lambda_N^{1/2}}{\pi\gamma\,b_N\sqrt{m}\,(1+\Lambda_N\mu_i)} \longrightarrow \frac{1}{\pi}\,m^{-\frac{\alpha+1}{2}}\,\frac{ym^\alpha}{1+ym^\alpha},$$

locally uniformly in $m > 0$ and in y on compact subsets of $(0,\infty)$. Near $m = 0$, lemma SG.6(v) and lemma SG.7 bound the normalized integrand by $C_y m^{\alpha/2-1}$ for $1/B \le m \le \delta$ and by $C_y\sqrt{B}\,i_{\mathrm{c}}^{-\alpha/2}m^{-1/2}$ for $m < 1/B$, so the range $m \le \delta$ contributes $C_y(\delta^{\alpha/2} + B^{-\alpha/2})$. For $m > M$, lemma SG.10 and $V_\alpha(p) \ge cp^\alpha$ on $p \le 1/2$ bound the normalized tail by

$$\frac{C\,a_N}{b_N}\Big(\frac{\gamma}{M}\Big)^{\frac{\alpha-1}{2}} = CK_\alpha^{-\alpha/2}M^{-\frac{\alpha-1}{2}},$$

using $a_N/b_N = K_\alpha^{-\alpha/2}\gamma^{(1-\alpha)/2}$. This proves the joint statement for y in compact subsets of $(0,\infty)$. The two joint endpoints use the head and the tail. If $y \to \infty$, then $m_N(1/\mu_i) = \gamma S_N(x^\alpha) \sim K_\alpha i/B \to 0$, so lemma SG.11 gives $\sqrt{2}\,\mathfrak{d}_i^{\le M} \sim \sqrt{\mu_i/(\gamma s_2)} = b_N\sqrt{y/s_2}$, while the tail is $O(b_N M^{-(\alpha-1)/2}) = o(b_N\sqrt{y})$. If $y \to 0$, the head is $O_M(\sqrt{\mu_i/\gamma}) = O_M(b_N\sqrt{y}) = o(b_N y^{p_\alpha})$. In the tail, the substitution $p = K_\alpha x u$ of lemma SG.9 has the lower cutoff $u_0 = M/(\gamma K_\alpha x) = MB/(K_\alpha i) \to 0$, so

$$a_N \Psi_\alpha^{[M/\gamma]}(x) \sim a_N d_\alpha \sqrt{K_\alpha}\, x^{-(\alpha-1)/2} = b_N d_\alpha y^{p_\alpha},$$

the identity being a direct computation from eq. (SG.23). With lemma SG.10 and $M \to \infty$ this gives the endpoint, and the three cases together prove the joint statement by passing to subsequences. □

Proof of the drift part of theorem SG.1 and of theorem SG.2(ii). Let $\alpha > 1$ and $c_N = 1$, and take a row sequence $i \to \infty$ along which i/N converges. If the limit is zero, lemma SG.13 applies, and the second factor of eq. (SG.30) tends to one by lemma SG.9. If the limit x_0 is positive, then $y = (i_c/i)^\alpha \to 0$, the head is $o(a_N)$ by lemma SG.11, and lemma SG.10 gives $\sqrt{2}\,\mathfrak{d}_i(0) \sim a_N \Psi_\alpha(x_0)$; by lemma SG.13, $b_N \Phi_{\alpha,\beta}(y) \sim b_N d_\alpha y^{p_\alpha} = a_N d_\alpha \sqrt{K_\alpha}(i/N)^{-(\alpha-1)/2}$, so the approximant is $a_N \Psi_\alpha(x_0)(1+o(1))$ as well. The sequential argument of the case $\alpha \le 1$ gives the uniform statement. The endpoint statements of theorem SG.2(ii) follow from lemma SG.13 and lemma SG.9, the case $i/i_c \to \infty$ through the uniform statement and $\Phi_{\alpha,\beta}(y) \sim d_\alpha y^{p_\alpha}$. □

The approximants are bounded below by $c\,a_N$, uniformly in the row: for $\alpha \le 1$ because $\Psi_\alpha \ge c$, and for $\alpha > 1$ because $\Phi_{\alpha,\beta}(y)/y^{p_\alpha}$ is positive and continuous on $(0,\infty)$ with limits d_α at 0 and ∞ at ∞, so that $\sqrt{2}\,\widehat{\mathfrak{d}}_i \ge c\, b_N y^{p_\alpha} \Psi_\alpha(x)/(d_\alpha\sqrt{K_\alpha}x^{-(\alpha-1)/2}) = c\, a_N \Psi_\alpha(x)/d_\alpha$.

SG.6 Proof of the volatility statements

Throughout this subsection the volatility is used in the participation form eq. (SG.22). For intervals $I, J \subset (0,\gamma)$ write

$$\mathfrak{v}_i[I,J] \stackrel{\text{def}}{=} \frac{1}{\pi^2}\sum_{\ell\in\mathbb{Z}} \int_I\!\int_J W_i(m_1)W_i(m_2)\,\frac{\mathsf{D}_\ell(m_1,m_2)}{\Delta_\ell^N(m_1,m_2)}\,\mathrm{d}m_2\,\mathrm{d}m_1,$$

so that $\mathfrak{v}_i(0) = \mathfrak{v}_i[(0,\gamma),(0,\gamma)]$, and $m_- \stackrel{\text{def}}{=} \min(m_1,m_2)$, $m_+ \stackrel{\text{def}}{=} \max(m_1,m_2)$.

Tools valid at every dimension.

Lemma SG.14 (Determinant)**.** *For every lag and all $m_1, m_2 \in (0,\gamma)$,*

$$0 \le \mathsf{M}_\ell \le m_-, \qquad \mathsf{D}_\ell^2 \le (m_1 - \mathsf{M}_\ell)(m_2 - \mathsf{M}_\ell),$$

$$\Delta_\ell^N \ge \Big(1 - \frac{m_-}{\gamma}\Big)\Big(1 - \frac{\mathsf{T}_N}{m_+}\Big) \ge c_\alpha\Big(1 - \frac{m_1}{\gamma}\Big)\Big(1 - \frac{m_2}{\gamma}\Big),$$

and $0 \le 1 - \Delta_\ell^N \le m_-/m_+ + 2m_-/\gamma$. If $m_1, m_2 \le \delta \le 1$, then $1 - \Delta_\ell^N \le C_\beta\delta + 2\delta/\gamma$.

Proof. The scalars are integrals over the shared ages. Take $\ell \ge 0$; negative lags exchange the arguments. For an age a and the shared scalar g put $X \stackrel{\text{def}}{=} q_1 w_a^2 g^2$ and $Y \stackrel{\text{def}}{=} \beta^{2\ell} q_2 w_a^2 g^2$. Then

$$\mathsf{M}_\ell = \sum_a \mathbb{E}\,\frac{XY}{(1+X)(1+Y)}, \qquad \mathsf{D}_\ell = \sum_a \mathbb{E}\,\frac{\sqrt{XY}}{(1+X)(1+Y)},$$

$$m_1 - \mathsf{M}_\ell = \sum_a \mathbb{E}\,\frac{X}{(1+X)(1+Y)},$$

and $m_2 - \mathsf{M}_\ell \ge \sum_a \mathbb{E}\,Y(1+X)^{-1}(1+Y)^{-1}$, since $m_2 = r_\beta(q_2)$ also contains the ℓ fresh ages of the later buffer, whose weights $w_0,\dots,w_{\ell-1}$ complete $\{\beta^\ell w_a\} = \{w_{a+\ell}\}$. This gives the first bound, and the Cauchy–Schwarz inequality gives the second.

The determinant is bounded below by an affine function of M_ℓ. Substituting the bound on D_ℓ^2,

$$\Delta_\ell^N \ge 1 - \frac{\mathsf{T}_N}{\gamma} - \mathsf{M}_\ell\,\mathsf{c}, \qquad \mathsf{c} \stackrel{\text{def}}{=} \frac{m_1 m_2 + \mathsf{T}_N(\gamma - m_1 - m_2)}{\gamma m_1 m_2} > 0,$$

where positivity uses $\mathsf{T}_N \le m_-$ when $\gamma < m_1 + m_2$, because $f_j(m_-) \le f_j(m_+)$ gives $\mathsf{T}_N \le B^{-1}\sum_j f_j(m_-) = m_-$. Evaluating at $\mathsf{M}_\ell = m_-$ gives $(1 - m_-/\gamma)(1 - \mathsf{T}_N/m_+)$. For the last factor,

$$\mathsf{T}_N \le \frac{1}{B}\sum_j f_j(m_+)^2 = m_+ - \Lambda m_N'(\Lambda)\big|_{\Lambda=\Lambda_N(m_+)} \le m_+ - c_\alpha m_+\Big(1 - \frac{m_+}{\gamma}\Big)$$

by lemma SG.7.

The upper bounds follow from the definition. Put $\mathsf{a} \stackrel{\text{def}}{=} \mathsf{T}_N\mathsf{M}_\ell/(m_1 m_2) \in [0, m_-/m_+]$ and $\mathsf{g} \stackrel{\text{def}}{=} \mathsf{M}_\ell/\gamma \in [0, m_-/\gamma]$; then

$$1 - \Delta_\ell^N = \mathsf{a} + \mathsf{g} - \mathsf{a}\mathsf{g} + \frac{\mathsf{T}_N\mathsf{D}_\ell^2}{\gamma m_1 m_2}, \qquad 0 \le \frac{\mathsf{T}_N\mathsf{D}_\ell^2}{\gamma m_1 m_2} \le \frac{\mathsf{T}_N}{\gamma} \le \frac{m_-}{\gamma}.$$

For $m_j \le \delta \le 1$, lemma SG.1(i) gives $q_j \le C m_j$ and $\mathsf{M}_\ell \le 3 s_4 q_1 q_2 \le C m_1 m_2$, so $\mathsf{a} \le C\mathsf{T}_N \le C\delta$. □

Lemma SG.15 (Age factorization)**.** *For real g and $q > 0$ put $\mathcal{Z}_g(q) \stackrel{\text{def}}{=} \sqrt{q}\sum_{a\ge 0} w_a g\,(1 + w_a^2 q g^2)^{-1}$.*

(i) $|\mathcal{Z}_g(q)| \le C_\beta$ and $\mathbb{E}\,\mathcal{Z}_g(q)^2 \le q$, for $g \sim \mathcal{N}(0,1)$.

(ii) For $m_0 > 0$ and $\mathcal{A}_g(m) \stackrel{\text{def}}{=} \mathcal{Z}_g(Q_m)/E(m)$,

$$\sup_{t \ge m_0}\Big|\int_{m_0}^t \big(\mathcal{A}_g(m) - \pi\,\mathrm{sign}(g)\big)\,\mathrm{d}m\Big| \le C_{\beta,m_0}\big(1 + |\log|g||\big) \qquad (g \ne 0).$$

(iii) $\sum_{\ell\in\mathbb{Z}} \mathsf{D}_\ell(m_1,m_2) = \mathbb{E}\big[\mathcal{Z}_g(Q_{m_1})\mathcal{Z}_g(Q_{m_2})\big]$.

(iv) The flat coefficient $\mathfrak{v}_i^{\text{flat}} \stackrel{\text{def}}{=} \mathbb{E}\big[\big(\frac{1}{\pi}\int_0^\gamma W_i(m)\mathcal{Z}_g(Q_m)\,\mathrm{d}m\big)^2\big]$ satisfies $0 \le \mathfrak{v}_i^{\text{flat}} \le \mathfrak{v}_i(0)$, and it is the integral of eq. (SG.22) summed over the lags with every Δ_ℓ^N replaced by one.

Proof. With $t \stackrel{\text{def}}{=} (1-\beta)\sqrt{q}|g|$ and $\phi(t) \stackrel{\text{def}}{=} t/(1+t^2)$, one has $\mathcal{Z}_g(q) = \mathrm{sign}(g)\sum_a \phi(t\beta^a)$. Splitting at the last a with $t\beta^a \ge 1$ and using $\phi(t) \le \min(t, t^{-1})$ bounds the sum by two geometric series, uniformly in t. The second bound of (i) is $|\mathcal{Z}_g(q)| \le \sqrt{q}|g|\sum_a w_a = \sqrt{q}|g|$.

For (ii), since $\mathrm{d}m/E(m) = \mathrm{d}q/q$ at $q = Q_m$ and $\int \phi(\tau)\,2\mathrm{d}\tau/\tau = 2\arctan\tau$,

$$\int_{m_0}^t \mathcal{A}_g(m)\,\mathrm{d}m = 2\,\mathrm{sign}(g)\sum_{a\ge 0}\Big[\arctan\big(w_a|g|\sqrt{Q_t}\big) - \arctan\big(w_a|g|\sqrt{Q_{m_0}}\big)\Big].$$

The same splitting, with $\arctan\tau \le \tau$ and $\pi/2 - \arctan\tau \le \tau^{-1}$, gives $2\sum_a \arctan(\tau\beta^a) = \pi\log_+\tau/|\log\beta| + O_\beta(1)$ uniformly in $\tau > 0$, and lemma SG.1(iii) gives $t = r_\beta(Q_t) = \log_+((1-\beta)\sqrt{Q_t})/|\log\beta| + O_{\beta,m_0}(1)$ for $t \ge m_0$. Since $|\log_+(|g|\tau) - \log_+\tau| \le |\log|g||$, subtraction proves (ii).

For (iii), expand $\mathcal{Z}_g(q_1)\mathcal{Z}_g(q_2)$ over ordered pairs of ages (a,b). A pair with $b = a + \ell \ge a$ has $w_b = \beta^\ell w_a$, and its term is the a-th term of $\mathsf{D}_\ell(m_1,m_2)$; a pair with $b < a$ gives the term of the negative lag. The scalar g is the same in each pair, and all terms are nonnegative.

For (iv), (iii) identifies the flat integral, and $0 < \Delta_\ell^N \le 1$ (lemma SG.14) with nonnegative terms gives the inequality. □

Lemma SG.16 (Spectral pieces)**.** *For intervals $I, J \subset (0,\gamma)$, $0 \le \mathfrak{v}_i[I,J] \le \big(\mathfrak{v}_i[I,I]\,\mathfrak{v}_i[J,J]\big)^{1/2}$.*

Proof. Nonnegativity holds termwise. Let first I, J be compact. Under the change of variables of the proof of proposition SG.4, they correspond to compact sets $U_I, U_J \subset (0,\infty)$ of the variable u with $u^2 = s_N(m)$, and the filters $f_I \stackrel{\text{def}}{=} \frac{2}{\pi}\int_{U_I} \iota_{u^2}\,\mathrm{d}u$, f_J are odd, bounded and Lipschitz. For real a, b the filter $f \stackrel{\text{def}}{=} af_I + bf_J$ has, in the independent model along the replications, row covariances whose finite-time sums $\frac{1}{nT}\sum_k \mathbb{E}\,\|\sum_{t'\le T} f^{\natural}(M^{\mathrm{g}}_{t'})^\top(u_i \otimes e_k)\|^2$ are nonnegative; the coupling of lemma SG.3, with $\mathrm{Lip}(f)$ in place of $\varrho^{-1/2}$, gives the geometric lag envelope, so these sums converge to the integrated covariance as in the proof of theorem SD.4(iii). In the limit of the replications, the integrated covariance of f is, by bilinearity and the integral form of the responses (proof of proposition SG.4(iii)),

$$a^2\mathfrak{v}_i[I,I] + 2ab\,\mathfrak{v}_i[I,J] + b^2\mathfrak{v}_i[J,J] \ge 0.$$

A nonnegative quadratic form has a nonnegative determinant, which is the claim. For general intervals, exhaust them by compact intervals; the three integrals of nonnegative kernels increase to their limits. □

Lemma SG.17 (Upper participation endpoint)**.** *For every $\eta \in (0,1)$ there is C_η such that, for all large N and all $1 \le i \le N$,*

$$\mathfrak{v}_i\big[[\eta\gamma,\gamma),[\eta\gamma,\gamma)\big] \le C_\eta\,\gamma\,(i/N)^\alpha.$$

Proof. Write $d_j \stackrel{\text{def}}{=} \gamma - m_j$ and $d \stackrel{\text{def}}{=} \max(d_1,d_2)$. For $m \ge \eta\gamma$, lemma SG.6(iii), whose proof gives $V_N(p) \asymp_\eta (1-p)^{-1}$ for $p \ge \eta$, and eq. (SG.42) give

$$\Lambda_N(m) \asymp_\eta \frac{N^\alpha}{c_N}\frac{\gamma}{\gamma-m}, \qquad \frac{\sqrt{\mu_i\Lambda_N(m)}}{1+\mu_i\Lambda_N(m)} \le \big(\mu_i\Lambda_N(m)\big)^{-1/2} \le C_\eta\Big(\frac{i}{N}\Big)^{\alpha/2}\sqrt{\frac{\gamma-m}{\gamma}}.$$

With $\vartheta_N(m) \le C(1+(\gamma-m)^{-1})$ (lemma SG.7) this gives

$$W_i(m) \le C_\eta\Big(\frac{i}{N}\Big)^{\alpha/2}\frac{\gamma-m+1}{\gamma^{3/2}} \qquad (m \ge \eta\gamma).$$

The determinant is bounded below by powers of the deficits. If $d \ge 1$,

$$\Delta^N_\ell \ge c_\eta\, d^2/\gamma^2.$$

Indeed, if the smaller deficit is at least $d/2$, the last bound of lemma SG.14 applies; otherwise $m_+ - m_- \ge d/2$, so $1-\mathsf{T}_N/m_+ \ge 1-m_-/m_+ \ge d/(2\gamma)$, and the first factor is $1-m_-/\gamma = d/\gamma$. If $0 < d \le 1$, then

$$\Delta^N_\ell \ge c\,d/\gamma^2$$

uniformly in the lag. To see this, note first that the affine bound of the proof of lemma SG.14, evaluated at M_ℓ instead of m_-, reads $\Delta^N_\ell \ge (1-m_-/\gamma)(1-\mathsf{T}_N/m_+) + (m_- - \mathsf{M}_\ell)\mathsf{c}$. Second, $\mathsf{T}_N = \gamma - d_1 - d_2 + \mathsf{R}$ with $\mathsf{R} \stackrel{\text{def}}{=} B^{-1}\sum_j(1-f_j(m_1))(1-f_j(m_2))$, and $1-f_j(m_k) \le Cd_k/\gamma$ gives $0 \le \mathsf{R} \le Cd_1d_2/\gamma$; inserting this in c gives $\mathsf{c} \ge (d_1+d_2)/(2\gamma^2)$ for large γ. Third, $m_- - \mathsf{M}_\ell \ge c_\beta$. For this, the ratio Q_{m_1}/Q_{m_2} lies in a fixed compact subset of $(0,\infty)$, because $\frac{\mathrm{d}}{\mathrm{d}m}\log Q_m = 1/E(m)$ is bounded on $m \ge 1$. If $m_- = m_2$ and $\ell \ge 1$, the fresh age 0 of the later buffer contributes $\mathbb{E}\,q_2w_0^2g^2(1+q_2w_0^2g^2)^{-1} \ge 1/2$ to $m_2 - \mathsf{M}_\ell$, since $q_2 \to \infty$. Otherwise choose a shared age a with $q_1w_a^2 \in [1,\beta^{-2}]$; on $1 \le |g| \le 2$ both X and Y are bounded above and X (or Y) below, so the term $X(1+X)^{-1}(1+Y)^{-1}$ (or the one with Y) contributes at least a constant.

The lag sum is bounded before integration. By lemma SG.15(i),(iii), $\sum_\ell \mathsf{D}_\ell \le C_\beta$. Outside the unit square $d \le 1$, the integrand of the lag sum is therefore at most

$$C_\eta\Big(\frac{i}{N}\Big)^\alpha\frac{(d_1+1)(d_2+1)}{\gamma\,d^2},$$

whose integral over $\{0 < d_j < (1-\eta)\gamma,\ d \ge 1\}$ is $O(\gamma(i/N)^\alpha)$: on $d_1 \le d_2$, integrating d_1 first leaves $\gamma^{-1}\int_1^\gamma O(t+1)\,\mathrm{d}t = O(\gamma)$. On the unit square the integrand is at most $C(i/N)^\alpha/(\gamma\max(d_1,d_2))$, whose integral is $O((i/N)^\alpha/\gamma)$. □

The mesoscopic crossover below and at one. Let $0 < \alpha \le 1$, let $i \to \infty$ with $x \overset{\text{def}}{=} i/N \to 0$, and put

$$b_{i,N} \overset{\text{def}}{=} \frac{\sqrt{\gamma}\, x^{\alpha/2}\, \Psi_\alpha(x)}{c_0}, \qquad b_{i,N}^2 \sim \begin{cases} \gamma x^\alpha \big[\log(x^{-\alpha})\big]^2/(1-\alpha), & \alpha < 1, \\ \frac{4}{9}\gamma x \big[\log(1/x)\big]^3, & \alpha = 1, \end{cases} \tag{SG.47}$$

the equivalence by lemma SG.9. Write $b = b_{i,N}$ and

$$\mathcal{J}_i(m) \overset{\text{def}}{=} \frac{\sqrt{\mu_i \Lambda_N(m)}}{\sqrt{m}\,(1+\mu_i\Lambda_N(m))}\sqrt{1-\frac{m}{\gamma}} = \frac{\mathcal{B}_i(m)}{\sqrt{\gamma\mu_i}}, \qquad \mathcal{V}_i(g) \overset{\text{def}}{=} \frac{1}{\pi}\int_0^\gamma W_i(m)\, \mathcal{Z}_g(Q_m)\, \mathrm{d}m,$$

so that $W_i = \mathcal{J}_i \vartheta_N$ and $\mathfrak{v}_i^{\text{flat}} = \mathbb{E}\ \mathcal{V}_i(g)^2$.

Lemma SG.18 (The flat response)**.** *Along every such row sequence,*

$$\Big\| \mathcal{V}_i(g) - \frac{g}{\sqrt{s_2}} - b\,\mathrm{sign}(g) \Big\|_{L^2(g)} = o(1+b),$$

and for fixed $0 < \delta < M$ the $L^2(g)$ norm of the part of $\mathcal{V}_i$ over $\delta < m < M$ is $o(1+b)$.

Proof. The tail is an averaged integral. Put $B_i^{[M]} \overset{\text{def}}{=} \int_M^\gamma \mathcal{J}_i\, \mathrm{d}m$. Since $W_i \mathcal{Z}_g = \mathcal{J}_i\, (E\vartheta_N)\, \mathcal{A}_g$, the proof of lemma SG.10, with $\mathcal{B}_i$ replaced by $\mathcal{J}_i$ and lemma SG.2 replaced by lemma SG.15(ii) with $m_0 = 1$, whose constant has a finite second moment in g, gives, through Minkowski's integral inequality,

$$\Big\| \frac{1}{\pi}\int_M^\gamma W_i \mathcal{Z}_g\, \mathrm{d}m - B_i^{[M]}\,\mathrm{sign}(g) \Big\|_{L^2(g)} \le \frac{C}{M} B_i^{[M]} + o\big(B_i^{[M]}\big),$$

uniformly over the rows; here $\mathcal{A}_g$ is bounded for $m \ge M$ by lemma SG.15(i). By lemma SG.10 and eq. (SG.45),

$$B_i^{[M]} = \frac{\pi\gamma}{\sqrt{2\pi}\sqrt{\gamma\mu_i}}\, a_N \Psi_\alpha^{[M/\gamma]}(x)\,(1+o(1)) = (1+o(1))\,\frac{\sqrt{\gamma}\, x^{\alpha/2}}{c_0}\,\Psi_\alpha^{[M/\gamma]}(x),$$

using $a_N/\sqrt{\mu_i} = x^{\alpha/2}$. This is $b + o(1+b)$. Indeed, if $b \to 0$ positivity bounds it by $b + o(1)$; if b is bounded below, then eq. (SG.47) gives $e^{T - \log\gamma} \le CT^2$, respectively CT^3, with T of lemma SG.12, so $T \le \log\gamma + C\log\log\gamma$ and lemma SG.12(iii) gives $\Psi_\alpha^{[M/\gamma]} \sim \Psi_\alpha$.

The head has a bounded norm. On $m \le M$, $\|\mathcal{Z}_g(Q_m)\|_{L^2} \le \sqrt{Q_m} \le C_M\sqrt{m}$ and $\vartheta_N \le C_M \Lambda_N'/\Lambda_N$, because $1/E(m) \le C_M/m \le C_M \Lambda_N'/\Lambda_N$. Hence, with $u = \mu_i \Lambda_N(m)$,

$$\Big\| \frac{1}{\pi}\int_0^M W_i \mathcal{Z}_g\, \mathrm{d}m \Big\|_{L^2} \le C_M \int_0^M \frac{\sqrt{\mu_i\Lambda_N}}{1+\mu_i\Lambda_N}\frac{\Lambda_N'}{\Lambda_N}\,\mathrm{d}m \le C_M \int_0^\infty \frac{\sqrt{u}}{1+u}\frac{\mathrm{d}u}{u} = C_M\pi. \tag{SG.48}$$

The head has a limit when $a_i \overset{\text{def}}{=} m_N(1/\mu_i) \to 0$: then it tends to $g/\sqrt{s_2}$ in L^2. The Riemann error of $m_N(r/\mu_i)$ relative to a_i is at most $1/(Ba_i) \le 2/i$, because the first i terms of Ba_i are at least $1/2$. With the small-V laws of lemma SG.5, this gives, locally uniformly in $r > 0$,

$$\mu_i \Lambda_N(a_i r) \to r, \qquad a_i \frac{\Lambda_N'}{\Lambda_N}(a_i r) \to \frac{1}{r}, \qquad Q_{a_i r} \sim \frac{a_i r}{s_2}, \qquad E(a_i r) \sim a_i r,$$

and $\mathcal{Z}_g(Q_m) = g\sqrt{Q_m}\,(1 + O(Q_m g^2))$, so $\mathcal{Z}_g(Q_{a_i r}) = g\sqrt{a_i r/s_2} + o_{L^2}(\sqrt{a_i})$. With $m = a_i r$ the head becomes

$$\frac{g}{\pi\sqrt{s_2}}\int_0^\infty \frac{\mathrm{d}r}{\sqrt{r}\,(1+r)} = \frac{g}{\sqrt{s_2}}.$$

The ranges of small and large r are controlled by eq. (SG.48) in the variable u, and on $\delta < m < M$ one has $\mu_i\Lambda_N(\delta) \to \infty$, as in the proof of lemma SG.11, so the bound eq. (SG.48) restricted to that range tends to zero.

The cases are combined by the size of b. By eq. (SG.47) and lemma SG.5, $a_i \sim \gamma x^\alpha/(1-\alpha)$ for $\alpha < 1$ and $a_i \sim \gamma x \log(1/x)$ for $\alpha = 1$, so $a_i \asymp b^2/T^2 \to 0$ when b is bounded. Then head, middle and tail give the claim after $M \to \infty$. If $b \to \infty$, the head and $g/\sqrt{s_2}$ are $O(1) = o(b)$. Passing to subsequences along which b converges in $[0,\infty]$ proves the first claim. For the middle range, lemma SG.6(ii) and lemma SG.5(iii) give $\Lambda_N(m) \asymp \Lambda_N(1)$ on $[\delta, M]$, so with $y \overset{\text{def}}{=} \mu_i\Lambda_N(1)$ its norm is at most $C_{\delta,M}\min(\sqrt{y}, y^{-1/2})$; this tends to zero unless y stays in a compact subset of $(0,\infty)$, and in that case $a_i \asymp 1$ and $b^2 \asymp a_i T^2 \to \infty$. □

Lemma SG.19 (Multiplicative diagonal strip)**.** *For fixed $R > 1$, $M \geq 1$ and $0 < \eta < 1/2$, let $\mathcal{D} \stackrel{\text{def}}{=} \{M < m_1, m_2 < \eta\gamma,\ R^{-1} \leq m_1/m_2 \leq R\}$. Along the row sequences above, the lag-summed integral of eq.* (SG.22) *over $\mathcal{D}$ is $o((1+b)^2)$.*

Proof. On $\mathcal{D}$, ϑ_N and $1/\Delta_\ell^N$ are bounded by lemmas SG.7 and SG.14, uniformly in lag and dimension, $\Lambda_N(m_1)/\Lambda_N(m_2)$ is bounded by a constant depending on (R, α, η), and $\sum_\ell \mathsf{D}_\ell \leq C_\beta$. Integrating over the second coordinate, whose range has length at most $C_R m_1$, bounds the strip by

$$C_{R,\eta} \int_M^{\eta\gamma} \frac{\mu_i \Lambda_N(m)}{(1+\mu_i\Lambda_N(m))^2}\,\mathrm{d}m. \tag{SG.49}$$

For $\alpha < 1$, $\Lambda_N(m) \asymp \Lambda_N(1) m$ on this range, and with $y \stackrel{\text{def}}{=} \mu_i \Lambda_N(1)$ the integral is at most $C y^{-1}\log(1+y\gamma)$. By proposition SG.6 with $c_N = 1$, $y\gamma \asymp x^{-\alpha}$ and $y^{-1} \asymp \gamma x^\alpha$, so the bound is $O(b^2/T) = o(b^2)$.

For $\alpha = 1$ put $\xi \stackrel{\text{def}}{=} B/i$ and $\mathcal{L}(m) \stackrel{\text{def}}{=} \log(e\gamma/m)$. The identity $S_1(V) = V\log(1+1/V)$ and lemma SG.6(ii) give $\mu_i\Lambda_N(m) \asymp \xi m/\mathcal{L}(m)$ on this range. Let m_0 solve $\xi m_0 = \mathcal{L}(m_0)$; then $m_0 \leq (1+\log(\gamma\xi))/\xi$ when $\gamma\xi > 1$. Bounding the integrand by one below m_0 and by $C\mathcal{L}(m)/(\xi m)$ above, the integral is at most $C(1+\log(\gamma\xi))^2/\xi$. Since $\gamma\xi = 1/x$ and $b^2 \sim \frac{4}{9}T^3/\xi$, this is $O(b^2/T)$ as well. □

Lemma SG.20 (Mesoscopic crossover)**.** *Let $0 < \alpha \leq 1$. Along every row sequence $i \to \infty$ with $i/N \to 0$,*

$$\mathfrak{v}_i(0) \sim \frac{1}{s_2} + \frac{2c_0}{\sqrt{s_2}}\, b_{i,N} + b_{i,N}^2.$$

Proof. The flat coefficient has the claimed form. By lemma SG.18, $\mathbb{E}\, g^2 = 1$, $\mathbb{E}\,\mathrm{sign}(g)^2 = 1$ and $\mathbb{E}[g\,\mathrm{sign}(g)] = c_0$,

$$\mathfrak{v}_i^{\text{flat}} = \frac{1}{s_2} + \frac{2c_0}{\sqrt{s_2}}\, b + b^2 + o\big((1+b)^2\big).$$

It remains to show that the nonnegative difference $\mathfrak{v}_i(0) - \mathfrak{v}_i^{\text{flat}}$ is $o((1+b)^2)$; the quadratic is bounded above and below by multiples of $(1+b)^2$, which then gives the relative equivalence.

Fix $0 < \delta < 1 < M$, $R > 1$ and $0 < \eta < 1/2$, and split $(0,\gamma)$ into $I_1 \stackrel{\text{def}}{=} (0,\delta)$, $I_2 \stackrel{\text{def}}{=} [\delta, M]$, $I_3 \stackrel{\text{def}}{=} (M, \eta\gamma)$ and $I_4 \stackrel{\text{def}}{=} [\eta\gamma, \gamma)$. The square $I_4 \times I_4$ is at most $C_\eta \gamma x^\alpha$ by lemma SG.17, which is $o((1+b)^2)$ by eq. (SG.47). On $(0,\eta\gamma)^2$, lemma SG.14 bounds $1/\Delta_\ell^N$ by C_η, so the exact square there is at most C_η times the flat square, which is $O((1+b)^2)$ by the estimates of lemma SG.18 restricted to $(0,\eta\gamma)$. Lemma SG.16 then makes every term meeting I_4 equal to $o((1+b)^2)$. In the same way the flat norm over I_2 is $o(1+b)$ by lemma SG.18, and the terms meeting I_2 are $o((1+b)^2)$.

The remaining products compare the exact and flat kernels. On $I_1 \times I_1$, $1/\Delta_\ell^N - 1 \leq C_\beta\delta$ by lemma SG.14, and the flat square is bounded by eq. (SG.48), so the change is at most $C\delta$. On $I_1 \times I_3$ and its transpose, $1 - \Delta_\ell^N \leq \delta/M + 2\delta/\gamma$, so the change is at most $C\delta M^{-1}(1+b)^2 + o((1+b)^2)$. On $I_3 \times I_3$, the strip of lemma SG.19 is $o((1+b)^2)$, and off the strip $1 - \Delta_\ell^N \leq R^{-1} + 2\eta$, which for $R^{-1} + 2\eta < 1/2$ changes the integral by at most $C(R^{-1}+\eta)$ times the flat total, that is by $O((R^{-1}+\eta)(1+b)^2)$. Letting $N \to \infty$ and then $\delta \downarrow 0$, $M \to \infty$, $R \to \infty$ and $\eta \downarrow 0$ proves the claim. □

Proof of theorem SG.3(ii). By lemma SG.4 we take $c_N = 1$; the indices do not depend on c_N. At $i/i_{\mathrm{c},\mathfrak{v}} \to y \in (0,\infty)$, eq. (SG.33) gives

$$\gamma x^\alpha = \frac{(1-\alpha)\,y^\alpha}{s_2 \log^2\gamma}(1+o(1)) \quad (\alpha < 1), \qquad \gamma x = \frac{9y}{4 s_2 \log^3\gamma}(1+o(1)) \quad (\alpha = 1),$$

and $\log(N/i) = \alpha^{-1}\log\gamma + O(\log\log\gamma)$, respectively $\log\gamma + O(\log\log\gamma)$. Hence $b_{i,N} \to y^{\alpha/2}/\sqrt{s_2}$ by eq. (SG.47), and lemma SG.20 gives $s_2\mathfrak{v}_i(0) \to 1 + 2c_0 y^{\alpha/2} + y^\alpha$. If the ratio tends to zero, then $b \to 0$, and if it tends to infinity with $i/N \to 0$, then $b \to \infty$: the functions $x^\alpha[\log(x^{-\alpha})]^2$ and $x[\log(1/x)]^3$ are increasing for small x, so comparison with a fixed multiple of

$i_{\mathrm{c},\mathfrak{v}}$ gives the extended limits. In the latter case $\mathfrak{v}_i(0) \sim b^2$, which is the displayed equivalent. For bulk rows the equivalent $\mathfrak{v}_i(0) \sim \gamma\Gamma_\alpha(i/N)$ is part (i). Uniformity on compact sets of y follows by passing to convergent subsequences of the ratio. □

Small rows of the bulk profile.

Lemma SG.21 (Continuum matching below and at one)**.** *For $0 < \alpha \le 1$, $\Gamma_\alpha(x) \sim x^\alpha \Psi_\alpha(x)^2/c_0^2$ as $x \downarrow 0$; with lemma SG.9 this is the second equivalence of eq. (SG.27).*

Proof. The bulk profile has a positive double-integral form. With $V_p \stackrel{\mathrm{def}}{=} V_\alpha(p)$ put

$$A_x(p) \stackrel{\mathrm{def}}{=} \sqrt{\frac{f_x(p)(1-f_x(p))(1-p)}{p}}, \qquad t(p,q) \stackrel{\mathrm{def}}{=} \int_0^1 \frac{V_p}{V_p + r^\alpha}\frac{V_q}{V_q + r^\alpha}\,\mathrm{d}r = \frac{V_q p - V_p q}{V_q - V_p},$$

the second form by partial fractions in r. For $0 < p < q < 1$ the identities $f_x(q) - f_x(p) = x^\alpha(V_q - V_p)/((V_q + x^\alpha)(V_p + x^\alpha))$ and $1 - t(p,q)/q = V_q(q-p)/(q(V_q - V_p))$ give

$$\frac{f_x(q) - f_x(p)}{q-p}\sqrt{\frac{U_\alpha(p)}{U_\alpha(q)}} = \frac{A_x(p)A_x(q)}{\Delta_{\mathrm{c}}(p,q)}, \qquad \Delta_{\mathrm{c}}(p,q) \stackrel{\mathrm{def}}{=} (1-p)\Big(1 - \frac{t(p,q)}{q}\Big),$$

so that, substituting $q = p + \tau$ in eq. (SG.24),

$$\Gamma_\alpha(x) = 2\int_{0<p<q<1} \frac{A_x(p)A_x(q)}{\Delta_{\mathrm{c}}(p,q)}\,\mathrm{d}p\,\mathrm{d}q, \qquad F_x \stackrel{\mathrm{def}}{=} \Big(\int_0^1 A_x\Big)^2 = \frac{x^\alpha \Psi_\alpha(x)^2}{c_0^2},$$

the last identity because $A_x = x^{\alpha/2}\psi_x$ with ψ_x of the proof of lemma SG.9. Without the denominator the double integral is F_x. The continuum elasticity gives $t(p,q) \le q - \mathsf{T}(V_q) \le q - cq(1-q)$, by lemma SG.5(ii), so $\Delta_{\mathrm{c}} \ge c(1-p)(1-q)$; also $t(p,q) \le p$.

The upper endpoint is of lower order. For fixed $\eta > 0$ and $p \ge \eta$, $A_x(p) \le C_\eta x^{\alpha/2}(1-p)$, since $V_\alpha \asymp (1-p)^{-1}$. Write $d_p = 1 - p \ge d_q = 1 - q$. If $d_q \ge d_p/2$, then $\Delta_{\mathrm{c}} \ge c d_p^2$; otherwise $1 - t/q \ge 1 - p/q \ge d_p/2$, with the same conclusion. So the square $[\eta, 1)^2$ contributes $O_\eta(x^\alpha)$. On $p < \eta/2 < \eta < q$, $\Delta_{\mathrm{c}} \ge 1/4$, and this rectangle contributes at most $C_\eta x^{\alpha/2}\sqrt{F_x}$; the rest of the crossed region lies in the square with cutoff $\eta/2$. Both bounds are $o(F_x)$, because $\Psi_\alpha(x) \to \infty$.

The interior diagonal strip is of lower order. On $p < q < \eta < 1/2$ with $q/p \le R$, $A_x(q) \le C_{\eta,R}A_x(p)$ and $\Delta_{\mathrm{c}} \ge c$, so the strip contributes at most

$$C_{\eta,R}\int_0^\eta pA_x(p)^2\,\mathrm{d}p \le C_{\eta,R}\int_0^\eta f_x(p)(1-f_x(p))\,\mathrm{d}p.$$

For $\alpha < 1$, $V_\alpha(p) \asymp p$ bounds this by $Cx^\alpha\int_0^\eta p(p+x^\alpha)^{-2}\mathrm{d}p = O(x^\alpha(1 + \log(1/x)))$. For $\alpha = 1$, $V_1(p) \asymp p/\mathcal{L}(p)$ with $\mathcal{L}(p) = 1 + \log(1/p)$; the bound one below $p_0 = x\log(1/x)$ and $f_x(1-f_x) \le Cx\mathcal{L}(p)/p$ above give $O(x[\log(1/x)]^2)$. Both are $o(F_x)$.

Off the strip, $p/q < 1/R$ and $p, q < \eta$ give $0 \le 1 - \Delta_{\mathrm{c}} \le \eta + 1/R$, and for $\eta + 1/R < 1/2$ the exact and flat integrals there differ by at most $C(\eta + 1/R)F_x$. Letting $x \downarrow 0$, then $\eta \downarrow 0$ and $R \to \infty$, proves $\Gamma_\alpha/F_x \to 1$. □

The bulk.

Lemma SG.22 (Localization of the lag kernel)**.** *(i) For $\ell \ge 0$, with $t \stackrel{\mathrm{def}}{=} \log(\beta^{2\ell}q_2/q_1)$,*

$$\mathsf{D}_\ell(m_1,m_2) = \frac{r_\beta(q_1e^t) - r_\beta(q_1)}{2\sinh(t/2)},$$

$$\mathsf{D}_\ell \le C\min(m_1,m_2,1)^{1/2}\big(1 + |m_1 - m_2 + \ell|\big)e^{-|\log\beta||m_1-m_2+\ell|}.$$

(ii) For every $\tau_0 > 0$ there is $c > 0$ such that $\Delta_\ell^N \ge c$ for all $\ell \ge \tau_0\gamma$, all $m_1, m_2 \in (0,\gamma)$ and all large γ.

(iii) Let $n \to \infty$ with $\gamma/n \to \rho_0 \in (0,\infty]$, and let ℓ be integers with $\ell/n \to \tau > 0$. For every continuous φ with compact support in $(0,\rho_0)^2$,

$$\iint \varphi\Big(\frac{m_1}{n},\frac{m_2}{n}\Big)\frac{\mathsf{D}_\ell(m_1,m_2)}{\pi^2 n\,E(m_1)E(m_2)}\,\mathrm{d}m_1\,\mathrm{d}m_2 \longrightarrow \int \varphi(r,r+\tau)\,\mathrm{d}r.$$

Proof. For (i), the divided-difference form eq. (SG.15) gives, with $p \overset{\text{def}}{=} \beta^{2\ell}q_2 = q_1 e^{t}$,

$$\mathsf{D}_\ell = \sqrt{q_1 q_2}\,\beta^\ell \frac{r_\beta(q_1)-r_\beta(p)}{q_1-p} = \sqrt{q_1 p}\,\frac{r_\beta(p)-r_\beta(q_1)}{p-q_1} = \frac{r_\beta(q_1e^{t})-r_\beta(q_1)}{e^{t/2}-e^{-t/2}}.$$

The function $qr'_\beta(q)$ is bounded on $(0,\infty)$ (lemma SG.1), so the numerator is at most $C|t|$. For $m_1,m_2 \ge 1$, lemma SG.1(iii) gives $t = 2|\log\beta|(m_2-m_1-\ell)+O(1)$, which proves the bound. If $m_1 < 1 \le m_2$, dropping the first denominator gives $\mathsf{D}_\ell \le \sqrt{q_1}\,\sqrt{p}\,F_\beta(p)$, and $\sqrt{q_1} \le C\sqrt{m_1}$, $\sqrt{p}\,F_\beta(p) \le C\min\{\sqrt{p},(1+\log_+ p)/\sqrt{p}\}$ with $\log p = 2|\log\beta|(m_2-\ell)+O(1)$, where $|m_2-\ell|$ differs from $|m_1-m_2+\ell|$ by less than one. The case $m_2<1\le m_1$ exchanges the roles, and if both $m_j<1$ then $\mathsf{D}_\ell \le \beta^\ell s_2\sqrt{q_1q_2} \le C\beta^\ell\sqrt{m_1m_2}$.

For (ii), let $\ell \ge \tau_0\gamma$. Since $\mathsf{M}_\ell \le \sum_a \mathbb{E}\,Y(1+Y)^{-1} = r_\beta(\beta^{2\ell}q_2)$, two cases arise. If $m_2 \ge \ell/2$, lemma SG.1(iii) gives $\mathsf{M}_\ell \le (m_2-\ell)_+ + C$, so $\mathsf{M}_\ell/m_2 \le 1-c\tau_0$. If $m_2 < \ell/2$, then $\mathsf{M}_\ell \le s_2\beta^{2\ell}Q_{m_2} \le Cm_2e^{-|\log\beta|\ell}$. In both cases $1-\mathsf{T}_N\mathsf{M}_\ell/(m_1m_2) \ge 1-\mathsf{M}_\ell/m_2 \ge c$ and $1-\mathsf{M}_\ell/\gamma \ge c$, using $\mathsf{T}_N \le m_1$. The subtracted term is at most $\gamma^{-1}\min(m_-, C^2/m_+) \le C/\gamma$, because $\mathsf{D}_\ell \le C$, $\mathsf{D}_\ell^2 \le m_1m_2$ and $\mathsf{T}_N \le m_-$.

For (iii), the large-argument behaviour of r_β is periodic in $\log q$. Put $\mathsf{f}(x) = xJ(x)$ and

$$\mathcal{H}(u) \overset{\text{def}}{=} \sum_{k\ge 0}\mathsf{f}\big((1-\beta)^2\beta^{2k}e^{u}\big) + \sum_{k<0}\Big(\mathsf{f}\big((1-\beta)^2\beta^{2k}e^{u}\big)-1\Big),$$

which converges by $\mathsf{f}(x)\le x$ and $1-\mathsf{f}(x) \le \sqrt{\pi/2}\,x^{-1/2}$. Shifting u by $2|\log\beta|$ shifts k by one, so $\mathcal{H}(u+2|\log\beta|) = \mathcal{H}(u)+1$, and $|r_\beta(e^u)-\mathcal{H}(u)| \le Ce^{-u/2}$. In the coordinates $a \overset{\text{def}}{=} \log Q_{m_1}$ and t, one has $\mathrm{d}m_1/E(m_1) = \mathrm{d}a$ and $\mathrm{d}m_2/E(m_2) = \mathrm{d}t$ at fixed m_1, so the measure of the statement is $\pi^{-2}n^{-1}\mathsf{D}_\ell\,\mathrm{d}a\,\mathrm{d}t$. By (i) the region $|m_2-m_1-\ell| > A$ carries mass at most Ce^{-cA} per unit of m_1, and is removed first. On the rest, $\mathcal{H}$ replaces r_β with an error that vanishes on the support. For each t,

$$\int_a^{a+2|\log\beta|}\big(\mathcal{H}(a'+t)-\mathcal{H}(a')\big)\,\mathrm{d}a' = t,$$

because the derivative in t of the left side is $\mathcal{H}(a+2|\log\beta|+t)-\mathcal{H}(a+t) = 1$. Hence one period of a, which is one unit of m_1, carries the mass

$$\frac{1}{\pi^2 n}\int_{\mathbb{R}}\frac{t}{2\sinh(t/2)}\,\mathrm{d}t = \frac{1}{n},$$

by $1/\sinh(t/2) = 2\sum_{k\ge0}e^{-(k+1/2)t}$ on $t>0$ and $\sum_{k\ge0}(k+\frac12)^{-2} = \pi^2/2$. On the support, $m_2 = m_1+\ell+O(A)$, so $m_2/n = m_1/n+\tau+o(1)$. A continuous compactly supported φ varies by $o(1)$ over each of the $O(n)$ periods in its support, which gives the limit; letting then $A\to\infty$ concludes. □

Proof of theorem SG.3(i). By lemma SG.4 we take $c_N = 1$. Write $g_i(m) \overset{\text{def}}{=} \sqrt{f_i(m)(1-f_i(m))}\sqrt{(\gamma-m)/m}$, so that $W_i = \gamma^{-1/2}g_i\vartheta_N$. Let $i/N \to x \in (0,1]$ and $\ell/\gamma \to \tau \in (0,1)$, and use the coordinates $m_1 = \gamma p$, $m_2 = \gamma p'$.

On compact sets of p the integrand converges. By lemma SG.6(i),(ii) and eq. (SG.42), $c_N\Lambda_N(\gamma p)N^{-\alpha} \to V_\alpha(p)$ together with its logarithmic derivative, so $f_i(\gamma p)\to f_x(p)$, $g_i(\gamma p)\to A_x(p)$ with A_x of lemma SG.21, and $\vartheta_N(\gamma p) = 1/E(\gamma p)+O(1/\gamma)$. On the shell $m_2 = m_1+\ell+O(1)$, the ratio $\beta^{2\ell}Q_{m_2}/Q_{m_1}$ stays in a compact subset of $(0,\infty)$, and the divided difference of r_β gives $m_1-\mathsf{M}_\ell = O(1)$. Consequently

$$\frac{\mathsf{T}_N}{\gamma} \to t(p,p'), \qquad \Delta_\ell^N \to \Delta_{\mathrm{c}}(p,p'), \qquad p' = p+\tau,$$

in the notation of lemma SG.21. With lemma SG.22(iii) for $n=\gamma$, the integral eq. (SG.22) restricted to compact sets of p tends to

$$\int_0^{1-\tau} \frac{A_x(p)A_x(p+\tau)}{\Delta_{\mathrm{c}}(p,p+\tau)}\,\mathrm{d}p = C_{\alpha,x}(\tau),$$

by the identity of lemma SG.21.

The compact restriction is removed by bounds uniform in $x \ge x_0 > 0$ and $\tau \ge \tau_0 > 0$. Since $\mu_j \le \mu_i$ for $j \ge i$ and $(i/j)^\alpha \ge x_0^\alpha$ for $j \le N$, the terms $N/2 \le j \le N$ of m_N give $m \ge c\,x_0^\alpha \gamma f_i(m)$, hence

$$g_i(m) \le C_{x_0}, \qquad g_i(m) \le C_{x_0}\big(1-m/\gamma\big)\ (m \ge \gamma/2), \qquad \vartheta_N(m) \le C\big(1+m^{-1}+(\gamma-m)^{-1}\big),$$

the second bound by $1-f_i(m) \le C(\gamma-m)/\gamma$ as in lemma SG.17. With lemma SG.22(i),(ii), the integrand is dominated by $C\gamma^{-1} g_i(m_1) g_i(m_2) \vartheta_N(m_1)\vartheta_N(m_2)\mathsf{D}_\ell$. The three bounds give $g_i\vartheta_N \le C_{x_0}(1+m^{-1})$ on $(0,\gamma)$ and $g_i\vartheta_N \le C_{x_0}$ on $m \ge \gamma/2$. Along the shell, this majorant has total $O(\delta + \gamma^{-1}\log\gamma)$ on $1 < m_1 < \delta\gamma$ and $O(\delta)$ on $m_2 > (1-\delta)\gamma$; on $m_1 \le 1$ the factor $\min(m_1,1)^{1/2}$ of D_ℓ compensates the factor m_1^{-1} of ϑ_N. Off the shell, on $|m_1 - m_2 + \ell| \ge \delta\gamma$, the exponential factor of lemma SG.22(i) gives $\mathsf{D}_\ell \le C\min(m_1,1)^{1/4}\min(m_2,1)^{1/4} e^{-|\log\beta|\delta\gamma/2}$, and with $g_i\vartheta_N \le C_{x_0}(1+m^{-1})$ the majorant has total at most

$$C\gamma\, e^{-|\log\beta|\delta\gamma/2} \longrightarrow 0.$$

The rest of the domain, $m_1 \ge \delta\gamma$, $m_2 \le (1-\delta)\gamma$, $|m_1 - m_2 + \ell| < \delta\gamma$, lies in a compact set of p, on which the integrand converges. This proves the first limit, locally uniformly.

Large lags are negligible. For $\ell \ge \gamma + C$, write $\mathsf{a} \overset{\text{def}}{=} \beta^\ell$. Stein's lemma [62, Lemma 1], $\mathbb{E}[g\phi(g)] = \mathbb{E}[\phi'(g)]$, applied to $\phi(g) = g^3(1+sg^2)^{-1}$ gives

$$\mathbb{E}\,\frac{g^4}{1+sg^2} = 3\,\mathbb{E}\,\frac{g^2}{1+sg^2} - 2s\,\mathbb{E}\,\frac{g^4}{(1+sg^2)^2} \le 3\,\mathbb{E}\,\frac{g^2}{1+sg^2} \qquad (s \ge 0),$$

and with $s = q_1 w_a^2$, $w_a \le w_0$ and $q_1 F_\beta(q_1) = m_1$ this gives $\mathsf{M}_\ell \le 3w_0^2\mathsf{a}^2 q_2 m_1$; also $\mathsf{D}_\ell \le \mathsf{a}\sqrt{q_1 q_2}\,F_\beta(q_1)$ gives $\mathsf{D}_\ell^2 \le \mathsf{a}^2 q_2 m_1 s_2$. Since $q_2/m_2 \le Q_\gamma/\gamma$, these give $1 - \Delta_\ell^N \le C\mathsf{a}^2 Q_\gamma \le 1/2$, and $\mathsf{D}_\ell \le C\mathsf{a}\sqrt{Q_\gamma}\min(m_1,1)^{1/2}\min(m_2,1)^{1/2}$. The bounds on g_i and ϑ_N give $\int_0^\gamma \min(m,1)^{1/2} g_i\vartheta_N\,\mathrm{d}m \le C\gamma$, so

$$\overline{C}_{i,\ell}(0) \le C\gamma\,\beta^\ell\sqrt{Q_\gamma} \le C\gamma\, e^{-|\log\beta|(\ell-\gamma)}.$$

For $\eta > 0$ the lags beyond $(1+\eta)\gamma$ therefore contribute $o(\gamma)$, and the lags in $[(1-\eta)\gamma,(1+\eta)\gamma]$ and in $[0,\eta\gamma]$ contribute at most $3\eta\gamma + O(1)$, since $\overline{C}_{i,\ell}(0) \le 1$ (lemma SG.3). The Riemann sum $\gamma^{-1}\sum_\ell \overline{C}_{i,\ell}(0)$ over the remaining lags converges to the integral of $C_{\alpha,x}$, which gives $\mathfrak{v}_i(0)/\gamma \to 2\int_0^1 C_{\alpha,x} = \Gamma_\alpha(x)$.

Positivity and continuity of Γ_α follow: f_x is strictly increasing, so $C_{\alpha,x} > 0$, and the bounds above are locally uniform in x. □

The supercritical crossover. Let $\alpha > 1$ and $X \overset{\text{def}}{=} i/i_{\mathrm{c}} = K_\alpha i/B$.

Lemma SG.23 (Supercritical volatility)**.** *The function $G_{\alpha,\beta}$ is finite, positive and continuous on $(0,\infty)$ and satisfies the two endpoint laws of eq. (SG.28). Under Assumption S4, for every row sequence $i \to \infty$: if $X \to X_0 \in (0,\infty)$ then $\mathfrak{v}_i(0) \to G_{\alpha,\beta}(X_0)$; if $X \to 0$ then $\mathfrak{v}_i(0) \to 1/s_2$; if $X \to \infty$ and $i/N \to 0$ then $\mathfrak{v}_i(0) \sim \frac{4K_\alpha}{\alpha-1}X$. Moreover $\Gamma_\alpha(x) \sim \frac{4K_\alpha^2}{\alpha-1}x$ as $x \downarrow 0$.*

Proof. The profile is dominated by a product. By lemma SG.15(i),(iii), $\sum_\ell \mathsf{D}_\ell(u,v) \le C\,\mathsf{w}(u)\mathsf{w}(v)$ with $\mathsf{w}(u) \overset{\text{def}}{=} \min(\sqrt{u},1)$. Since $t_\alpha(u,v) \le (1-\alpha^{-1})\max(u,v)$, which for $u \ge v$ is $\alpha r - r^\alpha \le \alpha - 1$ at $r = v/u$, and $\mathsf{M}_\ell \le \min(u,v)$, one has $\Delta_\ell^\infty \ge 1/\alpha$. Hence eq. (SG.26) is at most a constant times $I(X)^2$, with

$$I(X) \overset{\text{def}}{=} \int_0^\infty \frac{X^{\alpha/2}h^{(\alpha-1)/2}}{X^\alpha + h^\alpha}\Big(1+\frac{1}{h}\Big)\mathsf{w}(h)\,\mathrm{d}h < \infty,$$

finite at zero by $\mathsf{w}(h)=\sqrt{h}$ and at infinity because $\alpha>1$. This gives finiteness, continuity on compact sets of X, and positivity from the zero-lag term.

The finite-dimensional integral is dominated in the same way. On $m_1,m_2\le\gamma/2$, lemma SG.14 gives $\Delta_\ell^N\ge c_\alpha$, and lemma SG.6(v) gives $\mu_i\Lambda_N(m)\asymp(m/X)^\alpha$ for $1/B\le m\le\gamma/2$, so the one-variable factor is bounded by the integrand of $I(X)$; on $m<1/B$, $\mu_i\Lambda_N(m)\asymp Bm\,i^{-\alpha}$ and $\vartheta_N\le C/m$ bound its integral by $Ci^{-\alpha/2}$. On $m\ge\gamma/4$, the bounds of lemma SG.17 give a one-variable integral at most

$$C\sqrt{\gamma}\,(i/N)^{\alpha/2}=C_\alpha X^{\alpha/2}\gamma^{(1-\alpha)/2}.$$

The rectangle $m_1<\gamma/4$, $m_2>\gamma/2$ has $\Delta_\ell^N\ge c$ and is bounded by the product of the two one-variable bounds, and the square $m_1,m_2\ge\gamma/4$ is $O(\gamma(i/N)^\alpha)=O(X^\alpha\gamma^{1-\alpha})$ by lemma SG.17. These bounds hold for the full lag sum.

At $X\to X_0$ the integrand converges on compact sets. By lemma SG.6(iv), $\mu_i\Lambda_N(m)\to(m/X_0)^\alpha$ and $m\Lambda_N'/\Lambda_N\to\alpha$, so $f_i(m)\to\zeta_{X_0}(m)$ and $\vartheta_N\to\vartheta_\infty$. The normalized sum T_N is a Riemann sum in $X'=K_\alpha j/B$,

$$\mathsf{T}_N(u,v)\to\frac{1}{K_\alpha}\int_0^\infty\frac{u^\alpha}{X'^\alpha+u^\alpha}\frac{v^\alpha}{X'^\alpha+v^\alpha}\,\mathrm{d}X'=t_\alpha(u,v),$$

by partial fractions and $\int_0^\infty(X'^\alpha+u^\alpha)^{-1}\mathrm{d}X'=K_\alpha u^{1-\alpha}$, with the value at $u=v$ given by $\int_0^\infty(1+y^\alpha)^{-2}\mathrm{d}y=(1-\alpha^{-1})K_\alpha$ [21, (5.12.3), (5.5.3)]. Since $\mathsf{M}_\ell/\gamma\to0$ and $\mathsf{T}_N\mathsf{D}_\ell^2/(\gamma m_1m_2)\to0$ on compact sets, $\Delta_\ell^N\to\Delta_\ell^\infty$. On a compact square $\mathsf{D}_\ell\le C\beta^{|\ell|}$, so the lag series converges uniformly there. The majorants remove the truncations, which proves $\mathfrak{v}_i(0)\to G_{\alpha,\beta}(X_0)$.

At $X\to0$ with $i\to\infty$, rescale $m=Xr$. On compact sets of $r>0$, $Bm=K_\alpha ir\to\infty$ and $m\to0$, so

$$f_i(Xr)\to\frac{r^\alpha}{1+r^\alpha},\qquad X\vartheta_N(Xr)\to\frac{\alpha}{r},\qquad\mathsf{D}_\ell(Xr,Xs)\sim\beta^{|\ell|}X\sqrt{rs},\qquad\Delta_\ell^N\to1,$$

by lemma SG.1(i), $Q_m\sim m/s_2$ and $E(m)\sim m$. Each lag term tends to

$$\beta^{|\ell|}\Big[\frac{\alpha}{\pi}\int_0^\infty\frac{r^{\alpha/2-1}}{1+r^\alpha}\,\mathrm{d}r\Big]^2=\beta^{|\ell|},$$

the integral being π/α. The contributions of $m<\delta X$ and $m>RX$ to I are $O(\delta^{\alpha/2})$ and $O(R^{-\alpha/2}+X^{\alpha/2})$, the dimension errors vanish, and on compact rescaled squares the lag series is dominated by $C\beta^{|\ell|}$. Hence $\mathfrak{v}_i(0)\to\sum_\ell\beta^{|\ell|}=(1+\beta)/(1-\beta)=1/s_2$. The same computation without dimension errors gives $G_{\alpha,\beta}(X)\to1/s_2$ as $X\to0$.

At $X\to\infty$ with $X/\gamma\to0$, rescale $m=Xr$ and $\ell/X\to\tau>0$. On the shell $m_2=m_1+\ell+O(1)$ one has $\mathsf{M}_\ell=m_1+O(1)$ and $\Delta_\ell^N\to1-t_\alpha(r,r+\tau)/(r+\tau)$, and lemma SG.22(iii) with $n=X$ gives

$$\overline{C}_{i,\ell}(0)\to C_\alpha^\infty(\tau)\overset{\text{def}}{=}\int_0^\infty\frac{\zeta_1(r+\tau)-\zeta_1(r)}{\tau}\Big(\frac{r}{r+\tau}\Big)^{\frac{\alpha-1}{2}}\mathrm{d}r,\qquad\zeta_1(r)=\frac{r^\alpha}{1+r^\alpha},$$

by the algebra of lemma SG.21 with $V_\alpha(p)$ replaced by r^α. The integral I divided by $\sqrt{X}$ is bounded, its parts $m<\delta X$ and $m>RX$ are $O(\delta^{(\alpha+1)/2})+o(1)$ and $O(R^{-(\alpha-1)/2})$, and the top bounds divided by $\sqrt{X}$ and X tend to zero because $X/\gamma\to0$. Lags above $2RX$ are exponentially small on the remaining square by lemma SG.22(i), and lags below ηX contribute at most $2\eta X+O(1)$. Hence $\mathfrak{v}_i(0)\sim2X\int_0^\infty C_\alpha^\infty$, and, with $\nu\overset{\text{def}}{=}(\alpha-1)/2$ and $r=ts$,

$$2\int_0^\infty C_\alpha^\infty(\tau)\,\mathrm{d}\tau=2\int_0^1\frac{t^\nu}{1-t}\Big\{\int_0^\infty\big(\zeta_1(s)-\zeta_1(ts)\big)\mathrm{d}s\Big\}\mathrm{d}t=2K_\alpha\int_0^1t^{\nu-1}\,\mathrm{d}t=\frac{4K_\alpha}{\alpha-1},$$

because $\int_0^\infty(\zeta_1(s)-\zeta_1(ts))\,\mathrm{d}s=\int_0^\infty\big((1+t^\alpha s^\alpha)^{-1}-(1+s^\alpha)^{-1}\big)\mathrm{d}s=K_\alpha(t^{-1}-1)$. The same computation without dimension errors gives $G_{\alpha,\beta}(X)\sim\frac{4K_\alpha}{\alpha-1}X$.

For the bulk matching, $V_\alpha(p)\sim(p/K_\alpha)^\alpha$ at zero, and the substitution $p=K_\alpha xr$, $q=K_\alpha xs$ in the double-integral form of lemma SG.21 turns $\Gamma_\alpha(x)/(K_\alpha x)$ into the double integral just evaluated. On

$p, q \le 1/2$ the continuum elasticity gives $\Delta_c \ge c_\alpha$, and the one-variable tails of $\sqrt{f_x(1-f_x)}/\sqrt{p}$, divided by $\sqrt{x}$, are $O(\delta^{(\alpha+1)/2})$ and $O(R^{-(\alpha-1)/2})$; the parts with p or q above $1/4$ are $O(x^\alpha)$ and $O(x^{(\alpha+1)/2})$, both $o(x)$. □

Proof of the volatility part of theorem SG.1 and of theorem SG.3(iii). Take $c_N = 1$ and a row sequence $i \to \infty$ along which i/N converges; as for the drift, this suffices. Let $\alpha \le 1$. If $i/N \to x > 0$, part (i) gives $\mathfrak{v}_i(0) \sim \gamma\Gamma_\alpha(x)$ and $\widehat{\mathfrak{v}}_i \sim \gamma\Gamma_\alpha(x)$, since $\gamma \to \infty$. If $i/N \to 0$, lemma SG.21 gives $\sqrt{\gamma\Gamma_\alpha(i/N)} = b_{i,N}(1+o(1))$, so $\widehat{\mathfrak{v}}_i$ is the quadratic of lemma SG.20 up to a factor $1+o(1)$, and lemma SG.20 gives the claim.

Let $\alpha > 1$. If $i/N \to x > 0$, then $X \to \infty$, part (i) gives $\mathfrak{v}_i(0) \sim \gamma\Gamma_\alpha(x)$, and by lemma SG.23

$$\widehat{\mathfrak{v}}_i = G_{\alpha,\beta}(X)\frac{(\alpha-1)\Gamma_\alpha(x)}{4K_\alpha^2 x} \sim \frac{4K_\alpha X}{\alpha-1}\cdot\frac{(\alpha-1)\Gamma_\alpha(x)}{4K_\alpha^2 x} = \gamma\Gamma_\alpha(x),$$

since $X/(K_\alpha x) = \gamma$. If $i/N \to 0$, the second factor of $\widehat{\mathfrak{v}}_i$ tends to one, and after a further subsequence along which X tends to 0, to a positive number, or to infinity, lemma SG.23 gives the three cases. The endpoint statements of theorem SG.3(iii) are those of lemma SG.23, the case $X \to \infty$ without $i/N \to 0$ through the uniform statement. □

SG.7 Proofs of the remaining statements

Proof of theorem SG.1. The drift and volatility parts are proved in appendices SG.5 and SG.6; the maximum of two sequences tending to zero tends to zero. □

Proof of proposition SG.5. Part (i) is lemma SG.9, the proof of theorem SG.3(i) for the positivity and continuity of Γ_α, lemma SG.21 for $\alpha \le 1$ and lemma SG.23 for $\alpha > 1$. Part (ii) is lemmas SG.13 and SG.23. □

Proof of corollary SG.1. Part (i) is theorem SG.2(i) and theorem SG.3(ii) at $i/i_{c,\mathfrak{v}} \to y = c_0^{-2/\alpha}x$, for which $2c_0y^{\alpha/2} = 2x^{\alpha/2}$ and $y^\alpha = c_0^{-2}x^\alpha$. For part (ii), the left column is the resolved case of theorems SG.2 and SG.3, the two indices of the case $\alpha \le 1$ differing by a fixed factor. In the right column the drift is the unresolved case of theorem SG.2. For the volatility, theorem SG.1 gives $\mathfrak{v}_i(0) \sim \widehat{\mathfrak{v}}_i$, and $\widehat{\mathfrak{v}}_i \sim \gamma\Gamma_\alpha(i/N)$ when $i/i_\star \to \infty$: for $\alpha \le 1$ because $\gamma\Gamma_\alpha(i/N) \to \infty$ there (mesoscopic rows by the proof of theorem SG.3(ii), bulk rows because $\gamma \to \infty$), and for $\alpha > 1$ by the computation in the last proof.

For part (iii), eq. (SG.32) and $i_c = B/K_\alpha$ give

$$\frac{i_\star}{N} = \left(\gamma s_2 d^2 \log^2\gamma\right)^{-1/\alpha}\ (\alpha<1), \quad \frac{i_\star}{N} = \left(\gamma s_2 d^2 \log^3\gamma\right)^{-1}\ (\alpha=1), \quad \frac{i_\star}{N} = \frac{1}{K_\alpha\gamma}\ (\alpha>1),$$

all tending to zero. A row sequence $i \to \infty$ with $i/i_\star \to 0$ exists if and only if $i_\star \to \infty$, which is eq. (SG.35), and $i = \lfloor\sqrt{i_\star}\rfloor$ realizes it; $i = N$ realizes the other window. Part (iv) equates $\sqrt{\mu_i/(\gamma s_2)}$ with $d_\alpha\sqrt{c_N K_\alpha}N^{-1/2}i^{-(\alpha-1)/2}$, which gives $i = B/(K_\alpha s_2 d_\alpha^2)$, and $1/s_2$ with $\frac{4K_\alpha^2}{\alpha-1}\frac{i}{B}$, which gives $i = B(\alpha-1)/(4s_2K_\alpha^2)$; neither involves c_N, and neither do the indices of eqs. (SG.32) and (SG.33). □

Proof of corollary SG.2. Let ϵ_N be the supremum in eq. (SG.31) over $i \ge n_N$. Then $(1-\epsilon_N)\widehat{\mathfrak{d}}_i \le \mathfrak{d}_i(0) \le (1+\epsilon_N)\widehat{\mathfrak{d}}_i$ for $i \in I_N$; multiplying by $a_{i,N} \ge 0$ and summing bounds the first ratio between $1-\epsilon_N$ and $1+\epsilon_N$. The second ratio is the same with $\mathfrak{v}_i(0)$ and $\widehat{\mathfrak{v}}_i$. □

Proof of proposition SG.7. The drift error is uniform and small relative to the approximant. By eq. (SG.13) and $Q_\gamma \ge 1$,

$$\max_i|\mathfrak{d}_i(\varrho_N) - \mathfrak{d}_i(0)| \le \frac{C}{\gamma}\sqrt{\varrho_N Q_\gamma} = C\,a_N\left(\frac{\varrho_N Q_\gamma}{c_N N^{-\alpha}\gamma^2}\right)^{1/2} = o(a_N),$$

while $\widehat{\mathfrak{d}}_i \geq c\, a_N$ for all rows (end of appendix SG.5). The volatility error is uniformly small: by eq. (SG.21) with $H = H_N$, $D_i \leq D_\star$ and $E_i \leq E_\star$,

$$\max_i |\mathfrak{v}_i(\varrho_N) - \mathfrak{v}_i(0)| \leq 2(1+2H_N)\sqrt{D_\star}\,\varrho_N^{1/4} + \frac{4E_\star \beta^{H_N/3}}{1-\beta^{1/3}} \longrightarrow 0,$$

and the same bound holds for each lag coefficient by proposition SG.4(i). Since $\mathfrak{v}_i(0) \geq 1$, the additive errors preserve every relative statement, and they preserve the limits of theorem SG.3, which are positive or infinite. For existence, choose H_N with $E_\star \beta^{H_N/3} \leq 1/N$, and then $\varrho_N \in (0, \bar{z}_N]$ small enough for the first two quantities of eq. (SG.38) to be at most $1/N$. □

SH Large-batch drift for a general spectrum

This section computes the half-anisotropic drift coefficient $\mathfrak{d}_i = \mathfrak{d}(\mu_i, 1)$ of corollary SC.2 when the batch oversamples the input dimension, for an arbitrary output spectrum. The main result, theorem SH.1, shows that when $3(1-\beta^2)N_{\text{in}} < B$ the coefficient equals $\sqrt{B_{\text{eff}}/N_{\text{in}}}$ times a profile $\mathfrak{d}^{\text{os}}$ of the empirical spectral distribution of Σ_{out}, up to a factor in $[1-\epsilon_{\text{os}}, (1-\epsilon_{\text{os}})^{-1}]$ with $\epsilon_{\text{os}} = 3(1-\beta^2)N_{\text{in}}/B$. This holds at every normalized ridge, including zero ridge, for every eigenvalue, every spectrum and every pair of dimensions. Corollary SH.1 reads the result as the expected regularized update.

Every statement except corollary SH.1 concerns the deterministic coefficient at a fixed finite dimension. No statement concerns the algorithm at zero ridge, a joint limit in the aspect ratio and the dimension, or the volatility.

SH.1 The results in brief

This subsection states the results informally, in the square case $N_{\text{out}} = N_{\text{in}} = N$ at zero ridge, and points to the formal statements.

The square-root law and packed modes. When B_{eff} exceeds $3(1+\beta)^2 N$, the drift coefficient of every mode is at most the square-root law,

$$\mathfrak{d}_i \lesssim \sqrt{\frac{B_{\text{eff}}}{2N}}\,\sqrt{\mu_i},$$

and it is within a constant of this law whenever the mode is *packed*: a positive fraction δ of all eigenvalues lies below a fixed multiple $K\mu_i$ (definition SH.2 and corollary SH.2). The constant is $0.94\sqrt{\delta/(1+K)}$. Packing at the smallest eigenvalue implies packing at every mode, and every mode of a power law is packed (remark SH.3). In expectation the update is then $\sqrt{B_{\text{eff}}/(2N)}$ times a square-root preconditioner applied to the error, on every packed mode (corollary SH.1).

The exact law. The two-sided bound comes from an exact formula (theorem SH.1 and lemma SH.8),

$$\mathfrak{d}_i = \vartheta_i \sqrt{\frac{B_{\text{eff}}}{2N}}\,\sqrt{\mu_i}\,\chi(\mu_i), \qquad \chi(\mu) = \int_0^\infty \frac{2}{\pi}\frac{\sqrt{y}}{(1+y)^2}\sqrt{1-m(\mu y)}\,\mathrm{d}y, \qquad m(c) = \frac{1}{N}\sum_k \frac{\mu_k}{c+\mu_k},$$

with ϑ_i within a factor $1 \pm O(N/B_{\text{eff}})$ of one. The efficiency $\chi(\mu) \in (0,1]$ has a probabilistic meaning (remark SH.2): $1-m(c)$ is the fraction of the spectrum below c after each eigenvalue is multiplied by an independent random factor, and $\chi(\mu)$ is an average of its square root around the level μ. It is bounded by an explicit count,

$$\mathsf{s}(\mu) \leq \chi(\mu) \leq \sqrt{\mathsf{s}(\mu)}, \qquad \mathsf{s}(\mu) = \frac{1}{N}\sum_k \frac{1+2\sqrt{\mu_k/\mu}}{\left(1+\sqrt{\mu_k/\mu}\right)^2}$$

(lemma SH.11), where each eigenvalue contributes about 1 if it is below μ and about $2\sqrt{\mu/\mu_k}$ if it is far above.

Head and tail modes. The efficiency tends to one exactly on head modes, which lie above all but a vanishing fraction of the spectrum (proposition SH.1). A mode that lies far below most of the spectrum is a tail mode: its drift is of gradient-descent order up to a logarithm (proposition SH.2). For the slowest mode, the gain over gradient descent lies between the logarithm and the square root of the condition number, according to how much of the spectrum sits near the smallest eigenvalue (corollaries SH.3 and SH.4). For power laws every mode keeps the square-root law up to a constant depending only on the exponent, so the slowest mode gains the square root of the condition number (corollary SH.5).

SH.2 Standing setup

Data and scales. Throughout the section the input covariance is $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$, and the output covariance is $\Sigma_{\text{out}} = \sum_{k=1}^{N_{\text{out}}} \mu_k \, u_k \otimes u_k$, positive semidefinite with orthonormal eigenvectors (u_k). The integers $N_{\text{out}}, N_{\text{in}}, B \geq 1$ are arbitrary. The momentum $\beta \in [0, 1)$ is fixed, the weights are the EMA weights $w_j = (1-\beta)\beta^j$ of eq. (SA.18), and

$$s_2 = \frac{1-\beta}{1+\beta}, \qquad B_{\text{eff}} \stackrel{\text{def}}{=} \frac{B}{s_2}, \qquad \gamma_{\text{in}} = \frac{N_{\text{in}}}{B}, \qquad \gamma_{\text{out}} = \frac{N_{\text{out}}}{B}, \qquad r \stackrel{\text{def}}{=} \frac{\gamma_{\text{out}}}{\gamma_{\text{in}}} = \frac{N_{\text{out}}}{N_{\text{in}}}.$$

The effective batch B_{eff} is that of section 3. The largest weight is $w_0 = 1 - \beta$, and $w_0^2/s_2 = 1 - \beta^2$. For a function ϕ on $[0, \infty)$ we write $\phi(\Sigma_{\text{out}}) \stackrel{\text{def}}{=} \sum_k \phi(\mu_k) \, u_k \otimes u_k$.

The branch and the drift coefficient. For $s > 0$, $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)(-s)$ denotes the solution of the weighted system eq. (SB.2) for the EMA weights and the spectral measures $\nu_{\text{out}} = B^{-1} \sum_k \delta_{\mu_k}$ and $\nu_{\text{in}} = \gamma_{\text{in}} \, \delta_1$ of eq. (SA.41). It exists and is unique by theorem SB.1(i), applied with $C_\Sigma = \max(1, \|\Sigma_{\text{out}}\|)$ and $\gamma_\star = \max(\gamma_{\text{in}}, \gamma_{\text{out}})$, and it is continuous in s by theorem SB.1(ii). Since ν_{in} is a point mass, the system reads, with every quantity evaluated at $-s$,

$$\sigma = \int \frac{x \, \nu_{\text{out}}(\mathrm{d}x)}{s + \tilde{s}_1 x}, \qquad \tilde{\sigma} = \frac{\gamma_{\text{in}}}{s + s_1}, \qquad q = s\sigma\tilde{\sigma}, \qquad \tilde{s}_1 = s\tilde{\sigma} \, F_\beta(q), \qquad s_1 = s\sigma \, F_\beta(q). \tag{SH.1}$$

The functions J, F_β and L_β are those of eq. (SB.1) for the EMA weights. For $s > 0$ and $\mu \geq 0$ put

$$\mathsf{k}(s; \mu) \stackrel{\text{def}}{=} \frac{s \, L_\beta\big(q(-s)\big) \, \mu}{\big(s + \tilde{s}_1(-s)\mu\big)\big(s + s_1(-s)\big)}, \tag{SH.2}$$

and for $\varrho > 0$

$$\mathfrak{d}_\varrho(\mu) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi} \int_0^\infty \mathsf{k}(\varrho + u^2; \mu) \, \mathrm{d}u, \qquad \mathfrak{d}_i(\varrho) \stackrel{\text{def}}{=} \mathfrak{d}_\varrho(\mu_i). \tag{SH.3}$$

This is the kernel $\mathfrak{d}(\mu, 1)$ of eq. (SC.4) at the ridge ϱ, and $\mathfrak{d}_i(\varrho)$ is the row coefficient of corollary SC.2. Since $0 \leq J \leq 1$, one has $0 \leq L_\beta \leq \sum_j w_j = 1$, and the integrand satisfies $0 \leq \mathsf{k}(s; \mu) \leq \mu/s$; it is continuous in $s > 0$ because the branch is continuous and L_β is continuous (lemma SB.1(ii)). The zero-ridge coefficients $\mathfrak{d}_0(\mu)$ and $\mathfrak{d}_i(0)$ are defined in theorem SH.1(ii) as the limits as $\varrho \downarrow 0$. In the square case with $\beta > 0$ and a positive spectrum, proposition SG.2(ii) shows that $\mathfrak{d}_i(\varrho)$ converges to the coefficient $\mathfrak{d}_i(0)$ of eq. (SG.12), so the two definitions of $\mathfrak{d}_i(0)$ agree.

Conventions. A probability measure is a Borel probability measure on $[0, \infty)$; the generic letter is ν. The variable g is standard Gaussian. Every constant of the section is an explicit absolute number unless its dependence is stated.

SH.3 The limit profile

For a probability measure ν and $c > 0$ put

$$m_\nu(c) \stackrel{\text{def}}{=} \int \frac{x \, \nu(\mathrm{d}x)}{c + x}, \qquad 1 - m_\nu(c) = \int \frac{c \, \nu(\mathrm{d}x)}{c + x} \in (0, 1]. \tag{SH.4}$$

For the empirical measure of a spectrum, each eigenvalue x enters $m_\nu(c)$ with the weight $x/(c+x)$, which is close to one when $x \gg c$ and close to zero when $x \ll c$. The number $m_\nu(c)$ is therefore a

soft count of the share of the spectrum above c. It is nonincreasing in c. For $\hat{s} > 0$ and $b \geq 0$ we also write

$$m_\nu(\hat{s}/b) \stackrel{\text{def}}{=} \int \frac{bx\,\nu(\mathrm{d}x)}{\hat{s} + bx},$$

which agrees with eq. (SH.4) for $b > 0$ and vanishes at $b = 0$. In the square case $N_{\text{out}} = N_{\text{in}} = N$, the participation function of definition SG.1 is $m_N(\Lambda) = \gamma\, m_{\hat{\nu}}(1/\Lambda)$, where $\hat{\nu} = N^{-1}\sum_k \delta_{\mu_k}$ and $\gamma = N/B$.

Lemma SH.1 (The profile equation)**.** *Let ν be a probability measure and $r > 0$. For every $\hat{s} > 0$ the equation*

$$b + r\, m_\nu(\hat{s}/b) = 1 \tag{SH.5}$$

has exactly one solution $b = b_0(\hat{s})$ in $(0, 1]$. The function b_0 is continuous and nondecreasing on $(0, \infty)$.

Proof. The left side of eq. (SH.5) is a continuous strictly increasing function of $b \in [0, 1]$. Its integrand $bx/(\hat{s} + bx)$ lies in $[0, 1)$ and is nondecreasing in b, so for $0 \leq b < b' \leq 1$

$$\big(b' + r\, m_\nu(\hat{s}/b')\big) - \big(b + r\, m_\nu(\hat{s}/b)\big) \geq b' - b,$$

and the left side is continuous in b. At $b = 0$ the left side is 0, and at $b = 1$ it is at least 1. The intermediate value theorem gives exactly one root in $(0, 1]$.

The root is continuous in $\hat{s}$ by a compactness argument. The map $(\hat{s}, b) \mapsto m_\nu(\hat{s}/b)$ is jointly continuous on $(0, \infty) \times [0, 1]$. If $\hat{s}_n \to \hat{s} > 0$, every limit point $b^* \in [0, 1]$ of $b_0(\hat{s}_n)$ therefore satisfies $b^* + r\, m_\nu(\hat{s}/b^*) = 1$, and $b^* = b_0(\hat{s})$ by uniqueness.

The root is nondecreasing because $m_\nu(\hat{s}/b)$ is nonincreasing in $\hat{s}$. For $\hat{s}' \geq \hat{s}$, the left side of eq. (SH.5) at $\hat{s}'$ is at most the left side at $\hat{s}$, for every b, so its root is at least $b_0(\hat{s})$. □

Definition SH.1 (Oversampled objects)**.** *Let ν be a probability measure and $r > 0$.*

(i) The profile root $b_0 = b_0^{\nu,r}$ is the function of lemma SH.1.

(ii) For $\mu \geq 0$ and $\hat{\varrho} \geq 0$, the oversampled drift profile is

$$\mathfrak{d}^{\text{os}}_{\nu,r}(\mu; \hat{\varrho}) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi} \int_0^\infty \frac{b_0(\hat{s})\,\mu}{\hat{s} + b_0(\hat{s})\,\mu}\,\mathrm{d}v, \qquad \hat{s} = \hat{\varrho} + v^2. \tag{SH.6}$$

In the square case we write $\mathfrak{d}^{\text{os}}_\nu \stackrel{\text{def}}{=} \mathfrak{d}^{\text{os}}_{\nu,1}$ and $\mathfrak{d}^{\text{os}}_\nu(\mu) \stackrel{\text{def}}{=} \mathfrak{d}^{\text{os}}_{\nu,1}(\mu; 0)$.

(iii) The density ϖ on $(0, \infty)$ and the square-root efficiency χ_ν are

$$\varpi(y) \stackrel{\text{def}}{=} \frac{2}{\pi}\,\frac{\sqrt{y}}{(1+y)^2}, \qquad \chi_\nu(\mu) \stackrel{\text{def}}{=} \int_0^\infty \varpi(y)\sqrt{1 - m_\nu(\mu y)}\,\mathrm{d}y \qquad (\mu > 0). \tag{SH.7}$$

(iv) At a finite dimension, the empirical spectral probability measure of Σ_{out}, the rescaled ridge and the small parameter are

$$\begin{gathered}\hat{\nu}_{\text{out}} \stackrel{\text{def}}{=} \frac{1}{N_{\text{out}}}\sum_{k=1}^{N_{\text{out}}} \delta_{\mu_k} = \frac{\nu_{\text{out}}}{\gamma_{\text{out}}}, \qquad \hat{\varrho} \stackrel{\text{def}}{=} \frac{\varrho}{s_2\gamma_{\text{in}}} \quad (\varrho \geq 0), \\ \epsilon_{\text{os}} \stackrel{\text{def}}{=} 3(1-\beta^2)\gamma_{\text{in}} = 3(1+\beta)^2\,\frac{N_{\text{in}}}{B_{\text{eff}}}.\end{gathered} \tag{SH.8}$$

The profile depends on the spectrum only through the probability measure and the shape ratio, and on the ridge only through the rescaled ridge. The oversampled regime of this section is $\epsilon_{\text{os}} < 1$, that is $B > 3(1-\beta^2)N_{\text{in}}$. Lemma SH.8 below shows that $\mathfrak{d}^{\text{os}}_\nu(\mu) = \sqrt{\mu/2}\,\chi_\nu(\mu)$ in the square case.

Lemma SH.2 (Ceilings of the profile)**.** *Let ν be a probability measure, $r > 0$ and $\mu \geq 0$. The profile $\mathfrak{d}^{\text{os}}_{\nu,r}(\mu; \hat{\varrho})$ is finite, continuous in $\hat{\varrho} \in [0, \infty)$, positive if and only if $\mu > 0$, and*

$$\mathfrak{d}^{\text{os}}_{\nu,r}(\mu; \hat{\varrho}) \leq \sqrt{\mu/2}, \qquad \mathfrak{d}^{\text{os}}_{\nu,r}(\mu; \hat{\varrho}) \leq \frac{\mu}{\sqrt{2\hat{\varrho}}} \quad (\hat{\varrho} > 0). \tag{SH.9}$$

Proof. The bounds come from $b_0 \le 1$. The map $y \mapsto y\mu/(\hat{s}+y\mu)$ is nondecreasing, and $\hat{s} = \hat{\varrho} + v^2$, so the integrand of eq. (SH.6) satisfies

$$\frac{b_0(\hat{s})\,\mu}{\hat{s}+b_0(\hat{s})\,\mu} \le \frac{\mu}{\hat{s}+\mu} \le \min\Big\{\frac{\mu}{v^2+\mu},\ \frac{\mu}{\hat{\varrho}+v^2}\Big\}.$$

The two elementary integrals

$$\frac{\sqrt{2}}{\pi}\int_0^\infty \frac{\mu\,\mathrm{d}v}{v^2+\mu} = \sqrt{\mu/2}, \qquad \frac{\sqrt{2}}{\pi}\int_0^\infty \frac{\mu\,\mathrm{d}v}{\hat{\varrho}+v^2} = \frac{\mu}{\sqrt{2\hat{\varrho}}}$$

give eq. (SH.9). Continuity in $\hat{\varrho}$ is dominated convergence with the majorant $\mu/(v^2+\mu)$, using the continuity of b_0 (lemma SH.1). The integrand is positive when $\mu > 0$ because $b_0 > 0$, and it vanishes when $\mu = 0$. □

SH.4 Scalar bounds

The error constant of theorem SH.1 comes from two-sided bounds on F_β and L_β and from an a priori bound on the branch variable q.

Lemma SH.3 (Scalar bounds)**.** *(i) For $x \ge 0$,*

$$\frac{1}{1+3x} \le J(x) \le 1.$$

(ii) For $q \ge 0$,

$$\frac{s_2}{1+3(1-\beta)^2 q} \le F_\beta(q) \le s_2, \qquad \frac{1}{1+3(1-\beta)^2 q} \le L_\beta(q) \le 1.$$

(iii) Assume that $\beta > 0$, or that $\beta = 0$ and $\gamma_{\mathrm{in}} < 1$. Then there is a unique $q_\star > 0$ with $q_\star F_\beta(q_\star) = \gamma_{\mathrm{in}}$, and $qF_\beta(q) < \gamma_{\mathrm{in}}$ holds if and only if $q < q_\star$.

(iv) If $\epsilon_{\mathrm{os}} < 1$, then the hypothesis of (iii) holds and

$$1 + 3(1-\beta)^2 q_\star \le \frac{1}{1-\epsilon_{\mathrm{os}}}.$$

Proof. The lower bound in (i) is the Cauchy–Schwarz inequality with the Gaussian moments $\mathbb{E}\,g^2 = 1$ and $\mathbb{E}\,g^4 = 3$:

$$1 = \mathbb{E}\,g^2 = \mathbb{E}\left[\Big(\frac{g^2}{1+xg^2}\Big)^{1/2}\big(g^2(1+xg^2)\big)^{1/2}\right] \le J(x)^{1/2}\,(1+3x)^{1/2}.$$

The upper bound holds because $g^2/(1+xg^2) \le g^2$.

Part (ii) follows from (i) term by term. Every weight satisfies $w_j \le w_0 = 1-\beta$, so

$$w_j^2\,J(w_j^2 q) \ge \frac{w_j^2}{1+3(1-\beta)^2 q}, \qquad w_j\,J(w_j^2 q) \ge \frac{w_j}{1+3(1-\beta)^2 q}.$$

Summing with $\sum_j w_j^2 = s_2$ and $\sum_j w_j = 1$ gives the lower bounds, and $J \le 1$ gives the upper bounds.

Part (iii) is lemma SB.1(iii). The function $q \mapsto qF_\beta(q)$ increases strictly from 0 to the number of positive weights, which is ∞ for $\beta > 0$ and 1 for $\beta = 0$. In both cases γ_{in} lies in its range.

For part (iv), if $\beta = 0$ then $\gamma_{\mathrm{in}} = \epsilon_{\mathrm{os}}/3 < 1$, so the hypothesis of (iii) holds. The lower bound of (ii) at $q_\star$ gives

$$\frac{s_2\,q_\star}{1+3(1-\beta)^2 q_\star} \le q_\star F_\beta(q_\star) = \gamma_{\mathrm{in}}.$$

Multiplying by $(1-\beta)^2/s_2 = 1-\beta^2$ and writing $y = 3(1-\beta)^2 q_\star$, this reads $y \le \epsilon_{\mathrm{os}}(1+y)$, that is $y(1-\epsilon_{\mathrm{os}}) \le \epsilon_{\mathrm{os}}$. Hence

$$1 + y \le 1 + \frac{\epsilon_{\mathrm{os}}}{1-\epsilon_{\mathrm{os}}} = \frac{1}{1-\epsilon_{\mathrm{os}}}.$$

□

SH.5 The oversampled limit law

The proof of theorem SH.1 is exact rather than asymptotic. In the rescaled variables $\hat{s} = s/(s_2\gamma_{\text{in}})$ and $b = \tilde{s}_1(-s)/(s_2\gamma_{\text{in}})$, the branch eq. (SH.1) becomes the profile equation eq. (SH.5) with one extra factor $F_\beta(q)/s_2 \in (0,1]$ (lemma SH.4). A comparison lemma places b between this factor times b_0 and b_0 (lemma SH.5). The coupling identity bounds q by $q_\star$, and lemma SH.3 then bounds the two scalar factors uniformly in s.

Lemma SH.4 (Exact reduction of the half-anisotropic branch)**.** *Fix $s > 0$, write $(\sigma, \tilde{\sigma}, \tilde{s}_1, s_1, q)$ for the branch at $-s$, and put $F = F_\beta(q)$, $L = L_\beta(q)$ and $\widehat{F} \overset{\text{def}}{=} F/s_2 \in (0,1]$.*

(a) One has $s + s_1 = s(1+\sigma F)$ and $\tilde{s}_1 = \gamma_{\text{in}} F/(1+\sigma F) > 0$.

(b) One has $qF = \sigma\tilde{s}_1 = \gamma_{\text{in}}\sigma F/(1+\sigma F) < \gamma_{\text{in}}$, and $\tilde{s}_1 = F(\gamma_{\text{in}} - qF)$.

(c) With $\hat{s} \overset{\text{def}}{=} s/(s_2\gamma_{\text{in}})$ and $b \overset{\text{def}}{=} \tilde{s}_1/(s_2\gamma_{\text{in}})$, one has $b \in (0, \widehat{F}]$ and

$$\frac{b}{\widehat{F}} + r\, m_{\hat{\nu}_{\text{out}}}(\hat{s}/b) = 1.$$

(d) For every $\mu \geq 0$,

$$\mathsf{k}(s;\mu) = \frac{1}{s_2\gamma_{\text{in}}}\,\frac{L}{\widehat{F}}\,\frac{b\mu}{\hat{s}+b\mu}.$$

Proof. Part (a) is substitution in eq. (SH.1). From $s_1 = s\sigma F$ we get $s + s_1 = s(1+\sigma F)$. Hence $\tilde{\sigma} = \gamma_{\text{in}}/(s(1+\sigma F))$, and

$$\tilde{s}_1 = s\tilde{\sigma}F = \frac{\gamma_{\text{in}}F}{1+\sigma F},$$

which is positive because $F \geq s_2/(1+3(1-\beta)^2 q) > 0$ by lemma SH.3(ii). The same lemma gives $\widehat{F} \in (0,1]$.

Part (b) is the coupling eq. (SB.4) in this setting. By the definition of q and by (a),

$$qF = s\sigma\tilde{\sigma}F = \sigma\tilde{s}_1 = \frac{\gamma_{\text{in}}\sigma F}{1+\sigma F} < \gamma_{\text{in}}.$$

Rewriting $\tilde{s}_1(1+\sigma F) = \gamma_{\text{in}}F$ as $\tilde{s}_1 = \gamma_{\text{in}}F - F\sigma\tilde{s}_1$ gives $\tilde{s}_1 = F(\gamma_{\text{in}} - qF)$.

Part (c) is (b) in the rescaled variables. Since $\nu_{\text{out}} = \gamma_{\text{out}}\hat{\nu}_{\text{out}} = r\gamma_{\text{in}}\hat{\nu}_{\text{out}}$,

$$\sigma = r\gamma_{\text{in}}\int \frac{x\,\hat{\nu}_{\text{out}}(\mathrm{d}x)}{s_2\gamma_{\text{in}}(\hat{s}+bx)} = \frac{r}{s_2}\int \frac{x\,\hat{\nu}_{\text{out}}(\mathrm{d}x)}{\hat{s}+bx}, \qquad \text{so} \qquad qF = \sigma\tilde{s}_1 = r\gamma_{\text{in}}\, m_{\hat{\nu}_{\text{out}}}(\hat{s}/b).$$

The identity $\tilde{s}_1 = F(\gamma_{\text{in}} - qF)$ of (b) then reads $s_2\gamma_{\text{in}}\, b = F\gamma_{\text{in}}\big(1 - r\, m_{\hat{\nu}_{\text{out}}}(\hat{s}/b)\big)$, which is the displayed equation after division by $s_2\gamma_{\text{in}}\widehat{F}$. One has $b > 0$ by (a), and $b \leq \widehat{F}$ because $m_{\hat{\nu}_{\text{out}}} \geq 0$.

Part (d) uses (a) twice. By (a), $s + s_1 = s(1+\sigma F)$ and $1/(1+\sigma F) = \tilde{s}_1/(\gamma_{\text{in}}F)$, so

$$\mathsf{k}(s;\mu) = \frac{L\mu}{(s+\tilde{s}_1\mu)(1+\sigma F)} = \frac{L}{\gamma_{\text{in}}F}\,\frac{\tilde{s}_1\mu}{s+\tilde{s}_1\mu} = \frac{L}{s_2\gamma_{\text{in}}\,\widehat{F}}\,\frac{b\mu}{\hat{s}+b\mu}. \qquad \square$$

Lemma SH.5 (Comparison)**.** *Let ν be a probability measure, $r > 0$, $\hat{s} > 0$ and $\widehat{F} \in (0,1]$, and let $b > 0$ satisfy $b/\widehat{F} + r\, m_\nu(\hat{s}/b) = 1$. Then $\widehat{F}\, b_0(\hat{s}) \leq b \leq b_0(\hat{s})$, and for every $\mu \geq 0$*

$$\widehat{F}\,\frac{b_0(\hat{s})\,\mu}{\hat{s}+b_0(\hat{s})\,\mu} \leq \frac{b\mu}{\hat{s}+b\mu} \leq \frac{b_0(\hat{s})\,\mu}{\hat{s}+b_0(\hat{s})\,\mu}.$$

Proof. Both bounds on b come from the strictly increasing function $y \mapsto y/\widehat{F} + r\, m_\nu(\hat{s}/y)$, which equals 1 at $y = b$. At $y = b_0 = b_0(\hat{s})$, since $\widehat{F} \leq 1$,

$$\frac{b_0}{\widehat{F}} + r\, m_\nu(\hat{s}/b_0) \geq b_0 + r\, m_\nu(\hat{s}/b_0) = 1, \qquad \text{so} \qquad b \leq b_0.$$

At $y = \widehat{F} b_0$, since $m_\nu(\hat{s}/y)$ is nondecreasing in y and $\widehat{F} b_0 \le b_0$,

$$b_0 + r\, m_\nu\big(\hat{s}/(\widehat{F} b_0)\big) \le b_0 + r\, m_\nu(\hat{s}/b_0) = 1, \qquad \text{so} \qquad \widehat{F} b_0 \le b.$$

The map $y \mapsto y\mu/(\hat{s} + y\mu)$ is nondecreasing, which gives the right inequality of the display. The left one follows from

$$\frac{b\mu}{\hat{s} + b\mu} \ge \frac{\widehat{F} b_0 \mu}{\hat{s} + \widehat{F} b_0 \mu} \ge \widehat{F}\, \frac{b_0 \mu}{\hat{s} + b_0 \mu}. \qquad \square$$

Theorem SH.1 (Oversampled limit law)**.** *Let $N_{\text{out}}, N_{\text{in}}, B \ge 1$, let Σ_{out} be positive semidefinite, let $\beta \in [0, 1)$, and assume that $\beta > 0$, or that $\beta = 0$ and $\gamma_{\text{in}} < 1$. Let $q_\star$ be as in lemma SH.3(iii), and put $\Lambda \overset{\text{def}}{=} 1 + 3(1-\beta)^2 q_\star$. Write $\mathfrak{d}^{\text{os}} = \mathfrak{d}^{\text{os}}_{\hat{\nu}_{\text{out}}, r}$.*

(i) For every $\varrho > 0$ and $\mu \ge 0$,

$$\Lambda^{-1}\, \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho}) \le \sqrt{s_2 \gamma_{\text{in}}}\, \mathfrak{d}_\varrho(\mu) \le \Lambda\, \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho}).$$

(ii) For every $\mu \ge 0$ the limit

$$\mathfrak{d}_0(\mu) \overset{\text{def}}{=} \lim_{\varrho \downarrow 0} \mathfrak{d}_\varrho(\mu) = \frac{\sqrt{2}}{\pi} \int_0^\infty \mathsf{k}(u^2; \mu)\, \mathrm{d}u$$

exists in $[0, \infty)$, and it satisfies the bounds of (i) with $\varrho = \hat{\varrho} = 0$.

(iii) If $\epsilon_{\text{os}} < 1$, then $\Lambda \le (1 - \epsilon_{\text{os}})^{-1}$. Consequently, for every $\varrho \ge 0$ and $1 \le i \le N_{\text{out}}$,

$$\mathfrak{d}_i(\varrho) = \vartheta_i \sqrt{\frac{B_{\text{eff}}}{N_{\text{in}}}}\, \mathfrak{d}^{\text{os}}(\mu_i; \hat{\varrho}) \qquad \text{for some } \vartheta_i \in \big[1 - \epsilon_{\text{os}}, (1 - \epsilon_{\text{os}})^{-1}\big], \tag{SH.10}$$

and for every $\mu \ge 0$

$$\Big| \sqrt{s_2 \gamma_{\text{in}}}\, \mathfrak{d}_\varrho(\mu) - \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho}) \Big| \le \frac{\epsilon_{\text{os}}}{1 - \epsilon_{\text{os}}}\, \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho}) \le \frac{\epsilon_{\text{os}}}{1 - \epsilon_{\text{os}}} \sqrt{\frac{\mu}{2}}.$$

(iv) For $\varrho \ge 0$ let $\vartheta_\varrho(\mu) \overset{\text{def}}{=} \sqrt{s_2 \gamma_{\text{in}}}\, \mathfrak{d}_\varrho(\mu) / \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho})$ for $\mu > 0$ and $\vartheta_\varrho(0) \overset{\text{def}}{=} 1$. Then $\operatorname{spec} \vartheta_\varrho(\Sigma_{\text{out}}) \subset [\Lambda^{-1}, \Lambda]$ and

$$\mathfrak{d}_\varrho(\Sigma_{\text{out}}) = \sqrt{\frac{B_{\text{eff}}}{N_{\text{in}}}}\, \vartheta_\varrho(\Sigma_{\text{out}})\, \mathfrak{d}^{\text{os}}(\Sigma_{\text{out}}; \hat{\varrho}).$$

Proof. The proof starts from a uniform bound on the two scalar factors. For $s > 0$ let $b(s)$, $\widehat{F}(s)$ and $L(s)$ be as in lemma SH.4. By lemma SH.4(b), $q(-s) F_\beta(q(-s)) < \gamma_{\text{in}}$, so $q(-s) < q_\star$ by lemma SH.3(iii). Lemma SH.3(ii) then gives, for every $s > 0$,

$$L(s) \ge \frac{1}{1 + 3(1-\beta)^2 q(-s)} \ge \Lambda^{-1}, \qquad \widehat{F}(s) \ge \frac{1}{1 + 3(1-\beta)^2 q(-s)} \ge \Lambda^{-1}. \tag{SH.11}$$

Part (i) is a pointwise sandwich of the integrands. Substitute $u = \sqrt{s_2 \gamma_{\text{in}}}\, v$ in eq. (SH.3), so that $\varrho + u^2 = s_2 \gamma_{\text{in}}\, \hat{s}$ with $\hat{s} = \hat{\varrho} + v^2$. By lemma SH.4(d),

$$\sqrt{s_2 \gamma_{\text{in}}}\, \mathfrak{d}_\varrho(\mu) = \frac{\sqrt{2}}{\pi} \int_0^\infty I(\hat{s})\, \mathrm{d}v, \qquad I(\hat{s}) \overset{\text{def}}{=} \frac{L}{\widehat{F}}\, \frac{b\mu}{\hat{s} + b\mu}\Big|_{s = s_2 \gamma_{\text{in}} \hat{s}}.$$

Write $I_0(\hat{s}) \overset{\text{def}}{=} b_0(\hat{s})\mu/(\hat{s} + b_0(\hat{s})\mu)$ with $b_0 = b_0^{\hat{\nu}_{\text{out}}, r}$. Lemma SH.4(c) places b under the hypotheses of lemma SH.5, which gives

$$L\, I_0 \le I \le \widehat{F}^{-1} I_0,$$

using $L \le 1$ on the right. By eq. (SH.11), $\Lambda^{-1} I_0 \le I \le \Lambda I_0$ for every $\hat{s} > 0$. Integrating in v proves (i), since $\frac{\sqrt{2}}{\pi} \int_0^\infty I_0\, \mathrm{d}v = \mathfrak{d}^{\text{os}}(\mu; \hat{\varrho})$.

Part (ii) is dominated convergence with a majorant that does not depend on the ridge. By lemma SH.4(d), $b \le \widehat{F} \le 1$, $L \le 1$ and eq. (SH.11),

$$\mathsf{k}(s;\mu) \le \frac{\Lambda}{s_2\gamma_{\text{in}}} \min\Big\{1,\ \frac{b\mu}{\hat{s}}\Big\} \le \Lambda \min\Big\{\frac{1}{s_2\gamma_{\text{in}}},\ \frac{\mu}{s}\Big\},$$

and the right side is nonincreasing in s. Hence, for $\varrho > 0$ and $u > 0$,

$$\mathsf{k}(\varrho + u^2;\mu) \le \Lambda \min\Big\{\frac{1}{s_2\gamma_{\text{in}}},\ \frac{\mu}{u^2}\Big\},$$

which is integrable on $(0,\infty)$. The integrand $\mathsf{k}(\cdot;\mu)$ is continuous on $(0,\infty)$ (appendix SH.2), so $\mathsf{k}(\varrho+u^2;\mu) \to \mathsf{k}(u^2;\mu)$ for every $u > 0$, and dominated convergence gives the existence and the value of the limit. At $\varrho = 0$ the substitution and the pointwise sandwich of (i) apply without change, with $\hat{s} = v^2 > 0$.

Part (iii) combines (i) and (ii) with lemma SH.3(iv), which gives $\Lambda \le (1-\epsilon_{\text{os}})^{-1}$ and $\Lambda^{-1} \ge 1-\epsilon_{\text{os}}$. If $\mu_i = 0$, both sides of eq. (SH.10) vanish and $\vartheta_i = 1$ serves. If $\mu_i > 0$, the profile is positive by lemma SH.2, and $\vartheta_i \overset{\text{def}}{=} \sqrt{s_2\gamma_{\text{in}}}\,\mathfrak{d}_i(\varrho)/\mathfrak{d}^{\text{os}}(\mu_i;\hat{\varrho})$ lies in $[\Lambda^{-1},\Lambda]$. The identity $(s_2\gamma_{\text{in}})^{-1/2} = \sqrt{B/(s_2 N_{\text{in}})} = \sqrt{B_{\text{eff}}/N_{\text{in}}}$ gives eq. (SH.10). For the error bound, the two-sided bound gives

$$\Big|\sqrt{s_2\gamma_{\text{in}}}\,\mathfrak{d}_\varrho(\mu) - \mathfrak{d}^{\text{os}}(\mu;\hat{\varrho})\Big| \le \max\Big\{\epsilon_{\text{os}},\ \frac{1}{1-\epsilon_{\text{os}}} - 1\Big\}\mathfrak{d}^{\text{os}}(\mu;\hat{\varrho}) = \frac{\epsilon_{\text{os}}}{1-\epsilon_{\text{os}}}\,\mathfrak{d}^{\text{os}}(\mu;\hat{\varrho}),$$

and lemma SH.2 bounds $\mathfrak{d}^{\text{os}}(\mu;\hat{\varrho})$ by $\sqrt{\mu/2}$.

Part (iv) is (i) and (ii) eigenvalue by eigenvalue. At $\mu = 0$ both $\mathfrak{d}_\varrho(0)$ and $\mathfrak{d}^{\text{os}}(0;\hat{\varrho})$ vanish, and at $\mu > 0$ the ratio $\vartheta_\varrho(\mu)$ lies in $[\Lambda^{-1},\Lambda]$ by (i) and (ii). The functions of Σ_{out} act on each eigenvector u_k by the value at μ_k, which gives the operator identity. □

The relative error $\epsilon_{\text{os}}/(1-\epsilon_{\text{os}})$ in theorem SH.1(iii) depends only on ϵ_{os}. It does not depend on N_{out}, N_{in}, the spectrum or its norm, the eigenvalue or the ridge, and it is at most $2\epsilon_{\text{os}}$ when $\epsilon_{\text{os}} \le \frac{1}{2}$. Because the error is relative to the profile, the theorem is also accurate on modes where $\mathfrak{d}^{\text{os}}(\mu) \ll \sqrt{\mu}$. The shape ratio r is arbitrary in the theorem; appendices SH.6 to SH.8 specialize to $r = 1$.

Corollary SH.1 (Expected regularized update)**.** *Assume Assumption S3 with $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$, and assume that $\epsilon_{\text{os}} = 3(1-\beta^2)N_{\text{in}}/B \le \bar{\epsilon}$ along the sequence of dimensions for a fixed $\bar{\epsilon} < 1$. For $\varrho \in [\varrho_-,\varrho_+]$ let ϑ_ϱ be the function of theorem SH.1(iv) at the current dimension. Then*

$$\sup_{\varrho\in[\varrho_-,\varrho_+]} \Big\| \mathbb{E}\, Q_\varrho(M_t) - \sqrt{\frac{B_{\text{eff}}}{N_{\text{in}}}}\,\vartheta_\varrho(\Sigma_{\text{out}})\,\mathfrak{d}^{\text{os}}_{\hat{\nu}_{\text{out}},r}\Big(\Sigma_{\text{out}};\frac{\varrho}{s_2\gamma_{\text{in}}}\Big)\,\frac{\Delta_t}{\sqrt{\mathscr{R}(t)}}\Big\|_{\text{F}} \longrightarrow 0, \tag{SH.12}$$

and $\operatorname{spec}\vartheta_\varrho(\Sigma_{\text{out}}) \subset \big[1-\epsilon_{\text{os}},(1-\epsilon_{\text{os}})^{-1}\big]$ *at every dimension and for every ϱ.*

Proof. The candidate of theorem SC.2 is the drift operator applied to the state. With $\Sigma_{\text{in}} = \mathrm{Id}_{N_{\text{in}}}$, the deterministic equivalents of eq. (SB.3) are $\overline{R}(-s) = (s\,\mathrm{Id} + \tilde{s}_1(-s)\Sigma_{\text{out}})^{-1}$ and $\overline{\widetilde{R}}(-s) = (s + s_1(-s))^{-1}\,\mathrm{Id}$, so the candidate T of eq. (SC.2) is

$$T(s) = \frac{s\,L_\beta(q(-s))}{s+s_1(-s)}\,\Sigma_{\text{out}}\big(s\,\mathrm{Id} + \tilde{s}_1(-s)\Sigma_{\text{out}}\big)^{-1}\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}} = \mathsf{k}(s;\Sigma_{\text{out}})\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}}.$$

Integrating against $\frac{2}{\pi}\,\mathrm{d}u$ at $s = \varrho + u^2$, as in eq. (SC.3), and using $\frac{2}{\pi} = \sqrt{2}\cdot\frac{\sqrt{2}}{\pi}$,

$$\overline{\widetilde{G}}_\varrho = \sqrt{2}\,\mathfrak{d}_\varrho(\Sigma_{\text{out}})\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}} = \mathfrak{d}_\varrho(\Sigma_{\text{out}})\,\frac{\Delta_t}{\sqrt{\mathscr{R}(t)}}. \tag{SH.13}$$

The drift operator is then rewritten by theorem SH.1. At every dimension $\epsilon_{\text{os}} \le \bar{\epsilon} < 1$, so lemma SH.3(iv) gives the hypothesis of the theorem and $\Lambda \le (1-\epsilon_{\text{os}})^{-1}$. Theorem SH.1(iv) gives

$$\mathfrak{d}_\varrho(\Sigma_{\text{out}}) = \sqrt{\frac{B_{\text{eff}}}{N_{\text{in}}}}\,\vartheta_\varrho(\Sigma_{\text{out}})\,\mathfrak{d}^{\text{os}}_{\hat{\nu}_{\text{out}},r}(\Sigma_{\text{out}};\hat{\varrho}), \qquad \operatorname{spec}\vartheta_\varrho(\Sigma_{\text{out}}) \subset \big[1-\epsilon_{\text{os}},(1-\epsilon_{\text{os}})^{-1}\big].$$

Inserting this into eq. (SH.13), eq. (SC.8) of theorem SC.2 gives eq. (SH.12). □

Corollary SH.1 is an iterated statement. The dimension limit is taken at a fixed momentum, at ridges in a fixed compact subset of $(0,\infty)$ and at converging aspect ratios, as in Assumption S3. The factor $\vartheta_\varrho(\Sigma_{\text{out}})$ is within $\epsilon_{\text{os}}/(1-\epsilon_{\text{os}})$ of the identity at every dimension, uniformly in the spectrum, so no exchange of limits occurs. The zero-ridge coefficient $\mathfrak{d}_0$ of theorem SH.1(ii) is a deterministic limit, and the corollary makes no statement about the algorithm at zero ridge.

The relevant ridge for the profile is the rescaled ridge $\hat{\varrho} = \varrho/(s_2\gamma_{\text{in}})$. At a fixed normalized ridge $\varrho > 0$ and $\gamma_{\text{in}} \downarrow 0$, one has $\hat{\varrho} \to \infty$, and lemma SH.2 gives $\mathfrak{d}^{\text{os}}(\mu;\hat{\varrho}) \le \mu/\sqrt{2\hat{\varrho}}$, a bound linear in μ. The square-root law of the next subsection concerns bounded $\hat{\varrho}$, in particular $\hat{\varrho} = 0$.

Remark SH.1 (Scope of the linear response; heuristic)**.** *This remark is heuristic, and no statement of the appendix uses it. Theorem SC.2 requires the diffuseness $\|\widehat{A}_t\| \to 0$ of Assumption S3, which suffices at fixed aspect ratios. At the scale of theorem SH.1, the spectral parameter that carries the profile is $s = s_2\gamma_{\text{in}}\hat{s}$ with $\hat{s}$ of order one, so the squared singular values of the noise part of the buffer that the filter resolves are of order $s_2\gamma_{\text{in}}$. In the square case $N_{\text{out}} = N_{\text{in}} = N$, a delocalized mean gradient $\mathbb{E}\, G_{t+1} = {\Sigma_{\text{out}}}^{1/2}\widehat{A}_t$, of Frobenius norm of order one and of rank of order N, has operator norm of order $N^{-1/2}$. The response of the filter to the mean gradient should then be linear when $N^{-1/2} \ll \sqrt{s_2\gamma_{\text{in}}}$, that is when*

$$B_{\text{eff}} \ll N^2.$$

For larger batches the response is presumably nonlinear in Δ_t, and theorem SH.1 then describes the deterministic coefficient only.

SH.6 The square case: the square-root law and head modes

Throughout this subsection and the next two, $r = 1$, which is the profile that theorem SH.1 attaches to the square case $N_{\text{out}} = N_{\text{in}}$, and ν is an arbitrary probability measure. As in appendix SF, the Stieltjes transform on the negative axis is $S_\nu(-c) \overset{\text{def}}{=} \int \nu(\mathrm{d}x)/(c+x)$ for $c > 0$, and we put

$$\Upsilon_\nu(c) \overset{\text{def}}{=} c\,\big(1 - m_\nu(c)\big) = \int \frac{c^2\,\nu(\mathrm{d}x)}{c+x} = c^2 S_\nu(-c) \qquad (c > 0). \tag{SH.14}$$

Lemma SH.6 (Elementary integrals)**.** *One has*

$$\int_0^1 \frac{\sqrt{y}\,\mathrm{d}y}{(1+y)^2} = \frac{\pi}{4} - \frac{1}{2}, \qquad \int_1^\infty \frac{\sqrt{y}\,\mathrm{d}y}{(1+y)^2} = \frac{\pi}{4} + \frac{1}{2}, \qquad \int_0^1 \frac{y\,\mathrm{d}y}{(1+y)^2} = \log 2 - \frac{1}{2}.$$

In particular ϖ is a probability density on $(0,\infty)$, and for $\mu > 0$

$$\frac{\sqrt{2}}{\pi}\,\mu \int_0^\infty \frac{\sqrt{c}\,\mathrm{d}c}{(c+\mu)^2} = \sqrt{\mu/2}.$$

Proof. The first two integrals have the antiderivative $\arctan\sqrt{y} - \sqrt{y}/(1+y)$. Its derivative is

$$\frac{1}{2\sqrt{y}\,(1+y)} - \frac{1-y}{2\sqrt{y}\,(1+y)^2} = \frac{\sqrt{y}}{(1+y)^2},$$

and it takes the values 0, $\pi/4 - 1/2$ and $\pi/2$ at $y = 0$, $y = 1$ and $y = \infty$. The third integral uses $y(1+y)^{-2} = (1+y)^{-1} - (1+y)^{-2}$. The sum of the first two is $\pi/2$, which is the normalization of ϖ. The last identity is the substitution $c = \mu y$ together with $\int_0^\infty \sqrt{y}\,(1+y)^{-2}\,\mathrm{d}y = \pi/2$. □

Lemma SH.7 (The c-parametrization)**.** *The function Υ_ν is continuously differentiable on $(0,\infty)$, with*

$$\Upsilon_\nu'(c) = \int \Big(1 - \frac{x^2}{(c+x)^2}\Big)\nu(\mathrm{d}x) \in (0,1],$$

and $0 < \Upsilon_\nu(c) \le c$. It tends to 0 as $c \downarrow 0$ and to ∞ as $c \to \infty$, so it is an increasing bijection of $(0,\infty)$. With $c_\nu \overset{\text{def}}{=} \Upsilon_\nu^{-1}$ one has $c_\nu(\hat{s}) \ge \hat{s}$, and the profile root of the square case is

$$b_0^{\nu,1}(\hat{s}) = \frac{\hat{s}}{c_\nu(\hat{s})}, \qquad \frac{b_0^{\nu,1}(\hat{s})\,\mu}{\hat{s} + b_0^{\nu,1}(\hat{s})\,\mu} = \frac{\mu}{c_\nu(\hat{s}) + \mu}.$$

Proof. The derivative of the integrand of eq. (SH.14) in c is

$$\partial_c \frac{c^2}{c+x} = 1 - \frac{x^2}{(c+x)^2} \in (0,1] \qquad (c>0),$$

which gives the formula for Υ'_ν. The bounds $0 < \Upsilon_\nu(c) \le c$ hold because $0 < c/(c+x) \le 1$, and they give $\Upsilon_\nu(c) \to 0$ as $c \downarrow 0$. As $c \to \infty$, $1 - m_\nu(c) \uparrow 1$ by monotone convergence, so $\Upsilon_\nu(c) \ge c/2$ for large c.

The identification of b_0 is a change of unknown. For $b_0 = b_0^{\nu,1}(\hat{s})$ put $c \stackrel{\text{def}}{=} \hat{s}/b_0$. Then eq. (SH.5) with $r = 1$ reads

$$\frac{\hat{s}}{c} + m_\nu(c) = 1, \qquad \text{that is} \qquad \Upsilon_\nu(c) = \hat{s}.$$

Hence $c = c_\nu(\hat{s})$, and $c \ge \Upsilon_\nu(c) = \hat{s}$. Substituting $b_0 = \hat{s}/c$ gives the last identity. □

The next lemma is the central representation. It writes the profile as the square-root law times a ϖ-average of $\sqrt{1-m_\nu}$.

Lemma SH.8 (Representation of the profile)**.** *For every probability measure ν and every $\mu > 0$,*

$$\mathfrak{d}_\nu^{\mathrm{os}}(\mu) = \frac{\sqrt{2}}{\pi}\,\mu \int_0^\infty \frac{\sqrt{\Upsilon_\nu(c)}}{(c+\mu)^2}\,\mathrm{d}c = \sqrt{\frac{\mu}{2}}\,\chi_\nu(\mu), \qquad \chi_\nu(\mu) \in (0,1]. \tag{SH.15}$$

Proof. The first step changes the variable from v to c. By lemma SH.7, at $\hat{\varrho} = 0$,

$$\mathfrak{d}_\nu^{\mathrm{os}}(\mu) = \frac{\sqrt{2}}{\pi} \int_0^\infty \frac{\mu\,\mathrm{d}v}{c_\nu(v^2)+\mu}.$$

The map $c \mapsto \sqrt{\Upsilon_\nu(c)}$ is an increasing continuously differentiable bijection of $(0,\infty)$, by lemma SH.7. Substituting $v = \sqrt{\Upsilon_\nu(c)}$, so that $c_\nu(v^2) = c$,

$$\int_0^\infty \frac{\mu\,\mathrm{d}v}{c_\nu(v^2)+\mu} = \int_0^\infty \frac{\mu}{c+\mu}\,\frac{\mathrm{d}}{\mathrm{d}c}\sqrt{\Upsilon_\nu(c)}\,\mathrm{d}c.$$

The second step integrates by parts. The boundary term satisfies

$$0 \le \frac{\mu\sqrt{\Upsilon_\nu(c)}}{c+\mu} \le \frac{\mu\sqrt{c}}{c+\mu},$$

which tends to 0 as $c \downarrow 0$ and as $c \to \infty$. The derivative of $\mu/(c+\mu)$ is $-\mu/(c+\mu)^2$, which gives the first formula of eq. (SH.15).

The third step substitutes $c = \mu y$. Since $\Upsilon_\nu(\mu y) = \mu y\,(1 - m_\nu(\mu y))$,

$$\begin{aligned}\frac{\sqrt{2}}{\pi}\,\mu \int_0^\infty \frac{\sqrt{\Upsilon_\nu(c)}}{(c+\mu)^2}\,\mathrm{d}c &= \frac{\sqrt{2}}{\pi}\sqrt{\mu}\int_0^\infty \frac{\sqrt{y}\,\sqrt{1-m_\nu(\mu y)}}{(1+y)^2}\,\mathrm{d}y \\ &= \sqrt{\frac{\mu}{2}}\int_0^\infty \varpi(y)\sqrt{1-m_\nu(\mu y)}\,\mathrm{d}y.\end{aligned}$$

Finally, $0 < 1 - m_\nu \le 1$ and $\int \varpi = 1$ (lemma SH.6) give $\chi_\nu(\mu) \in (0,1]$. □

The ϖ-average of the ratio m_ν has a closed form, which makes the analysis of head modes exact.

Lemma SH.9 (The averaged ratio)**.** *For $\mu > 0$ and $x \ge 0$,*

$$\int_0^\infty \varpi(y)\,\frac{x}{\mu y + x}\,\mathrm{d}y = \frac{x}{(\sqrt{\mu}+\sqrt{x})^2}.$$

The right side is nondecreasing in x and at most $\min\{x/\mu,\ 1,\ x/(\mu+x)\}$. Consequently

$$m_\nu^\varpi(\mu) \stackrel{\text{def}}{=} \int_0^\infty \varpi(y)\,m_\nu(\mu y)\,\mathrm{d}y = \int \frac{x\,\nu(\mathrm{d}x)}{(\sqrt{\mu}+\sqrt{x})^2} \le m_\nu(\mu). \tag{SH.16}$$

Proof. For $x = 0$ both sides vanish. For $a, \xi > 0$ we first compute two integrals. The substitution $y = w^2$ and the partial fractions

$$\frac{w^2}{(w^2+a)(w^2+\xi)} = \frac{1}{\xi - a}\Big(\frac{\xi}{w^2+\xi} - \frac{a}{w^2+a}\Big) \qquad (a \neq \xi)$$

give, after letting $a \to \xi$ in the case $a = \xi$,

$$\int_0^\infty \frac{\sqrt{y}\,\mathrm{d}y}{(y+a)(y+\xi)} = \frac{\pi}{\sqrt{a}+\sqrt{\xi}}.$$

Differentiating in a gives

$$\int_0^\infty \frac{\sqrt{y}\,\mathrm{d}y}{(y+a)^2(y+\xi)} = \frac{\pi}{2\sqrt{a}\,(\sqrt{a}+\sqrt{\xi})^2}.$$

The identity follows at $a = 1$ and $\xi = x/\mu$. Since $x/(\mu y + x) = \xi/(y+\xi)$,

$$\int_0^\infty \varpi(y)\,\frac{\xi}{y+\xi}\,\mathrm{d}y = \frac{2}{\pi}\,\xi\,\frac{\pi}{2(1+\sqrt{\xi})^2} = \frac{\xi}{(1+\sqrt{\xi})^2} = \frac{x}{(\sqrt{\mu}+\sqrt{x})^2}.$$

The right side equals $(\sqrt{x}/(\sqrt{\mu}+\sqrt{x}))^2$, which is nondecreasing in x and at most 1 and x/μ. It is at most $x/(\mu+x)$ because $(\sqrt{\mu}+\sqrt{x})^2 \geq \mu + x$. Integrating the identity against $\nu(\mathrm{d}x)$ gives eq. (SH.16). □

The defect $1 - \chi_\nu$ is comparable to the averaged ratio within a factor of two.

Lemma SH.10 (Defect bounds)**.** *For every probability measure ν and every $\mu > 0$,*

$$\tfrac{1}{2}\, m_\nu^{\varpi}(\mu) \leq 1 - \chi_\nu(\mu) \leq m_\nu^{\varpi}(\mu).$$

Proof. For $m \in [0,1]$ one has

$$\frac{m}{2} \leq 1 - \sqrt{1-m} \leq m,$$

because $\sqrt{1-m} \geq 1 - m$ and $(1 - \frac{m}{2})^2 \geq 1 - m$. Apply this with $m = m_\nu(\mu y)$ and integrate against $\varpi(y)\,\mathrm{d}y$, using $\int \varpi = 1$ and the definitions eqs. (SH.7) and (SH.16). □

The complement of the averaged ratio is an explicit count of the spectrum, and it determines the efficiency up to a square root.

Lemma SH.11 (Smoothed count)**.** *For a probability measure ν and $\mu > 0$ put*

$$\mathsf{s}_\nu(\mu) \overset{\text{def}}{=} 1 - m_\nu^{\varpi}(\mu) = \int \frac{1 + 2\sqrt{x/\mu}}{\big(1+\sqrt{x/\mu}\big)^2}\,\nu(\mathrm{d}x) \in (0,1]. \tag{SH.17}$$

Then

$$\mathsf{s}_\nu(\mu) \leq \chi_\nu(\mu) \leq \sqrt{\mathsf{s}_\nu(\mu)}.$$

Proof. The formula for s_ν is eq. (SH.16), since $1 - \xi/(1+\sqrt{\xi})^2 = (1+2\sqrt{\xi})/(1+\sqrt{\xi})^2$ with $\xi = x/\mu$. The integrand is positive and at most 1, so $\mathsf{s}_\nu(\mu) \in (0,1]$.

The lower bound is the upper bound of lemma SH.10.

The upper bound is Jensen's inequality for the concave function $\sqrt{\cdot}$ and the probability density ϖ:

$$\chi_\nu(\mu) = \int_0^\infty \varpi(y)\sqrt{1 - m_\nu(\mu y)}\,\mathrm{d}y \leq \Big(\int_0^\infty \varpi(y)\big(1 - m_\nu(\mu y)\big)\,\mathrm{d}y\Big)^{1/2} = \sqrt{1 - m_\nu^{\varpi}(\mu)}.$$ □

Remark SH.2 (Probabilistic reading)**.** *The objects of definition SH.1 are probabilities. The density ϖ is the law of $Y = (g_1^2 + g_2^2 + g_3^2)/g_4^2$ with independent standard Gaussians g_j, since the ratio of independent χ_3^2 and χ_1^2 variables has density proportional to $y^{1/2}(1+y)^{-2}$. Let $L = E/E'$ with independent standard exponentials E, E', so that $\Pr(L \leq t) = t/(1+t)$, and let $X \sim \nu$ be independent of L and Y. Then $c/(c+x) = \Pr(xL \leq c)$ gives*

$$1 - m_\nu(c) = \Pr(XL \leq c), \qquad \chi_\nu(\mu) = \mathbb{E}_Y\Big[\Pr\big(XL \leq \mu Y \,\big|\, Y\big)^{1/2}\Big], \qquad \mathsf{s}_\nu(\mu) = \Pr(XL \leq \mu Y).$$

For the empirical measure $\nu = \hat{\nu}_{\mathrm{out}}$, X is a uniformly chosen eigenvalue, and $1 - m_\nu(c)$ is the fraction of the spectrum below c after each eigenvalue is multiplied by an independent random factor. The efficiency $\chi_\nu(\mu)$ is thus an averaged square root of the fraction of the spectrum lying below μ, on a logarithmic scale.

Theorem SH.2 (Square-root law). *Let* $C_* \stackrel{\mathrm{def}}{=} \frac{1}{2} + \frac{2\log 2}{\pi} \approx 0.9413$.

(i) For every probability measure ν and every $\mu > 0$,

$$\max\Big\{C_*\sqrt{1 - m_\nu(\mu)},\ 1 - m_\nu(\mu)\Big\}\sqrt{\frac{\mu}{2}} \ \le\ \mathfrak{d}_\nu^{\mathrm{os}}(\mu) \ \le\ \sqrt{\frac{\mu}{2}}.$$

(ii) For every probability measure ν, every $\mu > 0$ and every $K > 0$,

$$1 - m_\nu(\mu) \ge \frac{\nu([0, K\mu])}{1 + K}; \qquad \textit{in particular} \qquad 1 - m_\nu(\mu) \ge \tfrac{1}{2}\,\nu([0, \mu]).$$

(iii) In the square case $N_{\mathrm{out}} = N_{\mathrm{in}}$, let $\underline{\mathfrak{d}}(\mu) \stackrel{\mathrm{def}}{=} C_*\sqrt{1 - m_{\hat{\nu}_{\mathrm{out}}}(\mu)}\sqrt{\mu/2}$ for $\mu > 0$ and $\underline{\mathfrak{d}}(0) \stackrel{\mathrm{def}}{=} 0$. Then

$$\underline{\mathfrak{d}}(\Sigma_{\mathrm{out}}) \preceq \mathfrak{d}^{\mathrm{os}}_{\hat{\nu}_{\mathrm{out}}}(\Sigma_{\mathrm{out}}) \preceq \Big(\frac{\Sigma_{\mathrm{out}}}{2}\Big)^{1/2},$$

where, for $\mu_k > 0$,

$$1 - m_{\hat{\nu}_{\mathrm{out}}}(\mu_k) = \frac{1}{N_{\mathrm{out}}}\sum_{l=1}^{N_{\mathrm{out}}} \frac{\mu_k}{\mu_k + \mu_l}.$$

Proof. The upper bound in (i) is lemma SH.8, since $\chi_\nu \le 1$.

The first lower bound compares $1 - m_\nu(\mu y)$ with $1 - m_\nu(\mu)$ on each side of $y = 1$. For $y \ge 1$, monotonicity gives $1 - m_\nu(\mu y) \ge 1 - m_\nu(\mu)$. For $y \le 1$, since $\mu + x \ge \mu y + x$,

$$\frac{\mu y}{\mu y + x} \ge y\,\frac{\mu}{\mu + x}, \qquad \text{so} \qquad 1 - m_\nu(\mu y) \ge y\,\big(1 - m_\nu(\mu)\big).$$

Inserting both bounds into eq. (SH.7) and using lemma SH.6,

$$\begin{aligned}\chi_\nu(\mu) &\ge \sqrt{1 - m_\nu(\mu)}\ \frac{2}{\pi}\Big[\int_0^1 \frac{y\,\mathrm{d}y}{(1+y)^2} + \int_1^\infty \frac{\sqrt{y}\,\mathrm{d}y}{(1+y)^2}\Big] \\ &= \sqrt{1 - m_\nu(\mu)}\ \frac{2}{\pi}\Big(\log 2 + \frac{\pi}{4}\Big) = C_*\sqrt{1 - m_\nu(\mu)}.\end{aligned}$$

Lemma SH.8 turns this into the first lower bound of (i).

The second lower bound in (i) follows from the averaged ratio. By lemmas SH.9 and SH.10,

$$\chi_\nu(\mu) \ge 1 - m_\nu^{\varpi}(\mu) \ge 1 - m_\nu(\mu),$$

and lemma SH.8 again converts this into a bound on the profile.

Part (ii) holds because $\mu/(\mu + x) \ge 1/(1 + K)$ for $x \le K\mu$, so

$$1 - m_\nu(\mu) = \int \frac{\mu\,\nu(\mathrm{d}x)}{\mu + x} \ge \frac{\nu([0, K\mu])}{1 + K}.$$

Part (iii) is (i) applied to each eigenvalue, with $\nu = \hat{\nu}_{\mathrm{out}}$. On the kernel of Σ_{out} all three operators vanish. The formula for $1 - m_{\hat{\nu}_{\mathrm{out}}}(\mu_k)$ is eq. (SH.4). □

The factor $\sqrt{1 - m_\nu(\mu)}$ is the only loss against the square-root law, and by (ii) it is bounded below by the square root of the fraction of the spectrum lying within a bounded factor of μ. The next proposition determines when the loss disappears.

Proposition SH.1 (Head modes). *(i) For every probability measure ν, every $\mu > 0$ and every $\delta > 0$,*

$$\frac{\delta}{2(1 + \sqrt{\delta})^2}\,\nu\big([\delta\mu, \infty)\big) \le 1 - \chi_\nu(\mu) \le \delta + \nu\big([\delta\mu, \infty)\big).$$

(ii) For every sequence of probability measures ν_N and numbers $\mu_N > 0$, one has $\mathfrak{d}^{\mathrm{os}}_{\nu_N}(\mu_N)/\sqrt{\mu_N/2} \to 1$ if and only if $\nu_N([\delta\mu_N,\infty)) \to 0$ for every $\delta > 0$.

Proof. Part (i) bounds the averaged ratio of lemma SH.9 from both sides. By that lemma, $m^{\varpi}_{\nu}(\mu) = \int x(\sqrt{\mu}+\sqrt{x})^{-2}\,\nu(\mathrm{d}x)$ with a nondecreasing integrand, so

$$m^{\varpi}_{\nu}(\mu) \ge \frac{\delta\mu}{(\sqrt{\mu}+\sqrt{\delta\mu})^2}\,\nu\big([\delta\mu,\infty)\big) = \frac{\delta}{(1+\sqrt{\delta})^2}\,\nu\big([\delta\mu,\infty)\big).$$

The integrand is at most $x/\mu < \delta$ on $\{x < \delta\mu\}$ and at most 1 elsewhere, so

$$m^{\varpi}_{\nu}(\mu) \le \delta + \nu\big([\delta\mu,\infty)\big).$$

Lemma SH.10 converts the two bounds into (i).

Part (ii) follows from (i) and lemma SH.8, which gives $\mathfrak{d}^{\mathrm{os}}_{\nu_N}(\mu_N)/\sqrt{\mu_N/2} = \chi_{\nu_N}(\mu_N)$. If the masses vanish for every δ, the upper bound of (i) gives $\limsup_N(1-\chi_{\nu_N}(\mu_N)) \le \delta$ for every $\delta > 0$. Conversely, if $\chi_{\nu_N}(\mu_N) \to 1$, the lower bound of (i) forces $\nu_N([\delta\mu_N,\infty)) \to 0$ for each $\delta > 0$. □

A mode μ of a spectrum is a head mode when it lies above all but a vanishing fraction of the spectrum, in the sense of proposition SH.1(ii). On head modes the profile is the square-root law $\sqrt{\mu/2}$ to leading order.

The lower bound of theorem SH.2 needs only a count of the spectrum below a multiple of the mode. We name this condition and record the resulting two-sided bound on the drift coefficient itself.

Definition SH.2 (Packed mode)**.** *Let $K \ge 1$ and $\delta \in (0,1]$. In the square case $N_{\mathrm{out}} = N_{\mathrm{in}} = N$, an eigenvalue $\mu_i > 0$ of Σ_{out} is* (K,δ)-packed *if*

$$\#\{k \le N : \mu_k \le K\mu_i\} \ge \delta N.$$

Corollary SH.2 (Packed modes)**.** *Assume $N_{\mathrm{out}} = N_{\mathrm{in}} = N$ and $\epsilon_{\mathrm{os}} < 1$, and let $\mathfrak{d}_i(0)$ be the zero-ridge coefficient of theorem SH.1(ii). For every mode with $\mu_i > 0$,*

$$\mathfrak{d}_i(0) \le \frac{1}{1-\epsilon_{\mathrm{os}}}\sqrt{\frac{B_{\mathrm{eff}}}{2N}}\,\sqrt{\mu_i},$$

and if μ_i is (K,δ)-packed, then

$$\mathfrak{d}_i(0) \ge (1-\epsilon_{\mathrm{os}})\,C_*\sqrt{\frac{\delta}{1+K}}\,\sqrt{\frac{B_{\mathrm{eff}}}{2N}}\,\sqrt{\mu_i}, \qquad C_* = \frac{1}{2} + \frac{2\log 2}{\pi}.$$

Proof. Theorem SH.1(iii) at $\varrho = 0$ gives

$$\mathfrak{d}_i(0) = \vartheta_i\sqrt{\frac{B_{\mathrm{eff}}}{N}}\,\mathfrak{d}^{\mathrm{os}}_{\hat{\nu}_{\mathrm{out}}}(\mu_i), \qquad \vartheta_i \in \big[1-\epsilon_{\mathrm{os}},(1-\epsilon_{\mathrm{os}})^{-1}\big].$$

The upper bound of theorem SH.2(i) gives the first inequality. For the second, theorem SH.2(ii) with $\nu = \hat{\nu}_{\mathrm{out}}$ and the packing condition give

$$1 - m_{\hat{\nu}_{\mathrm{out}}}(\mu_i) \ge \frac{\hat{\nu}_{\mathrm{out}}([0,K\mu_i])}{1+K} = \frac{\#\{k : \mu_k \le K\mu_i\}}{N(1+K)} \ge \frac{\delta}{1+K},$$

and the first lower bound of theorem SH.2(i) gives the second inequality. □

Remark SH.3 (Where packing holds)**.** *The count $\#\{k : \mu_k \le K\mu\}$ is nondecreasing in μ, so if the smallest positive eigenvalue is (K,δ)-packed, then every mode is. If all eigenvalues lie in $[\mu_{\min},\kappa\mu_{\min}]$ with $\mu_{\min} > 0$, every mode is $(\kappa,1)$-packed, since $\mu_k \le \kappa\mu_{\min} \le \kappa\mu_i$ for all k. If the spectrum consists of clusters of equal eigenvalues, each holding at least pN of them, every mode is $(1,p)$-packed, since its own cluster lies in $[0,\mu_i]$. For the geometric spectrum $\mu_k = q^k$ with fixed $q \in (0,1)$, the smallest eigenvalue μ_N has only $1 + \lfloor \log K/\log(1/q) \rfloor$ eigenvalues in $[0,K\mu_N]$, a vanishing fraction as $N \to \infty$, so it is not (K,δ)-packed for any fixed K,δ once N is large. For the power law $\mu_k = c_N k^{-\alpha}$, $1 \le k \le N$, every mode is $(2,1-2^{-1/\alpha})$-packed, because $\mu_k \le 2\mu_i$ for every $k \ge 2^{-1/\alpha}i$, and there are at least $N - 2^{-1/\alpha}i \ge (1-2^{-1/\alpha})N$ such k. Conversely, packing is necessary for a lower bound of this form: by proposition SH.2(vi), along a sequence of spectra the efficiency $\chi_{\nu_N}(\mu_N)$ stays bounded below only if, for some K, the fraction of the spectrum in $[0,K\mu_N]$ stays bounded below, and by proposition SH.2(ii),(iii) a mode far below most of the spectrum receives drift of gradient-descent order up to a logarithm.*

SH.7 Tail modes and acceleration

For a probability measure ν and $\ell \geq 0$ put

$$\mathcal{I}_\nu(\ell) \stackrel{\text{def}}{=} \int_{(\ell,\infty)} \frac{\nu(\mathrm{d}x)}{x} \in [0,\infty], \qquad \mathcal{I}_\nu \stackrel{\text{def}}{=} \mathcal{I}_\nu(0).$$

When $\nu(\{0\}) = 0$, $\mathcal{I}_\nu = \int x^{-1}\nu(\mathrm{d}x)$ is the limit of $S_\nu(-c)$ as $c \downarrow 0$. Tail modes are controlled by the following integral.

Lemma SH.12 (The tail integral)**.** *For $\mu > 0$ and $\Gamma \geq 0$ put*

$$\mathcal{E}(\mu,\Gamma) \stackrel{\text{def}}{=} \frac{\sqrt{2}}{\pi}\,\mu \int_0^\infty \frac{\min\{\sqrt{c},\ c\sqrt{\Gamma}\}}{(c+\mu)^2}\,\mathrm{d}c.$$

Then $\mathcal{E}$ is nondecreasing in Γ, $\mathcal{E}(\mu,0) = 0$, and for $\Gamma > 0$

$$\mathcal{E}(\mu,\Gamma) \leq \frac{\sqrt{2}}{\pi}\sqrt{\Gamma}\,\mu\Big[\log\Big(1+\frac{1}{\mu\Gamma}\Big)+2\Big].$$

Proof. Monotonicity and the value at $\Gamma = 0$ are read off the integrand. For the bound, split the integral at $a = 1/\Gamma$, where the two terms of the minimum cross, and use $c\sqrt{\Gamma}$ below a and $\sqrt{c}$ above a. The two pieces are

$$\int_0^a \frac{c\,\mathrm{d}c}{(c+\mu)^2} = \log\Big(1+\frac{a}{\mu}\Big) - \frac{a}{a+\mu} \leq \log\Big(1+\frac{a}{\mu}\Big), \qquad \int_a^\infty \frac{\sqrt{c}\,\mathrm{d}c}{(c+\mu)^2} \leq \int_a^\infty c^{-3/2}\,\mathrm{d}c = 2\sqrt{\Gamma}.$$

Multiplying the first by $\sqrt{\Gamma}$ and adding the second gives the bound. □

The universal lower bound of proposition SH.2(v) rests on one scalar inequality.

Lemma SH.13 (A logarithmic inequality)**.** *For every $k > 0$,*

$$\log(1+k) - \frac{k}{1+k} \geq \frac{1}{2}\log k.$$

Proof. Let $D(k)$ be the left side minus the right side. Its derivative is

$$D'(k) = \frac{1}{1+k} - \frac{1}{(1+k)^2} - \frac{1}{2k} = \frac{k^2-2k-1}{2k(1+k)^2},$$

which is negative on $(0, 1+\sqrt{2})$ and positive on $(1+\sqrt{2},\infty)$. So D attains its minimum at $k_* = 1+\sqrt{2}$. There $1+k_* = \sqrt{2}\,k_*$ and $k_*/(1+k_*) = 1/\sqrt{2}$, so

$$D(k_*) = \tfrac{1}{2}\log 2 + \tfrac{1}{2}\log k_* - \tfrac{1}{\sqrt{2}} = \tfrac{1}{2}\log(2+2\sqrt{2}) - \tfrac{1}{\sqrt{2}}.$$

This is positive because $e^{\sqrt{2}} < e^{3/2} < 4.5 < 2+2\sqrt{2}$, where the middle inequality is $e^3 < 20.1 < 20.25$. □

Proposition SH.2 (Tail modes)**.** *Let ν be a probability measure and $\mu > 0$.*

(i) (Split bound.) For every $\ell \geq 0$ with $\mathcal{I}_\nu(\ell) < \infty$,

$$\mathfrak{d}^{\mathrm{os}}_\nu(\mu) \leq \sqrt{\nu([0,\ell])}\,\sqrt{\frac{\mu}{2}} + \mathcal{E}\big(\mu, \mathcal{I}_\nu(\ell)\big).$$

(ii) (Gradient-descent-like upper bound.) If $\nu(\{0\}) = 0$ and $\mathcal{I}_\nu < \infty$, then

$$\mathfrak{d}^{\mathrm{os}}_\nu(\mu) \leq \frac{\sqrt{2}}{\pi}\sqrt{\mathcal{I}_\nu}\,\mu\Big[\log\Big(1+\frac{1}{\mu\mathcal{I}_\nu}\Big)+2\Big].$$

(iii) (Asymptotics.) If $\nu(\{0\}) = 0$ and $\mathcal{I}_\nu < \infty$, then, for fixed ν,

$$\mathfrak{d}^{\mathrm{os}}_\nu(\mu) \sim \frac{\sqrt{2}}{\pi}\sqrt{\mathcal{I}_\nu}\,\mu\log\frac{1}{\mu} \qquad (\mu \downarrow 0).$$

(iv) (Counts.) For every $K > 0$,

$$C_*\sqrt{\frac{\nu([0,K\mu])}{1+K}} \le \chi_\nu(\mu) \le \sqrt{\nu([0,K\mu])} + \frac{2}{\pi\sqrt{K}}\big[\log(1+K)+2\big].$$

(v) (Universal lower bound.) If $\nu([0,X]) = 1$ for some $X > 0$, then

$$\mathfrak{d}_\nu^{\mathrm{os}}(\mu) \ge \frac{1}{\pi}\frac{\mu}{\sqrt{X}}\Big[\log\Big(1+\frac{X}{\mu}\Big) - \frac{X}{X+\mu}\Big] \ge \frac{1}{2\pi}\frac{\mu}{\sqrt{X}}\log\frac{X}{\mu}.$$

(vi) (Sequences.) For probability measures ν_N and numbers $\mu_N > 0$ put $\chi_N \stackrel{\mathrm{def}}{=} \chi_{\nu_N}(\mu_N)$. Then $\liminf_N \chi_N > 0$ if and only if $\liminf_N \nu_N([0,K\mu_N]) > 0$ for some $K > 0$; and $\chi_N \to 0$ if and only if $\nu_N([0,K\mu_N]) \to 0$ for every $K > 0$.

Proof. Part (i) bounds $\sqrt{\Upsilon_\nu}$ by splitting ν at ℓ. Since $c/(c+x) \le 1$ on $[0,\ell]$ and $c/(c+x) \le c/x$ on (ℓ,∞),

$$1 - m_\nu(c) \le \nu([0,\ell]) + c\,\mathcal{I}_\nu(\ell).$$

With $\Upsilon_\nu(c) = c(1-m_\nu(c)) \le c$, this gives

$$\sqrt{\Upsilon_\nu(c)} \le \min\Big\{\sqrt{c},\ \sqrt{c\,\nu([0,\ell])} + c\sqrt{\mathcal{I}_\nu(\ell)}\Big\} \le \sqrt{c\,\nu([0,\ell])} + \min\Big\{\sqrt{c},\ c\sqrt{\mathcal{I}_\nu(\ell)}\Big\}.$$

The last inequality is checked on the two cases of the inner minimum. Insert this into the first formula of eq. (SH.15). The first term gives $\sqrt{\nu([0,\ell])}\sqrt{\mu/2}$ by lemma SH.6, and the second term gives $\mathcal{E}(\mu,\mathcal{I}_\nu(\ell))$.

Part (ii) is (i) at $\ell = 0$, where $\nu([0,0]) = \nu(\{0\}) = 0$, followed by lemma SH.12.

Part (iii) matches the upper bound (ii) with a lower bound. The upper bound is (ii), since

$$\frac{\log(1+1/(\mu\mathcal{I}_\nu))+2}{\log(1/\mu)} \longrightarrow 1 \qquad (\mu \downarrow 0).$$

For the lower bound, fix $c_1 > 0$. The transform $S_\nu(-c)$ is nonincreasing in c, so for $c \le c_1$

$$\sqrt{\Upsilon_\nu(c)} = c\sqrt{S_\nu(-c)} \ge c\sqrt{S_\nu(-c_1)}.$$

By the first formula of eq. (SH.15),

$$\mathfrak{d}_\nu^{\mathrm{os}}(\mu) \ge \frac{\sqrt{2}}{\pi}\,\mu\sqrt{S_\nu(-c_1)}\int_0^{c_1}\frac{c\,\mathrm{d}c}{(c+\mu)^2} \ge \frac{\sqrt{2}}{\pi}\,\mu\sqrt{S_\nu(-c_1)}\Big[\log\Big(1+\frac{c_1}{\mu}\Big) - 1\Big].$$

Dividing by $\frac{\sqrt{2}}{\pi}\sqrt{\mathcal{I}_\nu}\,\mu\log(1/\mu)$ and letting $\mu \downarrow 0$ gives a $\liminf$ of at least $\sqrt{S_\nu(-c_1)/\mathcal{I}_\nu}$. As $c_1 \downarrow 0$, $S_\nu(-c_1) \uparrow \mathcal{I}_\nu$ by monotone convergence, and $\mathcal{I}_\nu > 0$, which gives the lower bound.

Part (iv) combines theorem SH.2 with (i). The lower bound is theorem SH.2(i) with (ii) and lemma SH.8. For the upper bound take $\ell = K\mu$ in (i). Then $\mathcal{I}_\nu(K\mu) \le 1/(K\mu)$, so by the monotonicity of $\mathcal{E}$ and lemma SH.12,

$$\frac{\mathcal{E}(\mu,\mathcal{I}_\nu(K\mu))}{\sqrt{\mu/2}} \le \frac{\mathcal{E}(\mu,1/(K\mu))}{\sqrt{\mu/2}} \le \frac{2}{\pi\sqrt{K}}\big[\log(1+K)+2\big].$$

Dividing (i) by $\sqrt{\mu/2}$ and using lemma SH.8 gives the upper bound.

Part (v) bounds $\sqrt{\Upsilon_\nu}$ from below near the origin. For $c \le X$, since $x \le X$ on the support,

$$S_\nu(-c) \ge \frac{1}{c+X} \ge \frac{1}{2X}, \qquad \text{so} \qquad \sqrt{\Upsilon_\nu(c)} \ge \frac{c}{\sqrt{2X}}.$$

By the first formula of eq. (SH.15),

$$\mathfrak{d}_\nu^{\mathrm{os}}(\mu) \ge \frac{\sqrt{2}}{\pi}\,\frac{\mu}{\sqrt{2X}}\int_0^X \frac{c\,\mathrm{d}c}{(c+\mu)^2} = \frac{1}{\pi}\frac{\mu}{\sqrt{X}}\Big[\log\Big(1+\frac{X}{\mu}\Big) - \frac{X}{X+\mu}\Big].$$

The second inequality is lemma SH.13 with $k = X/\mu$.

Part (vi) follows from (iv). Write $P_K(N) \overset{\text{def}}{=} \nu_N([0, K\mu_N])$ in this proof. If $\liminf_N P_K(N) > 0$ for some K, then

$$\liminf_N \chi_N \geq C_* \sqrt{\liminf_N P_K(N)/(1+K)} > 0.$$

Conversely, let $\liminf_N \chi_N = \underline{\chi} > 0$, and choose K with $\frac{2}{\pi\sqrt{K}}[\log(1+K)+2] \leq \underline{\chi}/4$. Then (iv) gives $\sqrt{P_K(N)} \geq \chi_N - \underline{\chi}/4 \geq \underline{\chi}/2$ for all large N. For the second statement, the lower bound of (iv) gives $P_K(N) \leq (1+K)\chi_N^2/C_*^2$, so $\chi_N \to 0$ forces $P_K(N) \to 0$ for every K. If $P_K(N) \to 0$ for every K, the upper bound of (iv) gives $\limsup_N \chi_N \leq \frac{2}{\pi\sqrt{K}}[\log(1+K)+2]$ for every K, and the right side tends to 0 as $K \to \infty$. □

The two-level spectrum shows that the crossover between square-root and gradient-descent-like behaviour sits at $p\varkappa \asymp \log^2 \varkappa$ and not at $p\varkappa \asymp 1$.

Corollary SH.3 (Two-level spectrum)**.** *Let* $\nu_{p,\varkappa} \overset{\text{def}}{=} (1-p)\delta_1 + p\,\delta_{1/\varkappa}$ *with* $p \in (0,1)$ *and* $\varkappa \geq 2$*, write* $\mathfrak{d}^{\text{os}} = \mathfrak{d}^{\text{os}}_{\nu_{p,\varkappa}}$*, and put*

$$\Omega_{p,\varkappa} \overset{\text{def}}{=} \max\Big\{\sqrt{\frac{p}{\varkappa}},\ \frac{\log\varkappa}{\varkappa}\Big\}, \qquad c_{\text{hi}} \overset{\text{def}}{=} \frac{2}{C_*}\Big(\frac{1}{\sqrt{2}} + \frac{\sqrt{2}}{\pi}\,\frac{\log 3 + 2}{\log 2}\Big) \leq 5.78.$$

Then

$$\mathfrak{d}^{\text{os}}(1) \in \Big[\frac{C_*}{2}, \frac{1}{\sqrt{2}}\Big],$$

$$\max\Big\{\frac{C_*}{2}\sqrt{\frac{p}{\varkappa}},\ \frac{1}{2\pi}\frac{\log\varkappa}{\varkappa}\Big\} \leq \mathfrak{d}^{\text{os}}(1/\varkappa) \leq \sqrt{\frac{p}{2\varkappa}} + \frac{\sqrt{2}}{\pi}\,\frac{\log(1+\varkappa)+2}{\varkappa},$$

and consequently

$$\frac{\Omega_{p,\varkappa}}{\sqrt{2}\,\pi} \leq \frac{\mathfrak{d}^{\text{os}}(1/\varkappa)}{\mathfrak{d}^{\text{os}}(1)} \leq c_{\text{hi}}\,\Omega_{p,\varkappa}.$$

Proof. The top mode is controlled by theorem SH.2(i), since

$$1 - m_{\nu_{p,\varkappa}}(1) = \frac{1-p}{2} + \frac{p\varkappa}{1+\varkappa} \geq \frac{1}{2}.$$

The upper bound for $\mathfrak{d}^{\text{os}}(1/\varkappa)$ is proposition SH.2(i) with $\ell = 1/\varkappa$. There $\nu_{p,\varkappa}([0,\ell]) = p$ and $\mathcal{I}_{\nu_{p,\varkappa}}(\ell) = 1 - p \leq 1$, so by lemma SH.12 at $\mu = 1/\varkappa$

$$\mathcal{E}(1/\varkappa, 1-p) \leq \mathcal{E}(1/\varkappa, 1) \leq \frac{\sqrt{2}}{\pi}\,\frac{\log(1+\varkappa)+2}{\varkappa}.$$

The two lower bounds for $\mathfrak{d}^{\text{os}}(1/\varkappa)$ are theorem SH.2(i) with $1 - m_{\nu_{p,\varkappa}}(1/\varkappa) \geq \frac{1}{2}\nu_{p,\varkappa}([0,1/\varkappa]) = \frac{p}{2}$ (theorem SH.2(ii)), and proposition SH.2(v) with $X = 1$.

The ratio bounds follow by dividing by $\mathfrak{d}^{\text{os}}(1)$. On the lower side, $\min\{C_*/2, 1/(2\pi)\} = 1/(2\pi)$ and $\mathfrak{d}^{\text{os}}(1) \leq 1/\sqrt{2}$. On the upper side, $\mathfrak{d}^{\text{os}}(1) \geq C_*/2$, and $(\log(1+\varkappa)+2)/\log\varkappa \leq (\log 3+2)/\log 2$ for $\varkappa \geq 2$. The last ratio is decreasing in $\varkappa$, because the numerator of its derivative is

$$\frac{\log\varkappa}{1+\varkappa} - \frac{\log(1+\varkappa)+2}{\varkappa} < 0.$$

The numerical value is $c_{\text{hi}} = 5.778\ldots$. □

The two terms of $\Omega_{p,\varkappa}$ cross at $p\varkappa = \log^2\varkappa$. The square-root term dominates when $p\varkappa \gg \log^2\varkappa$, and the gradient-descent term times a logarithm dominates when $p\varkappa \ll \log^2\varkappa$. In the window $1 \ll p\varkappa \ll \log^2\varkappa$ the slow mode is still gradient-descent-like up to the logarithm, although a non-negligible count of eigenvalues sits at $1/\varkappa$. Figure S2 shows the two regimes.

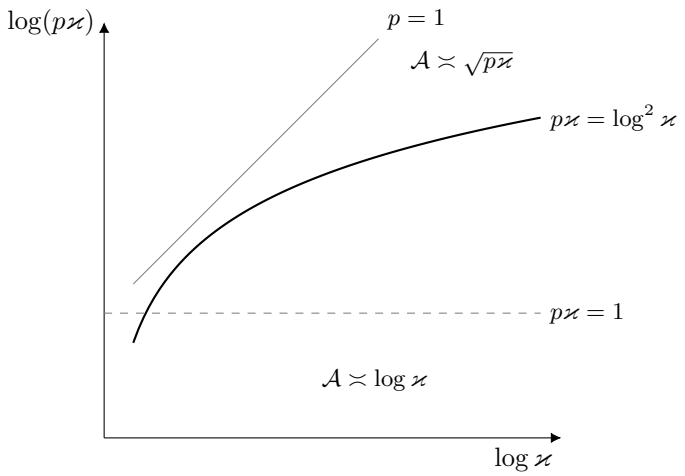


Figure S2: Regimes of the two-level spectrum $(1-p)\delta_1 + p\,\delta_{1/\varkappa}$ (corollaries SH.3 and SH.4). Admissible parameters lie below the line $p = 1$. The thick curve $p\varkappa = \log^2 \varkappa$ separates square-root acceleration from acceleration by a logarithm; the dashed line $p\varkappa = 1$ is not the boundary.

Acceleration. For a mode μ below the top μ_+ of the spectrum, gradient descent contracts the mode at the rate μ/μ_+ relative to the top mode. We measure the oversampled profile against this rate by

$$\mathcal{A} \stackrel{\text{def}}{=} \frac{\mathfrak{d}_\nu^{\text{os}}(\mu)/\mathfrak{d}_\nu^{\text{os}}(\mu_+)}{\mu/\mu_+}, \qquad \varkappa \stackrel{\text{def}}{=} \frac{\mu_+}{\mu}.$$

Corollary SH.4 (Acceleration of a slow mode)**.** *Let ν be a probability measure with $\nu([0,\mu_+]) = 1$ for some $\mu_+ > 0$, and let $0 < \mu \le \mu_+/2$.*

(i) One has

$$\chi_\nu(\mu)\sqrt{\varkappa} \le \mathcal{A} \le \frac{\sqrt{2}}{C_*}\,\chi_\nu(\mu)\sqrt{\varkappa}.$$

(ii) One has

$$\frac{1}{\sqrt{2}\,\pi}\log\varkappa \le \mathcal{A} \le \frac{\sqrt{2}}{C_*}\sqrt{\varkappa}.$$

(iii) For every $K > 0$,

$$C_*\sqrt{\frac{\nu([0,K\mu])}{1+K}}\,\sqrt{\varkappa} \le \mathcal{A} \le \frac{\sqrt{2}}{C_*}\Big[\sqrt{\nu([0,K\mu])} + \frac{2(\log(1+K)+2)}{\pi\sqrt{K}}\Big]\sqrt{\varkappa}.$$

(iv) If $\nu(\{0\}) = 0$ and $\mu_+\mathcal{I}_\nu \le \Gamma < \infty$, then

$$\mathcal{A} \le \frac{2\sqrt{2}}{\pi C_*}\sqrt{\Gamma}\,\big[\log(1+\varkappa)+2\big].$$

(v) For the two-level spectrum of corollary SH.3, with $\mu_+ = 1$ and $\mu = 1/\varkappa$,

$$\frac{1}{\sqrt{2}\,\pi}\max\big\{\sqrt{p\varkappa},\ \log\varkappa\big\} \le \mathcal{A} \le c_{\text{hi}}\max\big\{\sqrt{p\varkappa},\ \log\varkappa\big\}.$$

(vi) Along sequences $(\nu_N, \mu_{+,N}, \mu_N)$ satisfying the hypotheses, with acceleration $\mathcal{A}_N$ and ratio $\varkappa_N$, one has $\liminf_N \mathcal{A}_N/\sqrt{\varkappa_N} > 0$ if and only if $\liminf_N \nu_N([0,K\mu_N]) > 0$ for some $K > 0$.

Proof. The top mode is pinned by theorem SH.2(i). Since $x \le \mu_+$ on the support, $1 - m_\nu(\mu_+) \ge \frac{1}{2}$, so

$$\frac{C_*}{2}\sqrt{\mu_+} \le \mathfrak{d}^{\mathrm{os}}_\nu(\mu_+) \le \sqrt{\frac{\mu_+}{2}}.$$

By lemma SH.8, $\mathcal{A} = \varkappa\, \chi_\nu(\mu)\sqrt{\mu/2}/\mathfrak{d}^{\mathrm{os}}_\nu(\mu_+)$, and the two bounds give (i).

Part (ii) uses proposition SH.2(v) with $X = \mu_+$ for the lower bound,

$$\mathcal{A} \ge \varkappa\, \frac{\frac{1}{2\pi}\,\mu\,\mu_+^{-1/2}\log\varkappa}{\sqrt{\mu_+/2}} = \frac{\log\varkappa}{\sqrt{2}\,\pi},$$

and $\chi_\nu \le 1$ in (i) for the upper bound. Part (iii) is (i) with proposition SH.2(iv).

For part (iv), $\mathcal{I}_\nu \le \Gamma/\mu_+$, and $\mathcal{I}_\nu \ge 1/\mu_+$ because the support lies in $(0, \mu_+]$, so $1/(\mu\mathcal{I}_\nu) \le \varkappa$. Proposition SH.2(ii) and the lower bound for $\mathfrak{d}^{\mathrm{os}}_\nu(\mu_+)$ give

$$\mathcal{A} \le \varkappa\, \frac{\frac{\sqrt{2}}{\pi}\sqrt{\Gamma/\mu_+}\;\mu\,\big[\log(1+\varkappa)+2\big]}{\frac{C_*}{2}\sqrt{\mu_+}} = \frac{2\sqrt{2}}{\pi C_*}\sqrt{\Gamma}\,\big[\log(1+\varkappa)+2\big].$$

Part (v) is corollary SH.3 multiplied by $\varkappa$. Part (vi) is (i) together with proposition SH.2(vi). □

Full $\sqrt{\varkappa}$ acceleration of a slow mode, up to absolute constants, holds exactly when a fraction of the spectrum bounded below lies within a bounded factor of that mode (corollary SH.4(vi)). Acceleration is always at least of order $\log\varkappa$, and it is of order $\log\varkappa$ only when $\mu_+\mathcal{I}_\nu = O(1)$, for instance for a single small eigenvalue (corollary SH.4(ii),(iv)).

SH.8 Power-law spectra, the identity and the normalization

Lemma SH.14 (Scale invariance)**.** *For $a > 0$ let ν^a be the image of a probability measure ν under $x \mapsto ax$, and let $r > 0$. Then, for $\hat{s}, c, \mu > 0$ and $\hat{\varrho} \ge 0$,*

$$b_0^{\nu^a,r}(a\hat{s}) = b_0^{\nu,r}(\hat{s}), \qquad \mathfrak{d}^{\mathrm{os}}_{\nu^a,r}(a\mu; a\hat{\varrho}) = \sqrt{a}\,\mathfrak{d}^{\mathrm{os}}_{\nu,r}(\mu;\hat{\varrho}),$$
$$m_{\nu^a}(ac) = m_\nu(c), \qquad \chi_{\nu^a}(a\mu) = \chi_\nu(\mu).$$

Proof. The third identity is the substitution $x \mapsto ax$ in eq. (SH.4). It gives $m_{\nu^a}(a\hat{s}/b) = m_\nu(\hat{s}/b)$, so the equations eq. (SH.5) for $(\nu^a, a\hat{s})$ and for $(\nu, \hat{s})$ coincide, and so do their roots. The second identity follows by substituting $v = \sqrt{a}\,w$ in eq. (SH.6). Then $a\hat{\varrho} + v^2 = a(\hat{\varrho} + w^2)$, the root is unchanged by the first identity, and the integrand is unchanged. The fourth identity is the third inserted in eq. (SH.7). □

Corollary SH.5 (Power-law spectra)**.** *Let $\alpha > 0$, $N \ge 1$ and $c_N > 0$, let $\mu_k = c_N k^{-\alpha}$ for $1 \le k \le N$, and put $\nu_N \overset{\mathrm{def}}{=} N^{-1}\sum_{k=1}^N \delta_{\mu_k}$. For $0 < x \le 1$ put*

$$\Xi_\alpha(x) \overset{\mathrm{def}}{=} \int_0^\infty \varpi(y)\Big(\int_0^1 \frac{y\,u^\alpha}{y\,u^\alpha + x^\alpha}\,\mathrm{d}u\Big)^{1/2}\mathrm{d}y.$$

(i) For every $1 \le i \le N$,

$$0 \le \chi_{\nu_N}(\mu_i) - \Xi_\alpha(i/N) \le N^{-1/2}.$$

(ii) If $i \le \varepsilon N$ with $\varepsilon \in (0,1]$, then $1 - 2\varepsilon^{\alpha/(1+\alpha)} \le \chi_{\nu_N}(\mu_i) \le 1$. Consequently, for every sequence $\varepsilon_N \to 0$,

$$\max_{i \le \varepsilon_N N}\Big|\frac{\mathfrak{d}^{\mathrm{os}}_{\nu_N}(\mu_i)}{\sqrt{\mu_i/2}} - 1\Big| \longrightarrow 0.$$

(iii) The function Ξ_α is nonincreasing on $(0,1]$, and

$$1 - \Xi_\alpha(x) \le 2x^{\alpha/(1+\alpha)}, \qquad \Xi_\alpha(x) \ge \frac{C_*}{\sqrt{2(1+\alpha)}}.$$

(iv) For every $1 \le i \le N$,

$$\frac{C_*}{\sqrt{2(1+\alpha)}}\sqrt{\frac{\mu_i}{2}} \le \mathfrak{d}^{\mathrm{os}}_{\nu_N}(\mu_i) \le \sqrt{\frac{\mu_i}{2}}.$$

(v) If $N^\alpha \ge 2$*, the acceleration of corollary SH.4 for* $\nu = \nu_N$*,* $\mu_+ = \mu_1$ *and* $\mu = \mu_N$*, for which* $\varkappa = N^\alpha$*, satisfies*

$$\frac{C_*}{\sqrt{2(1+\alpha)}}\, N^{\alpha/2} \le \mathcal{A} \le \frac{\sqrt{2}}{C_*}\, N^{\alpha/2}.$$

Proof. By lemma SH.14 with $a = N^\alpha/c_N$, the efficiencies $\chi_{\nu_N}(\mu_i)$ do not change when μ_k is replaced by $(k/N)^{-\alpha}$, and neither does the mass $\nu_N([\delta\mu_i, \infty))$ for $\delta > 0$. Parts (i) and (ii) concern only these quantities, so in their proofs $\mu_k = (k/N)^{-\alpha}$.

Part (i) compares a Riemann sum with its integral. For $c > 0$,

$$1 - m_{\nu_N}(c) = \frac{1}{N}\sum_{k=1}^{N} \phi_c(k/N), \qquad \phi_c(u) \stackrel{\mathrm{def}}{=} \frac{c\,u^\alpha}{c\,u^\alpha + 1},$$

and ϕ_c is nondecreasing on $[0,1]$ with values in $[0,1)$. For a nondecreasing $\phi \ge 0$ and $k < N$,

$$\int_{(k-1)/N}^{k/N} \phi \le \frac{\phi(k/N)}{N} \le \int_{k/N}^{(k+1)/N} \phi.$$

Summing over k, with the last term $\phi_c(1)/N$ kept separately, gives

$$0 \le \frac{1}{N}\sum_{k=1}^{N} \phi_c(k/N) - \int_0^1 \phi_c \le \frac{\phi_c(1)}{N} \le \frac{1}{N}.$$

Since $|\sqrt{a} - \sqrt{b}| \le \sqrt{|a-b|}$ and $\int \varpi = 1$, the definition eq. (SH.7) gives

$$0 \le \chi_{\nu_N}(\mu) - \int_0^\infty \varpi(y)\Big(\int_0^1 \phi_{\mu y}\Big)^{1/2} \mathrm{d}y \le N^{-1/2}.$$

At $\mu = \mu_i = x^{-\alpha}$ with $x = i/N$ one has $\phi_{x^{-\alpha}y}(u) = y\,u^\alpha/(y\,u^\alpha + x^\alpha)$, which is (i).

Part (ii) is proposition SH.1(i). The mass above $\delta\mu_i$ is

$$\nu_N\big([\delta\mu_i, \infty)\big) = \frac{1}{N}\,\#\big\{k :\ k \le i\,\delta^{-1/\alpha}\big\} \le \varepsilon\,\delta^{-1/\alpha},$$

and the choice $\delta = \varepsilon^{\alpha/(1+\alpha)}$ gives $1 - \chi_{\nu_N}(\mu_i) \le 2\varepsilon^{\alpha/(1+\alpha)}$. The uniform statement follows with $\varepsilon = \varepsilon_N$ and lemma SH.8.

Part (iii) identifies Ξ_α with the efficiency of a limiting measure. Let ν_∞ be the law of $U^{-\alpha}$ with U uniform on $(0,1)$. Then

$$1 - m_{\nu_\infty}(c) = \int_0^1 \frac{c\,\mathrm{d}u}{c + u^{-\alpha}} = \int_0^1 \phi_c, \qquad \text{so} \qquad \Xi_\alpha(x) = \chi_{\nu_\infty}(x^{-\alpha}).$$

Since $1 - m_\nu$ is nondecreasing, χ_ν is nondecreasing in μ, and Ξ_α is nonincreasing in x. The mass of ν_∞ above $\delta x^{-\alpha}$ is

$$\nu_\infty\big([\delta x^{-\alpha}, \infty)\big) = \Pr\big(U \le x\,\delta^{-1/\alpha}\big) \le x\,\delta^{-1/\alpha},$$

and proposition SH.1(i) with $\delta = x^{\alpha/(1+\alpha)}$ gives $1 - \Xi_\alpha(x) \le 2x^{\alpha/(1+\alpha)}$. Finally, since $u^\alpha + x^\alpha \le 2$ for $u, x \le 1$,

$$1 - m_{\nu_\infty}(x^{-\alpha}) = \int_0^1 \frac{u^\alpha\,\mathrm{d}u}{u^\alpha + x^\alpha} \ge \int_0^1 \frac{u^\alpha}{2}\,\mathrm{d}u = \frac{1}{2(1+\alpha)},$$

and theorem SH.2(i) with lemma SH.8, which give $\chi_\nu(\mu) \ge C_*\sqrt{1 - m_\nu(\mu)}$, yield $\Xi_\alpha(x) \ge C_*/\sqrt{2(1+\alpha)}$.

Part (iv) combines (i), (iii) and lemma SH.8: one has $\chi_{\nu_N}(\mu_i) \ge \Xi_\alpha(i/N) \ge C_*/\sqrt{2(1+\alpha)}$ and $\chi_{\nu_N} \le 1$.

Part (v) is corollary SH.4(i). The support of ν_N lies in $[0, \mu_1]$, the slowest mode satisfies $\mu_N = \mu_1 N^{-\alpha} \le \mu_1/2$, and $\chi_{\nu_N}(\mu_N)$ lies between $C_*/\sqrt{2(1+\alpha)}$ and 1 by (iv). □

In the square case $N_{\rm out} = N_{\rm in} = N$ with the spectrum of corollary SH.5, the measure ν_N is $\hat{\nu}_{\rm out}$, and theorem SH.1(iii) at zero ridge reads $\sqrt{2}\,\mathfrak{d}_i(0) = \sqrt{\mu_i/(\gamma s_2)}\;\vartheta_i\,\chi_{\nu_N}(\mu_i)$ with $\gamma = N/B$, when $\epsilon_{\rm os} < 1$. This has the leading form of the resolved-window endpoint $\sqrt{2}\,\mathfrak{d}_i(0) \sim \sqrt{\mu_i/(\gamma s_2)}$ of corollary SG.1(ii). The two results hold in disjoint regimes, $\gamma < 1/(3(1-\beta^2))$ here and $\gamma \to \infty$ under Assumption S4 there, and neither implies the other.

Corollary SH.6 (Identity covariance)**.** *(i) One has* $\mathfrak{d}^{\rm os}_{\delta_1}(1) = \dfrac{4\sqrt{2}}{3\pi} = \dfrac{8}{3\pi\sqrt{2}}$.

(ii) If $\Sigma_{\rm out} = {\rm Id}_N$, $N_{\rm out} = N_{\rm in} = N$, $\gamma = N/B$ *and* $\epsilon_{\rm os} < 1$, *then*

$$\mathfrak{d}_0(1) = \frac{8\,\vartheta}{3\pi\sqrt{2s_2\gamma}} \qquad \text{for some } \vartheta \in \big[1-\epsilon_{\rm os}, (1-\epsilon_{\rm os})^{-1}\big].$$

Proof. Part (i) is an explicit integral. For $\nu = \delta_1$ one has $\Upsilon_{\delta_1}(c) = c^2/(1+c)$, so by the first formula of eq. (SH.15) and the substitution $w = 1 + c$,

$$\mathfrak{d}^{\rm os}_{\delta_1}(1) = \frac{\sqrt{2}}{\pi}\int_0^\infty \frac{c\,{\rm d}c}{(1+c)^{5/2}} = \frac{\sqrt{2}}{\pi}\int_1^\infty (w-1)\,w^{-5/2}\,{\rm d}w = \frac{\sqrt{2}}{\pi}\Big(2 - \frac{2}{3}\Big) = \frac{4\sqrt{2}}{3\pi}.$$

Part (ii) is theorem SH.1(ii)–(iii) at $\varrho = 0$, $r = 1$ and $\mu = 1$, where $\hat{\nu}_{\rm out} = \delta_1$ and $\sqrt{B_{\rm eff}/N} = (s_2\gamma)^{-1/2}$. □

Remark SH.4 (Agreement with the isotropic results)**.** *For* $\Sigma_{\rm out} = {\rm Id}_N$ *and* $N_{\rm out} = N_{\rm in} = N$, *the kernel* $\mathfrak{d}_\varrho(1)$ *is the isotropic kernel of appendix SF at the ridge* ϱ, *and its limit as* $\varrho \downarrow 0$ *is the zero-ridge constant* $C_\beta(\gamma)$ *of eq.* (SF.21), *by proposition SF.5(ii). Hence* $\mathfrak{d}_0(1) = C_\beta(\gamma)$, *and corollary SH.6(ii) is a second two-sided estimate of the square-case constant. It has the form of theorem SF.7(ii), which gives* $C_\beta(\gamma) = 8\vartheta/(3\pi\sqrt{2s_2\gamma})$ *with* $\vartheta \in [1-\varepsilon_\beta, (1-\varepsilon_\beta)^{-1/2}]$ *and* $\varepsilon_\beta = 6\gamma(1-\beta^2) \le \frac{1}{2}$. *The window of corollary SH.6,* $(1-\beta^2)\gamma < \frac{1}{3}$, *is four times wider than the window* $(1-\beta^2)\gamma \le \frac{1}{12}$ *of that theorem. Both estimates carry the momentum factor* $s_2^{-1/2}$, *which is the replacement of* B *by* $B_{\rm eff}$. *At* $\beta = 0$, *corollary SH.6(ii) gives the limit* $C(\gamma) \sim 8/(3\pi\sqrt{2\gamma})$ *of theorem SF.4(iii) with the rate* $|\vartheta - 1| \le 3\gamma/(1-3\gamma)$.

Remark SH.5 (Normalization of the oversampled drift)**.** *The drift coefficient enters the expected update through the identity* $\widetilde{\overline{G}}_\varrho = \mathfrak{d}_\varrho(\Sigma_{\rm out})\Delta_t/\sqrt{\mathscr{R}(t)}$ *of eq.* (SH.13), *with* $\mathscr{R}(t) = \frac{1}{2}\|{\Sigma_{\rm out}}^{1/2}\Delta_t\|_{\rm F}^2$. *In the row recursion eq.* (14), *the row* i *loses* $2\eta\,\mathfrak{d}_i\,\Omega_i/\sqrt{\mathcal{R}}$ *per step at first order. In the square case* $N_{\rm out} = N_{\rm in} = N$ *at zero ridge with* $\epsilon_{\rm os} < 1$, *theorem SH.1(iii) and lemma SH.8 give*

$$\mathfrak{d}_i(0) = \sqrt{\frac{B_{\rm eff}}{N}}\;\vartheta_i\,\chi_{\hat{\nu}_{\rm out}}(\mu_i)\,\sqrt{\frac{\mu_i}{2}}, \qquad \vartheta_i \in \big[1-\epsilon_{\rm os}, (1-\epsilon_{\rm os})^{-1}\big].$$

On head modes $\chi_{\hat{\nu}_{\rm out}}(\mu_i) \to 1$ *(proposition SH.1), so there* $\mathfrak{d}_i(0) \approx \sqrt{\mu_i B_{\rm eff}/(2N)}$, *and the expected update restricted to head modes is approximately*

$$\sqrt{\frac{B_{\rm eff}}{N}}\,\Big(\frac{\Sigma_{\rm out}}{2}\Big)^{1/2}\frac{\Delta_t}{\sqrt{\mathscr{R}(t)}} = \sqrt{\frac{B_{\rm eff}}{N}}\,{\Sigma_{\rm out}}^{1/2}\,\frac{\Delta_t}{\sqrt{2\mathscr{R}(t)}}.$$

The square-root law therefore carries the factor $2^{-1/2}$: *against* $\Delta_t/\sqrt{\mathscr{R}(t)}$ *it is* $(\Sigma_{\rm out}/2)^{1/2}$, *not* ${\Sigma_{\rm out}}^{1/2}$. *On the other modes the efficiency* $\chi_{\hat{\nu}_{\rm out}}(\mu_i) < 1$ *multiplies this law. The approximation is that of corollary SH.1 at a positive ridge; at zero ridge the display concerns the deterministic coefficient only.*

What the main text uses. The half-anisotropic section of the main text (section 4) uses theorem SH.1, in the form eq. (SH.10) at zero ridge, and corollary SH.1, in the form eq. (SH.12), for the drift of MUON when $N \le B_{\rm eff}$, together with the normalization of remark SH.5. Its power-law subsection uses theorem SH.2(i) and proposition SH.1(ii) for the square-root law and its scope, proposition SH.2 and corollaries SH.3 and SH.4 for the head and tail dichotomy and the acceleration of the slowest mode, and corollary SH.5(iv)–(v) for power-law spectra. No later appendix section depends on this one.

SI Deterministic model: Volterra analysis and the phase map

This section studies the row recursion eq. (14) of the main text in continuous time, with drift and volatility coefficients that are fixed numbers. The coefficients are adopted: they are inputs of a model and are not proved to be the coefficients of the algorithm at the times and ridges where the model is used. The one place in the appendix where the actual dynamics are proved is theorem SE.1 (with its row form corollary SE.1); it holds at a fixed positive normalized ridge, in proportional dimensions and on a finite interval of macroscopic time (remark SE.1). The model of this section is used for all times and with zero-ridge coefficients, so its conclusions are statements about the model.

The risk of the model solves a nonlinear Volterra equation. Appendix SI.2 proves that this equation has a unique solution, identifies its limit (the floor), reconstructs the row equations through a change of clock, and bounds first hitting times from below by the forcing and from above by a supersolution. Appendix SI.3 solves the model exactly when all drift coefficients are equal and proves the comparison of times behind the isotropic rate ratio of Sec. 3.2 of the main text in theorem SI.6. Appendix SI.4 collects the power-sum estimates. Appendix SI.5 introduces the adopted power-law arrays of SIGNSVD and SIGNSGD, states the hitting-time orders behind theorem 5.1 of the main text as theorem SI.7, and then states the sharper results on which it rests: the order of the hitting time in every phase with its joint window of dimension and accuracy, the exact constants on the two phase boundaries and in saturation, the accuracy-first law for fixed arrays, and the ratio of the two hitting times with its dimension factor. Appendix SI.6 removes the windows: it introduces the accuracy scale $\epsilon_N = N^{-(\alpha+\varsigma-1)}$, gives the hitting times for all joint (N, ϵ), shows that below ϵ_N the ratio freezes at its accuracy-first value, and states the resulting phase map in the accuracy exponent. Appendices SI.7 and SI.8 prove the comparison and kernel lemmas, appendix SI.9 proves the phase theorems, and appendix SI.10 proves the results of appendix SI.6.

Standing conventions. Time $t \geq 0$ is the continuous step count, and the step size $\eta > 0$ is constant. In the notation of appendix SE, $\eta = h/B$ and $t = B\tau$. The target exponent of the initial row energies is ς, written β in the main text; β is reserved for the momentum. For positive quantities, $x \sim y$ means $x/y \to 1$ along the stated limit, and $x \asymp y$ means that x/y lies between two positive constants whose dependence is stated. A first hitting time is the infimum of a set of times, and the infimum of the empty set is $+\infty$. For functions g and h on $[0, \infty)$ we write

$$(g * h)(u) \stackrel{\text{def}}{=} \int_0^u g(v)\, h(u - v)\, \mathrm{d}v.$$

A constant called fixed does not depend on the dimension N, the batch B or the accuracy ϵ.

SI.1 The deterministic model

Let (μ_i, u_i) be the eigenpairs of Σ_{out}. For an error matrix Δ the row energy of the eigenvector u_i is $\|\Delta^\top u_i\|^2$, and in the half-anisotropic case $\Sigma_{\text{in}} = \mathrm{Id}$ the risk is $\frac{1}{2}\sum_i \mu_i \|\Delta^\top u_i\|^2$ (section 4).

Definition SI.1 (Deterministic row model)**.** *Let the index set $\mathcal{I}$ be $\{1, \dots, N\}$ for an integer $N \geq 1$, or $\mathbb{N}$. The deterministic row model consists of the following four items.*

(M1) Adopted coefficients. *Numbers $\mu_i > 0$, $\mathfrak{d}_i > 0$, $\mathfrak{v}_i \geq 0$ and initial row energies $\Omega_i(0) \geq 0$, $i \in \mathcal{I}$, and a constant step $\eta > 0$ are fixed, with*

$$F_0 \stackrel{\text{def}}{=} \frac{1}{2}\sum_{i\in\mathcal{I}} \mu_i \Omega_i(0) \in (0, \infty), \qquad \mathcal{S} \stackrel{\text{def}}{=} \sum_{i\in\mathcal{I}} \frac{\mu_i \mathfrak{v}_i}{2\mathfrak{d}_i} < \infty. \tag{SI.1}$$

They do not depend on time or on the state. An array carrying the superscript s-SVD or s-SGD, such as $\mathfrak{d}_i^{\text{s-SVD}}$ or $\mathfrak{v}_i^{\text{s-SGD}}$, is an array adopted in this sense, and not a coefficient proved for an algorithm.

(M2) Kernel replacement. *For an update matrix S computed at a state Δ of positive risk $\mathscr{R}(\Delta)$, the conditional row pairing and the conditional raw second moment of the row are replaced by*

$$\mathbb{E}\left[\langle \Delta^\top u_i, \mathsf{S}^\top u_i\rangle \,\middle|\, \Delta\right] \rightsquigarrow \frac{\mathfrak{d}_i \|\Delta^\top u_i\|^2}{\sqrt{\mathscr{R}(\Delta)}}, \qquad \mathbb{E}\left[\|\mathsf{S}^\top u_i\|^2 \,\middle|\, \Delta\right] \rightsquigarrow \mathfrak{v}_i.$$

(M3) Dynamical closure. *The row energies* $\Omega_i(t)$ *solve*

$$\frac{\mathrm{d}\Omega_i}{\mathrm{d}t} = -\frac{2\eta\,\mathfrak{d}_i}{\sqrt{\mathcal{R}(t)}}\,\Omega_i(t) + \eta^2\mathfrak{v}_i, \qquad \mathcal{R}(t) \stackrel{\text{def}}{=} \frac{1}{2}\sum_{i\in\mathcal{I}}\mu_i\Omega_i(t), \tag{SI.2}$$

from the initial values of (M1), on an interval starting at 0 *on which* $\mathcal{R}$ *is continuous and positive.*

(M4) Transfer. *A statement about the stochastic algorithm is drawn from the model only through an approximation theorem with its own regime, horizon and observable.*

In this appendix the approximation theorem of (M4) is theorem SE.1.

Within this section the letters $\mathfrak{d}_i$ and $\mathfrak{v}_i$ denote the numbers of (M1). They are the coefficients of corollary SE.1 only when that identification is made, which is the content of (M2). The first time at which the model reaches a level $a > 0$ is

$$t_a \stackrel{\text{def}}{=} \inf\{t \geq 0 :\ \mathcal{R}(t) \leq a\},$$

and the main text's $t_{4\epsilon}$ is t_a at $a = 4\epsilon$.

The factors in eq. (SI.2) come from the following identity for one step.

Proposition SI.1 (One step of a row)**.** *Let* $\Delta \in \mathbb{R}^{N_{\text{out}}\times N_{\text{in}}}$ *be fixed, let* u *be a unit vector in* $\mathbb{R}^{N_{\text{out}}}$*, let* $\eta > 0$*, and let* S *be a random matrix with* $\mathbb{E}\,\|\mathsf{S}\|_{\text{F}}^2 < \infty$*. Then*

$$\mathbb{E}\,\|(\Delta - \eta\mathsf{S})^\top u\|^2 - \|\Delta^\top u\|^2 = -2\eta\ \mathbb{E}\langle\Delta^\top u, \mathsf{S}^\top u\rangle + \eta^2\ \mathbb{E}\,\|\mathsf{S}^\top u\|^2.$$

Proof. Expanding the square gives

$$\|(\Delta - \eta\mathsf{S})^\top u\|^2 = \|\Delta^\top u\|^2 - 2\eta\langle\Delta^\top u, \mathsf{S}^\top u\rangle + \eta^2\|\mathsf{S}^\top u\|^2.$$

Each term has finite expectation, because $|\langle\Delta^\top u, \mathsf{S}^\top u\rangle| \leq \|\Delta\|_{\text{F}}\|\mathsf{S}\|_{\text{F}}$ and $\|\mathsf{S}^\top u\|^2 \leq \|\mathsf{S}\|_{\text{F}}^2$. Taking expectations proves the identity. □

Applied at the state $\Delta = \Delta_t$ to the update S of step t, with the expectation conditional on the past, the first-order term of proposition SI.1 is the row pairing and the second-order term is the raw second moment of the row, not its variance. Under (M2) one step of the identity is one step of eq. (14), and (M3) is that recursion with a continuous step count. In the macroscopic time $\tau = t/B$ and with $\eta = h/B$, eq. (SI.2) reads

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\big(B\Omega_i\big) = -\frac{2h\,\mathfrak{d}_i}{\sqrt{\mathcal{R}}}\,B\Omega_i + h^2\mathfrak{v}_i,$$

which is eq. (SE.13). The identity itself does not prove (M2) or (M3).

Hypothesis (M2) is not implied by the positive-ridge theorems. Theorem SE.1 and corollary SE.1 prove that the algorithm follows eq. (SE.13) at a fixed normalized ridge $\varrho > 0$, in proportional dimensions and on the interval $[0, T_\star]$ of eq. (SE.10), with the coefficients $\mathfrak{d}(\mu_i, 1)$ and $\mathfrak{v}_i(\varrho)$. By remark SE.1 they say nothing about the zero ridge, about times beyond $T_\star$, or about an exchange of the ridge and dimension limits. The model of this section uses zero-ridge coefficients for all times, and therefore rests on (M1)–(M3).

Remark SI.1 (Where the adopted coefficients come from)**.** *The power-law arrays of definition SI.3 are modelled on the coefficients of the main text's half-anisotropic table in the oversampled regime* $B \geq N$ *(section 4), at zero momentum and zero ridge. For momentum* $\beta \in (0, 1)$*, theorem SG.1 computes the zero-ridge row coefficients in the thin-batch regime* $N/B \to \infty$*, and remark SG.3 inserts them into the row equations as adopted coefficients. The oversampled regime is not covered by that theorem. The resolved-window endpoint of corollary SG.1(ii),*

$$\sqrt{2}\,\mathfrak{d}_i(0) \sim \sqrt{\frac{\mu_i}{\gamma\, s_2}}, \qquad \mathfrak{v}_i(0) \to \frac{1}{s_2},$$

has the dependence on μ_i *and* γ *of the adopted* SIGNSVD *array, up to constant factors, and is the evidence for that array. In the oversampled regime itself, for* $B \gg N$ *and rows* $i \ll N$ *of the power law, the drift coefficient is* $\mathfrak{d}_i \sim \sqrt{\mu_i B/(2N)}$*, the value in the main text's table; the adopted*

SIGNSVD *array is larger by the uniform factor* $\sqrt{2}$*. A uniform factor* c *in the drift at a fixed volatility rescales the matched-step clock of* SIGNSVD *by* c^{-2}*, so with the factor removed every ratio of first hitting times below is halved; no exponent and no phase boundary changes. The adopted arrays with momentum reduce to it at a rescaled step (remark SI.5). The* SIGNSGD *array is compared with the corrected mode array of the* SIGNSGD *section in remark SI.4. No statement of this section identifies a trajectory of an algorithm.*

SI.2 The Volterra equation of the risk

Throughout this subsection the data of (M1) are fixed. The Volterra equation is written in an intrinsic time $s \geq 0$, related to the step count by the clock of theorem SI.2. Its forcing and kernel are

$$F(s) \stackrel{\text{def}}{=} \frac{1}{2} \sum_{i \in \mathcal{I}} \mu_i \mathcal{Q}_i(0)\, e^{-2\mathfrak{d}_i s}, \qquad k(s) \stackrel{\text{def}}{=} \frac{\eta}{2} \sum_{i \in \mathcal{I}} \mu_i \mathfrak{v}_i\, e^{-2\mathfrak{d}_i s} \quad (s > 0), \qquad \bar{k} \stackrel{\text{def}}{=} \frac{\eta \mathcal{S}}{2}. \tag{SI.3}$$

For a countable index set the series defining k may diverge at $s = 0$; it is finite for $s > 0$ by lemma SI.1. The equation is

$$Q = F + k * \sqrt{Q}, \tag{SI.4}$$

and a solution is a continuous nonnegative function on $[0, \infty)$ that satisfies it at every time.

Lemma SI.1 (Forcing and kernel)**.** *Assume (M1). The forcing* F *is continuous and strictly decreasing on* $[0, \infty)$*, with* $F(0) = F_0$*,* $F > 0$ *and* $F(s) \to 0$ *as* $s \to \infty$*. The kernel satisfies*

$$\int_0^\infty k(s)\, \mathrm{d}s = \bar{k}, \qquad k(s) \leq \frac{\bar{k}}{e\, s} \quad (s > 0).$$

For every $S' > 0$ *and every bounded function* f *on* $[0, S']$*, the convolution* $k * f$ *is continuous on* $[0, S']$*, vanishes at* 0*, and*

$$\sup_{0 \leq s \leq S'} |(k * f)(s)| \leq \bar{k} \sup_{0 \leq s \leq S'} |f(s)|. \tag{SI.5}$$

Proof. Each term of F is continuous and nonincreasing. The tail of the series beyond the index M is at most $\frac{1}{2} \sum_{i > M} \mu_i \mathcal{Q}_i(0)$ uniformly in s, which tends to zero because $F_0 < \infty$. The partial sums therefore converge uniformly, and F is continuous. Since $F_0 > 0$, some term has a positive coefficient, and that term is strictly decreasing; hence F is strictly decreasing and positive. Each term tends to zero and the tails are uniformly small, so $F(s) \to 0$.

Integrating each exponential gives

$$\int_0^\infty k(s)\, \mathrm{d}s = \frac{\eta}{2} \sum_i \frac{\mu_i \mathfrak{v}_i}{2\mathfrak{d}_i} = \bar{k}.$$

The scalar inequality $b e^{-bs} \leq 1/(es)$, valid for $b, s > 0$, applied with $b = 2\mathfrak{d}_i$, gives

$$k(s) = \sum_i \frac{\eta \mu_i \mathfrak{v}_i}{4\mathfrak{d}_i} (2\mathfrak{d}_i) e^{-2\mathfrak{d}_i s} \leq \frac{\bar{k}}{es}.$$

For bounded f, the convolution with the i-th term of k is

$$\frac{\eta \mu_i \mathfrak{v}_i}{2}\, e^{-2\mathfrak{d}_i s} \int_0^s e^{2\mathfrak{d}_i v} f(v)\, \mathrm{d}v,$$

which is continuous and vanishes at 0. The tail of the sum of these convolutions beyond the index M is at most $\sup |f| \cdot \frac{\eta}{2} \sum_{i > M} \mu_i \mathfrak{v}_i / (2\mathfrak{d}_i)$, which tends to zero. Uniform convergence gives continuity of $k * f$, and the same bound for the full sum gives eq. (SI.5). □

Lemma SI.2 (Tail of a convolution)**.** *Assume (M1), and let* $f \geq 0$ *be bounded on* $[0, \infty)$ *with* $\ell_- \stackrel{\text{def}}{=} \liminf_{s \to \infty} f(s)$ *and* $\ell_+ \stackrel{\text{def}}{=} \limsup_{s \to \infty} f(s)$*. Then*

$$\bar{k}\, \ell_- \leq \liminf_{s \to \infty} (k * f)(s) \leq \limsup_{s \to \infty} (k * f)(s) \leq \bar{k}\, \ell_+.$$

Proof. Fix $\varepsilon' > 0$ and choose s_0 such that $(\ell_- - \varepsilon')_+ \le f(v) \le \ell_+ + \varepsilon'$ for $v \ge s_0$. For $s > s_0$ split the convolution as

$$(k * f)(s) = \int_0^{s-s_0} k(v) f(s-v)\,\mathrm{d}v + \int_{s-s_0}^{s} k(v) f(s-v)\,\mathrm{d}v.$$

The first integral lies between $(\ell_- - \varepsilon')_+ \int_0^{s-s_0} k$ and $(\ell_+ + \varepsilon') \int_0^{s-s_0} k$. The second lies between 0 and $\sup f \cdot \int_{s-s_0}^{\infty} k$, which tends to zero because k is integrable. Letting $s \to \infty$ and then $\varepsilon' \to 0$ proves the claim. □

The next lemma is the comparison principle used throughout the section. It is stated for a general kernel, because the power-law analysis applies it to rescaled kernels.

Lemma SI.3 (Comparison)**.** *Let $S' > 0$, let $\mathsf{k} \ge 0$ be integrable on $(0, S')$, let $G \ge 0$ be continuous on $[0, S']$, and let $m > 0$. Let x and y be continuous and nonnegative on $[0, S']$, with $y \ge m$ and*

$$x \le G + \mathsf{k} * \sqrt{x}, \qquad y \ge G + \mathsf{k} * \sqrt{y} \qquad \text{on } [0, S'].$$

Then $x \le y$ on $[0, S']$.

Proof. Put $d \stackrel{\text{def}}{=} (x - y)_+$. Where $x > y$ one has

$$\sqrt{x} - \sqrt{y} = \frac{x - y}{\sqrt{x} + \sqrt{y}} \le \frac{d}{\sqrt{m}},$$

so $(\sqrt{x} - \sqrt{y})_+ \le d/\sqrt{m}$ everywhere. Subtracting the two inequalities and using $\mathsf{k} \ge 0$ gives

$$d \le \frac{1}{\sqrt{m}}\, \mathsf{k} * d \qquad \text{on } [0, S']. \tag{SI.6}$$

Since k is integrable, there is $\iota > 0$ with $m^{-1/2} \int_0^{\iota} \mathsf{k} \le \frac{1}{2}$. Suppose $d = 0$ on $[0, j\iota] \cap [0, S']$ for some integer $j \ge 0$. For s in $[j\iota, (j+1)\iota] \cap [0, S']$ the integrand of $\mathsf{k} * d$ vanishes for $v \le j\iota$, so eq. (SI.6) gives

$$d(s) \le \frac{1}{\sqrt{m}} \int_{j\iota}^{s} \mathsf{k}(s - v)\, d(v)\,\mathrm{d}v \le \frac{1}{2} \sup_{[j\iota, (j+1)\iota] \cap [0, S']} d.$$

Taking the supremum over s shows that this supremum vanishes. Finitely many steps cover $[0, S']$. □

Theorem SI.1 (Existence, uniqueness and floor)**.** *Assume (M1). The Volterra equation eq. (SI.4) has a unique solution Q on $[0, \infty)$. It satisfies*

$$0 < F(s) \le Q(s) \le M \stackrel{\text{def}}{=} \Big(\frac{\bar{k} + \sqrt{\bar{k}^2 + 4F_0}}{2} \Big)^2 \qquad (s \ge 0), \tag{SI.7}$$

and

$$\lim_{s \to \infty} Q(s) = \bar{k}^2 = \frac{\eta^2 \mathcal{S}^2}{4}.$$

If $\mathcal{S} > 0$ and $\epsilon > 0$, the unique step $\eta > 0$ with floor $\eta^2 \mathcal{S}^2/4 = \epsilon$ is $\eta = 2\sqrt{\epsilon}/\mathcal{S}$.

The theorem allows countably many modes and a kernel that is unbounded at the origin; only the two sums of eq. (SI.1) are assumed finite.

Proof. The solution is constructed by monotone iteration. Put $Q_0 \stackrel{\text{def}}{=} F$ and $Q_{n+1} \stackrel{\text{def}}{=} F + k * \sqrt{Q_n}$. Positivity of k gives $Q_1 \ge Q_0$, and induction gives $Q_{n+1} \ge Q_n$ for every n. The constant M is the positive root of

$$M = F_0 + \bar{k}\sqrt{M}.$$

If $Q_n \le M$, then eq. (SI.5) gives $Q_{n+1} \le F_0 + \bar{k}\sqrt{M} = M$. Each iterate is continuous by lemma SI.1. The iterates therefore increase to a limit Q with $F \le Q \le M$, and letting $n \to \infty$ in the iteration gives $Q = F + k * \sqrt{Q}$. The function $\sqrt{Q}$ is bounded, so $k * \sqrt{Q}$ is continuous by lemma SI.1, and Q is continuous.

Every solution is at least F, because $k * \sqrt{Q} \geq 0$. On a compact interval $[0, S']$ every solution is therefore at least $F(S') > 0$. Lemma SI.3, with $G = F$, $\mathsf{k} = k$ and $m = F(S')$, applied in both directions to two solutions, shows that they agree on $[0, S']$. This proves uniqueness.

For the limit, suppose first $\mathcal{S} = 0$. Then every $\mathfrak{v}_i$ vanishes, $k = 0$, and $Q = F \to 0 = \bar{k}^2$.

Suppose $\mathcal{S} > 0$. The limit inferior of Q is positive. To see this, choose an index j with $\mathfrak{v}_j > 0$, put $b_j \stackrel{\text{def}}{=} 2\mathfrak{d}_j$ and $c_j \stackrel{\text{def}}{=} \eta\mu_j\mathfrak{v}_j/2$, and let

$$z(s) \stackrel{\text{def}}{=} c_j \int_0^s e^{-b_j(s-v)} \sqrt{Q(v)}\, \mathrm{d}v.$$

Then $0 < z \leq k * \sqrt{Q} \leq Q$ for $s > 0$. Differentiating this single integral gives

$$z' = -b_j z + c_j \sqrt{Q} \geq -b_j z + c_j \sqrt{z}.$$

For a fixed $s_0 > 0$ the function $\mathsf{r} \stackrel{\text{def}}{=} \sqrt{z}$ is differentiable on $[s_0, \infty)$ and satisfies $\mathsf{r}' \geq -\frac{b_j}{2}\mathsf{r} + \frac{c_j}{2}$. Multiplying by $e^{b_j s/2}$ and integrating from s_0 gives

$$\mathsf{r}(s) \geq \frac{c_j}{b_j} + \Big(\mathsf{r}(s_0) - \frac{c_j}{b_j}\Big) e^{-b_j(s-s_0)/2}.$$

Hence $\liminf_{s\to\infty} Q(s) \geq (c_j/b_j)^2 > 0$.

Write $\ell_- \stackrel{\text{def}}{=} \liminf Q$ and $\ell_+ \stackrel{\text{def}}{=} \limsup Q$, both finite by eq. (SI.7). Since $F \to 0$, lemma SI.2 applied to $f = \sqrt{Q}$ in eq. (SI.4) gives

$$\ell_- \geq \bar{k}\sqrt{\ell_-}, \qquad \ell_+ \leq \bar{k}\sqrt{\ell_+}.$$

Because $\ell_- > 0$, the first inequality gives $\ell_- \geq \bar{k}^2$, and the second gives $\ell_+ \leq \bar{k}^2$. Thus $Q(s) \to \bar{k}^2$. Solving $\eta^2\mathcal{S}^2/4 = \epsilon$ for $\eta > 0$ gives the last assertion. □

The clock converts the intrinsic time s into the step count t.

Theorem SI.2 (Clock and row equations)**.** *Assume (M1) and let Q be the solution of theorem SI.1. For $i \in \mathcal{I}$ and $s \geq 0$ put*

$$E_i(s) \stackrel{\text{def}}{=} \mathcal{Q}_i(0) e^{-2\mathfrak{d}_i s} + \eta\mathfrak{v}_i \int_0^s e^{-2\mathfrak{d}_i(s-v)} \sqrt{Q(v)}\, \mathrm{d}v, \qquad t(s) \stackrel{\text{def}}{=} \frac{1}{\eta}\int_0^s \sqrt{Q(v)}\, \mathrm{d}v,$$

and $t_* \stackrel{\text{def}}{=} \lim_{s\to\infty} t(s) \in (0, \infty]$. Then $\frac{1}{2}\sum_i \mu_i E_i(s) = Q(s)$ for every s, and t is continuously differentiable and strictly increasing from $[0, \infty)$ onto $[0, t_*)$. Its inverse is the clock

$$\phi(t) = \int_0^t \frac{\eta}{\sqrt{\mathcal{R}(t')}}\, \mathrm{d}t', \tag{SI.8}$$

where $\mathcal{Q}_i(t) \stackrel{\text{def}}{=} E_i(\phi(t))$ and $\mathcal{R}(t) \stackrel{\text{def}}{=} Q(\phi(t))$ solve eq. (SI.2) on $[0, t_*)$ with continuous positive risk. Every solution of eq. (SI.2) with continuous positive risk on an interval $[0, t_1)$ agrees with this one there, and $[0, t_*)$ is the maximal such interval. If $\mathcal{S} > 0$, then $t_* = \infty$ and $\mathcal{R}(t) \to \eta^2\mathcal{S}^2/4$ as $t \to \infty$. If $\mathcal{S} = 0$, then $t_* = \eta^{-1}\int_0^\infty \sqrt{F(s)}\, \mathrm{d}s$, and $\mathcal{R}(t) \to 0$ as $t \uparrow t_*$ when $t_* < \infty$.

Proof. Summing the formulas for E_i with weights $\mu_i/2$ and using eq. (SI.3) gives

$$\frac{1}{2}\sum_i \mu_i E_i(s) = F(s) + \int_0^s k(s-v)\sqrt{Q(v)}\, \mathrm{d}v = Q(s).$$

Each E_i is differentiable, with

$$E_i' = -2\mathfrak{d}_i E_i + \eta\mathfrak{v}_i\sqrt{Q}.$$

Since Q is continuous and positive, $t' = \sqrt{Q}/\eta > 0$, so t is a continuously differentiable increasing bijection onto $[0, t_*)$. Its inverse satisfies $\phi'(t) = \eta/\sqrt{Q(\phi(t))}$ and $\phi(0) = 0$, which is eq. (SI.8). The chain rule applied to $E_i \circ \phi$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t} E_i(\phi(t)) = \frac{\eta}{\sqrt{\mathcal{R}(t)}}\big(-2\mathfrak{d}_i E_i(\phi(t)) + \eta\mathfrak{v}_i\sqrt{\mathcal{R}(t)}\big),$$

which is eq. (SI.2).

Conversely, let $(\mathfrak{Q}_i)$ solve eq. (SI.2) on $[0, t_1)$ with continuous positive risk. Define ϕ by eq. (SI.8); it is continuously differentiable with positive derivative. In the time $s = \phi(t)$ the chain rule gives $\frac{\mathrm{d}}{\mathrm{d}s}\mathfrak{Q}_i = -2\mathfrak{d}_i\mathfrak{Q}_i + \eta\mathfrak{v}_i\sqrt{\mathcal{R}}$. Multiplying by $e^{2\mathfrak{d}_i s}$ and integrating gives the formula for E_i with $\mathcal{R}$ in place of Q, and summing with weights $\mu_i/2$ gives eq. (SI.4) on the image interval. Uniqueness on compact intervals (theorem SI.1) identifies the risk and every row with the construction.

If $\mathcal{S} > 0$, the positive limit of theorem SI.1 gives $\sqrt{Q} \geq \bar{k}/2$ for large s, so $t_* = \infty$, and $\mathcal{R}(t) = Q(\phi(t))$ converges to $\bar{k}^2$. If $\mathcal{S} = 0$, then $Q = F$ and $t_* = \eta^{-1}\int_0^\infty \sqrt{F}$. When this is finite, $\phi(t) \to \infty$ as $t \uparrow t_*$ and $\mathcal{R}(t) = F(\phi(t)) \to 0$, so no solution with positive continuous risk extends to t_*. □

For a level $a > 0$ write

$$H_Q(a) \stackrel{\text{def}}{=} \inf\{s \geq 0 :\ Q(s) \leq a\}, \qquad H_F(a) \stackrel{\text{def}}{=} \inf\{s \geq 0 :\ F(s) \leq a\}$$

for the first intrinsic times at which the risk and the forcing reach a.

Theorem SI.3 (Hitting times)**.** *Assume (M1) and $\mathcal{S} > 0$, fix $\epsilon > 0$, and take $\eta = 2\sqrt{\epsilon}/\mathcal{S}$, so that $\bar{k} = \sqrt{\epsilon}$.*

(i) For every $a > 0$, $H_Q(a) \geq H_F(a)$ and

$$t_a \geq \frac{1}{\eta}\int_0^{H_F(a)} \sqrt{F(s)}\,\mathrm{d}s.$$

If $H_Q(a) < \infty$, then $t_a = t(H_Q(a))$ with $t(\cdot)$ of theorem SI.2.

(ii) Fix $c_0 > 1$ with $F_0 > c_0\epsilon$, let s_c be the time with $F(s_c) = c_0\epsilon$, and put

$$C_* \stackrel{\text{def}}{=} \max_{0 \leq s \leq s_c} \frac{(k * \sqrt{F})(s)}{\sqrt{\epsilon F(s)}}, \qquad m_* \stackrel{\text{def}}{=} \frac{C_*}{1 - 1/(2\sqrt{c_0})}.$$

Then $0 \leq C_* \leq \sqrt{F_0/(c_0\epsilon)}$ and

$$Q(s) \leq F(s) + m_*\sqrt{\epsilon F(s)} \qquad (0 \leq s \leq s_c). \tag{SI.9}$$

Consequently, for every $T \geq c_0 + m_*\sqrt{c_0}$,

$$\frac{1}{\eta}\int_0^{H_F(T\epsilon)} \sqrt{F(s)}\,\mathrm{d}s \ \leq\ t_{T\epsilon} \ \leq\ \frac{1}{\eta}\int_0^{s_c} \sqrt{F(s) + m_*\sqrt{\epsilon F(s)}}\,\mathrm{d}s. \tag{SI.10}$$

Neither bound assumes that the risk is monotone.

Proof. For (i), the convolution in eq. (SI.4) is nonnegative, so $Q \geq F$. Hence $\{Q \leq a\} \subseteq \{F \leq a\}$, and taking infima gives $H_Q(a) \geq H_F(a)$. If the first set is nonempty, it is closed by continuity and contains its infimum. The clock is a strictly increasing bijection, so the first time at which $\mathcal{R}$ reaches a is $t(H_Q(a))$. Using $\sqrt{Q} \geq \sqrt{F}$ in the integral defining $t(\cdot)$ and $H_Q(a) \geq H_F(a)$ gives the lower bound. If the first set is empty, then $t_a = \infty$ and the bound holds.

For (ii), the time s_c exists and is unique because F is continuous and strictly decreasing to zero. The ratio defining C_* is continuous on $[0, s_c]$, since $F \geq c_0\epsilon > 0$ there, and attains its maximum. Because $\sqrt{F} \leq \sqrt{F_0}$ and $\int k = \sqrt{\epsilon}$,

$$(k * \sqrt{F})(s) \leq \sqrt{F_0\epsilon}, \qquad \frac{(k * \sqrt{F})(s)}{\sqrt{\epsilon F(s)}} \leq \sqrt{\frac{F_0}{F(s)}} \leq \sqrt{\frac{F_0}{c_0\epsilon}}.$$

The function $\mathsf{B} \stackrel{\text{def}}{=} F + m_*\sqrt{\epsilon F}$ is a supersolution on $[0, s_c]$. The inequality $\sqrt{1+x} \leq 1 + x/2$ for $x \geq 0$ gives

$$\sqrt{\mathsf{B}(v)} \leq \sqrt{F(v)} + \frac{m_*}{2}\sqrt{\epsilon}.$$

Hence, for $0 \le s \le s_c$, using $F(s) \ge c_0\epsilon$ in the last step,

$$F(s) + (k * \sqrt{\mathsf{B}})(s) \le F(s) + (k * \sqrt{F})(s) + \frac{m_*\epsilon}{2} \le F(s) + \sqrt{\epsilon F(s)}\Big(C_* + \frac{m_*}{2\sqrt{c_0}}\Big) = \mathsf{B}(s).$$

Both Q and B are at least $F \ge c_0\epsilon$ on $[0, s_c]$, so lemma SI.3 with $m = c_0\epsilon$ gives eq. (SI.9). At s_c it gives

$$Q(s_c) \le \epsilon\,(c_0 + m_*\sqrt{c_0}) \le T\epsilon,$$

so $H_Q(T\epsilon) \le s_c$. Part (i) and the clock integral, bounded with eq. (SI.9), give eq. (SI.10). □

Remark SI.2 (The whole-interval ratio)**.** *The constant C_* of theorem SI.3(ii) is a maximum over $[0, s_c]$, not the value of the ratio at s_c. The endpoint value does not dominate the ratio in general. For two modes with $\eta = 1$, $\mu_1 = \mu_2 = 1$, $\mathfrak{d}_1 = 1$, $\mathfrak{d}_2 = \frac{1}{2}$, $\mathfrak{v}_1 = 2$, $\mathfrak{v}_2 = 0$, one has $k(s) = e^{-2s}$ and $\epsilon = \bar{k}^2 = \frac{1}{4}$. Choose the initial energies so that $F(s) = M'(e^{-2s} + 2^{-20}e^{-s})$, with M' fixed by $F(40\log 2) = 2\epsilon$. Bounding $\sqrt{F}$ between $\sqrt{M'}e^{-s}$ and $\sqrt{M'}(e^{-s} + 2^{-10}e^{-s/2})$ and integrating the exponentials shows that the quotient $(k * \sqrt{F})/\sqrt{F}$ exceeds $\frac{9}{10}$ at $s = 4\log 2$ and is below $\frac{7}{10}$ at $s = 40\log 2$; the ratio of theorem SI.3 is this quotient divided by $\sqrt{\epsilon} = \frac{1}{2}$. The upper bound eq. (SI.9) therefore needs the ratio on the whole interval.*

SI.3 Common drift and the isotropic comparison

When all drift coefficients are equal, the model closes on the risk alone and can be solved exactly. Throughout this subsection the index set is finite, $\mathfrak{d}_i = \mathfrak{d} > 0$ for every i, and

$$\mathfrak{v}_{\mathrm{tot}} \stackrel{\mathrm{def}}{=} \sum_{i=1}^{N} \mu_i\mathfrak{v}_i > 0,$$

which is the volatility $\mathfrak{v}$ of eq. (8).

Proposition SI.2 (Common drift)**.** *Assume (M1) with a finite index set, $\mathfrak{d}_i = \mathfrak{d} > 0$ for all i and $\mathfrak{v}_{\mathrm{tot}} > 0$. Then*

$$F(s) = F_0 e^{-2\mathfrak{d}s}, \qquad k(s) = \frac{\eta\mathfrak{v}_{\mathrm{tot}}}{2}e^{-2\mathfrak{d}s}, \qquad \mathcal{S} = \frac{\mathfrak{v}_{\mathrm{tot}}}{2\mathfrak{d}},$$

and the risk solves

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}t} = -2\eta\mathfrak{d}\sqrt{\mathcal{R}} + \frac{\eta^2\mathfrak{v}_{\mathrm{tot}}}{2}. \tag{SI.11}$$

Put $a_\circ \stackrel{\mathrm{def}}{=} \eta\mathfrak{d}$, $b_\circ \stackrel{\mathrm{def}}{=} \eta\mathfrak{v}_{\mathrm{tot}}/(4\mathfrak{d}) = \eta\mathcal{S}/2$ and $u_0 \stackrel{\mathrm{def}}{=} \sqrt{F_0}$.

(i) If $u_0 \ne b_\circ$, then for every $t \ge 0$

$$a_\circ t = u_0 - \sqrt{\mathcal{R}(t)} + b_\circ \log\frac{|u_0 - b_\circ|}{|\sqrt{\mathcal{R}(t)} - b_\circ|}, \tag{SI.12}$$

and $\sqrt{\mathcal{R}}$ is strictly decreasing with limit $b_\circ$ if $u_0 > b_\circ$, strictly increasing with limit $b_\circ$ if $u_0 < b_\circ$, and never equal to $b_\circ$. If $u_0 = b_\circ$, then $\mathcal{R} \equiv b_\circ^2$.

(ii) Let $a \ge 0$. If $F_0 \le a$, then $t_a = 0$. If $F_0 > a$, then $t_a < \infty$ if and only if $a > b_\circ^2$, and then t_a is given by eq. (SI.12) with $\sqrt{\mathcal{R}(t)} = \sqrt{a}$.

The closure needs only a common drift; the weights μ_i, the volatilities and the initial energies may differ from row to row.

Proof. Summing eq. (SI.2) with weights $\mu_i/2$ gives

$$\frac{\mathrm{d}\mathcal{R}}{\mathrm{d}t} = -\frac{\eta\mathfrak{d}}{\sqrt{\mathcal{R}}}\sum_i \mu_i\mathcal{Q}_i + \frac{\eta^2}{2}\sum_i \mu_i\mathfrak{v}_i = -2\eta\mathfrak{d}\sqrt{\mathcal{R}} + \frac{\eta^2\mathfrak{v}_{\mathrm{tot}}}{2}.$$

The formulas for F, k and $\mathcal{S}$ follow from eq. (SI.3) and eq. (SI.1) by finite summation. Since $\mathcal{S} > 0$, theorem SI.2 gives a global solution with positive risk and $\mathcal{R}(t) \to b_\circ^2$. In the variable $\mathsf{u} \stackrel{\mathrm{def}}{=} \sqrt{\mathcal{R}}$ the relation eq. (SI.11) reads

$$\mathsf{u}' = -\frac{a_\circ(\mathsf{u} - b_\circ)}{\mathsf{u}}.$$

The function $\mathsf{w} \stackrel{\text{def}}{=} \mathsf{u} - b_\circ$ therefore satisfies $\mathsf{w}' = -(a_\circ/\mathsf{u})\mathsf{w}$, so that

$$\mathsf{w}(t) = \mathsf{w}(0) \exp\Big(- a_\circ \int_0^t \frac{\mathrm{d}t'}{\mathsf{u}(t')}\Big).$$

Thus w keeps the sign of $\mathsf{w}(0)$, $|\mathsf{w}|$ is strictly decreasing when $\mathsf{w}(0) \neq 0$, and $\mathsf{w} \equiv 0$ when $\mathsf{w}(0) = 0$. When $\mathsf{w}(0) \neq 0$, the equation is separable:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\mathsf{u} + b_\circ \log|\mathsf{u} - b_\circ|\Big) = \frac{\mathsf{u}}{\mathsf{u} - b_\circ}\,\mathsf{u}' = -a_\circ,$$

and integrating from 0 to t gives eq. (SI.12). This proves (i).

For (ii), $\mathcal{R}(0) = F_0$ gives $t_a = 0$ when $F_0 \leq a$. Let $F_0 > a$. If $u_0 > b_\circ$, the risk decreases strictly to $b_\circ^2$ without reaching it, so it reaches a at a finite time exactly when $a > b_\circ^2$, and that time solves eq. (SI.12). If $u_0 \leq b_\circ$, the risk never falls below $F_0 > a$, so $t_a = \infty$, and in this case $a < F_0 \leq b_\circ^2$. □

Theorem SI.4 (Matched floors)**.** *In the setting of proposition SI.2, let $\epsilon > 0$, $T > 1$ and $F_0 > T\epsilon$, and take the matched step $\eta = 2\sqrt{\epsilon}/\mathcal{S} = 4\mathfrak{d}\sqrt{\epsilon}/\mathfrak{v}_{\mathrm{tot}}$. Put $X \stackrel{\text{def}}{=} \sqrt{F_0/\epsilon} > \sqrt{T}$. Then the floor is ϵ and*

$$t_{T\epsilon} = \frac{\mathfrak{v}_{\mathrm{tot}}}{4\mathfrak{d}^2}\Big[X - \sqrt{T} + \log\frac{X-1}{\sqrt{T}-1}\Big]. \tag{SI.13}$$

If $x \stackrel{\text{def}}{=} 1/X \leq \frac{1}{2}$, then

$$\Big|\frac{t_{T\epsilon}}{\frac{\mathfrak{v}_{\mathrm{tot}}}{4\mathfrak{d}^2}\sqrt{F_0/\epsilon}} - 1\Big| \leq x\Big[\sqrt{T} + |\log(\sqrt{T}-1)| + \log\frac{1}{x} + 2x\Big]. \tag{SI.14}$$

For two such models with the same F_0, ϵ and T,

$$\frac{t^{(1)}_{T\epsilon}}{t^{(2)}_{T\epsilon}} = \frac{\mathfrak{v}^{(1)}_{\mathrm{tot}}/(\mathfrak{d}^{(1)})^2}{\mathfrak{v}^{(2)}_{\mathrm{tot}}/(\mathfrak{d}^{(2)})^2}.$$

Proof. At the matched step $b_\circ = \sqrt{\epsilon}$ and $a_\circ = 4\mathfrak{d}^2\sqrt{\epsilon}/\mathfrak{v}_{\mathrm{tot}}$. Since $T\epsilon > b_\circ^2$ and $F_0 > T\epsilon$, proposition SI.2(ii) applies with $a = T\epsilon$, and dividing eq. (SI.12) by $a_\circ$ gives eq. (SI.13). Dividing eq. (SI.13) by its leading term $\frac{\mathfrak{v}_{\mathrm{tot}}}{4\mathfrak{d}^2}X$ and using $X - 1 = X(1-x)$ gives

$$\frac{t_{T\epsilon}}{\frac{\mathfrak{v}_{\mathrm{tot}}}{4\mathfrak{d}^2}X} - 1 = x\Big[-\sqrt{T} + \log\frac{1}{x} + \log(1-x) - \log(\sqrt{T}-1)\Big].$$

For $0 < x \leq \frac{1}{2}$,

$$|\log(1-x)| = \int_0^x \frac{\mathrm{d}y}{1-y} \leq 2x,$$

which proves eq. (SI.14). The bracket in eq. (SI.13) is positive and depends only on F_0, ϵ and T, so it cancels in the ratio. □

The isotropic arrays. Let $N, B \geq 1$ be integers, $\gamma \stackrel{\text{def}}{=} N/B$ and $m \stackrel{\text{def}}{=} \min(B, N)$. Let $\mathcal{N}_B \stackrel{\text{def}}{=} \mathbb{E}\left[(\sum_{b=1}^B x_b^2 y_b^2)^{1/2}\right]$ for independent standard Gaussian x_b, y_b, as in section 3. The isotropic arrays have N rows with $\mu_i = 1$, common initial energies, and

$$\mathfrak{d}_i^{\text{s-SVD}} = C(\gamma), \quad \sum_{i=1}^N \mathfrak{v}_i^{\text{s-SVD}} = m; \qquad \mathfrak{d}_i^{\text{s-SGD}} = \frac{\mathcal{N}_B}{\sqrt{\pi}}, \quad \sum_{i=1}^N \mathfrak{v}_i^{\text{s-SGD}} = N^2, \tag{SI.15}$$

where $C(\gamma) = C_0(\gamma) > 0$ is the zero-ridge drift constant eq. (SF.21) at $\beta = 0$; eq. (9) gives an approximate form of it. These are the kernels of the table following eq. (8) without momentum, and they are adopted in the sense of (M1). The arrays with momentum, with the pair $(C_\beta(\gamma), \mathfrak{v}_{\mathrm{iso},\beta})$ of eq. (SF.21) and definition SF.1 in place of $(C(\gamma), m)$, leave the Volterra algebra of proposition SI.2 and theorem SI.4 unchanged, because that algebra uses only a common drift and a total volatility. The bounds

$$\max\Big\{\frac{2}{\pi}\sqrt{B},\ \sqrt{B} - \frac{4}{\sqrt{B}}\Big\} \leq \mathcal{N}_B \leq \sqrt{B} \tag{SI.16}$$

are proposition SJ.1. Put

$$\rho(\gamma) \stackrel{\text{def}}{=} \pi\, C(\gamma)^2\, \gamma \max(1, \gamma).$$

Proposition SI.3 (The isotropic ratio)**.** *Take the arrays eq.* (SI.15) *with a common* F_0*, the matched steps* $\eta^\star = 2\sqrt{\epsilon}/\mathcal{S}$ *of each, and* $T > 1$ *with* $F_0 > T\epsilon$*. Then*

$$\mathcal{S}^{\text{s-SVD}} = \frac{m}{2C(\gamma)}, \qquad \mathcal{S}^{\text{s-SGD}} = \frac{N^2\sqrt{\pi}}{2\mathcal{N}_B}, \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} = \frac{\pi N^2 C(\gamma)^2}{m\,\mathcal{N}_B^2} = \rho(\gamma)\,\frac{B}{\mathcal{N}_B^2}.$$

Moreover $1 \le B/\mathcal{N}_B^2 \le \pi^2/4$ *for every* B*, and for* $B > 4$

$$0 \le \frac{t^{\text{s-SGD}}_{T\epsilon}/t^{\text{s-SVD}}_{T\epsilon}}{\rho(\gamma)} - 1 \le \Big(1 - \frac{4}{B}\Big)^{-2} - 1.$$

Proof. The noise constants are $\mathfrak{v}_{\text{tot}}/(2\mathfrak{d})$ by proposition SI.2. The exact ratio of theorem SI.4 is

$$\frac{N^2/(\mathcal{N}_B^2/\pi)}{m/C(\gamma)^2} = \frac{\pi N^2 C(\gamma)^2}{m\,\mathcal{N}_B^2}.$$

Since $N/B = \gamma$ and $N/m = \max(1,\gamma)$,

$$\frac{N^2}{mB} = \gamma\max(1,\gamma),$$

which gives the second form. Squaring and inverting eq. (SI.16) gives $1 \le B/\mathcal{N}_B^2 \le \pi^2/4$, and for $B > 4$ the lower bound $\mathcal{N}_B \ge \sqrt{B}(1 - 4/B) > 0$ gives $B/\mathcal{N}_B^2 \le (1 - 4/B)^{-2}$. □

The coefficient $C(\gamma)$ is now described through a scalar measure. Let Z be a standard Gaussian variable and

$$\Phi(v) \overset{\text{def}}{=} \mathbb{E}\,\frac{vZ^2}{1 + vZ^2} \qquad (v \ge 0).$$

Definition SI.2 (Scalar measure hypothesis)**.** *Let* $\gamma > 0$*. The scalar measure hypothesis at* γ *is the existence of a finite nonnegative measure* μ_γ *on* $[0,\infty)$ *of mass* γ *whose transform* $\sigma_\gamma(t) \overset{\text{def}}{=} \int (x+t)^{-1}\mu_\gamma(\mathrm{d}x)$*,* $t > 0$*, satisfies*

$$\psi_\gamma(t) \overset{\text{def}}{=} \gamma - t\,\sigma_\gamma(t) = \Phi\big(t\,\sigma_\gamma(t)^2\big) \qquad (t > 0). \tag{SI.17}$$

Under it, $C(\gamma) \overset{\text{def}}{=} (\sqrt{2}\,\gamma^2)^{-1}\int \sqrt{x}\,\mu_\gamma(\mathrm{d}x)$*.*

This is the square-case scalar equation of the isotropic analysis, with the same measure for both companions, and $C(\gamma)$ is its half-moment drift coefficient; the negative-axis form of the same quantity is eq. (SF.21) at $\beta = 0$, and proposition SF.7 identifies the two forms. The hypothesis holds at every $\gamma > 0$, with exactly one measure μ_γ: proposition SF.6 gives exactly one isotropic measure pair at (γ,γ), and the correspondence stated after definition SF.2 identifies both of its measures with μ_γ. The definition of $C(\gamma)$ therefore applies at every $\gamma > 0$.

Lemma SI.4 (An integral for the drift constant)**.** *The map* Φ *is a continuously differentiable increasing bijection from* $(0,\infty)$ *onto* $(0,1)$*; let* V *be its inverse and* $w(y) \overset{\text{def}}{=} V(y)^{-1/2}$ *for* $0 < y < 1$*. The function* $w_\circ(y) \overset{\text{def}}{=} \sqrt{y}\,w(y)$ *is positive, strictly decreasing, at most* 1*, and tends to* 1 *as* $y \downarrow 0$*. For every* $\gamma > 0$*,*

$$C(\gamma) = \frac{\sqrt{2}}{\pi\gamma^2}\,I(\gamma), \qquad I(\gamma) \overset{\text{def}}{=} \int_0^{\min(\gamma,1)} (\gamma - y)\,w(y)\,\mathrm{d}y. \tag{SI.18}$$

Moreover

$$\mathcal{A} \overset{\text{def}}{=} \int_0^1 w(y)\,\mathrm{d}y = \sqrt{\frac{\pi}{2}}, \qquad \mathcal{D} \overset{\text{def}}{=} \int_0^1 y\,w(y)\,\mathrm{d}y \in (0,\mathcal{A}), \qquad I(\gamma) = \gamma\mathcal{A} - \mathcal{D} \quad (\gamma \ge 1). \tag{SI.19}$$

In particular $C(\gamma)$ does not depend on the choice of the measure.

Proof. The map Φ is continuously differentiable with

$$\Phi'(v) = \mathbb{E}\,\frac{Z^2}{(1+vZ^2)^2} > 0.$$

The quotient $\Phi(v)/v = \mathbb{E}[Z^2/(1+vZ^2)]$ is strictly decreasing in v, tends to $\mathbb{E}\,Z^2 = 1$ as $v \downarrow 0$, and $\Phi(v) \to 1$ as $v \to \infty$. Hence Φ is an increasing bijection onto $(0,1)$ with a continuously differentiable inverse. Since

$$w_\circ(y)^2 = \frac{y}{V(y)} = \frac{\Phi(V(y))}{V(y)}$$

and V is increasing, $w_\circ$ is strictly decreasing, at most 1, and tends to 1 at 0. In particular $w(y) \le y^{-1/2}$, so every integral in the statement is finite.

The function ψ_γ is the inverse of an explicit function. For $0 < y < \min(\gamma, 1)$ put $\Theta_\gamma(y) \stackrel{\text{def}}{=} (\gamma - y)^2/V(y)$. It is strictly decreasing. It tends to ∞ as $y \downarrow 0$, because $V(y)/y \to 1$. It tends to 0 at the upper endpoint, because the numerator tends to 0 if $\gamma < 1$ and the denominator tends to ∞ if $\gamma \ge 1$. So Θ_γ is a bijection onto $(0,\infty)$. The quantity $t\sigma_\gamma(t) = \int t(x+t)^{-1}\mu_\gamma(\mathrm{d}x)$ lies in $(0,\gamma]$, and eq. (SI.17) gives $\psi_\gamma(t) = \Phi(\cdot) \in (0,1)$; hence $0 < \psi_\gamma(t) < \min(\gamma,1)$. By eq. (SI.17), $V(\psi_\gamma(t)) = t\sigma_\gamma(t)^2 = (\gamma - \psi_\gamma(t))^2/t$, that is,

$$t = \Theta_\gamma(\psi_\gamma(t)) \qquad (t > 0).$$

The half moment is an integral of ψ_γ. The half-moment formula eq. (SF.26) of lemma SF.3(v), applied to $\nu = \mu_\gamma$, expresses $\int \sqrt{x}\,\mu_\gamma(\mathrm{d}x)$ through $\int x(x+t)^{-1}\mu_\gamma(\mathrm{d}x)$. This inner integral is

$$\int \frac{x}{x+t}\,\mu_\gamma(\mathrm{d}x) = \gamma - t\sigma_\gamma(t) = \psi_\gamma(t),$$

so that

$$\int \sqrt{x}\,\mu_\gamma(\mathrm{d}x) = \frac{1}{\pi}\int_0^\infty t^{-1/2}\psi_\gamma(t)\,\mathrm{d}t.$$

The change of variables $t = \Theta_\gamma(y)$ and an integration by parts evaluate this integral:

$$\begin{aligned}\int_0^\infty t^{-1/2}\psi_\gamma(t)\,\mathrm{d}t &= \int_0^{\min(\gamma,1)} y\,\big(-\Theta_\gamma'(y)\big)\Theta_\gamma(y)^{-1/2}\,\mathrm{d}y \\ &= \Big[-2y\sqrt{\Theta_\gamma(y)}\Big]_0^{\min(\gamma,1)} + 2\int_0^{\min(\gamma,1)} \sqrt{\Theta_\gamma(y)}\,\mathrm{d}y.\end{aligned}$$

The boundary term vanishes at 0 because $\sqrt{\Theta_\gamma(y)} = (\gamma - y)w(y) \le \gamma y^{-1/2}$, and at the upper endpoint because $y \le 1$ and $\Theta_\gamma \to 0$. Hence the integral equals $2I(\gamma)$, and the definition of $C(\gamma)$ gives eq. (SI.18).

For eq. (SI.19), the substitution $y = \Phi(v)$ gives

$$\mathcal{A} = \int_0^\infty v^{-1/2}\Phi'(v)\,\mathrm{d}v = \mathbb{E}\int_0^\infty \frac{Z^2 v^{-1/2}}{(1+vZ^2)^2}\,\mathrm{d}v = \mathbb{E}\,|Z|\int_0^\infty \frac{y'^{-1/2}}{(1+y')^2}\,\mathrm{d}y' = \sqrt{\frac{2}{\pi}}\cdot\frac{\pi}{2},$$

with $y' = vZ^2$ in the third step. The last integral is the Beta integral $\Gamma(\frac12)\Gamma(\frac32)/\Gamma(2) = \pi/2$ [21, 5.12.3], and $\mathbb{E}\,|Z| = \sqrt{2/\pi}$. Since $0 < y < 1$ and $w > 0$, $0 < \mathcal{D} < \mathcal{A}$. For $\gamma \ge 1$ the range of integration in $I(\gamma)$ is $(0,1)$, which gives the last formula. □

Theorem SI.5 (Extrema of the isotropic ratio)**.** *The ratio ρ is continuous on $(0,\infty)$, strictly decreasing on $(0,1]$ and strictly increasing on $[1,\infty)$, and*

$$\min_{\gamma>0}\rho(\gamma) = \rho(1) = \pi C(1)^2, \qquad \sup_{\gamma>0}\rho(\gamma) = \lim_{\gamma\downarrow 0}\rho(\gamma) = \frac{32}{9\pi}, \qquad \lim_{\gamma\to\infty}\rho(\gamma) = 1.$$

The minimum is attained only at $\gamma = 1$ and the supremum is not attained. For $\gamma \ge 1$, $\rho(\gamma) = (1 - \mathsf{d}/\gamma)^2$ with $\mathsf{d} \stackrel{\text{def}}{=} \mathcal{D}/\mathcal{A} = 1 - \sqrt{\pi}\,C(1) \in (0,1)$.

Proof. For $0 < \gamma < 1$, eq. (SI.18) gives

$$\rho(\gamma) = \frac{2}{\pi}\,\gamma^{-3} I(\gamma)^2, \qquad I'(\gamma) = \int_0^\gamma w(y)\,\mathrm{d}y,$$

where the derivative has no boundary term because the integrand of I vanishes at $y = \gamma$. The derivative of $\gamma^{-3} I^2$ is $\gamma^{-4} I\,(2\gamma I' - 3I)$, and

$$2\gamma I'(\gamma) - 3I(\gamma) = \int_0^\gamma (3y - \gamma)\, y^{-1/2} w_\circ(y)\,\mathrm{d}y.$$

The same integral with $w_\circ(y)$ replaced by the constant $w_\circ(\gamma/3)$ vanishes, since $\int_0^\gamma (3y - \gamma) y^{-1/2}\mathrm{d}y = 2\gamma^{3/2} - 2\gamma^{3/2} = 0$. Subtracting it gives

$$2\gamma I'(\gamma) - 3I(\gamma) = \int_0^\gamma (3y - \gamma)\, y^{-1/2} \big(w_\circ(y) - w_\circ(\gamma/3)\big)\,\mathrm{d}y < 0,$$

because $w_\circ$ is strictly decreasing, so the integrand is negative except at $y = \gamma/3$. Hence ρ is strictly decreasing on $(0, 1)$.

For $\gamma \geq 1$, eq. (SI.19) and $\mathcal{A}^2 = \pi/2$ give

$$\rho(\gamma) = \frac{2}{\pi}\big(\mathcal{A} - \mathcal{D}/\gamma\big)^2 = \Big(1 - \frac{\mathsf{d}}{\gamma}\Big)^2,$$

which increases strictly to 1. The formula eq. (SI.18) is continuous in γ, which gives continuity at 1, and $\rho(1) = (1 - \mathsf{d})^2 = \pi C(1)^2$ identifies d.

As $\gamma \downarrow 0$, the substitution $y = \gamma y'$ gives

$$\gamma^{-3/2} I(\gamma) = \int_0^1 (1 - y')\, y'^{-1/2}\, w_\circ(\gamma y')\,\mathrm{d}y' \longrightarrow \int_0^1 (1 - y')\, y'^{-1/2}\,\mathrm{d}y' = \frac{4}{3},$$

since $0 < w_\circ \leq 1$ and $w_\circ(\gamma y') \to 1$ for each y'. Hence $\rho(\gamma) \to \frac{2}{\pi} \cdot \frac{16}{9} = \frac{32}{9\pi}$, and $\frac{32}{9\pi} > 1$ because $\pi < \frac{32}{9}$. Strict monotonicity on each side of 1 gives the remaining claims. □

Corollary SI.1 (The value at $\gamma = 1$)**.** *The function ρ of proposition SI.3 satisfies*

$$\rho(1) = \frac{1}{4}\Big(1 + \frac{\log(1 + \sqrt{2})}{\sqrt{2}}\Big)^2 \approx 0.6587.$$

Proof. By the definition of ρ at $\gamma = 1$, $\rho(1) = \pi\, C(1)^2$. Theorem SF.6 evaluates the scalar coefficient at aspect ratio one,

$$C(1) = \frac{1 + \log(1 + \sqrt{2})/\sqrt{2}}{2\sqrt{\pi}}.$$

Squaring and multiplying by π gives the display. □

The interpolation $[\pi\gamma(9\pi/32 + \gamma)]^{-1/2}$ of eq. (9) is an approximation of $C(\gamma)$ and is not used here.

The ratio of the times to a common floor quoted in Sec. 3.2 of the main text is, at $\beta = 0$, the following theorem; at $\beta > 0$ the same algebra applies with the momentum pair, as stated after eq. (SI.15).

Theorem SI.6 (Isotropic comparison)**.** *Let $N, B \geq 1$, $\gamma = N/B$, $m = \min(B, N)$, and take the isotropic arrays eq. (SI.15) with a common initial risk $F_0 = \mathcal{R}^{\text{s-SVD}}(0) = \mathcal{R}^{\text{s-SGD}}(0)$. Choose $\eta^\star_{\text{s-SVD}}$ and $\eta^\star_{\text{s-SGD}}$ by eq. (16), so that both models have floor ϵ, and fix $T > 1$ with $F_0 > T\epsilon$.*

(i) For every such ϵ, $t^{\text{s-SGD}}_{T\epsilon}/t^{\text{s-SVD}}_{T\epsilon} = \rho(\gamma)\, B/\mathcal{N}_B^2$. As $\epsilon/F_0 \to 0$, uniformly in N, B and $C(\gamma)$,

$$t^{\text{s-SVD}}_{T\epsilon} \sim \frac{m}{4C(\gamma)^2}\sqrt{\frac{F_0}{\epsilon}}, \qquad t^{\text{s-SGD}}_{T\epsilon} \sim \frac{\pi N^2}{4\mathcal{N}_B^2}\sqrt{\frac{F_0}{\epsilon}}.$$

(ii) If in addition $B \to \infty$, with N and γ allowed to vary, then

$$t^{\text{s-SGD}}_{T\epsilon} \sim \frac{\pi N^2}{4B}\sqrt{\frac{F_0}{\epsilon}}, \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}}\Big/\rho(\gamma) \longrightarrow 1,$$

and the ratio tends to $\rho(\gamma_0)$ if $\gamma \to \gamma_0 \in (0, \infty)$.

(iii) For every $\gamma > 0$,

$$\rho(1) \le \rho(\gamma) < \frac{32}{9\pi}, \qquad \rho(1) \le \frac{t^{s\text{-}SGD}_{T\epsilon}}{t^{s\text{-}SVD}_{T\epsilon}} \le \frac{\pi^2}{4} \cdot \frac{32}{9\pi},$$

with $\rho(\gamma) = \rho(1)$ only at $\gamma = 1$; and $\rho(\gamma) \to \frac{32}{9\pi}$ as $\gamma \downarrow 0$ and $\rho(\gamma) \to 1$ as $\gamma \to \infty$.

Proof. Part (i) is theorem SI.4, whose relative error eq. (SI.14) depends only on ϵ/F_0 and T, together with proposition SI.3. For (ii), write

$$\frac{t^{\text{s-SGD}}_{T\epsilon}}{\frac{\pi N^2}{4B}\sqrt{F_0/\epsilon}} = \frac{t^{\text{s-SGD}}_{T\epsilon}}{\frac{\pi N^2}{4\mathcal{N}_B^2}\sqrt{F_0/\epsilon}} \cdot \frac{B}{\mathcal{N}_B^2}.$$

The first factor tends to 1 by (i), and the second lies in $[1, (1-4/B)^{-2}]$ by proposition SI.3, so it tends to 1. The same bound gives the ratio statement, and continuity of ρ (theorem SI.5) gives the limit along $\gamma \to \gamma_0$. Part (iii) is theorem SI.5 combined with $1 \le B/\mathcal{N}_B^2 \le \pi^2/4$. □

The ratio exponent of ϵ is zero, and by (iii) and corollary SI.1 the ratio is below one near $\gamma = 1$ in the limit (ii), where SIGNSGD is faster, and above one for small γ, where SIGNSVD is faster. The main text's formula with B in place of $\mathcal{N}_B^2$ holds in the limit (ii); at fixed B the exact ratio carries the factor $B/\mathcal{N}_B^2 \in [1, \pi^2/4]$.

SI.4 Power sums

The forcing and the kernels of the power-law models are finite sums of decaying exponentials with power-law weights and rates. This subsection collects the estimates used for them. For $p, q > 0$ put

$$M_{p,q} \stackrel{\text{def}}{=} \Big(\frac{p}{eq}\Big)^{p/q},$$

and write $\Gamma(d) = \int_0^\infty y^{d-1}e^{-y}\,\mathrm{d}y$ for $d > 0$ [21, 5.2.1].

Lemma SI.5 (Sum and integral)**.** *Let $p, q, a > 0$ and $f_a(x) \stackrel{\text{def}}{=} x^{-p}e^{-ax^{-q}}$ for $x > 0$, $f_a(0) \stackrel{\text{def}}{=} 0$. The function f_a has a unique maximum, at $x = (qa/p)^{1/q}$, with value $M_{p,q}a^{-p/q}$. For every integer $N \ge 1$,*

$$\begin{aligned} \Big|\sum_{i=1}^{N} f_a(i) - \int_0^N f_a(x)\,\mathrm{d}x\Big| &\le 2M_{p,q}a^{-p/q}, \\ \int_0^N f_a(x)\,\mathrm{d}x &= \frac{a^{(1-p)/q}}{q}\int_{a/N^q}^{\infty} y^{(p-1)/q-1}e^{-y}\,\mathrm{d}y. \end{aligned} \tag{SI.20}$$

Proof. The derivative is

$$f_a'(x) = x^{-p-1}e^{-ax^{-q}}\big(-p + qa\,x^{-q}\big),$$

which changes sign once, from positive to negative, at $x = (qa/p)^{1/q}$. Evaluating there gives the maximum. Since f_a tends to 0 at both ends of $(0, \infty)$, its total variation is twice the maximum. On each interval $[i-1, i]$,

$$\Big|f_a(i) - \int_{i-1}^{i} f_a(x)\,\mathrm{d}x\Big| \le \int_{i-1}^{i} |f_a(i) - f_a(x)|\,\mathrm{d}x \le \int_{i-1}^{i} |f_a'(y)|\,\mathrm{d}y,$$

and summing over $i \le N$ bounds the difference by the total variation. The substitution $y = ax^{-q}$ gives the formula for the integral. □

Lemma SI.6 (Convergent power sums)**.** *Let $p > 1$, $q > 0$ and $d \stackrel{\text{def}}{=} (p-1)/q$. For every $a > 0$,*

$$\Big|\sum_{i \ge 1} i^{-p}e^{-ai^{-q}} - \frac{\Gamma(d)}{q}\,a^{-d}\Big| \le 2M_{p,q}a^{-p/q}, \tag{SI.21}$$

and for every integer $N \ge 1$, $0 \le \sum_{i>N} i^{-p} \le N^{1-p}/(p-1)$.

Proof. The series converges by comparison with $\sum_i i^{-p}$. Letting $N \to \infty$ in eq. (SI.20) gives

$$\int_0^\infty f_a(x)\,\mathrm{d}x = \frac{a^{-d}}{q}\,\Gamma(d),$$

and the error bound passes to the limit. The tail bound is the comparison $\sum_{i>N} i^{-p} \le \int_N^\infty x^{-p}\,\mathrm{d}x$. □

For $a > 1$ and $q > 0$ define

$$\begin{gathered} d(a,q) \stackrel{\text{def}}{=} \frac{a-1}{q}, \qquad Z_a \stackrel{\text{def}}{=} 1 + \frac{1}{a-1}, \qquad A_a \stackrel{\text{def}}{=} 2^{-2a}e^{-2}, \\ E_{a,q} \stackrel{\text{def}}{=} \max\Big\{2^{d(a,q)}Z_a,\ \frac{\Gamma(d(a,q))}{q} + 2^{1-1/q}M_{a,q}\Big\}, \end{gathered} \tag{SI.22}$$

and, for an integer $N \ge 1$ or $N = \infty$,

$$L_{N,a,q}(u) \stackrel{\text{def}}{=} \sum_{i=1}^{N} i^{-a}e^{-2ui^{-q}} \qquad (u \ge 0).$$

Lemma SI.7 (Finite power sums)**.** *Let $a > 1$ and $q > 0$.*

(i) For every $N \in \mathbb{N} \cup \{\infty\}$ and $u \ge 0$, $L_{N,a,q}(u) \le E_{a,q}(1+u)^{-d(a,q)}$.

(ii) For every integer $N \ge 4$ and $0 \le u \le (N/4)^q$, $\frac{1}{2}L_{N,a,q}(u) \ge A_a(1+u)^{-d(a,q)}$.

Proof. Write $d = d(a,q)$. Comparison with an integral gives $\sum_{i\ge1} i^{-a} \le 1 + \int_1^\infty x^{-a}\mathrm{d}x = Z_a$. For $0 \le u \le 1$, this and $(1+u)^{-d} \ge 2^{-d}$ give (i). For $u \ge 1$, lemma SI.6 with $2u$ in place of a gives

$$L_{N,a,q}(u) \le L_{\infty,a,q}(u) \le \frac{\Gamma(d)}{q}(2u)^{-d} + 2M_{a,q}(2u)^{-a/q} \le \Big[\frac{\Gamma(d)}{q}2^{-d} + 2^{1-a/q}M_{a,q}\Big]u^{-d},$$

where $u^{-a/q} \le u^{-d}$ because $a/q > d$. Since $u^{-d} \le 2^d(1+u)^{-d}$ for $u \ge 1$ and $d - a/q = -1/q$, this gives (i).

For (ii) and $0 \le u \le 1$, the first term gives $\frac{1}{2}L_{N,a,q}(u) \ge \frac{1}{2}e^{-2} \ge A_a$. For $1 \le u \le (N/4)^q$ put $j \stackrel{\text{def}}{=} \lceil u^{1/q} \rceil$. Then $j \le 2u^{1/q}$ and $2j \le N$. The j indices $j \le i < 2j$ satisfy $i^{-a} \ge (2j)^{-a}$ and $ui^{-q} \le 1$, so

$$\frac{1}{2}L_{N,a,q}(u) \ge \frac{1}{2}\,j\,(2j)^{-a}e^{-2} = 2^{-a-1}e^{-2}j^{1-a} \ge 2^{-2a}e^{-2}u^{-d} \ge A_a(1+u)^{-d}. \qquad \square$$

For the forcing exponent $p > 1$ and a rate exponent $q > 0$ put

$$f_{N,q}(u) \stackrel{\text{def}}{=} \frac{1}{2}L_{N,p,q}(u), \qquad r_q \stackrel{\text{def}}{=} \frac{p-1}{2q}, \qquad C_{p,q} \stackrel{\text{def}}{=} \sqrt{E_{p,q}/2}, \tag{SI.23}$$

for $N \in \mathbb{N} \cup \{\infty\}$. By lemma SI.7, $f_{N,q}(u) \le C_{p,q}^2(1+u)^{-2r_q}$ for all $u \ge 0$ and all N, and $f_{N,q}(u) \ge A_p(1+u)^{-2r_q}$ for $N \ge 4$ and $0 \le u \le (N/4)^q$. When $r_q > 1$ put

$$J(p,q) \stackrel{\text{def}}{=} \int_0^\infty \sqrt{f_{\infty,q}(u)}\,\mathrm{d}u, \tag{SI.24}$$

which is finite because $\sqrt{f_{\infty,q}} \le C_{p,q}(1+u)^{-r_q}$ and positive because $f_{\infty,q}(u) \ge \frac{1}{2}e^{-2u}$. For every integer N put $J_N(p,q) \stackrel{\text{def}}{=} \int_0^\infty \sqrt{f_{N,q}(u)}\,\mathrm{d}u$, which is finite for every $q > 0$ since every rate of the finite sum is at least $2N^{-q}$.

Lemma SI.8 (Harmonic sums)**.** *For $a > 0$ and $N \ge 1$ let $H_N(a) \stackrel{\text{def}}{=} \sum_{i=1}^N i^{-a}$. Then*

$$\int_1^{N+1} x^{-a}\,\mathrm{d}x \le H_N(a) \le 1 + \int_1^N x^{-a}\,\mathrm{d}x,$$

and $H_N(a) \ge N^{1-a}$ for $a \le 1$. Consequently, as $N \to \infty$, $H_N(a) \sim N^{1-a}/(1-a)$ for $0 < a < 1$, $H_N(1) \sim \log N$ with $H_N(1) \ge \log(N+1)$, and $0 \le \zeta(a) - H_N(a) \le N^{1-a}/(a-1)$ for $a > 1$, where $\zeta(a) = \sum_{i\ge1} i^{-a}$. Moreover $\zeta(2) < 2$.

Proof. The function $x \mapsto x^{-a}$ is decreasing, so i^{-a} lies between its integrals over $[i, i+1]$ and $[i-1, i]$; summing gives the two bounds, and evaluating the integrals gives the asymptotics. For $a \le 1$ every term is at least N^{-a}. The tail bound is lemma SI.6. Finally $\zeta(2) < 1 + \int_1^\infty x^{-2}\mathrm{d}x = 2$. □

SI.5 The adopted power-law arrays and the phase map

Definition SI.3 (Adopted power-law arrays)**.** *Fix $\alpha > 0$ and $\varsigma \in \mathbb{R}$ with $p \overset{\text{def}}{=} \alpha + \varsigma > 1$, and integers $B \ge N \ge 1$. Put $\gamma \overset{\text{def}}{=} N/B \in (0, 1]$ and $\bar{\mu} \overset{\text{def}}{=} H_N(\alpha)/N$, and for $1 \le i \le N$ let $\mu_i \overset{\text{def}}{=} i^{-\alpha}$ and $\mathcal{Q}_i(0) \overset{\text{def}}{=} i^{-\varsigma}$. The* SIGNSVD *and* SIGNSGD *arrays are*

$$\begin{aligned} \mathfrak{d}_i^{\text{s-SVD}} &\overset{\text{def}}{=} \gamma^{-1/2} i^{-\alpha/2} = \sqrt{\mu_i}\sqrt{B/N}, & \mathfrak{v}_i^{\text{s-SVD}} &\overset{\text{def}}{=} 1, \\ \mathfrak{d}_i^{\text{s-SGD}} &\overset{\text{def}}{=} \mu_i \sqrt{\frac{B}{\pi\bar{\mu}}}, & \mathfrak{v}_i^{\text{s-SGD}} &\overset{\text{def}}{=} N\Big[1 + \frac{2}{\pi}\Big(\frac{\mu_i}{\bar{\mu}} - 1\Big)\Big]. \end{aligned} \tag{SI.25}$$

For $j \in \{$s-SVD, s-SGD$\}$, the time scales, rate exponents and forcing exponents are

$$T_0^{\text{s-SVD}} \overset{\text{def}}{=} \sqrt{\gamma}, \quad T_0^{\text{s-SGD}} \overset{\text{def}}{=} \sqrt{\frac{\pi\bar{\mu}}{B}}, \qquad q_{\text{s-SVD}} \overset{\text{def}}{=} \frac{\alpha}{2}, \quad q_{\text{s-SGD}} \overset{\text{def}}{=} \alpha, \qquad r_j \overset{\text{def}}{=} r_{q_j} = \frac{p-1}{2q_j}.$$

For $\epsilon > 0$ each model is run at its matched step $\eta_j^\star \overset{\text{def}}{=} 2\sqrt{\epsilon}/\mathcal{S}_j$. Its risk Q_j solves eq. (SI.4) with forcing F_j and kernel k_j, and its dimensionless risk, kernel and first hit at a fixed multiplier T are

$$Y_j(u) \overset{\text{def}}{=} Q_j(T_0^j u), \qquad g_j(u) \overset{\text{def}}{=} \frac{T_0^j\, k_j(T_0^j u)}{\sqrt{\epsilon}}, \qquad \hat{u}_j \overset{\text{def}}{=} \inf\{u \ge 0 :\ Y_j(u) \le T\epsilon\}.$$

The SIGNSGD volatility equals $N(1 - \frac{2}{\pi}) + \frac{2}{\pi} N^2 i^{-\alpha}/H_N(\alpha)$, a positive constant plus a nonnegative spectral term, and its total is $\sum_i \mathfrak{v}_i^{\text{s-SGD}} = N^2$. The SIGNSVD array has $\mathfrak{d}_i^{\text{s-SVD}} \propto i^{-\alpha/2}$ and the SIGNSGD array has $\mathfrak{d}_i^{\text{s-SGD}} \propto i^{-\alpha}$; this difference of rate exponents moves the phase boundaries. In terms of ς,

$$r_{\text{s-SVD}} = \frac{\alpha + \varsigma - 1}{\alpha}, \qquad r_{\text{s-SGD}} = \frac{\alpha + \varsigma - 1}{2\alpha},$$

so $r_{\text{s-SVD}} = 1$ exactly at $\varsigma = 1$ and $r_{\text{s-SGD}} = 1$ exactly at $\varsigma = \alpha + 1$. For $r > 0$ and $x \ge 0$ put

$$I_r(x) \overset{\text{def}}{=} \int_0^x (1+w)^{-r}\,\mathrm{d}w = \begin{cases} \dfrac{(1+x)^{1-r} - 1}{1-r}, & r \ne 1, \\ \log(1+x), & r = 1. \end{cases}$$

Proposition SI.4 (Noise constants, kernels and clocks)**.** *The arrays of definition SI.3 satisfy (M1), with $F_0 = \frac{1}{2}H_N(p)$, and the following hold.*

(i) The noise constants, matched steps and time-scale products are

$$\begin{aligned} \mathcal{S}_{\text{s-SVD}} &= \tfrac{1}{2}\sqrt{\gamma}\, H_N(\alpha/2), & \eta^\star_{\text{s-SVD}} &= \frac{4\sqrt{\epsilon}}{\sqrt{\gamma}\, H_N(\alpha/2)}, \\ \mathcal{S}_{\text{s-SGD}} &= \frac{N^2}{2}\sqrt{\frac{\pi\bar{\mu}}{B}}, & \eta^\star_{\text{s-SGD}} &= \frac{4\sqrt{\epsilon}}{N^2}\sqrt{\frac{B}{\pi\bar{\mu}}}, \\ \mathcal{S}_{\text{s-SVD}} T_0^{\text{s-SVD}} &= \tfrac{\gamma}{2} H_N(\alpha/2), & \mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}} &= \frac{\pi N^2 \bar{\mu}}{2B} = \tfrac{\pi\gamma}{2} H_N(\alpha), \end{aligned} \tag{SI.26}$$

and the dimension factor is

$$D_N \overset{\text{def}}{=} \frac{\mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}}}{\mathcal{S}_{\text{s-SVD}} T_0^{\text{s-SVD}}} = \frac{\pi H_N(\alpha)}{H_N(\alpha/2)}. \tag{SI.27}$$

(ii) The dimensionless risks solve $Y_j = f_{N,q_j} + \sqrt{\epsilon}\, g_j * \sqrt{Y_j}$, with $f_{N,q}$ of eq. (SI.23) and

$$g_{\text{s-SVD}} = \frac{2}{H_N(\alpha/2)} L_{N,\alpha,\alpha/2}, \qquad g_{\text{s-SGD}} = \frac{2(1 - 2/\pi)}{N}\, L_{N,\alpha,\alpha} + \frac{4}{\pi H_N(\alpha)}\, L_{N,2\alpha,\alpha}. \tag{SI.28}$$

Both kernels are positive, continuous and decreasing. The kernel $g_{\text{s-SVD}}$ has integral 1, and the two terms of $g_{\text{s-SGD}}$ have integrals $1-2/\pi$ and $2/\pi$. Their maxima are

$$\begin{aligned} k_{N,\text{s-SVD}} &\stackrel{\text{def}}{=} g_{\text{s-SVD}}(0) = \frac{2H_N(\alpha)}{H_N(\alpha/2)}, \\ k_{N,\text{s-SGD}} &\stackrel{\text{def}}{=} g_{\text{s-SGD}}(0) = \frac{2(1-2/\pi)H_N(\alpha)}{N} + \frac{4H_N(2\alpha)}{\pi H_N(\alpha)}. \end{aligned} \tag{SI.29}$$

(iii) If $\hat{u}_j < \infty$, then

$$t^j_{T\epsilon} = \frac{\mathcal{S}_j T_0^j}{2\sqrt{\epsilon}} \int_0^{\hat{u}_j} \sqrt{Y_j(u)}\,\mathrm{d}u. \tag{SI.30}$$

(iv) As $N \to \infty$ at fixed α,

$$D_N \sim \begin{cases} \pi \dfrac{1-\alpha/2}{1-\alpha}\, N^{-\alpha/2}, & 0<\alpha<1, \\ \dfrac{\pi}{2} N^{-1/2} \log N, & \alpha = 1, \\ \pi(1-\alpha/2)\,\zeta(\alpha)\, N^{\alpha/2-1}, & 1<\alpha<2, \\ \pi\zeta(2)/\log N, & \alpha = 2, \\ \pi\zeta(\alpha)/\zeta(\alpha/2), & \alpha > 2. \end{cases} \tag{SI.31}$$

Proof. All coefficients are positive, since $\mu_i/\bar{\mu} = Ni^{-\alpha}/H_N(\alpha) > 0$ and $2/\pi < 1$, and the sums in eq. (SI.1) are finite. For SIGNSVD,

$$\mathcal{S}_{\text{s-SVD}} = \sum_{i=1}^{N} \frac{i^{-\alpha}}{2\gamma^{-1/2} i^{-\alpha/2}} = \frac{\sqrt{\gamma}}{2} H_N(\alpha/2).$$

For SIGNSGD the factor μ_i cancels and $\sum_i \mathfrak{v}_i^{\text{s-SGD}} = N^2$, so

$$\mathcal{S}_{\text{s-SGD}} = \frac{1}{2}\sqrt{\frac{\pi\bar{\mu}}{B}} \sum_{i=1}^{N} \mathfrak{v}_i^{\text{s-SGD}} = \frac{N^2}{2}\sqrt{\frac{\pi\bar{\mu}}{B}}.$$

The steps and the products follow, using $N\bar{\mu} = H_N(\alpha)$ and $N/B = \gamma$. This proves (i).

For (ii), substituting $v = T_0^j w$ in the convolution gives

$$Y_j(u) = F_j(T_0^j u) + T_0^j \int_0^u k_j\big(T_0^j(u-w)\big)\sqrt{Y_j(w)}\,\mathrm{d}w = F_j(T_0^j u) + \sqrt{\epsilon}\,(g_j * \sqrt{Y_j})(u).$$

Since $\mathfrak{d}_i^j T_0^j = i^{-q_j}$, the forcing is $F_j(T_0^j u) = \frac{1}{2}\sum_i i^{-p} e^{-2ui^{-q_j}} = f_{N,q_j}(u)$. At the matched step,

$$g_j(u) = \frac{T_0^j}{\mathcal{S}_j} \sum_{i=1}^{N} \mu_i \mathfrak{v}_i^j e^{-2ui^{-q_j}},$$

and inserting (i) and the arrays gives eq. (SI.28). Each exponential satisfies $\int_0^\infty e^{-2ui^{-q}}\mathrm{d}u = i^q/2$, which gives the three integrals. Each kernel is a positive combination of decreasing exponentials, and its value at 0 is eq. (SI.29).

For (iii), theorem SI.3(i) gives $t^j_{T\epsilon} = t(H_Q(T\epsilon))$ with $H_Q(T\epsilon) = T_0^j \hat{u}_j$, and the change of variables $v = T_0^j u$ in the clock gives

$$t(T_0^j \hat{u}_j) = \frac{T_0^j}{\eta_j^\star} \int_0^{\hat{u}_j} \sqrt{Y_j(u)}\,\mathrm{d}u,$$

which is eq. (SI.30). Part (iv) follows from eq. (SI.27) and lemma SI.8. □

Windows. The phase theorems hold jointly in the dimension N and the accuracy ϵ, on explicit windows. For $T \geq 4$ fixed, write $C_j \stackrel{\text{def}}{=} C_{p,q_j}$ and $A \stackrel{\text{def}}{=} A_p$, and put

$$\epsilon_0 \stackrel{\text{def}}{=} \min\Big\{\frac{1}{4},\ \frac{1}{8T},\ C_{\text{s-SVD}}^2,\ C_{\text{s-SGD}}^2,\ \Big(\frac{A}{T}\Big)^2,\ \min_{j:\, r_j<1} \frac{A}{T\, 2^{2r_j/(1-r_j)}}\Big\}, \tag{SI.32}$$

where a minimum over the empty set is 1. For $0 < \epsilon < C_j^2$ put

$$H_j(\epsilon) \stackrel{\text{def}}{=} \Big(\frac{C_j}{\sqrt{\epsilon}}\Big)^{1/r_j} - 1.$$

The low-exponent window is the set of (N, B, ϵ) with $N \geq 4$, $B \geq N$, $0 < \epsilon < \epsilon_0$ and, for both j,

$$k_{N,j}\, C_j\, I_{r_j}\big(H_j(\epsilon)\big) \leq \sqrt{\epsilon}, \qquad \text{and} \qquad H_j(\epsilon) \leq (N/4)^{q_j} \ \text{ if } r_j \leq 1. \tag{SI.33}$$

Finally, put

$$\Pi_r(\epsilon) \stackrel{\text{def}}{=} \begin{cases} \epsilon^{-1/(2r)}, & 0 < r < 1, \\ \epsilon^{-1/2}\log(1/\epsilon), & r = 1, \\ \epsilon^{-1/2}, & r > 1. \end{cases} \tag{SI.34}$$

The main text's theorem 5.1 states the exponents of the following theorem, and its ratio is corollary SI.6 with theorem SI.13.

Theorem SI.7 (Time to $T\epsilon$ in the power-law phases)**.** *Fix $\alpha > 0$ and ς with $\alpha + \varsigma > 1$ and $\varsigma \notin \{1, \alpha + 1\}$, and fix $T \geq 4$. Consider the arrays of definition SI.3 at their matched steps along a sequence of (N, B, ϵ) with $N \to \infty$, $\epsilon \downarrow 0$ and integers $B \geq N$, such that*

- *$\epsilon N^{\alpha+\varsigma-1} \to \infty$ if $\varsigma < 1$;*
- *$\epsilon N \to \infty$ if $\varsigma > 1$ and $\alpha > 1$;*
- *(N, B, ϵ) lies in the low-exponent window eq. (SI.33) for all large indices if $\varsigma > 1$ and $\alpha \leq 1$.*

Then there are constants $0 < c_- \leq c_+$, depending only on α, ς and T, such that for all large indices of the sequence, with $x \asymp y$ meaning $c_- \leq x/y \leq c_+$,

$$t_{T\epsilon}^{\text{s-SVD}} \asymp \gamma\, H_N(\alpha/2)\, \epsilon^{-\alpha/[2(\alpha+\varsigma-1)]} \ (\varsigma < 1), \qquad t_{T\epsilon}^{\text{s-SVD}} \asymp \sqrt{\gamma}\, \mathcal{S}_{\text{s-SVD}}\, \epsilon^{-1/2} \ (\varsigma > 1), \tag{SI.35}$$

$$t_{T\epsilon}^{\text{s-SGD}} \asymp \frac{N^2 \pi \bar{\mu}}{B}\, \epsilon^{-\alpha/(\alpha+\varsigma-1)} \ (\varsigma < \alpha + 1), \qquad t_{T\epsilon}^{\text{s-SGD}} \asymp \sqrt{\frac{\pi\bar{\mu}}{B}}\, \mathcal{S}_{\text{s-SGD}}\, \epsilon^{-1/2} \ (\varsigma > \alpha + 1). \tag{SI.36}$$

Moreover $H_N(\alpha/2) \asymp N^{1-\alpha/2}$ for $\alpha < 2$, $H_N(\alpha/2) \asymp \log N$ for $\alpha = 2$, and $H_N(\alpha/2) \asymp 1$ for $\alpha > 2$, with constants depending only on α.

The exponents eq. (18) of the main text select the branches of these displays: the maximum in e_{Muon} switches at $\varsigma = 1$ and the maximum in $e_{\text{s-SGD}}$ at $\varsigma = \alpha + 1$. The limit is joint in N and ϵ, on the stated windows; at fixed N the exponent of ϵ is $-\frac{1}{2}$ in every phase (corollary SI.2), and $\epsilon N \to \infty$ alone does not suffice when $\alpha < 1$ (proposition SI.7). The saturated prefactors carry the time scales T_0^j, that is, $\mathcal{S}_j T_0^j$ in place of $\mathcal{S}_j$. The phase boundaries $\varsigma = 1$ and $\varsigma = \alpha + 1$ carry a logarithm, given with its constant in theorems SI.9 and SI.10.

The theorem rests on the following sharper statements, proved in appendix SI.9.

Theorem SI.8 (The range $\varsigma < 1$)**.** *Fix $\alpha > 0$, $1 - \alpha < \varsigma < 1$ and $T \geq 4$. There are N_* and E_* and $0 < c_- \leq c_+$, all depending only on α, ς and T, such that for all $N \geq N_*$, $B \geq N$ and $0 < \epsilon < \epsilon_0$ with $\epsilon N^{p-1} \geq E_*$, and for both j, the first hit $\hat{u}_j$ is attained and*

$$c_- \leq \frac{t_{T\epsilon}^j}{\mathcal{S}_j T_0^j\, \epsilon^{-1/(2r_j)}} \leq c_+.$$

In particular $t_{T\epsilon}^{\text{s-SVD}} \asymp \mathcal{S}_{\text{s-SVD}} T_0^{\text{s-SVD}} \epsilon^{-\alpha/[2(p-1)]}$ and $t_{T\epsilon}^{\text{s-SGD}} \asymp \mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}} \epsilon^{-\alpha/(p-1)}$. Every sequence with $N \to \infty$, $\epsilon \downarrow 0$ and $\epsilon N^{p-1} \to \infty$ lies in this window for all large indices, uniformly over $B \geq N$; for example $\epsilon = N^{-\omega}$ with $0 < \omega < p - 1$.

Theorem SI.9 (Main covariance range). *Fix $\alpha > 1$, $\varsigma \geq 1$ and $T \geq 4$. There are N_*, $E_* \geq 1$ and $0 < c_- \leq c_+$, all depending only on α, ς and T, such that for all $N \geq N_*$, $B \geq N$ and $0 < \epsilon < \epsilon_0$ with $\epsilon N \geq E_*$, and for both j, the first hit $\hat{u}_j$ is attained and*

$$c_- \leq \frac{t^j_{T\epsilon}}{\mathcal{S}_j T_0^j \, \Pi_{r_j}(\epsilon)} \leq c_+.$$

Along every sequence with $N \to \infty$ and $\epsilon \downarrow 0$ in this window, for each j,

$$t^j_{T\epsilon} \sim \frac{\mathcal{S}_j T_0^j}{4\sqrt{8q_j}} \, \epsilon^{-1/2} \log \frac{1}{\epsilon} \quad (r_j = 1), \qquad t^j_{T\epsilon} \sim \frac{\mathcal{S}_j T_0^j \, J(p, q_j)}{2\sqrt{\epsilon}} \quad (r_j > 1). \tag{SI.37}$$

If $\alpha = 2$ and $\varsigma > 1$, the second equivalent holds for $j =$ s-SVD along every sequence with $N \to \infty$, $\epsilon \downarrow 0$ and $B \geq N$, with no relation between N and ϵ. Every sequence with $N \to \infty$, $\epsilon \downarrow 0$ and $\epsilon N \to \infty$ lies in the window for all large indices, uniformly over $B \geq N$.

Theorem SI.10 (Low covariance exponents). *Fix $0 < \alpha \leq 1$, $\varsigma \geq 1$ and $T \geq 4$. There are $0 < c_- \leq c_+$, depending only on α, ς and T, such that for every (N, B, ϵ) in the low-exponent window eq. (SI.33), and for both j, the first hit $\hat{u}_j$ is attained and*

$$c_- \leq \frac{t^j_{T\epsilon}}{\mathcal{S}_j T_0^j \, \Pi_{r_j}(\epsilon)} \leq c_+.$$

Along every sequence with $N \to \infty$ and $\epsilon \downarrow 0$ in this window, the equivalents eq. (SI.37) hold.

The low-exponent window is not empty.

Proposition SI.5 (The low-exponent window). *Fix $0 < \alpha \leq 1$. The kernel heights eq. (SI.29) satisfy, as $N \to \infty$,*

$$k_{N,\text{s-SVD}} \sim \begin{cases} 2(1-\alpha/2)N^{-\alpha/2}/(1-\alpha), & 0 < \alpha < 1, \\ N^{-1/2} \log N, & \alpha = 1, \end{cases}$$

$$k_{N,\text{s-SGD}} \sim \begin{cases} 2\big[(1-\frac{2}{\pi})/(1-\alpha) + \frac{2}{\pi}(1-\alpha)/(1-2\alpha)\big] N^{-\alpha}, & 0 < \alpha < \frac{1}{2}, \\ \frac{2}{\pi} N^{-1/2} \log N, & \alpha = \frac{1}{2}, \\ \frac{4}{\pi}(1-\alpha)\zeta(2\alpha) N^{\alpha-1}, & \frac{1}{2} < \alpha < 1, \\ \frac{4}{\pi}\zeta(2)/\log N, & \alpha = 1. \end{cases}$$

For fixed $\varsigma \geq 1$ and $T \geq 4$, the following sequences lie in the low-exponent window for all large N, uniformly over integers $B_N \geq N$: $\epsilon_N = N^{-\omega}$ with $0 < \omega < \min(\alpha, 1-\alpha)$ if $0 < \alpha < 1$, and $\epsilon_N = (\log N)^{-\omega}$ with $0 < \omega < 1$ if $\alpha = 1$.

At fixed dimension the accuracy limit has exponent $\frac{1}{2}$ in every phase.

Theorem SI.11 (Fixed arrays). *Assume (M1) with a finite index set and $\mathcal{S} > 0$, take $\eta = 2\sqrt{\epsilon}/\mathcal{S}$, and fix $T \geq 4$. As $\epsilon \downarrow 0$ with all other data fixed,*

$$t_{T\epsilon} \sim \frac{\mathcal{S}}{2\sqrt{\epsilon}} \int_0^\infty \sqrt{F(s)} \, \mathrm{d}s, \qquad 0 < \int_0^\infty \sqrt{F(s)} \, \mathrm{d}s < \infty.$$

Corollary SI.2 (Fixed dimension). *For the arrays of definition SI.3 at fixed N, B, α, ς and $T \geq 4$, as $\epsilon \downarrow 0$,*

$$t^j_{T\epsilon} \sim \frac{\mathcal{S}_j T_0^j \, J_N(p, q_j)}{2\sqrt{\epsilon}}, \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \longrightarrow D_N \, \frac{J_N(p, \alpha)}{J_N(p, \alpha/2)} \geq D_N.$$

If $\alpha \geq 4$, this limit exceeds $\pi/2$ for every N.

The ratio of the two hitting times follows by division.

Corollary SI.3 (Ratio of the hitting times). *In the windows of theorem SI.8 ($\varsigma < 1$), theorem SI.9 ($\alpha > 1$, $\varsigma \geq 1$) and theorem SI.10 ($\alpha \leq 1$, $\varsigma \geq 1$), with constants depending only on α, ς and T,*

$$\frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \asymp D_N \cdot \begin{cases} \epsilon^{-\alpha/[2(p-1)]}, & \varsigma < 1, \\ \epsilon^{-1/2}/\log(1/\epsilon), & \varsigma = 1, \\ \epsilon^{-(\alpha+1-\varsigma)/[2(p-1)]}, & 1 < \varsigma < \alpha + 1, \\ \log(1/\epsilon), & \varsigma = \alpha + 1, \\ 1, & \varsigma > \alpha + 1. \end{cases} \tag{SI.38}$$

Along every sequence with $N \to \infty$ and $\epsilon \downarrow 0$ in the window of theorem SI.9 or theorem SI.10,

$$\frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \sim \frac{D_N \log(1/\epsilon)}{2\sqrt{8\alpha}\, J(p,\alpha/2)} \quad (\varsigma = \alpha+1), \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \sim D_N \frac{J(p,\alpha)}{J(p,\alpha/2)} \quad (\varsigma > \alpha+1).$$

A ratio tending to infinity means that SIGNSVD reaches $T\epsilon$ first in the model; a ratio tending to zero means that SIGNSGD does. The sign of the exponent of ϵ does not decide the winner by itself, because D_N varies with N.

Proposition SI.6 (Limits of the ratio)**.** *Write $\mathsf{R} \overset{\text{def}}{=} t^{\text{s-SGD}}_{T\epsilon}/t^{\text{s-SVD}}_{T\epsilon}$, and consider sequences with $N \to \infty$, $\epsilon \downarrow 0$ and $B_N \geq N$ in the indicated windows. The constants in $\asymp$ and O depend only on α, ς and T.*

(i) The range $\varsigma < 1$ *(window of theorem SI.8). With $\Xi_N \overset{\text{def}}{=} \epsilon N^{p-1}$ and $r_{\text{s-SVD}} = (p-1)/\alpha$,*

$$\mathsf{R} \asymp \begin{cases} \Xi_N^{-1/(2r_{\text{s-SVD}})}, & 0<\alpha<1, \\ (\log N)\,\Xi_N^{-1/(2r_{\text{s-SVD}})}, & \alpha = 1, \\ N^{\alpha-1}\,\Xi_N^{-1/(2r_{\text{s-SVD}})}, & 1<\alpha<2, \\ (N/\log N)\,\Xi_N^{-1/(2r_{\text{s-SVD}})}, & \alpha = 2, \\ \epsilon^{-1/(2r_{\text{s-SVD}})}, & \alpha > 2. \end{cases}$$

Hence $\mathsf{R} \to 0$ for $0<\alpha<1$ along sequences with $\Xi_N \to \infty$, and $\mathsf{R} \to \infty$ for $\alpha > 2$. For $\alpha = 1$ and $\epsilon_N = N^{-r_{\text{s-SVD}}}(\log N)^b$ with $b > 0$, $\mathsf{R} \to \infty$ if $b < 2r_{\text{s-SVD}}$ and $\mathsf{R} \to 0$ if $b > 2r_{\text{s-SVD}}$. For $1<\alpha<2$ and $\epsilon_N = N^{-\omega}$ with $0<\omega<p-1$, put $\omega_c \overset{\text{def}}{=} r_{\text{s-SVD}}(2-\alpha) \in (0, p-1)$; then $\mathsf{R} \to 0$ if $\omega < \omega_c$ and $\mathsf{R} \to \infty$ if $\omega > \omega_c$. For $\alpha = 2$ and $\epsilon_N = (\log N)^{-\omega}$ with $\omega > 0$, $\mathsf{R} \to 0$ if $\omega < 2r_{\text{s-SVD}}$ and $\mathsf{R} \to \infty$ if $\omega > 2r_{\text{s-SVD}}$. All these sequences lie in the window.

(ii) $\alpha > 2$, $\varsigma \geq 1$ *(window of theorem SI.9).* $\mathsf{R} \to \infty$ *if* $\varsigma \leq \alpha+1$, *and* $\mathsf{R} \to \pi\zeta(\alpha)J(p,\alpha)/(\zeta(\alpha/2)J(p,\alpha/2)) \in (0,\infty)$ *if* $\varsigma > \alpha+1$; *this limit exceeds* $\pi/2$ *if* $\alpha \geq 4$.

(iii) $1 < \alpha \leq 2$, $\varsigma \geq 1$ *(window of theorem SI.9).* $\mathsf{R} \to 0$ *if* $\varsigma > \alpha+1$. *If* $\varsigma = \alpha+1$ *and* $1<\alpha<2$, *then* $\mathsf{R} \to 0$. *If* $\alpha = 2$, $\varsigma = 3$ *and* $\log(1/\epsilon)/\log N \to c_\infty$, *then* $0 \leq c_\infty \leq 1$ *and* $\mathsf{R} \to \pi\zeta(2)c_\infty/(8J(5,1))$.

(iv) $0 < \alpha \leq 1$, $\varsigma \geq 1$ *(low-exponent window).* $\mathsf{R} \to 0$; *more precisely* R *is* $O(\log(1/\epsilon)^{-2})$, $O(\epsilon^{(\varsigma-1)/(p-1)})$, $O(\sqrt{\epsilon}\log(1/\epsilon))$ *and* $O(\sqrt{\epsilon})$ *for* $\varsigma = 1$, $1 < \varsigma < \alpha+1$, $\varsigma = \alpha+1$ *and* $\varsigma > \alpha+1$ *respectively.*

(v) $1 < \alpha \leq 2$, $1 \leq \varsigma < \alpha+1$ *(window of theorem SI.9). Put*

$$e_\varsigma \overset{\text{def}}{=} \frac{\alpha+1-\varsigma}{2(p-1)}, \qquad \varsigma_c(\alpha) \overset{\text{def}}{=} \frac{\alpha^2-2\alpha+3}{3-\alpha},$$

so that $1 < \varsigma_c(\alpha) < \alpha+1$ for $1<\alpha<2$, and $\varsigma_c(2) = 3$.

- *(a) If $1<\alpha<2$ and $\varsigma_c(\alpha) \leq \varsigma < \alpha+1$, then $\mathsf{R} \to 0$ along every sequence with $N \to \infty$, $\epsilon \downarrow 0$ and $\epsilon N \to \infty$.*
- *(b) If $1<\alpha<2$ and $1 \leq \varsigma < \varsigma_c(\alpha)$, put $\omega_c \overset{\text{def}}{=} (2-\alpha)(p-1)/(\alpha+1-\varsigma) \in (0,1)$. For $\epsilon_N = N^{-\omega}$ with $0<\omega<1$, $\mathsf{R} \to 0$ if $\omega < \omega_c$ and $\mathsf{R} \to \infty$ if $\omega > \omega_c$. For $\epsilon_N = 1/\log N$, $\mathsf{R} \to 0$.*
- *(c) If $\alpha = 2$ and $1 \leq \varsigma < 3$, then $\mathsf{R} \to \infty$ for $\epsilon_N = N^{-\omega}$ with $0<\omega<1$, and $\mathsf{R} \to 0$ for $\epsilon_N = 1/\log N$.*

All the sequences named in (b) and (c) lie in the window for large N.

Proposition SI.7 (A family outside the windows)**.** *Fix $0<\alpha<1$, $\varsigma = 1$, $\alpha < \omega < 1$ and $T \geq 4$, let $\epsilon_N \overset{\text{def}}{=} N^{-\omega}$, and let $B_N \geq N$ be arbitrary integers. Then $\epsilon_N N \to \infty$, and*

$$\frac{t^{\text{s-SVD}}_{T\epsilon_N}}{\mathcal{S}_{\text{s-SVD}} T_0^{\text{s-SVD}}\, \epsilon_N^{-1/2} \log(1/\epsilon_N)/(4\sqrt{8 q_{\text{s-SVD}}})} \longrightarrow \frac{\alpha}{\omega} < 1,$$

$$\frac{t^{\text{s-SGD}}_{T\epsilon_N}}{\mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}}\, \epsilon_N^{-1}} \asymp N^{-(\omega-\alpha)/2} \longrightarrow 0,$$

with constants depending only on α, ω and T.

Remark SI.3 (The windows cannot be dropped)**.** *Corollary SI.2 shows that at fixed N the exponent of ϵ is $-\frac{1}{2}$ in every phase, so the exponents of theorem SI.7 require $N \to \infty$ jointly with $\epsilon \downarrow 0$. In the family of proposition SI.7, $\varsigma = 1$ and $\epsilon N \to \infty$. The order $\mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}} \epsilon^{-1}$, which theorem SI.10 gives for* SIGNSGD *at $\varsigma = 1$ inside its window, fails on this family by the factor $N^{-(\omega-\alpha)/2}$; the* SIGNSVD *time keeps its order but not the constant of eq. (SI.37). The family violates the height condition eq. (SI.33): by proposition SI.5, $k_{N,\text{s-SVD}} C_{\text{s-SVD}} \log(C_{\text{s-SVD}}/\sqrt{\epsilon_N})/\sqrt{\epsilon_N}$ grows like a constant times $N^{(\omega-\alpha)/2} \log N$. The cause is the finite cutoff: the hitting clock is governed by the finite forcing integral J_N, while the infinite-forcing accuracy scale lies beyond the cutoff. Within the windows, proposition SI.6 gives the following map of the ratio $t_{T\epsilon}^{\text{s-SGD}}/t_{T\epsilon}^{\text{s-SVD}}$: it tends to 0 for $\alpha < 1$ in every phase; it tends to ∞ for $\alpha > 2$ and $\varsigma \leq \alpha + 1$, and to a positive constant for $\alpha > 2$ and $\varsigma > \alpha + 1$; for $1 < \alpha \leq 2$ it tends to 0 on and above the curve $\varsigma = \varsigma_c(\alpha)$ (for $\alpha < 2$) and depends on the joint family below it, as it does in the range $\varsigma < 1$ for $1 \leq \alpha \leq 2$.*

Remark SI.4 (The adopted SIGNSGD array and the corrected modes)**.** *The* SIGNSGD *section computes the raw annealed row volatility of the actual sign update, $\mathfrak{v}_i^{\text{corr}}$, for a Haar eigenbasis independent of the batch, comparable dimensions and a covariance bounded below (theorem SJ.4). At $N_{\text{out}} = N_{\text{in}} = N$, with $\Sigma_{\text{out}} = U\Lambda U^\top$, the state $\Delta = UD$ and $\mathscr{R} = \frac{1}{2}\operatorname{Tr}(\Lambda D D^\top)$, its difference from the adopted array is*

$$\mathfrak{v}_i^{\text{corr}} - \mathfrak{v}_i^{\text{s-SGD}} = \frac{2}{\pi} N \Big[\frac{2\mu_i}{\operatorname{Tr}\Sigma_{\text{out}}} + \frac{(B + 2N + 3)\,(\Lambda D D^\top \Lambda)_{ii}}{2\mathscr{R}\,\operatorname{Tr}\Sigma_{\text{out}}} \Big] \geq 0,$$

where $\mathfrak{v}_i^{\text{s-SGD}} = N(1 - \frac{2}{\pi}) + \frac{2}{\pi} N^2 \mu_i / \operatorname{Tr}\Sigma_{\text{out}}$. The corrected array depends on the state, its total exceeds N^2, and the power-law family has no covariance lower bound fixed in N. The phase theorems of this subsection concern the adopted array eq. (SI.25). Replacing it by another nonnegative array with total N^2 and the same drift leaves $\mathcal{S}_{\text{s-SGD}}$ unchanged, but not the kernel $g_{\text{s-SGD}}$ on which the joint estimates rest.

Remark SI.5 (Momentum arrays)**.** *For $\beta \in [0, 1)$ put $B_{\text{eff}} \overset{\text{def}}{=} B(1+\beta)/(1-\beta) = B/s_2$. The adopted* MUON *array with momentum β has the μ_i and $\Omega_i(0)$ of definition SI.3, the batch B_{eff} in place of B in the drift, and the volatility multiplied by B_{eff}/B:*

$$\mathfrak{d}_i^\beta \overset{\text{def}}{=} \sqrt{\mu_i}\sqrt{\frac{B_{\text{eff}}}{N}} = \sqrt{\frac{B_{\text{eff}}}{B}}\,\mathfrak{d}_i^{\text{s-SVD}}, \qquad \mathfrak{v}_i^\beta \overset{\text{def}}{=} \frac{B_{\text{eff}}}{B} = \frac{B_{\text{eff}}}{B}\,\mathfrak{v}_i^{\text{s-SVD}}.$$

At $\beta = 0$ it is the SIGNSVD *array. The evidence for it is the resolved-window endpoint of corollary SG.1(ii), quoted in remark SI.1, which reads $\sqrt{2}\,\mathfrak{d}_i(0) \sim \sqrt{\mu_i B_{\text{eff}}/N}$ and $\mathfrak{v}_i(0) \to B_{\text{eff}}/B$ in this notation.*

The momentum array at a step η is the SIGNSVD *array at the step $\eta\sqrt{B_{\text{eff}}/B}$. For every $\eta > 0$, the right side of eq. (SI.2) satisfies*

$$-\frac{2\eta\,\mathfrak{d}_i^\beta}{\sqrt{\mathcal{R}}}\,\Omega_i + \eta^2 \mathfrak{v}_i^\beta = -\frac{2\eta'\,\mathfrak{d}_i^{\text{s-SVD}}}{\sqrt{\mathcal{R}}}\,\Omega_i + \eta'^2 \mathfrak{v}_i^{\text{s-SVD}}, \qquad \eta' \overset{\text{def}}{=} \eta\sqrt{\frac{B_{\text{eff}}}{B}}.$$

By eq. (SI.1) the noise constant of the momentum array is $\mathcal{S}^\beta = \sqrt{B_{\text{eff}}/B}\,\mathcal{S}_{\text{s-SVD}}$, so its matched step $2\sqrt{\epsilon}/\mathcal{S}^\beta$ is mapped to the matched step $\eta^\star_{\text{s-SVD}}$. At matched steps the two models therefore solve the same row equations from the same initial values, and they have the same solution by the uniqueness in theorem SI.2. In particular $t_{T\epsilon}^\beta = t_{T\epsilon}^{\text{s-SVD}}$, and theorem SI.7, corollary SI.3 and proposition SI.6 hold for the momentum array without change. Written with the noise constant of the momentum array, the saturated prefactor $\sqrt{\gamma}\,\mathcal{S}_{\text{s-SVD}}$ of eq. (SI.35) is $\sqrt{N/B_{\text{eff}}}\,\mathcal{S}^\beta$.

The oversampled isotropic case refines the volatility factor. There the volatility with momentum is $c_\beta B_{\text{eff}}/B$ times the volatility without momentum, with $c_\beta \overset{\text{def}}{=} s_2 V_\beta \in (s_2, 1)$, up to a factor $1 + O(\gamma)$ (remark SF.1 and lemma SF.10). A factor $c > 0$ on every volatility of an array multiplies every matched-step hitting time by c. Indeed, the noise constant becomes $c\,\mathcal{S}$ and the matched step becomes $\eta^\star/c$, so the right side of eq. (SI.2) becomes $1/c$ times its value for the original array at its matched step. The risk of the modified model at time t is therefore the original risk at time t/c.

SI.6 The accuracy scale

The phase theorems above hold in windows that keep the accuracy ϵ above a power of N. This subsection describes the two models of definition SI.3 at every accuracy, jointly in N and ϵ. The organizing quantity is the accuracy scale

$$\epsilon_N \stackrel{\text{def}}{=} N^{-(p-1)}. \tag{SI.39}$$

It bounds the risk carried by the modes beyond N. Indeed, by lemma SI.6, for every $q > 0$ and $u \geq 0$,

$$0 \leq f_{\infty,q}(u) - f_{N,q}(u) \leq \frac{1}{2} \sum_{i>N} i^{-p} \leq \frac{\epsilon_N}{2(p-1)}.$$

In this subsection ϵ_N always denotes the scale eq. (SI.39), and accuracies that vary with N are written $\epsilon = N^{-\omega}$.

Two scale variables organize the estimates. For each model put

$$x \stackrel{\text{def}}{=} \frac{\epsilon}{\epsilon_N} = \epsilon N^{p-1}, \qquad \mathsf{y}_j \stackrel{\text{def}}{=} \epsilon N^{2q_j}, \qquad \sqrt{\mathsf{y}_j} = \sqrt{x}\, N^{q_j(1-r_j)}, \tag{SI.40}$$

where the last identity holds because $2q_j r_j = p - 1$. The regime $x \to 0$, in which the accuracy falls below the accuracy scale, is called frozen. Neither f_{N,q_j} nor g_j depends on B (proposition SI.4(ii)). Hence the clock integral and the ratio of the two times,

$$\mathsf{I}_j \stackrel{\text{def}}{=} \int_0^{\hat{u}_j} \sqrt{Y_j(u)}\, \mathrm{d}u, \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} = D_N \frac{\mathsf{I}_{\text{s-SGD}}}{\mathsf{I}_{\text{s-SVD}}}, \tag{SI.41}$$

do not depend on B; the ratio follows from eq. (SI.30) for the two models and eq. (SI.27). In this subsection constants depend only on α, ς and T unless stated otherwise, and $a \lesssim b$ means $a \leq cb$ for such a constant c.

Forcing integrals. The finite forcing integral grows like a power of N when $r_q < 1$. For $0 < r < 1$ put

$$\mathcal{K}(r) \stackrel{\text{def}}{=} \int_0^\infty y^{-r}\, \Gamma(2r, y)^{1/2}\, \mathrm{d}y, \qquad J_\circ(p,q) \stackrel{\text{def}}{=} (8q)^{-1/2}\, \mathcal{K}(r_q) \quad (r_q < 1), \tag{SI.42}$$

where $\Gamma(a, y) \stackrel{\text{def}}{=} \int_y^\infty t^{a-1} e^{-t}\, \mathrm{d}t$ is the upper incomplete Gamma function [21, 8.2.2]. The integral converges. Indeed, $e^{-t} \leq e^{-y/2} e^{-t/2}$ for $t \geq y$, so

$$\Gamma(2r, y) \leq e^{-y/2} \int_0^\infty t^{2r-1} e^{-t/2}\, \mathrm{d}t = 2^{2r}\, \Gamma(2r)\, e^{-y/2},$$

and the integrand of $\mathcal{K}(r)$ is at most $2^r \Gamma(2r)^{1/2} y^{-r} e^{-y/4}$, which is integrable because $r < 1$.

Proposition SI.8 (Forcing integrals below the critical exponent)**.** *Let $p > 1$ and $q > 0$ with $r \stackrel{\text{def}}{=} r_q < 1$, and write $C = C_{p,q}$ and $A = A_p$.*

(i) For every $N \geq 1$, $J_N(p,q) \leq 2\Gamma(1-r)\, C\, N^{q(1-r)}$.

(ii) For every N with $(N/4)^{q(1-r)} \geq 2$,

$$J_N(p,q) \geq \frac{\sqrt{A}}{2(1-r)}\, 4^{-q(1-r)} N^{q(1-r)}.$$

(iii) As $N \to \infty$, $J_N(p,q)/N^{q(1-r)} \to J_\circ(p,q)$, and $0 < J_\circ(p,q) \leq 2\Gamma(1-r)\, C$.

(iv) $\mathcal{K}(\frac{1}{2}) = \sqrt{2\pi}$; consequently $J_\circ(q+1, q) = \frac{1}{2}\sqrt{\pi/q}$.

Together with the two other regimes, the growth of the forcing integral is

$$J_N(p,q) \begin{cases} \sim J_\circ(p,q)\, N^{q(1-r_q)}, & r_q < 1, \\ \sim \sqrt{q/8}\, \log N, & r_q = 1, \\ \to J(p,q) \in (0,\infty), & r_q > 1. \end{cases} \tag{SI.43}$$

The case $r_q = 1$ is lemma SI.12(i), since $a_q q = \sqrt{q/8}$. The case $r_q > 1$ is the bound $0 \leq J(p,q) - J_N(p,q) \leq [(2(p-1))^{-1/2} + C_{p,q}/(r_q - 1)] N^{-q(r_q-1)}$ proved in the proof of lemma SI.10(iii).

The accuracy-first ratio. At fixed dimension the accuracy limit is explicit. By corollary SI.2, for fixed N, B, α, ς and $T \geq 4$,

$$\lim_{\epsilon \downarrow 0} \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} = \rho_N \stackrel{\text{def}}{=} D_N \, \frac{J_N(p,\alpha)}{J_N(p,\alpha/2)}. \tag{SI.44}$$

The next theorem lets $N \to \infty$ in ρ_N. Write $s \stackrel{\text{def}}{=} r_{\text{s-SVD}} = (p-1)/\alpha$, so that $r_{\text{s-SGD}} = s/2$; thus $s < 1$, $s = 1$, $s > 1$ according as $\varsigma < 1$, $\varsigma = 1$, $\varsigma > 1$. Put

$$Q_A \stackrel{\text{def}}{=} \frac{\mathcal{K}(s/2)}{\sqrt{2}\,\mathcal{K}(s)} \;\; (\varsigma < 1), \qquad K_A \stackrel{\text{def}}{=} \frac{\pi(2-\alpha)}{2\sqrt{2}\,(1-\alpha)} \, \frac{\mathcal{K}(s/2)}{\mathcal{K}(s)} \;\; (0 < \alpha < 1,\ \varsigma < 1),$$

$$K_c \stackrel{\text{def}}{=} \frac{\pi(1-\alpha/2)\,\zeta(\alpha)\,\mathcal{K}\big(\frac{3\alpha-2}{2\alpha}\big)}{\sqrt{8\alpha}\,J(3\alpha-1,\alpha/2)} \;\; (1 < \alpha < 2).$$

Theorem SI.12 (Accuracy-first ratio)**.** *Fix $\alpha > 0$ and ς with $p = \alpha + \varsigma > 1$. As $N \to \infty$:*

(a) if $\varsigma < 1$, then $\rho_N \sim D_N Q_A N^{\alpha/2}$; hence $\rho_N \to K_A$ for $0 < \alpha < 1$, while ρ_N is asymptotic to $\frac{\pi}{2} Q_A \log N$, $\pi(1-\frac{\alpha}{2})\zeta(\alpha) Q_A N^{\alpha-1}$, $\pi\zeta(2) Q_A N/\log N$ and $\frac{\pi\zeta(\alpha)}{\zeta(\alpha/2)} Q_A N^{\alpha/2}$ for $\alpha = 1$, $1 < \alpha < 2$, $\alpha = 2$ and $\alpha > 2$;

(b) if $\varsigma = 1$, then $\rho_N \sim D_N \frac{2\sqrt{\pi}}{\alpha} N^{\alpha/2}/\log N$; hence $\rho_N \log N \to \pi^{3/2}(2-\alpha)/(\alpha(1-\alpha))$ for $0 < \alpha < 1$, $\rho_N \to \pi^{3/2}$ for $\alpha = 1$, and $\rho_N \to \infty$ for $\alpha > 1$, with order $N^{\alpha-1}/\log N$, $N/\log^2 N$ and $N^{\alpha/2}/\log N$ for $1 < \alpha < 2$, $\alpha = 2$ and $\alpha > 2$;

(c) if $1 < \varsigma < \alpha + 1$, then

$$\rho_N \sim D_N \, \frac{\mathcal{K}(s/2)}{\sqrt{8\alpha}\,J(p,\alpha/2)} \, N^{(\alpha+1-\varsigma)/2};$$

hence $\rho_N \to 0$ for $0 < \alpha \leq 1$, with order $N^{(1-\varsigma)/2}$ (times $\log N$ at $\alpha = 1$); $\rho_N \asymp N^{(2\alpha-1-\varsigma)/2}$ for $1 < \alpha < 2$, and on the chord $\varsigma = 2\alpha - 1$, $\rho_N \to K_c$; and $\rho_N \to \infty$ for $\alpha \geq 2$;

(d) if $\varsigma = \alpha + 1$, then $\rho_N \sim D_N \sqrt{\alpha/8}\,\log N / J(p,\alpha/2)$; hence $\rho_N \to 0$ for $0 < \alpha < 2$, $\rho_N \to \pi\zeta(2)/(2J(5,1))$ for $\alpha = 2$, and $\rho_N \to \infty$ with order $\log N$ for $\alpha > 2$;

(e) if $\varsigma > \alpha + 1$, then $\rho_N \sim D_N J(p,\alpha)/J(p,\alpha/2)$; hence $\rho_N \to 0$ for $0 < \alpha \leq 2$ (with order $1/\log N$ at $\alpha = 2$), and $\rho_N \to \pi\zeta(\alpha)J(p,\alpha)/(\zeta(\alpha/2)J(p,\alpha/2)) \in (0,\infty)$ for $\alpha > 2$.

The theorem partitions the half plane $\{p > 1\}$ into three sets.

Corollary SI.4 (Accuracy-first winners)**.** *On $\{(\alpha,\varsigma) :\ \alpha > 0,\ \alpha + \varsigma > 1\}$, as $N \to \infty$, $\rho_N \to \infty$ exactly on*

$$\mathsf{V} \stackrel{\text{def}}{=} \{\alpha = 1,\ \varsigma < 1\} \cup \{1 < \alpha < 2,\ \varsigma < 2\alpha - 1\} \cup \{\alpha = 2,\ \varsigma < 3\} \cup \{\alpha > 2,\ \varsigma \leq \alpha + 1\};$$

ρ_N converges to a finite positive limit exactly on

$$\mathsf{T} \stackrel{\text{def}}{=} \{0 < \alpha < 1,\ \varsigma < 1\} \cup \{(1,1)\} \cup \{1 < \alpha < 2,\ \varsigma = 2\alpha - 1\} \cup \{(2,3)\} \cup \{\alpha > 2,\ \varsigma > \alpha + 1\},$$

with limits K_A, $\pi^{3/2}$, K_c, $\pi\zeta(2)/(2J(5,1))$ and $\pi\zeta(\alpha)J(p,\alpha)/(\zeta(\alpha/2)J(p,\alpha/2))$ on the five parts; and $\rho_N \to 0$ exactly on

$$\mathsf{G} \stackrel{\text{def}}{=} \{0 < \alpha < 1,\ \varsigma \geq 1\} \cup \{\alpha = 1,\ \varsigma > 1\} \cup \{1 < \alpha < 2,\ \varsigma > 2\alpha - 1\} \cup \{\alpha = 2,\ \varsigma > 3\}.$$

Remark SI.6 (Boundary lines and tie constants)**.** *Theorem SI.12 decides every boundary line. On $\alpha = 1$: V for $\varsigma < 1$ (order $\log N$), the tie $\pi^{3/2}$ at $\varsigma = 1$, and G for $\varsigma > 1$. On $\alpha = 2$: V for $\varsigma < 3$, a tie at $\varsigma = 3$, and G for $\varsigma > 3$ (order $1/\log N$). On $\varsigma = 1$: G for $\alpha < 1$ (order $1/\log N$), a tie at $\alpha = 1$, and V for $\alpha > 1$. On $\varsigma = \alpha + 1$: G for $\alpha < 2$, a tie at $\alpha = 2$, and V for $\alpha > 2$ (order $\log N$). The chord $\varsigma = 2\alpha - 1$, $1 < \alpha < 2$, is a tie. It lies above the curve ς_c of proposition SI.6(v), because*

$$2\alpha - 1 - \varsigma_c(\alpha) = \frac{3(\alpha-1)(2-\alpha)}{3-\alpha} > 0 \qquad (1 < \alpha < 2).$$

Evaluating the closed forms by quadrature gives $K_A = 2.7107$ at $(\alpha,\varsigma) = (0.5, 0.7)$, $K_A = 5.5299$ at $(0.8, 0.5)$, $\pi^{3/2} = 5.5683$, $K_c = 3.4958$ at $\alpha = 1.5$, and $\pi\zeta(2)/(2J(5,1)) = 3.0077$.

The frozen regime. Below the accuracy scale the clock integral freezes at its accuracy-first value.

Theorem SI.13 (Frozen regime)**.** *Fix $\alpha > 0$, ς with $p = \alpha + \varsigma > 1$, and $T \geq 4$. Consider a sequence of (N, B, ϵ) with $N \to \infty$, integers $B \geq N$ and $x = \epsilon/\epsilon_N \to 0$. For all large indices both first hits are attained, and for both j*

$$\frac{\mathsf{I}_j}{J_N(p, q_j)} \longrightarrow 1, \qquad t^j_{T\epsilon} \sim \frac{\mathcal{S}_j T_0^j \, J_N(p, q_j)}{2\sqrt{\epsilon}}, \qquad \frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \sim \rho_N.$$

Moreover, if $r_j < 1$, there are N_2 and c such that

$$\Big| \frac{\mathsf{I}_j}{J_N(p, q_j)} - 1 \Big| \leq c \left(x^{1/4} + x^{1/2} \log\left(e + x^{-1/2}\right) \right)^{1/2}$$

for all $N \geq N_2$, $B \geq N$, $\epsilon < 1/(2T)$ and $x \leq 1$.

Since the ratio is asymptotic to ρ_N, every limit, order and winner of theorem SI.12 and corollary SI.4 holds for the actual ratio of hitting times along every frozen sequence. The family of proposition SI.7 is frozen, since $x = N^{\alpha-\omega} \to 0$ there; its ratio is therefore asymptotic to $\pi^{3/2}(2-\alpha)/(\alpha(1-\alpha)\log N)$ by theorem SI.12(b).

The unified clock. For a model with exponents $q = q_j$ and $r = r_j$ put

$$\Upsilon_j(N, \epsilon) \stackrel{\text{def}}{=} \begin{cases} \min\left\{\epsilon^{-(1-r)/(2r)},\ N^{q(1-r)}\right\}, & r < 1, \\ \min\left\{\log(1/\epsilon),\ \log N\right\}, & r = 1, \\ 1, & r > 1, \end{cases} \tag{SI.45}$$

and let $\mathcal{U}$ be the set of triples (α, ς, j) with $\alpha + \varsigma > 1$ such that either $r_j < 1$, or $j =$ s-SVD and $\alpha \neq 1$, or $j =$ s-SGD and $\alpha \notin \{\frac{1}{2}, 1\}$. Equivalently, $(\alpha, \varsigma, j) \notin \mathcal{U}$ exactly for SignSVD at $\alpha = 1$, $\varsigma \geq 1$ and for SignSGD at $\alpha \in \{\frac{1}{2}, 1\}$, $\varsigma \geq \alpha + 1$. For $r < 1$ the two branches of Υ_j cross at the accuracy scale, since $q(1-r) = (p-1)(1-r)/(2r)$:

$$\Upsilon_j(N, \epsilon) = \max\{\epsilon, \epsilon_N\}^{-(1-r_j)/(2r_j)} \qquad (r_j < 1). \tag{SI.46}$$

Theorem SI.14 (Unified clock)**.** *Fix $\alpha > 0$, ς with $\alpha + \varsigma > 1$, $T \geq 4$ and $j \in \{$s-SVD, s-SGD$\}$.*

(i) There are N_0, $\epsilon_0 > 0$ and $c_- > 0$ such that for all $N \geq N_0$, $B \geq N$ and $0 < \epsilon < \epsilon_0$ the first hit $\hat{u}_j$ is attained and $\mathsf{I}_j \geq c_- \Upsilon_j(N, \epsilon)$.

(ii) If $(\alpha, \varsigma, j) \in \mathcal{U}$, there are N_0, $\epsilon_0 > 0$ and c_+ such that for all $N \geq N_0$, $B \geq N$ and $0 < \epsilon < \epsilon_0$, $\mathsf{I}_j \leq c_+ \Upsilon_j(N, \epsilon)$.

In particular, for $(\alpha, \varsigma, j) \in \mathcal{U}$, $t^j_{T\epsilon} \asymp \mathcal{S}_j T_0^j \, \epsilon^{-1/2} \, \Upsilon_j(N, \epsilon)$ for all $N \geq N_0$, $B \geq N$ and $0 < \epsilon < \epsilon_0$.

The threshold ϵ_0 may be taken to be $\epsilon_0(p, q_j, T)$ of lemma SI.10, and N_0 the largest of the finitely many explicit thresholds of the proof; $c_\pm$ are the displayed expressions of the proof. The unified clock contains the main-range window of theorem SI.9 and extends it from $\epsilon N \geq E_*$ to $\epsilon \geq \epsilon_N$.

Corollary SI.5 (Extended main-range window)**.** *Let $\alpha > 1$, $\varsigma \geq 1$ and $T \geq 4$. There are N_0, $\epsilon_0 > 0$ and $0 < c_- \leq c_+$ such that for all $N \geq N_0$, $B \geq N$ and $0 < \epsilon < \epsilon_0$ with $\epsilon \geq \epsilon_N$, both first hits are attained and, for both j,*

$$c_- \leq \frac{t^j_{T\epsilon}}{\mathcal{S}_j T_0^j \, \Pi_{r_j}(\epsilon)} \leq c_+.$$

Since $\varsigma \geq 1$ gives $p - 1 \geq \alpha > 1$, the condition $\epsilon/\epsilon_N \to \infty$ is strictly weaker than the condition $\epsilon N \to \infty$ of theorem SI.9: for $1 < \omega < p - 1$ the family $\epsilon = N^{-\omega}$ satisfies the first and violates the second. Dividing the two orders, the ratio law eq. (SI.38) of corollary SI.3 holds on the whole extended window. Only orders are extended; the equivalents eq. (SI.37) are not.

The ratio at every accuracy. For the ratio we combine theorem SI.14 with the exponent of D_N. For each model put

$$e_j \stackrel{\text{def}}{=} \max\Big\{\frac{1}{2},\ \frac{1}{2r_j}\Big\}, \quad \text{that is} \quad e_{\text{s-SVD}} = \max\Big\{\frac{1}{2},\ \frac{\alpha}{2(p-1)}\Big\}, \quad e_{\text{s-SGD}} = \max\Big\{\frac{1}{2},\ \frac{\alpha}{p-1}\Big\}, \tag{SI.47}$$

and $\chi \stackrel{\text{def}}{=} e_{\text{s-SGD}} - e_{\text{s-SVD}} \geq 0$; these are the exponents of the main text, where χ is called κ. Put further

$$e_D \stackrel{\text{def}}{=} \begin{cases} -\frac{\alpha}{2}, & 0<\alpha<1, \\ -\frac{1}{2}, & \alpha=1, \\ \frac{\alpha}{2}-1, & 1<\alpha<2, \\ 0, & \alpha\geq 2, \end{cases} \qquad L \stackrel{\text{def}}{=} [\alpha=1]-[\alpha=2]+[\varsigma=\alpha+1]-[\varsigma=1],$$

where $[\cdot]$ is 1 if the condition holds and 0 otherwise. By eq. (SI.31), $D_N \asymp N^{e_D}(\log N)^{[\alpha=1]-[\alpha=2]}$, and

$$\chi\,(p-1) = \max\Big\{\frac{p-1}{2}, \alpha\Big\} - \max\Big\{\frac{p-1}{2}, \frac{\alpha}{2}\Big\} = \begin{cases} \frac{\alpha}{2}, & \varsigma<1, \\ \frac{\alpha+1-\varsigma}{2}, & 1\leq\varsigma\leq\alpha+1, \\ 0, & \varsigma\geq\alpha+1. \end{cases} \tag{SI.48}$$

Corollary SI.6 (Ratio at every accuracy)**.** *Fix $\alpha>0$, ς with $\alpha+\varsigma>1$, and $T\geq 4$, and let N_0 and ϵ_0 be thresholds of theorem SI.14 valid for both models. If $(\alpha,\varsigma,$ s-SVD$)$ and $(\alpha,\varsigma,$ s-SGD$)$ both lie in $\mathcal{U}$, then for $N\geq N_0$, $B\geq N$ and $0<\epsilon<\epsilon_0$,*

$$\frac{t^{\text{s-SGD}}_{T\epsilon}}{t^{\text{s-SVD}}_{T\epsilon}} \asymp D_N\,\frac{\Upsilon_{\text{s-SGD}}(N,\epsilon)}{\Upsilon_{\text{s-SVD}}(N,\epsilon)}, \tag{SI.49}$$

and if moreover $\varsigma\notin\{1,\alpha+1\}$, the right side equals $D_N\max\{\epsilon,\epsilon_N\}^{-\chi}$. If only $(\alpha,\varsigma,$ s-SGD$)\in\mathcal{U}$, the upper half of eq. (SI.49) holds; if only $(\alpha,\varsigma,$ s-SVD$)\in\mathcal{U}$, the lower half holds. For $\epsilon=N^{-\omega}$ with fixed $\omega>0$,

$$D_N\,\frac{\Upsilon_{\text{s-SGD}}(N,\epsilon)}{\Upsilon_{\text{s-SVD}}(N,\epsilon)} \asymp N^{e_D+\chi\min\{\omega,p-1\}}\,(\log N)^L,$$

with constants depending also on ω.

The accuracy exponent. Parametrize the accuracy by $\epsilon=N^{-\omega}$ with $\omega>0$. Then $\omega>p-1$ is the frozen regime, and $\omega<p-1$ lies above the accuracy scale. Put

$$S(\omega) \stackrel{\text{def}}{=} e_D+\chi\min\{\omega,p-1\}, \qquad \mathsf{E} \stackrel{\text{def}}{=} \{\alpha=1,\ \varsigma\geq 1\}\cup\{\alpha=\tfrac{1}{2},\ \varsigma\geq\tfrac{3}{2}\},$$

and let $\mathsf{R} \stackrel{\text{def}}{=} t^{\text{s-SGD}}_{T\epsilon}/t^{\text{s-SVD}}_{T\epsilon}$ at $\epsilon=N^{-\omega}$, with arbitrary integers $B_N\geq N$. By eq. (SI.48), $\chi\min\{\omega,p-1\}=\min\{\omega/(p-1),1\}\cdot\chi(p-1)$ is $\frac{\alpha}{2}$, $\frac{\alpha+1-\varsigma}{2}$ or 0 times $\min\{\omega/(p-1),1\}$.

Theorem SI.15 (Accuracy-exponent map)**.** *Fix $\alpha>0$, ς with $p=\alpha+\varsigma>1$, $\omega>0$ and $T\geq 4$. As $N\to\infty$:*

- *(i) if $\omega>p-1$, then $\mathsf{R}\sim\rho_N$, so every conclusion of theorem SI.12 and corollary SI.4 holds for R;*
- *(ii) if $(\alpha,\varsigma)\notin\mathsf{E}$, then $\mathsf{R}\asymp N^{S(\omega)}(\log N)^L$;*
- *(iii) if $\alpha=1$ and $1\leq\varsigma<2$, then $\mathsf{R}\lesssim N^{S(\omega)}(\log N)^L$.*

Reading off the sign of $S(\omega)$, and of L where $S(\omega)=0$, gives the winner. We say that SignSVD wins, that SignSGD wins, or that the two are balanced according as $\mathsf{R}\to\infty$, $\mathsf{R}\to 0$ or $\mathsf{R}\asymp 1$. For $1<\alpha<2$ and $\varsigma<2\alpha-1$ put

$$\omega_c \stackrel{\text{def}}{=} \begin{cases} \dfrac{(2-\alpha)(p-1)}{\alpha}, & \varsigma<1, \\ \dfrac{(2-\alpha)(p-1)}{\alpha+1-\varsigma}, & 1\leq\varsigma<2\alpha-1, \end{cases} \qquad \text{so that} \quad 0<\omega_c<p-1. \tag{SI.50}$$

Corollary SI.7 (Phases of the accuracy-exponent map)**.** *Under the hypotheses of theorem SI.15:*

(a) Phase A, $\alpha > 2$, $\varsigma \le \alpha + 1$*:* SIGNSVD *wins for every* ω.

(b) Phase B, $0 < \alpha < 1$ *and* $\varsigma \ge 1$ *with* $(\alpha, \varsigma) \notin \mathsf{E}$*, or* $1 < \alpha < 2$ *and* $\varsigma > 2\alpha - 1$*:* SIGNSGD *wins for every* ω.

(c) Phase C, $1 < \alpha < 2$, $\varsigma < 2\alpha - 1$*:* SIGNSGD *wins if* $\omega < \omega_c$ *and* SIGNSVD *wins if* $\omega > \omega_c$*; at* $\omega = \omega_c$ *the two are balanced if* $\varsigma \neq 1$*, and* SIGNSGD *wins, with order* $1/\log N$*, if* $\varsigma = 1$.

(d) Phase D, $0 < \alpha < 1$, $\varsigma < 1$*:* SIGNSGD *wins if* $\omega < p - 1$*, and the two are balanced if* $\omega \ge p - 1$*, with limit* K_A *if* $\omega > p - 1$.

(e) Tie, $\alpha > 2$, $\varsigma > \alpha + 1$*: balanced for every* ω.

(f) Chord, $1 < \alpha < 2$, $\varsigma = 2\alpha - 1$*:* SIGNSGD *wins if* $\omega < p - 1$*, and the two are balanced if* $\omega \ge p - 1$*, with limit* K_c *if* $\omega > p - 1$.

(g) $\alpha = 1$*: for* $\varsigma < 1$, SIGNSGD *wins if* $\omega < p - 1$ *and* SIGNSVD *wins, with order* $\log N$*, if* $\omega \ge p - 1$*; for* $\varsigma = 1$, SIGNSGD *wins if* $\omega < 1$*, and* $\mathsf{R} \to \pi^{3/2}$ *if* $\omega > 1$*; for* $1 < \varsigma < 2$, SIGNSGD *wins for every* ω.

(h) $\alpha = 2$*:* SIGNSVD *wins for* $\varsigma < 3$*, the two are balanced for* $\varsigma = 3$*, and* SIGNSGD *wins for* $\varsigma > 3$*, for every* ω.

The threshold ω_c is the critical exponent of the windowed map. For $\varsigma < 1$ it is the $\omega_c = r_{\text{s-SVD}}(2 - \alpha)$ of proposition SI.6(i), and for $1 \le \varsigma < 2\alpha - 1$ it is the formula of proposition SI.6(v)(b). For $\varsigma < 1$, $\omega_c < 2 - \alpha < 1$, because $p - 1 < \alpha$. For $1 \le \varsigma < 2\alpha - 1$, eq. (SI.58) shows that $\omega_c < 1$ exactly when $\varsigma < \varsigma_c(\alpha)$, and $\omega_c = 1$ exactly on the curve $\varsigma = \varsigma_c(\alpha)$. The curve of proposition SI.6(v) is therefore the cut of the map at $\epsilon = N^{-1}$, the accuracy at which the window of theorem SI.9 ends, while the chord $\varsigma = 2\alpha - 1$ is its cut at $\epsilon = \epsilon_N$. Since $\omega_c < p - 1$, SIGNSVD wins at every point of Phase C once $\omega \ge p - 1$; as $\omega \to \infty$ the boundary between Phases C and B becomes the chord.

What is not covered. Three points are not proved. First, the winner on $\{\alpha = 1,\ \varsigma \ge 2\} \cup \{\alpha = \frac{1}{2},\ \varsigma \ge \frac{3}{2}\}$ for $\omega \le p - 1$, outside the windows of proposition SI.5. There one of the two models lies outside $\mathcal{U}$: the kernel templates of lemma SI.25, on which the upper bound of theorem SI.14 rests, do not cover the SIGNSVD kernel at $\alpha = 1$ or the SIGNSGD kernel at $\alpha \in \{\frac{1}{2}, 1\}$. Second, at $\alpha = \varsigma = \omega = 1$ only $\mathsf{R} \lesssim 1$ is proved. Third, for $\omega \le p - 1$ on the balanced regions only orders are proved, not limit constants.

SI.7 Comparison through the first hit

The phase theorems are proved in two steps. A supersolution of square form, $(h + \sqrt{\epsilon})^2$, bounds the dimensionless risk up to a horizon at which it is at most 4ϵ; this places the first hit and bounds the contribution of the noise to the clock. The clock integral through the hit is then evaluated from the forcing alone. The first step needs a bound on the convolution of the kernel with h, which appendix SI.8 supplies for each kernel; the second step is common to all phases.

Lemma SI.9 (Square supersolution)**.** *Let* $\epsilon > 0$, $T \ge 4$ *and* $H > 0$*. Let* $g \ge 0$ *be continuous on* $[0, \infty)$ *with* $\int_0^\infty g \le 1$*, let* $f \ge 0$ *be continuous with* $f(0) > T\epsilon$*, and let* $Y > 0$ *be continuous on* $[0, \infty)$ *with* $Y = f + \sqrt{\epsilon}\, g * \sqrt{Y}$*. Let* h *be continuous on* $[0, H]$ *with* $h \ge \sqrt{\epsilon}$*,* $h(H) = \sqrt{\epsilon}$ *and* $f \le h^2$*. Assume that on* $[0, H]$ *either*

$$\textit{(a)}\ \ g * h \le 2h, \qquad \textit{or} \qquad \textit{(b)}\ \ g * h \le \sqrt{\epsilon}.$$

(i) On $[0, H]$*,* $f \le Y \le (h + \sqrt{\epsilon})^2$ *and* $Y - f \le 2\sqrt{\epsilon}\, h + \epsilon$*; under (b), moreover* $Y - f \le 2\epsilon$.

(ii) The first hit $\hat{u} \stackrel{\text{def}}{=} \inf\{u \ge 0 :\ Y(u) \le T\epsilon\}$ *satisfies* $0 < \hat{u} \le H$ *and* $Y(\hat{u}) = T\epsilon$*, and*

$$0 \le \mathcal{E} \stackrel{\text{def}}{=} \int_0^{\hat{u}} \left(\sqrt{Y} - \sqrt{f}\right) \mathrm{d}u \le \int_0^H \sqrt{2\sqrt{\epsilon}\, h + \epsilon}\, \mathrm{d}u;$$

under (b), moreover $\mathcal{E} \le \sqrt{2\epsilon}\, H$.

(iii) Let $C > 0$, $D \ge 1$, $0 < r_0 < 2$ and $0 < \epsilon < C^2$, and take $h(u) = C(1 + u/D)^{-r_0}$ and $H = D[(C/\sqrt{\epsilon})^{1/r_0} - 1]$. Then $h \ge \sqrt{\epsilon}$ on $[0, H]$, $h(H) = \sqrt{\epsilon}$, $1 + H \le D(C/\sqrt{\epsilon})^{1/r_0}$, and

$$\int_0^H \sqrt{2\sqrt{\epsilon}\, h + \epsilon}\, \mathrm{d}u \le D\, C^{1/r_0} \Big[\frac{\sqrt{2}}{1 - r_0/2} + 1\Big] \epsilon^{(r_0-1)/(2r_0)}. \tag{SI.51}$$

Proof. Put $U \stackrel{\text{def}}{=} (h + \sqrt{\epsilon})^2$, so that $U \ge \epsilon$. Since $\int g \le 1$, $g * (h + \sqrt{\epsilon}) \le g * h + \sqrt{\epsilon}$. Under (a),

$$f + \sqrt{\epsilon}\, g * \sqrt{U} \le h^2 + \sqrt{\epsilon}(2h + \sqrt{\epsilon}) = U.$$

Under (b), using $h \ge \sqrt{\epsilon}$,

$$f + \sqrt{\epsilon}\, g * \sqrt{U} \le h^2 + 2\epsilon \le h^2 + 2\sqrt{\epsilon}h + \epsilon = U.$$

Lemma SI.3, with kernel $\sqrt{\epsilon}g$, $G = f$ and $m = \epsilon$, gives $Y \le U$ on $[0, H]$. The equation gives $Y \ge f$ and

$$Y - f = \sqrt{\epsilon}\, g * \sqrt{Y} \le \sqrt{\epsilon}\, g * (h + \sqrt{\epsilon}),$$

which is at most $2\sqrt{\epsilon}h + \epsilon$ under (a) and at most $2\epsilon \le 2\sqrt{\epsilon}h + \epsilon$ under (b). This proves (i).

At H the bound gives $Y(H) \le (2\sqrt{\epsilon})^2 = 4\epsilon \le T\epsilon$, while $Y(0) = f(0) > T\epsilon$. The set $\{u \le H : Y(u) \le T\epsilon\}$ is closed, nonempty and does not contain 0; its least element $\hat{u}$ is the first hit, and continuity gives $Y(\hat{u}) = T\epsilon$. No monotonicity of Y is used. The inequality $\sqrt{x} - \sqrt{y} \le \sqrt{x - y}$ for $x \ge y \ge 0$ and (i) give the bounds on $\mathcal{E}$. This proves (ii).

For (iii), h is decreasing with $h(H) = \sqrt{\epsilon}$, and $1 + H/D = (C/\sqrt{\epsilon})^{1/r_0}$ gives the bound on $1 + H$ because $D \ge 1$. The inequality $\sqrt{x + y} \le \sqrt{x} + \sqrt{y}$ gives

$$\int_0^H \sqrt{2\sqrt{\epsilon}\, h + \epsilon}\, \mathrm{d}u \le \sqrt{2C}\, \epsilon^{1/4} D\, I_{r_0/2}(H/D) + \sqrt{\epsilon}\, H.$$

Since $r_0 < 2$, $I_{r_0/2}(x) \le (1 + x)^{1 - r_0/2}/(1 - r_0/2)$. Inserting $1 + H/D = (C/\sqrt{\epsilon})^{1/r_0}$,

$$\sqrt{2C}\, \epsilon^{1/4} D\, I_{r_0/2}(H/D) \le \frac{\sqrt{2}\, D\, C^{1/r_0}}{1 - r_0/2}\, \epsilon^{(r_0-1)/(2r_0)}, \qquad \sqrt{\epsilon}\, H \le D\, C^{1/r_0} \epsilon^{(r_0-1)/(2r_0)},$$

which proves eq. (SI.51). □

The clock integral through the hit is evaluated from the forcing alone. For $p > 1$ and $q > 0$ write $C = C_{p,q}$, $A = A_p$, $r = r_q$, and

$$a_q \stackrel{\text{def}}{=} (8q)^{-1/2}, \qquad \epsilon_0(p, q, T) \stackrel{\text{def}}{=} \min\Big\{\frac{1}{4},\ \frac{1}{8T},\ C^2,\ \Big(\frac{A}{T}\Big)^2,\ \frac{A}{T\, 2^{2r/(1-r)}}\Big\},$$

where the last entry is omitted if $r \ge 1$. The threshold ϵ_0 of eq. (SI.32) is the smaller of $\epsilon_0(p, q_j, T)$ over the two models.

Lemma SI.10 (The clock integral)**.** *Fix $p > 1$, $q > 0$ and $T \ge 4$, and let $N \ge 4$ and $0 < \epsilon < \epsilon_0(p, q, T)$. Let Y be continuous on $[0, \infty)$ with $Y \ge f_{N,q}$, and suppose that its first $T\epsilon$ hit $\hat{u}$ is attained, $0 < \hat{u} < \infty$ with $Y(\hat{u}) = T\epsilon$. Let $H \ge \hat{u}$ and $\mathcal{E} \stackrel{\text{def}}{=} \int_0^{\hat{u}} (\sqrt{Y} - \sqrt{f_{N,q}})\, \mathrm{d}u$.*

(i) If $r < 1$ and $H \le (N/4)^q$, then

$$\frac{\sqrt{A}}{2(1 - r)} \Big(\frac{A}{T}\Big)^{(1-r)/(2r)} \epsilon^{-(1-r)/(2r)} \le \int_0^{\hat{u}} \sqrt{Y}\, \mathrm{d}u \le C\, I_r(H) + \mathcal{E}.$$

(ii) If $r = 1$ and $H \le (N/4)^q$, then $\int_0^{\hat{u}} \sqrt{Y}\, \mathrm{d}u \ge \frac{\sqrt{A}}{4} \log \frac{1}{\epsilon}$, and

$$\Big| \int_0^{\hat{u}} \sqrt{Y}\, \mathrm{d}u - \frac{a_q}{2} \log \frac{1}{\epsilon} \Big| \le K_{p,q,T} + a_q \log_+ \big((1 + H)\sqrt{\epsilon}\big) + \frac{H N^{-q}}{\sqrt{2(p - 1)}} + \mathcal{E},$$

where $K_{p,q,T} \stackrel{\text{def}}{=} C + a_q \log 2 + q M_{p,q} 2^{-p/q}/a_q + \frac{a_q}{2} \log(T/A)$.

(iii) If $r > 1$, then $\frac{1}{2\sqrt{2}} \le \int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u \le J(p,q) + \mathcal{E}$, and

$$\Big| \int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u - J(p,q) \Big| \\ \le \Big[\frac{1}{\sqrt{2(p-1)}} + \frac{C}{r-1} \Big] N^{-q(r-1)} + \frac{C}{r-1} \Big(1 + \frac{1}{2} \log \frac{1}{2T\epsilon} \Big)^{1-r} + \mathcal{E}.$$

Proof. Write $f = f_{N,q}$. By definition,

$$\int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u = \int_0^{\hat{u}} \sqrt{f}\,\mathrm{d}u + \mathcal{E}. \tag{SI.52}$$

The level at the hit bounds the hit from below: $f(\hat{u}) \le Y(\hat{u}) = T\epsilon$.

For (i) and (ii), $\hat{u} \le H \le (N/4)^q$, so the lower envelope of eq. (SI.23) applies at $\hat{u}$ and on $[0, \hat{u}]$. At $\hat{u}$ it gives

$$1 + \hat{u} \ge \Big(\frac{A}{T\epsilon} \Big)^{1/(2r)}. \tag{SI.53}$$

For $r < 1$, integrating the lower envelope gives

$$\int_0^{\hat{u}} \sqrt{f}\,\mathrm{d}u \ge \sqrt{A}\, I_r(\hat{u}) \ge \frac{\sqrt{A}}{1-r} \Big[\Big(\frac{A}{T\epsilon} \Big)^{(1-r)/(2r)} - 1 \Big] \ge \frac{\sqrt{A}}{2(1-r)} \Big(\frac{A}{T} \Big)^{(1-r)/(2r)} \epsilon^{-(1-r)/(2r)},$$

where the last step uses $(A/(T\epsilon))^{(1-r)/(2r)} \ge 2$, which is the last entry of ϵ_0. The upper envelope gives $\int_0^{\hat{u}} \sqrt{f} \le C\, I_r(H)$, and eq. (SI.52) proves (i).

For $r = 1$, integrating the lower envelope and using eq. (SI.53) gives

$$\int_0^{\hat{u}} \sqrt{f}\,\mathrm{d}u \ge \sqrt{A} \log(1 + \hat{u}) \ge \frac{\sqrt{A}}{2} \log \frac{A}{T\epsilon} \ge \frac{\sqrt{A}}{4} \log \frac{1}{\epsilon},$$

where the last step uses $\epsilon \le (A/T)^2$. For the constant, $r = 1$ means $p = 2q + 1$, so $d(p,q) = 2$ and $\Gamma(2) = 1$. Lemma SI.6 with $2u$ in place of a, divided by 2, gives $|f_{\infty,q}(u) - a_q^2 u^{-2}| \le M_{p,q}(2u)^{-p/q}$. Dividing by $\sqrt{f_{\infty,q}} + a_q/u \ge a_q/u$ gives, for $u \ge 1$,

$$\Big| \sqrt{f_{\infty,q}(u)} - \frac{a_q}{u} \Big| \le \frac{M_{p,q} 2^{-p/q}}{a_q}\, u^{-1-1/q}.$$

The right side has integral $qM_{p,q}2^{-p/q}/a_q$ over $[1, \infty)$, and $\sqrt{f_{\infty,q}} \le C$ on $[0,1]$. Together with $0 \le \log(1+z) - \log z \le \log 2$ for $z \ge 1$, this gives, for every $z \ge 0$,

$$\Big| \int_0^z \sqrt{f_{\infty,q}}\,\mathrm{d}u - a_q \log(1+z) \Big| \le C + a_q \log 2 + \frac{qM_{p,q}2^{-p/q}}{a_q}.$$

The finite sum differs from the infinite one by $0 \le f_{\infty,q} - f \le N^{1-p}/(2(p-1))$ (lemma SI.6), and $N^{(1-p)/2} = N^{-q}$, so

$$0 \le \int_0^{\hat{u}} \big(\sqrt{f_{\infty,q}} - \sqrt{f} \big)\,\mathrm{d}u \le \frac{\hat{u} N^{-q}}{\sqrt{2(p-1)}} \le \frac{H N^{-q}}{\sqrt{2(p-1)}}.$$

Finally eq. (SI.53) and $\hat{u} \le H$ give

$$-\frac{1}{2} \log \frac{T}{A} \le \log(1+\hat{u}) - \frac{1}{2} \log \frac{1}{\epsilon} \le \log \big((1+H)\sqrt{\epsilon} \big).$$

Combining the last three displays with eq. (SI.52) proves (ii).

For (iii), $\sqrt{f_{\infty,q}} \le C(1+u)^{-r}$ is integrable and $J(p,q) > 0$. Since $f \ge \frac{1}{2} e^{-2u}$, the level at the hit gives $\hat{u} \ge l_\epsilon \stackrel{\text{def}}{=} \frac{1}{2} \log \frac{1}{2T\epsilon}$, and $l_\epsilon \ge \log 2$ because $\epsilon \le 1/(8T)$. Hence

$$\int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u \ge \int_0^{\log 2} \frac{e^{-u}}{\sqrt{2}}\,\mathrm{d}u = \frac{1}{2\sqrt{2}}.$$

The upper bound follows from eq. (SI.52) and $f \le f_{\infty,q}$. For the limit, split

$$J(p,q) - \int_0^{\hat{u}} \sqrt{f}\,\mathrm{d}u = \big(J(p,q) - J_N(p,q)\big) + \int_{\hat{u}}^{\infty} \sqrt{f}\,\mathrm{d}u.$$

On $[0, N^q]$ the pointwise bound $\sqrt{f_{\infty,q}} - \sqrt{f} \le N^{(1-p)/2}/\sqrt{2(p-1)} = N^{-qr}/\sqrt{2(p-1)}$ holds, and on $[N^q, \infty)$ the integrand is at most $C(1+u)^{-r}$; hence

$$0 \le J(p,q) - J_N(p,q) \le \Big[\frac{1}{\sqrt{2(p-1)}} + \frac{C}{r-1}\Big] N^{-q(r-1)}.$$

The upper envelope and $\hat{u} \ge l_\epsilon$ give

$$0 \le \int_{\hat{u}}^{\infty} \sqrt{f}\,\mathrm{d}u \le \frac{C}{r-1}(1+l_\epsilon)^{1-r}.$$

With eq. (SI.52) this proves (iii). □

The accuracy-first limit and the family outside the windows use the following comparison for finite exponential mixtures.

Lemma SI.11 (Finite exponential mixtures)**.** *Let $b_{\min} > 0$, let*

$$f(u) = \sum_{i=1}^{n} a_i e^{-b_i u}, \qquad g(u) = \sum_{l=1}^{n'} w_l\, c_l\, e^{-c_l u},$$

with $a_i \ge 0$, $F_0 \stackrel{\text{def}}{=} \sum_i a_i > 0$, $b_i, c_l \ge b_{\min}$, $w_l \ge 0$ and $\sum_l w_l = 1$. Let $T \ge 4$, $0 < \epsilon < F_0/T$, and let $Y > 0$ be the continuous solution of $Y = f + \sqrt{\epsilon}\, g * \sqrt{Y}$. Put

$$\mathsf{J} \stackrel{\text{def}}{=} \int_0^{\infty} \sqrt{f(u)}\,\mathrm{d}u, \qquad H \stackrel{\text{def}}{=} \frac{2}{b_{\min}} \log \frac{F_0}{\epsilon}, \qquad \mathsf{d} \stackrel{\text{def}}{=} \sqrt{\epsilon}\, H.$$

Then $0 < \mathsf{J} \le 2\sqrt{F_0}/b_{\min}$, the first $T\epsilon$ hit $\hat{u}$ satisfies $0 < \hat{u} \le H$ and $Y(\hat{u}) = T\epsilon$, and

$$\mathsf{J} - \frac{2\sqrt{T\epsilon}}{b_{\min}} \le \int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u \le \mathsf{J} + 2\sqrt{\mathsf{d}\,\mathsf{J}} + \mathsf{d}.$$

Proof. The equation has a unique positive continuous solution by theorem SI.1, applied with the rates b_i and c_l and zero coefficients where needed. Since $f \le F_0 e^{-b_{\min} u}$, the integral J is positive and at most $2\sqrt{F_0}/b_{\min}$.

Let $h(u) \stackrel{\text{def}}{=} \sqrt{F_0}\, e^{-b_{\min} u/4}$, so that $f \le h^2$ and $h(H) = \sqrt{\epsilon}$. Integrating each exponential gives

$$\frac{(g*h)(u)}{h(u)} = \sum_l w_l \frac{c_l}{c_l - b_{\min}/4}\big(1 - e^{-(c_l - b_{\min}/4)u}\big) \le \frac{4}{3}.$$

Hence $U \stackrel{\text{def}}{=} (h + \sqrt{\epsilon})^2$ satisfies

$$f + \sqrt{\epsilon}\, g * \sqrt{U} \le h^2 + \tfrac{4}{3}\sqrt{\epsilon}\, h + \epsilon \le U,$$

and lemma SI.3 with $m = \epsilon$ gives $Y \le U$ on every compact interval. At H, $Y(H) \le 4\epsilon \le T\epsilon$, while $Y(0) = F_0 > T\epsilon$, so the first hit is attained in $(0, H]$ with $Y(\hat{u}) = T\epsilon$.

For the lower bound, all rates are at least $b_{\min}$, so $f(v) \le f(\hat{u}) e^{-b_{\min}(v-\hat{u})}$ for $v \ge \hat{u}$, and $f(\hat{u}) \le Y(\hat{u}) = T\epsilon$. Hence

$$\int_{\hat{u}}^{\infty} \sqrt{f}\,\mathrm{d}v \le \frac{2\sqrt{T\epsilon}}{b_{\min}}, \qquad \int_0^{\hat{u}} \sqrt{Y}\,\mathrm{d}u \ge \int_0^{\hat{u}} \sqrt{f}\,\mathrm{d}u \ge \mathsf{J} - \frac{2\sqrt{T\epsilon}}{b_{\min}}.$$

For the upper bound, write $\mathsf{I} \stackrel{\text{def}}{=} \int_0^{\hat{u}} \sqrt{Y}$. The inequality $\sqrt{x+y} \le \sqrt{x} + \sqrt{y}$ and the Cauchy–Schwarz inequality give

$$\mathsf{I} \le \mathsf{J} + \epsilon^{1/4} \int_0^{\hat{u}} \sqrt{(g * \sqrt{Y})(u)}\,\mathrm{d}u \le \mathsf{J} + \epsilon^{1/4}\Big(\hat{u} \int_0^{\hat{u}} (g*\sqrt{Y})(u)\,\mathrm{d}u\Big)^{1/2}.$$

Exchanging the order of integration and using $\int g = 1$,

$$\int_0^{\hat{u}} (g * \sqrt{Y})(u)\,\mathrm{d}u = \int_0^{\hat{u}} \sqrt{Y(v)} \int_0^{\hat{u}-v} g(w)\,\mathrm{d}w\,\mathrm{d}v \le \mathsf{I}.$$

Since $\sqrt{\epsilon}\,\hat{u} \le \mathsf{d}$, this gives $\mathsf{I} \le \mathsf{J} + \sqrt{\mathsf{d}\,\mathsf{I}}$. The quadratic $x^2 - \sqrt{\mathsf{d}}\,x - \mathsf{J}$ is increasing beyond its positive root and nonnegative at $x = \sqrt{\mathsf{J}} + \sqrt{\mathsf{d}}$, so $\sqrt{\mathsf{I}} \le \sqrt{\mathsf{J}} + \sqrt{\mathsf{d}}$. Squaring proves the upper bound. □

Lemma SI.12 (Two finite forcing integrals)**.** *Let $p > 1$, $q > 0$, $C = C_{p,q}$ and $A = A_p$.*

(i) If $p = 2q + 1$, then for every $N \ge 1$

$$\left|J_N(p,q) - a_q\, q \log N\right| \le 2C + \frac{qM_{p,q}2^{-p/q}}{a_q} + \frac{1}{\sqrt{2(p-1)}}.$$

(ii) If $p = q + 1$, then for every $N \ge 4 \cdot 4^{1/q}$

$$\sqrt{A}\,4^{-q/2}N^{q/2} \le J_N(p,q) \le 3C\,N^{q/2}.$$

Proof. Put $L \stackrel{\text{def}}{=} N^q$. Every rate of $f_{N,q}$ is at least $2/L$, so $f_{N,q}(u) \le f_{N,q}(L)e^{-2(u-L)/L}$ for $u \ge L$, and

$$\int_L^\infty \sqrt{f_{N,q}}\,\mathrm{d}u \le L\sqrt{f_{N,q}(L)} \le L\,C(1+L)^{-r_q}. \tag{SI.54}$$

For (i), $r_q = 1$, so eq. (SI.54) is at most C. As in the proof of lemma SI.10(ii),

$$0 \le \int_0^L \left(\sqrt{f_{\infty,q}} - \sqrt{f_{N,q}}\right)\mathrm{d}u \le \frac{LN^{-q}}{\sqrt{2(p-1)}} = \frac{1}{\sqrt{2(p-1)}},$$
$$\left|\int_0^L \sqrt{f_{\infty,q}}\,\mathrm{d}u - a_q \log L\right| \le C + \frac{qM_{p,q}2^{-p/q}}{a_q},$$

where the second bound splits the integral at 1 and uses $\sqrt{f_{\infty,q}} \le C$ on $[0,1]$. Since $a_q \log L = a_q q \log N$, adding the three errors proves (i).

For (ii), $r_q = \frac{1}{2}$. The upper envelope gives

$$\int_0^L \sqrt{f_{N,q}}\,\mathrm{d}u \le C\int_0^L (1+u)^{-1/2}\,\mathrm{d}u \le 2C\sqrt{L},$$

and eq. (SI.54) is at most $C\sqrt{L}$, which proves the upper bound. For the lower bound put $V \stackrel{\text{def}}{=} (N/4)^q \ge 4$. The lower envelope on $[0,V]$ gives

$$J_N(p,q) \ge 2\sqrt{A}\left(\sqrt{1+V} - 1\right) \ge \sqrt{A}\sqrt{V} = \sqrt{A}\,4^{-q/2}N^{q/2},$$

using $\sqrt{V} \ge 2$. □

SI.8 Kernel estimates

This subsection proves the convolution bounds (a) and (b) of lemma SI.9 for the kernels eq. (SI.28). Three mechanisms appear. A decreasing kernel of small local mass contracts a slowly decaying envelope (lemma SI.13). A kernel with an integrable power tail contracts a dilated envelope (lemma SI.14). For the SignSVD kernel with $1 < \alpha \le 2$, whose tail is not integrable, the harmonic normalization $H_N(\alpha/2)$ makes the late part of the convolution small (lemmas SI.16 to SI.18).

Lemma SI.13 (Local mass)**.** *Let $g \ge 0$ be continuous and nonincreasing on $[0,\infty)$, and put $m_g(u) \stackrel{\text{def}}{=} \int_0^u g(v)\,\mathrm{d}v$. Let $C, D > 0$, $0 < r < 1$ and $h(u) \stackrel{\text{def}}{=} C(1+u/D)^{-r}$. Then*

$$(g * h)(u) \le \frac{m_g(u)}{1-r}\,h(u) \qquad (u \ge 0).$$

Proof. At $u = 0$ both sides vanish. Fix $u > 0$. On $[0, u]$ the function g is nonincreasing and $v \mapsto h(u - v)$ is nondecreasing, so

$$\int_0^u \int_0^u \big(g(v) - g(v')\big)\big(h(u - v) - h(u - v')\big)\,\mathrm{d}v\,\mathrm{d}v' \le 0.$$

Expanding the product gives

$$u\,(g * h)(u) \le m_g(u) \int_0^u h(v)\,\mathrm{d}v.$$

Direct integration gives

$$\frac{1}{u\,h(u)} \int_0^u h(v)\,\mathrm{d}v = \frac{D + u - D(1 + u/D)^r}{(1 - r)\,u} \le \frac{1}{1 - r},$$

since $(1 + u/D)^r \ge 1$. □

For $\upsilon > 1$, $0 < r_0 \le \upsilon$ and $K > 0$ put

$$\delta_\circ \stackrel{\text{def}}{=} 1 - (2/3)^{1/r_0}, \qquad b(r_0, \upsilon) \stackrel{\text{def}}{=} \begin{cases} (1 - r_0)^{-1}, & r_0 < 1, \\ (e(\upsilon - 1))^{-1}, & r_0 = 1, \\ (r_0 - 1)^{-1}, & r_0 > 1, \end{cases} \tag{SI.55}$$

$$D(K, r_0, \upsilon) \stackrel{\text{def}}{=} \max\Big\{1, \big[2K\delta_\circ^{-\upsilon} b(r_0, \upsilon)\big]^{1/(\upsilon - 1)}\Big\}.$$

Lemma SI.14 (Dilated envelope)**.** *Let $g \ge 0$ be continuous on $[0, \infty)$ with $\int_0^\infty g \le 1$, and suppose $g(v) \le K(1 + v)^{-\upsilon}$ for $0 \le v \le H_*$, where $H_* > 0$, $\upsilon > 1$ and $K > 0$. Let $0 < r_0 \le \upsilon$, $C > 0$, $D = D(K, r_0, \upsilon)$ and $h(u) \stackrel{\text{def}}{=} C(1 + u/D)^{-r_0}$. Then*

$$\frac{(g * h)(u)}{h(u)} \le \frac{3}{2} + K\delta_\circ^{-\upsilon} D^{1-\upsilon} b(r_0, \upsilon) \le 2 \qquad (0 \le u \le H_*).$$

Proof. Split the convolution at $v = \delta_\circ(D + u)$. For $0 \le v \le \min(u, \delta_\circ(D + u))$,

$$\frac{h(u - v)}{h(u)} = \Big(\frac{D + u}{D + u - v}\Big)^{r_0} \le (1 - \delta_\circ)^{-r_0} = \frac{3}{2},$$

and $\int g \le 1$ bounds this part by $\frac{3}{2}h(u)$. If $\delta_\circ(D + u) \ge u$ there is nothing else. Otherwise, on the remaining part $v \le u \le H_*$ and $g(v) \le K\delta_\circ^{-\upsilon}(D + u)^{-\upsilon}$. Enlarging the integral of $h(u - v)$ to $[0, u]$ and writing $y = u/D$ gives

$$\frac{1}{h(u)} \int_{\delta_\circ(D+u)}^u g(v) h(u - v)\,\mathrm{d}v \le K\delta_\circ^{-\upsilon} D^{1-\upsilon} (1 + y)^{r_0 - \upsilon} I_{r_0}(y).$$

It remains to bound $(1 + y)^{r_0 - \upsilon} I_{r_0}(y)$ by $b(r_0, \upsilon)$. For $r_0 < 1$, $I_{r_0}(y) \le (1 + y)^{1 - r_0}/(1 - r_0)$ and $(1 + y)^{1-\upsilon} \le 1$. For $r_0 = 1$, $I_1(y) = \log(1 + y)$ and $\sup_{x \ge 1} x^{1-\upsilon} \log x = 1/(e(\upsilon - 1))$. For $r_0 > 1$, $I_{r_0} \le 1/(r_0 - 1)$ and $(1 + y)^{r_0 - \upsilon} \le 1$. The choice of D makes the second term at most $\frac{1}{2}$. □

The next three lemmas concern the SIGNSVD kernel. Write $q \stackrel{\text{def}}{=} \alpha/2$ and $g \stackrel{\text{def}}{=} g_{\text{s-SVD}}$.

Lemma SI.15 (SIGNSVD kernel, $0 < \alpha < 2$, small times)**.** *Let $0 < \alpha < 2$ and $0 < r < 1$, and put*

$$\delta_1 \stackrel{\text{def}}{=} \Big[\frac{1 - r}{8}\Big]^{1/(1-q)}, \qquad \mathsf{c}_1 \stackrel{\text{def}}{=} \frac{(1 - r)\delta_1^q}{8},$$

$$N_1 \stackrel{\text{def}}{=} \Big\lceil \max\Big\{4,\ 2^{1/(1-q)},\ \Big[\frac{8(1 - q)}{1 - r}\Big]^{1/(1-q)}\Big\}\Big\rceil.$$

For $N \ge N_1$ and $0 \le u \le \mathsf{c}_1 N^q$, $m_g(u) \le \frac{3}{4}(1 - r)$. Consequently, for every $C > 0$ and $h(u) = C(1 + u)^{-r}$, $(g * h)(u) \le \frac{3}{4}h(u)$ on $[0, \mathsf{c}_1 N^q]$.

Proof. Integrating the kernel gives exactly

$$m_g(u) = \frac{1}{H_N(q)} \sum_{i=1}^{N} i^{-q}\big(1 - e^{-2ui^{-q}}\big).$$

By lemma SI.8 and $N^{1-q} \geq 2$, $H_N(q) \geq (N^{1-q}-1)/(1-q) \geq N^{1-q}/(2(1-q))$. Let $J \overset{\text{def}}{=} \lfloor \delta_1 N \rfloor$. The indices $i \leq J$ contribute at most $H_J(q)/H_N(q)$, with $H_0(q) = 0$, and for $J \geq 1$, $H_J(q) \leq 1 + J^{1-q}/(1-q)$; hence

$$\frac{H_J(q)}{H_N(q)} \leq 2(1-q)N^{-(1-q)} + 2\delta_1^{1-q} \leq \frac{1-r}{4} + \frac{1-r}{4}.$$

For $i > J$ one has $i > \delta_1 N$, and $1 - e^{-x} \leq x$ gives a contribution of at most

$$\frac{2u}{(\delta_1 N)^q} \cdot \frac{1}{H_N(q)} \sum_{i>J} i^{-q} \leq 2\mathsf{c}_1 \delta_1^{-q} = \frac{1-r}{4} \qquad (u \leq \mathsf{c}_1 N^q).$$

Adding gives $m_g(u) \leq \frac{3}{4}(1-r)$. The kernel is decreasing, so lemma SI.13 with $D = 1$ gives the last assertion. □

Lemma SI.16 (SIGNSVD kernel, $1 < \alpha < 2$, $\varsigma \geq 1$)**.** *Let $1 < \alpha < 2$ and $\varsigma \geq 1$, so that $r_{\text{s-SVD}} \geq 1$. Put $r_0 \overset{\text{def}}{=} 1$ if $r_{\text{s-SVD}} = 1$ and $r_0 \overset{\text{def}}{=} \min\{r_{\text{s-SVD}}, (1 + 2/\alpha)/2\}$ if $r_{\text{s-SVD}} > 1$. Put $\upsilon_\circ \overset{\text{def}}{=} 2 - 2/\alpha \in (0,1)$, $K_{\text{V}} \overset{\text{def}}{=} 2E_{\alpha,q}$, $\delta_\circ \overset{\text{def}}{=} 1 - (2/3)^{1/r_0}$, $d_\alpha \overset{\text{def}}{=} (\alpha-1)(2-\alpha)/(2\alpha)$,*

$$(J_{\text{V}}, d_{\text{V}}) \overset{\text{def}}{=} \begin{cases} \big(K_{\text{V}}\delta_\circ^{-\upsilon_\circ}/(e\, d_\alpha),\ d_\alpha/2\big), & r_0 = 1, \\ \big(K_{\text{V}}\delta_\circ^{-\upsilon_\circ}/(r_0 - 1),\ 1 - q - (r_0 - \upsilon_\circ)/(2r_0)\big), & r_0 > 1, \end{cases}$$

$$N_{\text{V}} \overset{\text{def}}{=} \big\lceil \max\{1, (2J_{\text{V}})^{1/d_{\text{V}}}\} \big\rceil.$$

Then $d_{\text{V}} > 0$. Let $C = C_{p,q}$, $N \geq N_{\text{V}}$, $0 < \epsilon < C^2$ with $\epsilon N \geq C^2$, $h(u) = C(1+u)^{-r_0}$ and $H = (C/\sqrt{\epsilon})^{1/r_0} - 1$. Then $1 + H \leq \sqrt{N}$ and $(g * h)(u) \leq 2h(u)$ on $[0, H]$.

Proof. The quantity $1 - q - (r_0 - \upsilon_\circ)/(2r_0) = \frac{1}{2} - q + \upsilon_\circ/(2r_0)$ decreases in r_0 and vanishes at $r_0 = \upsilon_\circ/(2q-1) = 2/\alpha$. Since $r_0 \leq (1 + 2/\alpha)/2 < 2/\alpha$ for $\alpha < 2$, $d_{\text{V}} > 0$. The window gives $1 + H = (C/\sqrt{\epsilon})^{1/r_0} \leq N^{1/(2r_0)} \leq \sqrt{N}$, because $r_0 \geq 1$.

By lemma SI.7(i) with $a = \alpha > 1$,

$$g(v) = \frac{2}{H_N(q)} L_{N,\alpha,q}(v) \leq \frac{K_{\text{V}}}{H_N(q)} (1+v)^{-\upsilon_\circ}.$$

Split the convolution at $v = \delta_\circ(1+u)$. As in the proof of lemma SI.14, the early part is at most $\frac{3}{2}h(u)$. On the late part $g(v) \leq K_{\text{V}}\delta_\circ^{-\upsilon_\circ}(1+u)^{-\upsilon_\circ}/H_N(q)$, and

$$\frac{1}{h(u)} \int_{\delta_\circ(1+u)}^{u} g(v)h(u-v)\,\mathrm{d}v \leq \frac{K_{\text{V}}\delta_\circ^{-\upsilon_\circ}}{H_N(q)} (1+u)^{r_0-\upsilon_\circ} I_{r_0}(u).$$

By lemma SI.8, $H_N(q) \geq N^{1-q}$. If $r_0 = 1$, then $(1+u)^{1-\upsilon_\circ}\log(1+u) \leq N^{(1-\upsilon_\circ)/2} \cdot \frac{1}{2}\log N$ for $u \leq H$, and $(1-\upsilon_\circ)/2 - (1-q) = -d_\alpha$; with $\sup_{x \geq 1} x^{-d_\alpha/2}\log x = 2/(e\, d_\alpha)$ the late part is at most

$$\frac{K_{\text{V}}\delta_\circ^{-\upsilon_\circ}}{2} N^{-d_\alpha} \log N \leq J_{\text{V}} N^{-d_{\text{V}}}.$$

If $r_0 > 1$, then $I_{r_0} \leq 1/(r_0 - 1)$ and $(1+u)^{r_0-\upsilon_\circ} \leq (1+H)^{r_0-\upsilon_\circ} \leq N^{(r_0-\upsilon_\circ)/(2r_0)}$, so the late part is at most $J_{\text{V}}N^{-d_{\text{V}}}$. In both cases $N \geq N_{\text{V}}$ makes the late part at most $\frac{1}{2}$, and the sum is at most 2. □

Lemma SI.17 (SIGNSVD kernel, $\alpha = 2$)**.** *Let $\alpha = 2$, so $q = 1$ and $g = 2L_{N,2,1}/H_N(1)$.*

(i) For every $v \geq 0$,

$$g(v) \leq \frac{1}{H_N(1)} \Big[\frac{1}{1+v} + \frac{16}{(1+v)^2}\Big].$$

(ii) For $0 < r < 1$, $C > 0$ and $h(u) = C(1+u)^{-r}$,

$$\frac{(g*h)(u)}{h(u)} \le \frac{\log(1+u) + A_r^{\log}}{H_N(1)}, \qquad A_r^{\log} \stackrel{\text{def}}{=} 2^r\Big(r + \frac{1}{1-r}\Big) + \frac{16}{1-r}.$$

(iii) For $C > 0$ and $h(u) = C(1+u)^{-1}$,

$$\frac{(g*h)(u)}{h(u)} \le \frac{2\log(1+u) + 64}{H_N(1)}.$$

Proof. For (i) and $v \ge 1$, lemma SI.6 with $p = 2$, $q = 1$, $a = 2v$ gives, since $\Gamma(1) = 1$ and $M_{2,1} = (2/e)^2 < 1$,

$$2L_{N,2,1}(v) \le 2L_{\infty,2,1}(v) \le \frac{1}{v} + \frac{M_{2,1}}{v^2} \le \frac{1}{1+v} + \frac{2}{(1+v)^2} + \frac{4}{(1+v)^2},$$

using $\frac{1}{v} - \frac{1}{1+v} = \frac{1}{v(1+v)} \le \frac{2}{(1+v)^2}$ and $v^{-2} \le 4(1+v)^{-2}$ for $v \ge 1$. For $0 \le v \le 1$, $2L_{N,2,1}(v) \le 2\zeta(2) \le 4 \le 16(1+v)^{-2}$ by lemma SI.8. Dividing by $H_N(1)$ proves (i).

For (ii), the kernel $(1+v)^{-1}$ gives

$$\frac{\big((1+\cdot)^{-1} * h\big)(u)}{h(u)} - \log(1+u) = \int_0^u \frac{\big((1+u)/(1+u-v)\big)^r - 1}{1+v}\,\mathrm{d}v \le \int_0^1 \frac{(1-w)^{-r} - 1}{w}\,\mathrm{d}w,$$

where $w = v/(1+u)$ and $\mathrm{d}v/(1+v) \le \mathrm{d}w/w$ with a nonnegative integrand. For $0 < w \le \frac{1}{2}$ the mean value theorem bounds the integrand by $r2^{r+1}$, and for $\frac{1}{2} < w < 1$ it is at most $2(1-w)^{-r}$; the two integrals are at most $r2^r$ and $2^r/(1-r)$. The kernel $(1+v)^{-2}$ is decreasing with integral at most 1, so lemma SI.13 bounds its ratio by $1/(1-r)$. Part (i) then gives (ii).

For (iii), partial fractions give

$$(1+u)\int_0^u \frac{\mathrm{d}v}{(1+v)(1+u-v)} = \frac{2(1+u)}{2+u}\log(1+u) \le 2\log(1+u).$$

For the second term split at $v = u/2$: on $[0, u/2]$, $1 + u - v \ge (1+u)/2$, and on $[u/2, u]$, $(1+v)^{-2} \le 4(1+u)^{-2}$. Hence

$$(1+u)\int_0^u \frac{\mathrm{d}v}{(1+v)^2(1+u-v)} \le 2 + \frac{4\log(1+u)}{1+u} \le 2 + \frac{4}{e} \le 4.$$

Part (i) then gives (iii). □

Lemma SI.18 (SIGNSVD kernel, $\alpha = 2, \varsigma > 1$)**.** *Let $\alpha = 2$ and $\varsigma > 1$, so $q = 1$ and $r \stackrel{\text{def}}{=} r_{\text{s-SVD}} > 1$. Put $C \stackrel{\text{def}}{=} C_{p,1}$, $K \stackrel{\text{def}}{=} 2E_{2,1}$, $L_N \stackrel{\text{def}}{=} H_N(1)$ and*

$$\mathsf{A}_0 \stackrel{\text{def}}{=} \frac{CK}{r-1}\Big(\frac{2^r}{e} + 2\Big), \qquad L_0 \stackrel{\text{def}}{=} \max\Big\{1, \frac{4K}{2-\sqrt{2}}\Big\},$$

$$h(u) \stackrel{\text{def}}{=} C(1+u)^{-r}, \qquad \mathsf{w}(u) \stackrel{\text{def}}{=} \sqrt{\mathsf{A}_0}\,\epsilon^{1/4} L_N^{-1/2}(1+u)^{-1/2}.$$

Let $N \ge e^{L_0}$, $B \ge N$ and $\epsilon > 0$. Then $f_{N,1} \le Y_{\text{s-SVD}} \le (h + \mathsf{w} + \sqrt{\epsilon})^2$ on $[0, \infty)$. If moreover $T \ge 4$ and $0 < \epsilon < \min\{1, C^2, 1/(8T)\}$, the first hit satisfies $0 < \hat{u} \le H \stackrel{\text{def}}{=} \max\{(2C/\sqrt{\epsilon})^{1/r}, 4\mathsf{A}_0/(L_N\sqrt{\epsilon})\} - 1$, and

$$0 \le \mathcal{E} \le \epsilon^{1/8}\sqrt{C + \sqrt{\mathsf{A}_0} + 1} + \frac{C}{r-1}\big(1 + \epsilon^{-1/8}\big)^{1-r} + 2\sqrt{\mathsf{A}_0}(2C)^{1/(2r)} L_N^{-1/2}\epsilon^{(r-1)/(4r)} + (2C)^{1/r}\epsilon^{(r-1)/(2r)} + \frac{8\mathsf{A}_0}{L_N}.$$

In particular $\mathcal{E} \to 0$ along every sequence with $N \to \infty$ and $\epsilon \downarrow 0$.

Proof. By lemma SI.7(i), $g(v) \le K/(L_N(1+v))$. The convolution with h is bounded by splitting at $u/2$. On $[0,u/2]$, $h(u-v) \le C2^r(1+u)^{-r}$, so this part is at most

$$\frac{CK2^r}{L_N}(1+u)^{-r}\log(1+u) \le \frac{CK2^r}{e(r-1)L_N(1+u)},$$

using $\sup_{x\ge 1} x^{-(r-1)}\log x = 1/(e(r-1))$. On $[u/2,u]$, $g(v) \le 2K/(L_N(1+u))$ and $\int h \le C/(r-1)$, so this part is at most $2CK/((r-1)L_N(1+u))$. Hence

$$\sqrt{\epsilon}\,(g*h)(u) \le \frac{\sqrt{\epsilon}\,\mathsf{A}_0}{L_N(1+u)} = \mathsf{w}(u)^2.$$

The convolution with w is bounded in the same way. On $[0,u/2]$, $\mathsf{w}(u-v) \le \sqrt{2}\,\mathsf{w}(u)$ and $\int g = 1$. On $[u/2,u]$, $g(v) \le 2K/(L_N(1+u))$ and $\int_0^{u/2}(1+z)^{-1/2}\mathrm{d}z \le 2\sqrt{1+u}$. Hence

$$(g*\mathsf{w})(u) \le \Big(\sqrt{2} + \frac{4K}{L_N}\Big)\mathsf{w}(u) \le 2\,\mathsf{w}(u),$$

because $L_N \ge \log(N+1) \ge L_0$. Put $U \stackrel{\text{def}}{=} (h+\mathsf{w}+\sqrt{\epsilon})^2 \ge \epsilon$. Using $\int g = 1$,

$$f_{N,1} + \sqrt{\epsilon}\,g*\sqrt{U} \le h^2 + \mathsf{w}^2 + 2\sqrt{\epsilon}\,\mathsf{w} + \epsilon \le U,$$

and lemma SI.3 with $m = \epsilon$ gives $Y \le U$ on every compact interval.

At H, $h(H) \le \sqrt{\epsilon}/2$ and $\mathsf{w}(H)^2 = \sqrt{\epsilon}\mathsf{A}_0/(L_N(1+H)) \le \epsilon/4$, so $Y(H) \le 4\epsilon \le T\epsilon$. Since $Y(0) \ge \frac{1}{2} > T\epsilon$, the first hit is attained in $(0,H]$ as in lemma SI.9(ii).

For $\mathcal{E}$, the kernel has integral 1 and $h+\mathsf{w}+\sqrt{\epsilon}$ is decreasing, so

$$0 \le Y - f_{N,1} \le \sqrt{\epsilon}\,\big(C + \sqrt{\mathsf{A}_0}\epsilon^{1/4}L_N^{-1/2} + \sqrt{\epsilon}\big) \le \sqrt{\epsilon}\,\big(C + \sqrt{\mathsf{A}_0} + 1\big).$$

Let $R_0 \stackrel{\text{def}}{=} \epsilon^{-1/8}$. On $[0, \min(\hat{u}, R_0)]$ bound $\sqrt{Y} - \sqrt{f}$ by $\sqrt{Y-f}$, and on $[R_0, \hat{u}]$ by $h+\mathsf{w}+\sqrt{\epsilon}$. This gives

$$\mathcal{E} \le R_0\,\epsilon^{1/4}\sqrt{C + \sqrt{\mathsf{A}_0} + 1} + \frac{C}{r-1}(1+R_0)^{1-r} + \int_0^H \mathsf{w}\,\mathrm{d}u + \sqrt{\epsilon}\,H.$$

Integrating $(1+u)^{-1/2}$ and using $\sqrt{\max(x,y)} \le \sqrt{x} + \sqrt{y}$,

$$\int_0^H \mathsf{w}\,\mathrm{d}u \le 2\sqrt{\mathsf{A}_0}(2C)^{1/(2r)}L_N^{-1/2}\epsilon^{(r-1)/(4r)} + \frac{4\mathsf{A}_0}{L_N}, \qquad \sqrt{\epsilon}\,H \le (2C)^{1/r}\epsilon^{(r-1)/(2r)} + \frac{4\mathsf{A}_0}{L_N}.$$

This proves the bound. Each term tends to zero, since $r > 1$ and $L_N \ge \log(N+1) \to \infty$. □

SI.9 Proofs of the phase theorems

All proofs below follow one scheme. For each model j an envelope $h_j(u) = C_j(1+u/D_j)^{-r_{0,j}}$ and a horizon H_j are fixed, with $D_j \ge 1$ and $r_{0,j} \le r_j$, so that $f_{N,q_j} \le C_j^2(1+u)^{-2r_j} \le h_j^2$. The window places H_j below the cutoff $(N/4)^{q_j}$, where the lower forcing envelope holds, before any kernel bound is used. A kernel lemma of appendix SI.8 gives hypothesis (a) or (b) of lemma SI.9, which places the first hit in $(0,H_j]$ and bounds the noise error $\mathcal{E}_j$. Since $f_{N,q_j}(0) = \frac{1}{2}H_N(p) \ge \frac{1}{2}$ and $\epsilon < 1/(8T)$, the hypothesis $f(0) > T\epsilon$ always holds. Lemma SI.10 then evaluates the clock integral, and eq. (SI.30) converts it into time.

Proof of theorem SI.8. Here $r_{\text{s-SVD}} = (p-1)/\alpha \in (0,1)$ and $r_{\text{s-SGD}} = r_{\text{s-SVD}}/2 < \frac{1}{2}$. Take $r_{0,j} = r_j$ for both models. For SIGNSGD put $D_{\text{s-SGD}} \stackrel{\text{def}}{=} 1$ and $\tau_{\text{s-SGD}} \stackrel{\text{def}}{=} 4^{-\alpha}$. For SIGNSVD put:

- if $0 < \alpha < 2$: $D_{\text{s-SVD}} \stackrel{\text{def}}{=} 1$, $\tau_{\text{s-SVD}} \stackrel{\text{def}}{=} \min\{4^{-q_{\text{s-SVD}}}, \mathsf{c}_1\}$ and $N_{\text{V}} \stackrel{\text{def}}{=} N_1$, with c_1, N_1 of lemma SI.15 at $r = r_{\text{s-SVD}}$;
- if $\alpha = 2$: $D_{\text{s-SVD}} \stackrel{\text{def}}{=} 1$, $\tau_{\text{s-SVD}} \stackrel{\text{def}}{=} \frac{1}{4}$ and $N_{\text{V}} \stackrel{\text{def}}{=} \lceil \exp(A^{\log}_{r_{\text{s-SVD}}}) \rceil$, with $A^{\log}_r$ of lemma SI.17;

- if $\alpha > 2$: $\upsilon_{\mathrm{V}} \stackrel{\text{def}}{=} 2 - 2/\alpha > 1$, $K_{\mathrm{V}} \stackrel{\text{def}}{=} 2E_{\alpha,\alpha/2}$, $D_{\text{s-SVD}} \stackrel{\text{def}}{=} D(K_{\mathrm{V}}, r_{\text{s-SVD}}, \upsilon_{\mathrm{V}})$ of eq. (SI.55), $\tau_{\text{s-SVD}} \stackrel{\text{def}}{=} 4^{-\alpha/2}$ and $N_{\mathrm{V}} \stackrel{\text{def}}{=} 4$.

Put $N_* \stackrel{\text{def}}{=} \max\{4, N_{\mathrm{V}}\}$ and

$$E_* \stackrel{\text{def}}{=} \max\Big\{1,\ C_{\text{s-SVD}}^2\Big(\frac{D_{\text{s-SVD}}}{\tau_{\text{s-SVD}}}\Big)^{2r_{\text{s-SVD}}},\ C_{\text{s-SGD}}^2\Big(\frac{D_{\text{s-SGD}}}{\tau_{\text{s-SGD}}}\Big)^{2r_{\text{s-SGD}}}\Big\},$$

and let $H_j \stackrel{\text{def}}{=} D_j[(C_j/\sqrt{\epsilon})^{1/r_j} - 1]$, which is positive since $\epsilon < C_j^2$.

The horizons lie below the cutoff. Since $(p-1)/(2r_j) = q_j$ and $C_j^2(D_j/\tau_j)^{2r_j} \le E_* \le \epsilon N^{p-1}$,

$$H_j \le D_j C_j^{1/r_j} \epsilon^{-1/(2r_j)} \le \tau_j N^{q_j} \le (N/4)^{q_j}.$$

The convolution bound (a) of lemma SI.9 holds on $[0, H_j]$. For SIGNSGD the kernel is decreasing with integral 1, so lemma SI.13 gives $g * h \le h/(1 - r_{\text{s-SGD}}) \le 2h$ on all of $[0, \infty)$. For SIGNSVD with $0 < \alpha < 2$, $H_{\text{s-SVD}} \le \mathsf{c}_1 N^{q_{\text{s-SVD}}}$ and lemma SI.15 gives $g * h \le \frac{3}{4}h$. For $\alpha = 2$, lemma SI.17(ii) gives, for $u \le H_{\text{s-SVD}} \le N/4$,

$$\frac{(g*h)(u)}{h(u)} \le \frac{\log(1+u) + A^{\log}_{r_{\text{s-SVD}}}}{H_N(1)} \le 2,$$

since $H_N(1) \ge \log(N+1) \ge \log(1+u)$ and $H_N(1) \ge \log N \ge A^{\log}_{r_{\text{s-SVD}}}$. For $\alpha > 2$, lemma SI.7(i) and $H_N(\alpha/2) \ge 1$ give $g(v) \le K_{\mathrm{V}}(1+v)^{-\upsilon_{\mathrm{V}}}$ for all v, and lemma SI.14 applies on every interval.

By lemma SI.9, the first hit is attained with $0 < \hat{u}_j \le H_j$, and by eq. (SI.51)

$$\mathcal{E}_j \le D_j C_j^{1/r_j}\Big[\frac{\sqrt{2}}{1 - r_j/2} + 1\Big]\epsilon^{-(1-r_j)/(2r_j)}.$$

Lemma SI.10(i) applies with $H = H_j \le (N/4)^{q_j}$. Its lower bound is a fixed multiple of $\epsilon^{-(1-r_j)/(2r_j)}$. For its upper bound, $1 + H_j \le D_j(C_j/\sqrt{\epsilon})^{1/r_j}$ gives

$$C_j I_{r_j}(H_j) \le \frac{C_j(1+H_j)^{1-r_j}}{1-r_j} \le \frac{D_j^{1-r_j} C_j^{1/r_j}}{1-r_j}\,\epsilon^{-(1-r_j)/(2r_j)}.$$

Hence $\int_0^{\hat{u}_j} \sqrt{Y_j}$ is between two fixed multiples of $\epsilon^{-(1-r_j)/(2r_j)}$. Multiplying by $\mathcal{S}_j T_0^j/(2\sqrt{\epsilon})$, by eq. (SI.30), gives the order $\mathcal{S}_j T_0^j \epsilon^{-1/(2r_j)}$, and $1/(2r_{\text{s-SVD}}) = \alpha/[2(p-1)]$, $1/(2r_{\text{s-SGD}}) = \alpha/(p-1)$.

All thresholds are fixed, so a sequence with $N \to \infty$, $\epsilon \downarrow 0$ and $\epsilon N^{p-1} \to \infty$ meets them for large indices, whatever $B \ge N$. For $\epsilon = N^{-\omega}$, $\epsilon N^{p-1} = N^{p-1-\omega} \to \infty$. □

Proof of theorem SI.9. Here $r_{\text{s-SVD}} = (p-1)/\alpha \ge 1$ and $r_{\text{s-SGD}} = r_{\text{s-SVD}}/2 \ge \frac{1}{2}$. For SIGNSGD put $\upsilon_{\mathrm{G}} \stackrel{\text{def}}{=} 2 - 1/\alpha > 1$, $K_{\mathrm{G}} \stackrel{\text{def}}{=} 4(1 - \frac{2}{\pi})E_{\alpha,\alpha} + \frac{4}{\pi}E_{2\alpha,\alpha}$,

$$r_{0,\text{s-SGD}} \stackrel{\text{def}}{=} \begin{cases} r_{\text{s-SGD}}, & r_{\text{s-SGD}} \le 1, \\ \min\{r_{\text{s-SGD}}, (1+\upsilon_{\mathrm{G}})/2\}, & r_{\text{s-SGD}} > 1, \end{cases} \qquad D_{\text{s-SGD}} \stackrel{\text{def}}{=} D(K_{\mathrm{G}}, r_{0,\text{s-SGD}}, \upsilon_{\mathrm{G}}).$$

For SIGNSVD put:

- if $\alpha > 2$: $\upsilon_{\mathrm{V}} \stackrel{\text{def}}{=} 2 - 2/\alpha > 1$, $K_{\mathrm{V}} \stackrel{\text{def}}{=} 2E_{\alpha,\alpha/2}$, $r_{0,\text{s-SVD}} \stackrel{\text{def}}{=} 1$ if $r_{\text{s-SVD}} = 1$ and $\min\{r_{\text{s-SVD}}, (1+\upsilon_{\mathrm{V}})/2\}$ otherwise, $D_{\text{s-SVD}} \stackrel{\text{def}}{=} D(K_{\mathrm{V}}, r_{0,\text{s-SVD}}, \upsilon_{\mathrm{V}})$ and $N_{\mathrm{V}} \stackrel{\text{def}}{=} \lceil 4^{\alpha/(\alpha-2)} \rceil$;
- if $1 < \alpha < 2$: $r_{0,\text{s-SVD}}$ and N_{V} of lemma SI.16, and $D_{\text{s-SVD}} \stackrel{\text{def}}{=} 1$;
- if $\alpha = 2$ and $\varsigma = 1$: $r_{0,\text{s-SVD}} \stackrel{\text{def}}{=} 1$, $D_{\text{s-SVD}} \stackrel{\text{def}}{=} 1$ and $N_{\mathrm{V}} \stackrel{\text{def}}{=} \lceil e^{64} \rceil$;

- if $\alpha = 2$ and $\varsigma > 1$: $D_{\text{s-SVD}} \stackrel{\text{def}}{=} 1$ and $N_{\text{V}} \stackrel{\text{def}}{=} \lceil e^{L_0} \rceil$ with L_0 of lemma SI.18.

Put $N_* \stackrel{\text{def}}{=} \lceil \max\{4, 4^{\alpha/(\alpha-1)}, N_{\text{V}}\} \rceil$ and

$$E_* \stackrel{\text{def}}{=} \max\left\{1,\ D_{\text{s-SVD}}^{2r_{0,\text{s-SVD}}} C_{\text{s-SVD}}^2,\ D_{\text{s-SGD}}^{2r_{0,\text{s-SGD}}} C_{\text{s-SGD}}^2\right\},$$

where the middle entry is $C_{\text{s-SVD}}^2$ when $D_{\text{s-SVD}} = 1$. In every case $r_{0,j} \le r_j$ and $r_{0,j} < 2$; moreover $\frac{1}{2} \le r_{0,\text{s-SGD}} \le \upsilon_{\text{G}}$, $r_{0,\text{s-SVD}} \ge 1$, $r_{0,\text{s-SVD}} \le \upsilon_{\text{V}}$ when $\alpha > 2$, and $r_{0,j} > 1$ exactly when $r_j > 1$. Except for SIGNSVD when $\alpha = 2$ and $\varsigma > 1$, let $h_j \stackrel{\text{def}}{=} C_j(1 + u/D_j)^{-r_{0,j}}$ and $H_j \stackrel{\text{def}}{=} D_j[(C_j/\sqrt{\epsilon})^{1/r_{0,j}} - 1]$.

The horizons lie below the cutoffs. Since $D_j^{2r_{0,j}} C_j^2 \le E_* \le \epsilon N$,

$$H_j \le D_j C_j^{1/r_{0,j}} \epsilon^{-1/(2r_{0,j})} \le N^{1/(2r_{0,j})}.$$

Hence $H_{\text{s-SGD}} \le N$, and $H_{\text{s-SVD}} \le \sqrt{N}$ because $r_{0,\text{s-SGD}} \ge 1$. The choice of N_* gives $N \le (N/4)^\alpha$ and $\sqrt{N} \le (N/4)^{\alpha/2}$, and for $\alpha > 2$ also $N \le (N/4)^{\alpha/2}$.

The convolution bound (a) holds on $[0, H_j]$. For SIGNSGD, lemma SI.7(i) with $a = \alpha > 1$ and $a = 2\alpha$ gives

$$g_{\text{s-SGD}}(u) \le \frac{2(1 - \frac{2}{\pi})E_{\alpha,\alpha}}{N}(1+u)^{1-\upsilon_{\text{G}}} + \frac{4}{\pi}E_{2\alpha,\alpha}(1+u)^{-\upsilon_{\text{G}}} \le K_{\text{G}}(1+u)^{-\upsilon_{\text{G}}} \qquad (0 \le u \le N),$$

using $H_N(\alpha) \ge 1$ and $(1 + u)/N \le 2$; lemma SI.14 with $H_* = N$ applies. For SIGNSVD with $\alpha > 2$ the envelope $g \le K_{\text{V}}(1 + u)^{-\upsilon_{\text{V}}}$ holds for all u and lemma SI.14 applies. For $1 < \alpha < 2$, lemma SI.16 applies, since $\epsilon N \ge E_* \ge C_{\text{s-SVD}}^2$. For $\alpha = 2$ and $\varsigma = 1$, lemma SI.17(iii) gives, for $u \le H_{\text{s-SVD}} \le \sqrt{N} - 1$,

$$\frac{(g * h)(u)}{h(u)} \le \frac{2\log(1 + u) + 64}{H_N(1)} \le 2,$$

because $2\log(1 + u) \le \log N \le H_N(1)$ and $H_N(1) \ge \log(N + 1) \ge 64$.

By lemma SI.9, each hit is attained with $\hat{u}_j \le H_j$, and eq. (SI.51) gives $\mathcal{E}_j \le D_j C_j^{1/r_{0,j}}[\sqrt{2}/(1 - r_{0,j}/2) + 1]\epsilon^{(r_{0,j}-1)/(2r_{0,j})}$. For SIGNSVD when $\alpha = 2$ and $\varsigma > 1$, lemma SI.18 gives the hit and a bound on $\mathcal{E}$ that is fixed for $\epsilon < 1$ and tends to zero along every sequence with $N \to \infty$ and $\epsilon \downarrow 0$. Now apply lemma SI.10 with $H = H_j$, in its three cases.

If $r_j < 1$, then j = s-SGD, $r_{0,j} = r_j$ and $H_j \le N \le (N/4)^\alpha$; the argument of the proof of theorem SI.8 gives the order $\epsilon^{-(1-r_j)/(2r_j)}$ for the clock integral.

If $r_j = 1$, then $r_{0,j} = 1$, $H_j \le \sqrt{N} \le (N/4)^{q_j}$, and $(1 + H_j)\sqrt{\epsilon} \le D_j C_j$. Since $q_j \ge \alpha/2 > \frac{1}{2}$, $H_j N^{-q_j} \le N^{1/2-q_j} \le 1$, and $\mathcal{E}_j \le (2\sqrt{2} + 1)D_j C_j$. Part (ii) of lemma SI.10 gives

$$\Big| \int_0^{\hat{u}_j} \sqrt{Y_j}\, \mathrm{d}u - \frac{a_{q_j}}{2} \log \frac{1}{\epsilon} \Big| \le K'$$

with K' fixed, together with the lower bound $\frac{\sqrt{A}}{4} \log \frac{1}{\epsilon}$. Since $\log(1/\epsilon) \ge \log 4$, the clock integral is between two fixed multiples of $\log(1/\epsilon)$, and it is $\sim \frac{a_{q_j}}{2} \log(1/\epsilon)$ as $\epsilon \downarrow 0$.

If $r_j > 1$, part (iii) of lemma SI.10 gives the lower bound $1/(2\sqrt{2})$ and the upper bound $J(p, q_j) + \mathcal{E}_j$, which is fixed because $\epsilon < 1$. Along a sequence with $N \to \infty$ and $\epsilon \downarrow 0$, the three error terms of part (iii) tend to zero, so the clock integral tends to $J(p, q_j)$. For SIGNSVD when $\alpha = 2$ and $\varsigma > 1$ this uses only lemma SI.18, $N \ge 4$ and $\epsilon < \epsilon_0$, which hold for large indices of every sequence with $N \to \infty$ and $\epsilon \downarrow 0$.

Multiplying by $\mathcal{S}_j T_0^j/(2\sqrt{\epsilon})$ gives the orders $\mathcal{S}_j T_0^j \Pi_{r_j}(\epsilon)$ and, with $a_q = (8q)^{-1/2}$,

$$\frac{\mathcal{S}_j T_0^j}{2\sqrt{\epsilon}} \cdot \frac{a_{q_j}}{2} \log \frac{1}{\epsilon} = \frac{\mathcal{S}_j T_0^j}{4\sqrt{8q_j}}\, \epsilon^{-1/2} \log \frac{1}{\epsilon}.$$

The thresholds are fixed, which proves the last assertion. □

Proof of theorem SI.10. Take $h_j \stackrel{\text{def}}{=} C_j(1+u)^{-r_j}$ and $H_j \stackrel{\text{def}}{=} H_j(\epsilon)$. Each kernel is decreasing with maximum $k_{N,j}$, so for $u \le H_j$

$$(g_j * h_j)(u) \le k_{N,j} \int_0^u h_j(v)\,\mathrm{d}v \le k_{N,j} C_j I_{r_j}(H_j) \le \sqrt{\epsilon}$$

by eq. (SI.33). This is hypothesis (b) of lemma SI.9, which gives the hit $0 < \hat{u}_j \le H_j$ and

$$\mathcal{E}_j \le \sqrt{2\epsilon}\, H_j \le \sqrt{2}\, C_j^{1/r_j} \epsilon^{(r_j-1)/(2r_j)}.$$

If $r_j < 1$, eq. (SI.33) gives $H_j \le (N/4)^{q_j}$ and lemma SI.10(i) applies; its upper bound is at most $[C_j^{1/r_j}/(1-r_j) + \sqrt{2}C_j^{1/r_j}]\epsilon^{-(1-r_j)/(2r_j)}$. If $r_j = 1$, then $H_j N^{-q_j} \le 4^{-q_j}$, $(1+H_j)\sqrt{\epsilon} = C_j$ and $\mathcal{E}_j \le \sqrt{2}C_j$, and lemma SI.10(ii) applies. If $r_j > 1$, $\mathcal{E}_j \le \sqrt{2}C_j^{1/r_j}$ for $\epsilon < 1$ and $\mathcal{E}_j \to 0$ as $\epsilon \downarrow 0$, and lemma SI.10(iii) applies. The conclusions follow as in the proof of theorem SI.9. □

Proof of proposition SI.5. The equivalents for the heights follow from eq. (SI.29) and lemma SI.8. For $\alpha < \frac{1}{2}$ both terms of $k_{N,\text{s-SGD}}$ are of order $N^{-\alpha}$ and their constants add. At $\alpha = \frac{1}{2}$ the second term is $\frac{2}{\pi}N^{-1/2}\log N$ and dominates. For $\frac{1}{2} < \alpha < 1$, $H_N(2\alpha) \to \zeta(2\alpha)$ and the second term dominates because $\alpha - 1 > -\alpha$. At $\alpha = 1$ the second term $\frac{4}{\pi}\zeta(2)/\log N$ dominates the first, which is of order $(\log N)/N$.

For $0 < \epsilon < C^2$ and $H = (C/\sqrt{\epsilon})^{1/r} - 1$, direct integration gives, for ϵ below a fixed threshold,

$$\frac{k\,C\,I_r(H)}{\sqrt{\epsilon}} \le K_{C,r}\,k \cdot \begin{cases} \epsilon^{-1/(2r)}, & 0 < r < 1, \\ \epsilon^{-1/2}\log(1/\epsilon), & r = 1, \\ \epsilon^{-1/2}, & r > 1, \end{cases} \tag{SI.56}$$

with $K_{C,r}$ depending only on C and r. Here $r_{\text{s-SVD}} \ge 1$ and $r_{\text{s-SGD}} \ge \frac{1}{2}$, and each right side is at most $K_{C,r}\,k\,\epsilon^{-1}\log(1/\epsilon)$.

Let $0 < \alpha < 1$, $\varpi_\alpha \stackrel{\text{def}}{=} \min(\alpha, 1-\alpha)$ and $\epsilon_N = N^{-\omega}$ with $0 < \omega < \varpi_\alpha$. The height equivalents give $k_{N,\text{s-SVD}} = O(N^{-\alpha/2})$ and $k_{N,\text{s-SGD}} = O(N^{-\varpi_\alpha}\log N)$. Since $r_{\text{s-SVD}} \ge 1$, the SIGNSVD side of eq. (SI.56) is $O(N^{-(\alpha-\omega)/2}\log N)$, and the SIGNSGD side is $O(N^{\omega-\varpi_\alpha}(\log N)^2)$; both tend to zero. Moreover $H_j(\epsilon_N) \le C_j^{1/r_j} N^{\omega/(2r_j)}$ and $\omega/(2r_j) < q_j$ because $2r_jq_j = p - 1 \ge \alpha > \omega$, so $H_j(\epsilon_N) \le (N/4)^{q_j}$ for large N.

Let $\alpha = 1$ and $\epsilon_N = (\log N)^{-\omega}$ with $0 < \omega < 1$. The SIGNSVD side of eq. (SI.56) is at most a fixed power of $\log N$ times $N^{-1/2}\log\log N$, and the SIGNSGD side is $O((\log N)^{\omega-1}\log\log N)$; both tend to zero. Each $H_j(\epsilon_N)$ is at most a fixed power of $\log N$, hence below $(N/4)^{q_j}$ for large N.

The comparisons of powers and logarithms used here follow from $e^{cx} \ge (cx)^n/n!$ for $c > 0$ and an integer n larger than the power in question. Finally $\epsilon_N < \epsilon_0$ and $N \ge 4$ for large N, and no condition involves B. □

Proof of theorem SI.11. At the matched step, $k = \sqrt{\epsilon}\,g$ with

$$g(s) = \frac{1}{\mathcal{S}}\sum_i \mu_i \mathfrak{v}_i e^{-2\mathfrak{d}_i s} = \sum_i w_i\, c_i\, e^{-c_i s}, \qquad c_i \stackrel{\text{def}}{=} 2\mathfrak{d}_i, \quad w_i \stackrel{\text{def}}{=} \frac{\mu_i \mathfrak{v}_i}{2\mathfrak{d}_i \mathcal{S}},$$

and $\sum_i w_i = 1$. The forcing has the rates $2\mathfrak{d}_i$, and $b_{\min} \stackrel{\text{def}}{=} 2\min_i \mathfrak{d}_i > 0$ is fixed. For $\epsilon < F_0/T$, lemma SI.11 applies to $Q = F + \sqrt{\epsilon}\,g * \sqrt{Q}$ in the intrinsic time s. With $\mathsf{J} = \int_0^\infty \sqrt{F}$ and $\mathsf{d}_\epsilon = (2/b_{\min})\sqrt{\epsilon}\log(F_0/\epsilon)$ it gives

$$\mathsf{J} - \frac{2\sqrt{T\epsilon}}{b_{\min}} \le \int_0^{H_Q(T\epsilon)} \sqrt{Q}\,\mathrm{d}s \le \mathsf{J} + 2\sqrt{\mathsf{d}_\epsilon \mathsf{J}} + \mathsf{d}_\epsilon.$$

Since $\mathsf{d}_\epsilon \to 0$, the clock integral tends to $\mathsf{J} \in (0, \infty)$. By theorem SI.3(i), $t_{T\epsilon} = \eta^{-1}\int_0^{H_Q(T\epsilon)} \sqrt{Q}$ and $\eta^{-1} = \mathcal{S}/(2\sqrt{\epsilon})$. □

Proof of corollary SI.2. The substitution $s = T_0^j u$ gives $\int_0^\infty \sqrt{F_j(s)}\,\mathrm{d}s = T_0^j J_N(p, q_j)$, and theorem SI.11 gives the first equivalent. Dividing the two equivalents and using eq. (SI.27) gives the limit of the ratio. Since $i^{-\alpha} \le i^{-\alpha/2}$, each term of $f_{N,\alpha}$ is at least the corresponding term of $f_{N,\alpha/2}$, so $J_N(p,\alpha) \ge J_N(p,\alpha/2)$. If $\alpha \ge 4$, then $H_N(\alpha) \ge 1$ and $H_N(\alpha/2) \le \zeta(2) < 2$ by lemma SI.8, so $D_N > \pi/2$. □

Proof of corollary SI.3. In each window both theorems apply to both models at once. Dividing the two orders and using eq. (SI.27) gives a factor D_N times $\Pi_{r_{\text{s-SGD}}}(\epsilon)/\Pi_{r_{\text{s-SVD}}}(\epsilon)$. For $\varsigma < 1$ this is $\epsilon^{-\alpha/(p-1)+\alpha/(2(p-1))}$. For $\varsigma = 1$, $p-1 = \alpha$ and it is $\epsilon^{-1}/(\epsilon^{-1/2}\log(1/\epsilon))$. For $1 < \varsigma < \alpha+1$ it is $\epsilon^{-\alpha/(p-1)+1/2} = \epsilon^{-(\alpha+1-\varsigma)/(2(p-1))}$. For $\varsigma = \alpha+1$ it is $\log(1/\epsilon)$, and for $\varsigma > \alpha+1$ it is 1. At $\varsigma = \alpha+1$ the SIGNSVD model is saturated, since $r_{\text{s-SVD}} = 2$, and dividing the equivalents eq. (SI.37) gives

$$\frac{\mathcal{S}_{\text{s-SGD}} T_0^{\text{s-SGD}}\, \epsilon^{-1/2}\log(1/\epsilon)/(4\sqrt{8\alpha})}{\mathcal{S}_{\text{s-SVD}} T_0^{\text{s-SVD}} J(p,\alpha/2)/(2\sqrt{\epsilon})} = \frac{D_N \log(1/\epsilon)}{2\sqrt{8\alpha}\, J(p,\alpha/2)}.$$

For $\varsigma > \alpha+1$ both are saturated and the quotient is $D_N J(p,\alpha)/J(p,\alpha/2)$. □

Proof of proposition SI.6. For (i), corollary SI.3 gives $\mathsf{R} \asymp D_N \epsilon^{-1/(2r_{\text{s-SVD}})}$, and $\epsilon = \Xi_N N^{-(p-1)}$ with $(p-1)/(2r_{\text{s-SVD}}) = \alpha/2$ gives

$$\mathsf{R} \asymp D_N\, N^{\alpha/2}\, \Xi_N^{-1/(2r_{\text{s-SVD}})}.$$

The equivalents eq. (SI.31) give the five rows; for $\alpha > 2$, D_N has a positive limit. For $0 < \alpha < 1$ the first row tends to 0 when $\Xi_N \to \infty$, and for $\alpha > 2$ the last row tends to infinity because $\epsilon \downarrow 0$. For $\alpha = 1$, $p - 1 = r_{\text{s-SVD}}$, so $\Xi_N = (\log N)^b \to \infty$ and $\mathsf{R} \asymp (\log N)^{1-b/(2r_{\text{s-SVD}})}$. For $1 < \alpha < 2$, $\Xi_N = N^{p-1-\omega} \to \infty$ and

$$\mathsf{R} \asymp N^{\alpha-1} N^{-(p-1-\omega)/(2r_{\text{s-SVD}})} = N^{(\omega-\omega_c)/(2r_{\text{s-SVD}})},$$

with $\omega_c/(p-1) = (2-\alpha)/\alpha \in (0,1)$. For $\alpha = 2$, $\Xi_N = N^{p-1}(\log N)^{-\omega} \to \infty$ and $\mathsf{R} \asymp (\log N)^{\omega/(2r_{\text{s-SVD}})-1}$. Positive powers of N dominate powers of $\log N$, which gives each limit and membership in the window.

For (ii), $D_N \to \pi\zeta(\alpha)/\zeta(\alpha/2) > 0$. The factor in eq. (SI.38) tends to infinity for $\varsigma \le \alpha+1$, and for $\varsigma > \alpha+1$ the equivalent of corollary SI.3 gives the limit. If $\alpha \ge 4$, $\zeta(\alpha) > 1$ and $\zeta(\alpha/2) \le \zeta(2) < 2$, so D_N has limit above $\pi/2$, and $J(p,\alpha) \ge J(p,\alpha/2)$ as in the proof of corollary SI.2.

For (iii), $D_N \to 0$ by eq. (SI.31), which gives the saturated limit. At $\varsigma = \alpha+1$, $\epsilon N \ge E_* \ge 1$ gives $\log(1/\epsilon) \le \log N$. For $1 < \alpha < 2$, $D_N \log N \asymp N^{\alpha/2-1}\log N \to 0$. For $\alpha = 2$, $2\sqrt{8\alpha} = 8$ and $D_N \sim \pi\zeta(2)/\log N$, which gives the limit, with $0 \le c_\infty \le 1$.

For (iv), eqs. (SI.27) and (SI.29) give the identity $D_N = \frac{\pi}{2} k_{N,\text{s-SVD}}$. At $\varsigma = 1$, $r_{\text{s-SVD}} = 1$ and the height condition reads $k_{N,\text{s-SVD}} C_{\text{s-SVD}} \log(C_{\text{s-SVD}}/\sqrt{\epsilon}) \le \sqrt{\epsilon}$, so $k_{N,\text{s-SVD}} = O(\sqrt{\epsilon}/\log(1/\epsilon))$ and $\mathsf{R} = O(\log(1/\epsilon)^{-2})$ by eq. (SI.38). For $\varsigma > 1$, $r_{\text{s-SVD}} > 1$ and $I_{r_{\text{s-SVD}}}(H) \ge I_{r_{\text{s-SVD}}}(1) > 0$ once $H \ge 1$, so $k_{N,\text{s-SVD}} = O(\sqrt{\epsilon})$. In the branch $1 < \varsigma < \alpha+1$ the remaining exponent is $\frac{1}{2} - \frac{\alpha+1-\varsigma}{2(p-1)} = \frac{\varsigma-1}{p-1} > 0$, and the last two branches give $O(\sqrt{\epsilon}\log(1/\epsilon))$ and $O(\sqrt{\epsilon})$.

For (v), the rows $1 \le \varsigma < \alpha+1$ of eq. (SI.38) have the common form

$$\mathsf{R} \asymp D_N\, \epsilon^{-e_\varsigma}\, \ell_\epsilon, \qquad \ell_\epsilon \stackrel{\text{def}}{=} \begin{cases} 1/\log(1/\epsilon), & \varsigma = 1, \\ 1, & 1 < \varsigma < \alpha+1, \end{cases} \tag{SI.57}$$

since $e_1 = \frac{1}{2}$. By eq. (SI.31), $D_N \asymp N^{\alpha/2-1}$ for $1 < \alpha < 2$ and $D_N \asymp 1/\log N$ for $\alpha = 2$. The position of ς relative to the curve ς_c is the sign of one exponent. Expanding,

$$(\alpha-2)(p-1) + (\alpha+1-\varsigma) = (\alpha-2)(\alpha-1) + (\alpha-2)\varsigma + \alpha + 1 - \varsigma = \alpha^2 - 2\alpha + 3 - (3-\alpha)\varsigma,$$

so that

$$\frac{\alpha}{2} - 1 + e_\varsigma = \frac{(3-\alpha)\big(\varsigma_c(\alpha) - \varsigma\big)}{2(p-1)}. \tag{SI.58}$$

The bounds on ς_c follow from

$$\varsigma_c(\alpha) - 1 = \frac{\alpha(\alpha-1)}{3-\alpha}, \qquad \alpha + 1 - \varsigma_c(\alpha) = \frac{2\alpha(2-\alpha)}{3-\alpha}.$$

For (a), $\varsigma \geq \varsigma_c(\alpha) > 1$, so $\ell_\epsilon = 1$, and eq. (SI.57) gives

$$\mathsf{R} \asymp N^{\alpha/2-1}\epsilon^{-e_\varsigma} = (\epsilon N)^{-e_\varsigma} N^{\alpha/2-1+e_\varsigma}.$$

By eq. (SI.58) the exponent of N is at most 0, and $(\epsilon N)^{-e_\varsigma} \to 0$ because $e_\varsigma > 0$ and $\epsilon N \to \infty$.

For (b) with $\epsilon_N = N^{-\omega}$, eq. (SI.57) gives

$$\mathsf{R} \asymp N^{\alpha/2-1+\omega e_\varsigma}\, \ell_{\epsilon_N}, \qquad \ell_{\epsilon_N} \in \Big\{1, \frac{1}{\omega \log N}\Big\}.$$

The exponent of N is negative exactly when $\omega < (1-\alpha/2)/e_\varsigma = \omega_c$. By eq. (SI.58), $\omega_c < 1$ exactly when $\varsigma < \varsigma_c(\alpha)$. A positive or negative power of N dominates the factor ℓ_{ϵ_N}, which gives both limits. For $\epsilon_N = 1/\log N$,

$$\mathsf{R} \asymp N^{\alpha/2-1}(\log N)^{e_\varsigma}\ell_{\epsilon_N} \longrightarrow 0.$$

For (c), $D_N \asymp 1/\log N$, and eq. (SI.57) gives

$$\mathsf{R} \asymp \frac{N^{\omega e_\varsigma}\ell_{\epsilon_N}}{\log N} \longrightarrow \infty \quad (\epsilon_N = N^{-\omega}), \qquad \mathsf{R} \asymp (\log N)^{e_\varsigma - 1}\ell_{\epsilon_N} \longrightarrow 0 \quad (\epsilon_N = 1/\log N),$$

where the second limit uses $e_\varsigma = (3-\varsigma)/(2(1+\varsigma)) \leq \frac{1}{2}$ for $\varsigma \geq 1$.

Every named sequence has $N \to \infty$, $\epsilon_N \downarrow 0$ and $\epsilon_N N \to \infty$ ($N^{1-\omega}$ or $N/\log N$), so it lies in the window of theorem SI.9 for large N. □

Proof of proposition SI.7. Both kernels are finite exponential mixtures with rates $2i^{-q_j}$ and weights summing to one: for SIGNSVD the weights are $i^{-q}/H_N(q)$ with $q = \alpha/2$, and for SIGNSGD they are $(1-\frac{2}{\pi})/N$ and $2i^{-\alpha}/(\pi H_N(\alpha))$. Every forcing and kernel rate is at least $b_{\min} \stackrel{\text{def}}{=} 2N^{-q_j}$, and $F_0 = \frac{1}{2}H_N(p) \in [\frac{1}{2}, \frac{1}{2}Z_p]$. For $N > (2T)^{1/\omega}$ one has $\epsilon_N < 1/(2T) \leq F_0/T$, so lemma SI.11 applies, with

$$H \leq N^{q_j}L_N^*, \qquad \mathsf{d} \leq N^{q_j-\omega/2}L_N^*, \qquad L_N^* \stackrel{\text{def}}{=} \omega \log N + \log Z_p.$$

Write $\mathsf{I}_j \stackrel{\text{def}}{=} \int_0^{\hat{u}_j}\sqrt{Y_j}$ and $\delta_1 \stackrel{\text{def}}{=} (\omega-\alpha)/2 > 0$; by eq. (SI.30), $t^j_{T\epsilon_N} = \mathcal{S}_j T_0^j \mathsf{I}_j/(2\sqrt{\epsilon_N})$.

For SIGNSVD, $q = \alpha/2$ and $p = \alpha + 1 = 2q + 1$. Lemma SI.11 gives

$$J_N(p,q) - \mathsf{I} \leq \sqrt{T}\, N^{-\delta_1}, \qquad \mathsf{I} - J_N(p,q) \leq 2\sqrt{N^{-\delta_1}L_N^* J_N(p,q)} + N^{-\delta_1}L_N^*,$$

and lemma SI.12(i) gives $J_N(p,q) = a_q q \log N + O(1)$. Hence $\mathsf{I} = a_q q \log N + O(1)$. The ratio in the statement equals $2\mathsf{I}/(a_q \log(1/\epsilon_N))$, and $\log(1/\epsilon_N) = \omega \log N$, so it tends to $2q/\omega = \alpha/\omega$.

For SIGNSGD, $q = \alpha$ and $p = q+1$. For $N \geq 4 \cdot 4^{1/\alpha}$, lemma SI.12(ii) gives $c_1' N^{\alpha/2} \leq J_N(p,\alpha) \leq c_2' N^{\alpha/2}$ with $c_1' = \sqrt{A}\, 4^{-\alpha/2}$ and $c_2' = 3C_{p,\alpha}$. Lemma SI.11 gives

$$\frac{J_N - \mathsf{I}}{J_N} \leq \frac{\sqrt{T}}{c_1'}N^{-\delta_1}, \qquad \frac{\mathsf{I} - J_N}{J_N} \leq 2\sqrt{\frac{N^{-\delta_1}L_N^*}{c_1'}} + \frac{N^{-\delta_1}L_N^*}{c_1'},$$

so $\mathsf{I}/J_N \to 1$ and $\mathsf{I} \asymp N^{\alpha/2}$. Then

$$\frac{t^{\text{s-SGD}}_{T\epsilon_N}}{\mathcal{S}_{\text{s-SGD}}T_0^{\text{s-SGD}}\epsilon_N^{-1}} = \frac{1}{2}\sqrt{\epsilon_N}\,\mathsf{I} \asymp N^{(\alpha-\omega)/2}.$$

Finally $\epsilon_N N = N^{1-\omega} \to \infty$. None of the bounds involves B. □

Proof of theorem SI.7. For $\varsigma < 1$, theorem SI.8 applies for large indices, with $\mathcal{S}_{\text{s-SVD}}T_0^{\text{s-SVD}} = \frac{\gamma}{2}H_N(\alpha/2)$ and $\mathcal{S}_{\text{s-SGD}}T_0^{\text{s-SGD}} = \pi N^2\bar{\mu}/(2B)$ from eq. (SI.26). For $\varsigma > 1$ and $\alpha > 1$, theorem SI.9 applies; for $\varsigma > 1$ and $\alpha \leq 1$, theorem SI.10 applies. In both cases $r_{\text{s-SVD}} > 1$, so $t^{\text{s-SVD}}_{T\epsilon} \asymp \mathcal{S}_{\text{s-SVD}}T_0^{\text{s-SVD}}\epsilon^{-1/2}$, and $r_{\text{s-SGD}} < 1$ or $r_{\text{s-SGD}} > 1$ according as $\varsigma < \alpha + 1$ or $\varsigma > \alpha + 1$, which gives eq. (SI.36). The orders of $H_N(\alpha/2)$ follow from lemma SI.8: for $a < 1$, $N^{1-a} \leq H_N(a) \leq 1 + N^{1-a}/(1-a) \leq (1+(1-a)^{-1})N^{1-a}$; for $a = 1$ and $N \geq 2$, $\log N \leq H_N(1) \leq 1 + \log N \leq 3\log N$; for $a > 1$, $1 \leq H_N(a) \leq \zeta(a)$. □

SI.10 Proofs for the accuracy scale

This subsection proves the results of appendix SI.6. The hitting estimates combine lemma SI.11 with a restart: after a first phase controlled by a supersolution, the memory term of the equation is itself a finite exponential mixture with the same rates, so the equation starts afresh from a small initial value (lemma SI.21). Both kernels must be controlled up to times of order N^{q_j}, which requires a kernel bound with a slow part (lemma SI.23).

The models as finite mixtures. As in the proof of proposition SI.7, both kernels are finite exponential mixtures with the forcing rates $c_{j,i} \stackrel{\text{def}}{=} 2i^{-q_j}$:

$$g_j(u) = \sum_{i=1}^{N} w_{j,i}\, c_{j,i}\, e^{-c_{j,i}u}, \qquad w_{\text{s-SVD},i} = \frac{i^{-\alpha/2}}{H_N(\alpha/2)}, \qquad w_{\text{s-SGD},i} = \frac{1-2/\pi}{N} + \frac{2i^{-\alpha}}{\pi H_N(\alpha)},$$

with nonnegative weights of sum one, and $f_{N,q_j}(u) = \sum_i \frac{1}{2} i^{-p} e^{-c_{j,i}u}$. Every forcing and kernel rate is at least $b_{\min} = 2N^{-q_j}$, so $2/b_{\min} = N^{q_j}$, and $F_0 = \frac{1}{2}H_N(p) \in [\frac{1}{2}, \frac{1}{2}\zeta(p)]$. Each g_j is positive, continuous and nonincreasing, with integral one. Hence lemma SI.11 applies whenever $\epsilon < 1/(2T)$; both first hits are then attained, and with $q = q_j$, $\mathsf{y} = \mathsf{y}_j$ and $J_N = J_N(p, q_j)$,

$$\begin{aligned} J_N - \sqrt{T\epsilon}\, N^q \le \mathsf{I}_j &\le \big(\sqrt{J_N} + \sqrt{\mathsf{d}}\big)^2, \\ \mathsf{d} = \sqrt{\mathsf{y}}\, \log\frac{F_0}{\epsilon} &\le \sqrt{\mathsf{y}}\Big[\log\frac{\zeta(p)}{2} + (p-1)\log N + \log\frac{1}{x}\Big]. \end{aligned} \tag{SI.59}$$

Here $\mathsf{J} + 2\sqrt{\mathsf{d}\mathsf{J}} + \mathsf{d} = (\sqrt{\mathsf{J}} + \sqrt{\mathsf{d}})^2$, and the last bound uses $1/\epsilon = N^{p-1}/x$. We use repeatedly that $\sqrt{z}\log(1/z) \le 2/e$ for $0 < z \le 1$.

Forcing integrals. The scaling argument for proposition SI.8 needs one majorant uniform in N, obtained by halving every rate at the cost of a cutoff factor.

Lemma SI.19 (Shift bound)**.** *Let $p > 1$, $q > 0$, $r = r_q$ and $N \ge 1$. For every $u > 0$,*

$$f_{N,q}(u) \le e^{-uN^{-q}} f_{N,q}(u/2) \le C_{p,q}^2\, 2^{2r}\, u^{-2r}\, e^{-uN^{-q}}.$$

Proof. For $1 \le i \le N$ one has $i^{-q} \ge N^{-q}$, so $2ui^{-q} \ge uN^{-q} + ui^{-q}$ and

$$e^{-2ui^{-q}} \le e^{-uN^{-q}}\, e^{-ui^{-q}}.$$

Summing against $\frac{1}{2}i^{-p}$ gives the first inequality. The second is the upper envelope $f_{N,q}(u/2) \le C_{p,q}^2(1+u/2)^{-2r}$ of eq. (SI.23) together with $(1+u/2)^{-2r} \le (u/2)^{-2r}$. □

Proof of proposition SI.8. The idea is to rescale time by the slowest rate N^{-q}. The rescaled forcing is a Riemann sum, and lemma SI.19 supplies a majorant that does not depend on N.

For (i), lemma SI.19 gives $\sqrt{f_{N,q}(u)} \le 2^r C u^{-r} e^{-u/(2N^q)}$, and

$$\int_0^\infty u^{-r} e^{-u/(2N^q)}\, \mathrm{d}u = (2N^q)^{1-r}\, \Gamma(1-r).$$

The product of the constants is $2\Gamma(1-r)CN^{q(1-r)}$.

For (ii), put $V \stackrel{\text{def}}{=} (N/4)^q$. The hypothesis gives $N > 4$ and $V^{1-r} \ge 2$. The lower envelope of eq. (SI.23) on $[0, V]$ gives

$$J_N(p,q) \ge \sqrt{A}\int_0^V (1+u)^{-r}\, \mathrm{d}u \ge \frac{\sqrt{A}}{1-r}\big(V^{1-r} - 1\big) \ge \frac{\sqrt{A}}{2(1-r)}\, V^{1-r},$$

and $V^{1-r} = 4^{-q(1-r)} N^{q(1-r)}$.

For (iii), substitute $u = N^q z$. Since $N^q \cdot N^{-(p-1)/2} = N^{q(1-r)}$,

$$\frac{J_N(p,q)}{N^{q(1-r)}} = \int_0^\infty \sqrt{\Phi_N(z)/2}\, \mathrm{d}z, \qquad \Phi_N(z) \stackrel{\text{def}}{=} 2N^{p-1} f_{N,q}(N^q z) = \frac{1}{N}\sum_{i=1}^{N} \varphi_z(i/N),$$

where $\varphi_z(t) \overset{\text{def}}{=} t^{-p} e^{-2zt^{-q}}$ for $0 < t \le 1$ and $\varphi_z(0) \overset{\text{def}}{=} 0$. For fixed $z > 0$ the function φ_z is continuous on $[0,1]$, so the right Riemann sums converge:

$$\Phi_N(z) \longrightarrow \Phi(z) \overset{\text{def}}{=} \int_0^1 \varphi_z(t)\,\mathrm{d}t.$$

Lemma SI.19 at $u = N^q z$, with $2qr = p - 1$, gives the majorant

$$\frac{\Phi_N(z)}{2} \le N^{p-1} e^{-z} C^2\, 2^{2r} (N^q z)^{-2r} = 4^r C^2 z^{-2r} e^{-z}.$$

Its square root $2^r C z^{-r} e^{-z/2}$ does not depend on N and has integral $2\Gamma(1-r)C$ over $(0,\infty)$, finite because $r < 1$. Hence the integrals converge,

$$\frac{J_N(p,q)}{N^{q(1-r)}} \longrightarrow \int_0^\infty \sqrt{\Phi(z)/2}\,\mathrm{d}z \le 2\Gamma(1-r)\,C.$$

To evaluate the limit, substitute $t = w^{-1/q}$ with $w \in [1,\infty)$. Then $\mathrm{d}t = q^{-1} w^{-1/q-1} \mathrm{d}w$, $t^{-p} = w^{p/q}$ and $(p-1)/q = 2r$, so

$$\Phi(z) = \frac{1}{q} \int_1^\infty w^{2r-1} e^{-2zw}\,\mathrm{d}w = \frac{(2z)^{-2r}}{q}\,\Gamma(2r, 2z).$$

The substitution $y = 2z$ then gives

$$\int_0^\infty \sqrt{\Phi(z)/2}\,\mathrm{d}z = (2q)^{-1/2} \int_0^\infty (2z)^{-r} \Gamma(2r,2z)^{1/2}\,\mathrm{d}z = (8q)^{-1/2}\,\mathcal{K}(r) = J_\circ(p,q),$$

which is positive because the integrand is.

For (iv), $\Gamma(1,y) = e^{-y}$, so

$$\mathcal{K}(\tfrac12) = \int_0^\infty y^{-1/2} e^{-y/2}\,\mathrm{d}y = \sqrt{2}\,\Gamma(\tfrac12) = \sqrt{2\pi}.$$

If $p = q + 1$, then $r = \frac12$ and $J_\circ(p,q) = \sqrt{2\pi}/\sqrt{8q} = \frac12\sqrt{\pi/q}$. □

Proof of theorem SI.12. The proof multiplies the asymptotics of three factors. For SIGNSVD, $q = \alpha/2$, $r = s$ and $q(1-r) = (1-\varsigma)/2$; for SIGNSGD, $q = \alpha$, $r = s/2$ and $q(1-r) = (\alpha+1-\varsigma)/2$. By eq. (SI.43),

$$J_N(p, \tfrac{\alpha}{2}) \begin{cases} \sim (4\alpha)^{-1/2} \mathcal{K}(s)\, N^{(1-\varsigma)/2}, & \varsigma < 1, \\ \sim \frac{\sqrt{\alpha}}{4} \log N, & \varsigma = 1, \\ \to J(p, \alpha/2), & \varsigma > 1, \end{cases}$$

and

$$J_N(p,\alpha) \begin{cases} \sim (8\alpha)^{-1/2} \mathcal{K}(\frac{s}{2})\, N^{(\alpha+1-\varsigma)/2}, & \varsigma < \alpha+1, \\ \sim \sqrt{\alpha/8}\, \log N, & \varsigma = \alpha+1, \\ \to J(p,\alpha), & \varsigma > \alpha+1. \end{cases}$$

At $\varsigma = 1$ one has $s/2 = \frac12$, and proposition SI.8(iv) gives $J_N(p,\alpha) \sim \frac12\sqrt{\pi/\alpha}\, N^{\alpha/2}$, whence

$$\frac{J_N(p,\alpha)}{J_N(p,\alpha/2)} \sim \frac{2\sqrt{\pi}}{\alpha}\,\frac{N^{\alpha/2}}{\log N} \qquad (\varsigma = 1).$$

In case (a) the quotient is

$$\frac{(8\alpha)^{-1/2}\mathcal{K}(s/2)}{(4\alpha)^{-1/2}\mathcal{K}(s)}\, N^{\alpha/2} = Q_A\, N^{\alpha/2}.$$

Products and quotients of $\sim$-equivalent positive sequences are $\sim$-equivalent, so in each case ρ_N has the stated equivalent. Multiplying by eq. (SI.31) gives the limits and orders. For $0 < \alpha < 1$ in (a), $D_N N^{\alpha/2} \to \pi(1-\frac{\alpha}{2})/(1-\alpha)$, which gives K_A. At $(\alpha,\varsigma) = (1,1)$,

$$\frac{\pi}{2} N^{-1/2} \log N \cdot 2\sqrt{\pi}\,\frac{N^{1/2}}{\log N} = \pi^{3/2}.$$

In (c) with $1 < \alpha < 2$ the exponent of N is

$$\frac{\alpha}{2} - 1 + \frac{\alpha + 1 - \varsigma}{2} = \frac{2\alpha - 1 - \varsigma}{2},$$

which vanishes on $\varsigma = 2\alpha - 1$, where $p = 3\alpha - 1$ and $s = (3\alpha - 2)/\alpha$; this gives K_c. In (d) at $\alpha = 2$,

$$\frac{\pi\zeta(2)}{\log N} \cdot \frac{\log N}{2} = \frac{\pi\zeta(2)}{2},$$

and $J(p, \alpha/2) = J(5, 1)$. Every remaining case is the same substitution, using that a positive power of N dominates every power of $\log N$. □

Proof of corollary SI.4. Collect the cases of theorem SI.12. For $1 < \alpha < 2$ the chord value $2\alpha - 1$ lies in $(1, \alpha + 1)$, so (a), (b) and the part $\varsigma < 2\alpha - 1$ of (c) give V, and the rest of (c) together with (d) and (e) gives G. For $\alpha = 2$, cases (a) to (c) give $\varsigma < 3$, case (d) gives the tie at $\varsigma = 3$, and case (e) gives G. The columns $0 < \alpha < 1$, $\alpha = 1$ and $\alpha > 2$ are read off directly. The three sets are disjoint and cover $\{p > 1\}$. □

Comparison with a tail term. Three comparison lemmas carry the hitting estimates. The first is a square supersolution with an extra tail term w, used when a slowly decaying part of the kernel meets the forcing envelope.

Lemma SI.20 (Square supersolution with a tail term)**.** *Let $g \geq 0$ be continuous with $\int_0^\infty g \leq 1$, let $\epsilon > 0$ and $u_0 > 0$, and let $f, h, w \geq 0$ be continuous on $[0, u_0]$ with*

$$f \leq h^2, \qquad g * h \leq 2h + \frac{w^2}{\sqrt{\epsilon}}, \qquad g * w \leq 2w \qquad \text{on } [0, u_0].$$

If $Y > 0$ is continuous and $Y = f + \sqrt{\epsilon}\, g * \sqrt{Y}$ on $[0, u_0]$, then on $[0, u_0]$

$$Y \leq (h + w + \sqrt{\epsilon})^2 \qquad \text{and} \qquad g * \sqrt{Y} \leq G \stackrel{\text{def}}{=} 2h + \frac{w^2}{\sqrt{\epsilon}} + 2w + \sqrt{\epsilon}.$$

Proof. The candidate $U \stackrel{\text{def}}{=} (h + w + \sqrt{\epsilon})^2$ is a supersolution. Indeed, by the three hypotheses and $\int g \leq 1$,

$$f + \sqrt{\epsilon}\, g * \sqrt{U} \leq h^2 + \sqrt{\epsilon}\Big(2h + \frac{w^2}{\sqrt{\epsilon}} + 2w + \sqrt{\epsilon}\Big) = h^2 + w^2 + \epsilon + 2\sqrt{\epsilon}\,(h + w) \leq U.$$

Since $U \geq \epsilon$, lemma SI.3 with kernel $\sqrt{\epsilon}\, g$, $G = f$ and $m = \epsilon$ gives $Y \leq U$ on $[0, u_0]$. Then $g * \sqrt{Y} \leq g * (h + w + \sqrt{\epsilon}) \leq G$ by the hypotheses and $\int g \leq 1$. □

The second lemma is the restart. After a first phase the memory term is a nonnegative finite exponential mixture with the same rates, so the problem restarts as a fresh instance of lemma SI.11, whose initial value is now small.

Lemma SI.21 (Restart)**.** *Assume the setting of lemma SI.11, with any $\epsilon > 0$. Let $u_0 \geq 0$, suppose that $Y > T\epsilon$ on $[0, u_0]$, and put $G_0 \stackrel{\text{def}}{=} (g * \sqrt{Y})(u_0)$. Then the first $T\epsilon$ hit $\hat{u}$ is attained in $\big(u_0, u_0 + \frac{2}{b_{\min}} \log \frac{Y(u_0)}{\epsilon}\big]$, and*

$$\int_{u_0}^{\hat{u}} \sqrt{Y}\, \mathrm{d}u \leq \big(\tilde{\mathsf{J}}^{1/2} + \tilde{\mathsf{d}}^{1/2}\big)^2,$$

$$\tilde{\mathsf{J}} \stackrel{\text{def}}{=} \int_{u_0}^{\infty} \sqrt{f}\, \mathrm{d}u + \frac{2}{b_{\min}} \epsilon^{1/4} G_0^{1/2}, \qquad \tilde{\mathsf{d}} \stackrel{\text{def}}{=} \frac{2}{b_{\min}} \sqrt{\epsilon} \log \frac{Y(u_0)}{\epsilon}.$$

Proof. The memory splits exactly at u_0. Put

$$y_l \stackrel{\text{def}}{=} c_l \int_0^{u_0} e^{-c_l(u_0 - v)} \sqrt{Y(v)}\, \mathrm{d}v \geq 0,$$

so that $\sum_l w_l y_l = G_0$. For $\tilde{Y}(z) \stackrel{\text{def}}{=} Y(u_0 + z)$, splitting the convolution integral at $v = u_0$ gives

$$\tilde{Y} = \tilde{f} + \sqrt{\epsilon}\, g * \sqrt{\tilde{Y}}, \qquad \tilde{f}(z) \stackrel{\text{def}}{=} \sum_i a_i e^{-b_i u_0} e^{-b_i z} + \sqrt{\epsilon} \sum_l w_l y_l e^{-c_l z}.$$

This $\tilde{f}$ is a finite exponential mixture with nonnegative coefficients and rates at least $b_{\min}$, and

$$\tilde{f}(0) = f(u_0) + \sqrt{\epsilon}\, G_0 = Y(u_0) > T\epsilon.$$

By theorem SI.1, applied with the union of the two rate lists and zero coefficients where needed, $\tilde{Y}$ is the unique positive continuous solution of its equation. So lemma SI.11 applies with initial value $Y(u_0)$ in place of F_0. Since $Y > T\epsilon$ on $[0, u_0]$, the first hit of Y is u_0 plus that of $\tilde{Y}$, and lemma SI.11 bounds both, with $\int_0^\infty \sqrt{\tilde{f}}$ in place of $\tilde{\mathsf{J}}$. Finally $\tilde{f}(z) \le f(u_0 + z) + \sqrt{\epsilon}\, G_0\, e^{-b_{\min} z}$ and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ give $\int_0^\infty \sqrt{\tilde{f}} \le \tilde{\mathsf{J}}$, and $x \mapsto (\sqrt{x} + \sqrt{\tilde{\mathsf{d}}})^2$ is increasing. □

The two lemmas are always used together, and we combine them once. In the models $2/b_{\min} = N^{q_j}$, so $(2/b_{\min})\sqrt{\epsilon} = \sqrt{\mathsf{y}_j}$.

Lemma SI.22 (Two-phase bound)**.** *Assume the setting of lemma SI.11. Let $u_0 > 0$, and let $h, w \ge 0$ be continuous on $[0, u_0]$ and satisfy the three hypotheses of lemma SI.20 there. Let G be as in that lemma and*

$$\Theta_0 \stackrel{\text{def}}{=} 1 + \frac{h(u_0) + w(u_0)}{\sqrt{\epsilon}}.$$

(a) On $[0, u_0]$, $Y \le (h + w + \sqrt{\epsilon})^2$ and $\sqrt{Y} \le \sqrt{f} + \epsilon^{1/4} G^{1/2}$.

(b) If $h(H) + w(H) \le \sqrt{\epsilon}$ for some $H \in [0, u_0]$, then $\hat{u} \le H$.

(c) With $\mathsf{I} \stackrel{\text{def}}{=} \int_0^{\hat{u}} \sqrt{Y}$,

$$\mathsf{I} \le \int_0^{u_0} \sqrt{Y}\, \mathrm{d}u + \left(\tilde{\mathsf{J}}^{1/2} + \tilde{\mathsf{d}}^{1/2}\right)^2,$$

$$\tilde{\mathsf{J}} \le \int_{u_0}^\infty \sqrt{f}\, \mathrm{d}u + \frac{2}{b_{\min}} \sqrt{\epsilon}\, \Theta_0, \qquad \tilde{\mathsf{d}} \le \frac{4}{b_{\min}} \sqrt{\epsilon}\, \log \Theta_0.$$

Proof. Part (a) is lemma SI.20, together with $Y - f = \sqrt{\epsilon}\, g * \sqrt{Y} \le \sqrt{\epsilon}\, G$ and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$.

For (b), part (a) gives $Y(H) \le 4\epsilon \le T\epsilon$.

For (c), the bound is immediate if $\hat{u} \le u_0$. Otherwise $Y > T\epsilon$ on $[0, u_0]$, and lemma SI.21 applies with $G_0 \le G(u_0)$. By (a),

$$Y(u_0) \le \left(h(u_0) + w(u_0) + \sqrt{\epsilon}\right)^2 = \epsilon\, \Theta_0^2,$$

which gives the bound on $\tilde{\mathsf{d}}$. Expanding the square,

$$\sqrt{\epsilon}\, \Theta_0^2 = \sqrt{\epsilon} + 2(h + w)(u_0) + \frac{(h + w)(u_0)^2}{\sqrt{\epsilon}} \ge G(u_0),$$

so $\epsilon^{1/4} G(u_0)^{1/2} \le \sqrt{\epsilon}\, \Theta_0$, which gives the bound on $\tilde{\mathsf{J}}$. □

Kernels with a slow part. The convolution hypotheses are verified by splitting each kernel into a fast integrable part and a slow part of height of order N^{-q}; the slow part is controlled up to times of order N^q. The next lemma extends lemma SI.14, which is its case $K_s = 0$.

Lemma SI.23 (Kernel with a slow part)**.** *Let $g \ge 0$ be continuous with $\int_0^\infty g \le 1$, and suppose that for $0 \le v \le U_0$*

$$g(v) \le K_f (1 + v)^{-\mathsf{v}} + K_s N^{-q} \Big(\frac{1 + v}{N^q}\Big)^{-\mathsf{b}}, \tag{SI.60}$$

with $K_f, K_s \ge 0$, $\upsilon > 1$, $0 \le \mathsf{b} < 1$, $N \ge 1$ *and* $q > 0$. *Let* $0 < r_0 \le \upsilon$, $D \ge 1$, $C > 0$, $h(u) \stackrel{\text{def}}{=} C(1+u/D)^{-r_0}$, $\delta_\circ = 1-(2/3)^{1/r_0}$ *and* $\mathsf{k}_f \stackrel{\text{def}}{=} K_f \delta_\circ^{-\upsilon} D^{1-\upsilon} b(r_0,\upsilon)$, *with* $b(r_0,\upsilon)$ *of eq.* (SI.55). *Then for* $0 \le u \le U_0$,

$$(g * h)(u) \le \Big(\frac{3}{2} + \mathsf{k}_f\Big) h(u) + K_s \delta_\circ^{-\mathsf{b}} N^{-q} \Big(\frac{D+u}{N^q}\Big)^{-\mathsf{b}} C\, D\, I_{r_0}(u/D),$$

and if $r_0 < 1$ *the last term is at most* $\frac{K_s \delta_\circ^{-\mathsf{b}}}{1-r_0}\big(\frac{D+u}{N^q}\big)^{1-\mathsf{b}} h(u)$.

Proof. Fix $u \in (0, U_0]$ and split the convolution at $v = \delta_\circ(D+u)$. For $0 \le v \le \min\{u, \delta_\circ(D+u)\}$ one has $D+u-v \ge (1-\delta_\circ)(D+u)$, so $h(u-v) \le (1-\delta_\circ)^{-r_0} h(u) = \frac{3}{2} h(u)$, and the mass of g bounds this part by $\frac{3}{2}h(u)$. For $\delta_\circ(D+u) < v \le u$ one has $1+v > \delta_\circ(D+u)$, so eq. (SI.60) gives

$$g(v) \le K_f \delta_\circ^{-\upsilon} (D+u)^{-\upsilon} + K_s \delta_\circ^{-\mathsf{b}} N^{-q} \Big(\frac{D+u}{N^q}\Big)^{-\mathsf{b}},$$

while $\int_0^u h = C\, D\, I_{r_0}(u/D)$. With $z = u/D$ the fast part is at most

$$K_f \delta_\circ^{-\upsilon} D^{1-\upsilon} (1+z)^{r_0-\upsilon} I_{r_0}(z)\, h(u) \le \mathsf{k}_f\, h(u),$$

by the bound $(1+z)^{r_0-\upsilon} I_{r_0}(z) \le b(r_0,\upsilon)$ of the proof of lemma SI.14. The slow part is the stated term. For $r_0 < 1$, finally,

$$D\, I_{r_0}(u/D) \le \frac{D(1+u/D)^{1-r_0}}{1-r_0} = \frac{(D+u)(1+u/D)^{-r_0}}{1-r_0},$$

which gives the last assertion. □

For the SIGNSGD model at a forcing exponent $r < 1$ we need the analogue of lemma SI.15, for every $\alpha > 0$.

Lemma SI.24 (SIGNSGD kernel below the critical exponent)**.** *Let* $\alpha > 0$ *and* $0 < r < 1$, *and write* $g = g_{\text{s-SGD}}$. *There are* $D_{\mathrm{G}} \ge 1$, $\tau_{\mathrm{G}} > 0$ *and* N_{G}, *depending only on* α *and* r, *such that for* $N \ge N_{\mathrm{G}}$, $C > 0$ *and* $h(u) = C(1+u/D_{\mathrm{G}})^{-r}$,

$$g * h \le 2h \qquad \text{on } [0, \tau_{\mathrm{G}} N^\alpha].$$

Proof. Write $g = g_1 + g_2$ for the two terms of eq. (SI.28). Integrating each exponential gives the exact masses

$$m_{g_1}(u) = \frac{1-2/\pi}{N} \sum_{i=1}^{N} \big(1 - e^{-2ui^{-\alpha}}\big), \qquad m_{g_2}(u) = \frac{2/\pi}{H_N(\alpha)} \sum_{i=1}^{N} i^{-\alpha} \big(1 - e^{-2ui^{-\alpha}}\big).$$

For $\delta \in (0,1)$ split at $J \stackrel{\text{def}}{=} \lfloor \delta N \rfloor$. The terms $i \le J$ contribute at most $(1-\frac{2}{\pi})\delta$ and $\frac{2}{\pi} H_J(\alpha)/H_N(\alpha)$, with $H_0 \stackrel{\text{def}}{=} 0$. For $i > J$, $i^{-\alpha} < (\delta N)^{-\alpha}$ and $1 - e^{-t} \le t$ give at most $2u\delta^{-\alpha} N^{-\alpha}$ in total. Hence

$$m_{g_1} \le \Big(1 - \frac{2}{\pi}\Big)\big[\delta + 2u\delta^{-\alpha} N^{-\alpha}\big], \qquad m_{g_2} \le \frac{2}{\pi}\Big[\frac{H_J(\alpha)}{H_N(\alpha)} + 2u\delta^{-\alpha} N^{-\alpha}\Big]. \tag{SI.61}$$

Case $0 < \alpha < 1$. By lemma SI.8, $H_J(\alpha) \le 1 + J^{1-\alpha}/(1-\alpha)$, and $H_N(\alpha) \ge N^{1-\alpha}/(2(1-\alpha))$ when $N^{1-\alpha} \ge 2$; hence

$$\frac{H_J(\alpha)}{H_N(\alpha)} \le 2(1-\alpha) N^{-(1-\alpha)} + 2\delta^{1-\alpha}.$$

Take $\delta \stackrel{\text{def}}{=} ((1-r)/16)^{1/(1-\alpha)}$, $\tau_{\mathrm{G}} \stackrel{\text{def}}{=} (1-r)\delta^\alpha/8$, $D_{\mathrm{G}} \stackrel{\text{def}}{=} 1$, and N_{G} with $N_{\mathrm{G}}^{1-\alpha} \ge \max\{2, 32(1-\alpha)/(1-r)\}$. Since $\delta \le \delta^{1-\alpha} = (1-r)/16$, on $[0, \tau_{\mathrm{G}} N^\alpha]$

$$m_g \le \delta + 2\delta^{1-\alpha} + 2(1-\alpha) N^{-(1-\alpha)} + 2u\delta^{-\alpha} N^{-\alpha} \le (1-r)\Big(\frac{1}{16} + \frac{1}{8} + \frac{1}{16} + \frac{1}{4}\Big) = \frac{1-r}{2}.$$

The kernel is nonincreasing, so lemma SI.13 gives $g * h \le \frac{1}{2} h$.

Case $\alpha = 1$. Take $\delta \stackrel{\text{def}}{=} (1-r)/8$ and $\tau_{\mathrm{G}} \stackrel{\text{def}}{=} \delta(1-r)/16$. By eq. (SI.61), $m_{g_1} \le (1-r)/4$ on $[0, \tau_{\mathrm{G}} N]$, so $g_1 * h \le h/4$ by lemma SI.13. For g_2, lemma SI.7(i) gives

$$g_2(v) \le \frac{4E_{2,1}}{\pi H_N(1)} (1+v)^{-1}.$$

Take $D_{\mathrm{G}} \stackrel{\text{def}}{=} 1$ and split at $\delta_\circ(1+u)$ with $\delta_\circ \stackrel{\text{def}}{=} 1-(2/3)^{1/r}$. The early part is at most $\frac{3}{2} \cdot \frac{2}{\pi} h(u)$, since g_2 has mass $\frac{2}{\pi}$. Since $\int_0^u h \le (1+u)h(u)/(1-r)$, the late part is at most

$$\frac{4E_{2,1}}{\pi H_N(1)\, \delta_\circ(1+u)} \int_0^u h \le \frac{4E_{2,1}}{\pi \delta_\circ (1-r) H_N(1)}\, h(u) \le \frac{h(u)}{4}$$

once $\log(N+1) \ge 16E_{2,1}/(\pi\delta_\circ(1-r))$, which defines N_{G}; here $H_N(1) \ge \log(N+1)$ by lemma SI.8. In total $g * h \le (\frac{1}{2} + \frac{3}{\pi})h < 2h$.

Case $\alpha > 1$. By lemma SI.7(i) and $H_N(\alpha) \ge 1$, the kernel satisfies eq. (SI.60) for all $v \ge 0$ with $q = \alpha$ and

$$K_f = \frac{4}{\pi} E_{2\alpha,\alpha}, \qquad \upsilon = 2 - \frac{1}{\alpha}, \qquad K_s = 2\Big(1 - \frac{2}{\pi}\Big) E_{\alpha,\alpha}, \qquad \mathsf{b} = 1 - \frac{1}{\alpha}, \tag{SI.62}$$

because $N^{-1}(1+v)^{-(\alpha-1)/\alpha} = N^{-\alpha}((1+v)/N^\alpha)^{-(\alpha-1)/\alpha}$. Apply lemma SI.23 with $r_0 = r$ and $D_{\mathrm{G}} \stackrel{\text{def}}{=} \max\{1, [4K_f\delta_\circ^{-\upsilon}/(1-r)]^{1/(\upsilon-1)}\}$, where $\delta_\circ = 1-(2/3)^{1/r}$. Then $\mathsf{k}_f \le \frac{1}{4}$ and

$$g * h \le \frac{7}{4} h + \frac{K_s \delta_\circ^{-\mathsf{b}}}{1-r} \Big(\frac{D_{\mathrm{G}} + u}{N^\alpha}\Big)^{1-\mathsf{b}} h \le 2h \quad \text{when } D_{\mathrm{G}} + u \le \tau' N^\alpha,$$

where

$$\tau' \stackrel{\text{def}}{=} \Big(\frac{(1-r)\delta_\circ^{\mathsf{b}}}{4K_s}\Big)^{1/(1-\mathsf{b})}.$$

Take $\tau_{\mathrm{G}} \stackrel{\text{def}}{=} \tau'/2$ and N_{G} with $N_{\mathrm{G}}^\alpha \ge 2D_{\mathrm{G}}/\tau'$. □

Phase-one data for $r_j < 1$. For a model with $r_j < 1$ fix (D, τ, N°) as follows. For SIGNSVD, where $r_j < 1$ means $\varsigma < 1$, take $D = D_{\text{s-SVD}}$, $\tau = \tau_{\text{s-SVD}}$ and $N^\circ = \max\{4, N_{\mathrm{V}}\}$ from the proof of theorem SI.8; that proof shows $g_{\text{s-SVD}} * h \le 2h$ on $[0, \tau N^{\alpha/2}]$ for $N \ge N^\circ$ and every $h = C(1+u/D)^{-r_j}$ with $C > 0$. For SIGNSGD take $(D, \tau, N^\circ) = (D_{\mathrm{G}}, \tau_{\mathrm{G}}, N_{\mathrm{G}})$ from lemma SI.24 with $r = r_j$. With $C = C_{p,q_j}$ and $h = C(1+u/D)^{-r_j}$ one has $f_{N,q_j} \le h^2$, by the upper envelope of eq. (SI.23) and $D \ge 1$, and $g_j * h \le 2h$ on $[0, \tau N^{q_j}]$ for $N \ge N^\circ$. So lemmas SI.9 and SI.22 apply with $w = 0$ and $u_0 = \tau N^{q_j}$.

For models with $r_j \ge 1$ the kernel is controlled against a slower envelope.

Lemma SI.25 (Kernel templates)**.** *In each of the following cases there are constants N_{T}, $K_f \ge 0$, $K_s > 0$, $\upsilon > 1$ and $\mathsf{b} \in (0,1)$, depending only on α, such that for $N \ge N_{\mathrm{T}}$ the kernel g_j satisfies eq. (SI.60) with $q = q_j$ on $0 \le v \le N^{q_j} - 1$:*

- *(i)* SIGNSVD, $0 < \alpha < 1$: $K_f = 0$, $K_s = 4(1-\frac{\alpha}{2})/(1-\alpha)$, $\mathsf{b} = \frac{1}{2}$;
- *(ii)* SIGNSVD, $1 < \alpha < 2$: $K_f = 0$, $K_s = 4(1-\frac{\alpha}{2})E_{\alpha,\alpha/2}$, $\mathsf{b} = 2 - \frac{2}{\alpha}$;
- *(iii)* SIGNSGD, $0 < \alpha < \frac{1}{2}$: $K_f = 0$, $K_s = 2(1-\frac{2}{\pi})/(1-\alpha) + \frac{4}{\pi}(1-\alpha)/(1-2\alpha)$, $\mathsf{b} = \frac{1}{2}$;
- *(iv)* SIGNSGD, $\frac{1}{2} < \alpha < 1$: $K_f = 0$, $K_s = 2(1-\frac{2}{\pi})/(1-\alpha) + \frac{4}{\pi}(1-\alpha)E_{2\alpha,\alpha}$, $\mathsf{b} = \max\{2 - \frac{1}{\alpha}, \frac{1}{2}\}$;
- *(v)* SIGNSGD, $\alpha > 1$: *the constants eq. (SI.62), for all $\upsilon \ge 0$.*

Proof. On $0 \le v \le N^q - 1$ one has $(1+v)/N^q \le 1$, so a bound $K' N^{-q}((1+v)/N^q)^{-\mathsf{b}'}$ with $0 \le \mathsf{b}' \le \mathsf{b}$ implies the same bound with b; in particular a constant bound $K' N^{-q}$ has template form with any b. Let N_{T} be such that $N^{1-a} \ge 2$ for the relevant $a < 1$, so that $H_N(a) \ge N^{1-a}/(2(1-a))$; also $H_N(a) \le N^{1-a}/(1-a)$ for $0 < a < 1$, by lemma SI.8.

(i) The kernel is nonincreasing, so

$$g_{\text{s-SVD}} \le g_{\text{s-SVD}}(0) = \frac{2H_N(\alpha)}{H_N(\alpha/2)} \le \frac{4(1-\alpha/2)}{1-\alpha}\, N^{-\alpha/2}.$$

(ii) By lemma SI.7(i),

$$g_{\text{s-SVD}}(v) \le 4\Big(1-\frac{\alpha}{2}\Big) E_{\alpha,\alpha/2}\, N^{-(1-\alpha/2)} (1+v)^{-(2-2/\alpha)},$$

which is the template with $q = \alpha/2$ and $\mathsf{b} = 2 - \frac{2}{\alpha} \in (0,1)$, since $-q + q\mathsf{b} = \frac{\alpha}{2} - 1$.

(iii) The first term satisfies

$$g_1 \le \frac{2(1-2/\pi)}{N}\, H_N(\alpha) \le \frac{2(1-2/\pi)}{1-\alpha}\, N^{-\alpha}.$$

For the second term, the two bounds of lemma SI.8, with $\int_1^{N+1} \ge \int_1^N$ in the lower one, give for $0 < a < 1$

$$(1-a)H_N(a) \ge N^{1-a} - 1, \qquad (1-a)H_N(a) \le N^{1-a} - a.$$

Enlarge N_{T} so that $N^{-\alpha} \le 2\alpha$ for $N \ge N_{\mathrm{T}}$. The upper bound with $a = 2\alpha$ and the lower bound with $a = \alpha$ then give

$$(1-2\alpha)H_N(2\alpha) \le N^{1-2\alpha} - N^{-\alpha} = N^{-\alpha}\big(N^{1-\alpha} - 1\big) \le (1-\alpha)N^{-\alpha} H_N(\alpha).$$

Since g_2 is nonincreasing and $L_{N,2\alpha,\alpha}(0) = H_N(2\alpha)$,

$$g_2 \le \frac{4H_N(2\alpha)}{\pi H_N(\alpha)} \le \frac{4(1-\alpha)}{\pi(1-2\alpha)}\, N^{-\alpha}.$$

(iv) The term g_1 is bounded as in (iii). Write $d \stackrel{\text{def}}{=} d(2\alpha,\alpha) = (2\alpha-1)/\alpha$. The envelope of lemma SI.7(i) holds with a strictly smaller constant: there is $E' < E_{2\alpha,\alpha}$, depending only on α, such that

$$L_{N,2\alpha,\alpha}(v) \le E'(1+v)^{-d} \qquad (N \ge 1,\ v \ge 0).$$

To see this, note that the term $i = 1$ of $L_{N,2\alpha,\alpha}(v)$ is e^{-2v}. The other terms sum to at most $Z_{2\alpha} - 1$ by the proof of lemma SI.7. Hence

$$L_{N,2\alpha,\alpha}(v) \le Z_{2\alpha} - 1 + e^{-2v}.$$

Since $Z_{2\alpha} - 1 + e^{-1} < Z_{2\alpha}$, continuity gives $r > 1$ with $(1+r)^d(Z_{2\alpha} - 1 + e^{-1}) < 2^d Z_{2\alpha}$. On $0 \le v \le \frac{1}{2}$ the last display gives $(1+v)^d L_{N,2\alpha,\alpha}(v) \le (3/2)^d Z_{2\alpha}$. On $\frac{1}{2} \le v \le r$ it gives $(1+v)^d L_{N,2\alpha,\alpha}(v) \le (1+r)^d(Z_{2\alpha} - 1 + e^{-1})$. Both bounds are below $2^d Z_{2\alpha} \le E_{2\alpha,\alpha}$. For $v \ge r > 1$, the display in the proof of lemma SI.7(i) and $\Gamma(d)/\alpha + 2^{1-1/\alpha} M_{2\alpha,\alpha} \le E_{2\alpha,\alpha}$ give $L_{N,2\alpha,\alpha}(v) \le E_{2\alpha,\alpha}(2v)^{-d}$, so

$$(1+v)^d L_{N,2\alpha,\alpha}(v) \le E_{2\alpha,\alpha}\Big(\frac{1+v}{2v}\Big)^d \le E_{2\alpha,\alpha}\Big(\frac{1+r}{2r}\Big)^d < E_{2\alpha,\alpha}.$$

Let E' be the largest of the three bounds.

Enlarge N_{T} so that $N^{1-\alpha}(E_{2\alpha,\alpha} - E') \ge E_{2\alpha,\alpha}$ for $N \ge N_{\mathrm{T}}$. The lower bound $(1-\alpha)H_N(\alpha) \ge N^{1-\alpha} - 1$ of (iii) then gives

$$\frac{E'}{H_N(\alpha)} \le \frac{(1-\alpha)E'}{N^{1-\alpha} - 1} \le (1-\alpha)E_{2\alpha,\alpha}\, N^{\alpha-1}.$$

Therefore

$$g_2(v) \le \frac{4E'}{\pi H_N(\alpha)}(1+v)^{-d} \le \frac{4}{\pi}(1-\alpha)E_{2\alpha,\alpha}\, N^{-\alpha}\Big(\frac{1+v}{N^\alpha}\Big)^{-(2\alpha-1)/\alpha},$$

since $-\alpha + (2\alpha - 1) = \alpha - 1$.

(v) This was shown in the proof of lemma SI.24. □

The cases not covered are SIGNSVD at $\alpha \in \{1,2\}$ and $\alpha > 2$, and SIGNSGD at $\alpha \in \{\frac{1}{2}, 1\}$ with $r_{\text{s-SGD}} \ge 1$. SIGNSVD at $\alpha \ge 2$ is treated directly in the proof of theorem SI.14; the remaining cases are the triples excluded from $\mathcal{U}$.

The frozen regime.

Proof of theorem SI.13. Drop the index j and write $q = q_j$, $r = r_j$, $\mathsf{y} = \mathsf{y}_j$, $\mathsf{I} = \mathsf{I}_j$ and $J_N = J_N(p,q)$. Along the sequence $\epsilon = x\epsilon_N \to 0$, so eventually $\epsilon < 1/(2T)$ and eq. (SI.59) holds; in particular both hits are attained. The lower bound in eq. (SI.59) costs $\sqrt{T\epsilon}\,N^q = \sqrt{T\mathsf{y}}$ in every case. The three regimes of r differ only in how the upper error is controlled.

Case $r > 1$. Here $\mathsf{y} = xN^{-2q(r-1)}$, so $\sqrt{T\epsilon}\,N^q = \sqrt{Tx}\,N^{-q(r-1)} \to 0$. By eq. (SI.59) and $\sqrt{x}\log(1/x) \le 2/e$,

$$\mathsf{d} \le N^{-q(r-1)}\Big[\sqrt{x}\Big(\log\frac{\zeta(p)}{2} + (p-1)\log N\Big) + \frac{2}{e}\Big] \longrightarrow 0.$$

Since $J_N \to J(p,q) \in (0,\infty)$ by eq. (SI.43), eq. (SI.59) gives $\mathsf{I}/J_N \to 1$.

Case $r = 1$. Here $p - 1 = 2q$, $\mathsf{y} = x$, and $J_N = \sqrt{q/8}\log N + O(1)$ by lemma SI.12(i). The lower error is $\sqrt{Tx} \to 0$, and

$$\mathsf{d} \le \sqrt{x}\Big(\log\frac{\zeta(p)}{2} + 2q\log N\Big) + \frac{2}{e} = o(\log N).$$

Thus $(\mathsf{I} - J_N)/J_N \le 2\sqrt{\mathsf{d}/J_N} + \mathsf{d}/J_N \to 0$.

Case $r < 1$. Lemma SI.11 alone would give a relative error of order $\sqrt{x}\log N$; the restart removes the logarithm. Use the phase-one data (D, τ, N°) with $u_0 = \tau N^q$ and $w = 0$, and take N so large that $N \ge N^\circ$ and $D \le u_0$. Put $C = C_{p,q}$ and $\mathsf{e}_r \stackrel{\text{def}}{=} q(1-r)$.

The first phase costs a fraction $x^{1/4}$. Lemma SI.22(a) with $G = 2h + \sqrt{\epsilon}$ gives, on $[0, u_0]$,

$$\sqrt{Y} \le \sqrt{f} + \sqrt{2}\,\epsilon^{1/4}\sqrt{h} + \sqrt{\epsilon}.$$

Integrating the envelope, $\int_0^{u_0}\sqrt{h} \le \sqrt{C}\,D^{r/2}(D+u_0)^{1-r/2}/(1-\frac{r}{2})$, and $D + u_0 \le 2\tau N^q$. Since $\frac{p-1}{4} = \frac{qr}{2}$, one has $\epsilon^{1/4}N^{q(1-r/2)} = x^{1/4}N^{\mathsf{e}_r}$, and by eq. (SI.40) $\sqrt{\epsilon}\,u_0 = \tau\sqrt{\mathsf{y}} = \tau x^{1/2}N^{\mathsf{e}_r}$. Hence

$$\int_0^{u_0}\sqrt{Y}\,\mathrm{d}u \le \int_0^{u_0}\sqrt{f}\,\mathrm{d}u + \big(c_1 x^{1/4} + \tau x^{1/2}\big)N^{\mathsf{e}_r}, \qquad c_1 \stackrel{\text{def}}{=} \frac{\sqrt{2C}\,D^{r/2}(2\tau)^{1-r/2}}{1-r/2}. \tag{SI.63}$$

The restart costs the same fraction. At the end of the first phase $h(u_0) \le C(D/\tau)^r N^{-qr}$ and $\sqrt{\epsilon} = \sqrt{x}\,N^{-qr}$, so

$$\frac{h(u_0)}{\sqrt{\epsilon}} \le c_2 x^{-1/2}, \qquad c_2 \stackrel{\text{def}}{=} C\Big(\frac{D}{\tau}\Big)^r, \qquad \Theta_0 \le 1 + c_2 x^{-1/2}.$$

In lemma SI.22(c), $G(u_0) = 2h(u_0) + \sqrt{\epsilon}$ and

$$N^q\epsilon^{1/4}\big(2h(u_0) + \sqrt{\epsilon}\big)^{1/2} \le \sqrt{\mathsf{y}}\,\big(\sqrt{2c_2}\,x^{-1/4} + 1\big).$$

With eq. (SI.40) this gives

$$\tilde{\mathsf{J}} \le \int_{u_0}^{\infty}\sqrt{f}\,\mathrm{d}u + \big(\sqrt{2c_2}\,x^{1/4} + x^{1/2}\big)N^{\mathsf{e}_r}, \qquad \tilde{\mathsf{d}} \le 2x^{1/2}\log\big(1 + c_2x^{-1/2}\big)N^{\mathsf{e}_r}.$$

Adding eq. (SI.63) and using $\int_0^{u_0}\sqrt{f} + \int_{u_0}^{\infty}\sqrt{f} = J_N$,

$$\mathsf{I} \le J_N + a_1(x)N^{\mathsf{e}_r} + 2\sqrt{\big(J_N + a_2(x)N^{\mathsf{e}_r}\big)\tilde{\mathsf{d}}} + \tilde{\mathsf{d}}, \tag{SI.64}$$

with $a_1(x) \stackrel{\text{def}}{=} (c_1 + \sqrt{2c_2})x^{1/4} + (\tau+1)x^{1/2}$ and $a_2(x) \stackrel{\text{def}}{=} \sqrt{2c_2}\,x^{1/4} + x^{1/2}$.

The relative error is of order the square root of $\mathfrak{e}(x) \stackrel{\text{def}}{=} x^{1/4} + x^{1/2}\log(e + x^{-1/2})$. Since $\log(1 + c_2t) \le (1 + \log\max\{1, c_2\})\log(e+t)$ for $t \ge 0$, each of $a_1(x)$, $a_2(x)$ and $\tilde{\mathsf{d}}/N^{\mathsf{e}_r}$ is at most $c\,\mathfrak{e}(x)$

for $x \le 1$. By proposition SI.8(i),(ii), $c_J N^{\mathsf{e}_r} \le J_N \le c'_J N^{\mathsf{e}_r}$ for large N, with fixed $c_J, c'_J > 0$. Since $\mathfrak{e}$ is bounded on $(0,1]$, eq. (SI.64) gives

$$\frac{\mathsf{I} - J_N}{J_N} \le c\,\mathfrak{e}(x)^{1/2} \qquad (x \le 1).$$

The lower bound of eq. (SI.59) gives

$$\frac{J_N - \mathsf{I}}{J_N} \le \frac{\sqrt{T}\mathsf{y}}{J_N} = \frac{\sqrt{T}\,x^{1/2} N^{\mathsf{e}_r}}{J_N} \le c\,x^{1/2} \le c\,\mathfrak{e}(x)^{1/2}.$$

Here N_2 is the largest of N°, the threshold of proposition SI.8(ii), and the threshold for $D \le \tau N^q$.

In all three cases $\mathsf{I}_j / J_N(p, q_j) \to 1$, and no bound depends on B. The two equivalents follow from eqs. (SI.30) and (SI.41), and dividing $D_N \mathsf{I}_{\text{s-SGD}} / \mathsf{I}_{\text{s-SVD}}$ by $\rho_N = D_N J_N(p,\alpha)/J_N(p,\alpha/2)$ gives the ratio. □

The unified clock.

Proof of theorem SI.14(i). Write $q = q_j$, $r = r_j$, $f = f_{N,q}$ and $A = A_p$, and let $\epsilon < \epsilon_0(p,q,T)$. Attainment is eq. (SI.59), since $\epsilon_0 \le 1/(8T)$. Since $Y_j \ge f$, the first hit is not before $\inf\{u : f(u) \le T\epsilon\}$, so I_j is at least the integral of $\sqrt{f}$ up to that time. For $N \ge 4$ the lower envelope of eq. (SI.23) gives $f > T\epsilon$ on $[0, m)$, where

$$m \stackrel{\text{def}}{=} \min\{\sigma_*, V\}, \qquad \sigma_* \stackrel{\text{def}}{=} \Big(\frac{A}{T\epsilon}\Big)^{1/(2r)} - 1, \qquad V \stackrel{\text{def}}{=} (N/4)^q,$$

so $\mathsf{I}_j \ge \sqrt{A}\, I_r(m)$.

If $r < 1$: the last entry of ϵ_0 gives $(1+\sigma_*)^{1-r} \ge 2$, and for $(N/4)^{q(1-r)} \ge 2$ also $V^{1-r} \ge 2$. Hence

$$\mathsf{I}_j \ge \frac{\sqrt{A}}{2(1-r)} \min\Big\{\Big(\frac{A}{T}\Big)^{(1-r)/(2r)} \epsilon^{-(1-r)/(2r)},\ 4^{-q(1-r)} N^{q(1-r)}\Big\}.$$

If $r = 1$: $\epsilon < (A/T)^2$ gives $\log(1+\sigma_*) = \frac{1}{2}\log\frac{A}{T\epsilon} \ge \frac{1}{4}\log\frac{1}{\epsilon}$, and $N \ge 16$ gives $\log(1+V) \ge \frac{q}{2}\log N$. If $r > 1$: $f(u) \ge \frac{1}{2}e^{-2u}$, so the first hit is at least $\frac{1}{2}\log\frac{1}{2T\epsilon} > \log 2$, and

$$\mathsf{I}_j \ge \int_0^{\log 2} \frac{e^{-u}}{\sqrt{2}}\,\mathrm{d}u = \frac{1}{2\sqrt{2}}.$$

In each case $\mathsf{I}_j \ge c_- \Upsilon_j(N,\epsilon)$. □

Proof of theorem SI.14(ii) when $r_j < 1$. Use the phase-one data (D, τ, N°) and $u_0 = \tau N^q$, with $q = q_j$, $r = r_j$, $C = C_{p,q}$ and $h = C(1+u/D)^{-r}$. Put

$$E_* \stackrel{\text{def}}{=} \Big(\frac{DC^{1/r}}{\tau}\Big)^{2r}, \qquad \Pi \stackrel{\text{def}}{=} \epsilon^{-(1-r)/(2r)}.$$

By eq. (SI.40), $N^{q(1-r)} = x^{(1-r)/(2r)}\,\Pi$, which separates two cases.

Window $x \ge E_*$. Let $H \stackrel{\text{def}}{=} D[(C/\sqrt{\epsilon})^{1/r} - 1]$, which is positive for $\epsilon < C^2$. Since $\epsilon^{-1/(2r)} = x^{-1/(2r)} N^q$,

$$H \le DC^{1/r}\epsilon^{-1/(2r)} = DC^{1/r} x^{-1/(2r)} N^q \le \tau N^q = u_0,$$

and $h(H) = \sqrt{\epsilon}$. Lemma SI.9, under its hypothesis (a) on $[0,H]$, gives $\hat{u}_j \le H$ and $Y_j \le (h + \sqrt{\epsilon})^2$ there, so

$$\mathsf{I}_j \le \int_0^H \big(h + \sqrt{\epsilon}\big)\,\mathrm{d}u \le DC^{1/r}\Big[\frac{1}{1-r} + 1\Big]\Pi.$$

Since $x \ge E_*$, $N^{q(1-r)} \ge E_*^{(1-r)/(2r)}\Pi$, and therefore $\Upsilon_j \ge \min\{1, E_*^{(1-r)/(2r)}\}\,\Pi$.

Band $x < E_*$. The bound eq. (SI.64) was derived for every $x > 0$, and $a_1(x)$, $a_2(x)$ and $\tilde{\mathsf{d}}/N^{\mathsf{e}_r}$ are bounded on $(0, E_*]$. With proposition SI.8(i) this gives $\mathsf{I}_j \le c(E_*) N^{q(1-r)}$. Here $N^{q(1-r)} < E_*^{(1-r)/(2r)}\Pi$, so $\Upsilon_j \ge \min\{1, E_*^{-(1-r)/(2r)}\}\,N^{q(1-r)}$. □

For $r_j \geq 1$ the forcing envelope decays too fast for the slow part of the kernel, and the tail term w of lemma SI.20 absorbs the difference.

Proof of theorem SI.14(ii) for the templated models with $r_j \geq 1$. These are SIGNSVD for $0 < \alpha < 1$ and $1 < \alpha < 2$, and SIGNSGD for $\alpha \in (0, \frac{1}{2}) \cup (\frac{1}{2}, 1) \cup (1, \infty)$, with the constants of lemma SI.25; put $\upsilon \stackrel{\text{def}}{=} 2$ if $K_f = 0$. Write $q = q_j$, $r = r_j$, $C = C_{p,q}$, $\mathsf{y} = \mathsf{y}_j$, $J_N = J_N(p,q)$ and $\mathsf{I} = \mathsf{I}_j$.

Set-up. Let $r_0 \stackrel{\text{def}}{=} 1$ if $r = 1$ and $r_0 \stackrel{\text{def}}{=} \min\{r, (1+\upsilon)/2\}$ if $r > 1$, so that $r_0 \leq r$ and $r_0 \leq \upsilon$. Let $\delta_\circ \stackrel{\text{def}}{=} 1 - (2/3)^{1/r_0}$, $\delta_w \stackrel{\text{def}}{=} 1 - (2/3)^{2/\mathsf{b}}$,

$$D \stackrel{\text{def}}{=} \max\Big\{1, \big[4K_f\delta_\circ^{-\upsilon} b(r_0,\upsilon)\big]^{\frac{1}{\upsilon-1}}, \big[4K_f\delta_w^{-\upsilon}(1-\tfrac{\mathsf{b}}{2})^{-1}\big]^{\frac{1}{\upsilon-1}}\Big\},$$

$$\tau_w \stackrel{\text{def}}{=} \min\Big\{\frac{1}{2}, \Big(\frac{(1-\mathsf{b}/2)\delta_w^{\mathsf{b}}}{4K_s}\Big)^{\frac{1}{1-\mathsf{b}}}\Big\},$$

$u_0 \stackrel{\text{def}}{=} \tau_w N^q - D$, which is positive and at most $N^q - 1$ for large N, and $h \stackrel{\text{def}}{=} C(1+u/D)^{-r_0}$. Put $\mathsf{L} \stackrel{\text{def}}{=} 1/(r_0 - 1)$ if $r > 1$ and $\mathsf{L} \stackrel{\text{def}}{=} \Lambda_0 \stackrel{\text{def}}{=} \log(1 + u_0/D)$ if $r = 1$, $A_e \stackrel{\text{def}}{=} K_s\delta_\circ^{-\mathsf{b}} CD\mathsf{L}$, and

$$w(u) \stackrel{\text{def}}{=} \epsilon^{1/4} A_e^{1/2} N^{-q/2}\Big(\frac{D+u}{N^q}\Big)^{-\mathsf{b}/2}, \qquad \frac{w(u)}{\sqrt{\epsilon}} = A_e^{1/2}\,\mathsf{y}^{-1/4}\Big(\frac{D+u}{N^q}\Big)^{-\mathsf{b}/2}.$$

Hypotheses of lemma SI.22. The upper envelope of eq. (SI.23), $D \geq 1$ and $r_0 \leq r$ give $f \leq h^2$. Lemma SI.23 for h has $\mathsf{k}_f \leq \frac{1}{4}$. Its remainder is at most

$$K_s\delta_\circ^{-\mathsf{b}} N^{-q}\Big(\frac{D+u}{N^q}\Big)^{-\mathsf{b}} CD\mathsf{L} = \frac{w^2}{\sqrt{\epsilon}} \qquad \text{on } [0, u_0],$$

because $I_{r_0} \leq 1/(r_0 - 1)$ for $r_0 > 1$ and $I_1(u/D) \leq \Lambda_0$; so $g * h \leq 2h + w^2/\sqrt{\epsilon}$. Lemma SI.23 for w, an envelope of the same form with exponent $\mathsf{b}/2 < 1$, gives on $[0, u_0]$

$$g * w \leq \frac{7}{4}w + \frac{K_s\delta_w^{-\mathsf{b}}}{1-\mathsf{b}/2}\,\tau_w^{1-\mathsf{b}}\,w \leq 2w,$$

since $D + u \leq \tau_w N^q$ there.

Elementary integrals and levels. For $0 \leq z \leq u_0$,

$$\int_0^z w\,\mathrm{d}u \leq \frac{A_e^{1/2}\mathsf{y}^{1/4}}{1-\mathsf{b}/2}\Big(\frac{D+z}{N^q}\Big)^{1-\mathsf{b}/2}, \qquad \sqrt{\epsilon}\,z \leq \sqrt{\mathsf{y}}\,\frac{D+z}{N^q}. \tag{SI.65}$$

Further, $h(u) \leq \sqrt{\epsilon}/2$ for $u \geq H_h \stackrel{\text{def}}{=} D[(2C/\sqrt{\epsilon})^{1/r_0} - 1]$, and $w(u) \leq \sqrt{\epsilon}/2$ once $D + u \geq N^q(4A_e)^{1/\mathsf{b}}\mathsf{y}^{-1/(2\mathsf{b})}$; let $H_w \geq 0$ be the least such u and $H \stackrel{\text{def}}{=} \max\{H_h, H_w\}$. If $H \leq u_0$, lemma SI.22(a),(b) give $\hat{u} \leq H$ and $\mathsf{I} \leq \int_0^H (h + w + \sqrt{\epsilon})$. In every case lemma SI.22(c) and $\int_{u_0}^\infty \sqrt{f} \leq J_N$ give

$$\mathsf{I} \leq \int_0^{u_0} \big(h + w + \sqrt{\epsilon}\big)\,\mathrm{d}u + \big(\tilde{\mathsf{J}}^{1/2} + \tilde{\mathsf{d}}^{1/2}\big)^2, \qquad \tilde{\mathsf{J}} \leq J_N + \sqrt{\mathsf{y}}\,\Theta_0, \qquad \tilde{\mathsf{d}} \leq 2\sqrt{\mathsf{y}}\,\log\Theta_0. \tag{SI.66}$$

Saturated case $r > 1$. Here $J_N \leq C/(r-1)$, $\int_0^\infty h = CD/(r_0 - 1)$ and $\Upsilon_j = 1$. Let

$$Y_* \stackrel{\text{def}}{=} \max\Big\{1,\ (2C)^2\Big(\frac{D}{\tau_w}\Big)^{2r_0},\ (4A_e)^2\tau_w^{-2\mathsf{b}}\Big\}.$$

(a) Suppose $\mathsf{y} \geq Y_*$. Since $r_0 > 1$ and $\epsilon = \mathsf{y}N^{-2q}$,

$$D + H_h = D(2C)^{1/r_0}\mathsf{y}^{-1/(2r_0)}N^{q/r_0} \leq D(2C)^{1/r_0}\mathsf{y}^{-1/(2r_0)}N^q \leq \tau_w N^q,$$

and similarly $D + H_w \leq \tau_w N^q$ when $H_w > 0$; so $H \leq u_0$. Then $\sqrt{\epsilon}\,H_h \leq D(2C)^{1/r_0}$ because $\epsilon \leq 1$, and $\sqrt{\epsilon}\,H_w \leq (4A_e)^{1/\mathsf{b}}\mathsf{y}^{-(1-\mathsf{b})/(2\mathsf{b})} \leq (4A_e)^{1/\mathsf{b}}$. By eq. (SI.65), $\int_0^{H_w} w \leq c\,A_e^{1/\mathsf{b}}\mathsf{y}^{-(1-\mathsf{b})/(2\mathsf{b})} \leq c$, and

$$\int_0^{H_h} w\,\mathrm{d}u \leq c\,\mathsf{y}^\eta N^{-q(1-1/r_0)(1-\mathsf{b}/2)}, \qquad \eta \stackrel{\text{def}}{=} \frac{1}{4} - \frac{1-\mathsf{b}/2}{2r_0}.$$

If $\eta \le 0$ this is at most c because $\mathsf{y} \ge 1$. If $\eta > 0$, the bound $\mathsf{y} \le N^{2q}$ bounds it by $cN^{2q\eta - q(1-1/r_0)(1-\mathsf{b}/2)} = cN^{-q(1-\mathsf{b})/2} \le c$.

(b) Suppose $\mathsf{y} \le N^{-q(r_0-1)}$. Then eq. (SI.59) applies directly, with

$$\mathsf{d} = \sqrt{\mathsf{y}}\Big[\log F_0 + 2q\log N + \log\frac{1}{\mathsf{y}}\Big] \le N^{-q(r_0-1)/2}\Big(\Big|\log\frac{\zeta(p)}{2}\Big| + 2q\log N\Big) + \frac{2}{e} \le c,$$

so $\mathsf{I} \le (\sqrt{C/(r-1)} + \sqrt{c})^2$.

(c) Suppose $N^{-q(r_0-1)} < \mathsf{y} < Y_*$. Now

$$\frac{h(u_0)}{\sqrt{\epsilon}} = CD^{r_0}\tau_w^{-r_0}\,\mathsf{y}^{-1/2}N^{-q(r_0-1)} \le CD^{r_0}\tau_w^{-r_0}N^{-q(r_0-1)/2} \le 1$$

for large N, and $w(u_0)/\sqrt{\epsilon} = A_e^{1/2}\tau_w^{-\mathsf{b}/2}\mathsf{y}^{-1/4}$, so $\Theta_0 \le 2 + A_e^{1/2}\tau_w^{-\mathsf{b}/2}\mathsf{y}^{-1/4}$. By eq. (SI.65) the first phase contributes at most $CD/(r_0-1) + cY_*^{1/4} + \tau_w Y_*^{1/2}$. In eq. (SI.66), $\log(2+t) \le \log 3 + t$ gives

$$\tilde{\mathsf{d}} \le 2\sqrt{Y_*}\log 3 + 2A_e^{1/2}\tau_w^{-\mathsf{b}/2}Y_*^{1/4}, \qquad \tilde{\mathsf{J}} \le \frac{C}{r-1} + 2\sqrt{Y_*} + A_e^{1/2}\tau_w^{-\mathsf{b}/2}Y_*^{1/4}:$$

the factor $\mathsf{y}^{-1/4}$ in Θ_0 always meets a factor $\sqrt{\mathsf{y}}$. In all three cases I is bounded by a constant c_+.

Critical case $r = 1$. Here $r_0 = 1$, $p - 1 = 2q$, $A_e = c_e\Lambda_0$ with $c_e \stackrel{\text{def}}{=} K_s\delta_\circ^{-\mathsf{b}}CD$, $\Lambda_0 \le q\log N + c$, and $\Upsilon_j \le \log N$. Let

$$Y_c \stackrel{\text{def}}{=} \max\Big\{\Big(\frac{2CD}{\tau_w}\Big)^2,\ (4c_e\Lambda_0)^2\tau_w^{-2\mathsf{b}}\Big\},$$

so that $Y_c^{1/2} \le c\Lambda_0$ for large N.

(a) Suppose $\mathsf{y} < 1$. Then $\log(1/\epsilon) > 2q\log N$, so $\Upsilon_j \ge \min\{1, 2q\}\log N$. By eq. (SI.59), $\mathsf{d} \le |\log(\zeta(p)/2)| + 2q\log N + 2/e$, and $J_N \le \sqrt{q/8}\log N + c$ by lemma SI.12(i). So $\mathsf{I} \le 2J_N + 2\mathsf{d} \le c\log N$.

(b) Suppose $1 \le \mathsf{y} < Y_c$. Then $\log(1/\epsilon) = 2q\log N - \log\mathsf{y} \ge q\log N$ for large N, so $\Upsilon_j \ge \min\{1, q\}\log N$. The first phase contributes

$$\int_0^{u_0}\big(h + w + \sqrt{\epsilon}\big)\,\mathrm{d}u \le CD\Lambda_0 + c\,(c_e\Lambda_0)^{1/2}\mathsf{y}^{1/4} + \tau_w\mathsf{y}^{1/2} \le c\log N.$$

For the restart, $h(u_0)/\sqrt{\epsilon} = (CD/\tau_w)\mathsf{y}^{-1/2} \le CD/\tau_w$, so $\Theta_0 \le 1 + CD/\tau_w + (c_e\Lambda_0)^{1/2}\tau_w^{-\mathsf{b}/2}\mathsf{y}^{-1/4}$. With $\log(a+t) \le \log a + t$ for $a \ge 1$,

$$\tilde{\mathsf{d}} \le 2\sqrt{\mathsf{y}}\log\Big(1 + \frac{CD}{\tau_w}\Big) + 2(c_e\Lambda_0)^{1/2}\tau_w^{-\mathsf{b}/2}\mathsf{y}^{1/4} \le c\log N,$$

and $\tilde{\mathsf{J}} \le J_N + c\log N$. Hence $\mathsf{I} \le c\log N$.

(c) Suppose $\mathsf{y} \ge Y_c$. Then $H_h, H_w \le u_0$ exactly as in the saturated case (a), so $\hat{u} \le H$ and $\mathsf{I} \le \int_0^H (h + w + \sqrt{\epsilon})$. Now $\int_0^H h = CD\log(1 + H/D)$, with $\log(1 + H_h/D) = \log(2C) + \frac{1}{2}\log(1/\epsilon)$ and, writing $\mathsf{y} = \epsilon N^{2q}$,

$$\log(1 + H_w/D) \le \log 2 + \frac{1}{2\mathsf{b}}\log\frac{1}{\epsilon} + \Big[\frac{1}{\mathsf{b}}\log(4c_e\Lambda_0) - \frac{q(1-\mathsf{b})}{\mathsf{b}}\log N - \log D\Big]_+,$$

whose bracket vanishes for large N. Also $\sqrt{\epsilon}\,H_h \le 2CD$, $\sqrt{\epsilon}\,H_w + \int_0^{H_w} w \le c\,(c_e\Lambda_0)^{1/\mathsf{b}}\mathsf{y}^{-(1-\mathsf{b})/(2\mathsf{b})}$, and $\int_0^{H_h} w \le c\,(c_e\Lambda_0)^{1/2}\mathsf{y}^{-(1-\mathsf{b})/4}$. If $\mathsf{y} \le N^q$, then $\log(1/\epsilon) \ge q\log N \ge c\Lambda_0$, and $\mathsf{y} \ge Y_c$ bounds the last two terms by $c\Lambda_0$ and $c\Lambda_0^{1/2}$. If $\mathsf{y} > N^q$, they are at most $c(\log N)^{1/\mathsf{b}}N^{-q(1-\mathsf{b})/(2\mathsf{b})} + c(\log N)^{1/2}N^{-q(1-\mathsf{b})/4} \le c$. In both cases $\mathsf{I} \le c\log(1/\epsilon) + c \le c'\log(1/\epsilon)$, because $\log(1/\epsilon) \ge \log(1/\epsilon_0) > 0$. Finally $\mathsf{y} \ge 1$ gives $\log(1/\epsilon) \le 2q\log N$, so $\mathsf{I} \le c''\Upsilon_j$. □

Proof of theorem SI.14(ii) for SIGNSVD *at* $\alpha \geq 2$, $\varsigma \geq 1$. Write $q = \alpha/2$, $s = r_{\text{s-SVD}} \geq 1$, $C = C_{p,q}$, $g = g_{\text{s-SVD}}$, $\mathsf{y} = \mathsf{y}_{\text{s-SVD}}$ and $\mathsf{I} = \mathsf{I}_{\text{s-SVD}}$.

Case $\alpha > 2$. By lemma SI.7(i) and $H_N(\alpha/2) \geq 1$, $g(v) \leq 2E_{\alpha,\alpha/2}(1+v)^{-\upsilon_{\mathrm{V}}}$ for all $v \geq 0$, with $\upsilon_{\mathrm{V}} = 2 - \frac{2}{\alpha} > 1$. Take $r_0 = 1$ if $s = 1$ and $r_0 = \min\{s, (1+\upsilon_{\mathrm{V}})/2\}$ if $s > 1$, and $D = D(2E_{\alpha,\alpha/2}, r_0, \upsilon_{\mathrm{V}})$ of eq. (SI.55). Lemma SI.14 gives $g * h \leq 2h$ on $[0,\infty)$ for $h = C(1+u/D)^{-r_0}$. Lemma SI.9(i),(ii), under its hypothesis (a) at $H = D[(C/\sqrt{\epsilon})^{1/r_0} - 1]$, give $\hat{u} \leq H$ and

$$\mathsf{I} \leq \int_0^H \left(h + \sqrt{\epsilon}\right) \mathrm{d}u = CD\, I_{r_0}(H/D) + \sqrt{\epsilon}\, H.$$

For $s > 1$ this is at most $CD/(r_0 - 1) + DC^{1/r_0}$. For $s = 1$ it is at most $CD\log(C/\sqrt{\epsilon}) + DC$. For $s = 1$ and $\mathsf{y} \geq 1$, $\log(1/\epsilon) \leq \alpha \log N$, so $\mathsf{I} \leq c\Upsilon_{\text{s-SVD}}$; for $s = 1$ and $\mathsf{y} < 1$, case (a) of the critical case above applies verbatim, since it uses only eq. (SI.59) and lemma SI.12(i).

Case $\alpha = 2$, $\varsigma > 1$. By lemma SI.18, for $N \geq e^{L_0}$ and $\epsilon < \epsilon_0$ the hit is attained and the noise error $\mathcal{E}$ there is bounded by a constant, since each of its five terms is. Hence $\mathsf{I} \leq J_N(p,1) + \mathcal{E} \leq J(p,1) + c$.

Case $\alpha = 2$, $\varsigma = 1$. Here $q = 1$; take $D = 1$ and $h = C/(1+u)$. Lemma SI.17(iii) gives, for $N \geq e^{64}$,

$$g * h \leq \frac{2\log(1+u) + 64}{H_N(1)}\, h \leq 2h \qquad \text{on } [0, u_0], \qquad u_0 \stackrel{\text{def}}{=} e^{-32}(N+1) - 1,$$

because $H_N(1) \geq \log(N+1) \geq \log(1+u) + 32$ there. If $\mathsf{y} < 1$, case (a) of the critical case applies. If $\mathsf{y} \geq 1$ and $H \stackrel{\text{def}}{=} C/\sqrt{\epsilon} - 1 \leq u_0$, lemma SI.9 gives $\hat{u} \leq H$ and

$$\mathsf{I} \leq C\log(C/\sqrt{\epsilon}) + C \leq c\log(1/\epsilon) \leq c\,\Upsilon_{\text{s-SVD}}.$$

If $\mathsf{y} \geq 1$ and $H > u_0$, then $C/\sqrt{\epsilon} > e^{-32}N$, so $\mathsf{y} < C^2 e^{64}$ and $\log(1/\epsilon) \geq 2\log N - c$. The first phase contributes at most $C\log(N+1) + 2e^{-32}\sqrt{\mathsf{y}}$, and $\Theta_0 \leq 1 + Ce^{32}\mathsf{y}^{-1/2} \leq 1 + Ce^{32}$ in lemma SI.22(c), with $w = 0$, gives $\tilde{\mathsf{d}} \leq c$ and $\tilde{\mathsf{J}} \leq J_N + c$. By lemma SI.12(i), $\mathsf{I} \leq c\log N \leq c'\Upsilon_{\text{s-SVD}}$. □

Consequences.

Proof of corollary SI.5. For $\alpha > 1$ every triple (α, ς, j) lies in $\mathcal{U}$. Let $x = \epsilon/\epsilon_N \geq 1$. If $r_j < 1$, then

$$\epsilon^{-(1-r_j)/(2r_j)} = x^{-(1-r_j)/(2r_j)} N^{q_j(1-r_j)} \leq N^{q_j(1-r_j)},$$

so $\Upsilon_j = \epsilon^{-(1-r_j)/(2r_j)}$ and $\epsilon^{-1/2}\Upsilon_j = \Pi_{r_j}(\epsilon)$. If $r_j = 1$, then $p - 1 = 2q_j$ gives $\mathsf{y}_j = x \geq 1$, so $\log(1/\epsilon) \leq 2q_j \log N$ and $\Upsilon_j \asymp \log(1/\epsilon)$. If $r_j > 1$, then $\Upsilon_j = 1$. Apply theorem SI.14 and eq. (SI.30). □

Proof of corollary SI.6. Divide the bounds of theorem SI.14, using eq. (SI.41); the lower bounds hold for every model. If $r_j < 1$, then $1/(2r_j) > \frac{1}{2}$, so $e_j = 1/(2r_j)$ and $(1-r_j)/(2r_j) = e_j - \frac{1}{2}$; by eq. (SI.46), $\Upsilon_j = \max\{\epsilon, \epsilon_N\}^{-(e_j - 1/2)}$. If $r_j > 1$, then $e_j = \frac{1}{2}$ and $\Upsilon_j = 1$, which is the same formula. For $\varsigma \notin \{1, \alpha+1\}$ neither r_j equals 1, so

$$\frac{\Upsilon_{\text{s-SGD}}}{\Upsilon_{\text{s-SVD}}} = \max\{\epsilon, \epsilon_N\}^{-(e_{\text{s-SGD}} - e_{\text{s-SVD}})} = \max\{\epsilon, \epsilon_N\}^{-\chi}.$$

For $\epsilon = N^{-\omega}$ and $r_j < 1$,

$$\Upsilon_j = N^{\min\{\omega, p-1\}(1-r_j)/(2r_j)},$$

because $q_j(1-r_j) = (p-1)(1-r_j)/(2r_j)$. For $r_j \geq 1$, $\Upsilon_j \asymp (\log N)^{[r_j=1]}$, with constants depending on ω. Hence

$$\frac{\Upsilon_{\text{s-SGD}}}{\Upsilon_{\text{s-SVD}}} \asymp N^{\chi \min\{\omega, p-1\}} (\log N)^{[\varsigma=\alpha+1]-[\varsigma=1]},$$

and multiplying by $D_N \asymp N^{e_D}(\log N)^{[\alpha=1]-[\alpha=2]}$ gives the last display. □

Proof of theorem SI.15. For $\omega > p-1$, $x = N^{-(\omega-(p-1))} \to 0$, and theorem SI.13 applies. Both models lie in $\mathcal{U}$ exactly when $(\alpha, \varsigma) \notin \mathsf{E}$: SIGNSVD fails only at $\alpha = 1$, $\varsigma \geq 1$, and SIGNSGD only at $\alpha \in \{\frac{1}{2}, 1\}$, $\varsigma \geq \alpha + 1$. For $\alpha = 1$ and $1 \leq \varsigma < 2$ only SIGNSGD lies in $\mathcal{U}$. Apply corollary SI.6; for large N, $N^{-\omega} < \epsilon_0$. □

Proof of corollary SI.7. For $(\alpha, \varsigma) \notin \mathsf{E}$ use theorem SI.15(ii): the winner is SIGNSVD if $S(\omega) > 0$ and SIGNSGD if $S(\omega) < 0$; if $S(\omega) = 0$, it is SIGNSVD if $L > 0$, SIGNSGD if $L < 0$, and the two are balanced if $L = 0$. Write $\omega' \stackrel{\text{def}}{=} \min\{\omega, p-1\}$ and use eq. (SI.48).

For $0 < \alpha < 1$, $e_D = -\frac{\alpha}{2}$. If $\varsigma < 1$, then $S(\omega) = \frac{\alpha}{2}(\omega'/(p-1) - 1)$ and $L = 0$, which gives Phase D, with the limit K_A from theorem SI.15(i). If $\varsigma \geq 1$, then $\alpha + 1 - \varsigma \leq \alpha$, so $S(\omega) \leq 0$, with equality only at $\varsigma = 1$, $\omega \geq p-1$, where $L = -1$; this gives the first part of Phase B.

For $\alpha = 1$, $e_D = -\frac{1}{2}$ and $[\alpha = 1] = 1$. If $\varsigma < 1$, then $S(\omega) = \frac{1}{2}(\omega'/(p-1) - 1)$ and $L = 1$. If $1 \leq \varsigma < 2$, use theorem SI.15(iii): $S(\omega) = -\frac{1}{2} + \frac{\omega'}{p-1} \cdot \frac{2-\varsigma}{2} < 0$ unless $\varsigma = 1$ and $\omega \geq p-1$, so the upper bound forces $\mathsf{R} \to 0$ for $\omega < p-1$ and for $1 < \varsigma < 2$. At $\varsigma = 1$ and $\omega > 1$ use theorem SI.15(i) and theorem SI.12(b).

For $1 < \alpha < 2$, $e_D = \frac{\alpha}{2} - 1 < 0$ and $L = -[\varsigma = 1]$, since $\varsigma = \alpha + 1$ gives $S(\omega) = e_D < 0$. If $\varsigma < 1$, then $S(\omega) > 0$ exactly when $\omega' > (2-\alpha)(p-1)/\alpha = \omega_c$, and $\omega_c < p-1$. If $1 \leq \varsigma \leq \alpha + 1$, then

$$S(\omega) = \frac{\alpha}{2} - 1 + \frac{\omega'}{p-1} \cdot \frac{\alpha + 1 - \varsigma}{2},$$

which vanishes at $\omega' = \omega_c$; and $\omega_c < p-1$ exactly when $\varsigma < 2\alpha - 1$. On the chord $S(\omega) < 0$ for $\omega < p-1$ and $S(\omega) = 0$, $L = 0$ for $\omega \geq p-1$, with the limit K_c from theorem SI.15(i). Above the chord and for $\varsigma > \alpha + 1$, $S(\omega) < 0$. This gives Phase C, the chord and the second part of Phase B.

For $\alpha = 2$, $e_D = 0$; $S(\omega) > 0$ for $\varsigma < 3$, and $S(\omega) = 0$ for $\varsigma \geq 3$ with $L = -1 + [\varsigma = 3]$. For $\alpha > 2$, $e_D = 0$; $S(\omega) > 0$ for $\varsigma < \alpha + 1$, and for $\varsigma \geq \alpha + 1$, $S(\omega) = 0$ with $L = [\varsigma = \alpha + 1]$. This gives Phase A and the Tie. □

SJ SIGNSGD: drift and volatility

This section computes the drift and the volatility of one step of SIGNSGD, the comparator of the main text, defined in eq. (5). SIGNSGD is ADAM at $\beta_1 = \beta_2 = 0$; it keeps no buffer, so this section has no momentum version, and the momentum β of the other sections of the appendix does not enter it. Every statement concerns one step from a given state with one fresh batch; none is a statement about a trajectory. Appendix SJ.1 fixes the update and its exact risk identity. Appendix SJ.2 proves the Gaussian sign tools, among them the bounds on the batch constant $\mathcal{N}_B$ and the exact isotropic volatility. Appendix SJ.3 proves the isotropic drift for the actual batch with an explicit error and deduces the isotropic recursion proposition SJ.4. Appendix SJ.4 introduces an explicit law for the state in a Haar eigenbasis (definition SJ.3) and proves under it the half-anisotropic drift (theorem SJ.2), the covariance-weighted volatility (theorem SJ.3) and the corrected row volatility (theorem SJ.4), and from them the half-anisotropic recursion proposition SJ.8.

Standing conventions. The sign of a real number is 1, -1 or 0 according as the number is positive, negative or zero, and sign of a matrix acts entrywise. The model, risk and moment identities are those of definition SA.1, Assumption S1 and proposition SA.1, and the conventions of appendix SA.1 apply. Outer products are written $x \otimes y$. For a matrix Δ we write $\Delta_{i\cdot}$ for its ith row and $\Delta_{\cdot j}$ for its jth column, both as vectors. The step $\eta \geq 0$ is deterministic. Unless a subscript says otherwise, an expectation integrates the fresh batch and, in appendix SJ.4, also the Haar matrix U; such expectations are called annealed. A constant is called universal if it does not depend on any parameter of the model.

SJ.1 The update and its risk identity

For a state $\Delta \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ and a batch $(x_{\text{in}}^{(a)}, x_{\text{out}}^{(a)})_{a \leq B}$ as in definition SA.1, the physical gradient is $G^\circ = \frac{1}{B} \sum_a d_a^\circ \, x_{\text{out}}^{(a)} \otimes x_{\text{in}}^{(a)}$ with $d_a^\circ = (x_{\text{out}}^{(a)})^\top \Delta x_{\text{in}}^{(a)}$, as in proposition SA.1(ii).

Definition SJ.1 (SIGNSGD step)**.** *The* SIGNSGD *step from the state* Δ *with step* $\eta \geq 0$ *is*

$$\Delta^+ \stackrel{\text{def}}{=} \Delta - \eta S, \qquad S \stackrel{\text{def}}{=} \operatorname{sign}(G^\circ). \tag{SJ.1}$$

The sign does not see positive scalars, so $S = \operatorname{sign}(G_{t+1})$ for the normalized gradient of eq. (SA.10) whenever the risk is positive. The update S is determined by the state and the batch, and $\|S\|_{\mathrm{F}}^2 \leq N_{\mathrm{out}} N_{\mathrm{in}}$. Hence proposition SA.4 applies to it, with S as the update, and gives

$$\mathbb{E}_t\left[\mathscr{R}(t+1)\right] = \mathscr{R}(t) - \eta\big\langle \Delta_t, \mathbb{E}_t\, S\big\rangle_\Sigma + \frac{\eta^2}{2}\, \mathbb{E}_t\, \|S\|_\Sigma^2, \tag{SJ.2}$$

$$\mathbb{E}_t\left[\mathscr{Q}_i(\Delta_{t+1})\right] = \mathscr{Q}_i(\Delta_t) - 2\eta\, \mathbb{E}_t\left\langle \Delta_t^\top u_i, S^\top u_i\right\rangle + \eta^2\, \mathbb{E}_t\, \|S^\top u_i\|^2. \tag{SJ.3}$$

The second line is eq. (SA.21) with $\sum_j \mathscr{D}_{ij}(t) = \mathbb{E}_t\langle \Delta_t^\top u_i, S^\top u_i\rangle$ and $\sum_j \mathcal{V}_{ij}(t) = \mathbb{E}_t\, \|S^\top u_i\|^2$. The row energy is unhalved, so its quadratic term carries η^2 and not $\eta^2/2$.

The batch constant of the main text is

$$\mathcal{N}_B \stackrel{\text{def}}{=} \mathbb{E}\left[\Big(\sum_{b=1}^{B} x_b^2 y_b^2\Big)^{1/2}\right], \qquad x_b, y_b \text{ independent standard Gaussian.} \tag{SJ.4}$$

SJ.2 Gaussian sign tools

Lemma SJ.1 (Arcsine law)**.** *Let* (g_1, g_2) *be a centered jointly Gaussian pair with variances* $v_1, v_2 > 0$ *and correlation* $\rho = \mathbb{E}[g_1 g_2]/\sqrt{v_1 v_2} \in [-1, 1]$*. Then*

$$\mathbb{E}\left[\operatorname{sign}(g_1)\operatorname{sign}(g_2)\right] = \frac{2}{\pi}\arcsin\rho. \tag{SJ.5}$$

If $v_1 = 0$ *or* $v_2 = 0$*, the left side is* 0*.*

Proof. Sheppard's formula [60], in the form of [64, Theorem 40], states that for ρ-correlated standard Gaussians g, h one has

$$\mathbb{P}\left(\operatorname{sign}(g) \neq \operatorname{sign}(h)\right) = \frac{\arccos\rho}{\pi}.$$

Its hypothesis holds for $g = g_1/\sqrt{v_1}$ and $h = g_2/\sqrt{v_2}$, and dividing by the positive standard deviations does not change the signs. Each g_i has a density, so the signs are ± 1 almost surely and

$$\mathbb{E}\left[\operatorname{sign}(g_1)\operatorname{sign}(g_2)\right] = 1 - 2\,\mathbb{P}\left(\operatorname{sign}(g_1) \neq \operatorname{sign}(g_2)\right) = 1 - \frac{2}{\pi}\arccos\rho = \frac{2}{\pi}\arcsin\rho.$$

A centered Gaussian of variance zero vanishes almost surely, and its sign is then 0. □

Lemma SJ.2 (Small shifts)**.** *Let* X *have an even density* f*, and put* $h(x) \stackrel{\text{def}}{=} \mathbb{E}\operatorname{sign}(X + x)$ *for real* x*.*

(i) For every real x*, with oriented integrals,*

$$h(x) = 2\int_0^x f(y)\,\mathrm{d}y.$$

(ii) If $a > 0$*,* $\alpha > 0$*,* $L_f \geq 0$ *and* $|f(y) - f(0)| \leq L_f |y|^\alpha$ *for* $|y| \leq a$*, then*

$$|h(x) - 2x f(0)| \leq \frac{2L_f}{\alpha+1}|x|^{\alpha+1} \qquad (|x| \leq a).$$

(iii) If f *is continuously differentiable on* $[-a, a]$ *with a* K_f*-Lipschitz derivative, then* $|h(x) - 2x f(0)| \leq \frac{K_f}{3}|x|^3$ *for* $|x| \leq a$*.*

(iv) If $X \sim \mathcal{N}(0, \sigma^2)$ *with* $\sigma > 0$*, then for every real* x

$$\left|h(x) - \sqrt{\frac{2}{\pi}}\,\frac{x}{\sigma}\right| \leq \frac{|x|^3}{3\sqrt{2\pi}\,\sigma^3}. \tag{SJ.6}$$

Proof. The identity (i) splits the law at the shifted origin. Symmetry puts mass $\frac{1}{2}$ on each half-line, so

$$h(x) = \mathbb{P}(X > -x) - \mathbb{P}(X < -x) = 1 - 2\,\mathbb{P}(X < -x) = 2\int_{-x}^{0} f(y)\,\mathrm{d}y = 2\int_{0}^{x} f(y)\,\mathrm{d}y.$$

The bound (ii) integrates the modulus of f in (i):

$$|h(x) - 2xf(0)| \le 2\Big|\int_0^x |f(y) - f(0)|\,\mathrm{d}y\Big| \le 2L_f \int_0^{|x|} y^\alpha\,\mathrm{d}y = \frac{2L_f}{\alpha+1}|x|^{\alpha+1}.$$

For (iii), evenness forces $f'(0) = 0$. The Lipschitz bound then gives $|f'(y)| \le K_f|y|$, and integrating once,

$$|f(y) - f(0)| \le \frac{K_f}{2}\,y^2 \qquad (|y| \le a).$$

This is (ii) with $\alpha = 2$ and $L_f = K_f/2$.

For (iv), the Gaussian density $f_\sigma(y) = f_\sigma(0)e^{-y^2/(2\sigma^2)}$ satisfies, by $0 \le 1 - e^{-u} \le u$ for $u \ge 0$,

$$|f_\sigma(y) - f_\sigma(0)| \le \frac{f_\sigma(0)}{2\sigma^2}\,y^2 \qquad (y \in \mathbb{R}).$$

This is (ii) with $\alpha = 2$, $L_f = f_\sigma(0)/(2\sigma^2)$ and every a, and $2f_\sigma(0) = \sqrt{2/\pi}/\sigma$. □

Remark SJ.1 (Why a regularity hypothesis)**.** *Symmetry alone gives a quadratic error at best. For the Laplace density $f(y) = \frac{1}{2}e^{-|y|}$, (i) gives $h(x) = \operatorname{sign}(x)(1 - e^{-|x|})$, and*

$$\frac{h(x) - 2xf(0)}{x^2} = \frac{1 - e^{-x} - x}{x^2} \longrightarrow -\frac{1}{2} \qquad (x \downarrow 0).$$

Proposition SJ.1 (The batch constant)**.** *For every integer $B \ge 1$, the constant eq. (SJ.4) satisfies $\mathcal{N}_1 = 2/\pi$ and*

$$\max\Big\{\frac{2}{\pi}\sqrt{B},\ \sqrt{B} - \frac{4}{\sqrt{B}}\Big\} \le \mathcal{N}_B \le \sqrt{B}, \qquad 0 \le 1 - \frac{\mathcal{N}_B}{\sqrt{B}} \le \frac{4}{B}. \tag{SJ.7}$$

In particular $\mathcal{N}_B/\sqrt{B} \to 1$ as $B \to \infty$.

Proof. Put $T_B \overset{\text{def}}{=} \sum_b x_b^2 y_b^2$, so that $\mathcal{N}_B = \mathbb{E}\sqrt{T_B}$. A standard Gaussian g has $\mathbb{E}|g| = \sqrt{2/\pi}$, and independence gives

$$\mathcal{N}_1 = \mathbb{E}|x_1|\ \mathbb{E}|y_1| = \frac{2}{\pi}.$$

The normalized sum T_B/B has mean 1 and variance $8/B$. Indeed the summands $x_b^2 y_b^2$ are independent with mean 1 and second moment $(\mathbb{E}\,x_b^4)(\mathbb{E}\,y_b^4) = 9$, so

$$\mathbb{E}\,\frac{T_B}{B} = 1, \qquad \mathbb{E}\Big(\frac{T_B}{B} - 1\Big)^2 = \frac{9-1}{B} = \frac{8}{B}.$$

The upper bound is Jensen's inequality: $\mathcal{N}_B \le (\mathbb{E}\,T_B)^{1/2} = \sqrt{B}$.

The Jensen deficit is at most half the variance. For every $s \ge 0$, rationalizing $\sqrt{s} - 1$ gives the identity

$$\sqrt{s} = 1 + \frac{s-1}{2} - \frac{(s-1)^2}{2(\sqrt{s}+1)^2}.$$

The last term lies between 0 and $(s-1)^2/2$. Take $s = T_B/B$ and expectations; the linear term has mean zero, so

$$0 \le 1 - \mathbb{E}\sqrt{T_B/B} \le \frac{1}{2}\cdot\frac{8}{B} = \frac{4}{B},$$

which is the second part of eq. (SJ.7) and gives $\mathcal{N}_B \ge \sqrt{B} - 4/\sqrt{B}$.

The remaining lower bound compares the Euclidean and the ℓ^1 norms. For real $z_1, \dots, z_B$,

$$\Big(\sum_b z_b^2\Big)^{1/2} \ge \frac{1}{\sqrt{B}}\sum_b |z_b|.$$

With $z_b = x_b y_b$ and expectations, $\mathcal{N}_B \ge B^{-1/2}\cdot B\cdot\frac{2}{\pi} = \frac{2}{\pi}\sqrt{B}$. □

The isotropic volatility is exact. It needs every gradient entry to be nonzero, which follows from the fact that the zero set of a real polynomial on $\mathbb{R}^d$ that is not identically zero has Lebesgue measure zero; this is the polynomial case of [46], since polynomials are real analytic.

Proposition SJ.2 (Isotropic volatility)**.** *Let $\Sigma_{\text{in}} = I_{\text{in}}$ and $\Sigma_{\text{out}} = I_{\text{out}}$ in Assumption S1, let $\Delta \in \mathbb{R}^{N_{\text{out}} \times N_{\text{in}}}$ be deterministic, and let $S = \operatorname{sign}(G^\circ)$ for a batch of size $B \geq 1$. Then*

$$\mathbb{E}\,\|S\|_\Sigma^2 = \mathbb{E}\,\|S\|_{\text{F}}^2 = \begin{cases} N_{\text{in}} N_{\text{out}}, & \Delta \neq 0, \\ 0, & \Delta = 0. \end{cases} \tag{SJ.8}$$

In particular the step eq. (SJ.1) satisfies

$$\mathbb{E}\,\mathscr{R}(\Delta^+) = \mathscr{R}(\Delta) - \eta\,\langle \Delta, \mathbb{E}\,S\rangle_{\text{F}} + \frac{\eta^2}{2}\, N_{\text{in}} N_{\text{out}}\, \mathbf{1}_{\{\Delta \neq 0\}}. \tag{SJ.9}$$

Proof. With identity covariances, $\|S\|_\Sigma^2 = \operatorname{Tr}(SS^\top) = \|S\|_{\text{F}}^2$ by eq. (SA.2), so the first equality holds pointwise.

If $\Delta = 0$ then $G^\circ = 0$ and $S = 0$.

If $\Delta \neq 0$, every entry of G° is nonzero almost surely. Indeed BG°_{ij} is a polynomial in the batch coordinates. Setting all samples but the first to zero leaves the polynomial

$$(z_1)_i\,(w_1)_j\,\big(z_1^\top \Delta w_1\big),$$

a product of three polynomials that are not identically zero, the last one because some $\Delta_{kl} \neq 0$. A product of polynomials that are not identically zero is not identically zero. The batch has a Gaussian density, so the zero set of this polynomial has probability zero [46]. A finite union over the entries shows that $\|S\|_{\text{F}}^2 = N_{\text{in}} N_{\text{out}}$ almost surely.

The risk identity is eq. (SJ.2) at a deterministic state, with $\langle \Delta, \mathbb{E}\,S\rangle_\Sigma = \langle \Delta, \mathbb{E}\,S\rangle_{\text{F}}$ for identity covariances. □

SJ.3 Isotropic drift

Throughout this subsection $\Sigma_{\text{in}} = I_{\text{in}}$, $\Sigma_{\text{out}} = I_{\text{out}}$, the state $\Delta \neq 0$ is deterministic, and

$$F \overset{\text{def}}{=} \|\Delta\|_{\text{F}} = \sqrt{2\mathscr{R}(\Delta)}, \qquad G^\circ = \frac{1}{B}\sum_{a=1}^{B} \big(z_a^\top \Delta w_a\big)\, z_a \otimes w_a.$$

Fix an entry (i, j). Write $\bar{\Delta}$ for Δ with row i and column j deleted, $r \overset{\text{def}}{=} (\Delta_{il})_{l \neq j}$ for the rest of row i and $q \overset{\text{def}}{=} (\Delta_{kj})_{k \neq i}$ for the rest of column j. For one sample (z, w) write $\xi \overset{\text{def}}{=} z_i$, $\zeta \overset{\text{def}}{=} w_j$, and z', w' for the remaining coordinates; the four blocks ξ, ζ, z', w' are independent standard Gaussian. The single-sample coordinate is

$$X \overset{\text{def}}{=} \big(z^\top \Delta w\big)\,\xi\zeta = \Delta_{ij}\,\xi^2\zeta^2 + \xi^2\zeta\,(r^\top w') + \xi\zeta^2\,(q^\top z') + \xi\zeta\,\big(z'^\top \bar{\Delta} w'\big), \tag{SJ.10}$$

and BG°_{ij} is the sum of B independent copies of X.

Lemma SJ.3 (Moments of one coordinate)**.** *The coordinate eq. (SJ.10) has mean $\mathbb{E}\,X = \Delta_{ij}$, variance*

$$s^2 \overset{\text{def}}{=} \operatorname{Var} X = F^2 + 2\|\Delta_{i\cdot}\|^2 + 2\|\Delta_{\cdot j}\|^2 + 3\Delta_{ij}^2, \qquad F^2 \leq s^2 \leq 8F^2,$$

and third centered moment

$$m_3 \overset{\text{def}}{=} \mathbb{E}(X - \Delta_{ij})^3 = 24\Delta_{ij}\|\bar{\Delta}\|_{\text{F}}^2 + 126\Delta_{ij}\big(\|r\|^2 + \|q\|^2\big) + 200\Delta_{ij}^3 + 54\, q^\top \bar{\Delta} r,$$

so that $|m_3| \leq 200|\Delta_{ij}|F^2 + 54F\|r\|\|q\|$. For every integer $p \geq 1$, $\mathbb{E}\,|X|^p \leq ((2p-1)!!)^2 F^p$, and the standardized coordinate $Y \overset{\text{def}}{=} (X - \Delta_{ij})/s$ satisfies $\mathbb{E}\,Y^4 \leq 88208$.

Proof. Call the four terms of eq. (SJ.10) $T_1, \ldots, T_4$. The moments are Gaussian pairings [29]; a product with an odd total degree in one of the independent blocks has mean zero. Only T_1 has even degree in every block, so $\mathbb{E}\,X = \Delta_{ij}\,\mathbb{E}\,\xi^2\,\mathbb{E}\,\zeta^2 = \Delta_{ij}$.

The second moment is the sum of the four squares, because every cross product has odd degree in z' or in w':

$$\mathbb{E}\, X^2 = 9\Delta_{ij}^2 + 3\|r\|^2 + 3\|q\|^2 + \|\bar{\Delta}\|_{\mathrm{F}}^2.$$

With $F^2 = \|\bar{\Delta}\|_{\mathrm{F}}^2 + \|r\|^2 + \|q\|^2 + \Delta_{ij}^2$, $\|\Delta_{i\cdot}\|^2 = \|r\|^2 + \Delta_{ij}^2$ and $\|\Delta_{\cdot j}\|^2 = \|q\|^2 + \Delta_{ij}^2$, subtracting Δ_{ij}^2 gives the formula for s^2. Each added term is at most F^2 times its coefficient, so $F^2 \le s^2 \le 8F^2$.

The third moment keeps T_1^3, the terms $3T_1T_k^2$ for $k = 2, 3, 4$, and $6T_2T_3T_4$. With $\mathbb{E}\,\xi^4 = 3$ and $\mathbb{E}\,\xi^6 = 15$,

$$\mathbb{E}\, X^3 = 225\Delta_{ij}^3 + 135\Delta_{ij}\big(\|r\|^2 + \|q\|^2\big) + 27\Delta_{ij}\|\bar{\Delta}\|_{\mathrm{F}}^2 + 54\, q^\top \bar{\Delta} r.$$

Here $54 = 6\ \mathbb{E}\,\xi^4\,\mathbb{E}\,\zeta^4$ and $\mathbb{E}[(r^\top w')(q^\top z')(z'^\top \bar{\Delta} w')] = q^\top \bar{\Delta} r$. The centered moment is $m_3 = \mathbb{E}\, X^3 - 3\Delta_{ij}\,\mathbb{E}\, X^2 + 2\Delta_{ij}^3$, which gives the stated formula. Its coefficients $24, 126, 200$ are at most 200, and $|q^\top \bar{\Delta} r| \le F\|q\|\|r\|$, which gives the bound.

For the absolute moments, put $d = z^\top \Delta w$. Conditionally on w, d is centered Gaussian with variance $\|\Delta w\|^2$, so

$$\mathbb{E}\, d^{2p} = (2p-1)!!\ \mathbb{E}\,\|\Delta w\|^{2p} \le \big((2p-1)!!\big)^2 F^{2p}.$$

The inequality holds because $\|\Delta w\|^2 = F^2 \sum_l \pi_l g_l^2$ for weights $\pi_l \ge 0$ summing to 1 (the normalized eigenvalues of $\Delta^\top \Delta$) and independent standard Gaussian g_l, and Jensen's inequality gives $(\sum_l \pi_l g_l^2)^p \le \sum_l \pi_l g_l^{2p}$. The Cauchy–Schwarz inequality, which needs no independence of d and $\xi\zeta$, gives

$$\mathbb{E}\,|X|^p \le \big(\mathbb{E}\, d^{2p}\ \mathbb{E}\,|\xi\zeta|^{2p}\big)^{1/2} \le \big((2p-1)!!\big)^2 F^p.$$

Finally $(a+b)^4 \le 8(a^4+b^4)$, $\mathbb{E}\, X^4 \le 105^2 F^4 = 11025F^4$ and $s \ge F$ give

$$\mathbb{E}\, Y^4 \le \frac{8\,(\mathbb{E}\, X^4 + \Delta_{ij}^4)}{s^4} \le 8\,(11025 + 1) = 88208. \qquad \square$$

The characteristic function of Y decays at a rate that depends only on the size of the deleted block $\bar{\Delta}$. Let ξ, ζ, g be independent standard Gaussian scalars and put

$$\psi(x) \stackrel{\text{def}}{=} \mathbb{E}\,\big(1 + x^2\xi^2\zeta^2\big)^{-1/2} \quad (x \ge 0), \qquad C_{\mathrm{ch}} \stackrel{\text{def}}{=} \sqrt{8}\,\big(\mathbb{E}\,|g|^{-1/2}\big)^2. \tag{SJ.11}$$

The function ψ is nonincreasing, $\psi(x) < 1$ for $x > 0$, and $\mathbb{E}\,|g|^{-1/2} = 2^{-1/4}\Gamma(1/4)/\sqrt{\pi}$ is finite.

Lemma SJ.4 (Decay of the characteristic function)**.** *If $\|\bar{\Delta}\|_{\mathrm{F}} \ge F/2$, then $\phi_Y(t) \stackrel{\text{def}}{=} \mathbb{E}\, e^{\mathrm{i}tY}$ satisfies*

$$|\phi_Y(t)| \le \psi\big(|t|/8\big) \le \min\big\{1,\ C_{\mathrm{ch}}|t|^{-1/2}\big\} \qquad (t \ne 0). \tag{SJ.12}$$

Proof. The coordinate is conditionally Gaussian in z'. Given ξ, ζ, w', the terms of eq. (SJ.10) containing z' are $\xi\zeta\, z'^\top(\bar{\Delta} w' + \zeta q)$, a centered Gaussian of variance $\xi^2\zeta^2\|\bar{\Delta} w' + \zeta q\|^2$. Integrating z' first and taking absolute values gives

$$|\,\mathbb{E}\, e^{\mathrm{i}tX}| \le \mathbb{E}\exp\Big(-\frac{t^2\xi^2\zeta^2}{2}\,\|\bar{\Delta} w' + \zeta q\|^2\Big).$$

The integral over w' is bounded by the Frobenius norm of $\bar{\Delta}$. For $\lambda \ge 0$ and a fixed vector h, write $\bar{\Delta} = \sum_l \sigma_l\, a_l \otimes b_l$ in singular value form. The coordinates $b_l^\top w'$ are independent standard Gaussian, and completing the square gives, for one of them,

$$\mathbb{E}\ e^{-\lambda(\sigma_l g + c)^2/2} = (1 + \lambda\sigma_l^2)^{-1/2}\, e^{-\lambda c^2/(2(1+\lambda\sigma_l^2))} \le (1 + \lambda\sigma_l^2)^{-1/2}.$$

Directions orthogonal to the a_l contribute a factor at most 1, so

$$\mathbb{E}_{w'}\, e^{-\lambda\|\bar{\Delta} w' + h\|^2/2} \le \prod_l (1 + \lambda\sigma_l^2)^{-1/2} \le \big(1 + \lambda\|\bar{\Delta}\|_{\mathrm{F}}^2\big)^{-1/2}.$$

With $\lambda = t^2\xi^2\zeta^2$ this gives

$$|\,\mathbb{E}\, e^{\mathrm{i}tX}| \le \psi\big(|t|\,\|\bar{\Delta}\|_{\mathrm{F}}\big).$$

Centering multiplies the characteristic function by a phase, so $|\phi_Y(t)| = |\,\mathbb{E}\, e^{\mathrm{i}tX/s}| \le \psi(|t|\|\bar{\Delta}\|_{\mathrm{F}}/s)$. By lemma SJ.3, $\|\bar{\Delta}\|_{\mathrm{F}}/s \ge (F/2)/(\sqrt{8}F) > 1/8$, and ψ is nonincreasing, which gives the first inequality. The second follows from $(1+x^2)^{-1/2} \le x^{-1/2}$:

$$\psi(x) \le x^{-1/2}\ \mathbb{E}\,|\xi|^{-1/2}\ \mathbb{E}\,|\zeta|^{-1/2},$$

evaluated at $x = |t|/8$. □

The following lemma compares a normalized batch sum with a Gaussian. It compares expected signs directly, uniformly in the shift, and uses only the characteristic function.

Lemma SJ.5 (Sign comparison)**.** *There is a universal constant C_2 such that the following holds with $C_1 \stackrel{\text{def}}{=} 4\sqrt{2\pi}/(3\pi)$. Let Y be centered with variance 1, $\mathbb{E}\,Y^4 \le 88208$, and characteristic function satisfying eq. (SJ.12). Let $Y_1, \dots, Y_B$ be independent copies of Y, $B \ge 4$, and g standard Gaussian. Then for every real x*

$$\Big|\,\mathbb{E}\,\mathrm{sign}\Big(\frac{1}{\sqrt{B}}\sum_{a=1}^{B} Y_a + x\Big) - \mathbb{E}\,\mathrm{sign}(g+x)\Big| \le \frac{C_1|\,\mathbb{E}\,Y^3|}{\sqrt{B}} + \frac{C_2}{B}. \tag{SJ.13}$$

Proof. Write $M_4 \stackrel{\text{def}}{=} 88208$, $\varkappa_3 \stackrel{\text{def}}{=} \mathbb{E}\,Y^3$, $\phi \stackrel{\text{def}}{=} \phi_Y$, $V \stackrel{\text{def}}{=} B^{-1/2}\sum_a Y_a$, and put

$$\delta \stackrel{\text{def}}{=} \min\big\{1, \sqrt{3/(2M_4)}\big\}, \qquad \rho_0 \stackrel{\text{def}}{=} \psi(\delta/8) \in (0,1), \qquad C_4 \stackrel{\text{def}}{=} \frac{M_4}{24} + \frac{1}{8}.$$

Expected signs through the characteristic function. For $\varepsilon > 0$ and real v,

$$\frac{2}{\pi}\int_0^\infty e^{-\varepsilon u}\,\frac{\sin(uv)}{u}\,\mathrm{d}u = \frac{2}{\pi}\arctan\frac{v}{\varepsilon},$$

since both sides vanish at $v = 0$ and have the same derivative $\frac{2}{\pi}\varepsilon/(\varepsilon^2+v^2)$ in v. Apply this at $v = V + x$ and at $v = g + x$, take expectations and subtract. The shift contributes the phase $e^{\mathrm{i}ux}$, so

$$\Big|\,\mathbb{E}\,\frac{2}{\pi}\arctan\frac{V+x}{\varepsilon} - \mathbb{E}\,\frac{2}{\pi}\arctan\frac{g+x}{\varepsilon}\Big| \le \frac{2}{\pi}\int_0^\infty \big|\phi(u/\sqrt{B})^B - e^{-u^2/2}\big|\,\frac{\mathrm{d}u}{u}.$$

As $\varepsilon \downarrow 0$ the arctangents tend to the signs, including the value 0 at 0, and dominated convergence turns the left side into the left side of eq. (SJ.13). It remains to bound the integral.

Small frequencies. Symmetrization bounds $|\phi|$ near the origin. For an independent copy Y', $|\phi(t)|^2 = \mathbb{E}\cos(t(Y-Y'))$, $\mathbb{E}(Y-Y')^2 = 2$ and $\mathbb{E}(Y-Y')^4 = 2\,\mathbb{E}\,Y^4 + 6 \le 8M_4$. The inequality $\cos y \le 1 - y^2/2 + y^4/24$ gives, for $|t| \le \delta$,

$$|\phi(t)|^2 \le 1 - t^2 + \frac{M_4}{3}t^4 \le 1 - \frac{t^2}{2}, \qquad \text{hence} \qquad |\phi(t)| \le e^{-t^2/4}.$$

Taylor's formula with remainder for $e^{\mathrm{i}tY}$ to third order and for $e^{-t^2/2}$ to second order gives

$$|\phi(t) - e^{-t^2/2}| \le \frac{|\varkappa_3|}{6}|t|^3 + C_4 t^4.$$

Telescoping the Bth powers, with both factors bounded by $e^{-t^2/4}$ and $(B-1)/(4B) \ge 1/8$, gives for $0 \le u \le \delta\sqrt{B}$

$$\big|\phi(u/\sqrt{B})^B - e^{-u^2/2}\big| \le e^{-u^2/8}\Big(\frac{|\varkappa_3|u^3}{6\sqrt{B}} + \frac{C_4 u^4}{B}\Big).$$

With $\int_0^\infty u^2 e^{-u^2/8}\mathrm{d}u = 4\sqrt{2\pi}$ and $\int_0^\infty u^3 e^{-u^2/8}\mathrm{d}u = 32$, the integral over $[0, \delta\sqrt{B}]$ against $\mathrm{d}u/u$ is at most

$$\frac{4\sqrt{2\pi}\,|\varkappa_3|}{6\sqrt{B}} + \frac{32C_4}{B}.$$

Large frequencies. For $|t| \geq \delta$, eq. (SJ.12) and the monotonicity of ψ give $|\phi(t)| \leq \rho_0$ and $|\phi(t)|^4 \leq \min\{1, C_{\mathrm{ch}}^4 t^{-2}\}$. Hence

$$\int_{\delta\sqrt{B}}^{\infty} |\phi(u/\sqrt{B})|^B \frac{\mathrm{d}u}{u} = \int_{\delta}^{\infty} |\phi(t)|^B \frac{\mathrm{d}t}{t} \leq \rho_0^{B-4} \int_{\delta}^{\infty} \min\{1, C_{\mathrm{ch}}^4 t^{-2}\} \frac{\mathrm{d}t}{t} \leq J \rho_0^{B-4},$$

where, with $S_0 \stackrel{\mathrm{def}}{=} \max\{\delta, C_{\mathrm{ch}}^2\}$, $J \stackrel{\mathrm{def}}{=} \log(S_0/\delta) + C_{\mathrm{ch}}^4/(2S_0^2)$. Maximizing $y \mapsto y\rho_0^y$ gives $B\rho_0^{B-4} \leq \rho_0^{-4}/(e|\log \rho_0|)$. The Gaussian tail satisfies

$$\int_{\delta\sqrt{B}}^{\infty} e^{-u^2/2} \frac{\mathrm{d}u}{u} \leq \frac{e^{-\delta^2 B/2}}{\delta^2 B} \leq \frac{1}{\delta^2 B}.$$

Conclusion. Multiplying the three bounds by $2/\pi$ gives eq. (SJ.13) with $C_1 = \frac{2}{\pi} \cdot \frac{4\sqrt{2\pi}}{6}$ and

$$C_2 \stackrel{\mathrm{def}}{=} \frac{2}{\pi}\Big(32C_4 + \frac{J\rho_0^{-4}}{e|\log \rho_0|} + \delta^{-2}\Big). \qquad \square$$

The coordinate drift follows by combining the comparison with the Gaussian shift eq. (SJ.6) and the batch constant.

Proposition SJ.3 (Drift of one coordinate)**.** *Let $B \geq 4$ and suppose $\|\bar{\Delta}\|_{\mathrm{F}} \geq F/2$ for the entry (i, j). With s, m_3 from lemma SJ.3 and C_1, C_2 from lemma SJ.5,*

$$\begin{aligned}\Big|\,\mathbb{E}\,\mathrm{sign}(G_{ij}^{\circ}) - \sqrt{\frac{2}{\pi}}\mathcal{N}_B \frac{\Delta_{ij}}{F}\Big| \leq{}& \frac{C_1|m_3|}{s^3\sqrt{B}} + \frac{C_2}{B} + \frac{B^{3/2}|\Delta_{ij}|^3}{3\sqrt{2\pi}\,s^3} \\ &+ \sqrt{\frac{2}{\pi}}\,\frac{\sqrt{B}\,|\Delta_{ij}|\,(s^2 - F^2)}{sF(s+F)} + \sqrt{\frac{2}{\pi}}\,\frac{4|\Delta_{ij}|}{F\sqrt{B}}.\end{aligned} \tag{SJ.14}$$

Proof. The sign of G_{ij}° is the sign of a shifted normalized sum. With Y_a the standardized copies of X,

$$\mathrm{sign}\,(G_{ij}^{\circ}) = \mathrm{sign}\,\Big(\frac{1}{\sqrt{B}}\sum_a Y_a + \frac{\sqrt{B}\,\Delta_{ij}}{s}\Big).$$

Lemmas SJ.3 and SJ.4 verify the hypotheses of lemma SJ.5, with $\mathbb{E}\,Y^3 = m_3/s^3$. The comparison at $x = \sqrt{B}\Delta_{ij}/s$ produces the first two terms. The Gaussian shift eq. (SJ.6) with $\sigma = 1$ replaces $\mathbb{E}\,\mathrm{sign}(g + x)$ by $\sqrt{2/\pi}\,x$ at the cost of the third term.

The last two terms replace $\sqrt{B}/s$ by $\mathcal{N}_B/F$. By the triangle inequality and proposition SJ.1,

$$\Big|\frac{\sqrt{B}}{s} - \frac{\mathcal{N}_B}{F}\Big| \leq \sqrt{B}\Big(\frac{1}{F} - \frac{1}{s}\Big) + \frac{\sqrt{B} - \mathcal{N}_B}{F} \leq \frac{\sqrt{B}\,(s^2 - F^2)}{sF(s+F)} + \frac{4}{F\sqrt{B}}.$$

Multiplying by $\sqrt{2/\pi}\,|\Delta_{ij}|$ gives the last two terms. $\square$

The isotropic regime is the regime in which every entry of the state is small, with explicit constants.

Definition SJ.2 (Isotropic regime)**.** *Fix $K \geq 1$, $0 < f_- \leq f_+$ and $C_\Delta > 0$. A deterministic state Δ with batch size B is in the isotropic regime if, with $k \stackrel{\mathrm{def}}{=} \min\{N_{\mathrm{out}}, N_{\mathrm{in}}, B\}$,*

$$\max\{N_{\mathrm{out}}, N_{\mathrm{in}}, B\} \leq Kk, \qquad f_- \leq \|\Delta\|_{\mathrm{F}} \leq f_+, \qquad \max_{i,j}|\Delta_{ij}| \leq \frac{C_\Delta}{k}. \tag{SJ.15}$$

Theorem SJ.1 (Isotropic drift)**.** *There is a constant C_r depending only on K, f_-, C_Δ such that, for every state in the isotropic regime with $k \geq k_0 \stackrel{\mathrm{def}}{=} \max\{4, 8KC_\Delta^2/f_-^2\}$,*

$$\Big|\,\mathbb{E}\,\mathrm{sign}(G_{ij}^{\circ}) - \sqrt{\frac{2}{\pi}}\mathcal{N}_B \frac{\Delta_{ij}}{F}\Big| \leq e_k \stackrel{\mathrm{def}}{=} \frac{C_2}{B} + \frac{C_r}{k^{3/2}} \qquad \textit{for all } (i,j). \tag{SJ.16}$$

With $\mathsf{D} \stackrel{\mathrm{def}}{=} \langle \Delta, \mathbb{E}\,S\rangle_{\mathrm{F}}$ and $\mathsf{D}_0 \stackrel{\mathrm{def}}{=} \sqrt{2/\pi}\,\mathcal{N}_B F = \mathcal{N}_B\sqrt{4\mathscr{R}(\Delta)/\pi}$,

$$|\mathsf{D} - \mathsf{D}_0| \leq KF\Big(C_2 + \frac{C_r}{\sqrt{k}}\Big), \qquad \Big|\frac{\mathsf{D}}{\mathsf{D}_0} - 1\Big| \leq \frac{K(C_2 + C_r/\sqrt{k})}{\sqrt{2/\pi}\,\mathcal{N}_B} \leq \frac{\pi K(C_2 + C_r)}{2\sqrt{2/\pi}\,\sqrt{k}}. \tag{SJ.17}$$

No hypothesis is made on the operator norm or the rank of Δ.

Proof. The entry bound makes the deleted block large. Every row and column has squared norm at most $KkC_\Delta^2/k^2 = KC_\Delta^2/k$, so for $k \geq k_0$

$$\|\bar{\Delta}\|_{\mathrm{F}}^2 \geq F^2 - \frac{2KC_\Delta^2}{k} \geq \frac{3}{4}F^2,$$

and proposition SJ.3 applies to every entry.

The four non-batch terms of eq. (SJ.14) are of order $k^{-3/2}$. With $V \overset{\text{def}}{=} (4K+3)C_\Delta^2$ and $K_3 \overset{\text{def}}{=} 200C_\Delta/f_- + 54KC_\Delta^2/f_-^2$, lemma SJ.3 gives

$$0 \leq s^2 - F^2 \leq \frac{V}{k}, \qquad \frac{|m_3|}{s^3} \leq \frac{K_3}{k}.$$

Insert these, $k \leq B \leq Kk$, $|\Delta_{ij}| \leq C_\Delta/k$ and $s \geq F \geq f_-$ into eq. (SJ.14). The result is eq. (SJ.16) with

$$C_r \overset{\text{def}}{=} C_1 K_3 + \frac{K^{3/2}C_\Delta^3}{3\sqrt{2\pi}\, f_-^3} + \sqrt{\frac{2}{\pi}}\,\frac{\sqrt{K}\, C_\Delta V}{2f_-^3} + \sqrt{\frac{2}{\pi}}\,\frac{4C_\Delta}{f_-}.$$

The sum over entries costs the ℓ^1 norm of Δ. Since $\sum_{ij}\Delta_{ij}^2/F = F$,

$$|\mathsf{D} - \mathsf{D}_0| \leq \sum_{i,j}|\Delta_{ij}|\, e_k \leq \sqrt{N_{\mathrm{out}}N_{\mathrm{in}}}\, F\, e_k \leq KkF\Big(\frac{C_2}{k} + \frac{C_r}{k^{3/2}}\Big).$$

Dividing by D_0 and using $\mathcal{N}_B \geq \frac{2}{\pi}\sqrt{B} \geq \frac{2}{\pi}\sqrt{k}$ from proposition SJ.1 gives the relative bound. □

The summed error is of order one against a leading term of order $\sqrt{k}$. The isotropic recursion of the main text follows.

Proposition SJ.4 (Isotropic SIGNSGD recursion)**.** *Let $\Sigma_{\mathrm{in}} = I_{\mathrm{in}}$ and $\Sigma_{\mathrm{out}} = I_{\mathrm{out}}$, and let Δ_t be the state at step t. Suppose that, almost surely, Δ_t with the batch size B is in the isotropic regime of definition SJ.2 and $k \geq k_0$. Then the* SIGNSGD *step satisfies*

$$\mathbb{E}_t\left[\mathscr{R}(t+1)\right] = \mathscr{R}(t) - \eta\,\mathcal{N}_B\sqrt{\frac{4\mathscr{R}(t)}{\pi}} + \frac{\eta^2}{2}\,N_{\mathrm{in}}N_{\mathrm{out}} + E_t, \qquad |E_t| \leq \eta\,K\sqrt{2\mathscr{R}(t)}\,\Big(C_2 + \frac{C_r}{\sqrt{k}}\Big), \tag{SJ.18}$$

with C_2 universal and C_r from theorem SJ.1. The error is at most $\pi K(C_2+C_r)/(2\sqrt{2/\pi}\sqrt{k})$ times the drift term.

Proof. Conditionally on the past, Δ_t is fixed and the batch has its unconditional law, because the batch is independent of the past. The identity eq. (SJ.9) of proposition SJ.2 at $\Delta = \Delta_t \neq 0$ gives

$$\mathbb{E}_t\left[\mathscr{R}(t+1)\right] = \mathscr{R}(t) - \eta\,\mathsf{D} + \frac{\eta^2}{2}\,N_{\mathrm{in}}N_{\mathrm{out}}.$$

Theorem SJ.1 replaces D by $\mathsf{D}_0 = \mathcal{N}_B\sqrt{4\mathscr{R}(t)/\pi}$ with the stated error, since $F = \sqrt{2\mathscr{R}(t)}$. □

In the main text's notation, eq. (SJ.18) is eq. (8) with $\mathfrak{d} = \mathcal{N}_B/\sqrt{\pi}$ and $\mathfrak{v} = N_{\mathrm{in}}N_{\mathrm{out}}$; the volatility is exact and the drift carries a relative error of order $k^{-1/2}$.

SJ.4 Half-anisotropic data

SIGNSGD depends on the basis in which the sign is taken. The main text draws the eigenbasis of Σ_{out} from the Haar measure (eq. (13)). The following definition adds the law of the state that the proofs need.

Definition SJ.3 (Haar state law)**.** *Fix $K \geq 1$, $0 < c \leq L$, $0 < f_- \leq f_+$ and $C_{\mathrm{col}} > 0$. Let $N_{\mathrm{out}}, N_{\mathrm{in}}, B$ be positive integers with $k \overset{\text{def}}{=} \min\{N_{\mathrm{out}}, N_{\mathrm{in}}, B\} \geq 2$ and $\max\{N_{\mathrm{out}}, N_{\mathrm{in}}, B\} \leq Kk$. Let $\Lambda = \mathrm{diag}(\mu_1, \ldots, \mu_{N_{\mathrm{out}}})$ and $D \in \mathbb{R}^{N_{\mathrm{out}}\times N_{\mathrm{in}}}$ be deterministic with*

$$c\, I_{\mathrm{out}} \preceq \Lambda \preceq L\, I_{\mathrm{out}}, \qquad f_- \leq \|D\|_{\mathrm{F}} \leq f_+, \qquad \max_j \|D_{\cdot j}\|^2 \leq \frac{C_{\mathrm{col}}}{k}. \tag{SJ.19}$$

Let U be Haar distributed on the orthogonal group $O(N_{\mathrm{out}})$ and independent of the whitened batch $(z_b, w_b)_{b\le B}$. The half-anisotropic Haar model is

$$\Sigma_{\mathrm{out}} = U\Lambda U^\top, \qquad \Sigma_{\mathrm{in}} = I_{\mathrm{in}}, \qquad \Delta = UD, \qquad x_{\mathrm{out}}^{(b)} = U\Lambda^{1/2} z_b, \qquad x_{\mathrm{in}}^{(b)} = w_b.$$

In this model $u_i = Ue_i$ is the eigenvector of μ_i, the row energies are $\mathscr{Q}_i(\Delta) = \|u_i^\top \Delta\|^2 = \|D_{i\cdot}\|^2$, and the risk is deterministic:

$$2\mathscr{R} \overset{\mathrm{def}}{=} \|\Delta\|_\Sigma^2 = \mathrm{Tr}\left(\Lambda D D^\top\right).$$

This is the whitened form of Assumption S1, and the residuals are $d_b^\circ = z_b^\top \Lambda^{1/2} D w_b$, which do not involve U. We use

$$\bar\mu \overset{\mathrm{def}}{=} \frac{\mathrm{Tr}\,\Lambda}{N_{\mathrm{out}}}, \qquad T_D \overset{\mathrm{def}}{=} \Lambda D D^\top \Lambda, \qquad L_B \overset{\mathrm{def}}{=} \sqrt{\frac{2}{\pi}}\,\frac{\mathcal{N}_B}{\sqrt{2\mathscr{R}\,\bar\mu}} = \frac{\mathcal{N}_B}{\sqrt{\pi\mathscr{R}\,\bar\mu}},$$

the fixed constants

$$P_- \overset{\mathrm{def}}{=} cf_-^2, \quad P_+ \overset{\mathrm{def}}{=} Lf_+^2, \quad H_0 \overset{\mathrm{def}}{=} L^2 f_+^2, \quad A_0 \overset{\mathrm{def}}{=} \frac{H_0}{c}, \quad A_1 \overset{\mathrm{def}}{=} \frac{L^2 C_{\mathrm{col}}}{c}, \quad A_2 \overset{\mathrm{def}}{=} LC_{\mathrm{col}}, \tag{SJ.20}$$

which satisfy $P_- \le 2\mathscr{R} \le P_+$, $\mathrm{Tr}\,T_D \le H_0$ and $c \le \bar\mu \le L$, and the batch mean of the update given the eigenbasis,

$$M(U) \overset{\mathrm{def}}{=} \mathbb{E}\left[S \,\middle|\, U\right], \qquad S = \mathrm{sign}(G^\circ),$$

whose entries lie in $[-1, 1]$. A constant is called admissible if it depends only on $K, c, L, f_-, f_+, C_{\mathrm{col}}$.

An entrywise bound of order $1/\min\{N_{\mathrm{out}}, N_{\mathrm{in}}, B\}$ on the state does not make the residual of a deleted coordinate asymptotically Gaussian, which is why definition SJ.3 prescribes the law of the state. For $\Delta = d^{-1}\mathbf{1}\otimes\mathbf{1} \in \mathbb{R}^{d\times d}$ every entry is $1/d$ and $\|\Delta\|_{\mathrm{F}} = 1$, but

$$z^\top \Delta w = \Big(d^{-1/2}\sum_i z_i\Big)\Big(d^{-1/2}\sum_j w_j\Big)$$

is a product of two independent standard Gaussians for every d.

The independence of D and U is used in every proof below. After one step the new state need not be of the form UD' with D' independent of U, so the results below do not iterate.

SJ.4.1 Haar tools

For $q \ge 1$ put $d_q(n) \overset{\mathrm{def}}{=} n(n+2)\cdots(n+2q-2)$.

Lemma SJ.6 (Spherical moments)**.** *Let v be uniform on the unit sphere of $\mathbb{R}^n$, $n \ge 2$. Then*

$$\mathbb{E}\,\|v\|_1^2 = 1 + \frac{2(n-1)}{\pi}. \tag{SJ.21}$$

For a vector a, a positive semidefinite T and an integer $1 \le q \le 4$,

$$\mathbb{E}\,|v^\top a|^{2q} = \frac{(2q-1)!!\,\|a\|^{2q}}{d_q(n)}, \qquad \mathbb{E}\left(v^\top T v\right)^q \le \frac{(2q-1)!!\,(\mathrm{Tr}\,T)^q}{d_q(n)}. \tag{SJ.22}$$

For symmetric H with $\mathrm{Tr}\,H = 0$,

$$\mathbb{E}\left(v^\top H v\right)^4 = \frac{12(\mathrm{Tr}\,H^2)^2 + 48\,\mathrm{Tr}\,H^4}{d_4(n)}. \tag{SJ.23}$$

In particular, if $c\,\mathrm{Id} \preceq \Lambda \preceq L\,\mathrm{Id}$ in dimension n and $\bar\mu = \mathrm{Tr}\,\Lambda/n$, then $\mathbb{E}(v^\top \Lambda v - \bar\mu)^4 \le 60L^4/n^2$.

Proof. A standard Gaussian vector g in $\mathbb{R}^n$ is $g = Rv$ with radius R and direction v independent, and $\mathbb{E}\,R^{2q} = d_q(n)$ by the Gaussian moments of $\|g\|^2$. Every identity follows by computing a Gaussian moment and dividing by $\mathbb{E}\,R^{2q}$.

For eq. (SJ.21), $\mathbb{E}\, R^2 = n$, and $\mathbb{E}\,|g_i g_j| = 2/\pi$ for $i \neq j$ gives $\mathbb{E}\,|v_i v_j| = 2/(\pi n)$. Expanding the square of the ℓ^1 norm,

$$\mathbb{E}\,\|v\|_1^2 = \sum_i \mathbb{E}\, v_i^2 + \sum_{i \neq j} \mathbb{E}\,|v_i v_j| = 1 + n(n-1)\frac{2}{\pi n}.$$

For eq. (SJ.22), rotation invariance gives $\mathbb{E}\,|g^\top a|^{2q} = (2q-1)!!\|a\|^{2q}$. For T, write $g^\top T g = \operatorname{Tr} T \sum_l \pi_l g_l^2$ with weights $\pi_l \geq 0$ summing to 1; Jensen's inequality gives $\mathbb{E}(\sum_l \pi_l g_l^2)^q \leq (2q-1)!!$.

For eq. (SJ.23), diagonalize H with eigenvalues h_l of sum zero. Then $g^\top H g = \sum_l h_l(g_l^2 - 1)$ is a sum of independent centered terms with $\mathbb{E}(g_l^2-1)^2 = 2$ and $\mathbb{E}(g_l^2-1)^4 = 60$. Only multiplicities four and two-two survive in the fourth power:

$$\mathbb{E}(g^\top H g)^4 = 60 \sum_l h_l^4 + 24 \sum_{l<l'} h_l^2 h_{l'}^2 = 12(\operatorname{Tr} H^2)^2 + 48 \operatorname{Tr} H^4.$$

For $H = \Lambda - \bar{\mu}\,\mathrm{Id}$, $\|H\| \leq L$, so $\operatorname{Tr} H^2 \leq nL^2$ and $\operatorname{Tr} H^4 \leq nL^4$, and $d_4(n) \geq n^4$ gives the last bound. □

Lemma SJ.7 (Haar sign matrix)**.** *Let U be Haar on $O(n)$, $n \geq 2$, let $a \in \mathbb{R}^n$ be nonzero, and put $Y \overset{\text{def}}{=} U^\top \operatorname{sign}(Ua)$. Then*

$$\mathbb{E}_U\left[Y \otimes Y\right] = \Big(1 - \frac{2}{\pi}\Big)\,\mathrm{Id} + \frac{2n}{\pi}\,\frac{a \otimes a}{\|a\|^2}. \tag{SJ.24}$$

The same holds conditionally on a random nonzero a independent of U.

Proof. Put $v = a/\|a\|$, so that $Y = U^\top \operatorname{sign}(Uv)$. The vector Uv is uniform on the sphere, so it has no zero coordinate almost surely, and $\|Y\|^2 = n$.

Invariance reduces the second moment to two numbers. If Q is orthogonal with $Qv = v$, then UQ is Haar and replacing U by UQ replaces Y by $Q^\top Y$. Hence $\mathbb{E}[Y \otimes Y]$ commutes with every such Q. In an orthonormal basis starting with v, reflections of single other basis vectors make the off-diagonal entries vanish, and permutations of the other basis vectors make their diagonal entries equal. Therefore

$$\mathbb{E}[Y \otimes Y] = c_n\, v \otimes v + d_n(\mathrm{Id} - v \otimes v).$$

The two numbers follow from two scalar identities. First, $v^\top Y = (Uv)^\top \operatorname{sign}(Uv) = \|Uv\|_1$, so $c_n = 1 + 2(n-1)/\pi$ by eq. (SJ.21). Second, the trace is $\mathbb{E}\,\|Y\|^2 = n$, so $c_n + (n-1)d_n = n$. Solving gives $d_n = 1 - 2/\pi$ and $c_n - d_n = 2n/\pi$. Conditioning on an independent a gives the last claim. □

The arcsine law and the Haar sign matrix give the row volatility exactly for one sample. At this batch size the per-row volatility does not depend on the state, which may even be a function of the eigenbasis.

Proposition SJ.5 (One sample)**.** *Let $B = 1$, $N_{\text{out}} \geq 2$, $\Sigma_{\text{out}} = U\Lambda U^\top$ with U Haar, $\Lambda = \operatorname{diag}(\mu_i) \succ 0$, $\Sigma_{\text{in}} = I_{\text{in}}$ and $u_i \overset{\text{def}}{=} Ue_i$, and let the sample be $x_{\text{out}} = U\Lambda^{1/2} z$, $x_{\text{in}} = w$ with (z, w) standard Gaussian and independent of U. Let Δ be any function of U with $\Delta \neq 0$ almost surely, and $S = \operatorname{sign}(G^\circ)$. Then*

$$\mathbb{E}\,\|S^\top u_i\|^2 = N_{\text{in}}\ \mathbb{E}\left[u_i^\top \Sigma_{\text{out}}^{\text{sign}} u_i\right] = N_{\text{in}}\Big(1 - \frac{2}{\pi} + \frac{2N_{\text{out}}}{\pi}\ \mathbb{E}\,\frac{\mu_i g_i^2}{\sum_l \mu_l g_l^2}\Big),$$

where $(\Sigma_{\text{out}}^{\text{sign}})_{kl} \overset{\text{def}}{=} \frac{2}{\pi}\arcsin\big((\Sigma_{\text{out}})_{kl}/\sqrt{(\Sigma_{\text{out}})_{kk}(\Sigma_{\text{out}})_{ll}}\big)$ and g is a standard Gaussian vector in $\mathbb{R}^{N_{\text{out}}}$.

Proof. For one sample the update factorizes. Given U, the vector $y = x_{\text{out}}$ has the nondegenerate law $\mathcal{N}(0, \Sigma_{\text{out}})$, so its coordinates are nonzero almost surely, and so are those of $x = x_{\text{in}}$. The

residual $y^\top \Delta x$ is, given U and x, Gaussian with variance $\|{\Sigma_{\text{out}}}^{1/2}\Delta x\|^2$, which is positive almost surely because $\Delta \neq 0$. Hence, entrywise,

$$S = \operatorname{sign}\left(y^\top \Delta x\right)\, \operatorname{sign}(y) \otimes \operatorname{sign}(x),$$

and therefore

$$\|S^\top u_i\|^2 = \left(u_i^\top \operatorname{sign}(y)\right)^2 \|\operatorname{sign}(x)\|^2 = N_{\text{in}}\left(u_i^\top \operatorname{sign}(y)\right)^2.$$

The residual sign has disappeared, so the state no longer enters.

Given U, lemma SJ.1 applies to every pair of coordinates of y, whose variances $(\Sigma_{\text{out}})_{kk}$ are positive, and gives $\mathbb{E}[\operatorname{sign}(y) \otimes \operatorname{sign}(y) \mid U] = \Sigma_{\text{out}}^{\text{sign}}$. This is the first equality.

For the second, write $y = Ua$ with $a = \Lambda^{1/2} g$, nonzero almost surely and independent of U. Then $u_i^\top \operatorname{sign}(y) = e_i^\top U^\top \operatorname{sign}(Ua)$, and lemma SJ.7 gives

$$\mathbb{E}\left[(u_i^\top \operatorname{sign}(y))^2 \,\middle|\, g\right] = 1 - \frac{2}{\pi} + \frac{2N_{\text{out}}}{\pi}\,\frac{\mu_i g_i^2}{\sum_l \mu_l g_l^2}. \qquad \square$$

SJ.4.2 Half-anisotropic drift

The coordinate drift of an anisotropic gradient is the isotropic one for a whitened state. Let $\Sigma_{\text{out}} = \Sigma$ be positive semidefinite in dimension N_{out} and $\Sigma_{\text{in}} = I_{\text{in}}$, let Δ be deterministic with $2\mathscr{R} = \operatorname{Tr}(\Sigma\Delta\Delta^\top) > 0$, and fix an entry (p, j) with $v_p \overset{\text{def}}{=} \Sigma_{pp} > 0$. Put

$$\mu_{pj} \overset{\text{def}}{=} \frac{(\Sigma\Delta)_{pj}}{\sqrt{v_p}}, \qquad R_p \overset{\text{def}}{=} \frac{\|(\Sigma\Delta)_{p\cdot}\|^2}{v_p}, \qquad C_j \overset{\text{def}}{=} \Delta_{\cdot j}^\top \Sigma \Delta_{\cdot j}. \tag{SJ.25}$$

Lemma SJ.8 (Whitening a coordinate)**.** *Let Q be orthogonal with first column $\Sigma^{1/2}e_p/\sqrt{v_p}$ and put $A \overset{\text{def}}{=} Q^\top \Sigma^{1/2}\Delta$. Then $\|A\|_{\text{F}}^2 = 2\mathscr{R}$, the entry A_{1j} equals μ_{pj}, row 1 of A has squared norm R_p, column j of A has squared norm C_j, and $\operatorname{sign}(G^\circ_{pj})$ has the law of $\operatorname{sign}(G^A_{1j})$, where G^A is the isotropic gradient of the state A. In particular proposition SJ.3 applies to (p, j) with $F^2 = 2\mathscr{R}$, Δ_{ij} replaced by μ_{pj}, and $\|\bar{A}\|_{\text{F}}^2 = 2\mathscr{R} - R_p - C_j + \mu_{pj}^2$, whenever this is at least $\mathscr{R}/2$ and $B \geq 4$; its comparison term is*

$$\sqrt{\frac{2}{\pi}}\,\mathcal{N}_B\,\frac{\mu_{pj}}{\sqrt{2\mathscr{R}}} = \frac{\mathcal{N}_B\,(\Sigma\Delta)_{pj}}{\sqrt{\pi\mathscr{R}\,v_p}}.$$

Proof. Represent the output feature as $y = \Sigma^{1/2}Qz$ with z standard Gaussian. Since $Q^\top\Sigma^{1/2}e_p = \sqrt{v_p}\,e_1$, the coordinate is $y_p = \sqrt{v_p}\,z_1$, and the residual is $y^\top\Delta x = z^\top A x$. Hence

$$\left(y^\top \Delta x\right) y_p x_j = \sqrt{v_p}\,\left(z^\top A x\right) z_1 x_j,$$

and the positive factor $\sqrt{v_p}$ does not change the sign of the batch sum. Row 1 of A is $e_p^\top\Sigma\Delta/\sqrt{v_p}$ and column j has squared norm $\Delta_{\cdot j}^\top\Sigma\Delta_{\cdot j}$, and $\|A\|_{\text{F}}^2 = \operatorname{Tr}(\Sigma\Delta\Delta^\top)$. The deleted block has the squared norm left after removing row 1 and column j. $\square$

In the Haar model, $\Sigma\Delta = U\Lambda D$. Write $\omega_p \overset{\text{def}}{=} U^\top e_p$ for the pth row of U, a uniform unit vector, so that

$$v_p = \omega_p^\top \Lambda \omega_p \in [c, L], \qquad \mu_{pj} = \frac{\omega_p^\top \Lambda D_{\cdot j}}{\sqrt{v_p}}, \qquad R_p = \frac{\omega_p^\top T_D \omega_p}{v_p}, \qquad C_j = D_{\cdot j}^\top \Lambda D_{\cdot j} \leq \frac{A_2}{k}.$$

The last quantity is deterministic.

Lemma SJ.9 (Deletion estimates)**.** *In the Haar model put*

$$Q_0 \overset{\text{def}}{=} 105\Big(\frac{4A_0}{P_-}\Big)^4, \qquad k_* \overset{\text{def}}{=} \max\{4, \lceil 4A_2/P_-\rceil\},$$

and let $k \geq k_*$. On the event $\mathcal{A}_p \overset{\text{def}}{=} \{R_p \leq P_-/4\}$, the deleted block of lemma SJ.8 has squared norm at least $\mathscr{R}$. For $1 \leq q \leq 4$,

$$\mathbb{E}\,R_p^q \leq \frac{(2q-1)!!\,A_0^q}{{N_{\text{out}}}^q}, \qquad \mathbb{E}\,|\mu_{pj}|^{2q} \leq \frac{(2q-1)!!\,A_1^q}{{N_{\text{out}}}^q k^q}, \qquad \mathbb{P}(\mathcal{A}_p^c) \leq \frac{Q_0}{{N_{\text{out}}}^4}. \tag{SJ.26}$$

Proof. Ellipticity removes the random denominator: $v_p \geq c$. Then eq. (SJ.22) with $T = T_D$, $\operatorname{Tr} T_D \leq H_0$, and with $a = \Lambda D_{\cdot j}$, $\|a\|^2 \leq L\, C_j \leq L^2 C_{\mathrm{col}}/k$, together with $d_q(N_{\mathrm{out}}) \geq {N_{\mathrm{out}}}^q$, gives the two moment bounds. On $\mathcal{A}_p$, $R_p + C_j \leq P_-/4 + A_2/k \leq P_-/2 \leq \mathscr{R}$, so the deleted block has squared norm $2\mathscr{R} - R_p - C_j + \mu_{pj}^2 \geq \mathscr{R}$. Markov's inequality for R_p^4 gives the probability bound. □

Proposition SJ.6 (Coordinate errors in Frobenius norm)**.** *In the Haar model with $k \geq k_*$, let $D_v \overset{\mathrm{def}}{=} \operatorname{diag}(v_1, \ldots, v_{N_{\mathrm{out}}})$ and*

$$E \overset{\mathrm{def}}{=} M(U) - \sqrt{\frac{2}{\pi}}\, \frac{\mathcal{N}_B}{\sqrt{2\mathscr{R}}}\, D_v^{-1/2} U \Lambda D.$$

There is an admissible constant C_E such that $\mathbb{E}_U \|E\|_{\mathrm{F}}^2 \leq C_E$.

Proof. The entry E_{pj} is the error of proposition SJ.3 for the whitened state of lemma SJ.8. We square and sum the five terms of eq. (SJ.14) before averaging over U; this keeps the batch remainder at the scale of the whole matrix.

Good events. On $\mathcal{A}_p$ the bound eq. (SJ.14) applies, with $F^2 = 2\mathscr{R} \geq P_-$. Call its five terms $t_1, \ldots, t_5$, so that $E_{pj}^2 \mathbf{1}_{\mathcal{A}_p} \leq 5 \sum_l t_l^2$. Here $s^3 \geq (2\mathscr{R})^{3/2}$,

$$m_3^2 \leq 80000\, \mu_{pj}^2 (2\mathscr{R})^2 + 5832\, (2\mathscr{R}) R_p C_j, \qquad s^2 - 2\mathscr{R} \leq 5(R_p + C_j),$$

the last because $\mu_{pj}^2 \leq R_p$. Summing over the $N_{\mathrm{out}} N_{\mathrm{in}}$ entries with eq. (SJ.26) and $N_{\mathrm{out}}, N_{\mathrm{in}}, B \in [k, Kk]$ gives

$$\sum_{p,j} \mathbb{E}\, t_1^2 \leq \frac{K C_1^2}{k P_-^3} \big(80000 P_+^2 A_1 + 5832 P_+ A_2 A_0\big), \qquad \sum_{p,j} t_2^2 \leq \frac{N_{\mathrm{out}} N_{\mathrm{in}} C_2^2}{B^2} \leq K^2 C_2^2,$$

$$\sum_{p,j} \mathbb{E}\, t_3^2 \leq \frac{15 K^4 A_1^3}{18 \pi P_-^3 k}, \qquad \sum_{p,j} \mathbb{E}\, t_5^2 \leq \frac{16 \cdot \frac{2}{\pi} K A_1}{P_- k}.$$

For t_4 the correlated moment is bounded by Cauchy–Schwarz, $\mathbb{E}[\mu_{pj}^2 R_p^2] \leq (\mathbb{E}\, \mu_{pj}^4\ \mathbb{E}\, R_p^4)^{1/2} \leq \sqrt{315} A_1 A_0^2/({N_{\mathrm{out}}}^3 k)$, and $C_j^2\, \mathbb{E}\, \mu_{pj}^2 \leq A_2^2 A_1/(N_{\mathrm{out}} k^3)$, so

$$\sum_{p,j} \mathbb{E}\, t_4^2 \leq \frac{25 \cdot \frac{2}{\pi} K^2 A_1}{2 P_-^3 k} \big(\sqrt{315} A_0^2 + A_2^2\big).$$

Only t_2 contributes a term that does not decay; it is bounded.

Bad events. On every event $|M_{pj}| \leq 1$ and $\mathcal{N}_B^2 \leq B$, so $E_{pj}^2 \leq 2 + \frac{4}{\pi} B \mu_{pj}^2/P_-$. Cauchy–Schwarz gives $\mathbb{E}[\mu_{pj}^2 \mathbf{1}_{\mathcal{A}_p^{\mathrm{c}}}] \leq \sqrt{3Q_0} A_1/({N_{\mathrm{out}}}^3 k)$, and

$$\sum_{p,j} \mathbb{E}\left[E_{pj}^2 \mathbf{1}_{\mathcal{A}_p^{\mathrm{c}}}\right] \leq \frac{2KQ_0}{k^2} + \frac{\frac{4}{\pi}\sqrt{3Q_0} A_1 K^2}{P_- k}.$$

No independence between different rows of U is used. Adding five times the good-event sums to the bad-event sum gives the claim with an admissible C_E, since $k \geq 1$. □

Theorem SJ.2 (Half-anisotropic drift)**.** *In the Haar model of definition SJ.3 with $k \geq k_*$, there is an admissible constant C_* such that*

$$\mathbb{E}_U \left\| U^\top M(U) - L_B\, \Lambda D \right\|_{\mathrm{F}}^2 \leq C_*, \tag{SJ.27}$$

while the leading term satisfies

$$b_- B \leq \|L_B \Lambda D\|_{\mathrm{F}}^2 \leq b_+ B, \qquad b_- \overset{\mathrm{def}}{=} \frac{8 c^2 f_-^2}{\pi^3 L^2 f_+^2}, \quad b_+ \overset{\mathrm{def}}{=} \frac{2 L^2 f_+^2}{\pi c^2 f_-^2}. \tag{SJ.28}$$

In particular the relative error of the mean update in $L^2(U)$ Frobenius norm is at most $\sqrt{C_*/(b_- B)}$.

Proof. The diagonal normalization of proposition SJ.6 is replaced by the constant $\bar{\mu}$. Since $U^\top U \Lambda D = \Lambda D$ and $\sqrt{2/\pi}\,\mathcal{N}_B/\sqrt{2\mathscr{R}\bar{\mu}} = L_B$,

$$U^\top M - L_B \Lambda D = U^\top E + U^\top N, \qquad N \stackrel{\text{def}}{=} \sqrt{\frac{2}{\pi}}\,\frac{\mathcal{N}_B}{\sqrt{2\mathscr{R}}}\big(D_v^{-1/2} - \bar{\mu}^{-1/2}\,\mathrm{Id}\,\big) U \Lambda D.$$

The correction N is controlled by the fluctuation of the diagonal of Σ_{out}. On $[c, L]$ the derivative of $x^{-1/2}$ is at most $\frac{1}{2}c^{-3/2}$ in absolute value, and the pth row of $U\Lambda D$ has squared norm $e_p \stackrel{\text{def}}{=} \omega_p^\top T_D \omega_p$. Hence

$$\mathbb{E}_U \,\|N\|_{\mathrm{F}}^2 \le \frac{\frac{2}{\pi}B}{4P_- c^3} \sum_p \mathbb{E}_U\left[(v_p - \bar{\mu})^2 e_p\right].$$

The two factors are dependent, and Cauchy–Schwarz with lemma SJ.6 gives

$$\mathbb{E}_U\left[(v_p - \bar{\mu})^2 e_p\right] \le \big(\,\mathbb{E}_U (v_p - \bar{\mu})^4\;\mathbb{E}_U\, e_p^2\big)^{1/2} \le \Big(\frac{60 L^4}{{N_{\text{out}}}^2} \cdot \frac{3 H_0^2}{{N_{\text{out}}}^2}\Big)^{1/2} = \frac{\sqrt{180}\, L^2 H_0}{{N_{\text{out}}}^2}.$$

Summing over N_{out} rows and using $B \le K N_{\text{out}}$ gives $\mathbb{E}_U \,\|N\|_{\mathrm{F}}^2 \le C_N \stackrel{\text{def}}{=} \frac{2}{\pi} K \sqrt{180} L^2 H_0/(4P_- c^3)$.

Orthogonal invariance of the Frobenius norm, $\|X+Y\|_{\mathrm{F}}^2 \le 2\|X\|_{\mathrm{F}}^2 + 2\|Y\|_{\mathrm{F}}^2$ and proposition SJ.6 give eq. (SJ.27) with $C_* \stackrel{\text{def}}{=} 2C_E + 2C_N$.

For eq. (SJ.28), write

$$\|L_B \Lambda D\|_{\mathrm{F}}^2 = \frac{2}{\pi}\,\frac{\mathcal{N}_B^2\,\|\Lambda D\|_{\mathrm{F}}^2}{2\mathscr{R}\,\bar{\mu}},$$

and use $c^2 f_-^2 \le \|\Lambda D\|_{\mathrm{F}}^2 \le L^2 f_+^2$, $c^2 f_-^2 \le 2\mathscr{R}\bar{\mu} \le L^2 f_+^2$, and $4B/\pi^2 \le \mathcal{N}_B^2 \le B$ from proposition SJ.1. □

Corollary SJ.1 (Row and risk drift)**.** *Under the hypotheses of theorem SJ.2,*

$$\sum_{i=1}^{N_{\text{out}}} \Big|\,\mathbb{E}\,\big\langle \Delta^\top u_i, S^\top u_i\big\rangle - L_B\,\mu_i\,\mathscr{Q}_i(\Delta)\Big| \le f_+\sqrt{C_*}, \tag{SJ.29}$$

$$\Big|\,\mathbb{E}\,\big\langle \Delta, S\big\rangle_\Sigma - L_B\,\|\Lambda D\|_{\mathrm{F}}^2\Big| \le L f_+\sqrt{C_*}, \tag{SJ.30}$$

and the leading total satisfies $L_B\|\Lambda D\|_{\mathrm{F}}^2 \ge d_-\sqrt{B}$ *with* $d_- \stackrel{\text{def}}{=} 2\sqrt{2/\pi}\,c^2 f_-^2/(\pi L f_+)$.

Proof. Put $T_U \stackrel{\text{def}}{=} U^\top M(U) - L_B \Lambda D$. The row pairing given U is $\langle D_{i\cdot}, (U^\top M)_{i\cdot}\rangle$, because $u_i^\top \Delta = D_{i\cdot}$ and $u_i^\top M = (U^\top M)_{i\cdot}$, and $L_B \mu_i \|D_{i\cdot}\|^2 = \langle D_{i\cdot}, (L_B \Lambda D)_{i\cdot}\rangle$. Cauchy–Schwarz in the row sum gives

$$\sum_i |\langle D_{i\cdot}, (T_U)_{i\cdot}\rangle| \le \|D\|_{\mathrm{F}} \|T_U\|_{\mathrm{F}},$$

and $\mathbb{E}_U\,\|T_U\|_{\mathrm{F}} \le \sqrt{C_*}$ by theorem SJ.2. Since D is deterministic, averaging over U gives eq. (SJ.29). For eq. (SJ.30), $\langle \Delta, S\rangle_\Sigma = \langle U\Lambda D, S\rangle_{\mathrm{F}}$, whose batch mean is $\langle \Lambda D, U^\top M\rangle_{\mathrm{F}}$, and $\|\Lambda D\|_{\mathrm{F}} \le L f_+$. The lower bound uses $\mathcal{N}_B \ge \frac{2}{\pi}\sqrt{B}$, $\|\Lambda D\|_{\mathrm{F}}^2 \ge c^2 f_-^2$ and $\sqrt{2\mathscr{R}\bar{\mu}} \le L f_+$. □

The relative error of the total drift is therefore at most $L f_+ \sqrt{C_*}/(d_- \sqrt{B})$. In the notation of the main text, $L_B \mu_i \mathscr{Q}_i = \mathfrak{d}_i \mathscr{Q}_i/\sqrt{\mathscr{R}}$ with $\mathfrak{d}_i = \mu_i \mathcal{N}_B/\sqrt{\pi\bar{\mu}}$, which is the tabulated drift $\mu_i\sqrt{B/(\pi\bar{\mu})}$ with $\sqrt{B}$ replaced by $\mathcal{N}_B$; by proposition SJ.1 the two differ by a factor in $[1 - 4/B, 1]$.

SJ.4.3 Half-anisotropic volatility

The update factors through the eigenbasis. With $A \stackrel{\text{def}}{=} \Lambda^{1/2} D$, which is $U^\top A_t$ for the rotated error of eq. (SA.9),

$$G^\circ = UH, \qquad H \stackrel{\text{def}}{=} \frac{1}{B} \sum_{b=1}^{B} \left(z_b^\top A w_b\right) \Lambda^{1/2} z_b \otimes w_b, \tag{SJ.31}$$

and H is independent of U. Its jth column is the mean $h_j = \frac{1}{B}\sum_b X_b$ of independent copies of

$$X \stackrel{\text{def}}{=} \left(z^\top A w\right) w_j \, \Lambda^{1/2} z \in \mathbb{R}^{N_{\text{out}}}, \qquad m_j \stackrel{\text{def}}{=} \Lambda D_{\cdot j}, \qquad Z \stackrel{\text{def}}{=} X - m_j.$$

Lemma SJ.10 (Column moments)**.** *There are admissible constants V_0, A_4, A_8 such that, for every column j,*

$$\mathbb{E}\, X = m_j, \qquad V_j \stackrel{\text{def}}{=} \mathbb{E}[Z \otimes Z] = (2\mathscr{R} + 2C_j)\Lambda + 2T_D + 3\, m_j \otimes m_j, \tag{SJ.32}$$

$$\mathbb{E}[h_j \otimes h_j] = \frac{2\mathscr{R} + 2C_j}{B}\,\Lambda + \frac{2}{B}\,T_D + \Big(1 + \frac{3}{B}\Big) m_j \otimes m_j, \tag{SJ.33}$$

$$\mathbb{E}[HH^\top] = \frac{2\mathscr{R}\,(N_{\text{in}} + 2)}{B}\,\Lambda + \Big(1 + \frac{2N_{\text{in}} + 3}{B}\Big) T_D, \tag{SJ.34}$$

with $C_j = D_{\cdot j}^\top \Lambda D_{\cdot j}$, and

$$V_j \succeq 2\mathscr{R}\,\Lambda, \qquad \|V_j\| \le V_0, \qquad \|m_j\|^2 \le \frac{L^2 C_{\text{col}}}{k}, \tag{SJ.35}$$

$$\mathbb{E}\,|a^\top Z|^4 \le A_4 \|a\|^4, \quad \mathbb{E}\,|a^\top Z|^8 \le A_8 \|a\|^8 \quad (a \in \mathbb{R}^{N_{\text{out}}}), \tag{SJ.36}$$

$$\mathbb{E}\,\|Z\|^4 \le A_4 {N_{\text{out}}}^2, \quad \mathbb{E}\,\|Z\|^8 \le A_8 {N_{\text{out}}}^4. \tag{SJ.37}$$

Every column h_j is nonzero almost surely.

Proof. Mean and covariance. Put $y = \Lambda^{1/2} z$. Integrating w first with the Gaussian identities $\mathbb{E}[(y^\top D w) w_j] = y^\top D_{\cdot j}$ and $\mathbb{E}[(y^\top D w)^2 w_j^2] = y^\top D D^\top y + 2(y^\top D_{\cdot j})^2$, and then z with the pairing identity $\mathbb{E}[y \otimes y\,(y^\top M y)] = \operatorname{Tr}(\Lambda M)\Lambda + 2\Lambda M \Lambda$ for symmetric M [29], gives

$$\mathbb{E}\, X = \Lambda D_{\cdot j}, \qquad \mathbb{E}[X \otimes X] = (2\mathscr{R} + 2C_j)\Lambda + 2T_D + 4\, m_j \otimes m_j.$$

Subtracting $m_j \otimes m_j$ gives eq. (SJ.32). Averaging B independent copies gives eq. (SJ.33), and summing over j with $\sum_j C_j = 2\mathscr{R}$ and $\sum_j m_j \otimes m_j = T_D$ gives eq. (SJ.34).

Bounds. All terms of V_j are positive semidefinite, so $V_j \succeq 2\mathscr{R}\Lambda$. With $C_j \le L C_{\text{col}}/k$, $\|m_j\|^2 \le L^2 C_{\text{col}}/k$ and $\|T_D\| \le \operatorname{Tr} T_D \le H_0$,

$$\|V_j\| \le V_0 \stackrel{\text{def}}{=} L P_+ + 2H_0 + 5L^2 C_{\text{col}}.$$

Higher moments. Put $r = z^\top A w$. Conditionally on w, r is Gaussian with variance $\|Aw\|^2$, and the convexity argument of lemma SJ.3 gives $\mathbb{E}\, r^{2q} \le ((2q-1)!!)^2 (2\mathscr{R})^q$. For $p = 4, 8$, Hölder's inequality on the three dependent factors of $a^\top X$ gives

$$\mathbb{E}\,|a^\top X|^p \le \left(\mathbb{E}\,|r|^{3p}\ \mathbb{E}\,|w_j|^{3p}\ \mathbb{E}\,|a^\top \Lambda^{1/2} z|^{3p}\right)^{1/3} \le \big((3p-1)!!\big)^2 P_+^{p/2} L^{p/2} \|a\|^p.$$

Centering costs a factor 2^p, which gives the directional bounds with $A_4 \stackrel{\text{def}}{=} 16(11!!)^2 P_+^2 L^2$ and $A_8 \stackrel{\text{def}}{=} 256(23!!)^2 P_+^4 L^4$. The inequalities $(\sum_l Z_l^2)^2 \le N_{\text{out}} \sum_l Z_l^4$ and $(\sum_l Z_l^2)^4 \le {N_{\text{out}}}^3 \sum_l Z_l^8$ give the vector bounds.

Nonvanishing. Some coordinate of h_j is a polynomial in the batch that is not identically zero. Indeed choose $w = e_j$ if $D_{\cdot j} \neq 0$ and $w = e_j + e_l$ with $D_{\cdot l} \neq 0$ otherwise, so that $w_j = 1$ and $Aw \neq 0$. Taking $z = Aw$ in the first sample and all other samples zero makes h_j a nonzero multiple of $\Lambda^{1/2} A w \neq 0$. The batch has a Gaussian density, so this coordinate is nonzero almost surely [46]. □

Proposition SJ.7 (Quadratic forms of a column)**.** *There are admissible constants V_* and D_* such that, for every symmetric $\Pi \in \mathbb{R}^{N_{\text{out}} \times N_{\text{out}}}$ and every column j,*

$$\mathbb{E}\left|h_j^\top \Pi h_j - \frac{2\mathscr{R}}{B}\operatorname{Tr}(\Pi\Lambda)\right| \le \|\Pi\|\Big(\sqrt{\frac{V_*}{k}} + \frac{D_*}{k}\Big). \tag{SJ.38}$$

Proof. Put $W \overset{\text{def}}{=} \frac{1}{B}\sum_b Z_b$, $V \overset{\text{def}}{=} V_j$ and $a \overset{\text{def}}{=} \|\Pi\|$, so that $h_j^\top \Pi h_j = m_j^\top \Pi m_j + 2m_j^\top \Pi W + W^\top \Pi W$.

The fluctuation. The variance of the Gram term is

$$\operatorname{Var}\left(W^\top \Pi W\right) = \frac{\operatorname{Var}(Z_1^\top \Pi Z_1)}{B^3} + \frac{2(B-1)}{B^3}\operatorname{Tr}(\Pi V \Pi V).$$

To see this, expand $B^2 W^\top \Pi W = \sum_b Z_b^\top \Pi Z_b + 2\sum_{b<b'} Z_b^\top \Pi Z_{b'}$. A covariance between two terms vanishes unless every sample index occurs in both, because an index that occurs once has a centered factor. The diagonal terms are independent, and each of the $B(B-1)/2$ cross pairs has second moment $\operatorname{Tr}(\Pi V \Pi V)$. The two pieces are bounded by

$$\operatorname{Var}(Z_1^\top \Pi Z_1) \le a^2\,\mathbb{E}\,\|Z\|^4 \le a^2 A_4 {N_{\text{out}}}^2, \qquad \operatorname{Tr}(\Pi V \Pi V) = \|V^{1/2}\Pi V^{1/2}\|_{\text{F}}^2 \le N_{\text{out}} a^2 V_0^2.$$

The linear term has $\operatorname{Var}(2m_j^\top \Pi W) = 4m_j^\top \Pi V \Pi m_j / B \le 4a^2 V_0 \|m_j\|^2/B$. With $\operatorname{Var}(X_1+X_2) \le 2\operatorname{Var}X_1 + 2\operatorname{Var}X_2$ and $N_{\text{out}} \le Kk \le KB$,

$$\operatorname{Var}\left(h_j^\top \Pi h_j\right) \le a^2\Big[\frac{2A_4 {N_{\text{out}}}^2}{B^3} + \frac{4N_{\text{out}}V_0^2}{B^2} + \frac{8V_0 L^2 C_{\text{col}}}{kB}\Big] \le \frac{a^2 V_*}{k},$$

with $V_* \overset{\text{def}}{=} 2A_4K^2 + 4KV_0^2 + 8V_0L^2C_{\text{col}}$.

The bias. The mean is $\operatorname{Tr}(\Pi V)/B + m_j^\top \Pi m_j$. By eq. (SJ.32), its difference from $(2\mathscr{R}/B)\operatorname{Tr}(\Pi\Lambda)$ is at most

$$a\Big[\frac{2C_j \operatorname{Tr}\Lambda + 2\operatorname{Tr}T_D + 3\|m_j\|^2}{B} + \|m_j\|^2\Big] \le \frac{a}{k}\big(2L^2 C_{\text{col}} K + 2H_0 + 4L^2 C_{\text{col}}\big) \overset{\text{def}}{=} \frac{aD_*}{k}.$$

Cauchy–Schwarz bounds the mean absolute fluctuation by the square root of the variance, and adding the bias gives eq. (SJ.38). □

The Haar sign matrix reduces the weighted volatility to the columns of H. Define the covariance-weighted volatility of the actual update and the corresponding sum for the main text's array:

$$\begin{aligned} \mathcal{Q} &\overset{\text{def}}{=} \mathbb{E}\,\|S\|_\Sigma^2 = \mathbb{E}\operatorname{Tr}\left(\Sigma_{\text{out}} S S^\top\right), \\ \mathcal{Q}_{\text{print}} &\overset{\text{def}}{=} N_{\text{in}}\Big[\Big(1-\frac{2}{\pi}\Big)\operatorname{Tr}\Lambda + \frac{2}{\pi}N_{\text{out}}\frac{\operatorname{Tr}\Lambda^2}{\operatorname{Tr}\Lambda}\Big] = \sum_i \mu_i\, \mathfrak{v}_i^{\text{s-SGD}}, \end{aligned} \tag{SJ.39}$$

where $\mathfrak{v}_i^{\text{s-SGD}} \overset{\text{def}}{=} N_{\text{in}}\big[1 + \frac{2}{\pi}\big(\frac{\mu_i}{\bar{\mu}} - 1\big)\big]$ is the main text's SIGNSGD row volatility, written for general $N_{\text{out}}, N_{\text{in}}$.

Lemma SJ.11 (Haar reduction)**.** *In the Haar model, put $R_j \overset{\text{def}}{=} h_j^\top \Lambda h_j / \|h_j\|^2$. Then $c \le R_j \le L$ almost surely, every entry of G° is nonzero almost surely, $\|S\|_{\text{F}}^2 = N_{\text{out}}N_{\text{in}}$, and*

$$\mathcal{Q} = N_{\text{in}}\Big(1-\frac{2}{\pi}\Big)\operatorname{Tr}\Lambda + \frac{2}{\pi}N_{\text{out}}\sum_{j=1}^{N_{\text{in}}}\mathbb{E}\,R_j, \qquad \mathbb{E}\,\|S^\top u_i\|^2 = N_{\text{in}}\Big(1-\frac{2}{\pi}\Big) + \frac{2}{\pi}N_{\text{out}}\sum_{j=1}^{N_{\text{in}}}\mathbb{E}\,\frac{(h_j)_i^2}{\|h_j\|^2}. \tag{SJ.40}$$

Proof. Condition on H, which is independent of U and has nonzero columns almost surely by lemma SJ.10. The jth column of $U^\top S$ is $U^\top \operatorname{sign}(Uh_j)$, and lemma SJ.7 applies to it. Taking the trace against Λ, and the (i,i) entry, gives

$$\begin{aligned} \mathbb{E}_U\left[\operatorname{sign}(Uh_j)^\top U\Lambda U^\top \operatorname{sign}(Uh_j)\right] &= \Big(1-\frac{2}{\pi}\Big)\operatorname{Tr}\Lambda + \frac{2}{\pi}N_{\text{out}}R_j, \\ \mathbb{E}_U\left[(e_i^\top U^\top \operatorname{sign}(Uh_j))^2\right] &= 1 - \frac{2}{\pi} + \frac{2}{\pi}N_{\text{out}}\frac{(h_j)_i^2}{\|h_j\|^2}. \end{aligned}$$

Summing over j and averaging over H gives eq. (SJ.40); no independence between columns is needed. A Haar rotation of a fixed nonzero vector has no zero coordinate almost surely, so every entry of UH is nonzero and $\|S\|_{\mathrm{F}}^2 = N_{\mathrm{out}}N_{\mathrm{in}}$. □

Theorem SJ.3 (Half-anisotropic weighted volatility)**.** *In the Haar model of definition SJ.3 there is an admissible constant C_{vol} such that*

$$|\mathcal{Q} - \mathcal{Q}_{\mathrm{print}}| \le C_{\mathrm{vol}} \frac{N_{\mathrm{out}}N_{\mathrm{in}}}{\sqrt{k}}, \tag{SJ.41}$$

and both $\mathcal{Q}$ and $\mathcal{Q}_{\mathrm{print}}$ lie in $[cN_{\mathrm{out}}N_{\mathrm{in}}, LN_{\mathrm{out}}N_{\mathrm{in}}]$. In particular $|\mathcal{Q}/\mathcal{Q}_{\mathrm{print}} - 1| \le C_{\mathrm{vol}}/(c\sqrt{k})$.

Proof. The ratio R_j is compared with $a_0 \overset{\mathrm{def}}{=} \operatorname{Tr}\Lambda^2/\operatorname{Tr}\Lambda \in [c, L]$ through a deterministic denominator. Put

$$x_0 \overset{\mathrm{def}}{=} \frac{2\mathscr{R}}{B}\operatorname{Tr}\Lambda, \qquad y_0 \overset{\mathrm{def}}{=} \frac{2\mathscr{R}}{B}\operatorname{Tr}\Lambda^2, \qquad x_- \overset{\mathrm{def}}{=} \frac{c^2 f_-^2}{K} \le x_0.$$

On the almost sure event $h_j \neq 0$,

$$R_j - a_0 = \frac{h_j^\top \Lambda h_j - y_0}{x_0} + R_j \frac{x_0 - \|h_j\|^2}{x_0}.$$

With $|R_j| \le L$ and eq. (SJ.38) for $\Pi = \Lambda$ and $\Pi = \mathrm{Id}$,

$$|\,\mathbb{E}\,R_j - a_0| \le \frac{2L}{x_-}\Big(\sqrt{\frac{V_*}{k}} + \frac{D_*}{k}\Big).$$

Inserting this in eq. (SJ.40), summing over the N_{in} columns and using $k^{-1} \le k^{-1/2}$ gives eq. (SJ.41) with $C_{\mathrm{vol}} \overset{\mathrm{def}}{=} \frac{4L}{\pi x_-}(\sqrt{V_*} + D_*)$, since $\mathcal{Q}_{\mathrm{print}} = N_{\mathrm{in}}(1 - \frac{2}{\pi})\operatorname{Tr}\Lambda + \frac{2}{\pi}N_{\mathrm{out}}N_{\mathrm{in}}a_0$.

For the range, $\|S\|_{\mathrm{F}}^2 = N_{\mathrm{out}}N_{\mathrm{in}}$ and $c\,\mathrm{Id} \preceq \Sigma_{\mathrm{out}} \preceq L\,\mathrm{Id}$ give the bounds on $\mathcal{Q}$. The quotient $\mathcal{Q}_{\mathrm{print}}/(N_{\mathrm{out}}N_{\mathrm{in}}) = (1 - \frac{2}{\pi})\bar{\mu} + \frac{2}{\pi}a_0$ is a convex combination of two numbers in $[c, L]$. □

Corollary SJ.2 (One step of the risk)**.** *In the Haar model with $k \ge k_*$, for every deterministic $\eta \ge 0$,*

$$\mathbb{E}\,\mathscr{R}(\Delta - \eta S) = \mathscr{R} - \eta L_B \|\Lambda D\|_{\mathrm{F}}^2 + \frac{\eta^2}{2}\mathcal{Q}_{\mathrm{print}} + \mathcal{E}_\eta, \qquad |\mathcal{E}_\eta| \le \eta L f_+ \sqrt{C_*} + \frac{\eta^2}{2} C_{\mathrm{vol}} \frac{N_{\mathrm{out}}N_{\mathrm{in}}}{\sqrt{k}}.$$

Proof. Given U, the data satisfy definition SA.1 with covariance $U\Lambda U^\top$, because the whitened batch is independent of U. The identity eq. (SJ.2) applied given U and averaged over U gives

$$\mathbb{E}\,\mathscr{R}(\Delta - \eta S) = \mathscr{R} - \eta\,\mathbb{E}\langle \Delta, S\rangle_\Sigma + \frac{\eta^2}{2}\mathcal{Q}.$$

Corollary SJ.1 and theorem SJ.3 give the two error terms. □

SJ.4.4 The corrected row volatility

The weighted total of the main text's row array is correct to leading order, but the individual rows are not: the actual row volatility depends on the state. The proof controls the random column norms in eq. (SJ.40) to fourth order.

Lemma SJ.12 (Fourth moments of sums)**.** *Let $Z \in \mathbb{R}^{N_{\mathrm{out}}}$ be centered with covariance V and satisfy eqs.* (SJ.36) *and* (SJ.37)*, let $Z_1, \dots, Z_B$ be independent copies, and put $S_s \overset{\mathrm{def}}{=} \sum_{b \le s} Z_b$,*

$$\mathcal{W} \overset{\mathrm{def}}{=} \sum_{b \neq b'} \langle Z_b, Z_{b'}\rangle, \qquad \mathcal{D}_B \overset{\mathrm{def}}{=} \sum_{b=1}^{B} \big(\|Z_b\|^2 - \operatorname{Tr} V\big).$$

Then, for $0 \le s \le B$ and $a \in \mathbb{R}^{N_{\mathrm{out}}}$,

$$\mathbb{E}\,\|S_s\|^4 \le 3A_4 s^2 {N_{\mathrm{out}}}^2, \qquad \mathbb{E}(a^\top S_s)^4 \le 3A_4 s^2 \|a\|^4,$$
$$\mathbb{E}\,\mathcal{W}^4 \le 144 A_4^2 B^4 {N_{\mathrm{out}}}^2, \qquad \mathbb{E}\,\mathcal{D}_B^4 \le (16A_8 + 3A_4^2) B^2 {N_{\mathrm{out}}}^4.$$

Proof. Sums. In the expansion of $\|S_s\|^4$, a term in which some sample index occurs once has mean zero. The remaining terms give

$$\mathbb{E}\,\|S_s\|^4 = s\;\mathbb{E}\,\|Z\|^4 + s(s-1)\big[(\operatorname{Tr} V)^2 + 2\operatorname{Tr} V^2\big].$$

With $\operatorname{Tr} V \le N_{\text{out}}\sqrt{A_4}$ and $\operatorname{Tr} V^2 \le N_{\text{out}} A_4$, which follow from $a^\top V a \le (\mathbb{E}\,|a^\top Z|^4)^{1/2}$, this is at most $3A_4 s^2 {N_{\text{out}}}^2$. The scalar case is $s\;\mathbb{E}(a^\top Z)^4 + 3s(s-1)(a^\top V a)^2 \le 3A_4 s^2 \|a\|^4$.

The off-diagonal sum. A random partition splits $\mathcal{W}$ into independent sums. Let $\epsilon_1, \dots, \epsilon_B$ be independent fair Bernoulli selectors, independent of the Z_b. For fixed vectors,

$$\mathcal{W} = 4\;\mathbb{E}_\epsilon \Big\langle \sum_b \epsilon_b Z_b,\; \sum_{b'} (1-\epsilon_{b'}) Z_{b'} \Big\rangle,$$

because diagonal coefficients vanish and each ordered off-diagonal pair has coefficient $\frac{1}{4}$ in mean. Jensen's inequality gives a factor 4^4. For a fixed partition into s and s' samples the two sums are independent; conditioning on the second and applying the two bounds above gives

$$\mathbb{E}\,|\langle S_s, S'_{s'}\rangle|^4 \le 3A_4 s^2\;\mathbb{E}\,\|S'_{s'}\|^4 \le 9A_4^2 s^2 s'^2 {N_{\text{out}}}^2.$$

With $ss' \le B^2/4$, the constant is $4^4 \cdot 9/16 = 144$.

The diagonal sum. For $Y \overset{\text{def}}{=} \|Z\|^2 - \operatorname{Tr} V$, Jensen's inequality gives $\mathbb{E}\,Y^4 \le 16\;\mathbb{E}\,\|Z\|^8 \le 16A_8 {N_{\text{out}}}^4$ and $\mathbb{E}\,Y^2 \le \mathbb{E}\,\|Z\|^4 \le A_4 {N_{\text{out}}}^2$, and for independent centered summands $\mathbb{E}\,\mathcal{D}_B^4 = B\;\mathbb{E}\,Y^4 + 3B(B-1)(\mathbb{E}\,Y^2)^2$. □

Lemma SJ.13 (Column norms)**.** *In the Haar model put $x_0 \overset{\text{def}}{=} 2\mathscr{R}\operatorname{Tr}\Lambda/B$. There are admissible constants C_{good}, C_{bad} such that, for every column j and row i,*

$$\mathbb{E}\,\Big|\frac{(h_j)_i^2}{\|h_j\|^2} - \frac{(h_j)_i^2}{x_0}\Big| \le \frac{C_{\text{good}}}{k^{3/2}} + \frac{C_{\text{bad}}}{k^2}. \tag{SJ.42}$$

Proof. The centered norm. Write $h_j = m_j + S_B/B$ with S_B as in lemma SJ.12 for $Z = X - m_j$. Then exactly

$$\|h_j\|^2 - \mathbb{E}\,\|h_j\|^2 = \frac{2m_j^\top S_B}{B} + \frac{\mathcal{D}_B}{B^2} + \frac{\mathcal{W}}{B^2}.$$

The first term has fourth moment at most $48A_4\|m_j\|^4/B^2 \le 48A_4 H_0^2/k^2$. Lemma SJ.12 and $N_{\text{out}} \le Kk \le KB$ bound the other two by $(16A_8 + 3A_4^2)K^4/k^2$ and $144A_4^2K^2/k^2$. With $|x+y+z|^4 \le 27(x^4+y^4+z^4)$,

$$\mathbb{E}\,\big(\|h_j\|^2 - \mathbb{E}\,\|h_j\|^2\big)^4 \le \frac{F_4}{k^2}, \qquad F_4 \overset{\text{def}}{=} 27\big[48A_4H_0^2 + (16A_8 + 3A_4^2)K^4 + 144A_4^2K^2\big].$$

The bias. Taking traces in eq. (SJ.33),

$$0 \le \mathbb{E}\,\|h_j\|^2 - x_0 = \frac{2C_j \operatorname{Tr}\Lambda + 2\operatorname{Tr} T_D}{B} + \Big(1 + \frac{3}{B}\Big)\|m_j\|^2 \le \frac{b_0}{k},$$

with $b_0 \overset{\text{def}}{=} 2L^2KC_{\text{col}} + 2LP_+ + 4L^2C_{\text{col}}$. Hence, with $Q_4 \overset{\text{def}}{=} 8(F_4 + b_0^4)$,

$$\mathbb{E}\,\big(\|h_j\|^2 - x_0\big)^4 \le \frac{Q_4}{k^2}, \qquad \mathbb{E}\,\big(\|h_j\|^2 - x_0\big)^2 \le \frac{\sqrt{Q_4}}{k}.$$

The same decomposition and lemma SJ.12 give $\mathbb{E}(h_j)_i^4 \le H_4/k^2$ with $H_4 \overset{\text{def}}{=} 8(L^4C_{\text{col}}^2 + 3A_4)$. Also $x_0 \ge a_0 \overset{\text{def}}{=} P_- c/K$.

The ratio. Put $\delta \overset{\text{def}}{=} \|h_j\|^2 - x_0$. On $\{\|h_j\|^2 \ge x_0/2\}$,

$$(h_j)_i^2\,\Big|\frac{1}{\|h_j\|^2} - \frac{1}{x_0}\Big| \le \frac{2}{a_0^2}\,(h_j)_i^2|\delta|,$$

whose mean is at most $2a_0^{-2}(H_4/k^2)^{1/2}(\sqrt{Q_4}/k)^{1/2} = C_{\text{good}}k^{-3/2}$ by Cauchy–Schwarz. Markov's inequality gives $\mathbb{P}(\|h_j\|^2 < x_0/2) \le 16Q_4/(a_0^4k^2) \stackrel{\text{def}}{=} b_1/k^2$. On that event $(h_j)_i^2/\|h_j\|^2 \le 1$, and Cauchy–Schwarz gives

$$\frac{1}{x_0}\,\mathbb{E}\left[(h_j)_i^2\,\mathbf{1}_{\{\|h_j\|^2<x_0/2\}}\right] \le \frac{\sqrt{H_4 b_1}}{a_0 k^2}.$$

Adding the three contributions gives eq. (SJ.42) with $C_{\text{bad}} \stackrel{\text{def}}{=} b_1 + \sqrt{H_4 b_1}/a_0$. □

Theorem SJ.4 (Corrected row volatility)**.** *In the Haar model of definition SJ.3, define the raw annealed row volatility of the update and the corrected array*

$$\mathfrak{v}_i^{(0)} \stackrel{\text{def}}{=} \mathbb{E}\,\|S^\top u_i\|^2, \tag{SJ.43}$$

$$\mathfrak{v}_i^{\text{corr}} \stackrel{\text{def}}{=} N_{\text{in}}\Big(1-\frac{2}{\pi}\Big) + \frac{2}{\pi}\,N_{\text{out}}\Big[\frac{(N_{\text{in}}+2)\,\mu_i}{\operatorname{Tr}\Lambda} + \frac{(B+2N_{\text{in}}+3)\,(T_D)_{ii}}{2\mathscr{R}\,\operatorname{Tr}\Lambda}\Big]. \tag{SJ.44}$$

There is an admissible constant C_{row} such that, for every i,

$$|\mathfrak{v}_i^{(0)} - \mathfrak{v}_i^{\text{corr}}| \le C_{\text{row}}\,\frac{N_{\text{in}}}{\sqrt{k}}, \qquad \Big|\frac{\mathfrak{v}_i^{(0)}}{\mathfrak{v}_i^{\text{corr}}} - 1\Big| \le \frac{C_{\text{row}}}{(1-2/\pi)\sqrt{k}}. \tag{SJ.45}$$

The actual total is exact, $\sum_i \mathfrak{v}_i^{(0)} = N_{\text{out}}N_{\text{in}}$, while

$$\sum_i \mathfrak{v}_i^{\text{corr}} - N_{\text{out}}N_{\text{in}} = \frac{2}{\pi}\,N_{\text{out}}\Big[2 + \frac{(B+2N_{\text{in}}+3)\operatorname{Tr}T_D}{2\mathscr{R}\,\operatorname{Tr}\Lambda}\Big] \in \Big(0,\ \frac{2}{\pi}N_{\text{out}}\big[2+(3K+3)L/c\big]\Big]. \tag{SJ.46}$$

The corrected array exceeds the main text's array row by row:

$$\mathfrak{v}_i^{\text{corr}} - \mathfrak{v}_i^{\text{s-SGD}} = \frac{2}{\pi}\,N_{\text{out}}\Big[\frac{2\mu_i}{\operatorname{Tr}\Sigma_{\text{out}}} + \frac{(B+2N_{\text{in}}+3)\,(\Lambda DD^\top\Lambda)_{ii}}{2\mathscr{R}\,\operatorname{Tr}\Sigma_{\text{out}}}\Big] \ge 0. \tag{SJ.47}$$

Proof. The row volatility. By lemma SJ.11,

$$\mathfrak{v}_i^{(0)} = N_{\text{in}}\Big(1-\frac{2}{\pi}\Big) + \frac{2}{\pi}\,N_{\text{out}}\sum_{j=1}^{N_{\text{in}}}\mathbb{E}\,\frac{(h_j)_i^2}{\|h_j\|^2}.$$

Replacing each $\|h_j\|^2$ by x_0 costs, by lemma SJ.13 and $N_{\text{out}} \le Kk$, at most

$$\frac{2}{\pi}\,N_{\text{out}}N_{\text{in}}\Big(\frac{C_{\text{good}}}{k^{3/2}} + \frac{C_{\text{bad}}}{k^2}\Big) \le C_{\text{row}}\frac{N_{\text{in}}}{\sqrt{k}}, \qquad C_{\text{row}} \stackrel{\text{def}}{=} \frac{2}{\pi}K(C_{\text{good}}+C_{\text{bad}}).$$

The replaced sum is $\sum_j \mathbb{E}(h_j)_i^2/x_0 = (\mathbb{E}\,HH^\top)_{ii}/x_0$, and eq. (SJ.34) with $x_0 = 2\mathscr{R}\operatorname{Tr}\Lambda/B$ identifies it:

$$\frac{(\mathbb{E}\,HH^\top)_{ii}}{x_0} = \frac{(N_{\text{in}}+2)\mu_i}{\operatorname{Tr}\Lambda} + \frac{(B+2N_{\text{in}}+3)(T_D)_{ii}}{2\mathscr{R}\operatorname{Tr}\Lambda}.$$

This proves the first bound in eq. (SJ.45). The relative bound follows from $\mathfrak{v}_i^{\text{corr}} \ge N_{\text{in}}(1-2/\pi)$.

Totals. By lemma SJ.11, $\|S\|_{\text{F}}^2 = N_{\text{out}}N_{\text{in}}$, and $\sum_i \|S^\top u_i\|^2 = \|S\|_{\text{F}}^2$ by orthogonality. Summing eq. (SJ.44) over i gives the identity in eq. (SJ.46). Its bracket is positive, and $\operatorname{Tr}T_D \le 2\mathscr{R}L$, $\operatorname{Tr}\Lambda \ge cN_{\text{out}}$ and $B+2N_{\text{in}}+3 \le (3K+3)N_{\text{out}}$ give the upper bound.

The difference. The main text's array is $\mathfrak{v}_i^{\text{s-SGD}} = N_{\text{in}}(1-\frac{2}{\pi}) + \frac{2}{\pi}N_{\text{out}}N_{\text{in}}\mu_i/\operatorname{Tr}\Lambda$. Subtracting it from eq. (SJ.44) gives eq. (SJ.47), with $\operatorname{Tr}\Sigma_{\text{out}} = \operatorname{Tr}\Lambda$ and $T_D = \Lambda DD^\top\Lambda$. Both terms are nonnegative. □

For $N_{\text{out}} = N_{\text{in}} = N$ the total excess is of order N against a total N^2, and the weighted totals agree to leading order by theorem SJ.3. For an individual row the state-dependent term can be of the same order as $\mathfrak{v}_i^{\text{s-SGD}}$: for $\Lambda = \text{Id}$, $B = N$ and $D = e_1 \otimes \mathbf{1}/\sqrt{N}$, the corrected coefficient of the first row is $(1+\frac{6}{\pi})N + \frac{10}{\pi}$, whereas $\mathfrak{v}_1^{\text{s-SGD}} = N$.

Proposition SJ.8 (Half-anisotropic SIGNSGD recursion)**.** *In the Haar model of definition SJ.3 with $k \geq k_*$, let $\eta \geq 0$ be deterministic and $\Delta^+ = \Delta - \eta S$. Then, with $\mathfrak{d}_i \overset{\text{def}}{=} \mu_i \mathcal{N}_B / \sqrt{\pi \bar{\mu}}$, there are real numbers $E_1, \ldots, E_{N_{\text{out}}}$ such that*

$$\mathbb{E}\ \mathscr{Q}_i(\Delta^+) = \mathscr{Q}_i(\Delta) - \frac{2\eta\, \mathfrak{d}_i}{\sqrt{\mathscr{R}}}\, \mathscr{Q}_i(\Delta) + \eta^2\, \mathfrak{v}_i^{\text{corr}} + E_i, \qquad \sum_i |E_i| \leq 2\eta f_+ \sqrt{C_*} + \eta^2 C_{\text{row}}\, \frac{N_{\text{out}} N_{\text{in}}}{\sqrt{k}}. \tag{SJ.48}$$

The risk satisfies

$$\mathbb{E}\ \mathscr{R}(\Delta^+) = \mathscr{R} - \eta \sum_i \frac{\mu_i\, \mathfrak{d}_i}{\sqrt{\mathscr{R}}}\, \mathscr{Q}_i(\Delta) + \frac{\eta^2}{2} \sum_i \mu_i\, \mathfrak{v}_i^{\textit{s-SGD}} + \mathcal{E}_\eta, \tag{SJ.49}$$

with $\mathcal{E}_\eta$ bounded as in corollary SJ.2.

Proof. Given U, the state is fixed and the whitened batch is independent of U, so eq. (SJ.3) applies given U. Averaging over U gives

$$\mathbb{E}\ \mathscr{Q}_i(\Delta^+) = \mathscr{Q}_i(\Delta) - 2\eta\ \mathbb{E}\langle \Delta^\top u_i, S^\top u_i \rangle + \eta^2 \mathfrak{v}_i^{(0)}.$$

The identity $L_B \mu_i = \mathfrak{d}_i / \sqrt{\mathscr{R}}$ holds by the definition of L_B. Define E_i as -2η times the error in eq. (SJ.29) plus $\eta^2(\mathfrak{v}_i^{(0)} - \mathfrak{v}_i^{\text{corr}})$. Corollary SJ.1 and theorem SJ.4 bound the sum of their absolute values. The risk identity is corollary SJ.2, since $L_B \|\Lambda D\|_{\text{F}}^2 = \sum_i L_B \mu_i^2 \|D_{i\cdot}\|^2$ and $\mathcal{Q}_{\text{print}} = \sum_i \mu_i \mathfrak{v}_i^{\text{s-SGD}}$. □

The proposition is the main text's recursion eq. (14) for SIGNSGD at one step, with three differences that are part of its statement: the drift uses $\mathcal{N}_B$, which differs from $\sqrt{B}$ by a factor in $[1 - 4/B, 1]$; the per-row volatility is the state-dependent $\mathfrak{v}_i^{\text{corr}}$, and the main text's array $\mathfrak{v}_i^{\text{s-SGD}}$ is correct only in the covariance-weighted total eq. (SJ.49); and the hypotheses of definition SJ.3, among them a covariance bounded below by a fixed $c > 0$ and a state independent of the eigenbasis, which hold at one step and are not shown to persist.

SK Numerical simulations for Figures 3, 4 and 6

This appendix describes the simulations behind Figs. 3, 4 and 6 and the checks of the approximations (9). Nothing in it is used in a proof.

Model and algorithms. All runs use the square isotropic model with $N_{\text{in}} = N_{\text{out}} = N = 512$ and $\Sigma_{\text{in}} = \Sigma_{\text{out}} = \text{Id}_N$. At an error $\Delta = W - W^\star$, a batch of size B gives the gradient

$$G = \frac{1}{B} \sum_{b=1}^{B} x_{\text{out}}^b \otimes x_{\text{in}}^b \left\langle x_{\text{out}}^b \otimes x_{\text{in}}^b, \Delta \right\rangle, \qquad \mathbb{E}\ G = \Delta, \qquad \mathscr{R} = \tfrac{1}{2}\|\Delta\|_{\text{F}}^2.$$

MUON keeps the buffer $M_t = \beta M_{t-1} + (1-\beta) G_{t+1}$ and steps $W_{t+1} = W_t - \eta S_t$. With the exact polar factor, $S_t = UV^\top$ is computed from a singular value decomposition of M_t over the singular values above 10^{-13} times the largest. With five Newton–Schulz steps (NS-5), S_t is the result of five iterations of

$$X \leftarrow aX + \big(bXX^\top + c(XX^\top)^2\big)X$$

from $X_0 = M_t / \|M_t\|_{\text{F}}$, with $(a, b, c) = (3.4445, -4.7750, 2.0315)$. SIGNSGD steps $W_{t+1} = W_t - \eta\, \text{sign}(G_{t+1})$, without momentum. All arithmetic is in double precision.

Panels (a) and (b): frozen error. The error is frozen at $\Delta = \text{Id}_N / \sqrt{N}$; by the invariance of the model under $\Delta \mapsto O_1 \Delta O_2$ with orthogonal O_1, O_2, this is equivalent in law to any Δ with the same singular values. The buffer is run to stationarity for $\lceil 37/\log(1/\beta) + N/B \rceil$ steps (none at $\beta = 0$), after which T steps are recorded, with $T = 100$ at $\beta = 0$ and $T = \max(600, 3N/B + 300)$ at $\beta = 0.9$, two seeds each. The drift estimate is the mean of $\langle S_t, \Delta \rangle / (\sqrt{2}\|\Delta\|_{\text{F}})$. The volatility estimate is the mean of $\|S_t\|_{\text{F}}^2$ plus twice the centred lag sum $\sum_{h \geq 1} \langle S_t - \bar{c}\hat{\Delta},\ S_{t-h} - \bar{c}\hat{\Delta} \rangle$, where

$\hat{\Delta} = \Delta/\|\Delta\|_{\mathrm{F}}$ and $\bar{c}$ is the mean of $\langle S_t, \hat{\Delta}\rangle$; it is averaged over $t \geq T/2$, so that every lag up to $T/2$ enters with full weight. Both estimates agree with the quadrature to within about 1% for $B_{\mathrm{eff}}/N \leq 4$. For larger B_{eff}/N the measured volatility lies below the $N \to \infty$ constant by a finite-N correction of relative size about $0.7\, B_{\mathrm{eff}}/N^2$: at $\beta = 0.9$ it is 2% at $B_{\mathrm{eff}}/N = 16$ and 9% at $B_{\mathrm{eff}}/N = 64$ for $N = 512$, and it roughly doubles each time N is halved (18% and 32% at $B_{\mathrm{eff}}/N = 64$ for $N = 256$ and 128). Quadrupling T does not change it. The drift is 1.4% low at $B_{\mathrm{eff}}/N = 64$.

Panel (c): rates. Each run starts from a Gaussian Δ_0 with $\mathscr{R}(0) = 6.25$ (or 4 at the smallest batches) and uses the constant step $\eta = 4\mathfrak{d}/\mathfrak{v}$ built from the theory constants, which puts the floor at $\mathscr{R}_\star = 1$ for full whitening. The recursion (8), $\mathscr{R}(t{+}1) = \mathscr{R}(t) - a\sqrt{\mathscr{R}(t)} + b$ with $a = 2\eta\mathfrak{d}$ and $b = \eta^2\mathfrak{v}/2$, is fitted to the seed-averaged risk by least squares in $\log \mathscr{R}$. Near the floor $\sqrt{\mathscr{R}_\star} = b/a$ the recursion contracts by the factor $1 - a^2/(2b) = 1 - 4\mathfrak{d}^2/\mathfrak{v}$ per step, which does not depend on η, and we report

$$1 - \text{rate} = \frac{a^2}{4b} = \frac{2\mathfrak{d}^2}{\mathfrak{v}},$$

the factor of (12). So the measured rate does not depend on the step, even when NS-5 reaches a floor different from the one it was tuned for. Runs last between $7/\kappa$ and $9/\kappa$ steps, $\kappa = 8\mathfrak{d}^2/\mathfrak{v}$.

The SIGNSGD drift is $\mathcal{N}_B/\sqrt{\pi}$ with $\mathcal{N}_B$ as in Sec. 3, and the main text writes $\sqrt{B}$ for $\mathcal{N}_B$. In panel (c) each measured ratio is multiplied by $\mathcal{N}_B^2/B$, which puts SIGNSGD on the $\sqrt{B}$ scale of the curves. We use $\mathcal{N}_1 = 2/\pi$, a Monte Carlo table with $2 \cdot 10^6$ samples for $2 \leq B \leq 64$, and

$$\mathcal{N}_B = \sqrt{B}\Big(1 - \frac{1}{B} + \frac{5}{B^2} + O(B^{-3})\Big)$$

for $B > 64$. The expansion follows from the cumulants 1, 8 and 200 of $(xy)^2$; at $B = 64$ it agrees with the table to $4 \cdot 10^{-5}$.

Grid. Panel (c) uses $B_{\mathrm{eff}}/N = 2^k$, $k = -4, \ldots, 4$, at $\beta \in \{0, 0.5, 0.9\}$, with SIGNSGD, NS-5 and the exact polar factor at each point and two seeds, between 10^2 and $6 \cdot 10^5$ steps. Full whitening inverts the oldest direction of the buffer, whose weight is $\beta^{N/B}$. Where this is below 10^{-11} the exact polar factor is not computable in double precision, and that marker is omitted: $\beta = 0.5$ at $B_{\mathrm{eff}}/N = 1/16$ ($0.5^{46.5} \approx 10^{-14}$) and $\beta = 0.9$ at $B_{\mathrm{eff}}/N = 1/16$ ($0.9^{256} \approx 2 \cdot 10^{-12}$). The same rule removes the frozen point at $\beta = 0.9$, $B = 2$ from panels (a) and (b).

Newton–Schulz. The quintic coefficients above do not converge to the polar factor: with many steps the singular values of S_t oscillate in a band near $[0.68, 1.13]$, so $\|S_t\|_{\mathrm{F}}^2 \approx 0.86N$. This is close to a scalar multiple of the polar factor; it changes the floor, but it cancels in $2\mathfrak{d}^2/\mathfrak{v}$. In frozen runs at $\beta = 0.9$, $B = 13$ ($B_{\mathrm{eff}}/N \approx 0.48$), the rate with K steps equals the exact rate within the seed spread for every $K \geq 7$. With $K = 5$ the drift is 16% lower and the volatility 38% lower, so the rate is 14% higher: five steps do not reach the oldest buffer directions, whose singular values are below $a^{-5}\|M_t\|_{\mathrm{F}} \approx 2 \cdot 10^{-3}\|M_t\|_{\mathrm{F}}$, and act as a small positive ridge. The effect grows as B_{eff}/N decreases; at $\beta = 0.5$ the NS-5 rate exceeds the full-whitening rate by factors 1.15, 1.42, 1.76 and 1.96 at $B_{\mathrm{eff}}/N = 1/2$, $1/4$, $1/8$ and $1/16$, and at $\beta = 0.9$ by 1.37, 1.79 and 1.92 at $B = 7$, 3 and 2 ($B_{\mathrm{eff}}/N \approx 0.26$, 0.11 and 0.07). For $B_{\mathrm{eff}} \geq 2N$ NS-5 and full whitening agree within 3%. Without momentum the update has rank $\min\{B, N\}$ and there is no stale direction to suppress, and the sign of the effect reverses: for $B < N$ the NS-5 rate is 5–8% below the exact one (6%, 8% and 5% at $B/N = 1/8$, $1/4$ and $1/2$), and at $B = N$ it is 7% above.

The approximations (9). The forms for $\beta > 0$ interpolate between the two batch limits: additively in C_β^2, because the thin-batch limit $1/(2\sqrt{\pi})$ is a positive constant, and additively in $\mathfrak{v}_{\mathrm{iso},\beta}B/B_{\mathrm{eff}}$. Inserted in the ratio of the factors $1 - \text{rate}$, they give

$$\frac{1 - \text{rate}_{\text{Muon}}}{1 - \text{rate}_{\text{s-SGD}}} \approx \frac{32/(9\pi) + N/(4B_{\mathrm{eff}})}{c_\beta + N/(2B_{\mathrm{eff}})},$$

which tends to $32/(9\pi c_\beta)$ for $B_{\mathrm{eff}} \gg N$ and to $\frac{1}{2}$ for $B_{\mathrm{eff}} \ll N$. Figure S3 compares them with the quadrature of the exact constants, and the table gives the largest relative error over $1/64 \leq B_{\mathrm{eff}}/N \leq 64$; all three approximations overestimate, most near $B_{\mathrm{eff}}/N \approx 0.3$.

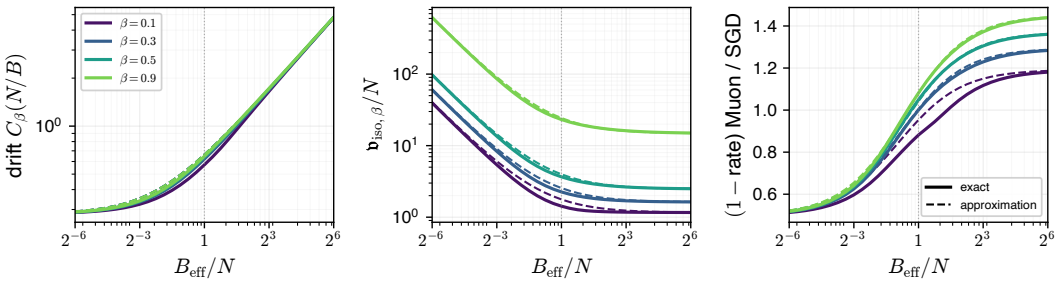


Figure S3: The algebraic forms (9) (dashed) against the exact zero-ridge constants (solid), for $\beta \in \{0.1, 0.3, 0.5, 0.9\}$. **Left:** drift constant. **Centre:** integrated volatility. **Right:** ratio of the factors $1 - \text{rate}$.

β	0.1	0.3	0.5	0.9	0.95
C_β	16%	8.0%	5.7%	4.6%	4.6%
$\mathfrak{v}_{\mathrm{iso},\beta}$	26%	16%	12%	8.8%	8.7%
ratio of $1 - \text{rate}$	8.7%	2.8%	2.1%	2.1%	2.1%

The accuracy degrades as $\beta \downarrow 0$ because the limit $\beta \downarrow 0$ at zero ridge is discontinuous for $N > B$ (proposition SF.12).

SL Finite-ridge effects

The theory of the paper is proved for the regularized SIGNSVD map Q_ϱ of definition SA.2, and its sharpest statements are made in the limit of vanishing ridge, which is exact whitening. Practical MUON does not whiten exactly: it applies a few Newton–Schulz steps, which amplify small singular values only up to a point. This appendix collects numerical evidence on what such imperfect whitening does, and on when a positive ridge describes it. Nothing in this appendix is used in a proof.

SL.1 Newton–Schulz as a ridge

Fig. 6 shows that MUON with five Newton–Schulz steps (NS-5) and momentum behaves like a small positive ridge. This subsection measures, for every number of steps $K \leq 7$, which ridge reproduces NS-K in the isotropic model, how accurately, and where the description fails. The calibration is made on a frozen error (Secs. SL.1.1 and SL.1.2) and then tested, without refitting, on the dynamics (Sec. SL.1.3).

SL.1.1 Frozen-error calibration: method

Model and updates. All runs use the square isotropic model of Appendix SK, with $N = 512$ unless stated otherwise, momentum buffer $M_t = \beta M_{t-1} + (1-\beta)G_t$ and $\beta \in \{0, 0.05, 0.1, 0.25, 0.5, 0.9\}$. Every update we compare is a spectral function of the buffer. If $M_t = \sum_k \sigma_k\, p_k \otimes r_k$ is a singular value decomposition, then the update is $\sum_k \phi(\sigma_k)\, p_k \otimes r_k$ with

$$\phi_{\mathrm{polar}}(\sigma) = 1, \qquad \phi_\varrho(\sigma) = \frac{\sigma}{\sqrt{\sigma^2 + \varrho}}, \qquad \phi_{\text{NS-}K}(\sigma) = p^{\circ K}\Big(\frac{\sigma}{\|M_t\|_{\mathrm{F}}}\Big),$$

where $p(x) = ax + bx^3 + cx^5$ with $(a, b, c) = (3.4445, -4.7750, 2.0315)$ and $p^{\circ K}$ is its K-fold composition. The last formula is exact: K Newton–Schulz iterations from $X_0 = M_t/\|M_t\|_{\mathrm{F}}$ act on each singular value separately.

Units of the ridge. Newton–Schulz acts on $M_t/\|M_t\|_{\mathrm{F}}$, so we parametrize the ridge in the same units,

$$\varrho = \varrho_X\, \|M_t\|_{\mathrm{F}}^2.$$

In the frozen state of Appendix SK, which has $2\mathscr{R} = 1$, the normalized and the physical ridge coincide, and the measured $\|M_t\|_{\mathrm{F}}^2$ agrees with $s_2 N^2/B = N^2/B_{\mathrm{eff}}$ to within 5% in every run

(median ratio 1.013). A fixed ϱ_X therefore corresponds to the normalized ridge $\varrho_X N^2/B_{\text{eff}}$, which is $\varrho_X N$ times the mean eigenvalue of $M_t M_t^\top$. The heuristic of section 1.2 matches the small-σ slope a^K of $\phi_{\text{NS-}K}$ with the slope $\varrho_X^{-1/2}$ of ϕ_ϱ, which gives $\varrho_X = a^{-2K}$.

The frozen filter bank. The error is frozen at $\Delta = \mathrm{Id}_N/\sqrt{N}$, and the buffer is burnt in and recorded exactly as in Appendix SK: $\lceil 37/\log(1/\beta) + N/B \rceil$ burn-in steps (none at $\beta = 0$), then $T = \max(600, 3N/B + 300)$ recorded steps ($T = 400$ at $\beta = 0$). Because the error is frozen, the sequence M_t does not depend on the update rule. One double-precision singular value decomposition per step therefore gives, on the same random numbers, the frozen statistics of the exact polar factor, of NS-K for $K = 1, \dots, 7$, and of exact ridges at the 16 values $\varrho_X \in \{10^{-10}, 10^{-9.33}, \dots, 1\}$. For each update we record the drift $\mathfrak{d}$, the integrated volatility $\mathfrak{v}$ and the same-time volatility $\mathfrak{v}_{\text{same}} = \mathbb{E}\,\|S_t\|_{\text{F}}^2$ with the estimators of Appendix SK.

Two scale-free statistics. NS-K does not output singular values equal to one, so its updates differ from any ridge by an overall amplitude. Two statistics do not change when the update is multiplied by a constant:

$$I_1 = \frac{\mathfrak{d}^2}{\mathfrak{v}}, \qquad I_2 = \frac{\mathfrak{v}_{\text{same}}}{\mathfrak{v}}.$$

The first sets the rate at the optimal step, $1 - \text{rate} = 2I_1$, and $\pi\gamma N I_1$ is the ratio plotted in Fig. 6 (the MUON speedup). The second is the share of the volatility that comes from a single step.

Calibration. For each configuration (N, B, β) and each K, the effective ridge $\varrho_X(K)$ is the value at which the exact-ridge bank has the I_1 of NS-K, found by linear interpolation in $\log \varrho_X$. This calibration uses no theory and no finite-N correction, because the bank and NS-K are evaluated on the same buffers. As a second, independent calibration we solve the same equation with the positive-ridge theory of Appendix SF, that is $\mathfrak{d}_{\text{iso}}(\varrho)$ and $\mathfrak{v}_{\text{iso}}(\varrho)$ at $\varrho = \varrho_X s_2 N^2/B$. When the I_1 of NS-K lies outside the range spanned by the ridges, we report that no ridge matches. The fitted ridge then predicts I_2, which was not used in the fit.

Grid. At $N = 512$: $\beta \in \{0, 0.05, 0.1, 0.25, 0.5, 0.9\}$ and $B_{\text{eff}}/N = 2^k$, $k = -4, \dots, 4$, two seeds, 108 runs. For the dependence on N: $N \in \{256, 1024\}$ at $\beta \in \{0.1, 0.9\}$ and $B_{\text{eff}}/N \in \{1/8, 1/2, 2\}$, two seeds, 24 runs.

SL.1.2 Frozen-error calibration: results

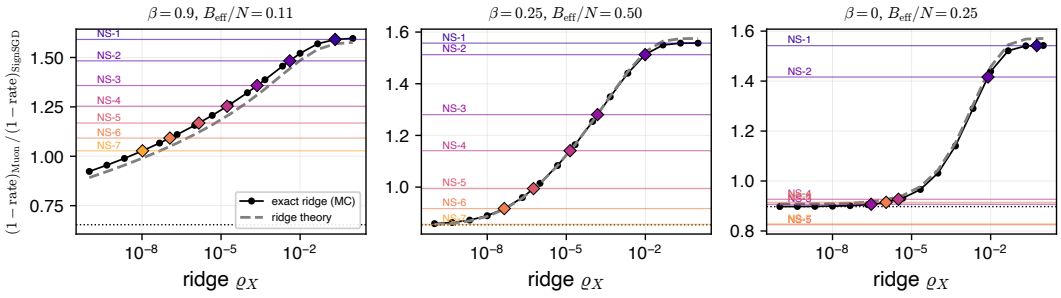


Figure S4: Rate statistic $\pi\gamma N I_1 = (1 - \text{rate})_{\text{Muon}}/(1 - \text{rate})_{\text{SignSGD}}$ (values above 1: MUON converges faster) of the exact ridge as a function of ϱ_X (black: frozen bank; dashed: ridge theory), the levels of NS-K (horizontal lines) and the calibrated ridges (diamonds), for three configurations. Dotted: exact polar factor. Right: at $\beta = 0$ the levels of NS-5 and NS-7 lie below every ridge, and no ridge matches.

Calibration curves. Fig. S4 shows the mechanism. The rate statistic of the ridge increases monotonically from the exact polar value at $\varrho_X \to 0$ to the value of a linear update at large ϱ_X, and each NS-K level crosses it once. The theory curve follows the bank to within a few percent; the gap at $\beta = 0.9$, $B = 3$ is the finite-N drift bias of the polar factor at large N/B noted in Appendix SK. At $\beta = 0$ the NS-K levels with $K \geq 5$ lie below the exact polar value, which no ridge reaches.

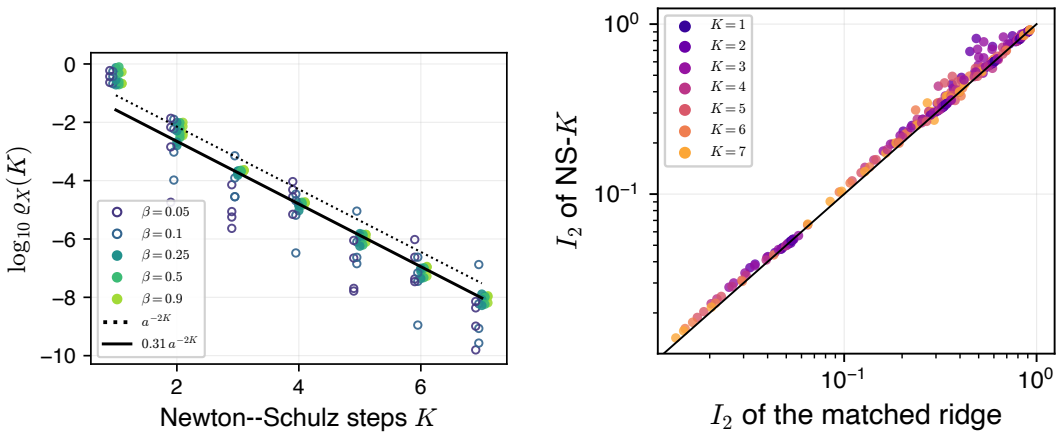


Figure S5: Left: calibrated $\log_{10} \varrho_X(K)$ for every configuration with $B_{\text{eff}} < N$ and $\beta > 0$ (filled: $\beta \geq 0.25$; open: $\beta \in \{0.05, 0.1\}$), with the heuristic a^{-2K} (dotted) and the fit $0.31\, a^{-2K}$ (solid). Right: the lag share I_2 of NS-K against that of the matched ridge, all $\beta > 0$; the line is equality.

K	bank: median [range]	theory: median	$\log_{10} a^{-2K}$	$\log_{10} 0.31 a^{-2K}$	I_2: NS/ridge-1
1	-0.58 $[-0.72, -0.10]$	-1.37	-1.07	-1.58	$+0.006$
2	-2.34 $[-2.80, -2.00]$	-2.36	-2.15	-2.65	$+0.11$
3	-3.67 $[-3.82, -3.63]$	-3.75	-3.22	-3.73	$+0.13$
4	-4.78 $[-5.03, -4.73]$	-4.84	-4.30	-4.80	$+0.10$
5	-5.92 $[-6.25, -5.81]$	-6.03	-5.37	-5.87	$+0.09$
6	-7.01 $[-7.35, -6.94]$	-7.14	-6.45	-6.95	$+0.07$
7	-8.01 $[-8.28, -7.89]$	-8.19	-7.52	-8.02	$+0.07$

Table S1: Calibrated $\log_{10} \varrho_X(K)$ at $N = 512$, $\beta \in \{0.25, 0.5, 0.9\}$, $B_{\text{eff}}/N \in \{1/16, \ldots, 1/2\}$ (twelve configurations per K, nine at $K = 7$ where the polar end of the bank is not reached).

The effective ridge. For $\beta \geq 0.25$ and $B_{\text{eff}} < N$ the calibrated ridges collapse onto one law (Fig. S5, Table S1),

$$\varrho_X(K) \approx 0.31\, a^{-2K}, \qquad a = 3.4445,$$

with a scatter of 0.13 in $\log_{10} \varrho_X$ over the 57 calibrations with $K \geq 3$. A fit with a free slope gives -1.11 decades per step, against $-2 \log_{10} a = -1.07$. The heuristic a^{-2K} has the right dependence on K and is a factor of about three too large. The theory calibration agrees with the bank calibration to within 0.2 in $\log_{10} \varrho_X$ for $K \geq 2$, and lies slightly below it. At $K = 1$ the rate statistic of the ridge is nearly flat at large ϱ_X (one Newton–Schulz step on $M_t/\|M_t\|_{\text{F}}$ is almost linear), so $\varrho_X(1)$ is only weakly determined.

Where the ridge describes Newton–Schulz. Fig. S6 compares every frozen NS-K measurement with the ridge theory at the calibrated $\varrho_X(K)$, with no further fitting.

(i) For $\beta \geq 0.25$ the calibrated ridge reproduces NS-3, NS-5 and NS-7 across the whole range $1/16 \leq B_{\text{eff}}/N \leq 16$, including the minimum near $B_{\text{eff}}/N \approx 1/2$ and the gain over the exact polar factor at small B_{eff}/N.

(ii) For $B_{\text{eff}} \geq N$ the effect of the ridge is small: for $K \geq 4$ the rate statistic of NS-K is within 2% of that of the exact polar factor, and the effective ridge is not determined there.

(iii) For $\beta \in \{0.05, 0.1\}$ a ridge still matches each NS-K, and Newton–Schulz still improves on the exact polar factor at small batch, but the matched ridge scatters over one to two decades and is not monotone in K at $\beta = 0.05$. With small momentum the weights β^j of old batches separate the singular values of the buffer into widely spaced groups, and any ridge between two groups has the same effect.

(iv) At $\beta = 0$ no ridge describes NS-K. For $K \geq 5$ and $B < N$, NS-K is slower than the exact polar factor (by 6% at $K = 5$), while every ridge is faster. The buffer is then a single

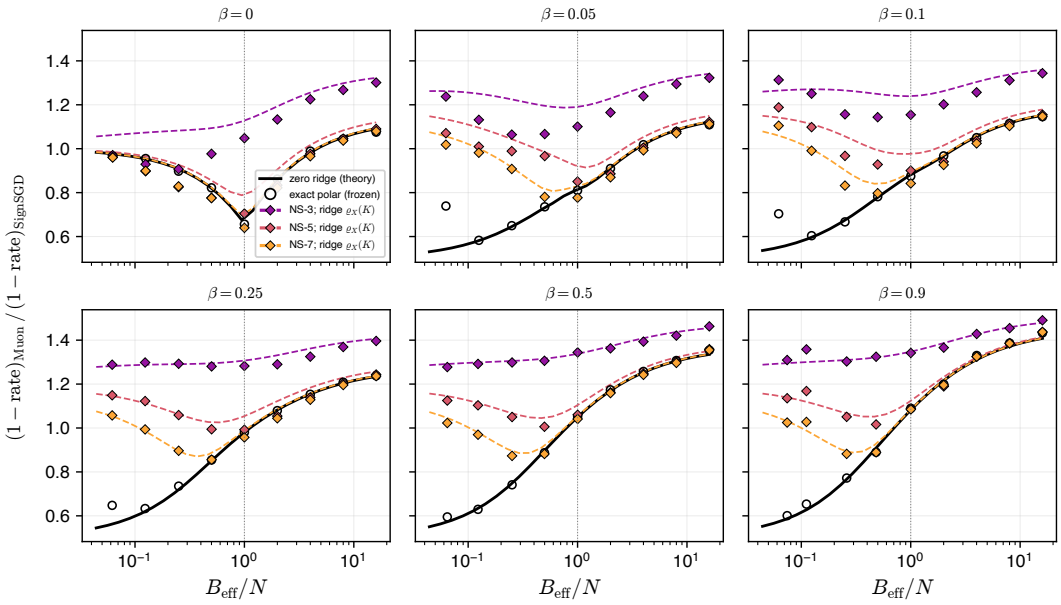


Figure S6: Frozen rate statistic $\pi\gamma N I_1 = (1-\text{rate})_{\text{Muon}}/(1-\text{rate})_{\text{SignSGD}}$ against B_{eff}/N for each momentum; values above 1 mean that MUON converges faster than SIGNSGD. Black: zero-ridge theory and frozen exact polar factor. Diamonds: frozen NS-3, NS-5, NS-7. Dashed: ridge theory at the single calibrated ridge $\varrho_X(K) = 0.31\, a^{-2K}$, the same for every panel.

minibatch gradient of rank $\min(B, N)$, with no small singular values for a ridge to suppress, and the remaining effect of Newton–Schulz is the spread of its outputs over $[0.68, 1.2]$, which adds volatility.

At the smallest batches with $\beta > 0$ the frozen exact polar factor lies above the zero-ridge curve. There the oldest directions of the buffer have weights below the rank threshold 10^{-13} of Appendix SK, so the computed polar factor is itself a polar factor with a tiny ridge.

One Newton–Schulz step is SGD. In every panel the rate statistic decreases with K, and NS-1 is the fastest. One Newton–Schulz step is nearly linear on $M_t/\|M_t\|_{\text{F}}$, whose bulk singular values are of order $N^{-1/2}$, so NS-1 is essentially momentum SGD with a normalized step. In the isotropic model there is nothing to precondition, and SGD is the most efficient of these updates: for nearly Gaussian gradient entries a linear update beats the sign update by the factor $\pi/2 \approx 1.57$, and NS-1 sits close to this value for every β and B_{eff}/N. In this model, taking the fewest steps therefore amounts to running SGD.

Consistency. The ridge fitted on I_1 predicts the lag share I_2 of NS-K to within about 10% (Fig. S5, right): NS-K has 7–13% more same-time volatility than the matched ridge for $2 \le K \le 7$, and the same at $K = 1$. The excess is the volatility added by the spread of the Newton–Schulz outputs, which a ridge does not have.

Dependence on the dimension. At $\beta = 0.9$ the calibrated ϱ_X does not depend on N: $\log_{10}\varrho_X(5) = -5.87, -5.84, -5.89$ and $\log_{10}\varrho_X(3) = -3.65, -3.63, -3.66$ at $N = 256, 512, 1024$ and $B_{\text{eff}}/N = 1/8$ (Fig. S7). The Frobenius-normalized ridge is therefore the right unit, and the normalized ridge $\varrho_X N^2/B_{\text{eff}}$ grows with N at a fixed B_{eff}/N. Accordingly the gain of NS-5 over the exact polar factor grows with N, from 1.09 to 1.14 to 1.16 at $B_{\text{eff}}/N = 1/2$. At $\beta = 0.1$ the calibrated ridge drifts with N, as the scatter in (iii) suggests.

SL.1.3 Dynamical validation

Protocol. The dynamical runs follow the protocol of Fig. 6, described in Appendix SK. Each run starts from a Gaussian Δ_0 with $\mathscr{R}(0) = 6.25$ (4 at the smallest batches), warms the buffer at W_0 for

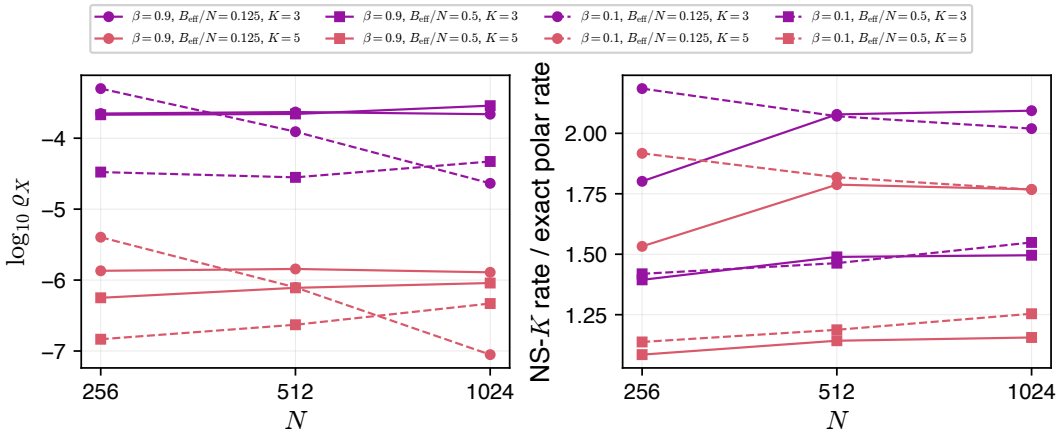


Figure S7: Dependence on N. Left: calibrated $\log_{10} \varrho_X$. Right: rate statistic of NS-K divided by that of the exact polar factor. Solid: $\beta = 0.9$; dashed: $\beta = 0.1$.

the burn-in of the frozen runs, and uses the constant step $\eta = 4\mathfrak{d}/\mathfrak{v}$ built from the frozen $(\mathfrak{d}, \mathfrak{v})$ of the same NS-K, so that its floor is near $\mathscr{R}_\star = 1$. Runs last $7/\kappa$ to $9/\kappa$ steps, $\kappa = 8\mathfrak{d}^2/\mathfrak{v}$, between $2 \cdot 10^2$ and $3.5 \cdot 10^5$ steps. Newton–Schulz is computed by matrix products in single precision, as in practice, not by the spectral formula. The grid is $N = 512$, $\beta \in \{0, 0.1, 0.25, 0.5, 0.9\}$, $B_{\text{eff}}/N = 2^k$ with $k = -4, \dots, 3$, $K = 1, \dots, 7$ and two seeds, 560 runs; the runs with different K share their initial errors. The rate is read from the recursion fit of Appendix SK, and we report the rate statistic $(1 - \text{rate})_{\text{Muon}}/(1 - \text{rate})_{\text{SignSGD}}$ on the $\sqrt{B}$ scale of the main text.

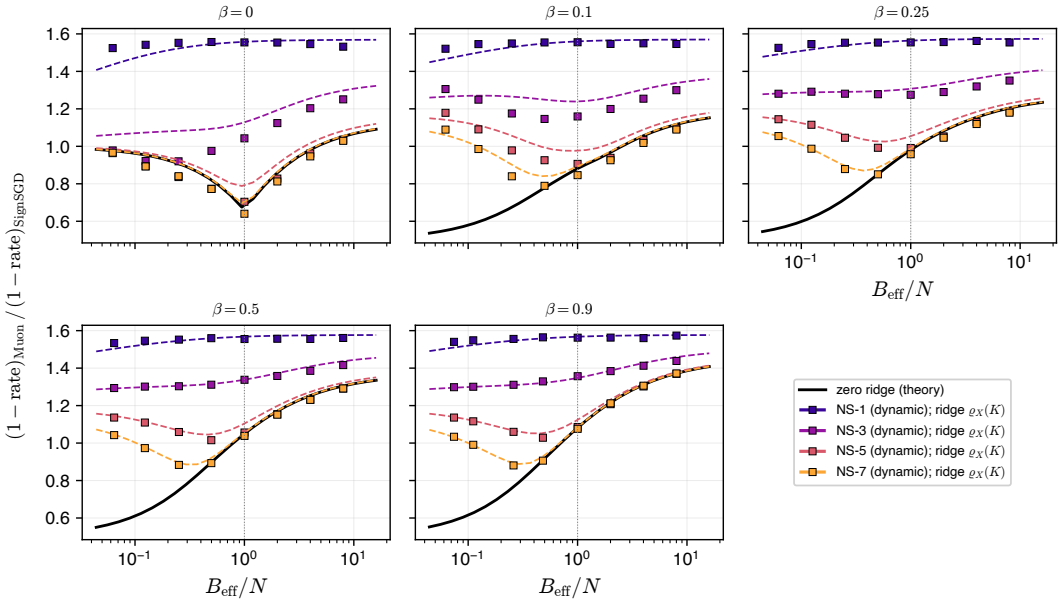


Figure S8: Dynamical rate statistic $(1-\text{rate})_{\text{Muon}}/(1-\text{rate})_{\text{SignSGD}}$ of NS-K (squares; values above 1: MUON converges faster) against the ridge theory at the frozen calibration $\varrho_X(K) = 0.31\, a^{-2K}$ (dashed), with no refitting. Solid: zero ridge.

Results. The dynamics reproduce the frozen calibration, and the calibrated ridge predicts the dynamical rates where the frozen calibration said it would (Fig. S8).

(i) The dynamical rate statistic agrees with the frozen $\pi\gamma N I_1$ of the same NS-K to 0.8% on average and 4.5% at most, over all 280 pairs of configuration and K; the two seeds

differ by 0.3% in the median. Single-precision Newton–Schulz therefore behaves as its double-precision spectral formula.

(ii) For $\beta \geq 0.25$ the ridge theory at $\varrho_X(K)$ predicts the dynamical rate of NS-K to 0.6% on average (2.7% at most) for $B_{\text{eff}}/N \leq 1/4$, and to 2.1% on average at $B_{\text{eff}}/N = 1/2$. For $B_{\text{eff}} \geq N$ it is 2.6% too fast on average (8.6% at most): there NS-K is close to the exact polar factor, and the spread of its outputs, which a ridge does not have, is the larger effect.

(iii) For $\beta = 0.1$ the error is 4.8% on average and up to 11%, and for $\beta = 0$ it is 6.5% on average and up to 15%, with the ridge predicting too fast a rate, as the frozen results anticipated.

SL.1.4 Learning rates

Sec. 3.1 gives the optimal learning rate $\eta_\star/\sqrt{\mathcal{R}} = 2\mathfrak{d}/\mathfrak{v}$ of exact whitening. Fig. S9 shows the same quantity for NS-K, measured on the frozen bank ($N = 512$), against the ridge theory at $\varrho_X(K)$.

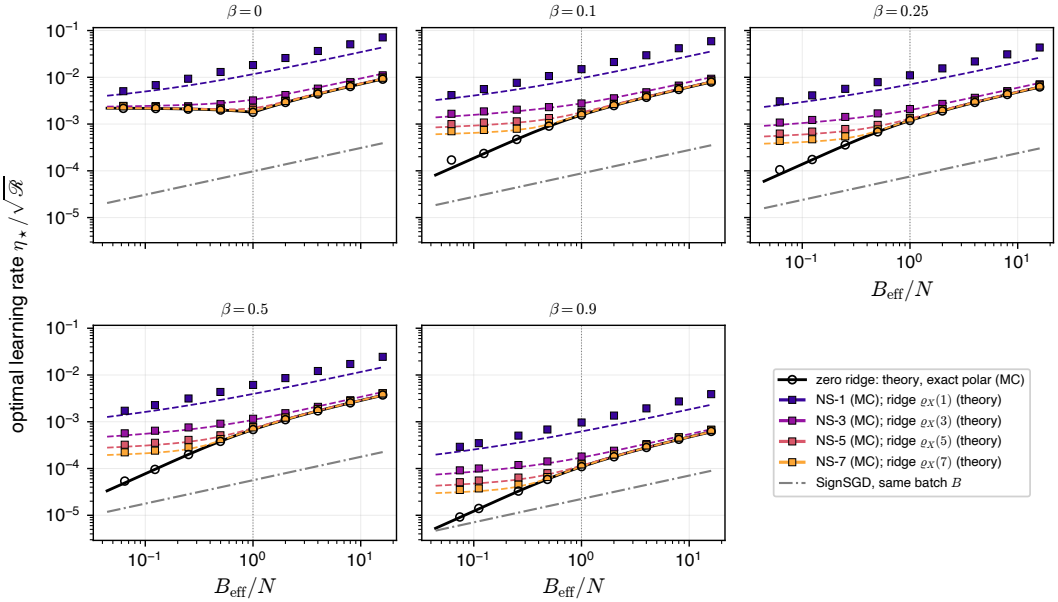


Figure S9: Optimal learning rate $\eta_\star/\sqrt{\mathcal{R}} = 2\mathfrak{d}/\mathfrak{v}$ against B_{eff}/N, $N = 512$. Black: zero ridge (theory) and exact polar factor (open circles, frozen bank). Dashed: ridge theory at the calibrated $\varrho_X(K) = 0.31\, a^{-2K}$; squares: NS-$K$ (frozen bank). Grey: SIGNSGD at the same batch B.

Finite Newton–Schulz flattens the small-batch learning rate. With momentum and exact whitening, the learning rate grows like B for $B_{\text{eff}} < N$ (slope 0.95 in $\log B_{\text{eff}}$ between $B_{\text{eff}}/N = 1/16$ and 1), because the volatility of the whitened weak directions of the buffer grows like N^2/B. A ridge caps the gain on those directions: the slope falls to 0.25–0.34 for NS-5 at every $\beta \in [0.1, 0.9]$, and at $B_{\text{eff}}/N = 1/16$ the NS-5 learning rate is 6–8 times that of exact whitening. The picture is then close to that at $\beta = 0$, where the rank-B gradient has no weak directions, NS-5 and exact whitening nearly coincide, and the learning rate is nearly flat in B for $B < N$. For $B_{\text{eff}} \geq N$ the ridge matters little, and every curve rises with slope between $\frac{1}{2}$ and 0.6, as in Sec. 3.1. NS-K stays above SIGNSGD at the same batch in every configuration.

Amplitude. The calibration matches the scale-free statistic $I_1 = \mathfrak{d}^2/\mathfrak{v}$, while the learning rate $2\mathfrak{d}/\mathfrak{v}$ also carries the overall amplitude of the update. For $K \geq 3$ the NS-K learning rate is between 0.92 and 1.16 times the ridge theory at $\varrho_X(K)$ over all 54 frozen configurations at $N = 512$. NS-1 is nearly linear, $S_t \approx a\, M_t/\|M_t\|_{\text{F}}$, a smaller update than the ridge map at $\varrho_X(1)$, and its learning rate is 1.15–1.64 times higher. The exact-polar points at the smallest B_{eff}/N for $\beta \in \{0.1, 0.25\}$ lie above the zero-ridge curve, for the rank-threshold reason of Sec. SL.1.2.

Scope. These measurements are in the isotropic model with the standard Muon coefficients. The calibration concerns the rate at the optimal step and the lag share; it does not claim that NS-K and the ridge have the same drift or volatility separately, since they differ by an overall amplitude.

SM Momentum in MUON: theory against an ablation

This appendix compares the momentum theory of the appendix with simulations of MUON at $\beta \in \{0, 0.9, 0.95, 0.99\}$ on the half-anisotropic power-law problem at $B = 2N$. The theory makes two predictions in this oversampled regime. The noise floor changes with β through an explicit noise constant $\mathcal{S}(\beta)$, which we compute and which matches the measurements to within 1–6% with momentum (Sec. SM.2). The ϵ-exponent of the time to reach a target does not change with β; the time constants with momentum are outside the theory at these settings, because the buffer is dominated by the signal early in the descent (Sec. SM.3).

Algorithms. Both updates are run on the half-anisotropic problem of Sec. 4, with $\Sigma_{\text{in}} = \text{Id}$ and $\Sigma_{\text{out}} = U \operatorname{diag}(\mu) U^\top$ for a Haar-random orthogonal U and $\mu_i = i^{-\alpha}$. Without momentum the update is $\Delta_{t+1} = \Delta_t - \eta \operatorname{orthog}(G_t)$, where orthog is five Newton–Schulz steps with the coefficients of [32]. With momentum a heavy-ball buffer $H_{t+1} = \beta H_t + G_t$ is orthogonalized instead, as in `optax.contrib.scale_by_muon` with `nesterov=False` [20]. The heavy-ball buffer is the normalized average M_t of definition SA.2 divided by $1 - \beta$ (remark SA.2); the polar factor, and Newton–Schulz applied to a Frobenius-normalized input, do not see this factor, so the simulated algorithm is the algorithm of the appendix at zero ridge, up to the Newton–Schulz approximation.

Setup. We use the matched-floor protocol of Sec. 5: for each target ϵ the step is $\eta^\star = 2\sqrt{\epsilon}/\mathcal{S}$, which puts the risk floor at ϵ. In the simulations $\mathcal{S} = \mathcal{S}_{\text{eff}}$ is calibrated per (β, ς) from a short probe at $\epsilon = 0.25$, as $2\sqrt{\mathscr{R}_\infty}/\eta$ with $\mathscr{R}_\infty$ the measured plateau. The problems are $\alpha = 1.5$ with $\varsigma \in \{0.7, 1.5, 3.0\}$, one in each ϵ-exponent regime of Thm. 5.1; $N = 1024$, $B = 2048$, 16 trials, and $\epsilon \in \{2^{-4}, 2^{-6}, 2^{-8}, 2^{-10}\}$. SIGNSGD at $\beta = 0$ is run for reference. Fig. S10 first compares MUON at $\beta = 0$ with the deterministic model of Sec. 5 over a wider range of ϵ.

SM.1 What the theory predicts at $B = 2N$

Drift. At $B = 2N$ the small parameter of theorem SH.1 is $\epsilon_{\text{os}} = 3(1 - \beta^2)N/B = 1.5(1 - \beta^2)$, which equals 0.29, 0.15 and 0.03 at $\beta = 0.9, 0.95, 0.99$. In this window the theorem gives

$$\mathfrak{d}_i = \vartheta_i \sqrt{\frac{B_{\text{eff}}}{N}}\, \mathfrak{d}^{\text{os}}(\mu_i), \qquad \vartheta_i \in \left[1 - \epsilon_{\text{os}}, (1 - \epsilon_{\text{os}})^{-1}\right],$$

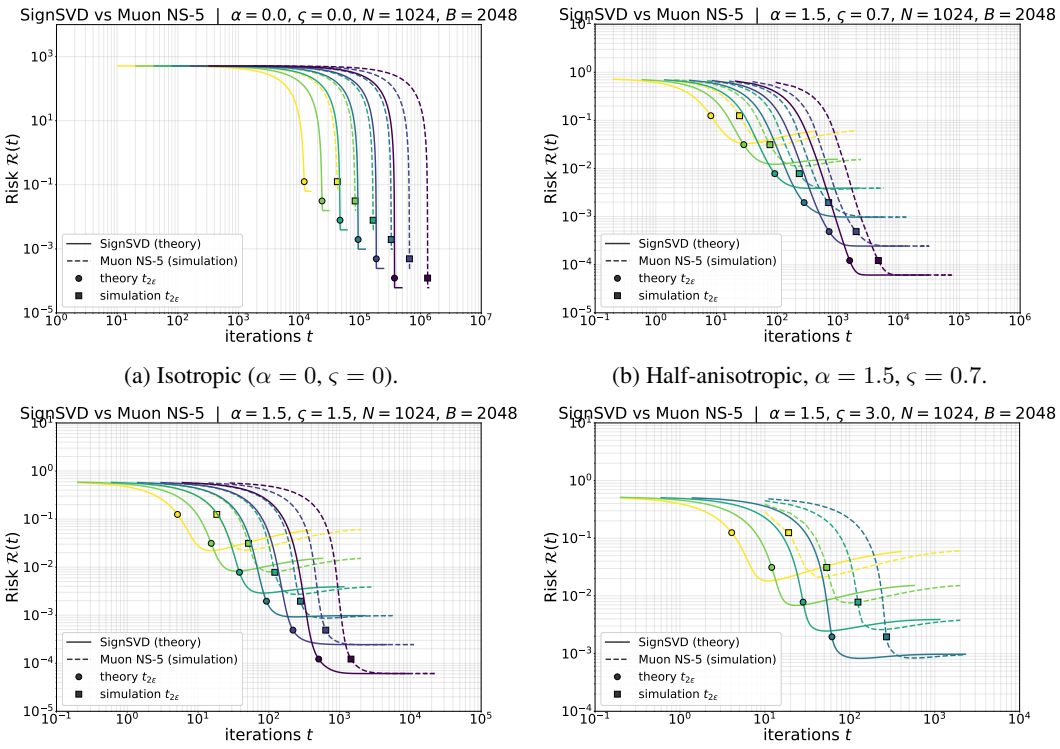
with a profile $\mathfrak{d}^{\text{os}}$ that does not depend on β: momentum enters the drift only through the effective batch $B_{\text{eff}} = B(1 + \beta)/(1 - \beta)$. We evaluate the zero-ridge drift kernel of each row exactly (eq. (SC.4), single integral over the branch). Relative to $\sqrt{B_{\text{eff}}/B}$ times the drift at $\beta = 0$, the ratio lies in $[1.00, 1.19]$, $[1.00, 1.14]$ and $[0.95, 1.02]$ over all rows at $\beta = 0.9, 0.95, 0.99$.

Volatility. The zero-ridge row volatility is explicit at a finite spectrum: it is 1 plus twice the sum over lags of double integrals of the two-time row kernel (propositions SG.3 and SG.4), and without momentum it equals 1, since the polar factor of a full-rank square gradient is orthogonal. With momentum the head rows have $\mathfrak{v}_1 = 19.0, 38.9, 198.2$, essentially the factor $B_{\text{eff}}/B = 19, 39, 199$ of a linear buffer, while the tail rows have less, $\mathfrak{v}_{1000} = 9.2, 16.7, 56.7$. Our code reproduces the isotropic constants of Fig. 4 to 0.2% when all eigenvalues are equal.

Noise constant. The noise constant of the matched-floor protocol is (eq. (16))

$$\mathcal{S}(\beta) = \sum_{i=1}^{N} \frac{\mu_i\, \mathfrak{v}_i}{2\, \mathfrak{d}_i},$$

and it does not depend on the target exponent ς, because neither coefficient does. The theory values are in Table S2.



(a) Isotropic ($\alpha = 0, \varsigma = 0$).

(b) Half-anisotropic, $\alpha = 1.5, \varsigma = 0.7$.

(c) Half-anisotropic, $\alpha = 1.5, \varsigma = 1.5$.

(d) Half-anisotropic, $\alpha = 1.5, \varsigma = 3.0$.

Figure S10: MUON without momentum (Newton–Schulz, dashed) against the deterministic time-to-ϵ model of Sec. 5 (solid), $N = 1024$, $B = 2048$, 16 trials per target, colour-coded by $\epsilon \in \{2^{-4}, \dots, 2^{-14}\}$. Circles: model $t_{2\epsilon}$; squares: simulated $t_{2\epsilon}$. The simulations follow the model's ϵ-exponent in every regime.

	$\beta = 0$	$\beta = 0.9$	$\beta = 0.95$	$\beta = 0.99$
theory $\mathcal{S}(\beta)$	11.99	39.10	54.79	116.56
measured $\mathcal{S}_{\text{eff}}$, $\varsigma = 0.7$	9.97	36.89	52.48	117.39
measured $\mathcal{S}_{\text{eff}}$, $\varsigma = 1.5$	9.97	36.91	52.49	117.38
measured $\mathcal{S}_{\text{eff}}$, $\varsigma = 3.0$	9.97	36.89	52.49	117.22
measured / theory	0.83	0.94	0.96	1.01

Table S2: Noise constant at $\alpha = 1.5$, $N = 1024$, $B = 2048$. Theory: exact polar factor, zero ridge, coefficients of Sec. SM.1. Measured: MUON with five Newton–Schulz steps, calibrated $\mathcal{S}_{\text{eff}}$; the three target exponents agree to four significant figures at every β.

SM.2 The noise floor

The theory predicts the noise floor with momentum (Table S2, Fig. S11). For $\beta \in \{0.9, 0.95, 0.99\}$ the measured noise constants are within 6%, 4% and 1% of $\mathcal{S}(\beta)$, and, as predicted, they do not depend on ς. The floor at a given step is $(\eta\,\mathcal{S}(\beta)/2)^2$, so the noise-floor adjustment for momentum is the ratio $\mathcal{S}(\beta)/\mathcal{S}(0)$, equal to 3.26, 4.57 and 9.72. It is smaller than the plain rescaling $\sqrt{B_{\text{eff}}/B} = 4.36$, 6.24, 14.1 by the factors 0.75, 0.73 and 0.69: the drift grows by $\sqrt{B_{\text{eff}}/B}$ on every row, but the volatility of the tail rows grows by less than B_{eff}/B, and the tail rows dominate $\mathcal{S}$.

Without momentum the measured constant is 17% below the theory. Here the gradient is a square matrix with aspect ratio $N/B = 1/2$ and spread-out singular values, and five Newton–Schulz steps differ from the exact polar factor by the spread of their outputs, the effect identified in Appendix SL at $\beta = 0$. With momentum the buffer averages many batches, its singular values are concentrated,

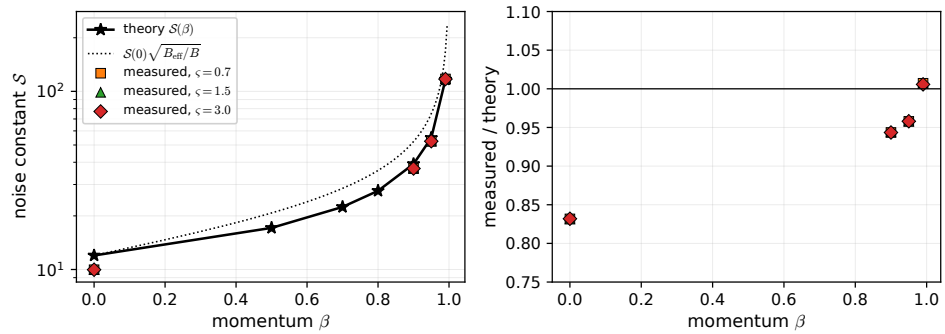


Figure S11: Noise constant against momentum. Left: theory $\mathcal{S}(\beta)$ (stars, line), the plain rescaling $\mathcal{S}(0)\sqrt{B_{\text{eff}}/B}$ (dotted), and the measured $\mathcal{S}_{\text{eff}}$ for the three target exponents, which coincide. Right: measured over theory.

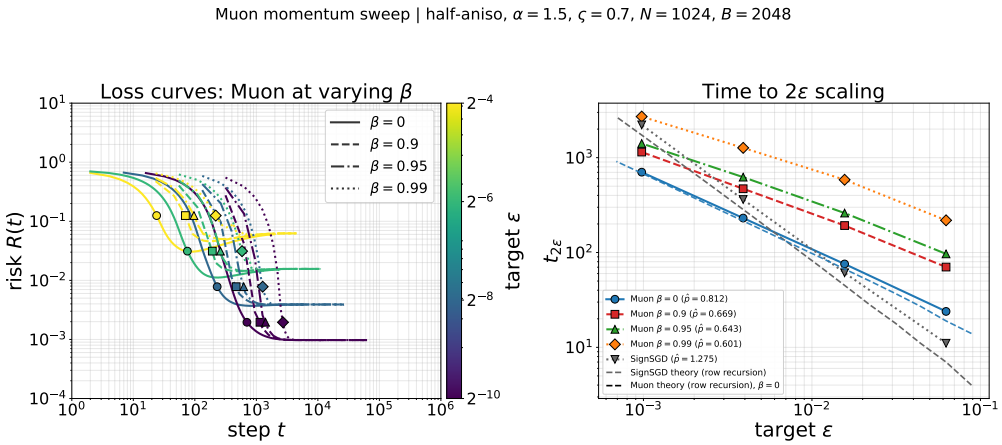


Figure S12: $\alpha = 1.5$, $\varsigma = 0.7$, $N = 1024$, $B = 2048$, 16 trials. Left: risk at the matched step for each target ϵ (colour) and β (linestyle); markers at the measured $t_{2\epsilon}$. Right: $t_{2\epsilon}$ against ϵ. Blue dashed: the row recursion with the theory coefficients of Sec. SM.1 at $\beta = 0$, at its own matched step (no line is drawn for $\beta > 0$; see Sec. SM.3). Grey dashed: the same recursion with the SIGNSGD kernels of Table 1. The asymptotic exponents are 0.625 (SIGNSVD) and 1.25 (SIGNSGD).

and the difference is small. The empirical prefactor $\hat{\rho} \approx 0.84$ of an earlier version of this appendix, defined relative to the measured $\beta = 0$ value, combines this 17% with the factors 0.69–0.75 above.

SM.3 Time to ϵ

The exponent. The ϵ-exponent of the time to reach a target is set by the power-law orders in i of $\mathfrak{d}_i$ and $\mathfrak{v}_i$ (Thm. 5.1). Momentum multiplies the drift of every row by $\sqrt{B_{\text{eff}}/B}$ up to a bounded factor, and the volatility by factors bounded above and below uniformly in i (between 9.1 and 19 at $\beta = 0.9$, between 55 and 198 at $\beta = 0.99$), so it does not change these orders, and the exponent is the same for every β. The measured slopes (Figs. S12–S14, right panels) agree: they lie within 10–30% of the SIGNSVD exponent for every β, do not depend systematically on β, and show the same finite-N deviation already present at $\beta = 0$.

The time constants. Without momentum the theory follows the simulations (blue dashed lines in the right panels: the row recursion with the coefficients of Sec. SM.1 at its own matched step): within 0.8–1.06 at $\varsigma = 0.7$ and 0.7–0.84 at $\varsigma = 1.5$, and about 0.55 at $\varsigma = 3.0$, where the gap includes the Newton–Schulz offset of Sec. SM.2. With momentum the theory predicts a curve close to the $\beta = 0$ one, while the simulations are slower by large factors (2–8 at $\beta = 0.9$, 5–30 at $\beta = 0.99$), so the panels show no theory line for $\beta > 0$. Fig. S15 shows the gap on the risk itself.

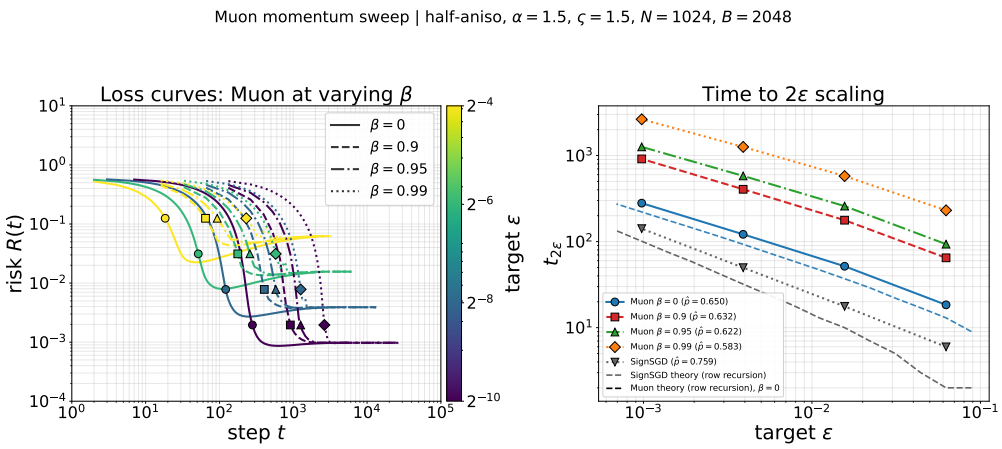


Figure S13: $\alpha = 1.5$, $\varsigma = 1.5$; conventions as in Fig. S12. Asymptotic exponents 0.5 (SIGNSVD) and 0.75 (SIGNSGD).

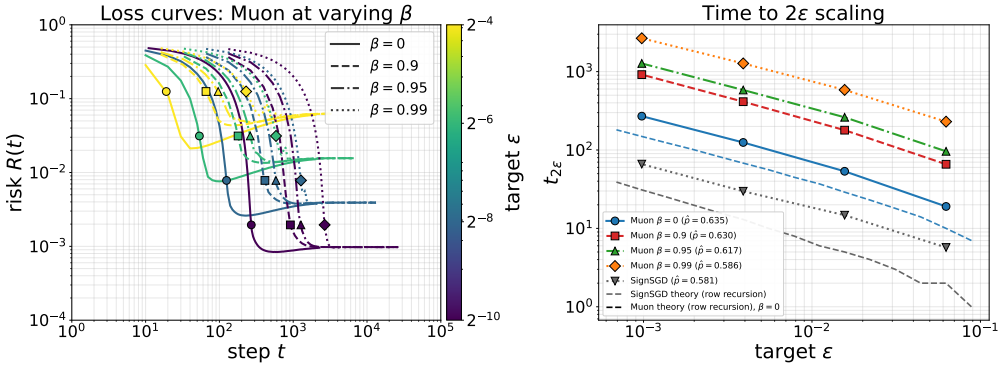


Figure S14: $\alpha = 1.5$, $\varsigma = 3.0$; conventions as in Fig. S12. Both algorithms are saturated, with asymptotic exponent 0.5.

Why the momentum transient is outside the theory. The gain of the drift by $\sqrt{B_{\text{eff}}/B}$ (Sec. SM.1) is a linear response: the polar factor of a buffer that is mostly noise responds linearly to the signal in it, and averaging more batches raises the signal-to-noise ratio. It requires the noise to dominate the buffer (remark SH.1, which asks for $B_{\text{eff}} \ll N^2$ for a spread-out error). Here $B_{\text{eff}}/N^2 = 0.04$ at $\beta = 0.9$ and 0.39 at $\beta = 0.99$, and the initial error is concentrated: at $\varsigma = 0.7$ the top mode carries about two thirds of the weighted risk. Early in the descent the buffer is therefore close to its signal $\Sigma_{\text{out}}\Delta_t$, whose polar factor has bounded entries, and the extra signal-to-noise ratio buys no extra speed. Near the floor the error is small and spread out, the noise dominates again, and the linear response holds; this is why the noise floor and the exponent are predicted.

A controlled comparison separates this effect from two others (Table S3). A Gaussian model of the gradient reproduces the simulations to within about 30%. In it, a fresh buffer at the current state with the exact polar factor, which has neither a stale buffer nor Newton–Schulz, is already 2.4 and 7 times slower than the theory at $\beta = 0.9$ and 0.99; this is the loss of linear response. Replacing the exact polar factor by five Newton–Schulz steps slows the descent by a further 1.3–1.35: the effective ridge $\varrho_X \|M_t\|_{\text{F}}^2$ of Appendix SL grows with the signal in $\|M_t\|_{\text{F}}$ and suppresses the noise directions through which the tail rows descend. Using the actual momentum buffer, whose signal lags the state, slows it by 1.1–1.2 at $\beta = 0.9$ and by 1.6–2.9 at $\beta = 0.99$; with this stale buffer Newton–Schulz is faster than the exact polar factor, because its ridge down-weights the old directions, as in Appendix SL.

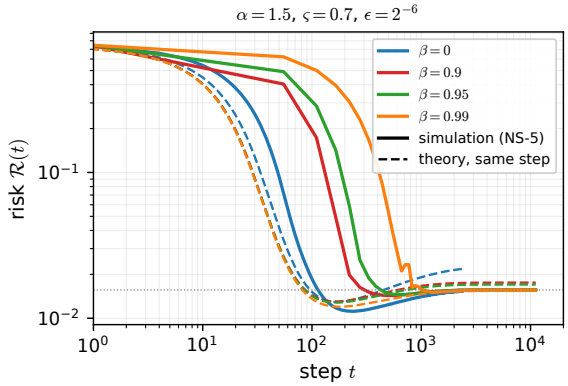


Figure S15: Risk at $\alpha = 1.5$, $\varsigma = 0.7$, $\epsilon = 2^{-6}$: simulations (solid) against the row recursion with the theory coefficients at the simulations' own step (dashed). With momentum the simulated descent is slower than the theory, increasingly with β; the floors agree to the accuracy of Table S2.

	theory	fresh buffer		momentum buffer		MUON run
		polar	NS-5	polar	NS-5	
$\beta = 0$	58	68	71	—	—	76
$\beta = 0.9$	51	120	160	141	179	220
$\beta = 0.99$	50	357	482	1039	787	605

Table S3: Steps to 2ϵ at $\alpha = 1.5$, $\varsigma = 0.7$, $\epsilon = 2^{-6}$, with the step and initial error of the ablation. Gaussian gradient model $G_t = \Sigma_{\text{out}}\Delta_t + \sqrt{2\mathscr{R}(t)/B}\,\Sigma_{\text{out}}{}^{1/2}W_t$ in the eigenbasis; the fresh buffer has the noise level of the momentum buffer but is formed at the current state. One run per entry.

Takeaway. At $B = 2N$ the theory predicts how momentum changes the noise floor of MUON, through the explicit noise constant $\mathcal{S}(\beta)$, to within a few percent, and it predicts that the ϵ-exponent of the time to target does not change, as observed. The transient with momentum is outside the theory at these settings: the buffer is signal-dominated early in the descent, so the linear-response gain of the drift is not realized, and the stale buffer and the Newton–Schulz ridge add smaller slowdowns. At matched floor, momentum therefore costs time in this ablation.

SN Additional Related Work

MUON & SIGNSVD. The original motivation for MUON [32] stems from steepest descent with respect to the spectral norm [6]. The method shares key features with related approaches, including stochastic spectral descent [14] and orthogonalized gradient methods [68], and can also be interpreted as a form of SIGNSGD [7]. Its empirical success [32, 41, 58, 73] has motivated a number of recent variants [42, 43, 52, 54]. Given the growing interest in MUON, the discussion below is necessarily non-exhaustive and highlights only a subset of recent theoretical and empirical developments.

On the theoretical side, several works have analyzed the performance of MUON. For instance, [15, 33, 59, 61] study MUON or closely related variants such as SIGNSVD using classical worst-case complexity arguments for non-convex optimization, showing that these methods achieve $\mathcal{O}(1/\sqrt{T})$ convergence rates comparable to SGD. More specifically, [61] analyzes MUON via an inexact linear minimization oracle, while [15] connects MUON to a broader class of algorithms known as Lion-$\mathcal{K}$. In [33], the authors further characterize the dependence on the number of Newton–Schultz steps and compare MUON with exact polarization. In addition, [31] establishes regret bounds for MUON.

Several works have studied SIGNSVD [70] and MUON [27] in simplified linear settings. In [70], explicit loss dynamics for SIGNSVD are derived in the gradient flow regime (vanishing learning rate and negligible stochasticity), enabling analysis of generalization for both SIGNSVD and standard gradient descent; see also [53] which studies gradient flow SIGNSVD but on more complicated loss functions. Meanwhile, [27] shows that even for simple quadratic objectives, MUON exhibits nontrivial behavior. However, these works do not operate in the high-dimensional regime considered here. In contrast, [72] studies SIGNSVD on a high-dimensional isotropic Gaussian least squares problem and derives explicit risk dynamics, but does not analyze the implications of the dynamics. Building on this line of work, this work moves beyond the isotropic setting and analyze the precise scaling behavior of both MUON and SIGNSVD as well as the explicit dynamics.

Beyond least squares, [23, 67] study the implicit bias of spectral methods in linear multiclass logistic regression with separable data. Additionally, [35, 38] analyze MUON in attention through associative memory models; in particular, [35] shows that the critical batch size scales as $\sqrt{N_{\text{in}}}$ under power-law covariance. To our knowledge, this is the first work to provide explicit training dynamics of spectral methods over the full course of optimization with anisotropic data.

Finally, several works aim to explain the effectiveness of MUON and related spectral optimizers in deep learning, primarily through the lens of a *single update*. For instance, [19] identifies a layerwise condition, known as low stable rank, under which a spectral update yields a larger loss decrease than a Euclidean gradient step in the deterministic MUON setting, and shows that intermediate-layer activations in deep networks often satisfy this property. Similarly, [63] analyzes a single deterministic MUON update under an isotropic assumption on the data and the loss, reducing the loss to a univariate function and showing that MUON is directionally optimal, though not strictly optimal. However, [27] demonstrates on simple quadratic problems that single-step analyses do not reliably predict overall performance. In contrast, our work provides explicit dynamics for *stochastic* spectral optimizers over the full course of training.

Random matrix theory and machine learning. This paper uses random matrix theory to analyze a matrix least squares problem. In particular, our analysis requires analysis of spectral properties of sample covariance matrices (e.g., power-law population covariance), and the use of deterministic equivalents for the resolvents of random matrices. Both these sets of tools have long predated modern machine learning literature, yet have found growing use in the field. See [18] for a modern introduction.

High-dimensional optimization literature for SGD and its variants. Most of the literature on high-dimensional optimization focuses on streaming SGD under certain settings such as least squares beyond isotropic settings [50]. A few works have expanded beyond least squares, see e.g., [5, 16, 45, 47], but again with focus on streaming SGD with many working on power-law covariances [4, 11, 12, 24, 40, 51]. There have been some extensions beyond SGD. Notably, [48, 50, 77] gave exact dynamics for multi-pass SGD on a high-dimensional empirical least squares problem with non-Gaussian covariances. Extensions for deterministic dynamics of momentum algorithms in small batch [39, 49, 69, 76] and large batch setting [37] as well as power-law covariances in [24] where Nesterov acceleration methods with time-dependent momentum parameters were explored on

least squares problems and single/multi-index [25]. Deterministic dynamics for high-dimensional optimization beyond the SGD/SGD with momentum framework have been considered, see [17, 30] for stochastic adaptive learning rates (e.g., AdaGrad-Norm), SIGNSGD [34, 75], and clipped SGD [44]. High-dimensional limits for spectral optimizers have not yet been explored in detail, with the only related result being the workshop paper [72].

SIGNSGD. SIGNSGD has been proposed as a simplified proxy for ADAM, of which it is a special case (setting $\beta_1 = \beta_2 = \varepsilon = 0$) [2, 3]. In [75], the authors study SIGNSGD on a high-dimensional linear regression problem, deriving explicit loss dynamics that precisely quantify the *diagonal preconditioning* effect of SIGNSGD and characterize how it reshapes the distribution of gradient noise.

SO Lean formalization

The statements of this supplement were formalized in Lean 4 (v4.34.1) with Mathlib. The formal development is published with the paper; its audit checks that no declaration depends on `sorry` and that only Lean's standard axioms (`propext`, `Classical.choice`, `Quot.sound`) are used.

Of the 366 labelled statements, 305 are proved, 40 are proved assuming one or more of the cited results listed in table S4, and 21 (remarks and numerical statements) are not formalized.

A cited result enters the formalization as an explicit hypothesis: a Lean structure stating the cited theorem exactly as it is used, which every consumer takes as an argument. No axiom is added.

Certificate	Source	Proved in the foundations library
`GaussianOpNorm`	[71, Thm. 4.4.5]	`GaussianFoundations.gaussianOpNorm`
`GaussianOpNormL6`	[71, Thm. 4.4.5]	`GaussianFoundations.gaussianOpNormL6`
`GaussianQuadFormLp`	[71, Thm. 6.2.1]	`GaussianFoundations.gaussianQuadFormLp`
`GaussianPoincare`	[13, Thm. 3.20]	`GaussianFoundations.gaussianPoincare`
`GoldenThompson`	[8, Thm. IX.3.7]	`MatrixConcentration.matrixMartingale_trace_mgf (different route)`

Table S4: Cited results used as explicit hypotheses in the formalization.

Statement	Status	Lean declaration	Certificates
definition SA.1	proved	`Muon.Setup.def_app_setup_model` (+8)	
Assumption S1	proved	`Muon.Setup.assum_data_target` (+3)	
proposition SA.1	proved	`Muon.Setup.prop_app_setup_moments_i` (+2)	
lemma SA.1	proved	`Muon.Setup.lem_app_setup_decomposition_i` (+3)	
proposition SA.2	proved	`Muon.Setup.prop_app_setup_normalization_i` (+1)	
lemma SA.2	proved	`Muon.Setup.lem_app_setup_filter_i` (+3)	
lemma SA.3	proved	`Muon.Setup.lem_app_setup_contour`	
proposition SA.3	proved	`Muon.Setup.prop_app_setup_scaling_i` (+2)	
definition SA.2	proved	`Muon.Setup.def_app_setup_algorithm` (+18)	
remark SA.1	not formalized	—	
remark SA.2	not formalized	—	
proposition SA.4	proved	`Muon.Setup.prop_app_setup_risk_recursion_mode` (+3)	
corollary SA.1	proved	`Muon.Setup.cor_app_setup_augmented_i` (+3)	
proposition SA.5	proved	`Muon.Setup.prop_app_setup_support_i` (+3)	
definition SA.3	proved	`Muon.Setup.def_app_setup_frozen` (+6)	
lemma SA.4	proved	`Muon.Setup.sampleX` (+6)	
lemma SA.5	proved	`Muon.Setup.SigmaG` (+12)	

Statement	Status	Lean declaration	Certificates
proposition SA.6	proved	`Muon.Setup.startup_weight_sum` (+7)	
remark SA.3	not formalized	—	
lemma SA.6	proved	`Muon.Setup.lem_app_setup_truncation_i` (+4)	
lemma SA.7	proved	`Muon.Setup.lem_app_setup_residual_i` (+2)	
lemma SA.8	proved	`Muon.Setup.lem_app_setup_intertwining_i` (+2)	
Assumption S2	proved	`Muon.Setup.assum_proportional_regime` (+1)	
Assumption S3	proved	`Muon.Setup.SetupRegime` (+3)	
definition SA.4	proved	`Muon.Setup.def_deterministic_equivalent` (+4)	
lemma SB.1	proved	`Muon.DetEquiv.lem_app_de_scalar_i` (+2)	
definition SB.1	proved	`Muon.DetEquiv.IsBranchSol` (+6)	
theorem SB.1	proved	`Muon.DetEquiv.thm_app_de_branch_i` (+4)	
lemma SB.2	proved	`Muon.DetEquiv.lem_app_de_compact_i` (+3)	
proposition SB.1	proved	`Muon.DetEquiv.prop_app_de_nomomentum`	
remark SB.1	proved	`Muon.DetEquiv.rem_app_de_nomomentum_a` (+2)	
proposition SB.2	proved	`Muon.DetEquiv.prop_app_de_normalization_i` (+1)	
remark SB.2	not formalized	—	
lemma SB.3	proved	`Muon.DetEquiv.lem_app_de_resolvent` (+3)	
lemma SB.4	proved	`Muon.DetEquiv.lem_app_de_deletion` (+8)	
lemma SB.5	proved	`Muon.DetEquiv.lem_app_de_coeff` (+1)	
lemma SB.6	proved	`Muon.DetEquiv.lem_app_de_moments_i` (+4)	
lemma SB.7	proved	`Muon.DetEquiv.lem_app_de_bessel_i` (+2)	
lemma SB.8	proved	`Muon.DetEquiv.lem_app_de_stein_i` (+7)	
lemma SB.9	proved	`Muon.DetEquiv.lem_app_de_lipschitz`	
theorem SB.2	proved	`Muon.DetEquiv.thm_app_de_gram`	
corollary SB.1	proved	`Muon.DetEquiv.cor_app_de_trace_i` (+4)	
lemma SB.10	proved	`Muon.DetEquiv.lem_app_de_variance_i` (+3)	
lemma SB.11	proved	`Muon.DetEquiv.lem_app_de_scalars`	
lemma SB.12	proved	`Muon.DetEquiv.lem_app_de_replace`	
lemma SB.13	proved	`Muon.DetEquiv.lem_app_de_gaussian`	
theorem SC.1	proved	`Muon.Drift.thm_app_drift_mixed_ii_T` (+4)	
theorem SC.2	proved	`Muon.Drift.gbar_integrable_bound` (+2)	
proposition SC.1	proved	`Muon.Drift.prop_app_drift_truncation_ii` (+4)	
corollary SC.1	proved	`Muon.Drift.Erho` (+6)	
proposition SC.2	proved	`Muon.Drift.prop_app_drift_onestep` (+1)	
lemma SC.1	proved	`Muon.Drift.lem_app_drift_column` (+2)	
lemma SC.2	proved	`Muon.Drift.lem_app_drift_coeffmoments` (+4)	
lemma SC.3	proved	`Muon.Drift.lem_app_drift_multiplier_i` (+3)	
lemma SC.4	proved	`Muon.Drift.lem_app_drift_rank` (+2)	
lemma SC.5	proved	`Muon.Drift.lem_app_drift_covariance` (+1)	

Statement	Status	Lean declaration	Certificates
lemma SC.6	proved	`Muon.Drift.lem_app_drift_exact` (+1)	
lemma SC.7	proved	`Muon.Drift.lem_app_drift_derivatives_i` (+5)	
lemma SC.8	proved	`Muon.Drift.rPrimeB` (+3)	
lemma SC.9	proved	`Muon.Drift.lem_app_drift_targets_i_weights` (+4)	
proposition SC.3	proved	`Muon.Drift.bB` (+8)	
corollary SC.2	proved	`Muon.Drift.dIdx` (+5)	
remark SC.1	not formalized	—	
lemma SD.1	proved	`Muon.Volatility.lem_app_vol_sharp_i` (+2)	
lemma SD.2	proved	`Muon.Volatility.lem_app_vol_finite_i` (+3)	
lemma SD.3	proved	`Muon.Volatility.lem_app_vol_square_i` (+2)	
definition SD.1	proved	`Muon.Volatility.def_app_vol_lag` (+8)	
lemma SD.4	proved	`Muon.Volatility.lem_app_vol_centering_i` (+1)	
theorem SD.1	proved	`Muon.Volatility.thm_app_vol_half_same_i_out` (+11)	
remark SD.1	not formalized	—	
definition SD.2	proved	`Muon.Volatility.def_app_vol_product` (+8)	
definition SD.3	proved	`Muon.Volatility.def_app_vol_replication` (+9)	
lemma SD.5	proved	`Muon.Volatility.lem_app_vol_scalar_weak`	
lemma SD.6	proved	`Muon.Volatility.lem_app_vol_replacement` (+1)	
corollary SD.1	proved	`Muon.Volatility.cor_app_vol_filtered_replacement_i` (+1)	
theorem SD.2	proved	`Muon.Volatility.thm_app_vol_lag_reduction_i` (+10)	
lemma SD.7	proved	`Muon.Volatility.lem_app_vol_lagsum`	
definition SD.4	proved	`Muon.Volatility.def_app_vol_tree` (+19)	
lemma SD.8	proved cited	mod. `Muon.Volatility.def_app_vol_tree_replication` (+1)	
lemma SD.9	proved cited	mod. `Muon.Volatility.lem_app_vol_norm_i`	`GaussianOpNorm` (+2)
lemma SD.10	proved cited	mod. `Muon.Volatility.lem_app_vol_tree_extension_i`	`GaussianOpNorm` (+3)
lemma SD.11	proved	`Muon.Volatility.lem_app_vol_cutoff_i` (+1)	
theorem SD.3	proved cited	mod. `Muon.Volatility.thm_app_vol_lag_kernel_error_i`	`GaussianOpNorm` (+4)
remark SD.2	proved cited	mod. `Muon.Volatility.lag_closure_generic`	`GaussianOpNorm`
theorem SD.4	proved cited	mod. `Muon.Volatility.thm_app_vol_integrated_i`	`GaussianOpNorm` (+4)
theorem SD.5	proved cited	mod. `Muon.Volatility.thm_app_vol_closed_kernel_i`	`GaussianOpNorm` (+2)
corollary SD.2	proved cited	mod. `Muon.Volatility.cor_app_vol_closed_bound_i`	`GaussianOpNorm` (+3)
proposition SD.1	proved cited	mod. `Muon.Volatility.prop_app_vol_closed_remainder_i`	`GaussianOpNorm` (+1)
theorem SD.6	proved cited	mod. `Muon.Volatility.thm_app_vol_closed_filtered_i`	`GaussianOpNorm` (+1)
corollary SD.3	proved cited	mod. `Muon.Volatility.cor_app_vol_closed_model_i`	`GaussianOpNorm` (+2)
proposition SD.2	proved cited	mod. `Muon.Volatility.prop_app_vol_closed_lag_i`	`GaussianOpNorm` (+1)
definition SD.5	proved	`Muon.Volatility.def_app_vol_closed_profile` (+14)	

Statement	Status	Lean declaration	Certificates
lemma SD.12	proved	`Muon.Volatility.lem_app_vol_closed_profile_i` (+4)	
lemma SD.13	proved	`Muon.Volatility.IsBranchPoint` (+6)	
lemma SD.14	proved	`Muon.Volatility.lem_app_vol_closed_perron`	
lemma SD.15	proved	`Muon.Volatility.toutT` (+10)	
lemma SD.16	proved	`Muon.Volatility.lem_app_vol_closed_contour` (+6)	
lemma SD.17	proved	`Muon.Volatility.lem_app_vol_closed_encoding_i` (+9)	
lemma SD.18	proved mod. cited	`Muon.Volatility.lem_app_vol_closed_one_sa` (+2)	GaussianOpNorm
lemma SD.19	proved mod. cited	`Muon.Volatility.lem_app_vol_closed_path` (+1)	GaussianOpNorm
theorem SD.7	proved	`Muon.Volatility.thm_app_vol_mode_kernels_i` (+9)	
theorem SD.8	proved	`Muon.Volatility.thm_app_vol_joint_forcing_i_bar` (+7)	
proposition SD.3	proved	`Muon.Volatility.prop_app_vol_disintegration_i` (+2)	
proposition SD.4	proved	`Muon.Volatility.prop_app_vol_half_recovery`	
corollary SD.4	proved	`Muon.Volatility.cor_app_vol_marginals_tests` (+1)	
corollary SD.5	proved mod. cited	`Muon.Volatility.thm_app_vol_double_sam` (+4)	GaussianOpNorm
remark SD.3	not formalized	—	
definition SE.1	proved	`Muon.Dynamics.odeRisk` (+5)	
lemma SE.1	proved	`Muon.Dynamics.lem_app_dyn_field_i` (+3)	
proposition SE.1	proved	`Muon.Dynamics.prop_app_dyn_ode_i` (+2)	
theorem SE.1	proved mod. cited	`Muon.Dynamics.thm_app_dyn_main_i` (+5)	GaussianOpNorm, GaussianOpNormL6, GaussianPoincare, GaussianQuadFormLp
corollary SE.1	proved	`Muon.Dynamics.cor_app_dyn_half_i` (+2)	
corollary SE.2	proved	`Muon.Dynamics.cor_app_dyn_continuum_i` (+4)	
remark SE.1	not formalized	—	
remark SE.2	not formalized	—	
definition SE.2	proved	`Muon.Dynamics.def_app_dyn_chains` (+8)	
lemma SE.2	proved	`Muon.Dynamics.lem_app_dyn_chains_i` (+3)	
lemma SE.3	proved	`Muon.Dynamics.lem_app_dyn_batch_i` (+5)	
lemma SE.4	proved	`Muon.Dynamics.lem_app_dyn_schatten_i` (+3)	
lemma SE.5	proved	`Muon.Dynamics.lem_app_dyn_second_i` (+3)	
lemma SE.6	proved	`Muon.Dynamics.lem_app_dyn_covariance`	
lemma SE.7	proved	`Muon.Dynamics.lem_app_dyn_mean_i` (+7)	
lemma SE.8	proved	`Muon.Dynamics.lem_app_dyn_tangent`	
lemma SE.9	proved	`Muon.Dynamics.lem_app_dyn_derivative_i` (+1)	
lemma SE.10	proved mod. cited	`Muon.Dynamics.lem_app_dyn_coefficients`	GaussianQuadFormLp
lemma SE.11	proved	`Muon.Dynamics.lem_app_dyn_raw`	
proposition SE.2	proved	`Muon.Dynamics.prop_app_dyn_reduction`	
lemma SE.12	proved	`Muon.Dynamics.lem_app_dyn_minor`	
proposition SE.3	proved	`Muon.Dynamics.prop_app_dyn_scalar`	
lemma SE.13	proved	`Muon.Dynamics.lem_app_dyn_ridge`	

Statement	Status		Lean declaration	Certificates
proposition SE.4	proved		`Muon.Dynamics.prop_app_dyn_stationary_tangent`	
lemma SE.14	proved		`Muon.Dynamics.lem_app_dyn_movement_i` (+1)	
lemma SE.15	proved		`Muon.Dynamics.lem_app_dyn_buffer_i` (+1)	
lemma SE.16	proved		`Muon.Dynamics.lem_app_dyn_coupling_i` (+3)	
lemma SE.17	proved		`Muon.Dynamics.lem_app_dyn_startup_i` (+1)	
lemma SE.18	proved		`Muon.Dynamics.lem_app_dyn_taylor`	
lemma SE.19	proved cited	mod.	`Muon.Dynamics.lem_app_dyn_poincare_i` (+1)	`GaussianPoincare`
lemma SE.20	proved		`Muon.Dynamics.lem_app_dyn_martingale`	
lemma SE.21	proved		`Muon.Dynamics.lem_app_dyn_prefix_i` (+3)	
proposition SE.5	proved		`Muon.Dynamics.prop_app_dyn_transport`	
proposition SE.6	proved		`Muon.Dynamics.prop_app_dyn_response_i` (+1)	
lemma SE.22	proved		`Muon.Dynamics.lem_app_dyn_quadratic`	
lemma SE.23	proved		`Muon.Dynamics.lem_app_dyn_matrix_martingale`	
lemma SE.24	proved		`Muon.Dynamics.lem_app_dyn_diffuse` (+2)	
lemma SE.25	proved		`Muon.Dynamics.lem_app_dyn_uniform` (+2)	
lemma SE.26	proved cited	mod.	`Muon.Dynamics.lem_app_dyn_block` (+4)	`GaussianOpNorm`, `GaussianOpNormL6`, `GaussianPoincare`, `GaussianQuadFormLp`
lemma SE.27	proved cited	mod.	`Muon.Dynamics.lem_app_dyn_interval_i` (+4)	`GaussianOpNorm`, `GaussianOpNormL6`, `GaussianPoincare`, `GaussianQuadFormLp`
proposition SF.1	proved		`Muon.Isotropic.prop_app_iso_closure_i` (+2)	
theorem SF.1	proved		`Muon.Isotropic.thm_app_iso_drift`	
theorem SF.2	proved cited	mod.	`Muon.Isotropic.thm_app_iso_volatility`	`GaussianOpNorm`
proposition SF.2	proved cited	mod.	`Muon.Isotropic.prop_app_iso_volatility_zero`	`GaussianOpNorm`
definition SF.1	proved cited	mod.	`Muon.Isotropic.def_app_iso_vol_zero` (+2)	`GaussianOpNorm`
proposition SF.3	proved cited	mod.	`Muon.Isotropic.prop_app_iso_vol_moments_i` (+2)	`GaussianOpNorm`
proposition SF.4	proved cited	mod.	`Muon.Isotropic.IsIsoScalarOde` (+3)	`GaussianOpNorm`
corollary SF.1	proved cited	mod.	`Muon.Isotropic.cor_app_iso_dynamics_scalar` (+4)	`GaussianOpNorm`, `GaussianOpNormL6`, `GaussianPoincare`, `GaussianQuadFormLp`
lemma SF.1	proved		`Muon.Isotropic.lem_app_iso_gaussian_i` (+4)	
proposition SF.5	proved		`Muon.Isotropic.prop_app_iso_zero_i` (+1)	
definition SF.2	proved		`Muon.Isotropic.IsIsoMeasurePair`	
lemma SF.2	proved		`Muon.Isotropic.lem_app_iso_transforms_i` (+1)	
lemma SF.3	proved		`Muon.Isotropic.lem_app_iso_moments_i` (+4)	
proposition SF.6	proved		`Muon.Isotropic.prop_app_iso_measure_unique` (+2)	
proposition SF.7	proved		`Muon.Isotropic.prop_app_iso_spectral_i` (+1)	

Statement	Status	Lean declaration	Certificates
lemma SF.4	proved	`Muon.Isotropic.lem_app_iso_weights_i` (+4)	
lemma SF.5	proved	`Muon.Isotropic.lem_app_iso_comparison_i` (+2)	
lemma SF.6	proved	`Muon.Isotropic.lem_app_iso_count_i` (+1)	
lemma SF.7	proved	`Muon.Isotropic.lem_app_iso_constants_i` (+3)	
lemma SF.8	proved	`Muon.Isotropic.lem_app_iso_participation_i` (+1)	
theorem SF.3	proved	`Muon.Isotropic.thm_app_iso_master_i` (+1)	
lemma SF.9	proved	`Muon.Isotropic.lem_app_iso_mp`	
theorem SF.4	proved	`Muon.Isotropic.thm_app_iso_small_i` (+2)	
proposition SF.8	proved	`Muon.Isotropic.prop_app_iso_reciprocity_ii` (+2)	
theorem SF.5	proved	`Muon.Isotropic.thm_app_iso_large_i` (+3)	
proposition SF.9	proved	`Muon.Isotropic.prop_app_iso_exact`	
theorem SF.6	proved	`Muon.Isotropic.thm_app_iso_one`	
theorem SF.7	proved	`Muon.Isotropic.thm_app_iso_small_momentum_i` (+3)	
lemma SF.10	proved	`Muon.Isotropic.lem_app_iso_V_i` (+3)	
lemma SF.11	proved cited	mod. `Muon.Isotropic.lem_app_iso_lag_i` (+1)	`GaussianOpNorm`
theorem SF.8	proved cited	mod. `Muon.Isotropic.thm_app_iso_small_volatility_i` (+1)	`GaussianOpNorm`
lemma SF.12	proved	`Muon.Isotropic.lem_app_iso_lag_lower_i` (+1)	
lemma SF.13	proved	`Muon.Isotropic.lem_app_iso_slope`	
lemma SF.14	proved	`Muon.Isotropic.lem_app_iso_shifts_i` (+4)	
lemma SF.15	proved	`Muon.Isotropic.lem_app_iso_kernel_bounds_i` (+4)	
proposition SF.10	proved	`Muon.Isotropic.prop_app_iso_vol_representation`	
lemma SF.16	proved	`Muon.Isotropic.lem_app_iso_averaging_i` (+2)	
lemma SF.17	proved	`Muon.Isotropic.lem_app_iso_low_levels_i` (+1)	
theorem SF.9	proved	`Muon.Isotropic.thm_app_iso_vol_rate`	
theorem SF.10	proved	`Muon.Isotropic.thm_app_iso_vol_profile_i` (+1)	
theorem SF.11	proved cited	mod. `Muon.Isotropic.thm_app_iso_zero_ridge_i` (+2)	`GaussianOpNorm`
lemma SF.18	proved	`Muon.Isotropic.lem_app_iso_ridge_excess`	
theorem SF.12	proved	`Muon.Isotropic.thm_app_iso_vol_ridge_i` (+1)	
theorem SF.13	proved	`Muon.Isotropic.thm_app_iso_fixed_ridge_i` (+1)	
corollary SF.2	proved	`Muon.Isotropic.cor_app_iso_fixed_ridge_i` (+3)	
theorem SF.14	proved	`Muon.Isotropic.thm_app_iso_crossover`	
proposition SF.11	proved	`Muon.Isotropic.prop_app_iso_limit_order_i` (+2)	
proposition SF.12	proved	`Muon.Isotropic.prop_app_iso_beta_zero_i` (+3)	
proposition SF.13	proved	`Muon.Isotropic.prop_app_iso_one_momentum_i` (+2)	
remark SF.1	not formalized	—	
definition SG.1	proved	`Muon.PowerLaw.gammaOf` (+11)	
proposition SG.1	proved	`Muon.PowerLaw.prop_app_pl_branch_i` (+3)	

Statement	Status	Lean declaration	Certificates
lemma SG.1	proved	`Muon.PowerLaw.lem_app_pl_weights_i_F` (+10)	
lemma SG.2	proved	`Muon.PowerLaw.lem_app_pl_average_i` (+2)	
proposition SG.2	proved	`Muon.PowerLaw.prop_app_pl_drift_zero_i` (+4)	
proposition SG.3	proved mod. cited	`Muon.PowerLaw.prop_app_pl_response_i` (+3)	`GaussianOpNorm`
lemma SG.3	proved mod. cited	`Muon.PowerLaw.lem_app_pl_two_ridge_i` (+2)	`GaussianOpNorm`
proposition SG.4	proved mod. cited	`Muon.PowerLaw.prop_app_pl_vol_zero_i` (+4)	`GaussianOpNorm`
definition SG.2	proved mod. cited	`Muon.PowerLaw.cbarZero` (+3)	`GaussianOpNorm`
lemma SG.4	proved mod. cited	`Muon.PowerLaw.lem_app_pl_amplitude_i` (+1)	`GaussianOpNorm`
Assumption S4	proved	`Muon.PowerLaw.assum_app_pl_regime` (+1)	
definition SG.3	proved	`Muon.PowerLaw.profU` (+11)	
proposition SG.5	proved	`Muon.PowerLaw.prop_app_pl_profiles_i` (+1)	
theorem SG.1	proved mod. cited	`Muon.PowerLaw.triple_vol_gt_one` (+1)	`GaussianOpNorm`
theorem SG.2	proved	`Muon.PowerLaw.thm_app_pl_drift_i` (+1)	
theorem SG.3	proved mod. cited	`Muon.PowerLaw.thm_app_pl_volatility_i_part` (+4)	`GaussianOpNorm`
corollary SG.1	proved mod. cited	`Muon.PowerLaw.cor_app_pl_endpoints_i` (+3)	`GaussianOpNorm`
proposition SG.6	proved	`Muon.PowerLaw.kappaOf` (+4)	
corollary SG.2	proved mod. cited	`Muon.PowerLaw.cor_app_pl_weighted`	`GaussianOpNorm`
proposition SG.7	proved mod. cited	`Muon.PowerLaw.prop_app_pl_ridge_exists` (+9)	`GaussianOpNorm`
remark SG.1	not formalized	—	
proposition SG.8	proved mod. cited	`Muon.PowerLaw.aAlpha` (+7)	`GaussianOpNorm`
remark SG.2	not formalized	—	
remark SG.3	not formalized	—	
lemma SG.5	proved	`Muon.PowerLaw.lem_app_pl_continuum_i_lt` (+8)	
lemma SG.6	proved	`Muon.PowerLaw.lem_app_pl_finite_inverse_i_a` (+8)	
lemma SG.7	proved	`Muon.PowerLaw.lem_app_pl_elasticity_i` (+1)	
lemma SG.8	proved	`Muon.PowerLaw.lem_app_pl_drift_participation_i`	
lemma SG.9	proved	`Muon.PowerLaw.lem_app_pl_Psi_i_integrable` (+6)	
lemma SG.10	proved	`Muon.PowerLaw.lem_app_pl_tail`	
lemma SG.11	proved	`Muon.PowerLaw.lem_app_pl_head_i` (+1)	
lemma SG.12	proved	`Muon.PowerLaw.lem_app_pl_cutoff_pre` (+3)	
lemma SG.13	proved	`Muon.PowerLaw.lem_app_pl_super_i` (+3)	
lemma SG.14	proved	`Muon.PowerLaw.lem_app_pl_determinant_i` (+4)	
lemma SG.15	proved	`Muon.PowerLaw.Zg` (+6)	
lemma SG.16	proved	`Muon.PowerLaw.lem_app_pl_pieces`	
lemma SG.17	proved	`Muon.PowerLaw.lem_app_pl_upper`	
lemma SG.18	proved	`Muon.PowerLaw.lem_app_pl_vector` (+6)	
lemma SG.19	proved	`Muon.PowerLaw.lem_app_pl_strip` (+1)	
lemma SG.20	proved	`Muon.PowerLaw.lem_app_pl_meso` (+2)	
lemma SG.21	proved	`Muon.PowerLaw.lem_app_pl_matching`	

Statement	Status	Lean declaration	Certificates
lemma SG.22	proved	`Muon.PowerLaw.lem_app_pl_localization_i` (+3)	
lemma SG.23	proved	`Muon.PowerLaw.lem_app_pl_supervol_i_finite` (+8)	
lemma SH.1	proved	`Muon.Oversampled.lem_app_os_root_i` (+2)	
definition SH.1	proved	`Muon.Oversampled.profileRoot` (+11)	
lemma SH.2	proved	`Muon.Oversampled.lem_app_os_profile_i` (+4)	
lemma SH.3	proved	`Muon.Oversampled.lem_app_os_scalar_i` (+3)	
lemma SH.4	proved	`Muon.Oversampled.lem_app_os_reduction_a` (+3)	
lemma SH.5	proved	`Muon.Oversampled.lem_app_os_compare_i` (+1)	
theorem SH.1	proved	`Muon.Oversampled.limit_generic_i` (+11)	
corollary SH.1	proved	`Muon.Oversampled.OsUpdateHyp` (+3)	
remark SH.1	not formalized	—	
lemma SH.6	proved	`Muon.Oversampled.lem_app_os_integrals_i` (+7)	
lemma SH.7	proved	`Muon.Oversampled.Upsilon` (+9)	
lemma SH.8	proved	`Muon.Oversampled.lem_app_os_repr_i` (+2)	
lemma SH.9	proved	`Muon.Oversampled.lem_app_os_average_i` (+4)	
lemma SH.10	proved	`Muon.Oversampled.lem_app_os_defect`	
lemma SH.11	proved	`Muon.Oversampled.smoothCount` (+3)	
remark SH.2	not formalized	—	
theorem SH.2	proved	`Muon.Oversampled.cStar` (+4)	
proposition SH.1	proved	`Muon.Oversampled.prop_app_os_head_i` (+1)	
definition SH.2	proved	`Muon.Oversampled.def_app_os_packed`	
corollary SH.2	proved	`Muon.Oversampled.cor_app_os_packed_i` (+1)	
remark SH.3	proved	`Muon.Oversampled.rem_app_os_packed_i` (+1)	
lemma SH.12	proved	`Muon.Oversampled.lem_app_os_tailint_i` (+4)	
lemma SH.13	proved	`Muon.Oversampled.lem_app_os_log`	
proposition SH.2	proved	`Muon.Oversampled.prop_app_os_tail_i` (+6)	
corollary SH.3	proved	`Muon.Oversampled.nuTL` (+9)	
corollary SH.4	proved	`Muon.Oversampled.accelA` (+6)	
lemma SH.14	proved	`Muon.Oversampled.lem_app_os_scale_i` (+3)	
corollary SH.5	proved	`Muon.Oversampled.plMu` (+7)	
corollary SH.6	proved	`Muon.Oversampled.cor_app_os_identity_i` (+1)	
remark SH.4	not formalized	—	
remark SH.5	proved	`Muon.Oversampled.rem_app_os_normalization`	
definition SI.1	proved	`Muon.Phase.RowModel` (+8)	
proposition SI.1	proved	`Muon.Phase.prop_app_phase_row_identity`	
remark SI.1	proved	`Muon.Phase.rem_app_phase_adoption_i`	
lemma SI.1	proved	`Muon.Phase.lem_app_phase_kernel_i` (+3)	
lemma SI.2	proved	`Muon.Phase.lem_app_phase_tail`	
lemma SI.3	proved	`Muon.Phase.lem_app_phase_comparison`	
theorem SI.1	proved	`Muon.Phase.thm_app_phase_volterra_i` (+4)	
theorem SI.2	proved	`Muon.Phase.thm_app_phase_clock`	
theorem SI.3	proved	`Muon.Phase.thm_app_phase_hitting_i` (+1)	

Statement	Status	Lean declaration	Certificates
remark SI.2	not formalized	—	
proposition SI.2	proved	`Muon.Phase.prop_app_phase_common_i` (+1)	
theorem SI.4	proved	`Muon.Phase.thm_app_phase_matched`	
proposition SI.3	proved	`Muon.Phase.prop_app_phase_iso_ratio`	
definition SI.2	proved	`Muon.Phase.def_app_phase_scalar`	
lemma SI.4	proved	`Muon.Phase.lem_app_phase_ratio_integral`	
theorem SI.5	proved	`Muon.Phase.thm_app_phase_iso_extrema`	
corollary SI.1	proved	`Muon.Phase.cor_app_phase_iso_one` (+1)	
theorem SI.6	proved	`Muon.Phase.thm_app_phase_iso`	
lemma SI.5	proved	`Muon.Phase.lem_app_phase_variation_i` (+2)	
lemma SI.6	proved	`Muon.Phase.lem_app_phase_laplace_i` (+1)	
lemma SI.7	proved	`Muon.Phase.lem_app_phase_sums_i` (+1)	
lemma SI.8	proved	`Muon.Phase.lem_app_phase_harmonic_i` (+6)	
definition SI.3	proved	`Muon.Phase.def_app_phase_arrays` (+7)	
proposition SI.4	proved	`Muon.Phase.prop_app_phase_prefactors_m1` (+5)	
theorem SI.7	proved	`Muon.Phase.thm_app_phase_map` (+1)	
theorem SI.8	proved	`Muon.Phase.thm_app_phase_A`	
theorem SI.9	proved	`Muon.Phase.thm_app_phase_B` (+2)	
theorem SI.10	proved	`Muon.Phase.thm_app_phase_C` (+1)	
proposition SI.5	proved	`Muon.Phase.prop_app_phase_C_window_i` (+2)	
theorem SI.11	proved	`Muon.Phase.thm_app_phase_fixed`	
corollary SI.2	proved	`Muon.Phase.cor_app_phase_fixed`	
corollary SI.3	proved	`Muon.Phase.cor_app_phase_ratio_i` (+4)	
proposition SI.6	proved	`Muon.Phase.prop_app_phase_winners_i` (+14)	
proposition SI.7	proved	`Muon.Phase.prop_app_phase_cutoff`	
remark SI.3	not formalized	—	
remark SI.4	not formalized	—	
remark SI.5	proved	`Muon.Phase.rem_app_phase_momentum_i` (+2)	
proposition SI.8	proved	`Muon.Phase.prop_app_phase_forcing_r_i` (+4)	
theorem SI.12	proved	`Muon.Phase.thm_app_phase_accfirst_i` (+28)	
corollary SI.4	proved	`Muon.Phase.cor_app_phase_accfirst_winners_i` (+3)	
remark SI.6	not formalized	—	
theorem SI.13	proved	`Muon.Phase.thm_app_phase_frozen_i` (+5)	
theorem SI.14	proved	`Muon.Phase.thm_app_phase_unified_i` (+2)	
corollary SI.5	proved	`Muon.Phase.cor_app_phase_extended`	
corollary SI.6	proved	`Muon.Phase.cor_app_phase_unified_ratio_i` (+4)	
theorem SI.15	proved	`Muon.Phase.thm_app_phase_omega_i` (+3)	
corollary SI.7	proved	`Muon.Phase.cor_app_phase_omega_winners_a` (+7)	
lemma SI.9	proved	`Muon.Phase.lem_app_phase_square_i` (+2)	
lemma SI.10	proved	`Muon.Phase.lem_app_phase_clock_integral_i` (+5)	
lemma SI.11	proved	`Muon.Phase.lem_app_phase_mixture` (+1)	

Statement	Status	Lean declaration	Certificates
lemma SI.12	proved	`Muon.Phase.lem_app_phase_forcing_integrals_i` (+1)	
lemma SI.13	proved	`Muon.Phase.lem_app_phase_mass`	
lemma SI.14	proved	`Muon.Phase.lem_app_phase_dilation`	
lemma SI.15	proved	`Muon.Phase.lem_app_phase_svd_small`	
lemma SI.16	proved	`Muon.Phase.lem_app_phase_svd_main`	
lemma SI.17	proved	`Muon.Phase.lem_app_phase_log_i` (+2)	
lemma SI.18	proved	`Muon.Phase.lem_app_phase_log_global_i` (+2)	
lemma SI.19	proved	`Muon.Phase.lem_app_phase_shift`	
lemma SI.20	proved	`Muon.Phase.lem_app_phase_square_tail`	
lemma SI.21	proved	`Muon.Phase.lem_app_phase_restart`	
lemma SI.22	proved	`Muon.Phase.lem_app_phase_twophase_i` (+2)	
lemma SI.23	proved	`Muon.Phase.lem_app_phase_split`	
lemma SI.24	proved	`Muon.Phase.lem_app_phase_G_small`	
lemma SI.25	proved	`Muon.Phase.lem_app_phase_templates_i` (+4)	
definition SJ.1	proved	`Muon.SignSGD.def_app_sgd_update` (+3)	
lemma SJ.1	proved	`Muon.SignSGD.lem_app_sgd_arcsine_i` (+1)	
lemma SJ.2	proved	`Muon.SignSGD.lem_app_sgd_shift_i` (+3)	
remark SJ.1	not formalized	—	
proposition SJ.1	proved	`Muon.SignSGD.prop_app_sgd_batch_coefficient_i` (+3)	
proposition SJ.2	proved	`Muon.SignSGD.prop_app_sgd_iso_volatility_i` (+1)	
lemma SJ.3	proved	`Muon.SignSGD.lem_app_sgd_moments_i` (+6)	
lemma SJ.4	proved	`Muon.SignSGD.lem_app_sgd_decay`	
lemma SJ.5	proved	`Muon.SignSGD.lem_app_sgd_comparison` (+1)	
proposition SJ.3	proved	`Muon.SignSGD.prop_app_sgd_coordinate`	
definition SJ.2	proved	`Muon.SignSGD.def_app_sgd_iso_regime` (+3)	
theorem SJ.1	proved	`Muon.SignSGD.thm_app_sgd_iso_drift`	
proposition SJ.4	proved	`Muon.SignSGD.prop_app_sgd_iso_filtered` (+1)	
definition SJ.3	proved	`Muon.SignSGD.def_app_sgd_state_law` (+14)	
lemma SJ.6	proved	`Muon.SignSGD.lem_app_sgd_sphere_i` (+4)	
lemma SJ.7	proved	`Muon.SignSGD.lem_app_sgd_haar_sign_i` (+1)	
proposition SJ.5	proved	`Muon.SignSGD.prop_app_sgd_one_sample`	
lemma SJ.8	proved	`Muon.SignSGD.lem_app_sgd_whitening_i` (+2)	
lemma SJ.9	proved	`Muon.SignSGD.lem_app_sgd_deletion_i` (+2)	
proposition SJ.6	proved	`Muon.SignSGD.prop_app_sgd_coordinate_sum`	
theorem SJ.2	proved	`Muon.SignSGD.thm_app_sgd_half_drift_i` (+1)	
corollary SJ.1	proved	`Muon.SignSGD.cor_app_sgd_half_pairings`	
lemma SJ.10	proved	`Muon.SignSGD.lem_app_sgd_column_i` (+4)	
proposition SJ.7	proved	`Muon.SignSGD.prop_app_sgd_quadratic`	
lemma SJ.11	proved	`Muon.SignSGD.lem_app_sgd_reduction_i` (+4)	
theorem SJ.3	proved	`Muon.SignSGD.thm_app_sgd_half_volatility`	
corollary SJ.2	proved	`Muon.SignSGD.cor_app_sgd_half_risk`	
lemma SJ.12	proved	`Muon.SignSGD.lem_app_sgd_partition`	
lemma SJ.13	proved	`Muon.SignSGD.lem_app_sgd_column_norm`	

Statement	Status	Lean declaration	Certificates
theorem SJ.4	proved	`Muon.SignSGD.thm_app_sgd_modes_i` (+3)	
proposition SJ.8	proved	`Muon.SignSGD.prop_signsgd_recursion_anisotropic`	

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